Bałaban's small-field renormalization group for four-dimensional lattice Yang–Mills theory

Green functions, the background field, the small-field flow and the one-loop coefficient

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Abstract

We study Bałaban's renormalization group for pure Yang–Mills theory on a four-dimensional lattice torus, with his symmetric block averaging, in the small-field region. We first prove coercivity, exponential decay and pointwise regularity for the linear operators of the method (Green functions with an averaging term, minimizer maps, constrained Green functions) at the trivial background and at locally regular backgrounds, uniformly in the number of averaging steps and in the size of the torus. We then prove Bałaban's variational theorem on the torus: for real coarse configurations of small curvature, the background field is unique up to the residual gauge group among configurations of small curvature and current, and it is locally regular, uniformly in the number of steps. Next we construct one small-field renormalization group step, with Bałaban's exponential gauge fixing, in polymer form. For a simple gauge group, iterating the step in Bałaban's generation form gives the flow of the small-field model, a modified model in which every step keeps only small fields. On a fixed torus, as long as the couplings stay below a fixed bound, the density of every scale is controlled uniformly in the scale, the couplings and the torus, and the couplings run by $g_{k+1}^{-2} = g_k^{-2} - \beta_{k+1}$. The one-loop coefficient summed over k steps is $k\beta_\infty + O(1)$, with $\beta_\infty = \frac{11 C_A}{24\pi^2 T_\rho} \log L$. Every small final coupling g_K is reached from a bare coupling, and then $g_k^{-2} = g_K^{-2} + (K-k)\beta_\infty$ with an error bounded by a power of $\log(K-k+2)$. Large fields and the continuum limit are not treated.

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1 Introduction

Main result. The note studies pure Yang–Mills theory on a four-dimensional lattice torus, with the Wilson action S, by Bałaban's renormalization group with his symmetric block averaging [18, 19]. It treats the small-field part of this renormalization group. Its results concern the *small-field model*, a modification of the lattice theory in which every step keeps only small fields: the curvature of the coarse configuration is at most a threshold h_k, and the fluctuation field and the white noise of the step are bounded by cutoffs. The densities of this model are

$$\rho_0 = \exp\big(-g_0^{-2}S - E_0\big), \qquad \rho_{k+1} = \mathbf{1}_{\mathfrak{U}_{k+1}(h_{k+1})}\, T^{\mathrm{gf}}\big(\chi^{\mathrm{w}}\chi_\xi\, \rho_k\big), \quad k = 0, 1, \dots, K-1,$$

with the bare coupling g_0, a constant E_0, the renormalization group map T^{gf} with Bałaban's exponential gauge fixing, and the cutoffs χ^{w} on the fluctuation field and χ_ξ on the white noise (Definition 48.1, Remark 52.6(c)). What a step discards is nonnegative, so the partition function of the small-field model is at most that of the lattice theory with the same E_0 (Remark 52.6(c)). For a compact, connected and simple gauge group the note proves two results about this model.

- *The flow* (Theorem N). The couplings run by $g_{k+1}^{-2} = g_k^{-2} - \beta_{k+1}$, where β_{k+1} is read off from the terms that the step creates. As long as the couplings stay in $(0, \gamma]$, with γ small as in Remark 53.5, the density of every scale has Bałaban's generation form

(Definition 53.6), each generation is bounded at the scale where it is created, and the β_k are bounded. For the counterterm E_0 that makes the vacuum energy of the last scale K vanish, the partition function is an explicit product of normalizations times a number between $e^{-c|\mathbb{T}^{(K)}|}$ and $e^{c|\mathbb{T}^{(K)}|}$, where $|\mathbb{T}^{(K)}|$ is the number of sites of the last lattice. The constants do not depend on the scale, the number of steps, the torus or the couplings, except that c depends on the last coupling g_K (Theorem N(vi)).

- *The ultraviolet behaviour of the couplings* (Theorem M(iv)). Let $p_0(g) = (\log g^{-1})^q$ be the white-noise cutoff. For every small final coupling g and every number K of steps there is a bare coupling g_0 for which the flow stays in $(0, \gamma]$ and ends at $g_K = g$, and

$$\frac{1}{g_k^2} = \frac{1}{g^2} + (K-k)\beta_\infty + O\Big(\big(\log(K-k+2)\big)^{6q+1}\Big) \quad (k \le K), \qquad \beta_\infty = \frac{11 C_A}{24\pi^2 T_\rho} \log L,$$

with a constant independent of K and of the torus. Here C_A is the Casimir of the adjoint representation, and T_ρ is the normalization $\mathrm{tr}_\rho(T^a T^b) = T_\rho \delta^{ab}$ of the trace of the representation ρ in the action (Theorem M). Here β_∞ is the one-loop coefficient per step (Theorem M(iii)). The coupling g of the action S is related to the standard one by $g^2_{\mathrm{std}} = g^2/T_\rho$ (Section 57, after Theorem 57.1; Section G1).

The torus is fixed at the last scale, with a side of at least $n_{\min}$ there (Section 52.1), and all bounds are uniform in the number of scales (Remark 53.14(d)). With $K = N$ steps on a torus of physical side ℓ, the finest spacing is L^{-N} (Section 52.1); so the bounds are uniform in the lattice spacing at fixed physical volume. Every hypothesis of Theorem L holds at every step of the flow (Theorem N(v)), and the hypothesis $g_k \le \gamma$ of Theorem N holds for the bare couplings of Theorem M(iv).

What is not proved. The large-field stage is not treated, so nothing is proved about the partition function or the correlation functions of the lattice theory itself (Remark 52.6(d)). No limit of the measure or of correlation functions as the lattice spacing tends to 0 is considered: Theorem M(iv) concerns the couplings of the small-field flow (Section 1.5).

How to read. The guide to the proof follows this introduction. Section G0 gives four reading paths, Section G1 states the result in more detail, Section G2 the scales, and Section G3 the strategy of the four-dimensional flow. Sections G4–G10 follow the Parts, and Sections G11–G14 give the order of the parameters, the notation (with Appendix B), the comparison with Bałaban and Dimock, and the places where the work is hardest. The companion volume *Details* expands the proofs of Parts VII–XI step by step (Section G0).

1.1 Bałaban's renormalization group

Bałaban's construction of lattice Yang–Mills theory [3, 4, 5, 6, 7, 8, 9, 18, 19] is a renormalization group (RG) analysis. A gauge field on a fine lattice of spacing $\eta = L^{-k}$ is averaged over blocks of side 1, and this averaging is performed in k steps, each enlarging the blocks by a factor L. After k steps one has to integrate over all fine configurations whose block averages equal a given coarse configuration. Following the usual Gaussian strategy, one expands the action around its minimizer subject to the averaging constraint. This minimizer is called the *background field* [8]. The fluctuations around it are then Gaussian to leading order, with an explicit covariance.

Every estimate in this programme rests on properties of a small family of *linear* operators:

- Green functions of the form $(\text{Laplace-type operator} + a\, Q^* Q)^{-1}$, where Q is the k-step averaging operator;
- the *minimizer map* H, which sends a coarse configuration B to the fine configuration of least action with average B;

- the *constrained Green function*, i.e. the covariance of the fluctuation field, which has vanishing averages;
- inverses of averaged Green functions, such as $(QGQ^*)^{-1}$, which live on the coarse lattice.

The properties needed are *exponential decay* on the unit scale and *local regularity*: bounds on values, first derivatives and Hölder quotients of first derivatives. They must be uniform in the number k of RG steps, which means uniform in the ratio η^{-1} between the unit scale and the lattice scale, and uniform in the size of the volume.

For the full non-abelian problem these operators depend on the background field U. Bałaban obtained them by perturbation from the case $U = 1$ [3], through random-walk expansions [4, 7]. The case $U = 1$ is the foundation of the whole analysis. It is also the case in which all the scales are already present, together with the gauge structure that distinguishes gauge fields from scalar fields.

In four dimensions the Wilson action carries no power of the lattice spacing, so a change of units does not change the coupling: the coupling is marginal, and the coupling of the next scale is produced by the fluctuation integral alone (Section 31). The terms that a step creates are of order one, and the iteration does not contract them. Bałaban keeps them as generations, each bounded at the scale where it is created, and cancels their marginal parts at the later scales by counterterms proportional to the action [18, 19] (Remark 49.1). Every step has a small-field part, in which the fields are close to the background field, and a large-field part. The note treats the small-field part.

1.2 What this note does

The note carries out the small-field part of Bałaban's analysis of four-dimensional lattice Yang–Mills theory on a single torus, without the nested domains Ω_j of [8]: from the linear operators to the flow over all scales and the one-loop coefficient. It has eleven Parts, in four layers.

- *The linear operators* (Parts I–III, Theorems A–G): Green functions with an averaging term, minimizer maps and constrained Green functions, with exponential decay and pointwise bounds uniform in the number of averaging steps and in the torus; at $U = 1$, at backgrounds close to 1 in a global gauge, and at locally regular backgrounds.
- *The background field* (Parts IV–VI, Theorems H–J): Bałaban's symmetric nonlinear averaging, the axial-to-Landau chart, and the variational problem. For a real coarse configuration of small curvature the minimizers of the action with prescribed averages form exactly one critical orbit among the configurations of small curvature and current, and they are locally regular, uniformly in the number of steps.
- *One step and its localization* (Parts VII–IX, Theorem K): one step of the renormalization group as an exact Gaussian integral around the background field, with a coercive fluctuation operator and an exponentially decaying covariance; the localization of its dependence on the background, of radius type (Part VIII) and of volume type, by random walks with weakening parameters (Part IX).
- *The flow and the one-loop coefficient* (Parts X–XI, Theorems L, N and M): the step in polymer form, its iteration over all scales in Bałaban's generation form, the one-loop coefficient of the running coupling, and the ultraviolet behaviour of the couplings.

Figure G1 shows how these results feed Theorems N and M(iv), and Table 2 lists the main results of each Part.

Scope. The dimension is $d = 4$ throughout, and the gauge group G is compact and connected; from Part VII on it is semisimple, and for the flow of Part X (Sections 50–53) and in Part XI simple. Semisimplicity gives the quadratic bound for normalized activities (Lemma 38.5). For simple G a quadratic form in the curvature that is invariant under conjugation by G and under B_4 is a multiple of a single isotropic form (Lemma 50.10(a)); so the period density of a generation is a multiple of that of S, and the running coupling is one number (Proposition 50.13). The flow results of Parts VII–X and Theorem M(iv) concern the small-field model on a fixed torus; Part XI also uses tori of growing side, through the torus independence of Proposition 41.14 (Part IX), and infinite-volume limits of the one-loop terms (Theorem M(ii), (iii)). The large-field stage is not treated (Section 1.5).

The trivial background (Part I). Part I treats the linear operators at the trivial background $U = 1$, on a single torus with k-fold averaging everywhere. It is self-contained, and it is meant to be readable by a PhD student who knows linear algebra, Fourier series and the basics of elliptic equations. All lattice computations are written out.

Let us make precise what "$U = 1$" means. The gauge field is $U = e^{i\eta A}$ with A taking values in the Lie algebra $\mathfrak{g}$ of the gauge group. Expanding the Wilson action to second order around $U = 1$ gives $\frac{1}{2}\|dA\|^2$, and infinitesimal gauge transformations act as $A \mapsto A + d\lambda$. The non-abelian structure enters only at higher orders. At $U = 1$ every operator acts componentwise in a basis of $\mathfrak{g}$. Hence it suffices to treat real-valued fields, and this is what we do. This is *not* abelian gauge theory. It is the linearization of non-abelian theory at its trivial background. The operators obtained are those which Bałaban perturbs around in [7, 8].

The operators of Part I. The precise definitions are given in Sections 2–9. Here is an overview. We use a fine torus $\mathbb{T}_\eta$ of spacing η and a unit torus $\mathbb{T} \subset \mathbb{T}_\eta$, the scalar block average Q_0, and the gauge-field average $\mathbf{Q}$ of [3]. The averaged gauge field $(\mathbf{Q}A)_\mu$ is an average of A_μ over the block and along lines of unit length in direction μ. It satisfies $\mathbf{Q}\, d = d_1 Q_0$, where d_1 is the unit-lattice gradient (gauge covariance). Since the blocks are centred, Q_0 and $\mathbf{Q}$ commute with the translations of $\mathbb{T}$ and with the hyperoctahedral group B_4 acting about the block centres (Lemma 2.6(f)). We study:

(i) the scalar Green functions $G_\sigma = (-\Delta + aQ_\sigma^* Q_\sigma)^{-1}$. Here Q_σ is either the block average Q_0 or one of the "tube" averages $Q_1, \dots, Q_d$ that make up $\mathbf{Q}$;

(ii) the operators derived from them: $(Q_\sigma G_\sigma Q_\sigma^*)^{-1}$, the minimizer map H_σ, the constrained Green function C_σ, and the effective action. We also study the projection R which defines Bałaban's Landau-type gauge;

(iii) the gauge-field operator $\mathbf{\Delta}_a = d^*d + dRd^* + a\mathbf{Q}^*\mathbf{Q}$ and its inverse $\mathbf{G}$;

(iv) the gauge-field minimizer map $\mathbf{H} = \mathbf{G}\mathbf{Q}^*(\mathbf{Q}\mathbf{G}\mathbf{Q}^*)^{-1}$, the effective action $\mathbf{D}$, and the constrained Green function $\mathbf{C}$, which is the operator appearing in the Euler–Lagrange equation of the variational problem [8, (110)–(111)].

1.3 Main results

The theorems of Parts I and II are stated with the conventions of Part I: $d = 4$, $L \geq 3$ is odd, $k \geq 1$, $\eta = L^{-k}$, and the unit torus has side $n \geq 8$. Since L is odd, $N = L^k$ is odd. The parameter $a > 0$ in front of the averaging term is fixed. Bałaban takes $a = 1$. Every constant in Part I depends only on a and a Hölder exponent $\alpha \in (0, 1)$; constants that depend only on the dimension are absolute (Section 2). In Part II it depends in addition on $\dim \mathfrak{g}$ and on the decay rate γ_W of the potential term W of Theorem F. In particular, the constants of Parts I and II do not depend on L, k or n. The constants of Theorems 18.2,

19.6 and 19.7 (on the torus) and 19.9 of Part III do not depend on L, k, n or $M \geq M_0$ either (Lemma 31.3(a)); the random-walk expansion and Theorem 19.10(b) have constants that depend on M (Remark 31.4(b)). From Part IV on, constants may depend on L (Section 21), and the flow needs L large (Remark 49.8(2)); the order in which all parameters are chosen is in Section G11.

For a fine-lattice field u (scalar or vector-valued) and a set X we write

$$|u|_X = \sup_{x\in X} |u(x)|, \qquad |\partial u|_X = \max_{\mu} \sup_{x\in X} |\partial_\mu u(x)|, \qquad [v]_{\alpha,X} = \sup_{x\neq x'\in X} \frac{|v(x)-v(x')|}{|x-x'|^\alpha}.$$

Here ∂_μ is the forward lattice derivative, and the maximum also runs over components. $B(y)$ is the unit block centred at $y \in \mathbb{T}$, the cube of N^d fine points with centre y, and $|y-y'|$ is the distance on the torus. For a gauge field B on $\mathbb{T}$, $|B|_{y'}$ is the maximum of $|B|$ over the unit bonds starting at y'.

Theorem A (Coercivity; Corollary 3.4, Theorem 9.6)**.** *There is $c > 0$ such that for all scalar fields f and all $\sigma \in \{0,\dots,d\}$,*

$$\|\partial f\|^2 + a\|Q_\sigma f\|^2 \;\geq\; c\,(\|f\|^2 + \|\partial f\|^2),$$

and for all gauge fields A,

$$\langle A, \mathbf{\Delta}_a A\rangle = \|dA\|^2 + \|Rd^*A\|^2 + a\|\mathbf{Q}A\|^2 \;\geq\; c\,(\|A\|^2 + \|\partial A\|^2).$$

Theorem B (Scalar Green functions; Theorem 7.1)**.** *There are $C, \gamma > 0$ such that for all $y, y' \in \mathbb{T}$ and all f supported in $B(y')$:*

$$|G_\sigma f|_{B(y)} + |\partial G_\sigma f|_{B(y)} + [\partial G_\sigma f]_{\alpha,B(y)} \leq Ce^{-\gamma|y-y'|}\,|f|_{B(y')},$$
$$|G_\sigma \partial_\mu^* f|_{B(y)} + [G_\sigma \partial_\mu^* f]_{\alpha,B(y)} \leq Ce^{-\gamma|y-y'|}\,|f|_{B(y')}.$$

Corresponding L^2 bounds hold for G_σ, ∂G_σ, $G_\sigma\partial^*$ and $\partial G_\sigma \partial^*$ between blocks.

Theorem C (Averaged scalar operators; Section 8)**.** *The operator $Q_\sigma G_\sigma Q_\sigma^*$ on the unit torus satisfies $c \leq Q_\sigma G_\sigma Q_\sigma^* \leq C$. Its inverse has a kernel bounded by $Ce^{-\gamma|y-y'|}$. The minimizer map H_σ satisfies*

$$|H_\sigma g|_{B(y)} + |\partial H_\sigma g|_{B(y)} + [\partial H_\sigma g]_{\alpha,B(y)} \leq C\sum_{y'} e^{-\gamma|y-y'|}|g(y')|,$$

and the derivative bounds hold with $|g(y')|$ replaced by the unit-lattice gradient $|d_1 g(y')|$. The constrained Green function C_σ satisfies the bounds of Theorem B. The effective action $\mathcal{D}_\sigma$, defined by $\langle g, \mathcal{D}_\sigma g\rangle = \min\{\|\partial f\|^2 : Q_\sigma f = g\}$, satisfies $c\|d_1 g\|^2 \leq \langle g, \mathcal{D}_\sigma g\rangle \leq C\|d_1 g\|^2$ and has an exponentially decaying kernel. The projection R is I minus an operator with exponential decay.

Theorem D (Gauge-field Green function; Theorem 10.1)**.** *The operator $\mathbf{G} = \mathbf{\Delta}_a^{-1}$ satisfies, for J supported in $B(y')$,*

$$|\mathbf{G}J|_{B(y)} + |\partial \mathbf{G}J|_{B(y)} + [\partial \mathbf{G}J]_{\alpha,B(y)} \leq Ce^{-\gamma|y-y'|}\,|J|_{B(y')},$$

together with L^2 block bounds for $\mathbf{G}$, $\partial\mathbf{G}$, $\mathbf{G}\partial^*$ and $\partial\mathbf{G}\partial^*$.

Theorem E (Gauge-field minimizer and fluctuation operators; Section 11)**.** *One has $c \leq \mathbf{Q}\mathbf{G}\mathbf{Q}^* \leq C$, and $(\mathbf{Q}\mathbf{G}\mathbf{Q}^*)^{-1}$ has an exponentially decaying kernel. The minimizer map $\mathbf{H}$ sends B to the minimizer of $\|dA\|^2$ subject to $\mathbf{Q}A = B$ and $Rd^*A = 0$, and it satisfies*

$$|\mathbf{H}B|_{B(y)} + |\partial \mathbf{H}B|_{B(y)} + [\partial \mathbf{H}B]_{\alpha,B(y)} \leq C\sum_{y'} e^{-\gamma|y-y'|}|B|_{y'}.$$

The gauge-field effective action satisfies $c\|d_1 B\|^2 \le \langle B, \mathbf{D}B\rangle = \|d\mathbf{H}B\|^2 \le C\|d_1 B\|^2$. *The constrained Green function* $\mathbf{C} = \mathbf{G} - \mathbf{G}dRd^*\mathbf{G} - \mathbf{HQG}$ *is the Green function of* $\|dA\|^2$ *on* $\{\mathbf{Q}A = 0,\ Rd^*A = 0\}$, *and, for* J *supported in* $B(y')$,

$$|\mathbf{C}J|_{B(y)} + |\partial\mathbf{C}J|_{B(y)} + [\mathbf{C}J]_{\alpha,B(y)} \le Ce^{-\gamma|y-y'|}\,|J|_{B(y')}.$$

These are, at $U = 1$ and on a single torus, the bounds that Bałaban states in [3, Sect. F] and [7, Theorems 3.12, 3.13] and uses in [8].

Backgrounds (Part II). In Part II the fields take values in the complexified Lie algebra $\mathfrak{g}^c$. The background $U = e^{i\eta A}$ satisfies $|A|, |\partial A|, [\partial A]_\alpha \le \varepsilon$ in a global gauge (hypothesis (B)). Hypothesis (B) is restrictive. It forces all Wilson loops to obey a perimeter bound, so the background must be close to flat on scales larger than $O(1)$ (Section 12). The operators now involve parallel transport. Part II is stated for a model averaging pair $(Q'(U), \mathbf{Q}(U))$, with transports along a single coordinate-ordered contour from the block centre and only the main term of Bałaban's linearized averaging. It transforms exactly under gauge transformations, but it intertwines the covariant derivatives only up to a defect: $\mathbf{Q}(U)D_U = D_V Q'(U) + E(U)$ with $E(U) = O(\varepsilon)$. For the linearization of Bałaban's symmetric averaging $E = 0$ (Part IV), and the results apply to it (Remark 19.12): the real statements under properties (i)–(iv) of that remark, the analytic ones with property (v) in addition.

Theorem F (Backgrounds; Sections 14–16)**.** *Let* U *satisfy (B) with parameter* ε, *and let* $W \in \mathcal{W}(\varepsilon, \gamma_W)$ *in this gauge (Section 12). For* ε *small enough, with constants independent of* L, k, n *and* U*:*

(i) $G'(U) = (D^T D + aQ'^T Q')^{-1}$ *satisfies the bounds of Theorem B. The inverse* $\Gamma'(U)$ *of* $Q'G'Q'^T$, *the minimizer map* $H'(U)$, *the constrained Green function* $C'(U)$ *and the projection* $R(U)$ *satisfy the decay and pointwise bounds of Theorem C (Theorem 14.1, Proposition 14.2). The bounds in terms of* $|d_1 g|$ *and the effective-action bounds of Theorem C are stated only at* $U = 1$.

(ii) The gauge-field Green function $\mathbf{G}(U)$ *satisfies the bounds of Theorem D (Theorem 15.4).*

(iii) The inverse of $\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T$, *the minimizer map* $\mathbf{H}(U)$ *and the constrained Green function* $\mathbf{C}(U)$ *satisfy the decay and pointwise bounds of Theorem E (Theorem 16.1). For every* J, $\mathbf{C}(U)J$ *is the unique* $A \in \mathcal{X}(U)$ *with* $q_U(A', A) = \mathrm{b}(A', J)$ *for all* $A' \in \mathcal{X}(U)$. *For real* U, *real symmetric* W *and real data* B, $\mathbf{H}(U)B$ *is the unique minimizer of* $q_U(A, A)$ *among real* A *with* $\mathbf{Q}(U)A = B$ *and* $R(U)D_U^T A = 0$. *The effective-action bounds of Theorem E are stated only at* $U = 1$.

Here $\mathrm{b}(A, A') = \eta^d \sum_{x,\mu} \mathrm{tr}\,(A_\mu(x) A'_\mu(x))$ *is the bilinear form,* $q_U(A, A') = \mathrm{b}(D_U A, D_U A') + \mathrm{b}(A, WA')$ *is the quadratic form, with the small potential term* W, $\mathcal{X}(U) = \{A : \mathbf{Q}(U)A = 0,\ R(U)D_U^T A = 0\}$, *and* $\mathbf{Q}_\pi = \mathbf{Q}(U)\Pi(U)$. *For real* U, *real symmetric* W *and real* A, $q_U(A, A) = \|D_U A\|^2 + \langle A, WA\rangle$. *The projection* $\Pi(U)$ *onto Bałaban's gauge slice* $\{R(U)D_U^T A = 0\}$ *along the residual gauge directions is Bałaban's gauge-invariant extension [7, (3.118)–(3.119)]. All these operators depend analytically on* A *and* W *and transform covariantly under* G*-valued gauge transformations, with* W *transformed accordingly. The identities* $\mathbf{G}DR = DG'R$, $RD^T\mathbf{G} = RG'D^T$ *and* $\mathbf{Q}_\pi \mathbf{G} DR = 0$ *hold exactly.*

The operators are covariant under G-valued gauge transformations u, with W replaced by $R(u)WR(u)^{-1}$; the latter need not belong to $\mathcal{W}(\varepsilon, \gamma_W)$ in the new gauge. Hence Theorem F transfers to every background gauge equivalent to one satisfying (B), as follows (Proposition 16.3): the bounds on values and the L^2 block bounds hold with ∂ replaced by the covariant derivative, the identities hold, and the Hölder bounds hold in the gauge in which (B) holds.

The comparisons with the operators at $U = 1$ proved in Sections 14–16 hold only in that gauge; they fail for a pure gauge $U = 1^u$.

The proofs are perturbative: every operator is written as its $U = 1$ counterpart plus a perturbation of a specific structure (Section 13), and three tools of Part I, the Combes–Thomas method, inversion with decay and local regularity, are shown to be stable under such perturbations. For the pointwise bounds a further regularity estimate is needed, a lattice Schauder estimate for divergence-form equations (Theorem 6.3(d)).

Local regularity (Part III). In Part III the global hypothesis (B) is replaced by a local one. A real background is (M, ε)-*regular* if on every cube of side of order M there is a gauge in which (B) holds; the gauges on different cubes are unrelated. This is the form of Bałaban's regularity conditions on the background.

Theorem G (Regular backgrounds; Sections 18–20 and 41)**.** *There is M_0 such that for $M \geq M_0$ and (M, ε)-regular backgrounds with ε small (independent of M) the following hold.*

(i) The statements of Theorem F remain true, with local potentials $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$ (Section 19). The comparisons with the operators at $U = 1$ of Sections 14–16 need a global gauge and fail for a pure gauge $U = 1^u$; they are stated only under (B), in those sections. The scalar operators need only $(1, \varepsilon)$-regularity. Sup norms of derivatives are taken with the covariant derivative, and Hölder quotients in local gauges.

(ii) The operators whose inverses are $G'(U)$ and $\mathbf{G}(U)$ have Dirichlet versions on arbitrary unions X of blocks. These depend only on the background on the bonds of $\hat{X}$, the set of points at ∞-distance at most 1 from X, and they are coercive (the gauge-field version with a mass term near the boundary and Dirichlet conditions for all components of the gauge field). Their inverses, the inverse of the averaged scalar Green function on X, and $\mathbf{1}_X - R_X$, with R_X the Dirichlet version of the projection R, have exponential decay uniformly in the domain.

(iii) Locality in the background: *if two regular backgrounds agree on a region, the propagators, the minimizer map and the fluctuation operator agree deep inside it up to errors that are exponentially small in the distance to the complement.*

(iv) Bałaban's random-walk expansion converges for M large: for the scalar Green function $G'(U)$ as a sum of products of local propagators (Theorem 20.2(a)); for the gauge-field Green function $\mathbf{G}(U)$ in the factorized form of Section 41, with the domains Λ_j^v of Section 40 for a fixed $\rho \geq 15$, so that (41.1) holds with $w = 10M$, for M large depending on ρ, on tori of side $n \geq C(\rho + C_v)M$, with C_v the margin of level (L3) there (Theorem 20.2(b)). Under the same hypotheses, $\Gamma'(U)$, $H'(U)$ and $R(U)$ are expanded at levels (L1) and (L2) there. Section 41 is written for $E = 0$; the additional terms for $E \neq 0$ are treated in the proof of Theorem 20.2(b).

(v) All results extend analytically to complex perturbations $Ue^{\mathrm{i}\eta\delta A}$ of a regular background by a field δA that is small in the local gauges, for potentials W that depend analytically on the background, with coercivity replaced by Re*-coercivity.*

The key new step is a local proof of coercivity (Section 19). An IMS localization at scale M reduces it to cubes. On each cube one works in the local gauge, perturbs around $U = 1$, and uses the coercivity of Part I. Fourier analysis is used only at $U = 1$.

The nonlinear problem (Parts IV and V). Bałaban's averaging of a gauge field is nonlinear and is performed one scale at a time. We use his symmetric four-dimensional averaging [18, (0.3)–(0.12)] (Section 21). Its blocks are centred at the coarse points. The

transport from a block centre y to a point x of the block is the group average (Lemma 21.5) of the transports along the coordinate-ordered contours from y to x, one for each order of the directions (Definition 21.1), and the coarse bond variables are built from these averaged contour variables. The k-fold composition $U \mapsto U^{(k)}$ is covariant with respect to the values of gauge transformations at the points of $\mathbb{T}$, and under the group $\mathfrak{S}_k$ generated by the translations of $\mathbb{T}$ and by B_4 about its points (Lemma 21.10). So the residual gauge group of the variational problem is $\mathfrak{G}_0 = \{u : u|_{\mathbb{T}} = 1\}$. Part IV constructs the bridge to the linear theory. Following [5, 6] and [18, pp. 253–254], we introduce the averaged axial gauge, defined through the averaged contour variables, and the gauge-adapted averaging $\mathcal{Q}$. We show (Proposition 22.10, Corollary 22.11) that the linearization of $\mathcal{Q}$ at a regular reference U_0 is an averaging pair $(Q'_k(U_0), \mathbf{Q}_k(U_0))$ to which Parts II and III apply, with *exact* covariance ($E = 0$), and that it is covariant under $\mathfrak{S}_k$. We then prove Bałaban's axial-to-Landau theorem [6, Theorem 2] on a single torus.

Theorem H (Axial-to-Landau chart; Theorem 24.2)**.** *Let U_0 be (M, ε)-regular with current at most ε, where ε is small. Every real configuration U with small curvature and current, and with k-fold averages close to those of U_0, has exactly one small representative $e^{\mathrm{i}\eta A}$ relative to U_0 that lies in Bałaban's Landau gauge $R(U_0)D^T_{U_0}A = 0$ and is related to the averaged axial representative of U by the correspondence transformation. It obeys $|A| + |\nabla A| + |\Delta_{U_0}A| + [\nabla A]_\alpha \le C(\varepsilon + \text{curvature} + \text{current} + \text{difference of averages})$, uniformly in k, n and η. The curvature and the current in this bound are those of U.*

Theorem I (The variational problem relative to a regular reference; Theorem 25.1)**.** *Let U_0 be (M, ε)-regular with current at most ε, and let V be a real configuration on $\mathbb{T}$ with $V(U_0^{(k)})^{-1} = e^{\mathrm{i}B}$, $|B| \le \beta$. If $\varepsilon + \beta$ is small, then the Wilson action restricted to $\{U : U^{(k)} = V\}$ has exactly one critical $\mathfrak{G}_0$-orbit among configurations of small curvature and current. The orbit consists of the minimizers of the action among these configurations. Its Landau representative relative to U_0 satisfies*

$$|A_*| + |\nabla A_*| + |\Delta_{U_0}A_*| + [\nabla A_*]_\alpha \le C(\varepsilon + \beta), \qquad A_* = \mathbf{H}B - \mathbf{C}J_{U_0} + O((\varepsilon + \beta)^2).$$

The minimizer is again regular, it depends analytically on B, and the derivative with respect to B decays exponentially. The critical orbit, the Landau representative and the multiplier are covariant under $\mathfrak{S}_k$.

This is the content of Bałaban's variational theorem [8, Theorem 1] for one step k of his induction, given the reference configuration U_0, on a single torus. Along the way we reprove Bałaban's results on the averaging [5] and on gauge fixing [6] in the form needed.

The induction (Part VI). Bałaban proves his theorem by induction over k. The reference at step k is the solution of the $(k-1)$-step problem for the trivial extension of V; it has exactly the prescribed averages. Theorem I then gives the solution at step k. Its regularity bound is worse by a power of L, because units change by a factor L. Bałaban restores the bound by a local analysis on nested domains [8, Sect. F]. We stay on the torus and use the uniformity of Theorem I in the reference. Near a given point we replace the reference by a configuration that is trivial, in a local gauge, near the point and equals the solution far away. Relative to it, the solution near the point is determined by the coarse field in the gauge of the coordinate-ordered contours from the point, which is controlled by the curvature of V alone; this gauge is a device of the proof (Section 28). The errors are exponentially small in the splicing radius, or quadratic.

Theorem J (Bałaban's variational theorem; Theorem 30.3, Corollary 30.4)**.** *There are a_1, B_3, n_0 such that the following holds on tori of side $n \ge n_0$, for every k and every real*

configuration V on $\mathbb{T}$ whose curvature is at most $\varepsilon_1 \le a_1$. The Wilson action on $\{U : U^{(k)} = V\}$ has exactly one critical $\mathfrak{G}_0$-orbit among configurations of small curvature and current. It consists of the minimizers of the action among these configurations, which have curvature and current at most $B_3\varepsilon_1$ and are $(M, B_3\varepsilon_1)$-regular. For every $r \in \mathfrak{S}_k$ the minimal orbit of rV is the image under r of the minimal orbit of V. The constants do not depend on k, n or η. Moreover, every real configuration whose curvature and current are at most θ, with θ small, is $(M, C\theta)$-regular.

The last statement is the torus version of Bałaban's derivation of local regularity from gauge-invariant conditions [6, Sect. F]. So the hypothesis of Part III holds for all configurations of Bałaban's small-field class.

One renormalization group step (Part VII). The renormalization group transformation T integrates over the configurations on $\mathbb{T} = \mathbb{T}^{(k)}$ with prescribed averages on $\mathbb{T}' = \mathbb{T}^{(k+1)}$ (Section 31). In four dimensions the Wilson action carries no power of the lattice spacing, so a change of units does not change the coupling. The coupling of the next step is produced by the fluctuation integral, through the coefficient β of the action in the effective density [18, (1.20)–(1.22)]. The integrand is invariant under the gauge transformations $\mathcal{G}'$ of $\mathbb{T}$ that are trivial on $\mathbb{T}'$. Following Bałaban [18, (0.14)–(0.19)], we fix this invariance by an exponential gauge-fixing function gf_W: a product over the blocks of weights of the form $e^{-|u-1|^2/2g^2}$ in the averaged contour variables relative to the base configuration W of the step (Definition 32.2). By a Faddeev–Popov identity without determinant (Lemma 32.3(a)), the gauge-fixed map T^{gf} agrees with T on $\mathcal{G}'$-invariant integrands supported on regular configurations. The small-field step is $T_{\mathrm{sf}}(\rho) = T^{\mathrm{gf}}(\chi^{\mathrm{w}}\chi_\xi\rho)$, with bondwise cutoffs on the fluctuation field and on the white noise, as in [18, (2.9), (2.10)] and [15, (206)]. These cutoffs are not $\mathcal{G}'$-invariant, so the small-field steps define a modified model (Remark 52.6). We write the gauge-fixed step exactly as a Gaussian integral:

- the background of the step is the $(k+1)$-step background field U' (Lemma 31.6);
- the constraint is eliminated on the middle bond $b_0(c)$ of each coarse bond c, with local Jacobians (Lemma 32.7). The fluctuation field lives on the set $\mathfrak{F}$ of the other bonds, which is invariant under the lattice symmetries (Definition 32.1);
- the action becomes a function of the coarse fluctuation through the coarse effective action $V \mapsto S(U_k(V))$, whose gradient is the Lagrange multiplier and whose derivatives decay along trees (Proposition 33.1). The gauge fixing adds the quadratic form $\frac{1}{2}\|\mathfrak{G}_W Z\|^2$, with the averaged transports $\mathfrak{G}_W$ of Definition 22.6, and a remainder of third order that is a sum of terms localized in single blocks.

Theorem K (One renormalization group step; Theorems 34.9 and 34.4, Corollary 34.6, Lemma 32.6, Proposition 31.7)**.** *Let G be compact, connected and semisimple, let $M \ge \max(M_0, 5L)$, and let the side of the torus satisfy the conditions of Section 31. There are $a_{\mathrm{step}}, r_2 > 0$, depending only on L, G, a, α and M, and $c_L > 0$, depending only on L, such that the following holds for every $k \ge 1$. Let V' be a real configuration on $\mathbb{T}'$ with curvature at most $\varepsilon_1' \le a_{\mathrm{step}}$, let U' be its background field and W the base configuration on $\mathbb{T}$, and let $w \le r_2$. Let ρ be a gauge-invariant density of the form (31.2), with coupling $g = g_k$ and threshold $\tilde{h}$, where $a_1 \ge \tilde{h} \ge 2(C_* + 1)L^{-2}\varepsilon_1' + 4w$.*

(i) (Exact representation.) *$T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(V')$ is $\mathcal{N}_k\, e^{-g^{-2}S(U')}(\det \Delta_k)^{-1/2} \prod_c \mathfrak{j}_c$ times the integral, with respect to the Gaussian measure with covariance $C_k = \Delta_k^{-1}$ on the fields on $\mathfrak{F}$, of χ^{w} times the exponential of an interaction of third order, of local Jacobian terms and of the old activities. The constant $\mathcal{N}_k$ is explicit and does not depend on the background;*

the $\mathfrak{j}_c$ *are local Jacobians, and* $\log\det\Delta_k$ *is a sum of quasi-local, gauge-invariant densities, which depend on the whole background with exponential decay.*

(ii) (Coercivity.) $\frac{1}{4}c_L\|Z\|^2 \le \Delta_k(Z,Z) \le C_L\|Z\|^2$ *for real fields* Z *on* $\mathfrak{F}$*, and* C_k *has exponential decay; at complex bases close to* W*,* $\operatorname{Re}\Delta_k \ge \frac{1}{8}c_L$*. The constant* c_L *comes from a Poincaré inequality for one step, in which the gauge-fixing term is necessary (Lemma 34.2).*

(iii) (Gauge fixing.) $T^{\mathrm{gf}}(f) = T(f)$ *for every* $\mathcal{G}'$*-invariant* f *that vanishes outside Bałaban's set* $\mathfrak{M}$ *of regular configurations. Let* $\chi = \chi^{\mathrm{w}}\chi'$ *with a covariant cutoff* χ'*, for instance* $\chi' = 1$ *or* $\chi' = \chi_\xi$*. Then* $T^{\mathrm{gf}}(\chi\rho)(V')$ *does not depend on the choice of* U' *in the minimal orbit of* V'*, and it is invariant under real gauge transformations of* V'*. If* $\rho \ge 0$*, the part discarded by the cutoffs,* $T(\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(\tilde h)}\rho) - T^{\mathrm{gf}}(\chi\rho)$*, is nonnegative.*

(iv) (Lattice symmetry.) T *commutes with the group* $\mathfrak{S}_{k+1}$ *generated by the translations of* $\mathbb{T}'$ *and by* B_4 *about the points of* $\mathbb{T}'$*. If* $\rho\circ r = \rho$ *for some* $r \in \mathfrak{S}_{k+1}$*, then* $T^{\mathrm{gf}}(\chi\rho)(rV') = T^{\mathrm{gf}}(\chi\rho)(V')$ *for the cutoffs* χ *of (iii). Replacing* (W,U') *by* (rW,rU') *conjugates* Δ_k *and* C_k *by an orthogonal map of the fields on* $\mathfrak{F}$*.*

The constants do not depend on k*,* n*,* η *or the coupling.*

Theorem K in words. The step integrates over the configurations V on $\mathbb{T}$ with prescribed averages V' on $\mathbb{T}'$. Its base W is the k-step average of the background field $U' = U_{k+1}(V')$; then W has the averages V', and U' is also its k-step background field (Lemma 31.6). On the support of χ^{w}, $V = e^{\mathrm{i}Z}W$ with $|Z| \le w$ on every bond (Definition 32.4). The fluctuation variables are the values of Z on $\mathfrak{F}$, and Δ_k is the quadratic part in Z of the action with the gauge-fixing exponent (Lemma 34.1). The density ρ is $\exp(-g^{-2}S(U_k(V)) + \mathfrak{E}_k(V))$ on the small-field set $\mathfrak{U}_k(\tilde h)$ (31.2). So the step is an explicit prefactor, with $e^{-g^{-2}S(U')}$ and $(\det\Delta_k)^{-1/2}$, times a Gaussian integral with covariance $C_k = \Delta_k^{-1}$ of the cutoff and of the exponential of an interaction of third order, of Jacobian terms and of the old activities (i). The form Δ_k is coercive and C_k decays exponentially, uniformly in k (ii). The gauge fixing does not change the step on invariant integrands supported on regular configurations, and, for $\rho \ge 0$, the cutoffs discard a nonnegative part (iii). The precise statements are Theorems 34.9 and 34.4.

The lattice symmetry rests only on intrinsic objects: the Haar measure, the averaging, the coarse action and gf_W. The gauge-field operators of Part II at curved backgrounds are covariant under translations and permutations of the axes, but in general not under reflections (Remark 31.8). Part VII also shows that the curvature of background fields is bounded by $(C_* + \delta)$ times the curvature of the coarse configuration, with C_* independent of L (Proposition 31.5). This makes the small-field sets of consecutive steps compatible for $L^2 \ge 8(C_*+1)$, provided the couplings satisfy $g_{k+1} \le 2g_k$ and are small enough that the fluctuation cutoff obeys $4w \le \frac{1}{2}h_k$ (Remark 49.8, the condition (Q2)). Along the flow both hold at every step (Theorem N(v)).

Localization (Parts VIII and IX). The renormalization group needs the background dependence of the step as a sum of local pieces. Part VIII obtains this with Dirichlet approximants and telescoping, using interior pointwise bounds with derivatives for the Dirichlet operators (Theorem 35.2). The pieces are of *radius type*: a piece depending on a ball of radius m is bounded by $e^{-\gamma m}$. A polymer expansion needs *volume type*, a small factor for every cube of the dependence region. Part IX provides it, as Bałaban and Dimock do, by random walks with weakening parameters $s \in \mathbb{D}_\kappa$. It constructs them for the covariance, its square root, the coarse Hessian and the determinant in L^2 norms (Section 41). It proves interior estimates

with L^p data for the local propagators (Section 43). With them, regularity tiers give pointwise $C^{1,\beta}$ bounds for the weakened propagators $\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$ (Section 44). In four dimensions L^2 data lie at the endpoint $p = \frac{d}{2}$ of the interior estimates, and the tiers pass through block L^4 (Lemma 44.6). With these bounds Part IX builds the weakened background map $A_*(s; B)$, analytic in s and factorizing over polymers (Theorem 45.5).

The small-field step (Part X). The activities are analytic functions on complex domains. In four dimensions the constants of these domains do not depend on the coupling, the step, the torus or the polymer (Remark 38.8), as in [18, (1.11)–(1.16)]. Part X first extends Part IX, and the averaged backgrounds of Part IV, to complex references (Section 46). Then it follows Dimock's scheme [15]:

- the step quantities (interaction, gauge-fixing remainder, Jacobians, old activities composed with the background map, Hessian) become functions of the weakening parameters that factorize over polymers (Section 47);
- with the cutoffs on the fluctuation field and on the white noise, the fluctuation integral is evaluated by a Mayer and polymer expansion (Section 48). By parity in the coupling there is no term of order g; the pure sources are of order $g^2 p_0^6$.

Here $p_0 = p_0(g)$ is the white-noise cutoff, which grows more slowly than every power of g^{-1}, for instance as a power of $\log g^{-1}$ (Section 49).

Theorem L (The small-field step in polymer form; Theorem 48.9, Propositions 52.2 and 52.5)**.** *Let G be compact, connected and semisimple, and let the Hölder exponent satisfy $\alpha \le \frac{1}{2}$. Let the parameters be chosen as in Remark 49.8, and assume the conditions of Theorem 48.9 on the coupling, the thresholds and the old activities.*

(i) For a density ρ of the form (31.2) at step k, the small-field step is, at real critical references with V' small field,

$$T_{\mathrm{sf}}(\rho)(V') = \mathcal{N}_k N_\chi \exp\Big(-\frac{1}{g^2} S(U') - \frac{1}{2}\sum_b \ell_k(b;1) + \sum_X \mathfrak{e}(X, U') + \sum_c \log \mathfrak{j}_c + \mathfrak{E}(U') + \sum_Y E^{\#}(Y, U') \Big),$$

with explicit constants $\mathcal{N}_k$, N_χ and densities $\ell_k(b;1)$ at the trivial base. Every term depends on U' only near its polymer, is invariant under real gauge transformations, is real at real references, and is analytic in the reference on complex domains. On the domains of Remark 38.8 the one-loop activities are bounded by $Ce^{-(\kappa_2-1)|X|}$ and the Jacobian terms by C, with C independent of the coupling; their quadratic parts are marginal in four dimensions. The new sources $E^{\#} = E^{\#}_{\mathrm{p}} + E^{\#}_{\mathrm{m}}$ consist of a pure part of order $g^2 p_0^6$ and a mixed part bounded by the old activities $\mathfrak{E}$. If ρ is invariant under some $r \in \mathfrak{S}_{k+1}$, so is the sum over polymers in the exponent; single activities are covariant at most under the translations of their grids.

(ii) The first step, from $\rho_0 = \exp(-g_0^{-2} S - E_0)$, has the same form, with $\mathfrak{E} = 0$ and no mixed sources.

(iii) (Vacuum energy.) *If the steps $k < K$ are given by (i) and (ii), with the hypotheses of (i) at every step, their constants split into explicit normalizations and a vacuum energy ε_k per site of $\mathbb{T}^{(k)}$, with $\varepsilon_{k+1} = L^d \varepsilon_k + \delta_k$, where δ_k does not depend on E_0, its pure part is at most $CL^d M^{-d} g_k^2 p_0(g_k)^6$, and its mixed part is bounded by the old activities. There is exactly one counterterm E_0 with $\varepsilon_K = 0$. If $|\delta_{k+m}| \le D_k L^{dm/2}$ for $0 \le m < K - k$, then*

$|\varepsilon_k| \le 2L^{-d}D_k$. *Along the flow of Theorem N the hypotheses of (i) hold at every step, and the growth bound holds with* $D_k = Cg_k p_0(g_k)$ *(Theorem N(vi)).*

In words, Theorem L says the following. The small-field step maps a density of the form (31.2) to $e^{-g^{-2}S(U')}$ times explicit normalizations and densities and the exponential of a sum of activities on polymers. Each activity depends on the reference only near its polymer, is gauge invariant, and is analytic in the reference on the complex domains of Remark 38.8, whose constants do not depend on the coupling (i). The new activities are of three kinds. The one-loop activities and the Jacobian terms are bounded independently of the coupling, and their quadratic parts are marginal. The pure sources are of order $g^2 p_0^6$; by parity in the coupling there is no term of order g (Proposition 48.8). The mixed sources are bounded by the old activities. The hypotheses on the coupling, the thresholds and the old activities hold at every step of the flow (Theorem N(v)). The small-field steps define a modified model: at every step the integration is restricted to small fields, and the rest is discarded. The discarded parts are nonnegative, so the partition function of the small-field model is at most that of the lattice theory with the same counterterm (Remark 52.6). Theorem L is a statement about one step; it is not a flow theorem. The iteration needs the four-dimensional organization of the flow [18, 19]: generations of activities, each created by one step and bounded at its creation scale; the cancellation of the marginal quadratic parts of old generations by counterterms $-\beta_j S$, with $g_{j+1}^{-2} = g_j^{-2} - \beta_{j+1}$; and an iteration without contraction. Part X carries this out: the generations, their symmetrization and their coefficient β_j (Section 50), the extraction of the marginal part of an old generation at later scales (Section 51), one step in generation form (Theorem 53.7), and the flow over all scales at which the couplings stay in $(0, \gamma]$ (Theorem 53.12). The flow supplies, at every step, the bounds $g_{k+1} \le 2g_k$ and the bounds on the old activities that Theorem L assumes, and the growth bound on the δ_k in (iii). It is summarized in Theorem N below: Theorem L is the step, Theorem N the flow. Controlling the discarded parts is the large-field stage, the analogue of [16, 17], which is not treated here. Parts VII–X write out the small-field part of Bałaban's four-dimensional analysis [18, 19], for one step and for its iteration over all scales on a fixed torus, as long as the couplings stay in $(0, \gamma]$, in Dimock's scheme, on the operator theory of Parts I–IX.

Boundary conditions. The gauge-field Dirichlet operators contain an exterior divergence term, so that all components of the gauge field satisfy Dirichlet conditions (Section 19). With these conditions the local commutators of the weakened walks are bounded uniformly in η on proper domains (Remark 41.5). On the torus the exterior term vanishes, so the torus operators, and with them Theorems F, H–J and the torus statements of Theorem G, do not involve it.

The one-loop coefficient (Part XI). In four dimensions the coupling is marginal, and its flow is driven by the coefficient β. Part XI computes the one-loop part of this coefficient, summed over the steps. Let $\Gamma_k^{(1)}(V)$ be the one-loop term of the composite k-step map, obtained by Laplace's method from the integral over the fine configurations with averages V near the minimal orbit (Proposition 54.3), and Γ_j^{step} the one-loop term of the step from scale j to scale $j+1$ (Section 54). For $j \ge 1$ it is given by Theorem K(i). For $j = 0$ it is given by Lemma 52.3, the formula of Theorem K(i) for the step from scale 0 with the fluctuation operator of Proposition 52.2(b), the *formula for the first step*. For a gauge-invariant function Φ of the coarse configuration, $\Pi_\Phi(p)$ is its vacuum polarization, the second variation at $V = 1$ in momentum space, and T_Φ its marginal tensor, the part of $\Pi_\Phi(p)$ of second order in p (Definition 54.7). Theorem M is stated before Theorem N, and its part (iv) uses the flow of Theorem N; the generations and the couplings of the flow are introduced in the paragraph on Part X before Theorem N.

Theorem M (The one-loop coefficient; Lemma 54.5, Proposition 54.6, Lemma 52.3, Theorem 55.4, Lemmas 54.9, 56.2 and 58.5, Theorems 57.1, 58.1 and 58.9, Proposition 58.6)**.**

Let G be compact, connected and simple, with the trace of a representation ρ in the action, $\mathrm{tr}_\rho(T^aT^b) = T_\rho \delta^{ab}$, and C_A the Casimir of the adjoint representation.

(i) (Composition and chart.) *For every small-field V and $k \geq 2$, up to a constant independent of V, $\Gamma_k^{(1)}(V) = \Gamma_{k-1}^{(1)}(U_k(V)^{(k-1)}) + \Gamma_{k-1}^{\mathrm{step}}(U_k(V))$. It follows that $\Gamma_k^{(1)}(V) = \sum_{j<k} \Gamma_j^{\mathrm{step}}(U_k(V))$ for $k \geq 1$, up to constants independent of V. Moreover, $\Gamma_k^{(1)}(V)$ is given in the Landau chart at the background field $U_k(V)$ by the determinants of $\mathbf{\Delta}_*$, of $\mathbf{Q}_k \mathbf{G}_* \mathbf{Q}_k^T$ and of a ghost operator.*

(ii) (Symmetry.) *$\Gamma_k^{(1)}$ and the functions $V' \mapsto \Gamma_j^{\mathrm{step}}(U_{j+1}(V'))$, $j \geq 0$, are exactly invariant under the translations and under B_4. As the side of the torus tends to infinity along the admissible sides (Lemmas 56.2 and 58.5), their second variations at 1 converge, with exponential decay uniform in the side. In infinite volume, $\Pi(Rp) = R\,\Pi(p) R^T$ for $R \in B_4$, $\Pi(p) = O(|p|^2)$, and the marginal tensor is $\beta(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$ with a single coefficient β; in particular there is no term linear in p. These functions are invariant under single reflections, so no topological density can occur in their local expansions.*

(iii) (The cumulative coefficient.) *The marginal tensors $T_{j+1}^{(1)}$ of the steps satisfy, uniformly in k,*

$$\sum_{j<k} T_{j+1}^{(1)} = \big(k\beta_\infty + O(1)\big)\big(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu\big), \qquad \beta_\infty = \frac{11 C_A}{24\pi^2 T_\rho} \log L.$$

(iv) (The ultraviolet limit.) *Let $p_0(g) = (\log g^{-1})^q$, let the parameters and the torus be as in Theorem N, and let γ be as in Theorem 58.9. Then there are $g_{\mathrm{uv}} > 0$ and C, independent of K and of the torus, such that for $0 < g \leq g_{\mathrm{uv}}$ and every K the small-field flow of Theorem N has a bare coupling g_0, depending on K, g and the torus, with $g_k \in (0, \gamma]$ for $k \leq K$, $g_K = g$, and*

$$\Big|\frac{1}{g_k^2} - \frac{1}{g^2} - (K-k)\beta_\infty\Big| \leq C\big(\log(K-k+2)\big)^{6q+1} \qquad \text{for every } k \leq K.$$

The properties (F1)–(F3) of the flow that this uses are proved in Proposition 58.6. Property (F2), the comparison of the one-loop parts of the coefficients β_{j+1} of the flow with the coefficients of (iii), rests on the torus independence of the one-loop pieces (Proposition 41.14, Lemma 58.3). Here g is the coupling of the last scale; in Parts VII and X, $g = g_0$ denotes the bare coupling.

Theorem M in words. The one-loop term of k steps is the sum of the one-loop terms of the single steps, up to constants, and it is given in the Landau chart by three determinants (i). These functions of the coarse configuration are invariant under the lattice symmetries. Hence the infinite-volume limits of their second variations at 1 have no term linear in p, and their marginal tensors are multiples of $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$ (ii). Summed over k steps, the coefficient is $k\beta_\infty + O(1)$ (iii). With the flow of Theorem N this gives (iv): for a small final coupling g and any number K of steps there is a bare coupling that leads to $g_K = g$ through couplings in $(0, \gamma]$, and $g_k^{-2} - g^{-2}$ is $(K-k)\beta_\infty$ up to a power of $\log(K-k+2)$. Since $\beta_\infty > 0$, at $k = 0$ this says that g_0^{-2} grows like $K\beta_\infty$: at a fixed g the bare coupling tends to 0 as the number of steps grows. Part (iv) concerns the couplings of the small-field flow (Remark 58.10(a)).

Part (ii) rests on the symmetry of Bałaban's averaging, the invariance of the Haar measure and the covariance of the minimal orbits. It does not use the covariance of the gauge-field operators of Part II, which are covariant under reflections only at flat backgrounds. For the step functions it has two proofs: one through the invariance of $\Gamma_k^{(1)}$, which uses no covariance property of the gauge fixing, and a direct one, which uses only the covariance of the quadratic

part of the gauge fixing, that of the averaged contour variables. In particular there is no term linear in p; such a term would produce relevant terms of order 1 in every step (Remark 54.8, Remark 58.10(b)). The theorems of Part XI have no hypothesis on a linear term. Parts (iii) and (iv) are our version, for the couplings of the small-field flow of Part X, of Bałaban's Theorem 2 [18], which he states for $SU(2)$. Bałaban's couplings come from his own generations, a different construction (Remark 53.14(a) and the statement of Proposition 58.8). Part (iii) is the growth $k\beta_\infty + O(1)$, in particular the positivity for large k, of the one-loop coefficient summed over the steps, and (iv) derives from it the behaviour of the couplings at all scales, with the flow of Part X (Theorem 53.12, Proposition 58.6) in place of his Theorem 3. In particular (iv) gives the analogue of his bounds (0.31) for these couplings at the scales k with $K - k \geq m_1$, for some m_1 independent of K, g and the torus; for the last m_1 steps it gives the upper bound, and the lower bound up to an additive constant (the paragraph after Theorem 58.9). The main input of (iii) is the coefficient $\beta_{\mathrm{F}} = k\beta_\infty + O(1)$ of the vacuum polarization on the fine lattice (Theorem 57.1). The infinite-volume objects of Part XI are constructed from the tori: the operators at $U = 1$ by folding (Section 54.2), and the one-loop pieces of the steps through their torus independence (Proposition 41.14). Remark 58.10 lists what remains.

The small-field flow (Part X). The flow iterates the step of Theorem L in Bałaban's generation form [18, (0.22)–(0.30)], on a torus whose side at the last scale K is $n_K \geq n_{\min}$ (Lemma 52.1). A *generation* of scale j is a family $E = \{E(X, \cdot)\}$ of normalized, gauge-invariant analytic activities on the polymers X of the grid of scale j. They are real at real configurations and covariant under the group $\mathfrak{S}_j^{\mathrm{grid}}$ of the symmetries of that grid, and their sum at the background fields of scale j is invariant under $\mathfrak{S}_j$ on the small-field set (Definition 50.1). A generation is bounded only at its own scale, on domains $\mathcal{U}^b(X)$ whose constants do not depend on the coupling, the step, the torus or the polymer. Its coefficient β_E is read off from its period density $\bar{T}_E$, the second variation of its activities at the linear fields of constant curvature, per unit volume, as a multiple of that of S (Definition 50.11, Proposition 50.13). The new terms of a step, i.e. the one-loop activities, the Jacobian terms and the new sources of Theorem L(i), reblocked, form a raw generation (Lemma 53.4); the new generation is its average over $\mathfrak{S}_{k+1}^{\mathrm{grid}}$ (Definition 50.7). The old generations enter the next step through the old activities of Definition 53.1: in every grid cube $\square$ the counterterm $-\beta_j S_\square$ is grouped with the small polymers of the generation j in $\square$. We write $(E - \beta S)(U) = \sum_X E(X, U) - \beta S(U)$, and $h_k = p(g_k)g_k$, $p_{0,k} = p_0(g_k)$ with the small-field parameters of Section 49.1.

Theorem N (The four-dimensional small-field flow; Theorem 53.12, Propositions 50.13, 51.5 and 53.10, Corollaries 53.11 and 53.13)**.** *Let G be compact, connected and simple, let the parameters be chosen as in Remark 49.8, with the bound γ on the couplings subject to (Γ1)–(Γ3) of Remark 53.5, and let $n_K \geq n_{\min}$. Let $g_0 \in (0, \gamma]$, $E_0 \in \mathbb{R}$ and $\rho_0 = \exp(-g_0^{-2}S - E_0)$. For $k \geq 0$, as long as $g_k \leq \gamma$, let $E^{(k+1)}$ be the new generation of the step from scale k to scale $k+1$ applied to ρ_k, let $\beta_{k+1} = \beta_{E^{(k+1)}}$ and*

$$\frac{1}{g_{k+1}^2} = \frac{1}{g_k^2} - \beta_{k+1},$$

and, if $g_{k+1} \leq \gamma$, let $\rho_{k+1} = \mathbf{1}_{\mathfrak{U}_{k+1}(h_{k+1})} T_{\mathrm{sf}}(\rho_k)$. There are constants K, C_β, C and C', which depend only on the parameters fixed before γ (C' also on the function p_0), and not on k, K, the torus or the couplings, such that (i)–(v) hold for every $1 \leq k \leq K$ with $g_1, \dots, g_k \leq \gamma$.

(i) (Generation form.) *For $V \in \mathfrak{U}_k(h_k)$, with $U_k = U_k(V)$,*

$$\rho_k(V) = \mathcal{Z}_k \exp\Big(-\frac{1}{g_k^2} S(U_k) - \varepsilon_k |\mathbb{T}^{(k)}| + \sum_{j=1}^{k} \big(E^{(j)} - \beta_j S\big)(U_k) \Big),$$

and $\rho_k(V) = 0$ for the other configurations V. The density ρ_k is invariant under real gauge transformations and under $\mathfrak{S}_k$.

(ii) (Creation scale.) *Each $E^{(j)}$, $j \le k$, is a generation of scale j, and $\sup_{\mathcal{U}^\flat(X)} |E^{(j)}(X,\cdot)| \le \mathsf{K} e^{-\kappa_n d_j(X)}$ for every polymer X of scale j.*

(iii) (Coefficients.) *$\bar{T}_{E^{(j)}}(F) = \beta_j \bar{T}_S(F) = \frac{1}{2}\beta_j \sum_{\mu<\nu} \operatorname{tr}(F_{\mu\nu}^2)$ for all curvature data F: the period density has no topological and no anisotropic part. The number β_j is real, and $|\beta_j| \le C_\beta \mathsf{K}$.*

(iv) (Later scales.) *Let $j \le k$, $m = k - j$, and let $\square$ be a grid cube of scale k. The sum of $E^{(j)}(X,U)$ over the small polymers X of scale j based in $\square$ (Definition 49.5) differs from $\beta_j S_\square(U)$ by at most $C\mathsf{K}L^{-\alpha m}$ if $m \ge m_0$, and by at most $C\mathsf{K}$ if $m < m_0$, for $U \in \mathcal{U}^\flat(\square^+)$ of scale k. The sum over the other polymers of scale j whose closure at scale k is Y is at most $C\mathsf{K}L^{-m}e^{-\kappa_n d_k(Y)}$ on $\mathcal{U}^\flat(Y)$. In particular $|\sum_{j=1}^k (E^{(j)} - \beta_j S)(U_k(V))| \le C\mathsf{K}|\mathbb{T}^{(k)}|$ for $V \in \mathfrak{U}_k(h_k)$.*

(v) (Couplings.) *For $1 \le j \le k$, $|g_j^{-2} - g_{j-1}^{-2}| = |\beta_j| \le C_\beta \mathsf{K}$ and $\frac{3}{4} g_{j-1} \le g_j \le 2g_{j-1}$, and the step from scale $j-1$ to scale j satisfies the hypotheses of Theorem L(i) (of Theorem L(ii) if $j = 1$).*

(vi) (Vacuum and stability.) *Let $g_1, \dots, g_K \le \gamma$. The constants $\mathcal{Z}_k$ are explicit products of normalizations. The vacuum energies satisfy $\varepsilon_{k+1} = L^4 \varepsilon_k + \delta_k$ for $k < K$, where δ_k does not depend on E_0 and $|\delta_k| \le C(g_k^2 p_{0,k}^6 + g_k p_{0,k})$. There is exactly one E_0 with $\varepsilon_K = 0$, and for it $|\varepsilon_k| \le C' g_k p_{0,k}$ for $k \le K$. The partition function of the small-field model is $\int \rho_K = \mathcal{Z}_K e^{-\varepsilon_K |\mathbb{T}^{(K)}|} I_K$, where I_K does not depend on E_0 and*

$$e^{-c|\mathbb{T}^{(K)}|} \le I_K \le e^{c|\mathbb{T}^{(K)}|},$$

with c depending on g_K and on the parameters, but not on K, the torus or $g_0, \dots, g_{K-1}$.

(vii) (Continuity.) *Let $0 \le j < K$. For every sequence $(g_0, \dots, g_j) \in (0,\gamma]^{j+1}$ with $g_{i+1} \le 2g_i$ for $i < j$, the generation $E^{(j+1)}$ and the coefficient β_{j+1} are defined by the steps above, with the couplings taken from the sequence and not from the recursion, as in Proposition 53.10. They depend only on $(g_0, \dots, g_j)$, and continuously, at every point of the domains of the activities. For the couplings of the recursion they are those of the preamble. The set of the $g_0 \in (0,\gamma]$ with $g_1, \dots, g_K \le \gamma$ is closed in $(0,\gamma]$ and contains $(0, (\gamma^{-2} + KC_\beta \mathsf{K})^{-1/2}]$, and $g_0 \mapsto (g_1, \dots, g_K)$ is continuous on it.*

Theorem N in words. As long as the couplings stay in $(0,\gamma]$, the density of every scale keeps Bałaban's generation form (i). Each step creates one generation, a family of activities that is bounded once, at its creation scale, on domains whose constants do not depend on the coupling (ii). Its coefficient β_j is read off from its period density, which is exactly isotropic (iii). At a later scale, $m \ge m_0$ scales after its creation, the small polymers of a generation in a grid cube add up to $\beta_j S_\square$ up to $C\mathsf{K}L^{-\alpha m}$, and its other polymers carry a factor L^{-m}; so the counterterm $-\beta_j S$ cancels its marginal part, and the sum over all generations is bounded per unit volume, uniformly in the scale (iv). The iteration uses no contraction (Remark 49.1). The couplings change by bounded amounts per step, and each step satisfies the hypotheses of Theorem L (v). Item (vi) writes, for the counterterm E_0 with $\varepsilon_K = 0$, the partition function of the small-field model as the explicit product $\mathcal{Z}_K$ of normalizations times a number between $e^{-c|\mathbb{T}^{(K)}|}$ and $e^{c|\mathbb{T}^{(K)}|}$, with c depending on g_K but not on the number of steps or the torus. Item (vii) gives the continuity in the couplings that the exit argument in the proof of Theorem 58.9 uses. Section G1 restates the theorem in more detail.

Concordance with Section 53. Items (i)–(vi) of Theorem N are items (a)–(f) of Theorem 53.12, in this order. The partition function in (vi) is Corollary 53.13, and item (vii) is Proposition 53.10 with Corollary 53.11. Item (iii) is also Proposition 50.13, and item (iv) is Proposition 51.5, Lemma 51.7(a) and Remark 51.8. The constant C of Theorem N stands for C_{ex}, C_{ne} and C_{lat} in (iv) and for C_{vac} in (vi), and C' is C'_{vac} (Theorem 53.12, Lemma 51.7).

Theorem N is the analogue, for the small-field model on the torus, of Bałaban's Theorem 3 [18], with his hypothesis $0 < g_k \le \gamma$ for $k \le K$; as there, no asymptotic behaviour of the couplings is assumed (Remark 53.14(a)). Unlike there, the dependence on the couplings, which is a dependence on the whole sequence $(g_0, \dots, g_{j-1})$, is only shown to be continuous, whereas his is smooth, and the small-field domain shrinks with the coupling, as in Dimock's scheme (Remark 53.14(a)). Items (i), (ii) and (iv) correspond to his (0.22)–(0.24), (0.25) and (0.28)–(0.30); the later-scale bounds (0.29) are asserted by his Theorem 1, and they are not among the inductive assumptions listed in his Theorem 3. The proof is an induction over the scales with the step in generation form (Theorem 53.7). Its two four-dimensional ingredients are the symmetrization of every new generation over the symmetries of its grid, which makes the period density exactly isotropic (Proposition 50.13), and the extraction of the marginal part of an old generation at every later scale. For the latter, a configuration of a later scale is written, in a suitable gauge on each small polymer of the generation, as a linear field of constant curvature plus a smaller remainder (Lemma 51.1, Proposition 51.5). The old activities are then bounded uniformly in k (Proposition 53.2), and the iteration needs no contraction. The coefficients β_j are defined on the given torus. Their one-loop parts are compared with the coefficients of Theorem M(iii) in (F2) of Proposition 58.6 (Theorem M(iv)). Bałaban's definition [18, (1.20)–(1.22)] assigns a coefficient to the family of the generations $E^{(j)}$ of this note built with fixed couplings on tori of growing side. Proposition 58.8 shows that this coefficient is defined and that, at fixed couplings, β_j converges to it as the torus grows. This rests on the generation locality of Section 59; no rate is proved. Theorem M does not use it.

Reading guide. Section G0 of the guide gives four reading paths, Figure G1 the main line, and the introduction of each Part its imports and exports. The main line is as follows. For the small-field step it is Parts I–VII, Section 35 and the interior estimates of Section 36, Section 37 up to Proposition 37.6, Section 38 except Proposition 38.1 and Corollary 38.9, Part IX and Part X. The radius-type results of Part VIII (Theorem 37.7, Corollary 37.8, Proposition 38.1, Corollary 38.9, and Section 39) are superseded on the main line by Parts IX and X. Each of them is marked as superseded where it appears, in the status paragraphs at the head of Sections 37–39. The flow (Theorem N) is Sections 49–53 of Part X. Sections 54–57 of Part XI use Parts I–VII and, from Part X, mainly Propositions 47.8 and 52.2 and Lemma 52.3; Section 58, which combines Part XI with the flow, uses Part X throughout, and the torus independence of the one-loop pieces (Section 41.5). Section 59 extends it to whole steps at fixed old data, at flat configurations, and proves that the generations converge as the torus grows, at fixed couplings; it uses Parts IV–X, Lemma 54.10, and Sections 58.2 and 58.3.

1.4 Methods

Table 1 lists the main tools of the note, Part by Part, with the place where each is introduced and what it gives. Section G14 of the guide names the places where the work is hardest.

Part I uses four tools. Each is introduced in its own section, before it is used.

1. Coercivity (Section 3 and Section 9). A Poincaré inequality on unit blocks shows that the averaging term controls exactly those modes that the Laplacian does not control. This gives the bounds of Theorem A for scalars. For gauge fields the situation is subtler, because

Part	Tool	Introduced in	What it gives
I	block Poincaré inequality; for gauge fields the near separation of gradient and co-closed parts by **Q**	Section 3, Lemma 9.3	coercivity with an averaging term (Theorem A)
I	the Combes–Thomas method	Section 5	exponential decay of the inverses, on the fine and on the coarse lattice (Theorems B–E)
I	a lattice Schauder estimate	Theorem 6.3, Appendix A	pointwise bounds on values, derivatives and Hölder quotients from L^2 bounds
I	one Fourier lower bound	Corollary 4.3	the lower bound on QGQ^*, uniformly in k
II	perturbation in the background	Section 13	the operators at backgrounds close to 1 in a global gauge (Theorem F)
III	IMS localization at scale M; Dirichlet operators; comparison of domains	Theorems 19.6 and 19.10	coercivity at locally regular backgrounds, and locality in the background (Theorem G)
IV	the averaged axial gauge and a recursive correspondence map	Theorems 22.5 and 24.2	the axial-to-Landau chart (Theorem H)
V	a contraction around a regular reference	Proposition 26.3	the variational problem relative to a reference (Theorem I)
VI	a change of the reference near a point; patching of local gauges	Lemmas 29.2 and 28.3	the induction over k on the torus (Theorem J)
VII	exponential gauge fixing without a Faddeev–Popov determinant; a Poincaré inequality for one step	Lemmas 32.3 and 34.2	the step as a Gaussian integral, and its coercivity (Theorem K)
VIII	Dirichlet approximants and telescoping; absorption inside a cube	Theorem 35.2, Section 36, Lemma 37.1	interior bounds; localization of radius type (the telescoping is off the main line)
IX, X	random walks with weakening parameters; the localization lemma	Theorem 41.4, Lemma 47.2	localization of volume type
IX	interior L^p estimates and regularity tiers	Sections 43 and 44	pointwise $C^{1,\beta}$ bounds for the weakened propagators
X	complex references	Proposition 46.7	analyticity of the activities in the background
X	Mayer and polymer expansion with two cutoffs; parity in the coupling	Theorem 48.6, Proposition 48.8	the step in polymer form, with pure sources of order $g^2 p_0^6$ (Theorem L)
X	symmetrization of a generation; a normal form at later scales	Lemma 50.8, Lemma 51.1, Proposition 51.5	one coefficient β_j per generation; old activities bounded uniformly in the scale, without contraction (Theorem N, Proposition 53.2)
XI	Laplace's method; the lattice symmetry of the averaging	Propositions 54.3 and 54.6, Lemma 54.9	the one-loop term as a sum over the steps, with no term linear in p (Theorem M(i), (ii))
XI	folding of the torus operators; torus independence of the one-loop pieces	Section 54.2, Proposition 41.14 (Part IX)	the operators in infinite volume; the comparison (F2) of the flow with Part XI (Proposition 58.6)
XI	reduction to the lattice Feynman gauge	Proposition 56.8, Theorem 57.1	the coefficient β_∞ (Theorem M(iii))
XI	an exit argument	Theorem 58.9, Corollary 53.11	the bare coupling of Theorem M(iv)

Table 1: The main tools of the note.

$\|dA\|^2 + a\|\mathbf{Q}A\|^2$ is invariant under gauge transformations $A \mapsto A + d\lambda$ with $Q_0\lambda = 0$. The gauge-fixing term $\|Rd^*A\|^2$ removes this degeneracy. We prove the coercivity in position space. We split A into a gradient part and a co-closed part, and show that the averaging operator nearly separates them: $d_1^*\mathbf{Q}A$ differs from an average of d^*A by the unit-lattice divergence of a quantity bounded by $\|\partial A\|$ (Lemma 9.3). Bałaban's original proof [3] is a computation in momentum space.

2. The Combes–Thomas method (Section 5). If an operator is coercive and conjugation by $e^{\alpha\rho}$, for Lipschitz ρ, perturbs it by $O(\alpha)$, then its inverse decays exponentially [11]. This works for the fine-lattice operators G_σ and $\mathbf{G}$, which contain the large term $-\Delta$ of norm $\sim \eta^{-2}$, because the perturbation is controlled by the quadratic form itself. It also works for the coarse operators $(QGQ^*)^{-1}$. Bałaban and [26] use Fourier analysis or random-walk expansions instead. We use neither: on a single torus at $U = 1$, Combes–Thomas suffices, and it also survives perturbation in U.

3. Local regularity (Section 6). Exponential decay is first obtained in L^2 between unit blocks. To pass to pointwise bounds on values and derivatives, uniformly in η, we prove a discrete interior regularity estimate. If $-\Delta u = F$ holds on a cube of side 4 (with u defined on a slightly larger cube), then u, ∂u and the Hölder quotients of ∂u on the concentric cube of side 2 are bounded by $|F|$ and the L^2 norm of u. The proof uses a cutoff and the decay of the Green function of the fine lattice $\eta\mathbb{Z}^d$ and its derivatives. The latter is proved in Appendix A.

4. One Fourier lower bound (Section 4). Averaged Green functions QGQ^* must be bounded *below* uniformly in k. We deduce this from the operator inequality $G \geq (-\Delta + 2a)^{-1}$ and an elementary computation in momentum space. The same computation gives one more estimate needed in the gauge-field coercivity proof.

1.5 What is not done here

The variational theorem is proved on a single torus with the small-field condition everywhere (Theorem J). The renormalization group is carried out for one small-field step (Theorems K and L) and iterated over all scales in the small-field model, on a fixed torus, as long as the couplings stay in $(0, \gamma]$ (Theorem N). Two omissions matter most.

- *The large-field stage.* At every step the small-field model discards the large fields, and what it discards is nonnegative (Remark 52.6(c)). Controlling the discarded parts, and so removing the modification, is the large-field stage, the analogue of [16, 17]; it is not treated (Remark 52.6(d)). Hence nothing is proved about the partition function or the correlation functions of the lattice theory itself.

- *The continuum limit.* No limit of the measure or of correlation functions as the lattice spacing tends to 0 is considered. Theorem M(iv) is a statement about the couplings of the small-field flow, uniform in the number of steps at a fixed last torus (Theorem 58.9). With $K = N$ steps on a torus of physical side ℓ the bounds of Theorems N and M(iv) are uniform in the spacing L^{-N} (Remark 53.14(d)); they are not statements about a limit of the measure.

In addition, we do not treat:

- a rate in the comparison of the β_j with the coefficient that Bałaban's (1.20)–(1.22) assigns to the generations of this note: Proposition 58.8 proves the convergence as the torus grows, at fixed couplings, without a rate and without uniformity in the couplings (Remark 59.13). Theorem M does not use it;

- the exact lower half of the analogue of Bałaban's (0.31) for the last m_1 steps, which Theorem M(iv) gives up to an additive constant (Remark 58.10(c));
- the nested domains Ω_j of [8], which belong to the large-field stage;
- gauges on cubes of size up to $c\,\varepsilon_1^{-1}$ with bounds linear in the size [8, (9), (10)]; we obtain polynomial bounds (Remark 30.6);
- gauge groups with abelian factors in Part X: the second-order bound for normalized activities (Lemma 38.5) uses semisimplicity;
- dimensions other than 4.

Section 16.2 lists the input of the variational theorem and where it is provided.

Relation to [26]. The paper [26] gives a self-contained account of exponential decay for the *scalar* operators of Bałaban's approach on general domains with Neumann boundary conditions, using the method of images and analytic continuation of Fourier transforms. Here the domain is a torus, the methods are different, and the emphasis is on gauge fields. The scalar part (Sections 7–8) is included because the gauge-field operators are built from scalar ones: gauge transformations are scalar fields, and the projection R is defined in terms of G_0.

Status. The note was written with the assistance of Claude (Anthropic). It has not been checked by a human expert.

A guide to the proof

This guide describes how the proof is organized. It states no result of its own and proves nothing: every informal statement names the result it paraphrases, and that result is authoritative. Sections G0–G3 give the reading paths, the result, the scales and the strategy of the four-dimensional flow. Sections G4–G10 follow the Parts. Sections G11–G14 give the order of the parameters, the comparison with Bałaban and Dimock, and the places where the work is hardest. Appendix B is the notation index.

G0 How to read this note

The note has eleven Parts. Table 2 lists their main results, and Figure G1 shows how these results feed Theorems N and M(iv). The main theorems of Section 1 are attached to the Parts: Theorems A–E to Part I, F to Part II, G to Part III, H to Part IV, I to Part V, J to Part VI, K to Part VII, L and N to Part X, and M to Part XI. The introduction of each Part lists what it imports and what it exports.

Four reading paths.

(a) *An overview in one day.* The statements of Theorems A–N; Sections G0–G3 of this guide; the introductions of the Parts; and the remarks that explain the flow: Remarks 49.1, 49.8, 52.6, 53.5 and 53.14.

(b) *The main line to Theorem N.* Parts I–VII. From Part VIII: Section 35, the interior estimates of Section 36, Section 37 up to Proposition 37.6, and Section 38 except Proposition 38.1 and Corollary 38.9. Then Parts IX and X. The radius-type results of Part VIII, Theorem 37.7, Corollary 37.8, Proposition 38.1, Corollary 38.9 and Section 39,

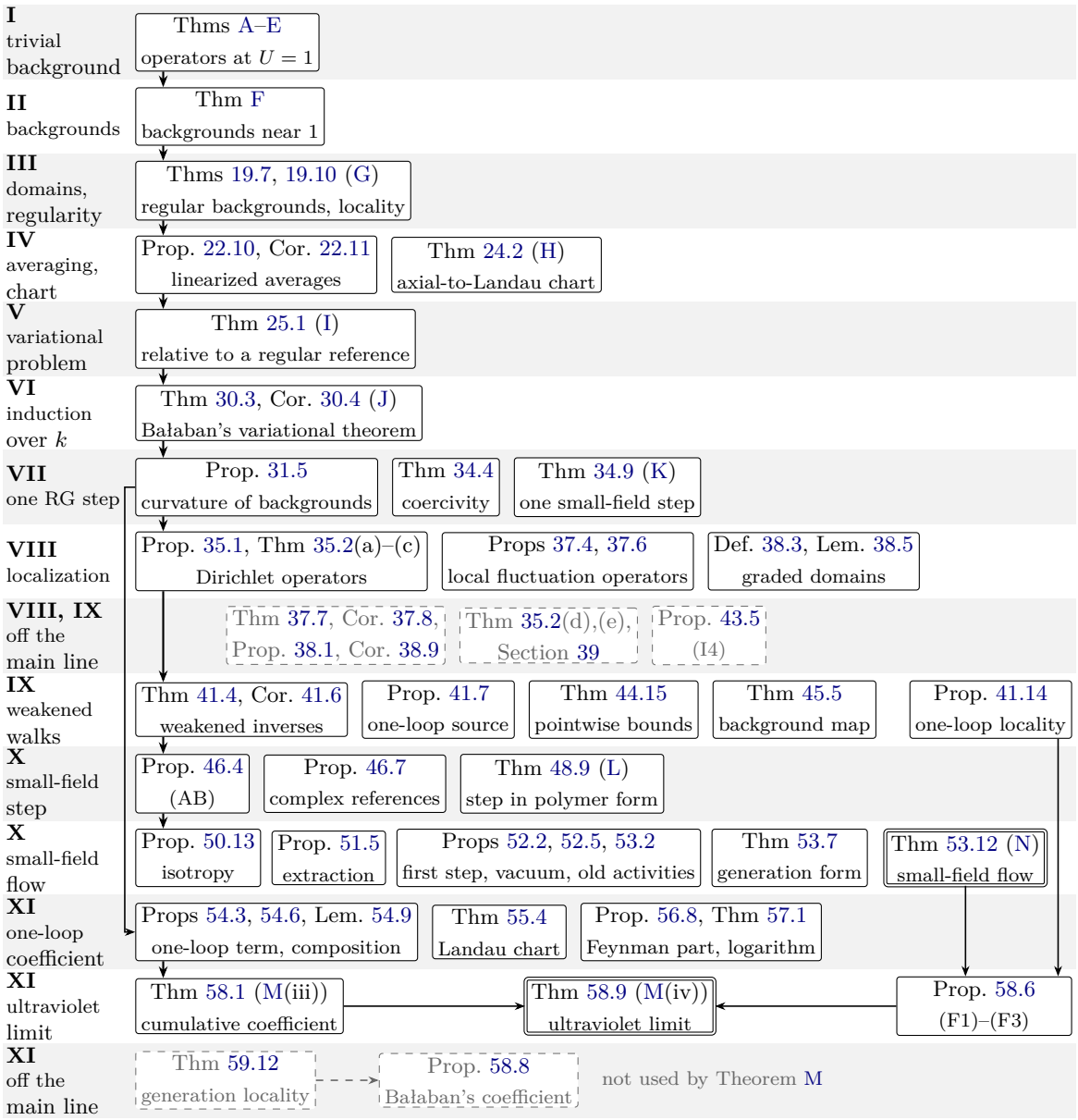


Figure G1: The main line of the proof. Each band is a Part; Part X is split into the step (Sections 46–48) and the flow (Sections 49–53), and Part XI into Sections 54–57, Section 58 and Section 59. A box names results by their printed numbers; the letter in parentheses is the theorem of Section 1 that collects them. A thick arrow: the next band of the main line uses the band above it. The thin arrow on the left: Sections 54–57 use Parts I–VII and, from Part X, mainly Propositions 47.8, 52.2 and Lemma 52.3. The thin arrows on the right: Proposition 58.6, the properties (F1)–(F3), uses the flow and the torus independence of the one-loop pieces (the reading guide in Section 1). Double frames: Theorem 53.12, the small-field flow (Theorem N), and Theorem 58.9, the ultraviolet limit (Theorem M(iv)). Grey dashed frames: results off the main line. These are the radius-type results of Part VIII, superseded by Parts IX and X (the status paragraphs of Sections 37–39); Theorem 35.2(d),(e), used only in Section 39; Proposition 43.5, not used outside Section 43; and Theorem 59.12, which gives Proposition 58.8. Theorem M does not use Proposition 58.8.

Part	Main results	On the main line
I, §§2–11	Theorems A–E: at $U = 1$, coercivity (Theorem 9.6), decay and $C^{1,\alpha}$ bounds for the Green functions and the minimizer maps, and decay, C^1 and $C^{0,\alpha}$ bounds for the constrained Green function $\mathbf{C}$ (Theorems 7.1, 10.1, 11.3, 11.7); interior regularity (Theorem 6.3)	yes; Part XI uses it in infinite volume (Corollary 54.11)
II, §§12–16	Theorem F: the same operators at complex backgrounds close to 1 in a global gauge (Theorems 14.1, 15.4, 16.1)	yes
III, §§17–20	Theorem G: Dirichlet operators and coercivity at (M, ε)-regular backgrounds (Theorem 19.6), decay (Theorems 19.7, 19.9), locality (Theorem 19.10), the scalar random walk (Theorem 20.2)	yes
IV, §§21–24	Theorem H: Bałaban's symmetric averaging; its linearized gauge-adapted form is an averaging pair with $E = 0$ (Proposition 22.10, Corollary 22.11); the axial-to-Landau chart (Theorem 24.2)	yes
V, §§25–27	Theorem I: the variational problem relative to a regular reference (Theorem 25.1)	yes
VI, §§28–30	Theorem J: the background field $U_k(V)$ for every k (Theorem 30.3); regularity from gauge-invariant conditions (Corollary 30.4)	yes
VII, §§31–34	Theorem K: one step as a Gaussian integral (Theorem 34.9), coercivity (Theorem 34.4), the curvature of background fields (Proposition 31.5), lattice symmetry (Proposition 31.7)	yes
VIII, §§35–39	Dirichlet averaged operators (Proposition 35.1), interior bounds (Theorem 35.2), Dirichlet Hessians and local fluctuation operators (Propositions 37.4, 37.6), graded domains (Definition 38.3), the quadratic bound (Lemma 38.5)	partly: Theorem 37.7, Corollary 37.8, Proposition 38.1, Corollary 38.9 and Section 39 are superseded
IX, §§40–45	weakened inverses (Theorem 41.4, Corollary 41.6), the one-loop source in polymer form (Proposition 41.7), torus independence (Proposition 41.14), pointwise bounds by tiers (Theorems 44.13, 44.15), the weakened background map (Theorem 45.5)	yes; Proposition 41.14 serves Part XI
X, §§46–53	complex references (Propositions 46.4, 46.7); Theorem L: the step in polymer form (Theorem 48.9); generations and extraction (Propositions 50.13, 51.5); Theorem N: the flow (Theorem 53.12, Corollary 53.13)	yes
XI, §§54–59	Theorem M: composition and symmetry of the one-loop terms (Proposition 54.6, Lemma 54.9), the Landau-chart formula (Theorem 55.4), the logarithm (Theorem 57.1), the cumulative coefficient (Theorem 58.1), (F1)–(F3) (Proposition 58.6), the ultraviolet limit (Theorem 58.9)	yes, up to Theorem 58.9; Section 59 and Proposition 58.8 are off it

Table 2: The Parts of the note and their main results. "On the main line" refers to the line that ends at Theorems 53.12 and 58.9, as in the reading guide of Section 1.

are replaced on this line by Parts IX and X, and each is marked where it appears (the reading guide of Section 1).

(c) *The one-loop line to Theorem M(iv).* Sections 54–57 use Parts I–VII and, from Part X, mainly Propositions 47.8 and 52.2 and Lemma 52.3. Section 58 combines them with the flow; it uses Part X throughout and the torus independence of Section 41.5. Section 59 and Proposition 58.8 are not used by Theorem 58.9 (Remark 58.10(c)).

(d) *The operator theory alone.* Parts I–III, with Theorems A–G. Part I is self-contained; for the Green function of the fine lattice it uses Appendix A (Section 6). The gauge-field random walk of Theorem G(iv) is completed in Section 41 of Part IX.

The companion. The companion volume *Details* works out the steps of the proofs of Parts VII–XI, and its chapter on the parameter ledger records the constants of these Parts with their dependencies and the order in which they are chosen. Its chapters follow the Parts: Part VII, Part VIII, Part IX, the two halves of Part X and the two halves of Part XI. A proof of this note that the companion expands carries the marginal link Details. The link opens the companion at the first Detail on that result; conversely, the companion's references to the note are links into this file. Both kinds of link are relative. They work when the note, `YM4-small-field-flow.pdf`, and the companion, `Details/details.pdf`, keep their relative places, and they break if the two files are separated. Acrobat, Skim, Okular and Evince follow links to a named place in another file; macOS Preview follows them poorly. The companion's index of the note's results serves when a link does not work.

Where in the note. The reading guide in Section 1; the status paragraphs at the head of Sections 37–39.

In the companion. The chapter "How to read this volume", and the index.

G1 The result

The lattice model. The gauge group G is compact and connected, and $d = 4$ (Section 1). The torus has physical side ℓ; its finest lattice $\mathbb{T}^{(0)}$ has spacing $\epsilon = L^{-N}$, with the odd block factor $L \geq 3$ and the number N of scales (Section 52.1). The Wilson action of a configuration U is

$$S(U) = \sum_p \Big(\mathsf{N} - \tfrac{1}{2} \operatorname{tr} \big(U(\partial p) + U(\partial p)^{-1} \big) \Big),$$

summed over the positively oriented plaquettes, with no power of the spacing in front, since in four dimensions the action is dimensionless (Section 23). The trace is that of a representation ρ by $\mathsf{N} \times \mathsf{N}$ matrices, with $\operatorname{tr}_\rho(T^a T^b) = T_\rho \delta^{ab}$ for Hermitian generators T^a (Section 57). The lattice measure is $\rho_0(V) \prod_b dV(b)$ on $\mathbb{T}^{(0)}$, with the Haar measure on each positively oriented bond, $\rho_0 = \exp(-g_0^{-2} S - E_0)$, the bare coupling g_0 and a constant E_0 (Theorem N).

Normalization of the coupling. The period density of S, its second-order part per unit volume at the linear field of constant curvature $F = F^a T^a$, is $\frac{1}{2} \sum_{\mu<\nu} \operatorname{tr}(F_{\mu\nu}^2) = \frac{1}{2} T_\rho \sum_{\mu<\nu} F_{\mu\nu}^a F_{\mu\nu}^a$ (Theorem 53.12(c)). So $g^{-2} S$ has the period density of the continuum action $\frac{1}{2 g_{\mathrm{std}}^2} \sum_{\mu<\nu} \int F_{\mu\nu}^a F_{\mu\nu}^a$ with $g_{\mathrm{std}}^2 = g^2 / T_\rho$; for $SU(\mathsf{N})$ in the fundamental representation $T_\rho = \frac{1}{2}$ and $g_{\mathrm{std}}^2 = 2 g^2$ (Lemma 57.7; Section 57, the paragraph "Normalization of the coupling"). The one-loop coefficient per step is $\beta_\infty = \frac{11 C_A}{24 \pi^2 T_\rho} \log L$ for g^{-2}, i.e. $\frac{11 C_A}{24 \pi^2} \log L$ for g_{std}^{-2}, and $\beta_\infty = \frac{11 \mathsf{N}}{12 \pi^2} \log L$ for $SU(\mathsf{N})$ in the fundamental representation (Theorem 57.1). Its sign is

Cutoff	Condition	Size	Defined at
small-field set $\mathfrak{U}_k(h_k)$	the curvature of V on $\mathbb{T}^{(k)}$ is at most h_k at every plaquette	$h_k = p(g_k)\, g_k$	Definition 21.4, Section 49.1
fluctuation cutoff χ^{w}	$V(b) = e^{\mathrm{i}Z(b)}W(b)$ with $\lvert Z(b)\rvert \le w$ on every bond, W the base of the step	$w = C_w L^d C_T\, g\, p_0(g)$	Definition 32.4, Remark 53.5(iii)
white-noise cutoff χ_ξ	$\lvert\xi(b)\rvert \le p_0$ on the bonds of $\mathfrak{F}$, where $Z = g\, C_k^{1/2}\xi$ on $\mathfrak{F}$	$p_0 = p_0(g)$	Definition 48.1, Section 49.1

Table 3: The cutoffs of the step from scale k to scale $k+1$, at $g = g_k$. The functions p, p_0 grow more slowly than every power of g^{-1}, and $p_0/p \to 0$ as $g \to 0$, for instance $p_0 = (\log g^{-1})^q$, $p = (\log g^{-1})^{q'}$, $0 < q < q'$ (Section 49.1); for γ small, (Γ2) gives $8C_w L^d C_T p_0 \le p$, hence $4w \le \frac{1}{2}h_k$ (Remark 53.5, Consequences). The cutoffs χ^{w} and χ_ξ are not invariant under $\mathcal{G}'$ (Remark 52.6(c)).

that of asymptotic freedom (proof of Lemma 57.7), and its value agrees with the heat-kernel coefficient of the continuum one-loop term (Remark 57.8).

The gauge group, Part by Part. The hypotheses on G grow along the note (Section 1).

- *Parts I–VI:* G compact and connected.
- *Parts VII–X:* G also semisimple; Part VII does not use it (Section 31). An $\mathrm{Ad}(G)$-invariant vector of $\mathfrak{g}^c$ then vanishes, so a normalized, gauge-invariant activity has no term of first order and is of second order in the curvature, the current and the complex part (Lemma 38.5 and its proof). This makes the one-loop source marginal rather than relevant (Remark 41.8(b)), and it is why groups with abelian factors are not treated.
- *The flow (Sections 50–53) and Part XI:* G simple. A quadratic form in a constant curvature that is invariant under conjugation by G and under B_4 is then a single multiple of the isotropic form (Lemma 50.10(a)). So the period density of a generation is β_E times that of S (Proposition 50.13), and the running coupling is one number. Part XI uses the same fact in (57.1).

The step and the small-field model. The renormalization group map T sends a density of the configurations V on $\mathbb{T} = \mathbb{T}^{(k)}$ to the image density of V' on $\mathbb{T}' = \mathbb{T}^{(k+1)}$ under Bałaban's one-step averaging $\mathfrak{a}'$ (31.1). The note iterates the *small-field step* $T_{\mathrm{sf}}(\rho) = T^{\mathrm{gf}}(\chi^{\mathrm{w}}\chi_\xi\,\rho)$ (Definition 48.1). Here T^{gf} is T with Bałaban's exponential gauge-fixing function gf_W (Definition 32.5); on the coarse configurations of Lemma 31.6 it agrees with T on integrands that are invariant under the gauge transformations $\mathcal{G}'$ of the blocks and vanish outside Bałaban's regular set $\mathfrak{M}$ (Lemma 32.6(iv)). The functions χ^{w}, χ_ξ are the cutoffs of Table 3. The densities of the small-field model are $\rho_{k+1} = \mathbf{1}_{\mathfrak{U}_{k+1}(h_{k+1})} T_{\mathrm{sf}}(\rho_k)$ (Remark 52.6(c), Theorem N). The threshold h_k and the bound w tend to 0 with the coupling, while p_0 grows; so the small-field domain shrinks with the coupling, unlike Bałaban's (Remark 53.14(a)).

The model as one integral. With $V_0 = V$, $V_{k+1} = \mathfrak{a}_{k+1}(V_k)$ and W_k the base of V_{k+1}, the small-field model unrolls to one integral over the configurations of scale 0 (Remark 52.6(c)):

$$\int \rho_K\, dV_K = \int \prod_b dV(b)\ \rho_0(V) \prod_{k<K} \Big[\mathrm{gf}_{W_k}(V_k)\, \chi_k^{\mathrm{w}}(V_k)\, \chi_{\xi,k}(V_k)\, \mathbf{1}_{\mathfrak{U}_{k+1}(h_{k+1})}(V_{k+1})\Big].$$

The cutoffs of step k depend on V_{k+1} as well as on V_k, and they are not invariant under $\mathcal{G}'$, so the small-field steps define a modified model. What a step discards is nonnegative

(Lemma 32.6(iv)); hence the partition function of the small-field model is at most that of the lattice theory with the same counterterm, $Z_{\mathrm{sf}} = \int \rho_K \le \int \rho_0 = Z_{\mathrm{lat}}$ (Remark 52.6(c)).

The flow in words: Theorem N. Theorem N summarizes Theorem 53.12, Propositions 50.13, 51.5, 53.10 and Corollaries 53.11, 53.13. Hypotheses: G simple; the parameters as in Remark 49.8 (Section G11); γ subject to (Γ1)–(Γ3) of Remark 53.5; the last torus of side $n_K \ge n_{\min}$, with ℓL^N a power of L (Section G2); $g_0 \in (0, \gamma]$. Then, at every scale k such that $g_1, \dots, g_k \le \gamma$:

- on the small-field set, $\rho_k(V) = \mathcal{Z}_k \exp\big(- g_k^{-2} S(U_k) - \varepsilon_k |\mathbb{T}^{(k)}| + \sum_{j \le k} (E^{(j)} - \beta_j S)(U_k)\big)$ at the background field $U_k = U_k(V)$: Bałaban's generation form (Theorem N(i));
- each generation $E^{(j)}$ is bounded once, at its creation scale, by $\mathsf{K} e^{-\kappa_n d_j(X)}$ on domains whose constants do not depend on the coupling, with the rate κ_n of Remark 49.8(1) and the tree length $d_j(X)$ of the polymer X (Section 49.2; Theorem N(ii));
- β_j is read off from the period density of $E^{(j)}$, which is exactly isotropic; $|\beta_j| \le C_\beta \mathsf{K}$, $g_j^{-2} = g_{j-1}^{-2} - \beta_j$ and $\frac{3}{4} g_{j-1} \le g_j \le 2 g_{j-1}$ (Theorem N(iii), (v));
- at a later scale $j + m$, the small polymers of $E^{(j)}$ in a grid cube $\square$ add up to $\beta_j S_\square$ up to $C \mathsf{K} L^{-\alpha m}$ for $m \ge m_0$, the age from which the small polymers are extracted (Definition 49.5), and the other polymers carry a factor L^{-m} (Theorem N(iv));
- every step satisfies the hypotheses of Theorem L (Theorem N(v)), and the generations and coefficients depend continuously on the couplings (Theorem N(vii)).

The constants depend only on the parameters fixed before γ (one also on p_0), not on k, K, the torus or the couplings (Theorem N). No behaviour of the couplings beyond $g_k \le \gamma$ is assumed (Remark 53.14(a)).

What Theorem N(vi) says. Let $g_1, \dots, g_K \le \gamma$, and fix the counterterm E_0 by $\varepsilon_K = 0$; exactly one E_0 does this. The partition function of the small-field model is then $\mathcal{Z}_K I_K$, with an explicit product $\mathcal{Z}_K$ of normalizations and $e^{-c|\mathbb{T}^{(K)}|} \le I_K \le e^{c|\mathbb{T}^{(K)}|}$, where c depends on the last coupling g_K and on the parameters, not on N, K, the torus or $g_0, \dots, g_{K-1}$ (Corollary 53.13). So, at a fixed last coupling, $I_K = e^{O(|\mathbb{T}^{(K)}|)}$ uniformly in the number of steps, with $|\mathbb{T}^{(K)}| = n_K^4$ the volume in the units of the last scale. The bound is crude, per unit volume, which suffices because the final torus is fixed (Remark 52.6(a)); $\mathcal{Z}_K$ is not compared with the normalizations of a Gaussian reference (Remark 52.6(b)).

The ultraviolet limit in words: Theorem M(iv). This is Theorem 58.9. Let G be simple, the parameters and the torus as in Theorem N, $p_0(g) = (\log g^{-1})^q$, and γ subject to (Γ1)–(Γ4) and the two further conditions of Theorem 58.9. There are $g_{\mathrm{uv}} > 0$ and C, independent of K and of the torus, such that for $0 < g \le g_{\mathrm{uv}}$ and every K some bare coupling g_0, depending on K, g and the torus, gives a flow in $(0, \gamma]$ with $g_K = g$ and

$$\Big| \frac{1}{g_k^2} - \frac{1}{g^2} - (K - k)\beta_\infty \Big| \le C \big(\log(K - k + 2) \big)^{6q+1} \qquad (k \le K).$$

At $k = 0$: $g_0^{-2} = g^{-2} + K\beta_\infty$ up to a power of $\log(K+2)$, so the bare coupling tends to 0 as the number of steps grows. With $K = N$ the last lattice is the unit lattice of the torus of physical side ℓ, so $n_K = \ell$ and the torus conditions read $\ell \ge n_{\min}$, ℓ a power of L; letting $N \to \infty$ at fixed ℓ shrinks $\epsilon = L^{-N}$, and the bound is uniform in N (Section 52.1). The theorem concerns the couplings of the small-field model (Remark 58.10(a)). Theorem M(i)–(iii) compute β_∞ (Section G10).

No step of the main line is conditional. The hypotheses of Theorems K and L on the density, the coupling and the old activities hold at every step of the flow (Theorem 53.7(a), Theorem 53.12(e)). The hypothesis $g_k \le \gamma$ of Theorem N holds, up to the last scale, for the bare coupling of Theorem 58.9. The properties (F1)–(F3) that Theorem 58.9 uses are proved in Proposition 58.6. Proposition 58.8 is proved without a rate, and Theorem 58.9 does not use it (Remark 58.10(c)).

What is not proved. The large-field stage and the removal of the small-field modification (Remark 52.6(d)), hence any statement on the partition function or the correlation functions of the lattice theory itself (Section 1, "What is not done here"). Any limit of the measure or of correlation functions as $\epsilon \to 0$: Theorem 58.9 concerns the couplings only. The exact lower half of the analogue of Bałaban's (0.31) for the last m_1 steps (Remark 58.10(c)). A rate, or uniformity in the couplings, in Proposition 58.8 (Remark 59.13). Smoothness in the couplings; only continuity is proved (Proposition 53.10). Bounds linear in the size for gauges on cubes of size $c\,\varepsilon_1^{-1}$; the bounds are polynomial (Remark 30.6). Gauge groups with abelian factors, and dimensions other than 4 (Section 1).

Bałaban's theorems. Four of the theorems that the note cites by number concern the variational problem and the flow.

- [8, Theorem 1], the variational theorem: Theorem I is its content for one step, given a reference, and Theorem J its version on the torus for every k, without nested domains (Section 1, after Theorems I and J).
- [18, Theorem 1] asserts the generation form (0.22)–(0.24) with terms that satisfy the later-scale bounds (0.29). [18, Theorem 3] is the inductive statement of the flow under $0 < g_k \le \gamma$; Theorem 53.12 is its analogue for the small-field model on the torus (Remark 53.14(a)).
- [18, Theorem 2], stated for $SU(2)$, gives a bare coupling with $\frac{1}{g^2} + \beta \log(L^k\varepsilon)^{-1} \le \frac{1}{g_k^2} \le \frac{1}{g^2} + \beta' \log(L^k\varepsilon)^{-1}$, $0 < \beta \le \beta'$: his (0.31), where $\log(L^k\varepsilon)^{-1} = (K-k)\log L$ for $\varepsilon = L^{-K}$. Theorem 58.9 gives the analogue for the couplings of the small-field flow when $K - k \ge m_1$, and its upper half for all k. Bałaban's couplings come from his own generations, a different construction (Proposition 58.8(b); Section 1, after Theorem M).

Where in the note. Section 1; Remarks 52.6, 53.14 and 58.10; Theorems 53.12 and 58.9, Corollary 53.13.

In the companion. The chapters on the second halves of Parts X and XI.

G2 Scales, units and where the fields live

The tower and the units of scale k. On the torus of physical side ℓ live the lattices $\mathbb{T}^{(0)}, \dots, \mathbb{T}^{(K)}$ of the physical spacings $\epsilon, L\epsilon, \dots, L^K\epsilon$, with $\epsilon = L^{-N}$; the steps go from $\mathbb{T}^{(k)}$ to $\mathbb{T}^{(k+1)}$, $k < K \le N$, and for $K = N$ the last lattice is the physical unit lattice (Section 52.1). Figure G2 shows the tower. Statements about scale k are written in its units. Then $\mathbb{T}^{(k)}$ is the unit lattice, the torus has side $n_k = \ell L^{N-k}$, a lattice $\mathbb{T}^{(i)}$ with $i \ge k$ has spacing L^{i-k}, and the fine lattice $\mathbb{T}_\eta$ has spacing $\eta = L^{-k}$ (Sections 52.1 and 49.2). The fine lattice is $\mathbb{T}^{(0)}$ at every scale; only its spacing in the units changes. A change of units changes the domains and the norms in which an activity is measured, not the activity (Section 49.2). Read in the units of scale $j < k$, a point of a domain of scale k lies in a domain of scale j whose bound

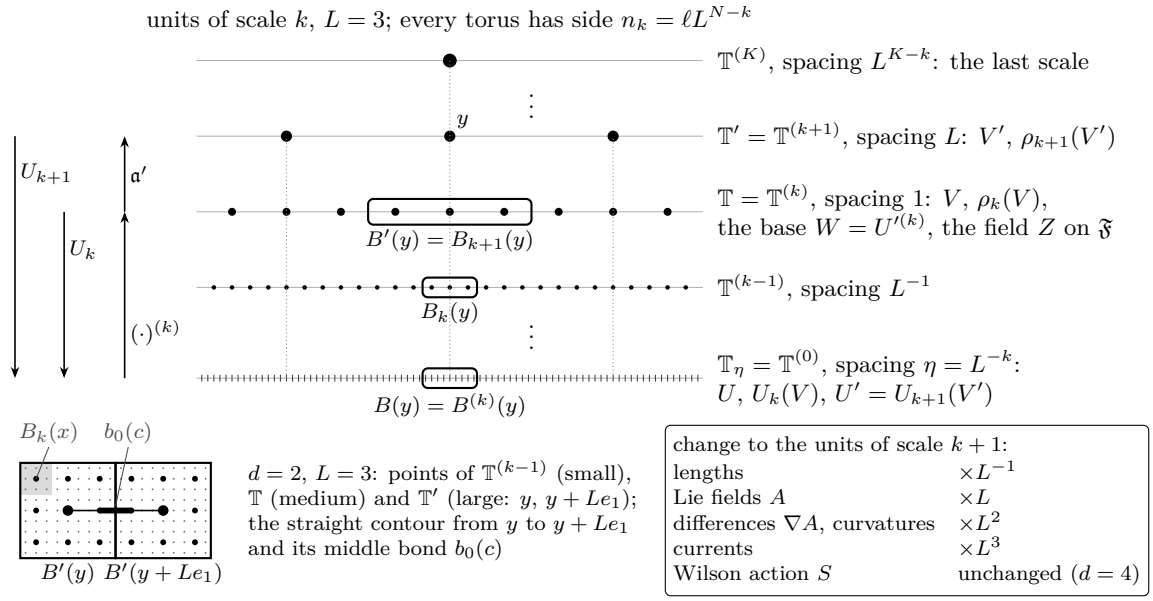


Figure G2: Scales and units. Everything is drawn in the units of scale k, for $L = 3$: the tower in a one-dimensional section, the inset for $d = 2$. In these units $\mathbb{T}^{(j)}$ has spacing L^{j-k}, and all lattices of the tower lie on one torus, of side $n_k = \ell L^{N-k}$ (Section 52.1). In its own units, $\mathbb{T}^{(j)}$ has spacing 1 and side $n_j = \ell L^{N-j}$; in Part VII, n denotes the side of $\mathbb{T}'$ in its own units, $n = n_{k+1} = n_k/L$. The lattices are nested, and the blocks are centred: $B_j(y)$ is the cube of L^d points of $\mathbb{T}^{(j-1)}$ centred at $y \in \mathbb{T}^{(j)}$, and $B(y) = B^{(k)}(y)$ is the block of Part I (Section 21). The configurations of step k live on $\mathbb{T} = \mathbb{T}^{(k)}$, the background fields on $\mathbb{T}_\eta$. Arrows on the left: the background fields $V' \mapsto U_{k+1}(V')$ and $V \mapsto U_k(V)$ (Theorem 30.3); the k-fold average $U \mapsto U^{(k)}$ and the one-step averaging $\mathfrak{a}' = \mathfrak{a}_{k+1}$ (Definition 21.8). The base of the step is $W = U'^{(k)}$, and $\mathfrak{a}'(W) = V'$ (Lemma 31.6). Inset: the blocks $B'(y)$ and $B'(y + Le_1)$, one block $B_k(x)$ shaded, and the middle bond $b_0(c)$ of the straight contour of the coarse bond c from y to $y + Le_1$ (Definition 32.1 and Figure 4). Box: the change of units of the proof of Lemma 30.2, from scale k to scale $k + 1$; the Wilson action carries no power of the spacing in four dimensions (Section 31).

Symbol	Meaning	In the units of scale k	Where
$\epsilon = L^{-N}$	spacing of the finest lattice $\mathbb{T}^{(0)}$, physical units; N is the number of scales	$\eta = L^{-k}$	§52.1
ℓ, n_k	physical side of the torus; its side in the units of scale k	$n_k = \ell L^{N-k}$	§52.1
$\mathbb{T}^{(i)}$	lattice of scale i; side n_i in its own units	spacing L^{i-k}	§49.2
$\mathbb{T}_\eta$	fine lattice, $\mathbb{T}^{(0)}$; background fields and activities live on it	spacing η	§49.2
$\mathbb{T}$, $\mathbb{T}'$, n	in Part VII: $\mathbb{T}^{(k)}$, $\mathbb{T}^{(k+1)}$, and $n = n_{k+1} = n_k/L$ the side of $\mathbb{T}'$; in Parts VI, VIII, IX, $n = n_k$	spacings 1, L	§31, §52.1
$N = L^k$	in Parts I–VII, except Section 31.1: the number of fine points per unit length, with $\eta = 1/N$; unrelated to the number of scales	—	§2
$B(y)$, $B'(y)$	block of $\mathbb{T}_\eta$ centred at $y \in \mathbb{T}$; block of L^4 points of $\mathbb{T}$ centred at $y \in \mathbb{T}'$	sides 1, L	§2, Def. 32.1
grid cube $\square$	union of M^4 blocks of scale j; polymers of scale j are $*$-connected unions of them	side $L^{j-k}M$	§49.2

Table 4: Scales and units. The fourth column gives the place of the definition.

on the curvature and the current of the real part, and whose bound on the gradient of the complex part, are smaller by $L^{-2(k-j)}$, and whose bound on the complex part is smaller by $L^{-(k-j)}$ (Definition 38.3, Remark 38.8(i)). The Wilson action carries no power of the spacing, so a change of units does not change the coupling (Section 23).

Where the fields live. Table 4 collects the lattices.

- The density $\rho_k(V)$ is a function of a configuration V on $\mathbb{T}^{(k)}$ (Theorem N(i)). It is evaluated at the background field $U_k(V)$, a configuration on the fine lattice: an element of the minimal orbit of the k-step variational problem with averages V (Theorem 30.3). So $S(U_k(V))$ and every activity are functions on the fine lattice.
- One step goes from V on $\mathbb{T} = \mathbb{T}^{(k)}$ to $V' = \mathfrak{a}'(V)$ on $\mathbb{T}' = \mathbb{T}^{(k+1)}$. Its reference is the background field $U' = U_{k+1}(V')$, and its base is $W = U'^{(k)}$, the average of U' on $\mathbb{T}^{(k)}$. For V' of small curvature, $\mathfrak{a}'(W) = V'$ and $U_k(W) = U'$ (Lemma 31.6).
- On the support of χ^{w}, $V = e^{\mathrm{i}Z}W$ bond by bond (Definition 32.4). The values of Z on the middle bonds $b_0(c)$ of the coarse bonds c are fixed by the constraint $\mathfrak{a}'(V) = V'$, and the fluctuation variables are the values on the other bonds of $\mathbb{T}^{(k)}$, the set $\mathfrak{F}$ (Definition 32.1, Lemma 32.7). The white noise is $\xi = g^{-1}C_k^{-1/2}Z$, with the fluctuation covariance C_k at the reference U' (Definition 48.1).
- A generation $E^{(j)}$ is a family of activities on the polymers of scale j, bounded on the domains of scale j only. At a later scale k it is evaluated at $U_k(V)$, which is again a background field of scale j, of the average of $U_k(V)$ on $\mathbb{T}^{(j)}$ (Section 50, Lemma 50.3(b)).

The torus conditions. At each scale k the side n_k satisfies (Section 52.1):

- (T0) $n_k \geq n_*(L, M, \rho)$, the largest of the torus conditions of the results used; here ρ is the scale of the walk domains of Part IX (Section 40), not a density;
- (T1) $LM \mid n_k$;

- (T2) $n_k \geq 8LM$, so that the polymers with $d_M(X) \leq L$ do not wrap around the torus.

All three hold at every scale $k \leq K$ if the last torus satisfies $n_K \geq n_{\min} = \max(8LM, n_*)$ and ℓL^N is a power of L (Section 52.1, Lemma 52.1(a)). Polymers of diameter larger than $\frac{1}{2}n_k - 8$ are *wrapping*; they are bounded at all later scales by Lemma 52.1(c)–(e).

What "uniform" means. The constants of the flow depend only on the parameters of Remark 49.8(1)–(5) (one of them also on p_0), not on k, K, N, the torus or the couplings (Theorem 53.12), and all bounds are uniform in N and K, given $n_K \geq n_{\min}$ and ℓL^N a power of L (Remark 53.14(d), Section 52.1); in particular as $N \to \infty$ at a fixed physical side $\ell \geq n_{\min}$, a power of L, with $K = N$.

Where in the note. Sections 52.1 and 49.2; the conventions of Section 31; Lemmas 31.6, 52.1 and 50.3; Remark 38.8(i).

In the companion. The chapter "How to read this volume" (its notation table), and the parameter ledger (the torus).

G3 Four-dimensional power counting and the strategy of the flow

The coupling is marginal. In four dimensions the Wilson action is dimensionless, so a change of units does not change the coupling. The coupling of the next scale is produced by the fluctuation integral, through the coefficient β of S in the effective density (Section 1, the paragraph on Part VII; Section 23). Each step creates terms of order one: the one-loop activities and the Jacobian terms are bounded independently of the coupling, and they are not small (Theorem 48.9). The iteration does not contract them (Remark 49.1).

Power counting of an activity. A normalized, gauge-invariant activity has no constant and no linear term: it vanishes at $U = 1$ by normalization, and its first order vanishes for semisimple G (Lemma 38.5). Its orders in the curvature behave as follows at the later scales (Remark 41.8(b), Section 51).

- *Order* 0. The constants are removed by the normalization and go into the vacuum energy ε_k. It obeys $\varepsilon_{k+1} = L^4\varepsilon_k + \delta_k$, and a single counterterm E_0 in the bare density makes $\varepsilon_K = 0$ (Remark 48.10(a), Proposition 52.5).
- *Order* 2*: marginal.* Read in the units of a polymer of scale j, the curvature of a configuration of scale $k = j + m$ is of order L^{-2m}, so the quadratic part of one polymer is of order L^{-4m}. A grid cube of scale k contains L^{4m} grid cubes of scale j. So the quadratic part of a generation is of order one per cube at every later scale. It must be extracted as a coupling constant, not bounded (Remark 41.8(b)).
- *Orders* ≥ 3*: irrelevant.* The orders ≥ 3, together with the non-constant part of the curvature in the quadratic order, contribute $O(L^{-(4+\alpha)m})$ per polymer (Lemma 51.1(d)), hence $O(L^{-\alpha m})$ per cube of scale k (Section 51).

In the momentum expansion of a vacuum polarization a term linear in p would be relevant. It is excluded by the symmetry of Bałaban's averaging (Lemma 54.9, Remark 58.10(b)); see Section G14.

Generations. The note follows Bałaban's organization of the flow [18, (0.22)–(0.30)]. The new terms of the step that creates scale j are kept, as a family on the polymers of scale j, in a separate generation $E^{(j)}$ (Section 50). A generation is bounded only at its creation scale, on domains whose constants do not depend on the coupling (Theorem N(ii)). At every later scale only its argument changes: it is evaluated at the background field of that scale (Section 50, Lemma 50.3). Figure G4 shows the life of one generation.

The counterterm and the isotropy. The generation $E^{(j)}$ enters the density together with the counterterm $-\beta_j S$, and the coupling runs by $g_j^{-2} = g_{j-1}^{-2} - \beta_j$ (Theorem 53.7(c)). The number β_j is read off from the period density of $E^{(j)}$, the second variation of its activities at the linear fields of constant curvature, per unit volume (Definition 50.11). A step does not produce a covariant family. The new generation is the average of the raw one over the symmetries of its grid, which keeps the sum and the bound, on a slightly smaller domain (Definition 50.7, Lemma 50.8). For simple G its period density is then exactly β_j times that of S, with no anisotropic and no topological part (Proposition 50.13). Exactness is what the extraction uses: at a later scale the cell sums of the generation are cancelled by those of $\beta_j S$, because the two period densities are proportional (Section 51, "Cancellation").

Extraction at later scales. Let $k = j + m$ and let $\square$ be a grid cube of scale k (Proposition 51.5).

- On a small polymer X of scale j, in a complex gauge chosen for X, a configuration of scale k is a linear field of constant curvature plus a remainder that is smaller by $L^{-\alpha m}$ (Lemma 51.1). So $E^{(j)}(X, U)$ is a quadratic form in the curvature at one point of X, up to $O(L^{-(4+\alpha)m})$.
- By the grid covariance these forms add up, cell by cell, to the period density, and so does $S_\square$ (Lemmas 51.3 and 51.4). The two cancel. Over the L^{4m} cells the error is $O(L^{-\alpha m})$, for $m \geq m_0$.
- The recent polymers ($m < m_0$), the polymers too large for this gauge and the wrapping ones are not expanded; they carry the factor L^{-m} (Lemma 51.7). For $m < m_0$ no polymer is small, and the counterterm $-\beta_j S_\square$ alone is bounded (Proposition 51.5).

In each grid cube of scale k the counterterm $-\beta_j S_\square$ is grouped with the small polymers of $E^{(j)}$ in $\square$. The resulting old activities $\mathcal{E}_k$ are bounded by $C_{\text{old}}\mathsf{K}$, uniformly in k: the sum over the generations converges because of the gains $L^{-\alpha m}$ and L^{-m} (Definition 53.1, Proposition 53.2(b)). No contraction is used (Remark 49.1).

Where smallness comes from. The activities are of order one. What is small is the following.

- *The fluctuation cutoff.* The fluctuation field is at most $w \propto g\, p_0(g)$. The old activities enter the new generation only through the mixed sources, of size τ times their bound, with $\tau \leq C_P L^d g_k p_0(g_k)/\delta_1$ (Proposition 53.2(d), Remark 58.7(a)). Here τ is the relative change of an old activity under the background map of the step, $|f| \leq 4\tau \|\mathcal{E}(X)\|$ (Proposition 47.10), C_P a constant of the step that depends on κ, δ, L, M, ρ, α, but not on k, n, η or s (Section 47, Setting), and δ_1 the gradient budget of the domains (Definition 38.3). Remark 49.1 attributes the smallness to this cutoff, as in [19, (2.41)].
- *Parity.* The pure sources are even in the coupling and vanish at coupling 0, so they are $O(g^2 p_0^6)$; there is no term of order g (Proposition 48.8). Without parity the bound (F3) on the coefficients would be only $O(g\, p_0^3)$, which is not summable along the flow (Remark 58.7(b)).

- *The gains at later scales:* $L^{-\alpha m}$ after the extraction and L^{-m} for the other polymers (Proposition 51.5, Lemma 51.7).

For the couplings the same sources give: the part of β_{j+1} beyond one loop is $O(g_j^2 p_0 (g_j)^6)$ (Proposition 58.6, (F3)); the one-loop part equals the infinite-volume coefficient up to $e^{-c_2 n_{j+1}/M}$ (Proposition 58.6, (F2)); and the one-loop parts add up to $k\beta_\infty + O(1)$ (Theorem 58.1). Along the flow of Theorem 58.9 the errors add up to $O\big((\log(K-k+2))^{6q+1}\big)$.

What weakening buys. A polymer expansion needs each quantity of the step as a sum over polymers, with a small factor for every box of the region on which a term depends: localization of volume type. Dirichlet approximants and telescoping (Part VIII) give only radius type, a factor $e^{-\gamma m}$ for a piece that depends on a ball of radius m (Section 1, the paragraph on Parts VIII and IX). Part IX writes the operators of the step as analytic functions of weakening parameters $s = (s_\square)$ in $\mathbb{D}_\kappa = \{|s_\square| \le e^\kappa\}$, one parameter per box (Sections 40 and 41). Lemma 47.2 then localizes: if f is analytic on $\mathbb{D}_\kappa$, bounded by F, and, when $s_\square = 0$ outside a union of boxes, depends only on the parameters and the data of the component of that union that contains the polymer Y, then

$$f(1) = \sum_{Z \supset Y} f_{\mathrm{loc}}(Y, Z), \qquad |f_{\mathrm{loc}}(Y, Z)| \le F\, e^{-(\kappa-1)|Z \setminus Y|},$$

and $f_{\mathrm{loc}}(Y, Z)$ depends only on the data near Z. The proof is the fundamental theorem of calculus in each $s_\square$ outside Y, followed by a Cauchy estimate in each differentiated variable: every box of $Z \setminus Y$ costs $e^{-(\kappa-1)}$.

The budget of one step. The step from scale k to scale $k+1$ produces four kinds of new terms (Theorem 48.9): one-loop activities $\mathfrak{e}(X)$, bounded by $Ce^{-(\kappa_2-1)|X|}$ with C independent of the coupling for the polymers that do not wrap, with marginal quadratic parts (Proposition 41.7); Jacobian terms $\log \mathfrak{j}_c$, bounded by a constant (Proposition 47.8); pure sources $E_{\mathrm{p}}^{\#}$ of order $g^2 p_0^6$ (Proposition 48.8); and mixed sources $E_{\mathrm{m}}^{\#}$ of order τ times the bound on the old activities (Proposition 53.2(d)). Reblocked, they form the raw new generation, with $\|E'\|_{k+1}^{\mathrm{raw}} \le B_{\mathrm{new}} + C_{\mathrm{new}}(\epsilon_{\mathrm{p}} + \epsilon_{\mathrm{m}})$ (Lemma 53.4(b)). Here ϵ_{p} and ϵ_{m} bound the pure and the mixed pieces of the cluster expansion (Proposition 48.5), C_{new} is a constant of the parameters (Remark 53.5(ii)), and $B_{\mathrm{new}}^{(0)}$ is the value of B_{new} for the first step. The first term comes from the one-loop and Jacobian bounds; it does not depend on the domain constants, on m_0, on the couplings or on the torus (Remark 53.5(i)). The bound of every generation is $\mathsf{K} = 2\max(B_{\mathrm{new}}, B_{\mathrm{new}}^{(0)})$, and condition ($\Gamma$3) on γ makes the sources fit into it (Remark 53.5, Theorem 53.7(b)).

Where in the note. Remark 49.1; the openings of Sections 50 and 51; Remark 41.8; Lemma 47.2; Propositions 50.13, 51.5, 53.2 and 58.6; Remark 58.7.

In the companion. The chapters on the two halves of Part X.

G4 The linear theory (Parts I–III)

Every step of the renormalization group uses a few linear operators on the fine lattice $\mathbb{T}_\eta$ of spacing $\eta = L^{-k}$: a Green function with an averaging term, the minimizer map, the constrained Green function and the inverse of an averaged Green function. Parts I–III prove that these operators decay exponentially and satisfy pointwise bounds with one derivative and a Hölder quotient, on one torus after k averaging steps, with constants independent of k and of the side

n of the torus. The settings are the trivial background $U = 1$ (Part I), backgrounds close to 1 in one global gauge (Part II), and Bałaban's locally regular backgrounds (Part III). The results are Theorems A–G.

Part I. Theorems A–E hold with constants that depend only on a and the Hölder exponent α, not on L, k or n (the conventions before Theorem A). Three mechanisms carry Part I.

- *The averaging term supplies what the Laplacian misses.* A Poincaré inequality on unit blocks shows that $a\|Q_\sigma f\|^2$ controls exactly the modes that $\|\partial f\|^2$ does not (Lemma 3.1, Corollary 3.4). For gauge fields, $\|dA\|^2 + a\|\mathbf{Q}A\|^2$ is invariant under $A \mapsto A + d\lambda$ with $Q_0\lambda = 0$. The term $\|Rd^*A\|^2$ of Bałaban's Landau-type gauge $Rd^*A = 0$ removes this degeneracy (Lemma 9.1, Theorem 9.6).
- *Combes–Thomas.* If an operator is coercive and conjugation by $e^{\alpha\rho}$, with ρ Lipschitz, changes its form by $O(\alpha)$, then its inverse decays exponentially (Lemmas 5.5 and 5.7).
- *Interior regularity on the fine lattice.* A lattice Schauder estimate (Theorem 6.3) turns the L^2 decay between unit blocks into bounds on values, derivatives and Hölder quotients, uniformly in η.

Part II. Here $U = e^{i\eta A}$ with $|A|, |\partial A|, [\partial A]_\alpha \le \varepsilon$ in one global gauge, hypothesis (B) of Section 12. Part II uses a model averaging pair $(Q'(U), \mathbf{Q}(U))$, which intertwines the covariant derivatives up to a defect $E(U) = O(\varepsilon)$ (Lemma 12.5). Every operator is its $U = 1$ counterpart plus an admissible perturbation (Definitions 13.1 and 13.4). Combes–Thomas, the Neumann series and the regularity bootstrap are stable under such perturbations (Lemmas 13.2, 13.3 and 13.5). The result is Theorem F, with constants independent of L, k, n and U.

Part III. Part III replaces (B) by Bałaban's local condition: U is (M, ε)-regular if every cube of side of order M carries its own gauge in which (B) holds (Definition 18.1).

- *Dirichlet operators on domains.* The scalar and the gauge-field operator have versions on any union X of blocks (Definition 17.1), which depend only on the background on the bonds of $\hat{X}$, the points at ∞-distance at most 1 from X. They are coercive, the gauge-field version with a mass term near the boundary and Dirichlet conditions for all components. Their inverses, the inverse of the averaged scalar Green function and $\mathbf{1}_X - R_X$ decay exponentially, uniformly in X (Theorems 18.2, 19.6 and 19.7). The scalar coercivity is proved block by block. The gauge-field coercivity comes from an IMS localization at scale M (Lemma 17.4), which reduces it to single cubes, the local gauge and $U = 1$.
- *Random walks.* Bałaban's random-walk expansion converges (Theorem 20.2); for the gauge field it is completed in Part IX. Parts I–III prove their bounds without it (Section 20, after Theorem 20.2); in Part IX the weakened walk gives both the localized representation and the bounds (Theorem 41.4).

The result is Theorem G; the threshold M_0 and the smallness of ε do not depend on M (Theorem 19.6).

The model pair as a proof device. The proofs of Parts II and III use the averaging pair only through five properties (Remark 19.12). The linearization of Bałaban's symmetric averaging has (i)–(iv) with $E = 0$ (Proposition 22.10), and (v) at small complex perturbations (Remark 19.12). So the real results of Parts II and III, and Theorem 20.2, hold for it (Corollary 22.11).

Where in the note. Part I: Sections 2–11 and Appendix A. Part II: Sections 12–16. Part III: Sections 17–20. The companion has no chapter on Parts I–III.

G5 The background field (Parts IV–VI)

Let V be a real configuration of small curvature on the unit lattice $\mathbb{T}$ of scale k. Its background field $U_k(V)$ minimizes the Wilson action S on the fine lattice $\mathbb{T}_\eta$ among the configurations of small curvature and current whose k-fold average is V. Parts IV–VI construct it. They prove that its orbit is the only critical orbit among these configurations, and that it is regular, analytic in V and covariant under the lattice symmetries, uniformly in k (Theorem J, which is Theorem 30.3 with Corollary 30.4; analyticity: Corollary 30.5).

The averaging and the minimal orbit. Part IV uses Bałaban's symmetric averaging. Its blocks are centred at the coarse points. The transport from a block centre y to a point x of the block is the group average Av (Lemma 21.5) of the transports along the coordinate-ordered contours from y to x, one for each order of the directions (Definitions 21.1 and 21.6). The k-fold average $U \mapsto U^{(k)}$ is covariant under the values of gauge transformations at the points of $\mathbb{T}$, and under the group $\mathfrak{S}_k$ generated by the translations of $\mathbb{T}$ and by B_4 about its points (Lemma 21.10). So the residual gauge group of the variational problem is $\mathfrak{G}_0 = \{u : u|_{\mathbb{T}} = 1\}$, and the minimizer is unique only as a $\mathfrak{G}_0$-orbit, the *minimal orbit*. One writes $U_k(V)$ for any element of it.

The bridge to the linear theory (Section 22). The linear theory applies to an averaging pair with the properties of Remark 19.12. Part IV produces one, with exact covariance, in three moves.

- The *averaged axial gauge* asks that the averaged contour variables, relative to the background, be 1 at every scale (Definition 22.2).
- The *gauge-adapted averaging* $\mathcal{Q}$ passes to this gauge before each averaging step, through Bałaban's correspondence transformation (Definition 22.4, Theorem 22.5).
- The linearization at a regular reference U_0 is an averaging pair $(Q'_k(U_0), \mathbf{Q}_k(U_0))$ with exact covariance, $\mathbf{Q}_k(U_0)D_{U_0} = D_{W_k}Q'_k(U_0)$, that is $E = 0$, and with covariance under $\mathfrak{S}_k$ (Proposition 22.10). So Parts II and III apply to it (Corollary 22.11).

With the expansion of the Wilson action near a regular configuration (Proposition 23.3) this gives the chart, Bałaban's axial-to-Landau theorem (Theorem H, which is Theorem 24.2). So the $\mathfrak{G}_0$-orbits near U_0 are parametrized by fields in the Landau gauge (Corollary 24.3).

The relative problem (Part V). Theorem I (Theorem 25.1) solves the variational problem near an (M, ε)-regular reference U_0 with small current, for real coarse configurations V whose averages differ from those of U_0 by $e^{\mathrm{i}B}$, $|B| \le \beta$. In the Landau chart the constraint is nonlinear. The linearizing transformation $A = A' - \mathbf{H}\mathcal{D}(A')$ turns it into the linear constraint $\mathbf{Q}A' = B$ (Lemma 26.1). A critical point is then a fixed point $A_1 = -\mathbf{C}(J + \nabla\mathcal{V}(A_1 + \mathbf{H}B))$ of a contraction (Proposition 26.3). Hence the Landau representative is $A_* = \mathbf{H}B - \mathbf{C}J_{U_0} + O((\varepsilon + \beta)^2)$, and its orbit minimizes the action (Proposition 26.6). The gauge covariance of $\mathcal{Q}$ (Lemma 27.1) makes the critical point critical on the original constraint set, with a Lagrange multiplier ω (Proposition 27.2). The bounds are uniform in the reference, which is what Part VI uses.

The induction over k (Part VI). At step k the reference is the solution of step $k-1$ for the trivial extension V_0 of V, whose averages are exactly V (Lemma 30.1(a)). In the units

of step k, the curvature of V_0 is up to L^2 times that of V (Lemma 30.1(b)); in its own units, those of step $k-1$, V_0 satisfies the same curvature bound as V does in the units of step k. The solution of step $k-1$, returned to the units of step k, has a regularity bound that is worse by a power of L (Lemma 30.2), and Theorem 25.1 keeps this loss. The note stays on the torus and improves the bound locally (Lemma 29.2). Near a point y, the reference is spliced to the trivial configuration (Definition 28.5, Lemma 28.6), in a local gauge adapted to the coordinate-ordered contours from y (Lemma 28.4). In that gauge the coarse field is bounded by the curvature of V alone. Theorem 25.1, applied with the spliced reference, gives the solution near y to first order as $\mathbf{H}$ applied to that coarse field. Since $\mathbf{H}$ decays, the regularity parameter improves from ε_0 to $C_{\mathrm{imp}}\varepsilon_1 + \frac{1}{2}\varepsilon_0$, where ε_1 bounds the curvature of V (Lemma 29.2). Iterating the improvement gives Theorem 30.3: for every k and every real V of curvature at most $\varepsilon_1 \le a_1$, a unique minimal orbit, of curvature and current at most $B_3\varepsilon_1$ and $(M, B_3\varepsilon_1)$-regular, covariant under $\mathfrak{S}_k$. The constants a_1, B_3 depend on L, G, a, α and M, and not on k, n or η. Small curvature and current imply local regularity, so the hypothesis of Part III follows from gauge-invariant conditions (Corollary 30.4).

Gauge	Condition or weight	Role, and where
Bałaban's Landau-type gauge	$Rd^*A = 0$; at a background $R(U_0)D^T_{U_0}A = 0$	Removes the directions $d\lambda$, $Q_0\lambda = 0$, left free by the averaging term (Lemma 9.1). Defines the operators of Parts I–III, the chart and the variational problem (Theorems E, 24.2, 25.1), and the one-loop formula of Part XI (Theorem 55.4).
Averaged axial gauge	averaged contour variables $= 1$ relative to the background, at every scale	Defines the gauge-adapted averaging $\mathcal{Q}$, whose linearization has $E = 0$ (Definitions 22.2, 22.4, Proposition 22.10).
Coordinate-ordered contour gauge Γ^{co}	the coarse field in the gauge of the coordinate-ordered contours from a point y	A device of the proofs of Lemma 29.2 and Proposition 31.5 (Lemma 28.4).
Exponential gauge fixing gf_W	Gaussian weights in the averaged contour variables relative to the base W	Fixes the gauge transformations of $\mathbb{T}$ that are trivial on $\mathbb{T}'$ in one step, with a Faddeev–Popov identity without determinant (Definition 32.2, Lemma 32.3). Parts VII–X.
Background Feynman gauge	gauge-fixing term $D_U D_U^T$	The reference computation of the one-loop coefficient (Section 56, Lemma 57.2).

Table 5: The gauge conditions of the note.

Where in the note. Part IV: Sections 21–24. Part V: Sections 25–27. Part VI: Sections 28–30. Table 5 lists the gauge conditions of the note. The companion has no chapter on Parts IV–VI.

G6 One renormalization group step (Part VII)

Part VII writes one step of the small-field renormalization group exactly as a Gaussian integral around the background field (Theorem 34.9), with a fluctuation covariance that is coercive and decays uniformly in k (Theorem 34.4, Corollary 34.6), a gauge fixing that changes no integral of a gauge-invariant integrand supported on regular configurations, a nonnegative discarded part (Lemmas 32.3 and 32.6), and lattice symmetry (Proposition 31.7). Theorem K collects these results; its constants depend on L, G, a, α and M, and not on k, n, η or the coupling. Figure G3 shows the step and its cutoffs.

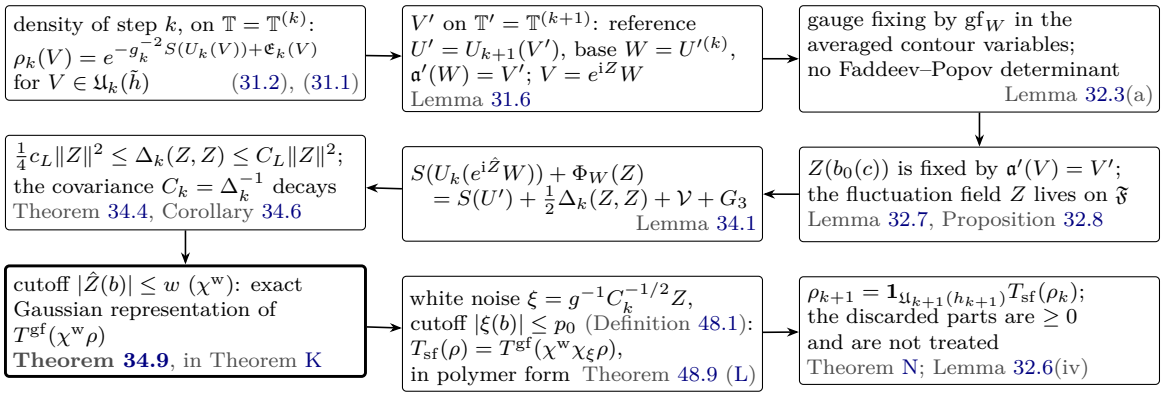


Figure G3: One small-field step, from scale k to scale $k+1$, for $k \ge 1$ (for $k = 0$, Lemma 52.3 and Proposition 52.2); follow the arrows. The gauge fixing and the elimination of the constraint are exact. Theorem 34.9 (thick frame) holds for V' of curvature $\varepsilon_1' \le a_{\mathrm{step}}$, $w \le r_2$ and $a_1 \ge \tilde{h} \ge 2(C_* + 1)L^{-2}\varepsilon_1' + 4w$. The cutoffs act on the variables of the gauge fixing relative to the base, so the steps define a modified model. Along the flow $w = C_w L^d C_T g_k p_0(g_k)$ and $4w \le \frac{1}{2} h_k$ (Remark 53.5). The blocks and the bonds $b_0(c)$ are drawn in Figures 4 and G2.

The step. Let $\mathbb{T} = \mathbb{T}^{(k)}$ and $\mathbb{T}' = \mathbb{T}^{(k+1)}$, both in the units of step k. The map T sends a density ρ of configurations V on $\mathbb{T}$ to the density of the image measure under the one-step averaging $\mathfrak{a}'$ ((31.1)). Fix V' on $\mathbb{T}'$ with small curvature, and let $U' = U_{k+1}(V')$ be its background field, the *reference* of the step. The *base* is $W = U'^{(k)}$, the k-fold average of the reference, a configuration on $\mathbb{T}$ with $\mathfrak{a}'(W) = V'$. The configurations of the step are written $V = e^{\mathrm{i}Z} W$.

Gauge fixing and coordinates. Following Bałaban, gf_W is a product over the blocks of Gaussian weights $e^{-|u-1|^2/2g^2}$ in the averaged contour variables relative to W (Definition 32.2). By a Faddeev–Popov identity without determinant, inserting it changes no integral of an invariant integrand supported on regular configurations (Lemma 32.3(a)). The constraint $\mathfrak{a}'(e^{\mathrm{i}Z} W) = V'$ is solved for Z on the middle bond $b_0(c)$ of each coarse bond c, with local Jacobians $\mathfrak{j}_c$ (Lemma 32.7); the free variables live on the set $\mathfrak{F}$ of the other bonds (Definition 32.1). With these coordinates the gauge-fixed step is, for ρ of the form (31.2) with threshold $\tilde{h}$ and $g = g_k$ (Theorem 34.9),

$$T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(V') = \mathcal{N}_k \exp\Big(-\frac{1}{g^2} S(U') - \frac{1}{2} \sum_{b \in \mathfrak{F}} \ell_k(b) + \sum_c \log \mathfrak{j}_c \Big)$$
$$\times \int d\mu_{C_k}(Y)\, \chi^{\mathrm{w}}(\hat{Z}(gY)) \exp\Big(-\frac{(\mathcal{V} + G_3)(gY)}{g^2} + \mathfrak{J}(gY) + \mathfrak{E}(e^{\mathrm{i}\hat{Z}(gY)} W) \Big).$$

Here μ_{C_k} is the Gaussian measure on the fields on $\mathfrak{F}$ with covariance $C_k = \Delta_k^{-1}$; $\mathcal{V} + G_3$ is an interaction of third order; and $\mathfrak{E}$ is the old part of the density.

Coercivity. The covariance exists and decays because $\frac{1}{4} c_L \|Z\|^2 \le \Delta_k(Z,Z) \le C_L \|Z\|^2$ (Theorem K(ii)). The constant c_L depends only on d and L. It comes from a Poincaré inequality for one step (Lemma 34.2). The gauge-fixing term is necessary in it.

A curvature bound independent of L. The curvature of a background field is at most $(C_* + \delta)$ times the curvature of the coarse configuration, where C_* depends only on a, α and G (Proposition 31.5). This makes the small-field sets of consecutive steps compatible as soon as $L^2 \ge 8(C_* + 1)$, given $h_{k+1} \le 2h_k$ (Lemma 31.6(a), Remark 49.8(2)).

The cutoffs and the modified model. The small-field step is $T_{\mathrm{sf}}(\rho) = T^{\mathrm{gf}}(\chi^{\mathrm{w}}\chi_\xi\rho)$, with two bondwise cutoffs: χ^{w} bounds the field Z relative to the base by w (Definition 32.4), and χ_ξ bounds the white noise $\xi = g^{-1}C_k^{-1/2}Z$ by p_0 (Definition 48.1). The cutoffs χ^{w} and χ_ξ are not invariant under the gauge transformations of the blocks, so the small-field steps define a modified model (Remark 52.6(c)). What a step discards is nonnegative (Theorem K(iii)), so the partition function of the small-field model is at most that of the lattice theory with the same counterterm (Remark 52.6(c)). The large-field stage, which starts from the discarded parts, is not treated (Remark 52.6(d)).

Where in the note. Sections 31–34. In the companion: the chapter on Part VII.

G7 Localization (Parts VIII–IX)

The step of Part VII depends on the background on the whole torus: through the covariance C_k, the coarse Hessian, the determinant $\det \Delta_k$, and the background field of the configuration that is integrated. A polymer expansion needs each of these as a sum of local pieces, analytic and gauge invariant, with a small factor for every cube of the region on which a piece depends. Parts VIII and IX provide this.

- *Radius type.* A piece that depends on the background in a ball of radius m is bounded by $e^{-\gamma m}$. Dirichlet approximants on growing domains, followed by telescoping, give this form (Theorem 37.7).
- *Volume type.* A piece that depends on a polymer X carries a small factor for every box of X. It comes from random walks with weakening parameters (Definition 40.2, Theorem 41.4), and the localization lemma turns it into pieces with a factor $e^{-(\kappa-1)}$ per box (Lemma 47.2).

What Part VIII keeps. Part VIII is written for radius-type locality, which Parts IX and X supersede. On the main line stay the Dirichlet averaged operators (Proposition 35.1), their interior bounds (Theorem 35.2), the Dirichlet Hessians (Propositions 37.4 and 37.6) and the graded domains of the activities (Definition 38.3, Lemma 38.5); the reading guide in Section 1 lists the results off the main line.

Weakened walks (Sections 40–41). Each box $\square$ of a paving carries a weakening parameter $s_\square$ in a disc $\mathbb{D}_\kappa$. A *weakenable family* is a sum $O(s) = \sum_\omega s^{n_\omega} O_\omega$ over polymers X_ω, where O_ω depends on the data only near X_ω and the monomial s^{n_ω} involves exactly the boxes of X_ω (Definition 40.2). The inverse on a domain X is a Neumann series around a parametrix made of local Dirichlet inverses on the domains Λ_j^v, the unions of the blocks within distance ρM of the grid points z_j (Section 40), and each step of the walk carries the monomial $s^{X(j)}$ of a union $X(j)$ of boxes. Under the hypotheses (W1)–(W3) and that of Lemma 41.3, and for M large, this series is a weakenable family that represents the inverse (Theorem 41.4). The walks are nested at five levels, (L1)–(L5), from the scalar Green function to the resolvents of the fluctuation operator, because the non-local part of each operator is built from inverses of earlier levels (Section 41). Each walk costs at most 2δ in the rate κ, and each product at most δ (Corollary 41.6).

Outputs. The weakened square root $C_k^{1/2}(s)$ of the covariance and the determinant densities $\ell_k(b)$ are weakenable families (Corollary 41.6). The one-loop source is a sum over polymers, $-\frac{1}{2}\log\det\Delta_k(U) = -\frac{1}{2}\log\det\Delta_k(1) + \sum_X \mathfrak{e}(X,U)$, of normalized graded activities with volume decay, quadratically small in the curvature (Proposition 41.7). On the domain of the

next step this quadratic part is marginal in four dimensions, so it must be extracted as a coupling constant, not bounded (Remark 41.8(b)).

Tiers and the background map (Sections 42–45). The weakened background map needs pointwise $C^{1,\alpha}$ bounds for the weakened $\mathbf{H}(s)$, $\mathbf{C}(s)$ and $\mathbf{C}(s)D^T$, uniformly on $\mathbb{D}_\kappa$ ((42.1)). In four dimensions L^2 data sit at the endpoint $p = \frac{d}{2}$, so the last interior factors of a walk term raise them along $L^2 \to L^4 \to L^8 \to C^{1,\beta}$ (Lemmas 44.5 and 44.6). The results are Theorems 44.13 and 44.15, at level (L3) with $W = 0$ and with $\beta = \min(\alpha, \frac{1}{2})$. With these bounds, the background map becomes the fixed point $A_*(s; B)$ of a weakened contraction (Theorem 45.5). It is analytic in (s, B). At $s = 1$ and a critical reference it is the background map of Part V. If $s_\square = 0$ outside a union of boxes, it factorizes over the components of that union.

Where in the note. Part VIII: Sections 35–39. Part IX: Sections 40–45. In the companion: the chapters on Part VIII and on Part IX.

G8 The step in polymer form (Sections 46–48)

Sections 46–48 turn the exact Gaussian form of one step (Theorem 34.9) into polymer form. The result is Theorem 48.9, which is Theorem L(i). At real critical references, with V' small field,

$$T_{\mathrm{sf}}(\rho)(V') = \mathcal{N}_k N_\chi \exp\Big(-\frac{1}{g^2} S(U') - \frac{1}{2}\sum_b \ell_k(b;1) + \sum_X \mathfrak{e}(X, U') + \sum_c \log \mathfrak{j}_c + \mathfrak{E}(U') + \sum_Y \big(E^{\#}_{\mathrm{p}}(Y, U') + E^{\#}_{\mathrm{m}}(Y, U')\big)\Big).$$

Every term depends on U' only near its polymer, is invariant under real gauge transformations, is real at real references, and is analytic in the reference on complex domains. On the domains of Remark 38.8 the one-loop activities are bounded by $Ce^{-(\kappa_2-1)|X|}$ and the Jacobian terms by C, with C independent of the coupling; their quadratic parts are marginal. The pure sources $E^{\#}_{\mathrm{p}}$ are of order $g^2 p_0^6$, and the mixed sources $E^{\#}_{\mathrm{m}}$ are bounded by the old activities. The hypotheses of Theorem 48.9 on the coupling, the thresholds and the old activities are supplied along the flow by Theorem N(v) (Table 6).

Complex references (Section 46). The activities of the flow must be analytic in the background on the graded domains (Definition 38.3). So the step evaluates the operators of Part IX, the interior estimates and the weakened background map at complex references $U = e^{\mathrm{i}\eta A'} U_r$, with U_r real and regular and A' small. Three mechanisms carry them over (Proposition 46.7).

- *Identities from the real points.* An exact identity proved at real backgrounds extends to a convex ball of complex references by the identity theorem, since the real points of the ball are real backgrounds (Remark 19.12, last paragraph).
- *Fixed coercivity constants and rates.* The coercivity constants and the decay rates do not depend on the size δ_c of the complex part.
- *Monotone majorants.* At complex references of size $\delta_c < \delta_P$ the other constants are at most twice their real values.

The thresholds are fixed in an order that avoids circularity: the cap $\bar{\delta}$ and the threshold $\varepsilon_{\max}$ first, then δ_P (Remark 49.8(4),(5)). In four dimensions the constants θ_r, δ_1, δ_0 of the graded domains do not depend on the coupling, the step, the torus or the polymer (Remark 38.8).

Step quantities and the fluctuation integral (Sections 47–48). The ingredients of the step become functions of the weakening parameters that factorize over polymers (Propositions 47.5, 47.7, 47.8, 47.10 and 47.12). The localization lemma then writes each of them, at $s = 1$, as a sum of pieces with a factor $e^{-(\kappa-1)}$ for every box beyond the polymer (Lemma 47.2). The fluctuation integral follows Dimock's small-field scheme [15].

- *White noise.* The fluctuation is written $Z = gC_k^{1/2}\xi$, and the small-field condition is imposed on ξ: $|\xi(b)| \le p_0$ on every bond of $\mathfrak{F}$ (Definition 48.1). On this set the fluctuation cutoff χ^{w} equals 1, every step quantity is defined, and the composed configurations stay in the domains of the old activities (Lemma 48.2).
- *Mayer and polymer expansion.* The localized pieces (Proposition 48.5) are integrated by a Mayer expansion with the Kotecký–Preiss criterion (Theorem 48.6). The output splits into a pure class, from the interaction, gauge-fixing and Jacobian pieces alone, and a mixed class, which involves the old activities (Corollary 48.7).
- *Parity.* Replace g by a complex parameter t in the fluctuation. The pure pieces are even under $(t, \xi) \mapsto (-t, -\xi)$ and vanish at $t = 0$, so the pure output divided by t^2 is analytic. Cauchy's estimate gives $E_{\mathrm{p}}^{\#} = O(g^2 p_0^6)$, with no term of order g (Proposition 48.8).

Section 48 assumes three conditions (Q1)–(Q3) on the small-field parameters. Along the flow, (Q1) and (Q3) hold by condition (Γ2) on the bound γ of the couplings, and (Q2) holds for the next threshold as long as $g_{k+1} \le \min(2g_k, \gamma)$ (the paragraph after (Q1)–(Q3) in Section 48; Remark 53.5).

Where in the note. Sections 46–48; the first step and the vacuum energy are in Section 52 (Theorem L(ii),(iii)). In the companion: the chapter on the first half of Part X.

G9 The four-dimensional flow (Sections 49–53)

Sections 49–53 iterate the step of Theorem 48.9 over all scales at which the couplings stay in $(0, \gamma]$, in Bałaban's generation form, on a fixed torus. The result is Theorem 53.12, summarized as Theorem N. It is a theorem about the small-field model of Remark 52.6(c) on the given torus; the large-field stage is not treated (Remark 53.14(d)).

Why generations. The new activities of a step are of order one, not small in the coupling, and their quadratic part in the curvature is marginal in four dimensions (Remark 41.8(b)). So the effective action of a step is not turned into a single activity of the next scale and contracted. Following Bałaban, the new terms of each step are kept as a separate *generation*, bounded only at its creation scale (Remark 49.1, the opening of Section 50). At every later scale its marginal part is cancelled by a counterterm $-\beta_j S$, with $g_{j+1}^{-2} = g_j^{-2} - \beta_{j+1}$, and what remains is small by explicit gains. The iteration needs no contraction. The smallness of what a step adds beyond the one-loop and Jacobian terms comes from the fluctuation cutoff and from parity (Remark 49.1, Proposition 48.8). Figure G4 shows the life of a generation.

Polymer grids (Section 49). Activities are functions on the fine lattice, the same lattice at every scale (Section 49.2). The polymer grid of scale j consists of Bałaban's cubes of side M; polymers are unions of grid cubes, and $d_j(X)$ is their tree length. The grids of different scales

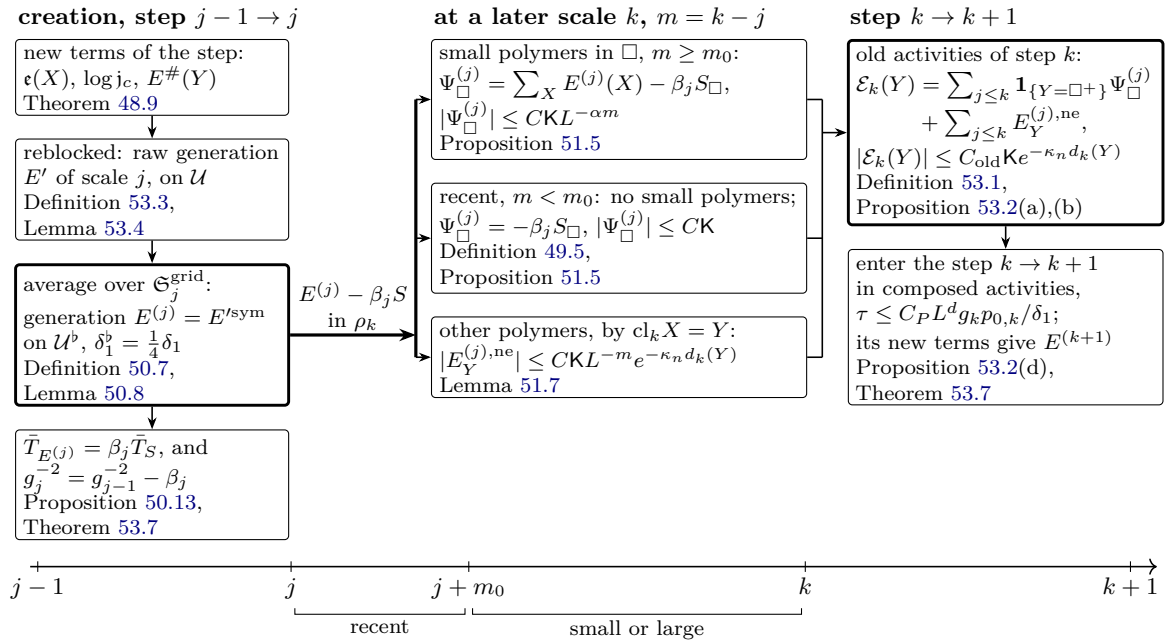


Figure G4: The life of a generation in the flow; the axis is the scale. Left, the step from scale $j-1$ to scale j: its new terms, reblocked, form the raw generation E'; its average over the symmetries of the grid of scale j is the generation $E^{(j)}$, bounded at its creation scale by $\mathsf{K}e^{-\kappa_n d_j(X)}$ on the domains $\mathcal{U}^\flat(X)$ (Theorem N(ii)); its period density gives the coefficient β_j and the coupling g_j. The density ρ_k of every later scale contains $E^{(j)} - \beta_j S$ (Theorem N(i)). Middle, a later scale k and a grid cube $\square$ of scale k: a polymer of scale j is recent, small or large at scale k (Definition 49.5). The counterterm $-\beta_j S_\square$ is grouped with the small polymers of $E^{(j)}$ in $\square$, and the sum is bounded on $\mathcal{U}^\flat(\square^+)$, by $C\mathsf{K}L^{-\alpha m}$ for $m \geq m_0$; the other polymers gain L^{-m} (Theorem N(iv)). Right, the old activities collect the generations $j \leq k$ and enter the step from scale k to scale $k+1$, whose new terms form the next generation. The bounds hold for couplings in $(0, \gamma]$; C, C_{old} and K do not depend on j, k, the torus or the couplings (Theorem N, Proposition 53.2).

are aligned (Lemma 49.2), and reblocking keeps the decay rate (Lemma 49.3, Remark 49.8). Seen from a later scale $k = j + m$, a polymer of scale j is *recent* if $m < m_0$, *small* if $m \geq m_0$ and $d_j(X) \leq c'm$, and *large* otherwise (Definition 49.5).

A generation (Section 50). A generation of scale j is a family $E = \{E(X, \cdot)\}$ of normalized, gauge-invariant graded activities on the polymers of scale j, on the domains $\mathcal{U}^\flat(X)$, real at real configurations (G1), covariant under the translations (G2) and all symmetries $\mathfrak{S}_j^{\text{grid}}$ (G4) of the grid, whose sum at the background fields of scale j is invariant under $\mathfrak{S}_j$ (G3) (Definition 50.1). Its norm is $\|E\|_j = \sup_X e^{\kappa_n d_j(X)} \sup_{\mathcal{U}^\flat(X)} |E(X, \cdot)|$. At a later scale k the generation is evaluated at $U_k(V)$, which is a background field of scale j of a configuration in the small-field set of scale j (Lemma 50.3). So the sum and its symmetry persist.

Symmetrization and the coefficient β. The new terms of a step, reblocked, form a raw generation: it has (G1), on the larger domains $\mathcal{U}(X)$, and (G3), but the step does not make it covariant under the grid symmetries (Definition 53.3, Lemma 53.4, the opening of Section 50). Its average over $\mathfrak{S}_j^{\text{grid}}$ is a generation with the same sum at the background fields and no larger norm (Definition 50.7, Lemma 50.8). The price is a smaller domain, $\delta_1^\flat = \frac{1}{4}\delta_1$: a reflection turns forward differences into backward ones, which at a curved real part costs a factor and a plaquette term (Lemma 50.6, Remark 50.9(d)). The coefficient of a generation is read off from its *period density* $\bar{T}_E$, the second variation of its activities at the linear fields of constant curvature F, per unit volume (Definition 50.11). By (G4) and the gauge invariance of the activities it is a quadratic form invariant under B_4 and under conjugation by G. For simple G such a form is a multiple of the action's (Lemma 50.10): $\bar{T}_E = \beta_E \bar{T}_S$, with β_E real and $|\beta_E| \leq C_\beta \|E\|_j$, and there is no topological and no anisotropic part (Proposition 50.13). The flow uses the simplicity of G here.

Extraction at later scales (Section 51). Let $m = k - j \geq m_0$. On a small polymer X of scale j, a configuration of the domain of scale k is, in a complex gauge built in three stages (real, then affine, then quadratic), a linear field of constant curvature F_X plus a remainder of order $L^{-(2+\alpha)m}$ (Lemma 51.1(a),(b)). So $E(X, U)$ is a quadratic form in F_X up to $O(L^{-(4+\alpha)m})$ (Lemma 51.1(d)). Summed over the small polymers based in a grid cube $\square$ of scale k, which contains L^{4m} grid cubes of scale j, these quadratic forms add up to $M^4 \sum_Q \bar{T}_E(F_Q)$ over the cubes Q of scale j in $\square$ (Lemma 51.3), and the counterterm $\beta_E S_\square$ cancels them. The error is $C_{\text{ex}} \|E\|_j L^{-\alpha m}$ (Proposition 51.5). The polymers that are not small carry a gain L^{-m} (Lemma 51.7), and the wrapping ones are bounded at all later scales (Lemma 52.1(c)–(e)).

Old activities and one step (Section 53). In every grid cube $\square$ of scale k the counterterm $-\beta_j S_\square$ is grouped with the small polymers of generation j based in $\square$; the other polymers are grouped by their closure at scale k (Definition 53.1). For a recent generation the whole counterterm enters the old activity. The resulting *old activities* $\mathcal{E}_k$ are bounded by $C_{\text{old}}\mathsf{K}$ with decay, uniformly in k, K, the torus and the couplings. Their sum at a background field is $\sum_{j \leq k}(E^{(j)} - \beta_j S)$, and they satisfy the hypothesis of Theorem 48.9 on old activities (Proposition 53.2). One step then preserves the generation form, with $g_{k+1}^{-2} = g_k^{-2} - \beta_{k+1}$ and $g_{k+1} \leq 2g_k$ (Theorem 53.7; the first step is Proposition 53.8). The generations and the coefficients depend only on the earlier couplings $(g_0, \dots, g_{j-1})$, and continuously (Proposition 53.10, Corollary 53.11).

Theorem 53.12 in words. Let G be compact, connected and simple. Choose the parameters in the order of Remark 49.8, a bound γ on the couplings with (Γ1)–(Γ3) of Remark 53.5, and a torus whose side at the last scale is $n_K \geq n_{\min}$, with ℓL^N a power of L (Section 52.1). Start

from $\rho_0 = \exp(-g_0^{-2}S - E_0)$ with $g_0 \in (0,\gamma]$. Let the generations, the coefficients and the couplings be defined by the recursion of the steps, and put $\rho_j = \mathbf{1}_{\mathfrak{U}_j(h_j)} T_{\mathrm{sf}}(\rho_{j-1})$ as long as $g_1,\dots,g_j \le \gamma$. Then, at every scale $k \le K$ that the recursion reaches in this way, whatever the later couplings:

(a) the density has the generation form $\mathcal{Z}_k \exp(-g_k^{-2}S(U_k) - \varepsilon_k|\mathbb{T}^{(k)}| + \sum_{j\le k}(E^{(j)} - \beta_j S)(U_k))$ on the small-field set, and vanishes outside it;

(b) every generation is bounded at its creation scale by $\mathsf{K}e^{-\kappa_n d_j(X)}$, on domains whose constants do not depend on the coupling;

(c) its period density is exactly isotropic, and $|\beta_j| \le C_\beta \mathsf{K}$;

(d) at every later scale its small polymers in a grid cube differ from $\beta_j S_\square$ by at most $C_{\mathrm{ex}}\mathsf{K}L^{-\alpha m}$ ($C_{\mathrm{ex}}\mathsf{K}$ for recent generations), and the whole sum $\sum_j (E^{(j)} - \beta_j S)$ at a background field is at most a constant times K per site;

(e) consecutive couplings satisfy $|g_j^{-2} - g_{j-1}^{-2}| \le C_\beta \mathsf{K}$ and $\frac{3}{4}g_{j-1} \le g_j \le 2g_{j-1}$, and every step satisfies the hypotheses of Theorem 48.9 (of Proposition 52.2 for the first step);

(f) if all couplings up to scale K stay below γ, there is exactly one E_0 with $\varepsilon_K = 0$, and then $|\varepsilon_k| \le C'_{\mathrm{vac}} g_k p_{0,k}$.

The constants depend only on the parameters fixed before γ (C'_{vac} also on the function p_0), not on k, K, the torus or the couplings. The stability bound adds that, for the counterterm with $\varepsilon_K = 0$, the partition function of the small-field model is, up to the explicit normalization $\mathcal{Z}_K$, between $e^{-c|\mathbb{T}^{(K)}|}$ and $e^{c|\mathbb{T}^{(K)}|}$; the constant c depends on g_K and on the parameters, not on K, the torus or the earlier couplings (Corollary 53.13). Bounds per volume suffice because the final torus is fixed (Remark 52.6(a)).

Concordance. Theorem N(i)–(v) are Theorem 53.12(a)–(e), with Proposition 50.13 in (iii) and Proposition 51.5 and Lemma 51.7 in (iv); (vi) is (f) with Corollary 53.13; (vii) is Proposition 53.10 with Corollary 53.11. Items (a), (b) and (d) correspond to Bałaban's (0.22)–(0.24), (0.25) and (0.28)–(0.30) [18], and Theorem N is the analogue, for the small-field model on the torus, of his Theorem 3 (the paragraph after Theorem N).

What the theorem does not say. No asymptotic behaviour of the couplings is assumed. The theorem bounds $|\beta_j|$ but does not control its sign; Part XI controls the sums of the β_j through the one-loop coefficient (Section G10). The dependence on the couplings is a dependence on the whole sequence $(g_0,\dots,g_{j-1})$, and only its continuity is proved. The small-field domain shrinks with the coupling, as in Dimock's scheme (Remark 53.14(a)).

Who supplies the hypotheses. The main theorems take their hypotheses from later Parts. Table 6 lists, for Theorems K, L and N, each hypothesis and the result that supplies it along the flow. All conditions on γ are upper bounds by quantities fixed before γ, so they hold together for γ small (Remark 49.8(6), Remark 53.5, Feasibility).

Where in the note. Section 49: parameters, grids, reblocking, polymer classes, the order of the parameters (Remark 49.8). Section 50: generations, symmetrization, period density. Section 51: extraction. Section 52: the torus, the first step, the vacuum, the modified model. Section 53: old activities, one step in generation form, the flow. Remarks 49.1 and 53.14 summarize the organization and the comparison with Bałaban.

In the companion. The chapter on the second half of Part X, and the parameter ledger.

Theorem	Hypothesis	Supplied by
K (one step)	G compact, connected, semisimple	Theorem N assumes G simple
	$M \geq \max(M_0, 5L)$	Remark 49.8(4)
	side of the torus	(T0)–(T2), from $n_K \geq n_{\min}$ and ℓL^N a power of L (Section 52.1, Lemma 52.1(a))
	curvature of V' at most $\varepsilon_1' \leq a_{\text{step}}$	$\varepsilon_1' = h_{k+1} \leq \gamma p(\gamma) \leq a_{\text{step}}$, by (Γ2)
	$w \leq r_2$	(Q1), by (Γ2)
	density of the form (31.2), with $a_1 \geq \tilde{h} \geq 2(C_* + 1)L^{-2}\varepsilon_1' + 4w$	Theorem 53.7(a) with $\tilde{h} = h_k$; the inequality is (Q2), from $g_{k+1} \leq \min(2g_k, \gamma)$ (Theorem 53.7(c),(d))
L (step in polymer form)	$\alpha \leq \frac{1}{2}$; parameters as in Remark 49.8	Remark 49.8(4); assumption of Theorem N
	(Q1), (Q3), $2g \leq r_t$	(Γ2) of Remark 53.5
	(Q2) with $\varepsilon_1' = h_{k+1}$	Theorem 53.7(d)
	(DC5)	the domain constants, Remark 49.8(5)
	decay and bound of the old activities	Proposition 53.2(b),(d)
	rate condition of Proposition 48.5	Remark 49.8(3)
	smallness conditions of Corollary 48.7	(Γ2), with the bound $C_{\text{old}}\mathsf{K}$ of Proposition 53.2
N (the flow)	G simple; parameters in the order of Remark 49.8	choices; each parameter depends only on earlier ones
	(Γ1)–(Γ3)	γ small (Remark 53.5, Feasibility)
	$n_K \geq n_{\min}$, ℓL^N a power of L (Section 52.1)	choice of the torus
M(iv) (UV limit)	(Γ4), $\gamma \leq e^{-3q}$, $C_3\gamma^2 p_0(\gamma)^6 \leq \frac{1}{2}\beta_\infty$	γ small (Remark 49.8(6))
	(F1)–(F3)	Proposition 58.6

Table 6: The hypotheses of the main theorems and where they are supplied along the flow (Theorem N(v), Theorem 53.7, Remark 49.8).

G10 The one-loop coefficient and the ultraviolet limit (Part XI)

Part XI computes the one-loop part of the coefficient β, summed over the steps, and derives from the flow of Part X the ultraviolet behaviour of the couplings. Its results are Theorem M: (i) the composition of the one-loop terms and their form in the Landau chart; (ii) their exact lattice symmetry, hence no term linear in the momentum; (iii) the cumulative coefficient $k\beta_\infty + O(1)$; (iv) the ultraviolet limit (Theorem 58.9). Separately, generation locality shows that, at fixed couplings, β_j converges as the torus grows to the coefficient that Bałaban's definition assigns to the generations of this note, without a rate (Proposition 58.8); Theorem M does not use it. Figure G5 shows the chain of reductions.

Normalization. The action is $S = \sum_p (\mathsf{N} - \frac{1}{2}\operatorname{tr}(U(\partial p) + U(\partial p)^{-1}))$ (Section 23), with the trace of a representation ρ, $\operatorname{tr}_\rho(T^a T^b) = T_\rho \delta^{ab}$ (Theorem M). Its continuum limit is $\frac{T_\rho}{4}\sum_{\mu\nu} F^a_{\mu\nu} F^a_{\mu\nu}$ (Remark 57.8). So $g^{-2}S$ corresponds to the standard $\frac{1}{4g^2_{\text{std}}}\sum F^a F^a$ with

$$g^2_{\text{std}} = g^2/T_\rho .$$

The coefficient is $\mathsf{b} = \frac{11 C_A}{24\pi^2 T_\rho}$ and $\beta_\infty = \mathsf{b}\log L$ (Theorem 57.1); along the flow, $g_k^{-2} - g_K^{-2} = (K-k)\beta_\infty$ up to the error of Theorem 58.9. For $SU(N)$ in the fundamental representation, $T_\rho = \frac{1}{2}$ and $\beta_\infty = \frac{11N}{12\pi^2}\log L$. In Theorem 58.9, g is the coupling of the last scale and g_0 the bare coupling (Section 58.2).

The chain of reductions (Sections 54–57).

- *The composite one-loop term.* $\Gamma_k^{(1)}(V)$ is defined by Laplace's method on the fibre of the fine configurations with averages V, near the minimal orbit (Proposition 54.3). Integrating the fibre scale by scale gives $\Gamma_k^{(1)}(V) = \sum_{j<k} \Gamma_j^{\text{step}}(U_k(V))$ up to constants (Lemma 54.5, Proposition 54.6).

- *Exact symmetry.* $\Gamma_k^{(1)}$ and the step functions $V' \mapsto \Gamma_j^{\text{step}}(U_{j+1}(V'))$ are invariant under the translations and under B_4. So their marginal tensors, the parts of order p^2 of their vacuum polarizations in infinite volume, are $\beta(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$ with a single coefficient, and there is no term linear in p (Lemma 54.9). The symmetry comes from Bałaban's symmetric averaging, the Haar measure and the covariance of the minimal orbits (the paragraph after Theorem M); a linear term would produce relevant terms of order 1 in every step (Remark 54.8).

- *Infinite volume and the change of scale.* The operators at $U = 1$ are constructed in infinite volume by folding the operators on the tori (Section 54.2). In four dimensions the change of scale leaves marginal tensors unchanged (Lemma 54.17). Hence $\sum_{j<k} T^{(1)}_{j+1} = T_{\Gamma^{(1)}_k}$ (Corollary 54.18).

- *The Landau chart.* Up to a constant, $\Gamma_k^{(1)}(V) = -\frac{1}{2}\log\det \mathbf{\Delta}_* - \frac{1}{2}\log\det(\mathbf{Q}_k \mathbf{G}_* \mathbf{Q}_k^T)$ plus the logarithm of a ghost determinant, at the background $U_0 = U_k(V)$, where $\mathbf{\Delta}_* = q_2 + D_{U_0} R D_{U_0}^T + a\mathbf{Q}_k^T\mathbf{Q}_k$ is built from the Hessian form q_2 of Proposition 33.2 and $\mathbf{G}_* = \mathbf{\Delta}_*^{-1}$ (Theorem 55.4, Corollary 55.5). Its marginal tensor is the p^2-tensor T_{fine} of a fine-lattice vacuum polarization at $U = 1$ (Lemma 56.3).

- *Reduction to the Feynman gauge.* The vector and ghost operators are compared with the lattice operators of the background Feynman gauge with mass 1. Power counting with the weights $\langle\partial\rangle$ bounds the trace per cell of every product of degree $D > 4$, and of every product with a factor of finite rank per cell, uniformly in η (Lemmas 56.6 and 56.7).

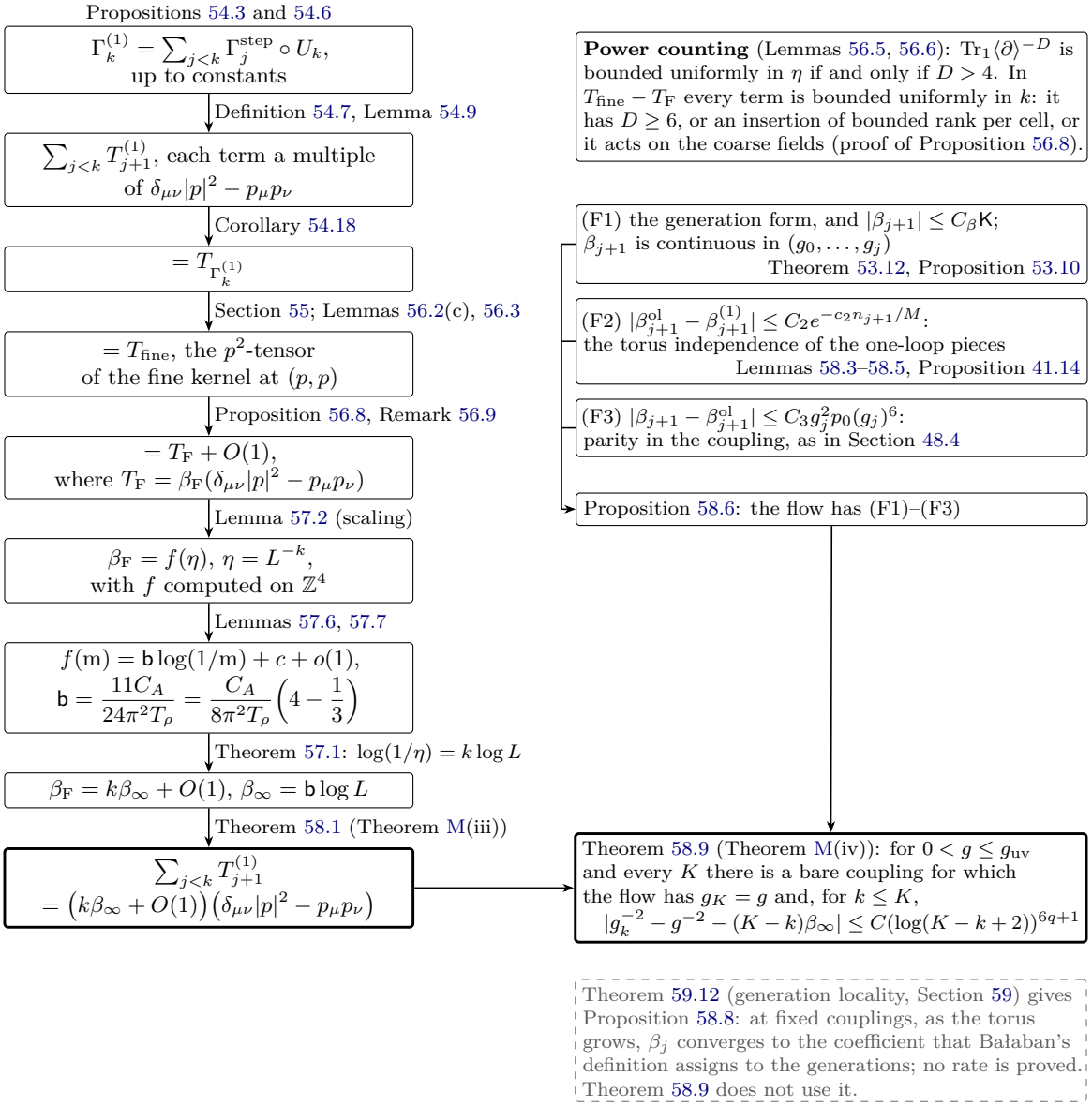


Figure G5: Part XI: the chain of reductions to the one-loop coefficient (left) and the ultraviolet limit (right). Left: each arrow carries the results that give the next line; together they are Theorem 58.1, which is Theorem M(iii), with $O(1)$ bounded uniformly in k. The marginal tensors have no term linear in p (Lemma 54.9). The coefficient is $\mathsf{b} = \frac{11C_A}{24\pi^2 T_\rho}$; in Lemma 57.7 the factor $4 - \frac{1}{3} = \frac{11}{3}$ is the sum of the spin part 4 and the orbital part $-\frac{1}{3}$, which is $+\frac{1}{3}$ from the scalar determinant and $-\frac{2}{3}$ from the vector determinant. Right: the properties (F1)–(F3) of the flow (Section 58.2), proved in Proposition 58.6, and Theorem M(iii) give Theorem M(iv), under the conditions on γ stated there; for the last m_1 steps see the paragraph after Theorem 58.9. Dashed: off the line of Theorem M(iv).

Hence $T_{\text{fine}} = T_{\text{F}} + O(1)$ uniformly in k, where T_{F} is the tensor of the Feynman-gauge one-loop term (Proposition 56.8).

- *The lattice integral.* By scaling, the isotropic coefficient of T_{F} is $\beta_{\text{F}} = f(\eta)$, the p^2-coefficient on $\mathbb{Z}^4$ with mass η (Lemma 57.2). As $\mathrm{m} \to 0$, $f(\mathrm{m}) = \mathsf{b}\log(1/\mathrm{m}) + c + o(1)$ (Lemma 57.6). The coefficient comes from the spherical average of the continuum Feynman-gauge integrand (Lemma 57.7 and its proof):

$$\mathsf{b} = \frac{C_A}{8\pi^2 T_\rho}\Big(4 - \frac{1}{3}\Big) = \frac{11 C_A}{24\pi^2 T_\rho},$$

with the spin part 4 and the orbital part $-\frac{1}{3}$, itself $+\frac{1}{3}$ from the scalar determinant and $-\frac{2}{3}$ from the vector determinant. Since $\log(1/\eta) = k \log L$, $\beta_{\text{F}} = k\beta_\infty + O(1)$ (Theorem 57.1).

Together: $\sum_{j<k} T^{(1)}_{j+1} = (k\beta_\infty + O(1))(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, uniformly in k (Theorem 58.1).

Transfer to the flow (Section 58). The flow reads β_{j+1} off on the torus; Part XI computes $\beta^{(1)}_{j+1}$ in infinite volume. Three properties of the flow connect them, all proved in Proposition 58.6 under (Γ1)–(Γ4).

(F1) The densities have the generation form, and β_{j+1} is a continuous function of $(g_0, \dots, g_j)$ with $|\beta_{j+1}| \le C_\beta \mathsf{K}$ (Proposition 53.10).

(F2) The one-loop part β^{ol}_{j+1} of β_{j+1} differs from $\beta^{(1)}_{j+1}$ by at most $C_2 e^{-c_2 n_{j+1}/M}$. This rests on the torus independence of the one-loop pieces (Proposition 41.14, Lemma 58.3) and on the infinite-volume limits of the second variations (Lemma 58.5). These errors are summable over the scales (Remark 58.7(b)).

(F3) The rest is $|\beta_{j+1} - \beta^{\text{ol}}_{j+1}| \le C_3 g_j^2 p_0(g_j)^6$, by parity in the coupling and Cauchy's estimate. Without parity it would be of order $g p_0^3$, which is not summable along the flow (Remark 58.7(b)).

The exit argument and Theorem 58.9. The set of bare couplings g_0 for which $g_1, \dots, g_K$ stay in $(0, \gamma]$ is closed and contains a small interval, and $g_0 \mapsto g_K$ is continuous on it (Corollary 53.11). At the end of its first component some g_j equals γ, and the lower bound from (F2) and (F3) forces g_K above a fixed number; the intermediate value theorem gives g_0 with $g_K = g$ (proof of Theorem 58.9). With $p_0(g) = (\log g^{-1})^q$ and γ subject to (Γ1)–(Γ4), $\gamma \le e^{-3q}$ and $C_3 \gamma^2 p_0(\gamma)^6 \le \frac{1}{2}\beta_\infty$, Theorem 58.9 states: there are $g_{\text{uv}} > 0$ and C, independent of K and of the torus, such that for $0 < g \le g_{\text{uv}}$ and every K some bare coupling gives $g_k \in (0, \gamma]$ for $k \le K$, $g_K = g$, and

$$\Big|\frac{1}{g_k^2} - \frac{1}{g^2} - (K-k)\beta_\infty\Big| \le C\big(\log(K-k+2)\big)^{6q+1} \qquad (k \le K).$$

For $K - k \ge m_1$, with m_1 independent of K, g and the torus, this is the analogue of Bałaban's (0.31) for the couplings of the small-field flow. For the last m_1 steps it gives the upper half, and the lower half up to an additive constant (the paragraph after Theorem 58.9, Remark 58.10(c)). The theorem concerns the small-field model (Remark 58.10(a)).

Generation locality (Section 59). For polymers that do not wrap, and at flat configurations, the new terms of a step are a single functional, defined on the infinite lattice, of the old activities (Proposition 59.8). By induction over the scales, at fixed couplings the generations converge as the torus grows, and so do the coefficients (Theorem 59.12). Hence β_j converges to the coefficient that Bałaban's definition [18, (1.20)–(1.22)] assigns to the generations of this note (Proposition 58.8). No rate is proved. This line is not used by Theorem 58.9.

The β's and the T's. Table 7 lists the coefficients and tensors of Part XI and of the flow.

Symbol	Meaning	Where
$\beta_j = \beta_{E^{(j)}}$	coefficient of the generation $E^{(j)}$, read off from its period density on the torus; $g_j^{-2} = g_{j-1}^{-2} - \beta_j$	Proposition 50.13, Theorem 53.12
$\beta_{j+1}^{\mathrm{ol}}$	one-loop part of β_{j+1}, through the volume coefficient (58.1); independent of the couplings	Section 58.2
$\beta_{j+1}^{(1)}$	coefficient of $T_{j+1}^{(1)}$, in infinite volume	Theorem 58.1
β_{F}	isotropic coefficient of T_{F}; $\beta_{\mathrm{F}} = f(\eta)$	Theorem 57.1, Lemma 57.2
b, β_∞	$\mathsf{b} = \frac{11 C_A}{24\pi^2 T_\rho}$, $\beta_\infty = \mathsf{b} \log L$	Lemma 57.7, Theorem 57.1
β_j^{B}, β_j^∞	Bałaban's coefficient of the generations of this note, the limit of β_j as the torus grows	Proposition 58.8, Theorem 59.12
T_Φ; $T_{j+1}^{(1)}$, $T_{\Gamma_k^{(1)}}$	marginal tensor of a function Φ, the p^2-part of its vacuum polarization; those of the step functions and of the composite one-loop term	Definition 54.7, Corollary 54.18
T_{fine}, T_{F}	p^2-tensors at $U = 1$ of the fine-lattice one-loop term and of its Feynman-gauge counterpart	Lemma 56.3, Proposition 56.8
$\bar{T}_E$, $\bar{T}_S$	period densities: quadratic forms in a constant curvature	Definition 50.11
T_ρ	a number, the normalization of the trace: $\mathrm{tr}_\rho(T^a T^b) = T_\rho \delta^{ab}$	Theorem M

Table 7: Coefficients and tensors of the flow and of Part XI.

Where in the note. Section 54: composition, symmetry, infinite volume, change of scale. Section 55: the Landau chart. Section 56: reduction to the Feynman gauge. Section 57: the lattice integral and the coefficient b. Section 58: (F1)–(F3) and the ultraviolet limit. Section 59: generation locality.

In the companion. The chapters on the first and on the second half of Part XI.

G11 The order of the parameters

The parameters of Parts VII–XI are chosen in a fixed order, and each may depend only on those chosen before it (Remark 49.8). The order is stated in Remark 49.8, completed by Remarks 38.8, 34.7 and 53.5, and, for the constants of Parts V and VI, by the proofs of Theorems 25.1 and 30.3. The parameter ledger of the companion collects the constants with their dependencies; Table 8 follows it, and Figure G6 shows the chain. Three facts make the order acyclic.

- Every lower bound on M uses only quantities fixed before M (Remark 49.8(4)). Several constants of Part VII do not depend on M (Remark 34.7).
- The domain constants are fixed after M and M', and the cap $\bar{\delta}$ on the size of the complex references comes before the threshold $\varepsilon_{\max}$ and the radius δ_P (Remark 38.8).
- Every condition on γ is an upper bound by quantities fixed before γ, and remains true when γ decreases (Remark 49.8(6)). The constants of the flow do not depend on the couplings or on the torus (Theorem 53.12).

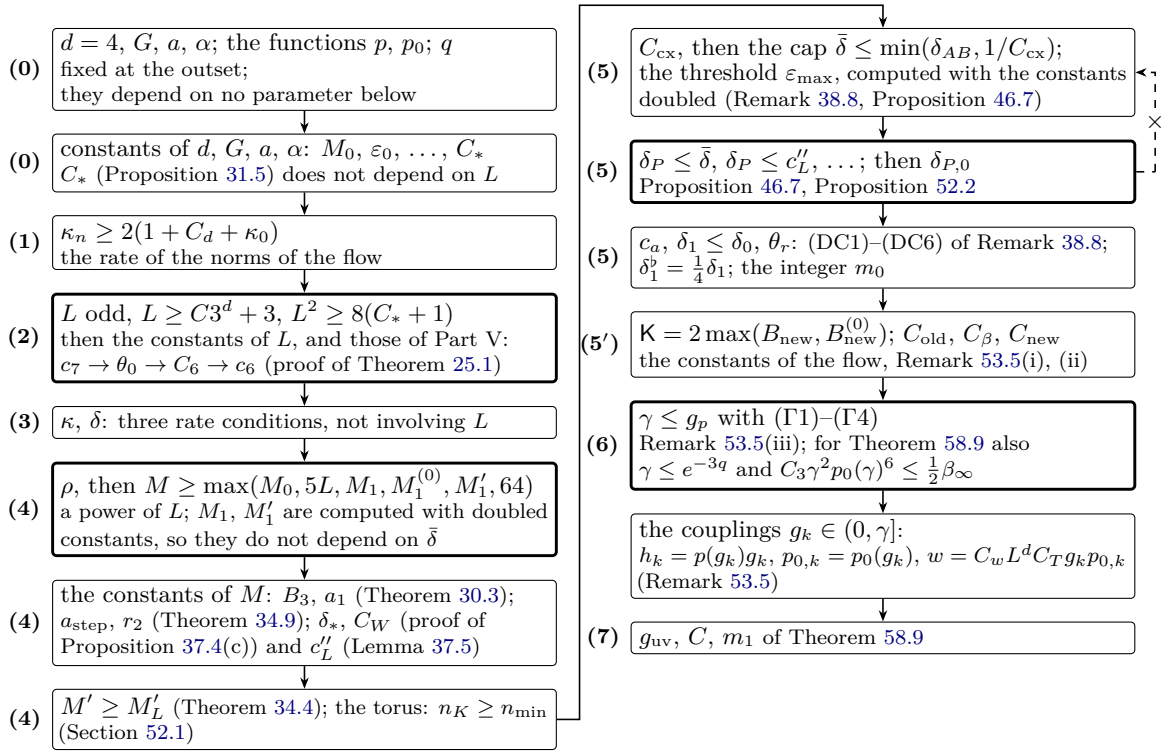


Figure G6: The order in which the parameters are chosen. Each box may depend on the boxes before it, in the order of the arrows: the left column, then the right one (Remark 49.8). A tag is the stage of the box: (1)–(6) are the items of Remark 49.8, (0) the data fixed at the outset and their constants, (5′) the constants of the flow, which Remark 53.5 places between items (5) and (6), and (7) the constants of Theorem 58.9. Thick frames: the block size L, the scale M, the size δ_P of the complex references, and the bound γ on the couplings. The radius δ_* and the constant C_W are fixed in the proof of Proposition 37.4(c), and c_L'' with $C_{\text{cmp}}c_L'' \leq \delta_*$ in the proof of Lemma 37.5; they depend on M, and $\delta_P \leq c_L''$ (Proposition 46.7). Crossed dashed arrow: $\varepsilon_{\max}$ does not depend on δ_P, since its thresholds are computed with every multiplicative constant doubled (Remark 38.8). Every condition on γ is an upper bound by quantities fixed before it, so the order has no cycle (Remark 49.8(6)).

The torus satisfies (T0)–(T2) at every scale once $n_K \geq n_{\min}$ and ℓL^N is a power of L, and $n_{\min}$ depends on the parameters up to stage (4) (Remark 49.8, Section 52.1).

The named constants. The constants that recur in the statements are the following.

- a_1, B_3: background fields exist for real coarse configurations of curvature at most $\varepsilon_1 \leq a_1$, and they have curvature and current at most $B_3\varepsilon_1$ (Theorem 30.3).
- C_*: the curvature of a background field is at most $(C_* + \delta)$ times that of the coarse configuration, with C_* independent of L (Proposition 31.5). Hence the condition $L^2 \geq 8(C_* + 1)$.
- θ_0, c_6, C_6: the class $\mathfrak{U}(\theta_0)$ of configurations of small curvature and current in which the critical orbit is unique, the threshold $\varepsilon + \beta \leq c_6$ on the reference and the data, and the constant of the bounds (Theorem 25.1).
- a_{step}, r_2, c_L: the curvature threshold of the coarse configuration and the bound on the fluctuation cutoff in Theorem K, and the coercivity constant of the step, from a Poincaré inequality for one step (Theorem K(ii), Lemma 34.2).
- $h_k = p(g_k)g_k$, w, p_0: the three cutoffs of the small-field step (Table 3).
- κ_2: the decay rate of the one-loop pieces, $|\mathfrak{e}(X)| \leq Ce^{-(\kappa_2 - 1)|X|}$ (Proposition 41.7).
- K, C_β, C_{old}: the bound of every generation, the bound $|\beta_j| \leq C_\beta \mathsf{K}$, and the bound $C_{\text{old}}\mathsf{K}$ of the old activities (Theorem N, Proposition 53.2).
- m_0: the age after which the marginal part of a generation is extracted (Definition 49.5); m_1: the number of last steps for which Theorem 58.9 gives the lower half of the analogue of (0.31) only up to a constant.

Where in the note. Remarks 49.8, 38.8, 34.7 and 53.5; the proofs of Theorems 25.1 and 30.3; Section 52.1.

In the companion. The chapter on the parameter ledger. Its tables list the constants of Parts VII–XI, and those of Parts I–VI that these Parts use, with their definitions, dependencies, conditions and uses, and it checks that the order has no cycle.

G12 Notation

The notation index is Appendix B. It lists the symbols used across several Parts with their meanings and the places of their definitions. Letters with several meanings are listed with all of them, and the index says how the meanings are kept apart. Appendix B also lists the named conditions, such as (A1)–(A11), (L1)–(L5), (DC1)–(DC6) and (Γ1)–(Γ4), and the operators, and it gives the former names of the renamed symbols. Remark 47.1 collects the notation of Sections 47–53.

G13 Comparison with Bałaban and Dimock

The note follows Bałaban's renormalization group for four-dimensional lattice Yang–Mills theory [3, 4, 5, 6, 7, 8, 18, 19], and, for the small-field step in polymer form, Dimock's account of Bałaban's method [15]. Table 9 lists where it follows them and where it differs. The comparison is restricted to what the note states about these papers.

Stage	Parameters and their role	Main conditions	Fixed in
(0)	$d = 4$; the group G; the averaging coefficient a; the Hölder exponent α; the functions p, p_0 and, in Theorem 58.9, the exponent q. Then the constants of Parts I–III that do not depend on L or M (Remark 31.4(b)) and of the interior estimates, among them M_0, and C_*	$\alpha \le \frac{1}{2}$ from Section 45 on; C_* does not depend on L	§49.1, Thm. 19.6, Prop. 31.5
(1)	the rate κ_n of the activity norms, after the constants κ_0, K_0 of d_M and C_d of the polymer expansion	$\kappa_n \ge 2(1 + C_d + \kappa_0)$	Rem. 49.8(1)
(2)	the block factor L, odd. Then the constants of L alone, among them c_L and β_∞; the constants of Part V (θ_0, c_6, C_6, γ'); the constants of Part VII that do not depend on M (C_L, c'_L)	$L \ge C3^d + 3$ (reblocking); $L^2 \ge 8(C_* + 1)$ (thresholds of consecutive steps)	Rem. 49.8(2), Thm. 25.1, Rem. 34.7
(3)	the weakening rate κ and the loss δ of Part IX; $\kappa'_w = \kappa - C\delta$ and $\kappa_2 = 3^{-d}\kappa'_w$	three rate conditions, which make every source decay at the rate κ_n after reblocking	Rem. 49.8(3)
(4)	ρ; the thresholds M_1, $M_1^{(0)}$, M'_1; the grid size M. Then the constants of M: a_1, B_3, a_2, B_5, a_δ, r_2, a_{step}, the constants of Theorem 45.5; then δ_*, C_W, and c''_L. Then the scale M' of the step; then n_*, $n_{\min}$	$M > 32(L+1)$, a power of L, $M \ge \max(M_0, 5L, M_1, M_1^{(0)}, M'_1, 64)$, $M \ge 16r$; $C_{\text{AB}}c''_L \le c'_L$, $C_{\text{cmp}}c''_L \le \delta_*$; $M' \ge M'_L$; (T0)–(T2)	Rem. 49.8(4), Thm. 30.3, proofs of Prop. 37.4(c) and Lemma 37.5, §52.1
(5)	the domain constants: the cap $\bar\delta$, the threshold $\varepsilon_{\max}$, the radii δ_P, $\delta_{P,0}$ of the complex references, c_a, $\delta_1 \le \delta_0$, θ_r; $\delta_1^\flat = \frac{1}{4}\delta_1$; then m_0; the constants of the step quantities	$\bar\delta \le \min(\delta_{AB}, 1/C_{\text{cx}})$; $\delta_P \le \bar\delta$, $\delta_P \le c''_L$; (DC1)–(DC6)	Rem. 38.8, Rem. 49.8(5), Prop. 46.7
(5′)	the constants of the flow: B_{new}, $B_{\text{new}}^{(0)}$, then $\mathsf{K} = 2\max(B_{\text{new}}, B_{\text{new}}^{(0)})$; C_{old}, C_β, C_{ex}, C_{new}, $C_{\text{new}}^{(0)}$; c_{F}; C_2, C_3 of (F2), (F3)	none; they do not depend on the couplings or the torus	Rem. 53.5(i),(ii), Prop. 58.6
(6)	the bound $\gamma \le g_p$ on the couplings	(Γ1)–(Γ4); for Theorem 58.9 also $\gamma \le e^{-3q}$ and $C_3\gamma^2 p_0(\gamma)^6 \le \frac{1}{2}\beta_\infty$	Rem. 49.8(6), Rem. 53.5
(7)	g_{uv}, C and m_1 of the ultraviolet limit	g_{uv} small	Thm. 58.9

Table 8: The order of the parameters. Stage (5′) lies between stages (5) and (6) of Remark 49.8. The constants δ_*, C_W and c''_L depend on M and are fixed after it (proofs of Proposition 37.4(c) and Lemma 37.5); c''_L enters the choice of δ_P (Proposition 46.7).

Item	Bałaban, Dimock	This note	Where
coercivity at $U = 1$	momentum space [3]	position space; one Fourier lower bound	Thm. 9.6, Cor. 4.3
positivity at regular backgrounds	[7, Theorem 3.11] and local axial-gauge estimates [4, Lemma 2.4]	IMS localization and the coercivity of Part I	Thm. 19.6, §60
Dirichlet operators	Dirichlet operators on domains [7]	a mass term near the boundary (§60); in the note, Dirichlet conditions for all components (Rem. 41.5)	Thm. G(ii), Rem. 41.5
averaging	symmetric averaging [18, (0.3)–(0.12)]	the same; its linearization has $E = 0$	§21, Cor. 22.11
axial-to-Landau chart	[6, Theorem 2]	torus version, with a recursive correspondence map	Thm. 22.5, Thm. 24.2
induction over k	nested domains Ω_j [8, Sect. F]	on the torus, by a change of the reference (splicing)	Lemma 29.2, Thm. 30.3
gauges on large cubes	bounds linear in the size [8, (9), (10)]	polynomial bounds	Rem. 30.6
gauge fixing of the step	exponential gauge fixing [18, (0.14)–(0.19)], Faddeev–Popov identity [18, (0.15)–(0.16)], weights relative to 1	the same, relative to the base W	Lemma 32.3 and after it
small-field cutoffs	fixed ε_0 for the domain; cutoff [18, (2.9)] with fixed ε_1 or depending on the coupling; white noise [18, (2.10)], [15, (206)]	$h_k = p(g_k)g_k$ and $p_0(g_k)$ depend on the coupling, as in [15]; the domain shrinks with the coupling	Rem. 53.14(a)
volume-type localization	random walks with weakening parameters (Bałaban, Dimock); boundary estimates for Dirichlet propagators [2, 3, 7]	the same walks; pointwise bounds through interior L^p estimates and tiers, without boundary estimates	§§41–44
step in polymer form	Dimock's scheme [15]: weakened step quantities, Mayer and polymer expansion with the two cutoffs; domain constants independent of the coupling [18, (1.11)–(1.16)]	the same	§§47–48, Rem. 38.8
generations	[18, (0.22)–(0.30)]; activities and β_{j+1} smooth in one coupling on $[0, \gamma]$	the same form; they depend on $(g_0, \dots, g_{j-1})$, continuously	Rem. 53.14(a), Prop. 53.10
extraction	a gauge common to a cube of scale k; fifth order, Ward identities; gain exponential in L^{k-j} [18, (0.27)]	a complex gauge per polymer; second order; gain L^{-m} for the large polymers	§51
later-scale bounds (0.29)	asserted in his Theorem 1, not among the inductive assumptions of his Theorem 3	proved in the induction	Thm. 53.12(d), Rem. 53.14(a)
the coefficient β_j	vacuum polarization of the generations [18, (1.20)–(1.22)]	period density of the symmetrized generation on the torus; as the torus grows it converges, at fixed couplings and without a rate, to the coefficient that Bałaban's definition assigns to these generations	Prop. 50.13, Prop. 58.8
ultraviolet limit	[18, Theorem 2], for $SU(2)$: (0.31)	compact simple G; the analogue of (0.31), for the couplings of the small-field flow, for $K - k \geq m_1$; the lower half up to a constant for the last m_1 steps	Thm. 58.9
vacuum energy	[15, Theorem 24]	the same bound; one counterterm E_0	Prop. 52.5
large fields	[16, 17]	not treated	Rem. 52.6(d)

Table 9: The note compared with Bałaban's and Dimock's papers, as the note describes them.

The generations and the couplings. Two differences in the flow need a word. First, Bałaban's generation of scale j depends only on the coupling g_{j-1}, while the generation of the note depends on the sequence $(g_0, \dots, g_{j-1})$ (Proposition 58.8(b)). The note builds each generation from the step that creates it, applied to the density of the flow, whose old activities depend on all earlier couplings (Theorem 53.7, Proposition 53.10). Second, only continuity in the couplings is proved; it is what the exit argument of Theorem 58.9 needs (Corollary 53.11).

What Dimock's scheme contributes. From [15] the note takes the organization of the small-field step: the step quantities as functions of weakening parameters that factorize over polymers, the Mayer and polymer expansion of the fluctuation integral with cutoffs on the fluctuation field and on the white noise (Section 1, the paragraph on Part X), the tree length d_M of the polymer grid (Section 49.2), the coupling-dependent small-field domain (Remark 53.14(a)), and the bound on the vacuum energy (proof of Proposition 52.5). Dimock's papers treat the large-field stage in [16, 17]; the note does not (Section 1, the paragraph after Theorem L).

Where in the note. The paragraph "Relation to Bałaban's construction" of Section 60; Remark 53.14(a); the opening of Section 51; Proposition 58.8; the paragraph after Theorem 58.9.

In the companion. The chapters on the second half of Part X and on the second half of Part XI.

G14 Where the hard parts are

These are the places to check first. Each rests on an idea specific to the problem, or on a long estimate on which much depends.

(1) *Dirichlet conditions for all components* (Section 19, Remark 41.5). The weakened walks need bounded local commutators. With only the interior divergence in the gauge-field Dirichlet operators, the normal components on the crossing bonds would be free, and the commutator term would be of order $\eta^{-1/2}M^{-1}$; the hypothesis of Lemma 41.3 would fail uniformly in η (Remark 41.5).

(2) *Pointwise bounds for the weakened operators* (Sections 42–44). The background map needs pointwise $C^{1,\beta}$ bounds for $\mathbf{H}(s)$ and $\mathbf{C}(s)$, while the walks give L^2 bounds. In four dimensions L^2 data lie at the endpoint $p = \frac{d}{2}$ of the interior estimates, so the regularity is raised in tiers, through block L^4 (Lemma 44.6, Theorems 44.13 and 44.15).

(3) *Complex references* (Section 46). The activities must be analytic in the background. So the results of Parts VIII and IX that the step uses, the expansion of the action (Proposition 23.3) and the averaged backgrounds of Part IV are extended to complex references; the coercivity constants and the decay rates are fixed numbers, independent of the size of the complex part, and the other constants change by at most a factor 2 (Propositions 46.4 and 46.7). The order $\bar{\delta}$, $\varepsilon_{\max}$, δ_P of the constants avoids a circle (Remark 38.8).

(4) *The curvature constant C_** (Proposition 31.5). That C_* does not depend on L is what makes the small-field sets of consecutive steps compatible, through $L^2 \geq 8(C_* + 1)$ (Remark 49.8(2)).

(5) *No term linear in p* (Lemma 54.9, Remark 58.10(b)). A term linear in the momentum in the vacuum polarization is of Chern–Simons type; under the change of scale it would

produce relevant terms of order one per step (Remark 54.8). The one-loop terms are exactly invariant under the translations and under B_4, by the symmetry of Bałaban's averaging; so their vacuum polarizations have no linear term and their marginal tensors are isotropic.

(6) *Exact isotropy of β* (Proposition 50.13). The extraction cancels a generation against a multiple of S, so the period density must be exactly proportional to that of S. The step does not produce covariant families; the symmetrization over the grid symmetries does, at the price of a smaller domain (Lemma 50.8).

(7) *The extraction* (Lemma 51.1, Proposition 51.5). The normal form of a later-scale configuration on a small polymer, in a complex gauge chosen per polymer, with a remainder smaller by $L^{-\alpha m}$.

(8) *(F2) and (F3)* (Proposition 58.6). (F2) rests on the torus independence of the one-loop pieces and of the Jacobian terms (Proposition 41.14, Lemma 58.3), on the infinite-volume minimizer (Proposition 54.15) and on the infinite-volume limits of the second variations (Lemma 58.5) (Remark 58.10(a)). (F3) rests on parity and on Cauchy estimates in a complex coupling inserted in the fluctuation only (proof of Proposition 58.6).

Where in the note. The results cited in the list.

In the companion. The chapters on Parts IX, X and XI expand these places step by step.

Part I: The trivial background

Part I studies the linear operators of Bałaban's renormalization group at the trivial background $U = 1$, on a single fine torus $\mathbb{T}_\eta$ of spacing $\eta = L^{-k}$ and side n, with averages over the unit blocks $B(y)$, $y \in \mathbb{T}$ (Section 2). At $U = 1$ every operator acts componentwise in a basis of the Lie algebra, so the fields are real-valued, and every constant of Part I depends only on the coefficient a of the averaging term and on the Hölder exponent α, not on L, k or n (Section 1). Parts II and III write every operator at a background as its counterpart at $U = 1$ plus a perturbation (Section 13), so Part I is the base of the whole operator theory.

What this Part proves. Bounds uniform in k: coercivity, exponential decay between unit blocks, and, for the Green functions, the minimizer maps and the constrained Green functions, pointwise bounds on values, first derivatives and Hölder quotients. They hold for the scalar Green functions $G_\sigma = (-\Delta + aQ_\sigma^* Q_\sigma)^{-1}$ (Theorem 7.1); for the averaged operators $(Q_\sigma G_\sigma Q_\sigma^*)^{-1}$, the minimizer maps, the effective actions, the constrained Green functions and the projection R of Bałaban's Landau-type gauge (Section 8); for the gauge-field operator $\mathbf{\Delta}_a = d^*d + dRd^* + a\mathbf{Q}^*\mathbf{Q}$ and its inverse $\mathbf{G}$ (Theorems 9.6 and 10.1); and for the gauge-field minimizer map $\mathbf{H}$, effective action $\mathbf{D}$ and constrained Green function $\mathbf{C}$ (Section 11). Theorems A–E of the Introduction collect these results. The tools are a block Poincaré inequality (Section 3), one Fourier lower bound (Corollary 4.3), the Combes–Thomas method (Section 5) and a lattice Schauder estimate (Theorem 6.3). The coercivity of $\mathbf{\Delta}_a$ is proved in position space; its key step is Lemma 9.3, by which $\mathbf{Q}$ nearly separates exact and co-closed fields.

Imports. Appendix A gives the decay of the Green function of $\mathbb{Z}^d$ and of its differences, hence that of $\eta\mathbb{Z}^d$ (Proposition 6.1). The averages $\mathbf{Q}$ and the projection R are those of [3].

No proof in Part I uses a later result. Section 2 points ahead only to say where $\mathbf{Q}$ comes from: it is the linearization at $U = 1$ of the composite of k gauge-adapted one-step averagings (Proposition 22.10(c)).

Exports.

- Theorem 6.3 (interior regularity) is used in Parts II–X, and Lemma 2.6 (properties of the averages, among them $\mathbf{Q}\, d = d_1 Q_0$ and the lattice symmetry) in Parts II–IV, VII, IX and XI.
- The decay calculus and the Combes–Thomas lemmas (Lemmas 5.2, 5.5 and 5.7) are used in Parts II, III, VII, VIII, X and XI.
- Proposition 11.5, the comparison of $\mathbf{D}$ with $\|d_1 B\|^2$, is used in Parts VII and XI.
- The operator theorems (Theorems 7.1, 9.6, 10.1, 11.3, 11.7, Propositions 8.1, 8.8, 11.2) are the unperturbed operators of Parts II and III, and Part XI uses them directly, in infinite volume (Section 54.2).

The constants of Lemmas 3.1, 4.4 and 9.3 and Corollary 3.3 are tracked in the proof of Lemma 31.3, on the uniformity of the constants of Parts II and III. All of Part I is on the main line.

Details for this Part. The companion has no chapter on Part I; its parameter ledger lists the constants of Part I that Parts VII–XI use.

2 Lattices, fields and averaging operators

This section fixes notation and collects the elementary identities used throughout. Nothing here is deep, but the identities of Lemma 2.4 and the restriction estimate of Lemma 2.7 are used repeatedly.

2.1 Lattices and blocks

Fix $d = 4$, an odd integer $L \geq 3$, and integers $k \geq 1$ and $n \geq 8$, and put

$$N = L^k, \qquad \eta = \frac{1}{N}.$$

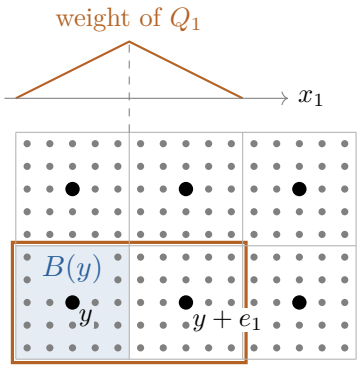
Thus $N \geq 3$ is odd. We keep the letter d in formulas, but $d = 4$ throughout the paper. A constant is called *absolute* if it is a number, independent of L, k, n and all other parameters; since d is fixed, constants that depend only on the dimension are absolute. The *unit torus* and the *fine torus* are

$$\mathbb{T} = (\mathbb{Z}/n\mathbb{Z})^d, \qquad \mathbb{T}_\eta = (\eta\mathbb{Z}/n\mathbb{Z})^d.$$

Thus $\mathbb{T}_\eta$ is a torus of side n with lattice spacing η, and $\mathbb{T} \subset \mathbb{T}_\eta$ is the sublattice of points with integer coordinates. We write $e_1, \dots, e_d$ for the unit coordinate vectors, and $|x - x'|$ for the Euclidean distance on the torus (the minimum over representatives). Since $n \geq 8$, every subset of a cube of side at most 7 can be identified with a subset of $\eta\mathbb{Z}^d$, and we do this tacitly. On cubes of side at most $4 \leq n/2$ the torus distance coincides with the distance in $\eta\mathbb{Z}^d$. On larger cubes it can only be smaller, which makes Hölder quotients computed with the torus distance larger; this direction is harmless wherever such quotients appear as hypotheses.

For $y \in \mathbb{T}$ the *block* centred at y is

$$B(y) = \Big\{ x \in \mathbb{T}_\eta : \ |x_\mu - y_\mu| \leq \frac{1-\eta}{2}, \ \mu = 1, \dots, d \Big\}.$$



fine points $\mathbb{T}_\eta$ (spacing η)
unit points $\mathbb{T}$ (large dots): block centres
block $B(y)$: support of Q_0 at y
tube $B(y) \cup B(y + e_1)$: support of Q_1 at y

Figure 1: A two-dimensional picture of the two lattices, for $N = 5$. The blocks are centred at the unit points. The block average $(Q_0 f)(y)$ is the mean of f over $B(y)$. The tube average $(Q_1 f)(y)$ is the mean over $x \in B(y)$ of the averages of f along the unit segments from x to $x + e_1$. Its weight is the triangle shown above the tube. It is peaked at the face between $B(y)$ and $B(y + e_1)$.

Since N is odd, $\frac{1-\eta}{2} = \frac{N-1}{2}\,\eta$ is a multiple of η. So $B(y)$ is the cube of N^d fine points with centre y. The blocks partition $\mathbb{T}_\eta$, and we write $y(x)$ for the centre of the block containing x. The parameter L enters only through $N = L^k$, which is odd. Our constants never depend on L, and the reader may think of N as an arbitrary odd integer $N \geq 3$.

Remark 2.1 (Relation to Bałaban's notation). In [3] the fine lattice is T_η, the unit lattice is $T_1^{(k)}$, and the fields have been rescaled so that the averaging blocks have side 1. Our $\mathbb{T}_\eta$ and $\mathbb{T}$ are these lattices, with the torus side denoted by n. All of Bałaban's operators after k renormalization steps live on this pair of lattices. Our blocks are Bałaban's cubes centred at the points of the coarse lattice [18, (0.1)–(0.3)].

Lattice symmetries. Let $B_4 = (\mathbb{Z}_2)^4 \rtimes S_4$ be the hyperoctahedral group, i.e. the group of the $2^4 \cdot 4! = 384$ signed permutation matrices of $\mathbb{R}^4$. Let $\mathfrak{S}_k$ be the group of the maps $r(x) = Rx + v$ of $\mathbb{T}_\eta$ with $R \in B_4$ and $v \in \mathbb{T}$. The subscript records that $\mathbb{T}$ is the lattice reached after k averaging steps. The group $\mathfrak{S}_k$ is generated by the translations by vectors of $\mathbb{T}$ and by the maps $x \mapsto y_0 + R(x - y_0)$ with $y_0 \in \mathbb{T}$, i.e. by B_4 acting about the block centres. Every $r \in \mathfrak{S}_k$ is affine and preserves the Euclidean and the ∞-distance on $\mathbb{T}_\eta$. It maps $\mathbb{T}$ onto $\mathbb{T}$, and $B(y)$ onto $B(ry)$. It acts on scalar fields on $\mathbb{T}_\eta$ and on $\mathbb{T}$ by $(rf)(x) = f(r^{-1}x)$. Throughout, B_4 denotes this group; a block always appears together with its point, as in $B(y)$, $B_j(y)$ or $B'(y)$.

2.2 Fields, derivatives and the Hodge identity

A *scalar field* is a function $f : \mathbb{T}_\eta \to \mathbb{R}$. A *gauge field* (a 1-form) is a d-tuple $A = (A_1, \dots, A_d)$ of scalar fields. Here $A_\mu(x)$ is the value of A on the oriented bond $(x, x + \eta e_\mu)$, and the value on the reversed bond is $-A_\mu(x)$. We write $A(b)$ for the value on an oriented bond $b = (b_-, b_+)$, so that $A(-b) = -A(b)$. A map $r \in \mathfrak{S}_k$ maps oriented bonds to oriented bonds, $rb = (rb_-, rb_+)$, and it acts on gauge fields by $(rA)(b) = A(r^{-1}b)$. A 2-form is a collection $F = (F_{\mu\nu})$ with $F_{\nu\mu} = -F_{\mu\nu}$. Here $F_{\mu\nu}(x)$ is the value on the plaquette spanned by x, ηe_μ and ηe_ν. The inner products are

$$\langle f, g \rangle = \eta^d \sum_{x \in \mathbb{T}_\eta} f(x) g(x), \qquad \langle A, B \rangle = \sum_\mu \langle A_\mu, B_\mu \rangle, \qquad \langle F, E \rangle = \sum_{\mu < \nu} \langle F_{\mu\nu}, E_{\mu\nu} \rangle,$$

and $\|\cdot\|$ denotes the corresponding norms. For a set $X \subset \mathbb{T}_\eta$, $\|f\|_X$ is the norm of $\mathbf{1}_X f$. Unit-lattice fields are defined in the same way on $\mathbb{T}$, with inner product $\sum_{y\in\mathbb{T}}$ and no factor η^d, and $\mathfrak{S}_k$ acts on them in the same way. With these normalizations a block has volume $\eta^d N^d = 1$, so constant fields have comparable norms on both lattices.

The forward and backward derivatives are

$$(\partial_\mu f)(x) = \frac{f(x+\eta e_\mu) - f(x)}{\eta}, \qquad (\partial_\mu^* f)(x) = \frac{f(x-\eta e_\mu) - f(x)}{\eta}.$$

Then ∂_μ^* is the adjoint of ∂_μ. All ∂_μ and ∂_ν^* commute with one another and with translations of $\mathbb{T}_\eta$. The exterior derivative and its adjoint are

$$(df)_\mu = \partial_\mu f, \qquad d^*A = \sum_\mu \partial_\mu^* A_\mu,$$

$$(dA)_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad (d^*F)_\nu = \sum_\mu \partial_\mu^* F_{\mu\nu}.$$

One checks directly that d^* is the adjoint of d and that $d(df) = 0$. The (non-positive) Laplacian is $\Delta = -\sum_\mu \partial_\mu^*\partial_\mu = -d^*d$ on scalar fields. It acts on gauge fields componentwise. We use the shorthand

$$\|\partial f\|^2 = \sum_\mu \|\partial_\mu f\|^2 = \langle f, -\Delta f\rangle, \qquad \|\partial A\|^2 = \sum_{\mu,\nu} \|\partial_\nu A_\mu\|^2.$$

Gauge fields of the form $d\lambda$ are called *exact*, and those with $d^*A = 0$ *co-closed*.

Lemma 2.2 (Hodge identities)**.** *For every gauge field A,*

$$(d^*d + dd^*)A = -\Delta A \quad \textit{(componentwise)}, \qquad \|dA\|^2 + \|d^*A\|^2 = \|\partial A\|^2.$$

Moreover $-\Delta$ is invertible on scalar fields with mean zero, and every gauge field decomposes orthogonally as $A = d\lambda + A'$ with $d^*A' = 0$ and $\lambda = (-\Delta)^{-1}d^*A$. The same statements hold on the unit torus, with ∂_μ replaced by the unit difference $D_\mu g(y) = g(y+e_\mu) - g(y)$.

Proof. For the first identity,

$$(d^*dA)_\mu = \sum_\nu \partial_\nu^*(\partial_\nu A_\mu - \partial_\mu A_\nu), \qquad (dd^*A)_\mu = \partial_\mu \sum_\nu \partial_\nu^* A_\nu,$$

and the mixed terms cancel because ∂_μ and ∂_ν^* commute. Taking the inner product with A gives the second identity. Next, $-\Delta$ is symmetric and $\langle f, -\Delta f\rangle = \|\partial f\|^2$. This vanishes only for constant f on the connected torus. Hence $-\Delta$ is invertible on the orthogonal complement of the constants. Since d^*A has mean zero ($\langle 1, d^*A\rangle = \langle d1, A\rangle = 0$), λ is well defined. Then $d^*(A - d\lambda) = d^*A + \Delta\lambda = 0$, and $\langle d\lambda, A'\rangle = \langle \lambda, d^*A'\rangle = 0$. The proof on $\mathbb{T}$ is identical. □

On the unit torus we write d_1 for the exterior derivative built from D_μ, and d_1^* for its adjoint, with $(D_\mu^* g)(y) = g(y - e_\mu) - g(y)$.

2.3 Directional convolutions

The flat block and tube averages introduced in this section are obtained by restricting products of one-dimensional averages to the unit lattice, so we introduce a convenient language for those products.

Definition 2.3. Let $\mu \in \{1, \dots, d\}$ and let $\kappa : \eta\mathbb{Z} \to \mathbb{R}$ vanish outside an interval of length at most 6. The *directional convolution* in direction μ with kernel κ is

$$(Kf)(x) = \sum_{s\in\eta\mathbb{Z}} \eta\,\kappa(s)\, f(x + se_\mu),$$

where $x + se_\mu$ is taken modulo the torus. Products of directional convolutions in different directions are called *product convolutions*.

Directional convolutions commute with each other and with all translations of $\mathbb{T}_\eta$. The adjoint of K has kernel $s \mapsto \kappa(-s)$. The composition of two convolutions in the same direction has kernel $(\kappa_1 \star \kappa_2)(s) = \sum_t \eta\, \kappa_1(t)\kappa_2(s-t)$. Moreover $|\kappa_1 \star \kappa_2| \le \|\kappa_1\|_\infty \sum_t \eta |\kappa_2(t)|$, and the support lengths add. The box averages a_μ, a_μ^*, a_μ^c and the triangle average c_μ defined below have nonnegative kernels of mass 1, bounded by 1. Hence every composition of such averages, possibly composed with one further convolution whose kernel is bounded by M, has a kernel bounded by 1 (respectively M). Its support length is the sum of the support lengths.

We use the following operators in direction μ:

- the forward and backward *box averages* a_μ, a_μ^*, with kernels $\mathbf{1}_{[0,1)}$ and $\mathbf{1}_{(-1,0]}$:

$$(a_\mu f)(x) = \eta \sum_{j=0}^{N-1} f(x + j\eta e_\mu), \qquad (a_\mu^* f)(x) = \eta \sum_{j=0}^{N-1} f(x - j\eta e_\mu);$$

- the *centred box average* a_μ^c, with kernel $\mathbf{1}_{[-(1-\eta)/2,(1-\eta)/2]}$:

$$(a_\mu^c f)(x) = \eta \sum_{|j| \le (N-1)/2} f(x + j\eta e_\mu).$$

  Its kernel is even, so a_μ^c is self-adjoint;

- the unit difference $D_\mu f(x) = f(x + e_\mu) - f(x)$, the shift $(\tau_\mu f)(x) = f(x + (1-\eta)e_\mu)$ and the half shift $(\tau_\mu^{1/2} f)(x) = f(x + \frac{1-\eta}{2} e_\mu)$. The half shift is a shift of $\mathbb{T}_\eta$, because N is odd, and $(\tau_\mu^{1/2})^2 = \tau_\mu$;

- the *triangle average* $c_\mu = a_\mu a_\mu^*$.

Lemma 2.4 (One-dimensional identities)**.** *For each μ:*

(a) $D_\mu = a_\mu \partial_\mu = \partial_\mu a_\mu$ *and* $D_\mu^* = a_\mu^* \partial_\mu^*$;

(b) $a_\mu = \tau_\mu a_\mu^*$;

(b′) $a_\mu^c = \tau_\mu^{1/2} a_\mu^*$, *and hence* $a_\mu = \tau_\mu^{1/2} a_\mu^c$;

(c) c_μ *has kernel* $c(s) = (1 - |s|)_+$, *and* $\sum_s \eta\, c(s) = 1$;

(d) $1 - c_\mu = e_\mu \partial_\mu^* \partial_\mu$, *where* e_μ *has a kernel* k *with* $0 \le k \le \frac{1}{2}$, $k(s) = 0$ *for* $|s| \ge 1$, *and* $|k(s) - k(s')| \le |s - s'|$. *Consequently* $e_\mu \partial_\mu$ *is a directional convolution with kernel* $k_1(s) = (k(s-\eta) - k(s))/\eta$, *which satisfies* $|k_1| \le 1$ *and vanishes outside* $[-1, 1+\eta]$;

(e) $a_\mu - 1 = g_\mu \partial_\mu$, *where* g_μ *has kernel* $(1 - \eta - s)\mathbf{1}_{[0,1)}(s) \in [0, 1]$.

Proof. All identities are between convolution operators, so it suffices to check them on $\eta\mathbb{Z}$, i.e. for functions of one variable t. The torus versions follow by periodization.

(a) $(a\partial f)(t) = \eta \sum_{j=0}^{N-1} \frac{f(t+(j+1)\eta) - f(t+j\eta)}{\eta} = f(t+1) - f(t)$, since $N\eta = 1$. Taking adjoints gives the second identity.

(b) $(\tau a^* f)(t) = \eta \sum_{j=0}^{N-1} f(t + 1 - \eta - j\eta) = \eta \sum_{j'=0}^{N-1} f(t + j'\eta)$, with $j' = N - 1 - j$.

(b′) $(\tau^{1/2} a^* f)(t) = \eta \sum_{j=0}^{N-1} f(t + \frac{N-1}{2}\eta - j\eta) = \eta \sum_{|j'| \le (N-1)/2} f(t + j'\eta)$, with $j' = \frac{N-1}{2} - j$. The second identity follows from the first and (b), since $\tau^{1/2} a^c = \tau a^* = a$.

(c) The kernel of aa^* at $s \in \eta\mathbb{Z}$ is $\eta \cdot \#\{t \in \eta\mathbb{Z} : t \in [0,1),\ s - t \in (-1, 0]\}$. For $0 \le s < 1$ the admissible t are $s, s+\eta, \dots, 1-\eta$, which gives $\eta \cdot (1-s)/\eta = 1 - s$. The case $s < 0$ is symmetric. Finally $\sum_s \eta\, c(s) = \eta(1 + 2\sum_{j=1}^{N-1}(1 - j\eta)) = \eta(2N - 1 - (N-1)) = 1$.

(d) Let $-\Delta_1 h(t) = \eta^{-2}(2h(t) - h(t+\eta) - h(t-\eta))$ denote the one-dimensional operator $\partial^*\partial$. A direct check shows $-\Delta_1(-|t|/2) = \eta^{-1}\delta_{t,0}$. Put

$$k(t) = -\frac{|t|}{2} + \frac{1}{2}\sum_s \eta\,|t-s|\,c(s).$$

Then $-\Delta_1 k = \eta^{-1}\delta_{\cdot,0} - c$. Hence the convolution e with kernel k satisfies $e\,\partial^*\partial f = (-\Delta_1 k)\star f = f - cf$. If $|t| \geq 1$, then $|t-s| = |t| - \mathrm{sgn}(t)s$ for all s in the support of c. Since c has mass 1 and mean 0, this gives $k(t) = 0$. For $|t| < 1$, the triangle inequality and $\sum \eta c(s)(t-s) = t$ give $k(t) \geq 0$, while $\sum_s \eta|t-s|c(s) \leq |t| + 1$ gives $k(t) \leq \frac{1}{2}$. Each of the two terms defining k is Lipschitz with constant $\frac{1}{2}$. The formula for the kernel of $e\partial$ follows by summation by parts: $\sum_s \eta k(s)\frac{f(t+s+\eta)-f(t+s)}{\eta} = \sum_s \eta \frac{k(s-\eta)-k(s)}{\eta} f(t+s)$.

(e) $(af)(t) - f(t) = \eta \sum_{j=0}^{N-1}\sum_{i=0}^{j-1} \eta\,\partial f(t+i\eta) = \sum_{i=0}^{N-1} \eta\,\eta(N-1-i)\,\partial f(t+i\eta)$, and $\eta(N-1-i) = 1-\eta-i\eta$. □

2.4 Averaging operators

Let ι denote restriction to the unit torus, $(\iota f)(y) = f(y)$ for $y \in \mathbb{T}$. Its adjoint is $(\iota^* g)(x) = \eta^{-d} g(x)$ if $x \in \mathbb{T}$ and 0 otherwise. Put $b = a_1^c a_2^c \cdots a_d^c$, the centred average over the cube of side $1-\eta$ about the point.

Definition 2.5 (Averaging operators)**.** The *block average* Q_0 and the *tube averages* Q_μ, $\mu = 1, \ldots, d$, map scalar fields on $\mathbb{T}_\eta$ to scalar fields on $\mathbb{T}$:

$$(Q_0 f)(y) = (\iota b f)(y) = \eta^d \sum_{x \in B(y)} f(x),$$

$$(Q_\mu f)(y) = (\iota a_\mu b f)(y) = \eta^d \sum_{x \in B(y)} \eta \sum_{j=0}^{N-1} f(x + j\eta e_\mu).$$

The *gauge-field average* maps gauge fields on $\mathbb{T}_\eta$ to gauge fields on $\mathbb{T}$:

$$(\mathbf{Q}A)_\mu = Q_\mu A_\mu, \qquad \text{i.e.} \qquad (\mathbf{Q}A)_\mu(y) = \eta^d \sum_{x \in B(y)} \eta \sum_{j=0}^{N-1} A_\mu(x + j\eta e_\mu).$$

We write Q_σ, $\sigma \in \{0, 1, \ldots, d\}$, when a statement holds for all of them, and $\mathcal{K}_\sigma$ for the product convolution with $Q_\sigma = \iota \mathcal{K}_\sigma$ ($\mathcal{K}_0 = b$, $\mathcal{K}_\mu = a_\mu b$).

The gauge-field average is the linearization at $U = 1$ of Bałaban's averaging. In four dimensions the one-step averaging is [18, (0.12)], and we construct it in Section 21. Besides the straight segments from the points x of a block to their translates $x + e_\mu$, it contains contour terms, built from averages over the coordinate-ordered contours inside the blocks. At $U = 1$ these terms vanish in the averaged axial gauge of Section 22. Dropping them is the "second point of view" of [3, (1.13)], which we adopt. What remains is $\sum_{x \in B(y)} \eta^{d+1} A([x, x+e_\mu])$, as in [3, (1.11), (1.18)], where $A([x, x+e_\mu])$ is the sum of A over the unit segment from x to $x + e_\mu$. This is exactly $(\mathbf{Q}A)_\mu(y)$. That the composition of k gauge-adapted one-step averagings (Definition 22.4) linearizes to $\mathbf{Q}$ is Proposition 22.10(c); the plain composition of the averagings (21.3) linearizes to $\mathbf{Q}$ minus coarse gradients (Lemma 22.8(f)).

Lemma 2.6 (Properties of the averages)**.** *(a)* $(Q_\sigma f)(y) = \eta^d \sum_x w_\sigma(y,x) f(x)$ *with* $0 \leq w_\sigma \leq 1$, $\eta^d \sum_x w_\sigma(y,x) = 1$ *and* $\sum_y w_\sigma(y,x) = 1$. *Moreover* $w_0(y,x) = \mathbf{1}_{B(y)}(x)$, *and* $w_\mu(y,\cdot)$ *vanishes outside* $B(y) \cup B(y + e_\mu)$.

(b) $Q_\sigma 1 = 1$ *and* $Q_\sigma^* 1 = 1$. *Further,* $(Q_0^* g)(x) = g(y(x))$, $Q_0 Q_0^* = I$, *and* $Q_0^* Q_0$ *is the orthogonal projection onto fields that are constant on each block.*

(c) $|(Q_\sigma f)(y)|^2 \le \eta^d \sum_x w_\sigma(y,x)|f(x)|^2$. *Hence* $\|Q_\sigma\| \le 1$, $\|\mathbf{Q}\| \le 1$, *and* $|(Q_\sigma^* g)(x)| \le \max\{|g(y)| : w_\sigma(y,x) \ne 0\}$, *where at most two* y *occur.*

(d) (Gauge covariance.) $D_\mu Q_0 = Q_\mu \partial_\mu$, *i.e.* $\mathbf{Q}\, d = d_1 Q_0$. *More generally,* $D_\mu Q_\sigma = \iota\, a_\mu \mathcal{K}_\sigma \partial_\mu$.

(e) (Curvature.) $d_1 \mathbf{Q} = \mathbf{Q}^{(2)} d$, *where* $(\mathbf{Q}^{(2)} F)_{\mu\nu} = \iota\, a_\mu a_\nu b\, F_{\mu\nu}$.

(f) (Lattice symmetry.) *For every* $r \in \mathfrak{S}_k$, $Q_0 r = r Q_0$ *on scalar fields and* $\mathbf{Q} r = r \mathbf{Q}$ *on gauge fields.*

Proof. (a) For Q_0 this is the definition. For Q_μ, the coefficient of $f(x)$ in $(Q_\mu f)(y)$ is $\eta^d \cdot \eta \,\#\{j \in \{0,\dots,N-1\} : x - j\eta e_\mu \in B(y)\}$. So $w_\mu(y,x) \le \eta N = 1$, and $w_\mu(y,x) \ne 0$ only if $x \in B(y) \cup B(y+e_\mu)$. The normalization $\eta^d \sum_x w_\sigma = Q_\sigma 1 = 1$ is clear. The identity $\sum_y w_\sigma(y,x) = 1$ is $Q_\sigma^* 1 = 1$, which follows from $a_\mu^* 1 = 1$ and $(b\,\iota^* g)(x) = g(y(x))$. Here $b = b^*$, since the kernels of the a_ν^c are even, and $(b\,\iota^* g)(x)$ is the sum of $g(y)$ over the $y \in \mathbb{T}$ with $x \in B(y)$; there is exactly one such y, because the blocks partition $\mathbb{T}_\eta$.

(b) This follows from (a). Moreover $(Q_0 Q_0^* g)(y) = \eta^d \sum_{x \in B(y)} g(y) = g(y)$. Hence $Q_0^* Q_0$ is a self-adjoint idempotent, and its range consists of the block-constant fields.

(c) The first bound is Cauchy–Schwarz with the probability weights $\eta^d w_\sigma(y,\cdot)$. Summing over y and using $\sum_y w_\sigma(y,x) = 1$ gives $\|Q_\sigma f\|^2 \le \|f\|^2$. (Below we sometimes use the weaker bound $\|Q_\sigma\|^2 \le 2$.) The bound on $Q_\sigma^* g$ follows from $(Q_\sigma^* g)(x) = \sum_y w_\sigma(y,x) g(y)$ and $\sum_y w_\sigma(y,x) = 1$.

(d) ι commutes with unit translations, so $D_\mu Q_\sigma = \iota D_\mu \mathcal{K}_\sigma = \iota \mathcal{K}_\sigma a_\mu \partial_\mu$ by Lemma 2.4(a). For $\sigma = 0$ this equals $\iota a_\mu b\, \partial_\mu = Q_\mu \partial_\mu$.

(e) $(d_1 \mathbf{Q} A)_{\mu\nu} = D_\mu Q_\nu A_\nu - D_\nu Q_\mu A_\mu = \iota\, a_\nu b\, a_\mu \partial_\mu A_\nu - \iota\, a_\mu b\, a_\nu \partial_\nu A_\mu$, by (a) of Lemma 2.4.

(f) Let $r \in \mathfrak{S}_k$. Since $r^{-1} B(y) = B(r^{-1} y)$,

$$(Q_0 r f)(y) = \eta^d \sum_{x \in B(y)} f(r^{-1}x) = \eta^d \sum_{x' \in B(r^{-1}y)} f(x') = (r Q_0 f)(y).$$

For $\mathbf{Q}$ we use oriented unit bonds $c = (c_-, c_+)$, and write $(\mathbf{Q}A)(c)$ for the value of $\mathbf{Q}A$ on c. For fine points x, x' that differ by a unit vector $\pm e_\mu$, let $A([x,x'])$ be the sum of A over the N bonds of the straight segment from x to x', oriented from x to x'. We claim that for every oriented unit bond c,

$$(\mathbf{Q}A)(c) = \eta^{d+1} \sum_{x \in B(c_-)} A([x, x + c_+ - c_-]).$$

For $c = (y, y+e_\mu)$ this is Definition 2.5, because $\eta \sum_{j=0}^{N-1} A_\mu(x + j\eta e_\mu) = \eta A([x, x+e_\mu])$. For the reversed bond $-c$, use $B(y+e_\mu) = B(y) + e_\mu$ and $A([x+e_\mu, x]) = -A([x, x+e_\mu])$: the right-hand side for $-c$ is $-(\mathbf{Q}A)(c) = (\mathbf{Q}A)(-c)$. This proves the claim. Now r^{-1} is affine, so it maps the segment from x to $x + c_+ - c_-$, with its orientation, onto the segment from $r^{-1}x$ to $r^{-1}x + (r^{-1}c)_+ - (r^{-1}c)_-$. Hence $(rA)([x, x+c_+-c_-]) = A([r^{-1}x, r^{-1}x + (r^{-1}c)_+ - (r^{-1}c)_-])$. Substituting $x' = r^{-1}x$, which runs over $r^{-1}B(c_-) = B((r^{-1}c)_-)$, the claim gives

$$(\mathbf{Q} r A)(c) = \eta^{d+1} \sum_{x' \in B((r^{-1}c)_-)} A([x', x' + (r^{-1}c)_+ - (r^{-1}c)_-]) = (\mathbf{Q}A)(r^{-1}c) = (r\mathbf{Q}A)(c). \quad \square$$

Part (d) is the reason for the particular form of $\mathbf{Q}$. A gauge transformation $A \mapsto A + d\lambda$ changes the average by the unit-lattice gradient of the block average of λ:

$$\mathbf{Q}(A + d\lambda) = \mathbf{Q}A + d_1 Q_0 \lambda.$$

In particular, the gauge transformations that preserve all averages are those with $Q_0\lambda$ constant. Modulo constants, these are those with $Q_0\lambda = 0$. This is the residual gauge freedom of the variational problem, cf. [3, (1.13), (1.20)].

Part (f) holds because the blocks are centred at the unit points. With blocks having the unit point at a corner, Q_0 would commute with the translations of $\mathbb{T}$ but not with B_4 about the unit points. Geometrically, (f) says that the tube of c, the union of the segments from the points of $B(c_-)$ in the direction of c, is mapped onto itself by every $r \in \mathfrak{S}_k$ with $rc = \pm c$, for instance by the reflection that fixes the midpoint of c and reverses c.

2.5 Restriction of product convolutions

The averaging operators restrict a smoothed field to the unit lattice. The following estimate is used whenever we need to bound such restrictions in ℓ^2.

Lemma 2.7 (Restriction lemma)**.** *Let $\mathcal{K} = K_1 \cdots K_d$ be a product convolution, where K_μ has a kernel bounded by M_μ and vanishing outside an interval of length $\ell_\mu \le 6$. Then for every scalar field f,*

$$\|\iota \mathcal{K} f\|_{\ell^2(\mathbb{T})} \le \prod_{\mu=1}^{d} M_\mu(\ell_\mu + 1)\ \|f\|.$$

Proof. Write $(\mathcal{K}f)(y) = \sum_t \eta^d \kappa(t) f(y+t)$ with $\kappa(t) = \prod_\mu \kappa_\mu(t_\mu)$. By Cauchy–Schwarz,

$$|(\mathcal{K}f)(y)|^2 \le \Big(\sum_t \eta^d |\kappa(t)|\Big)\Big(\sum_t \eta^d |\kappa(t)|\,|f(y+t)|^2\Big).$$

The first factor is at most $\prod_\mu M_\mu(\ell_\mu + \eta)$. Summing the second factor over $y \in \mathbb{T}$ gives $\sum_x \eta^d |f(x)|^2 \sum_y |\kappa(x-y)|$. For fixed x and μ, at most $\ell_\mu + 1$ integers y_μ satisfy $\kappa_\mu(x_\mu - y_\mu) \neq 0$. Since $\ell_\mu + 1 \le 7 < n$, no point is counted twice on the torus. Hence $\sum_y |\kappa(x-y)| \le \prod_\mu M_\mu(\ell_\mu + 1)$. □

When several convolutions act in the same direction, they are first merged into one directional convolution, using the composition rule above. For example, Q_σ itself, $\iota\, a_\mu a_\nu b$, and $\iota\, b\, g_\mu$ satisfy the hypotheses with absolute constants. A shift τ_μ or $\tau_\mu^{1/2}$ composed with a convolution only moves the support of the kernel, so it does not change the bound.

3 Poincaré inequalities and scalar coercivity

The Laplacian $-\Delta$ on $\mathbb{T}_\eta$ has eigenvalues as small as $(2\pi/n)^2$. So it controls a field only up to modes that vary on the scale of the whole torus. The averaging term $a\|Q_\sigma f\|^2$ supplies the missing control on the unit scale. The precise statement is a Poincaré inequality on unit blocks, uniform in η.

Lemma 3.1 (Block Poincaré inequality)**.** *For every block $B = B(y)$ and every scalar field f,*

$$\|f - (Q_0 f)(y)\|_B^2 \le \frac{d}{2}\,\eta^d \sum_{b \subset B} |\partial f(b)|^2.$$

Here the sum runs over all fine bonds $b = (x, x + \eta e_\mu)$ with both endpoints in B, and $\partial f(b) = \partial_\mu f(x)$. Consequently, for every scalar field f,

$$\|f - Q_0^* Q_0 f\|^2 \le \frac{d}{2}\|\partial f\|^2, \qquad \|f\|^2 \le \frac{d}{2}\|\partial f\|^2 + \|Q_0 f\|^2. \tag{3.1}$$

Proof. Let $\bar f = (Q_0 f)(y)$ be the mean of f over B. Since B has N^d points and volume 1,

$$\|f - \bar f\|_B^2 = \eta^d \sum_{x \in B} |f(x) - \bar f|^2 = \frac{\eta^{2d}}{2} \sum_{x,x' \in B} |f(x) - f(x')|^2.$$

For $x, x' \in B$ we join x' to x by the lattice path that first moves in direction 1 from x'_1 to x_1, then in direction 2 from x'_2 to x_2, and so on. It stays in B and has at most $d(N-1)$ steps. Then $f(x) - f(x')$ is a sum of at most dN terms $\pm\eta\,\partial f(b)$, and by Cauchy–Schwarz

$$|f(x) - f(x')|^2 \le dN\eta^2 \sum_{b \in \mathrm{path}(x',x)} |\partial f(b)|^2 = d\eta \sum_{b \in \mathrm{path}(x',x)} |\partial f(b)|^2.$$

A bond $b = (z, z + \eta e_\mu)$ lies on the μ-th segment of the path only if $z_\nu = x_\nu$ for $\nu < \mu$ and $z_\nu = x'_\nu$ for $\nu > \mu$. The remaining free coordinates are x'_ν for $\nu < \mu$, x_ν for $\nu > \mu$, and the pair (x_μ, x'_μ). Hence at most $N^{\mu-1} \cdot N^{d-\mu} \cdot N^2 = N^{d+1}$ pairs (x, x') use b. Therefore

$$\|f - \bar f\|_B^2 \le \frac{\eta^{2d}}{2}\, d\eta\, N^{d+1} \sum_{b \subset B} |\partial f(b)|^2 = \frac{d}{2}\, \eta^d \sum_{b \subset B} |\partial f(b)|^2,$$

using $\eta N = 1$. Summing over blocks gives the first inequality in (3.1), since bonds between different blocks only add to the right-hand side. The second follows from the orthogonal decomposition $f = (f - Q_0^* Q_0 f) + Q_0^* Q_0 f$ and $\|Q_0^* Q_0 f\| = \|Q_0 f\|$ (Lemma 2.6(b)). □

Lemma 3.2 (Comparison of averages)**.** *For $\mu = 1, \dots, d$ and every scalar field f,*

$$\|Q_\mu f - Q_0 f\| \le 3 \cdot 2^d\, \|\partial_\mu f\|, \qquad \|D_\nu Q_\sigma f\| \le 6^d\, \|\partial_\nu f\| \quad (\sigma = 0, \dots, d).$$

Proof. By Lemma 2.4(e), $Q_\mu f - Q_0 f = \iota\, b(a_\mu - 1) f = \iota\, b\, g_\mu\, \partial_\mu f$. In direction μ the operator $a_\mu^c g_\mu$ has a kernel bounded by $1 + \eta \le 2$ and supported in an interval of length 2. In every other direction the kernel is $\mathbf{1}_{[-(1-\eta)/2,(1-\eta)/2]}$. Lemma 2.7 gives the constant $2 \cdot 3 \cdot 2^{d-1}$. Similarly, $D_\nu Q_\sigma f = \iota\, a_\nu \mathcal{K}_\sigma \partial_\nu f$ by Lemma 2.6(d). This is a product of at most three box averages in each direction, with kernels bounded by 1 and supported in intervals of length at most 3. Lemma 2.7 gives a factor at most 4 in direction ν and at most 3 in each other direction. □

Corollary 3.3 (Poincaré inequality for all averages)**.** *There is an absolute constant C_P such that for every $\sigma \in \{0, \dots, d\}$ and every scalar field f,*

$$\|f\|^2 \le C_P(\|\partial f\|^2 + \|Q_\sigma f\|^2).$$

For gauge fields, $\|A\|^2 \le C_P(\|\partial A\|^2 + \|\mathbf{Q} A\|^2)$.

Proof. By (3.1) and Lemma 3.2, $\|f\|^2 \le \frac{d}{2}\|\partial f\|^2 + 2\|Q_\sigma f\|^2 + 2 \cdot 9 \cdot 4^d \|\partial f\|^2$. Apply this to each component A_μ with $\sigma = \mu$ and sum over μ. □

Corollary 3.4 (Scalar coercivity)**.** *Let $a > 0$ and $c_0 = \min(1, a)/(2 + 2C_P)$. For every σ and every scalar field f,*

$$\langle f, (-\Delta + a Q_\sigma^* Q_\sigma) f \rangle = \|\partial f\|^2 + a\|Q_\sigma f\|^2 \ge c_0(\|f\|^2 + \|\partial f\|^2).$$

In particular $-\Delta + a Q_\sigma^* Q_\sigma$ is invertible, and its inverse G_σ satisfies $0 < G_\sigma \le c_0^{-1}$.

Proof. $\|\partial f\|^2 + a\|Q_\sigma f\|^2 \ge \min(1, a)(\frac{1}{2}\|\partial f\|^2 + \frac{1}{2}(\|\partial f\|^2 + \|Q_\sigma f\|^2))$. Then use Corollary 3.3 on the last bracket. □

This is the scalar half of Theorem A. The next lemma is used in the gauge-field part. It concerns fields with vanishing block averages, i.e. generators of the residual gauge transformations.

Lemma 3.5 (Fields with vanishing averages)**.** *If $Q_0\lambda = 0$, then*

$$\|\lambda\| \le \sqrt{d/2}\,\|\partial\lambda\|, \qquad \|\partial\lambda\| \le \sqrt{d/2}\,\|\Delta\lambda\|.$$

In particular $-\Delta$ is injective on $\mathcal{N}(Q_0) = \{\lambda : Q_0\lambda = 0\}$.

Proof. The first inequality is (3.1) with $Q_0\lambda = 0$. For the second, $\|\partial\lambda\|^2 = \langle\lambda, -\Delta\lambda\rangle \le \|\lambda\|\|\Delta\lambda\| \le \sqrt{d/2}\,\|\partial\lambda\|\,\|\Delta\lambda\|$. □

4 Two Fourier estimates

Fourier analysis plays only a small role in this paper. Its main use is to show that averaged Green functions such as $Q_\sigma G_\sigma Q_\sigma^*$ are bounded *below* uniformly in η. Upper bounds and decay are proved in position space. We also need one estimate on "biharmonic extensions" in the gauge-field coercivity proof.

4.1 Fourier transform and Bloch decomposition

The dual of the fine torus is $\mathbb{T}_\eta^* = \big((2\pi/n)\mathbb{Z}/(2\pi/\eta)\mathbb{Z}\big)^d$. We represent it by the momenta p with $-\pi/\eta < p_\mu \le \pi/\eta$. For a scalar field f we put

$$\hat f(p) = \eta^d \sum_{x\in\mathbb{T}_\eta} e^{-\mathrm{i}p\cdot x} f(x), \qquad f(x) = n^{-d} \sum_{p\in\mathbb{T}_\eta^*} e^{\mathrm{i}p\cdot x}\hat f(p), \qquad \|f\|^2 = n^{-d}\sum_p |\hat f(p)|^2.$$

On the unit torus, $\hat g(q) = \sum_{y\in\mathbb{T}} e^{-\mathrm{i}q\cdot y} g(y)$ for $q \in \mathbb{T}^*$, which we represent by $-\pi < q_\mu \le \pi$, and $\|g\|^2 = n^{-d}\sum_q |\hat g(q)|^2$.

Every $p \in \mathbb{T}_\eta^*$ can be written uniquely as $p = q + 2\pi l$ with $q \in \mathbb{T}^*$ and $l \in \mathbb{Z}^d$. For fixed q exactly N^d vectors l occur. This is the *Bloch decomposition*.

A directional convolution with kernel κ in direction μ acts in momentum space as multiplication by $\tilde\kappa(p_\mu)$, where $\tilde\kappa(\theta) = \sum_s \eta\,\kappa(s)e^{\mathrm{i}\theta s}$. Its adjoint has symbol $\overline{\tilde\kappa}$. In particular

$$\widehat{\partial_\mu f}(p) = \frac{e^{\mathrm{i}\eta p_\mu} - 1}{\eta}\,\hat f(p), \qquad \widehat{-\Delta f}(p) = \hat\Delta(p)\hat f(p), \quad \hat\Delta(p) = \sum_\mu \frac{4}{\eta^2}\sin^2\frac{\eta p_\mu}{2},$$

and a_μ has symbol $\alpha(p_\mu)$, with

$$\alpha(\theta) = \eta\sum_{j=0}^{N-1} e^{\mathrm{i}j\eta\theta} = \eta\,\frac{e^{\mathrm{i}\theta}-1}{e^{\mathrm{i}\eta\theta}-1}, \qquad |\alpha(\theta)| = \frac{|\sin(\theta/2)|}{N|\sin(\eta\theta/2)|}.$$

The centred box average a_μ^c has the real symbol

$$\alpha_c(\theta) = \eta\sum_{|j|\le(N-1)/2} e^{\mathrm{i}j\eta\theta} = \frac{\sin(\theta/2)}{N\sin(\eta\theta/2)} = e^{-\mathrm{i}(1-\eta)\theta/2}\alpha(\theta),$$

by Lemma 2.4(b′). In particular $|\alpha_c| = |\alpha|$, so every bound below on $|\alpha|$ holds for $|\alpha_c|$. Restriction to the unit lattice and its adjoint satisfy the *Poisson summation formula*:

$$\widehat{\iota f}(q) = \sum_l \hat f(q + 2\pi l), \qquad \widehat{\iota^* g}(p) = \hat g(q) \quad \text{for } p = q + 2\pi l. \tag{4.1}$$

The first identity follows by inserting the inversion formula for f into $\sum_y e^{-\mathrm{i}q\cdot y} f(y)$ and using $\sum_{y\in\mathbb{T}} e^{\mathrm{i}(p-q)\cdot y} = n^d\mathbf{1}[p - q \in 2\pi\mathbb{Z}^d]$. The second is immediate from the definition of ι^*.

Lemma 4.1 (Symbol bounds)**.** *Let $|\theta| \le \pi/\eta$.*

(a) $\frac{4}{\pi^2}\theta^2 \le \frac{4}{\eta^2}\sin^2\frac{\eta\theta}{2} \le \theta^2$. *Hence* $\frac{4}{\pi^2}|p|^2 \le \hat\Delta(p) \le |p|^2$ *on the fundamental domain.*

(b) $|\alpha(\theta)| \le 1$, *and* $|\alpha(\theta)| \ge 2/\pi$ *if* $|\theta| \le \pi$.

(c) If $\theta = q + 2\pi l$ *with* $|q| \le \pi$ *and* $l \in \mathbb{Z}\setminus\{0\}$, *then* $|\theta| \ge \pi|l|$ *and* $|\alpha(\theta)| \le |q|/(2|l|)$.

Proof. (a) uses $\frac{2}{\pi}t \le \sin t \le t$ for $0 \le t \le \pi/2$. (b) The first bound is the triangle inequality. For $|\theta| \le \pi$, $|\sin(\theta/2)| \ge |\theta|/\pi$ and $N|\sin(\eta\theta/2)| \le |\theta|/2$. (c) $|\theta| \ge 2\pi|l| - |q| \ge \pi|l|$. Moreover $|\sin(\theta/2)| = |\sin(q/2)| \le |q|/2$, and $N|\sin(\eta\theta/2)| \ge N\frac{2}{\pi}\frac{\eta|\theta|}{2} = |\theta|/\pi \ge |l|$. □

4.2 The lower bound

Lemma 4.2 (Fourier lower bound)**.** *Let K be a translation-invariant operator on scalar fields of $\mathbb{T}_\eta$ with symbol $\hat K(p) \ge 0$ on $\mathbb{T}^*_\eta$, and let $\mathcal{K}$ be a product convolution with symbol $\kappa(p)$. Represent each $q \in \mathbb{T}^*$ by $-\pi < q_\mu \le \pi$; this is a point of $\mathbb{T}^*_\eta$. Then for every field g on $\mathbb{T}$,*

$$\langle \mathcal{K}^*\iota^* g, K\mathcal{K}^*\iota^* g\rangle \;\ge\; m\,\|g\|^2, \qquad m = \min_{q\in\mathbb{T}^*} \hat K(q)|\kappa(q)|^2.$$

Proof. By (4.1), $h = \mathcal{K}^*\iota^* g$ has $\hat h(q+2\pi l) = \overline{\kappa(q+2\pi l)}\,\hat g(q)$. Therefore

$$\langle h, Kh\rangle = n^{-d}\sum_q |\hat g(q)|^2 \sum_l \hat K(q+2\pi l)|\kappa(q+2\pi l)|^2 \;\ge\; n^{-d}\sum_q |\hat g(q)|^2\,\hat K(q)|\kappa(q)|^2,$$

where we kept only the term $l = 0$. The claim follows from Parseval on $\mathbb{T}$. □

Corollary 4.3. *For every $\sigma \in \{0,\dots,d\}$ and $a > 0$,*

$$Q_\sigma(-\Delta + 2a)^{-1}Q_\sigma^* \;\ge\; c_F, \qquad c_F = \Big(\frac{2}{\pi}\Big)^{2d+2}\frac{1}{d\pi^2 + 2a}.$$

Proof. Apply Lemma 4.2 with $K = (-\Delta+2a)^{-1}$ and $\mathcal{K} = \mathcal{K}_\sigma$. The symbol of $\mathcal{K}_\sigma$ is a product of at most $d+1$ factors $\alpha(p_\nu)$ or $\alpha_c(p_\nu)$, each of modulus at least $2/\pi$ on $[-\pi,\pi]^d$, which contains the representatives of $\mathbb{T}^*$. Moreover $\hat\Delta(p) \le |p|^2 \le d\pi^2$ there. □

4.3 Biharmonic extensions

The following estimate is used in Section 9. Given a unit-lattice field ω with mean zero, the field $\lambda = (-\Delta)^{-2}Q_0^*\omega$ is the smoothest (in the sense of $\|\Delta\lambda\|$) fine field with prescribed block averages $Q_0\lambda$, which then have mean zero. We will see that this is exactly the gradient part of a gauge field in Bałaban's gauge. Its energy is controlled by the unit-lattice gradient of its averages.

Lemma 4.4 (Biharmonic extensions)**.** *There is an absolute constant C_B with the following property. Let ω be a field on $\mathbb{T}$ with mean zero, and let $\lambda = (-\Delta)^{-2}Q_0^*\omega$ (the inverse taken on mean-zero fields). Then*

$$\|\partial\lambda\| \le C_B\,\|d_1 Q_0\lambda\|.$$

Proof. Let $\beta(p) = \prod_\nu \alpha_c(p_\nu)$ be the symbol of b. By (4.1), $\hat\lambda(p) = \overline{\beta(p)}\,\hat\omega(q)\,\hat\Delta(p)^{-2}$ for $p = q + 2\pi l \ne 0$, and $\hat\lambda(0) = 0$. For $j \in \{2,3\}$ and $q \in \mathbb{T}^*\setminus\{0\}$ put $S_j(q) = \sum_l |\beta(q+2\pi l)|^2\hat\Delta(q+2\pi l)^{-j}$. Then

$$\|\partial\lambda\|^2 = \langle\lambda, -\Delta\lambda\rangle = n^{-d}\sum_{q\ne 0}|\hat\omega(q)|^2 S_3(q).$$

The term $q = 0$ is absent, because $\hat\omega(0) = 0$, and because for $l \neq 0$ some factor $\alpha_c(2\pi l_\nu)$ vanishes, since $0 < |l_\nu| < N$ for some ν on the fundamental domain. Similarly, $g = Q_0\lambda = \iota b\lambda$ has $\hat g(q) = \sum_l \beta(q+2\pi l)\hat\lambda(q+2\pi l) = \hat\omega(q)S_2(q)$, and

$$\|d_1 g\|^2 = n^{-d}\sum_{q\neq 0} \hat\Delta_1(q)\, S_2(q)^2\, |\hat\omega(q)|^2, \qquad \hat\Delta_1(q) = \sum_\nu 4\sin^2\frac{q_\nu}{2}.$$

It therefore suffices to show $S_3(q) \le C_B^2 \hat\Delta_1(q) S_2(q)^2$ for $q \in [-\pi,\pi]^d \setminus \{0\}$.

By Lemma 4.1, keeping only the term $l = 0$ in S_2, we get $S_2(q) \ge (2/\pi)^{2d}|q|^{-4}$, and $\hat\Delta_1(q) \ge \frac{4}{\pi^2}|q|^2$. Hence

$$\hat\Delta_1(q) S_2(q)^2 \ge \frac{4}{\pi^2}\Big(\frac{2}{\pi}\Big)^{4d}|q|^{-6}.$$

In S_3, the term $l = 0$ is at most $(\pi/2)^6|q|^{-6}$. For $l \neq 0$, Lemma 4.1(a),(c) gives $\hat\Delta(q+2\pi l) \ge \frac{4}{\pi^2}|q+2\pi l|^2 \ge 4|l|^2 \ge 4$ and $|\beta(q+2\pi l)|^2 \le \prod_{\nu:\, l_\nu\neq 0} \frac{\pi^2}{4l_\nu^2}$. Hence

$$\sum_{l\neq 0} \frac{|\beta(q+2\pi l)|^2}{\hat\Delta(q+2\pi l)^3} \le \frac{1}{64}\prod_{\nu=1}^{d}\Big(1 + 2\sum_{m\ge 1}\frac{\pi^2}{4m^2}\Big) = \frac{1}{64}\Big(1+\frac{\pi^4}{12}\Big)^d.$$

Since $|q|^2 \le d\pi^2$, this constant is at most $\frac{1}{64}(1+\pi^4/12)^d(d\pi^2)^3|q|^{-6}$, and the lemma follows. □

5 Operators with exponential decay and the Combes–Thomas method

Exponential decay will first be established in L^2, between unit blocks. This section develops the bookkeeping for this notion and the basic mechanism that produces it: if K is coercive and conjugation by $e^{\alpha\rho}$ changes K only by $O(\alpha)$, then K^{-1} decays exponentially.

5.1 Block decay

We work with four kinds of field spaces: scalar fields and gauge fields on $\mathbb{T}_\eta$, and scalar fields and gauge fields on $\mathbb{T}$. On each of them we have the *localization projections* $\mathbf{1}_y$, $y \in \mathbb{T}$. On the fine lattice, $\mathbf{1}_y$ is multiplication by the indicator of $B(y)$ (for gauge fields: of the bonds $(x, x+\eta e_\mu)$ with $x \in B(y)$). On the unit lattice, $\mathbf{1}_y$ keeps the value(s) at y. In all cases $\sum_y \mathbf{1}_y = I$.

Definition 5.1. A linear operator K between two such spaces has *decay* (C,γ) if

$$\|\mathbf{1}_y K \mathbf{1}_{y'}\| \le Ce^{-\gamma|y-y'|} \qquad \text{for all } y, y' \in \mathbb{T}.$$

For operators between unit-lattice spaces, $\|\mathbf{1}_y K \mathbf{1}_{y'}\|$ is the absolute value (or matrix norm) of the kernel $K(y,y')$, so decay is a pointwise kernel bound. We put

$$S_\gamma = \sum_{z\in\mathbb{Z}^d} e^{-\gamma|z|}.$$

Then $\sum_{y'\in\mathbb{T}} e^{-\gamma|y-y'|} \le S_\gamma$ for every y, uniformly in n, because distinct y' have distinct nearest representatives of $y'-y$ in $\mathbb{Z}^d$.

Lemma 5.2 (Calculus of decay)**.** *(a) If K has decay (C,γ), then $\|K\| \le CS_\gamma$.*

(b) If K_1 and K_2 have decay (C_1,γ) and (C_2,γ), then K_1K_2 has decay $(C_1C_2S_{\gamma/2},\gamma/2)$.

(c) *If K has decay (C,γ) then K^* has decay (C,γ). Sums behave in the obvious way, and decay (C,γ) implies decay (C,γ') for $\gamma' \le \gamma$.*

(d) *If $\mathbf{1}_y K \mathbf{1}_{y'} = 0$ for $|y-y'| > r$ and $\|K\| \le M$, then K has decay $(Me^{\gamma r},\gamma)$ for every $\gamma > 0$.*

Proof. (a) $\|\mathbf{1}_y K f\| \le \sum_{y'} C e^{-\gamma|y-y'|}\|\mathbf{1}_{y'} f\|$. The matrix $(e^{-\gamma|y-y'|})$ has row and column sums at most S_γ, so by the Schur test it defines an operator on $\ell^2(\mathbb{T})$ of norm at most S_γ. Hence $\|Kf\|^2 = \sum_y \|\mathbf{1}_y K f\|^2 \le C^2 S_\gamma^2 \sum_{y'} \|\mathbf{1}_{y'} f\|^2$.

(b) $\mathbf{1}_y K_1 K_2 \mathbf{1}_{y'} = \sum_{y''} \mathbf{1}_y K_1 \mathbf{1}_{y''} K_2 \mathbf{1}_{y'}$. By the triangle inequality $|y-y''| + |y''-y'| \ge \frac{1}{2}|y-y'| + \frac{1}{2}|y-y''|$, so the norm is at most $C_1 C_2 e^{-\gamma|y-y'|/2} \sum_{y''} e^{-\gamma|y-y''|/2}$.

(c) and (d) are immediate. □

5.2 Conjugation by exponential weights

A function $\rho : \mathbb{T}_\eta \to \mathbb{R}$ is 1-*Lipschitz* if $|\rho(x) - \rho(x')| \le |x - x'|$ for all x, x'. A typical example is $\rho(x) = \operatorname{dist}(x, B(y'))$, the torus distance to a block. For $\alpha \in \mathbb{R}$, multiplication by $e^{\alpha\rho}$ acts on all four field spaces. On unit-lattice fields it multiplies the value at y by $e^{\alpha\rho(y)}$, which is consistent because $\mathbb{T} \subset \mathbb{T}_\eta$. For an operator K we write

$$K_\alpha = e^{\alpha\rho} K e^{-\alpha\rho}.$$

Then $(K_1K_2)_\alpha = (K_1)_\alpha (K_2)_\alpha$, $(K^*)_\alpha = (K_{-\alpha})^*$, and $(K^{-1})_\alpha = (K_\alpha)^{-1}$. Since ι commutes with multiplication operators, $(\iota)_\alpha = \iota$.

Lemma 5.3 (Conjugation of decaying operators)**.** *Let K have decay (C,γ) and let ρ be 1-Lipschitz. For $|\alpha| \le \gamma/2$,*

$$\|K_\alpha - K\| \le C\, c_1(\gamma)\, |\alpha|, \qquad c_1(\gamma) = e^{\gamma\sqrt{d}}(2\sqrt{d} + 4/\gamma) S_{\gamma/4}.$$

Proof. Fix y, y', put $X = \mathbf{1}_y K \mathbf{1}_{y'}$, and write $e^{\alpha\rho} = e^{\alpha\rho(y)} E$ on the support of $\mathbf{1}_y$. Here E is multiplication by $e^{\alpha(\rho-\rho(y))}$. Since points of $B(y)$ are within distance $\sqrt{d}$ of y, $\|E - 1\| \le |\alpha|\sqrt{d}\, e^{|\alpha|\sqrt{d}}$ and $\|E\| \le e^{|\alpha|\sqrt{d}}$. Define E' similarly at y'. Then $\mathbf{1}_y(K_\alpha - K)\mathbf{1}_{y'} = sEXE'^{-1} - X$ with $s = e^{\alpha(\rho(y)-\rho(y'))}$. Since $|s-1| \le |\alpha||y-y'|e^{|\alpha||y-y'|}$, the decomposition

$$sEXE'^{-1} - X = (s-1)EXE'^{-1} + (E-1)XE'^{-1} + X(E'^{-1} - 1)$$

gives

$$\|\mathbf{1}_y(K_\alpha - K)\mathbf{1}_{y'}\| \le C|\alpha|\, e^{2|\alpha|\sqrt{d}}(|y-y'| + 2\sqrt{d}) e^{-(\gamma-|\alpha|)|y-y'|}.$$

For $|\alpha| \le \gamma/2$ we have $e^{2|\alpha|\sqrt{d}} \le e^{\gamma\sqrt{d}}$ and $(t + 2\sqrt{d})e^{-\gamma t/2} \le (2\sqrt{d} + 4/\gamma)e^{-\gamma t/4}$. Hence $K_\alpha - K$ has decay $(C|\alpha|e^{\gamma\sqrt{d}}(2\sqrt{d} + 4/\gamma), \gamma/4)$, and Lemma 5.2(a) concludes. □

Lemma 5.4 (Conjugated derivatives)**.** *Let ρ be 1-Lipschitz and $|\alpha| \le 1$. Then $(\partial_\mu)_\alpha = \partial_\mu + Y$ and $(\partial_\mu^*)_\alpha = \partial_\mu^* + Y'$, where Y, Y' are operators of norm at most $e|\alpha|$. On gauge fields the same holds componentwise.*

Proof. $(e^{\alpha\rho}\partial_\mu e^{-\alpha\rho} u)(x) = \partial_\mu u(x) + \eta^{-1}(e^{\alpha(\rho(x)-\rho(x+\eta e_\mu))} - 1)u(x + \eta e_\mu)$. Moreover $|e^t - 1| \le |t|e^{|t|}$ with $|t| \le |\alpha|\eta$. The computation for ∂_μ^* is the same. □

5.3 The Combes–Thomas method

We give two versions. The first is for bounded operators, typically on the unit lattice. The second is for fine-lattice operators containing $-\Delta$, whose norm is of order η^{-2}.

Lemma 5.5 (Combes–Thomas, bounded version)**.** *Let K act on one of the four field spaces, have decay (C,γ), and satisfy $\langle f, Kf\rangle \geq c\|f\|^2$ for all (real) fields f. Put $\gamma' = \min\{\gamma/2,\ c/(2Cc_1(\gamma))\}$. Then K is invertible and K^{-1} has decay $(2c^{-1}e^{2\gamma'\sqrt{d}}, \gamma')$.*

Proof. Fix y', let $\rho(x) = \operatorname{dist}(x, B(y'))$ and $\alpha = \gamma'$. By Lemma 5.3, $\langle f, K_\alpha f\rangle \geq c\|f\|^2 - \frac{c}{2}\|f\|^2$. Hence K_α is invertible with $\|K_\alpha^{-1}\| \leq 2/c$. Here we use that in finite dimensions $\langle f, Xf\rangle \geq \delta\|f\|^2$ implies $\|Xf\| \geq \delta\|f\|$. Now $K^{-1} = e^{-\alpha\rho}K_\alpha^{-1}e^{\alpha\rho}$. On the support of $\mathbf{1}_{y'}$ we have $\rho = 0$. On the support of $\mathbf{1}_y$ we have $\rho \geq |y-y'| - 2\sqrt{d}$. Therefore $\|\mathbf{1}_y K^{-1}\mathbf{1}_{y'}\| \leq e^{-\alpha(|y-y'|-2\sqrt{d})}\, 2/c$. □

Lemma 5.6 (Neumann series)**.** *Let $\mathfrak{N}$ have decay (C,γ) and $\|\mathfrak{N}\| \leq \theta < 1$. Then $(I+\mathfrak{N})^{-1}$ has decay $(2(1-\theta)^{-1}e^{2\gamma'\sqrt{d}}, \gamma')$ with $\gamma' = \min\{\gamma/2, (1-\theta)/(2Cc_1(\gamma))\}$.*

Proof. As above, with $\|\mathfrak{N}_\alpha\| \leq \theta + (1-\theta)/2$, so that $\|(I+\mathfrak{N}_\alpha)^{-1}\| \leq 2/(1-\theta)$. □

This lemma is the form in which exponential decay is used in the variational problem, cf. [8, (68)–(71)].

For the fine-lattice operators we write ∂ for the collection $(\partial_\mu)_\mu$ acting on scalar fields, or componentwise on gauge fields, and ∂^* for its adjoint.

Lemma 5.7 (Combes–Thomas, form version)**.** *Let $\mathfrak{A}$ act on scalar or gauge fields on $\mathbb{T}_\eta$. Suppose that:*

(i) $\langle u, \mathfrak{A}u\rangle \geq c\,(\|u\|^2 + \|\partial u\|^2)$ for all u;

(ii) there are $C_1, \alpha_1 > 0$ such that for every 1-Lipschitz ρ, every $|\alpha| \leq \alpha_1$ and every u,

$$|\langle u, (\mathfrak{A}_\alpha - \mathfrak{A})u\rangle| \leq C_1|\alpha|\,(\|u\|^2 + \|\partial u\|^2).$$

Let $\gamma_0 = \min\{\alpha_1, 1/3, c/(2C_1)\}$. Then the four operators $\mathfrak{A}^{-1}$, $\partial\mathfrak{A}^{-1}$, $\mathfrak{A}^{-1}\partial^*$ and $\partial\mathfrak{A}^{-1}\partial^*$ have decay (C_0, γ_0), with $C_0 = 2(1+\sqrt{d})^2c^{-1}e^{2\gamma_0\sqrt{d}}$.

Proof. Let $|\alpha| \leq \gamma_0$. By (i) and (ii), $\langle u, \mathfrak{A}_\alpha u\rangle \geq \frac{c}{2}(\|u\|^2 + \|\partial u\|^2)$, so $\mathfrak{A}_\alpha$ is invertible. Let $u = \mathfrak{A}_\alpha^{-1}h$. Then $\frac{c}{2}(\|u\|^2 + \|\partial u\|^2) \leq \|u\|\|h\|$. This gives $\|u\| \leq \frac{2}{c}\|h\|$ and then $\|\partial u\|^2 \leq \frac{2}{c}\|u\|\|h\| \leq \frac{4}{c^2}\|h\|^2$. If instead $h = \partial^* g$, then $\frac{c}{2}(\|u\|^2 + \|\partial u\|^2) \leq \langle \partial u, g\rangle \leq \|\partial u\|\|g\|$, which gives $\|\partial u\| \leq \frac{2}{c}\|g\|$ and $\|u\| \leq \frac{2}{c}\|g\|$. Thus

$$\|\mathfrak{A}_\alpha^{-1}\|,\ \|\partial\mathfrak{A}_\alpha^{-1}\|,\ \|\mathfrak{A}_\alpha^{-1}\partial^*\|,\ \|\partial\mathfrak{A}_\alpha^{-1}\partial^*\| \leq \frac{2}{c}.$$

Now fix y', let $\rho = \operatorname{dist}(\cdot, B(y'))$ and $\alpha = \gamma_0$. By Lemma 5.4, $\partial e^{-\alpha\rho} = e^{-\alpha\rho}(\partial + Y)$ and $e^{\alpha\rho}\partial^* = (\partial^* + Y'^*)e^{\alpha\rho}$, where Y' comes from $(\partial)_{-\alpha}$. Here $Y = (Y_\mu)_\mu$ and Y' consist of d operators, each of norm at most $e\alpha$ by Lemma 5.4, so $\|Y\|, \|Y'\| \leq \sqrt{d}\,e\alpha$. Hence, for instance,

$$\mathbf{1}_y\partial\mathfrak{A}^{-1}\partial^*\mathbf{1}_{y'} = \mathbf{1}_y e^{-\alpha\rho}(\partial + Y)\mathfrak{A}_\alpha^{-1}(\partial^* + Y'^*)\mathbf{1}_{y'},$$

where we used $e^{\alpha\rho}\mathbf{1}_{y'} = \mathbf{1}_{y'}$. The middle operator has norm at most $\frac{2}{c}(1+\sqrt{d}\,e\alpha)^2 \leq \frac{2}{c}(1+\sqrt{d})^2$, because $e\alpha \leq 1$, and $\|\mathbf{1}_y e^{-\alpha\rho}\| \leq e^{-\alpha(|y-y'|-2\sqrt{d})}$. The other three operators are treated in the same way. □

The hypothesis (ii) is verified with the help of the following observation.

Lemma 5.8 (Conjugation of quadratic forms)**.** *Let $X = X_0 - Z$, where $X_0 = \sum_j c_j \, \partial_{\mu_j}^{(*)}$ is a finite linear combination of the operators ∂_μ, ∂_μ^* (acting componentwise on scalar or gauge fields, with values in the appropriate form degree) with scalar coefficients $\sum_j |c_j| \le M$, and Z is bounded with $\|Z\| \le M$ and $\|Z_\alpha - Z\| \le M|\alpha|$ for $|\alpha| \le \alpha_1 \le 1$. Then for $|\alpha| \le \alpha_1$,*

$$|\langle u, ((X^*X)_\alpha - X^*X)u\rangle| \le C_M|\alpha| \, (\|u\|^2 + \|\partial u\|^2),$$

with C_M depending only on M and d.

Proof. By Lemma 5.4, applied to each term of X_0, $X_\alpha = X + W_\alpha$ with $\|W_\alpha\| \le (eM + M)|\alpha|$. Also $\|X_0 u\| \le M\|\partial u\|$. Since $(X^*X)_\alpha = (X_{-\alpha})^* X_\alpha$,

$$\langle u, ((X^*X)_\alpha - X^*X)u\rangle = \langle W_{-\alpha}u, Xu\rangle + \langle Xu, W_\alpha u\rangle + \langle W_{-\alpha}u, W_\alpha u\rangle.$$

Finally, $\|Xu\| \le M(\|\partial u\| + \|u\|)$. □

6 Local regularity on the fine lattice

The Combes–Thomas method gives L^2 bounds between unit blocks. To obtain bounds on values, derivatives and Hölder quotients that are uniform in η, we use an interior regularity estimate for the lattice Laplacian. This is the lattice analogue of the classical fact that a solution of $-\Delta u = F$ with bounded F is $C^{1,\alpha}$ in the interior for every $\alpha < 1$.

6.1 The Green function of the fine lattice

Let g_1 be the Green function of $-\Delta$ on the infinite unit lattice $\mathbb{Z}^d$:

$$g_1(x) = \frac{1}{(2\pi)^d} \int_{[-\pi,\pi]^d} \frac{e^{\mathrm{i}p\cdot x}}{\sum_\mu 4\sin^2(p_\mu/2)} \, dp,$$

which is well defined, since $|p|^{-2}$ is integrable near 0 in four dimensions, and satisfies $-\Delta g_1 = \delta_0$. On $\eta\mathbb{Z}^d$ put $g_\eta(x) = \eta^{2-d} g_1(x/\eta)$. Then the convolution

$$(\mathcal{G}h)(x) = \sum_{x' \in \eta\mathbb{Z}^d} \eta^d \, g_\eta(x - x') \, h(x')$$

satisfies $-\Delta\mathcal{G}h = h$ and $\mathcal{G}(-\Delta h) = h$ for every finitely supported h on $\eta\mathbb{Z}^d$. For the second identity, sum by parts. Since $\mathcal{G}$ is a convolution, it commutes with all lattice derivatives: $\partial_\mu \mathcal{G}h = (\partial_\mu g_\eta) \star h$, and similarly for ∂_μ^*.

Proposition 6.1 (Decay of derivatives)**.** *For every word ∂^β of length $|\beta| \le 3$ in the operators $\partial_\mu, \partial_\mu^*$,*

$$|\partial^\beta g_\eta(x)| \le C_\beta \, (|x| + \eta)^{2-d-|\beta|}, \qquad x \in \eta\mathbb{Z}^d,$$

with C_β depending only on β.

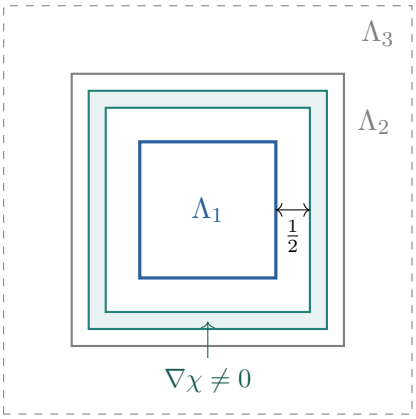
Proof. By scaling, $\partial^\beta g_\eta(x) = \eta^{2-d-|\beta|}(\nabla^\beta g_1)(x/\eta)$, where ∇^β is the corresponding word of unit-lattice differences. By Theorem A.1 in Appendix A, $|\nabla^\beta g_1(z)| \le C_\beta(1 + |z|)^{2-d-|\beta|}$. □

We also need two elementary lattice sums.

Lemma 6.2 (Lattice sums)**.** *Let $x \in \eta\mathbb{Z}^d$ and $R \ge \eta$.*

(a) For $0 \le s < d$: $\sum_{|x'-x| \le R} \eta^d (|x - x'| + \eta)^{-s} \le C_{d,s} R^{d-s}$.

(b) For $\sqrt{d}\,\eta \le r \le R$: $\sum_{r < |x'-x| \le R} \eta^d |x - x'|^{-d} \le C_d \, (1 + \log(R/r))$.



$\chi = 1$ inside the inner green square,
$\chi = 0$ outside the outer one.
The commutator supports lie in the green annulus enlarged by at most 2η.
For $\eta \le \frac{1}{16}$ this enlarged annulus is at distance $\ge \frac{1}{4}$ from $\Lambda_{1+2\eta}$.

Figure 2: Geometry of the proof of Theorem 6.3.

(c) For $s > d$ and $r \ge \sqrt{d}\,\eta$: $\sum_{|x'-x|>r} \eta^d |x-x'|^{-s} \le C_{d,s}\, r^{d-s}$.

Proof. Let $W(x') = x' + [-\eta/2, \eta/2)^d$. For $z \in W(x')$ we have $|z-x| + \eta \le (1+\sqrt{d})(|x-x'|+\eta)$. Hence each term in (a) is at most $(1+\sqrt{d})^s \int_{W(x')} (|z-x|+\eta)^{-s} dz$. The cubes $W(x')$ are disjoint and contained in the ball of radius $R + \sqrt{d}\eta \le (1+\sqrt{d})R$, which gives (a). In (b), $|x-x'| > r \ge \sqrt{d}\eta$ implies $\frac{1}{2}|x-x'| \le |z-x| \le \frac{3}{2}|x-x'|$ for $z \in W(x')$. Hence the sum is at most $(3/2)^d \int_{r/2 \le |z-x| \le 3R/2} |z-x|^{-d} dz = (3/2)^d |S^{d-1}| \log(3R/r)$. The same comparison gives (c). □

6.2 The regularity estimate

For $z \in \mathbb{R}^d$ and $r > 0$ let $\Lambda_r = \{x \in \mathbb{T}_\eta : |x-z|_\infty \le r\}$. Since $n \ge 8$, Λ_3 can be identified with a subset of $\eta\mathbb{Z}^d$.

Theorem 6.3 (Interior regularity)**.** *Let $\alpha \in (0,1)$. There is C, depending only on α, such that the following holds for every lattice spacing $\eta \le 1$, in particular $\eta = 1/N$ with $N \ge 1$, and all z; for a spacing not of the form $1/N$, Λ_r is the set of $x \in \eta\mathbb{Z}^d$ with $|x-z|_\infty \le r$. Let $u, F : \Lambda_3 \to \mathbb{R}$ and $g = (g_\mu) : \Lambda_3 \to \mathbb{R}^d$ satisfy*

$$-\Delta u = F + \sum_\mu \partial^*_\mu g_\mu \qquad \text{at every point of } \Lambda_2.$$

(a) If $g = 0$: $|u|_{\Lambda_1} + |\partial u|_{\Lambda_1} + [\partial u]_{\alpha,\Lambda_1} \le C(|F|_{\Lambda_2} + \|u\|_{\Lambda_2})$.

(b) In general: $|u|_{\Lambda_1} + [u]_{\alpha,\Lambda_1} \le C(|F|_{\Lambda_2} + |g|_{\Lambda_3} + \|u\|_{\Lambda_2})$.

(c) If $g = 0$: $\max_{\mu,\nu} |\partial_\mu \partial_\nu u|_{\Lambda_1} \le C(|F|_{\Lambda_2} + [F]_{\alpha,\Lambda_2} + \|u\|_{\Lambda_2})$.

(d) (Divergence form with Hölder data.) $|u|_{\Lambda_1} + |\partial u|_{\Lambda_1} + [\partial u]_{\alpha,\Lambda_1} \le C(|F|_{\Lambda_2} + |g|_{\Lambda_3} + [g]_{\alpha,\Lambda_3} + \|u\|_{\Lambda_2})$.

Proof. Step 0: coarse lattices. Let $\eta > \frac{1}{16}$ (for $\eta = 1/N$: $N < 16$). Then $|u(x)| \le \eta^{-d/2}\|u\|_{\Lambda_2} \le 16^{d/2}\|u\|_{\Lambda_2}$ for $x \in \Lambda_2$, and $[v]_{\alpha,X} \le 2\eta^{-\alpha}|v|_X$. A first difference at a point of Λ_1 involves only points of $\Lambda_{1+\eta} \subset \Lambda_2$ and costs a factor at most $2/\eta \le 32$. This proves (a), (b) and (d). For (c) with $\eta \le \frac{1}{2}$, a second difference at a point of Λ_1 involves only points of $\Lambda_{1+2\eta} \subset \Lambda_2$ and costs a factor at most $4/\eta^2$. If $\eta > \frac{1}{2}$ (for $\eta = 1/N$: $N = 1$), the difference $\partial_\mu\partial_\mu u(x)$ involves $u(x + 2\eta e_\mu)$, which may lie outside Λ_2. Put $v = x + \eta e_\mu$. Since $g = 0$ in (c), the equation at $v \in \Lambda_2$ gives

$$\eta^2 \partial_\mu \partial_\mu u(x) = (2d-2)\, u(v) - \sum_{\nu \ne \mu} \big(u(v+\eta e_\nu) + u(v-\eta e_\nu)\big) - \eta^2 F(v),$$

and all points on the right lie in Λ_2. The mixed differences $\partial_\mu\partial_\nu u(x)$, $\mu\neq\nu$, involve only points of Λ_2. We assume $\eta\le\frac{1}{16}$ from now on.

Step 1: cutoff. Fix $\chi\in C^\infty(\mathbb{R}^d)$ with $0\le\chi\le1$, $\chi=1$ on $\{|x-z|_\infty\le\frac32\}$ and $\chi=0$ on $\{|x-z|_\infty\ge\frac74\}$. Its lattice differences up to order three are bounded by absolute constants. Let $v=\chi u$, extended by zero to $\eta\mathbb{Z}^d$. Write $S_\mu u(x)=u(x+\eta e_\mu)$. The discrete Leibniz rules

$$\partial_\mu(\chi u)=\chi\,\partial_\mu u+(\partial_\mu\chi)S_\mu u,\qquad \partial_\mu^*(\varphi w)=\varphi\,\partial_\mu^* w+(\partial_\mu^*\varphi)S_\mu^{-1}w,$$

and $S_\mu^{-1}\partial_\mu u=-\partial_\mu^* u$ give, after a short computation,

$$-\Delta(\chi u)=\chi(-\Delta u)+\sum_\mu(\partial_\mu^* m_\mu+n_\mu),\qquad \begin{aligned} m_\mu&=(\partial_\mu\chi)S_\mu u-(\partial_\mu^*\chi)u,\\ n_\mu&=(\partial_\mu^*\partial_\mu^*\chi)S_\mu^{-1}u.\end{aligned}\tag{6.1}$$

The functions m_μ, n_μ vanish outside the annulus $A=\Lambda_{7/4+2\eta}\setminus\Lambda_{3/2-2\eta}$, where the differences of χ (up to second order, involving points at distance up to 2η) can be non-zero. They involve u only at points of Λ_2, and $|m_\mu|+|n_\mu|\le C|u|$ at neighbouring points. Similarly $\chi\,\partial_\mu^* g_\mu=\partial_\mu^*(\chi g_\mu)-(\partial_\mu^*\chi)S_\mu^{-1}g_\mu$. Hence

$$-\Delta v=\Phi_1+\Phi_2+\sum_\mu\partial_\mu^*(\chi g_\mu+m_\mu),\qquad \Phi_1=\chi F-\sum_\mu(\partial_\mu^*\chi)S_\mu^{-1}g_\mu,\quad \Phi_2=\sum_\mu n_\mu.$$

All terms vanish outside Λ_2. Since v has finite support,

$$v=\mathcal{G}\Phi_1+\mathcal{G}\Phi_2+\sum_\mu(\partial_\mu^* g_\eta)\star(\chi g_\mu)+\sum_\mu(\partial_\mu^* g_\eta)\star m_\mu.\tag{6.2}$$

For $x\in\Lambda_{1+2\eta}$ we have $v(x)=u(x)$. So u, ∂u and $\partial\partial u$ on Λ_1 can be computed from (6.2).

Step 2: the terms supported in the annulus. Points of $\Lambda_{1+2\eta}$ have distance at least $\frac12-4\eta\ge\frac14$ from A. By Proposition 6.1, every derivative of order at most three of g_η is bounded by a constant on $\{|x|\ge\frac14\}$. Therefore, for $x\in\Lambda_{1+2\eta}$ and $|\beta|\le2$,

$$|\partial^\beta\mathcal{G}\Phi_2(x)|+\sum_\mu|\partial^\beta(\partial_\mu^* g_\eta)\star m_\mu(x)|\le C\sum_{x'\in A}\eta^d\Big(|\Phi_2(x')|+\sum_\mu|m_\mu(x')|\Big)\le C\|u\|_{\Lambda_2},$$

by Cauchy–Schwarz. For $|\beta|\le1$ the derivatives of order one more are also bounded (they involve at most three derivatives of g_η), so these terms and their first derivatives are Lipschitz on Λ_1; for (c) only the sup bounds with $|\beta|=2$ are needed. A Lipschitz function on Λ_1 is Hölder: join x,x' by a monotone lattice path inside Λ_1, of length at most $\sqrt{d}|x-x'|\le2d$.

Step 3: the near terms in (a) and (b). Here Φ_1 and χg are bounded by $C(|F|_{\Lambda_2}+|g|_{\Lambda_3})$ and supported in Λ_2. By Lemma 6.2(a) with $s=d-2$ and $s=d-1$, for $x\in\Lambda_{1+2\eta}$,

$$|\mathcal{G}\Phi_1(x)|+|\partial\mathcal{G}\Phi_1(x)|+|(\partial^* g_\eta)\star(\chi g)(x)|\le C(|F|_{\Lambda_2}+|g|_{\Lambda_3}).$$

This proves the sup bounds in (a) and (b), and shows that $\mathcal{G}\Phi_1$ is Lipschitz on Λ_1. It remains to estimate the Hölder quotient of a convolution $k\star h$ with $|h|\le M$ supported in Λ_2. Here $k=\partial_\mu g_\eta$ in (a), and $k=\partial_\mu^* g_\eta$ in (b). In both cases $|k(y)|\le C(|y|+\eta)^{1-d}$, and every first difference of k is bounded by $C(|y|+\eta)^{-d}$. Let $x,x'\in\Lambda_1$, $\delta=|x-x'|\ge\eta$. Split the sum over x'' at $|x-x''|=4\sqrt{d}\,\delta$. On the near part, by Lemma 6.2(a) with $s=d-1$ (note $|x'-x''|\le(4\sqrt{d}+1)\delta$),

$$\sum_{|x-x''|\le4\sqrt{d}\delta}\eta^d\big(|k(x-x'')|+|k(x'-x'')|\big)\le C\delta,$$

On the far part we join x to x' by a monotone lattice path of at most $\sqrt{d}\delta/\eta$ steps. Every point w of the path, and its neighbours, satisfies $|w-x''| \geq \frac{1}{2}|x-x''|$. Hence $|k(x-x'')-k(x'-x'')| \leq C\sqrt{d}\,\delta\,|x-x''|^{-d}$, and by Lemma 6.2(b)

$$\sum_{4\sqrt{d}\delta<|x-x''|\leq 5\sqrt{d}} \eta^d |k(x-x'')-k(x'-x'')| \leq C\delta(1+\log(1/\delta)) \leq C_\alpha \delta^\alpha.$$

Here we may assume $\delta \leq 1$, the case $\delta > 1$ being covered by the sup bound. Multiplying by M proves the Hölder bounds in (a) and (b).

Step 4: second derivatives, (c). Fix $x \in \Lambda_1$ and let $\tilde\chi \in C_c^\infty$, with $0 \leq \tilde\chi \leq 1$, be equal to 1 on $\{|x'-z|_\infty \leq \frac{7}{4}\}$ and supported in $\{|x'-z|_\infty < \frac{15}{8}\}$. Write $\chi F(x'') = [\chi F(x'') - \chi F(x)\tilde\chi(x'')] + \chi F(x)\tilde\chi(x'')$. Then

$$\partial_\mu\partial_\nu \mathcal{G}(\chi F)(x) = \sum_{x''} \eta^d (\partial_\mu\partial_\nu g_\eta)(x-x'')\,[\chi F(x'') - \chi F(x)\tilde\chi(x'')] + \chi F(x)\,\mathcal{G}(\partial_\mu\partial_\nu\tilde\chi)(x),$$

because convolution commutes with differences. The last term is bounded by $C|F|_{\Lambda_2}$, since $\partial_\mu\partial_\nu\tilde\chi$ is bounded and supported in Λ_2: a second difference at x' involves only points within ∞-distance $2\eta \leq \frac{1}{8}$ of x', and $\frac{15}{8}+\frac{1}{8}=2$. In the sum, the term $x''=x$ vanishes. For $x'' \in \Lambda_{7/4}$ the bracket is at most $[\chi F]_{\alpha,\Lambda_2}|x-x''|^\alpha$. For $x'' \notin \Lambda_{7/4}$ we have $|x-x''| \geq \frac{3}{4}$, and the bracket is at most $|F|_{\Lambda_2}$. Using $|\partial_\mu\partial_\nu g_\eta(y)| \leq C|y|^{-d}$ for $y \neq 0$ and Lemma 6.2(a) with $s = d-\alpha$, the sum is at most $C([\chi F]_{\alpha,\Lambda_2} + |F|_{\Lambda_2})$. Finally $[\chi F]_{\alpha,\Lambda_2} \leq [F]_{\alpha,\Lambda_2} + C|F|_{\Lambda_2}$. The second derivatives of $\mathcal{G}\Phi_2$ and of the m-terms were estimated in Step 2.

Step 5: divergence form with Hölder data, (d). By Steps 2 and 3, all terms of (6.2) except $w = \sum_\mu (\partial_\mu^* g_\eta) \star G_\mu$, $G_\mu = \chi g_\mu$, already satisfy the bounds of (d); note that Φ_1 is bounded by $C(|F|_{\Lambda_2} + |g|_{\Lambda_3})$. The field G vanishes outside Λ_2, $|G| \leq |g|_{\Lambda_3}$, and $[G]_\alpha := [G]_{\alpha,\eta\mathbb{Z}^d} \leq [g]_{\alpha,\Lambda_3} + C|g|_{\Lambda_3}$. Here $[G]_{\alpha,\eta\mathbb{Z}^d}$ is the Hölder quotient over all pairs of points. For $z, z' \in \Lambda_3$ we use $|\chi(z)g(z) - \chi(z')g(z')| \leq |g(z)-g(z')| + |\chi(z)-\chi(z')|\,|g(z')|$. If $z \in \Lambda_3$ and $z' \notin \Lambda_3$, then $G(z') = \chi(z') = 0$ and $|G(z)| = |\chi(z)-\chi(z')|\,|g(z)|$; if both points lie outside Λ_3, the difference is zero. The bounded Lipschitz norm of χ gives the stated bound. We have $\partial_\nu w = \sum_\mu K_{\nu\mu} \star G_\mu$ with $K = K_{\nu\mu} = \partial_\nu \partial_\mu^* g_\eta$. By Proposition 6.1, $|K(y)| \leq C|y|^{-d}$ for $y \neq 0$ and every first difference of K at y is bounded by $C(|y|+\eta)^{-d-1}$.

Box cancellation. Let B be a *lattice box*, i.e. the set of lattice points z with $|z-c|_\infty \leq r$, and let ∂B be the set of points of B with a lattice neighbour outside B. Let v be a lattice point with $\operatorname{dist}_\infty(v, \partial B) \geq s \geq \eta$. Write $h = \partial_\mu^* g_\eta$, so that $K(v-z)$ is a difference quotient of $h(v-z)$ in the variable z_ν. Summing over $z \in B$ telescopes along the lines parallel to e_ν, so that $\sum_{z\in B} \eta^d K(v-z)$ equals η^{d-1} times a sum of at most $2(2r/\eta+1)^{d-1}$ values $h(v-z')$. Here z' is either a point of ∂B or a lattice point just outside B, one lattice step from a point of ∂B. So $|v-z'|_\infty + \eta \geq s$; if $v \notin B$, then $z' = v$ is possible. The set of points z' depends only on B, not on v. By Proposition 6.1, $|h(y)| \leq C(|y|+\eta)^{1-d}$, so each value is at most Cs^{1-d}. Since $\eta \leq s$, the number of values times η^{d-1} is at most $2(2r+\eta)^{d-1} \leq 2^d(r+s)^{d-1}$. Hence

$$\Big|\sum_{z\in B} \eta^d K(v-z)\Big| \leq C\,(r/s+1)^{d-1}. \tag{6.3}$$

Sup bound. In this step z denotes a summation variable; the centre of the cubes Λ_r is fixed. Let Q_0 be the lattice box of radius $\frac{5}{2}$ with the same centre as Λ_1; it contains the support of G, and ∂Q_0 has ∞-distance at least $\frac{3}{2}-\eta$ from Λ_1. For $x \in \Lambda_1$, $\sum_z \eta^d K(x-z)G(z) = \sum_{z\in Q_0} \eta^d K(x-z)(G(z)-G(x)) + G(x)\sum_{z\in Q_0}\eta^d K(x-z)$. The first sum is at most $C[G]_\alpha \sum_{z\in Q_0\setminus\{x\}} \eta^d |x-z|^{\alpha-d} \leq C[G]_\alpha$ by Lemma 6.2(a), the second at most $C|G|$ by (6.3).

Hölder bound. Let $x, x' \in \Lambda_1$ and $\delta = |x-x'|_\infty$; we may assume $\eta \leq \delta \leq \frac{1}{8}$. With $S(v) = \sum_{z\in Q_0} \eta^d K(v-z)$,

$$(K\star G)(x) - (K\star G)(x') = \sum_{z\in Q_0} \eta^d \big(K(x-z)-K(x'-z)\big)\big(G(z)-G(x)\big) + G(x)\big(S(x)-S(x')\big).$$

By the telescoping argument, $S(v)$ is η^{d-1} times a sum of $O(\eta^{1-d})$ values $h(v-z')$, over a set of points z' near ∂Q_0 that does not depend on v. These points are at distance ≥ 1 from Λ_1, where h has bounded derivatives. Hence $|S(x)-S(x')| \leq C\delta$. Let $B = \{z : |z-x|_\infty \leq 2\delta\}$. On B,

$$\sum_{z\in B} \eta^d |K(x-z)||G(z)-G(x)| \leq C[G]_\alpha \sum_{0<|z-x|\leq 2\sqrt{d}\delta} \eta^d |x-z|^{\alpha-d} \leq C[G]_\alpha\, \delta^\alpha,$$

and

$$\sum_{z\in B} \eta^d K(x'-z)(G(z)-G(x)) = \sum_{z\in B} \eta^d K(x'-z)(G(z)-G(x')) + (G(x')-G(x)) \sum_{z\in B} \eta^d K(x'-z).$$

The first term on the right is at most $C[G]_\alpha \delta^\alpha$, since $|z-x'| \leq 3\sqrt{d}\delta$ on B. The second is at most $[G]_\alpha (\sqrt{d}\delta)^\alpha \cdot C$ by (6.3), because x' has ∞-distance at least δ from ∂B. Outside B we have $|x-z|_\infty > 2\delta$; a monotone path from x to x' stays at ∞-distance at least $\frac{1}{2}|x-z|_\infty$ from z, so $|K(x-z)-K(x'-z)| \leq C\delta |x-z|^{-d-1}$. Let $r = \max(2\delta, \sqrt{d}\,\eta)$. By Lemma 6.2(c),

$$\sum_{|x-z|>r} \eta^d\, C\delta |x-z|^{-d-1}\, [G]_\alpha |x-z|^\alpha \leq C[G]_\alpha\, \delta \cdot r^{\alpha-1} \leq C[G]_\alpha\, \delta^\alpha.$$

If $r > 2\delta$, the remaining points with $2\delta < |x-z|_\infty$ and $|x-z| \leq \sqrt{d}\eta$ are at most C in number. Each contributes at most $\eta^d \cdot C\eta^{-d} \cdot [G]_\alpha (\sqrt{d}\eta)^\alpha \leq C[G]_\alpha \delta^\alpha$. Collecting the terms gives $[\partial w]_{\alpha,\Lambda_1} \leq C([G]_\alpha + |G|)$, and (d) follows. □

7 Scalar Green functions

We now combine the three tools for the simplest operators,

$$G_\sigma = (-\Delta + aQ_\sigma^* Q_\sigma)^{-1}, \qquad \sigma \in \{0, 1, \dots, d\}.$$

For $\sigma = 0$ this is the scalar Green function of [2] and [26] at the k-th step on a single torus. In [3] it is denoted G'_k. The tube versions $\sigma \geq 1$ are the componentwise building blocks of the gauge-field average $\mathbf{Q}$. They are needed in Section 11.

Theorem 7.1 (Scalar Green functions)**.** *Let $a > 0$. There are $C, \gamma > 0$, depending only on a, such that for every σ:*

(a) (L^2 decay) The operators G_σ, ∂G_σ, $G_\sigma \partial^*$ and $\partial G_\sigma \partial^*$ have decay (C, γ).

(b) (Pointwise bounds) Let $\alpha \in (0,1)$. There is C_α, depending only on a and α, such that for all y, y' and every f supported in $B(y')$,

$$|G_\sigma f|_{B(y)} + |\partial G_\sigma f|_{B(y)} + [\partial G_\sigma f]_{\alpha,B(y)} \leq C_\alpha e^{-\gamma|y-y'|} |f|_{B(y')},$$
$$|G_\sigma \partial_\mu^* f|_{B(y)} + [G_\sigma \partial_\mu^* f]_{\alpha,B(y)} \leq C_\alpha e^{-\gamma|y-y'|} |f|_{B(y')}.$$

By linearity, (b) implies for arbitrary bounded f

$$|G_\sigma f|_{B(y)} + |\partial G_\sigma f|_{B(y)} + [\partial G_\sigma f]_{\alpha,B(y)} \leq C_\alpha \sum_{y'} e^{-\gamma|y-y'|} |f|_{B(y')}. \tag{7.1}$$

The Hölder quotient is only taken inside a single block, so it is subadditive under such decompositions. Hence (7.1) follows from the triangle inequality.

Proof. (a) We apply Lemma 5.7 to $\mathfrak{A} = -\Delta + aQ_\sigma^* Q_\sigma$. Hypothesis (i) is Corollary 3.4. For (ii), write $-\Delta = d^*d$ and apply Lemma 5.8 with $X = d$, $Z = 0$. This gives $|\langle u, ((-\Delta)_\alpha + \Delta)u\rangle| \le C|\alpha|(\|u\|^2 + \|\partial u\|^2)$ for $|\alpha| \le 1$. The operator $Q_\sigma^* Q_\sigma$ has norm at most 2 and satisfies $\mathbf{1}_y Q_\sigma^* Q_\sigma \mathbf{1}_{y'} = 0$ for $|y - y'| > 2$. By Lemma 5.2(d) it has decay $(2e^2, 1)$, so by Lemma 5.3, $\|(Q_\sigma^* Q_\sigma)_\alpha - Q_\sigma^* Q_\sigma\| \le C|\alpha|$ for $|\alpha| \le \frac{1}{2}$. Lemma 5.7 gives (a).

(b) Let $u = G_\sigma f$ with f supported in $B(y')$. Then

$$-\Delta u = F, \qquad F = f - aQ_\sigma^* Q_\sigma u.$$

Fix y, and let $\Lambda_r = \Lambda_r(y)$ be the cubes centred at the centre y of $B(y)$, as in Section 6. Then $B(y) \subset \Lambda_1$. Let Y be the set of y'' with $B(y'')$ within distance 4 of Λ_3. It contains at most C points, all with $|y'' - y| \le C$. By Lemma 2.6(c) and (a),

$$|F|_{\Lambda_2} + \|u\|_{\Lambda_2} \le |f|_{\Lambda_2} + C \max_{y'' \in Y} \|u\|_{B(y'')} \le |f|_{\Lambda_2} + C \max_{y'' \in Y} e^{-\gamma|y'' - y'|} \|f\|.$$

Here $\|f\| \le |f|_{B(y')}$ because $B(y')$ has volume 1. Moreover $|f|_{\Lambda_2} = 0$ unless $|y - y'| \le C$. Since $|y'' - y'| \ge |y - y'| - C$, the right-hand side is at most $Ce^{-\gamma|y-y'|}|f|_{B(y')}$. Theorem 6.3(a) gives the first bound. For the second, $u = G_\sigma \partial_\mu^* f$ satisfies $-\Delta u = \partial_\mu^* f - aQ_\sigma^* Q_\sigma u$. Its L^2 norms on blocks are controlled by the decay of $G_\sigma \partial^*$ from part (a), and Theorem 6.3(b) applies with $g = f e_\mu$. □

Remark 7.2 (Cubes instead of blocks). Nothing in the proof of (b) uses the fact that $B(y)$ is a block. For every $z \in \mathbb{R}^d$ the same argument, applied to the cube $\Lambda_1(z)$, gives the bounds of (b) on $\Lambda_1(z)$, with $|y - y'|$ replaced by $\operatorname{dist}(z, B(y'))$. Consequently the Hölder bounds also hold on cubes $\Lambda_2(z)$, with a larger constant. Indeed, two points $x, x' \in \Lambda_2(z)$ with $|x - x'|_\infty \le \frac{1}{2}$ lie in a common cube $\Lambda_1(z_i)$ with $z_i \in z + \frac{1}{2}\mathbb{Z}^d$, $|z_i - z|_\infty \le \frac{9}{4}$, and for $|x - x'|_\infty > \frac{1}{2}$ the Hölder quotient is bounded by 4 times the supremum. The same remark applies to all pointwise bounds proved below.

Remark 7.3 (L^2 regularity). On the torus, $\sum_{\mu,\nu} \|\partial_\mu \partial_\nu u\|^2 = \|\Delta u\|^2$, because the differences commute. Hence $u = G_\sigma f$ satisfies $\sum_{\mu\nu} \|\partial_\mu \partial_\nu u\|^2 \le (\|f\| + 2a\|u\|)^2 \le (1 + 2a/c_0)^2 \|f\|^2$. This is the analogue of [3, Prop. 1.1]. Pointwise, second derivatives of $G_\sigma f$ are *not* bounded uniformly in η for merely bounded f. The Hölder bound on $\partial G_\sigma f$ is the substitute.

8 Averaged scalar operators, minimizers and the projection R

Throughout this section $\sigma \in \{0, \dots, d\}$ is fixed, $G = G_\sigma$, $Q = Q_\sigma$, and C, γ denote constants depending only on a (and on α in Hölder bounds). Their values may change from line to line.

8.1 The averaged Green function and its inverse

Proposition 8.1. *The operator QGQ^* on fields on $\mathbb{T}$ satisfies*

$$c_F \le QGQ^* \le 2c_0^{-1},$$

with c_F from Corollary 4.3. Both QGQ^* and $\Gamma = (QGQ^*)^{-1}$ have decay (C, γ), i.e. $|\Gamma(y, y')| \le Ce^{-\gamma|y-y'|}$.

Proof. The upper bound follows from $\|Q\|^2 \le 2$ and $G \le c_0^{-1}$. For the lower bound, $aQ^*Q \le 2a$ gives $-\Delta + aQ^*Q \le -\Delta + 2a$. Inversion reverses inequalities between positive definite operators: if $0 < X \le Y$, then $\langle h, Y^{-1}h\rangle = \|Y^{-1/2}X^{1/2}X^{-1/2}h\|^2 \le \|X^{-1/2}h\|^2$, because $\|X^{1/2}Y^{-1/2}\| \le 1$. Hence $G \ge (-\Delta + 2a)^{-1}$ and $QGQ^* \ge Q(-\Delta + 2a)^{-1}Q^* \ge c_F$ by Corollary 4.3.

For $\sigma = 0$, $\mathbf{1}_y Q_0 \mathbf{1}_{y'} = 0$ unless $y' = y$; for $\sigma \geq 1$, $\mathbf{1}_y Q_\sigma \mathbf{1}_{y'} = 0$ unless $y' \in \{y, y + e_\sigma\}$. Thus Q, viewed as a map from fine to unit fields, satisfies $\mathbf{1}_y Q \mathbf{1}_{y'} = 0$ for $|y - y'| > 1$. So it has decay $(\sqrt{2}e^\gamma, \gamma)$ for every γ, and the same holds for Q^*. With Theorem 7.1(a) and Lemma 5.2(b), QGQ^* has decay. Lemma 5.5 then gives the decay of Γ. □

8.2 The minimizer map

Definition 8.2. The *minimizer map* is $H = GQ^*\Gamma = GQ^*(QGQ^*)^{-1}$. It maps fields on $\mathbb{T}$ to fields on $\mathbb{T}_\eta$.

Lemma 8.3 (Variational characterization)**.** *For every field g on $\mathbb{T}$, Hg is the unique minimizer of $\|\partial f\|^2$ subject to $Qf = g$. In particular $QH = I$, H does not depend on a, and $H1 = 1$.*

Proof. $QHg = QGQ^*\Gamma g = g$. Let $Qf = g$ and write $f = Hg + h$ with $Qh = 0$. With $\mathfrak{A} = G^{-1}$,

$$\langle f, \mathfrak{A} f\rangle = \langle Hg, \mathfrak{A} Hg\rangle + 2\langle h, Q^*\Gamma g\rangle + \langle h, \mathfrak{A} h\rangle = \langle Hg, \mathfrak{A} Hg\rangle + \langle h, \mathfrak{A} h\rangle,$$

since $\langle h, Q^*\Gamma g\rangle = \langle Qh, \Gamma g\rangle = 0$. Thus Hg is the unique minimizer of $\langle f, \mathfrak{A} f\rangle$ on $\{Qf = g\}$. On this set $\langle f, \mathfrak{A} f\rangle = \|\partial f\|^2 + a\|g\|^2$, so Hg also minimizes $\|\partial f\|^2$, which does not involve a. The constant field 1 satisfies $Q1 = 1$ and $\|\partial 1\| = 0$, so $H1 = 1$. □

For a field g on $\mathbb{T}$ write $|d_1 g(y)| = \max_\mu \max\left(|D_\mu g(y)|, |D_\mu g(y - e_\mu)|\right)$, the largest unit difference over the $2d$ bonds containing y.

Theorem 8.4 (Bounds on the minimizer map)**.** *(a) H and ∂H have decay (C, γ).*

(b) For every field g on $\mathbb{T}$ and every y,

$$|Hg|_{B(y)} + |\partial Hg|_{B(y)} + [\partial Hg]_{\alpha, B(y)} \leq C \sum_{y'} e^{-\gamma|y-y'|} |g(y')|,$$
$$|\partial Hg|_{B(y)} + [\partial Hg]_{\alpha, B(y)} \leq C \sum_{y'} e^{-\gamma|y-y'|} |d_1 g(y')|.$$

(c) $\|\partial Hg\| \leq C\|d_1 g\|$.

Proof. (a) follows from Proposition 8.1, Theorem 7.1(a) and Lemma 5.2.

(b) $Hg = Gf$ with $f = Q^*\Gamma g$. By Lemma 2.6(c), $|f|_{B(y'')} \leq \max|\Gamma g|$ over at most two points near y'', which is bounded by $C\sum_{y'} e^{-\gamma|y''-y'|}|g(y')|$. Inserting this in (7.1) and using Lemma 5.2(b) for the convolution of exponentials gives the first bound. For the second, $H1 = 1$ gives $\partial Hg = \partial H(g - g(y))$, so we may replace $|g(y')|$ by $|g(y') - g(y)|$. Join y to y' by a monotone path on $\mathbb{T}$ with at most $\sqrt{d}|y - y'|$ steps. Every step starts at a point y_b with $|y_b - y| \leq \sqrt{d}|y - y'|$. Then

$$\sum_{y'} e^{-\gamma|y-y'|}|g(y') - g(y)| \leq \sum_{y'} e^{-\gamma|y-y'|} \sum_{b \in \text{path}} |d_1 g(y_b)| \leq S_{\gamma/2} \sum_{y_b} e^{-\gamma|y_b - y|/(2\sqrt{d})} |d_1 g(y_b)|,$$

after exchanging the sums and splitting $e^{-\gamma|y-y'|} \leq e^{-\gamma|y-y'|/2} e^{-\gamma|y_b-y|/(2\sqrt{d})}$. Each y_b is counted at most d times per y'; this factor is absorbed in C.

(c) The second bound in (b), without the Hölder term and in ℓ^2, follows by the same argument from the L^2 decay in (a). Indeed, $\|\partial Hg\|_{B(y)} \leq \sum_{y'} \|\mathbf{1}_y \partial H \mathbf{1}_{y'}\| \, |g(y') - g(y)|$. Summing the squares over y and using the Schur test as in Lemma 5.2(a) gives the claim. □

8.3 The effective action

Proposition 8.5 (Effective action)**.** *Let $\mathcal{D} = \Gamma - a$. Then for every field g on $\mathbb{T}$,*

$$\langle g, \mathcal{D} g\rangle = \|\partial H g\|^2 = \min\{\|\partial f\|^2 : Qf = g\},$$

and

$$6^{-2d}\|d_1 g\|^2 \le \langle g, \mathcal{D} g\rangle \le C\|d_1 g\|^2.$$

Moreover $\mathcal{D}1 = 0$ and $\mathcal{D}$ has decay (C, γ).

Proof. $\langle Hg, G^{-1}Hg\rangle = \langle \Gamma g, QGQ^*\Gamma g\rangle = \langle g, \Gamma g\rangle$. On the other hand it equals $\|\partial Hg\|^2 + a\|QHg\|^2 = \|\partial Hg\|^2 + a\|g\|^2$. The upper bound is Theorem 8.4(c). For the lower bound, $D_\mu g = D_\mu QHg$, so Lemma 3.2 gives $\|D_\mu g\| \le 6^d \|\partial_\mu Hg\|$. Since $H1 = 1$ and $\partial 1 = 0$, $\langle 1, \mathcal{D}1\rangle = 0$ with $\mathcal{D} \ge 0$, hence $\mathcal{D}1 = 0$. Decay follows from Proposition 8.1. □

In Bałaban's language, $\frac{1}{2}\langle g, \mathcal{D} g\rangle$ is the quadratic action after k RG steps with δ-function averaging (the scalar analogue of the gauge-field action of [3, (1.65)–(1.67)]). The two-sided bound says that it is comparable to the unit-lattice Dirichlet form, uniformly in k.

8.4 The constrained Green function

Proposition 8.6 (Constrained Green function)**.** *Let $C_\sigma = G - HQG = G - GQ^*\Gamma QG$. Then:*

(a) C_σ is symmetric, $0 \le C_\sigma \le G$, $QC_\sigma = 0$ and $C_\sigma Q^* = 0$;

(b) for every f, $u = C_\sigma f$ is the unique field with $Qu = 0$ and $\langle \partial v, \partial u\rangle = \langle v, f\rangle$ for all v with $Qv = 0$. In particular C_σ does not depend on a, and $-\Delta C_\sigma f - f \in \operatorname{ran} Q^*$;

(c) $G = C_\sigma + H(QGQ^*)H^*$;

(d) C_σ, ∂C_σ, $C_\sigma \partial^*$ and $\partial C_\sigma \partial^*$ have decay (C, γ). The pointwise bounds of Theorem 7.1(b) hold with G_σ replaced by C_σ.

Proof. (a) $C_\sigma = G^{1/2}(I - \Pi)G^{1/2}$ with $\Pi = G^{1/2}Q^*\Gamma QG^{1/2}$, an orthogonal projection. The identities $QC_\sigma = 0 = C_\sigma Q^*$ are direct.

(b) We have $(-\Delta + aQ^*Q)u = f - Q^*\Gamma QGf$, and $Q^*Qu = 0$. So $-\Delta u = f - Q^*\omega$ with $\omega = \Gamma QGf$, and $\langle \partial v, \partial u\rangle = \langle v, f\rangle - \langle Qv, \omega\rangle = \langle v, f\rangle$. Uniqueness holds because $\|\partial v\| > 0$ for $v \ne 0$ with $Qv = 0$.

(c) $H(QGQ^*)H^* = GQ^*\Gamma QG$.

(d) $C_\sigma f = Gf - H(QGf)$, where QGf is a unit-lattice field whose values decay away from the support of f. Combine Theorem 7.1 and Theorem 8.4. For $C_\sigma\partial^*$, the field $QG\partial^* f$ is bounded by the L^2 decay of $G\partial^*$. □

Remark 8.7. Probabilistically, C_σ is the covariance of the Gaussian field with covariance G conditioned on $Q\phi = 0$. Identity (c) is the decomposition of ϕ into the fluctuation field and H applied to the block averages. Iterating it over the RG steps gives the renormalization group formula for Green functions of [26, Sect. 2.5].

8.5 The projection R

From now on $\sigma = 0$. Bałaban's gauge condition involves the orthogonal projection onto the space

$$\mathcal{W} = -\Delta\,\mathcal{N}(Q_0) = \{-\Delta\nu :\ Q_0\nu = 0\}$$

of scalar fields, cf. [3, (1.38)–(1.44)]. By Lemma 3.5, $-\Delta$ is injective on $\mathcal{N}(Q_0)$. Since Q_0 is onto, $\dim \mathcal{W} = \dim \mathcal{N}(Q_0) = n^d(N^d - 1)$. Let

$$\mathcal{V} = \operatorname{ran}(G_0 Q_0^*), \qquad M = Q_0 G_0^2 Q_0^*, \qquad P_{\mathcal{V}} = G_0 Q_0^* M^{-1} Q_0 G_0.$$

Proposition 8.8 (The projection R). *(a) $M \geq c_F^2$. Hence $P_{\mathcal{V}}$ is well defined; it is the orthogonal projection onto $\mathcal{V}$, and $\dim \mathcal{V} = n^d$.*

(b) $\mathcal{V} = \mathcal{W}^{\perp}$. Thus $R := I - P_{\mathcal{V}}$ is the orthogonal projection onto $\mathcal{W}$, and R does not depend on a.

(c) $\mathcal{V}$ consists of the constants plus the fields $(-\Delta)^{-1}Q_0^*\omega$ with $\sum_y \omega(y) = 0$. Here $(-\Delta)^{-1}$ is the inverse on mean-zero fields.

(d) For $h \in \mathcal{W}$: $G_0 h = C_0 h$, $Q_0 G_0 h = 0$ and $-\Delta G_0 h = h$.

(e) $P_{\mathcal{V}}$, $\partial P_{\mathcal{V}}$ and $P_{\mathcal{V}}\partial^*$ have decay (C, γ). In particular $\|P_{\mathcal{V}} d^*\| \leq C$ on gauge fields.

Proof. (a) By Cauchy–Schwarz and Proposition 8.1, $\|G_0 Q_0^*\omega\|\,\|Q_0^*\omega\| \geq \langle Q_0^*\omega, G_0 Q_0^*\omega\rangle \geq c_F\|\omega\|^2$. Moreover $\|Q_0^*\omega\| = \|\omega\|$, so $\langle \omega, M\omega\rangle = \|G_0 Q_0^*\omega\|^2 \geq c_F^2\|\omega\|^2$. Thus $G_0 Q_0^*$ is injective, and $T(T^*T)^{-1}T^*$ with $T = G_0 Q_0^*$ is the orthogonal projection onto $\operatorname{ran} T$.

(b) For $\nu \in \mathcal{N}(Q_0)$,

$$\langle G_0 Q_0^*\omega, -\Delta\nu\rangle = \langle \omega, Q_0 G_0(-\Delta + aQ_0^*Q_0)\nu\rangle = \langle \omega, Q_0\nu\rangle = 0.$$

So $\mathcal{V} \perp \mathcal{W}$, and $\dim \mathcal{V} + \dim \mathcal{W} = n^d N^d$ is the dimension of the whole space.

(c) If $\varphi = G_0 Q_0^*\omega$, then $-\Delta\varphi = Q_0^*(\omega - aQ_0\varphi)$. The left side has mean zero, so $\omega' = \omega - aQ_0\varphi$ has mean zero and $\varphi - \bar{\varphi} = (-\Delta)^{-1}Q_0^*\omega'$. Hence $\mathcal{V}$ is contained in the space described, which has dimension $(n^d - 1) + 1 = n^d$.

(d) Let $h = -\Delta\nu$ with $Q_0\nu = 0$. Then $(-\Delta + aQ_0^*Q_0)\nu = h$, so $G_0 h = \nu$. The other statements follow, and $C_0 h = G_0 h - H_0 Q_0 G_0 h = \nu$.

(e) Compositions of operators with decay (Lemma 5.2), using Theorem 7.1(a), Proposition 8.1 and Lemma 5.5 for M^{-1}. □

Remark 8.9. Bałaban defines $R = I - \Delta^{-1}Q_0^*(Q_0\Delta^{-2}Q_0^*)^{-1}Q_0\Delta^{-1}$ on mean-zero fields [3, (1.38)]. It involves the non-local operator Δ^{-1}. He then observes that the same projection is given by the formula with G_0 [3, (1.44)]. Part (c) is the equivalence of the two descriptions. The G_0-formula is the one that makes the exponential decay of $I - R$ evident.

9 Gauge fields: Bałaban's gauge and coercivity

9.1 The quadratic action and its gauge invariance

The Wilson action of a lattice gauge field $U = e^{i\eta A}$ near $U = 1$ is $\frac{1}{2}\|dA\|^2$ plus terms of order three and higher in A ([3, (1.3)], [8, (26)]). The quadratic part is invariant under gauge transformations

$$A \mapsto A + d\lambda,$$

with λ an arbitrary scalar field. By Lemma 2.6(d), $\mathbf{Q}(A + d\lambda) = \mathbf{Q}A + d_1 Q_0\lambda$. Hence the constraint $\mathbf{Q}A = B$ is preserved exactly when $d_1 Q_0\lambda = 0$, that is, when $Q_0\lambda$ is constant. Constant λ act trivially, so the residual gauge group may be taken to be $\mathcal{N}(Q_0)$. Consequently the variational problem

$$\text{minimize } \|dA\|^2 \quad \text{subject to} \quad \mathbf{Q}A = B$$

has, at best, a minimizer that is unique up to $A \mapsto A + d\lambda$ with $\lambda \in \mathcal{N}(Q_0)$. A gauge condition is needed to select one representative.

9.2 Bałaban's gauge

Bałaban uses the condition

$$Rd^*A = 0, \tag{9.1}$$

with R the orthogonal projection onto $\mathcal{W} = -\Delta\mathcal{N}(Q_0)$ (Proposition 8.8). He calls it the Landau gauge [3, Sect. D], [8, (21)].

Lemma 9.1 (Gauge fixing)**.** *For every gauge field A there is exactly one $\lambda \in \mathcal{N}(Q_0)$ such that $A+d\lambda$ satisfies (9.1). Moreover $\|d\lambda\| \le \sqrt{d/2}\,\|Rd^*A\|$, and $d(A+d\lambda) = dA$, $\mathbf{Q}(A+d\lambda) = \mathbf{Q}A$.*

Proof. For $\lambda \in \mathcal{N}(Q_0)$ we have $d^*d\lambda = -\Delta\lambda \in \mathcal{W}$, hence $Rd^*d\lambda = -\Delta\lambda$. The condition $Rd^*(A + d\lambda) = 0$ becomes $-\Delta\lambda = -Rd^*A$. The right-hand side lies in $\mathcal{W}$, and $-\Delta$ is a bijection from $\mathcal{N}(Q_0)$ onto $\mathcal{W}$. The bound is Lemma 3.5. □

By Proposition 8.8(c), condition (9.1) says that d^*A is a constant plus $(-\Delta)^{-1}Q_0^*\omega$ for some unit-lattice ω. So d^*A is the potential of a block-constant charge density. It is a $C^{1,\alpha}$ function on the unit scale, not a rough one. By contrast, the gauge that minimizes $\|A\|$ on the orbit would require $d^*A \in \operatorname{ran} Q_0^*$, a field with jumps at block boundaries. This regularity is one reason for Bałaban's choice.

Remark 9.2 (Origin of the operator $\mathbf{\Delta}_a$). Bałaban arrives at (9.1) by a Faddeev–Popov argument [3, (1.21)–(1.39)]. It replaces the axial gauge fixing by the density $\exp(-\frac{1}{2}\|Rd^*A\|^2)$ and then takes the Landau limit. Adding the gauge-fixing term and the averaging term to the action gives the quadratic form

$$\langle A, \mathbf{\Delta}_a A\rangle = \|dA\|^2 + \|Rd^*A\|^2 + a\|\mathbf{Q}A\|^2, \qquad \mathbf{\Delta}_a = d^*d + dRd^* + a\mathbf{Q}^*\mathbf{Q} = -\Delta - dP_{\mathcal{V}}d^* + a\mathbf{Q}^*\mathbf{Q},$$

where we used Lemma 2.2 and $R = I - P_{\mathcal{V}}$. Up to the term $dP_{\mathcal{V}}d^*$, which has exponential decay, this is a Feynman-type gauge: the principal part is the componentwise Laplacian. On the constraint set $\{\mathbf{Q}A = B,\ Rd^*A = 0\}$ the form reduces to $\|dA\|^2 + a\|B\|^2$.

9.3 The averaging nearly separates exact and co-closed fields

The heart of the coercivity proof is the following identity. On the unit lattice, gradients $d_1 g$ and co-closed fields are orthogonal. We show that $\mathbf{Q}$ maps fine co-closed fields to unit fields that are co-closed up to a divergence controlled by derivatives.

Lemma 9.3. *There is an absolute constant C_W with the following property. For every gauge field A on $\mathbb{T}_\eta$,*

$$d_1^*\mathbf{Q}A = \iota\, b\, \bar{c}\, d^*A + d_1^*\mathsf{W}A, \qquad \bar{c} = c_1 c_2 \cdots c_d,$$

where $\mathsf{W}A$ is a gauge field on $\mathbb{T}$ with $\|\mathsf{W}A\| \le C_W\|\partial A\|$.

Proof. By Lemma 2.4(a), $D_\mu^* = a_\mu^*\partial_\mu^*$, and all operators involved commute. Hence

$$d_1^*\mathbf{Q}A = \sum_\mu D_\mu^*\iota\, a_\mu b A_\mu = \iota \sum_\mu b\, c_\mu\, \partial_\mu^* A_\mu.$$

For fixed μ, the telescoping identity $1 - x_1\cdots x_m = \sum_i x_1\cdots x_{i-1}(1 - x_i)$ for commuting x_i gives

$$c_\mu - \bar{c} = c_\mu\Big(1 - \prod_{\nu\neq\mu} c_\nu\Big) = \sum_{\nu\neq\mu} c_\mu \Pi_{\mu\nu}(1 - c_\nu), \qquad \Pi_{\mu\nu} = \prod_{\nu'<\nu,\ \nu'\neq\mu} c_{\nu'}.$$

By Lemma 2.4(d), $1 - c_\nu = e_\nu\partial_\nu^*\partial_\nu$. By Lemma 2.4(a),(b′), $a_\nu^c\partial_\nu^* = \tau_\nu^{1/2}a_\nu^*\partial_\nu^* = \tau_\nu^{1/2}D_\nu^*$, so that $b\,\partial_\nu^* = \big(\prod_{\nu'\neq\nu} a_{\nu'}^c\big)\tau_\nu^{1/2}D_\nu^*$. Since D_ν^* commutes with ι,

$$\iota\, b\,(c_\mu - \bar{c})\,\partial_\mu^* A_\mu = \sum_{\nu\neq\mu} D_\nu^*\,\iota\Big(\prod_{\nu'\neq\nu} a_{\nu'}^c\Big)\tau_\nu^{1/2}\, c_\mu\Pi_{\mu\nu}\,(e_\nu\partial_\nu)\,\partial_\mu^* A_\mu.$$

Summing over μ and collecting the coefficient of D_ν^* defines

$$(\mathsf{W}A)_\nu = \sum_{\mu\neq\nu} \iota\Big(\prod_{\nu'\neq\nu} a_{\nu'}^c\Big)\tau_\nu^{1/2}\, c_\mu \Pi_{\mu\nu}\,(e_\nu\partial_\nu)\,\partial_\mu^* A_\mu,$$

and the identity follows. Each operator acting on $\partial_\mu^* A_\mu$ is ι composed with a product convolution. In direction ν it is $\tau_\nu^{1/2} e_\nu \partial_\nu$, whose kernel is bounded by 1 and supported in an interval of length at most 3 (Lemma 2.4(d)); the half shift only moves the support. In every other direction it is $a_{\nu'}^c$ times at most one triangle average $c_{\nu'}$, whose kernel is an average of c and therefore bounded by 1, with support of length at most 3. By Lemma 2.7 the norm is at most $4^d\|\partial_\mu^* A_\mu\| = 4^d\|\partial_\mu A_\mu\|$. Hence, by Cauchy–Schwarz over $\mu\neq\nu$, $\|\mathsf{W}A\| \le 4^d d\,\|\partial A\|$. □

Lemma 9.4 (Separation)**.** *There is an absolute constant $C_\star$ such that for every co-closed gauge field B' on $\mathbb{T}_\eta$ and every scalar field g on $\mathbb{T}$,*

$$\|B'\|^2 + \|d_1 g\|^2 \le C_\star(\|dB'\|^2 + \|\mathbf{Q}B' + d_1 g\|^2).$$

Proof. Expand the square:

$$\|\mathbf{Q}B' + d_1 g\|^2 = \|\mathbf{Q}B'\|^2 + \|d_1 g\|^2 + 2\langle d_1^*\mathbf{Q}B', g\rangle.$$

By Lemma 9.3 and $d^*B' = 0$, $\langle d_1^*\mathbf{Q}B', g\rangle = \langle \mathsf{W}B', d_1 g\rangle$. This is bounded in absolute value by $C_W\|\partial B'\|\|d_1 g\| \le \frac14\|d_1 g\|^2 + C_W^2\|\partial B'\|^2$. By Lemma 2.2, $\|\partial B'\| = \|dB'\|$. Hence

$$\|\mathbf{Q}B' + d_1 g\|^2 \ge \|\mathbf{Q}B'\|^2 + \tfrac12\|d_1 g\|^2 - 2C_W^2\|dB'\|^2.$$

On the other hand, Corollary 3.3 gives $\|B'\|^2 \le C_P(\|dB'\|^2 + \|\mathbf{Q}B'\|^2)$. Therefore

$$\begin{aligned}\|B'\|^2 + \|d_1 g\|^2 &\le C_P\|dB'\|^2 + (C_P+2)(\|\mathbf{Q}B'\|^2 + \tfrac12\|d_1 g\|^2)\\ &\le (C_P + 2C_W^2(C_P+2))\|dB'\|^2 + (C_P+2)\|\mathbf{Q}B' + d_1 g\|^2.\end{aligned}$$ □

9.4 Coercivity

Lemma 9.5 (Gradient part in Bałaban's gauge)**.** *Let λ be a scalar field with mean zero and $R(-\Delta\lambda) = 0$. Then $\|d\lambda\| \le C_B\|d_1 Q_0\lambda\|$, with C_B from Lemma 4.4.*

Proof. $-\Delta\lambda$ lies in $\mathcal{W}^\perp = \mathcal{V}$ and has mean zero. By Proposition 8.8(c), $-\Delta\lambda = (-\Delta)^{-1}Q_0^*\omega$ with ω of mean zero. Since λ has mean zero, $\lambda = (-\Delta)^{-2}Q_0^*\omega$, and Lemma 4.4 applies. □

Theorem 9.6 (Coercivity of the gauge-field operator)**.** *There is $c_g > 0$, depending only on a, such that for every gauge field A on $\mathbb{T}_\eta$*

$$\langle A, \mathbf{\Delta}_a A\rangle = \|dA\|^2 + \|Rd^*A\|^2 + a\|\mathbf{Q}A\|^2 \;\ge\; c_g(\|A\|^2 + \|\partial A\|^2).$$

Proof. Step 1 (gauge fixing). Let $\lambda_1 \in \mathcal{N}(Q_0)$ be given by Lemma 9.1 and $B = A + d\lambda_1$. Then $Rd^*B = 0$, $dB = dA$, $\mathbf{Q}B = \mathbf{Q}A$, and $\|d\lambda_1\| \le \sqrt{d/2}\|Rd^*A\|$.

Step 2 (Hodge decomposition). By Lemma 2.2, $B = d\lambda + B'$ with $\lambda = (-\Delta)^{-1}d^*B$ of mean zero and $d^*B' = 0$. Then $R(-\Delta\lambda) = Rd^*B = 0$. By Lemma 9.5, $\|d\lambda\| \le C_B\|d_1 g\|$ with $g = Q_0\lambda$.

Step 3 (separation). By gauge covariance, $\mathbf{Q}B = \mathbf{Q}B' + d_1 g$, and $dB' = dB = dA$. Lemma 9.4 gives

$$\|B'\|^2 + \|d_1 g\|^2 \le C_\star(\|dA\|^2 + \|\mathbf{Q}A\|^2).$$

Step 4. Since $A = -d\lambda_1 + d\lambda + B'$,

$$\|A\| \le \sqrt{d/2}\,\|Rd^*A\| + (C_B+1)\sqrt{2C_\star}\,(\|dA\|^2 + \|\mathbf{Q}A\|^2)^{1/2}.$$

Hence $\|A\|^2 \le C'(\|dA\|^2 + \|Rd^*A\|^2 + \|\mathbf{Q}A\|^2)$.

Step 5 (derivatives). By Lemma 2.2, $\|\partial A\|^2 = \|dA\|^2 + \|d^*A\|^2$. Moreover $\|d^*A\| \le \|Rd^*A\| + \|P_{\mathcal{V}}d^*A\| \le \|Rd^*A\| + C\|A\|$ by Proposition 8.8(e). Combining with Step 4 proves the theorem, with c_g depending on $\min(1,a)$ and the constants above. □

Corollary 9.7. *On the space* $\mathcal{X} = \{A : \mathbf{Q}A = 0,\ Rd^*A = 0\}$ *one has* $\|dA\|^2 \ge c_g\|A\|^2$.

This completes the proof of Theorem A.

10 The gauge-field Green function

Let $\mathbf{G} = \mathbf{\Delta}_a^{-1}$ with $\mathbf{\Delta}_a = d^*d + dRd^* + a\mathbf{Q}^*\mathbf{Q}$. By Theorem 9.6 it is well defined and $0 < \mathbf{G} \le c_g^{-1}$. In [3] this operator is denoted G_k (with $a = 1$). In [7, 8] its covariant version G_1, $G(U)$ is the basic propagator.

Theorem 10.1 (Gauge-field Green function)**.** *There are* $C, \gamma > 0$*, depending only on* a*, such that:*

(a) $\mathbf{G}$, $\partial\mathbf{G}$, $\mathbf{G}\partial^*$ *and* $\partial\mathbf{G}\partial^*$ *have decay* (C,γ)*;*

(b) for $\alpha \in (0,1)$ *there is* C_α *such that for all* y, y' *and every gauge field* J *supported in* $B(y')$*,*

$$|\mathbf{G}J|_{B(y)} + |\partial\mathbf{G}J|_{B(y)} + [\partial\mathbf{G}J]_{\alpha,B(y)} \le C_\alpha e^{-\gamma|y-y'|}\,|J|_{B(y')}.$$

As in Remark 7.2, the bounds of (b) also hold on arbitrary cubes $\Lambda_1(z)$ *and, for the Hölder quotient, on cubes* $\Lambda_2(z)$*.*

Proof. (a) We verify the hypotheses of Lemma 5.7 for $\mathfrak{A} = \mathbf{\Delta}_a$ acting on gauge fields. Hypothesis (i) is Theorem 9.6. For (ii), write

$$\mathbf{\Delta}_a = d^*d + (Rd^*)^*(Rd^*) + a\mathbf{Q}^*\mathbf{Q},$$

which holds because R is an orthogonal projection. The first term is X^*X with $X = d$ (from gauge fields to 2-forms). Lemma 5.8 applies with $Z = 0$. The second term is X^*X with $X = d^* - Z$ and $Z = P_{\mathcal{V}}d^*$. By Proposition 8.8(e), Z is bounded and has decay. By Lemma 5.3, $\|Z_\alpha - Z\| \le C|\alpha|$ for $|\alpha| \le \gamma/2$. So Lemma 5.8 applies again. The last term is bounded and local, as in the proof of Theorem 7.1.

(b) Let $u = \mathbf{G}J$. Since $d^*d + dd^* = -\Delta$ componentwise (Lemma 2.2) and $dRd^* = dd^* - dP_{\mathcal{V}}d^*$,

$$-\Delta u = F, \qquad F = J - a\mathbf{Q}^*\mathbf{Q}u + dP_{\mathcal{V}}d^*u.$$

Each component of u satisfies a scalar equation of the form treated in Theorem 6.3(a). Fix y and let $\Lambda_r = \Lambda_r(y)$ be centred at the centre y of $B(y)$. As in the proof of Theorem 7.1, part (a) gives

$$\|u\|_{\Lambda_2} + |J - a\mathbf{Q}^*\mathbf{Q}u|_{\Lambda_2} \le Ce^{-\gamma|y-y'|}|J|_{B(y')}.$$

For the last term, write $dP_{\mathcal{V}}d^*u = (dG_0)\,Q_0^*\omega$ with $\omega = M^{-1}Q_0G_0d^*u$. The operator $M^{-1}Q_0G_0d^*\mathbf{G}$ has decay by Proposition 8.8 and part (a). Hence $|\omega(y'')| \le Ce^{-\gamma|y''-y'|}\|J\|$, with a smaller γ. By (7.1) for G_0 and Lemma 5.2(b),

$$|dP_{\mathcal{V}}d^*u|_{B(y'')} = |\partial G_0Q_0^*\omega|_{B(y'')} \le C\sum_{y'''} e^{-\gamma|y''-y'''|}|\omega(y''')| \le Ce^{-\gamma|y''-y'|}|J|_{B(y')}.$$

Thus $|F|_{\Lambda_2} \le Ce^{-\gamma|y-y'|}|J|_{B(y')}$, and Theorem 6.3(a) applied to each component concludes the proof. □

By linearity, for an arbitrary bounded gauge field J,

$$|\mathbf{G}J|_{B(y)} + |\partial\mathbf{G}J|_{B(y)} + [\partial\mathbf{G}J]_{\alpha,B(y)} \le C_\alpha \sum_{y'} e^{-\gamma|y-y'|}|J|_{B(y')}. \tag{10.1}$$

This completes the proof of Theorem D.

Remark 10.2. The operator $\mathbf{\Delta}_a$ is not of the form "Laplacian plus a local term": $dP_{\mathcal{V}}d^*$ is non-local. It is nevertheless exponentially decaying, and it is a bounded operator up to the two derivatives, which is all that Combes–Thomas and the regularity estimate need. In the covariant case this term becomes $D_U P_{\mathcal{V}}(U) D_U^*$, and the same structure persists [7].

11 The gauge-field minimizer and the fluctuation operators

11.1 Identities

The following identities express how $\mathbf{G}$ interacts with gauge directions. They reduce the gauge part of every operator below to scalar operators. Recall from Proposition 8.8(d) that for h in $\mathcal{W} = \operatorname{ran} R$, $\nu = G_0 h$ is the unique field with $Q_0\nu = 0$ and $-\Delta\nu = h$.

Proposition 11.1. *For all scalar fields h:*

$$\mathbf{G}dRh = dG_0Rh, \qquad \mathbf{Q}\mathbf{G}dR = 0, \qquad Rd^*\mathbf{G}dR = R.$$

Taking adjoints, $Rd^*\mathbf{G} = RG_0d^*$ and $Rd^*\mathbf{G}\mathbf{Q}^* = 0$.

Proof. Let $\nu = G_0Rh$, so $Q_0\nu = 0$ and $-\Delta\nu = Rh$. Then

$$\mathbf{\Delta}_a d\nu = d^*dd\nu + dRd^*d\nu + a\mathbf{Q}^*\mathbf{Q}d\nu = 0 + dR(-\Delta\nu) + a\mathbf{Q}^*d_1Q_0\nu = dRh.$$

Hence $\mathbf{G}dRh = d\nu$. Further, $\mathbf{Q}d\nu = d_1Q_0\nu = 0$ and $Rd^*d\nu = R(-\Delta\nu) = Rh$. □

These are Bałaban's identities [3, (1.95), (1.97)] and [7, (3.152)].

11.2 The averaged gauge Green function

Proposition 11.2. *$c_F \le \mathbf{Q}\mathbf{G}\mathbf{Q}^* \le 2c_g^{-1}$. Both $\mathbf{Q}\mathbf{G}\mathbf{Q}^*$ and $\mathbf{\Gamma} = (\mathbf{Q}\mathbf{G}\mathbf{Q}^*)^{-1}$ have decay (C,γ).*

Proof. For every gauge field, $\|Rd^*A\| \le \|d^*A\|$ and $\|\mathbf{Q}A\|^2 \le 2\|A\|^2$. With Lemma 2.2, $\mathbf{\Delta}_a \le -\Delta + 2a$, hence $\mathbf{G} \ge (-\Delta + 2a)^{-1}$ as in Proposition 8.1. The operator $-\Delta$ acts componentwise and $(\mathbf{Q}A)_\mu = Q_\mu A_\mu$, so $\mathbf{Q}(-\Delta+2a)^{-1}\mathbf{Q}^*$ is the direct sum of the operators $Q_\mu(-\Delta+2a)^{-1}Q_\mu^*$. Each of these is at least c_F by Corollary 4.3. The decay statements follow as in Proposition 8.1, using Theorem 10.1(a) and Lemma 5.5. □

11.3 The minimizer map

Theorem 11.3 (Gauge-field minimizer)**.** *Let $\mathbf{H} = \mathbf{G}\mathbf{Q}^*\mathbf{\Gamma}$.*

(a) $\mathbf{Q}\mathbf{H} = I$ *and* $Rd^*\mathbf{H} = 0$.

(b) For every gauge field B on $\mathbb{T}$, $\mathbf{H}B$ is the unique minimizer of $\|dA\|^2$ on $\{A : \mathbf{Q}A = B,\ Rd^*A = 0\}$. Moreover $\|d\mathbf{H}B\|^2 = \min\{\|dA\|^2 : \mathbf{Q}A = B\}$, and $\mathbf{H}$ does not depend on a. If B is constant, then $\mathbf{H}B = B$ (the constant fine field).

(c) $\mathbf{H}$ and $\partial\mathbf{H}$ have decay (C,γ).

(d) For every B and y,

$$|\mathbf{H}B|_{B(y)} + |\partial\mathbf{H}B|_{B(y)} + [\partial\mathbf{H}B]_{\alpha,B(y)} \le C_\alpha \sum_{y'} e^{-\gamma|y-y'|}|B|_{y'}.$$

In (d), as in Section 1: for a gauge field B on $\mathbb{T}$, $|B|_{y'}$ is the maximum of $|B|$ over the unit bonds starting at y'.

Proof. (a) $\mathbf{Q}\mathbf{H} = \mathbf{Q}\mathbf{G}\mathbf{Q}^*\mathbf{\Gamma} = I$, and $Rd^*\mathbf{H} = 0$ by Proposition 11.1.

(b) Let $\mathbf{Q}A = B$ and $Rd^*A = 0$. Then $h = A - \mathbf{H}B$ lies in $\mathcal{X}$. As in Lemma 8.3,

$$\langle A, \mathbf{\Delta}_a A\rangle = \langle \mathbf{H}B, \mathbf{\Delta}_a \mathbf{H}B\rangle + \langle h, \mathbf{\Delta}_a h\rangle,$$

because $\langle h, \mathbf{\Delta}_a\mathbf{H}B\rangle = \langle h, \mathbf{Q}^*\mathbf{\Gamma}B\rangle = 0$. On the constraint set $\langle A, \mathbf{\Delta}_a A\rangle = \|dA\|^2 + a\|B\|^2$. Moreover $\langle h, \mathbf{\Delta}_a h\rangle = \|dh\|^2 > 0$ for $h \neq 0$ by Corollary 9.7. This proves the first claim and the independence of a. If only $\mathbf{Q}A = B$ is assumed, Lemma 9.1 provides a gauge transform of A that satisfies $Rd^*A = 0$ and has the same dA and $\mathbf{Q}A$, which gives the second claim. A constant fine field B satisfies $\mathbf{Q}B = B$, $dB = 0$ and $d^*B = 0$, so it is the minimizer.

(c) Composition of operators with decay.

(d) $\mathbf{H}B = \mathbf{G}J$ with $J = \mathbf{Q}^*\mathbf{\Gamma}B$. By Lemma 2.6(c) and Proposition 11.2, $|J|_{B(y'')} \le C\sum_{y'} e^{-\gamma|y''-y'|}|B|_{y'}$. Apply (10.1). □

Remark 11.4 (Comparison with [8]). In [8, (45)–(46)] the operator H is defined exactly by (b), in the covariant setting, and the bounds $|HB| \le B_0|B|$ and $|\nabla HB| \le B_0|B|$ are quoted from [7, Theorem 3.12]. Part (d) is their $U = 1$ version on a single torus, with the additional Hölder bound.

11.4 The gauge-field effective action

Proposition 11.5. *Let $\mathbf{D} = \mathbf{\Gamma} - a$. Then $\langle B, \mathbf{D}B\rangle = \|d\mathbf{H}B\|^2$, $\mathbf{D}$ has decay (C, γ), and*

$$c_D\|d_1B\|^2 \le \langle B, \mathbf{D}B\rangle \le C_D\|d_1B\|^2,$$

where d_1B is the unit-lattice curl and $c_D, C_D > 0$ are absolute constants. (The operators $\mathbf{H}$ and H_μ do not depend on a, so we may take $a = 1$ when estimating them.)

Proof. The identity is proved as in Proposition 8.5, using $Rd^*\mathbf{H}B = 0$.

Lower bound. By Lemma 2.6(e), $d_1B = d_1\mathbf{Q}\mathbf{H}B = \mathbf{Q}^{(2)}d\mathbf{H}B$. Each component of $\mathbf{Q}^{(2)}$ is $\iota a_\mu a_\nu b$, which by Lemma 2.7 has norm at most $9\cdot 2^{d-2}$. Hence $\|d_1B\| \le 9\cdot 2^{d-2}\|d\mathbf{H}B\|$.

Upper bound. Decompose B on the unit torus as $B = d_1h + B''$ with $d_1^*B'' = 0$ (Lemma 2.2 on $\mathbb{T}$). Then $d_1B'' = d_1B$, and by the unit-lattice Hodge identity $\sum_{\mu,\nu}\|D_\nu B''_\mu\|^2 = \|d_1B''\|^2 = \|d_1B\|^2$. Define a fine gauge field A by

$$A_\mu = \partial_\mu Q_0^*h + H_\mu B''_\mu,$$

where H_μ is the scalar minimizer map for the tube average Q_μ (Section 8 with $\sigma = \mu$). Then $\mathbf{Q}A = d_1Q_0Q_0^*h + (Q_\mu H_\mu B''_\mu)_\mu = d_1h + B'' = B$. By Theorem 11.3(b), Lemma 2.2 and Theorem 8.4(c),

$$\|d\mathbf{H}B\|^2 \le \|dA\|^2 = \|d(H_\mu B''_\mu)_\mu\|^2 \le \sum_\mu \|\partial H_\mu B''_\mu\|^2 \le C\sum_\mu \|d_1B''_\mu\|^2 = C\|d_1B\|^2.$$

Here $d_1B''_\mu$ is the unit gradient of the scalar field B''_μ. The decay of $\mathbf{D}$ follows from Proposition 11.2. □

This is the $U = 1$ form of [3, (1.66)–(1.67)]: after k steps the gauge-field action is comparable to the unit-lattice Wilson action $\|d_1B\|^2$, uniformly in k. Note how the upper bound uses the tube averages Q_μ. This is why the scalar theory was developed for all σ.

11.5 The constrained Green function

The Euler–Lagrange equation of the variational problem is posed on the tangent space

$$\mathcal{X} = \{A : \ \mathbf{Q}A = 0, \ Rd^*A = 0\}$$

of the constraint set, cf. [8, (100)–(111)]. The relevant Green function is defined as follows.

Definition 11.6. $\mathbf{C} = \mathbf{G} - \mathbf{G}dRd^*\mathbf{G} - \mathbf{HQG} = \mathbf{G} - dG_0RG_0d^* - \mathbf{GQ}^*\mathbf{\Gamma QG}$.

The two expressions agree by Proposition 11.1: $\mathbf{G}dRd^*\mathbf{G} = (\mathbf{G}dR)(Rd^*\mathbf{G}) = dG_0R\,RG_0d^*$. In [7, 8], $\mathbf{C}$ is the operator $\mathfrak{G} = G_1\mathfrak{P}^*$ of [7, (3.148)–(3.153)]. Here $\mathfrak{P}$ is the orthogonal projection onto $\mathcal{X}$ in the inner product $\langle \cdot, \mathbf{\Delta}_a \cdot \rangle$.

Theorem 11.7 (Constrained Green function)**.** *(a)* $\mathbf{C}$ *is symmetric,* $\mathbf{QC} = 0$ *and* $Rd^*\mathbf{C} = 0$.

(b) For every gauge field J, $A = \mathbf{C}J$ *is the unique element of* $\mathcal{X}$ *such that*

$$\langle d\,\delta A, dA\rangle = \langle \delta A, J\rangle \qquad \text{for all } \delta A \in \mathcal{X}.$$

In particular $\mathbf{C}$ *does not depend on* a, *and* $\langle J, \mathbf{C}J\rangle \le c_g^{-1}\|J\|^2$.

(c) $\mathbf{C}$ *and* $\partial\mathbf{C}$ *have decay* (C, γ).

(d) For all y, y' *and every* J *supported in* $B(y')$,

$$|\mathbf{C}J|_{B(y)} + |\partial\mathbf{C}J|_{B(y)} + [\mathbf{C}J]_{\alpha,B(y)} \le C_\alpha e^{-\gamma|y-y'|}\,|J|_{B(y')}.$$

Proof. (a) Each of the three terms is symmetric. By Proposition 8.8(d), $Q_0G_0R = 0$, so $\mathbf{Q}dG_0RG_0d^* = d_1Q_0G_0RG_0d^* = 0$. Together with $\mathbf{QH} = I$ this gives $\mathbf{QC} = \mathbf{QG} - 0 - \mathbf{QG} = 0$. Next,

$$Rd^*\mathbf{C} = RG_0d^* - R(-\Delta)G_0RG_0d^* - 0 = RG_0d^* - RRG_0d^* = 0,$$

using Proposition 11.1 and $-\Delta G_0R = R$ (Proposition 8.8(d)).

(b) Let $\delta A \in \mathcal{X}$. By (a), $\mathbf{C}J \in \mathcal{X}$, so $\langle \delta A, \mathbf{\Delta}_a\mathbf{C}J\rangle = \langle d\,\delta A, d\mathbf{C}J\rangle$. On the other hand, the computation in the proof of Proposition 11.1 shows $\mathbf{\Delta}_a dG_0Rh = dRh$. Hence

$$\mathbf{\Delta}_a\mathbf{C}J = J - dRG_0d^*J - \mathbf{Q}^*\mathbf{\Gamma QG}J,$$

and the last two terms are orthogonal to $\mathcal{X}$. Uniqueness and the bound follow from Corollary 9.7.

(c) The first and third terms are compositions of operators with decay, also after applying ∂: $\mathbf{G}$, $\partial\mathbf{G}$, $\mathbf{H}$, $\partial\mathbf{H}$, $\mathbf{Q}$. For the middle term, Proposition 11.1 and $R^2 = R$ give $dG_0RG_0d^* = \mathbf{G}\,dR\,G_0d^*$. On scalar fields $\partial_\mu = -S_\mu\partial_\mu^*$, with S_μ the shift by η. So $d = \partial^*T$ with $(Th)_{\mu\nu} = -\delta_{\mu\nu}S_\mu h$, a local operator of norm $\sqrt{d}$, and $\partial\mathbf{G}d = (\partial\mathbf{G}\partial^*)T$ has decay by Theorem 10.1(a) and Lemma 5.2(b),(d). Together with the decay of $R = I - P_{\mathcal{V}}$ and $G_0\partial^*$, this gives the decay of $\partial(dG_0RG_0d^*)$ without second derivatives of G_0.

(d) Write $\mathbf{C}J = \mathbf{G}J - d\sigma - \mathbf{H}(\mathbf{QG}J)$ with $\sigma = G_0RG_0d^*J$. The first and third terms are controlled by Theorems 10.1 and 11.3: the unit field $\mathbf{QG}J$ decays away from y' by Theorem 10.1(a). For σ we use the scalar regularity theory. By Proposition 8.8(d), $\sigma \in \mathcal{N}(Q_0)$ and

$$-\Delta\sigma = h, \qquad h = RG_0d^*J = G_0d^*J - P_{\mathcal{V}}G_0d^*J.$$

There is no averaging term in this equation, and h is Hölder continuous. Indeed, G_0d^*J is bounded and Hölder on cubes $\Lambda_2(z)$ by Theorem 7.1(b) and Remark 7.2. Moreover $P_{\mathcal{V}}G_0d^*J = G_0Q_0^*\omega$ with $\omega = M^{-1}Q_0G_0G_0d^*J$, which is bounded with bounded first derivatives. All these bounds carry the factor $e^{-\gamma\,\mathrm{dist}(z,B(y'))}|J|_{B(y')}$, and so does $\|\sigma\|_{\Lambda_2(z)}$, because G_0, R and $G_0\partial^*$ have decay and $\Lambda_2(z)$ meets a bounded number of blocks. Theorem 6.3(a) and (c) now bound $|\partial\sigma|$, its Hölder quotient, and $|\partial\partial\sigma|$ on $\Lambda_1(z)$. This controls $|d\sigma|$ and $|\partial d\sigma|$. The Hölder bound on $\mathbf{C}J$ follows from the bound on $|\partial\mathbf{C}J|$. □

Remark 11.8 (Use in the variational problem). In [8, (110)–(111)], the critical points of the action on the constraint set, after the linearizing change of variables, are the solutions of

$$A_1 + \mathbf{C}J + \mathbf{C}\,\tfrac{\delta V}{\delta A}(A_1 + \mathbf{H}B) = 0$$

(in the covariant version). The solution is found by the contraction mapping principle. This requires exactly the bounds $|\mathbf{C}J|, |\partial\mathbf{C}J| \le C \sup|J|$ in the weighted norms, i.e. part (d), together with the bounds on $\mathbf{H}$ in Theorem 11.3. Bałaban notes [8, Sect. E] that the regularity theorem for $\mathfrak{G}$ [7, Theorem 3.13] gives no bounds on second-order derivatives, and he supplies the needed second-order bounds by a separate argument. Part (d) likewise controls only $\mathbf{C}J$ and its first derivatives.

Remark 11.9 (A renormalization-group decomposition). Combining the definitions,

$$\mathbf{G} = \mathbf{C} + dG_0RG_0d^* + \mathbf{H}(\mathbf{Q}\mathbf{G}\mathbf{Q}^*)\mathbf{H}^*.$$

The three terms are, respectively, the fluctuation covariance on $\mathcal{X}$, a pure-gauge part, and the part generated by the averages. This is the gauge-field analogue of Proposition 8.6(c).

This completes the proof of Theorem E.

Part II: Backgrounds

Part II extends the operators of Part I to a background $U = e^{i\eta A}$ that is close to 1 in a single global gauge: $|A|, |\partial A|, [\partial A]_\alpha \le \varepsilon$ (hypothesis (B), Section 12). The fields take values in the complexified Lie algebra, and the operators involve parallel transport. The constants depend in addition on $\dim \mathfrak{g}$ and on the decay rate γ_W of a small potential term W, but not on L, k, n or U. Part III applies these results in a local gauge on each cube, at backgrounds that are only locally regular (Theorem G(i)). Through Part III they give the minimizer map and the fluctuation operator of the variational problem; Section 16.2 lists that input item by item.

What this Part proves. The main result is Theorem F. For ε small, the scalar operators $G'(U)$, $\Gamma'(U)$, $H'(U)$, $C'(U)$ and $R(U)$ satisfy the decay and pointwise bounds stated in Theorem F(i) (Theorem 14.1, Proposition 14.2). The gauge-field Green function $\mathbf{G}(U)$, built with Bałaban's gauge-invariant extension through the slice projection $\Pi(U)$ (Lemma 15.1), satisfies the bounds of Theorem D (Theorem 15.4). The minimizer map $\mathbf{H}(U)$ and the constrained Green function $\mathbf{C}(U)$ satisfy the decay and pointwise bounds of Theorem E, and have variational characterizations (Theorem 16.1). All these operators are analytic in A and W and covariant under G-valued gauge transformations, and the identities $\mathbf{G}DR = DG'R$, $RD^T\mathbf{G} = RG'D^T$ and $\mathbf{Q}_\pi\mathbf{G}DR = 0$ are exact. Proposition 16.3 transfers the bounds to every background obtained from one that satisfies (B) by a G-valued gauge transformation, with covariant derivatives, the Hölder bounds in the gauge of (B); the comparisons with the operators at $U = 1$ hold only in the gauge of (B). The proofs write every operator as its counterpart at $U = 1$ plus an admissible perturbation (Definition 13.1). Combes–Thomas, inversion with decay and local regularity are stable under such perturbations (Lemmas 13.2, 13.3 and 13.5).

Which averaging. Part II is stated for a model averaging pair $(Q'(U), \mathbf{Q}(U))$, with transport along one coordinate-ordered contour and only the main term of Bałaban's linearized averaging. It intertwines the covariant derivatives up to a defect: $\mathbf{Q}(U)D_U = D_V Q'(U) + E(U)$, with $\|E(U)\| \le C\varepsilon$ (Lemma 12.5(c)). The model pair is a proof device. For the linearization of Bałaban's symmetric averaging $E = 0$, and Corollary 22.11 applies the real results of Parts II and III to it, through Remark 19.12. A first reading may set $E = 0$.

Imports. From Part I: the operators at $U = 1$ (Theorems 7.1 and 10.1, Propositions 8.6, 8.8 and 11.2), Lemmas 2.6 and 5.2–5.7, and Theorem 6.3. No proof in Part II uses a later result.

Exports. Part III uses all of Part II. The expansion of covariant derivatives and the curvature operator (Lemmas 12.2 and 12.3), the perturbed inverses and the bootstrap (Lemmas 13.3 and 13.5), and the slice projection (Lemma 15.1) are used throughout Parts III–X. Theorem 16.1 is used again in Parts V–IX and XI. All of Part II is on the main line. Section 16.2 is a list, not a result: it maps the input of Bałaban's variational theorem to the places where this note proves it, mostly in Parts III–VI.

Details for this Part. The companion has no chapter on Part II; its parameter ledger lists the constants of Part II that Parts VII–XI use.

12 Background fields: setting and hypotheses

In this part of the paper we extend the results of Sections 7–11 to operators depending on a non-trivial background field U. These are the operators $G(U)$, $H(U)$, $\mathfrak{G}(U)$, $R(U)$, $G'(U)$ of [7], which enter the variational problem [8]. Two features distinguish this part from the case $U = 1$.

- The fields take values in the (complexified) Lie algebra, and the operators involve parallel transport. Everything is proved for complex backgrounds near real ones. Analyticity in the background then comes for free.
- The gauge directions no longer decouple from the action. Bałaban handles this by replacing the quadratic form by its gauge-invariant extension, built with the projection $\Pi(U)$ onto the gauge slice along the residual gauge directions [7, (3.118)–(3.119)]. We use the same construction (Section 15).
- We work with a simplified averaging operator: only the main term of Bałaban's linearized averaging. For it, the covariance identity $\mathbf{Q}(U)D_U = D_V Q'(U)$, which is exact for Bałaban's full operator [7, (3.114)–(3.115)], holds only up to a small "defect" $E(U)$. We correct for the defect by one additional term in the averaging part of the operator. With this correction all exact identities of Section 11 are restored.

The global small-field hypothesis. We assume that the background is close to 1 *in a single global gauge* (hypothesis (B) below). Bałaban's conditions are local. They consist of a gauge-invariant bound on the curvature [8, (2)] and, for the propagators, of the requirement that on each cube of size M there be a gauge in which U is close to 1, with bounds proportional to M [7, (3.35)–(3.36)]. Hypothesis (B) is a condition on the background U as given; a real gauge transformation can destroy it. The existence of a G-valued gauge transformation that carries U to a background satisfying (B) is a condition on the real gauge orbit of U. All operators below are defined covariantly, and they transform covariantly under G-valued gauge transformations (Proposition 16.3). Hence their bounds hold for every background that is gauge equivalent, by a G-valued gauge transformation, to one satisfying (B), with derivatives replaced by covariant derivatives and Hölder quotients taken in the gauge in which (B) holds. The comparisons with the operators at $U = 1$ are different: they hold in that gauge, and they transfer only as comparisons with the transformed reference 1^u. Apart from these comparisons, the special gauge is only a device in the proofs. For G^c-valued gauge transformations the operator $R(u)$ need not be an isometry, and the bounds do not transfer.

How restrictive (B) is. Hypothesis (B) is much stronger than Bałaban's local conditions, and it is not a local condition. The basic constraint is an isoperimetric bound. Let S be a planar lattice region and ∂S its boundary contour. If (B) holds, then $|U(\partial S) - 1| \le e^{\varepsilon|\partial S|} - 1$, where $|\partial S|$ is the length of ∂S in unit lengths. So all Wilson loops obey a perimeter bound with constant ε.

A background whose field strength is of size ε and *coherent* over a region of side ℓ has $U(\partial S) \approx e^{\mathrm{i}\varepsilon\ell^2}$ for squares S of side ℓ. In the abelian case, for $\varepsilon\ell^2 < \pi$, the perimeter bound then forces $\ell \lesssim 4$. Thus (B) essentially requires the background to be close to flat on all scales larger than $O(1)$. Field strengths of size ε are allowed only if they average out on the unit scale. Backgrounds with non-zero flux through a 2-cycle of the torus are the extreme case. In the abelian case, every gauge with $\eta|A| < \pi/4$ has vanishing total flux, because each plaquette angle is then the principal one and the flux is a sum of differences. So non-zero flux forces $|A| \ge \pi/(4\eta)$ somewhere, whatever the size of the torus.

Flat backgrounds, on the other hand, can satisfy (B). For a real flat background, if the holonomies h_μ around the cycles lie in a common maximal torus of G (always the case for $U(\mathsf{N})$ and $SU(\mathsf{N})$), one writes $h_\mu = e^{\mathrm{i}\theta_\mu}$ with commuting θ_μ and spreads them evenly, $A_\mu = \theta_\mu/n$. This gives (B) once $n \ge C_G/\varepsilon$. For other groups (e.g. $SO(3)$) commuting holonomies need not lie in a common torus.

Coherent field strengths of size ε over many blocks are natural inputs for the variational problem and typical configurations in the renormalization group. Hypothesis (B) excludes them. Part III therefore replaces (B) by the local condition of (M, ε)-regularity (Definition 18.1), which every background satisfying (B) obeys. Part VI derives it from Bałaban's gauge-invariant conditions (Theorem 30.3, Corollary 30.4).

12.1 Lie algebra valued fields

Let $G \subset U(\mathsf{N})$ be a compact connected Lie group, and let $\mathfrak{g}$ be the real Lie algebra of hermitian matrices X with $e^{\mathrm{i}tX} \in G$ for every $t \in \mathbb{R}$, following the convention of [3, 7]. Connectedness is used through $\det R(g) = 1$ on $\mathfrak{g}$, for $g \in G$ and for $g \in G^c$ (Propositions 32.8 and 47.8). For connected G this holds: $\det R(e^{\mathrm{i}X}) = e^{\mathrm{i}\,\mathrm{tr}\,\mathrm{ad}\,X} = 1$, since $\mathrm{ad}\,X$ is antisymmetric for the trace form, and G and G^c are generated by exponentials. For disconnected groups it can fail: on $(SU(2) \times SU(2)) \rtimes \mathbb{Z}_2$ the exchange of the factors has $\det R = -1$. Let $\mathfrak{g}^c = \mathfrak{g} + \mathrm{i}\mathfrak{g}$ be its complexification. On $\mathfrak{g}^c$ we use the bilinear form $\mathrm{b}(X, Y) = \mathrm{tr}(XY)$ and the Hilbert–Schmidt norm $|X| = (\mathrm{tr}\, X^*X)^{1/2}$; on $\mathfrak{g}$ the two are related by $\mathrm{b}(X, X) = |X|^2$. Then $|[X, Y]| \le 2|X||Y|$, and for $Z \in \mathfrak{g}^c$ the operator $R(e^Z)Y = e^Z Y e^{-Z} = e^{\mathrm{ad}\,Z}Y$ satisfies

$$\|R(e^Z) - 1\| \le e^{2|Z|} - 1. \tag{12.1}$$

Scalar fields, gauge fields and 2-forms are now $\mathfrak{g}^c$-valued. The inner products of Section 2 are replaced by

$$\mathrm{b}(f, g) = \eta^d \sum_x \mathrm{tr}\,(f(x)g(x)) \quad \text{(bilinear)}, \qquad \langle f, g\rangle = \eta^d \sum_x \mathrm{tr}\,(f(x)^* g(x)) \quad \text{(hermitian)},$$

with the analogous expressions for gauge fields, 2-forms and unit-lattice fields; norms $\|\cdot\|$ always refer to $\langle\cdot,\cdot\rangle$. The *transpose* X^T of a linear operator is taken with respect to b. If X maps $\mathfrak{g}$-valued fields to $\mathfrak{g}$-valued fields, X^T is the complex-linear extension of its adjoint on real fields.

Choosing an orthonormal basis (t_a) of $\mathfrak{g}$, every $\mathfrak{g}^c$-valued field is a $\dim \mathfrak{g}$-tuple of complex scalar fields. All operators of Part I act componentwise and are real. Hence every result of Sections 2–11 holds for $\mathfrak{g}^c$-valued fields. The L^2 statements hold with the same constants. For the pointwise statements, the kernel arguments of Section 6 go through with Hilbert–Schmidt norms; applying them componentwise instead costs a factor depending on $\dim \mathfrak{g}$. In particular, for real symmetric K the coercivity bounds of Theorem A hold for $\mathrm{Re}\langle u, Ku\rangle$ with complex u.

12.2 Backgrounds and the hypothesis (B)

A *background* is an assignment of $U(b) \in G^c$ to the oriented bonds b of $\mathbb{T}_\eta$, with $U(-b) = U(b)^{-1}$. We write $U_\mu(x) = U(x, x+\eta e_\mu)$, and $U(\Gamma)$ for the ordered product along a contour Γ. A gauge transformation is a map $u : \mathbb{T}_\eta \to G^c$; it is *real* if u takes values in G. It acts by $U^u(b) = u(b_-)U(b)u(b_+)^{-1}$ on backgrounds and by $\lambda \mapsto R(u)\lambda$ on scalar and gauge fields.

Fix $\alpha \in (0,1)$ and $\varepsilon > 0$.

Definition 12.1 (Hypothesis (B))**.** A background satisfies (B) with parameter ε if $U_\mu(x) = e^{\mathrm{i}\eta A_\mu(x)}$ for a $\mathfrak{g}^c$-valued gauge field A with

$$|A|_{\mathbb{T}_\eta} \le \varepsilon, \qquad |\partial A|_{\mathbb{T}_\eta} \le \varepsilon, \qquad \sup_z [\partial A]_{\alpha,\Lambda_3(z)} \le \varepsilon.$$

These bounds have the form of Bałaban's regularity conditions on the background in a cube [7, (3.35)–(3.36)], and of the regularity bounds [8, (9)] that the minimizers satisfy, but here they are imposed in one global gauge. The set of A satisfying (B) with strict inequalities is open and convex. The uniform bounds and the analyticity statements below hold on the smallness domains specified in the respective results. Operators that involve W are analytic in (A, W); they are analytic in A alone for fixed W or for an analytic choice $W = W(A)$.

12.3 Covariant derivatives and curvature

For scalar fields,

$$(D_\mu\lambda)(x) = \frac{R(U_\mu(x))\lambda(x+\eta e_\mu) - \lambda(x)}{\eta}, \qquad (D\lambda)_\mu = D_\mu\lambda,$$

and for gauge fields $(DA)_{\mu\nu} = D_\mu A_\nu - D_\nu A_\mu$, where D_μ acts on the scalar field A_ν. This is [7, (3.3), (3.4)]. We write $D = D_U$ when the background must be indicated. The transposes D^T play the role of d^*. At $U = 1$ we recover d and d^*.

Lemma 12.2 (Expansion of covariant derivatives)**.** *Assume (B) with $\varepsilon \le 1$. Then $D_\mu = \partial_\mu + T_\mu$ with $(T_\mu\lambda)(x) = \tau_\mu(x)\lambda(x+\eta e_\mu)$, where $\tau_\mu(x) = \eta^{-1}(R(e^{\mathrm{i}\eta A_\mu(x)}) - 1)$ is a linear map of $\mathfrak{g}^c$ satisfying*

$$|\tau_\mu| \le 2e^2\varepsilon, \qquad |\partial_\nu\tau_\mu| \le C\varepsilon, \qquad [\partial_\nu\tau_\mu]_{\alpha,\Lambda_3(z)} \le C\varepsilon.$$

Consequently, on scalar fields and componentwise on gauge fields,

$$D^T D = -\Delta + \sum_\mu \partial_\mu^*(a_\mu \cdot) + b, \tag{12.2}$$

where a_μ and b are multiplication operators composed with shifts by at most η, with $|a_\mu| + |\partial a_\mu| + |b| \le C\varepsilon$ and $[\partial a_\mu]_{\alpha,\Lambda_3(z)} + [b]_{\alpha,\Lambda_3(z)} \le C\varepsilon$. The same normal form holds for $D^TD + DD^T$ on gauge fields, which equals $-\Delta$ componentwise when $U = 1$ (Lemma 2.2).

Proof. The formula for D_μ is immediate. The bounds on τ follow from (12.1). Moreover $\tau_\mu(x) = \phi(\eta A_\mu(x))\,\mathrm{i}\,\mathrm{ad}\,A_\mu(x)$, where $\phi(Z) = (e^{\mathrm{i}\,\mathrm{ad}\,Z} - 1)/(\mathrm{i}\,\mathrm{ad}\,Z)$ is an entire function of Z; differentiating gives the bounds on $\partial\tau$ and its Hölder quotient. Next, $D_\mu^T = \partial_\mu^* + T_\mu^T$ with $(T_\mu^T w)(x) = \tau_\mu(x-\eta e_\mu)^T w(x-\eta e_\mu)$. Write

$$D_\mu^T D_\mu = \partial_\mu^*\partial_\mu + \partial_\mu^* T_\mu + T_\mu^T\partial_\mu + T_\mu^T T_\mu.$$

The second term is already in divergence form with $a_\mu u = \tau_\mu S_\mu u$. For the third, $T_\mu^T\partial_\mu u = -\tilde\tau_\mu\,\partial_\mu^* u$ with $\tilde\tau_\mu(x) = \tau_\mu(x-\eta e_\mu)^T$, and the discrete Leibniz rule gives $\tilde\tau_\mu\partial_\mu^* u = \partial_\mu^*(\tilde\tau_\mu u) - (\partial_\mu^*\tilde\tau_\mu)S_\mu^{-1}u$. The last term is bounded. Collecting terms gives (12.2). The computation for gauge fields is the same, term by term. □

On scalar fields $DD \neq 0$ in general. Let $\mathfrak{F} = D \circ D$ (scalar fields to 2-forms) denote the *curvature operator*.

Lemma 12.3 (Curvature operator)**.** $(\mathfrak{F}\lambda)_{\mu\nu}(x) = c_{\mu\nu}(x)\,\lambda(x + \eta e_\mu + \eta e_\nu)$ *with*

$$c_{\mu\nu}(x) = \eta^{-2}\Big(R(U_\mu(x)U_\nu(x + \eta e_\mu)) - R(U_\nu(x)U_\mu(x + \eta e_\nu))\Big).$$

Under (B) with $\varepsilon \leq 1$, $|c_{\mu\nu}| \leq C\varepsilon$ *and* $[c_{\mu\nu}]_{\alpha,\Lambda_3(z)} \leq C\varepsilon$.

Proof. Expanding $D_\mu D_\nu \lambda(x)$ gives four terms. After antisymmetrization in (μ, ν), the three terms $-R(U_\mu(x))\lambda(x + \eta e_\mu)$, $-R(U_\nu(x))\lambda(x + \eta e_\nu)$ and $\lambda(x)$ cancel. This leaves the formula. Write $c = \eta^{-2}R(X_2)(R(X_2^{-1}X_1) - 1)$. Then $X_2^{-1}X_1 = e^{-\mathrm{i}a'}e^{-\mathrm{i}b}e^{\mathrm{i}a}e^{\mathrm{i}b'}$, where

$$a = \eta A_\mu(x), \quad a' = \eta A_\mu(x + \eta e_\nu), \quad b = \eta A_\nu(x), \quad b' = \eta A_\nu(x + \eta e_\mu).$$

Let $\Psi(a, a', b, b') = e^{-\mathrm{i}a'}e^{-\mathrm{i}b}e^{\mathrm{i}a}e^{\mathrm{i}b'} - 1$, an entire function. Then

$$\Psi(a, a', b, b') = \big[\Psi(a, a', b, b') - \Psi(a, a, b, b)\big] + \Psi(a, a, b, b).$$

By Hadamard's lemma the bracket equals $(a' - a) \cdot \Psi_1 + (b' - b) \cdot \Psi_2$, with Ψ_i analytic functions of (a, a', b, b'). Moreover $a' - a = \eta^2 \partial_\nu A_\mu(x)$ and $b' - b = \eta^2 \partial_\mu A_\nu(x)$. The group commutator $\Psi(a, a, b, b)$ vanishes when $a = 0$ or $b = 0$, so it equals $\sum_{i,j} a_i b_j \Psi_{ij}(a, b)$ with analytic Ψ_{ij}. Hence $\eta^{-2}\Psi$ is a sum of products of ∂A, of $A \otimes A$, and of analytic functions of the four arguments $\eta A(\cdot)$. Each argument is bounded by $\eta\varepsilon \leq 1$ and Lipschitz in x with constant $\eta\varepsilon$, and $|A \otimes A| \leq \varepsilon^2 \leq \varepsilon$. Each factor is therefore bounded by $C\varepsilon$ or C, and is Hölder on $\Lambda_3(z)$ with seminorm $C\varepsilon$. Finally $(R(X_2^{-1}X_1) - 1)Y = [\Psi, Y](1 + \Psi)^{-1}$ for every Y, and $(1 + \Psi)^{-1} = X_1^{-1}X_2$ is a product of four exponentials of arguments of size at most $\eta\varepsilon \leq 1$, hence bounded and Lipschitz in x. Also $R(X_2)$ is Lipschitz. □

At leading order $c_{\mu\nu} \approx \mathrm{i}\,\mathrm{ad}\,F_{\mu\nu}$, where F is the curvature on the unit scale.

12.4 The quadratic form of the action

Definition 12.4 (Admissible potential terms)**.** For $\varepsilon, \gamma > 0$, let $\mathcal{W}(\varepsilon, \gamma)$ be the set of linear operators W on gauge fields such that $W^T = W$ and, for all y, y' and all u,

$$\|\mathbf{1}_y W \mathbf{1}_{y'}\| \leq \varepsilon e^{-\gamma|y-y'|}, \qquad |Wu|_{\Lambda_1(c_y)} + [Wu]_{\alpha,\Lambda_1(c_y)} \leq \varepsilon \sum_{y'} e^{-\gamma|y-y'|}\big(|u|_{\Lambda_1(c_{y'})} + [u]_{\alpha,\Lambda_1(c_{y'})}\big).$$

Here and below c_y is the centre of the block $B(y)$, so $c_y = y$.

The class $\mathcal{W}(\varepsilon, \gamma)$, with two arguments, is not related to the subspace $\mathcal{W} = -\Delta\mathcal{N}(Q_0)$ of Section 8.5 or to the subspace $\mathcal{W}(U)$ of Section 14.1. The quadratic form that replaces $\|dA\|^2$ is

$$q_U(A, A') = \mathrm{b}(DA, DA') + \mathrm{b}(A, WA'), \qquad W \in \mathcal{W}(\varepsilon, \gamma).$$

Two potentials motivate this definition. For $W = \Delta'(U)$, the curvature term of [7, (3.10)], $q_U(A, A)$ is the second-order term of the Wilson action at U. It is expected to be a local operator whose coefficients are, up to bounded analytic factors, the curvature coefficients $c_{\mu\nu}$ of Lemma 12.3. The linearizing transformation of the variational problem [8, (79)], [7, (3.127)–(3.128)] produces a further quadratic term $-2\langle HC^{(2)}(A), J\rangle$, proportional to the source J. We do not verify that these two terms belong to $\mathcal{W}$, and nothing below uses it. The results of Part II are stated for general W. Later parts use $W = 0$, except for the potential W_2 of Section 37, for which only L^2 statements are used; they are proved there.

12.5 Averaging operators with a background

For $y \in \mathbb{T}$ and $x \in B(y)$, let $\Gamma_{y,x}$ be the contour from the centre y of $B(y)$ to x that changes the coordinates in the order $d, d-1, \dots, 1$, as in [3, (1.7)]. It stays in $B(y)$ and has at most $\frac{d}{2}(N-1)$ bonds. For $x_j = x + j\eta e_\mu$, let $[x, x_j]$ be the straight contour. Define

$$(Q'(U)\lambda)(y) = \eta^d \sum_{x \in B(y)} R(U(\Gamma_{y,x}))\lambda(x),$$

$$(\mathbf{Q}(U)A)_\mu(y) = \eta^d \sum_{x \in B(y)} \eta \sum_{j=0}^{N-1} R(U(\Gamma_{y,x} \cup [x, x_j]))A_\mu(x_j).$$

Both average after parallel transport to the centre y. At $U = 1$ they reduce to Q_0 and $\mathbf{Q}$. $Q'(U)$ has the form of the linear scalar averaging of [7, (3.18), (3.19)], and $\mathbf{Q}(U)$ that of the main term [5, (125)] of the linearization [5, (124)] of the one-step gauge-field averaging; the remaining small terms of [5, (124)] are dropped.

We call $(Q'(U), \mathbf{Q}(U))$ the *model pair*. It is a proof device. A single ordered contour breaks the lattice symmetry of Lemma 2.6(f), and Bałaban's averaging transports instead with the group average over the coordinate-ordered contours in all orders (Section 21). Parts II and III are stated for the model pair, and they are applied to Bałaban's averaging through Remark 19.12: the real statements under properties (i)–(iv) of that remark, the analytic ones with property (v) in addition. There the model pair supplies only the trial fields of Theorem 18.2(c) and Lemma 19.8; it enters no statement about the renormalization group map. Averaged contours would serve equally well here, at the cost of a small-curvature hypothesis, which is needed to define the group average.

On the unit lattice let $V(y, y+e_\mu) = U([y, y+e_\mu])$ (straight contour), and $(D_V g)_\mu(y) = R(V(y, y+e_\mu))g(y+e_\mu) - g(y)$.

Lemma 12.5 (Properties of the averages)**.** *Assume (B) with $\varepsilon \le 1$.*

(a) $Q'(U)Q'(U)^T = I$, *and* $(Q'(U)^T g)(x) = R(U(\Gamma_{y(x),x}))^{-1} g(y(x))$.

(b) $Q'(U) - Q_0$ *and* $\mathbf{Q}(U) - \mathbf{Q}$ *act within distance* 2 *(that is,* $\mathbf{1}_y(\cdot)\mathbf{1}_{y'} = 0$ *for* $|y - y'| > 2$*), have block norms at most* $C\varepsilon$*, and satisfy the pointwise bounds* $|(Q'(U) - Q_0)\lambda(y)| \le C\varepsilon|\lambda|_{B(y)}$, $|(\mathbf{Q}(U) - \mathbf{Q})A(y)| \le C\varepsilon|A|_{B(y)\cup B(y+e_\mu)}$. *The same holds for the transposes.*

(c) (Covariance defect.) $\mathbf{Q}(U)D_U = D_V Q'(U) + E(U)$, *where*

$$(E(U)\lambda)_\mu(y) = \eta^d \sum_{x' \in B(y+e_\mu)} \Big[R(U(\Gamma_{y,x'-e_\mu} \cup [x'-e_\mu, x'])) - R(V(y, y+e_\mu)U(\Gamma_{y+e_\mu,x'})) \Big] \lambda(x').$$

In particular $E(U)$ acts within distance 2, $|E(U)\lambda(y)| \le C\varepsilon|\lambda|_{B(y+e_\mu)}$, *and* $\|E(U)\| \le C\varepsilon$. *Moreover* $|E(U)^T\omega|_\infty \le C\varepsilon|\omega|_\infty$*: by the formula, $(E(U)^T\omega)(x')$ is a sum of d terms, one for each μ, each the transpose of a bracket of norm at most $C\varepsilon$ applied to $\omega_\mu(y)$ with $x' \in B(y+e_\mu)$.*

(d) (Gauge covariance.) *For every gauge transformation u, $Q'(U^u)R(u) = R(u|_{\mathbb{T}})Q'(U)$, $\mathbf{Q}(U^u)R(u) = R(u|_{\mathbb{T}})\mathbf{Q}(U)$, $D_{U^u}R(u) = R(u)D_U$ and $E(U^u)R(u) = R(u|_{\mathbb{T}})E(U)$.*

Proof. (a) uses $R(X)^T = R(X^{-1})$ with respect to b, together with the fact that the blocks partition $\mathbb{T}_\eta$. (b) Every contour involved, including the products of two contours in (c), has at most $\frac{d}{2}(N-1) + N \le (\frac{d}{2}+1)N$ bonds. So $|U(\Gamma) - 1| \le e^{(d/2+1)\varepsilon} - 1$ and similarly for $U(\Gamma)^{-1}$, by submultiplicativity. Then $R(g)Y - Y = (g-1)Yg^{-1} + Y(g^{-1} - 1)$ gives

$\|R(g)-1\| \le |g-1||g^{-1}| + |g^{-1}-1| \le C\varepsilon$ for $g = U(\Gamma)$. (c) Along each line the sum over j telescopes:

$$\eta \sum_{j=0}^{N-1} R\big(U(\Gamma \cup [x, x_j])\big)(D_\mu \lambda)(x_j) = R\big(U(\Gamma \cup [x, x+e_\mu])\big)\lambda(x+e_\mu) - R\big(U(\Gamma)\big)\lambda(x),$$

with $\Gamma = \Gamma_{y,x}$. Summing over $x \in B(y)$, the second terms give $Q'(U)\lambda(y)$, and the first terms differ from $R(V)Q'(U)\lambda(y+e_\mu)$ exactly by $E(U)\lambda$. Both transports in the bracket are within $C\varepsilon$ of 1, by the bound in (b). (d) is the usual transformation law of contour products, $U^u(\Gamma) = u(\Gamma_-)U(\Gamma)u(\Gamma_+)^{-1}$. □

Remark 12.6 (Which averaging?). Only properties (a)–(d) of $Q'(U)$ and $\mathbf{Q}(U)$ are used below. Only the trial fields of Theorem 18.2(c) and of Lemma 19.8 use the explicit formula for $Q'(U)^T$ in (a) (Remark 19.12). The averaging of the renormalization group is Bałaban's symmetric averaging [18, (0.12)], constructed in Section 21. In it the transport from a block centre y to a point x of the block is the group average of the transports along the coordinate-ordered contours from y to x, one for each order in which the nonzero coordinates of $x-y$ are changed, and the coarse bond variables are formed with these averaged transports. Its k-step version is a composition of nonlinear one-step averagings with intermediate backgrounds, as in [7, (3.15)]. For the linearization of its gauge-adapted version (Definition 22.4) the covariance identity $\mathbf{Q}(U)D_U = D_V Q'(U)$ is exact, i.e. $E = 0$, as for the averaging of [7, (3.114)–(3.115)]; the identities [7, (3.124), (3.152)] are consequences. The model pair keeps a single contour per point and only the main term [5, (125)], and the defect $E(U)$ is the price of this simplification. Our treatment does not require $E = 0$. Its real statements apply to every averaging with properties (i)–(iv) of Remark 19.12, which correspond to (a)–(d) above; the analytic statements need in addition property (v) of that remark. In Part IV we construct Bałaban's averaging and show that its linearization has these properties with $E = 0$ (Proposition 22.10, Corollary 22.11).

13 Perturbation lemmas

Under hypothesis (B), every operator of Part II will be written as the corresponding operator of Part I plus a perturbation of a specific structure. This section shows that the three tools of Part I (Combes–Thomas, inversion with decay, local regularity) are stable under such perturbations.

13.1 Complex fields

Lemmas 5.5, 5.6 and 5.7 hold for operators on complex ($\mathfrak{g}^c$-valued) fields that are not necessarily symmetric, provided the lower bounds are imposed on $\mathrm{Re}\langle u, Ku\rangle$ for all complex fields u (not only real ones). The proofs are unchanged: $\mathrm{Re}\langle u, K_\alpha u\rangle \ge \delta\|u\|^2$ implies $\|K_\alpha u\| \ge \delta\|u\|$, and in the proof of Lemma 5.7 one uses $|\langle u, h\rangle| \ge \mathrm{Re}\langle u, h\rangle$. We use these lemmas in this form without further comment.

13.2 Admissible perturbations

Let $\mathfrak{D}$ be the set of operators consisting of the identity and all ∂_μ, ∂_μ^*, D_μ, D_μ^T, acting on scalar fields, on gauge fields (componentwise), or between gauge fields and 2-forms. Under (B) with $\varepsilon \le 1$, every $X \in \mathfrak{D}$ satisfies $\|Xu\| \le 2\|\partial u\| + C\|u\|$ and $\|X^*u\| \le 2\|\partial u\| + C\|u\|$. By Lemma 5.4 and Lemma 12.2, $X_\alpha - X$ is bounded by $C|\alpha|$ for $|\alpha| \le 1$ and 1-Lipschitz ρ.

Definition 13.1 (Admissible perturbations)**.** An operator δ is an *admissible perturbation of size ε and rate γ* if $\delta = \sum_{i=1}^m X_i M_i Y_i$ with $m \le m_0$, $X_i, Y_i \in \mathfrak{D}$, and each M_i has decay (ε, γ). Here m_0 is a fixed absolute number.

Lemma 13.2 (Stability of Combes–Thomas)**.** *Let $\mathfrak{A}_0$ satisfy the hypotheses of Lemma 5.7 with constants c, C_1, α_1, and let δ be admissible of size ε and rate γ. There is $\varepsilon_1 > 0$, depending only on c, γ and m_0, such that for $\varepsilon \le \varepsilon_1$ the operator $\mathfrak{A} = \mathfrak{A}_0 + \delta$ satisfies these hypotheses with constants $c/2$, $C_1 + 1$ and $\min(\alpha_1, \gamma/2, 1)$. In particular $\mathfrak{A}^{-1}$, $\partial\mathfrak{A}^{-1}$, $\mathfrak{A}^{-1}\partial^*$ and $\partial\mathfrak{A}^{-1}\partial^*$ have decay (C, γ_0) with C, γ_0 depending only on $c, C_1, \alpha_1, \gamma, m_0$. The same holds with ∂ replaced by D.*

Proof. By Lemma 5.2(a), $|\langle u, X_i M_i Y_i u\rangle| \le \|X_i^* u\|\, \varepsilon S_\gamma\, \|Y_i u\| \le C\varepsilon(\|u\|^2 + \|\partial u\|^2)$. This gives (i) with $c/2$ if $C\varepsilon \le c/2$. For (ii), note that $(XMY)_\alpha - XMY$ is a sum of three terms, each containing one of the differences $X_\alpha - X$, $M_\alpha - M$, $Y_\alpha - Y$. The first and last have norm at most $C|\alpha|$, and $\|M_\alpha - M\| \le \varepsilon c_1(\gamma)|\alpha|$ by Lemma 5.3. Every term is therefore bounded in form sense by $C\varepsilon|\alpha|(\|u\|^2 + \|\partial u\|^2)$. Replacing ∂ by D in the conclusions costs a bounded operator of size $C\varepsilon$, by Lemma 12.2. □

Lemma 13.3 (Perturbed inverses)**.** *In the situation of Lemma 13.2, let r and r_0 be decay rates of $\mathfrak{A}^{-1}$ and $\mathfrak{A}_0^{-1}$, respectively (with derivatives), and set $\gamma_0 = \min(\gamma, r, r_0)$. Then $\mathfrak{A}^{-1} - \mathfrak{A}_0^{-1} = -\mathfrak{A}^{-1}\delta\mathfrak{A}_0^{-1}$. This operator, as well as its compositions with ∂ on the left and with ∂^* on the right, has decay $(C\varepsilon, \gamma_0/4)$. The same statements hold, with ∂ replaced by 1, for bounded operators $K = K_0 + \delta$ on unit-lattice fields, where K_0 has decay, $\mathrm{Re}\langle f, K_0 f\rangle \ge c\|f\|^2$, and δ has decay (ε, γ) with $\varepsilon S_\gamma \le c/2$. In this bounded case, set $\gamma_0 = \min(\gamma, r_K, r_{K_0})$, where r_K, r_{K_0} are chosen decay rates of K^{-1}, K_0^{-1}.*

Proof. The identity is algebraic. Each term $\mathfrak{A}^{-1}X_i M_i Y_i \mathfrak{A}_0^{-1}$ is a product of the operators $\mathfrak{A}^{-1}X_i$ and $Y_i\mathfrak{A}_0^{-1}$ and of M_i, which has decay (ε, γ). Lemma 13.2 gives decay of $\mathfrak{A}^{-1}\partial^*$ and $\partial\mathfrak{A}_0^{-1}$. The remaining combinations reduce to these: $\partial_\mu = -\partial_\mu^* S_\mu$ and $\partial_\mu^* = -S_\mu^{-1}\partial_\mu$, where the shifts $S_\mu^{\pm 1}$ by η are local isometries, and $D_\mu = \partial_\mu + T_\mu$ with T_μ local and bounded. All three factors have decay at the common rate γ_0, by Lemma 5.2(c). Apply Lemma 5.2(b) twice; each application halves the rate. In the bounded case use Lemma 5.5 for K. □

13.3 Pointwise admissibility and the bootstrap

For pointwise bounds we need perturbations whose action is controlled locally. For a field u on $\mathbb{T}_\eta$ and $y \in \mathbb{T}$ put

$$N_y^0(u) = |u|_{\Lambda_1(c_y)}, \qquad N_y^1(u) = |u|_{\Lambda_1(c_y)} + |\partial u|_{\Lambda_1(c_y)}.$$

Definition 13.4 (Pointwise admissibility)**.** Let $i \in \{0, 1\}$. A linear operator δ is *pointwise admissible of type i*, with constants $(\varepsilon, C_\delta, \gamma)$, if for every field u there are fields F_u, g_u with $\delta u = F_u + \sum_\mu \partial_\mu^* g_{u,\mu}$ and, for every y,

$$|F_u|_{\Lambda_2(c_y)} + |g_u|_{\Lambda_3(c_y)} + i\,[g_u]_{\alpha,\Lambda_3(c_y)} \le \sum_{y'} e^{-\gamma|y-y'|}\Big(\varepsilon N_{y'}^i(u) + C_\delta\|u\|_{B(y')}\Big).$$

The smallness is required only for the terms controlled by N^i. Terms controlled by local L^2 norms, such as $dP_{\mathcal{V}}d^*$ in Section 10, may be large.

Lemma 13.5 (Bootstrap)**.** *Let δ be pointwise admissible of type i with constants $(\varepsilon, C_\delta, \gamma)$. There is $\varepsilon_* > 0$, depending only on α and γ, such that the following holds if $\varepsilon \le \varepsilon_*$; it does not depend on C_δ, M or γ_2. Let*

$$-\Delta u = J + \delta u \qquad \text{on } \mathbb{T}_\eta,$$

with $J = J_F + \sum_\mu \partial_\mu^* J_{g,\mu}$. Suppose that for some y_0, $M > 0$ and $\gamma_2 > 0$, and for all y,

$$\|u\|_{B(y)} + |J_F|_{\Lambda_2(c_y)} + |J_g|_{\Lambda_3(c_y)} + i\,[J_g]_{\alpha,\Lambda_3(c_y)} \le Me^{-\gamma_2|y-y_0|}.$$

Let $\gamma' = \frac{1}{2}\min(\gamma, \gamma_2)$. Then, with C depending only on $\alpha, \gamma, \gamma_2, C_\delta$,

$$\begin{aligned} &\textit{type } 1: && |u|_{B(y)} + |\partial u|_{B(y)} + [\partial u]_{\alpha,B(y)} \le CMe^{-\gamma'|y-y_0|}, \\ &\textit{type } 0: && |u|_{B(y)} + [u]_{\alpha,B(y)} \le CMe^{-\gamma'|y-y_0|}. \end{aligned}$$

Proof. We give the proof for type 1; type 0 is identical, with Theorem 6.3(b) in place of (d). The lattice is finite, so

$$\mathcal{N} = \max_y e^{\gamma'|y-y_0|} N_y^1(u) < \infty.$$

Fix y and apply Theorem 6.3(d) on the cubes $\Lambda_r(c_y)$, with $F = J_F + F_u$ and $g = J_g + g_u$:

$$N_y^1(u) + [\partial u]_{\alpha,\Lambda_1(c_y)} \le C\big(|F|_{\Lambda_2} + |g|_{\Lambda_3} + [g]_{\alpha,\Lambda_3} + \|u\|_{\Lambda_2}\big).$$

The cube $\Lambda_2(c_y)$ meets at most C blocks, all within distance C of y, so $\|u\|_{\Lambda_2} \le CMe^{-\gamma_2|y-y_0|}$. By the definition of pointwise admissibility and $N_{y'}^1(u) \le e^{-\gamma'|y'-y_0|}\mathcal{N}$,

$$|F_u|+|g_u|+[g_u]_\alpha \le \sum_{y'} e^{-\gamma|y-y'|}\Big(\varepsilon e^{-\gamma'|y'-y_0|}\mathcal{N} + C_\delta M e^{-\gamma_2|y'-y_0|}\Big) \le S_{\gamma/2}\, e^{-\gamma'|y-y_0|}(\varepsilon\mathcal{N} + C_\delta M),$$

where we used $\gamma|y-y'| + \gamma'|y'-y_0| \ge \gamma'|y-y_0| + \frac{\gamma}{2}|y-y'|$. Multiplying by $e^{\gamma'|y-y_0|}$ and maximizing over y gives $\mathcal{N} \le C_0 M + C_{\mathrm{reg}} S_{\gamma/2}\,\varepsilon\mathcal{N}$. Here C_{reg} is the constant of Theorem 6.3, which depends only on α, and C_0 depends on $\alpha, \gamma, \gamma_2, C_\delta$. For $\varepsilon \le \varepsilon_* = 1/(2C_{\mathrm{reg}}S_{\gamma/2})$ this yields $\mathcal{N} \le 2C_0 M$, and the Hölder bound follows from the same inequality. □

13.4 Building blocks

The perturbations in Sections 14–16 are assembled from the following pieces.

Lemma 13.6 (Building blocks)**.** *Assume (B) with $\varepsilon \le 1$.*

(a) The differential parts $D^TD + \Delta$ (on scalar fields) and $D^TD + DD^T + \Delta$ (on gauge fields) are admissible of size $C\varepsilon$ and pointwise admissible of both types with constants $(C\varepsilon, 0, 1)$.

(b) Every $W \in \mathcal{W}(\varepsilon, \gamma)$ is admissible of size ε and pointwise admissible of type 1 with constants $(C\varepsilon, 0, \gamma)$, since $[u]_{\alpha,\Lambda_1} \le C(|u| + |\partial u|)_{\Lambda_1}$.

(c) The operators $Q'(U)^TQ'(U) - Q_0^*Q_0$ and $\mathbf{Q}(U)^T\mathbf{Q}(U) - \mathbf{Q}^*\mathbf{Q}$ are admissible of size $C\varepsilon$ and pointwise admissible of both types with constants $(C\varepsilon, 0, 1)$. The operators $Q_0^*Q_0$ and $\mathbf{Q}^*\mathbf{Q}$ themselves are pointwise admissible with constants $(0, C, 1)$.

(d) If K is an operator with values in fine fields such that $u \mapsto Ku$ satisfies $|Ku|_{B(y)} + |\partial Ku|_{B(y)} + [\partial Ku]_{\alpha,B(y)} \le C_K \sum_{y'} e^{-\gamma|y-y'|}\|u\|_{B(y')}$, then $\partial_\mu K$ is pointwise admissible with constants $(0, C\,C_K, \gamma)$ (it contributes only to F_u).

Proof. (a) By (12.2), $(D^TD + \Delta)u = \sum_\mu \partial_\mu^*(a_\mu u) + bu$, and $[a_\mu u]_\alpha \le C\varepsilon(|u| + |\partial u|)$ on each cube. For type 0 the Hölder quotient is not needed. Admissibility in the sense of Definition 13.1 follows from the same formula: $\partial^* \circ a_\mu \circ 1$ and $1 \circ b \circ 1$ with a_μ, b local of size $C\varepsilon$. (b) is the definition of $\mathcal{W}$. (c) By Lemma 12.5(b), the differences are local, bounded by $C\varepsilon$, and their values are bounded pointwise by $C\varepsilon|u|$ on neighbouring blocks. They contribute only to F_u. The second statement follows from $|Q_0^*Q_0u| \le \max|Q_0u| \le \|u\|$ on blocks (Lemma 2.6). (d) is immediate. □

14 Scalar operators with a background

Throughout Sections 14–16 we assume (B) with parameter ε, fix $a > 0$ and $\alpha \in (0,1)$, and let C, γ denote constants depending only on a, α, $\dim \mathfrak{g}$ and, from Section 15 on, the decay rate $\gamma_W > 0$ of the potential term W. The rate γ_W is fixed first. The statement "for $\varepsilon \le \varepsilon_0$" means for ε below a threshold depending on the same quantities. It cannot be chosen uniformly as $\gamma_W \to 0$ (Remark 15.2).

The scalar operators are built from

$$\mathfrak{A}'(U) = D^T D + a\, Q'(U)^T Q'(U), \qquad \mathfrak{A}'(1) = -\Delta + a Q_0^* Q_0.$$

At real U, $\langle \lambda, \mathfrak{A}'(U)\lambda\rangle = \|D\lambda\|^2 + a\|Q'(U)\lambda\|^2$. This is the operator Δ'_a of [7, (3.24)] on a single torus. Its inverse $G'(U)$ is [7, G', (3.25)].

Theorem 14.1 (Scalar operators with a background)**.** *For $\varepsilon \le \varepsilon_0$:*

(a) $\mathfrak{A}'(U) = \mathfrak{A}'(1) + \delta'$, where δ' is admissible of size $C\varepsilon$ and pointwise admissible of both types with constants $(C\varepsilon, 0, 1)$. Hence $G'(U) = \mathfrak{A}'(U)^{-1}$ exists and is analytic in A. Moreover G', DG', $G'D^T$, $DG'D^T$ have decay (C,γ), and $G'(U) - G_0$ composed with derivatives has decay $(C\varepsilon,\gamma)$.

(b) For f supported in $B(y')$,

$$|G'f|_{B(y)} + |\partial G' f|_{B(y)} + [\partial G' f]_{\alpha,B(y)} \le Ce^{-\gamma|y-y'|}|f|_{B(y')},$$
$$|G'D_\mu^T f|_{B(y)} + [G'D_\mu^T f]_{\alpha,B(y)} \le Ce^{-\gamma|y-y'|}|f|_{B(y')}.$$

If f is merely bounded and Hölder, then

$$|\partial G' D_\mu^T f|_{B(y)} + [\partial G' D_\mu^T f]_{\alpha,B(y)} \le C\sum_{y'} e^{-\gamma|y-y'|}\big(|f|_{\Lambda_3(c_{y'})} + [f]_{\alpha,\Lambda_3(c_{y'})}\big).$$

(c) $Q'G'Q'^T$ differs from $Q_0 G_0 Q_0^*$ by an operator with decay $(C\varepsilon,\gamma)$. Its inverse $\Gamma'(U)$ exists and has decay (C,γ).

(d) The minimizer map $H'(U) = G'Q'^T\Gamma'$ satisfies $Q'H' = I$. For real U, $H'g$ is the unique minimizer of $\|D\lambda\|^2$ subject to $Q'(U)\lambda = g$. Moreover $|H'g|_{B(y)} + |\partial H' g|_{B(y)} + [\partial H' g]_{\alpha,B(y)} \le C\sum_{y'} e^{-\gamma|y-y'|}|g(y')|$.

(e) The constrained Green function $C'(U) = G' - H'Q'G'$ satisfies $Q'C' = 0 = C'Q'^T$. For every f, $C'f$ is the unique $\nu \in \mathcal{N}(Q'(U))$ with $\mathrm{b}(D\nu', D\nu) = \mathrm{b}(\nu', f)$ for all $\nu' \in \mathcal{N}(Q'(U))$. The operators C', DC', $C'D^T$ and $DC'D^T$ have decay (C,γ), $C'(U) - C'(1)$ composed with derivatives has decay $(C\varepsilon,\gamma)$, and the pointwise bounds of (b) hold with G' replaced by C'.

Proof. (a) $\delta' = (D^T D + \Delta) + a(Q'^T Q' - Q_0^* Q_0)$, so Lemma 13.6(a),(c) applies. The operator $\mathfrak{A}'(1)$ satisfies the hypotheses of Lemma 5.7 by the proof of Theorem 7.1. Lemmas 13.2 and 13.3 give the decay statements. Analyticity holds because $\mathfrak{A}'(U)$ is a polynomial in the entries of $R(e^{i\eta A})$, and its inverse exists with a uniform bound on the open set where (B) holds.

(b) Let $u = G'f$. Then $-\Delta u = J - \delta' u - aQ_0^* Q_0 u$ with $J = f$. The term $aQ_0^* Q_0 u$ is pointwise admissible with constants $(0, C, 1)$, and $\|u\|_{B(y)} \le Ce^{-\gamma|y-y'|}\|f\|$ by (a). Lemma 13.5 of type 1 gives the first bound. For $u = G'D_\mu^T f$ we have $J = \partial_\mu^* f + T_\mu^T f$, with $J_g = f e_\mu$ bounded. Lemma 13.5 of type 0 applies. For bounded Hölder f that is not localized, split $f = \sum_{y'} \chi_{y'} f$ with a smooth partition of unity $(\chi_{y'})$ subordinate to the cubes $\Lambda_1(c_{y'})$, with bounded derivatives. Then $[\chi_{y'} f]_\alpha \le C(|f| + [f]_\alpha)_{\Lambda_3(c_{y'})}$. Apply Lemma 13.5 of type 1 to each piece, with $J_g = \chi_{y'} f e_\mu$, and sum; this explains the Λ_3 norms on the right-hand side.

(c) $Q'G'Q'^T - Q_0G_0Q_0^* = (Q'-Q_0)G'Q'^T + Q_0(G'-G_0)Q'^T + Q_0G_0(Q'^T - Q_0^*)$. By Lemma 12.5(b) and (a), this has decay $(C\varepsilon,\gamma)$. Since $Q_0G_0Q_0^* \geq c_F$, the bounded version of Lemma 13.3 applies.

(d) $Q'H' = Q'G'Q'^T\Gamma' = I$. For real U the variational characterization is proved exactly as in Lemma 8.3, with $\langle f, \mathfrak{A}'f\rangle = \|Df\|^2 + a\|g\|^2$ on $\{Q'f = g\}$. The pointwise bound follows from (b) applied to $f = Q'^T\Gamma' g$, whose values on $B(y'')$ are bounded by $(1+C\varepsilon)|\Gamma' g(y'')|$, by Lemma 12.5(a),(b). The factor $1+C\varepsilon$ appears because $R(U(\Gamma))^{-1}$ need not be an isometry for complex U.

(e) The identities are algebraic. If $u = C'f$, then $\mathfrak{A}'u = f - Q'^T\Gamma'Q'G'f$ and $Q'u = 0$. Hence $D^TDu = f - Q'^T\omega$, which gives the characterization. For uniqueness, let $\nu \in \mathcal{N}(Q')$ with $\mathrm{b}(D\nu', D\nu) = 0$ for all $\nu' \in \mathcal{N}(Q')$. Then $D^TD\nu \in \mathcal{N}(Q')^{\perp_{\mathrm{b}}} = \operatorname{ran} Q'^T$, say $D^TD\nu = Q'^T\omega$. Hence $\mathfrak{A}'\nu = Q'^T\omega$, i.e. $\nu = G'Q'^T\omega$. Then $0 = Q'\nu = Q'G'Q'^T\omega$ gives $\omega = 0$ by (c), and so $\nu = 0$. The bounds follow from $C'f = G'f - H'(Q'G'f)$ and from (a), (c) and (d); at $U = 1$, $C'(1)$ is the operator C_σ of Proposition 8.6 with $\sigma = 0$. □

14.1 The gauge-fixing projection

Let

$$\mathcal{W}(U) = D^TD\,\mathcal{N}(Q'(U)), \qquad \mathcal{V}(U) = \operatorname{ran}\left(G'(U)Q'(U)^T\right), \qquad M'(U) = Q'G'^2Q'^T,$$

$$P_{\mathcal{V}}(U) = G'Q'^TM'^{-1}Q'G', \qquad R(U) = I - P_{\mathcal{V}}(U).$$

At $U = 1$, $\mathcal{W}(1)$ is the subspace $\mathcal{W} = -\Delta\,\mathcal{N}(Q_0)$ of Section 8.5, for $\mathfrak{g}^c$-valued fields. The subspace $\mathcal{W}(U)$ is not the class $\mathcal{W}(\varepsilon,\gamma_W)$ of potentials (Definition 12.4). The projection $R(U)$ has a background as its argument; for a group element or a gauge transformation u, $R(u)$ is the adjoint action of Section 12.1, applied pointwise.

Proposition 14.2 (The projection $R(U)$)**.** *For $\varepsilon \leq \varepsilon_0$:*

(a) $M'(U)$ is invertible and M'^{-1} has decay (C,γ).

(b) $R(U)^2 = R(U) = R(U)^T$, $\operatorname{ran} R(U) = \mathcal{W}(U)$ and $\ker R(U) = \mathcal{V}(U)$. In particular $R(U)$ does not depend on a. For real U, it is the orthogonal projection onto $\mathcal{W}(U)$, as in [7, (3.21), (3.25)].

(c) For $h \in \mathcal{W}(U)$: $G'h = C'h \in \mathcal{N}(Q'(U))$ and $D^TDG'h = h$. Equivalently, $Q'G'R = 0$ and $D^TDG'R = R$.

(d) $P_{\mathcal{V}}$, $DP_{\mathcal{V}}$, $P_{\mathcal{V}}D^T$ and $DP_{\mathcal{V}}D^T$ have decay (C,γ), and $P_{\mathcal{V}}(U) - P_{\mathcal{V}}(1)$ has decay $(C\varepsilon,\gamma)$.

(e) Pointwise: $P_{\mathcal{V}}h = G'Q'^T\omega$ with $|\omega(y)| \leq C\sum_{y'} e^{-\gamma|y-y'|}\|h\|_{B(y')}$. Hence $|P_{\mathcal{V}}h|_{B(y)} + |\partial P_{\mathcal{V}}h|_{B(y)} + [\partial P_{\mathcal{V}}h]_{\alpha,B(y)} \leq C\sum_{y'} e^{-\gamma|y-y'|}\|h\|_{B(y')}$. The same holds for $P_{\mathcal{V}}D^Tu$, with $\|h\|$ replaced by $\|u\|$.

Proof. (a) $M'(U) - M'(1)$ has decay $(C\varepsilon,\gamma)$ by Theorem 14.1, and $M'(1) = M \geq c_F^2$ (Proposition 8.8). Apply Lemma 13.3.

(b) With $T = G'Q'^T$ we have $T^T = Q'G'$, because $\mathfrak{A}'^T = \mathfrak{A}'$, and $P_{\mathcal{V}} = T(T^TT)^{-1}T^T$. Hence $P_{\mathcal{V}}^2 = P_{\mathcal{V}} = P_{\mathcal{V}}^T$, $\operatorname{ran} P_{\mathcal{V}} = \mathcal{V}$, and $\ker P_{\mathcal{V}} = \ker(Q'G')$. For $\nu \in \mathcal{N}(Q')$,

$$Q'G'D^TD\nu = Q'G'(\mathfrak{A}' - aQ'^TQ')\nu = Q'\nu = 0,$$

so $\mathcal{W} \subset \ker(Q'G')$. The map D^TD is injective on $\mathcal{N}(Q')$: if $D^TD\nu = 0$ with $Q'\nu = 0$, then $\mathfrak{A}'\nu = 0$, so $\nu = 0$. Since Q' is onto ($Q'Q'^T = I$), $\dim\mathcal{W} = \dim\mathcal{N}(Q')$. On the other hand $Q'G'$ is onto, because $Q'G'Q'^T$ is invertible, so $\dim\ker(Q'G') = \dim\mathcal{N}(Q')$ as well.

Therefore $\mathcal{W} = \ker P_{\mathcal{V}} = \operatorname{ran} R$. Finally $\mathrm{b}(G'Q'^T\omega, D^T D\nu) = \mathrm{b}(\omega, Q'G'D^T D\nu) = 0$, so $\mathcal{V}$ is the b-annihilator of $\mathcal{W}$, which does not involve a.

(c) If $h = D^T D\nu$ with $\nu \in \mathcal{N}(Q')$, then $\mathfrak{A}'\nu = h$, so $G'h = \nu$. Moreover $C'h = G'h - H'Q'G'h = \nu$.

(d) and (e) are compositions of operators with decay, using Theorem 14.1(b) for the pointwise statement. □

15 The gauge-field Green function with a background

15.1 The slice projection

The residual gauge directions are $D\mathcal{N}(Q'(U))$, and Bałaban's gauge slice is

$$\mathcal{S}(U) = \{A : \ R(U)D^T A = 0\}.$$

At $U = 1$ the identities $dd = 0$ and $\mathbf{Q}d = d_1 Q_0$ made the slice and the gauge directions interact trivially with the action and the averaging. With a background these identities fail by the curvature operator $\mathfrak{F} = DD$ and the defect $E(U)$. Following Bałaban [7, (3.118)–(3.119)], we correct for this with the projection onto the slice along the gauge directions. Let

$$S(U) = G'(U)R(U)D^T, \qquad \Pi(U) = I - DS(U).$$

Lemma 15.1 (Slice projection)**.** *For $\varepsilon \le \varepsilon_0$:*

(a) $S\,D\nu = \nu$ for $\nu \in \mathcal{N}(Q'(U))$, and $\operatorname{ran} S \subset \mathcal{N}(Q'(U))$. Hence $SDS = S$ and $\Pi^2 = \Pi$.

(b) $\Pi D\nu = 0$ for $\nu \in \mathcal{N}(Q'(U))$, and $RD^T\Pi = 0$. Thus Π is the projection onto $\mathcal{S}(U)$ along $D\mathcal{N}(Q'(U))$, and $\Pi A = A$ for $A \in \mathcal{S}(U)$.

(c) $D\Pi = D - \mathfrak{F}S$ and $\mathbf{Q}(U)\Pi = \mathbf{Q}(U) - E(U)S$.

(d) At $U = 1$: $\Pi(1) = I - dG_0 Rd^*$, $d\Pi(1) = d$ and $\mathbf{Q}\Pi(1) = \mathbf{Q}$.

Proof. (a) For $\nu \in \mathcal{N}(Q')$ we have $D^T D\nu \in \mathcal{W}$, so $RD^T D\nu = D^T D\nu$ and $G'D^T D\nu = \nu$ by Proposition 14.2(c). Also $\operatorname{ran} S \subset \operatorname{ran} G'R \subset \mathcal{N}(Q')$ by the same proposition.

(b) The first claim follows from (a). Next, $RD^T\Pi = RD^T - RD^T DG'RD^T = RD^T - R\,RD^T = 0$, by $D^T DG'R = R$.

(c) $D\Pi = D - DDS$. Moreover $\mathbf{Q}DS = (D_V Q' + E)S = ES$, because $Q'S = Q'G'RD^T = 0$.

(d) At $U = 1$, $\mathfrak{F} = 0$ and $E = 0$. □

15.2 The operator and its inverse

Let $W \in \mathcal{W}(\varepsilon, \gamma_W)$ and $\mathfrak{B} = D^T D + W$, so that $q_U(A, A') = \mathrm{b}(A, \mathfrak{B}A')$. Define

$$\mathfrak{A}_\pi(U) = \Pi^T \mathfrak{B}\,\Pi + DR\,D^T + a\,\Pi^T \mathbf{Q}(U)^T \mathbf{Q}(U)\,\Pi.$$

It is transpose-symmetric. By Lemma 15.1(d), at $U = 1$ and $W = 0$ it reduces to the operator $\mathbf{\Delta}_a$ of Section 9. The first term is Bałaban's gauge-invariant extension of the form q_U [7, (3.118)–(3.120), (3.128)]. Bałaban's averaging term is $a\,\mathbf{Q}^T\mathbf{Q}$ rather than $a\,\Pi^T\mathbf{Q}^T\mathbf{Q}\Pi$. The two coincide when $E = 0$, which holds for his averaging. With $E \neq 0$ the version with Π is needed, since it is what makes $\mathbf{Q}_\pi \mathbf{G} DR = 0$ in Theorem 15.4(d). The difference is the term $\mathcal{K}_2$ below, which is the only genuinely new ingredient.

Remark 15.2 (The threshold depends on γ_W). The threshold ε_0 below cannot be chosen uniformly as $\gamma_W \to 0$. Let $U = 1$, fix μ and $t \in \mathfrak{g}$ with $|t| = 1$, let e be the constant gauge field with $e_\mu = t$ and $e_\nu = 0$ for $\nu \neq \mu$, and let $W = -aP$, where P is the orthogonal projection onto the multiples of e. Since $\|e\|^2 = n^d$ and every block has volume 1, $\|\mathbf{1}_y W \mathbf{1}_{y'}\| = an^{-d}$. Moreover Wu is constant, with $|Wu| \le an^{-d} \sum_{y'} |u|_{B(y')}$. Torus distances are at most $dn/2$, so $W \in \mathcal{W}(\varepsilon_n, 1/n)$ with $\varepsilon_n = ae^{d/2} n^{-d}$. On the other hand $de = 0$ and $d^* e = 0$, so $\Pi(1)e = e$ by Lemma 15.1(d). Also $\mathbf{Q}^*\mathbf{Q}e = e$, because $Q_\mu 1 = 1$ and $Q_\mu^* 1 = 1$ (Lemma 2.6(b)). Hence $\mathfrak{A}_\pi(1)e = \Pi(1)^T(-ae + ae) = 0$, although $\varepsilon_n \to 0$ as $n \to \infty$.

Lemma 15.3 (Identities). *(a) $\mathfrak{A}_\pi D\nu = D(D^T D\nu)$ for $\nu \in \mathcal{N}(Q'(U))$. Consequently $\mathfrak{A}_\pi DG'Rh = DRh$ for every scalar field h.*

(b) For $A, A' \in \mathcal{S}(U)$: $\mathrm{b}(A, \mathfrak{A}_\pi A') = q_U(A, A') + a\,\mathrm{b}(\mathbf{Q}A, \mathbf{Q}A')$.

Proof. (a) $\Pi D\nu = 0$ kills the first and last terms, and $RD^T D\nu = D^T D\nu$. Apply this with $\nu = G'Rh$, for which $D^T D\nu = Rh$. (b) For $A \in \mathcal{S}$, $\Pi A = A$ and $RD^T A = 0$. □

Theorem 15.4 (Gauge-field Green function with a background). *For $\varepsilon \le \varepsilon_0$:*

(a) $\mathfrak{A}_\pi(U) = \mathbf{\Delta}_a + \delta_\pi$, where δ_π is admissible of size $C\varepsilon$ and pointwise admissible of type 1 with constants $(C\varepsilon, C, \gamma)$.

(b) $\mathbf{G}(U) = \mathfrak{A}_\pi(U)^{-1}$ exists and is analytic in A and W. The operators $\mathbf{G}$, $D\mathbf{G}$, $\mathbf{G}D^T$, $D\mathbf{G}D^T$ have decay (C, γ), and $\mathbf{G}(U) - \mathbf{G}$ (with derivatives) has decay $(C\varepsilon, \gamma)$.

(c) For all y, y' and J supported in $B(y')$,

$$|\mathbf{G}J|_{B(y)} + |\partial\mathbf{G}J|_{B(y)} + [\partial\mathbf{G}J]_{\alpha, B(y)} \le Ce^{-\gamma|y-y'|}|J|_{B(y')}.$$

(d) $\mathbf{G}DR = DG'R$, $RD^T\mathbf{G} = RG'D^T$, and $\mathbf{Q}_\pi \mathbf{G} DR = 0$, where $\mathbf{Q}_\pi = \mathbf{Q}(U)\Pi$.

Proof. (a) Using $DRD^T = DD^T - DP_{\mathcal{V}}D^T$ and $\mathbf{\Delta}_a = -\Delta - dP_{\mathcal{V}}(1)d^* + a\mathbf{Q}^*\mathbf{Q}$, we get

$$\delta_\pi = (D^T D + DD^T + \Delta) + W - (DP_{\mathcal{V}}(U)D^T - dP_{\mathcal{V}}(1)d^*) + a(\mathbf{Q}^T\mathbf{Q} - \mathbf{Q}^*\mathbf{Q}) + \mathcal{K}_1 + \mathcal{K}_2,$$

with $\mathcal{K}_1 = \Pi^T \mathfrak{B}\Pi - \mathfrak{B}$ and $\mathcal{K}_2 = a(\Pi^T\mathbf{Q}^T\mathbf{Q}\Pi - \mathbf{Q}^T\mathbf{Q})$. By Lemma 15.1(c), $\mathfrak{B}D = D^T\mathfrak{F} + WD$ and $D^T D^T = \mathfrak{F}^T$. Hence

$$\mathcal{K}_1 = -\mathfrak{B}DS - (\mathfrak{B}DS)^T + S^T(\mathfrak{F}^T\mathfrak{F} + D^T W D)S, \qquad \mathfrak{B}DS = D^T\mathfrak{F}S + WDS,$$
$$\mathcal{K}_2 = -a(\mathbf{Q}^T ES + (\mathbf{Q}^T ES)^T) + aS^T E^T ES.$$

Admissibility. Every term is of the form XMY with $X, Y \in \mathfrak{D}$ and M a composition of decaying operators containing at least one factor of size $C\varepsilon$: $\mathfrak{F}$ or W or E, the differences in Lemma 12.5(b), $T = D - d$, or $P_{\mathcal{V}}(U) - P_{\mathcal{V}}(1)$. For example $D^T\mathfrak{F}S = D^T(\mathfrak{F}G'R)D^T$, and

$$DP_{\mathcal{V}}(U)D^T - dP_{\mathcal{V}}(1)d^* = D\big(P_{\mathcal{V}}(U) - P_{\mathcal{V}}(1)\big)D^T + TP_{\mathcal{V}}(1)D^T + dP_{\mathcal{V}}(1)T^T.$$

The decay of the factors comes from Theorem 14.1 and Proposition 14.2.

Pointwise admissibility. The first, second and fourth terms are covered by Lemma 13.6(a)–(c). By Proposition 14.2(e) and Lemma 13.6(d), $DP_{\mathcal{V}}(U)D^T u$ and $dP_{\mathcal{V}}(1)d^* u$ contribute to F_u with a bound $C\sum e^{-\gamma|y-y'|}\|u\|_{B(y')}$. For $\mathcal{K}_1, \mathcal{K}_2$ we use the following facts. Write $Su = G'D^T u - G'P_{\mathcal{V}}D^T u$. By Theorem 14.1(b) and Proposition 14.2(e),

$$|Su| + [Su]_\alpha \le C\sum e^{-\gamma|\cdot|}(|u| + \|u\|), \qquad |DSu| + [DSu]_\alpha \le C\sum e^{-\gamma|\cdot|}(|u| + [u]_\alpha + \|u\|).$$

Moreover $S^T = DRG'$, and $DRG'f = DG'f - DP_{\mathcal{V}}G'f$ is bounded and Hölder whenever f is bounded, again by Theorem 14.1(b). With these:

- $D^T\mathfrak{F}Su = \sum_\mu \partial^*_\mu g_\mu$ + (bounded) with $g = \mathfrak{F}Su$. By Lemma 12.3, $|g| + [g]_\alpha \le C\varepsilon(|Su| + [Su]_\alpha)$.
- $WDSu$ and its Hölder quotient are bounded by $\varepsilon(|DSu| + [DSu]_\alpha)$, by the definition of $\mathcal{W}$.
- $(\mathfrak{B}DS)^T u = DRG'(\mathfrak{F}^T Du + D^T Wu)$. Here $\mathfrak{F}^T Du$ is bounded by $C\varepsilon(|u| + |\partial u|)$. Moreover $DG'D^T(Wu)$ is bounded by $C(|Wu| + [Wu]_\alpha)$ by the Hölder part of Theorem 14.1(b), and $DP_{\mathcal{V}}G'D^T(Wu)$ by $C\|Wu\|$ on nearby blocks, by Proposition 14.2(e).
- $S^T\mathfrak{F}^T\mathfrak{F}Su = DRG'(\mathfrak{F}^T\mathfrak{F}Su)$, and the terms of $\mathcal{K}_2$ are $\mathbf{Q}^T ESu$, $DRG'(E^T\mathbf{Q}u)$ and $DRG'(E^T ESu)$. They are of the form DRG'(bounded) or $\mathbf{Q}^T$(bounded), with bounds containing a factor ε.
- $S^T D^T WDSu = DRG'D^T f$ with $f = WDSu$. Here a derivative of f appears, so boundedness of f is not enough. By the second item, f is bounded and Hölder, with a factor ε. Write $DRG'D^T f = DG'D^T f - DP_{\mathcal{V}}G'D^T f$. The first term is bounded by $C(|f| + [f]_\alpha)$ by the Hölder part of Theorem 14.1(b), and the second by $C\|f\|$ on nearby blocks, by Proposition 14.2(e).

All bounds are summed with exponential weights, local in the sense of Definition 13.4. Using $[u]_\alpha \le C(|u| + |\partial u|)$ on cubes, this proves pointwise admissibility of type 1.

(b) The operator $\mathbf{\Delta}_a$ satisfies the hypotheses of Lemma 5.7 (proof of Theorem 10.1). Apply Lemmas 13.2 and 13.3. Analyticity follows as in Theorem 14.1.

(c) Let $u = \mathbf{G}J$. Then $-\Delta u = J + dP_{\mathcal{V}}(1)d^*u - a\mathbf{Q}^*\mathbf{Q}u - \delta_\pi u$, where the middle terms are pointwise admissible with constants $(0, C, \gamma)$, with $\gamma > 0$ chosen no larger than 1, the rate in part (a), and a rate of Proposition 14.2(e) at $U = 1$. We fix this rate before reducing ε_0 to satisfy the threshold of Lemma 13.5. The L^2 decay of u is given by (b). Apply Lemma 13.5 of type 1.

(d) The first identity follows from Lemma 15.3(a), and the second by transposition, since $\mathbf{G}^T = \mathbf{G}$, $G'^T = G'$ and $R^T = R$. The third follows from $\mathbf{Q}_\pi \mathbf{G} DR = \mathbf{Q}\Pi DG'R = 0$, by Lemma 15.1(b), because $G'R$ maps into $\mathcal{N}(Q')$. □

16 Minimizer and fluctuation operators with a background

We keep the assumptions of Section 15 and write $\mathbf{Q}_\pi = \mathbf{Q}(U)\Pi$. Let

$$\mathcal{X}(U) = \{A : \ \mathbf{Q}(U)A = 0, \ R(U)D^T A = 0\} = \mathcal{S}(U) \cap \ker \mathbf{Q}_\pi,$$

the tangent space of the constraint set in Bałaban's gauge [8, (100)].

Theorem 16.1 (Minimizer and constrained Green function)**.** *For $\varepsilon \le \varepsilon_0$:*

(a) $\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T$ differs from $\mathbf{Q}\mathbf{G}\mathbf{Q}^*$ by an operator with decay $(C\varepsilon, \gamma)$. Its inverse $\mathbf{\Gamma}_\pi$ exists and has decay (C, γ).

(b) The minimizer map $\mathbf{H}(U) = \mathbf{G}\mathbf{Q}_\pi^T \mathbf{\Gamma}_\pi$ satisfies $\mathbf{Q}(U)\mathbf{H} = I$, $RD^T\mathbf{H} = 0$ and $\Pi\mathbf{H} = \mathbf{H}$. For real U, real symmetric W and real data B, $\mathbf{H}B$ is real, and it is the unique minimizer of $q_U(A, A)$ among real A with $\mathbf{Q}(U)A = B$ and $R(U)D^T A = 0$. $\mathbf{H}$ and $\partial\mathbf{H}$ have decay (C, γ), and

$$|\mathbf{H}B|_{B(y)} + |\partial\mathbf{H}B|_{B(y)} + [\partial\mathbf{H}B]_{\alpha, B(y)} \le C \sum_{y'} e^{-\gamma|y-y'|} |B|_{y'}.$$

(c) The constrained Green function

$$\mathbf{C}(U) = \mathbf{G} - DG'RG'D^T - \mathbf{H}\mathbf{Q}_\pi\mathbf{G}$$

is transpose-symmetric and maps into $\mathcal{X}(U)$. For every J, $\mathbf{C}J$ is the unique $A \in \mathcal{X}(U)$ with

$$q_U(A', A) = \mathrm{b}(A', J) \qquad \text{for all } A' \in \mathcal{X}(U).$$

(d) $\mathbf{C}$ and $\partial\mathbf{C}$ have decay (C, γ), and for J supported in $B(y')$,

$$|\mathbf{C}J|_{B(y)} + |\partial\mathbf{C}J|_{B(y)} + [\mathbf{C}J]_{\alpha,B(y)} \le Ce^{-\gamma|y-y'|}|J|_{B(y')}.$$

Proof. (a) $\mathbf{Q}_\pi - \mathbf{Q} = (\mathbf{Q}(U) - \mathbf{Q}) - E(U)S$, and $ES = EG'RD^T$ has decay $(C\varepsilon, \gamma)$. Together with Theorem 15.4(b), every term of the difference has decay $(C\varepsilon, \gamma)$. Since $\mathbf{Q}\mathbf{G}\mathbf{Q}^* \ge c_F$ (Proposition 11.2), Lemma 13.3 applies.

(b) $\mathbf{Q}_\pi\mathbf{H} = I$. By Theorem 15.4(d), $RD^T\mathbf{H} = (\mathbf{Q}_\pi\mathbf{G}DR)^T\mathbf{\Gamma}_\pi = 0$. Hence $\mathbf{H}B \in \mathcal{S}$, so $\Pi\mathbf{H} = \mathbf{H}$ and $\mathbf{Q}\mathbf{H} = \mathbf{Q}_\pi\mathbf{H} = I$.

For the minimization, let U, W and B be real, let $A \in \mathcal{S}$ be real with $\mathbf{Q}A = B$, and put $h = A - \mathbf{H}B \in \mathcal{X}(U)$. All operators are real, so $\mathbf{H}B$ and h are real. Then $\mathrm{b}(h, \mathfrak{A}_\pi\mathbf{H}B) = \mathrm{b}(\mathbf{Q}\Pi h, \mathbf{\Gamma}_\pi B) = 0$. Lemma 15.3(b) gives

$$q_U(A, A) = q_U(\mathbf{H}B, \mathbf{H}B) + q_U(h, h), \qquad q_U(h, h) = \langle h, \mathfrak{A}_\pi h\rangle \ge \tfrac{c_g}{2}\|h\|^2,$$

for real fields. The bound on $q_U(h, h)$ is the coercivity of $\mathfrak{A}_\pi$ from Lemma 13.2.

Decay is a composition. For the pointwise bound, $\mathbf{H}B = \mathbf{G}J$ with $J = \mathbf{Q}_\pi^T\mathbf{\Gamma}_\pi B$. Now

$$\mathbf{Q}_\pi^T = \Pi^T\mathbf{Q}^T = \mathbf{Q}^T - DRG'\,(D^T\mathbf{Q}^T), \qquad D^T\mathbf{Q}(U)^T = (\mathbf{Q}D)^T = Q'^T D_V^T + E^T$$

by Lemma 12.5(c). So $D^T\mathbf{Q}^T\omega$ is bounded on each block by $C\max|\omega|$ over neighbouring points. By Theorem 14.1(b), DRG' maps bounded fields to bounded fields. Hence $|J|_{B(y'')} \le C\sum e^{-\gamma|y''-y'|}|B|_{y'}$, and the sum form of Theorem 15.4(c) concludes.

(c) Symmetry is clear. Since $\mathbf{Q}_\pi DG'R = \mathbf{Q}\Pi DG'R = 0$,

$$\mathbf{Q}_\pi\mathbf{C} = \mathbf{Q}_\pi\mathbf{G} - 0 - \mathbf{Q}_\pi\mathbf{G} = 0.$$

By Theorem 15.4(d) and $D^TDG'R = R$,

$$RD^T\mathbf{C} = RG'D^T - R\,RG'D^T - 0 = 0.$$

So $\mathbf{C}J \in \mathcal{S} \cap \ker\mathbf{Q}_\pi = \mathcal{X}$. For $A' \in \mathcal{X}$, Lemma 15.3(b) gives $q_U(A', \mathbf{C}J) = \mathrm{b}(A', \mathfrak{A}_\pi\mathbf{C}J)$. By Lemma 15.3(a) with $h = G'D^TJ$,

$$\mathfrak{A}_\pi\mathbf{C}J = J - DR\,G'D^TJ - \mathbf{Q}_\pi^T\mathbf{\Gamma}_\pi\mathbf{Q}_\pi\mathbf{G}J.$$

The last two terms are b-orthogonal to $\mathcal{X}$: $\mathrm{b}(A', DRh) = \mathrm{b}(RD^TA', h) = 0$ and $\mathrm{b}(A', \mathbf{Q}_\pi^T\omega) = \mathrm{b}(\mathbf{Q}A', \omega) = 0$.

For uniqueness, suppose $A \in \mathcal{X}$ and $q_U(A', A) = 0$ for all $A' \in \mathcal{X}$. Then $\mathfrak{A}_\pi A$ annihilates $\mathcal{X} = \ker(RD^T) \cap \ker\mathbf{Q}_\pi$, so $\mathfrak{A}_\pi A = DR\theta + \mathbf{Q}_\pi^T\omega$ for some θ, ω. Hence $A = DG'R\theta + \mathbf{G}\mathbf{Q}_\pi^T\omega$. Applying RD^T gives $0 = R\theta$, so $A = \mathbf{G}\mathbf{Q}_\pi^T\omega$. Applying $\mathbf{Q}_\pi$ gives $0 = \mathbf{\Gamma}_\pi^{-1}\omega$, so $\omega = 0$.

(d) For the decay, the first and third terms of $\mathbf{C}$ are compositions of operators with decay. For the middle term, Theorem 15.4(d) and $R^2 = R$ give $DG'RG'D^T = \mathbf{G}\,DR\,G'D^T$. Hence $\partial DG'RG'D^T = (\partial\mathbf{G}D)\,R\,(G'D^T)$, and $\partial\mathbf{G}D$ has decay: on scalar fields $D_\mu = -S_\mu\partial_\mu^* + T_\mu$, with S_μ the shift by η, so $\partial\mathbf{G}D$ is $\partial\mathbf{G}\partial^*$ up to local bounded factors. No second derivatives of G' are needed. For the pointwise bounds write $\mathbf{C}J = \mathbf{G}J - D\sigma - \mathbf{H}(\mathbf{Q}_\pi\mathbf{G}J)$ with $\sigma = G'RG'D^TJ$. The first and third terms are controlled by Theorem 15.4(c) and part (b), since $\mathbf{Q}_\pi\mathbf{G}J$ is a bounded unit-lattice field decaying away from y'.

By Proposition 14.2(c), $\sigma \in \mathcal{N}(Q'(U))$ and $D^TD\sigma = h := RG'D^TJ$. By Theorem 14.1(b) and Proposition 14.2(e), h is bounded and Hölder on cubes, with bounds $Ce^{-\gamma\,\mathrm{dist}}|J|_{B(y')}$. The

local L^2 norms of σ decay by composition. First, $-\Delta\sigma = h - (D^TD + \Delta)\sigma$, and Lemma 13.5 of type 1 bounds $|\sigma|$, $|\partial\sigma|$ and $[\partial\sigma]_\alpha$ on blocks. Next, by (12.2) and the Leibniz rule,

$$(D^TD + \Delta)\sigma = \sum_\mu \left(a_\mu \partial_\mu^* \sigma + (\partial_\mu^* a_\mu) S_\mu^{-1}\sigma\right) + b\sigma.$$

The right-hand side is bounded and Hölder on $\Lambda_2(z)$, with constants $C\varepsilon$ times the bounds just obtained. This uses the Hölder bounds on cubes (Remark 7.2) and $[\partial a_\mu]_\alpha + [b]_\alpha \le C\varepsilon$ from Lemma 12.2; this uses the Hölder condition on ∂A in (B), as does the curvature bound used in the pointwise admissibility argument of Theorem 15.4. Theorem 6.3(c) now bounds $|\partial\partial\sigma|$, and hence $|\partial D\sigma| \le |\partial\partial\sigma| + |\partial(T\sigma)|$. The Hölder bound on $\mathbf{C}J$ follows from the bound on $|\partial\mathbf{C}J|$. □

Remark 16.2 (Comparison with [7]). Bałaban's operators G_1, H_1 and $\mathfrak{G}$ [7, (3.128), (3.129), (3.153)] correspond to $\mathbf{G}(U)$, $\mathbf{H}(U)$ and $\mathbf{C}(U)$. His gauge-invariant extension Δ_π, with correction Δ'_π, corresponds to $\Pi^T\mathfrak{B}\Pi$, i.e. to the term $\mathcal{K}_1$. Our W plays the role of his curvature term Δ' [7, (3.10)] and of the term $-2\langle HC^{(2)}(A), J\rangle$ of [7, (3.128)]; we do not use that these terms belong to $\mathcal{W}$ (Section 12). For his averaging $E = 0$, so there is no $\mathcal{K}_2$.

16.1 Analyticity and gauge covariance

Proposition 16.3. *(a) All operators of Sections 14–16 are analytic functions of A on the open set where (B) holds with $\varepsilon < \varepsilon_0$, and of W in the ball of $\mathcal{W}(\varepsilon_0, \gamma_W)$. All bounds are uniform there. By the Cauchy estimates, on the smaller set where (B) holds with $\varepsilon \le \varepsilon_0/2$, the derivatives of order m with respect to A, in directions δA obeying (B) with $\varepsilon = 1$, satisfy the decay and pointwise bounds of the operators themselves, with constants multiplied by $m!\,(C/\varepsilon_0)^m$. The factor ε in the comparisons with $U = 1$ is not preserved. If $W = W(A)$ is an analytic function of A with values in the ball of $\mathcal{W}(\varepsilon_0, \gamma_W)$, the operators are analytic functions of A; otherwise only the joint analyticity in (A, W) is claimed.*

(b) Let $u : \mathbb{T}_\eta \to G$ be a gauge transformation, and suppose $W(U^u) = R(u)W(U)R(u)^{-1}$. Then

$$\begin{aligned} G'(U^u) &= R(u)G'(U)R(u)^{-1}, & R(U^u) &= R(u)R(U)R(u)^{-1}, \\ \Pi(U^u) &= R(u)\Pi(U)R(u)^{-1}, & \mathbf{G}(U^u) &= R(u)\mathbf{G}(U)R(u)^{-1}, \\ \mathbf{H}(U^u) &= R(u)\mathbf{H}(U)R(u|_{\mathbb{T}})^{-1}, & \mathbf{C}(U^u) &= R(u)\mathbf{C}(U)R(u)^{-1}. \end{aligned}$$

Consequently, let U satisfy (B), let $W \in \mathcal{W}(\varepsilon, \gamma_W)$ in this gauge, and let u be G-valued. At U^u, with the potential $R(u)WR(u)^{-1}$, the L^2 block bounds and the pointwise bounds of Sections 14–16 hold with ∂ replaced by D_{U^u}, and the Hölder bounds hold after transforming back by $R(u)^{-1}$, i.e. in the gauge in which (B) holds. A bound comparing an operator with its value at $U = 1$ transfers only as a comparison with the transformed reference: for example, $G'(U^u) - G'(1^u) = R(u)(G'(U) - G'(1))R(u)^{-1}$ has decay $(C\varepsilon, \gamma)$. It does not transfer as a comparison with $G'(1)$.

In (b), $R(u)$ is the adjoint action of the gauge transformation u (Section 12.1), applied pointwise, while $R(U)$ and $R(U^u)$ are the projections of Proposition 14.2.

Proof. (a) Each operator is built from $R(e^{i\eta A})$ by sums, products and inverses. The inverses exist, with uniform bounds, on the indicated set, so the operators are analytic there. (b) follows from Lemma 12.5(d) and the definitions, which are built from D, Q', $\mathbf{Q}$, W by gauge-covariant operations. For G-valued u, $R(u)$ is an isometry, and it acts pointwise, so it commutes with

the localization projections $\mathbf{1}_y$. The derivative bounds transfer because $D_{U^u} R(u) = R(u) D_U$ (Lemma 12.5(d)) and $D_U = \partial + T$ with T local of size $C\varepsilon$ in the gauge of (B) (Lemma 12.2). For the last claim, let G be non-abelian, and let x and $z = x + \eta e_\mu$ lie in different blocks with $R(u(x)u(z)^{-1}) \neq 1$. The coefficient of $\lambda(z)$ in $(\mathfrak{A}'(1^u)\lambda)(x)$ is $-\eta^{-2} R(u(x)u(z)^{-1})$, and in $(\mathfrak{A}'(1)\lambda)(x)$ it is $-\eta^{-2}$, because $Q'^T Q'$ has no matrix elements between different blocks (Lemma 12.5(a)). So $G'(1^u) \neq G'(1)$, although 1^u is gauge equivalent to 1, which satisfies (B) for every $\varepsilon > 0$. □

16.2 What the variational problem needs

We list the input of Bałaban's proof of the variational theorem [8], specialized to a single torus with the small-field condition everywhere, and indicate where it is provided in this paper.

Linear input (Parts II and III).

- The minimizer map $H(U_0)$ and its bounds [8, (45), (46)]: Theorem 16.1(b), and Theorem 19.9 at regular backgrounds.
- The fluctuation operator $\mathfrak{G} = G_1 \mathfrak{P}^*$ and the bounds of [7, Theorem 3.13] used in [8, (110)–(120)]: Theorem 16.1(c),(d) and Theorem 19.9. In Part V the additional quadratic terms of [8, (79)] are not put into the operator; they are treated as part of the nonlinearity.
- The Neumann series for operators with exponential decay [8, (68)–(71)]: Lemma 5.6.
- The scalar operators $G'(U_0)$, $H'(U_0)$ and the projection $R(U_0)$ entering the Landau gauge condition [8, (21)] and the gauge-fixing theorem [6, (1.91)–(1.100)]: Theorem 14.1, Proposition 14.2, Theorem 18.2.
- The Feynman-gauge operator $G(U_0) = \mathfrak{A}_F^{-1} = (D^T D + D R D^T + a \mathbf{Q}^T \mathbf{Q})^{-1}$ of [6, (1.58)]: Lemma 24.1(c).

Nonlinear input (Part IV).

- The nonlinear averaging, the regularity of averaged configurations [5, Props. 1, 2], in the symmetric four-dimensional form [18, (0.3)–(0.12)]: Section 21.
- The axial gauge, in the averaged form [18, pp. 253–254], the modified averaging and its composition [5, Sect. C], the expansion $\mathcal{Q}(e^{\mathrm{i}\eta A}) = e^{\mathrm{i}(\mathbf{Q}A + C(A))}$ with quadratic remainder [5, Props. 3–5]: Section 22. That the linearization of Bałaban's averaging satisfies $E = 0$ [7, (3.114)–(3.115)] is Proposition 22.10(b).
- The expansion of the Wilson action [8, (25)–(40)]: Proposition 23.3.
- The passage from the axial gauge to the Landau gauge [6, Theorem 2]: Theorem 24.2, from the averaged axial gauge.

The variational theorem relative to a reference (Part V). The linearizing transformation, the contraction, uniqueness, minimality, the second-order bounds, analyticity and the expansion [8, Sects. C–E, G]: Theorem 25.1.

The induction (Part VI). The trivial extension and the induction over k [8, Sect. A, (11)–(14)], and the local improvement of the regularity constants [8, Sect. F]: Lemma 30.1, Lemma 29.2 and Theorem 30.3. The derivation of local regularity from gauge-invariant conditions on the curvature [6, Sect. F, Prop. 6]: Corollary 30.4. Analyticity around the background field [8, Sect. G]: Corollary 30.5.

Not provided. The nested domains Ω_j and large fields, and gauges on cubes of size of order ε_1^{-1} with linear bounds (Section 60).

Part III: Domains, local regularity and random walks

Part III replaces the global hypothesis (B) of Part II by Bałaban's local one. A real background is (M, ε)-regular if on every cube of side $8M + 4$ there is a gauge in which $U = e^{i\eta A}$ with $|A|$, $|\partial A|$ and the Hölder quotients of ∂A at most ε; the gauges on different cubes are unrelated (Definition 18.1). Part VI shows that the background fields, and all real configurations with small curvature and current, are regular (Theorem 30.3, Corollary 30.4). The variational problem of Part V and the step of Part VII work at such backgrounds, and Parts VIII and IX build their localized operators from the Dirichlet operators of this Part.

What this Part proves. The main result is Theorem G. For $M \geq M_0$ and ε small, independently of M:

- the operators of Part II exist at regular backgrounds and keep their bounds, without the comparisons with $U = 1$, with covariant derivatives and with Hölder quotients in the local gauges (Theorems 18.2, 19.7 and 19.9);
- the scalar and the gauge-field operator have Dirichlet versions on unions X of blocks, which depend only on the background on the bonds of $\hat{X}$, the points at ∞-distance at most 1 from X, and are coercive; their inverses, the inverse of the averaged scalar Green function and $\mathbf{1}_X - R_X$ decay exponentially, uniformly in the domain (Theorems 18.2, 19.6 and 19.7); the gauge-field version carries a mass term near the boundary and Dirichlet conditions for all components;
- if two regular backgrounds agree on a region, the operators agree deep inside it, up to errors exponentially small in the distance to the complement (Theorem 19.10);
- all of this extends analytically to small complex perturbations $Ue^{i\eta\,\delta A}$, with Re-coercivity in place of coercivity (Proposition 19.11), and, for M large, Bałaban's random-walk expansion converges for the scalar Green function (Theorem 20.2(a)).

The key step is a local proof of the coercivity of the gauge-field operator (Theorem 19.6): an IMS localization at scale M (Lemma 17.4) reduces it to cubes, where one works in the local gauge and uses the coercivity of Part I. The statements are proved for the model pair of Part II; Remark 19.12 lists the properties of an averaging pair that the proofs use, and Corollary 22.11 verifies them for Bałaban's averaging.

Imports. Part II throughout, and from Part I the coercivity (Corollary 3.4, Theorem 9.6), the Combes–Thomas lemmas and Theorem 6.3. Four passages point ahead. The proof of Lemma 19.3 uses the covariant Hodge inequality of Lemma 26.4, a summation by parts that uses only the curvature bound. Theorem 20.2(b), the walk for the gauge-field Green function, is Theorem 41.4(a) at level (L3) with $s = 1$; it is proved with Section 41 and is best read after it. The second paragraph of Proposition 19.11, and its proof, concern the potentials of Section 35 (Part VIII). The last paragraph of Remark 19.12 uses Propositions 46.4 and 46.7 of Part X.

Exports. Definition 18.1 and Theorem 18.2 are used in Parts IV–XI, and Theorems 19.6, 19.7 and 19.9 in Parts IV–IX. Locality and the comparison of domains (Theorem 19.10, Lemma 17.7) are the base of the localization in Parts VII–IX, and Proposition 19.11 is used in Parts IV and VIII–X. Lemma 31.3(a) shows that the constants of Theorems 18.2, 19.6 and 19.7 (on the torus) and 19.9 do not depend on M, L, k or n; the random-walk expansion

and Theorem 19.10(b) have constants that depend on M (Remark 31.4(b)). The bounds of Parts I–VIII do not use the random-walk expansion (Remark 20.3); the weakened walks of Part IX are built on the same kind of parametrix (Lemma 41.2), and there the walk series also gives the bounds (Theorem 41.4).

Details for this Part. The companion has no chapter on Part III; its parameter ledger lists the constants of Part III that Parts VII–XI use, among them M_0.

17 Domains, partitions of unity and two general lemmas

Part II needs the background to be close to 1 in a single global gauge. Bałaban's conditions are local: on every cube there is a gauge in which the background is close to 1. To work with local conditions we need operators that live on bounded regions and depend only on the background in a neighbourhood of width 1 of these regions, together with a mechanism that turns local information into global estimates. This part provides both.

- *Dirichlet operators* on unions of blocks (this section and Section 18). Their decay and regularity follow as in Part I, because the Poincaré inequality, Combes–Thomas and local regularity are local.
- *Coercivity from local information* (Section 19). The scalar operators are coercive block by block. For the gauge-field operator we use an IMS localization at a large scale M: on each cube of size M we work in the local gauge, perturb around $U = 1$, and use the coercivity of Part I. Near the boundary of a domain we add a mass term. This makes the local operators coercive without any further gauge-theoretic input. The gauge-field operators also get Dirichlet conditions for all components, through the divergence at the exterior points next to the domain.
- *Comparison of domains and locality in the background* (Sections 18, 19). If two backgrounds agree on a region, the corresponding operators agree deep inside it up to exponentially small errors.
- *Bałaban's random-walk expansion* (Section 20), which writes the scalar Green function as a convergent sum of products of local propagators. For the gauge-field Green function the walk is completed in Section 41.

Unlike Part II, no Fourier analysis is used for backgrounds $U \neq 1$.

17.1 Domains and field spaces

Definition 17.1 (Domains)**.** A *domain* is a non-empty union of blocks $X = \bigcup_{y \in X_1} B(y)$, $X_1 \subset \mathbb{T}$. We write:

- $\ell^2_0(X)$ for the scalar fields vanishing outside X;
- $\ell^2_1(X)$ for the gauge fields vanishing on all bonds with no endpoint in X;
- $\ell^2(X_1)$ and $\ell^2(\mathcal{B}_1(X))$ for unit-lattice scalar fields supported in X_1 and unit-lattice gauge fields supported on unit bonds with an endpoint in X_1.

In all cases $\mathbf{1}_X$ denotes the orthogonal projection onto the corresponding space. For an operator $\mathfrak{A}$ on torus fields, $\mathfrak{A}_X = \mathbf{1}_X \mathfrak{A} \mathbf{1}_X$, regarded as an operator on $\ell^2(X)$, is its *Dirichlet restriction* to X. We write $\hat{X}$ for the set of points at ∞-distance at most 1 from X. The torus $\mathbb{T}_\eta$ is itself a domain, with $\hat{\mathbb{T}}_\eta = \mathbb{T}_\eta$.

The exterior derivative maps $\ell^2_0(X)$ into $\ell^2_1(X)$: if λ vanishes outside X, then $d\lambda$ vanishes on bonds without an endpoint in X. The adjoint of $d : \ell^2_0(X) \to \ell^2_1(X)$ is $d^*_X = \mathbf{1}_X d^*$. The same holds for the covariant derivative D and $D^T_X = \mathbf{1}_X D^T$. On $\ell^2_1(X)$ we use the norm

$$|\!|\!| A |\!|\!|^2_X = \|A\|^2 + \|DA\|^2 + \|D^T_X A\|^2,$$

and on $\ell^2_0(X)$ the norm $|\!|\!|\lambda|\!|\!|^2_X = \|\lambda\|^2 + \|D\lambda\|^2$. For $X = \mathbb{T}_\eta$ and $U = 1$ these are equivalent to $\|A\|^2 + \|\partial A\|^2$ and $\|\lambda\|^2 + \|\partial\lambda\|^2$, by Lemma 2.2. Fields in $\ell^2(X)$ are extended by zero, so DA includes the values on bonds and plaquettes that cross the boundary of X.

Remark 17.2 (Why Dirichlet restrictions). For a local operator such as $-\Delta + aQ^*_\sigma Q_\sigma$, the quadratic form of $\mathfrak{A}_X$ is the restriction of the torus form to fields supported in X. Coercivity is therefore inherited. Moreover $\mathfrak{A}_X$ depends only on the coefficients of $\mathfrak{A}$ inside $\hat{X}$. For operators built from inverses, such as the projection R, the Dirichlet version is defined by applying the construction to Dirichlet restrictions of local operators (Sections 18, 19). It is not the Dirichlet restriction of the torus operator. This is what makes the local operators depend only on the background in $\hat{X}$. For the gauge-field operator, the local part is the Dirichlet restriction of the torus operator $D^T D + DD^T$. This needs the divergence at the exterior points next to X, which $D^T_X = \mathbf{1}_X D^T$ omits (Section 19).

17.2 Partitions of unity at scale M

Let $M \geq 1$ be an integer, and assume $n \geq 8M$. (If $n < 8M$, the torus is covered by a single cube of the size considered below. The local hypothesis of Section 18 then gives the global hypothesis (B) with parameter 2ε (remark after Definition 18.1), and Part II applies. In this case we use the trivial partition: a single index $j = 0$, with any point z_0 of the torus as its centre, $\chi_0 = 1$ and $K_0 = \Lambda_0 = \mathbb{T}_\eta$ in place of the sets defined below. Lemmas 17.3–17.6 hold for it trivially, since every commutator with χ_0 vanishes.) Let $\check{M} = n/\lceil n/M \rceil \in [M/2, M]$, and let z_j run over the points of the grid $\check{M}\mathbb{Z}^d$ on the torus. When the partition is taken at the scale M of the polymer grid (Section 49.2), $M \mid n$, so $\check{M} = M$ and the z_j are the centres of its cubes.

Lemma 17.3 (Partition of unity)**.** *There are smooth functions χ_j on the torus with the following properties:*

(a) $\sum_j \chi_j^2 = 1$*;*

(b) χ_j *vanishes outside* $K_j = \{x : |x - z_j|_\infty \leq M\}$*;*

(c) $|\partial^\beta \chi_j| \leq C_\beta M^{-|\beta|}$ *for all lattice derivatives;*

(d) every point lies in at most 3^d *of the sets* K_j*.*

The constants are absolute.

Proof. Let ϕ be a smooth function on $\mathbb{R}$ with $\phi = 1$ on $[-\frac{1}{2}, \frac{1}{2}]$, $\phi = 0$ outside $(-1, 1)$ and $0 \leq \phi \leq 1$. Put $\phi_j(x) = \prod_\mu \phi((x_\mu - z_{j,\mu})/M)$, periodized. Then $\sum_k \phi_k^2 \geq 1$, since every point is within ∞-distance $\check{M}/2 \leq M/2$ of some z_k. Take $\chi_j = \phi_j / (\sum_k \phi_k^2)^{1/2}$. Property (c) follows from $\check{M} \geq M/2$. For (d), $n \geq 8M$ gives $\lceil n/M \rceil \leq n/M + 1 \leq 9n/(8M)$, so $\check{M} \geq 8M/9 > 2M/3$, and an interval of length $2M$ contains at most three points of $\check{M}\mathbb{Z}$. □

For each j let Λ_j be the union of the blocks that meet $\{x : |x - z_j|_\infty \leq 3M\}$. Thus K_j lies at ∞-distance at least $2M - 1$ from $\mathbb{T}_\eta \setminus \Lambda_j$.

Lemma 17.4 (IMS localization)**.** *Let $\mathfrak{A}$ be a real symmetric operator on $\ell^2_1(X)$ (or $\ell^2_0(X)$). Then*

$$\mathfrak{A} = \sum_j \chi_j \mathfrak{A} \chi_j + \frac{1}{2} \sum_j [\chi_j, [\chi_j, \mathfrak{A}]].$$

Proof. Expand $[\chi,[\chi,\mathfrak{A}]] = \chi^2\mathfrak{A} - 2\chi\mathfrak{A}\chi + \mathfrak{A}\chi^2$, sum over j, and use $\sum_j \chi_j^2 = 1$. □

Here χ_j acts on scalar fields by multiplication, on gauge fields by $(\chi_j A)(b) = \chi_j(b_-)A(b)$, and on 2-forms by $(\chi_j F)(p) = \chi_j(p_-)F(p)$, with p_- the base point of the plaquette p. The double commutators are small whenever $\mathfrak{A}$ is built from derivatives and operators with exponential decay.

Lemma 17.5 (Double commutators)**.** *Let $\mathfrak{A} = \sum_i X_i^T M_i Y_i$ be a finite sum of operators on $\ell^2_1(X)$. Here each X_i and each Y_i is 1, the derivative D from gauge fields to 2-forms, or the divergence D_X^T, and each M_i has decay (C_M,γ), as an operator between the corresponding field spaces. Then*

$$\Big|\Big\langle A, \sum_j [\chi_j,[\chi_j,\mathfrak{A}]]\,A\Big\rangle\Big| \le \frac{C\,C_M}{M^2}\,|\!|\!|A|\!|\!|_X^2,$$

with C depending only on γ and the number of terms.

Proof. Derivatives. Let $Z \in \{D, D_X^T\}$. The value of Zg at a point or a plaquette is a sum of transported values of g at distance at most 2η, divided by η. So $[\chi_j, Z]$ is a sum of multiplications by differences of χ_j over distances at most 2η, divided by η, composed with shifts and unitary transports. The coefficients of $[\chi_j,[\chi_j,Z]]$ are the squares of these differences, divided by η. By Lemma 17.3(c),(d),

$$\|[\chi_j,Z]\| \le CM^{-1}, \qquad \sum_j \|[\chi_j,Z]g\|^2 \le CM^{-2}\|g\|^2, \qquad \Big\|\sum_j [\chi_j,[\chi_j,Z]]\Big\| \le C\eta M^{-2}.$$

Decaying operators. Let K have decay (C_M,γ), fix blocks y, y', and write $\chi_j = \chi_j(c_y) + \psi_j$. Constants commute with K, so $\mathbf{1}_y[\chi_j,K]\mathbf{1}_{y'} = \mathbf{1}_y[\psi_j,K]\mathbf{1}_{y'}$ and $\mathbf{1}_y[\chi_j,[\chi_j,K]]\mathbf{1}_{y'} = \mathbf{1}_y[\psi_j,[\psi_j,K]]\mathbf{1}_{y'}$. Moreover $|\psi_j| \le CM^{-1}$ on $B(y)$ and $|\psi_j| \le CM^{-1}(1+|y-y'|)$ on $B(y')$, and only boundedly many j have $\chi_j \ne 0$ on $B(y)\cup B(y')$. Hence $\sum_j [\chi_j,[\chi_j,K]]$ has decay $(CC_M M^{-2}, \gamma/2)$, and its norm is at most $CC_M M^{-2}$ by Lemma 5.2(a). In the same way $\sum_j \|[\chi_j,K]g\|^2 \le CC_M^2 M^{-2}\|g\|^2$. Decay is a bound on blocks, so no bound on the individual entries of K is used. For operators between unit-lattice spaces, where decay is a kernel bound, the argument is the kernel bound $\sum_j(\chi_j(x)-\chi_j(x'))^2|K(x,x')| \le CM^{-2}|x-x'|^2|K(x,x')|$ together with the Schur test.

The form. Since $[\chi_j,\cdot\,]$ is a derivation,

$$\begin{aligned}[\chi_j,[\chi_j,X^TMY]] = {}& [\chi_j,[\chi_j,X^T]]MY + X^T[\chi_j,[\chi_j,M]]Y + X^TM[\chi_j,[\chi_j,Y]] \\ & + 2[\chi_j,X^T][\chi_j,M]Y + 2[\chi_j,X^T]M[\chi_j,Y] + 2X^T[\chi_j,M][\chi_j,Y].\end{aligned}$$

Pair with A, sum over j, and use the Cauchy–Schwarz inequality in the sums over j. Every term is bounded by $CC_M M^{-2}(\|A\| + \|XA\| + \|YA\|)^2$. For example, the fourth term gives $2\sum_j |\langle [\chi_j,X]A, [\chi_j,M]YA\rangle| \le 2\big(\sum_j \|[\chi_j,X]A\|^2\big)^{1/2}\big(\sum_j \|[\chi_j,M]YA\|^2\big)^{1/2} \le CC_M M^{-2}\|A\|\,\|YA\|$. Since $\|XA\|, \|YA\| \le |\!|\!|A|\!|\!|_X$ for $X,Y \in \{1, D, D_X^T\}$, the claim follows. No derivative of M_i enters. □

Lemma 17.6 (Localized norms)**.** *For every $A \in \ell^2_1(X)$,*

$$\sum_j \|\chi_j A\|^2 = \|A\|^2, \qquad \sum_j |\!|\!|\chi_j A|\!|\!|_X^2 \ge \tfrac{1}{2}|\!|\!|A|\!|\!|_X^2 - CM^{-2}\|A\|^2.$$

Proof. The first identity is Lemma 17.3(a). For the second, the Leibniz rule gives $D(\chi_j A) = \chi_j^\sharp DA + (\text{differences of } \chi_j)\cdot A$, where $\chi_j^\sharp$ denotes χ_j evaluated at shifted points. Hence $|D(\chi_j A)|^2 \ge \frac{1}{2}(\chi_j^\sharp)^2|DA|^2 - C|\partial\chi_j|^2|A|^2$. Summing over j and using $\sum_j(\chi_j^\sharp)^2 = 1$ (the partition identity at the shifted point) gives the bound for D. The term D_X^T is treated in the same way. □

17.3 An algebraic comparison formula

Lemma 17.7 (Comparison of inverses)**.** *Let $X' \subset X$ be domains, and let $\mathfrak{A}$ and $\mathfrak{A}'$ be invertible operators on $\ell^2(X)$ and $\ell^2(X')$. Define $\Delta = \mathfrak{A}\mathbf{1}_{X'} - \mathbf{1}_{X'}\mathfrak{A}'\mathbf{1}_{X'}$, as an operator from $\ell^2(X')$ to $\ell^2(X)$. Then*

$$\mathbf{1}_{X'}\mathfrak{A}^{-1}\mathbf{1}_{X'} - \mathfrak{A}'^{-1} = -\mathbf{1}_{X'}\mathfrak{A}^{-1}\Delta\,\mathfrak{A}'^{-1}.$$

Proof. For $f \in \ell^2(X')$ let $u = \mathfrak{A}'^{-1}f \in \ell^2(X')$. Then $\mathfrak{A}u = \mathbf{1}_{X'}\mathfrak{A}'u + \Delta u = f + \Delta u$, so $u = \mathfrak{A}^{-1}f + \mathfrak{A}^{-1}\Delta u$. Apply $\mathbf{1}_{X'}$. □

In applications, Δ is the sum of two kinds of terms. First, terms that couple X' across its boundary; for scalar second-order local operators these include "cut-bond" terms of the form $\mathbf{1}_{X\setminus X'}D^T\mathbf{1}_{\mathrm{cut}}D\mathbf{1}_{X'}$. On gauge-field spaces the complementary projection is $\mathbf{1}_X - \mathbf{1}_{X'}$, which need not equal $\mathbf{1}_{X\setminus X'}$. Second, differences of domain-dependent ingredients, including mass profiles and non-local terms, which are supported near the cut or exponentially small away from $X \setminus X'$. Derivative factors may occur at one or both ends of these terms. The cut-bond terms, however, contain one-sided derivatives at the cut, which are not bounded uniformly in η, so the sharp-cut form of the identity is not used for the bounds. The comparisons are proved with a smooth cutoff instead (Theorem 18.2(e)); this gives exponential smallness of the difference away from $X \setminus X'$, uniformly in η.

18 Regular backgrounds and scalar operators on domains

18.1 Local regularity

Fix $\alpha \in (0,1)$. Backgrounds in this part are real (G-valued); complex perturbations are treated at the end of Section 19.

Definition 18.1 (Regular backgrounds)**.** Let $M \geq 1$ be an integer and $\varepsilon \in (0,1]$. A real background U is (M,ε)*-regular* if the following holds for every $z \in \mathbb{T}_\eta$. On the cube $\hat{\Lambda} = \{x : |x - z|_\infty \leq 4M + 2\}$ there is a G-valued gauge transformation u such that $U^u = e^{i\eta A}$ on all bonds with both endpoints in $\hat{\Lambda}$, with

$$|A| \leq \varepsilon, \qquad |\partial A| \leq \varepsilon, \qquad [\partial A]_{\alpha,\Lambda_3(z')} \leq \varepsilon \quad \text{whenever } \Lambda_3(z') \subset \hat{\Lambda}.$$

Here $|\partial A|$ is the maximum of $|\partial_\nu A_\mu(x)|$ over the differences that involve only points of $\hat{\Lambda}$, and $[\partial A]_{\alpha,\Lambda_3(z')}$ is the supremum of $|\partial_\nu A_\mu(x) - \partial_\nu A_\mu(x')|/|x - x'|^\alpha$ over μ, ν and $x \neq x'$ such that both differences involve only points of $\Lambda_3(z')$.

A difference $\partial_\nu A_\mu(x)$ involves the points x, $x + \eta e_\mu$, $x + \eta e_\nu$ and $x + \eta e_\mu + \eta e_\nu$. Those that involve only points of $\hat{\Lambda}$ are exactly the differences determined by A on the bonds of $\hat{\Lambda}$. The same convention is used in Definition 28.1. It makes the definition invariant under the lattice symmetries, reflections included (proof of Theorem 25.1(h)). Compared with the quotients over all pairs of differences based in $\Lambda_3(z')$ that involve only points of $\hat{\Lambda}$, it changes the Hölder bound by at most a factor 2: if the points of two such differences do not lie in a common cube $\Lambda_3(z'') \subset \hat{\Lambda}$, then their base points are at distance greater than $6 - 2\eta \geq 4$, and their quotient is at most $2|\partial A|4^{-\alpha}$. So the two conventions are equivalent up to replacing ε by 2ε.

Every background satisfying (B) with parameter ε is (M,ε)-regular for every M. Regularity is a local condition: the gauges on different cubes are unrelated. In Bałaban's framework it is a consequence of gauge-invariant local conditions on the curvature. On cubes of size M one obtains gauges with bounds proportional to M [7, (3.35)–(3.36)], via the axial gauge and a change to a Landau-type gauge [6, Theorem 2]. In Parts II–V we take regularity as a hypothesis. Part VI shows that it follows from Bałaban's gauge-invariant conditions: the

background fields are regular (Theorem 30.3), and so is every real configuration with small curvature and current (Corollary 30.4).

Two consequences are used repeatedly. If U is $(1,\varepsilon)$-regular, then on every block and its neighbours the covariant derivative is, in a suitable gauge, $\partial + T$ with $\|T\| \le C\varepsilon$ (Lemma 12.2). The curvature operator $\mathfrak{F} = DD$ satisfies $|(\mathfrak{F}\lambda)_{\mu\nu}(x)| \le C\varepsilon|\lambda(x+\eta e_\mu+\eta e_\nu)|$; this bound is gauge covariant, and it holds by Lemma 12.3 in the local gauge. Hence $|\mathfrak{F}\lambda|_\infty \le C\varepsilon|\lambda|_\infty$ and $\|\mathfrak{F}\lambda\| \le C\varepsilon\|\lambda\|$.

Change of local gauge. Let u_1, u_2 be gauges in which (B) holds with parameter ε on a box Λ, with $U^{u_i} = e^{\mathrm{i}\eta A_i}$ there; for instance, Λ is the intersection of two cubes of Definition 18.1. Put $L = R(u_2u_1^{-1})$, so that $R(u_2)f = L\,R(u_1)f$ for every field f. For a bond $(x, x+\eta e_\mu)$ of Λ let $V_i(x) = R(e^{\mathrm{i}\eta A_{i,\mu}(x)})$. Since $U^{u_2} = (U^{u_1})^{u_2u_1^{-1}}$, we have $V_2(x)L(x+\eta e_\mu) = L(x)V_1(x)$, hence

$$\partial_\mu L(x) = V_2(x)^{-1}\Big(L(x)\,\frac{V_1(x)-1}{\eta} - \frac{V_2(x)-1}{\eta}\,L(x)\Big).$$

The matrices $L(x)$ and $V_i(x)$ are orthogonal. Since $|A_i|, |\partial A_i| \le \varepsilon$, the factors $(V_i-1)/\eta$ are bounded by $C\varepsilon$ and change by at most $C\eta\varepsilon$ between neighbouring points. So $|\partial L| \le C\varepsilon$. The second difference of L is bounded by $C\varepsilon$ wherever its complete stencil lies in Λ. Hence $|\partial L(x) - \partial L(x')| \le C\varepsilon|x-x'|$ on a common lifted box, provided the relevant coordinate paths and stencils lie in that lift, with distance measured there. For a field f put $\tilde f_i = R(u_i)f$. The product rule $\partial_\mu \tilde f_2(x) = \partial_\mu L(x)\,\tilde f_1(x+\eta e_\mu) + L(x)\,\partial_\mu \tilde f_1(x)$ gives, on every common lifted box $Q \subset \Lambda$ of side at most 8, with all relevant stencils and coordinate paths contained in that lift, sup norms over Q, and distances measured in the lift,

$$|\partial \tilde f_2| \le |\partial \tilde f_1| + C\varepsilon|f|, \qquad [\tilde f_2]_{\alpha,Q} \le [\tilde f_1]_{\alpha,Q} + C\varepsilon|f|,$$
$$[\partial \tilde f_2]_{\alpha,Q} \le [\partial \tilde f_1]_{\alpha,Q} + C\varepsilon(|f| + |\partial \tilde f_1|).$$

For gauge fields the same holds, with $L(b_-)$ in place of $L(x)$. So Hölder quotients of a field and of its first differences, computed in two local gauges, differ by at most $C\varepsilon$ times the sup norms of the field and of its first differences. For the applications below, fix a gauge on $\Lambda_4(c_y)$ and consider data on $\Lambda_3(c_y)$ and source pieces supported strictly inside $\Lambda_1(c_{y'})$. The required Hölder bounds transfer between these local gauges with a fixed larger constant. To prove this, use radius-one cubes on a half-unit grid. A pair at torus ∞-distance less than $\frac12$, together with its required stencils, fits in a common such cube; its coordinate paths stay inside the good-gauge cubes. If $n = 8$, $\Lambda_4(c_y)$ is the whole torus. If $n \ge 9$, every omitted coordinate interval has width at least one, so such a close pair cannot cross it. The other quotients are bounded by four times the supremum. For a source piece supported inside $\Lambda_1(c_{y'})$, pairs with a point outside $\Lambda_3(c_{y'})$ are controlled by this fixed support margin. No assertion on an arbitrary proper periodic box without such a collar is used.

18.2 Scalar operators on domains

For a domain X and a regular background U define, on $\ell^2_0(X)$ and on $\ell^2(X_1)$:

$$\begin{aligned}
&\mathfrak{A}'_X = \mathbf{1}_X(D^TD + aQ'(U)^TQ'(U))\mathbf{1}_X, && G'_X = (\mathfrak{A}'_X)^{-1}, && \Gamma'_X = (Q'G'_XQ'^T)^{-1},\\
&H'_X = G'_XQ'^T\Gamma'_X, && C'_X = G'_X - H'_XQ'G'_X, && M'_X = Q'(G'_X)^2Q'^T,\\
&P_{\mathcal V,X} = G'_XQ'^T(M'_X)^{-1}Q'G'_X, && R_X = \mathbf{1}_X - P_{\mathcal V,X}, && S_X = G'_XR_XD_X^T.
\end{aligned}$$

Here $Q' = Q'(U)$ maps $\ell^2_0(X)$ to $\ell^2(X_1)$, because the block averages of a field supported in X vanish on blocks outside X. All these operators depend only on the restriction of U to bonds in $\hat X$. They are covariant under gauge transformations on $\hat X$, as in Proposition 16.3. For $X = \mathbb{T}_\eta$ they are the operators of Section 14. Let $\mathcal N_X = \{\lambda \in \ell^2_0(X) : Q'\lambda = 0\}$ and $\mathcal W_X = D_X^TD\,\mathcal N_X$.

Theorem 18.2 (Scalar operators at regular backgrounds)**.** *There are $\varepsilon_0, c, C, \gamma > 0$, depending only on a, α and $\dim \mathfrak{g}$, such that the following holds for every $(1,\varepsilon)$-regular background with $\varepsilon \le \varepsilon_0$ and every domain X.*

(a) Coercivity: $\langle \lambda, \mathfrak{A}'_X \lambda\rangle \ge c\|\!|\lambda|\!\|_X^2$ *for* $\lambda \in \ell_0^2(X)$.

(b) Decay: G'_X, DG'_X, $G'_X D_X^T$ *and* $DG'_X D_X^T$ *have decay* (C,γ).

(c) Lower bounds: $Q' G'_X Q'^T \ge c$ *and* $M'_X \ge c^2$. *Moreover* Γ'_X *and* $(M'_X)^{-1}$ *have decay* (C,γ).

(d) The projection: R_X *is the orthogonal projection onto* $\mathcal{W}_X$. *For* $h \in \mathcal{W}_X$ *one has* $G'_X h \in \mathcal{N}_X$ *and* $D_X^T D G'_X h = h$; *equivalently* $Q' G'_X R_X = 0$ *and* $D_X^T D G'_X R_X = R_X$. *The operators* $P_{\mathcal{V},X}$, $DP_{\mathcal{V},X}$, $P_{\mathcal{V},X} D_X^T$, $DP_{\mathcal{V},X} D_X^T$, S_X *and* DS_X *have decay* (C,γ).

(e) Comparison of domains: *let* $X' \subset X$, *let* $\Sigma = \Sigma(X, X')$ *be the set of unit points of* X_1 *at distance at most* 2 *from* $X \setminus X'$ *(the* seam*), and let* F_X *be any of the operators in (b)–(d), with* $F_{X'}$ *its analogue on* X', *both extended by zero. Then for all unit points* y, y',

$$\|\mathbf{1}_y (F_X - F_{X'}) \mathbf{1}_{y'}\| \le C \sum_{w \in \Sigma} e^{-\gamma(|y-w|+|w-y'|)}.$$

The bound covers the blocks of the initial points of all bonds on which the gauge-field versions DG', $G'D^T$, $DG'D^T$, $DP_{\mathcal{V}}$, $P_{\mathcal{V}} D^T$, $DP_{\mathcal{V}} D^T$, S and DS act or take values, including the bonds that cross $\partial X'$, whose initial points may lie outside X'_1. In particular $\|\mathbf{1}_y (F_X - F_{X'}) \mathbf{1}_{y'}\| \le C' e^{-\gamma(\operatorname{dist}(y,\Sigma)+\operatorname{dist}(y',\Sigma))/2}$, and $\|\mathbf{1}_K (F_X - F_{X'})\| \le C' e^{-\gamma \operatorname{dist}(K,\Sigma)/2}$ for every set K of blocks, by a Schur bound.

(f) Locality in the background: *if U and U' are $(1,\varepsilon)$-regular and coincide on the bonds of $\hat{X}'$, $X' \subset X$, then $F_X(U) - F_X(U')$ satisfies the same bound.*

(g) Pointwise bounds ($X = \mathbb{T}_\eta$): *the bounds of Theorem 14.1(b), (d) and Proposition 14.2(e) hold. Here $|\partial \cdot|$ is replaced by $|D \cdot|$ in the sup norms, and the Hölder quotients of $\partial(\cdot)$ on $B(y)$ are computed in a gauge in which (B) holds near $B(y)$.*

Proof. (a) Fix a block $B = B(y) \subset X$ and a gauge u in which (B) holds on a neighbourhood of B. Put $\tilde{\lambda} = R(u)\lambda$, so $|\tilde{\lambda}| = |\lambda|$. On bonds with both endpoints in B, $|D_U \lambda| = |(\partial + T)\tilde{\lambda}|$ with $\|T\| \le C\varepsilon$, where T only involves values in B. By Lemma 3.1, $\|\tilde{\lambda} - (Q_0\tilde{\lambda})(y)\|_B^2 \le \frac{d}{2}\|\partial\tilde{\lambda}\|_{\mathrm{in}(B)}^2$, where $\mathrm{in}(B)$ denotes the bonds inside B. Moreover $R(U(\Gamma_{y,x}))\lambda(x) = R(u(y))^{-1} R(U^u(\Gamma_{y,x}))\tilde{\lambda}(x)$, and $|U^u(\Gamma_{y,x}) - 1| \le C\varepsilon$. Hence $|(Q'\lambda)(y)| \ge |(Q_0\tilde{\lambda})(y)| - C\varepsilon\|\lambda\|_B$. Combining,

$$\|\lambda\|_B^2 \le d\,\|\partial\tilde{\lambda}\|_{\mathrm{in}(B)}^2 + 2|(Q_0\tilde{\lambda})(y)|^2 \le 2d\,\|D\lambda\|_{\mathrm{in}(B)}^2 + 4|(Q'\lambda)(y)|^2 + C\varepsilon^2\|\lambda\|_B^2.$$

Summing over the blocks of X and absorbing the last term for $C\varepsilon^2 \le \frac{1}{2}$ gives $\|\lambda\|^2 \le C(\|D\lambda\|^2 + \|Q'\lambda\|^2)$. This implies (a), as in Corollary 3.4, since $\langle \lambda, \mathfrak{A}'_X \lambda\rangle = \|D\lambda\|^2 + a\|Q'\lambda\|^2$.

(b) Apply Lemma 5.7, with ∂ replaced by D. Its proof uses ∂ only through Lemma 5.4. For D the same computation gives $(D_\mu)_\alpha = D_\mu + Y$ with $\|Y\| \le e|\alpha|$, since the transports $R(U_\mu(x))$ are isometries. Conjugation preserves $\ell_0^2(X)$. Hypothesis (ii) follows as in Lemma 5.8, because $Q'^T Q'$ is block-diagonal and bounded.

(c) Let $g \in \ell^2(X_1)$ and $\psi(x) = \prod_\mu \cos^2(\pi x_\mu)$ for $x \in B(0)$. Then $m = \eta^d \sum_{x \in B(0)} \psi(x) = 2^{-d}$. Indeed, m is the product of the d one-dimensional sums $\eta \sum_{|j| \le (N-1)/2} \cos^2(\pi j/N)$. Since N is odd, j runs over a complete system of residues modulo N, and $\cos^2(\pi j/N)$ has period N in j. Since $N \ge 2$, $\sum_{j=0}^{N-1} \cos(2\pi j/N) = 0$. So each sum equals $\eta \sum_{j=0}^{N-1} \frac{1}{2}(1 + \cos(2\pi j/N)) = \frac{1}{2}$. Define $\varphi(x) = m^{-1}\psi(x-y)\, R(U(\Gamma_{y,x}))^{-1} g(y)$ for $x \in B(y)$, $y \in X_1$. Then $Q'\varphi = g$ and

$\langle Q'^T g, \varphi\rangle = \|g\|^2$. In the local gauge, φ is $m^{-1}\psi$ times a transport of $g(y)$ along $\Gamma_{y,x}$. The contours $\Gamma_{y,x}$ and $\Gamma_{y,x+\eta e_\mu}$ consist of at most dN pairs of parallel bonds at distance η, plus at most one further bond. So, by (B), the transports at neighbouring points differ by at most $C\eta\varepsilon$. At the points of $B(0)$ next to its faces, $|x_\mu| = \frac{1-\eta}{2}$ for some μ, and $\psi(x) \le \cos^2(\pi\frac{1-\eta}{2}) = \sin^2\frac{\pi\eta}{2} \le \frac{\pi^2}{4}\eta^2$. So across the faces of blocks φ is of order $\eta^2|g|$ on both sides. Hence $\|D\varphi\| \le C\|g\|$, and $\langle\varphi, \mathfrak{A}'_X\varphi\rangle \le C\|g\|^2$. Since $\mathfrak{A}'_X$ is positive, Cauchy–Schwarz in the inner product $\langle\cdot, \mathfrak{A}'_X\cdot\rangle$ gives

$$\langle Q'^T g, G'_X Q'^T g\rangle \ge \frac{\langle Q'^T g, \varphi\rangle^2}{\langle\varphi, \mathfrak{A}'_X\varphi\rangle} \ge \frac{\|g\|^2}{C}.$$

Then $\|G'_X Q'^T g\|\|g\| \ge \langle Q'^T g, G'_X Q'^T g\rangle$ and $\|Q'^T g\| = \|g\|$ give $M'_X \ge c^2$. Both operators have decay, by (b) and the locality of Q'. Lemma 5.5 gives the decay of the inverses.

(d) The proof of Proposition 14.2(b),(c) is purely algebraic. It uses $\mathfrak{A}'^T_X = \mathfrak{A}'_X$, the invertibility of $\mathfrak{A}'_X$ (for the injectivity of $D^T_X D$ on $\mathcal{N}_X$), and the surjectivity of Q' and $Q'G'_X$. All of these hold on X. The decay statements are compositions of (b) and (c).

(e) We use a smooth cutoff instead of the sharp cut, which would produce boundary terms. Let χ be a smooth function, at unit scale, of a mollified torus distance to $X \setminus X'$, with $|\partial^k\chi| \le C$ for $k \le 2$. It vanishes at distance at most 1 and equals 1 at distance at least 3. In particular it vanishes on $X \setminus X'$ and on the blocks of X' adjacent to $X \setminus X'$, and it is smooth also across ∂X. Thus $1-\chi$ and $\partial\chi$, restricted to X', and the one-mesh enlargements needed for their commutator stencils, are supported in blocks $B(y)$ whose unit labels satisfy $\mathrm{dist}(y, \Sigma) \le R_d$, where one may take $R_d = 6 + 3\sqrt{d}$. This is a distance between unit labels, and R_d depends only on the fixed dimension. This changes only the constant in the seam bound.

Let $f \in \ell^2_0(X')$ and $u = G'_X f$. The field χu lies in $\ell^2_0(X')$ and vanishes near the cut. On such fields $\mathfrak{A}'_X$ and $\mathfrak{A}'_{X'}$ agree: $D^T D(\chi u)$ vanishes at the points of $X \setminus X'$, the values outside X are projected out, and $Q'^T Q'$ is block-local. Hence $\mathfrak{A}'_{X'}(\chi u) = \chi f + [\mathfrak{A}'_X, \chi]u$, and

$$G'_X f - G'_{X'} f = (1-\chi)G'_X f - G'_{X'}(1-\chi)f + G'_{X'}[\mathfrak{A}'_X, \chi]G'_X f$$

as fields on $\mathbb{T}_\eta$. The commutator is $[D^T D, \chi] = D^T[D, \chi] + [D^T, \chi]D$, a sum of first-order operators with bounded coefficients supported where $\partial\chi \ne 0$, plus the block-local $a[Q'^T Q', \chi]$. Since $G'_{X'}$ contains the projection $\mathbf{1}_{X'}$, $G'_{X'}D^T$ acts as $G'_{X'}D^T_{X'}$. By the decay of G'_X, DG'_X and $G'_{X'}D^T_{X'}$ from (b), each term has a kernel bounded by $C\sum_{w\in\Sigma} e^{-\gamma(|y-w|+|w-y'|)}$, for all unit points y, y': the identity holds as an identity of torus fields, and the decay bounds of (b) hold at all blocks, also at those of the initial points of the bonds crossing $\partial X'$. If an input or output block has its label in $(X \setminus X')_1$, that label belongs to Σ. The individual decay bounds of both F_X and $F_{X'}$ then bound their difference by the corresponding seam summand. For gauge-field input or output, $F_{X'}$ need not vanish there: a bond crossing $\partial X'$ can have its initial point in such a block. For DG' apply D to the identity; $G'D^T$ follows by transposition. For $DG'D^T$ use the analogous identity with the cutoff on the input,

$$G'_X D^T_X - G'_{X'}D^T_{X'} = (1-\chi)G'_X D^T_X - G'_{X'}D^T_{X'}(1-\chi) - G'_{X'}[D^T, \chi] + G'_{X'}[\mathfrak{A}'_X, \chi]G'_X D^T_X,$$

in which every factor is bounded, and apply D to it. The cutoff is smooth because the sharp-cut identity contains the factor $G'_X\mathbf{1}_{X\setminus X'}D^T\mathbf{1}_{\mathrm{cut}}$, which is of order $\eta^{-1/2}$. That factor cannot be replaced by $G'_X D^T_X$: the replacement adds the term $G'_X\mathbf{1}_{X'}D^T\mathbf{1}_{\mathrm{cut}}DG'_{X'}$. For the composite operators in (c) and (d), write the difference of two products as a telescoping sum of products in which exactly one factor is replaced by its difference. Composing a kernel of the form $\sum_{w\in\Sigma} e^{-\gamma(|y-w|+|w-y'|)}$ with kernels that decay exponentially gives a kernel of the same form, with a smaller γ. For the unit-lattice inverses use Lemma 17.7 on $\ell^2(X_1)$.

(f) $F_{X'}(U) = F_{X'}(U')$, because these operators depend only on U on $\hat{X}'$. Write $F_X(U) - F_X(U') = [F_X(U) - F_{X'}(U)] - [F_X(U') - F_{X'}(U')]$ and apply (e) twice.

(g) In a gauge in which (B) holds on $\Lambda_4(c_y)$, the equation $\mathfrak{A}'u = f$ takes on $\Lambda_3(c_y)$ the form treated in Theorem 14.1(b). Lemma 18.3 below therefore applies, of type 1 for $G'f$ and of type 0 for $G'D_\mu^T f$ with bounded f, with the L^2 decay from (b). The perturbation is local and block-local, so its small inputs are bounded by the invariant norms $\bar{N}_z^i$ of that lemma. For bounded Hölder f split f with a partition of unity at unit scale, as in the proof of Theorem 14.1(b). Here the Hölder quotient of f on $\Lambda_3(c_{y'})$ is computed in a gauge in which (B) holds on $\Lambda_4(c_{y'})$. The piece $\chi_{y'}f$ is supported in $\Lambda_1(c_{y'})$. On $\Lambda_3(c_y)$ it is moved to the gauge of $\Lambda_4(c_y)$ by the change of local gauge after Definition 18.1, in the collared form stated at the end of that paragraph. This costs only a larger constant, so Lemma 18.3 applies to each piece. The statements for H', C', $P_{\mathcal{V}}$ and S follow from those for G' exactly as in Section 14. □

Lemma 18.3 (Bootstrap in local gauges)**.** *Let $i \in \{0,1\}$ and let U be $(1,\varepsilon)$-regular. For each $y \in \mathbb{T}$ let u_y be a real gauge in which (B) holds on $\Lambda_4(c_y)$, and $\tilde{u}_y = R(u_y)u$. Let ∇ be the covariant gradient: $\nabla v = Dv$ for scalar fields, and ∇v has the components $D_\nu v_\mu$ for gauge fields. Put $\bar{N}_z^0(v) = |v|_{\Lambda_1(c_z)}$ and $\bar{N}_z^1(v) = |v|_{\Lambda_1(c_z)} + |\nabla v|_{\Lambda_1(c_z)}$; these norms are invariant under real gauge transformations. Suppose that for each y the field $\tilde{u}_y$ satisfies on $\Lambda_2(c_y)$ an equation $-\Delta\tilde{u}_y = J_y + \delta_y\tilde{u}_y$, with $J_y = J_{F,y} + \sum_\mu \partial_\mu^* J_{g,y,\mu}$ and, for some y_0, $M > 0$, $\gamma_2 > 0$ and all y,*

$$\|u\|_{B(y)} + |J_{F,y}|_{\Lambda_2(c_y)} + |J_{g,y}|_{\Lambda_3(c_y)} + i\,[J_{g,y}]_{\alpha,\Lambda_3(c_y)} \le Me^{-\gamma_2|y-y_0|}.$$

Suppose moreover that for every field v there are fields $F_y[v]$ and $g_y[v]$ with $\delta_y R(u_y)v = F_y[v] + \sum_\mu \partial_\mu^* g_y[v]_\mu$ on $\Lambda_2(c_y)$ and

$$|F_y[v]|_{\Lambda_2(c_y)} + |g_y[v]|_{\Lambda_3(c_y)} + i\,[g_y[v]]_{\alpha,\Lambda_3(c_y)} \le \sum_z e^{-\gamma|y-z|}\big(\varepsilon\,\bar{N}_z^i(v) + C_\delta\|v\|_{B(z)}\big).$$

Let $\gamma' = \frac{1}{2}\min(\gamma,\gamma_2)$. If $\varepsilon \le \varepsilon_*$, then for $i = 1$

$$|u|_{B(y)} + |\nabla u|_{B(y)} + [\partial\tilde{u}_y]_{\alpha,B(y)} \le CMe^{-\gamma'|y-y_0|},$$

and for $i = 0$, $|u|_{B(y)} + [\tilde{u}_y]_{\alpha,B(y)} \le CMe^{-\gamma'|y-y_0|}$. For gauge fields the bound on $|\nabla u|$ contains that on $|Du|$.

In this lemma, as in Lemma 13.5, M is an amplitude, not the integer M of Definition 18.1; the background is $(1,\varepsilon)$-regular.

Proof. Put $\mathcal{N} = \max_y e^{\gamma'|y-y_0|}\bar{N}_y^i(u)$. The lattice is finite, so $\mathcal{N} < \infty$, and $\mathcal{N}$ is invariant under real gauge transformations. On the stencils based in $\Lambda_3(c_y)$, $R(u_y)$ is orthogonal, and by (B) on $\Lambda_4(c_y)$ the transports of U^{u_y} differ from 1 by at most $C\eta\varepsilon$; so there $|\tilde{u}_y| = |u|$, $|\partial\tilde{u}_y| \le |\nabla u| + C\varepsilon|u|$ and $|\nabla u| \le |\partial\tilde{u}_y| + C\varepsilon|u|$. Apply Theorem 6.3(d) for $i = 1$, and (b) for $i = 0$, to the equation on $\Lambda_2(c_y)$, with $F = J_{F,y} + F_y[u]$ and $g = J_{g,y} + g_y[u]$ on $\Lambda_3(c_y)$. The cube $\Lambda_2(c_y)$ meets at most C blocks. As in the proof of Lemma 13.5, the two hypotheses give

$$\mathcal{N} \le (1 + C\varepsilon)C_0M + (1 + C\varepsilon)^2 C_{\mathrm{reg}} S_{\gamma/2}\,\varepsilon\,\mathcal{N}.$$

So $\varepsilon_* = \min\{1, [2(1+C)^2 C_{\mathrm{reg}} S_{\gamma/2}]^{-1}\}$ suffices. It does not depend on M, C_δ or γ_2. The Hölder bound follows from the same local estimate. The small inputs of δ_y are the gauge-invariant norms $\bar{N}_z^i$; ordinary differences of $R(u_y)v$ at blocks far from y, where the gauge u_y is not controlled, do not enter. The Hölder hypotheses on J_y refer to the gauge u_y; data with Hölder bounds in another local gauge are moved to it by the change of local gauge after Definition 18.1. For $i = 0$ only values and block L^2 norms enter, and these are invariant under real gauge transformations; no Hölder bound on the divergence data is used. □

19 Gauge-field operators at regular backgrounds

19.1 Local potential terms and operators on domains

In this part we restrict the potential term to local operators, so that the Dirichlet versions on a domain X depend only on the background on the bonds of $\hat{X}$ (Definition 17.1).

Definition 19.1. $\mathcal{W}_{\mathrm{loc}}(\varepsilon)$ is the set of maps $U \mapsto W(U)$ with the following properties: $W(U)^T = W(U)$; $W(U)$ is real for real U, i.e. it maps real fields to real fields; $W(U^u) = R(u)W(U)R(u)^{-1}$; $(W(U)A)(b)$ depends only on the values of A, and of U, on bonds of plaquettes that contain b; and in every gauge in which (B) holds on a cube, the coefficients of W are bounded by ε and have Hölder quotients bounded by ε on that cube.

$\mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}(\varepsilon)$ is the set of $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$ that extend holomorphically to the complex perturbations U_δ of Section 19.6, for $\delta \le \delta_W$ with some $\delta_W > 0$, with $W(U_\delta) - W(U)$ a local operator of size $C\delta$ of the same form.

The conditions defining $\mathcal{W}_{\mathrm{loc}}(\varepsilon)$ are spatial; they do not imply analytic dependence on the background. They suffice for the real results of this section. The results on complex perturbations (Proposition 19.11) are stated for $W \in \mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}(\varepsilon)$. The only local potential used in the later parts is $W = 0$. The curvature term $\Delta'(U)$ of the Wilson action motivates these classes; we do not verify that it belongs to them (Section 12), and nothing below uses it. The potential W_2 and its Dirichlet versions W_2^X (Sections 37 and 41) are not local in this sense: $(W_2^X A)(b)$ depends on A only within distance 3 of b, but on U on all of $\hat{X}$, through ω_0^X. For them only existence, Re-coercivity and decay are taken from Proposition 19.11. Their comparison of domains is Proposition 35.1(d), which allows different potentials on the two domains, together with Proposition 37.4(d).

In a global gauge in which (B) holds, every $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$ belongs to $\mathcal{W}(C_\gamma\varepsilon, \gamma)$ for each fixed $\gamma > 0$. At a regular background the corresponding Hölder estimates hold in each local gauge in which (B) holds; the L^2 block-decay estimates hold in every gauge.

For a domain X let $\mathfrak{B}_X = \mathbf{1}_X(D^TD + W)\mathbf{1}_X$ and $\Pi_X = \mathbf{1}_X - DS_X$ on $\ell^2_1(X)$. Define

$$\mathfrak{A}^X = \Pi_X^T \mathfrak{B}_X \Pi_X + DR_X D_X^T + a\,\Pi_X^T \mathbf{Q}^T\mathbf{Q}\,\Pi_X \qquad \text{on } \ell^2_1(X).$$

Here $\mathbf{Q} = \mathbf{Q}(U)$, and all operators are restricted to $\ell^2_1(X)$. Let $J(X) = \{j : \Lambda_j \subset X\}$ and $m_X = \sum_{j \notin J(X)} \chi_j^2$ (as a multiplication operator on $\ell^2_1(X)$). For $\kappa \ge 0$ put

$$\mathfrak{A}^{X,\kappa} = \mathfrak{A}^X + \mathfrak{A}^{\mathrm{ext}}_X + \kappa\, m_X, \qquad \mathfrak{A}^{\mathrm{ext}}_X = D\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T \quad \text{on } \ell^2_1(X).$$

In Part IX, where κ denotes the weakening rate, this mass parameter is written κ_m. For $X = \mathbb{T}_\eta$ every j belongs to $J(\mathbb{T}_\eta)$, so $m_{\mathbb{T}_\eta} = 0$, and $\mathfrak{A}^{\mathbb{T}_\eta}$ is the operator $\mathfrak{A}_\pi(U)$ of Section 15. Its scalar ingredients are now those of Section 18. If $n < 8M$, the partition is the trivial one of Section 17.2; then $J(X)$ is empty and $m_X = 1$ for every proper domain X. For a proper domain the mass term is supported within distance about $8M$ of $\mathbb{T}_\eta \setminus X$. It plays the role of a boundary condition that is harmless for the gauge structure: it makes the local operators coercive near the boundary, where nothing else does.

Dirichlet conditions for all components. In $\mathfrak{A}^{\mathrm{ext}}_X$, D^T is the torus divergence of the zero extension, so $\langle A, \mathfrak{A}^{\mathrm{ext}}_X A\rangle = \|\mathbf{1}_{\mathbb{T}_\eta\setminus X} D^T A\|^2$. At a point $x \notin X$, $(D^TA)(x)$ involves only the bonds from x into X. So $\mathfrak{A}^{\mathrm{ext}}_X$ acts only on the bonds crossing ∂X, and $\mathfrak{A}^{\mathrm{ext}}_{\mathbb{T}_\eta} = 0$. With this term the divergence part of the form is the full $\|D^TA\|^2$. The local part of $\mathfrak{A}^{X,\kappa}$ is then the Dirichlet restriction $\mathbf{1}_X(D^TD + DD^T + W + a\mathbf{Q}^T\mathbf{Q})\mathbf{1}_X$ of the torus operator: all components of the gauge field satisfy Dirichlet conditions. Without $\mathfrak{A}^{\mathrm{ext}}_X$ the normal components on the bonds crossing ∂X would be free; these are relative boundary conditions, as for 1-forms in the

continuum. The local commutators of the walks of Section 41 would then not be bounded uniformly in η (Remark 41.5). On $\ell^2_1(X)$ we use, besides $\|\!|\cdot|\!\|_X$, the norm

$$\|\!|A|\!\|^2_{X,+} = \|\!|A|\!\|^2_X + \|\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T A\|^2 = \|A\|^2 + \|DA\|^2 + \|D^T A\|^2,$$

with the torus operators applied to the zero extension.

The operators $\mathfrak{A}^{X,\kappa}$ are transpose-symmetric, real for real U, depend only on U on $\hat{X}$ (the crossing bonds lie in $\hat{X}$), and are gauge covariant.

Lemma 19.2 (Identities on domains)**.** *For a regular background and a domain X, the statements of Lemmas 15.1 and 15.3 hold with Π, S, R, G', $\mathcal{N}(Q')$, $\mathfrak{A}_\pi$ replaced by Π_X, S_X, R_X, G'_X, $\mathcal{N}_X$, $\mathfrak{A}^X$, and with D^T replaced by D^T_X.*

Proof. The proofs of Lemmas 15.1 and 15.3 use only the identities $Q'G'R = 0$, $D^T D G' R = R$, $\mathbf{Q}D = D_V Q' + E$ and the definitions. The first two hold on X by Theorem 18.2(d). The third holds for all fields. □

The identities hold for $\mathfrak{A}^X$. For $\mathfrak{A}^{X,\kappa}$ on a proper domain, the analogue of Lemma 15.3(a) holds only up to the terms $\kappa m_X D\nu$ and $\mathfrak{A}^{\mathrm{ext}}_X D\nu$. These are supported within distance about $8M$ of $\mathbb{T}_\eta \setminus X$, respectively on the bonds crossing ∂X.

Lemma 19.3 (The full gradient on domains)**.** *Let U be real with curvature at most ε; (M,ε)-regular backgrounds have curvature at most $C\varepsilon$ (Lemma 12.3). Let ∇ be the covariant gradient, with components $D_\nu A_\mu$. Then for every domain X and every real $A \in \ell^2_1(X)$, extended by zero,*

$$\|\nabla A\|^2 \le \|\!|A|\!\|^2_{X,+} + C\varepsilon \|A\|^2.$$

For complex A the same holds, by splitting into real and imaginary parts. At the complex perturbations U_δ of Proposition 19.11, $\|\nabla_{U_\delta} A\|^2 \le (1 + C\delta)\|\!|A|\!\|^2_{X,+} + C(\varepsilon+\delta)\|A\|^2$, with the norm taken at U_δ.

Proof. The covariant Hodge inequality $\big|\|Dh\|^2 + \|D^T h\|^2 - \|\nabla h\|^2\big| \le C\varepsilon\|h\|^2$ holds for every gauge field h on the torus (Lemma 26.4). Its proof is a summation by parts and uses only the curvature bound. Apply it to the zero extension of A: its torus norms $\|DA\|$ and $\|D^T A\|$ are those in $\|\!|A|\!\|_{X,+}$. At U_δ the operators D, D^T and ∇ change by local operators of size $C\delta$. □

This is the gain from the Dirichlet conditions. With $\|\!|\cdot|\!\|_X$ alone, the full gradient is not controlled up to ∂X, because the exterior divergence is missing.

19.2 Coercivity from local information

Lemma 19.4 (Gårding inequality)**.** *There are ε_0, C_1 such that for $(1,\varepsilon)$-regular U with $\varepsilon \le \varepsilon_0$, every domain X and every real $B \in \ell^2_1(X)$,*

$$\langle B, \mathfrak{A}^X B\rangle \ge \tfrac{1}{2}\|\!|B|\!\|^2_X - C_1\|B\|^2.$$

Proof. For real B,

$$\langle B, \mathfrak{A}^X B\rangle = \|D\Pi_X B\|^2 + \langle \Pi_X B, W \Pi_X B\rangle + \|R_X D^T_X B\|^2 + a\|\mathbf{Q}\Pi_X B\|^2.$$

By Lemma 19.2, $D\Pi_X B = DB - \mathfrak{F} S_X B$, where $\|\mathfrak{F} S_X B\|$ is at most $C\varepsilon\|B\|$. Moreover $|\langle \Pi_X B, W\Pi_X B\rangle| \le C\varepsilon\|B\|^2$, because Π_X is bounded by Theorem 18.2(d). Finally, $\|R_X D^T_X B\| \ge \|D^T_X B\| - \|P_{\mathcal{V},X} D^T_X B\|$, and the last term is at most $C\|B\|$. Use $(s-t)^2 \ge \frac{1}{2}s^2 - t^2$. □

The Gårding inequality controls everything except $\|B\|$ itself. That control comes from the averaging and the gauge fixing, and this is where Part I enters.

Lemma 19.5 (Local coercivity)**.** *There is M_0 with the following property. Let $M \geq M_0$, and let U be (M,ε)-regular with $\varepsilon \leq \varepsilon_0$, where ε_0 does not depend on M. Let X be a domain and $j \in J(X)$. Then every real $B \in \ell^2_1(X)$ vanishing on bonds that do not start in K_j satisfies*

$$\langle B, \mathfrak{A}^X B\rangle \geq \tfrac{c_g}{2} \|\!| B |\!\|_X^2,$$

with c_g the constant of Theorem 9.6.

Proof. Step 1: from X to Λ_j. Compare $\mathfrak{A}^X$ with $\mathfrak{A}^{\Lambda_j}$ on such B. The local parts agree, since B and its derivatives are supported inside Λ_j. The quadratic forms differ through $\Pi_X - \Pi_{\Lambda_j} = -D(S_X - S_{\Lambda_j})$ and $R_X - R_{\Lambda_j} = P_{\mathcal{V},\Lambda_j} - P_{\mathcal{V},X}$. By Theorem 18.2(e), applied to DS and to $DP_{\mathcal{V}}D^T$, and since K_j has distance at least $2M-3$ from the seam $\Sigma(X,\Lambda_j)$, these differences, acting on fields supported in K_j, are bounded by $Ce^{-\gamma M}$. On blocks meeting $X \setminus \Lambda_j$, differentiated operators on Λ_j can have values on crossing bonds. Applied to fields supported in K_j, both these operators and the operators on X are of size $Ce^{-\gamma M}$ on such blocks, by their individual decay bounds. So

Details

$$|\langle B, (\mathfrak{A}^X - \mathfrak{A}^{\Lambda_j})B\rangle| \leq Ce^{-\gamma M} \|\!| B |\!\|_X^2.$$

Step 2: the local gauge. Let u be a gauge in which (B) holds on $\hat{\Lambda}_j$, and $\tilde{B} = R(u)B$. By covariance, $\langle B, \mathfrak{A}^{\Lambda_j}(U)B\rangle = \langle \tilde{B}, \mathfrak{A}^{\Lambda_j}(U^u)\tilde{B}\rangle$, where $\mathfrak{A}^{\Lambda_j}(U^u)$ carries the potential $W(U^u) = R(u)W(U)R(u)^{-1}$. Here and in Step 3, $\mathfrak{A}^{\Lambda_j}(1)$ is the operator at $U=1$ with the potential $W=0$. Since U^u satisfies (B) on $\hat{\Lambda}_j$, the proof of Theorem 15.4(a) applies on the domain Λ_j without change: all operators are Dirichlet operators on Λ_j, and their scalar ingredients are controlled by Theorem 18.2. That proof puts the potential into the perturbation: δ_π contains W, and $\mathcal{K}_1$ contains the terms with W. The potential enters only through $\langle \Pi_{\Lambda_j}\tilde{B}, W(U^u)\Pi_{\Lambda_j}\tilde{B}\rangle$; since $\|W(U^u)\| = \|W(U)\| \leq C\varepsilon$ and Π_{Λ_j} is bounded (Theorem 18.2(d)), this is at most $C\varepsilon\|\tilde{B}\|^2$. So the proof shows that $\mathfrak{A}^{\Lambda_j}(U^u) - \mathfrak{A}^{\Lambda_j}(1)$ is an admissible perturbation of size $C\varepsilon$, with C independent of M. Hence $|\langle \tilde{B}, (\mathfrak{A}^{\Lambda_j}(U^u) - \mathfrak{A}^{\Lambda_j}(1))\tilde{B}\rangle| \leq C\varepsilon \|\!| \tilde{B} |\!\|^2_{\Lambda_j,1}$. Here the subscript 1 indicates the norm at $U=1$. The admissible perturbation contains componentwise differences, so this bound needs $\|\partial\tilde{B}\|$. Since $\tilde{B}$ vanishes on the bonds that do not start in K_j, which has distance at least $2M-1$ from $\mathbb{T}_\eta \setminus \Lambda_j$, the exterior divergence of $\tilde{B}$ vanishes. So $\|\!| \tilde{B} |\!\|_{\Lambda_j,1}$ is the torus norm of $\tilde{B}$ at $U=1$, which controls $\|\partial\tilde{B}\|$ by Lemma 2.2.

Step 3: $U=1$ and $W=0$. At $U=1$ we have $d\Pi_{\Lambda_j}(1) = d$ and $\mathbf{Q}\Pi_{\Lambda_j}(1) = \mathbf{Q}$, as in Lemma 15.1(d), and $d^*_{\Lambda_j}\tilde{B} = d^*\tilde{B}$. The operator $\mathfrak{A}^{\Lambda_j}(1)$ has no potential term; the potential is part of the perturbation of Step 2. Therefore

$$\langle \tilde{B}, \mathfrak{A}^{\Lambda_j}(1)\tilde{B}\rangle - \langle \tilde{B}, \mathbf{\Delta}_a\tilde{B}\rangle = \langle \tilde{B}, d(P_{\mathcal{V}}(1) - P_{\mathcal{V},\Lambda_j}(1))d^*\tilde{B}\rangle,$$

where $\mathbf{\Delta}_a$ is the torus operator of Part I. By Theorem 18.2(e) with $X = \mathbb{T}_\eta$, $X' = \Lambda_j$, this is bounded by $Ce^{-\gamma M}\|\tilde{B}\|^2$. By Theorem 9.6 and Lemma 2.2, $\langle \tilde{B}, \mathbf{\Delta}_a\tilde{B}\rangle \geq c_g \|\!| \tilde{B} |\!\|^2_{\Lambda_j,1}$.

Conclusion. The norms $\|\!| \tilde{B} |\!\|_{\Lambda_j,1}$ and $\|\!| B |\!\|_X$ differ by a factor $1 + C\varepsilon$, because $D_{U^u} = \partial + T$ with $\|T\| \leq C\varepsilon$. Collecting the three steps,

$$\langle B, \mathfrak{A}^X B\rangle \geq (c_g - Ce^{-\gamma M} - C\varepsilon)(1 + C\varepsilon)^{-1} \|\!| B |\!\|_X^2.$$

The constants C and γ do not depend on M. Choose M_0 with $Ce^{-\gamma M_0} \leq \frac{c_g}{8}$, and ε_0, not larger than the thresholds of Theorems 18.2 and 15.4, with $C\varepsilon_0 \leq \min(\frac{c_g}{8}, \frac{1}{2})$. Then the right side is at least $\frac{c_g}{2}\|\!| B |\!\|_X^2$ for $M \geq M_0$ and $\varepsilon \leq \varepsilon_0$, and ε_0 does not depend on M. □

Theorem 19.6 (Coercivity at regular backgrounds)**.** *There are M_0, κ, $c_* > 0$ and $\varepsilon_0 > 0$, depending only on a, α, $\dim \mathfrak{g}$, such that the following holds for every $M \geq M_0$, every (M, ε)-regular background with $\varepsilon \leq \varepsilon_0$ and every domain X. In particular ε_0 does not depend on M:*

$$\langle A, \mathfrak{A}^{X,\kappa} A\rangle \geq c_* \|\!| A |\!\|^2_{X,+} \qquad \text{for all real } A \in \ell^2_1(X).$$

In particular, for $X = \mathbb{T}_\eta$ the operator $\mathfrak{A}_\pi(U)$ of Part II is coercive under the local hypothesis of regularity, without hypothesis (B).

Proof. Since $\langle A, \mathfrak{A}^{\mathrm{ext}}_X A\rangle = \|\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T A\|^2$, it suffices to show $\langle A, (\mathfrak{A}^X + \kappa m_X) A\rangle \geq c'_* \|\!| A |\!\|^2_X$, and to take $c_* = \min(c'_*, 1)$. Let $\kappa = C_1 + \frac{1}{2}$, with C_1 from Lemma 19.4. By Lemma 17.4 applied to $\mathfrak{A}^X$, and since $\langle A, m_X A\rangle = \sum_{j \notin J(X)} \|\chi_j A\|^2$,

$$\begin{aligned}\langle A, (\mathfrak{A}^X + \kappa m_X) A\rangle = &\sum_{j \in J(X)} \langle \chi_j A, \mathfrak{A}^X \chi_j A\rangle + \sum_{j \notin J(X)} \Big(\langle \chi_j A, \mathfrak{A}^X \chi_j A\rangle + \kappa \|\chi_j A\|^2 \Big) \\ &+ \frac{1}{2} \sum_j \langle A, [\chi_j, [\chi_j, \mathfrak{A}^X]] A\rangle.\end{aligned}$$

For $j \in J(X)$, Lemma 19.5 bounds the term below by $\frac{c_g}{2} \|\!| \chi_j A |\!\|^2_X$. For $j \notin J(X)$, Lemma 19.4 gives at least $\frac{1}{2} \|\!| \chi_j A |\!\|^2_X$. The operator $\mathfrak{A}^X$ is a finite sum of terms $X^T M Y$ as in Lemma 17.5. Indeed, $\Pi_X = \mathbf{1}_X - D S_X$ and $D\Pi_X = D - \mathfrak{F} S_X$ (Lemma 19.2) give

$$\Pi_X^T \mathfrak{B}_X \Pi_X = D^T D - D^T \mathfrak{F} S_X - (D^T \mathfrak{F} S_X)^T + (\mathfrak{F} S_X)^T \mathfrak{F} S_X + \Pi_X^T W \Pi_X,$$

and $D R_X D_X^T = D D_X^T - D P_{\mathcal{V},X} D_X^T$. The local terms $D^T D$ and $D D_X^T$ have $M = 1$. The terms $(\mathfrak{F} S_X)^T \mathfrak{F} S_X$, $\Pi_X^T W \Pi_X$, $D P_{\mathcal{V},X} D_X^T$ and $a \Pi_X^T \mathbf{Q}^T \mathbf{Q} \Pi_X$ have $X = Y = 1$ and an M with decay, by Theorem 18.2(d). The term $D^T \mathfrak{F} S_X$ has $X = D$, $M = \mathfrak{F} S_X$ and $Y = 1$, and its transpose has $X = 1$ and $Y = D$. No derivative of the coefficients of $\mathfrak{F}$ or W enters. Hence the double commutators are bounded by $C M^{-2} \|\!| A |\!\|^2_X$. With Lemma 17.6,

$$\langle A, (\mathfrak{A}^X + \kappa m_X) A\rangle \geq \tfrac{1}{2} \min(c_g, 1) \Big(\tfrac{1}{2} \|\!| A |\!\|^2_X - C M^{-2} \|A\|^2 \Big) - C M^{-2} \|\!| A |\!\|^2_X \geq c'_* \|\!| A |\!\|^2_X$$

for $M \geq M_0$.

If $n < 8M$, the partition is trivial (Section 17.2), and the double commutators vanish. For $X = \mathbb{T}_\eta$, the proof of Lemma 19.5 applies with $j = 0$ and $\Lambda_0 = \mathbb{T}_\eta$, in the global gauge: Step 1 and the error of Step 3 are absent. So $\langle A, \mathfrak{A}^{\mathbb{T}_\eta} A\rangle \geq \frac{c_g}{2} \|\!| A |\!\|^2_{\mathbb{T}_\eta}$. For a proper domain, $m_X = 1$, and Lemma 19.4 with $\kappa = C_1 + \frac{1}{2}$ gives $\langle A, (\mathfrak{A}^X + \kappa m_X) A\rangle \geq \frac{1}{2} \|\!| A |\!\|^2_X$. By the last display $c'_* \leq \frac{1}{4} \min(c_g, 1)$, so both bounds are at least $c'_* \|\!| A |\!\|^2_X$, and the theorem holds with the same κ and c_*. □

19.3 Green functions at regular backgrounds

Theorem 19.7 (Gauge-field Green functions at regular backgrounds)**.** *Let $M \geq M_0$ and let U be (M, ε)-regular with $\varepsilon \leq \varepsilon_0$.*

(a) For every domain X, $\mathbf{G}^X = (\mathfrak{A}^{X,\kappa})^{-1}$ exists, and the following operators have decay (C, γ), uniformly in X:

- *$\mathbf{G}^X$, $D\mathbf{G}^X$, $\mathbf{G}^X D^T$, $D\mathbf{G}^X D^T$, with D the derivative from gauge fields to 2-forms;*
- *$D^T \mathbf{G}^X$, $D^T \mathbf{G}^X D^T$, $\mathbf{G}^X D$, $D\mathbf{G}^X D$ and $D^T \mathbf{G}^X D$, with D^T on the left the torus divergence of the zero extension (which contains $D_X^T = \mathbf{1}_X D^T$), and D on the right the gradient of torus scalar fields followed by the projection onto $\ell^2_1(X)$; in $D^T \mathbf{G}^X D^T$ the right factor is the codifferential of 2-forms, and in $D\mathbf{G}^X D$ the left factor is the derivative into 2-forms;*

- $\nabla \mathbf{G}^X$, $\nabla \mathbf{G}^X D^T$ *and* $\nabla \mathbf{G}^X D$, *with* ∇ *the covariant gradient.*

(b) For $X = \mathbb{T}_\eta$, *the operator* $\mathbf{G}(U) = \mathfrak{A}_\pi(U)^{-1}$ *has the decay of (a) and satisfies Theorem 15.4(c), (d): the pointwise bounds and the identities. In the pointwise bounds* $|\partial \cdot|$ *is replaced by* $|\nabla \cdot|$, *with* ∇ *the covariant gradient of (a), and Hölder quotients are taken in local gauges. The comparison of Theorem 15.4(b) with the operator at* $U = 1$ *is not claimed: it needs a global gauge in which (B) holds. For a pure gauge* $U = 1^u$ *one has* $\mathbf{G}(U) = R(u)\mathbf{G}(1)R(u)^{-1}$, *which is not close to* $\mathbf{G}(1)$ *unless* $R(u)$ *is nearly constant.*

Proof. (a) Theorem 19.6 gives hypothesis (i) of Lemma 5.7, with the norm $\|\!| \cdot |\!\|_{X,+}$; the proof of that lemma goes through for this norm. For hypothesis (ii) note that conjugating $\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T$ by $e^{\alpha\rho}$ changes it by an operator of norm at most $C|\alpha|$. Since $|\!|\!| A |\!|\!|^2_{X,+} = \|A\|^2 + \|DA\|^2 + \|D^T A\|^2$, Combes–Thomas in this norm controls all blocks of $\mathcal{D}\mathbf{G}^X\mathcal{D}^T$ with $\mathcal{D} = (D, D^T)$. This gives the operators of the first two items; $\mathbf{G}^X D$ and $D\mathbf{G}^X D$ are transposes of $D^T\mathbf{G}^X$ and $D^T\mathbf{G}^X D^T$. For the last item apply Lemma 19.3 to φv, where $v = \mathbf{G}^X g$, $\mathbf{G}^X D^T g$ or $\mathbf{G}^X D g$ and φ is a cutoff at unit scale around a block. The commutators of D, D^T with φ are bounded, so $\|\nabla v\|_{B(y)}$ is bounded by $\|v\| + \|Dv\| + \|D^T v\|$ near $B(y)$, and these decay. Hypothesis (ii) follows from the structure of $\mathfrak{A}^X$ as a sum of terms $X^T M Y$ with decaying M, by Lemmas 5.3 and 5.8. The mass term commutes with $e^{\alpha\rho}$.

(b) Decay is (a) with $X = \mathbb{T}_\eta$. The identities follow from Lemma 19.2. For the pointwise bounds, write the equation $\mathfrak{A}_\pi u = J$ on $\Lambda_3(c_y)$ in a gauge in which (B) holds on $\Lambda_4(c_y)$. The local parts of $\mathfrak{A}_\pi$ have the normal form of Lemma 12.2. The non-local parts ($DP_\nu D^T$ and the terms involving S) were shown in the proof of Theorem 15.4(a) to be pointwise admissible. That proof used only the pointwise bounds of the scalar operators, which hold at regular backgrounds by Theorem 18.2(g). These scalar bounds hold on the cubes $\Lambda_1(z)$, in their local gauges (Remark 7.2). By the change of local gauge after Definition 18.1, they hold, with a larger constant, in the gauge of $\Lambda_4(c_y)$ on $\Lambda_3(c_y)$. The small terms of the non-local parts involve only values, Du and Hölder quotients in local gauges, so they are bounded by the invariant norms $|u| + |\nabla u|$ on the cubes $\Lambda_1(c_{y'})$, with weights $e^{-\gamma|y-y'|}$. Lemma 18.3 concludes. □

19.4 The averaged Green function, the minimizer and the fluctuation operator

At $U = 1$ the lower bound on $\mathbf{Q}\mathbf{G}\mathbf{Q}^*$ came from Fourier analysis. At regular backgrounds we use explicit trial fields instead.

Lemma 19.8 (Trial fields for the gauge average)**.** *Let* U *be* $(1, \varepsilon)$*-regular with* $\varepsilon \le \varepsilon_0$. *There is a linear map* $b \mapsto \Phi_b$ *from unit-lattice gauge fields to fine gauge fields such that* $\mathbf{Q}(U)\Phi_b = b$, $|\!|\!| \Phi_b |\!|\!|_{\mathbb{T}_\eta} \le C\|b\|$, *and* Φ_b *has exponential decay away from the support of* b. *Moreover there is* $\Phi_b^\pi \in \mathcal{S}(U)$ *with* $\mathbf{Q}_\pi \Phi_b^\pi = b$ *and* $|\!|\!| \Phi_b^\pi |\!|\!|_{\mathbb{T}_\eta} \le C\|b\|$.

Proof. Fix a unit bond (y, μ). Let $\phi(t) = \cos^2(\pi(t - \frac{1}{2}))$ for $t \in [0, 1]$ and $\phi(t) = 0$ otherwise. Let $\psi_\perp(z) = \prod_{\nu \neq \mu} \cos^2(\pi z_\nu)$ if $|z_\nu| \le \frac{1-\eta}{2}$ for all $\nu \neq \mu$, and $\psi_\perp(z) = 0$ otherwise. As in the proof of Theorem 18.2(c), $\psi_\perp \le \frac{\pi^2}{4}\eta^2$ at the points next to the transverse faces, i.e. where $|z_\nu| = \frac{1-\eta}{2}$ for some $\nu \neq \mu$, and

$$m_\perp := \eta^{d-1} \sum_{z:\, z_\mu = 0} \psi_\perp(z) = 2^{1-d}.$$

For a fine point x with $|x_\nu - y_\nu| \le \frac{1-\eta}{2}$ for $\nu \neq \mu$ and $0 \le x_\mu - y_\mu \le 1$, let x_0 be its projection onto the hyperplane $\{x_\mu = y_\mu\}$ through the centre of $B(y)$. Then $x_0 \in B(y)$, and $x = x_0 + j\eta e_\mu$

with $0 \le j \le N$. Let $\Gamma^{(\mu)}_{y,x} = \Gamma_{y,x_0} \cup [x_0, x]$. This canonical contour is single-valued in x. Define the μ-component

$$\Phi_{(y,\mu)}(x) = \psi_\perp(x_0 - y)\, \phi(x_\mu - y_\mu)\, R\big(U(\Gamma^{(\mu)}_{y,x})\big)^{-1},$$

acting on a vector $\beta \in \mathfrak{g}^c$, and $\Phi_{(y,\mu)}(x) = 0$ for all other x; all other components are 0. In $\mathbf{Q}(U)$ the value at x is transported along $\Gamma_{y,x'} \cup [x', x]$, for each base point $x' \in B(y)$ on the line of x with $0 \le x_\mu - x'_\mu < 1$. This contour and the canonical one have the same endpoints, lie in $B(y) \cup B(y + e_\mu)$, and have at most CN bonds. Since U is $(1, \varepsilon)$-regular, (B) holds in some gauge on a cube containing $B(y) \cup B(y + e_\mu)$. There every bond variable is $e^{\mathrm{i}\eta A}$ with $|A| \le \varepsilon$, so both transports are within $C\varepsilon$ of 1, and so is the product of the one with the inverse of the other. This statement is gauge covariant. Hence

$$\big(\mathbf{Q}(U)\Phi_{(y,\mu)}\beta\big)(y,\mu) = m_\perp \sigma_0 \beta + r_{(y,\mu)}\beta, \qquad \|r_{(y,\mu)}\| \le C\varepsilon,$$

with $\sigma_0 = \sum_t \eta\, w(t)\phi(t)$. Here t runs over $\eta\mathbb{Z}$, and $w(t)$ is the weight of the tube average of (y,μ) at the bonds $(z, z+\eta e_\mu)$ with $z_\mu - y_\mu = t$ (Lemma 2.6(a)). Explicitly, $w(t) = 1 - |t - \frac{1-\eta}{2}|$ for $|t - \frac{1-\eta}{2}| \le 1 - \eta$, and $w(t) = 0$ otherwise: a triangle with its peak at the bond that crosses the face between $B(y)$ and $B(y + e_\mu)$. With exact cancellation of the transports the error term would vanish; it does not, since the transports depend on the base point x' when the background is curved. The only other unit bonds where $\mathbf{Q}(U)\Phi_{(y,\mu)}\beta$ does not vanish are $(y \pm e_\mu, \mu)$. There the value has norm at most $m_\perp \sigma_\pm (1 + C\varepsilon)|\beta|$, with $\sigma_\pm = \sum_t \eta\, w(t \mp 1)\phi(t)$.

The margins. The tube weights of the bonds $(y + me_\mu, \mu)$, $m \in \mathbb{Z}$, add up to 1 at every bond in direction μ (Lemma 2.6(a)), and for $t \in [0,1]$ only $m \in \{-1, 0, 1\}$ contribute. Hence, with $\phi(i\eta) = \sin^2(\pi i/N)$,

$$\sigma_0 + \sigma_+ + \sigma_- = \sum_t \eta\, \phi(t) = \eta \sum_{i=0}^{N-1} \sin^2(\pi i/N) = \frac{1}{2}.$$

Next, put $h = \frac{N-1}{2}$, so that $w(i\eta) = 1 - |i - h|\eta$ for $0 \le i \le N$. Then

$$\sigma_0 = \frac{1}{2} - \eta^2 \sum_{i=0}^{N-1} |i - h| \sin^2 \frac{\pi i}{N}.$$

Write $i = h + m$, so that $\sin^2 \frac{\pi i}{N} = \cos^2 \frac{\pi(2m-1)}{2N}$, and pair m with $-m$. The sum becomes $\sum_{m=1}^{h} m(1 + \cos\frac{\pi}{N} \cos\frac{2\pi m}{N})$. Summing the arithmetico-geometric series gives $\sum_{m=1}^{h} m \cos\frac{2\pi m}{N} = -\big(8 \sin^2 \frac{\pi}{2N}\big)^{-1}$, and $\sum_{m=1}^{h} m = (N^2 - 1)/8$. Therefore

$$\sigma_0 = \frac{3}{8} + \frac{\eta^2}{8} + \frac{\eta^2 \cos(\pi\eta)}{8 \sin^2(\pi\eta/2)} \;\ge\; \frac{3}{8} + \frac{1}{4\pi^2} > \frac{2}{5},$$

using $\sin s \le s$ and $\cos(\pi\eta) \ge \frac{1}{2}$. So, for every odd $N \ge 3$,

$$\sigma_0 > \tfrac{2}{5}, \qquad \sigma_+ + \sigma_- < \tfrac{1}{10}, \qquad \sigma_0 - \sigma_+ - \sigma_- = 2\sigma_0 - \tfrac{1}{2} > \tfrac{3}{10}.$$

The continuum values are $\sigma_0 = \frac{3}{8} + \frac{1}{2\pi^2}$ and $\sigma_+ + \sigma_- = \frac{1}{8} - \frac{1}{2\pi^2}$, and the errors are $O(\eta^2)$.

Thus $\mathbf{Q}(U) \sum_{(y,\mu)} \Phi_{(y,\mu)}\beta_{(y,\mu)} = T_U \beta$. The diagonal of T_U is $m_\perp \sigma_0 + O(\varepsilon)$, and the sum of the off-diagonal entries of a row is at most $m_\perp(\sigma_+ + \sigma_-)(1 + C\varepsilon)$. Since $\sigma_0 - \sigma_+ - \sigma_- > \frac{3}{10}$, T_U is diagonally dominant uniformly in $\varepsilon \le \varepsilon_0$, for ε_0 small. So T_U^{-1} is bounded with exponentially decaying kernel, by the Neumann series around the diagonal. Put $\Phi_b = \sum \Phi_{(y,\mu)}(T_U^{-1} b)_{(y,\mu)}$.

The derivative bound follows as in the proof of Theorem 18.2(c). In a local gauge the canonical transports vary by $O(\eta\varepsilon)$ between neighbouring points. For a step in direction μ the contour grows by one bond. For a step ηe_ν, $\nu \ne \mu$, the two canonical contours agree up to

the segment in direction ν, where their lengths differ by one bond, and then run parallel at distance η. So, together with that bond, they bound a ladder of at most CN plaquettes, each with holonomy $O(\eta^2\varepsilon)$ in the gauge where (B) holds. The holonomy of the loop is a product of conjugates of these plaquette holonomies. Since U is real, $|g_1 \cdots g_m - 1| \le \sum_i |g_i - 1|$ and $|hgh^{-1} - 1| = |g - 1|$ for G-valued g_i, g, h. So the holonomy of the loop is $O(N\eta^2\varepsilon) = O(\eta\varepsilon)$. Moreover ϕ and $\psi_\perp$ are smooth on their supports, ϕ vanishes at $t = 0$ and $t = 1$, and $\psi_\perp$ is $O(\eta^2)$ next to the transverse faces. So $\Phi_{(y,\mu)}$ jumps by $O(\eta^2)$ across the boundary of its support. For the slice version, let $L : b \mapsto \Phi_b$. Then $\mathbf{Q}_\pi L = \mathbf{Q}\Pi L = I - ESL$, by Lemma 15.1(c), and $\|ESL\| \le C\varepsilon \le \frac{1}{2}$. Put $\Phi_b^\pi = \Pi L(I - ESL)^{-1}b$; then $\mathbf{Q}_\pi \Phi_b^\pi = b$, since $\mathbf{Q}_\pi \Pi = \mathbf{Q}_\pi$. Finally $\Pi = I - DS$ is bounded in $\|\!|\!| \cdot |\!|\!|$, because DS is bounded, $D(DS) = \mathfrak{F}S$, and $D^T DS = RD^T$ by Theorem 18.2(d). □

Theorem 19.9 (Minimizer and fluctuation operator at regular backgrounds)**.** *Let $M \ge M_0$ and let U be (M, ε)-regular with $\varepsilon \le \varepsilon_0$. Then $\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T \ge c$. Its inverse $\mathbf{\Gamma}_\pi$ has decay (C, γ). Theorem 16.1(b)–(d) hold for $\mathbf{H}(U)$ and $\mathbf{C}(U)$, with $|\partial \cdot|$ replaced by $|\nabla \cdot|$ (the covariant gradient of Theorem 19.7(a)) in the pointwise bounds and Hölder quotients taken in local gauges. The comparison of Theorem 16.1(a) with the operator $\mathbf{QGQ}^*$ at $U = 1$ is not claimed; the lower bound above takes its place.*

Proof. Since $\mathfrak{A}_\pi$ is positive (Theorem 19.6), the minimum of $\langle A, \mathfrak{A}_\pi A\rangle$ over $\{A : \mathbf{Q}_\pi A = b\}$ equals $\langle b, (\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T)^{-1} b\rangle$. With the trial field of Lemma 19.8 this is at most $\langle \Phi_b^\pi, \mathfrak{A}_\pi \Phi_b^\pi\rangle \le C\|\!|\!|\Phi_b^\pi|\!|\!|^2 \le C\|b\|^2$. Hence $(\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T)^{-1} \le C$. Decay of the inverse follows from Lemma 5.5. The rest of the proof of Theorem 16.1 is algebraic, or uses only the bounds now available: Theorems 19.7 and 18.2, and Lemma 18.3 in place of Lemma 13.5. □

19.5 Comparison of domains and locality in the background

Theorem 19.10 (Locality)**.** *Let $M \ge M_0$ and $\varepsilon \le \varepsilon_0$.*

(a) Let $X' \subset X$ be domains, U an (M, ε)-regular background, and Σ_M the set of unit points of X within distance $10M$ of $X \setminus X'$. For all unit points y, y', with the operators extended by zero,

$$\|\mathbf{1}_y (\mathbf{G}^X - \mathbf{G}^{X'}) \mathbf{1}_{y'}\| \le C \sum_{w \in \Sigma_M} e^{-\gamma(|y-w| + |w-y'|)}.$$

The same holds for $P\mathbf{G}^X Q - P\mathbf{G}^{X'} Q$ with $P \in \{1, D, D^T, \nabla\}$ and $Q \in \{1, D^T, D\}$, as in Theorem 19.7(a). The bound covers the blocks of the initial points of the bonds of $\ell_1^2(X')$ that cross $\partial X'$, which may lie outside X_1', and, for $P = D^T$, the torus divergence, the output points within distance η of X'.

(b) Let U and U' be (M, ε)-regular backgrounds that coincide on the bonds of $\hat{X}'$. Then each of the torus operators $\mathbf{G}$, $D\mathbf{G}$, $\mathbf{G}D^T$, $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$ satisfies

$$\begin{aligned}\|\mathbf{1}_y (F(U) - F(U')) \mathbf{1}_{y'}\| &\le C \sum_{w \in \Sigma_M} e^{-\gamma(|y-w| + |w-y'|)} \\ &\le C' e^{CM} e^{-\frac{\gamma}{2}(\mathrm{dist}(y, \mathbb{T}_\eta \setminus X') + \mathrm{dist}(y', \mathbb{T}_\eta \setminus X'))},\end{aligned}$$

where Σ_M is defined with $X = \mathbb{T}_\eta$.

Proof. (a) We use a smooth cutoff. A sharp cut would produce one-sided derivatives at the cut, which are not bounded uniformly in η. Let χ be a smooth function, at unit scale, of a mollified torus distance to $X \setminus X'$, with $|\partial^k \chi| \le C$ for $k \le 2$. It vanishes at distance at most 1 and equals 1 at distance at least 3. Gauge fields are multiplied by $(\chi A)(b) = \chi(b_-) A(b)$. Then $\chi(b_-) = 0$ for every bond b of $\ell_1^2(X)$ that is not in $\ell_1^2(X')$. So χ maps $\ell_1^2(X)$ into $\ell_1^2(X')$, and $P'\chi = \chi P$, with P, P' the projections onto $\ell_1^2(X)$, $\ell_1^2(X')$.

Write $\mathfrak{A}^{X,\kappa} = PLP + \mathfrak{N}^X + \kappa m_X$. Here $L = D^TD + DD^T + W + a\mathbf{Q}^T\mathbf{Q}$ is the torus operator: its local part includes $\mathfrak{A}_X^{\mathrm{ext}}$. And $\mathfrak{N}^X$ is the non-local part, built from $P_{\mathcal{V},X}$, S_X and R_X. The same holds on X' with the same L. Let $u = \mathbf{G}^X g$. Then $P'LP'(\chi u) = \chi PLu + P'[L,\chi]u$, so

$$\mathfrak{A}^{X',\kappa}(\chi u) = \chi g + P'[L,\chi]u + (\mathfrak{N}^{X'} - P'\mathfrak{N}^X P')\chi u + P'[\mathfrak{N}^X,\chi]u + \kappa(m_{X'} - m_X)\chi u.$$

Since $\chi u \in \ell^2_1(X')$, this gives, as torus fields,

$$\begin{aligned}\mathbf{G}^X g - \mathbf{G}^{X'} g = (1-\chi)u - \mathbf{G}^{X'}(1-\chi)g \\ + \mathbf{G}^{X'}\Big(P'[L,\chi] + (\mathfrak{N}^{X'} - P'\mathfrak{N}^X P')\chi + P'[\mathfrak{N}^X,\chi] + \kappa(m_{X'} - m_X)\chi\Big)u.\end{aligned}$$

Each term has the seam form:

- $1-\chi$ is supported within distance 3 of $X \setminus X'$.
- $[L,\chi]$ is a first-order operator with bounded coefficients, supported where $\partial\chi \neq 0$. There are no boundary terms on $\partial X \cap \partial X'$: both local parts are restrictions of the same torus operator, and the projections agree there. $\mathbf{G}^{X'}P'[L,\chi]u$ is bounded by $\mathbf{G}^{X'}D^T$, $\mathbf{G}^{X'}D$ and $\|\!|u|\!\|_{X,+}$ (Theorem 19.7(a)).
- $\mathfrak{N}^{X'} - P'\mathfrak{N}^X P'$ is a sum of products in which one scalar factor is replaced by its difference between X and X'. It is of seam form by Theorem 18.2(e), with the derivatives standing outside, where they are absorbed by $\mathbf{G}^{X'}$ and by u.
- $\mathfrak{N}^X$ is a finite sum of terms $X_i^T M_i Y_i$ as in Lemma 17.5, with $X_i, Y_i \in \{1, D, D_X^T\}$ and M_i with decay (proof of Theorem 19.6). Some terms carry a derivative at an end, for instance $-D^T\mathfrak{F}S_X$ and its transpose, so we do not bound the kernel of $[\mathfrak{N}^X,\chi]$ alone. We use
$$[X_i^T M_i Y_i, \chi] = [X_i^T,\chi]M_iY_i + X_i^T[M_i,\chi]Y_i + X_i^T M_i[Y_i,\chi].$$
The commutators of D and D^T with χ are bounded and supported where $\partial\chi \neq 0$. If M_i has decay (C,γ), then $\|\mathbf{1}_y[M_i,\chi]\mathbf{1}_{y'}\| \leq 2Ce^{-\gamma|y-y'|}$, and it vanishes unless y or y' lies within distance $3+\sqrt{d}$ of $X \setminus X'$; so $[M_i,\chi]$ has the seam form. The derivative X_i^T on the left is absorbed by $\mathbf{G}^{X'}$, since $\mathbf{G}^{X'}D^T$ and $\mathbf{G}^{X'}D$ decay (Theorem 19.7(a)). The derivative Y_i on the right falls on u, and $D\mathbf{G}^X$, $D^T\mathbf{G}^X$ decay. With P on the left and Q on the right, as below, the products $P\mathbf{G}^{X'}X_i^T$ and $Y_i\mathbf{G}^X Q$ also decay.
- $m_{X'} - m_X$ is supported within distance about $8M$ of $X \setminus X'$, inside Σ_M.

Composing with the decay of $\mathbf{G}^X$ and $\mathbf{G}^{X'}$ (Theorem 19.7(a)) gives the bound. The identity holds as an identity of torus fields, and these decay bounds hold at all blocks, so the bound holds for all unit points y, y', including the blocks of the initial points of the bonds crossing $\partial X'$. For the derivatives, apply $P \in \{D, D^T, \nabla\}$ to the identity, using Lemma 19.3 for ∇. For $Q \in \{D^T, D\}$ on the right, move the cutoff onto the input as in the proof of Theorem 18.2(e); this produces the commutators $[Q,\chi]$, which are bounded.

The cutoff is smooth, as in the proof of Theorem 18.2(e), because the cut terms of the sharp-cut identity of Lemma 17.7 contain one-sided derivatives at the cut. The term $\mathfrak{A}_X^{\mathrm{ext}}$ is what removes the boundary terms on $\partial X \cap \partial X'$: with the interior divergence alone, that is, with relative boundary conditions, the cutoff argument would leave boundary terms there.

(b) Apply (a) with $X = \mathbb{T}_\eta$ to U and to U', and note that $\mathbf{G}^{X',\kappa}(U) = \mathbf{G}^{X',\kappa}(U')$. For $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$ use the telescoping argument of Theorem 18.2(e), together with Lemma 17.7 on the unit lattice for $\mathbf{\Gamma}_\pi$. The factor e^{CM} comes from the width $10M$ of Σ_M. □

This is the property needed to localize the effective actions in the renormalization group. If the background is changed far away, the propagators deep inside change by an exponentially

small amount. The proof does not compare covariant derivatives of different backgrounds, so the η^{-1} problem discussed in Section 60 does not arise. It compares each background with a Dirichlet operator that does not see the change.

19.6 Complex perturbations

In the variational problem and in the renormalization group, the fields are expanded around a real regular background U, and the deviation is small in a global gauge relative to U. We therefore consider $U_\delta(b) = U(b)\, e^{i\eta\, \delta A(b)}$, where δA is $\mathfrak{g}^c$-valued with $|\delta A|, |\partial \delta A|, [\partial \delta A]_\alpha \le \delta$, the derivatives being taken in the local gauges of U.

Proposition 19.11. *Let $M \ge M_0$, $\varepsilon \le \varepsilon_0$, let U be (M,ε)-regular, and let $W \in \mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}(\varepsilon)$. There is $\delta_0 \in (0, \delta_W]$ such that for $\delta \le \delta_0$ all operators of Sections 18 and 19 exist at U_δ, depend analytically on δA, and satisfy the same bounds, with coercivity replaced by* Re*-coercivity. The* Re*-coercivity constants and the decay rates do not depend on $\delta \le \delta_0$: by Lemma 13.2 the operators at U_δ satisfy the hypotheses of Lemma 5.7 with the constants $c/2$, $C_1 + 1$ and $\min(\alpha_1, \gamma/2, 1)$, where c, C_1, α_1 are those at U and γ is the rate of the perturbation. Theorem 19.10 holds for pairs U_δ, $U'_{\delta'}$ that coincide on $\hat{X}'$. The L^2 statements (existence,* Re*-coercivity, decay, comparison of domains and locality) also hold, with $\hat{X}$ replaced by $X^{(r_W+1)}$, for potentials of range $r_W \le 3$ as in Section 35 that are real for real U, covariant, of norm at most ε, and extend holomorphically to the U_δ with $W(U_\delta) - W(U)$ of norm at most $C\delta$ and the same range. Here the range bounds the dependence on A and on U. The pointwise statements are not claimed for such potentials.*

Here δ_0 is a radius for the perturbations δA. It is not the spacing $\delta_j = L^j \eta$ of Section 21.1 at $j = 0$, nor the domain constant δ_0 of Definition 38.3.

Proof. $D_{U_\delta} = D_U + T_\delta$ with T_δ local and of size $C\delta$ in the sense of Lemma 12.2. Similarly, the averages change by local operators of size $C\delta$, and so do W, by the definition of $\mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}$, and $\mathfrak{A}^{\mathrm{ext}}_X$, in the form $\|\!| \cdot |\!\|_{X,+}$. Hence every operator at U_δ is the corresponding operator at U plus an admissible perturbation of size $C\delta$. Lemmas 13.2 and 13.3, with $\mathfrak{A}_0$ the operator at U, give existence and the bounds. For the gauge-field operators on domains, Lemma 5.7 is used in the norm $\|\!| \cdot |\!\|_{X,+}$, as in Theorem 19.7(a). The constants of the statement, and the rate $\gamma_0/4$ of the differences in Lemma 13.3, do not depend on $\delta \le \delta_0$. The lower bounds of Theorem 18.2(c) and Theorem 19.9 persist by perturbation. Analyticity follows as in Proposition 16.3. The bootstrap of Lemma 18.3 applies with ε replaced by $\varepsilon + C\delta$; we take $\varepsilon_0 \le \frac{1}{2}\varepsilon_*$ and $C\delta_0 \le \frac{1}{2}\varepsilon_*$, so that $\varepsilon + C\delta \le \varepsilon_*$. Locality follows as in Theorem 19.10, since the Dirichlet operators on X' at U_δ depend only on U_δ on $\hat{X}'$. For the potentials of range $r_W \le 3$, existence, Re-coercivity and decay use W only through its reality, its norm, its range in A, its covariance and the size of $W(U_\delta) - W(U)$; comparison of domains uses the same properties, and locality uses in addition its range in U, as in Section 35. □

Remark 19.12 (General averaging operators). The proofs of Parts II and III use the averaging pair $(Q'(U), \mathbf{Q}(U))$ only through the following properties, required in every local gauge. Property (v) is used only for the analytic statements.

(i) $Q'Q'^T = I$ for real U, and at real U the maps Q' and $\mathbf{Q}$ take real fields to real fields;

(ii) the closeness and locality bounds of Lemma 12.5(b);

(iii) a covariance identity $\mathbf{Q} D_U = D_V Q' + E$, with a real unit-lattice configuration V and a defect E satisfying the bounds of Lemma 12.5(c), including the bound on E^T;

(iv) gauge covariance as in Lemma 12.5(d), and dependence on U only on the bonds of $B(y)$, respectively $B(y) \cup B(y + e_\mu)$;

(v) the pair extends holomorphically to the complex backgrounds of the analytic statements (in Part II the complex A with (B), in Section 19.6 the U_δ), and (i)–(iii) hold there, uniformly, with the bilinear transpose in (i) and with V replaced by its holomorphic continuation in (iii).

The explicit transports along the contours $\Gamma_{y,x}$ enter only in two places: the trial fields of Theorem 18.2(c) and of Lemma 19.8. For a general pair use the scalar block profiles and the raw gauge tube profiles from those proofs, constructed in the local gauges. For the scalar profiles the trial-average map is $I + O(\varepsilon)$. For the gauge profiles, let L_0 be the raw tube map and let $T_{\text{model}} = \mathbf{Q}_{\text{model}} L_0$. By (ii), $T_{\text{general}} - T_{\text{model}}$ is a local operator of size $O(\varepsilon)$ in the block norm. The diagonal row and column margin of Lemma 19.8 persists, so its Neumann series gives the exact right inverse $L_0 T_{\text{general}}^{-1}$; then apply the slice correction from that lemma. The resulting exact gauge trial fields have exponential localization, rather than support in a single tube. Hence the real results of Parts II and III hold for every averaging pair with properties (i)–(iv), with constants depending only on upper bounds for the parameters in (i)–(iv), once these are below a fixed threshold. The analytic results (Proposition 16.3(a) and Proposition 19.11) hold if in addition (v) holds.

Properties (i)–(iv) alone do not give analyticity. Let G be non-abelian, and let $h_y(U) \in G$ be close to 1, covariant, depending only on U on the bonds of $B(y)$, and not analytic in U; for instance $h_y(U) = e^{\mathrm{i}\theta|F_y|F_y}$, with F_y the curvature of a fixed plaquette of $B(y)$ transported to y and $\theta > 0$ small. Put $L_y = R(h_y(U))$, let L_1 act by L_y at the initial point y of a unit bond, and replace $V(y,z)$ by $h_y V(y,z) h_z^{-1}$. The pair $(LQ', L_1\mathbf{Q})$ then has properties (i)–(iv), with the defect $L_1 E$, and the same operator $\mathfrak{A}'$, since L is orthogonal. Its minimizer map is $H'L^{-1}$, which is not analytic.

By Proposition 22.10, Bałaban's linearized averages $(Q'_k(U_0), \mathbf{Q}_k(U_0))$ have properties (i)–(iv) with $E = 0$ and $V = U_0^{(k)}$. Property (v) holds for them at the perturbations U_δ of Section 19.6, for δ small. Holomorphy, $Q'Q'^T = I$ and the closeness bounds are Proposition 46.4(iv)–(vi). The identity (iii) with $E = 0$ holds there by the identity theorem, as in step (iii) of the proof of Proposition 46.7, since for real δA the U_δ are $(M, \varepsilon + C\delta)$-regular. This needs $\varepsilon + C\delta \le \min(\varepsilon_0, c_3')$. Proposition 46.4 alone does not give $E = 0$ on $\{|A'|_\infty < \delta_{AB}\}$ (Remark 46.6).

The transfer argued in Remark 19.12 is checked use by use in Step 2 of the proof of Lemma 31.3, which lists every use of the averaging pair in the proofs of the results in its part (a), with the parameter of Definition 31.1 in place of the model bound $C\varepsilon$. That lemma is stated in Part VII, but it uses only Parts I–IV. The companion, chapter on Part VII, expands its proof.

20 Bałaban's random-walk expansion

The estimates of Sections 18 and 19 were obtained with Combes–Thomas, IMS localization and comparison of domains. In the renormalization group one also needs an *explicit* representation of the global operators as convergent sums of products of local propagators, each term depending on the background only in a bounded region. This is Bałaban's random-walk expansion [2, 4, 7]. We give it here for the scalar Green function G', with the local propagators of the previous sections. For the gauge-field Green function $\mathbf{G}$ the remainder is kept in factorized form, with the derivatives outside. That walk is constructed in Section 41; part (b) of Theorem 20.2 states what it gives.

20.1 The parametrix

Let $M \geq M_0$ and let U be (M,ε)-regular with $\varepsilon \leq \varepsilon_0$. For each j let Λ_j^w be the union of the blocks meeting $\{x : |x - z_j|_\infty \leq 9M\}$. If $|z_k - z_j|_\infty \leq 2M$, then $\Lambda_k \subset \Lambda_j^w$, so $k \in J(\Lambda_j^w)$. Since the sets K_k with $|z_k - z_j|_\infty \leq 2M$ cover K_j, the mass profile $m_{\Lambda_j^w}$ vanishes on K_j. Let $\mathfrak{A}$ be either the scalar operator $\mathfrak{A}'_{\mathbb{T}_\eta}$ or the gauge-field operator $\mathfrak{A}_\pi = \mathfrak{A}^{\mathbb{T}_\eta}$, and define the local propagators

$$G_j^w = (\mathfrak{A}'_{\Lambda_j^w})^{-1} \quad \text{(scalar case)}, \qquad G_j^w = (\mathfrak{A}^{\Lambda_j^w,\kappa})^{-1} \quad \text{(gauge case)}.$$

They depend on U only on $\hat\Lambda_j^w$ and have the bounds of Theorems 18.2(b) and 19.7(a). With the partition of unity $h_j = \chi_j^2$ ($\sum_j h_j = 1$) put

$$K = \sum_j h_j G_j^w, \qquad \mathcal{R}_j = -\Big([\mathfrak{A}, h_j] + h_j(\mathfrak{A}\mathbf{1}_{\Lambda_j^w} - \mathbf{1}_{\Lambda_j^w}\mathfrak{A}^{\Lambda_j^w}\mathbf{1}_{\Lambda_j^w})\Big)G_j^w, \qquad \mathcal{R} = \sum_j \mathcal{R}_j.$$

In the scalar case $\mathfrak{A}^{\Lambda_j^w}$ means $\mathfrak{A}'_{\Lambda_j^w}$.

Lemma 20.1. $\mathfrak{A}K = I - \mathcal{R}$.

Proof. $\mathfrak{A}h_jG_j^w = h_j\mathfrak{A}G_j^w + [\mathfrak{A}, h_j]G_j^w$. Since h_j is supported in $K_j \subset \Lambda_j^w$, $h_j m_{\Lambda_j^w} = 0$ and $h_j\mathfrak{A}^{\mathrm{ext}}_{\Lambda_j^w} = 0$, we have $h_j\mathfrak{A}^{\Lambda_j^w,\kappa}G_j^w = h_j$. Hence $h_j\mathfrak{A}G_j^w = h_j + h_j(\mathfrak{A}\mathbf{1}_{\Lambda_j^w} - \mathbf{1}_{\Lambda_j^w}\mathfrak{A}^{\Lambda_j^w}\mathbf{1}_{\Lambda_j^w})G_j^w$. Sum over j. □

Theorem 20.2 (Random-walk expansion)**.** *There is $M_1 \geq M_0$ such that for $M \geq M_1$ and $\varepsilon \leq \varepsilon_0$ the following holds; in (b), M_1 depends also on ρ.*

(a) The scalar Green function. *Let $\mathfrak{A} = \mathfrak{A}'_{\mathbb{T}_\eta}$, so that $\mathfrak{A}^{-1} = G'$ and $\mathcal{R}_j = -[\mathfrak{A}', h_j]G_j^w$, and let $\hat K_j = \{x : \mathrm{dist}_\infty(x, K_j) \leq 1\}$. The operator $\mathcal{R}_j$ vanishes unless it acts from Λ_j^w to $\hat K_j$, and $\mathcal{R}_j$ and $\mathcal{R}_jD^T$ have decay (CM^{-1}, γ). Hence $\|\mathcal{R}\| + \|\mathcal{R}D^T\| \leq C/M \leq \frac12$, and*

$$G' = K\sum_{n\geq 0}\mathcal{R}^n = \sum_\omega G_\omega, \qquad G_\omega = h_{j_0}G_{j_0}^w\mathcal{R}_{j_1}\cdots\mathcal{R}_{j_n},$$

where $\omega = (j_0, \dots, j_n)$ runs over all finite sequences of indices. The series converges in norm, also after applying D on the left and D^T on the right. The term G_ω vanishes unless $|z_{j_{i+1}} - z_{j_i}|_\infty \leq 10M+2$ for all i. It depends on U only on $\bigcup_i \hat\Lambda_{j_i}^w$, and $\|G_\omega\| \leq C(C/M)^n$. For fixed j_0 the sum of $\|G_\omega\|$ over all ω converges. The expansion reproduces the exponential decay of G', with a rate of order $M^{-1}\log M$.

(b) The gauge-field Green function. *Let $\rho \geq 15$, so that (41.1) holds with $w = 10M$, let Λ_j^v be the domains of Section 40 for this ρ, and let $n \geq C(\rho + C_v)M$, with C_v the margin of level (L3) of Section 41. Lemma 41.2 with $X = \mathbb{T}_\eta$ gives $\mathfrak{A}_\pi\mathcal{K}^v = I - \mathcal{R}^v$, where $\mathcal{K}^v = \sum_j h_j(\mathfrak{A}^{\Lambda_j^v,\kappa})^{-1}$ and $\mathcal{R}^v = D^T\mathcal{A} + \mathcal{B}$, with $\mathcal{A}$, $\mathcal{A}D^T$, $\mathcal{B}$ and $\mathcal{B}D^T$ of norm at most C/M (Lemma 41.3 at $s = 1$). Every D^T in front of $\mathcal{A}$ is absorbed by the factor on its left. The series $\mathcal{K}^v\sum_n(\mathcal{R}^v)^n$ converges in norm, also with D on the left and D^T on the right, and it equals $\mathbf{G} = \mathfrak{A}_\pi^{-1}$. This is Theorem 41.4(a) at level (L3), with $X = \mathbb{T}_\eta$ and $s = 1$. The scalar ingredients G', Γ' and $(M')^{-1}$ of $\mathfrak{A}_\pi$ are expanded at levels (L1) and (L2). The expansion reproduces the exponential decay of $\mathbf{G}$, with some rate $\gamma > 0$.*

Part (b) is Theorem 41.4(a) at level (L3), with $X = \mathbb{T}_\eta$ and $s = 1$. Its statement and its proof use the domains of Section 40 and the nested walks of Section 41, in Part IX; read them after Section 41. Part (a) does not depend on (b).

Proof. (a), the commutator. We have $[\mathfrak{A}', h_j] = D^T[D, h_j] + [D^T, h_j]D + a[Q'^T Q', h_j]$. The first two terms form a first-order operator whose coefficients are differences of h_j, bounded by CM^{-1} and supported where $\partial h_j \neq 0$, as in the proof of Theorem 18.2(e). The last term is block-local, with norm at most CM^{-1}, and it reaches every point of a unit block that meets K_j. So $[\mathfrak{A}', h_j]$ maps into fields supported in $\hat{K}_j$. By the decay of G_j^w, DG_j^w, $G_j^w D^T$ and $DG_j^w D^T$ (Theorem 18.2(b)), $\mathcal{R}_j$ and $\mathcal{R}_j D^T$ have decay (CM^{-1}, γ).

(a), the second term. It vanishes: $\mathfrak{A}'$ is local and h_j is supported at distance $\geq 8M - 1$ from $\mathbb{T}_\eta \setminus \Lambda_j^w$, so $h_j \mathfrak{A}' \mathbf{1}_{\Lambda_j^w} = h_j \mathbf{1}_{\Lambda_j^w} \mathfrak{A}' \mathbf{1}_{\Lambda_j^w}$.

(a), summation. Every point lies in at most 3^d sets K_j, so a unit block meets boundedly many sets $\hat{K}_j$. Hence $\mathcal{R}$ and $\mathcal{R}D^T$ have decay (CM^{-1}, γ), and Lemma 5.2(a) gives $\|\mathcal{R}\| + \|\mathcal{R}D^T\| \leq C/M$. Since $\mathfrak{A}'K = I - \mathcal{R}$ with $\|\mathcal{R}\| \leq \frac{1}{2}$, $K(I - \mathcal{R})^{-1}$ is a right inverse of $\mathfrak{A}'$. It equals G', which exists by coercivity. With derivatives, D falls on K and D^T on the last factor. The operators DK, KD^T and DKD^T are bounded, since $[D, h_j]$ is bounded, and $\mathcal{R}D^T$ is bounded. No derivative on the left of $\mathcal{R}$ is needed.

(a), localization. Since $\Lambda_j^w \subset \{x : |x - z_j|_\infty \leq 9M + 1\}$ and $\hat{K}_j \subset \{x : |x - z_j|_\infty \leq M + 1\}$, $G_{j_i}^w \mathcal{R}_{j_{i+1}}$ vanishes unless $|z_{j_{i+1}} - z_{j_i}|_\infty \leq 10M + 2$. The dependence on U is that of the local propagators. Since $\check{M} \geq 8M/9$ (proof of Lemma 17.3), each step has at most 28^d choices, so the sum of $\|G_\omega\| \leq C(C/M)^n$ over the ω with fixed j_0 converges for M large. Finally $\mathbf{1}_y G_\omega \mathbf{1}_{y'} = 0$ unless $n \geq |y - y'|/(CM) - 1$. Hence $\|\mathbf{1}_y G' \mathbf{1}_{y'}\| \leq C \sum_{n \geq |y-y'|/(CM)-1} (C/M)^n$, which decays like $e^{-c|y-y'|M^{-1} \log M}$. If the side of $\mathbb{T}$ is less than $8M$, the partition is trivial (Section 17.2): $\Lambda_0^w = \mathbb{T}_\eta$, $K = G'$ and $\mathcal{R} = 0$. Then (a) holds with the single term G', whose decay is Theorem 18.2(b).

(b). For the parametrix of Lemma 20.1 with $\mathfrak{A} = \mathfrak{A}_\pi$, the argument of (a) gives no bound on $\mathcal{R}_j$ uniform in η: $[\mathfrak{A}_\pi, h_j]$ contains $D^T \mathfrak{F} [S, h_j]$, and the coefficients of $\mathfrak{F}$ are only Hölder continuous. This is why the parametrix $\mathcal{K}^v$ of Lemma 41.2 and the factorized remainder are used. Section 41 is written in the conventions of Part VIII, with Bałaban's averaging pair and $E = 0$. This is used only in the formula for the non-local part $\mathfrak{N}^X$ at level (L3). For a pair with $E \neq 0$, $\mathbf{Q}\Pi_X = \mathbf{Q} - (D_V Q' + E)S_X = \mathbf{Q} - ES_X$, since $Q' S_X = Q' G'_X R_X D_X^T = 0$ (Theorem 18.2(d)). So $\mathfrak{N}^X$ contains in addition

$$-a(\mathbf{Q}^T E S_X + S_X^T E^T \mathbf{Q}) + a S_X^T E^T E S_X.$$

Since $S_X = G'_X R_X D_X^T$, these are factorized terms with $X_i^T = 1$, $Y_i \in \{1, D_X^T\}$ and middle factors $\mathbf{Q}^T E G'_X R_X$, $D R_X G'_X E^T \mathbf{Q}$ and $D R_X G'_X E^T E G'_X R_X$. They are bounded and compatible by (L1), (L2), Lemma 40.5(a) and the locality of E (Lemma 12.5(c)). So (W3) holds, and the rest of the proof of Theorem 41.4 is unchanged. The inputs of (L1)–(L3) are Theorems 18.2, 19.7(a), 19.10(a) and Lemma 19.3. Their thresholds and constants do not depend on $M \geq M_0$ (Theorem 19.6). Theorem 19.10(b), whose bound contains e^{CM}, is not used. So (b) holds for $\varepsilon \leq \varepsilon_0$, with ε_0 independent of M. This threshold is that of Part III for the model pair, whose averaging parameter is $C_{\mathrm{mod}}\varepsilon$ (Section 31). For Bałaban's pair one needs in addition $\varepsilon \leq \varepsilon_{\mathrm{B}}$ (Lemma 31.3(c)); by Lemma 31.2 it may be taken at $M = M_0$, so this threshold does not depend on M either (compare Section 41). □

Remark 20.3 (What the expansion adds). For the bounds of Parts I–VIII the random-walk expansion is not needed; in Part IX its weakened form also gives the bounds on the weakened inverses (Theorem 41.4, Corollary 41.6). Decay comes from Combes–Thomas, and locality in the background from Theorem 19.10. What it adds is a representation of G' and of $\mathbf{G}$ by terms that are local in the background, with explicit smallness: $\|G_\omega\| \leq C(C/M)^{|\omega|}$ in (a), and, in (b), $(C/M)^n$ for each of the 2^n regrouped products of bounded factors in $\mathcal{K}^v(\mathcal{R}^v)^n$. For $W \in \mathcal{W}_{\mathrm{loc}}^{\mathrm{an}}(\varepsilon)$ and averaging pairs with property (v) of Remark 19.12, the terms are analytic in

the complex perturbations of Proposition 19.11. Bałaban's cluster expansions of the fluctuation integrals and of the effective actions are built on this representation [7].

In the gauge case a fully local representation needs walk expansions of the scalar ingredients as well, including the averaged inverses Γ' and $(M')^{-1}$. They are constructed in Section 41, levels (L1)–(L3), with bounds on the sum of the norms of all terms. The averaged operators $\mathbf{\Gamma}_\pi$, $\mathbf{H}$ and $\mathbf{C}$ are expanded there at level (L4), in the setting of Part VIII.

The local propagators used here differ from Bałaban's. Bałaban works with Dirichlet operators and proves their positivity from local estimates in the axial gauge [4, Lemma 2.4]. We add a mass term near the boundary of the localization domain instead, and impose Dirichlet conditions on all components of the gauge field (Section 19). Both additions act only near the boundary: the mass term within distance about $8M$ of the complement of the domain, and the exterior divergence on the bonds that cross its boundary. So the local expressions of the two operators coincide away from the boundary. Their inverses differ in general, also away from the boundary. The parametrix uses the local expressions only on $\operatorname{supp} h_j$, where they agree with those of the operator being inverted (proofs of Lemmas 20.1 and 41.2), and it uses the local propagators only through their own properties (Lemma 20.1, hypothesis (W2) of Section 41), never through a comparison with Bałaban's.

Part IV: Symmetric averaging, averaged axial gauge and the Landau chart

Parts I–III treat linear operators. Part IV constructs the nonlinear input of the variational problem and of the renormalization group: Bałaban's symmetric four-dimensional averaging [18, (0.3)–(0.12)], the averaged axial gauge, the gauge-adapted averaging $\mathcal{Q}$, the expansion of the Wilson action, and the axial-to-Landau chart. Part V solves the variational problem in this chart, and the averaging defines the renormalization-group map of Part VII. From this Part on, constants may depend on L (conventions in Section 21).

What this Part proves.

- *The averaging* (Section 21). The blocks are centred at the coarse points, and the transport from a block centre to a point of the block is the group average (Lemma 21.5) of the transports along the coordinate-ordered contours, one for each order of the directions (Definitions 21.6 and 21.8). The averaging is covariant under the values of gauge transformations at the coarse points and under the lattice symmetries (Lemma 21.10), and it preserves small curvature (Proposition 21.12).

- *The bridge to the linear theory* (Section 22). The gauge-adapted averaging $\mathcal{Q}$ is related to the averaging by the correspondence theorem (Theorem 22.5), and $\mathcal{Q}_j(e^{\mathrm{i}\eta A}) = e^{\mathrm{i}\delta_j B_j}$ with $B_j = \mathbf{Q}_j A + C_j^{\mathrm{av}}(A)$ (Proposition 22.9). The linearized pair $(Q_k'(U_0), \mathbf{Q}_k(U_0))$ is exactly covariant ($E = 0$), local, close to the averages of Part I, and covariant under $\mathfrak{S}_k$ (Proposition 22.10). So the real results of Parts II and III apply to it (Corollary 22.11).

- *The action* (Section 23). Near a regular U_0, $S(e^{\mathrm{i}\eta A}U_0) = S(U_0) + \langle A, J\rangle + \frac{1}{2}\|D_{U_0}A\|^2 + \mathcal{N}(A)$, with the current $J = J_{U_0}$ and $|\nabla\mathcal{N}(A)|_\infty \le C(\varepsilon + |A|_\infty)|A|_{C^1}$ (Proposition 23.3).

- *The chart* (Section 24). Let U_0 be (M,ε)-regular with current at most ε, where ε is small. Every real configuration with small curvature and current whose averages are close to those of U_0 has exactly one small representative $e^{\mathrm{i}\eta A}$ in Bałaban's Landau gauge relative to U_0; A, its covariant gradient, its Hodge Laplacian and the Hölder quotients of

its gradient are bounded by ε, the curvature and the current of the configuration, and the difference of the averages (Theorem 24.2). So the constraint set $\{U : U^{(k)} = V\}$ is parametrized by Lie fields in the Landau gauge (Corollary 24.3).

The main theorem is Theorem H, which is Theorem 24.2: Bałaban's axial-to-Landau theorem [6, Theorem 2] on a single torus.

Imports. Lemma 2.6 and Theorem 6.3 of Part I, the perturbation lemmas and Lemmas 12.2–12.5 of Part II, and the regular-background theorems of Part III with Remark 19.12. Three passages point ahead: the constants of Lemma 24.1 are made uniform by Lemma 31.3, whose proof uses only Parts I–IV, Definition 31.1 and Lemma 31.2; the clause of Corollary 22.11 on complex perturbations uses Proposition 46.4; and one step of the proof of Proposition 22.10(f) is written out in the proof of Lemma 54.1(a).

Exports. Propositions 22.9, 22.10 and 23.3 and Corollary 22.11 are used throughout Parts V–XI. Theorem 24.2 and Corollary 24.3 go to the variational problem (Sections 25–27), the induction (Section 30), the step quantities (Section 47) and the Landau chart of Part XI (Sections 54 and 55). The averaged relative transports $\mathfrak{G}_W$ (Definition 22.6, Lemma 22.7) are the linearized averaged contour variables of the gauge fixing in Part VII. The lattice symmetry of the averaging enters Theorems K(iv), L(i) and M(ii). All of Part IV is on the main line, except Remark 22.12, a numerical check that no proof uses.

Details for this Part. The companion has no chapter on Part IV; its parameter ledger lists the constants of Part IV that Parts VII–XI use.

21 Configurations and the nonlinear averaging

In Parts I–III the averaging was linear. Bałaban's renormalization group uses a nonlinear block averaging of lattice gauge fields. It is defined one scale at a time, with blocks of L^d points, and the k-step averaging is the composition of k one-step averagings. We use the four-dimensional version [18, (0.3)–(0.12)], which is symmetric under the translations and the hyperoctahedral group of the coarse lattice. Its blocks are centred at the coarse points, and its contour variables are group averages over families of contours, one contour for each order of the coordinate directions. The earlier averaging of [5], with blocks that have the coarse point at a corner and a single ordered contour, is not symmetric [18, p. 252]. This part develops the nonlinear input of the variational problem:

- the group average, the averaging and the regularity of averaged configurations (this section);
- the averaged axial gauge and the *gauge-adapted averaging*, whose linearization is the averaging operator to which Parts II and III apply (Section 22);
- the expansion of the Wilson action (Section 23);
- the axial-to-Landau chart: every real configuration with small curvature and prescribed averages has a representative in Bałaban's Landau gauge relative to a regular reference configuration, with bounds (Section 24).

Part V uses these results to solve the variational problem relative to a regular reference configuration.

Conventions for Parts IV and V. Recall that $d = 4$ and that $L \geq 3$ is odd (Section 2); we keep the letter d in formulas. From now on constants may depend on L, in contrast to Parts I–III. Unless stated otherwise, $C, c > 0$ depend only on d, L, the group $G \subset U(\mathsf{N})$, including the radius c_G of its logarithm chart (defined below), the averaging parameter a of Part I, the Hölder exponent α, and the integer $M \geq M_0$ of Part III, which we fix once and for all; in Part VI we also assume $M \geq 5L$. The letter a also denotes curvature bounds, as in the classes $\mathfrak{U}_j(a)$ of Definition 21.4; there it is unrelated to the averaging parameter. The radius r_M of the group average (Lemma 21.5) depends only on G. The smallness constants c_1, c_2, c_3 of Parts IV and V satisfy $c_3 \leq c_2 \leq c_1 \leq r_M/C$, with C depending only on G; this makes all averaged contour variables used in Parts IV–VI defined. The constants c_1 and c_2 do not depend on L, by the proofs of Lemmas 21.11 and 22.8. The constant c_3 may depend on L, through the kernel bounds of Propositions 22.9(c) and 22.10(e). All Lie algebra valued fields are measured in the units of the unit lattice $\mathbb{T}$. A configuration on a lattice of spacing δ is written $X = e^{\mathrm{i}\delta B}$, and B is called its Lie field.

21.1 Scales, blocks and contour families

For $j = 0, \dots, k$ let $\delta_j = L^j \eta$ and $\mathbb{T}^{(j)} = (\delta_j \mathbb{Z}/n\mathbb{Z})^d$. Thus $\mathbb{T}^{(0)} = \mathbb{T}_\eta$, $\mathbb{T}^{(k)} = \mathbb{T}$, and $\mathbb{T}^{(j)} \subset \mathbb{T}^{(j-1)}$. For $j \geq 1$ and $y \in \mathbb{T}^{(j)}$ the *block*

$$B_j(y) = \Big\{ x \in \mathbb{T}^{(j-1)} : \ |x_\mu - y_\mu| \leq \tfrac{L-1}{2}\, \delta_{j-1}, \ \mu = 1, \dots, d \Big\}$$

is the cube of L^d points of $\mathbb{T}^{(j-1)}$ centred at y [18, (0.3)]. Every integer m has exactly one representation $m = Lq + r$ with $q \in \mathbb{Z}$ and $|r| \leq \frac{L-1}{2}$. Hence the blocks $B_j(y)$, $y \in \mathbb{T}^{(j)}$, partition $\mathbb{T}^{(j-1)}$.

The points of $\mathbb{T}_\eta$ in the scale-j block of y form $B^{(j)}(y) = \bigcup \{ B_1(x_1) : x_1 \in B_2(x_2), \dots, x_{j-1} \in B_j(y) \}$. Every integer m with $|m| \leq \frac{L^j - 1}{2}$ has exactly one representation $m = \sum_{i<j} m_i L^i$ with *balanced digits* $|m_i| \leq \frac{L-1}{2}$: the L^j such sums are distinct, by reduction modulo L, and they lie in an interval of L^j integers. Applied in each coordinate to $(x_\mu - y_\mu)/\eta$, this shows

$$B^{(j)}(y) = \Big\{ x \in \mathbb{T}_\eta : \ |x_\mu - y_\mu| \leq \tfrac{\delta_j - \eta}{2}, \ \mu = 1, \dots, d \Big\},$$

and that every $x \in B^{(j)}(y)$ lies in exactly one chain $x = x_0 \in B_1(x_1)$, $x_1 \in B_2(x_2)$, $\dots$, $x_{j-1} \in B_j(y)$. So the centred blocks of all scales are nested: $B^{(j)}(y)$ is the disjoint union of the $B^{(j-1)}(x)$, $x \in B_j(y)$. In particular $B^{(k)}(y) = B(y)$ is the block of Part I.

Definition 21.1 (Contour families [18, p. 252])**.** Let $j \geq 1$, $y \in \mathbb{T}^{(j)}$ and $x \in B_j(y)$, and write $x - y = \delta_{j-1} \sum_\mu n_\mu e_\mu$ with $|n_\mu| \leq \frac{L-1}{2}$. Let $S = \{\mu : n_\mu \neq 0\}$ and $s = |S|$. For an order $\pi = (\pi(1), \dots, \pi(s))$ of S let Γ_π be the contour on $\mathbb{T}^{(j-1)}$ that starts at y, takes $|n_{\pi(1)}|$ steps in the direction $\operatorname{sgn}(n_{\pi(1)}) e_{\pi(1)}$, then $|n_{\pi(2)}|$ steps in the direction $\operatorname{sgn}(n_{\pi(2)}) e_{\pi(2)}$, and so on. The *contour family* of (y, x) is

$$G(y, x) = \{ \Gamma_\pi : \ \pi \text{ an order of } S \}.$$

For $x = y$ it consists of the empty contour.

The family $G(y, x)$ has $s! \leq d! = 24$ elements. Each of them has $\sum_\mu |n_\mu| \leq 2(L-1)$ bonds and lies in the box spanned by y and x, i.e. in $\{z : z_\mu \text{ between } y_\mu \text{ and } x_\mu\} \subset B_j(y)$. If $s \leq 1$, $G(y, x)$ consists of the straight contour from y to x. The family is not, in general, the family of all shortest contours from y to x.

Lattice symmetries. For $0 \le j \le k$ let $\mathfrak{S}_j$ be the group of the maps $r(x) = Rx + v$ of $\mathbb{T}_\eta$ with $R \in B_4$ and $v \in \mathbb{T}^{(j)}$; for $j = k$ this is the group $\mathfrak{S}_k$ of Section 2, and $\mathfrak{S}_k \subset \mathfrak{S}_j$. Every $r \in \mathfrak{S}_j$ maps $\mathbb{T}^{(i)}$ onto itself and $B_i(y)$ onto $B_i(ry)$, for $i \le j$ and $y \in \mathbb{T}^{(i)}$, because the blocks are cubes centred at the points of $\mathbb{T}^{(i)}$. It maps coordinate-ordered contours to coordinate-ordered contours, with the order of the directions permuted. Since $G(y, x)$ contains all orders,

$$rG(y,x) = G(ry, rx), \qquad r \in \mathfrak{S}_j,\ y \in \mathbb{T}^{(i)},\ x \in B_i(y),\ 1 \le i \le j.$$

Likewise $G(y+v, x+v) = G(y,x) + v$ for $v \in \mathbb{T}^{(i)}$. For configurations and functions on $\mathbb{T}^{(i)}$ we put $(rY)(b) = Y(r^{-1}b)$ and $(rg)(x) = g(r^{-1}x)$. Here $r^{-1}b = (r^{-1}b_-, r^{-1}b_+)$ may be negatively oriented, and $Y(-b) = Y(b)^{-1}$. Then $(rY)(\Gamma) = Y(r^{-1}\Gamma)$ and $(rY)^{rg} = r(Y^g)$.

21.2 Configurations, curvature and lattice Stokes

A *configuration* on $\mathbb{T}^{(j)}$ assigns $Y(b) \in G^c$ to the oriented bonds, with $Y(-b) = Y(b)^{-1}$. It is *real* if it is G-valued. Products along contours are written $Y(\Gamma)$. A gauge transformation $g : \mathbb{T}^{(j)} \to G^c$ acts by $Y^g(b) = g(b_-)Y(b)g(b_+)^{-1}$, so that $Y^g(\Gamma) = g(\Gamma_-)Y(\Gamma)g(\Gamma_+)^{-1}$.

For a matrix h with $|h-1| < 1$ let $\ell(h) = -\mathrm{i}\log h$ be the principal logarithm, defined by the power series, so that $e^{\mathrm{i}\ell(h)} = h$. Norms of matrices are Hilbert–Schmidt norms, as in Part II, and $\|u\|$ denotes the operator norm. For $u \in G^c$ put $\kappa(u) = \|u\|\|u^{-1}\|$; then $|uZu^{-1}| \le \kappa(u)|Z|$, and $\kappa(u) = 1$ for $u \in G$. We fix $c_G \in (0, \frac14]$ so small that ℓ maps $\{g \in G : |g-1| \le c_G\}$ into $\mathfrak{g}$, and $\{g \in G^c : |g-1| \le c_G\}$ into $\mathfrak{g}^c$; all group elements below satisfy such a bound. The radius c_G depends on the group $G \subset U(\mathsf{N})$, not only on N. For $m \ge 7$ the circle $G_m = \{\mathrm{diag}(e^{\mathrm{i}t}, e^{\mathrm{i}mt}) : t \in \mathbb{R}\} \subset U(2)$ contains $g = \mathrm{diag}(e^{2\pi\mathrm{i}/m}, 1)$, with $|g-1| = 2\sin(\pi/m) < 1$ and $\ell(g) = \mathrm{diag}(2\pi/m, 0) \notin \mathfrak{g}_m = \{\mathrm{diag}(s, ms) : s \in \mathbb{R}\}$. So every threshold below c_G depends on G, including the radius c_G of its logarithm chart, and not only on N.

Lemma 21.2 (Elementary estimates)**.** *There is C, depending only on N, such that:*

(a) For $|X| \le \frac12$: $|e^{\mathrm{i}X} - 1| \le 2|X|$. For $|h-1| \le \frac12$: $|\ell(h)| \le 2|h-1|$.

(b) For G-valued $g_1, \dots, g_m$ with $\sum_i |\ell(g_i)| \le c_G$: $|\ell(g_1 \cdots g_m)| \le \sum_i |\ell(g_i)|$. Moreover $\ell(hgh^{-1}) = h\ell(g)h^{-1}$ and $\ell(g^{-1}) = -\ell(g)$.

(c) For $X_1, \dots, X_m \in \mathfrak{g}^c$ with $s = \sum_i |X_i| \le \frac14$: $\ell(e^{\mathrm{i}X_1} \cdots e^{\mathrm{i}X_m}) = \sum_i X_i + \beta$, where β is an analytic function of the X_i with $|\beta| \le Cs^2$.

(d) For $X \in \mathfrak{g}^c$ with $|X| < \log 2$: $\ell(e^{\mathrm{i}X}) = X$. For $h, u \in G^c$ with $|h-1| \le \frac14$ and $|uhu^{-1} - 1| \le \frac12$: $\ell(uhu^{-1}) = u\,\ell(h)\,u^{-1}$ and $\ell(h^{-1}) = -\ell(h)$.

Proof. (a) and (c) follow from the power series of exp and log; (c) is the Baker–Campbell–Hausdorff formula with the elementary bound on its remainder [5, (30)]. For (b), the Hilbert–Schmidt inner product defines a bi-invariant Riemannian metric on $U(\mathsf{N})$, and the distance from 1 to g is $|\ell(g)|$ when this is smaller than π. The triangle inequality and bi-invariance give $\mathrm{dist}(1, g_1 \cdots g_m) \le \sum_i \mathrm{dist}(1, g_i)$. (d) For $|X| < \log 2$ the power series of $\log(1+w)$ converges absolutely at $w = e^{\mathrm{i}X} - 1$, since $|w| \le e^{|X|} - 1 < 1$. So $\ell(e^{\mathrm{i}X})$ is a convergent power series in the single matrix X, whose coefficients are those of the scalar identity $-\mathrm{i}\log e^{\mathrm{i}z} = z$. Next, $(uhu^{-1} - 1)^m = u(h-1)^m u^{-1}$, and both power series converge. Finally $|\ell(h)| \le \frac12 < \log 2$ by (a), and $h^{-1} = e^{-\mathrm{i}\ell(h)}$, so $\ell(h^{-1}) = -\ell(h)$. □

A closed contour Γ based at z is *spanned by m plaquettes* if, in the groupoid of contours (with backtracking cancelled), $\Gamma = \prod_{i=1}^m \gamma_i (\partial p_i)^{\pm 1} \gamma_i^{-1}$ for plaquettes p_i and contours γ_i from z to the base point of p_i. Then $Y(\Gamma) = \prod_i Y(\gamma_i) Y(\partial p_i)^{\pm 1} Y(\gamma_i)^{-1}$ for every configuration Y.

Fix an order of the d directions. A contour from z to x is *coordinate-ordered* (for this order) if it changes the coordinates one after the other in this order, each one monotonically, as the contours of Definition 21.1.

Lemma 21.3 (Lattice Stokes)**.** *Let Y be real and let Γ be spanned by $p_1, \dots, p_m$ with $\sum_i |\ell(Y(\partial p_i))| \le c_G$. Then $|\ell(Y(\Gamma))| \le \sum_i |\ell(Y(\partial p_i))|$. On a lattice of spacing δ:*

(i) let Γ, Γ' be coordinate-ordered contours from z to $x = z + \delta \sum_\kappa n_\kappa e_\kappa$, for two orders that differ by the exchange of two consecutive directions μ, ν; then $\Gamma\Gamma'^{-1}$ is spanned by $|n_\mu||n_\nu|$ plaquettes of the box spanned by z and x;

(ii) let Γ, Γ' be the coordinate-ordered contours, for the same order, from a point z to neighbouring points x and x'; then the loop $\Gamma \cup [x, x'] \cup \Gamma'^{-1}$ is spanned by at most m plaquettes, where m is the number of bonds of Γ;

(iii) a closed contour of m bonds contained in a box of side s lattice spacings, at most half the side of the torus, is spanned by at most dsm plaquettes.

Proof. The bound follows from Lemma 21.2(b). (i) The two contours agree before and after their segments in the directions μ, ν. In between they run along the two pairs of sides of a rectangle of $|n_\mu| \times |n_\nu|$ plaquettes. (ii) Let $x' = x \pm \delta e_\nu$. The two contours agree up to their segments in the direction ν. These start at the same point, and one of them is the other followed by one bond parallel to $[x, x']$. After them the contours run parallel at distance δ, through the remaining bonds of Γ, and the strip between them consists of at most m plaquettes. (iii) Join every vertex of the contour to its first vertex by the coordinate-ordered path inside the box. Each bond of the contour, together with the two paths to its endpoints, bounds a surface of at most ds plaquettes. □

Definition 21.4 (Curvature and regularity classes)**.** For a plaquette $p = p_{\mu\nu}(x)$, $\mu < \nu$, of $\mathbb{T}^{(j)}$ write $Y(\partial p) = Y(b_1)Y(b_2)Y(b_3)^{-1}Y(b_4)^{-1}$ with $b_1 = (x, x+\delta_j e_\mu)$, $b_2 = (x+\delta_j e_\mu, x+\delta_j e_\mu+\delta_j e_\nu)$, $b_3 = (x + \delta_j e_\nu, x + \delta_j e_\mu + \delta_j e_\nu)$, $b_4 = (x, x + \delta_j e_\nu)$. If $|Y(\partial p) - 1| \le c_G$, the *curvature* is $F_Y(p) = \delta_j^{-2}\ell(Y(\partial p))$. For $a > 0$ let $\mathfrak{U}_j(a)$ be the set of real configurations on $\mathbb{T}^{(j)}$ with $|F_Y(p)| \le a$ for all plaquettes.

The curvature transforms by $F_{Y^g}(p) = g(x)F_Y(p)g(x)^{-1}$, so $\mathfrak{U}_j(a)$ is invariant under real gauge transformations; it is also invariant under $\mathfrak{S}_j$. The bound $|F_Y| \le a$ is Bałaban's condition $|Y(\partial p) - 1| \le a\delta_j^2$ up to a factor $1 + O(a\delta_j^2)$ [6, (1.7)]. In unit-lattice units it says that the field strength is at most a. For fine configurations we add a condition on the current in Section 23.

21.3 The group average

For $n \ge 1$ and a family $\mathbf{U} = (U_1, \dots, U_n) \in (G^c)^n$ put

$$\operatorname{diam} \mathbf{U} = \max_{i,i'} |U_i U_{i'}^{-1} - 1|, \qquad \mathcal{D}_r = \{\mathbf{U} : \operatorname{diam} \mathbf{U} < r\}.$$

The set $\mathcal{D}_r$ is invariant under permutations and under right multiplication $U_i \mapsto U_i v$ by any $v \in G^c$, and $\operatorname{diam}(u\mathbf{U}v) \le \kappa(u) \operatorname{diam} \mathbf{U}$ for $u, v \in G^c$. In particular $\mathcal{D}_r$ is invariant under left multiplication by G. Let $\mathfrak{f}(\operatorname{ad} H)$ denote the derivative of $Z \mapsto \ell(e^{\mathrm{i}Z}e^{\mathrm{i}H})$ at $Z = 0$. The function $\mathfrak{f}$, which is $z \mapsto \mathrm{i}z/(e^{\mathrm{i}z} - 1)$, is analytic for $|z| < 2\pi$, with $\mathfrak{f}(0) = 1$; it is used only for $\|\operatorname{ad} H\| \le 2|H|$ small, where $|\mathfrak{f}(\operatorname{ad} H) - 1| \le C|H|$.

Lemma 21.5 (Group average)**.** *There are $r_M \in (0, \frac{1}{4}c_G]$ and $C_M \ge 1$, depending only on N and c_G, such that the following holds for every $n \ge 1$.*

(a) (Existence and uniqueness.) *For* $\mathbf{U} \in \mathcal{D}_{r_M}$ *the equation*

$$\sum_{i=1}^{n} \ell(U_i U^{-1}) = 0 \tag{21.1}$$

has exactly one solution $U \in G^c$ *with* $\max_i |U_i U^{-1} - 1| \le 32 r_M$. *We call it the* group average $\mathrm{Av}(\mathbf{U}) = \mathrm{Av}(\{U_i\})$. *It satisfies* $\max_i |\ell(U_i \,\mathrm{Av}(\mathbf{U})^{-1})| \le 8 \,\mathrm{diam}\, \mathbf{U}$, *and* $|\ell(\mathrm{Av}(\mathbf{U}) U_m^{-1})| \le 2 \max_i |\ell(U_i U_m^{-1})|$ *for every* m. *The map* Av *is holomorphic on* $\mathcal{D}_{r_M}$.

(b) (Symmetries.)

- Av *is invariant under permutations of the* U_i, *and* $\mathrm{Av}(\{U, \dots, U\}) = U$.
- *If all* U_i *lie in* G, *then* $\mathrm{Av}(\mathbf{U}) \in G$.
- $\mathrm{Av}(\{u U_i v\}) = u \,\mathrm{Av}(\mathbf{U})\, v$ *for* $\mathbf{U} \in \mathcal{D}_{r_M}$, $u \in G$ *and* $v \in G^c$.
- $\mathrm{Av}(\{u U_i v\}) = u \,\mathrm{Av}(\mathbf{U})\, v$ *for* $\mathbf{U} \in \mathcal{D}_{r_M/2}$, $v \in G^c$ *and* $u \in G^c$ *with* $\kappa(u) \le 2$. *Here* $u\mathbf{U}v \in \mathcal{D}_{r_M}$.
- $\mathrm{Av}(\{U_i^{-1}\}) = \mathrm{Av}(\mathbf{U})^{-1}$, *with* $\{U_i^{-1}\} \in \mathcal{D}_{r_M}$, *if* $\mathbf{U} \in \mathcal{D}_{r_M}$ *is real, or if* $\mathbf{U} \in \mathcal{D}_{r_M/8}$ *and* $\kappa(\mathrm{Av}(\mathbf{U})) \le 2$.

(c) (Expansion.) *For* $Z_1, \dots, Z_n \in \mathfrak{g}^c$ *with* $\max_i |Z_i| \le r_M/8$,

$$\ell \,\mathrm{Av}(\{e^{\mathrm{i}Z_i}\}) = \frac{1}{n} \sum_i Z_i + \mathfrak{r}(Z),$$

with $\mathfrak{r}$ *holomorphic and* $|\mathfrak{r}(Z)| \le C_M \max_i |Z_i|^2$.

(d) (Weights.) *Let* $\mathbf{U} \in \mathcal{D}_{r_M/2}$ *and* $H_i = \ell(U_i \,\mathrm{Av}(\mathbf{U})^{-1})$. *For* $Z_1, \dots, Z_n \in \mathfrak{g}^c$,

$$\frac{d}{ds}\Big|_{s=0} \ell\big(\,\mathrm{Av}(\{e^{\mathrm{i}sZ_i} U_i\})\,\mathrm{Av}(\mathbf{U})^{-1}\big) = \sum_i \omega_i Z_i, \qquad \omega_i = \Big(\sum_m \mathfrak{f}(\mathrm{ad}\, H_m)\Big)^{-1} \mathfrak{f}(\mathrm{ad}\, H_i).$$

The weights ω_i *are linear maps of* $\mathfrak{g}^c$ *with* $\sum_i \omega_i = I$ *and* $|\omega_i - \frac{1}{n} I| \le \frac{C_M}{n} \max_m |H_m|$. *They are holomorphic in* $\mathbf{U}$, *and for real* $\mathbf{U}$ *they map* $\mathfrak{g}$ *into* $\mathfrak{g}$.

(e) (Lipschitz bound.) *For* $\mathbf{U} \in \mathcal{D}_{r_M/2}$ *and* $\xi_1, \dots, \xi_n \in \mathfrak{g}^c$ *with* $\max_i |\xi_i| \le r_M/16$,

$$\big|\ell\big(\,\mathrm{Av}(\{e^{\mathrm{i}\xi_i} U_i\})\,\mathrm{Av}(\mathbf{U})^{-1}\big)\big| \le C_M \max_i |\xi_i|.$$

This is the average of [18, (0.10)]. Bałaban defines averages by the properties (0.5)–(0.9) of [18], which are part (b) and the first-order part of (c), and attributes the definition by (21.1) to Federbush [20]. For real families, (21.1) says that U is a critical point of $U \mapsto \sum_i |\ell(U_i U^{-1})|^2$, i.e. Av is the Riemannian centre of mass for the bi-invariant metric of Lemma 21.2(b); we do not use this. Two-sided equivariance holds with complex u only up to the distortion $\kappa(u)$ of the domain. This is enough for the gauge transformations used below, which are real or close to 1.

Proof. For $Z, Y \in \mathfrak{g}^c$ with $|Z| + |Y| \le \frac{1}{2} c_G$ put $\beta(Z, Y) = \ell(e^{\mathrm{i}Z} e^{-\mathrm{i}Y}) - Z + Y$. Since $|e^{\mathrm{i}Z} e^{-\mathrm{i}Y} - 1| \le e^{c_G/2} - 1 \le c_G$, β takes values in $\mathfrak{g}^c$, and in $\mathfrak{g}$ if $Z, Y \in \mathfrak{g}$. By Lemma 21.2(c), β is holomorphic with $|\beta(Z, Y)| \le C_0 (|Z| + |Y|)^2$. It vanishes to second order at 0, so the Cauchy estimates give $|\partial_Y \beta(Z, Y)| \le C_0' (|Z| + |Y|)$ for $|Z| + |Y| \le \frac{1}{4} c_G$, where ∂_Y is the derivative in Y. Fix $\rho_0 \le \frac{1}{20} c_G$ with $25 C_0 \rho_0 \le 1$ and $10 C_0' \rho_0 \le 1$, and put $r_M = \rho_0/32$.

Step 1: a fixed point. Let $V \in G^c$ and $Z_1, \dots, Z_n \in \mathfrak{g}^c$ with $m = \max_i |Z_i| \le \rho_0$, and let $U_i = e^{\mathrm{i}Z_i}V$. For $U = e^{\mathrm{i}Y}V$ we have $U_iU^{-1} = e^{\mathrm{i}Z_i}e^{-\mathrm{i}Y}$. So, for $|Y| \le 4\rho_0$, U solves (21.1) if and only if $Y = T(Y)$, where

$$T(Y) = \frac{1}{n}\sum_i Z_i + \frac{1}{n}\sum_i \beta(Z_i, Y).$$

On the ball $\{|Y| \le 4\rho_0\}$ we have $|Z_i| + |Y| \le 5\rho_0 \le \frac{1}{4}c_G$, hence $|T(Y)| \le \rho_0 + 25C_0\rho_0^2 \le 2\rho_0$ and $|T(Y) - T(Y')| \le 5C_0'\rho_0|Y - Y'| \le \frac{1}{2}|Y - Y'|$. So T has exactly one fixed point Y_* in this ball. It is the uniform limit of the iterates $T^p(0)$, which are holomorphic in (Z_i), and which lie in $\mathfrak{g}$ if all $Z_i \in \mathfrak{g}$. Hence Y_* is holomorphic in (Z_i), and lies in $\mathfrak{g}$ in that case. Moreover $|Y_*| \le |T(Y_*) - T(0)| + |T(0)| \le \frac{1}{2}|Y_*| + m + C_0m^2$, so $|Y_*| \le 3m$. Inserting this into the definition of T gives $|Y_*| \le m + 16C_0m^2 \le 2m$, $|Y_* - \frac{1}{n}\sum_i Z_i| \le 16C_0m^2$, and

$$|\ell(U_iU_*^{-1})| = |Z_i - Y_* + \beta(Z_i, Y_*)| \le 3m + 9C_0m^2 \le 4m, \qquad U_* = e^{\mathrm{i}Y_*}V.$$

(a). Let $\mathbf{U} \in \mathcal{D}_{r_M}$, fix m_0, and apply Step 1 with $V = U_{m_0}$ and $Z_i = \ell(U_iU_{m_0}^{-1})$. By Lemma 21.2(a), $m \le 2\operatorname{diam}\mathbf{U} < 2r_M \le \rho_0$. This gives a solution $U_* = e^{\mathrm{i}Y_*}U_{m_0}$ with $|\ell(U_*U_{m_0}^{-1})| = |Y_*| \le 2m$, $|\ell(U_iU_*^{-1})| \le 4m \le 8\operatorname{diam}\mathbf{U}$, and hence $|U_iU_*^{-1} - 1| \le 16\operatorname{diam}\mathbf{U} < 32r_M$. Conversely, let $U \in G^c$ be a solution with $\max_i |U_iU^{-1} - 1| \le 32r_M = \rho_0$. Then $h = U_{m_0}U^{-1}$ satisfies $|h - 1| \le \rho_0$, and $Y = -\ell(h)$ satisfies $e^{\mathrm{i}Y} = h^{-1} = UU_{m_0}^{-1}$ and $|Y| \le 2\rho_0$. So Y is a fixed point of T in the ball, and $U = U_*$. This proves uniqueness. In particular U_* does not depend on m_0, which gives the bound on $\ell(\operatorname{Av}(\mathbf{U})U_m^{-1})$. With $m_0 = 1$, $\operatorname{Av}(\mathbf{U}) = e^{\mathrm{i}Y_*}U_1$ is holomorphic, since $Z_i = \ell(U_iU_1^{-1})$ is.

(b). Permutation invariance holds because (21.1), the neighbourhood $\{\max_i |U_iU^{-1} - 1| \le 32r_M\}$ and $\mathcal{D}_{r_M}$ are symmetric. The constant family is solved by U. For real $\mathbf{U}$ all Z_i lie in $\mathfrak{g}$, so $Y_* \in \mathfrak{g}$ and $\operatorname{Av}(\mathbf{U}) \in G$. For $v \in G^c$, $U \mapsto Uv$ maps the solutions for $\mathbf{U}$ to those for $\{U_iv\}$, and preserves the neighbourhood, since $(U_iv)(Uv)^{-1} = U_iU^{-1}$. For $u \in G^c$, $(uU_i)(uU)^{-1} = uh_iu^{-1}$ with $h_i = U_iU^{-1}$. If $u \in G$, then $|uh_iu^{-1} - 1| = |h_i - 1|$ and $\ell(uh_iu^{-1}) = u\ell(h_i)u^{-1}$, so $u\operatorname{Av}(\mathbf{U})$ solves (21.1) for $u\mathbf{U}$ and lies in the neighbourhood. If $\kappa(u) \le 2$ and $\mathbf{U} \in \mathcal{D}_{r_M/2}$, then $\operatorname{diam}(u\mathbf{U}) < r_M$, and for $U = \operatorname{Av}(\mathbf{U})$ we have $|h_i - 1| \le 16\operatorname{diam}\mathbf{U} < 8r_M$ and $|uh_iu^{-1} - 1| < 16r_M$. By Lemma 21.2(d), $\ell(uh_iu^{-1}) = u\ell(h_i)u^{-1}$, so again $u\operatorname{Av}(\mathbf{U})$ is the solution in the neighbourhood.

For the inverses let $\operatorname{Av} = \operatorname{Av}(\mathbf{U})$ and $h_i = U_i\operatorname{Av}^{-1}$. Then $U_i^{-1}\operatorname{Av} = \operatorname{Av}^{-1}h_i^{-1}\operatorname{Av}$ and $U_i^{-1}U_{i'} = \operatorname{Av}^{-1}h_i^{-1}(U_{i'}U_i^{-1})h_i\operatorname{Av}$. If $\mathbf{U}$ is real, all these matrices are unitary, so $\operatorname{diam}\{U_i^{-1}\} = \operatorname{diam}\mathbf{U}$ and $|U_i^{-1}\operatorname{Av} - 1| = |h_i - 1|$. If $\mathbf{U} \in \mathcal{D}_{r_M/8}$ and $\kappa(\operatorname{Av}) \le 2$, then $|h_i - 1| \le 2r_M$, $|h_i^{-1} - 1| \le 3r_M$, and therefore $|U_i^{-1}\operatorname{Av} - 1| \le 6r_M$ and $\operatorname{diam}\{U_i^{-1}\} \le 2(1 + 2r_M)(1 + 3r_M)\operatorname{diam}\mathbf{U} < r_M$. In both cases Lemma 21.2(d) gives $\ell(U_i^{-1}\operatorname{Av}) = \operatorname{Av}^{-1}\ell(h_i^{-1})\operatorname{Av} = -\operatorname{Av}^{-1}\ell(h_i)\operatorname{Av}$. Summing over i, Av^{-1} solves (21.1) for $\{U_i^{-1}\}$, and it lies in the neighbourhood. So $\operatorname{Av}(\{U_i^{-1}\}) = \operatorname{Av}^{-1}$.

(c). Apply Step 1 with $V = 1$, so that $U_* = e^{\mathrm{i}Y_*}$ solves (21.1) for $\{e^{\mathrm{i}Z_i}\}$, with $m \le r_M/8$. By Lemma 21.2(a),(c) the family has diameter at most $5m < r_M$, and $|e^{\mathrm{i}Z_i}U_*^{-1} - 1| \le 8m < 32r_M$. So $\operatorname{Av}(\{e^{\mathrm{i}Z_i}\}) = e^{\mathrm{i}Y_*}$, and $\ell\operatorname{Av}(\{e^{\mathrm{i}Z_i}\}) = Y_*$ by Lemma 21.2(d). The bound on $\mathfrak{r} = Y_* - \frac{1}{n}\sum_i Z_i$ is that of Step 1.

(d). Let $\operatorname{Av} = \operatorname{Av}(\mathbf{U})$. Then $|H_i| \le 8\operatorname{diam}\mathbf{U} < 4r_M$, and $\sum_i H_i = 0$ by (21.1). For small complex s the family $\{e^{\mathrm{i}sZ_i}U_i\}$ lies in $\mathcal{D}_{r_M}$, and by (a) and (b), $Y(s) = \ell(\operatorname{Av}(\{e^{\mathrm{i}sZ_i}U_i\})\operatorname{Av}^{-1})$ is holomorphic, with $Y(0) = 0$ and $e^{\mathrm{i}Y(s)} = \operatorname{Av}(\{e^{\mathrm{i}sZ_i}e^{\mathrm{i}H_i}\})$. Now $e^{\mathrm{i}sZ_i}e^{\mathrm{i}H_i}e^{-\mathrm{i}Y(s)} = e^{\mathrm{i}sZ_i}e^{-\mathrm{i}R(e^{\mathrm{i}H_i})Y(s)}e^{\mathrm{i}H_i}$, and $R(e^{\mathrm{i}H}) = e^{\mathrm{i}\operatorname{ad}H}$. Differentiating (21.1) at $s = 0$ therefore gives

$$\sum_i \mathfrak{f}(\operatorname{ad}H_i)(Z_i - e^{\mathrm{i}\operatorname{ad}H_i}Y'(0)) = 0.$$

Since $\mathfrak{f}(z)(e^{\mathrm{i}z} - 1) = \mathrm{i}z$ and $\sum_i H_i = 0$, we have $\sum_i \mathfrak{f}(\operatorname{ad}H_i)e^{\mathrm{i}\operatorname{ad}H_i} = \sum_i \mathfrak{f}(\operatorname{ad}H_i) + \mathrm{i}\operatorname{ad}(\sum_i H_i) = \sum_i \mathfrak{f}(\operatorname{ad}H_i)$. This operator is $n(1 + O(r_M))$, hence invertible, and

$Y'(0) = \sum_i \omega_i Z_i$. The identity $\sum_i \omega_i = I$ is clear. Since $|\mathfrak{f}(\operatorname{ad} H) - 1| \le C|H|$, we get $|(\frac{1}{n}\sum_m \mathfrak{f}(\operatorname{ad} H_m))^{-1} - 1| \le C \max |H_m|$, and hence the bound on $\omega_i - \frac{1}{n} I$. The weights are holomorphic because the H_i are. For real $\mathbf{U}$ the H_i lie in $\mathfrak{g}$. Then $\mathfrak{f}(\operatorname{ad} H) = \phi(\mathrm{i} \operatorname{ad} H)$ with $\phi(w) = w/(e^w - 1)$, which has real Taylor coefficients, and $\mathrm{i} \operatorname{ad} H$ maps $\mathfrak{g}$ into $\mathfrak{g}$.

(e). Let $x = \max_i |\xi_i|$, $\mathrm{Av} = \mathrm{Av}(\mathbf{U})$, and $\mathbf{V}_t = \{e^{\mathrm{i}t\xi_i} U_i\}$ for $t \in \mathbb{C}$ with $|t| \le r_M/(16x)$. Then $|e^{\mathrm{i}t\xi_i} - 1| \le \frac{1}{8} r_M$, so $\operatorname{diam} \mathbf{V}_t \le \frac{3}{4} r_M + C r_M^2 < r_M$. By (a), $f(t) = \ell(\mathrm{Av}(\mathbf{V}_t)\,\mathrm{Av}^{-1})$ is holomorphic, with $f(0) = 0$. Write $\mathrm{Av}(\mathbf{V}_t)\,\mathrm{Av}^{-1} = [\mathrm{Av}(\mathbf{V}_t) V_{t,1}^{-1}][V_{t,1}\,\mathrm{Av}^{-1}]$, where $V_{t,1} = e^{\mathrm{i}t\xi_1} U_1$. By (a) the first factor has logarithm of size at most $4 \operatorname{diam} \mathbf{V}_t \le 4 r_M$, and the second is $e^{\mathrm{i}t\xi_1} e^{\mathrm{i}H_1}$, with logarithm of size at most $\frac{1}{16} r_M + 4 r_M + C r_M^2$. Hence $|f| \le 9 r_M$, and the Schwarz lemma gives $|f(1)| \le 9 r_M \cdot 16x / r_M = 144x$. □

21.4 Averaged contour variables and averaged loops

Definition 21.6 (Averaged contour variables [18, (0.11)])**.** Let Y be a configuration on $\mathbb{T}^{(j-1)}$, $y \in \mathbb{T}^{(j)}$ and $x \in B_j(y)$. If the family $\{Y(\Gamma)\}_{\Gamma \in G(y,x)}$ lies in $\mathcal{D}_{r_M}$, the *averaged contour variable* is

$$Y(y,x) = \mathrm{Av}\left(\{Y(\Gamma)\}_{\Gamma \in G(y,x)}\right), \qquad \text{and we put} \quad Y(x,y) = Y(y,x)^{-1}.$$

For a bond c of $\mathbb{T}^{(j)}$, of either orientation, and $x \in B_j(c_-)$ put $x(c) = x + c_+ - c_- \in B_j(c_+)$. Let $[x, x(c)]$ be the straight contour of L bonds of $\mathbb{T}^{(j-1)}$ from x to $x(c)$, and $[c] = [c_-, c_+]$. The *averaged loops* are

$$\Lambda_{c,x}(Y) = Y(c_-, x)\, Y([x, x(c)])\, Y(x(c), c_+)\, Y([c])^{-1}, \qquad x \in B_j(c_-).$$

We have $Y(y,y) = 1$. If $x - y$ is parallel to a coordinate axis, $Y(y,x) = Y([y,x])$; in particular $Y(y,x) = Y(b)$ for a bond $b = (y,x)$, so the notation is consistent. If x lies on the axis of c, i.e. $x = c_- + m\delta_{j-1} e$ with $c_+ - c_- = \delta_j e$, then $\Lambda_{c,x}(Y) = 1$, since $[c_-, x][x, x(c)][x(c), c_+] = [c]$ in the groupoid of contours. The averaged loop $\Lambda_{c,x}(Y)$ is based at c_-, but it is not the holonomy of a single closed contour. Averaged contour variables are not multiplicative under concatenation. The only products of them used below are the averaged loops, which are products by definition.

The averaged contour variables inherit the symmetries of Av. For a gauge transformation g on $\mathbb{T}^{(j-1)}$ we have $Y^g(\Gamma) = g(y) Y(\Gamma) g(x)^{-1}$ for $\Gamma \in G(y,x)$. By Lemma 21.5(b),

$$Y^g(y,x) = g(y)\, Y(y,x)\, g(x)^{-1}, \tag{21.2}$$

for real g whenever $Y(y,x)$ is defined, and for G^c-valued g with $\kappa(g(y)) \le 2$ if the family of Y lies in $\mathcal{D}_{r_M/2}$. For $r \in \mathfrak{S}_j$, $(rY)(\Gamma) = Y(r^{-1}\Gamma)$ and $r^{-1} G(y,x) = G(r^{-1}y, r^{-1}x)$, so by permutation invariance $(rY)(y,x) = Y(r^{-1}y, r^{-1}x)$, and both are defined together. Moreover $Y(y,x)$ is real for real Y, and depends only on Y on the bonds of the box spanned by y and x.

Lemma 21.7 (Stokes bounds for averaged contour variables)**.** *Let Y be a real configuration on $\mathbb{T}^{(j-1)}$, and let $\varepsilon > 0$ with $8L^2\varepsilon \le r_M$.*

(i) Let $y \in \mathbb{T}^{(j)}$, and suppose that $|\ell(Y(\partial p))| \le \varepsilon$ for all plaquettes p with corners in $B_j(y)$. Let $x \in B_j(y)$. For $\Gamma, \Gamma' \in G(y,x)$ the loop $\Gamma\Gamma'^{-1}$ is spanned by at most $\frac{3}{2}(L-1)^2$ plaquettes of the box spanned by y and x; so $|\ell(Y(\Gamma)Y(\Gamma')^{-1})| \le \frac{3}{2}(L-1)^2\varepsilon$, and $A(\Gamma) - A(\Gamma')$ is a sum of at most $\frac{3}{2}(L-1)^2$ values $\pm A(\partial p)$ for every Lie field A. The family $\{Y(\Gamma)\}_{\Gamma \in G(y,x)}$ lies in $\mathcal{D}_{3L^2\varepsilon} \subset \mathcal{D}_{r_M/2}$. The averaged contour variable $Y(y,x)$ is defined and real, and $|\ell(Y(y,x)Y(\Gamma)^{-1})| \le 3(L-1)^2\varepsilon$ for $\Gamma \in G(y,x)$.

(ii) Under the hypothesis of (i), $|\ell(Y(y,x)Y(b)Y(y,x')^{-1})| \le 6(L-1)^2\varepsilon$ for every bond $b = (x,x')$ of $\mathbb{T}^{(j-1)}$ with $x, x' \in B_j(y)$.

(iii) *Let c be a bond of $\mathbb{T}^{(j)}$, and suppose that $|\ell(Y(\partial p))| \le \varepsilon$ for all plaquettes p with corners in $B_j(c_-) \cup B_j(c_+)$. Then $|\ell(\Lambda_{c,x}(Y))| \le 8L^2\varepsilon$ for $x \in B_j(c_-)$.*

(iv) (Complex configurations.) *Let Y_r be real, let $Y = e^{\mathrm{i}Z}Y_r$ bondwise with Z $\mathfrak{g}^c$-valued and $|Z| \le \zeta$, and let $64L\zeta \le r_M$. Under the hypothesis of (i) for Y_r, with Z given on the bonds of $B_j(y)$, the family $\{Y(\Gamma)\}_{\Gamma\in G(y,x)}$ lies in $\mathcal{D}_{r_M}$, the averaged contour variable $Y(y,x)$ is defined and holomorphic in Z, and $|\ell(Y(y,x)Y_r(y,x)^{-1})| \le 4C_ML\zeta$. Under the hypothesis of (iii) for Y_r, with Z given on the bonds of $B_j(c_-) \cup B_j(c_+)$ and of the straight contours $[x,x(c)]$ and $[c]$, $|\ell(\Lambda_{c,x}(Y))| \le 16L^2\varepsilon + 32C_ML\zeta$, provided $s := 8L^2\varepsilon + 16C_ML\zeta \le \min(s_0, \frac{1}{8}c_G)$, where $s_0 \in (0, \frac{1}{4}]$ depends only on N. Then also $|\Lambda_{c,x}(Y) - 1| \le \frac{1}{2}c_G$.*

Proof. (i) Two orders of S differ by at most $\binom{|S|}{2} \le 6$ exchanges of consecutive directions. Write $\Gamma\Gamma'^{-1}$ as the product of the corresponding loops $\Gamma_{\pi_{t-1}}\Gamma_{\pi_t}^{-1}$, based at y. By Lemma 21.3(i) each is spanned by at most $(\frac{L-1}{2})^2$ plaquettes of the box, which gives the first two claims and the abelian identity. By Lemma 21.2(a), $|Y(\Gamma)Y(\Gamma')^{-1} - 1| \le 3(L-1)^2\varepsilon < \frac{3}{8}r_M$. Lemma 21.5(a),(b), with $U_m = Y(\Gamma)$, gives the rest.

(ii) Let $x' = x \pm \delta_{j-1}e_\nu$. Choose $\Gamma_0 \in G(y,x)$ whose order of the directions has ν last, if ν occurs. Let $\Gamma_0' \in G(y,x')$ have the same order of the other directions, followed by ν if it occurs. Then $\Gamma_0' = \Gamma_0 b$ in the groupoid of contours, so $Y(\Gamma_0)Y(b)Y(\Gamma_0')^{-1} = 1$, and

$$Y(y,x)Y(b)Y(y,x')^{-1} = \big[Y(y,x)Y(\Gamma_0)^{-1}\big]\big[Y(\Gamma_0')Y(y,x')^{-1}\big].$$

Both factors lie in G and have logarithms of size at most $3(L-1)^2\varepsilon$, by (i). Lemma 21.2(b) gives the claim.

(iii) Let $v = c_+ - c_-$, choose $\Gamma_0 \in G(c_-,x)$, and note that $\Gamma_0 + v \in G(c_+, x(c))$. Then

$$\begin{aligned}\Lambda_{c,x}(Y) = \big[Y(c_-,x)Y(\Gamma_0)^{-1}\big]\, Y\big(\Gamma_0\,[x,x(c)]\,(\Gamma_0+v)^{-1}[c]^{-1}\big)\\ \times Y([c])\big[Y(\Gamma_0+v)Y(c_+,x(c))^{-1}\big]Y([c])^{-1}.\end{aligned}$$

The closed contour in the middle bounds the surface swept by Γ_0 under the translation by v. Each bond of Γ_0 contributes at most L plaquettes, so it is spanned by at most $2L(L-1)$ plaquettes with corners in $B_j(c_-) \cup B_j(c_+)$. The two brackets have logarithms of size at most $3(L-1)^2\varepsilon$ by (i), and conjugation by $Y([c]) \in G$ does not change this. By Lemma 21.2(b), $|\ell(\Lambda_{c,x}(Y))| \le (6(L-1)^2 + 2L(L-1))\varepsilon \le 8L^2\varepsilon$.

(iv) For a contour $\gamma = b_1 \cdots b_m$, $Y(\gamma)Y_r(\gamma)^{-1} = \prod_i \big(Y_r(\gamma_{<i})\, e^{\mathrm{i}Z(b_i)}\, Y_r(\gamma_{<i})^{-1}\big)$, and conjugation by the real $Y_r(\gamma_{<i})$ preserves the norm. The contours of $G(y,x)$ have at most $2(L-1)$ bonds, so $\xi_\Gamma := \ell(Y(\Gamma)Y_r(\Gamma)^{-1})$ satisfies $|\xi_\Gamma| \le 4L\zeta \le \frac{1}{16}r_M$ by Lemma 21.2(c). By (i), $\{Y_r(\Gamma)\} \in \mathcal{D}_{3L^2\varepsilon} \subset \mathcal{D}_{r_M/2}$, and $Y(\Gamma) = e^{\mathrm{i}\xi_\Gamma}Y_r(\Gamma)$. So the family lies in $\mathcal{D}_{r_M}$, its average is defined and holomorphic in Z by Lemma 21.5(a), and Lemma 21.5(e) gives $|\ell(Y(y,x)Y_r(y,x)^{-1})| \le 4C_ML\zeta$. For the loop, each of the four factors of $\Lambda_{c,x}(Y)$ is $e^{\mathrm{i}\eta}$ times the corresponding factor of $\Lambda_{c,x}(Y_r)$, with $|\eta| \le 4C_ML\zeta$ for the averaged contour variables and $|\eta| \le 2L\zeta$ for the straight contours. Moving the factors $e^{\mathrm{i}\eta}$ to the left conjugates them by real elements, so $\Lambda_{c,x}(Y) = e^{\mathrm{i}\eta_1}\cdots e^{\mathrm{i}\eta_4}\Lambda_{c,x}(Y_r)$ with $|\eta_i| \le 4C_ML\zeta$. By (iii), $\Lambda_{c,x}(Y_r) = e^{\mathrm{i}H}$ with $H = \ell(\Lambda_{c,x}(Y_r)) \in \mathfrak{g}$ and $|H| \le 8L^2\varepsilon$. Hence $\Lambda_{c,x}(Y) = e^{\mathrm{i}\eta_1}\cdots e^{\mathrm{i}\eta_4}e^{\mathrm{i}H}$ with $\sum_i|\eta_i| + |H| \le s$. The η_i are complex, so we use Lemma 21.2(c): $|\ell(\Lambda_{c,x}(Y))| \le s + Cs^2 \le 2s$ for $s \le s_0 := \min(\frac{1}{4}, C^{-1})$, with C of that lemma. This is the bound. If also $s \le \frac{1}{8}c_G$, then $|\ell(\Lambda_{c,x}(Y))| \le \frac{1}{4}c_G$, and $|\Lambda_{c,x}(Y) - 1| \le \frac{1}{2}c_G$ by Lemma 21.2(a). □

Consequently, all averaged contour variables and averaged loops of a real configuration Y on $\mathbb{T}^{(j-1)}$ are defined, with $|\ell(\Lambda_{c,x}(Y))| \le r_M$, if $|\ell(Y(\partial p))| \le \varepsilon_M$ for all plaquettes, where

$$\varepsilon_M = \frac{r_M}{8L^2}.$$

This is a quantitative form of Bałaban's restriction to regular configurations, $|U(\partial p)-1|<\varepsilon_0$ with ε_0 small [18, p. 254]. The set of these configurations is invariant under real gauge transformations and under $\mathfrak{S}_j$. In the units of the scale, $Y\in\mathfrak{U}_{j-1}(b)$ gives $\varepsilon=b\delta_{j-1}^2$ and $8L^2\varepsilon=8b\delta_j^2$: the condition $8b\delta_j^2\le r_M$ does not depend on L.

21.5 The one-step averaging

Definition 21.8 (Bałaban's averaging [18, (0.12)])**.** For a configuration Y on $\mathbb{T}^{(j-1)}$ and a bond c of $\mathbb{T}^{(j)}$ put

$$\mathfrak{a}_j(Y)(c)=\exp\Big(\mathrm{i}L^{-d}\sum_{x\in B_j(c_-)}\ell(\Lambda_{c,x}(Y))\Big)\,Y([c]),\tag{21.3}$$

whenever the averaged contour variables involved are defined and $|\Lambda_{c,x}(Y)-1|\le c_G$ for all $x\in B_j(c_-)$. Put $\mathfrak{a}^{(0)}=\mathrm{id}$, $\mathfrak{a}^{(j)}=\mathfrak{a}_j\circ\mathfrak{a}^{(j-1)}$, and $U^{(j)}=\mathfrak{a}^{(j)}(U)$ for configurations U on $\mathbb{T}_\eta$.

The averaged loops are small loops based at c_-, built from the straight segment $[x,x(c)]$, the translate of $[c]$ through x. The average of their logarithms corrects the straight-line transport $Y([c])$. For x on the axis of c the loop is 1. The definition is made for both orientations of c, and the two are consistent.

Lemma 21.9 (Reversal)**.** *Let c be a bond of $\mathbb{T}^{(j)}$ and $x\in B_j(c_-)$. Then $\Lambda_{-c,x(c)}(Y)=Y([c])^{-1}\Lambda_{c,x}(Y)^{-1}\,Y([c])$. Hence $\mathfrak{a}_j(Y)(-c)=\mathfrak{a}_j(Y)(c)^{-1}$ whenever both sides are defined. If Y is real, $\mathfrak{a}_j(Y)(-c)$ is defined whenever $\mathfrak{a}_j(Y)(c)$ is. So $\mathfrak{a}_j(Y)$ is a configuration on $\mathbb{T}^{(j)}$.*

Proof. For $-c$ the initial point is c_+, and $x\mapsto x(c)$ maps $B_j(c_-)$ onto $B_j(c_+)$, with inverse $\tilde x\mapsto\tilde x(-c)$. Moreover $[x(c),x]=[x,x(c)]^{-1}$ and $[-c]=[c]^{-1}$. With the convention $Y(x,y)=Y(y,x)^{-1}$,

$$\Lambda_{-c,x(c)}(Y)=Y(c_+,x(c))\,Y([x,x(c)])^{-1}\,Y(c_-,x)^{-1}\,Y([c])=Y([c])^{-1}\Lambda_{c,x}(Y)^{-1}\,Y([c]).$$

Put $u=Y([c])^{-1}$ and $\Lambda=\Lambda_{c,x}$, so that $u\Lambda u^{-1}=\Lambda_{-c,x(c)}^{-1}$. If both sides are defined, then $|\Lambda-1|\le c_G\le\frac14$ and $|\Lambda_{-c,x(c)}-1|\le c_G$, hence $|u\Lambda u^{-1}-1|\le\|\Lambda_{-c,x(c)}^{-1}\|\,|\Lambda_{-c,x(c)}-1|\le c_G/(1-c_G)\le\frac12$. Lemma 21.2(d), applied first to $h=\Lambda$ with this u and then to $h=\Lambda_{-c,x(c)}$, gives $\ell(\Lambda_{-c,x(c)})=-\ell(u\Lambda u^{-1})=-Y([c])^{-1}\ell(\Lambda_{c,x})Y([c])$. No reality is used here. Summing over x, the exponent $\mathrm{i}E_{-c}$ in (21.3) for $-c$ satisfies $E_{-c}=-Y([c])^{-1}E_cY([c])$. Hence $\mathfrak{a}_j(Y)(-c)=e^{\mathrm{i}E_{-c}}Y([c])^{-1}=Y([c])^{-1}e^{-\mathrm{i}E_c}=\mathfrak{a}_j(Y)(c)^{-1}$. Both formulas involve the same averaged contour variables, and for real Y, $|\Lambda_{-c,x(c)}-1|=|\Lambda_{c,x}-1|$. □

Only the convention $Y(x,y)=Y(y,x)^{-1}$ is used here, not the inverse property of Av.

Lemma 21.10 (Basic properties)**.** *(a) If $Y\in\mathfrak{U}_{j-1}(b)$ with $8b\delta_j^2\le r_M$, then $\mathfrak{a}_j(Y)$ is defined and real, and*

$$\big|\ell(\Lambda_{c,x}(Y))\big|\le 8b\delta_j^2.$$

(b) (Covariance.) *Let $\mathfrak{a}_j(Y)$ be defined. For every real gauge transformation g on $\mathbb{T}^{(j-1)}$, of any size, $\mathfrak{a}_j(Y^g)$ is defined and $\mathfrak{a}_j(Y^g)=\mathfrak{a}_j(Y)^{g|_{\mathbb{T}^{(j)}}}$. The same holds for $G^{\mathrm c}$-valued g with $\kappa(g(y))\le2$ for $y\in\mathbb{T}^{(j)}$, provided all families $\{Y(\Gamma)\}_{\Gamma\in G(y,x)}$ lie in $\mathcal{D}_{r_M/2}$ and $|\Lambda_{c,x}(Y)-1|\le\frac12 c_G$ for all c and x.*

(c) (Locality.) *$\mathfrak{a}_j(Y)(c)$ depends only on Y on the bonds of $B_j(c_-)\cup B_j(c_+)$, i.e. bonds with both endpoints in this set.*

(d) (Lattice symmetry.) *Let $\mathfrak{a}_j(Y)$ be defined and $r \in \mathfrak{S}_j$. Then $\mathfrak{a}_j(rY)$ is defined and* $\mathfrak{a}_j(rY) = r\,\mathfrak{a}_j(Y)$.

Consequently, wherever the averages are defined, $\mathfrak{a}^{(k)}(U^u) = \mathfrak{a}^{(k)}(U)^{u|_{\mathbb{T}}}$ for all real gauge transformations u on $\mathbb{T}_\eta$, and $\mathfrak{a}^{(k)}(rU) = r\,\mathfrak{a}^{(k)}(U)$ for $r \in \mathfrak{S}_k$. Moreover $U^{(k)}(c)$ depends only on U on the bonds of $B(c_-) \cup B(c_+)$.

Proof. (a) Apply Lemma 21.7 with $\varepsilon = b\delta_{j-1}^2$, so that $8L^2\varepsilon = 8b\delta_j^2 \le r_M$. Then $|\Lambda_{c,x} - 1| \le 2r_M \le c_G$. The averaged contour variables are real, and the average of elements of $\mathfrak{g}$ lies in $\mathfrak{g}$.

(b) By (21.2), $\Lambda_{c,x}(Y^g) = g(c_-)\Lambda_{c,x}(Y)g(c_-)^{-1}$ and $Y^g([c]) = g(c_-)Y([c])g(c_+)^{-1}$. For real g the families of Y^g have the same diameters as those of Y, and $|\Lambda_{c,x}(Y^g) - 1| = |\Lambda_{c,x}(Y) - 1|$. For complex g, $|\Lambda_{c,x}(Y^g) - 1| \le 2|\Lambda_{c,x}(Y) - 1| \le c_G$. In both cases $\ell(g\Lambda g^{-1}) = g\ell(\Lambda)g^{-1}$ by Lemma 21.2(d), and $\exp(\mathrm{i}gEg^{-1}) = ge^{\mathrm{i}E}g^{-1}$. So the exponential in (21.3) is conjugated by $g(c_-)$.

(c) The contours of $G(c_\pm, \cdot)$ stay in $B_j(c_\pm)$, and $[x, x(c)]$ and $[c]$ stay in $B_j(c_-) \cup B_j(c_+)$.

(d) The map r maps $B_j(c_-)$ onto $B_j((rc)_-)$, $x(c)$ to $(rx)(rc)$, $[x, x(c)]$ to $[rx, (rx)(rc)]$, and $[c]$ to $[rc]$. With $(rY)(y, x) = Y(r^{-1}y, r^{-1}x)$, this gives $\Lambda_{c,x}(rY) = \Lambda_{r^{-1}c, r^{-1}x}(Y)$ and $\mathfrak{a}_j(rY)(c) = \mathfrak{a}_j(Y)(r^{-1}c)$. Here $r^{-1}c$ may be negatively oriented; this is allowed, since (21.3) holds for both orientations by Lemma 21.9.

The consequences follow by induction over the scales. For $r \in \mathfrak{S}_k$ we use $\mathfrak{S}_k \subset \mathfrak{S}_j$. For locality we use that the bonds of $B_j(c_-) \cup B_j(c_+)$ have their scale-$(j-1)$ blocks in $B^{(j)}(c_-) \cup B^{(j)}(c_+)$. □

The covariance in (b) is with respect to the values of g at the points of the coarse lattice. Exact covariance with respect to an *average* of g cannot be expected. A blockwise rule ϕ with $\phi(1) = 1$ satisfying $\mathfrak{a}(Y^g) = \mathfrak{a}(Y)^{\phi(g)}$ for all g must be a homomorphism from $G^{B_j(y)}$ to G. For simple G, the nontrivial continuous homomorphisms of this kind are, up to automorphisms, evaluations at a single point. This is why the residual gauge group of the variational problem is

$$\mathfrak{G}_0 = \{u : \mathbb{T}_\eta \to G : \ u(y) = 1 \text{ for } y \in \mathbb{T}\},$$

and not the group of transformations with vanishing block averages. Section 22 shows how Bałaban nevertheless arrives at an averaging whose linearization is covariant with respect to averaged gauge transformations.

21.6 Regularity of averaged configurations

Lemma 21.11 (One step)**.** *There are $c_1, C_1 > 0$, depending only on G, such that, if $Y \in \mathfrak{U}_{j-1}(b)$ with $b\delta_j^2 \le c_1$, then $\mathfrak{a}_j(Y) \in \mathfrak{U}_j(b(1 + C_1 b\delta_j^2))$.*

Proof. Write $\delta = \delta_{j-1}$, $\delta' = \delta_j$, and let $c_1 \le r_M/128$ (the constant of the statement). Fix a plaquette p' of $\mathbb{T}^{(j)}$ in the (μ, ν)-plane with corners y, $y_1 = y + \delta' e_\mu$, $y_2 = y_1 + \delta' e_\nu$, $y_3 = y + \delta' e_\nu$. The four averaged bonds of $\partial p'$ depend only on Y on the box $R = B_j(y) \cup B_j(y_1) \cup B_j(y_2) \cup B_j(y_3)$, by Lemma 21.10(c). By Lemma 21.10(b) and the gauge invariance of $|F|$ we may replace Y by $\tilde{Y} = Y^v$, where v is a real gauge transformation with $\tilde{Y}(\Gamma'_{y,x}) = 1$ for all $x \in R$. Here $\Gamma'_{y,x}$ is the coordinate-ordered contour from y to x, for a fixed order of the directions; it stays in R. This single ordered contour is a proof device only.

Size of $\tilde{Y}$. In this proof b is the curvature bound of the statement, $Y \in \mathfrak{U}_{j-1}(b)$, and bonds are written b'. The contour $\Gamma'_{y,x}$ has fewer than $4L$ bonds. For a bond $b' = (x, x')$ of R, the holonomy of $\Gamma'_{y,x} \cup b' \cup (\Gamma'_{y,x'})^{-1}$ is $\tilde{Y}(b')$. By Lemma 21.3(ii) this loop is spanned by at most $4L$ plaquettes of R. Hence $\tilde{Y}(b') = e^{\mathrm{i}\delta A(b')}$ with $|\delta A(b')| \le 4Lb\delta^2 = 4b\delta\delta'$.

Averages. The contours of the families $G(y_i, \cdot)$, the segments $[x, x(c)]$ and the straight contours $[c]$ have at most $2L$ bonds. So the sum of $|\delta A|$ along each of them is at most $\sigma = 8b\delta'^2$,

and $16\sigma \le r_M$. For such a contour γ, $\ell(\tilde Y(\gamma)) = \delta A(\gamma) + O(\sigma^2)$ by Lemma 21.2(c), where $A(\gamma) = \sum_{b\in\gamma} \pm A(b)$ is the signed sum. By Lemma 21.5(c),

$$\ell(\tilde Y(y_i, x)) = \delta \bar A(y_i, x) + O(\sigma^2), \qquad \bar A(y_i, x) = \frac{1}{|G(y_i, x)|} \sum_{\Gamma \in G(y_i,x)} A(\Gamma).$$

By Lemma 21.2(c) applied to the averaged loops and to (21.3),

$$\mathfrak{a}_j(\tilde Y)(c) = e^{\mathrm{i}\delta \bar A_c + O(\sigma^2)}, \qquad \bar A_c = L^{-d} \sum_{x \in B_j(c_-)} \Big(\bar A(c_-, x) + A([x, x(c)]) - \bar A(c_+, x(c)) \Big).$$

Multiplying the four averaged bonds of $\partial p'$, again by Lemma 21.2(c),

$$\ell(\mathfrak{a}_j(\tilde Y)(\partial p')) = \delta(\bar A_{c^{(1)}} + \bar A_{c^{(2)}} - \bar A_{c^{(3)}} - \bar A_{c^{(4)}}) + O(\sigma^2),$$

where $c^{(1)} = (y, y_1)$, $c^{(2)} = (y_1, y_2)$, $c^{(3)} = (y_3, y_2)$, $c^{(4)} = (y, y_3)$. Write the sums over $B_j(y_1)$, $B_j(y_3)$ as sums over $x \in B_j(y)$, using $B_j(y_1) = B_j(y) + \delta' e_\mu$ and $B_j(y_3) = B_j(y) + \delta' e_\nu$. Then the terms $\bar A(y_i, \cdot)$ cancel in pairs: each occurs once as the initial term of one bond and once as the final term of the next. What remains is $\delta L^{-d} \sum_{x \in B_j(y)} A(\partial S_x)$, where S_x is the square of side δ' with corner x in the (μ, ν)-plane. By the abelian Stokes formula $A(\partial S_x) = \sum_{p \subset S_x} A(\partial p)$, a sum over L^2 plaquettes. By Lemma 21.2(c) again, $\delta A(\partial p) = \ell(\tilde Y(\partial p)) + O((b\delta\delta')^2)$, and $|\ell(\tilde Y(\partial p))| \le b\delta^2$. Altogether

$$|\ell(\mathfrak{a}_j(\tilde Y)(\partial p'))| \le L^2 b\delta^2 + CL^2(b\delta\delta')^2 + C\sigma^2 \le b\delta'^2(1 + C_1 b\delta'^2). \qquad \square$$

The cancellation uses the translation invariance of the blocks only; the constant C_1 does not depend on L.

Proposition 21.12 (Regularity of averages [5, Prop. 2])**.** *Possibly after decreasing c_1, so that $2c_1$ satisfies the hypothesis of Lemma 21.11, for $U \in \mathfrak{U}_0(a)$ with $a \le c_1$ and all $j \le k$, $U^{(j)} \in \mathfrak{U}_j(a(1 + Ca)) \subset \mathfrak{U}_j(2a)$.*

Proof. Let $b_0 = a$ and $b_j = b_{j-1}(1 + C_1 b_{j-1}\delta_j^2)$. As long as $b_i \le 2a$ for $i < j$, Lemma 21.11 applies, and

$$b_j \le a \exp\Big(2C_1 a \sum_{i \le j} \delta_i^2\Big) \le a\, e^{3C_1 a},$$

because $\sum_{i\le k} \delta_i^2 \le \sum_{m\ge 0} L^{-2m} \le \frac{4}{3}$. For a small this is at most $a(1 + Ca) \le 2a$, which closes the induction. $\square$

The proposition is local: the bound for a plaquette of $\mathbb{T}^{(j)}$ only uses the curvature of U on the scale-j blocks containing its corners. By Lemma 21.7 and Lemma 21.10(a), all averaged contour variables and averaged loops of $U^{(j-1)}$ are then defined, with $|\ell(\Lambda_{c,x}(U^{(j-1)}))| \le 16a\delta_j^2$. We also need the size of averaged configurations in a gauge in which U is small.

Lemma 21.13 (Averages in a local gauge)**.** *Let $U \in \mathfrak{U}_0(a)$ with $a \le c_1$. Let Λ be a union of blocks $B(y)$, and suppose $U = e^{\mathrm{i}\eta A}$ on the bonds of Λ with $|A| \le s$, where $s + 32a \le \frac{1}{2}c_G$. Then for every j and every bond c of $\mathbb{T}^{(j)}$ with $B^{(j)}(c_-) \cup B^{(j)}(c_+) \subset \Lambda$ we have $U^{(j)}(c) = e^{\mathrm{i}\delta_j A_j(c)}$ with $|A_j(c)| \le s + 32a$.*

Proof. By Lemma 21.10(a) and Proposition 21.12, the exponent in (21.3) is at most $16a\delta_j^2$. By Lemma 21.2(b), $|\ell(U^{(j)}(c))| \le 16a\delta_j^2 + \sum_{b\in[c]} |\ell(U^{(j-1)}(b))|$. The straight contour $[c]$ has L bonds, and their scale-$(j-1)$ blocks lie in Λ. Hence $\max|A_j| \le \max|A_{j-1}| + 16a\delta_j$, and $\sum_j \delta_j \le 2$. $\square$

22 The averaged axial gauge and the gauge-adapted averaging

The linear theory of Parts II and III is formulated in Bałaban's Landau gauge, with the residual gauge transformations $\mathcal{N}(Q'(U))$, i.e. those with vanishing *averages*. The nonlinear averaging of Section 21 is covariant with respect to the *values* of gauge transformations at the points of $\mathbb{T}$. Bałaban bridges the two by an intermediate gauge, the axial gauge, and by a modified averaging [5, Sect. C]. In four dimensions the axial gauge is defined through the averaged contour variables; this is Bałaban's Euclidean invariant axial gauge [18, pp. 253–254]. Bałaban uses it through exponential gauge-fixing functions in the integral. Here it is an algebraic condition, the *averaged axial gauge*. This section develops the construction in a form adapted to our needs. The main results are the correspondence theorem (Theorem 22.5), the expansion of the gauge-adapted averaging (Proposition 22.9), and the properties of its linearization (Proposition 22.10). The last one shows that Parts II and III apply to Bałaban's averaging.

All algebraic statements of this section follow from the transformation law (22.1) of averaged relative contour variables, from the fact that every point lies in exactly one block, and from the reality of the averaged contour variables of real configurations. They use no tree and no multiplicativity of contour variables under concatenation.

22.1 The reference configuration and relative configurations

Fix a real configuration $U_0 \in \mathfrak{U}_0(a)$ on $\mathbb{T}_\eta$ with $a \le c_1$, and put $W_j = U_0^{(j)}$. By Proposition 21.12, $W_j \in \mathfrak{U}_j(2a)$. By Lemma 21.7, with $\varepsilon = 2a\delta_j^2$, all averaged contour variables $W_j(y,x)$, $y \in \mathbb{T}^{(j+1)}$, $x \in B_{j+1}(y)$, are defined and real, their families lie in $\mathcal{D}_{6a\delta_{j+1}^2}$, and $|\ell(W_j(y,x)W_j(\Gamma)^{-1})| \le 6a\delta_{j+1}^2$ for $\Gamma \in G(y,x)$. A *relative configuration* at scale j is a configuration X on $\mathbb{T}^{(j)}$ close to 1; it represents the configuration XW_j (bondwise product). We use the following notation.

- *Relative transports.* $X_W(\Gamma) = (XW_j)(\Gamma)\, W_j(\Gamma)^{-1}$. For $\Gamma = b_1 \cdots b_m$ this is $\prod_i R(W_j(\Gamma_{<i}))X(b_i)$, with $\Gamma_{<i} = b_1 \cdots b_{i-1}$ and $R(Y)Z = YZY^{-1}$.
- *Averaged relative contour variables.* For X at scale $j-1$, $y \in \mathbb{T}^{(j)}$ and $x \in B_j(y)$,

$$\begin{aligned} X_W(y,x) &= (XW_{j-1})(y,x)\, W_{j-1}(y,x)^{-1} \\ &= \mathrm{Av}\,\big(\{X_W(\Gamma)\, W_{j-1}(\Gamma)\, W_{j-1}(y,x)^{-1}\}_{\Gamma \in G(y,x)}\big), \end{aligned}$$

  whenever $(XW_{j-1})(y,x)$ is defined; the second form is Lemma 21.5(b). In particular $1_W(y,x) = 1$, and $X_W(y,x) = X_W(\Gamma)$ if $G(y,x) = \{\Gamma\}$.
- *Relative gauge action.* $X^{(g)} = (XW_j)^g W_j^{-1}$, i.e. $X^{(g)}(b) = g(b_-)X(b)\, R(W_j(b))\big(g(b_+)^{-1}\big)$. Then $(X^{(g)})^{(h)} = X^{(hg)}$, and $X_W^{(g)}(\Gamma) = g(\Gamma_-)\, X_W(\Gamma)\, R(W_j(\Gamma))\big(g(\Gamma_+)^{-1}\big)$. The configurations $1^{(g)}$ are the *pure gauges*.
- *Relative averages.* For X at scale $j-1$ put $\overline{X} = \mathfrak{a}_j(XW_{j-1})W_j^{-1}$. By Lemma 21.10(b), $\overline{X^{(g)}} = \overline{X}^{(g|_{\mathbb{T}^{(j)}})}$ for real g. For a fine relative configuration U' put $U'^{[j]} = (U'U_0)^{(j)}W_j^{-1}$, the iterated relative average.
- *Lie fields.* We write $X = e^{\mathrm{i}\delta_j B}$, and $D_{W_j}\lambda(b) = \delta_j^{-1}(R(W_j(b))\lambda(b_+) - \lambda(b_-))$. At $j = 0$ this is the covariant derivative D_{U_0} of Part II; at $j = k$ it is the unit-lattice derivative D_V with $V = W_k$. To first order, $(e^{\mathrm{i}\delta_j B})^{(e^{\mathrm{i}\lambda})} = e^{\mathrm{i}\delta_j(B - D_{W_j}\lambda)}$. On reversed bonds the relative configuration is $X(-b) = R(W_j(b))^{-1}(X(b)^{-1})$, since $(XW_j)(-b) = (XW_j)(b)^{-1}$; correspondingly $B(-b) = -R(W_j(b))^{-1}B(b)$. Only positively oriented bonds carry independent variables.

Applying (21.2) to $(XW_{j-1})^g = X^{(g)}W_{j-1}$ gives the *transformation law*

$$X_W^{(g)}(y,x) = g(y)\, X_W(y,x)\, R(W_{j-1}(y,x))\,(g(x)^{-1}). \tag{22.1}$$

It holds for real g of any size whenever $(XW_{j-1})(y,x)$ is defined, and for G^c-valued g with $\kappa(g(y)) \le 2$ if the family $\{(XW_{j-1})(\Gamma)\}_{\Gamma\in G(y,x)}$ lies in $\mathcal{D}_{r_M/2}$. It has the form of the law for single contours, with $W_{j-1}(y,x)$ in place of $W_{j-1}(\Gamma)$. The unitarity of $R(W_{j-1}(y,x))$ comes from the reality of $W_{j-1}(y,x)$.

Definition 22.1 (Averaging domain)**.** A relative configuration X at scale $j-1$ lies in the *averaging domain* at scale j if for all $y \in \mathbb{T}^{(j)}$ and $x \in B_j(y)$ the family $\{(XW_{j-1})(\Gamma)\}_{\Gamma\in G(y,x)}$ lies in $\mathcal{D}_{r_M/2}$, and $|\Lambda_{c,x}(XW_{j-1}) - 1| \le \frac{1}{2}c_G$ for all bonds c of $\mathbb{T}^{(j)}$, with both orientations, and $x \in B_j(c_-)$. A fine relative configuration U' lies in the averaging domain up to scale j if, for $i = 1,\dots,j$, $U'^{[i-1]}$ is defined and lies in the averaging domain at scale i.

If X lies in the averaging domain at scale j, then $X_W(y,x)$ and $\overline{X}$ are defined. For real g, $X^{(g)}$ lies in the averaging domain too. For G^c-valued g with $\kappa(g(y)) \le 2$ at the points $y \in \mathbb{T}^{(j)}$, the averaged contour variables of $X^{(g)}W_{j-1}$ and $\overline{X^{(g)}}$ are defined, (22.1) holds, and $\overline{X^{(g)}} = \overline{X}^{(g|_{\mathbb{T}^{(j)}})}$, by Lemma 21.10(b). A real X lies in the averaging domain at scale j if $|\ell((XW_{j-1})(\partial p))| \le \varepsilon_M$ for all plaquettes p, by Lemma 21.7 and $r_M \le \frac{1}{4}c_G$; in particular if $XW_{j-1} \in \mathfrak{U}_{j-1}(b)$ with $8b\delta_j^2 \le r_M$. A real fine U' with $U'U_0 \in \mathfrak{U}_0(c_1)$ lies in the averaging domain up to scale k, since $U'^{[i]}W_i = (U'U_0)^{(i)} \in \mathfrak{U}_i(2c_1)$ and $16c_1 \le r_M$. For real U' and real g, $U'^{(g)}$ lies in the averaging domain up to scale j if and only if U' does, since $(U'^{(g)})^{[i]} = (U'^{[i]})^{(g|_{\mathbb{T}^{(i)}})}$. The conditions are local, by Lemma 21.10(c): a statement about $\overline{X}(c)$ or $X_W(y,x)$ needs them only for the families and loops of the blocks involved, and this is how the domain, and Lemma 21.10(b), are used below.

22.2 The averaged axial gauge

Definition 22.2 (Averaged axial gauge [6, (1.19)], [18, pp. 253–254])**.** A relative configuration X at scale $j-1$ is *axial at scale j* if the averaged contour variables $(XW_{j-1})(y,x)$ are defined and

$$X_W(y,x) = 1, \quad \text{i.e.} \quad (XW_{j-1})(y,x) = W_{j-1}(y,x), \qquad y \in \mathbb{T}^{(j)},\ x \in B_j(y).$$

A fine relative configuration U' belongs to Ax_j if $U'^{[i-1]}$ is defined and axial at scale i for $i = 1,\dots,j$.

In the rest of the note, "axial" means averaged axial. The condition is invariant under the lattice symmetries (Proposition 22.10(g)). The axial gauge of a single coordinate-ordered contour per point is not invariant under the permutations of the axes.

For h on $\mathbb{T}^{(j)}$ let $\mathcal{E}_j(h)$ be the function on $\mathbb{T}^{(j-1)}$ given by

$$\mathcal{E}_j(h)(x) = R\big(W_{j-1}(y,x)\big)^{-1} h(y), \qquad x \in B_j(y),$$

and put $\mathcal{E}^{(j)} = \mathcal{E}_1 \circ \cdots \circ \mathcal{E}_j$. This is the extension of h from $\mathbb{T}^{(j)}$ to $\mathbb{T}_\eta$ by parallel transport with the averaged contour variables of the background. It is multiplicative, $\mathcal{E}^{(j)}(hh') = \mathcal{E}^{(j)}(h)\mathcal{E}^{(j)}(h')$, and it does not change values at the points of $\mathbb{T}^{(j)}$, since $W_{j-1}(y,y) = 1$.

Lemma 22.3 (Axial transformations)**.** *(a) Let X be a relative configuration at scale $j-1$ in the averaging domain at scale j, and h a function on $\mathbb{T}^{(j)}$ with $\kappa(h(y)) \le 2$ for all y, e.g. a real one. There is exactly one G^c-valued g on $\mathbb{T}^{(j-1)}$ with $g|_{\mathbb{T}^{(j)}} = h$ and $X^{(g)}$ axial at scale j, namely*

$$g(x) = R(W_{j-1}(y,x))^{-1}\big(h(y)\, X_W(y,x)\big), \qquad x \in B_j(y).$$

If X is axial at scale j, this is $g = \mathcal{E}_j(h)$. If X and h are real, so is g.

(b) *Let U' be a real fine relative configuration in the averaging domain up to scale j, and h a real function on $\mathbb{T}^{(j)}$. There is exactly one real g on $\mathbb{T}_\eta$ with $g|_{\mathbb{T}^{(j)}} = h$ and $U'^{(g)} \in \mathrm{Ax}_j$. If $U' \in \mathrm{Ax}_j$, this is $g = \mathcal{E}^{(j)}(h)$.*

(c) *Every orbit of $\mathfrak{G}_0$ in $\mathfrak{U}_0(c_1)$ contains exactly one U with $UU_0^{-1} \in \mathrm{Ax}_k$.*

Proof. (a) Every g with $g|_{\mathbb{T}^{(j)}} = h$ satisfies $\kappa(g(y)) \le 2$ at the points of $\mathbb{T}^{(j)}$. By (22.1), $X_W^{(g)}(y,x) = 1$ if and only if

$$R(W_{j-1}(y,x))g(x) = h(y)\, X_W(y,x).$$

Since the blocks partition $\mathbb{T}^{(j-1)}$, this determines g; for $x = y$ it gives $g(y) = h(y)$. For axial X the right side is $h(y)$. (b) For real g, $(U'^{(g)})^{[i-1]} = (U'^{[i-1]})^{(g|_{\mathbb{T}^{(i-1)}})}$, and it lies in the averaging domain at scale i. By (a), its axiality at scale i determines $g|_{\mathbb{T}^{(i-1)}}$ uniquely from $g|_{\mathbb{T}^{(i)}}$, and the result is real. Proceed from $i = j$ down to $i = 1$. (c) is (b) with $j = k$, $h = 1$ and $U' = UU_0^{-1}$, which lies in the averaging domain up to scale k. □

Part (a) needs the averaged contour variables of XW_{j-1} to be defined, i.e. a smallness condition at the scale. For real configurations it holds on the regular configurations of Section 21, and this domain is invariant under real gauge transformations and under the lattice symmetries.

22.3 The gauge-adapted averaging and the correspondence theorem

For a relative configuration X at scale $j-1$ let

$$\tau_X(y) = \exp\Big(\mathrm{i}L^{-d} \sum_{x \in B_j(y)} \ell\big(X_W(y,x)\big)\Big), \qquad y \in \mathbb{T}^{(j)},$$

be the block average of the averaged relative contour variables. Let g_X be the transformation of Lemma 22.3(a) with $h = \tau_X^{-1}$.

Definition 22.4 (Gauge-adapted averaging)**.** The *modified one-step average* is $\tilde{\mathfrak{a}}_j(X) = \overline{X^{(g_X)}} = \overline{X}^{(\tau_X^{-1})}$, i.e.

$$\tilde{\mathfrak{a}}_j(X)(c) = \tau_X(c_-)^{-1}\, \overline{X}(c)\, R(W_j(c))\big(\tau_X(c_+)\big).$$

The *gauge-adapted averaging* is $\mathcal{Q}_0 = \mathrm{id}$, $\mathcal{Q}_j = \tilde{\mathfrak{a}}_j \circ \mathcal{Q}_{j-1}$, and $\mathcal{Q} = \mathcal{Q}_k$. For a fine relative configuration U_1 put $X_i = \mathcal{Q}_i(U_1)$ and define the *correspondence transformation*

$$\mathsf{w}_0(U_1) = 1, \qquad \mathsf{w}_j(U_1) = \mathcal{E}^{(j-1)}(g_{X_{j-1}})\, \mathsf{w}_{j-1}(U_1).$$

Finally let $\mathfrak{m}_j(g)(y) = g(y) \exp\big(\mathrm{i}L^{-d} \sum_{x \in B_j(y)} \ell(g(y)^{-1} R(W_{j-1}(y,x))g(x))\big)$ and $\mathfrak{M}^{(j)} = \mathfrak{m}_j \circ \cdots \circ \mathfrak{m}_1$; these are Bałaban's averages of gauge transformations [5, (78)–(80)], with averaged contour variables.

This is Bałaban's construction [5, (58)–(63), (88)–(92)], with the averaged contour variables of [18, (0.11)] in place of the contour variables: the modified average first moves X to the axial gauge at scale j, with a normalization chosen at the coarse points, and then averages. All maps are defined when the configurations involved are sufficiently close to 1: X lies in the averaging domain, the logarithms in τ_X and $\mathfrak{m}_j$ are defined, and $\kappa(\tau_X(y)) \le 2$. Precise domains are given in Lemma 22.8. We say that the maps are defined at U_1 up to scale j if this holds for $X_0, \dots, X_{j-1}$.

Theorem 22.5 (Correspondence)**.** *Let U_1 be a real fine relative configuration in the averaging domain up to scale j, at which the maps above are defined up to scale j. In (b), (e) and (f) the functions h and g are real.*

(a) $U_1^{(\mathsf{w}_j(U_1))} \in \mathrm{Ax}_j$ *and* $(U_1^{(\mathsf{w}_j(U_1))})^{[j]} = \mathcal{Q}_j(U_1)$.

(b) The transformations g with $U_1^{(g)} \in \mathrm{Ax}_j$ *are exactly* $g = \mathcal{E}^{(j)}(h)\,\mathsf{w}_j(U_1)$*, with h a function on* $\mathbb{T}^{(j)}$.

(c) At the points $y \in \mathbb{T}^{(j)}$*:* $\mathsf{w}_j(U_1)(y) = \tau_{X_{j-1}}(y)^{-1}\tau_{X_{j-2}}(y)^{-1}\cdots\tau_{X_0}(y)^{-1}$.

(d) If X_{j-1} is axial at scale j, then $\mathsf{w}_j(U_1) = \mathsf{w}_{j-1}(U_1)$ *and* $\mathcal{Q}_j(U_1) = \overline{X_{j-1}}$.

(e) (Pure gauges.) *If the maps are defined at* $1^{(g)}$*, then* $\mathcal{Q}_j(1^{(g)}) = 1^{(\mathfrak{M}^{(j)}(g))}$ *and* $\mathsf{w}_j(1^{(g)}) = \mathcal{E}^{(j)}(\mathfrak{M}^{(j)}(g))\,g^{-1}$.

(f) (Normalization.) $\mathfrak{m}_j(\mathcal{E}_j(h)g) = h\,\mathfrak{m}_j(g)$*, and* $\mathfrak{M}^{(j)}(\mathsf{w}_j(U_1)) = 1$.

Proof. (a), (b) By induction on j; the case $j = 0$ is trivial. All configurations and transformations below are real, so the averaging domain is preserved, and the relative averages are covariant. Let $U'' = U_1^{(\mathsf{w}_{j-1})} \in \mathrm{Ax}_{j-1}$, so that $U''^{[j-1]} = X_{j-1} =: X$. By Lemma 22.3(b), the transformations preserving Ax_{j-1} for U'' are $\mathcal{E}^{(j-1)}(h')$ with h' on $\mathbb{T}^{(j-1)}$, and $(U''^{(\mathcal{E}^{(j-1)}(h'))})^{[j-1]} = X^{(h')}$. Hence $U_1^{(g)} \in \mathrm{Ax}_j$ iff $g = \mathcal{E}^{(j-1)}(h')\mathsf{w}_{j-1}$ with $X^{(h')}$ axial at scale j. By Lemma 22.3(a), the latter h' are exactly $\mathcal{E}_j(h'')\,g_X$, h'' on $\mathbb{T}^{(j)}$, because $X^{(g_X)}$ is axial and the transformations preserving its axiality are the $\mathcal{E}_j(h'')$. By multiplicativity $g = \mathcal{E}^{(j)}(h'')\,\mathcal{E}^{(j-1)}(g_X)\,\mathsf{w}_{j-1} = \mathcal{E}^{(j)}(h'')\,\mathsf{w}_j$. For $g = \mathsf{w}_j$,

$$(U_1^{(\mathsf{w}_j)})^{[j]} = \overline{(U''^{(\mathcal{E}^{(j-1)}(g_X))})^{[j-1]}} = \overline{X^{(g_X)}} = \tilde{\mathfrak{a}}_j(X) = \mathcal{Q}_j(U_1).$$

(c) $\mathcal{E}^{(i)}$ does not change values at points of $\mathbb{T}^{(i)} \supset \mathbb{T}^{(j)}$, and $g_{X_i} = \tau_{X_i}^{-1}$ on $\mathbb{T}^{(i+1)}$. (d) If X is axial, then $\tau_X = 1$ and $g_X = \mathcal{E}_j(1) = 1$.

(e) One step: let $X = 1^{(g)}$ at scale $j-1$. By (22.1) and $1_W(y,x) = 1$, $X_W(y,x) = g(y)R(W(y,x))(g(x))^{-1}$. Since 1 is axial and $X^{(g_X)} = 1^{(g_X g)}$, Lemma 22.3(a) gives $g_X g = \mathcal{E}_j(h)$ with $h = (g_X g)|_{\mathbb{T}^{(j)}} = \tau_X^{-1} g$. Using $\ell(k^{-1}) = -\ell(k)$ and $\ell(RkR^{-1}) = R\ell(k)R^{-1}$,

$$\begin{aligned} h(y) &= \exp\Big(- \mathrm{i}L^{-d}\sum_x \ell\big(g(y)R(W(y,x))g(x)^{-1}\big)\Big)g(y) \\ &= g(y)\exp\Big(\mathrm{i}L^{-d}\sum_x \ell\big(g(y)^{-1}R(W(y,x))g(x)\big)\Big) = \mathfrak{m}_j(g)(y). \end{aligned}$$

Hence $\tilde{\mathfrak{a}}_j(1^{(g)}) = \overline{1^{(\mathcal{E}_j(h))}} = 1^{(h)}$ and $g_X = \mathcal{E}_j(\mathfrak{m}_j(g))g^{-1}$. Iterating, with $X_{j-1} = 1^{(\mathfrak{M}^{(j-1)}(g))}$,

$$\mathsf{w}_j(1^{(g)}) = \mathcal{E}^{(j-1)}(\mathcal{E}_j(\mathfrak{M}^{(j)}g)(\mathfrak{M}^{(j-1)}g)^{-1})\,\mathcal{E}^{(j-1)}(\mathfrak{M}^{(j-1)}g)\,g^{-1} = \mathcal{E}^{(j)}(\mathfrak{M}^{(j)}g)\,g^{-1}.$$

(f) For $g' = \mathcal{E}_j(h)g$, $g'(y)^{-1}R(W(y,x))g'(x) = g(y)^{-1}h(y)^{-1}h(y)R(W(y,x))g(x)$, which gives the first identity. For the second, by induction and the first identity, $\mathfrak{M}^{(j)}(\mathsf{w}_j) = \mathfrak{m}_j(g_X\,\mathfrak{M}^{(j-1)}(\mathsf{w}_{j-1})) = \mathfrak{m}_j(g_X)$. By the defining relation of g_X, $g_X(y)^{-1}R(W(y,x))g_X(x) = X_W(y,x)$, so $\mathfrak{m}_j(g_X) = \tau_X^{-1}\tau_X = 1$. □

Part (f) shows that $\mathsf{w}_k(U_1)$ is the transformation singled out in [6, (1.29)] by Bałaban's normalization $\mathfrak{M}^{(k)}(g) = 1$. We never need that normalization explicitly: the transformation is given by the recursion, and its values at the coarse points by (c). Parts (c) and (d) use only the definitions and Lemma 22.3(a), so they hold for complex U_1 as well, whenever the maps are defined at U_1 up to scale j.

22.4 Expansion of the modified average

We now expand the modified average around the trivial relative configuration. At scale j write $\delta = \delta_{j-1}$, $\delta' = \delta_j$, $W = W_{j-1}$. For a relative Lie field B at scale $j-1$ and a contour $\gamma = b_1 \cdots b_m$ on $\mathbb{T}^{(j-1)}$ let

$$B^W(\gamma) = \sum_i R(W(\gamma_{<i}))B(b_i),$$

so that $\ell((e^{\mathrm{i}\delta B})_W(\gamma)) = \delta B^W(\gamma) + O((\delta \sum_i |B(b_i)|)^2)$ by Lemma 21.2(c). The averaged contour variables are linearized by the following transports.

Definition 22.6 (Averaged relative transport)**.** Let W be a configuration on $\mathbb{T}^{(j-1)}$ whose families $\{W(\Gamma)\}_{\Gamma \in G(y,x)}$ lie in $\mathcal{D}_{r_M/2}$ for all $y \in \mathbb{T}^{(j)}$ and $x \in B_j(y)$. For $\Gamma \in G(y,x)$ let $\omega_\Gamma = \omega_\Gamma(W; y, x)$ be the weights of Lemma 21.5(d) at this family. For a relative Lie field B at scale $j-1$ put

$$\lambda_W(y,x)[B] = \sum_{\Gamma \in G(y,x)} \omega_\Gamma \, B^W(\Gamma), \qquad \mathfrak{G}_W B(y,x) = \lambda_W(y,x)[B].$$

The definition applies to any pair of lattices of spacings δ and $L\delta$ with centred blocks; in particular $\mathfrak{G}_W$ is the linearized averaged contour variable of the step in Part VII.

Lemma 22.7 (Averaged relative transports)**.** *Let W be as in Definition 22.6, $y \in \mathbb{T}^{(j)}$ and $x \in B_j(y)$.*

(a) $\lambda_W(y,x)[B] = \delta^{-1} \frac{d}{ds}\big|_{s=0} \ell((e^{\mathrm{i}s\delta B}W)(y,x)\, W(y,x)^{-1})$. *It is linear in B, depends only on B and W on the bonds of the box spanned by y and x, and is holomorphic in W. For real W and real B it is real.*

(b) If $W = 1$ on the bonds of this box, then $\lambda_1(y,x)[B] = |G(y,x)|^{-1} \sum_{\Gamma \in G(y,x)} B(\Gamma)$ is the mean of the line integrals $B(\Gamma) = \sum_{b \in \Gamma} \pm B(b)$.

(c) Let W be real. Then $\sum_\Gamma |\omega_\Gamma| \le 1 + C_M \max_\Gamma |\ell(W(\Gamma)W(y,x)^{-1})|$ and $|\lambda_W(y,x)[B]| \le 2(L-1)\sum_\Gamma |\omega_\Gamma| \max |B|$, the maximum over the bonds of the box. If $W \in \mathfrak{U}_{j-1}(b)$ with $8b\delta_j^2 \le r_M$, then $\sum_\Gamma |\omega_\Gamma| \le 1 + 3C_M b\delta_j^2$.

(d) (Covariance.) *For a real gauge transformation u,*

$$\lambda_{W^u}(y,x)[R(u)B] = R(u(y))\, \lambda_W(y,x)[B], \qquad (R(u)B)(b) = R(u(b_-))B(b).$$

For $r \in \mathfrak{S}_j$, $\lambda_{rW}(y,x)[r_W B] = \lambda_W(r^{-1}y, r^{-1}x)[B]$, where $(r_W B)(b) = B(r^{-1}b)$, with $B(-b) = -R(W(b))^{-1}B(b)$.

(e) (Local comparison.) *Suppose that $W = e^{\mathrm{i}\delta B_W}$ on the bonds of $B_j(y)$, with a $\mathfrak{g}^c$-valued B_W such that $|B_W| \le \varepsilon$ and $L\delta\varepsilon \le c$. Then*

$$|\lambda_W(y,x)[B] - \lambda_1(y,x)[B]| \le CL^2\delta\varepsilon \max |B|,$$

the maximum over the bonds of $B_j(y)$. The constants C, c depend only on G.

(f) (Complex bases.) *Let W be real with $W \in \mathfrak{U}_{j-1}(b)$ and $8b\delta_j^2 \le \frac{1}{4} r_M$, and let $\hat{W} = e^{\mathrm{i}\delta\beta}W$ on the bonds of the box spanned by y and x, with a $\mathfrak{g}^c$-valued β such that $L\delta \max|\beta| \le c$. Then $\hat{W}$ satisfies the hypothesis of Definition 22.6 at (y,x), and*

$$|\lambda_{\hat{W}}(y,x)[B] - \lambda_W(y,x)[B]| \le CL^2\delta \max|\beta| \max|B|.$$

Proof. (a) By Lemma 21.5(b), $(e^{\mathrm{i}s\delta B}W)(y,x)W(y,x)^{-1} = \mathrm{Av}(\{(e^{\mathrm{i}s\delta B})_W(\Gamma)\,U_\Gamma\})$ with $U_\Gamma = W(\Gamma)W(y,x)^{-1}$ and $\mathrm{Av}(\{U_\Gamma\}) = 1$, and $(e^{\mathrm{i}s\delta B})_W(\Gamma) = e^{\mathrm{i}s\delta B^W(\Gamma)+O(s^2)}$. The quantities H_Γ of Lemma 21.5(d) are the same for $\{U_\Gamma\}$ and for $\{W(\Gamma)\}$, so the weights are the same, and the formula follows from Lemma 21.5(d). Locality is clear, holomorphy follows from that of the weights, and reality from Lemma 21.5(d) and the reality of $R(W)$. (b) At $W = 1$ the family is $\{1,\dots,1\}$, so $H_\Gamma = 0$ and $\omega_\Gamma = |G(y,x)|^{-1}I$. (c) The first bound is Lemma 21.5(d). The second holds because $R(W(\cdot))$ is unitary and Γ has at most $2(L-1)$ bonds. For the third, $|\ell(W(\Gamma)W(y,x)^{-1})| \le 3(L-1)^2 b\delta^2 \le 3b\delta_j^2$ by Lemma 21.7(i). (d) $(e^{\mathrm{i}s\delta B}W)^u = e^{\mathrm{i}s\delta R(u)B}W^u$, and $r(e^{\mathrm{i}s\delta B}W) = e^{\mathrm{i}s\delta r_W B}\,rW$. Apply (a), (21.2), $(rY)(y,x) = Y(r^{-1}y, r^{-1}x)$, and $\ell(uhu^{-1}) = u\ell(h)u^{-1}$.

(e) The contours γ of the families have at most $2L$ bonds, so $|\ell(W(\gamma'))| \le 4L\delta\varepsilon$ for all initial pieces γ', by Lemma 21.2(c), and $\|R(W(\gamma')) - 1\| \le e^{8L\delta\varepsilon} - 1 \le 16L\delta\varepsilon$ by (12.1). Hence $|B^W(\Gamma) - B(\Gamma)| \le 32L^2\delta\varepsilon \max|B|$, and $|B^W(\Gamma)| \le 4L\max|B|$. The family $\{W(\Gamma)\}$ has diameter at most $CL\delta\varepsilon < r_M/2$, so by Lemma 21.5(a),(d), $|H_\Gamma| \le CL\delta\varepsilon$ and $|\omega_\Gamma - |G(y,x)|^{-1}I| \le |G(y,x)|^{-1}CL\delta\varepsilon$. Therefore

$$\begin{aligned}|\lambda_W(y,x)[B] - \lambda_1(y,x)[B]| &\le \sum_\Gamma |\omega_\Gamma - |G(y,x)|^{-1}I|\,|B^W(\Gamma)| + \max_\Gamma |B^W(\Gamma) - B(\Gamma)| \\ &\le CL^2\delta\varepsilon \max|B|.\end{aligned}$$

(f) Let $c_0 > 0$ be small, depending only on G, and take $c \le \frac14 c_0$ in the hypothesis. Put $\beta_* = \max|\beta|$, $R = c_0/(2L\delta\beta_*) \ge 2$, and $W_t = e^{\mathrm{i}t\delta\beta}W$ for complex t with $|t| \le R$. Then $|t\delta\beta| \le c_0/(2L)$ on every bond, so $W_t = e^{\mathrm{i}Z_t}W$ with $64L|Z_t| \le 32c_0 \le r_M$ for c_0 small. By Lemma 21.7(iv), with $\varepsilon = b\delta^2$ and $8L^2\varepsilon = 8b\delta_j^2 \le \frac14 r_M$, the families $\{W_t(\Gamma)\}$ lie in $\mathcal{D}_{r_M}$; by Lemma 21.7(i),(iv) and Lemma 21.5(a),(e) their diameter is at most $\frac38 r_M + Cc_0 < \frac12 r_M$ for c_0 small. So $\lambda_{W_t}(y,x)[B]$ is defined, and it is holomorphic in t by (a). On $|t| \le R$, $\sum_\Gamma |\omega_\Gamma| \le 1 + 4C_M r_M$ by Lemma 21.5(a),(d), and every transport $R(W_t(\gamma'))$ along an initial piece of a contour has norm at most $e^{4L|Z_t|} \le 2$, because $R(W(\gamma'))$ is unitary. Hence $|\lambda_{W_t}(y,x)[B]| \le 4(1+4C_M r_M)L\max|B|$ there. The Cauchy estimate on the disc $|t| \le R$ gives $|\frac{d}{dt}\lambda_{W_t}| \le 8(1+4C_M r_M)L\max|B|/R$ for $|t| \le 1$, and integrating from $t = 0$ to $t = 1$ gives the claim, since $1/R = 2L\delta\beta_*/c_0$. □

Define the *transported tube average*

$$(q_jB)(c) = L^{-d}\sum_{x\in B_j(c_-)} L^{-1}R(W(c_-,x))B^W([x,x(c)])$$

[5, (125)] and the *contour average* $(\gamma_jB)(y) = L^{-d}\sum_{x\in B_j(y)} L^{-1}\lambda_W(y,x)[B]$. The transports $R(W(c_-,x))$ are unitary, since $W(c_-,x)$ is real.

Lemma 22.8 (One-step expansion)**.** *There are $c_2, C > 0$, depending only on d and G, such that the following holds for $a \le c_2$ and complex B with $|B| \le c_2/\delta'$ on the bonds of $B_j(c_-)\cup B_j(c_+)$ (respectively $B_j(y)$). Put $X = e^{\mathrm{i}\delta B}$. Then the conditions of Definition 22.1 hold on the relevant blocks, and*

$$\tilde{\mathfrak{a}}_j(X)(c) = e^{\mathrm{i}\delta'(l_jB + c_j(B))(c)}, \qquad \tau_X(y) = e^{\mathrm{i}\delta'(\gamma_jB + c_j^\tau(B))(y)},$$

where $l_j = q_j + r_j$ is linear and c_j, c_j^τ are analytic, with:

(a) $(l_jB)(c)$ and $c_j(B)(c)$ depend only on B on the bonds of $B_j(c_-)\cup B_j(c_+)$; $\gamma_jB(y)$ and $c_j^\tau(B)(y)$ only on B on the bonds of $B_j(y)$;

(b) $|(q_jB)(c)|^2 \le L^{-d}\sum_x L^{-1}\sum_{b\in[x,x(c)]}|B(b)|^2$, $|(r_jB)(c)| \le Ca\delta'^2\max|B|$, and $|\gamma_jB| \le d\max|B|$;

(c) $|c_j(B)|+|c_j^\tau(B)| \le C\delta' \max|B|^2$ *and* $|c_j(B)-c_j(\tilde B)| \le C\delta'(\max|B|+\max|\tilde B|)\max|B-\tilde B|$ *for* $\max|B|, \max|\tilde B| \le c_2/(2\delta')$*, the maxima being over the relevant bonds;*

(d) if $W=1$ *on the relevant bonds, then* $r_j=0$ *and* q_j *is the tube average of Part I at scale* j*,* $(q_jB)(c) = L^{-d-1}\sum_{x\in B_j(c_-)} B([x,x(c)])$*;*

(e) real B *gives* $\mathfrak{g}$*-valued* l_jB*,* $c_j(B)$*,* γ_jB*,* $c_j^\tau(B)$*;*

(f) (The plain average.) *the relative average is*

$$\overline{X}(c) = e^{\mathrm{i}\delta'(p_jB+c_j^p(B))(c)}, \qquad p_j = l_j - \delta_j D_{W_j}\gamma_j,$$

where c_j^p *is analytic, real for real* B*, depends only on* B *on the bonds of* $B_j(c_-)\cup B_j(c_+)$*, and satisfies the bounds of (c). If* $W=1$ *on the relevant bonds, then* $p_j = q_j - \delta_j D_1\gamma_j$*, and the coefficient of* $B(b_0(c))$ *in* $(p_jB)(c)$ *is* L^{-d}*, where* $b_0(c)$ *is the middle bond of* $[c]$*. Equivalently,* $\delta B(b_0(c))$ *enters* $\ell(\overline{X}(c))$ *to first order with the coefficient* L^{1-d}*. For every* W*, the coefficient of* $B(b_0(c))$ *in* $(q_jB)(c)$ *is* $L^{-d}R(W([c_-,b_0(c)_-]))$*,* $\gamma_jB(c_\pm)$ *does not depend on* $B(b_0(c))$*, and* $(p_jB)(c')$ *does not depend on* $B(b_0(c))$ *unless* $c'=\pm c$*.*

Proof. Analyticity. The contours of the families, the segments $[x,x(c)]$ and $[c]$ have at most $2L$ bonds, so $\delta\sum|B| \le 2c_2$ along each of them. Since W is real, $R(W(\cdot))$ is unitary on $\mathfrak{g}^c$, and every relative transport $X_W(\gamma)$ along such a contour is a product of at most $2L$ factors $e^{\mathrm{i}\delta R(W(\cdot))B(b)}$, with $|\ell(X_W(\gamma))| \le 3c_2$. The families $\{W(\Gamma)\}$ lie in $\mathcal{D}_{6a\delta'^2}$. Hence the families $\{(XW)(\Gamma)\} = \{X_W(\Gamma)W(\Gamma)\}$ lie in $\mathcal{D}_{C(c_2+a\delta'^2)} \subset \mathcal{D}_{r_M/2}$, the averaged contour variables $(XW)(y,x)$ are defined and holomorphic in B, and $|\ell(X_W(y,x))| \le 3C_Mc_2$ by Lemma 21.5(e). By Lemma 21.10(a) the background loops $h_x = \Lambda_{c,x}(W)$ satisfy $|\ell(h_x)| \le 16a\delta'^2$. Hence the arguments of all logarithms in τ_X, $\overline{X}$ and $\tilde{\mathfrak{a}}_j(X)$, including the averaged loops of XW, are within $C(c_2+a\delta'^2)$ of 1, and $\kappa(\tau_X(y)) \le 2$. So the conditions of Definition 22.1 hold on the relevant blocks, g_X is defined there, and $\tilde{\mathfrak{a}}_j(X)(c) = \overline{X^{(g_X)}}(c) = \overline{X}^{(\tau_X^{-1})}(c)$ by Lemma 21.10(b). For c_2 small, these are analytic functions of the finitely many variables $B(b)$, bounded by C in the sense that $|\ell(\tilde{\mathfrak{a}}_j(X)(c))| \le C(c_2+a\delta'^2)$.

Remainders. Let $f(B) = \delta'^{-1}\ell(\tilde{\mathfrak{a}}_j(e^{\mathrm{i}\delta B})(c)) - (l_jB)(c)$, with l_j the derivative at $B=0$. Then $f(0)=0$ and $df(0)=0$, and $t\mapsto f(tB)$ is analytic for $|t| \le c_2/(\delta'\max|B|)$ with $|f| \le C/\delta'$ there. Cauchy's estimate for the second Taylor coefficient gives $|f(B)| \le C\delta'\max|B|^2$ for $\max|B| \le c_2/(2\delta')$. The Lipschitz bound follows from the Cauchy estimate for df. The same argument applies to τ_X, and to $\overline{X}$ in (f).

The linear part. Let $X = e^{\mathrm{i}s\delta B}$ and expand to first order in s. Write $H_x = \ell(h_x)$ and $E_{\mathrm{loop}} = L^{-d}\sum_x H_x$, so that $|H_x|, |E_{\mathrm{loop}}| \le 16a\delta'^2$ and $W_j(c) = e^{\mathrm{i}E_{\mathrm{loop}}}W([c])$. We have $(XW)(c_-,x) = X_W(c_-,x)W(c_-,x)$ and $(XW)(x(c),c_+) = W(x(c),c_+)X_W(c_+,x(c))^{-1}$, and $W(c_-,x)W([x,x(c)])W(x(c),c_+) = h_xW([c])$. Moving the background factors to the right,

$$\begin{aligned}\Lambda_{c,x}(XW) = X_W(c_-,x)\; &R(W(c_-,x))(X_W([x,x(c)]))\\ &\times R(h_xW([c]))(X_W(c_+,x(c))^{-1})\; R(h_x)(X_W([c])^{-1})\; h_x.\end{aligned}$$

By Lemma 22.7(a), $\ell(X_W(y,x)) = s\delta\lambda_W(y,x)[B] + O(s^2)$. Hence $\ell(\Lambda_{c,x}(XW)) = H_x + s\delta\,\mathfrak{f}(\operatorname{ad}H_x)\Xi_x + O(s^2)$, with $\mathfrak{f}$ as in Section 21 and

$$\begin{aligned}\Xi_x = \lambda_W(c_-,x)[B] &+ R(W(c_-,x))B^W([x,x(c)])\\ &- R(h_xW([c]))\lambda_W(c_+,x(c))[B] - R(h_x)B^W([c]).\end{aligned}$$

Let $\mathfrak{e}(\operatorname{ad}E_{\mathrm{loop}})$ be the derivative of $Z\mapsto \ell(e^{\mathrm{i}(E_{\mathrm{loop}}+Z)}e^{-\mathrm{i}E_{\mathrm{loop}}})$ at $Z=0$; the function $\mathfrak{e}$ is entire with $\mathfrak{e}(0)=1$. Then $\ell(\overline{X}(c)) = s\delta\big[\mathfrak{e}(\operatorname{ad}E_{\mathrm{loop}})\Xi + R(e^{\mathrm{i}E_{\mathrm{loop}}})B^W([c])\big] + O(s^2)$, with $\Xi =$

$L^{-d}\sum_x \mathfrak{f}(\operatorname{ad} H_x)\Xi_x$. Moreover $\ell(\tau_X(y)) = s\delta L^{-d}\sum_x \lambda_W(y,x)[B] + O(s^2)$. Since $\delta/\delta' = L^{-1}$,

$$l_j B(c) = L^{-1}\Big\{\mathfrak{e}(\operatorname{ad} E_{\mathrm{loop}})\Xi + R(e^{\mathrm{i}E_{\mathrm{loop}}})B^W([c]) \\ - L^{-d}\sum_x \lambda_W(c_-,x)[B] + R(W_j(c))L^{-d}\sum_x \lambda_W(c_+,x(c))[B]\Big\}.$$

If all factors $\mathfrak{e},\mathfrak{f}$ were 1, $R(h_x) = R(e^{\mathrm{i}E_{\mathrm{loop}}})$ and $R(h_x W([c])) = R(W_j(c))$, the averaged relative transports would cancel exactly, those at c_- against $\tau_X(c_-)$ and those at c_+ against $\tau_X(c_+)$, and only $q_j B$ would remain. Precisely,

$$L\, r_j B(c) = (\mathfrak{e}(\operatorname{ad} E_{\mathrm{loop}}) - 1)\Xi + L^{-d}\sum_x (\mathfrak{f}(\operatorname{ad} H_x) - 1)\Xi_x \\ + L^{-d}\sum_x (R(e^{\mathrm{i}E_{\mathrm{loop}}}) - R(h_x))B^W([c]) \\ + L^{-d}\sum_x (R(W_j(c)) - R(h_x W([c])))\lambda_W(c_+,x(c))[B].$$

The differences are controlled by $|\mathfrak{f}(\operatorname{ad} H_x) - 1| \le C|H_x|$, $|\mathfrak{e}(\operatorname{ad} E_{\mathrm{loop}}) - 1| \le C|E_{\mathrm{loop}}|$, and $|R(e^{\mathrm{i}E_{\mathrm{loop}}}) - R(h_x)| = |R(W_j(c)) - R(h_x W([c]))| \le C(|E_{\mathrm{loop}}| + |H_x|)$, since $R(W([c]))$ is unitary. By Lemma 22.7(c), $|\lambda_W(y,x)[B]| \le 2(L-1)(1+Ca\delta'^2)\max|B|$, so $|\Xi_x| \le (4(L-1)(1+Ca\delta'^2) + 2L)\max|B| \le 8L\max|B|$ for c_2 small. This gives the bound on r_j. The weights ω_Γ enter it only through $\sum_\Gamma |\omega_\Gamma| \le 1 + Ca\delta'^2$; their deviation from the mean does not enter r_j at leading order. The bound on q_j is the Cauchy–Schwarz inequality, since all transports in q_j are unitary. The bound on γ_j follows from Lemma 22.7(c): $|\gamma_j B| \le 2L^{-1}(L-1)(1+Ca\delta'^2)\max|B| \le 2(1+Ca\delta'^2)\max|B| \le d\max|B|$. If $W = 1$ on the relevant bonds, then $h_x = \Lambda_{c,x}(1) = 1$ and $E_{\mathrm{loop}} = 0$, so $r_j = 0$; this is (d). (e) follows from the formulas.

The constants. The constants c_2 and C depend only on d and G; they do not depend on L. Indeed, L enters the proof only through the number of bonds of the contours and segments, at most $2L$, and through the averages $L^{-d}\sum_x$. With $|B| \le c_2/\delta'$ and $\delta = \delta'/L$, every contour carries $\delta\sum|B| \le 2c_2$, so the bounds of *Analyticity* and *Remainders* are uniform in L. In *The linear part*, the bounds $|H_x|, |E_{\mathrm{loop}}| \le 16a\delta'^2$ do not contain L, and the factor L in $|\lambda_W(y,x)[B]| \le 2(L-1)(1+Ca\delta'^2)\max|B|$ and in $|\Xi_x| \le 8L\max|B|$ is compensated by the factor $\delta/\delta' = L^{-1}$ in l_j, r_j and γ_j. The remaining constants come from Lemmas 21.2, 21.5 and 21.10(a), and depend only on G (and d).

(f) By Definition 22.4, $\overline{X}(c) = \tau_X(c_-)\,\tilde{\mathfrak{a}}_j(X)(c)\,R(W_j(c))(\tau_X(c_+)^{-1})$, a product of three factors $e^{\mathrm{i}\delta' Z}$ with $|Z| \le C\max|B|$. Their linear parts add, and $(\gamma_j B)(c_-) - R(W_j(c))(\gamma_j B)(c_+) = -\delta' D_{W_j}\gamma_j B(c)$. So the derivative of $\delta'^{-1}\ell(\overline{X}(c))$ at $B = 0$ is $p_j B(c)$, and the remainder c_j^p is bounded as in *Remainders*. Locality and reality are clear. The straight contour $[c]$ consists of L bonds, and $b_0(c)$ is the only one that joins $B_j(c_-)$ to $B_j(c_+)$. The segment $[x, x(c)]$ contains $b_0(c)$ if and only if x lies on the axis of c, which holds for L points $x \in B_j(c_-)$. For such x, $G(c_-,x)$ consists of the straight contour $[c_-,x]$, so $W(c_-,x) = W([c_-,x])$, and the transport of $B(b_0(c))$ in $R(W(c_-,x))B^W([x,x(c)])$ is $R(W([c_-,b_0(c)_-]))$ for every W. Hence the coefficient of $B(b_0(c))$ in $(q_j B)(c)$ is $L^{-d-1}\cdot L\,R(W([c_-,b_0(c)_-]))$. The contour averages $\gamma_j B(c_\pm)$ involve only bonds of $B_j(c_\pm)$, so they do not contain $B(b_0(c))$. At $W = 1$ we have $W_j = 1$, $D_{W_j} = D_1$ and $r_j = 0$, so the coefficient in $(p_j B)(c)$ is L^{-d}, and that of $\delta B(b_0(c))$ in $\ell(\overline{X}(c)) = \delta'(p_j B)(c) + O(B^2)$ is $L\cdot L^{-d}$. Finally, $(p_j B)(c')$ depends only on the bonds of $B_j(c'_-)\cup B_j(c'_+)$. Since the blocks partition $\mathbb{T}^{(j-1)}$, $b_0(c)$ is such a bond only if $\{c'_-, c'_+\} = \{c_-, c_+\}$. □

Other references. The proofs of Lemmas 22.7 and 22.8 use only that $W = W_{j-1}$ is real with $W \in \mathfrak{U}_{j-1}(2a)$, that $W_j = \mathfrak{a}_j(W)$, and, in Lemma 22.8, that $a\delta'^2 \le c_2$, which keeps the background families in $\mathcal{D}_{6a\delta'^2}$ and the background loops at $|\ell(h_x)| \le 16a\delta'^2$ small. So both

lemmas hold for every such W with $a\delta'^2 \le c_2$, on any pair of lattices of spacings δ and $\delta' = L\delta$ with centred blocks, with $W_j := \mathfrak{a}_j(W)$. For $\delta' \le 1$ the condition $a\delta'^2 \le c_2$ follows from $a \le c_2$. Part VII uses the lemmas in this form for one step of the renormalization group, with $\delta' = 1$.

At $W = 1$, $\gamma_j B(y) = L^{-d-1} \sum_{x \in B_j(y)} \mathfrak{G}_1 B(y, x)$. So the plain average linearizes to the tube average minus the coarse gradient of an average of $\mathfrak{G}_1 B$, while the gauge-adapted average linearizes to the tube average itself.

22.5 Expansion of the gauge-adapted averaging

For a fine field h and a set $S \subset \mathbb{T}_\eta$ let $\|h\|_S$ be the ℓ^2 norm over the bonds starting in S, as in Part III. For a bond c of $\mathbb{T}$ put $S_c = B(c_-) \cup B(c_+)$.

Proposition 22.9 (Expansion of $\mathcal{Q}$)**.** *There are $c_3, C > 0$ such that for $a \le c_3$ and complex fine Lie fields A with $|A|_\infty \le c_3$ the following holds. For $j \le k$, $\mathcal{Q}_j(e^{\mathrm{i}\eta A}) = e^{\mathrm{i}\delta_j B_j}$ with*

$$B_j = \mathbf{Q}_j A + C_j^{\mathrm{av}}(A), \qquad \mathbf{Q}_j = l_j \cdots l_1,$$

where C_j^{av} is analytic, real for real A, and:

(a) $|B_j|_\infty \le 2|A|_\infty$, and $B_j(c)$ depends only on A on the bonds of $B^{(j)}(c_-) \cup B^{(j)}(c_+)$.

(b) $|C_j^{\mathrm{av}}(A)|_\infty \le C|A|_\infty^2$ and $|C_j^{\mathrm{av}}(A) - C_j^{\mathrm{av}}(\tilde A)|_\infty \le C(|A|_\infty + |\tilde A|_\infty)|A - \tilde A|_\infty$.

(c) (Kernels.) *For $j = k$ the linear maps $\mathbf{Q}_k$ and $dC_k^{\mathrm{av}}(A)$ have kernels, $\mathbf{Q}_k h(c) = \eta^d \sum_b K(c,b)h(b)$ and $dC_k^{\mathrm{av}}(A)h(c) = \eta^d \sum_b K_A(c,b)h(b)$, supported on bonds b starting in S_c, with*

$$|K(c,b)| \le 1 + Ca, \qquad |K_A(c,b)| \le C|A|_\infty.$$

Moreover $|d^2 C_k^{\mathrm{av}}(A)[h,h](c)| \le C\|h\|_{S_c}^2$. Consequently $|\mathbf{Q}_k h(c)| \le C\|h\|_{S_c}$ and $|dC_k^{\mathrm{av}}(A)h(c)| \le C|A|_\infty \|h\|_{S_c}$, and the transposes satisfy $|\mathbf{Q}_k^T \omega|_\infty \le C|\omega|_\infty$ and $|dC_k^{\mathrm{av}}(A)^T \omega|_\infty \le C|A|_\infty |\omega|_\infty$.

(d) The values of the correspondence transformation at the points of $\mathbb{T}$ are $\mathsf{w}_k(e^{\mathrm{i}\eta A})(y) = \exp\big(-\mathrm{i}\,\mathbf{\Gamma}_k A(y) + \mathrm{i} C_k^{\mathsf{w}}(A)(y)\big)$, with

$$\mathbf{\Gamma}_k A = \sum_{i=0}^{k-1} \delta_{i+1}\,\gamma_{i+1} \mathbf{Q}_i A, \qquad |\mathbf{\Gamma}_k A| \le C|A|_\infty, \qquad |C_k^{\mathsf{w}}(A)| \le C|A|_\infty^2,$$

and the analogous Lipschitz bound. At every fine point, $|\ell(\mathsf{w}_k(e^{\mathrm{i}\eta A}))| \le C|A|_\infty$.

Proof. (a), (b) By Lemma 22.8, $B_j = l_j B_{j-1} + c_j(B_{j-1})$ with $|l_j B|_\infty \le (1 + Ca\delta_j^2)|B|_\infty$ and $|c_j(B)|_\infty \le C\delta_j |B|_\infty^2$. By induction, as long as $|B_i|_\infty \le 2|A|_\infty$ for $i < j$,

$$|B_j|_\infty \le |A|_\infty \prod_{i \le j} (1 + Ca\delta_i^2 + 2C\delta_i |A|_\infty) \le |A|_\infty e^{C(a + |A|_\infty)} \le 2|A|_\infty,$$

using $\sum_i \delta_i \le 2$. The hypothesis $|B_{j-1}| \le c_2/\delta_j$ of Lemma 22.8 holds since $\delta_j \le 1$. Moreover $C_j^{\mathrm{av}}(A) = \sum_{i \le j} l_j \cdots l_{i+1} c_i(B_{i-1})$, so $|C_j^{\mathrm{av}}(A)| \le 2\sum_i C\delta_i (2|A|)^2 \le C|A|^2$. The Lipschitz bound is proved in the same way from the Lipschitz bound in Lemma 22.8(c). Locality follows from Lemma 22.8(a) by induction: the bonds of $B_j(c_-) \cup B_j(c_+)$ have both endpoints in this set, and their scale-$(j-1)$ blocks lie in $B^{(j)}(c_-) \cup B^{(j)}(c_+)$, by the nesting of the blocks.

(c) For $i \le j$, call a linear map K from scale-i fields to scale-j fields *of type M* if $Kv(c') = \delta_i^d \sum_{b'} K(c', b')v(b')$ with $|K(c', b')| \le M\delta_j^{-d}$, and $K(c', b') = 0$ unless both endpoints of b' lie in the union of the scale-j blocks of c'_- and c'_+. This support is preserved under composition, because the scale-j blocks are unions of scale-m blocks for $m \le j$. At $W = 1$ the

composition of tube averages from scale i to scale j is the tube average from scale i to scale j (proof of Proposition 22.10(c)). By Lemma 2.6(a), after rescaling, it is of type 1: its segments $[x, x(c')]$ lie in the two blocks, so both endpoints of each of their bonds do.

Transports. For real W, $q_j \cdots q_{i+1}$ is a sum over chains of bonds of the $W = 1$ weights times products of unitary transports; the averaged contour variables $W(c_-, x)$ are real. Hence its kernel is bounded in norm by the $W = 1$ kernel, and it is of type 1.

Insertions. Write $l_m = q_m + r_m$. By the formula for r_m in the proof of Lemma 22.8, $r_m v(c')$ is a combination of at most C_L values $v(b'')$, with b'' a bond of $B_m(c'_-) \cup B_m(c'_+)$ and coefficients bounded by $Ca\delta_m^2$. Indeed, a value $v(b'')$ enters $\lambda_W(y, x)[v]$ through at most 24 contours of $G(y, x)$, with total weight at most $\sum_\Gamma |\omega_\Gamma| \le 2$. Expand $l_j \cdots l_{i+1}$ as a sum over the sets I of scales at which r_m is chosen. For $I = \{m\}$ the term is $(q_j \cdots q_{m+1})\, r_m\, (q_{m-1} \cdots q_{i+1})$. In the normalization above, the left factor has kernel at most $\delta_m^d \delta_j^{-d}$ per scale-m bond, and the right factor at most $\delta_i^d \delta_{m-1}^{-d}$ per scale-i bond. For fixed c', b' at most C_L scale-m bonds contribute, so this term is of type $C_L a \delta_m^2 L^d$. For several insertions, write the term as $A_0 r_{m_1} A_1 \cdots r_{m_p} A_p$, with $m_1 > \cdots > m_p$ and transported tube averages A_q of type 1, and compose from the right. If T is of type M', with the support property above, then $A\, r_m\, T$ is of type $C_L a \delta_m^2 L^d M'$, by the count for one insertion: for fixed output and input bonds at most C_L (indeed $(2d)^2$) pairs of intermediate bonds of scales m and $m-1$ contribute. Hence the term is of type $\prod_{m \in I} C_L a \delta_m^2 L^d$. Each counting factor is attached to one insertion; the composition of the factors A_q alone needs none, by the exactness at $W = 1$. Altogether $l_j \cdots l_{i+1}$ is of type $\prod_m (1 + C_L a \delta_m^2) \le 1 + Ca$. For $i = 0$ and $j = k$ this is the bound on K.

First derivatives. Differentiating the recursion gives $dB_j = (l_j + dc_j(B_{j-1})) \, dB_{j-1}$. By the Cauchy estimate in Lemma 22.8(c), $dc_j(B)v(c')$ is a combination of at most C_L values of v, with coefficients bounded by $C\delta_j |B|_\infty \le 2C\delta_j |A|_\infty$. So dc_j plays the role of r_j, with $a\delta_j^2$ replaced by $C\delta_j |A|_\infty$. The same expansion shows that every product $\prod_m (l_m + dc_m)$ is of type $1 + C(a + |A|_\infty)$. The terms with at least one insertion of some dc_m form a map of type $C|A|_\infty$; for $i = 0$, $j = k$ they make up $dC_k^{\mathrm{av}}(A)$. This gives the bound on K_A.

Second derivatives. The recursion gives

$$d^2 B_j[h,h] = (l_j + dc_j)\, d^2 B_{j-1}[h,h] + d^2 c_j(B_{j-1})[dB_{j-1}h, dB_{j-1}h].$$

Since dB_{j-1} is of type C from scale 0, the Cauchy–Schwarz inequality gives $|dB_{j-1}h(b')| \le C\delta_{j-1}^{-d/2} \|h\|_{S_{b'}^{(j-1)}}$, where $S_{b'}^{(j-1)} = B^{(j-1)}(b'_-) \cup B^{(j-1)}(b'_+)$. By the Cauchy estimate, the source term at a scale-j bond c' is therefore bounded by $C_L \delta_j \delta_{j-1}^{-d} \|h\|^2_{S_{c'}^{(j)}}$. It is propagated to $\mathbb{T}$ by a map of type C from scale j, so its contribution at c is at most

$$C \sum_{c'} \delta_j^d\, C_L \delta_j \delta_{j-1}^{-d} \|h\|^2_{S_{c'}^{(j)}} \le C_L \delta_j \|h\|^2_{S_c},$$

because each fine bond lies in a bounded number of the sets $S_{c'}^{(j)}$, all of which lie in S_c. Summing over j gives the bound on $d^2 C_k^{\mathrm{av}}$.

The remaining consequences follow from the kernel bounds. The bounds on $\mathbf{Q}_k h(c)$ and $dC_k^{\mathrm{av}}(A)h(c)$ follow by the Cauchy–Schwarz inequality, since S_c has volume 2. The bounds on the transposes follow by summing the kernel bounds over c, since each fine bond lies in at most $2d$ sets S_c.

(d) By Theorem 22.5(c) and Lemma 22.8, $\ell(\tau_{X_i}) = \delta_{i+1}(\gamma_{i+1} B_i + c_{i+1}^\tau(B_i))$, of size at most $C\delta_{i+1}|A|$. Lemma 21.2(c) applied to the product gives the formula and the bounds, since $\sum_i \delta_{i+1} \le 2$. At a fine point x, $\mathsf{w}_k = \prod_i \mathcal{E}^{(i)}(g_{X_i})$. For $x \in B_{i+1}(y)$, $g_{X_i}(x) = R(W_i(y,x))^{-1} (\tau_{X_i}(y)^{-1}(X_i)_W(y,x))$, and $|\ell((X_i)_W(y,x))| \le C_M \max_\Gamma |\ell((X_i)_W(\Gamma))| \le C\delta_{i+1}|A|$ by Lemma 21.5(e). So $g_{X_i}(x)$ has logarithm at most $C\delta_{i+1}|A|$. Conjugation by $R(W)$ does not change the norm. □

22.6 The linearized averages

Let $q'_j\lambda(y) = L^{-d}\sum_{x\in B_j(y)} R(W_{j-1}(y,x))\lambda(x)$ be the transported block average of scalar fields, and put

$$Q'_k(U_0) = q'_k\cdots q'_1, \qquad \mathbf{Q}_k(U_0) = l_k\cdots l_1.$$

Explicitly, $(Q'_k\lambda)(y) = \eta^d\sum_{x\in B(y)} T(y,x)\lambda(x)$, where

$$T(y,x) = R(W_{k-1}(y,x_{k-1}))\,R(W_{k-2}(x_{k-1},x_{k-2}))\cdots R(W_0(x_1,x)),$$

and $x = x_0 \in B_1(x_1)$, $x_1 \in B_2(x_2)$, …, $x_{k-1} \in B_k(y)$ is the chain of x (Section 21). This is the transport along the chain of averaged contour variables, with the averaged backgrounds at each scale. In particular $Q'^T_k = \mathcal{E}^{(k)}$. These are Bałaban's linearized averages [7, (3.15)–(3.19)], with averaged contour variables.

Proposition 22.10 (Linearized averages)**.** *Let $U_0 \in \mathfrak{U}_0(a)$ be real with $a \le c_3$.*

(a) $Q'_kQ'^T_k = I$.

(b) (Exact covariance.) $\mathbf{Q}_k(U_0)D_{U_0} = D_{W_k}Q'_k(U_0)$. *In the notation of Part II, the defect vanishes: $E = 0$, with $V = W_k$.*

(c) At $U_0 = 1$: $Q'_k = Q_0$ and $\mathbf{Q}_k = \mathbf{Q}$, the operators of Part I.

(d) (Locality.) $(\mathbf{Q}_kA)(c)$ *depends only on A and U_0 on the bonds of $B(c_-)\cup B(c_+)$, and $(Q'_k\lambda)(y)$ only on λ and U_0 on $B(y)$.*

(e) (Closeness.) *Suppose that on a cube containing $B(c_-)\cup B(c_+)$ we have $U_0 = e^{\mathrm{i}\eta A_0}$ (in some real gauge) with $|A_0| \le \varepsilon$, and $a \le \varepsilon$, where $\varepsilon \le c_3$. Then, computed in this gauge, the kernels satisfy $|K_{\mathbf{Q}_k(U_0)}(c,b) - K_{\mathbf{Q}}(c,b)| \le C\varepsilon$ and $|K_{Q'_k(U_0)}(y,x) - K_{Q_0}(y,x)| \le C\varepsilon$. Hence the differences are bounded by $C\varepsilon$ from sup norms to sup norms and in the block norms of Lemma 12.5(b), and so are their transposes.*

(f) (Covariance.) *For every real gauge transformation u on $\mathbb{T}_\eta$: $W_k(U_0^u) = W_k(U_0)^{u|_{\mathbb{T}}}$, $Q'_k(U_0^u)R(u) = R(u|_{\mathbb{T}})Q'_k(U_0)$ and $\mathbf{Q}_k(U_0^u)R(u) = R(u|_{\mathbb{T}})\mathbf{Q}_k(U_0)$. Moreover, for $j \le k$ and relative configurations U' where the maps are defined, $\mathcal{Q}_j(R(u)U';U_0^u) = R(u)\mathcal{Q}_j(U';U_0)$, $\mathsf{w}_j(R(u)U';U_0^u) = u\,\mathsf{w}_j(U';U_0)\,u^{-1}$, and $U' \in \mathrm{Ax}_j(U_0)$ if and only if $R(u)U' \in \mathrm{Ax}_j(U_0^u)$. Here the second argument is the reference, $R(u)$ acts on relative configurations by $(R(u)X)(b) = u(b_-)X(b)u(b_-)^{-1}$, on every lattice $\mathbb{T}^{(i)}$ through the restriction of u, and $u\,\mathsf{w}_ju^{-1}$ is the pointwise conjugate.*

(g) (Lattice covariance.) *Let $r \in \mathfrak{S}_k$. Then $W_j(rU_0) = rW_j(U_0)$ for $j \le k$, and*

$$Q'_k(rU_0)\,r = r\,Q'_k(U_0), \qquad \mathbf{Q}_k(rU_0)\,r_{U_0} = r_{W_k}\,\mathbf{Q}_k(U_0).$$

Here r acts on scalar fields by $(r\lambda)(x) = \lambda(r^{-1}x)$, and $(r_{U_0}A)(b) = A(r^{-1}b)$ on relative Lie fields, with $A(-b) = -R(U_0(b))^{-1}A(b)$ on negatively oriented bonds; likewise r_{W_k} on $\mathbb{T}$. Moreover $\mathcal{Q}_j(rU';rU_0) = r\,\mathcal{Q}_j(U';U_0)$, $\mathsf{w}_j(rU';rU_0) = r\,\mathsf{w}_j(U';U_0)$, and $U' \in \mathrm{Ax}_j(U_0)$ if and only if $rU' \in \mathrm{Ax}_j(rU_0)$. Here the second argument is the reference configuration, and $(rX)(b) = X(r^{-1}b)$ for relative configurations, with $X(-b) = R(W(b))^{-1}(X(b)^{-1})$, so that $(rX)(rW) = r(XW)$.

Proof. (a) $q'^T_j g(x) = R(W_{j-1}(y,x))^{-1}g(y)$, because the volume factors $L^{-d}\delta_j^d/\delta_{j-1}^d$ equal 1. Hence $q'_jq'^T_j = I$, since $R(W_{j-1}(y,x))$ is unitary.

(b) Let λ be real and $g_t = e^{\mathrm{i}t\lambda}$. Then $1^{(g_t)} = e^{-\mathrm{i}t\eta D_{U_0}\lambda + O(t^2)}$ on $\mathbb{T}_\eta$. By Theorem 22.5(e), $\mathcal{Q}_k(1^{(g_t)}) = 1^{(\mathfrak{M}^{(k)}(g_t))}$ on $\mathbb{T}$. Differentiating the definition of $\mathfrak{m}_j$ at $g = 1$ gives $\mathfrak{M}^{(k)}(g_t) =$

$e^{\mathrm{i}tQ'_k\lambda+O(t^2)}$, hence $1^{(\mathfrak{M}^{(k)}(g_t))} = e^{-\mathrm{i}tD_{W_k}Q'_k\lambda+O(t^2)}$. Comparing the terms of first order in t gives $\mathbf{Q}_k D_{U_0}\lambda = D_{W_k}Q'_k\lambda$.

(c) At $U_0 = 1$ all W_j and all averaged contour variables are 1. So q'_j is the centred block average at scale j, and $l_j = q_j$ is the centred tube average at scale j, by Lemma 22.8(d). By the nesting of the blocks (Section 21), $B^{(j)}(y)$ is the disjoint union of the $B^{(j-1)}(x)$, $x \in B_j(y)$; hence $q'_j \cdots q'_1$ is the average over $B^{(j)}(y)$, and $Q'_k = Q_0$. For the tube averages, let $A([x, x+w])$ denote the sum of A over the fine bonds of the straight segment from x to $x+w$. Assume by induction that $v = q_{j-1}\cdots q_1 A$ is the tube average from scale 0 to scale $j-1$: $v(b) = L^{-(j-1)(d+1)} \sum_{x\in B^{(j-1)}(b_-)} A([x, x+b_+-b_-])$. Let c be a bond of $\mathbb{T}^{(j)}$, $w = (c_+ - c_-)/L$, and for $z \in B_j(c_-)$ let $b_{z,m} = (z+mw, z+(m+1)w)$, $0 \le m < L$, be the bonds of $[z, z(c)]$. Since $B^{(j-1)}(z+mw) = B^{(j-1)}(z) + mw$,

$$\begin{aligned}(q_j v)(c) &= L^{-d-1} \sum_{z\in B_j(c_-)} \sum_{m=0}^{L-1} v(b_{z,m}) \\ &= L^{-j(d+1)} \sum_{z\in B_j(c_-)} \sum_{x\in B^{(j-1)}(z)} \sum_{m=0}^{L-1} A([x+mw, x+(m+1)w]).\end{aligned}$$

By the nesting, x runs through $B^{(j)}(c_-)$ exactly once, and for each x the L segments concatenate to $[x, x+Lw]$. So $q_j v$ is the tube average from scale 0 to scale j, and $\mathbf{Q}_k$ is the tube average $\mathbf{Q}$ of Definition 2.5.

(d) follows from Lemma 22.8(a) and Lemma 21.10(c).

(e) By (f) we may compute in the given gauge. By Lemma 21.13, $W_j = e^{\mathrm{i}\delta_j A_{0,j}}$ on the relevant bonds with $|A_{0,j}| \le C\varepsilon$. Hence all transports $R(W_{j-1}(\gamma))$ along contours of at most $2L$ bonds of $\mathbb{T}^{(j-1)}$, and the transports $R(W_{j-1}(y,x))$, differ from 1 by at most $C\varepsilon\delta_j$, by Lemma 21.5(e). By Lemma 22.7(e), $|\lambda_{W_{j-1}}(y,x)[B] - \lambda_1(y,x)[B]| \le CL\varepsilon\delta_j \max|B|$. Together with the bound on r_j in Lemma 22.8(b), this shows that $l_j - q_j(1)$ is a combination of at most C_L values with coefficients $C_L\varepsilon\delta_j$. Telescope:

$$\mathbf{Q}_k(U_0) - \mathbf{Q} = \sum_j l_k \cdots l_{j+1}(l_j - q_j(1))q_{j-1}(1)\cdots q_1(1).$$

By the bookkeeping of Proposition 22.9(c), the kernel of the difference is bounded by $\sum_j C_L\varepsilon\delta_j \le C\varepsilon$. The same argument applies to Q'_k. Kernel bounds give the sup-norm bounds, the block norms (by the Schur test), and the transposes.

(f) follows from Lemma 21.10(b), the definitions, and (22.1) for real u. The nonlinear identities follow as in (g), with u in place of r: $(R(u)X)W_j^u = (XW_j)^u$ bondwise, so, by (21.2), Lemma 21.10(b) and the uniqueness in Lemma 22.3(a), the averaged relative contour variables, τ_X, g_X and the relative averages are conjugated by u at the points of each lattice, and the induction over j applies. It is written out in the proof of Lemma 54.1(a), paragraph "The averaging".

(g) Since $\mathfrak{S}_k \subset \mathfrak{S}_j$, Lemma 21.10(d) gives $W_j(rU_0) = rW_j(U_0)$. For a relative configuration X at scale $j-1$, $(rX)(rW_{j-1}) = r(XW_{j-1})$, so $(rX)_{rW}(y,x) = X_W(r^{-1}y, r^{-1}x)$ by the lattice covariance of the averaged contour variables, and $\overline{rX} = r\overline{X}$ by Lemma 21.10(d). Since r maps $B_j(y)$ onto $B_j(ry)$, $\tau_{rX} = r\tau_X$, $\mathcal{E}_j(rh) = r\mathcal{E}_j(h)$ and $\mathfrak{m}_j(rg) = r\mathfrak{m}_j(g)$, all computed with rW. Moreover $(rX)^{(rg)} = r(X^{(g)})$, so $g_{rX} = rg_X$ by the formula of Lemma 22.3(a), and $\tilde{\mathfrak{a}}_j(rX) = r\tilde{\mathfrak{a}}_j(X)$. By induction $\mathcal{Q}_j(rU') = r\mathcal{Q}_j(U')$ and $\mathsf{w}_j(rU') = r\mathsf{w}_j(U')$, and axiality is preserved. Since $r(e^{\mathrm{i}s\eta A}) = e^{\mathrm{i}s\eta\, r_{U_0}A}$ as relative configurations, differentiating $\mathcal{Q}_k$ at $s=0$ gives the identity for $\mathbf{Q}_k$. Finally $(q'_j(rW)\, r\lambda)(y) = L^{-d}\sum_{x\in B_j(y)} R((rW)(y,x))\lambda(r^{-1}x) = (q'_j(W)\lambda)(r^{-1}y)$. □

Corollary 22.11 (Parts II and III apply to Bałaban's averaging)**.** *Let U_0 be real and (M,ε)-regular with ε small. Replace the averaging pair $(Q'(U),\mathbf{Q}(U))$ by $(Q'_k(U_0),\mathbf{Q}_k(U_0))$, the defect E by 0, and the unit-lattice background V by W_k. Then the real results of Sections 14–19 hold, and so does Theorem 20.2. Proposition 19.11 holds for $\delta \le \min(\delta_0, \delta_{AB}/C)$ with $\varepsilon + C\delta \le \min(\varepsilon_0, c'_3)$; at the perturbations U_δ the averages are those of Proposition 46.4.*

Proof. An (M,ε)-regular background has curvature at most $C\varepsilon$, by Lemma 12.3 in the local gauges. Proposition 22.10 gives the properties (a)–(d) of Lemma 12.5, in each local gauge, with ε replaced by $C\varepsilon$. By Remark 19.12, these are the only properties of the averaging used in the real results of Parts II and III. Property (v) of that remark at the U_δ is shown there, from Proposition 46.4(iv)–(vi) and the identity theorem. Proposition 46.4 uses Part IV only at the real reference, so there is no circularity. It controls only the sup norm of the perturbation; the derivative and Hölder bounds on δA are hypotheses of Proposition 19.11. □

From now on $Q' = Q'_k(U_0)$, $\mathbf{Q} = \mathbf{Q}_k(U_0)$, and all operators of Parts II and III (G', R, H', Γ', $\mathbf{G}$, $\mathbf{H}$, $\mathbf{C}$, ...) are taken at the background U_0 with this averaging pair. Since $E = 0$, the slice projection satisfies $\mathbf{Q}\Pi = \mathbf{Q}$ (Lemma 15.1(c)), and the correction $\mathcal{K}_2$ of Section 15 vanishes.

Remark 22.12 (A numerical check). The algebra of Sections 21 and 22 was checked numerically for $G = SU(2)$ and $L = 3$, in $d = 2$ on small tori and in $d = 4$ on a single coarse bond. Among the statements checked, to rounding error, are Lemma 21.5(a), (c) and (b) for real arguments, (22.1) for real gauge transformations, Theorem 22.5(e),(f) and Proposition 22.10(a),(b),(f). Two controls fail, as they should: with a single contour order per family the covariance under the permutations of the axes fails, and with the mean of the logarithms of the contour variables in place of Av, (22.1) and $E = 0$ fail. No proof uses these checks.

23 The Wilson action near a regular configuration

23.1 The action and the current

The Wilson action of a configuration U on $\mathbb{T}_\eta$ is

$$S(U) = \sum_p \Big(\mathsf{N} - \tfrac{1}{2}\operatorname{tr}\big(U(\partial p) + U(\partial p)^{-1}\big)\Big),$$

the sum running over the positively oriented plaquettes. For real U the summand is $\operatorname{Re}\operatorname{tr}(1 - U(\partial p))$. The formula defines an analytic function of G^c-valued configurations. There is no power of η in front of the sum. This uses $d = 4$: the Wilson action is dimensionless, so it does not change when all lengths are rescaled, and in the expansions below the sums over plaquettes come with the factor $\eta^4 = \eta^d$ of the pairing. The action is invariant under real gauge transformations. It is also invariant under $\mathfrak{S}_k$: a map $r \in \mathfrak{S}_k$ permutes the plaquettes, $(rU)(\partial p)$ is conjugate to $U(\partial p')^{\pm 1}$ with $p' = r^{-1}p$, and $\operatorname{tr} H + \operatorname{tr} H^{-1}$ is invariant under conjugation and inversion. Throughout this section we use the real bilinear pairing $\langle A, B\rangle = \eta^d \sum_b \operatorname{tr}(A(b)B(b))$ on $\mathfrak{g}^c$-valued gauge fields, and $\|A\|^2 = \langle A, A\rangle$ for real fields; the gradient $\nabla\Phi$ of a function Φ of fields is taken with respect to this pairing. For a real configuration U and a fine Lie field A put $\Phi_U(A) = S(e^{\mathrm{i}\eta A}U)$, where $e^{\mathrm{i}\eta A}U$ is the bondwise product.

Definition 23.1 (Current and the class $\mathfrak{U}$)**.** For real U the *current* is $J_U = \nabla\Phi_U(0)$, a real $\mathfrak{g}$-valued gauge field. For $\theta > 0$ let

$$\mathfrak{U}(\theta) = \{U \in \mathfrak{U}_0(\theta) : \ |J_U|_\infty \le \theta\}.$$

Lemma 23.2 (The current)**.** *Let U be real. Put $F^s_U(p) = \eta^{-2} P_{\mathfrak{g}}\big((U(\partial p) - U(\partial p)^{-1})/2\mathrm{i}\big)$, where $P_{\mathfrak{g}}$ is the orthogonal projection onto $\mathfrak{g}$ with respect to the trace form. Then:*

(a) $J_U = D_U^T F_U^s$;

(b) $D_U^T J_U = 0$, $J_{U^u} = R(u)J_U$ *for real gauge transformations* u, *and* $J_{rU} = r_U J_U$ *for* $r \in \mathfrak{S}_k$, *with* r_U *as in Proposition 22.10(g);*

(c) if $U \in \mathfrak{U}_0(\theta)$, *then* $|F_U^s - F_U| \le C\theta^3\eta^4$, *hence* $|J_U - D_U^T F_U| \le C\theta^3\eta^3$.

So $\mathfrak{U}(\theta)$ is Bałaban's class $\mathfrak{U}_k(\{\Omega_j\}, \theta)$ for $\Omega_j = \mathbb{T}_\eta$: small curvature and small covariant divergence of the curvature [6, (1.7)–(1.9)], [8, (2)].

Proof. (a) For a plaquette $p = p_{\mu\nu}(x)$ with bonds $b_1, \dots, b_4$ as in Definition 21.4, $H_p = U(\partial p)$, and a fine field A,

$$(e^{\mathrm{i}\eta A}U)(\partial p) = e^{\mathrm{i}\eta a_1} e^{\mathrm{i}\eta a_2} e^{-\mathrm{i}\eta a_3} e^{-\mathrm{i}\eta a_4}\, H_p, \tag{23.1}$$

with $a_1 = A(b_1)$, $a_2 = R(U(b_1))A(b_2)$, $a_3 = R(H_p U(b_4))A(b_3)$, $a_4 = R(H_p)A(b_4)$. To first order the summand of S changes by $\operatorname{Im}\operatorname{tr}\big(\eta(a_1 + a_2 - a_3 - a_4)H_p\big)$. Since $\operatorname{tr}((R(H)X)H) = \operatorname{tr}(XH)$, the factors $R(H_p)$ drop out, and $\eta(a_1 + a_2 - a_3 - a_4)$ may be replaced by $\eta^2(D_U A)(p)$, with $(D_U A)(p) = (D_\mu A_\nu - D_\nu A_\mu)(x)$ as in Part II. Hence the first-order term is $\langle D_U A, F_U^s\rangle$.

(b) S is gauge invariant, and $e^{\mathrm{i}\eta A_t}U = U^{e^{\mathrm{i}t\lambda}}$ with $A_t = -tD_U\lambda + O(t^2)$. Hence $\langle D_U\lambda, J_U\rangle = 0$ for all λ. Covariance under gauge transformations is clear. For $r \in \mathfrak{S}_k$ we have $e^{\mathrm{i}\eta\, r_U A}\, rU = r(e^{\mathrm{i}\eta A}U)$, by the convention $A(-b) = -R(U(b))^{-1}A(b)$ on reversed bonds, and r_U preserves the pairing, since $\operatorname{tr}(R(h)XR(h)Y) = \operatorname{tr}(XY)$. Hence $\Phi_{rU}(r_U A) = \Phi_U(A)$, and differentiating at $A = 0$ gives $J_{rU} = r_U J_U$.

(c) $F_U(p) = \eta^{-2}\ell(H_p)$ and $(H - H^{-1})/2\mathrm{i} = \sin(\ell(H))$, and $\sin X - X = O(|X|^3)$. The second bound follows because D_U^T applied to a field bounded by s is bounded by $4ds/\eta$. □

23.2 Expansion

For fine fields we use the covariant gradient $\nabla A = \nabla_{U_0} A$, whose components are $D_{U_0,\nu}A_\mu$ for all μ, ν, the sup norm $|A|_\infty$, and $|A|_{C^1} = |A|_\infty + |\nabla A|_\infty$. On the lattice, trivially,

$$\eta|\nabla A|_\infty \le 2|A|_\infty. \tag{23.2}$$

Proposition 23.3 (Expansion of the action)**.** *There are* c_5, C *such that the following holds. Let* U_0 *be real and* $(1, \varepsilon)$*-regular with* $\varepsilon \le c_5$, *and put* $J = J_{U_0}$. *For complex fine Lie fields* A *with* $|A|_\infty \le c_5$,

$$\Phi(A) := S(e^{\mathrm{i}\eta A}U_0) = S(U_0) + \langle A, J\rangle + \tfrac{1}{2}\|D_{U_0}A\|^2 + \mathcal{N}(A),$$

where $\mathcal{N}$ *is analytic, real for real* A, *and:*

(a) $|\nabla\mathcal{N}(A)|_\infty \le C(\varepsilon + |A|_\infty)|A|_{C^1}$;

(b) if $|A|_{C^1}, |A'|_{C^1} \le r \le c_5$, *then* $|\nabla\mathcal{N}(A) - \nabla\mathcal{N}(A')|_\infty \le C(\varepsilon + r)|A - A'|_{C^1}$;

(c) for real A *with* $|A|_{C^1} \le c_5$ *and all real* h: $|d^2\mathcal{N}(A)[h,h]| \le C(\varepsilon + |A|_{C^1})(\|h\|^2 + \|\nabla h\|^2)$;

(d) for real A: $\nabla\Phi(A) = \Psi(A)^T J_{e^{\mathrm{i}\eta A}U_0}$, *where* $\Psi(A)$ *acts bondwise and* $|\Psi(A) - 1| \le C\eta|A|_\infty$.

Proof. Plaquette variables. Let $Z_p(A) = \ell(e^{\mathrm{i}\eta a_1} e^{\mathrm{i}\eta a_2} e^{-\mathrm{i}\eta a_3} e^{-\mathrm{i}\eta a_4})$, with the a_i of (23.1) for $U = U_0$, and $\zeta_p = \eta^{-2}Z_p$. By Lemma 21.2(c), Z_p is analytic for $|A| \le c_5$ and

$$\zeta_p(A) = (D_{U_0}A)(p) + \rho_p(A), \qquad \rho_p = \rho_p^{(1)} + \rho_p^{(2)},$$

where $\rho_p^{(1)} = \eta^{-1}(a_1 + a_2 - a_3 - a_4) - (D_{U_0}A)(p)$ is linear and $\rho_p^{(2)}$ collects the commutator terms of order two and higher. Since $|R(H_p) - 1| \le C\varepsilon\eta^2$, $|\rho_p^{(1)}(A)| \le C\varepsilon\eta|A|$. By the Cauchy

estimates, $|\rho_p^{(2)}(A)| \le C|A|^2$, $|d\rho_p^{(2)}(A)h| \le C|A||h|$ and $|d^2\rho_p^{(2)}(A)[h,h]| \le C|h|^2$, with sup norms over the bonds of p.

The summand. With $\mathfrak{s}(Z,H) = \mathsf{N} - \frac{1}{2}\operatorname{tr}(e^{\mathrm{i}Z}H + H^{-1}e^{-\mathrm{i}Z})$,

$$\mathfrak{s}(Z,H) - \mathfrak{s}(0,H) = \operatorname{tr}\left(Z\,\tfrac{H-H^{-1}}{2\mathrm{i}}\right) + \tfrac{1}{2}\operatorname{tr}\left(Z^2\,\tfrac{H+H^{-1}}{2}\right) + \mathfrak{s}_3(Z,H), \qquad |\mathfrak{s}_3| \le C|Z|^3.$$

Summing this over p, with $Z = \eta^2\zeta_p$ and $\frac{1}{2}(H+H^{-1}) = 1 + O(\varepsilon^2\eta^4)$,

$$\Phi(A) - S(U_0) = \langle \zeta, F^s\rangle + \tfrac{1}{2}\|\zeta\|^2 + \mathcal{R}(A), \qquad \mathcal{R}(A) = \eta^d \sum_p O(\varepsilon^2\eta^4|\zeta_p|^2 + \eta^2|\zeta_p|^3),$$

with $F^s = F^s_{U_0}$ and $\|\zeta\|^2 = \eta^d\sum_p \operatorname{tr}\zeta_p^2$. By the proof of Lemma 23.2(a), $\langle \rho^{(1)}(A), F^s\rangle = 0$, so the linear term is $\langle D_{U_0}A, F^s\rangle = \langle A, J\rangle$. Hence

$$\mathcal{N}(A) = \langle \rho^{(2)}(A), F^s\rangle + \langle D_{U_0}A, \rho(A)\rangle + \tfrac{1}{2}\|\rho(A)\|^2 + \mathcal{R}(A).$$

(a) We bound the gradient of each term. All of them are local, so it suffices to bound them in a gauge in which $U_0 = e^{\mathrm{i}\eta A_0}$ near the bond, with $|A_0|, |\partial A_0| \le \varepsilon$.

- $\nabla\langle \rho^{(2)}(A), F^s\rangle = d\rho^{(2)}(A)^T F^s$ is bounded by $C\varepsilon|A|$, since $|F^s| \le C\varepsilon$.
- $\nabla\langle D_{U_0}A, \rho(A)\rangle = D_{U_0}^T\rho(A) + d\rho(A)^T D_{U_0}A$. The second term is bounded by $C(|A| + \varepsilon\eta)|DA|$. For the first, $D_{U_0}^T$ takes differences of ρ between neighbouring plaquettes, divided by η. The function $\rho_p^{(2)}$ vanishes to second order in the values of A on the bonds of p, and its coefficients are analytic functions of the transports $R(U_0(b_1))$, $R(H_pU_0(b_4))$ and $R(H_p)$ of (23.1). On neighbouring parallel plaquettes the bond variables of U_0 differ by at most $\eta^2\varepsilon$, since $|\partial A_0| \le \varepsilon$, and so do the holonomies, up to a constant, since each is $1 + O(\eta^2\varepsilon)$. The field values differ by $\eta|\partial A| \le \eta(|\nabla A| + C\varepsilon|A|)$. So the change of the field values costs $C\eta|A|(|\nabla A| + \varepsilon|A|)$, and the change of the coefficients costs $C\eta^2\varepsilon|A|^2$. The transport in $D_{U_0}^T$ differs from 1 by $O(\eta\varepsilon)$. After division by η, these differences are bounded by $C|A|(|\nabla A| + \varepsilon|A|)$. The differences of $\rho^{(1)}$ are bounded by $C\varepsilon|A| + C\varepsilon\eta|\nabla A|$; here we used $|F_p - F_{p'}| \le 2\varepsilon$ for the curvature factors.
- $\nabla\frac{1}{2}\|\rho\|^2 = d\rho^T\rho$ is bounded by $C(|A| + \varepsilon\eta)(|A|^2 + \varepsilon\eta|A|)$.
- In $\mathcal{R}$, each derivative of ζ_p with respect to a bond variable costs at most $C\eta^{-1}$. Hence $|\nabla\mathcal{R}| \le C(\varepsilon^2\eta^3|\zeta| + \eta|\zeta|^2)$. Finally $|\zeta| \le C(|\nabla A| + \varepsilon\eta|A| + |A|^2)$ and, by (23.2), $\eta|\zeta| \le C|A|$.

(b) The bound (a) holds for complex A in the ball $|A|_{C^1} \le 2r$, since $2r \le 2c_5$ and we may assume (a) on that larger ball, after decreasing c_5. Apply Cauchy's estimate to $t \mapsto \nabla\mathcal{N}(A' + t(A - A'))$, as in Lemma 22.8.

(c) Differentiate the formula for $\mathcal{N}$ twice. The terms $\langle d^2\rho^{(2)}[h,h], F^s\rangle$, $2\langle D_{U_0}h, d\rho[h]\rangle + \langle D_{U_0}A, d^2\rho[h,h]\rangle$, $\|d\rho[h]\|^2 + \langle \rho, d^2\rho[h,h]\rangle$ are bounded by $C(\varepsilon + |A|_{C^1})(\|h\|^2 + \|\nabla h\|^2)$, by the pointwise bounds on ρ and Cauchy–Schwarz. For $d^2\mathcal{R}$ write $\mathcal{R}(A) = \eta^d\sum_p f_p(\zeta_p)$ with $f_p(\zeta) = \frac{1}{2}\operatorname{tr}\left(\zeta^2(\frac{1}{2}(H_p + H_p^{-1}) - 1)\right) + \eta^{-4}\mathfrak{s}_3(\eta^2\zeta, H_p)$. By the Cauchy estimates, $|f_p'(\zeta)| \le C(\varepsilon^2\eta^4|\zeta| + \eta^2|\zeta|^2)$ and $|f_p''(\zeta)| \le C(\varepsilon^2\eta^4 + \eta^2|\zeta|)$. On the bonds of p, $|d\zeta_p[h]| \le C(|\nabla h| + (\varepsilon\eta + |A|)|h|)$ and $|d^2\zeta_p[h,h]| \le C|h|^2$. The chain rule gives two kinds of terms. In $f_p''(\zeta_p)[d\zeta_p[h], d\zeta_p[h]]$ the main term is $\eta^2|\zeta_p||\nabla h|^2 \le C|A||h||\nabla h|$, by (23.2) applied to ζ and to h; the other terms are bounded by $C(\varepsilon + |A|)(|h|^2 + |\nabla h|^2)$, since $\eta|\nabla h| \le 2|h|$. In $f_p'(\zeta_p)\,d^2\zeta_p[h,h]$ we have $|f_p'(\zeta_p)| \le C(\varepsilon^2 + |A|)|A|$, since $\eta|\zeta_p| \le C|A|$. Summing over p with the factor η^d, every bond lying in boundedly many plaquettes, gives the bound.

(d) Bondwise, $e^{\mathrm{i}\eta(A+sh)} = e^{\mathrm{i}\eta s\Psi(A)h + O(s^2)}e^{\mathrm{i}\eta A}$, where $\Psi(A)h(b) = \mathfrak{d}(\eta\operatorname{ad}A(b))h(b)$ for an entire function $\mathfrak{d}$ with $\mathfrak{d}(0) = 1$. Hence $\frac{d}{ds}\Phi(A + sh)|_{s=0} = \langle \Psi(A)h, J_{e^{\mathrm{i}\eta A}U_0}\rangle$. □

24 The axial-to-Landau chart

In this section U_0 is a real reference configuration that is (M,ε)-regular and satisfies $|J_{U_0}|_\infty \le \varepsilon$. In particular $U_0 \in \mathfrak{U}(C\varepsilon)$. All operators of Parts II and III are taken at U_0 with the averaging pair $(Q'_k(U_0), \mathbf{Q}_k(U_0))$ (Corollary 22.11). Theorem 24.2 below is Bałaban's [6, Theorem 2] on a single torus: every real configuration with small curvature and current and with averages close to those of U_0 has a unique representative in the Landau gauge relative to U_0, and this representative is small and regular. As in Section 22, "axial" means averaged axial (Definition 22.2); the chart passes from the averaged axial representative of a $\mathfrak{G}_0$-orbit to its Landau representative.

24.1 Linear input

Lemma 24.1 (Sup-norm bounds at the reference)**.** *There are ε_*, C such that for $\varepsilon \le \varepsilon_*$:*

(a) $|G'f|_{C^1} \le C|f|_\infty$, $|H'g|_{C^1} \le C|g|_\infty$, and $D^TDH'g = Q'^T(\Gamma' - a)g$, so $|D^TDH'g|_\infty \le C|g|_\infty$.

(b) $|P_{\mathcal{V}}f|_{C^1} \le C|f|_\infty$ and $|P_{\mathcal{V}}D^TA|_{C^1} \le C|A|_\infty$. In particular $|Rf|_\infty \le C|f|_\infty$.

(c) The Feynman-gauge operator $\mathfrak{A}_F = D^TD + DRD^T + a\mathbf{Q}^T\mathbf{Q}$ is invertible, and $|\mathfrak{A}_F^{-1}f|_{C^1} \le C|f|_\infty$, $|\mathfrak{A}_F^{-1}\mathbf{Q}^T\omega|_{C^1} \le C|\omega|_\infty$.

(d) $|\mathbf{H}B|_{C^1} \le C|B|_\infty$ and $|\mathbf{C}f|_{C^1} \le C|f|_\infty$.

Here $|\cdot|_{C^1}$ uses the covariant gradient $\nabla = \nabla_{U_0}$; for scalar fields $\nabla = D_{U_0}$. The same bounds hold with exponential weights, provided $\gamma' \le \gamma/2$: if $|f(x)| \le e^{-\gamma'|x-z|}$ for some z, then $|Kf(x)| + |\nabla Kf(x)| \le Ce^{-\gamma'|x-z|}$ for each operator K above with a C^1 bound, and $|Kf(x)| \le Ce^{-\gamma'|x-z|}$ for $K = R$ and $K = D^TDH'$. For these two only sup bounds hold uniformly in η: at $U_0 = 1$, if f is a unit vector of $\mathfrak{g}$ at one fine point and 0 elsewhere, $|\nabla Rf|$ is of order η^{-1}. The constants ε_* and C may depend on L, G, a, α and M_0, but not on $M \ge M_0$, on k, on η or on the side n of $\mathbb{T}$. Indeed, the proof uses the theorems of Parts II and III at U_0, whose constants are uniform in this sense by Lemma 31.3, applied at $M = M_0$ as in Theorem 25.1; an (M,ε)-regular reference is (M_0,ε)-regular (Lemma 31.2).

Proof. By Corollary 22.11, the pointwise bounds of Theorems 18.2(g), 19.7(b) and 19.9 hold. Each has the form $|Kf|_{B(y)} + |\nabla Kf|_{B(y)} \le C\sum_{y'} e^{-\gamma|y-y'|}|f|_{B(y')}$. Summing over y' gives the sup-norm bounds, and the weighted bounds follow because $\sum_{y'} e^{-\gamma|y-y'|}e^{-\gamma'|y'-z|} \le Ce^{-\gamma'|y-z|}$. The identity in (a) follows from $\mathfrak{A}'H' = Q'^T\Gamma'$ and $Q'H' = I$. With it, the plain and weighted sup bounds on $D^TDH'g$ follow from the decay of Γ' and the sup bound on Q'^T, and those on $Rf = f - P_{\mathcal{V}}f$ from the bounds on $P_{\mathcal{V}}$. For (c), $\mathfrak{A}_F = \mathfrak{A}_\pi - \mathcal{K}_1$ with $W = 0$ and $\mathcal{K}_2 = 0$, because $E = 0$. By the proof of Theorem 15.4(a), $\mathcal{K}_1$ is admissible of size $C\varepsilon$ and pointwise admissible of type 1. Lemmas 13.2, 13.3 and 18.3 give the bounds, as for $\mathbf{G}$; in Lemma 18.3 the small inputs of $\mathcal{K}_1$ are bounded by the invariant norms $|v| + |\nabla v|$ on the cubes $\Lambda_1(c_{y'})$, with weights $e^{-\gamma|y-y'|}$. Finally $|\mathbf{Q}^T\omega| \le C|\omega|$ by Proposition 22.10(e). □

24.2 The theorem

Theorem 24.2 (Axial-to-Landau chart)**.** *There are $c_4, r_u, B_1 > 0$, with $B_1c_4 \le r_u/2$, with the following property. Let U_0 be as above, and let U be real with $U \in \mathfrak{U}(\theta)$. Suppose that $U' = UU_0^{-1} \in \mathrm{Ax}_k$ and $|\ell(U'^{[k]}(c))| \le \beta$ for all bonds c of $\mathbb{T}$. If $\varepsilon + \theta + \beta \le c_4$, there is exactly one fine relative configuration $\tilde{U} = e^{i\eta A}$ with A real and $|A|_\infty \le r_u$ such that*

(i) $R(U_0)D_{U_0}^TA = 0$ (Bałaban's Landau gauge relative to U_0), and

(ii) $\tilde{U}^{(\mathsf{w}_k(\tilde{U}))} = U'$.

Moreover $\mathcal{Q}(\tilde{U}) = U'^{[k]}$, *and*

$$|A|_\infty + |\nabla A|_\infty + |\Delta_{U_0} A|_\infty + \sup_y [\nabla \tilde{A}_y]_{\alpha, B(y)} \le B_1(\varepsilon + \theta + \beta),$$

where $\Delta_{U_0} = D^T D + D D^T$ *is the covariant Hodge Laplacian and* $\tilde{A}_y$ *is* A *in a local gauge of* U_0 *near* $B(y)$.

All maps in the statement are defined. Since $U \in \mathfrak{U}_0(c_4)$ with $c_4 \le c_1$, the relative configuration U' lies in the averaging domain up to scale k (Section 22), so Ax_k and $U'^{[k]}$ make sense. For $\tilde{U} = e^{i\eta A}$ with $|A|_\infty \le r_u \le c_3$ the maps of Definition 22.4 are defined, by Lemma 22.8 and Proposition 22.9. Condition (ii) says that $\tilde{U}$ and U' lie in the same gauge orbit, and that the transformation relating them is the correspondence transformation of Theorem 22.5. By Theorem 22.5(f), this transformation satisfies Bałaban's normalization. The bounds are Bałaban's [6, (1.36)–(1.39)] on a single torus. Note the sign: Δ_{U_0} is non-negative, and at $U_0 = 1$ it is $d^*d + dd^* = -\Delta$ (Lemma 2.2), with the non-positive Laplacian Δ of Section 2.2.

Corollary 24.3 (Parametrization of the constraint set)**.** *Let* V *be a real configuration on* $\mathbb{T}$ *with* $|\ell(V(c)W_k(c)^{-1})| \le \beta$ *for all* c, *and let* $\varepsilon + \theta + \beta \le c_4$. *Let* $\mathcal{S}_V$ *be the set of real* A *with* $|A|_\infty \le r_u$, $RD^T A = 0$, $e^{i\eta A} U_0 \in \mathfrak{U}(\theta)$ *and* $\mathcal{Q}(e^{i\eta A}) = V W_k^{-1}$. *Then*

$$A \longmapsto \mathsf{U}(A) = \big((e^{i\eta A})^{(\mathsf{w}_k(e^{i\eta A}))}\big) U_0$$

is a bijection from $\mathcal{S}_V$ *onto the set of averaged axial representatives* ($UU_0^{-1} \in \mathrm{Ax}_k$) *of the* $\mathfrak{G}_0$*-orbits in* $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$, *where* $\mathfrak{B}(V) = \{U : U^{(k)} = V\}$. *Moreover* $S(\mathsf{U}(A)) = \Phi(A)$.

Proof. $\mathsf{U}(A) = (e^{i\eta A} U_0)^{\mathsf{w}_k}$ is gauge equivalent to $e^{i\eta A} U_0$, so it lies in $\mathfrak{U}(\theta)$ and has the same action. By Theorem 22.5(a), $\mathsf{U}(A) U_0^{-1} \in \mathrm{Ax}_k$ and $\mathsf{U}(A)^{(k)} = \mathcal{Q}(e^{i\eta A}) W_k = V$. Conversely, for an averaged axial representative U of an orbit in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$ we have $U'^{[k]} = V W_k^{-1}$, and Theorem 24.2 gives exactly one preimage. The orbits have unique averaged axial representatives by Lemma 22.3(c), since $\mathfrak{U}(\theta) \subset \mathfrak{U}_0(c_1)$. □

24.3 Axial configurations are small

Lemma 24.4 ([6, Lemma 1, (1.65)])**.** *Let* $C_A = 45L$. *Let* $U_0, U \in \mathfrak{U}_0(\theta)$ *be real, with* $\theta \le c_1$, *and let* $U' = UU_0^{-1} \in \mathrm{Ax}_k$. *Suppose that* $|\ell(U'^{[k]}(c))| \le \beta$ *for all bonds* c *of* $\mathbb{T}$, *where* $\beta + C_A\theta \le \frac{1}{2} c_G$. *Then* $|\ell(U'^{[j]}(b))| \le \beta + C_A\theta$ *for all* $j \le k$ *and all bonds* b *of* $\mathbb{T}^{(j)}$.

Proof. One step. Fix $j \in \{1, \dots, k\}$, and write $W = W_{j-1}$, $\delta = \delta_{j-1}$, $\delta' = \delta_j$. Let X be a real relative configuration at scale $j-1$, axial at scale j, such that $Y = XW$ and W lie in $\mathfrak{U}_{j-1}(2\theta)$, and such that $|\ell(\overline{X}(c))| \le b'$ for all bonds c of $\mathbb{T}^{(j)}$, where $b' + 40L\theta\delta'^2 \le \frac{1}{2} c_G$. We claim that

$$|\ell(X(b))| \le b' + 40L\theta\delta'^2 \qquad \text{for every bond } b \text{ of } \mathbb{T}^{(j-1)}.$$

Since $X(-b) = R(W(b))^{-1}(X(b)^{-1})$ and $R(W(b))$ is unitary, it suffices to consider positively oriented bonds. We apply Lemma 21.7 to Y and to W with $\varepsilon = 2\theta\delta^2$. Then $8L^2\varepsilon = 16\theta\delta'^2 \le r_M$, because $\theta \le c_1 \le r_M/128$. All configurations below are real. So, by Lemma 21.2(b), the logarithm is subadditive under products and invariant under conjugation by elements of G, as long as the sums involved are at most c_G. All sums below are at most $b' + 40L\theta\delta'^2$.

Bonds inside a block. Let $b = (x, x')$ with $x, x' \in B_j(y)$. For a configuration Z on $\mathbb{T}^{(j-1)}$ put $T_b(Z) = Z(y,x) Z(b) Z(y,x')^{-1}$. Axiality gives $Y(y,x) = W(y,x)$ and $Y(y,x') = W(y,x')$. Hence

$$T_b(Y)\, T_b(W)^{-1} = W(y,x)\, Y(b) W(b)^{-1}\, W(y,x)^{-1} = R(W(y,x)) X(b).$$

Since $W(y,x) \in G$, Lemma 21.7(ii) gives

$$|\ell(X(b))| \le |\ell(T_b(Y))| + |\ell(T_b(W))| \le 12(L-1)^2\varepsilon \le 24\theta\delta'^2.$$

Bonds between blocks. Every other positively oriented bond b joins two blocks: b has direction μ, $x = b_-$ lies in $B_j(c_-)$ on the face $\{x_\mu - c_{-,\mu} = \frac{L-1}{2}\delta\}$, and $b_+ \in B_j(c_+)$, where $c = (c_-, c_+)$ and $c_+ = c_- + \delta' e_\mu$. The straight segment $[x, x(c)]$ consists of b followed by a contour γ' of $L-1$ bonds inside $B_j(c_+)$. (Every segment $[x, x(c)]$ with $x \in B_j(c_-)$ contains exactly one bond between the two blocks; here it is the first one.) By the definition of relative transports, $X_W([x,x(c)]) = X(b)\, R(W(b))(X_W(\gamma'))$, and $X_W(\gamma')$ is a product of $L-1$ factors $R(W(\gamma'_{<i}))X(b_i)$ with bonds b_i inside $B_j(c_+)$. By the previous step,

$$|\ell(X(b))| \le |\ell(X_W([x,x(c)]))| + 24(L-1)\theta\delta'^2.$$

Now we use the averaged loop of Y through x. By axiality, $Y(c_-,x) = W(c_-,x)$ and $Y(x(c),c_+) = W(x(c),c_+)$. Inserting $Y([x,x(c)]) = X_W([x,x(c)])W([x,x(c)])$ and $Y([c]) = X_W([c])W([c])$ into Definition 21.6 gives

$$\Lambda_{c,x}(Y) = R(W(c_-,x))\big(X_W([x,x(c)])\big)\; h_x\; X_W([c])^{-1}, \qquad h_x = \Lambda_{c,x}(W).$$

By Lemma 21.7(iii), $|\ell(\Lambda_{c,x}(Y))| \le 16\theta\delta'^2$ and $|\ell(h_x)| \le 16\theta\delta'^2$. Next, by (21.3) for Y and for W, and since $W_j = \mathfrak{a}_j(W)$,

$$\overline{X}(c) = \mathfrak{a}_j(Y)(c)\, W_j(c)^{-1} = e^{\mathrm{i}E_Y}\, X_W([c])\, e^{-\mathrm{i}E},$$

where $E_Y = L^{-d}\sum_{x'} \ell(\Lambda_{c,x'}(Y))$ and $E = L^{-d}\sum_{x'} \ell(h_{x'})$, the sums running over $x' \in B_j(c_-)$. Here $|E_Y|, |E| \le 16\theta\delta'^2$, so $|\ell(X_W([c]))| \le b' + 32\theta\delta'^2$. Since $W(c_-,x) \in G$, the formula for $\Lambda_{c,x}(Y)$ gives

$$|\ell(X_W([x,x(c)]))| \le |\ell(\Lambda_{c,x}(Y))| + |\ell(h_x)| + |\ell(X_W([c]))| \le b' + 64\theta\delta'^2.$$

Altogether $|\ell(X(b))| \le b' + (24L+40)\theta\delta'^2 \le b' + 40L\theta\delta'^2$, since $L \ge 3$. This proves the claim.

Iteration. Let $b_j = \max_c |\ell(U'^{[j]}(c))|$, the maximum over the bonds of $\mathbb{T}^{(j)}$. By Proposition 21.12, $U^{(j-1)}, W_{j-1} \in \mathfrak{U}_{j-1}(2\theta)$. The relative configuration $X = U'^{[j-1]}$ is real and axial at scale j, with $XW_{j-1} = U^{(j-1)}$ and $\overline{X} = U'^{[j]}$. By the claim, $b_{j-1} \le b_j + 40L\theta\delta_j^2$, as long as $b_j + 40L\theta\delta_j^2 \le \frac{1}{2}c_G$. Since $b_k \le \beta$ and $\delta_i = L^{i-k}$, downward induction gives

$$b_j \le \beta + 40L\theta \sum_{i=j+1}^{k} \delta_i^2 \le \beta + 40L\theta\, \frac{L^2}{L^2-1} \le \beta + 45L\theta,$$

and the condition of the claim holds at every step, because $\beta + C_A\theta \le \frac{1}{2}c_G$. □

The proof uses only the straight segments $[x,x(c)]$ and the averaged loops of (21.3): a bond between two blocks is compared with the averaged loop through its segment, and the other $L-1$ bonds of the segment lie inside a block. This costs the factor L in C_A. The constant C_A enters only the proof of Theorem 24.2, through c_4 and B_1, which depend on L anyway.

24.4 The a priori bound

Proposition 24.5 (A priori bound [6, Prop. 3])**.** *There are $r_*, B_0 > 0$ such that the following holds for $\varepsilon \le r_*$. Let $\tilde{U} = e^{\mathrm{i}\eta A}$ with A real, $|A|_\infty \le r_*$ and $RD^TA = 0$. Suppose $|J_{\tilde{U}U_0}|_\infty \le \theta$ and $\mathcal{Q}(\tilde{U}) = e^{\mathrm{i}B}$ with $|B|_\infty \le \beta$. Then*

$$|A|_{C^1} + |\Delta_{U_0}A|_\infty + \sup_y [\nabla \tilde{A}_y]_{\alpha,B(y)} \le B_0(\varepsilon + \theta + \beta).$$

Proof. By Proposition 23.3(d), $|\nabla\Phi(A)|_\infty \le 2\theta$. By Proposition 23.3 and $|J_{U_0}| \le \varepsilon$,

$$D^T DA = \nabla\Phi(A) - J_{U_0} - \nabla\mathcal{N}(A), \qquad |D^T DA|_\infty \le 2\theta + \varepsilon + C(\varepsilon + r_*)|A|_{C^1}.$$

By Proposition 22.9, $\mathcal{Q}(\tilde U) = e^{\mathrm{i}(\mathbf{Q}A + C_k^{\mathrm{av}}(A))}$ with $|C_k^{\mathrm{av}}(A)| \le C|A|_\infty^2 \le Cr_*|A|_\infty$, and $\mathbf{Q}A + C_k^{\mathrm{av}}(A)$ is real and small. On each bond, $e^{\mathrm{i}B} = e^{\mathrm{i}(\mathbf{Q}A + C_k^{\mathrm{av}}(A))}$, so the eigenvalues of B, counted with algebraic multiplicity, differ by multiples of 2π from those of the Hermitian matrix $\mathbf{Q}A + C_k^{\mathrm{av}}(A)$, which lie in $(-\pi, \pi)$; by Schur triangularization the squared moduli of the eigenvalues of B sum to at most $|B|^2$. So $|\mathbf{Q}A + C_k^{\mathrm{av}}(A)| \le |B|$ bondwise, and we may replace B by $\mathbf{Q}A + C_k^{\mathrm{av}}(A)$, still with $|B|_\infty \le \beta$. Then $\mathbf{Q}A = B - C_k^{\mathrm{av}}(A)$. Since $RD^TA = 0$,

$$\mathfrak{A}_F A = D^T DA + a\mathbf{Q}^T(B - C_k^{\mathrm{av}}(A)),$$

and Lemma 24.1(c) gives $|A|_{C^1} \le C_F(2\theta + \varepsilon + \beta) + C_F C(\varepsilon + r_*)|A|_{C^1}$. For $C_F C(\varepsilon + r_*) \le \frac12$ this gives the bound on $|A|_{C^1}$. Next, $RD^TA = 0$ means $D^TA = P_{\mathcal{V}} D^T A$, so

$$\Delta_{U_0} A = D^T DA + DP_{\mathcal{V}} D^T A,$$

which is bounded by Lemma 24.1(b). Finally, in a gauge in which U_0 is close to 1 on $\Lambda_3(c_y)$, Δ_{U_0} has the normal form $-\Delta + \sum_\mu \partial_\mu^*(a_\mu\,\cdot) + b$ of Lemma 12.2, componentwise. Theorem 6.3(d) gives the Hölder bound. □

The proposition needs only a bound on $|A|_\infty$, not on derivatives. The derivatives are controlled by the gauge condition and the action, because the derivative terms in $\nabla\mathcal{N}$ come with the small factor $\varepsilon + |A|_\infty$. This is why the construction below only has to control the sup norm of the new representative.

24.5 One scale of gauge transformation

For a fine relative configuration $\tilde U_1 = e^{\mathrm{i}\eta A_1}$ and a scalar field λ write $\tilde U_\lambda = \tilde U_1^{(e^{-\mathrm{i}\lambda})} = e^{\mathrm{i}\eta\mathcal{A}(\lambda)}$. A direct computation from the definition of the relative action gives

$$\mathcal{A}(\lambda)(b) = e^{-\mathrm{i}\,\mathrm{ad}\,\lambda(b_-)} A_1(b) + \mathfrak{d}(\,\mathrm{ad}\,\lambda(b_-)) D_{U_0}\lambda(b) + \eta\,\mathfrak{b}(\lambda(b_-), A_1(b), D_{U_0}\lambda(b)), \tag{24.1}$$

where $\mathfrak{d}$ is entire with $\mathfrak{d}(0) = 1$, and $\mathfrak{b}$ is analytic with $|\mathfrak{b}| \le C(|A_1| + |D\lambda|)|D\lambda|$ [6, (1.81)–(1.84)]. For $|\lambda|_{C^1} = |\lambda|_\infty + |D\lambda|_\infty$ small, $|\mathcal{A}(\lambda)|_\infty \le C(|A_1|_\infty + |\lambda|_{C^1})$.

Lemma 24.6 (One scale)**.** *There are $r_0, \rho_0, C_2 > 0$ such that for $\varepsilon \le r_0$ the following holds. Let A_1 be real with $|A_1|_{C^1} \le r \le r_0$. Then there is exactly one real λ with $|\lambda|_{C^1} \le \rho_0$ such that*

$$\textit{(E1)} \quad RD^T\mathcal{A}(\lambda) = 0, \qquad \textit{(E2)} \quad \mathsf{w}_k(\tilde U_\lambda) = \mathsf{w}_k(\tilde U_1)\, e^{\mathrm{i}\lambda}.$$

It satisfies $|\lambda|_{C^1} \le C_2 r$ and $|\mathcal{A}(\lambda)|_\infty \le C_2 r$.

Proof. Reformulation of (E2). Both $\mathsf{w}_k(\tilde U_\lambda)$ and $\mathsf{w}_k(\tilde U_1)e^{\mathrm{i}\lambda}$ move $\tilde U_\lambda$ into the axial gauge: $\tilde U_\lambda^{(\mathsf{w}_k(\tilde U_1)e^{\mathrm{i}\lambda})} = \tilde U_1^{(\mathsf{w}_k(\tilde U_1))}$. Here A_1 and λ are real, so all configurations and transformations are real, as Theorem 22.5(b) requires. By that theorem the two transformations coincide iff they agree at the points of $\mathbb{T}$. With

$$\Upsilon(\lambda) = \ell\big(\mathsf{w}_k(\tilde U_1)^{-1}\mathsf{w}_k(\tilde U_\lambda)\big) \quad \text{on } \mathbb{T},$$

(E2) is therefore equivalent to $\Upsilon(\lambda) = \lambda|_{\mathbb{T}}$.

Expansion of Υ. Υ is an analytic function of (A_1, λ) on $\{|A_1|_\infty \le 2r_0,\ |\lambda|_{C^1} \le 4(1 + C_H)\rho_0\}$, where C_H is the constant in $|H'X|_{C^1} \le C_H|X|_{,}$ with $|\Upsilon| \le C(r_0 + \rho_0)$ by Proposition 22.9(d), and $\Upsilon(A_1, 0) = 0$. At $A_1 = 0$, Theorem 22.5(e) gives $\mathsf{w}_k(1^{(g)})|_{\mathbb{T}} = \mathfrak{M}^{(k)}(g)\, g^{-1}|_{\mathbb{T}}$. With

$g = e^{-\mathrm{i}\lambda}$, the linear part of $\Upsilon(0,\cdot)$ is $\lambda|_{\mathbb{T}} - Q'_k\lambda$. This uses Theorem 22.5(e) for real λ only: $\Upsilon(0,\cdot)$ is analytic, so its linear part is determined by its values at real λ. Write

$$\Upsilon(A_1,\lambda) = \lambda|_{\mathbb{T}} - Q'\lambda - \mathcal{C}'(\lambda).$$

By the Cauchy estimates, applied separately to $\Upsilon(0,\lambda)$ (quadratic remainder) and to $\Upsilon(A_1,\lambda) - \Upsilon(0,\lambda)$ (which vanishes at $A_1 = 0$ and at $\lambda = 0$),

$$|\mathcal{C}'(\lambda)| \le C(|A_1|_\infty + |\lambda|_{C^1})|\lambda|_{C^1}, \qquad |\mathcal{C}'(\lambda) - \mathcal{C}'(\lambda')| \le C(r+\rho)|\lambda - \lambda'|_{C^1}$$

for $|\lambda|_{C^1}, |\lambda'|_{C^1} \le \rho \le 2(1+C_H)\rho_0$. So (E2) is equivalent to $Q'\lambda + \mathcal{C}'(\lambda) = 0$.

Linearizing (E2). Decompose $\lambda = \lambda' + H'X$ with $\lambda' \in \mathcal{N}(Q')$ and $X = Q'\lambda$. Then (E2) becomes $X = -\mathcal{C}'(\lambda' + H'X)$. Since $|H'X|_{C^1} \le C_H|X|$ (Lemma 24.1(a)), the right-hand side is a contraction in X on $|X|_\infty \le \rho_0$, for $|\lambda'|_{C^1} \le (1+C_H)\rho_0$. Its unique fixed point $X = \mathcal{X}(\lambda')$ satisfies $|\mathcal{X}(\lambda')| \le C(r + |\lambda'|_{C^1})|\lambda'|_{C^1}$ and has Lipschitz constant $C(r+\rho)$ in λ'. This is the analogue of [6, Sect. E].

Solving (E1). Apply D^T to (24.1). By the discrete Leibniz rule,

$$D^T\mathcal{A}(\lambda) = e^{-\mathrm{i}\operatorname{ad}\lambda}D^TA_1 + (1+V(\lambda))D^TD\lambda + \mathfrak{g}_3,$$

where $V(\lambda)$ is multiplication by $\mathfrak{d}(\operatorname{ad}\lambda) - 1$, so $|V| \le C|\lambda|_\infty$, and $|\mathfrak{g}_3| \le C|D\lambda|(|A_1| + |D\lambda|)$. Now $D^TD\lambda = D^TD\lambda' + Q'^T(\Gamma' - a)X$ by Lemma 24.1(a). Moreover $D^TD\lambda' \in \operatorname{ran} R$, so $R(1+V)D^TD\lambda' = (1+RVR)D^TD\lambda'$. Hence (E1) is equivalent to

$$(1+RVR)D^TD\lambda' = -R\mathfrak{h}(\lambda'), \qquad \mathfrak{h} = e^{-\mathrm{i}\operatorname{ad}\lambda}D^TA_1 + (1+V)Q'^T(\Gamma' - a)\mathcal{X}(\lambda') + \mathfrak{g}_3.$$

For $|V|$ small, $1 + RVR$ is invertible on bounded fields (Lemma 24.1(b)), and its inverse maps $\operatorname{ran} R$ into $\operatorname{ran} R$. For $h \in \operatorname{ran} R = \mathcal{W}$ the unique $\lambda' \in \mathcal{N}(Q')$ with $D^TD\lambda' = h$ is $G'h$ (Proposition 14.2(c)). Therefore (E1) and $\lambda' \in \mathcal{N}(Q')$ are equivalent to the fixed point equation

$$\lambda' = \mathcal{T}(\lambda') := -G'(1+RVR)^{-1}R\,\mathfrak{h}(\lambda').$$

We have $|D^TA_1| \le Cr$, so $|\mathfrak{h}|_\infty \le C(r+\rho^2)$ on $|\lambda'|_{C^1} \le \rho$, and $\mathcal{T}$ maps this ball into itself for $\rho = C'r$ and r small. The λ'-dependence of $\mathcal{T}$ enters through V, $\mathcal{X}$, $e^{-\mathrm{i}\operatorname{ad}\lambda}$ and $\mathfrak{g}_3$. Each of these is multiplied by a quantity of size $C(r+\rho)$, or has Lipschitz constant $C(r+\rho)$. Hence $\mathcal{T}$ is a contraction on $|\lambda'|_{C^1} \le (1+C_H)\rho_0$, for r_0 and ρ_0 small. Its unique fixed point, together with $\lambda = \lambda' + H'\mathcal{X}(\lambda')$, is the solution. It is real because all maps preserve reality, and $|\lambda|_{C^1} \le C_2r$. The bound on $\mathcal{A}(\lambda)$ follows from (24.1).

Uniqueness. If λ solves (E1), (E2) with $|\lambda|_{C^1} \le \rho_0$, then $X = Q'\lambda$ and $\lambda' = \lambda - H'X$ satisfy $|X| \le \rho_0$, $|\lambda'|_{C^1} \le (1+C_H)\rho_0$, $X = \mathcal{X}(\lambda')$ and $\lambda' = \mathcal{T}(\lambda')$. Since $\mathcal{T}$ is a strict contraction on this ball, λ', and hence λ, is unique. □

24.6 Proof of Theorem 24.2

By Lemma 24.4, applied with θ replaced by $\theta + C\varepsilon$ (we have $U \in \mathfrak{U}(\theta) \subset \mathfrak{U}_0(\theta)$, and the curvature of U_0 is at most $C\varepsilon$), the hypothesis implies $|\ell(U'^{[j]})| \le \beta_* := \beta + C_A(\theta + C\varepsilon)$ for all j. We prove the following statement $\mathcal{P}(k)$ by induction on k: *under the hypotheses of the theorem, with $|\ell(U'^{[j]})| \le \beta_*$ for all $j \le k$ and $\varepsilon + \theta + \beta_* \le c_4^\sharp$, there is a solution $\tilde U$ of (i), (ii) obeying the bounds with B_0' and β_* in place of B_1 and β.* Here B_0' and $c_4^\sharp$ are fixed in the induction step below. Then c_4 is chosen with $(1 + C_A(1+C))c_4 \le \min(c_4^\sharp, \frac12 c_G)$ and $(1+C)c_4 \le c_1$, so that Lemma 24.4 applies; this gives existence with $B_1 = B_0'(1 + C_A(1+C))$. Finally the uniqueness radius is $r_u = \min(r_*, r_0, \rho_0/(4C_w))$, with ρ_0 from Lemma 24.6 and C_w the constant of the uniqueness step below; here $r_* \le c_3$, since Proposition 24.5 uses Proposition 22.9. We decrease

c_4 further until $B_1 c_4 \le r_u/2$ and $C_w B_0 c_4 \le \rho_0/2$, so that the solution lies in the uniqueness ball. Since $C_A = 45L$, the constants c_4 and B_1 depend on L; so does $c_4^\sharp$, by the conventions of Part IV.

For the induction we use scale covariance. Rescaling all lengths by δ_j^{-1} maps the structure on $(\mathbb{T}_\eta, \mathbb{T}^{(j)})$ to the j-step structure on a torus of side n/δ_j with fine spacing η/δ_j. It multiplies Lie fields by δ_j, gradients by δ_j^2, curvatures by δ_j^2 and currents by δ_j^3. The action does not change, since $d = 4$ (Section 23). Hence the hypotheses of $\mathcal{P}(k)$ imply those of $\mathcal{P}(k-1)$ for the averaging up to scale $k-1$, in units of δ_{k-1}, with the same β_* and smaller ε, θ. Indeed, the cubes of Definition 18.1 in the units of δ_{k-1} lie in cubes of the same kind in the units of δ_k, so the gauge transformations restrict as in the proof of Lemma 31.2; the quantities $|A|$, $|\partial A|$ and $[\partial A]_{\alpha,\Lambda_3(z')}$ of that definition are multiplied by L^{-1}, L^{-2} and $L^{-2-\alpha}$, and $|J_{U_0}|$ by L^{-3}. So U_0 is $(M, L^{-1}\varepsilon)$-regular in the new units, $U \in \mathfrak{U}(L^{-2}\theta)$, and $\mathrm{Ax}_k \subset \mathrm{Ax}_{k-1}$ by Definition 22.2. All constants are uniform in k and n.

$k = 0$. There is no averaging: $R = 0$, $\mathsf{w}_0 = 1$, and $\tilde U = U'$. Since $\delta_0 = 1$ in these units, $|A| = |\ell(U')| \le \beta_*$, and the gradient and the Laplacian are bounded by $C|A|$.

From $k-1$ to k. By $\mathcal{P}(k-1)$, in units of δ_{k-1}, there is $\tilde U_1 = e^{\mathrm{i}\eta A_1}$ with $R_{k-1} D^T A_1 = 0$ (condition (i) at scale $k-1$: R_{k-1} is the projection $R(U_0)$ built with the pair $(Q'_{k-1}(U_0), \mathbf{Q}_{k-1}(U_0))$ in place of $(Q'_k(U_0), \mathbf{Q}_k(U_0))$), $\tilde U_1^{(\mathsf{w}_{k-1}(\tilde U_1))} = U'$, and, back in units of $\delta_k = 1$,

$$|A_1|_\infty \le L B'_0(\varepsilon + \theta + \beta_*), \qquad |\nabla A_1|_\infty \le L^2 B'_0(\varepsilon + \theta + \beta_*).$$

Since $U' \in \mathrm{Ax}_k$, the configuration $\mathcal{Q}_{k-1}(\tilde U_1) = U'^{[k-1]}$ is axial at scale k. By Theorem 22.5(d), $\mathsf{w}_k(\tilde U_1) = \mathsf{w}_{k-1}(\tilde U_1)$, so $\tilde U_1^{(\mathsf{w}_k(\tilde U_1))} = U'$. Apply Lemma 24.6 with $r = 2L^2 B'_0(\varepsilon+\theta+\beta_*) \ge |A_1|_{C^1}$; we need $r \le r_0$ and $C_2 r \le r_*$, which holds if $c_4^\sharp \le \min(r_0, r_*/C_2)/(2L^2 B'_0)$. We obtain λ such that $\tilde U = \tilde U_\lambda$ is in the Landau gauge at scale k, and

$$\tilde U^{(\mathsf{w}_k(\tilde U))} = \tilde U_1^{(\mathsf{w}_k(\tilde U_1) e^{\mathrm{i}\lambda} e^{-\mathrm{i}\lambda})} = U'.$$

Its Lie field satisfies $|\mathcal{A}(\lambda)|_\infty \le C_2 r$. By Theorem 22.5(a), $\mathcal{Q}(\tilde U) = U'^{[k]}$, so $|B| \le \beta_*$ with $B = \ell(U'^{[k]})$. Moreover $J_{\tilde U U_0}$ is the current of a configuration gauge equivalent to U, so $|J_{\tilde U U_0}| \le \theta$. If $C_2 r \le r_*$, Proposition 24.5 gives the bounds with $B'_0 := \max(B_0, C)$, where C is the constant of the case $k = 0$. The constant does not depend on k, which closes the induction.

Uniqueness. Let $\tilde U = e^{\mathrm{i}\eta A}$ and $\tilde U^\sharp = e^{\mathrm{i}\eta A^\sharp}$ both satisfy (i), (ii) and $|A|, |A^\sharp| \le r_u \le r_*$. By Proposition 24.5, both have $|\cdot|_{C^1} \le B_0(\varepsilon + \theta + \beta)$. Let $g = \mathsf{w}_k(\tilde U^\sharp)^{-1} \mathsf{w}_k(\tilde U)$. Then $\tilde U^\sharp = \tilde U^{(g)}$. By Proposition 22.9(d), $|\ell(g)| \le C(|A| + |A^\sharp|)$ pointwise, so $g = e^{-\mathrm{i}\lambda}$ with λ real. We take $C_w \ge 4C$ for this constant C, and $\rho_0 \le \frac12$, as we may in Lemma 24.6. Since $r_u \le \rho_0/(4C_w)$, this gives $|\lambda|_\infty \le 2Cr_u \le \frac12 C_w r_u \le \frac18 \rho_0 \le \frac{1}{16}$. The derivative is bounded by the bond equation. For a bond b put $\xi = \lambda(b_-)$ and $\xi' = R(U_0(b))\lambda(b_+)$, so that $\eta D_{U_0}\lambda(b) = \xi' - \xi$ and $|\xi|, |\xi'| \le \frac14$. The relation $\tilde U^\sharp = \tilde U^{(g)}$ reads $e^{\mathrm{i}\xi'} = e^{-\mathrm{i}\eta A(b)} e^{\mathrm{i}\xi} e^{\mathrm{i}\eta A^\sharp(b)}$. The factors are unitary, so Lemma 21.2(a) gives $|e^{\mathrm{i}\xi'} - e^{\mathrm{i}\xi}| \le |e^{-\mathrm{i}\eta A(b)} - 1| + |e^{\mathrm{i}\eta A^\sharp(b)} - 1| \le 2\eta(|A(b)| + |A^\sharp(b)|)$. Both $e^{\mathrm{i}\xi}$ and $e^{\mathrm{i}\xi'}$ lie in $\{|h-1| \le \frac12\}$, where ℓ is analytic with bounded derivative, and $\ell(e^{\mathrm{i}\xi}) = \xi$, $\ell(e^{\mathrm{i}\xi'}) = \xi'$ by Lemma 21.2(d). Hence $|D\lambda|_\infty \le C'(|A|_\infty + |A^\sharp|_\infty) \le 2C' r_u$, and we take $C_w \ge 4C'$ too. Thus $|\lambda|_{C^1} \le C_w(r_u + B_0(\varepsilon+\theta+\beta)) \le \rho_0$, by the choice of r_u and c_4, and λ solves (E1), (E2) of Lemma 24.6 with base $\tilde U_1 = \tilde U$. The same holds for $\lambda = 0$. By the uniqueness part of the lemma, $\lambda = 0$ and $\tilde U^\sharp = \tilde U$. □

Part V: The variational problem

Part V solves Bałaban's variational problem relative to a given reference: minimize the Wilson action on the set $\mathfrak{B}(V)$ of fine configurations of small curvature whose k-fold average is a

given real coarse configuration V (Section 25). The reference U_0 is real, (M,ε)-regular, with current $|J_{U_0}|_\infty \le \varepsilon$, and the averages of V are close to those of U_0: $V(c)W_k(c)^{-1} = e^{\mathrm{i}B(c)}$ with $|B|_\infty \le \beta$. The critical points are Bałaban's background fields. Part VI supplies the reference by an induction over k, and Part VII expands the renormalization-group step around the background field, with its multiplier as the gradient of the coarse action.

What this Part proves. The main result is Theorem 25.1, which is Theorem I. If $\varepsilon+\beta$ is small, then:

- the action on $\mathfrak{B}(V)$ has exactly one critical $\mathfrak{G}_0$-orbit among configurations of small curvature and current, and it consists of the minimizers there (Theorem 25.1(a),(b));
- its Landau representative A_* relative to U_0 is small with its derivatives, and $A_* = \mathbf{H}B - \mathbf{C}J_{U_0} + O((\varepsilon+\beta)^2)$ (Theorem 25.1(c));
- there is a unique Euler–Lagrange multiplier ω on the unit lattice (Theorem 25.1(d));
- the minimizer is $(M, \varepsilon + C_6(\varepsilon+\beta))$-regular, A_* is analytic in complex B, and dA_*/dB decays exponentially (Theorem 25.1(e),(f));
- the orbit is covariant under gauge transformations and under the lattice symmetries $\mathfrak{S}_k$ (Theorem 25.1(g),(h)).

The constants do not depend on k, n or η; they depend on L, and may be taken independent of $M \ge M_0$. The proof has four steps (Section 25): reduction to the Landau chart of Part IV; a linearizing transformation of the constraint and a contraction for $A_1 = -\mathbf{C}(J + \nabla\mathcal{V}(A_1 + \mathbf{H}B))$ (Lemma 26.1, Proposition 26.3); convexity, which gives minimality (Proposition 26.6); and the Euler–Lagrange equation, through the gauge covariance of the gauge-adapted averaging (Lemma 27.1, Proposition 27.2).

Imports. From Part IV: Theorem 24.2, Corollary 24.3, Propositions 22.9, 22.10 and 23.3, and Lemma 24.1. Parts II and III enter only through the pointwise bounds of Lemma 24.1, the identities for $\mathbf{H}$ and $\mathbf{C}$ (Theorem 16.1) and the coercivity on the tangent space (Lemma 26.5); the random-walk expansion is not used (Remark 27.6). No proof in Part V uses a later result. The constants are fixed at the end of Section 27; one condition there serves only the Dirichlet version of the contraction in Section 39, which is off the main line.

Exports. Theorem 25.1 goes to the induction (Lemma 29.2, Theorem 30.3), to Part VII, and to Parts IX–XI. The linearizing transformation, the bounds on $\mathcal{V}$ and the contraction (Lemmas 26.1, 26.2, Proposition 26.3) are used again for the coarse action in Part VII and for the weakened background map in Part IX (Section 45). The Euler–Lagrange equation (Proposition 27.2) goes to Parts VI, VII and XI. Lemma 26.7, Proposition 26.6 and Lemma 25.2 are used again only in Part XI.

Details for this Part. The companion has no chapter on Part V; its parameter ledger lists the constants of Part V that Parts VII–XI use, with their order of choice.

25 The variational theorem

25.1 The problem

Let V be a real configuration on $\mathbb{T}$. The *variational problem* is to minimize the Wilson action S on

$$\mathfrak{B}(V) = \{U \in \mathfrak{U}_0(c_1) \text{ real} : \ U^{(k)} = V\},$$

the set of fine configurations of small curvature whose k-fold average is V [8, (3), (5), (6)]. All averages in this definition are defined: for $U \in \mathfrak{U}_0(c_1)$ the configurations $U^{(j)}$ lie in $\mathfrak{U}_j(2c_1)$, by Proposition 21.12, and for a real reference $U_0 \in \mathfrak{U}_0(c_1)$ the relative configuration UU_0^{-1} lies in the averaging domain up to scale k (Section 22). So the domain of the group average is implied by the curvature bound c_1, through the condition $c_1 \le r_M/C$ of Part IV. Likewise, the small relative configurations $e^{i\eta A}$ of the Landau chart lie in the domain of the gauge-adapted averaging by Lemma 22.8, through $c_2, c_3 \le r_M/C$. Both S and $\mathfrak{B}(V)$ are invariant under the residual gauge group $\mathfrak{G}_0$. A *critical orbit* is a $\mathfrak{G}_0$-orbit of critical points of S restricted to $\mathfrak{B}(V)$. Its elements are Bałaban's *background fields* (minimal configurations) $U_k(V)$.

Bałaban's Theorem 1 [8] solves this problem by induction over the number k of averaging steps. The k-th step starts from a reference configuration U_0, the solution of the $(k-1)$-step problem for an extension of V. It solves the problem near U_0 by a contraction, and then re-establishes the regularity of the solution with constants independent of k, by a local analysis on cubes. The theorem below is the first part of this scheme: the problem relative to a given regular reference configuration [8, Sects. A–E, G]. The induction over k, with the local improvement of [8, Sect. F], is carried out in Part VI (Theorem 30.3).

25.2 Statement

Throughout this part, U_0 is a real reference configuration that is (M,ε)-regular with $|J_{U_0}|_\infty \le \varepsilon$. As before, $W_k = U_0^{(k)}$, and the operators of Parts II and III are taken at U_0 with the averaging pair $(Q'_k(U_0), \mathbf{Q}_k(U_0))$. We fix once and for all a small constant $\theta_0 > 0$; the uniqueness statements refer to the class $\mathfrak{U}(\theta_0)$.

Theorem 25.1 (The variational problem relative to a regular reference)**.** *There are $\theta_0, c_6, C_6, \gamma' > 0$ such that the following holds. Let U_0 be as above, and let V be a real configuration on $\mathbb{T}$ with*

$$V(c)W_k(c)^{-1} = e^{iB(c)}, \qquad |B|_\infty \le \beta.$$

If $\varepsilon + \beta \le c_6$, then:

(a) (Existence and uniqueness.) *$S|_{\mathfrak{B}(V)}$ has exactly one critical orbit in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$.*

(b) (Minimality.) *This orbit is the set of minimizers of S on $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$.*

(c) (Landau representative.) *The orbit contains $U_* = \mathsf{U}(A_*) = (e^{i\eta A_*}U_0)^{\mathsf{w}_k(e^{i\eta A_*})}$, where A_* is real, $R(U_0)D_{U_0}^T A_* = 0$, $\mathcal{Q}(e^{i\eta A_*}) = e^{iB}$, and*

$$|A_*|_{C^1} + |\Delta_{U_0} A_*|_\infty + \sup_y [\nabla \tilde{A}_{*,y}]_{\alpha,B(y)} \le C_6(\varepsilon+\beta).$$

The proof gives the Hölder bound on the unit cubes $\Lambda_1(c_y)$ in place of $B(y)$ (Theorem 6.3), and, by Remark 7.2, on every cube $\Lambda_1(z)$. This is the form used in the proof of (e) and in Corollary 30.4. To first order, $A_* = \mathbf{H}B - \mathbf{C}J_{U_0} + O((\varepsilon+\beta)^2)$ in $|\cdot|_{C^1}$.

(d) (Euler–Lagrange equation.) *There is a unique unit-lattice field ω with $\nabla\Phi(A_*) = d\mathcal{Q}(A_*)^T\omega$, where $\Phi(A) = S(e^{i\eta A}U_0)$ and $d\mathcal{Q}(A)$ is the derivative of $A \mapsto \ell(\mathcal{Q}(e^{i\eta A}))$. It satisfies $|\omega|_\infty \le C_6(\varepsilon+\beta)$.*

(e) (Regularity of the minimizer.) *U_* is $(M, \varepsilon + C_6(\varepsilon+\beta))$-regular and $U_* \in \mathfrak{U}(C_6(\varepsilon+\beta))$.*

(f) (Analyticity and decay.) *$B \mapsto A_*(B)$ extends to an analytic map on complex unit-lattice fields with $\varepsilon + |B|_\infty < c_6$, with $|A_*(B)|_{C^1} \le C_6(\varepsilon + |B|_\infty)$. For complex B the C^1 bound and the decay below are claimed, but not the second-order and Hölder bounds of (c). For real and complex B, if δB is supported on the bond c, then*

$$|\,dA_*(B)\delta B\,(x)| + |\nabla(dA_*(B)\delta B)(x)| \le C_6 e^{-\gamma'|x-c|}|\delta B|.$$

(g) (Covariance.) *For a real v on $\mathbb{T}$, the critical orbit for V^v is the image of the critical orbit for V under any real u with $u|_{\mathbb{T}} = v$.*

(h) (Lattice symmetry.) *Let $r \in \mathfrak{S}_k$. Then rU_0 satisfies the hypotheses above with the same ε, $W_k(rU_0) = rW_k$, and $(rV)(c)\,(rW_k)(c)^{-1} = e^{\mathrm{i}(r_{W_k}B)(c)}$ with $|r_{W_k}B|_\infty = |B|_\infty$. The critical orbit for rV is the image under r of the critical orbit for V. The Landau representatives and the multipliers satisfy*

$$A_*(r_{W_k}B; rU_0) = r_{U_0}A_*(B; U_0), \qquad \omega(r_{W_k}B; rU_0) = r_{W_k}\,\omega(B; U_0),$$

where the second argument is the reference. The first identity also holds for complex B as in (f).

Parts (a)–(c) and (e) are Bałaban's Theorem 1 [8], relative to a reference and on a single torus; the regularity bounds are [8, (9), (10)] with the reference-dependent constant. Parts (d) and (f) are [8, (171), (178)] and [8, Prop. 9, (190)]. Part (h) is the lattice symmetry of the problem; here r acts on configurations by $(rU)(b) = U(r^{-1}b)$, and r_{U_0}, r_{W_k} act on Lie fields relative to U_0 and W_k as in Proposition 22.10(g). The constants do not depend on k, n or η. They depend on L. The order in which they are fixed is given in the proof of the theorem, in Section 27.6. They may be taken independent of $M \geq M_0$, for ε below the threshold at $M = M_0$; this threshold includes ε_{B} of Lemma 31.3(c), taken at $M = M_0$. Lemma 31.3 is stated in Part VII, but it, its proof and the two items before it (Definition 31.1, Lemma 31.2) use only Parts I–IV. Indeed, an (M, ε)-regular reference is (M_0, ε)-regular (Lemma 31.2), the torus operators of Parts II and III do not depend on the regularity scale (Step 0 of the proof of Lemma 31.3), and the proof of the regularity statement in (e) uses only local estimates on the cubes of Definition 18.1. The thresholds that depend on M are those of Parts VI–VIII, for instance a_1 of Theorem 30.3 (whose constant B_3 also depends on M), a_δ of Proposition 31.5 and $\varepsilon_L \leq \varepsilon_{\mathrm{cmp}}(M'_L)$ with $M'_L \geq M$ (Theorem 34.4); the constants of Part VII do not (Remark 34.7).

In (c), A_* is the Landau representative of the orbit relative to U_0, and $U_* = \mathsf{U}(A_*)$ is its averaged axial representative. The configuration $e^{\mathrm{i}\eta A_*}U_0$ is gauge equivalent to U_*, but it need not lie in $\mathfrak{B}(V)$, since $\mathsf{w}_k(e^{\mathrm{i}\eta A_*})$ need not be 1 on $\mathbb{T}$.

Part (g) does not need (a): for real u with $u|_{\mathbb{T}} = v$, the map $U \mapsto U^u$ is a bijection from $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ onto $\mathfrak{B}(V^v) \cap \mathfrak{U}(\theta_0)$, because $\mathfrak{a}^{(k)}(U^u) = \mathfrak{a}^{(k)}(U)^{u|_{\mathbb{T}}}$. It preserves S and critical points, and it maps $\mathfrak{G}_0$-orbits to $\mathfrak{G}_0$-orbits, since $u\mathfrak{G}_0u^{-1} = \mathfrak{G}_0$. Part (h) is proved at the end of Section 27. The rest of this part proves (a)–(f). The proof has four steps:

1. Reduction to the Landau chart, by Corollary 24.3 (Section 25.3).
2. Linearization of the constraint and solution by contraction (Section 26).
3. Convexity, which gives minimality (Section 26).
4. The Euler–Lagrange equation, which shows that the constructed point is critical on $\mathfrak{B}(V)$ and gives the second-order bounds (Section 27).

25.3 Reduction to the Landau chart

Let $\theta_0 \leq c_4/3$ and $\varepsilon + \beta \leq c_6 \leq c_4/3$, so that Theorem 24.2 and Corollary 24.3 apply with $\theta = \theta_0$. Let

$$\mathcal{M}_V = \{A \text{ real}: \ RD^TA = 0,\ \ell(\mathcal{Q}(e^{\mathrm{i}\eta A})) = B,\ |A|_{C^1} < \rho_1\}, \qquad \rho_1 = 2B_1(\varepsilon + \theta_0 + \beta).$$

Lemma 25.2. *(a) $\mathcal{M}_V$ is a smooth manifold, and $\Phi = S \circ \mathsf{U}$ on $\mathcal{M}_V$, with U as in Corollary 24.3.*

(b) Every U in a critical orbit in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ has an averaged axial representative $\mathsf{U}(A)$ with $A \in \mathcal{M}_V$, and A is a critical point of $\Phi|_{\mathcal{M}_V}$.

(c) For every $U \in \mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ there is $A \in \mathcal{M}_V$ with $S(U) = \Phi(A)$.

Proof. (a) The map $A \mapsto \ell(\mathcal{Q}(e^{i\eta A}))$, restricted to the slice $\{RD^T A = 0\}$, has derivative $\mathbf{Q} + dC_k^{\mathrm{av}}(A)$. This is onto: $\mathbf{QH} = I$, $\mathbf{H}$ maps into the slice, and $|dC_k^{\mathrm{av}}(A)\mathbf{H}| \le C\rho_1 < 1$ in the sup norm by Proposition 22.9(b). Hence $\mathcal{M}_V$ is a submanifold. The identity $\Phi = S \circ \mathsf{U}$ holds by gauge invariance of S.

(b), (c) The orbit of U has a unique averaged axial representative (Lemma 22.3(c); note $U \in \mathfrak{U}(\theta_0) \subset \mathfrak{U}_0(c_1)$). By Corollary 24.3 it equals $\mathsf{U}(A)$ for some $A \in \mathcal{S}_V$, and by Theorem 24.2, $|A|_{C^1} \le B_1(\varepsilon + \theta_0 + \beta) < \rho_1$, so $A \in \mathcal{M}_V$. For (b), let $A(t)$ be a smooth curve in $\mathcal{M}_V$ with $A(0) = A$. Since $\rho_1 \le \frac{4}{3}B_1 c_4 \le \frac{2}{3}r_u$, the configurations $\mathsf{U}(A(t))$ are defined, real and smooth in t, and $\mathsf{U}(A(t))^{(k)} = \mathcal{Q}(e^{i\eta A(t)})W_k = V$. At $t = 0$ the curvature is that of U, at most $\theta_0 \le c_4/3 < c_1$; by continuity, $\mathsf{U}(A(t)) \in \mathfrak{U}_0(c_1)$ for small $|t|$. So $\mathsf{U}(A(t))$ is a curve in $\mathfrak{B}(V)$ through the critical point $\mathsf{U}(A)$, and $\frac{d}{dt}\Phi(A(t))\big|_{t=0} = \frac{d}{dt}S(\mathsf{U}(A(t)))\big|_{t=0} = 0$. □

By Lemma 25.2(b), uniqueness in (a) follows once we show that $\Phi|_{\mathcal{M}_V}$ has at most one critical point. Existence requires the converse: the critical point of $\Phi|_{\mathcal{M}_V}$ that we construct gives a critical point of $S|_{\mathfrak{B}(V)}$. The converse is not a formal consequence of (b), because U involves the gauge transformation w_k, which depends on A. It is proved in Section 27 via the Euler–Lagrange equation.

26 Solution in the Landau chart

We keep the setting of Theorem 25.1. Write $\mathbf{H} = \mathbf{H}(U_0)$ and $\mathbf{C} = \mathbf{C}(U_0)$ for the minimizer map and the constrained Green function of the quadratic form $q(A, A') = \langle DA, DA' \rangle$ (Theorem 19.9, with $W = 0$), and

$$\mathcal{X} = \{A : \ RD^T A = 0, \ \mathbf{Q}A = 0\}$$

for the tangent space of the constraint in Bałaban's gauge. By Theorem 16.1, $\mathbf{QH} = I$, $RD^T\mathbf{H} = 0$, $\mathbf{C}$ maps into $\mathcal{X}$, and for every f, $\mathbf{C}f$ is the unique $A \in \mathcal{X}$ with $q(k, A) = \langle k, f \rangle$ for all $k \in \mathcal{X}$. Moreover $q(k, \mathbf{H}B) = 0$ for $k \in \mathcal{X}$.

26.1 Linearizing the constraint

Lemma 26.1 (Linearizing transformation [8, Sect. C])**.** *There are $r_2^{\mathrm{V}}, C > 0$ such that for complex A' with $|A'|_{C^1} \le r_2^{\mathrm{V}}$ the equation $X = C_k^{\mathrm{av}}(A' - \mathbf{H}X)$ has exactly one solution $X = \mathcal{D}(A')$ with $|X|_\infty \le r_2^{\mathrm{V}}$. The map $\mathcal{D}$ is analytic, real for real A', and*

$$|\mathcal{D}(A')|_\infty \le C|A'|_\infty^2, \qquad |d\mathcal{D}(A')h(c)| \le C|A'|_\infty \|h\|_{S'_c}, \qquad |d^2\mathcal{D}(A')[h,h](c)| \le C\|h\|_{S'_c}^2,$$

where S'_c is a weighted neighbourhood of c: $\|h\|_{S'_c}^2 = \sum_y e^{-\gamma|y-c_-|}\|h\|_{B(y)}^2$. Moreover $d\mathcal{D}(A')$ has a kernel, $d\mathcal{D}(A')h(c) = \eta^d \sum_b K_{\mathcal{D}}(c,b)h(b)$, with $|K_{\mathcal{D}}(c,b)| \le C|A'|_\infty e^{-\gamma|c_- - b_-|}$; in particular $|d\mathcal{D}(A')^T\omega|_\infty \le C|A'|_\infty|\omega|_\infty$. For every unit-lattice field ω and every fine field h,

$$\big|(d^2\mathcal{D}(A')[h,\cdot])^T\omega\big|_{B(y)} \le C|\omega|_\infty \sum_{y'} e^{-\gamma|y-y'|}|h|_{B(y')} \qquad \text{for all } y \in \mathbb{T}, \tag{26.1}$$

where $|f|_{B(y)}$ is the maximum of $|f(b)|$ over the bonds b starting in $B(y)$, and transposes refer to the bilinear pairings of Section 2: with the factor η^d on $\mathbb{T}_\eta$ and without it on $\mathbb{T}$. The map $A' \mapsto A = A' - \mathbf{H}\mathcal{D}(A')$ satisfies

$$\ell(\mathcal{Q}(e^{i\eta A})) = \mathbf{Q}A', \qquad RD^T A = RD^T A'.$$

It is a bijection from $\{A' : |A'|_{C^1} < r_2^{\mathrm{V}}\}$ *onto a set containing* $\{A : |A|_{C^1} < r_2^{\mathrm{V}}/2\}$*, with analytic inverse.*

Proof. By Proposition 22.9(b) and $|\mathbf{H}X|_{C^1} \le C_H|X|_\infty$ (Lemma 24.1(d)),

$$|C_k^{\mathrm{av}}(A' - \mathbf{H}X) - C_k^{\mathrm{av}}(A' - \mathbf{H}X')| \le C(|A'| + C_H|X| + C_H|X'|)C_H|X - X'|.$$

So $X \mapsto C_k^{\mathrm{av}}(A' - \mathbf{H}X)$ is a contraction on $|X| \le r_2^{\mathrm{V}}$ for r_2^{V} small, mapping this ball into itself. Its fixed point is analytic in A', being a uniform limit of analytic iterates. The bound $|\mathcal{D}| \le C|A'|^2$ follows from $|C_k^{\mathrm{av}}(A)| \le C|A|^2$.

Derivatives. Put $A = A' - \mathbf{H}\mathcal{D}(A')$, $P = dC_k^{\mathrm{av}}(A)$, $T = (I + P\mathbf{H})^{-1}$ and $L = I - \mathbf{H}TP$. Differentiating the fixed-point equation once and twice gives

$$d\mathcal{D} = TP, \qquad dA = L, \qquad d^2\mathcal{D}(A')[h,k] = T\, d^2C_k^{\mathrm{av}}(A)[Lh, Lk].$$

Here T is a unit-lattice operator that is a small perturbation of I with decaying kernel, since $P\mathbf{H}$ has decay $(C|A|_\infty, \gamma)$. Proposition 22.9(c), together with the decay of $\mathbf{H}$ and T, gives the bounds in terms of $\|h\|_{S_c'}$ and the kernel bound.

The mixed bound (26.1). For r_2^{V} small, $|A|_\infty \le (1 + C_H)r_2^{\mathrm{V}} \le c_3/2$. By Proposition 22.9(a),(c), the kernel $K_A(c,b)$ of $dC_k^{\mathrm{av}}(A)$ is an analytic function of A on $\{|A|_\infty \le c_3\}$, bounded there by Cc_3, and $K_A(c,b) = 0$ unless b starts in S_c. Moreover $C_k^{\mathrm{av}}(A)(c)$ depends only on the values of A on the bonds starting in S_c. Let v be a fine field. By the Cauchy estimate in $t \mapsto K_{A+tv}(c,b)$, the kernel of $d^2C_k^{\mathrm{av}}(A)[v,\cdot]$, defined by $d^2C_k^{\mathrm{av}}(A)[v,k](c) = \eta^d \sum_b \partial_v K_A(c,b)k(b)$, satisfies $|\partial_v K_A(c,b)| \le C \max_{S_c}|v|$, with the same support; here $\max_{S_c}$ is over the bonds starting in S_c. The transpose has no factor η^d, so $(d^2C_k^{\mathrm{av}}(A)[v,\cdot])^T\omega(b) = \sum_c \partial_v K_A(c,b)^T\omega(c)$. Every fine bond starts in S_c for at most $2d$ bonds c of $\mathbb{T}$. Hence

$$\big|(d^2C_k^{\mathrm{av}}(A)[v,\cdot])^T\omega\big|_{B(y)} \le C|\omega|_\infty \max_{|y'-y|_\infty \le 1} |v|_{B(y')}.$$

Now $(d^2\mathcal{D}(A')[h,\cdot])^T\omega = L^T\big(d^2C_k^{\mathrm{av}}(A)[Lh,\cdot]\big)^T T^T\omega$. The operators T and T^T have decaying kernels, so $|T^T\omega|_\infty \le C|\omega|_\infty$. The operators P and P^T are local and bounded by $C|A|_\infty$. The operators $\mathbf{H}$ and $\mathbf{H}^T$ satisfy block bounds of the form $|\mathbf{H}X|_{B(y)} \le C\sum_c e^{-\gamma|y-c_-|}|X(c)|$ and $|\mathbf{H}^T f(c)| \le C\sum_{y'} e^{-\gamma|c_- - y'|}|f|_{B(y')}$, by the block form of the pointwise bounds in the proof of Lemma 24.1; the second follows from the first, since $\mathbf{H}^T f(c) = \langle \mathbf{H}\delta_c, f\rangle$ componentwise and a block has volume 1. Hence $|Lh|_{B(y)} \le C\sum_{y'} e^{-\gamma|y-y'|}|h|_{B(y')}$, and $L^T = I - P^TT^T\mathbf{H}^T$ preserves bounds of this form, with a smaller rate. This gives (26.1), after decreasing γ.

The identities follow from $\ell(\mathcal{Q}(e^{\mathrm{i}\eta A})) = \mathbf{Q}A + C_k^{\mathrm{av}}(A)$, $\mathbf{QH} = I$ and $RD^T\mathbf{H} = 0$. The inverse is explicit: at the fixed point $\mathcal{D}(A') = C_k^{\mathrm{av}}(A)$ with $A = A' - \mathbf{H}\mathcal{D}(A')$, so $A' = A + \mathbf{H}C_k^{\mathrm{av}}(A)$. Conversely, for $|A|_{C^1} < r_2^{\mathrm{V}}/2$ the field $A' = A + \mathbf{H}C_k^{\mathrm{av}}(A)$ satisfies $|A'|_{C^1} < r_2^{\mathrm{V}}$, and $X = C_k^{\mathrm{av}}(A)$ is a fixed point for this A'; by uniqueness $\mathcal{D}(A') = C_k^{\mathrm{av}}(A)$. □

By Lemma 26.1, the manifold $\mathcal{M}_V$ is mapped analytically onto (a subset of) the affine space

$$\mathcal{K}_B = \{A' :\ RD^TA' = 0,\ \mathbf{Q}A' = B\} = \mathbf{H}B + \mathcal{X}.$$

Critical points of $\Phi|_{\mathcal{M}_V}$ correspond to critical points of

$$\mathcal{J}(A') = \Phi\big(A' - \mathbf{H}\mathcal{D}(A')\big) \quad \text{on } \mathcal{K}_B \cap \{|A'|_{C^1} < r_2^{\mathrm{V}}\},$$

provided $\rho_1 \le r_2^{\mathrm{V}}/2$, which we assume.

26.2 The functional

By Proposition 23.3, with $J = J_{U_0}$,

$$\mathcal{J}(A') = S(U_0) + \langle A', J\rangle + \tfrac{1}{2}\|DA'\|^2 + \mathcal{V}(A'), \tag{26.2}$$

$$\mathcal{V}(A') = -\langle \mathbf{H}\mathcal{D}(A'), J\rangle - \langle DA', D\mathbf{H}\mathcal{D}(A')\rangle + \tfrac{1}{2}\|D\mathbf{H}\mathcal{D}(A')\|^2 + \mathcal{N}(A' - \mathbf{H}\mathcal{D}(A')).$$

Lemma 26.2 (Bounds on $\mathcal{V}$ [8, Prop. 4])**.** *There are $r_3 \le r_2^{\mathrm{V}}$ and C such that for complex A' with $|A'|_{C^1} \le r \le r_3$ (in the proof below, A' on the slice $RD^TA' = 0$; off the slice use $\mathbf{H}^TD^TDA' = (D\mathbf{H})^TDA'$, bounded by $C|DA'|_\infty$ by the kernel decay of $D\mathbf{H}$):*

(a) $|\nabla\mathcal{V}(A')|_\infty \le C(\varepsilon + r)|A'|_{C^1}$, and $|\nabla\mathcal{V}(A') - \nabla\mathcal{V}(A'')|_\infty \le C(\varepsilon + r)|A' - A''|_{C^1}$ if also $|A''|_{C^1} \le r$;

(b) for real A' and real h: $|d^2\mathcal{V}(A')[h,h]| \le C(\varepsilon + r)(\|h\|^2 + \|\nabla h\|^2)$.

Proof. We use two identities. First, $\mathfrak{A}_\pi\mathbf{H} = \mathbf{Q}^T\mathbf{\Gamma}_\pi$, $\Pi\mathbf{H} = \mathbf{H}$ and $\mathbf{Q}\Pi = \mathbf{Q}$ give $\Pi^TD^TD\mathbf{H} = \mathbf{Q}^T(\mathbf{\Gamma}_\pi - a)$. Hence

$$D^TD\mathbf{H} = \mathbf{Q}^T(\mathbf{\Gamma}_\pi - a) + S^T\mathfrak{F}^TD\mathbf{H},$$

since $\Pi = I - DS$ and $D^TD^T = \mathfrak{F}^T$. Here $S = G'RD^T$ and $\Pi = I - DS$ are the slice operators of Section 15.1 at U_0, not the action $S(U)$, and $\mathfrak{F} = D \circ D$ is the curvature operator of Lemma 12.3. By Lemma 24.1 and $|\mathfrak{F}| \le C\varepsilon$, $|D^TD\mathbf{H}X|_\infty \le C|X|_\infty$. Second, transposing the first identity, $\mathbf{H}^TD^TD = (\mathbf{\Gamma}_\pi - a)\mathbf{Q} + \mathbf{H}^TD^T\mathfrak{F}S$. On the slice $SA' = G'RD^TA' = 0$, so $\mathbf{H}^TD^TDA' = (\mathbf{\Gamma}_\pi - a)\mathbf{Q}A'$ for $A' \in \mathcal{K}_B$, which is bounded by $C|A'|_\infty$. Moreover $|\mathbf{H}^Tf|_\infty \le C|f|_\infty$, since $\mathbf{H}^T = \mathbf{\Gamma}_\pi\mathbf{Q}\mathbf{G}$ has a decaying kernel.

(a) The gradient is

$$\nabla\mathcal{V} = -d\mathcal{D}^T\mathbf{H}^TJ - D^TD\mathbf{H}\mathcal{D} - d\mathcal{D}^T\mathbf{H}^TD^TDA' + d\mathcal{D}^T\mathbf{H}^TD^TD\mathbf{H}\mathcal{D} + (I - \mathbf{H}d\mathcal{D})^T\nabla\mathcal{N}(A' - \mathbf{H}\mathcal{D}).$$

Each term is bounded using the identities above, $|J| \le \varepsilon$, Lemma 26.1, Proposition 23.3(a), and the kernel bound $|d\mathcal{D}(A')^T\omega|_\infty \le C|A'|_\infty|\omega|_\infty$ of Lemma 26.1. The Lipschitz bound follows from the Cauchy estimate on the complex ball of radius $2r$, as in Proposition 23.3(b).

(b) $d^2\mathcal{V}(A')[h,h]$ consists of three kinds of terms:

- terms containing $d\mathcal{D}[h]$ once or twice, bounded by $C|A'|\|h\|^2$ or $C|A'|\|h\|\|\nabla h\|$ by Lemma 26.1 and the L^2 boundedness of $\mathbf{H}$, $D\mathbf{H}$;
- terms $\langle d^2\mathcal{D}[h,h], \omega'\rangle$ with $\omega' = -\mathbf{H}^T(J + D^TDA + \nabla\mathcal{N}(A))$, $A = A' - \mathbf{H}\mathcal{D}$. By the identities above $|\omega'|_\infty \le C(\varepsilon + r)$, and $\sum_c |d^2\mathcal{D}[h,h](c)| \le C\|h\|^2$ because the weights $e^{-\gamma|y-c_-|}$ are summable;
- $d^2\mathcal{N}(A)[(I - \mathbf{H}d\mathcal{D})h, (I - \mathbf{H}d\mathcal{D})h]$, bounded by Proposition 23.3(c).

Altogether this gives (b). □

26.3 The equation and its solution

Proposition 26.3 ([8, Props. 5, 6])**.** *Let C_H, C_C be constants with $|\mathbf{H}B|_{C^1} \le C_H|B|_\infty$ and $|\mathbf{C}f|_{C^1} \le C_C|f|_\infty$ (Lemma 24.1(d)), and $K = 4C_C(1 + C_H)$. There is $c_7 > 0$ such that, if $\varepsilon + \beta \le c_7$, $\theta_0 \le c_7$ and $\rho_2 = 2\rho_1 + C_H\beta + K(\varepsilon + \beta)$ (with ρ_1 from Section 25.3), the following holds. A point $A' = A_1 + \mathbf{H}B$ with $A_1 \in \mathcal{X}$ and $|A'|_{C^1} < r_3$ is a critical point of $\mathcal{J}|_{\mathcal{K}_B}$ iff*

$$A_1 = \mathcal{T}(A_1) := -\mathbf{C}\big(J + \nabla\mathcal{V}(A_1 + \mathbf{H}B)\big). \tag{26.3}$$

The map $\mathcal{T}$ is a contraction on $\{A_1 \in \mathcal{X} : |A_1|_{C^1} \le \rho_2\}$. Its unique fixed point $A_{1,*}$ satisfies $|A_{1,*}|_{C^1} \le K(\varepsilon + \beta)$, and $A_{1,*} = -\mathbf{C}J + O((\varepsilon + \beta)^2)$. In particular $\mathcal{J}|_{\mathcal{K}_B}$ has exactly one critical point $A'_* = A_{1,*} + \mathbf{H}B$ with $|A'_* - \mathbf{H}B|_{C^1} \le \rho_2$.

Proof. The tangent space of $\mathcal{K}_B$ is $\mathcal{X}$. By (26.2), A' is critical iff $\langle k, J + D^T DA' + \nabla\mathcal{V}(A')\rangle = 0$ for all $k \in \mathcal{X}$. Since $\langle k, D^T D\mathbf{H}B\rangle = q(k, \mathbf{H}B) = 0$ for $k \in \mathcal{X}$, this is equivalent to $q(k, A_1) = -\langle k, J + \nabla\mathcal{V}(A')\rangle$ for all $k \in \mathcal{X}$, i.e. to (26.3), by the characterization of $\mathbf{C}$.

By Lemma 24.1(d), $|\mathbf{C}f|_{C^1} \le C_C|f|_\infty$. For $|A_1|_{C^1} \le \rho$ we have $|A'|_{C^1} \le \rho + C_H\beta$. By Lemma 26.2(a),

$$|\mathcal{T}(A_1)|_{C^1} \le C_C\varepsilon + C_C C(\varepsilon + \rho + C_H\beta)(\rho + C_H\beta),$$
$$|\mathcal{T}(A_1) - \mathcal{T}(A_1')|_{C^1} \le C_C C(\varepsilon + \rho + C_H\beta)|A_1 - A_1'|_{C^1}.$$

For $\rho \le \rho_2$, where $\rho_2 \le C(\varepsilon + \theta_0 + \beta)$, the Lipschitz constant is at most $\frac{1}{2}$ if c_7 is small. At $A_1 = 0$ the first bound gives $|\mathcal{T}(0)|_{C^1} \le C_C\varepsilon + C_C C(\varepsilon + C_H\beta)C_H\beta \le \frac{1}{2}K(\varepsilon + \beta)$ if c_7 is small. Hence, for $K(\varepsilon + \beta) \le \rho \le \rho_2$ and $|A_1|_{C^1} \le \rho$,

$$|\mathcal{T}(A_1)|_{C^1} \le |\mathcal{T}(0)|_{C^1} + \tfrac{1}{2}|A_1|_{C^1} \le \rho.$$

With $\rho = \rho_2$, which is at least $K(\varepsilon + \beta)$ by its definition, $\mathcal{T}$ maps the ball of radius ρ_2 into itself, and it is a contraction there. With $\rho = K(\varepsilon + \beta)$, its fixed point lies in the ball of radius $K(\varepsilon + \beta)$. The first-order formula follows from $|\nabla\mathcal{V}(A')| \le C(\varepsilon + \beta)^2$ at the fixed point. □

Put $A_* = A'_* - \mathbf{H}\mathcal{D}(A'_*)$. By Lemma 26.1, A_* is the unique critical point of $\Phi|_{\mathcal{M}_V}$; if $|A|_{C^1} < \rho_1$, then $|A'|_{C^1} \le \rho_1 + C\rho_1^2 \le 2\rho_1$, so $A_1 = A' - \mathbf{H}B$ satisfies $|A_1|_{C^1} \le \rho_2$. Moreover $|A_*|_{C^1} \le K'(\varepsilon + \beta)$ with $K' = K + C_H + 1$, and we assume $c_6 \le B_1\theta_0/K'$, so that $|A_*|_{C^1} < \rho_1$ and $A_* \in \mathcal{M}_V$. Also $A_* = \mathbf{H}B - \mathbf{C}J + O((\varepsilon + \beta)^2)$. This proves the first-order formula in Theorem 25.1(c), and, by Lemma 25.2(b), the uniqueness in (a).

26.4 Convexity and minimality

Lemma 26.4 (Covariant Hodge inequality)**.** *Let U_0 be real with curvature at most ε. For every real gauge field h, $|\|Dh\|^2 + \|D^T h\|^2 - \|\nabla h\|^2| \le C\varepsilon\|h\|^2$.*

Proof. Let D_ν act on the components h_μ as on scalar fields. Then $\|Dh\|^2 = \|\nabla h\|^2 - \sum_\mu \|D_\mu h_\mu\|^2 - \sum_{\mu\neq\nu}\langle D_\mu h_\nu, D_\nu h_\mu\rangle$ and $\|D^T h\|^2 = \sum_\mu \|D_\mu^T h_\mu\|^2 + \sum_{\mu\neq\nu}\langle D_\mu^T h_\mu, D_\nu^T h_\nu\rangle$. Since $R(U_0)$ is an isometry, $|D_\mu^T f(x)| = |D_\mu f(x - \eta e_\mu)|$, so the diagonal terms cancel. Summation by parts turns the remaining terms into $\sum_{\mu\neq\nu}\langle h_\mu, [D_\mu, D_\nu^T]h_\nu\rangle$. Expanding both products, all terms cancel except those at $x + \eta e_\mu - \eta e_\nu$:

$$[D_\mu, D_\nu^T]f(x) = \eta^{-2}\Big(R(U_\mu(x)U_\nu(x + \eta e_\mu - \eta e_\nu)^{-1}) \\ - R(U_\nu(x - \eta e_\nu)^{-1}U_\mu(x - \eta e_\nu))\Big)f(x + \eta e_\mu - \eta e_\nu).$$

The two transports go around the two halves of a plaquette, so their quotient is conjugate to its holonomy. Hence the operator norm is at most $C\varepsilon$, by Lemma 21.2. □

Lemma 26.5 (Coercivity on the tangent space)**.** *There is $c_X > 0$ such that $\|Dh\|^2 \ge c_X(\|h\|^2 + \|\nabla h\|^2)$ for all real $h \in \mathcal{X}$.*

Proof. For $h \in \mathcal{X}$, $\Pi h = h$, $RD^T h = 0$ and $\mathbf{Q}h = 0$, so $\langle h, \mathfrak{A}_\pi h\rangle = \|Dh\|^2$. By Theorem 19.6, $\langle h, \mathfrak{A}_\pi h\rangle \ge c_*(\|h\|^2 + \|Dh\|^2 + \|D^T h\|^2)$. Finally $\|Dh\|^2 + \|D^T h\|^2 \ge \|\nabla h\|^2 - C\varepsilon\|h\|^2$ by Lemma 26.4. □

Proposition 26.6 (Minimality)**.** *If $\varepsilon + \beta \le c_7$ and $\theta_0 \le c_7$ (after decreasing c_7; Lemma 26.7 below decreases it once more), then $S(U) \ge S(U_*)$ for all $U \in \mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$, where $U_* = \mathsf{U}(A_*)$. Equality holds iff U lies in the $\mathfrak{G}_0$-orbit of U_*.*

Proof. By Lemma 25.2(c) and Lemma 26.1, $S(U) = \mathcal{J}(A')$ for some $A' = A_1 + \mathbf{H}B \in \mathcal{K}_B$ with $|A_1|_{C^1} \le \rho_2$. The set $\{A_1 + \mathbf{H}B : \ A_1 \in \mathcal{X}, \ |A_1|_{C^1} \le \rho_2\}$ is convex and contains A'_*; on it $|A'|_{C^1} \le \rho_2 + C_H\beta \le r_3$. For $h = A' - A'_* \in \mathcal{X}$ and $A'_t = A'_* + th$,

$$\frac{d^2}{dt^2}\mathcal{J}(A'_t) = \|Dh\|^2 + d^2\mathcal{V}(A'_t)[h,h] \ge (c_X - C(\varepsilon + \rho_2))(\|h\|^2 + \|\nabla h\|^2),$$

by Lemmas 26.5 and 26.2(b). This is positive for c_7 small and $h \ne 0$. Since A'_* is critical, $\mathcal{J}(A') > \mathcal{J}(A'_*)$ unless $A' = A'_*$. Finally $\mathcal{J}(A'_*) = \Phi(A_*) = S(U_*)$. If $\mathcal{J}(A') = \mathcal{J}(A'_*)$, then $A' = A'_*$, and by Corollary 24.3 the axial representative of U is U_*. □

The convexity argument requires L^2 bounds on the second derivatives of the nonlinear averaging, Proposition 22.9(c). These hold because the averaging is built one scale at a time. At each scale the contours have only $O(L)$ bonds, and the field entering them is already an average over the finer scales. For a one-shot nonlinear averaging over blocks of side 1, the quadratic terms would contain integrals over single lattice lines, and the corresponding bounds would fail as $\eta \to 0$.

Proposition 26.6 proves (b) of Theorem 25.1, once we know that U_* lies in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ and is critical there. Membership in $\mathfrak{B}(V)$ holds by construction; the remaining two facts are proved in the next section.

At a critical reference the Hessian of the Lagrangian is coercive on all fine fields, not only on $\mathcal{X}$, once the gauge directions are fixed by the Landau and averaging terms. This is used only in Part XI (Sections 54 and 55); the proof of Theorem 25.1 does not use it, and the next lemma may be read with Part XI. For Theorem 25.1 it only decreases c_7 once more (Proposition 26.6). Its hypothesis uses the multiplier ω_0 of Proposition 27.2, from the next section.

Lemma 26.7 (The Lagrangian Hessian on all fields)**.** *In the setting of Theorem 25.1, suppose that $\beta = 0$ and $A_* = 0$, i.e. $V = W_k$ and U_0 is critical on $\mathfrak{B}(V)$. Let ω_0 be the multiplier of Proposition 27.2 at $A_* = 0$, so that $J_{U_0} = \nabla\Phi(0) = \mathbf{Q}_k^T\omega_0$. For real fine fields h put*

$$q_2(h,h) = \|Dh\|^2 + d^2\mathcal{N}(0)[h,h] - \langle \omega_0, d^2C_k^{\mathrm{av}}(0)[h,h]\rangle,$$
$$\langle h, \mathbf{\Delta}_* h\rangle = q_2(h,h) + \|RD^Th\|^2 + a\|\mathbf{Q}_k h\|^2,$$

with $D = D_{U_0}$, $R = R(U_0)$, $\mathcal{N}$ of Proposition 23.3 and C_k^{av} of Proposition 22.9. There is C', depending only on the constants of Propositions 23.3, 22.9 and 27.2, Theorem 18.2 and Lemmas 12.3 and 26.4, such that

$$\langle h, \mathbf{\Delta}_* h\rangle \ge (c_* - C'\varepsilon)(\|h\|^2 + \|Dh\|^2 + \|D^Th\|^2) \qquad \text{for all real } h, \tag{26.4}$$

with c_* of Theorem 19.6; more precisely, $|\langle h, (\mathbf{\Delta}_* - \mathfrak{A}_\pi)h\rangle| \le C'\varepsilon(\|h\|^2 + \|Dh\|^2 + \|D^Th\|^2)$, with $\mathfrak{A}_\pi$ of Section 15 at U_0. We decrease c_7 so that c_7 is at most the thresholds of Theorems 18.2 and 19.6 and of Proposition 23.3, and $C'c_7 \le \frac{1}{2}c_*$. Then, for $\varepsilon \le c_7$, $\langle h, \mathbf{\Delta}_* h\rangle \ge \frac{1}{2}c_*(\|h\|^2 + \|Dh\|^2 + \|D^Th\|^2)$ for all real h, and $q_2(h,h) \ge \frac{1}{2}c_*(\|h\|^2 + \|Dh\|^2)$ for $h \in \mathcal{X}$.

Proof. Since $E = 0$ (Proposition 22.10(b)), and since the potential of Part II is $W = 0$ here, as in the proof of Lemma 26.5, Lemma 15.1(c) gives $\mathbf{Q}_k\Pi = \mathbf{Q}_k$ and $D\Pi = D - \mathfrak{F}S$, with $S = G'RD^T$. So

$$\langle h, \mathfrak{A}_\pi h\rangle = \|Dh - \mathfrak{F}Sh\|^2 + \|RD^Th\|^2 + a\|\mathbf{Q}_k h\|^2,$$

and

$$\langle h, \mathbf{\Delta}_* h\rangle = \langle h, \mathfrak{A}_\pi h\rangle + 2\langle Dh, \mathfrak{F}Sh\rangle - \|\mathfrak{F}Sh\|^2 + d^2\mathcal{N}(0)[h,h] - \langle \omega_0, d^2C_k^{\mathrm{av}}(0)[h,h]\rangle.$$

Theorems 18.2 and 19.6 apply to the averaging of Part IV, by Corollary 22.11. We bound the last four terms.

- The coefficients $c_{\mu\nu}$ of $\mathfrak{F}$ (Lemma 12.3) change under a real gauge transformation by multiplication with orthogonal matrices on both sides. In a local gauge in which (B) holds, which exists by regularity, $|c_{\mu\nu}| \le C\varepsilon$. So $|(\mathfrak{F}\lambda)_{\mu\nu}(x)| \le C\varepsilon|\lambda(x+\eta e_\mu+\eta e_\nu)|$, by the exact formula of Lemma 12.3. Summing over the periodic lattice gives $\|\mathfrak{F}\lambda\| \le C\varepsilon\|\lambda\|$. Moreover $\|Sh\| \le C\|h\|$ (Theorem 18.2(d) with $X = \mathbb{T}_\eta$). Hence $|2\langle Dh, \mathfrak{F}Sh\rangle - \|\mathfrak{F}Sh\|^2| \le C\varepsilon(\|h\|^2 + \|Dh\|^2)$ for $\varepsilon \le 1$.
- By Proposition 23.3(c) at $A = 0$, $|d^2\mathcal{N}(0)[h,h]| \le C\varepsilon(\|h\|^2+\|\nabla h\|^2)$, and by Lemma 26.4, $\|\nabla h\|^2 \le \|Dh\|^2 + \|D^T h\|^2 + C\varepsilon\|h\|^2$.
- By Proposition 27.2 with $\beta = 0$, $|\omega_0|_\infty \le C\varepsilon$; by Proposition 22.9(c), $|d^2 C_k^{\mathrm{av}}(0)[h,h](c)| \le C\|h\|^2_{S_c}$. Every fine bond starts in exactly one block, and every block lies in S_c for $2d$ bonds c, so $\sum_c \|h\|^2_{S_c} = 2d\|h\|^2$. Hence $|\langle\omega_0, d^2 C_k^{\mathrm{av}}(0)[h,h]\rangle| \le C\varepsilon\|h\|^2$.

By Theorem 19.6 with $X = \mathbb{T}_\eta$, where $m_{\mathbb{T}_\eta} = 0$, $\mathfrak{A}^{\mathbb{T}_\eta} = \mathfrak{A}_\pi$ and $|\!|\!|h|\!|\!|^2_{\mathbb{T}_\eta,+} = \|h\|^2 + \|Dh\|^2 + \|D^T h\|^2$, we have $\langle h, \mathfrak{A}_\pi h\rangle \ge c_*(\|h\|^2 + \|Dh\|^2 + \|D^T h\|^2)$. This gives the bound on $\boldsymbol{\Delta}_* - \mathfrak{A}_\pi$ and (26.4). The constants of the cited results do not depend on c_7, θ_0 or c_6, so the additional condition on c_7 is compatible with the order of the constants at the end of the proof of Theorem 25.1. For $h \in \mathcal{X}$, $\mathbf{Q}_k h = 0$ and $RD^T h = 0$, so $\langle h, \boldsymbol{\Delta}_* h\rangle = q_2(h,h)$. □

27 The Euler–Lagrange equation and regularity of the minimizer

Let A_* be the critical point of $\Phi|_{\mathcal{M}_V}$ constructed in Section 26, and $U_* = \mathsf{U}(A_*)$. In this section we show that U_* is a critical point of $S|_{\mathfrak{B}(V)}$, derive its Euler–Lagrange equation, and prove the second-order bounds and the regularity of U_*. The key input is a gauge covariance property of the gauge-adapted averaging.

27.1 Covariance of the gauge-adapted averaging

Lemma 27.1. *Let $\tilde U$ be a small real fine relative configuration and g a real gauge transformation close to 1.*

(a) $\mathcal{Q}(\tilde U) = ((\tilde U U_0)^{(k)})^{\mathsf{w}_k(\tilde U)|_{\mathbb{T}}}\, W_k^{-1}$.

(b) $\mathcal{Q}(\tilde U^{(g)}) = \mathcal{Q}(\tilde U)^{(h_{\tilde U}(g))}$, *with* $h_{\tilde U}(g) = (\mathsf{w}_k(\tilde U^{(g)})\, g\, \mathsf{w}_k(\tilde U)^{-1})|_{\mathbb{T}}$.

(c) $h_1(g) = \mathfrak{M}^{(k)}(g)$.

Proof. (a) By Theorem 22.5(a), $\mathcal{Q}(\tilde U) = (\tilde U^{(\mathsf{w}_k)})^{[k]} = ((\tilde U U_0)^{\mathsf{w}_k})^{(k)} W_k^{-1}$, and $\mathfrak{a}^{(k)}$ is covariant (Lemma 21.10). (b) Both $\mathsf{w}_k(\tilde U^{(g)})$ and $\mathsf{w}_k(\tilde U)g^{-1}$ move $\tilde U^{(g)}$ into the axial gauge. Everything is real, so Theorem 22.5(b) applies, and $\mathsf{w}_k(\tilde U)g^{-1} = \mathcal{E}^{(k)}(h')\,\mathsf{w}_k(\tilde U^{(g)})$ for some h' on $\mathbb{T}$. Hence

$$(\tilde U^{(g)})^{(\mathsf{w}_k(\tilde U^{(g)}))} = (\tilde U^{(\mathsf{w}_k(\tilde U))})^{(\mathcal{E}^{(k)}(h')^{-1})},$$

and taking k-fold relative averages gives $\mathcal{Q}(\tilde U^{(g)}) = \mathcal{Q}(\tilde U)^{(h'^{-1})}$ with $h'^{-1} = h_{\tilde U}(g)$. (c) By Theorem 22.5(e), $\mathsf{w}_k(1^{(g)}) = \mathcal{E}^{(k)}(\mathfrak{M}^{(k)}(g))g^{-1}$, and $\mathsf{w}_k(1) = 1$. □

For a real fine field A with $|A|_{C^1}$ small, and scalar fields λ on $\mathbb{T}_\eta$ and μ on $\mathbb{T}$, define the linear maps

$$\Xi_A\lambda = \tfrac{d}{dt}\Big|_0 \mathcal{A}_t, \qquad e^{\mathrm{i}\eta\mathcal{A}_t} = (e^{\mathrm{i}\eta A})^{(e^{\mathrm{i}t\lambda})},$$
$$\Theta_A\lambda = -\mathrm{i}\tfrac{d}{dt}\Big|_0 h_{e^{\mathrm{i}\eta A}}(e^{\mathrm{i}t\lambda}), \qquad \Lambda_A\mu = \tfrac{d}{dt}\Big|_0 \ell(\mathcal{Q}(e^{\mathrm{i}\eta A})^{(e^{\mathrm{i}t\mu})}).$$

Differentiating Lemma 27.1(b) gives

$$d\mathcal{Q}(A)\,\Xi_A\lambda = \Lambda_A\Theta_A\lambda. \tag{27.1}$$

By (24.1), $\Xi_A\lambda = -D_{U_0}\lambda + \mathrm{i}[\lambda(b_-), A(b)] + \eta\,\mathfrak{b}'$ with $|\mathfrak{b}'| \le C|A||D\lambda|$. By Lemma 27.1(c) and the proof of Proposition 22.10(b), $\Theta_0 = Q'$. The definitions of Ξ_A, Θ_A, Λ_A make sense for complex A with $|A|_\infty$ small, by the same formulas, and they are analytic in A (see below). Lemma 27.1 is stated for real configurations only, because Theorem 22.5(b) is. Both sides of (27.1) are analytic in A and agree for real A, so (27.1) also holds for complex A, by analytic continuation. By the definition of $h_{\tilde U}$ in Lemma 27.1(b), with $\mathsf{w} = \mathsf{w}_k(e^{\mathrm{i}\eta A})$ evaluated at $y \in \mathbb{T}$,

$$\Theta_A\lambda(y) = R(\mathsf{w}(y))\lambda(y) - \mathrm{i}\,(d\mathsf{w}_k(A)[\Xi_A\lambda])(y)\,\mathsf{w}(y)^{-1}.$$

By Proposition 22.9(d), $\mathsf{w}_k(e^{\mathrm{i}\eta A})(y)$ is analytic and bounded for complex A with $|A|_\infty \le c_3$. So, by the Cauchy estimate, $|d\mathsf{w}_k(A)[h]| \le C|h|_\infty$ for $|A|_\infty \le c_3/2$. By (24.1), $\Xi_A\lambda$ is analytic in A with $|\Xi_A\lambda|_\infty \le C|\lambda|_{C^1}$. Hence Θ_A is analytic in A on $\{|A|_\infty \le c\}$, with $|\Theta_A\lambda|_\infty \le C|\lambda|_{C^1}$ there. The Cauchy estimate in A gives $|\Theta_A\lambda - Q'\lambda|_\infty \le C|A|_\infty|\lambda|_{C^1}$.

27.2 The Euler–Lagrange equation

Proposition 27.2 ([8, (171), (178)])**.** *There is a unique unit-lattice field ω with*

$$\nabla\Phi(A_*) = d\mathcal{Q}(A_*)^T\omega.$$

Moreover $\Lambda_{A_*}^T\omega = 0$, and $|\omega|_\infty \le C(\varepsilon+\beta)$.

Proof. Transversality. Consider the map $\mathcal{O}\lambda = (\Theta_{A_*}\lambda,\, RD^T\Xi_{A_*}\lambda)$. For $A_* = 0$ it is $\mathcal{O}_0\lambda = (Q'\lambda, -RD^TD\lambda)$, which is invertible with inverse $(g,f) \mapsto H'g - G'(f + RD^TDH'g)$, for $f \in \operatorname{ran} R$. Indeed, $\lambda = H'g + \lambda_0$ with $\lambda_0 \in \mathcal{N}(Q')$ and $D^TD\lambda_0 = -f - RD^TDH'g \in \operatorname{ran} R$, and $\lambda_0 = G'D^TD\lambda_0$ by Proposition 14.2(c). By Lemma 24.1, the inverse is bounded from sup norms to $|\cdot|_{C^1}$. Moreover $|\mathcal{O}\lambda - \mathcal{O}_0\lambda| \le C|A_*|_{C^1}|\lambda|_{C^1}$ by the bounds above, since $|D^T[\lambda, A_*]| \le C(|D\lambda||A_*| + |\lambda||\nabla A_*|)$. Hence $\mathcal{O}$ is invertible.

Orthogonality. The tangent space of $\mathcal{M}_V$ at A_* is $\ker RD^T \cap \ker d\mathcal{Q}(A_*)$. Let $k \in \ker d\mathcal{Q}(A_*)$, and let $\lambda = \mathcal{O}^{-1}(0, RD^Tk)$. Then $\Theta_{A_*}\lambda = 0$, so $d\mathcal{Q}(A_*)\Xi_{A_*}\lambda = 0$ by (27.1), and $RD^T(k - \Xi_{A_*}\lambda) = 0$. Thus $k - \Xi_{A_*}\lambda$ is tangent to $\mathcal{M}_V$, and $\langle\nabla\Phi(A_*), k - \Xi_{A_*}\lambda\rangle = 0$ since A_* is critical. Moreover $\langle\nabla\Phi(A_*), \Xi_{A_*}\lambda\rangle = 0$, because Φ is gauge invariant: $\Phi(\mathcal{A}_t) = \Phi(A_*)$. Therefore $\nabla\Phi(A_*) \perp \ker d\mathcal{Q}(A_*)$. Since $d\mathcal{Q}(A_*)$ is onto, $\nabla\Phi(A_*) = d\mathcal{Q}(A_*)^T\omega$ for exactly one ω.

The multiplier. For every λ, gauge invariance and (27.1) give $0 = \langle\nabla\Phi(A_*), \Xi\lambda\rangle = \langle\omega, \Lambda\Theta\lambda\rangle = \langle\Lambda^T\omega, \Theta\lambda\rangle$. Since Θ_{A_*} is onto (it is a small perturbation of Q', which has the right inverse H'), $\Lambda^T\omega = 0$.

Bound on ω. For a bond c of $\mathbb{T}$ and $\beta_0 \in \mathfrak{g}$ with $|\beta_0| = 1$, let

$$\varphi = \mathbf{H}\big(I + dC_k^{\mathrm{av}}(A_*)\mathbf{H}\big)^{-1}\delta_c\beta_0,$$

so that $d\mathcal{Q}(A_*)\varphi = \delta_c\beta_0$. By Proposition 22.9 and Lemma 24.1 (weighted version), $|\varphi(x)| + |\nabla\varphi(x)| \le Ce^{-\gamma'|x-c|}$. Then, by Proposition 23.3,

$$\operatorname{tr}(\beta_0\,\omega(c)) = \langle\varphi, \nabla\Phi(A_*)\rangle = \langle\varphi, J + \nabla\mathcal{N}(A_*)\rangle + \langle D\varphi, DA_*\rangle.$$

Hence $|\omega(c)| \le C\big(\varepsilon + C(\varepsilon+\beta)^2 + C(\varepsilon+\beta)\big)$, since $|A_*|_{C^1} \le C(\varepsilon+\beta)$. □

27.3 The constructed point is critical

Proposition 27.3. *U_* is a critical point of $S|_{\mathfrak{B}(V)}$.*

Proof. Let $\mathsf{w}_* = \mathsf{w}_k(e^{i\eta A_*})$ and $U_\flat = e^{i\eta A_*}U_0$, so that $U_* = U_\flat^{\mathsf{w}_*}$. Let $U(t)$ be a smooth curve in $\mathfrak{B}(V)$ with $U(0) = U_*$. For small t write $U(t)^{\mathsf{w}_*^{-1}} = e^{i\eta\mathcal{A}(t)}U_0$ with $\mathcal{A}(0) = A_*$. Then $S(U(t)) = \Phi(\mathcal{A}(t))$. Moreover $(e^{i\eta\mathcal{A}(t)}U_0)^{(k)} = V^{\mathsf{w}_*|_{\mathbb{T}}^{-1}}$ does not depend on t. By Lemma 27.1(a),

$$\mathcal{Q}(e^{i\eta\mathcal{A}(t)}) = V^{g(t)}W_k^{-1} = \mathcal{Q}(e^{i\eta A_*})^{(g(t))}, \qquad g(t) = \mathsf{w}_k(e^{i\eta\mathcal{A}(t)})|_{\mathbb{T}}\,\mathsf{w}_*|_{\mathbb{T}}^{-1},$$

with $g(0) = 1$. Differentiating at $t = 0$ gives $d\mathcal{Q}(A_*)\mathcal{A}'(0) = \Lambda_{A_*}\mu$ with $\mu = -ig'(0)$. By Proposition 27.2,

$$\tfrac{d}{dt}S(U(t))\big|_{t=0} = \langle\nabla\Phi(A_*), \mathcal{A}'(0)\rangle = \langle\omega, \Lambda_{A_*}\mu\rangle = \langle\Lambda_{A_*}^T\omega, \mu\rangle = 0. \qquad \square$$

27.4 Second-order bounds and regularity

Proposition 27.4. *(a)* $|\Delta_{U_0}A_*|_\infty + \sup_y[\nabla\tilde{A}_{*,y}]_{\alpha,B(y)} \le C(\varepsilon+\beta)$.

(b) $U_* \in \mathfrak{U}(C(\varepsilon+\beta))$.

(c) U_* *is* $(M, \varepsilon + C(\varepsilon+\beta))$*-regular.*

Proof. (a) By Proposition 23.3 and Proposition 27.2,

$$D^TDA_* = d\mathcal{Q}(A_*)^T\omega - J - \nabla\mathcal{N}(A_*).$$

Here $d\mathcal{Q}(A_*)^T\omega = \mathbf{Q}^T\omega + dC_k^{\mathrm{av}}(A_*)^T\omega$ is bounded by $C|\omega|$, by the kernel bounds of Proposition 22.9(c). Hence $|D^TDA_*|_\infty \le C(\varepsilon+\beta)$. Since $RD^TA_* = 0$, $\Delta_{U_0}A_* = D^TDA_* + DP_{\mathcal{V}}D^TA_*$, and the second term is bounded by $C|A_*|_\infty$ (Lemma 24.1(b)). The Hölder bound follows as in Proposition 24.5.

(b) U_* is gauge equivalent to $U_\flat = e^{i\eta A_*}U_0$. By (23.1), $U_\flat(\partial p) = e^{i\eta^2\zeta_p(A_*)}U_0(\partial p)$ with $|\zeta_p(A_*)| \le C|A_*|_{C^1}$. Hence $|F_{U_\flat}| \le C\varepsilon + C(\varepsilon+\beta)$. By Proposition 23.3(d), $J_{U_\flat} = \Psi(A_*)^{-T}\nabla\Phi(A_*)$, which is bounded by $C|\omega| \le C(\varepsilon+\beta)$.

(c) Let $\hat{\Lambda}$ be a cube as in Definition 18.1, and u a real gauge transformation with $U_0^u = e^{i\eta A_0}$ on $\hat{\Lambda}$, where $|A_0|, |\partial A_0|, [\partial A_0]_\alpha \le \varepsilon$. Then

$$U_\flat^u = e^{i\eta R(u)A_*}e^{i\eta A_0} = e^{i\eta(A_0 + R(u)A_* + \eta\mathfrak{c})} \quad \text{on } \hat{\Lambda},$$

where $\mathfrak{c} = \mathfrak{c}(a,b) = \eta^{-2}(\ell(e^{i\eta a}e^{i\eta b}) - \eta a - \eta b)$ with $a = R(u)A_*$ and $b = A_0$. In this gauge, ∂a differs from $R(u)\nabla A_*$ by at most $C\varepsilon|A_*|$. Since $|A_*|_{C^1} \le K'(\varepsilon+\beta)$ (Section 26), $|a| + |\partial a| \le C(\varepsilon+\beta)$; moreover $|b| + |\partial b| \le 2\varepsilon$.

The term $\eta\mathfrak{c}$. The function $(X,Y) \mapsto \ell(e^{iX}e^{iY}) - X - Y$ is analytic near 0 and vanishes if $X = 0$ or $Y = 0$. So $\mathfrak{c}(a,b) = \mathfrak{m}(\eta a, \eta b)[a,b]$, where $\mathfrak{m}(X,Y)$ is a bilinear map depending analytically on (X,Y). Hence $|\mathfrak{c}| \le C|a||b|$, and the chain rule for lattice differences gives $|\partial\mathfrak{c}| \le C(|a| + |\partial a|)(|b| + |\partial b|) \le C\varepsilon(\varepsilon+\beta)$. So $|\eta\mathfrak{c}| + |\partial(\eta\mathfrak{c})| \le C\eta\varepsilon(\varepsilon+\beta)$. Distinct points of $\mathbb{T}_\eta$ have distance at least η, so the Hölder quotients of $\partial(\eta\mathfrak{c})$ are at most $2\eta^{-\alpha}\cdot C\eta\varepsilon(\varepsilon+\beta) \le C\varepsilon(\varepsilon+\beta)$. The bound (23.2) alone gives only $|\partial(\eta\mathfrak{c})| \le 2|\mathfrak{c}|$, without the factor η.

The term a. Let $\Lambda_3(z') \subset \hat{\Lambda}$, and let $x \ne x'$ be points of $\Lambda_3(z')$. If $|x - x'|_\infty \ge \frac{1}{2}$, the Hölder quotient of ∂a at x, x' is at most $4\sup|\partial a| \le C(\varepsilon+\beta)$. If $|x - x'|_\infty < \frac{1}{2}$, the points of the two differences lie in a common cube $\Lambda_1(z_i)$ with z_i on a grid of spacing $\frac{1}{2}$, as in Remark 7.2. Let u'' be a gauge of Definition 18.1 on a cube $\Lambda_{4M+2}(z'') \supset \Lambda_3(z_i)$, with $U_0^{u''} = e^{i\eta A''}$ and $|A''|, |\partial A''| \le \varepsilon$. By (a), in the form on unit cubes (Theorem 25.1(c) and Remark 7.2), the

Hölder quotient of $\partial(R(u'')A_*)$ on $\Lambda_1(z_i)$ is at most $C(\varepsilon+\beta)$. On the box $O=\Lambda_1(z_i)\cap\hat\Lambda$ put $h=u(u'')^{-1}$. Then

$$h(b_+)=e^{-\mathrm{i}\eta A_0(b)}\,h(b_-)\,e^{\mathrm{i}\eta A''(b)}\qquad\text{for the bonds } b \text{ of } O.$$

So $\partial h(b)=\eta^{-1}(e^{-\mathrm{i}\eta A_0(b)}h(b_-)e^{\mathrm{i}\eta A''(b)}-h(b_-))$ is linear in $A_0(b)$ and $A''(b)$, with coefficients that are analytic functions of $h(b_-)$, $\eta A_0(b)$ and $\eta A''(b)$, since $e^{\mathrm{i}X}-1=X\int_0^1 \mathrm{i}e^{\mathrm{i}tX}\,dt$. Hence $|\partial h|\le C\varepsilon$. One more difference brings in ∂A_0, $\partial A''$ and ∂h, each with a bounded factor, so $|\partial\partial h|\le C\varepsilon$; no factor η^{-1} appears. The transformation h is unitary, so $R(h)$ is an isometry, and no smallness of h itself is used. Since $a=R(h)R(u'')A_*$ on O, the discrete product rule along a coordinate path from x to x' in O gives

$$|\partial a(x)-\partial a(x')|\le\Big([\partial(R(u'')A_*)]_{\alpha,\Lambda_1(z_i)}+C\varepsilon\big(|A_*|_\infty+|\nabla A_*|_\infty\big)\Big)|x-x'|^\alpha\le C(\varepsilon+\beta)|x-x'|^\alpha.$$

All points used here lie in the box spanned by the points of the two differences, which lies in O.

So $A_0+a+\eta\mathfrak{c}$ satisfies the three bounds of Definition 18.1 on $\hat\Lambda$, with $\varepsilon+C(\varepsilon+\beta)$ in place of ε. Hence $U_\flat$, and therefore U_*, is $(M,\varepsilon+C(\varepsilon+\beta))$-regular. □

27.5 Analyticity and decay

Proposition 27.5. *The map $B\mapsto A_*(B)$ extends analytically to complex B with $\varepsilon+|B|_\infty<c_6$, with the C^1 bound of Theorem 25.1(c). The second-order and Hölder bounds are claimed for real B only. For real and complex B, if δB is supported on the bond c, then $|dA_*(B)\delta B(x)|+|\nabla(dA_*(B)\delta B)(x)|\le Ce^{-\gamma'|x-c|}|\delta B|$.*

Proof. Lemma 26.1, Lemma 26.2(a) and Proposition 26.3 hold verbatim for complex B and complex A_1; reality was used only in the minimality argument. This gives the bound $|A_*(B)|_{C^1}\le C(\varepsilon+|B|)$. The fixed point of the contraction $\mathcal{T}$ is the uniform limit of iterates that are analytic in B, hence analytic. For real B it is the real solution, by uniqueness.

For the decay we first prove a block bound. The argument holds for complex $A'=A_1+\mathbf{H}B$ with $|A_1|_{C^1}\le\rho_2$. For a fine field h put $a_y(h)=|h|_{B(y)}+|\nabla h|_{B(y)}$, with $|\cdot|_{B(y)}$ as in (26.1). The map $h\mapsto d\nabla\mathcal{V}(A')h$ is obtained by differentiating the formula for $\nabla\mathcal{V}$ in the proof of Lemma 26.2. Its ingredients are the following.

- *The terms with $d^2\mathcal{D}$.* They add up to $(d^2\mathcal{D}(A')[h,\cdot])^T\omega'$, with $\omega'=-\mathbf{H}^T(J+D^TDA+\nabla\mathcal{N}(A))$ and $A=A'-\mathbf{H}\mathcal{D}(A')$, as in the proof of Lemma 26.2(b). There $|\omega'|_\infty\le C(\varepsilon+\rho_2)$, and (26.1) bounds these terms. The ℓ^2 bound on $d^2\mathcal{D}$ in Lemma 26.1 would not suffice: it gives no bound on this fine-lattice field that is pointwise and uniform in η.
- *The local term $d\nabla\mathcal{N}(A)[Lh]$*, with $L=I-\mathbf{H}\,d\mathcal{D}(A')$. The value $\nabla\mathcal{N}(A)(b)$ depends on A only on the bonds of the plaquettes that contain b (proof of Proposition 23.3). So, for b starting in $B(y)$, $d\nabla\mathcal{N}(A)k(b)=d\nabla\mathcal{N}(A)(\chi_y k)(b)$, where χ_y is a smooth function equal to 1 on $\Lambda_{3/2}(c_y)$ and to 0 outside $\Lambda_2(c_y)$; its differences are bounded by an absolute constant. Proposition 23.3(b) gives $|d\nabla\mathcal{N}(A)g|_\infty\le C(\varepsilon+\rho_2)|g|_{C^1}$ for every fine field g. Hence $|d\nabla\mathcal{N}(A)k|_{B(y)}\le C(\varepsilon+\rho_2)\max_{|y'-y|_\infty\le3}a_{y'}(k)$.
- *The operators $\mathbf{H}$, $D\mathbf{H}$, $\mathbf{H}^T=\mathbf{\Gamma}_\pi\mathbf{Q}\mathbf{G}$ and $(D\mathbf{H})^T$.* They satisfy block bounds with exponential decay, by the block form of the pointwise bounds in the proof of Lemma 24.1, as in the proof of (26.1).
- *The operator $D^TD\mathbf{H}$.* By the first identity in the proof of Lemma 26.2, $D^TD\mathbf{H}=\mathbf{Q}^T(\mathbf{\Gamma}_\pi-a)+S^T\mathfrak{F}^TD\mathbf{H}$, with $S^T=DRG'$ bounded by Lemma 24.1(a),(b). So it satisfies a block bound of the same kind.
- *The operators $d\mathcal{D}$ and $d\mathcal{D}^T$.* Their kernel $K_{\mathcal{D}}$ is bounded by $C\rho_2e^{-\gamma|c_--b_-|}$ (Lemma 26.1).

Every term of $d\nabla\mathcal{V}(A')h$ is a composition of these ingredients, and it contains at least one factor of size $C(\varepsilon+\rho_2)$: ω', $d\mathcal{D}$, $d\mathcal{D}^T$ or $d\nabla\mathcal{N}$. Hence

$$|d\nabla\mathcal{V}(A')h|_{B(y)} \le C(\varepsilon+\rho_2)\sum_{y'} e^{-\gamma|y-y'|}a_{y'}(h) \qquad \text{for all } y\in\mathbb{T}. \tag{27.2}$$

Now let $|f|_c = \sup_x e^{\gamma'|x-c|}|f(x)|$ and $|f|_{c,1} = |f|_c + |\nabla f|_c$. By Lemma 24.1 (weighted version), $|\mathbf{C}f|_{c,1} \le C|f|_c$ and $|\mathbf{H}\delta B|_{c,1} \le C|\delta B|$ for δB supported on c. Block bounds pass to the weighted norms when $\gamma' \le \gamma/2$, by $e^{\gamma'|x-c|} \le e^{\gamma'|x-x'|}e^{\gamma'|x'-c|}$. So (27.2) gives $|d\nabla\mathcal{V}(A')h|_c \le C(\varepsilon+\rho_2)|h|_{c,1}$. Differentiating (26.3),

$$\delta A_1 = -\mathbf{C}\,d\nabla\mathcal{V}(A')[\delta A_1 + \mathbf{H}\delta B].$$

The operator on the right has norm at most $\frac12$ in $|\cdot|_{c,1}$. Hence $|\delta A_1|_{c,1} \le C|\delta B|$, and $\delta A_* = (I - \mathbf{H}\,d\mathcal{D})(\delta A_1 + \mathbf{H}\delta B)$ obeys the same bound. □

27.6 Proof of Theorem 25.1

(a) Uniqueness was shown after Proposition 26.3. Existence: $U_* \in \mathfrak{B}(V)$ by construction; it is critical by Proposition 27.3; and it lies in $\mathfrak{U}(\theta_0)$ by Proposition 27.4(b), provided $C(\varepsilon+\beta) \le \theta_0$, i.e. $c_6 \le \theta_0/C$. (b) is Proposition 26.6. (c) follows from Proposition 26.3 and Proposition 27.4(a). (d) is Proposition 27.2. (e) is Proposition 27.4(b), (c). (f) is Proposition 27.5. (g) was noted after the statement, and (h) is proved in Section 27.7 below. The constants are fixed in the following order. First B_1, C_H, C_C, K, r_2^{V}, r_3 and c_7, which do not depend on θ_0. Then θ_0, subject to $\theta_0 \le \min(c_4/3, c_7)$, $\rho_1 \le r_2^{\mathrm{V}}/2$, $C\rho_1 < 1$ (Lemma 25.2(a)) and $\rho_2 + C_H\beta \le r_3$, all for $\varepsilon+\beta \le \theta_0$; these are conditions on θ_0 alone, since ρ_1 and ρ_2 are then at most $C\theta_0$. For the Dirichlet version of the contraction in Section 39 (Proposition 39.3(a), where $\varepsilon+|B|_\infty \le r_4 \le \theta_0$), θ_0 also satisfies $4C_{\mathrm{map}}S_{\gamma_{\mathrm{map}}/4}(1+8B_1+C_H+K)\theta_0 \le 1$, with $S_\gamma = \sum_{z\in\mathbb{Z}^d} e^{-\gamma|z|}$ as in Section 5. Here C_{map} and γ_{map} are the constant and the rate of the kernel $K(y,y') = C_{\mathrm{map}}(\varepsilon+\rho_2)e^{-\gamma_{\mathrm{map}}|y-y'|/2}$ in the proof of that proposition; they come from Theorem 35.2 and (27.2), and depend neither on θ_0 nor on M. Since $\varepsilon+\rho_2 \le (1+8B_1+C_H+K)\theta_0$ for $\varepsilon+|B|_\infty \le \theta_0$, this gives $\sum_{y'} K(y,y')e^{\gamma'|y-y'|} \le \frac14$ for $\gamma' \le \gamma_{\mathrm{map}}/4$. This condition is not used on the main line. Finally $c_6 \le \min(c_4/3,\, c_7,\, \theta_0,\, \theta_0/C,\, B_1\theta_0/K',\, \min(\theta_0,c_1)/(20d),\, \min(c_3,c_5)/C_6)$. The condition with $20d$ is used for $k=0$ in Corollary 30.5. The last one is used in Proposition 33.1: there $|A_*(B)|_{C^1} \le C_6(\varepsilon+|B|_\infty) \le \min(c_3,c_5)$, so that Propositions 23.3 and 22.9 apply at $A_*(B)$ for complex B. The constant C_6 of (c)–(f) holds for all $\varepsilon+\beta$ below thresholds fixed before c_6, so it does not depend on c_6. □

27.7 Lattice symmetry

Proof of Theorem 25.1(h). Let $r \in \mathfrak{S}_k$. Recall that r acts on configurations by $(rU)(b) = U(r^{-1}b)$, on scalar fields by $(r\lambda)(x) = \lambda(r^{-1}x)$, and on Lie fields relative to U_0 by r_{U_0}, with $A(-b) = -R(U_0(b))^{-1}A(b)$ on reversed bonds (Proposition 22.10(g)). With this convention $e^{\mathrm{i}\eta\, r_{U_0}A}\, rU_0 = r(e^{\mathrm{i}\eta A}U_0)$, as in the proof of Lemma 23.2(b). Details

The hypotheses at rU_0. Let u be a gauge on the cube of z as in Definition 18.1, with $U_0^u = e^{\mathrm{i}\eta A}$. Then ru is such a gauge on the cube of rz, and $(rU_0)^{ru} = r(U_0^u) = e^{\mathrm{i}\eta\, rA}$, where $(rA)(b) = A(r^{-1}b)$ and $A(-b) = -A(b)$. The map r is an isometry of $\mathbb{T}_\eta$. It maps the cube $\hat\Lambda$ of z onto the cube of rz, and the cubes $\Lambda_3(z')$ onto the cubes $\Lambda_3(rz')$. It maps a difference of A, together with the points it involves, onto a difference of rA and the points it involves, up to sign. More precisely, for each pair (μ,ν) there are a pair (μ',ν'), a sign and a vector w such that $\partial_{\nu'}(rA)_{\mu'}(rx+w) = \pm\partial_\nu A_\mu(x)$ for all x; for instance, for the reflection in the direction ν, $\partial_\nu(rA)_\mu(rx-\eta e_\nu) = -\partial_\nu A_\mu(x)$ if $\mu \ne \nu$, and $\partial_\nu(rA)_\nu(rx-2\eta e_\nu) = \partial_\nu A_\nu(x)$.

Since $|rx - rx'| = |x - x'|$, and the norms of Definition 18.1 are taken over the differences that involve only points of the cube in question, rA obeys the bounds of Definition 18.1 with the same ε. So rU_0 is (M, ε)-regular. By Lemma 23.2(b), $|J_{rU_0}|_\infty = |J_{U_0}|_\infty \le \varepsilon$. By Proposition 22.10(g), $W_k(rU_0) = rW_k$. Write $V = XW_k$ with the relative configuration $X = e^{\mathrm{i}B}$. Then $rV = (rX)(rW_k)$ and $rX = e^{\mathrm{i}\, r_{W_k} B}$, by the convention for reversed bonds, and $|r_{W_k} B|_\infty = |B|_\infty$, since $R(W_k(c))$ is unitary.

The critical orbit. The map $U \mapsto rU$ maps $\mathfrak{B}(V)$ onto $\mathfrak{B}(rV)$, since $\mathfrak{a}^{(k)}(rU) = r\,\mathfrak{a}^{(k)}(U)$ (Lemma 21.10) and $\mathfrak{U}_0(c_1)$ is invariant under $\mathfrak{S}_k$. It maps $\mathfrak{U}(\theta_0)$ onto itself, because $|F_{rU}(p)| = |F_U(r^{-1}p)|$ and $|J_{rU}|_\infty = |J_U|_\infty$. It preserves S (Section 23), so it maps critical points of $S|_{\mathfrak{B}(V)}$ to critical points of $S|_{\mathfrak{B}(rV)}$. Since r maps $\mathbb{T}$ onto $\mathbb{T}$, $r(U^u) = (rU)^{ru}$ with $ru \in \mathfrak{G}_0$ for $u \in \mathfrak{G}_0$; so r maps $\mathfrak{G}_0$-orbits onto $\mathfrak{G}_0$-orbits. By (a), it maps the critical orbit for V onto the critical orbit for rV.

Scalar operators. On scalar fields, $D_{rU_0} r = r_{U_0} D_{U_0}$. Indeed, by the convention for reversed bonds, $(D_{U_0}\lambda)(c) = \eta^{-1}(R(U_0(c))\lambda(c_+) - \lambda(c_-))$ holds for both orientations of a bond c, and therefore $(D_{rU_0} r\lambda)(c) = (D_{U_0}\lambda)(r^{-1}c)$. The maps r and r_{U_0} preserve the bilinear pairings, so $r^T = r^{-1}$, $r_{U_0}^T = r_{U_0}^{-1}$, and $D_{rU_0}^T r_{U_0} = r D_{U_0}^T$. Together with $Q'_k(rU_0) r = r Q'_k(U_0)$ (Proposition 22.10(g)), this shows that every operator built from D, D^T, Q', Q'^T by sums, products, transposes and inverses is covariant. In particular $G'(rU_0) r = r G'(U_0)$, $P_{\mathcal{V}}(rU_0) r = r P_{\mathcal{V}}(U_0)$ and $R(rU_0) r = r R(U_0)$.

The Landau representative. Let $\mathcal{S}_V(U_0)$ and U_{U_0} be as in Corollary 24.3, with $\theta = \theta_0$; the subscript and the second argument indicate the reference. Let $A \in \mathcal{S}_V(U_0)$. Then $r_{U_0} A$ is real, $|r_{U_0} A|_\infty = |A|_\infty$, and $R(rU_0) D_{rU_0}^T r_{U_0} A = r\, R(U_0) D_{U_0}^T A = 0$. Moreover $e^{\mathrm{i}\eta\, r_{U_0} A}\, rU_0 = r(e^{\mathrm{i}\eta A} U_0) \in \mathfrak{U}(\theta_0)$, and, by Proposition 22.10(g),

$$\mathcal{Q}(e^{\mathrm{i}\eta\, r_{U_0} A}; rU_0) = r\, \mathcal{Q}(e^{\mathrm{i}\eta A}; U_0) = r(V W_k^{-1}) = (rV)(rW_k)^{-1}.$$

So r_{U_0} maps $\mathcal{S}_V(U_0)$ into $\mathcal{S}_{rV}(rU_0)$, and onto it, by the same argument for r^{-1}. By Proposition 22.10(g) and its proof, $\mathsf{w}_k(e^{\mathrm{i}\eta\, r_{U_0} A}; rU_0) = r\, \mathsf{w}_k(e^{\mathrm{i}\eta A}; U_0)$ and $(rX)^{(rg)} = r(X^{(g)})$. Hence $\mathsf{U}_{rU_0}(r_{U_0} A) = r\, \mathsf{U}_{U_0}(A)$. Now $A_* = A_*(B; U_0)$ lies in $\mathcal{S}_V(U_0)$: it lies in $\mathcal{M}_V$, so $|A_*|_\infty < \rho_1 \le 2B_1 c_4 \le r_u$, and $e^{\mathrm{i}\eta A_*} U_0$ is gauge equivalent to $U_* \in \mathfrak{U}(\theta_0)$. The configuration $\mathsf{U}_{U_0}(A_*) = U_*$ is the averaged axial representative of the critical orbit for V. By Corollary 24.3, A_* is the only element of $\mathcal{S}_V(U_0)$ that U_{U_0} maps to it, and the same holds at (rU_0, rV). The configuration rU_* lies in the critical orbit for rV, by the previous step, and it is averaged axial relative to rU_0, by Proposition 22.10(g). So it is the averaged axial representative of that orbit, and it equals $\mathsf{U}_{rU_0}(r_{U_0} A_*)$ with $r_{U_0} A_* \in \mathcal{S}_{rV}(rU_0)$. Hence $A_*(r_{W_k} B; rU_0) = r_{U_0} A_*(B; U_0)$ for real B. For complex B, both sides are analytic in B on the connected set $\{\varepsilon + |B|_\infty < c_6\}$, by (f), and this set is invariant under r_{W_k}. They agree for real B, hence everywhere.

The multiplier. By $e^{\mathrm{i}\eta\, r_{U_0} A}\, rU_0 = r(e^{\mathrm{i}\eta A} U_0)$ and $S \circ r = S$, $\Phi_{rU_0}(r_{U_0} A) = \Phi_{U_0}(A)$. Since r_{U_0} preserves the pairing, $\nabla\Phi_{rU_0}(r_{U_0} A) = r_{U_0} \nabla\Phi_{U_0}(A)$. On relative configurations at scale k, $\ell((rX)(c)) = (r_{W_k} \ell X)(c)$, by Lemma 21.2(d). So differentiating the identity for $\mathcal{Q}$ gives $d\mathcal{Q}_{rU_0}(r_{U_0} A)\, r_{U_0} = r_{W_k}\, d\mathcal{Q}_{U_0}(A)$. Therefore

$$\nabla\Phi_{rU_0}(r_{U_0} A_*) = r_{U_0}\, d\mathcal{Q}_{U_0}(A_*)^T \omega = d\mathcal{Q}_{rU_0}(r_{U_0} A_*)^T\, r_{W_k} \omega,$$

and the uniqueness in (d) gives $\omega(r_{W_k} B; rU_0) = r_{W_k}\, \omega(B; U_0)$. □

The proof uses only the action, the averaging, the Landau gauge condition and the correspondence transformation. It does not use the covariance of the operators of Parts II and III that are built from the covariant derivative on gauge fields; see Proposition 31.7.

Remark 27.6 (What is used, and what is not). The proof uses Parts II and III only through the pointwise bounds of Lemma 24.1, the identities for **H** and **C**, and the coercivity on $\mathcal{X}$ (Lemma 26.5). All of these hold at (M, ε)-regular backgrounds. The random-walk expansion

is not used. The reference U_0 enters only through its regularity and through $|J_{U_0}| \le \varepsilon$. The theorem does not use the induction over k. In Bałaban's scheme the reference comes from that induction, and the bound on the regularity of the minimizer, $\varepsilon + C_6(\varepsilon + \beta)$, has to be improved to a bound depending only on the curvature of V. This is the local improvement of [8, Sect. F], carried out in Part VI (Lemma 29.2).

Part VI: The induction over the number of steps

Part VI proves Bałaban's variational theorem on the torus for every number k of averaging steps, with constants that do not depend on k. Theorem 25.1 solves the problem relative to a regular reference, but its regularity bound for the minimizer, $\varepsilon + C_6(\varepsilon + \beta)$, depends on the reference. In an induction over k the reference at step k is the solution at step $k-1$, and the change of units multiplies its regularity parameter by a power of L (Lemma 30.2). Without an improvement the constants would grow geometrically in k. Part VI supplies the improvement on the torus, without the nested domains of [8, Sect. F].

What this Part proves. The main result is Theorem 30.3, which with Corollary 30.4 is Theorem J. There are a_1, B_3 and n_0 such that, for all $k \ge 0$, all tori of side $n \ge n_0$ and every real coarse configuration V with curvature at most $\varepsilon_1 \le a_1$: the action on $\{U : U^{(k)} = V\}$ has exactly one critical orbit among configurations of small curvature and current; it consists of the minimizers there; the background fields $U_k(V)$ have curvature and current at most $B_3\varepsilon_1$ and are $(M, B_3\varepsilon_1)$-regular; and the minimal orbit of rV is the image of that of V for $r \in \mathfrak{S}_k$. The constants depend on L, G, a, α and M, and not on k, n or η. Corollary 30.4 shows that every real configuration with curvature and current at most $\theta \le a_2$ is $(M, B_5\theta)$-regular, so the hypothesis of Part III holds on Bałaban's small-field class. Corollary 30.5 gives analyticity of the background map around a background field.

How. The reference at step k is the solution of the $(k-1)$-step problem for the trivial extension of V, which has exactly the prescribed averages (Lemma 30.1), and Theorem 25.1 applies to it. The local improvement (Lemma 29.2) restores the regularity bound: near a point y, the reference is replaced by a configuration that is trivial near y in a local gauge and equals the minimizer far away (Definition 28.5, Lemma 28.6). Relative to it, the first-order formula of Theorem 25.1(c) shows that the minimizer near y is determined by the coarse field in the gauge of the coordinate-ordered contours from y (Lemma 28.4), which the curvature of V controls. The errors are exponentially small in the splicing radius, or quadratic (Section 28, Proposition 29.1). This gives $\varepsilon_0' = C_{\mathrm{imp}}\varepsilon_1 + \frac{1}{2}\varepsilon_0$, and the induction closes. Gauges on cubes larger than the regularity scale come from Lemma 28.3, with bounds that grow polynomially in the size.

Imports. Theorem 25.1 and its Euler–Lagrange equation (Proposition 27.2); from Part IV, the regularity of averages and the averages in a local gauge (Proposition 21.12, Lemma 21.13). One condition on a_1 in the proof of Theorem 30.3, $B_3 a_1 \le c_3'$ with c_3' of Section 46.1, serves only Part XI (proof of Theorem 55.4). No other proof in Part VI uses a later result.

Exports. The background fields $U_k(V)$ of Theorem 30.3, their regularity and their symmetry are used in Parts VII and IX–XI. Corollary 30.4 goes to Parts VIII–X, Corollary 30.5 to Parts VII and IX–XI, and Lemma 28.3 to Parts VII–X. The coordinate-ordered contours

and the splicing of Section 28 are used again for the curvature bound of Proposition 31.5. Lemmas 28.2, 28.4 and Proposition 29.1 serve only this Part. Linear bounds on cubes of size of order ε_1^{-1}, as in [8, (9), (10)], are not proved; Theorem 30.3 and Lemma 28.3 give polynomial ones (Remark 30.6).

Details for this Part. The companion has no chapter on Part VI; its parameter ledger lists a_1, B_3, n_0 and the other constants of Part VI that Parts VII–XI use.

28 Local gauges and splicing

Theorem 25.1 solves the variational problem relative to a regular reference configuration U_0. It bounds the regularity of the minimizer by $\varepsilon + C_6(\varepsilon + \beta)$, where ε is the regularity parameter of U_0. Bałaban's Theorem 1 [8] is proved by induction over k. The reference at step k is the solution at step $k-1$, and passing from units of δ_{k-1} to units of δ_k multiplies its regularity parameter by a power of L. Without improvement the constants would grow geometrically in k. Bałaban's Section F [8] closes the induction with a local argument. On a cube around a given point the minimizer is compared with the solution of a variational problem relative to the trivial configuration. That problem is posed on nested domains: the configuration is frozen outside the cube, and the averaging becomes finer towards its boundary.

We stay on the torus. The observation is that Theorem 25.1 holds uniformly for *all* regular references. Near a point y we replace the induction reference by a configuration that is trivial (in a local gauge) near y and coincides with the minimizer far from y. Relative to this *spliced* reference, the first-order formula of Theorem 25.1(c) shows that the minimizer near y is determined by the coarse field near y. The errors are exponentially small in the splicing radius, or quadratic in the a priori bound. The coarse field near y, in the gauge of the coordinate-ordered contours from y, is controlled by the curvature of V alone. This gives the improvement (Section 29), and the induction closes (Section 30).

This section provides the geometric input: gauges on large cubes (Lemma 28.3), a local gauge adapted to a coarse contour gauge (Lemma 28.4), and the splicing (Lemma 28.6).

Conventions for Part VI. We keep the conventions of Parts IV and V and assume in addition $M \geq 5L$. The torus has side $n \geq n_0$, where n_0 is fixed in Lemma 29.2. All boxes below have sides smaller than n, so they are identified with boxes in $\eta\mathbb{Z}^d$. Side lengths and shifts of boxes are tacitly rounded to multiples of η; this changes the constants below only by bounded factors. A *box* is a set $\{x \in \mathbb{T}_\eta : \ a_\mu \leq x_\mu \leq b_\mu\}$, and the bonds of a box are the bonds with both endpoints in it. We write $\Lambda_r(z) = \{x : |x - z|_\infty \leq r\}$, as in Section 6.

28.1 Regularity on a box

Definition 28.1 (Regularity on a box)**.** Let Λ be a box, and let A be a fine Lie field on the bonds of Λ. Put

$$\|A\|_{1,\alpha;\Lambda} = |A|_\Lambda + |\partial A|_\Lambda + \sup\{[\partial A]_{\alpha,\Lambda_3(z')} : \ \Lambda_3(z') \subset \Lambda\},$$

where $|\partial A|_\Lambda$ is the maximum of $|\partial_\nu A_\mu(x)|$ over all differences that involve only points of Λ, and the Hölder quotient on $\Lambda_3(z')$ is taken over the pairs of differences that involve only points of $\Lambda_3(z')$, as in Definition 18.1. A real configuration U is *σ-regular on Λ* if there is a real gauge transformation u on Λ with $U^u = e^{i\eta A}$ on the bonds of Λ and $\|A\|_{1,\alpha;\Lambda} \leq \sigma$.

This property is invariant under real gauge transformations of U, and it passes to subboxes. By Definition 18.1, an (M, ε)-regular configuration is 3ε-regular on every cube $\Lambda_{4M+2}(z)$.

Conversely, if $0 < \sigma \le 1$, a configuration that is σ-regular on every cube $\Lambda_{4M+2}(z)$ is (M,σ)-regular.

Two facts on gauge transformations of configurations of the form $e^{i\eta A}$ are used repeatedly. Let λ be a real scalar field on a box with $|\lambda| \le 1$. Then $(e^{i\eta A})^{e^{i\lambda}} = e^{i\eta A^\lambda}$, where, by (24.1) with $U_0 = 1$,

$$A^\lambda(b) = e^{i\,\mathrm{ad}\,\lambda(b_-)}A(b) - \mathfrak{d}\big(-\mathrm{ad}\,\lambda(b_-)\big)\partial\lambda(b) + \eta\,\mathfrak{b}\big(-\lambda(b_-), A(b), -\partial\lambda(b)\big). \tag{28.1}$$

Consequently

$$\|A^\lambda\|_{1,\alpha;\Lambda} \le C\big(\|A\|_{1,\alpha;\Lambda} + |\partial\lambda|_\Lambda + |\partial\partial\lambda|_\Lambda + \sup_{\Lambda_3(z')\subset\Lambda}[\partial\partial\lambda]_{\alpha,\Lambda_3(z')}\big), \tag{28.2}$$

provided the right-hand side is at most 1. Indeed, $A^\lambda(b)$ is an analytic function of $\lambda(b_-)$, $A(b)$ and $\partial\lambda(b)$, and its first differences and their Hölder quotients are controlled by the chain rule for lattice differences, as in the proof of Lemma 12.3. The term with the factor η is handled by (23.2).

28.2 Gauges on large cubes

Lemma 28.2 (Patching). *There is $C_P^{\mathrm{patch}} \ge 1$, depending only on d, α and G, with the following property. Let Λ be a box, μ a direction, $a \in \mathbb{R}$ and $w \ge 18$. Put $\Lambda' = \{x \in \Lambda : x_\mu \le a + w\}$ and $\Lambda'' = \{x \in \Lambda : x_\mu \ge a\}$, and suppose that all sides of Λ are at most $3w$. If U is σ-regular on Λ' and on Λ'', and $C_P^{\mathrm{patch}} w\sigma \le 1$, then U is $C_P^{\mathrm{patch}}\sigma$-regular on Λ.*

Proof. Let u', u'' be the gauges, with $U^{u'} = e^{i\eta A'}$ on Λ' and $U^{u''} = e^{i\eta A''}$ on Λ''. On the slab $O = \Lambda' \cap \Lambda''$ the transformation $g = u'(u'')^{-1}$ satisfies

$$e^{i\eta A'(b)} = g(b_-)\, e^{i\eta A''(b)}\, g(b_+)^{-1} \qquad \text{for the bonds } b \text{ of } O.$$

Fix $x_0 \in O$. Replacing u'' by $g(x_0)u''$, which replaces A'' by $R(g(x_0))A''$ and does not change its norm, we may assume $g(x_0) = 1$.

Size of g. By the relation above, $g(b_+) = e^{-i\eta A'(b)} g(b_-) e^{i\eta A''(b)}$. Every point of O is joined to x_0 by a lattice path in O of length at most $3dw$. By the bi-invariance of the metric (Lemma 21.2(b)), $g = e^{i\lambda}$ on O with λ real and $|\lambda| \le 6dw\sigma \le c_G$; here we take $C_P^{\mathrm{patch}} \ge 6d/c_G$.

Derivatives of λ. The relation gives $\lambda(b_+) = \ell(e^{-i\eta A'(b)} e^{i\lambda(b_-)} e^{i\eta A''(b)})$. Hence $\partial\lambda(b) = \mathfrak{p}(\lambda(b_-), \eta A'(b), \eta A''(b); A'(b), A''(b))$, where $\mathfrak{p}$ is analytic, linear in the last two arguments, and bounded by $C(|A'(b)| + |A''(b)|)$. Taking one more difference and using the chain rule, $|\partial\lambda|_O + |\partial\partial\lambda|_O + [\partial\partial\lambda]_{\alpha,\Lambda_3(z')} \le C\sigma$ for every $\Lambda_3(z') \subset O$.

Gluing. Let χ_1 be a smooth function on $\mathbb{R}$ with $\chi_1 = 1$ on $(-\infty, \frac13]$ and $\chi_1 = 0$ on $[\frac23, \infty)$, and put $\chi(x) = \chi_1((x_\mu - a)/w)$. Define

$$u = u' \ \text{ on } \{x_\mu \le a + \tfrac{w}{3}\}, \qquad u = e^{i\chi\lambda}u'' \ \text{ on } O, \qquad u = u'' \ \text{ on } \{x_\mu \ge a + \tfrac{2w}{3}\}.$$

The definitions agree on the overlaps, because $e^{i\lambda}u'' = gu'' = u'$. Hence $U^u = e^{i\eta A}$ on the bonds of Λ, with $A = A'$, $A = A''$, or $A = (A'')^{\chi\lambda}$ as in (28.1), on the three regions. Since $|\partial^\beta\chi| \le Cw^{-|\beta|}$ and $|\lambda| \le 6dw\sigma$, the field $\chi\lambda$ satisfies $|\partial(\chi\lambda)| + |\partial\partial(\chi\lambda)| + [\partial\partial(\chi\lambda)]_{\alpha,\Lambda_3(z')} \le C\sigma$; the terms in which derivatives fall on χ carry factors $w^{-1}\cdot w\sigma$ or $w^{-2}\cdot w\sigma$. By (28.2), $\|(A'')^{\chi\lambda}\|_{1,\alpha;O} \le C\sigma$.

Finally, since $w \ge 18$, every cube $\Lambda_3(z') \subset \Lambda$ (of side $6 \le w/3$) and every difference lie entirely in one of the three regions. Hence $\|A\|_{1,\alpha;\Lambda} \le C_P^{\mathrm{patch}}\sigma$. □

Lemma 28.3 (Large cubes). *There are $C_E \ge 1$ and $p \ge 1$, depending only on d, α and G, with the following property. Let $18 \le r \le R$, and let U be σ-regular on every cube $\Lambda_r(z)$, $z \in \mathbb{T}_\eta$. If $C_E(R/r)^p R\sigma \le 1$, then U is $C_E(R/r)^p\sigma$-regular on every cube $\Lambda_R(z)$.*

Proof. We enlarge boxes in rounds. Let $T = 2\eta\lfloor R/\eta\rfloor$ be the side of the cubes $\Lambda_R(z)$; $T < n$ by the conventions of Part VI, under which the cubes $\Lambda_R(z)$ are boxes. At the beginning of a round all sides equal some ℓ. If $\ell < T$, put

$$\delta = \min\Big(T-\ell,\ \eta\Big\lfloor\frac{\ell}{2\eta}\Big\rfloor\Big),$$

and, for $\mu = 1,\dots,d$ in turn, pass to the boxes whose side in direction μ is $\ell+\delta$; each of them is the union of two boxes of the current shape, shifted against each other by δ in direction μ. The overlap has width $w = \ell - \delta \ge \frac{\ell}{2}$, and during the round all sides lie between ℓ and $\ell+\delta \le \frac{3}{2}\ell \le 3w$. Initially $\ell = \ell_0 = 2\eta\lfloor r/\eta\rfloor \ge 36$, since 18 is a multiple of η. Hence $w \ge 18$, all sides of the unions are at most $3w$, and all boxes have sides at most $T < n$, so none of them wraps around the torus. By Lemma 28.2 and induction, after j steps U is $(C_P^{\mathrm{patch}})^j\sigma$-regular on every box of the current shape, as long as $C_P^{\mathrm{patch}}\cdot 3R\cdot(C_P^{\mathrm{patch}})^j\sigma \le 1$. A round with $\delta < T-\ell$ gives the side $\ell+\delta \ge \frac{3}{2}\ell - \eta$, so after i such rounds the side is at least $(\frac{3}{2})^i(\ell_0-2\eta)+2\eta \ge (\frac{3}{2})^i\cdot\frac{16}{9}r$, since $\ell_0 - 2\eta \ge 2r - 4\eta \ge \frac{16}{9}r$. Since a round with $\delta < T-\ell$ ends with a side less than $T \le 2R$, the number i of such rounds satisfies $(\frac{3}{2})^i < \frac{9}{8}\cdot\frac{R}{r}$; as $\frac{9}{8} < \frac{3}{2}$, this gives $i \le \lceil \log_{3/2}(R/r)\rceil$. So after at most $\lceil\log_{3/2}(R/r)\rceil + 1$ rounds, the last one with $\delta = T-\ell$, all sides equal T, and every cube $\Lambda_R(z)$ is a box of the current shape. The number of steps is $m \le d(\log_{3/2}(R/r)+2)$, so $(C_P^{\mathrm{patch}})^m \le (C_P^{\mathrm{patch}})^{2d}(R/r)^{d\log C_P^{\mathrm{patch}}/\log(3/2)}$. Hence the lemma holds with $p = d\log C_P^{\mathrm{patch}}/\log(3/2)$ and $C_E = 3(C_P^{\mathrm{patch}})^{2d+1}$. The rounding to multiples of η costs at most one round, so it changes C_E but not p. □

The loss is only polynomial in R/r. This is what makes the splicing below possible: the polynomial loss is beaten by the exponential decay of the operators of Part III.

28.3 A gauge adapted to a coarse contour gauge

For $y, y' \in \mathbb{T}$ let $\Gamma^{\mathrm{co}}_{y,y'}$ be the coordinate-ordered contour on $\mathbb{T}$ from y to y' that changes the coordinates in the order $d, d-1, \dots, 1$. It has at most $d|y'-y|_\infty$ bonds. The contours Γ^{co} are a proof device. The gauge they define on $\mathbb{T}$ is not invariant under the permutations of the axes, and it is not the averaged axial gauge of Section 22. Apart from Lemma 28.4, it enters only proofs: those of Lemma 29.2, Proposition 31.5 and the case $k=0$ of Theorem 30.3. Their statements do not refer to it. A single contour suffices here, and it gives the bound (ii) below, which is linear in $|c_- - y|_\infty$ with an absolute constant. Let $\phi_1 \in C^\infty(\mathbb{R})$ be supported in $[-1,1]$, with $\phi_1(0) = 1$ and $\sum_{m\in\mathbb{Z}}\phi_1(t-m) = 1$. For instance, $\phi_1(t) = \vartheta(1-|t|)$, where $\vartheta : [0,1] \to [0,1]$ is smooth, vanishes with all its derivatives at 0, and satisfies $\vartheta(s)+\vartheta(1-s) = 1$. Put $\phi(x) = \prod_\mu \phi_1(x_\mu)$. Then $\phi(y'-y'') = \delta_{y'y''}$ for $y', y'' \in \mathbb{T}$, and $\sum_{y'\in\mathbb{T}}\phi(x-y') = 1$.

Lemma 28.4 (Adapted local gauge)**.** *There are $C_8, q \ge 1$, depending only on d, L, G, α and M, with the following property. Let $U \in \mathfrak{U}_0(\varepsilon_0)$ be real and (M,ε_0)-regular, with $\varepsilon_0 \le c_1$. Suppose that $V = U^{(k)} \in \mathfrak{U}_k(\varepsilon_1)$ with $\varepsilon_1 \le 2\varepsilon_0$. Let $R \ge 4M+2$ be an integer and $y \in \mathbb{T}$, and assume $C_8R^q\varepsilon_0 \le 1$. Then there is a real gauge transformation u on $\Lambda_R(y)$ such that:*

(i) $U^u = e^{\mathrm{i}\eta A}$ on the bonds of $\Lambda_R(y)$, with $\|A\|_{1,\alpha;\Lambda_R(y)} \le C_8R^q\varepsilon_0$;

(ii) for every bond c of $\mathbb{T}$ with both endpoints in $\Lambda_R(y)$,

$$V^u(c) := u(c_-)V(c)u(c_+)^{-1} = e^{\mathrm{i}B_y(c)}, \qquad |B_y(c)| \le d\,(|c_- - y|_\infty + 1)\,\varepsilon_1.$$

Proof. Step 1: a gauge on a large cube. By Lemma 28.3 with $r = 4M+2 \ge 18$, $\Lambda_{R+3}(y)$ carries a real gauge u_1 with $U^{u_1} = e^{\mathrm{i}\eta A_1}$ and $\|A_1\|_{1,\alpha;\Lambda_{R+3}(y)} \le \sigma := 3C_E(R+3)^p\varepsilon_0$.

Step 2: the coarse contour gauge. Let c be a bond of $\mathbb{T}$ with both endpoints in $\Lambda_{R+1}(y)$. Then $B(c_-)\cup B(c_+) \subset \Lambda_{R+3}(y)$. By Lemma 21.13, applied to U^{u_1} with the union of blocks

$B(c_-) \cup B(c_+)$ (the averages are local, so u_1 may be extended arbitrarily outside the cube), $V^{u_1}(c) = e^{\mathrm{i}B_1(c)}$ with $|B_1(c)| \le \sigma_1 := \sigma + 32\varepsilon_0$. For unit points $y' \in \Lambda_{R+1}(y)$ put $h(y') = V^{u_1}(\Gamma^{\mathrm{co}}_{y,y'})$, so that $h(y) = 1$ and $|\lambda(y')| \le d(R+1)\sigma_1$ for $\lambda(y') = \ell(h(y'))$. For a bond $c = (y', y'')$ as above,

$$(V^{u_1})^h(c) = h(y')V^{u_1}(c)h(y'')^{-1} = V^{u_1}(\Gamma^{\mathrm{co}}_{y,y'} \cup c \cup (\Gamma^{\mathrm{co}}_{y,y''})^{-1})$$

is the holonomy of a closed contour based at y. The two contours are coordinate-ordered for the same order, from y to the neighbouring points y' and y''. By Lemma 21.3(ii) the loop is spanned by at most $d|y'-y|_\infty \le d(|y'-y|_\infty + 1)$ plaquettes of $\mathbb{T}$. Holonomies of closed contours are conjugated by gauge transformations, so Lemma 21.3 applied to V gives $|\ell((V^{u_1})^h(c))| \le d(|y'-y|_\infty + 1)\varepsilon_1$. Moreover $h(y'') = (V^{u_1})^h(c)^{-1}\, h(y')\, V^{u_1}(c)$. By bi-invariance, the distance between $h(y'')$ and $h(y')$ is at most $|\ell((V^{u_1})^h(c))| + |B_1(c)|$. Near 1 the logarithm is Lipschitz with respect to this distance. Hence

$$|\lambda(y'') - \lambda(y')| \le C(\sigma_1 + R\varepsilon_1) \le C\sigma' \quad \text{for neighbours } y', y'', \qquad \sigma' = \sigma_1 + R\varepsilon_0.$$

Step 3: smooth extension. For fine points $x \in \Lambda_R(y)$ put $\tilde\lambda(x) = \sum_{y'} \phi(x-y')\lambda(y')$; only points y' with $|y'-x|_\infty < 1$ contribute. Then $\tilde\lambda = \lambda$ at the points of $\mathbb{T}$ and $|\tilde\lambda| \le \max|\lambda|$. Since the lattice differences of $\sum_{y'} \phi(x-y') = 1$ vanish, every difference of order 1, 2 or 3 of $\tilde\lambda$ is $\sum_{y'} \partial^\beta\phi(x-y')(\lambda(y') - \lambda(y_x))$, where y_x is a unit point with $|x-y_x|_\infty < 1$. Hence these differences are bounded by $C\sigma'$. The same holds for the Hölder quotients of $\partial\partial\tilde\lambda$, which are bounded by the third differences on short distances and by twice the second differences on distances at least 1.

Let $u = e^{\mathrm{i}\tilde\lambda}u_1$ on $\Lambda_R(y)$. At the unit points, $u = hu_1$, which gives (ii). By (28.2), $U^u = e^{\mathrm{i}\eta A}$ with $\|A\|_{1,\alpha;\Lambda_R(y)} \le C(\sigma + \sigma') \le C_8 R^q \varepsilon_0$, for suitable C_8 and $q = p+1$. The smallness conditions used ($\sigma \le 1$, $|\tilde\lambda| \le c_G$, and the hypotheses of Lemmas 21.13 and 21.3) follow from $C_8 R^q \varepsilon_0 \le 1$ if C_8 is large. □

28.4 Splicing

Fix a smooth function χ_0 on $\mathbb{R}^d$ with $\chi_0 = 1$ on $\{|x|_\infty \le \frac14\}$, $\chi_0 = 0$ on $\{|x|_\infty \ge \frac12\}$ and $0 \le \chi_0 \le 1$. For $R \ge 1$ and $y \in \mathbb{T}$ put $\chi(x) = \chi_0((x-y)/R)$, so that $|\partial^\beta\chi| \le C_\beta R^{-|\beta|}$. For a positively oriented bond b write $\chi(b) = \chi(b_-)$. Definition 28.5 below prescribes $U^\sharp$ on positively oriented bonds, and $U^\sharp(-b) = U^\sharp(b)^{-1}$.

Definition 28.5 (Spliced configuration)**.** Let U be real, $y \in \mathbb{T}$, and let u be a real gauge transformation on $\Lambda_R(y)$ with $U^u = e^{\mathrm{i}\eta A}$ on the bonds of $\Lambda_R(y)$. The *spliced configuration* $U^\sharp$ is defined by

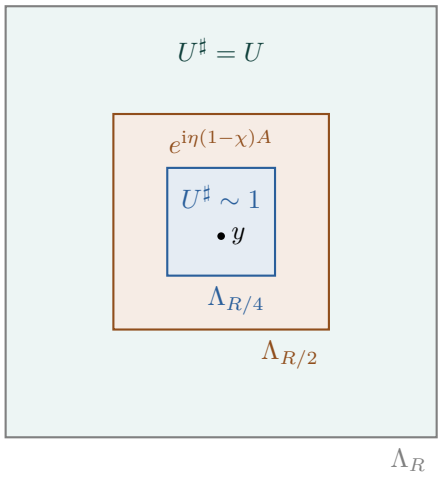
$$(U^\sharp)^u = e^{\mathrm{i}\eta(1-\chi)A} \text{ on the bonds of } \Lambda_R(y), \qquad U^\sharp = U \text{ on all other bonds.}$$

This is consistent: a bond of $\Lambda_R(y)$ with no endpoint in $\Lambda_{R/2}(y)$ has $\chi(b) = 0$, so the two prescriptions agree on it, and every bond with an endpoint in $\Lambda_{R/2}(y)$ is a bond of $\Lambda_R(y)$. See Figure 3.

Lemma 28.6 (Splicing)**.** *There is $C_9 \ge 1$, depending only on d, L, G, α and M, with the following property. Let $R \ge 16M+16$, and let U be real and (M,ε_0)-regular with $U \in \mathfrak{U}(\varepsilon_0)$. Let y, u and A be as in Definition 28.5, with $\|A\|_{1,\alpha;\Lambda_R(y)} \le s$ and $C_9(\varepsilon_0+s) \le c_1$. Then:*

(a) $U^\sharp = U$ on all bonds with no endpoint in $\Lambda_{R/2}(y)$, and $(U^\sharp)^u = 1$ on the bonds of $\Lambda_{R/4}(y)$.

(b) $U^\sharp$ is $(M, C_9(\varepsilon_0+s))$-regular, $|F_{U^\sharp}| \le C_9(\varepsilon_0+s)$ and $|J_{U^\sharp}| \le \varepsilon_0 + C_9 s$. Moreover $J_{U^\sharp} = J_U$ on bonds with no endpoint in $\Lambda_{R/2+1}(y)$, and $J_{U^\sharp} = 0$ on the bonds of $\Lambda_{R/4-1}(y)$.



In the gauge u of Lemma 28.4:

- if $B(c_-) \cup B(c_+) \subset \Lambda_{R/4}$, the reference average is 1, the data are the coarse field in the gauge of Γ^{co}, $|B^\sharp(c)| \le d(|c_- - y| + 1)\varepsilon_1$;
- in the transition region the bounds are $O(R^q \varepsilon_0)$;
- $B^\sharp(c) = 0$ if $B(c_-) \cup B(c_+)$ does not meet $\Lambda_{R/2}$;
- $J_{U^\sharp} = J_U$ on bonds with no endpoint in $\Lambda_{R/2+1}$.

Figure 3: The spliced reference configuration of Definition 28.5.

(c) Let $W^\sharp = (U^\sharp)^{(k)}$ and $V = U^{(k)}$. For every bond c of $\mathbb{T}$, $|\ell(V(c)W^\sharp(c)^{-1})| \le C_9(\varepsilon_0 + s)$. If $B(c_-) \cup B(c_+)$ does not meet $\Lambda_{R/2}(y)$, then $W^\sharp(c) = V(c)$. If $B(c_-) \cup B(c_+) \subset \Lambda_{R/4}(y)$, then $u(c_-)W^\sharp(c)u(c_+)^{-1} = 1$.

Proof. (a) follows from the definition, since $\chi = 1$ on $\Lambda_{R/4}(y)$.

(b) *Regularity.* If $\Lambda_{4M+2}(z) \subset \Lambda_R(y)$, the gauge u gives $(U^\sharp)^u = e^{i\eta(1-\chi)A}$ there, and $\|(1-\chi)A\|_{1,\alpha} \le Cs$ by the Leibniz rule, since $R \ge 1$. Otherwise $|z - y|_\infty > R - 4M - 2$. Then every point of $\Lambda_{4M+2}(z)$ has distance more than $R - 8M - 4 \ge R/2$ from y, so $U^\sharp = U$ on $\Lambda_{4M+2}(z)$, and the gauges of U apply. The curvature is bounded in the same way, using $|F_{e^{i\eta B}}| \le C(|\partial B| + |B|^2)$.

Current. The current at a bond depends only on the configuration on the plaquettes containing it and on their neighbours. This gives the two identities. In general, compute in the gauge u, which does not change $|J|$. For a configuration $e^{i\eta B}$ and a plaquette p with bonds $b_1, \dots, b_4$ as in Definition 21.4,

$$F^s_{e^{i\eta B}}(p) = \eta^{-2}\mathfrak{S}(\eta B(b_1), \eta B(b_2), \eta B(b_3), \eta B(b_4)), \qquad \mathfrak{S}(z) = P_{\mathfrak{g}} \sin \ell(e^{iz_1}e^{iz_2}e^{-iz_3}e^{-iz_4}),$$

with F^s as in Lemma 23.2. The function $\mathfrak{S}$ is analytic near 0, and $\mathfrak{S}(z) = z_1 + z_2 - z_3 - z_4 + \mathfrak{S}_2(z)$ with $|\mathfrak{S}_2(z)| \le C|z|^2$ and $|\nabla \mathfrak{S}_2(z)| \le C|z|$. Let $t = 1 - \chi$ and $A^\sharp = tA$, and let x_p be the base point of p. Then

$$F^s_{e^{i\eta A^\sharp}}(p) = t(x_p) F^s_{e^{i\eta A}}(p) + E(p),$$

$$E(p) = \sum_{i=1}^{4} \pm \frac{t(b_i) - t(x_p)}{\eta} A(b_i) + \eta^{-2}\Big(\mathfrak{S}_2(\eta A^\sharp(b_1), \dots) - t(x_p)\,\mathfrak{S}_2(\eta A(b_1), \dots)\Big).$$

Since $|\partial^\beta t| \le CR^{-|\beta|}$ and $|A| + |\partial A| \le s$, we get $|E| \le Cs$. Between neighbouring plaquettes, the arguments $\eta A^\sharp(b_i)$, $\eta A(b_i)$ and $t(x_p)$ change by at most $C\eta^2 s$ and $C\eta/R$. With $|\nabla \mathfrak{S}_2(z)| \le C|z|$ and the product rule, the difference quotients of E are bounded by Cs. By Lemma 12.2, $D^T_{e^{i\eta B}} = \partial^* + T^T_B$ with T_B local and bounded by $C|B|$. Using $J_{e^{i\eta A}} = D^T_{e^{i\eta A}} F^s_{e^{i\eta A}} = R(u)J_U$ (Lemma 23.2(a)),

$$J_{e^{i\eta A^\sharp}} = t\,R(u)J_U + [\partial^*, t]F^s_{e^{i\eta A}} + (T^T_{A^\sharp} t - t T^T_A) F^s_{e^{i\eta A}} + D^T_{e^{i\eta A^\sharp}} E.$$

Each term after the first is bounded by Cs, because $|F^s_{e^{i\eta A}}| \le Cs$ and $|\partial t| \le C/R$. Hence $|J_{U^\sharp}| \le \varepsilon_0 + Cs$.

(c) The average $U^{(k)}(c)$ depends only on the bonds of $B(c_-)\cup B(c_+)$ (Lemma 21.10). This gives the first identity. The second follows from (a) and covariance, since $\mathfrak{a}^{(k)}(1)=1$; the gauge u may be extended arbitrarily outside $\Lambda_R(y)$. In the remaining case $B(c_-)\cup B(c_+)\subset\Lambda_{R/2+2}(y)\subset\Lambda_R(y)$. Lemma 21.13, applied to U^u and to $(U^\sharp)^u$ on the union of blocks $B(c_-)\cup B(c_+)$, gives $V^u(c)=e^{\mathrm{i}B'}$ and $(W^\sharp)^u(c)=e^{\mathrm{i}B''}$ with $|B'|\le s+32\varepsilon_0$ and $|B''|\le s+32C'(\varepsilon_0+s)$. Here $C'\ge 1$ is the constant obtained in (b) for the curvature of $U^\sharp$. So (c) holds with $C_9\ge 33(1+C')$; the same bound applies in the second case. By covariance and Lemma 21.2(b), $|\ell(VW^{\sharp-1})(c)|=|\ell(e^{\mathrm{i}B'}e^{-\mathrm{i}B''})|\le|B'|+|B''|$. □

29 Localized bounds and the local improvement

29.1 Localized bounds for the relative problem

Theorem 25.1 bounds the solution in sup norms over the whole torus. Its proof gives more. By the first-order formula and the Euler–Lagrange equation, the solution near a point is controlled by the data near that point, up to exponentially decaying contributions from far away and a quadratic error.

In the setting of Theorem 25.1, let A_* and ω be as there. For $y\in\mathbb{T}$ let $c_y=y$ be the centre of $B(y)$, and let u_y be any real gauge transformation in which U_0 satisfies (B) on $\Lambda_4(c_y)$ with parameter $C\varepsilon$; write $\tilde A_{*,y}=R(u_y)A_*$. The bounds below hold uniformly in the choice of u_y. Put

$$\mathsf{b}(y)=\max\{|B(c)|:\ |c_--y|_\infty\le 1\},\qquad \mathsf{j}(y)=\max\{|J_{U_0}(b)|:\ b_-\in B(y)\},$$

$$\mathsf{N}(y)=|A_*|_{\Lambda_1(c_y)}+|\nabla A_*|_{\Lambda_1(c_y)}+|\Delta_{U_0}A_*|_{\Lambda_1(c_y)}\\ +[\partial\tilde A_{*,y}]_{\alpha,\Lambda_1(c_y)}+\max\{|\omega(c)|:\ |c_--y|_\infty\le 2\},$$

where c runs over the bonds of $\mathbb{T}$.

Proposition 29.1 (Localized bounds)**.** *There are $C_7,\gamma_7>0$, independent of k, n, η and U_0, such that, under the hypotheses of Theorem 25.1, for every $y\in\mathbb{T}$:*

(a) $\mathsf{N}(y)\le C_7\sum_{y'\in\mathbb{T}}e^{-\gamma_7|y-y'|}(\mathsf{b}(y')+\mathsf{j}(y'))+C_7(\varepsilon+\beta)^2$;

(b) the configuration $U_\flat=e^{\mathrm{i}\eta A_*}U_0$, which is gauge equivalent to the minimizer U_*, satisfies

$$|F_{U_\flat}(p)|\le|F_{U_0}(p)|+C_7\mathsf{N}(y),\qquad |J_{U_\flat}(b)|\le C_7\mathsf{N}(y)$$

for all plaquettes p and bonds b with base point in $B(y)$.

Proof. Throughout, sums of products of exponentially decaying kernels are again exponentially decaying, with a smaller rate; γ_7 is the rate obtained at the end. We use the pointwise bounds of Parts II and III at U_0, which hold by Corollary 22.11, on cubes $\Lambda_1(z)$ instead of blocks (Remark 7.2).

Step 1: first derivatives. By Theorem 25.1(c), $A_*=\mathbf{H}B-\mathbf{C}J_{U_0}+\mathcal{R}$ with $|\mathcal{R}|_{C^1}\le C(\varepsilon+\beta)^2$. By the pointwise bounds of Theorem 19.9 (Theorem 16.1(b),(d)),

$$|\mathbf{H}B|_{\Lambda_1(c_y)}+|\nabla\mathbf{H}B|_{\Lambda_1(c_y)}\le C\sum_{y'}e^{-\gamma|y-y'|}\mathsf{b}(y'),$$
$$|\mathbf{C}J_{U_0}|_{\Lambda_1(c_y)}+|\nabla\mathbf{C}J_{U_0}|_{\Lambda_1(c_y)}\le C\sum_{y'}e^{-\gamma|y-y'|}\mathsf{j}(y').$$

Let $\mathsf{N}_1(y) = |A_*|_{\Lambda_1(c_y)} + |\nabla A_*|_{\Lambda_1(c_y)}$. Then $\mathsf{N}_1(y)$ is bounded by the right-hand side of (a).

Step 2: the multiplier. By the proof of Proposition 27.2, for every bond c of $\mathbb{T}$ and $\beta_0 \in \mathfrak{g}$ with $|\beta_0| = 1$,

$$\operatorname{tr}(\beta_0\, \omega(c)) = \langle \varphi, J_{U_0} + \nabla \mathcal{N}(A_*) \rangle + \langle D\varphi, DA_* \rangle,$$

with $|\varphi(x)| + |\nabla\varphi(x)| \le Ce^{-\gamma'|x - c_-|}$. Moreover $|\nabla \mathcal{N}(A_*)|_\infty \le C(\varepsilon + |A_*|_\infty)|A_*|_{C^1} \le C(\varepsilon+\beta)^2$ by Proposition 23.3(a). Hence

$$|\omega(c)| \le C \sum_{y'} e^{-\gamma'|c_- - y'|}(\mathsf{j}(y') + \mathsf{N}_1(y')) + C(\varepsilon + \beta)^2.$$

Step 3: second order. By Propositions 23.3 and 27.2,

$$D^T D A_* = d\mathcal{Q}(A_*)^T \omega - J_{U_0} - \nabla \mathcal{N}(A_*).$$

By Proposition 22.9(c), the kernel of $d\mathcal{Q}(A_*)^T = \mathbf{Q}^T + dC_k^{\mathrm{av}}(A_*)^T$ connects a bond b only to bonds c with $b_- \in S_c$. Hence $|d\mathcal{Q}(A_*)^T\omega|_{\Lambda_1(c_y)} \le C \max\{|\omega(c)| : |c_- - y|_\infty \le 2\}$. Since $RD^T A_* = 0$, we have $\Delta_{U_0} A_* = D^T D A_* + DP_{\mathcal{V}} D^T A_*$. The second term is bounded on $\Lambda_1(c_y)$ by $C\sum_{y'} e^{-\gamma|y-y'|}|A_*|_{B(y')}$, by the pointwise bound on $\partial P_{\mathcal{V}} D^T$ in Proposition 14.2(e) (Theorem 18.2(g)) and $D = \partial + T$. This bounds $|\Delta_{U_0} A_*|_{\Lambda_1(c_y)}$. For the Hölder quotient, write Δ_{U_0} in the gauge u_y in the normal form of Lemma 12.2. Then $-\Delta \tilde{A}_{*,y} = R(u_y)\Delta_{U_0} A_* - \sum_\mu \partial_\mu^*(a_\mu \tilde{A}_{*,y}) - b\tilde{A}_{*,y}$ on $\Lambda_3(c_y)$, with $|a_\mu| + [a_\mu]_\alpha + |b| \le C\varepsilon$. Theorem 6.3(d) gives

$$[\partial \tilde{A}_{*,y}]_{\alpha, \Lambda_1(c_y)} \le C\Big(|\Delta_{U_0} A_*|_{\Lambda_2(c_y)} + \max_{|y'-y|_\infty \le 3} \mathsf{N}_1(y')\Big).$$

The cube $\Lambda_2(c_y)$ is covered by the cubes $\Lambda_1(c_{y'})$ with $|y' - y|_\infty \le 1$; the values of ω that enter there are bounded by Step 2 for all bonds. Combining Steps 1–3 gives (a).

(b) By (23.1), $U_\flat(\partial p) = e^{\mathrm{i}\eta^2 \zeta_p(A_*)} U_0(\partial p)$ with $|\zeta_p(A_*)| \le C(|A_*| + |\nabla A_*|)$ on the bonds of p. Lemma 21.2(b) gives the bound on the curvature. By Proposition 23.3(d) and Proposition 27.2, $J_{U_\flat} = \Psi(A_*)^{-T} d\mathcal{Q}(A_*)^T \omega$. As in Step 3, this is bounded at b by $C \max\{|\omega(c)| : |c_- - y|_\infty \le 2\}$. □

29.2 The local improvement

Lemma 29.2 (Local improvement)**.** *There are $\varepsilon_{\mathrm{imp}} > 0$, $C_{\mathrm{imp}} \ge 1$ and an integer n_0, depending only on d, L, G, a, α and M, with the following property. Let $k \ge 1$, $n \ge n_0$, and let $V \in \mathfrak{U}_k(\varepsilon_1)$ be a real configuration on $\mathbb{T}$. Let $U \in \mathfrak{B}(V)$ be a critical point of $S|_{\mathfrak{B}(V)}$. Suppose that U is (M, ε_0)-regular and $U \in \mathfrak{U}(\varepsilon_0)$, with $\varepsilon_0 \le \varepsilon_{\mathrm{imp}}$. Then U is (M, ε_0')-regular and $U \in \mathfrak{U}(\varepsilon_0')$, where*

$$\varepsilon_0' = C_{\mathrm{imp}}\varepsilon_1 + \tfrac{1}{2}\varepsilon_0.$$

If $\varepsilon_0' > 1$, the regularity conclusion is read as $(M, 1)$-regularity; it gives the three bounds of Definition 18.1 with ε_0' in place of ε.

The point is that C_{imp} does not depend on ε_0. Iterating the lemma brings the regularity parameter down to $2C_{\mathrm{imp}}\varepsilon_1$, whatever the a priori bound, as long as the latter is at most $\varepsilon_{\mathrm{imp}}$.

Proof. If $\varepsilon_1 > 2\varepsilon_0$ there is nothing to prove, since then $\varepsilon_0' \ge \varepsilon_0$. So let $\varepsilon_1 \le 2\varepsilon_0$.

Choice of constants. Let C_7, γ_7 be as in Proposition 29.1, C_8, q as in Lemma 28.4, and C_9 as in Lemma 28.6. Put

$$K_R = 2C_9(1 + C_8 R^q), \qquad S_1 = \sum_{w \in \mathbb{Z}^d} e^{-\gamma_7|w|}(1 + |w|), \qquad T(t) = \sum_{w \in \mathbb{Z}^d,\, |w|_\infty > t} e^{-\gamma_7|w|},$$

and $K_1 = \max(5, C_7)$. Fix an integer $R \geq 64(M+1)$ with

$$K_1 C_7 K_R\, T(R/8) \leq \tfrac{1}{4};$$

this is possible because K_R grows polynomially and T decays exponentially. Let $n_0 = 4R + 16$. Choose $\varepsilon_{\mathrm{imp}}$ so small that $K_1 C_7 K_R^2 \varepsilon_{\mathrm{imp}} \leq \frac{1}{4}$, $K_R \varepsilon_{\mathrm{imp}} \leq c_6$, $\varepsilon_{\mathrm{imp}} \leq \theta_0$, $C_8 R^q \varepsilon_{\mathrm{imp}} \leq 1$ and $C_9(1 + C_8 R^q)\varepsilon_{\mathrm{imp}} \leq c_1$.

Step 1: the adapted gauge. Fix $y \in \mathbb{T}$. By Lemma 28.4 there is a real gauge transformation u on $\Lambda_R(y)$ with $U^u = e^{\mathrm{i}\eta A}$, $\|A\|_{1,\alpha;\Lambda_R(y)} \leq s := C_8 R^q \varepsilon_0$, and $V^u(c) = e^{\mathrm{i}B_y(c)}$ with $|B_y(c)| \leq d(|c_- - y|_\infty + 1)\varepsilon_1$ for the bonds of $\mathbb{T}$ in $\Lambda_R(y)$.

Step 2: the spliced reference. Let $U^\sharp$ be the spliced configuration of Definition 28.5, built with this u. By Lemma 28.6, $U^\sharp$ is $(M, \varepsilon^\sharp)$-regular with $|J_{U^\sharp}| \leq \varepsilon^\sharp$, where $\varepsilon^\sharp = C_9(\varepsilon_0 + s)$. Moreover $V(c)W^\sharp(c)^{-1} = e^{\mathrm{i}B^\sharp(c)}$ with $|B^\sharp|_\infty \leq \beta^\sharp := C_9(\varepsilon_0 + s)$, where $W^\sharp = (U^\sharp)^{(k)}$. Then $\varepsilon^\sharp + \beta^\sharp \leq K_R \varepsilon_0 \leq c_6$. So Theorem 25.1 applies with the reference $U^\sharp$ and the coarse configuration V.

Step 3: identification. By Theorem 25.1(a), $S|_{\mathfrak{B}(V)}$ has exactly one critical orbit in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$. Since U is critical and lies in $\mathfrak{U}(\varepsilon_0) \subset \mathfrak{U}(\theta_0)$, it belongs to this orbit. Hence $U = (e^{\mathrm{i}\eta A^\sharp_*} U^\sharp)^v$ for a real gauge transformation v, where $A^\sharp_*$ is the Landau representative relative to $U^\sharp$. Applying u, we get on the bonds of $\Lambda_{R/4}(y)$, where $(U^\sharp)^u = 1$,

$$U^{uv^{-1}} = e^{\mathrm{i}\eta \tilde{A}}, \qquad \tilde{A} = R(u) A^\sharp_*.$$

In this gauge $\nabla_{U^\sharp}$ is the ordinary difference, and u is a gauge in which $U^\sharp$ satisfies (B) near every block in $\Lambda_{R/4-4}(y)$.

Step 4: the data near y. Apply Proposition 29.1 with reference $U^\sharp$. Let $y' \in \mathbb{T}$ with $|y' - y|_\infty \leq R/4 - 3$. Every bond c with $|c_- - y'|_\infty \leq 1$ has $B(c_-) \cup B(c_+) \subset \Lambda_{R/4}(y)$. So, by Lemma 28.6(c), $(W^\sharp)^u(c) = 1$, and

$$|B^\sharp(c)| = |\ell(V^u(c))| = |B_y(c)| \leq d(|y' - y|_\infty + 2)\varepsilon_1.$$

Moreover $\mathsf{j}(y') = 0$ by Lemma 28.6(b). For the remaining y' we only use $\mathsf{b}(y') + \mathsf{j}(y') \leq K_R \varepsilon_0$. Let $y_1 \in \mathbb{T}$ with $|y_1 - y|_\infty \leq 4M + 4$. For $|y' - y|_\infty > R/4 - 3$ we have $|y_1 - y'|_\infty > R/8$, because $R \geq 64(M+1)$. Therefore

$$\mathsf{N}(y_1) \leq C_7 d S_1 (4M + 7)\, \varepsilon_1 + C_7 K_R\, T(R/8)\, \varepsilon_0 + C_7 K_R^2 \varepsilon_0^2.$$

Step 5: conclusion. Let $z \in \mathbb{T}_\eta$, and choose $y \in \mathbb{T}$ with $|z - y|_\infty \leq 1$. Then $\Lambda_{4M+2}(z) \subset \Lambda_{4M+3}(y) \subset \Lambda_{R/4-4}(y)$. By Step 3, $U^{uv^{-1}} = e^{\mathrm{i}\eta\tilde{A}}$ on $\Lambda_{4M+2}(z)$. On it, $|\tilde{A}|$, $|\partial\tilde{A}|$ and the Hölder quotients of $\partial\tilde{A}$ on cubes $\Lambda_1(c_{y_1})$ are bounded by $\mathsf{N}(y_1)$, with the gauge $u_{y_1} = u$. Put $H = \max_{|y_1 - y|_\infty \leq 4M+4} \mathsf{N}(y_1)$. Let $x \neq x'$ be points of a cube $\Lambda_3(z') \subset \Lambda_{4M+2}(z)$. If $|x - x'|_\infty < \frac{1}{2}$, then in each coordinate x and x' differ by a multiple of $\eta = 1/N$ less than $\frac{1}{2}$, hence by at most $\frac{N-1}{2N}$, since N is odd. A difference involves points at most 2η beyond its base point in each coordinate (for $\partial_\mu \tilde{A}_\mu$, the point $x + 2\eta e_\mu$). So the points of the two differences span, in each coordinate, an interval of length at most $\frac{N-1}{2N} + \frac{2}{N} = \frac{N+3}{2N} \leq 1$, since $N \geq 3$. The centres c with $|p - c|_\infty \leq 1$ for all these points p form, in each coordinate, a closed interval of length at least 1. The block centres form a grid of spacing 1, so one of them lies in it: these points lie in a common cube $\Lambda_1(c_{y_1})$, and then $|y_1 - y|_\infty \leq 4M + 4$. If $|x - x'|_\infty \geq \frac{1}{2}$, the Hölder quotient is at most $4 \sup |\partial\tilde{A}|$. Hence $|\tilde{A}| \leq H$, $|\partial\tilde{A}| \leq H$, and the Hölder quotients of $\partial\tilde{A}$ on the cubes $\Lambda_3(z') \subset \Lambda_{4M+2}(z)$ are at most $4H$. These are the three bounds of Definition 18.1, and each is at most $K_1 H$, since $K_1 \geq 5$. By Proposition 29.1(b), with $F_{U^\sharp} = 0$ near y, the curvature and the current of U on the plaquettes and bonds based in $B(y)$ are bounded by $C_7 \mathsf{N}(y)$. All these quantities are gauge invariant. Every plaquette and

bond is based in some block, and every z is close to some y. With the choice of R and $\varepsilon_{\mathrm{imp}}$, Step 4 gives the lemma with $C_{\mathrm{imp}} = K_1 C_7 d S_1 (4M+7)$:

$$K_1 \max \mathsf{N} \le C_{\mathrm{imp}} \varepsilon_1 + \tfrac{1}{4}\varepsilon_0 + \tfrac{1}{4}\varepsilon_0. \qquad \square$$

Remark 29.3 (Comparison with [8, Sect. F]). The structure of the argument is Bałaban's: near a point the minimizer equals, up to small errors, $\mathbf{H}$ applied to the coarse field in a gauge centred at the point (for us the gauge of the contours Γ^{co} of Section 28); the latter grows only linearly with the distance, with slope given by the curvature of V, while $\mathbf{H}$ decays exponentially. Compare [8, (159)–(167)]. The difference is in how the problem is localized. Bałaban formulates a new variational problem on a nested sequence of cubes around the point. The configuration is frozen outside, and the problem is solved relative to the trivial configuration. Here the problem is not changed. Only the reference configuration is changed, and Theorem 25.1 is applied once more on the torus. This is possible because that theorem is uniform in the reference. The splicing is a purely geometric operation (Lemma 28.6). As in [8, (166)], the quadratic error forces the a priori bound to be small in terms of the localization radius.

30 The induction over the number of steps

We now prove Bałaban's variational theorem on the torus by induction over k. Each step has three parts:

- the solution of the $(k-1)$-step problem for the trivial extension of V is a reference with *exactly* the right averages [8, Sect. A];
- Theorem 25.1 solves the problem near this reference, with an a priori regularity bound that is worse by a power of L;
- Lemma 29.2, iterated, restores the regularity bound with a constant independent of k.

30.1 The trivial extension

Let $k \ge 1$. A bond b of $\mathbb{T}^{(k-1)}$ in direction μ is a *crossing bond* if its endpoints lie in different blocks, $b_- \in B_k(y)$ and $b_+ \in B_k(y+e_\mu)$; we then write $c(b) = (y, y+e_\mu)$. For a configuration V on $\mathbb{T}$ define V_0 on $\mathbb{T}^{(k-1)}$ by

$$V_0(b) = V(c(b)) \ \text{ for crossing bonds}, \qquad V_0(b) = 1 \ \text{ for bonds inside a block}$$

[8, (11)].

Lemma 30.1 (Trivial extension). *(a)* $\mathfrak{a}_k(V_0) = V$.

(b) Let p be a plaquette of $\mathbb{T}^{(k-1)}$. If p has crossing bonds in both of its directions, then $V_0(\partial p) = V(\partial p')$, where p' is the plaquette of $\mathbb{T}$ spanned by the four blocks containing the corners of p. Otherwise $V_0(\partial p) = 1$. Hence $F_{V_0}(p) = L^2 F_V(p')$ or 0, and $V \in \mathfrak{U}_k(\varepsilon_1)$ implies $V_0 \in \mathfrak{U}_{k-1}(L^2\varepsilon_1)$.

Proof. (a) Let $c = (y, y+e_\mu)$ and $x \in B_k(y)$. Every contour of the family $G(y,x)$ lies in the box spanned by y and x, which is contained in $B_k(y)$. So it consists of bonds inside a block, on which $V_0 = 1$. The family $\{V_0(\Gamma)\}_{\Gamma \in G(y,x)}$ is therefore the constant family $\{1, \dots, 1\}$, and $V_0(y,x) = 1$ by Lemma 21.5(b). Likewise $V_0(y+e_\mu, x(c)) = 1$. Write $x_\mu = y_\mu + m\delta_{k-1}$ with $|m| \le \frac{L-1}{2}$. The straight contour $[x, x(c)]$ has L bonds in direction μ, starting at the points with μ-coordinates $y_\mu + i\delta_{k-1}$, $m \le i \le m+L-1$. Exactly one of them, the one with $i = \frac{L-1}{2}$,

joins $B_k(y)$ to $B_k(y+e_\mu)$; it is a crossing bond b with $c(b)=c$, and the others lie inside a block. The same holds for $[c]$, which is the case $m=0$. Hence $V_0([x,x(c)])=V(c)=V_0([c])$. So every averaged loop is $\Lambda_{c,x}(V_0)=1\cdot V(c)\cdot 1\cdot V(c)^{-1}=1$, all logarithms in (21.3) vanish, and $\mathfrak{a}_k(V_0)(c)=V_0([c])=V(c)$. For the reversed bond $-c$ the claim follows from Lemma 21.9.

(b) The two bonds of p in direction μ are crossing or not simultaneously, and they connect the same pair of blocks if the bonds in direction ν do not cross. In that case $V_0(\partial p)$ is $V(c)V(c)^{-1}=1$ or 1. If both pairs cross, the four bonds carry the values of V on the four bonds of $\partial p'$, in the order of Definition 21.4. Finally $F_{V_0}(p)=\delta_{k-1}^{-2}\ell(V_0(\partial p))=L^2\ell(V(\partial p'))=L^2F_V(p')$. □

In units of δ_{k-1}, curvatures are multiplied by $\delta_{k-1}^2=L^{-2}$. So in its own units, V_0 satisfies the same curvature bound ε_1 as V.

30.2 Change of units

As in the proof of Theorem 24.2, the $(k-1)$-step structure on $(\mathbb{T}_\eta,\mathbb{T}^{(k-1)})$, in units of δ_{k-1}, is the $(k-1)$-step structure on a torus of side Ln with fine spacing $L\eta$. The Wilson action does not change, since $d=4$ (Section 23); so critical points and minimizers are the same.

Lemma 30.2 (Change of units)**.** *There is C_L^{units}, depending only on d, α, G, L and M, with the following property. Let U_0 be a real configuration on $\mathbb{T}_\eta$ which, in units of δ_{k-1}, lies in $\mathfrak{U}(\theta)$ and is (M,θ)-regular. If $C_L^{\mathrm{units}}\theta\le 1$, then in units of $\delta_k=1$, $U_0\in\mathfrak{U}(L^3\theta)$ and U_0 is $(M,C_L^{\mathrm{units}}\theta)$-regular.*

Proof. Passing to units of δ_k divides lengths by L and multiplies Lie fields by L, their differences and curvatures by L^2, currents by L^3, and Hölder quotients of differences by $L^{2+\alpha}$. This gives the first claim. For the second, let $r'=(4M+2)/L$. Every cube $\Lambda_{r'}(z)$ carries a gauge with $|A|\le L\theta$ and $|\partial A|\le L^2\theta$, and with Hölder quotients of ∂A at most $L^{2+\alpha}\theta$ on cubes of radius $3/L$. Let $r=r'-2$. Two points of $\Lambda_r(z)$ at distance at most $3/L$ lie in a common cube of radius $3/L$ contained in $\Lambda_{r'}(z)$. For points at larger distance the quotient is at most $2|\partial A|(L/3)^\alpha$. Hence U_0 is $3L^3\theta$-regular on every cube $\Lambda_r(z)$. Since $M\ge 5L$, $r\ge 18$. Lemma 28.3 with $R=4M+2$ gives the claim, with $C_L^{\mathrm{units}}=3C_E(4M+2)^{p+1}L^3$. □

30.3 The theorem

Theorem 30.3 (Bałaban's variational theorem on the torus)**.** *Assume the conventions of Part VI, in particular $M\ge\max(M_0,5L)$. Let θ_0 be as in Theorem 25.1 and n_0 as in Lemma 29.2. There are $a_1,B_3>0$, depending only on d, L, G, a, α and M, such that the following holds for all $k\ge 0$, all $n\ge n_0$ and every real configuration V on $\mathbb{T}$ with $V\in\mathfrak{U}_k(\varepsilon_1)$, $\varepsilon_1\le a_1$.*

(a) $S|_{\mathfrak{B}(V)}$ has exactly one critical orbit in $\mathfrak{B}(V)\cap\mathfrak{U}(\theta_0)$. This orbit is the set of minimizers of S on $\mathfrak{B}(V)\cap\mathfrak{U}(\theta_0)$.

(b) Every element $U_k(V)$ of this orbit lies in $\mathfrak{U}(B_3\varepsilon_1)$ and is $(M,B_3\varepsilon_1)$-regular.

(c) (Lattice symmetry.) *For $r\in\mathfrak{S}_k$, $rV\in\mathfrak{U}_k(\varepsilon_1)$, and the minimal orbit of rV is the image under r of the minimal orbit of V. In particular $S(U_k(rV))=S(U_k(V))$.*

The constants do not depend on k, n or η.

This is [8, Theorem 1] for a single torus with the small-field condition everywhere. Part (b) is the regularity statement [8, (8)–(10)] for cubes of the fixed size M. As in [8], the elements of the orbit are called *background fields* or *minimal configurations*.

Proof. Constants. Let C_{base} be the constant of the case $k=0$ below, and put Details

$$B_3=\max(4C_{\mathrm{imp}},C_{\mathrm{base}}),\qquad K=(1+C_6)C_L^{\mathrm{units}}.$$

Choose a_1 so small that $C_L^{\mathrm{units}}B_3a_1\le\min(c_6,1)$, $KB_3a_1\le\varepsilon_{\mathrm{imp}}$, $B_3a_1\le\theta_0$, $a_1\le c_1$, $d(8M+6)a_1\le c_G$, and $B_3a_1\le c_3'$, with c_3' of Section 46.1, which depends only on d, L and G. The last condition is used only in Part XI, where Proposition 46.4 is applied at the background fields of arbitrary small-field configurations (proof of Theorem 55.4). These choices do not depend on k.

The case $k=0$. Then $\eta=1$, $\mathbb{T}_\eta=\mathbb{T}$, $\mathfrak{B}(V)=\{V\}$ and $\mathfrak{G}_0=\{1\}$. We first prove (b); then $V\in\mathfrak{U}(B_3\varepsilon_1)\subset\mathfrak{U}(\theta_0)$, and (a) is trivial. The current $J_V=D_V^TF_V^s$ is a combination of at most $4d$ values of F_V^s, so $|J_V|\le C\varepsilon_1$. For regularity, fix z and a point $z_0\in\mathbb{T}$ with $|z-z_0|_\infty\le1$, and let u be the gauge on $\Lambda_{4M+2}(z)$ with $V^u(\Gamma^{\mathrm{co}}_{z_0,x})=1$, with the contours Γ^{co} of Section 28 (here the fine and the unit lattice coincide). As in the proof of Lemma 28.4, $|\ell(V^u(b))|\le d(4M+4)\varepsilon_1$ for every bond b of $\Lambda_{4M+2}(z)$. On the unit lattice, for $A=\ell(V^u)$, $|\partial A|\le2|A|$, and the Hölder quotients of ∂A are at most $2|\partial A|\le4|A|$, since distinct points have distance at least 1. So V is $(M,C_{\mathrm{base}}\varepsilon_1)$-regular and lies in $\mathfrak{U}(C_{\mathrm{base}}\varepsilon_1)$, with C_{base} depending only on d, G and M.

From $k-1$ to k. Let $V\in\mathfrak{U}_k(\varepsilon_1)$ with $\varepsilon_1\le a_1$, and let V_0 be its trivial extension. In units of δ_{k-1} the torus has side $Ln\ge n_0$, and V_0 has curvature at most ε_1 (Lemma 30.1). By the induction hypothesis there is a minimal configuration $U_0=U_{k-1}(V_0)$. In units of δ_{k-1} it lies in $\mathfrak{U}(B_3\varepsilon_1)$ and is $(M,B_3\varepsilon_1)$-regular. By Lemma 30.2, in units of δ_k,

$$U_0\in\mathfrak{U}(L^3B_3\varepsilon_1),\qquad U_0\text{ is }(M,\varepsilon)\text{-regular},\qquad\varepsilon:=C_L^{\mathrm{units}}B_3\varepsilon_1\le c_6.$$

Since $U_0^{(k-1)}=V_0$, Lemma 30.1(a) gives

$$W_k=U_0^{(k)}=\mathfrak{a}_k(V_0)=V,$$

so the hypotheses of Theorem 25.1 hold with $\beta=0$; note $|J_{U_0}|\le L^3B_3\varepsilon_1\le\varepsilon$. Parts (a) and (b) of that theorem give (a).

For (b), let U be an element of the orbit. By Theorem 25.1(e), U is $(M,\varepsilon_0^{(0)})$-regular and lies in $\mathfrak{U}(\varepsilon_0^{(0)})$, with $\varepsilon_0^{(0)}=(1+C_6)\varepsilon=KB_3\varepsilon_1\le\varepsilon_{\mathrm{imp}}$. If $\varepsilon_0^{(0)}\le B_3\varepsilon_1$ there is nothing to prove, so let $\varepsilon_0^{(0)}>B_3\varepsilon_1\ge4C_{\mathrm{imp}}\varepsilon_1$. Define $\varepsilon_0^{(m+1)}=C_{\mathrm{imp}}\varepsilon_1+\frac12\varepsilon_0^{(m)}$. If $\varepsilon_0^{(m)}\ge2C_{\mathrm{imp}}\varepsilon_1$, then $\varepsilon_0^{(m+1)}\le\varepsilon_0^{(m)}$. By Lemma 29.2 and induction over m, U is $(M,\varepsilon_0^{(m)})$-regular and lies in $\mathfrak{U}(\varepsilon_0^{(m)})$ for all m. Since

$$\varepsilon_0^{(m)}-2C_{\mathrm{imp}}\varepsilon_1=2^{-m}(\varepsilon_0^{(0)}-2C_{\mathrm{imp}}\varepsilon_1)\le2^{-m}KB_3\varepsilon_1,$$

we have $\varepsilon_0^{(m)}\le2C_{\mathrm{imp}}\varepsilon_1+\frac12B_3\varepsilon_1\le B_3\varepsilon_1$ once $2^m\ge2K$. This proves (b) with the same B_3, and closes the induction.

Part (c). The set $\mathfrak{U}_k(\varepsilon_1)$ is invariant under $\mathfrak{S}_k$ (Section 21). As in the proof of Theorem 25.1(h), the map $U\mapsto rU$ maps $\mathfrak{B}(V)\cap\mathfrak{U}(\theta_0)$ onto $\mathfrak{B}(rV)\cap\mathfrak{U}(\theta_0)$, preserves S and its critical points, and maps $\mathfrak{G}_0$-orbits onto $\mathfrak{G}_0$-orbits. So it maps the critical orbit of (a) for V onto that for rV. Since S is constant on this orbit and $S\circ r=S$, the last claim follows. □

The induction uses Theorem 25.1 with $\beta=0$: the reference has exactly the prescribed averages. On the nested domains of [8] this is not the case, and β is of order ε_1 [8, (14)]. Theorem 25.1 allows $\beta>0$, which is used in Lemma 29.2 and in Corollary 30.5 below.

30.4 Consequences

Corollary 30.4 (Local regularity from gauge-invariant conditions)**.** *There are $a_2, B_5 > 0$, depending only on d, L, G, a, α and M, such that for all $k \geq 0$ and $n \geq n_0$ every real configuration $U \in \mathfrak{U}(\theta)$ on $\mathbb{T}_\eta$ with $\theta \leq a_2$ is $(M, B_5\theta)$-regular. More precisely, U is gauge equivalent, by a real gauge transformation, to $e^{\mathrm{i}\eta A} U_k(U^{(k)})$. Here A lies in Balaban's Landau gauge relative to $U_k(U^{(k)})$, and A obeys the bounds of Theorem 24.2 with $\varepsilon + \theta + \beta = (2B_3 + 1)\theta$.*

Proof. Let $k = 0$. Then $\eta = 1$, $\mathbb{T}_\eta = \mathbb{T}$ and $U^{(0)} = U$. By the case $k = 0$ of the proof of Theorem 30.3, $\mathfrak{B}(U) = \{U\}$, so $U_0(U) = U$, and the representation holds with $A = 0$, for which the gauge condition and the bounds are trivial. The curvature of U is at most $\theta \leq a_1$, so Theorem 30.3(b), with $V = U$ and $\varepsilon_1 = \theta$, shows that U is $(M, B_3\theta)$-regular.

Let $k \geq 1$ and $V = U^{(k)}$. By Proposition 21.12, $V \in \mathfrak{U}_k(2\theta)$. Let $2\theta \leq a_1$ and $(2B_3 + 1)\theta \leq c_4$. By Theorem 30.3, $U_0 = U_k(V)$ exists. It is $(M, 2B_3\theta)$-regular, satisfies $|J_{U_0}| \leq 2B_3\theta$, and has $U_0^{(k)} = V$. Since $U \in \mathfrak{U}_0(c_1)$ and $U_0 \in \mathfrak{U}_0(2B_3\theta) \subset \mathfrak{U}_0(c_1)$, Lemma 22.3(c) applies: the $\mathfrak{G}_0$-orbit of U contains its averaged axial representative U^{ax}, with $U' = U^{\mathrm{ax}} U_0^{-1} \in \mathrm{Ax}_k$. Transformations in $\mathfrak{G}_0$ do not change the averages, so $U'^{[k]} = (U^{\mathrm{ax}})^{(k)} V^{-1} = 1$. Theorem 24.2, with $\varepsilon = 2B_3\theta$ and $\beta = 0$, gives $\tilde{U} = e^{\mathrm{i}\eta A}$ in the Landau gauge with $(\tilde{U} U_0)^{\mathsf{w}_k(\tilde{U})} = U^{\mathrm{ax}}$ and with the stated bounds. Finally, $e^{\mathrm{i}\eta A} U_0$ is $(M, 2B_3\theta + CB_1(2B_3 + 1)\theta)$-regular, by the argument in the proof of Proposition 27.4(c). That argument uses the Hölder bound on the unit cubes $\Lambda_1(z)$. For the field A of Theorem 24.2 this bound is given by the proof of Proposition 24.5 (Theorem 6.3(d)) and Remark 7.2. So the corollary holds with $B_5 = 2B_3 + CB_1(2B_3 + 1)$, which is at least the constant B_3 of the case $k = 0$. □

Corollary 30.4 is the torus version of [6, Sect. F, Prop. 6] and of the regularity conditions [7, (3.35)–(3.36)]. It removes the hypothesis of Part III for all configurations of Bałaban's small-field class: on tori of side $n \geq n_0$, regularity follows from the gauge-invariant bounds on curvature and current. The proof uses the background field as a reference: the local gauges are the Landau gauge relative to $U_k(U^{(k)})$, composed with the local gauges of the background field. The latter come out of the induction.

Corollary 30.5 (Analyticity around a background field [8, Sect. G])**.** *Let V be as in Theorem 30.3, and let U_0 be an element of its minimal orbit. For $k \geq 1$, Theorem 25.1 applies with this reference and $\varepsilon = B_3\varepsilon_1$. For real B on $\mathbb{T}$ with $|B|_\infty \leq \beta$ and $B_3\varepsilon_1 + \beta \leq c_6$, the critical orbit for $e^{\mathrm{i}B} V$ in $\mathfrak{B}(e^{\mathrm{i}B} V) \cap \mathfrak{U}(\theta_0)$ is unique, and its Landau representative $A_*(B)$ relative to U_0 satisfies $A_*(0) = 0$. The map $B \mapsto A_*(B)$ extends analytically to complex B, and $dA_*(B)\delta B$ decays exponentially away from the support of δB, as in Theorem 25.1(f). For $r \in \mathfrak{S}_k$, rU_0 is an element of the minimal orbit of rV, and $A_*(r_V B; rU_0) = r_{U_0} A_*(B; U_0)$, the second argument being the reference.*

Proof. U_0 is $(M, B_3\varepsilon_1)$-regular with $|J_{U_0}| \leq B_3\varepsilon_1$, and $W_k = V$. For $B = 0$, the configuration $U_0 = \mathsf{U}(0)$ is critical, and $A = 0$ lies in the Landau gauge with $\mathcal{Q}(1) = 1$; by uniqueness, $A_*(0) = 0$. For $k \geq 1$, the symmetry statement follows from Theorem 30.3(c) and Theorem 25.1(h), since $W_k = V$, and the rest is Theorem 25.1.

Let $k = 0$. The operators of Parts II–V need $k \geq 1$ (Section 2), but the conclusions hold directly. Here $\eta = 1$, $\mathbb{T}_\eta = \mathbb{T}$, $\mathfrak{G}_0 = \{1\}$ and $U_0 = V$, by the case $k = 0$ of the proof of Theorem 30.3. The averaging, the correspondence transformation and $\mathcal{Q}$ are the identity, so $\mathfrak{B}(e^{\mathrm{i}B} V) = \{e^{\mathrm{i}B} V\}$, and the relative field of $e^{\mathrm{i}B} V$ is $A_*(B) = B$; the Landau gauge condition is void. By Lemma 21.2(b), the curvature of $e^{\mathrm{i}B} V$ is at most $\varepsilon_1 + 4|B|_\infty$, and its current, a combination of at most $4d$ values of F^s, is at most $4d$ times this. Since $\varepsilon_1 \leq B_3\varepsilon_1$, both are at most $20d\, c_6 \leq \min(\theta_0, c_1)$, by the choice of c_6 in the proof of Theorem 25.1. So the configuration $e^{\mathrm{i}B} V$ lies in $\mathfrak{U}(\theta_0) \cap \mathfrak{U}_0(c_1)$, and its orbit is the unique critical orbit. The map $B \mapsto B$ is entire, $A_*(0) = 0$, and $|A_*(B)|_{C^1} \leq 3|B|_\infty$ by (23.2). Finally $dA_*(B)\delta B = \delta B$ is

supported on the bond c, and its covariant gradient on the bonds within distance 1 of c. This is the decay of Theorem 25.1(f), with $C_6 \geq 3e^{\gamma'}$. For $r \in \mathfrak{S}_0$, $rU_0 = rV$ is the element of the minimal orbit of rV (Theorem 30.3(c)), and $A_*(r_V B; rU_0) = r_V B = r_{U_0} A_*(B; U_0)$, since $U_0 = V$. □

Remark 30.6 (Cubes of size of order ε_1^{-1}). Bałaban also proves gauges on cubes of side $2M' \leq c\varepsilon_1^{-1}$, with bounds $B_3 M' \varepsilon_1$ linear in the size [8, (9), (10)]. Theorem 30.3 and Lemma 28.3 give gauges on cubes of side $2R$ with bounds $C(R/M)^p B_3 \varepsilon_1$, polynomial in R, for $R^{p+1}\varepsilon_1$ small. Linear bounds should follow by running Lemma 29.2 with a splicing radius proportional to R, with R limited to a power of ε_1^{-1} by the quadratic error. We do not prove this. The regularity at the fixed scale M is all that Parts II–V require.

Part VII: One step of the renormalization group

Part VII constructs one step of the small-field renormalization group and writes it exactly as a Gaussian integral around the background field. The step T integrates over the configurations V on $\mathbb{T} = \mathbb{T}^{(k)}$ whose one-step average is a given configuration V' on $\mathbb{T}' = \mathbb{T}^{(k+1)}$ (Section 31). In four dimensions a change of units does not change the coupling; the coupling of the next step comes from the fluctuation integral, and Part X reads it off from the new generation of activities (Section 31, paragraph "The couplings"). Part VII proves the exact representation and the linear estimates on which Parts VIII–X rest (Remark 34.11). The gauge group is semisimple from here on.

What this Part proves. The main theorem is Theorem K, assembled from Theorems 34.9 and 34.4, Corollary 34.6, Lemma 32.6 and Proposition 31.7. The anatomy of the step is as follows.

- *Gauge fixing.* Bałaban's exponential gauge-fixing function gf_W fixes the invariance under the gauge transformations of $\mathbb{T}$ that are trivial on $\mathbb{T}'$, in the averaged contour variables relative to the base $W = U'^{(k)}$, the k-step average of the background field U' of V' (Definition 32.2, Lemma 31.6). By a Faddeev–Popov identity without determinant (Lemma 32.3), the gauge-fixed step agrees with T on invariant integrands supported on regular configurations; for $\rho \geq 0$ the part removed by the cutoffs is nonnegative (Lemma 32.6).
- *Elimination.* The constraint is solved on the middle bond $b_0(c)$ of each coarse bond c, with local Jacobians, and the fluctuation field lives on the set $\mathfrak{F}$ of the other bonds (Lemma 32.7, Proposition 32.8).
- *The exponent.* The coarse effective action $V \mapsto S(U_k(V))$ is analytic, its gradient is the Lagrange multiplier, and its derivatives decay along trees (Proposition 33.1). After composition with the eliminated field $\hat{Z}(Z)$ and addition of the gauge-fixing exponent $\Phi_W(Z)$, the quadratic part is $\frac{1}{2}\Delta_k(Z, Z)$; the remainder is the sum of the third-order coarse-action interaction $\mathcal{V}(Z)$ and the block-local gauge-fixing remainder $G_3(Z)$ (Lemma 34.1).
- *Coercivity.* The fluctuation operator satisfies $\frac{1}{4}c_L\|Z\|^2 \leq \Delta_k(Z, Z) \leq C_L\|Z\|^2$, and its inverse C_k decays exponentially (Theorem 34.4, Corollary 34.6). The input is a Poincaré inequality for one step, in which the gauge-fixing term cannot be dropped (Lemma 34.2).

- *The representation.* For $k \geq 1$, the gauge-fixed step $T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(V')$ equals an explicit constant times $e^{-g^{-2}S(U')}(\det \Delta_k)^{-1/2}\prod_c \mathfrak{j}_c$ times a Gaussian integral with covariance C_k (Theorem 34.9), and $\log\det\Delta_k$ is a sum of quasi-local, gauge-invariant densities (Proposition 34.8).

The curvature of the background field of a coarse configuration of small curvature is at most $(C_* + \delta)$ times that curvature, with C_* independent of L (Proposition 31.5, with Lemma 31.3). So the small-field thresholds of consecutive steps are compatible for $L^2 \geq 8(C_* + 1)$, provided $g_{k+1} \leq 2g_k$ and the fluctuation cutoff is small, as in condition (Q2) of Remark 49.8. The step commutes with the lattice symmetries (Proposition 31.7). The constants do not depend on k, n, η or the coupling.

Imports and exports. Part VII uses the background fields and the variational problem (Theorems 25.1 and 30.3, Corollary 30.5), the averaging (Propositions 22.9 and 22.10, Lemma 22.7), and the operators of Parts II and III at regular backgrounds. No proof uses a later result; the white-noise cutoff of the small-field step is Definition 48.1, and the step from scale 0 is Lemma 52.3. Theorem 34.9, Corollary 34.6, Proposition 34.8, Lemma 32.7 and Proposition 32.8 are the inputs of Parts VIII–XI. Proposition 33.2 goes to Parts VIII, X and XI, Proposition 31.5 to Parts VIII–X, Lemma 32.6 to Parts X and XI, and Remark 34.10 to Part XI. All of Part VII is on the main line.

Details for this Part. The companion, chapter on Part VII, expands each step of this Part, from the uniformity of the constants to Theorem 34.9, and the order of choice of its constants. A proof with an expansion there carries the marginal link "Details".

31 The renormalization group transformation

This part carries out one step of Bałaban's renormalization group for pure Yang–Mills theory in four dimensions in the small-field region [18, 19]; the three-dimensional version is [9]. Here we derive an exact representation of one step as a Gaussian integral around the background field, and we prove the linear estimates on which the analysis of that integral rests:

- the coercivity and decay of the fluctuation covariance;
- the analyticity of the coarse effective action and the decay of its derivatives;
- the structure of the background-dependent normalization.

The polymer expansion of the interaction, its renormalization and the iteration are the subject of Parts VIII–X.

What one step does. The step integrates out the fluctuation of V around the base configuration $W = U'^{(k)}$, the k-step average of the background field U' of V' (Section 31.3). After Bałaban's gauge fixing relative to W and the elimination of the constraint on the middle bond $b_0(c)$ of each coarse bond c, it is an integral over fields Z on the set $\mathfrak{F}$ of the other bonds (Section 32). For a density of the form (31.2) the integrand contains the factor $e^{-g_k^{-2}F(Z)}$, with $F(Z) = S(U') + \frac{1}{2}\Delta_k(Z,Z) + \mathcal{V}(Z) + G_3(Z)$, where $\mathcal{V}$ and G_3 are of third order (Lemma 34.1). Section 33 studies the coarse action that enters Δ_k, and Section 34 proves that Δ_k is coercive and that $C_k = \Delta_k^{-1}$ decays (Corollary 34.6). For $k \geq 1$, Theorem 34.9 then writes the step as an explicit prefactor times a Gaussian integral with covariance C_k, and Theorem K collects the results of this part. The introduction to Part VII lists the stages of the step with their results.

Conventions for Part VII. We keep the conventions of Parts IV–VI. In particular $d = 4$ and $L \geq 3$ is odd (Section 2). In addition G is semisimple. Semisimplicity is not used in this part; it is needed for the renormalization in the next one. We work in the *units of step k*: the unit lattice is $\mathbb{T} = \mathbb{T}^{(k)}$, the fine lattice is $\mathbb{T}_\eta$ with $\eta = L^{-k}$, and the next coarse lattice is $\mathbb{T}' = \mathbb{T}^{(k+1)}$, of spacing L. Its blocks are $B'(y) = B_{k+1}(y)$, with L^d points of $\mathbb{T}$, and the one-step averaging is $\mathfrak{a}' = \mathfrak{a}_{k+1}$. The torus $\mathbb{T}$ has side Ln, where n is the side of $\mathbb{T}'$ in its own units. We assume $n \geq n_{\delta=1}$, with n_δ from Proposition 31.5, and $n \geq 80M'_L$, with M'_L from Theorem 34.4. Fields on $\mathbb{T}$ carry the unweighted norm $\|Z\|^2 = \sum_b |Z(b)|^2$.

31.1 The model and the transformation

Let $\epsilon = L^{-N}$ be the finest lattice spacing and g the bare coupling. Here N is the total number of scales (Section 52.1); it is not the integer $N = L^k = \eta^{-1}$ of Section 2, which appears in Section 31.2. The initial density on the ϵ-lattice is $\rho_0(U) = \exp(-g^{-2}S_{\mathrm{lat}}(U) - E_0)$, where $S_{\mathrm{lat}}(U) = \sum_p (\mathsf{N} - \operatorname{Re}\operatorname{tr} U(\partial p))$. In the units of step k the ϵ-lattice has spacing $\eta = L^{-k}$, and $S_{\mathrm{lat}} = S$, the action of Section 23: in four dimensions the Wilson action carries no power of the lattice spacing. So $\rho_0 = \exp(-g^{-2}S - E_0)$ in the units of every step, and a change of units alone does not change the coupling.

The averaging $\mathfrak{a}'$ uses group averages and logarithms, so it is defined only on part of the configuration space. Let $\operatorname{dom}\mathfrak{a}'$ be the set of real configurations V on $\mathbb{T}$ for which Definition 21.8 defines $\mathfrak{a}'(V)$: for every positively oriented bond c of $\mathbb{T}'$ and every $x \in B'(c_-)$, the averaged contour variables in $\Lambda_{c,x}(V)$ are defined, i.e. their families lie in $\mathcal{D}_{r_M}$ (Definition 21.6), and $|\Lambda_{c,x}(V) - 1| \leq c_G$. For real V these averaged contour variables are real (Lemma 21.5(b)), so the logarithms $\ell(\Lambda_{c,x}(V))$ lie in $\mathfrak{g}$, and $\mathfrak{a}'(V)$ is defined and real, also on the negatively oriented bonds (Lemma 21.9). Outside this set an average or a logarithm may be undefined, or a logarithm may leave $\mathfrak{g}$. The set $\operatorname{dom}\mathfrak{a}'$ is a relatively closed subset of the open set on which the averaged contour variables are defined, and $\mathfrak{a}'$ is continuous on it (Lemma 21.5(a)). It is invariant under real gauge transformations and under the lattice symmetries $\mathfrak{S}_{k+1}$ (Lemma 21.10(b),(d)). It contains $\mathfrak{U}_k(b)$ in its interior whenever $8bL^2 \leq r_M$: then, by Lemma 21.7(i),(iii) and the proof of Lemma 21.10(a), with $\delta_{k+1} = L$, the families lie in $\mathcal{D}_{r_M/2}$ and $|\Lambda_{c,x}(V) - 1| \leq 2r_M \leq \frac{1}{2}c_G$.

For an integrable density ρ on configurations V on $\mathbb{T}$ that vanishes outside $\operatorname{dom}\mathfrak{a}'$, let $T\rho$ be the image of the measure $\rho(V)\prod_b dV(b)$ under $\mathfrak{a}'$:

$$\int \varphi\, d(T\rho) = \int \prod_b dV(b)\ \varphi(\mathfrak{a}'(V))\, \rho(V)$$

for bounded Borel functions φ of configurations on $\mathbb{T}'$. Here b runs over the positively oriented bonds of $\mathbb{T}$, and $dV(b)$ is the normalized Haar measure. With $\varphi = 1$, the total mass of $T\rho$ is $\int \prod_b dV(b)\, \rho(V)$. We write

$$(T\rho)(V') = \int \prod_b dV(b)\ \prod_c \delta(\mathfrak{a}'(V)(c)\, V'(c)^{-1})\ \rho(V), \tag{31.1}$$

where c runs over the positively oriented bonds of $\mathbb{T}'$ and δ is the δ-function of G at 1 with respect to the Haar measure. This formula is a notation for the image measure. The left side denotes a density of $T\rho$ with respect to $\prod_c dV'(c)$, and it is used only where such a density exists; the remark at the end of this subsection says in what sense the formula holds, and when the density exists. Passing to the units of step $k+1$ relabels the configurations, without any Jacobian since they are group valued, and leaves S unchanged, since $L^{4-d} = 1$.

The couplings. Unlike in three dimensions, the coupling of step $k+1$ is therefore not produced by the change of units. It is produced only by the integration over the fluctuation field.

The effective action of step $k+1$ contains a term proportional to S, with a coefficient β of order one for fixed L, which Bałaban reads off from the vacuum polarization [18, (1.20)–(1.22)], and the inverse square of the coupling of step $k+1$ is that of step k corrected by β. The sequence of couplings is defined together with the generations of the flow, in Part X: β_{k+1} is read off from the period density of the new generation, and $g_{k+1}^{-2} = g_k^{-2} - \beta_{k+1}$ (Proposition 50.13, Theorems 53.7 and 53.12). Part XI computes the one-loop value of β and compares it with the one-loop parts of the coefficients of the flow (Section 58). In this part g_k is a parameter: the coupling of the density at step k, with $g_0 = g$. The statements about couplings and thresholds below hold for every such parameter that satisfies their hypotheses; along the flow these hypotheses hold at every step (Theorem 53.12(e)).

If ρ vanishes outside $\operatorname{dom}\mathfrak{a}'$ and $\rho(V^u) = \rho(V)$ for all real gauge transformations u on $\mathbb{T}$, then $T\rho$ is gauge invariant on $\mathbb{T}'$: it is invariant under $V' \mapsto V'^{u'}$ for every real gauge transformation u' on $\mathbb{T}'$. This follows from $\mathfrak{a}'(V^u) = \mathfrak{a}'(V)^{u|_{\mathbb{T}'}}$ for real u of any size (Lemma 21.10(b)), applied to an extension u of u', from the invariance of $\operatorname{dom}\mathfrak{a}'$, and from the invariance of the Haar measure under $V \mapsto V^u$. All densities below are gauge invariant. The map T also commutes with the lattice symmetries of $\mathbb{T}'$ (Proposition 31.7(a)).

The small-field step. In the small-field region the density at step k has, by assumption, the form

$$\rho_k(V) = \exp\Big(-\frac{1}{g_k^2} S(U_k(V)) + \mathfrak{E}_k(V) \Big) \qquad \text{for } V \in \mathfrak{U}_k(\tilde{h}), \tag{31.2}$$

with $U_k(V)$ the background field of Theorem 30.3, a gauge-invariant function $\mathfrak{E}_k$, measurable and bounded on $\mathfrak{U}_k(\tilde{h})$, and a threshold $\tilde{h} \le a_1$. The renormalization group step studied here is the *small-field step*

$$T_{\mathrm{sf}}(\rho) := T^{\mathrm{gf}}(\chi^{\mathrm{w}} \chi_\xi \, \rho)$$

(Definitions 32.5, 32.4 and 48.1). Here T^{gf} is the map (31.1) with Bałaban's gauge-fixing function in the integrand, χ^{w} restricts the fluctuation field around the background of the step to $|\cdot| \le w$, and χ_ξ imposes the small-field condition on the white noise, as in [15, (206)]. The gauge-fixing function and the cutoffs depend on V' through the base configuration and the reference of the step. In T^{gf} they are taken at $V' = \mathfrak{a}'(V)$. The gauge-fixing function is set to 0 unless $V \in \operatorname{dom}\mathfrak{a}'$ and $\mathfrak{a}'(V)$ is as in Lemma 31.6, so that the base is defined, and the cutoffs are set to 0 unless $\mathfrak{a}'(V)$ satisfies the hypotheses of Proposition 32.8. That proposition shows that, for such V', χ^{w} vanishes outside Bałaban's set $\mathfrak{M}$ of regular configurations (Section 32), which lies in the interior of $\operatorname{dom}\mathfrak{a}'$, so that the restriction to $\operatorname{dom}\mathfrak{a}'$ removes nothing, and that $\mathfrak{a}'$ is a submersion wherever $\chi^{\mathrm{w}} = 1$. Regular configurations in this sense are not the (M,ε)-regular backgrounds of Definition 18.1. These functions also depend on the choice of U' in the minimal orbit of V'. We fix a Borel choice; it exists by the remark at the end of this subsection. The density at each V' does not depend on the choice, by Lemma 32.6(i). The cutoffs act bondwise on the gauge-fixed variables, as Bałaban's cutoff does [18, (2.9), (2.10)]. In particular χ^{w} is not invariant under the gauge transformations of $\mathbb{T}$ that are trivial on $\mathbb{T}'$, so T_{sf} is a modified model: it is not of the form $T(\chi\rho)$ with an invariant cutoff χ. On integrands that are invariant under these gauge transformations and supported on regular configurations (Section 21), $T^{\mathrm{gf}} = T$ (Lemma 32.3(a)). The step is the analogue of [15, (172), (206)]; Bałaban's version is [9, (7)–(13)] in three dimensions and [18, (2.9), (2.10)] in four. The large-field complement is not treated here.

Remark (The image measure and the Borel choice). *The formula* (31.1). If δ is replaced by a continuous function $\delta_n \ge 0$ on G with $\int \delta_n = 1$, supported in a neighbourhood of 1 that shrinks as $n \to \infty$, the right side is a continuous density, and the measures with these densities converge weakly to $T\rho$: for continuous φ this follows from Fubini's theorem, the invariance of the Haar measure and the uniform continuity of φ. A sufficient condition for the density

$(T\rho)(V')$ to exist is that ρ vanishes outside an open subset of $\operatorname{dom}\mathfrak{a}'$ on which $\mathfrak{a}'$ is a submersion, i.e. its derivative has rank $\dim\mathfrak{g}$ times the number of positively oriented bonds of $\mathbb{T}'$. Then, near each point of this set, $\mathfrak{a}'$ is a coordinate projection in suitable analytic coordinates, and for almost every V' the density at V' is the integral of ρ over the fibre $\{\mathfrak{a}'(V) = V'\}$ with respect to the measure that the two Haar measures induce on it. Integrability of ρ alone does not make this fibre integral finite at every V'. If ρ is a bounded Borel function and vanishes outside a compact subset of the open set, the fibre integral converges absolutely at every V'; it is then taken as the density at every V', and the usual rules for δ-functions compute it. In that case $\int \prod_c dV'(c)\,(T\rho)(V') = \int \prod_b dV(b)\,\rho(V)$. The density is used for the small-field step. There the integrand is bounded, and the cutoff confines it to a compact subset of the region of Lemma 32.7, which gives the submersion, and Proposition 32.8 computes the density. Part XI uses it also for integrands supported in the gauge orbits of that region (proof of Lemma 54.5), where the same argument applies.

The Borel choice of U'. A Borel choice exists: near each V', Corollary 30.5 gives a choice that depends continuously on V', and finitely many such neighbourhoods cover the compact set of the V' as in Lemma 31.6. For such a choice the gauge-fixing function and the cutoffs are built from continuous maps ($\mathfrak{a}'$, the base, the averaged contour variables, ℓ and $C_k^{-1/2}$, which depends continuously on the base by Corollary 34.6), continuous functions and indicator functions of closed sets. So they are Borel functions of V.

31.2 Curvature of background fields

Theorem 30.3 bounds the curvature of $U_k(V)$ by $B_3\varepsilon_1$, where B_3 depends on L. The small-field thresholds of consecutive steps are compatible only if the curvature of the background is bounded with a constant that does *not* grow with L. That is proved here.

The constants of the pointwise bounds of Parts II and III must therefore be independent of L and of M; note that $M \geq 5L$ in Part VI. We make this explicit. Bałaban's averaging pair enters only through one parameter.

Definition 31.1 (Parameter of an averaging pair)**.** Let U be a real $(1,\varepsilon)$-regular background and $(Q', \mathbf{Q})$ an averaging pair with properties (i), (iii) and (iv) of Remark 19.12. Let $\varepsilon_Q \geq 0$. The pair has *parameter ε_Q at U* if the following holds for every $y \in \mathbb{T}$ and every real gauge transformation u in which (B) holds with parameter ε on $\Lambda_4(c_y)$. Here (B) on $\Lambda_4(c_y)$ refers to the bonds and differences that involve only points of $\Lambda_4(c_y)$, and the Hölder quotient on a cube $\Lambda_3(z'') \subset \Lambda_4(c_y)$ is taken over the pairs of differences that involve only points of $\Lambda_3(z'')$. If U is (M,ε)-regular, every gauge of Definition 18.1 on a cube containing $\Lambda_4(c_y)$ satisfies (B) with parameter ε on $\Lambda_4(c_y)$ in this sense. Computed in the gauge u, on the blocks and tubes contained in $\Lambda_4(c_y)$:

(a) $Q'(U^u) - Q_0$ and $\mathbf{Q}(U^u) - \mathbf{Q}$ satisfy the bounds of Lemma 12.5(b) with $C\varepsilon$ replaced by ε_Q: block norms, the bounds from sup norms to values, and the same for the transposes;

(b) the defect E satisfies the bounds of Lemma 12.5(c) with $C\varepsilon$ replaced by ε_Q, and its transpose satisfies $|E^T\omega|_\infty \leq \varepsilon_Q|\omega|_\infty$.

By Lemma 12.5, the model pair $(Q'(U), \mathbf{Q}(U))$ of Section 12 has parameter $C_{\mathrm{mod}}\varepsilon$, with an absolute constant C_{mod}.

Lemma 31.2 (Monotonicity of regularity)**.** *If U is (M,ε)-regular and $1 \leq M' \leq M$, then U is (M',ε)-regular.*

Proof. The cube $\{x : |x-z|_\infty \leq 4M'+2\}$ lies in $\{x : |x-z|_\infty \leq 4M+2\}$. Restrict the gauge transformation of Definition 18.1 to it. The bonds with both endpoints in the smaller cube have both endpoints in the larger one, and every cube $\Lambda_3(z')$ in the smaller cube lies in the larger one. So the three bounds of Definition 18.1 persist. □ Details

Lemma 31.3 (Uniformity of the constants of Parts II–III)**.** *Fix $a > 0$ and $\alpha \in (0,1)$. Let M_0 be the constant of Part III (Theorem 19.6). There are ε_{U}, θ_Q, C_H, $\gamma_H > 0$, depending only on a, α and $\dim \mathfrak{g}$, with the following property. Let $k \geq 1$ and let the side of $\mathbb{T}$ be at least $8M_0$. Let $M \geq M_0$, let U_0 be real and (M,ε)-regular with $\varepsilon \leq \varepsilon_{\mathrm{U}}$, and let $(Q', \mathbf{Q})$ be an averaging pair with properties (i)–(iv) of Remark 19.12 and parameter $\varepsilon_Q \leq \theta_Q$ at U_0. Then:*

(a) Theorem 18.2 (with $X = \mathbb{T}_\eta$), Theorems 19.6 and 19.7 (with $X = \mathbb{T}_\eta$), and Theorem 19.9 hold at U_0 with this pair, in the form stated there: existence, covariance, identities, coercivity, decay and the pointwise bounds in local gauges, with no comparison with the operators at $U = 1$. Their constants c, C, γ, c_* depend only on a, α, $\dim \mathfrak{g}$: neither on M, nor on $N = L^k$, n, k, nor on the pair beyond $\varepsilon_Q \leq \theta_Q$.

(b) Let $\mathbf{H} = \mathbf{H}(U_0)$ be the minimizer map of Theorem 19.9 (with $W = 0$) and ∇ the covariant difference at U_0. For every unit-lattice field B,

$$|\mathbf{H}B|_{B(y)} + |\nabla \mathbf{H}B|_{B(y)} \leq C_H \sum_{y'} e^{-\gamma_H |y-y'|} |B|_{y'}. \tag{31.3}$$

(c) For Bałaban's pair $(Q'_k(U_0), \mathbf{Q}_k(U_0))$ we have $E = 0$, and there are $c_Q, C_Q > 0$ such that the pair has parameter $\varepsilon_Q \leq C_Q \varepsilon$ for $\varepsilon \leq c_Q$. Here c_Q, C_Q may depend on d, L, G, a, α and M (the conventions of Part IV). Hence (a) and (b) hold for this pair whenever $\varepsilon \leq \varepsilon_{\mathrm{B}} = \min(\varepsilon_{\mathrm{U}}, \theta_Q / C_Q, c_Q)$. In particular (31.3) holds with C_H, γ_H independent of L and M, for ε small depending on L and M.

If the side of $\mathbb{T}$ is less than $8M_0$, (a)–(c) hold as well, with the same constants.

Proof. Step 0: reduction to $M = M_0$. For $X = \mathbb{T}_\eta$ every j lies in $J(\mathbb{T}_\eta)$, so $m_{\mathbb{T}_\eta} = 0$. Hence the torus operators $\mathfrak{A}'_{\mathbb{T}_\eta}$, G', Γ', $P_{\mathcal{V}}$, R, S, Π, $\mathfrak{A}_\pi = \mathfrak{A}^{\mathbb{T}_\eta}$, $\mathbf{G}$, $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$ are defined without reference to M. By Lemma 31.2, U_0 is (M_0,ε)-regular. Apply every result of Part III with $M = M_0$; their hypotheses are "$M \geq M_0$ and (M,ε)-regular". Their constants then depend on M_0, which depends only on $a, \alpha, \dim \mathfrak{g}$. Coercivity and decay are properties of the fixed torus operators, so proving them at scale M_0 proves them for U_0. This proves the independence of M. The scale M_0 also enters through the partition of unity (Lemma 17.3), which needs the side of $\mathbb{T}$ to be at least $8M_0$. If the side is smaller, the partition at scale M_0 is the trivial one of Section 17.2. The cube of Definition 18.1 at scale M_0 then covers $\mathbb{T}_\eta$, so one real gauge gives the bounds of that definition on all of $\mathbb{T}_\eta$. The proof of Theorem 19.6 treats this case, and the other results in (a) do not use the partition. In Steps 1–4 this gauge serves as the local gauge of every cube, and (a)–(c) follow with the same constants. Details

Step 1: Part I inputs. The model pair at $U = 1$ is $(Q_0, \mathbf{Q})$ on centred blocks of side 1 with N^d fine points. The following constants are used, and none depends on the odd integer $N = L^k \geq 3$ (Section 2: L enters only through N, and $k \geq 1$). In particular the margins of the trial fields of Theorem 18.2(c) and Lemma 19.8, built from products of $\cos^2$, hold for every odd $N \geq 3$ with the same constants.

- the block Poincaré constants (Lemma 3.1, Corollary 3.3), which are absolute;
- the gauge-field coercivity constant c_g (Theorem 9.6), depending on a, on the absolute constants C_P, C_W (Lemma 9.3) and C_B (Lemma 4.4), and on the bound in Proposition 8.8(e);
- the decay of $P_{\mathcal{V}}(1)$ and $dP_{\mathcal{V}}(1)d^*$ (Proposition 8.8), depending on a;
- the Combes–Thomas constants (Lemmas 5.5, 5.6, 5.7), explicit in the coercivity and conjugation constants;
- the interior regularity constant C_{reg} (Theorem 6.3), depending on α, for all $N \geq 1$.

All these are statements on the fine lattice with unit blocks. Distances, decay rates and the norms $|\cdot|_{B(y)}$ are measured in unit-lattice units. The supports of the averaging kernels, $B(y)$ and $S_c = B(c_-) \cup B(c_+)$, consist of a fixed number of unit blocks, whatever L is.

Step 2: where the averaging pair enters. We list every use of the pair in the proofs of the results in (a). In each, the model bound $C\varepsilon$ is replaced by ε_Q, and the conclusion holds once $\varepsilon_Q \le \theta_Q$, with θ_Q a threshold of the same kind as ε_{U}.

(A1) Theorem 18.2(a): in the local gauge, $|(Q'\lambda)(y) - (Q_0\tilde{\lambda})(y)| \le \varepsilon_Q \|\lambda\|_{B(y)}$, by Definition 31.1(a) and the locality (iv). The error term becomes $C\varepsilon_Q^2\|\lambda\|_B^2$ and is absorbed.

(A2) Theorem 18.2(b): hypothesis (ii) of Lemma 5.7 uses that $Q'^T Q'$ is block-diagonal, by (iv), and bounded by $(1+\varepsilon_Q)^2$.

(A3) Theorem 18.2(c): the trial field φ built with the model transports in the local gauge satisfies $\|D\varphi\| \le C\|g\|$, as before. For the general pair, $Q'\varphi = g + r$ with $\|r\| \le C(\varepsilon+\varepsilon_Q)\|g\|$, so $\langle Q'^T g, \varphi\rangle \ge (1 - C(\varepsilon+\varepsilon_Q))\|g\|^2$. The bound $M'_X \ge c^2$ uses $\|Q'^T g\| = \|g\|$, which is (i).

(A4) Theorem 18.2(d),(e): algebraic. They use (i) for the surjectivity of Q', and (iv) for the facts that Q' maps $\ell^2_0(X)$ to $\ell^2(X_1)$ and that $Q'^T Q'$ does not couple blocks.

(A5) Theorem 18.2(g) and Theorem 14.1(b),(d): $a(Q'^T Q' - Q_0^* Q_0)$ is pointwise admissible with constants $(C\varepsilon_Q, 0, 1)$ (Lemma 13.6(c) with $C\varepsilon$ replaced by ε_Q). The pointwise bound on H' uses $|Q'^T g|_{B(y)} \le (1+\varepsilon_Q)|g(y)|$, from the transposed bound in Definition 31.1(a).

(A6) Lemmas 15.1, 15.3 and 19.2: (iii), with the bounds on E from Definition 31.1(b). The real configuration V enters only through D_V, whose transport factors are unitary.

(A7) Lemma 19.5, Step 2: $a(\mathbf{Q}^T\mathbf{Q} - \mathbf{Q}^*\mathbf{Q})$ and the terms of $\mathcal{K}_2$ are admissible perturbations of size $C(\varepsilon_Q + \varepsilon_Q^2)$. The comparison of the scalar ingredients uses (A1)–(A5).

(A8) Theorem 19.6 and Theorem 19.7(a): $\mathbf{Q}^T\mathbf{Q}$ is local and bounded by $(1+\varepsilon_Q)^2$, by (ii) and (iv). This is all that the double commutators (Lemma 17.5) and hypothesis (ii) of Lemma 5.7 use.

(A9) Theorem 19.7(b): in the local gauge, $a(\mathbf{Q}^T\mathbf{Q} - \mathbf{Q}^*\mathbf{Q})$ is pointwise admissible of type 1 with constants $(C\varepsilon_Q, 0, 1)$, and $a\mathbf{Q}^*\mathbf{Q}$ with $(0, C, 1)$. The terms $\mathbf{Q}^T ESu$, $DRG'(E^T\mathbf{Q}u)$ and $DRG'(E^T ESu)$ of $\mathcal{K}_2$ are of the form $\mathbf{Q}^T$(bounded) or DRG'(bounded) with a factor ε_Q; this uses the transposed bound on E of Definition 31.1(b). Only sup bounds of the averages enter. No Hölder bound on their kernels is needed, since they contribute to F_u only.

(A10) Lemma 19.8: the model trial fields in the local gauge, built from products of $\cos^2$, with the Neumann series around the diagonally dominant matrix T_U. The margin is $m_\perp(\sigma_0 - \sigma_+ - \sigma_-) > \frac{3}{10} m_\perp$, with $m_\perp = 2^{1-d}$, for every odd $N \ge 3$. The general pair adds a local error of size $C(\varepsilon + \varepsilon_Q)$, which is absorbed for $\varepsilon \le \varepsilon_{\mathrm{U}}$ and $\varepsilon_Q \le \theta_Q$. The slice correction uses $\|ESL\| \le C\varepsilon_Q$.

(A11) Theorem 19.9: $\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T$ has decay, with constants controlled by (ii) and (iv). The pointwise bound on $\mathbf{H}B = \mathbf{G}J$, $J = \mathbf{Q}_\pi^T \mathbf{\Gamma}_\pi B$, uses three facts:

- $|\mathbf{Q}^T\omega|_{B(y)} \le (1 + C\varepsilon_Q) \max|\omega|$, the maximum over the bonds c with S_c meeting $B(y)$ (Lemma 2.6(c) for $\mathbf{Q}^*$, and the transposed bound of Definition 31.1(a));
- the identity $D^T\mathbf{Q}^T = Q'^T D_V^T + E^T$, from (iii), with the transposed bound on E of Definition 31.1(b);

- $|Q'^T g| \le (1+\varepsilon_Q)|g|$.

The transposed sup bounds of the pair and of E enter here, and also in (A5) and (A9). The pointwise bound on $\mathbf{C}$ uses in addition the forward bounds from sup norms to values of $\mathbf{Q}$ and of E (Definition 31.1(a),(b)), through $\mathbf{Q}_\pi \mathbf{G}J = \mathbf{Q}\mathbf{G}J - ES\mathbf{G}J$. In each case it enters only as a factor $1 + C\varepsilon_Q$, so it affects the smallness condition, not the constants.

The explicit contour transports $\Gamma_{y,x}$ of the model pair enter only in (A3) and (A10), as stated in Remark 19.12. The reality in (i) is used in the Gårding inequality (Lemma 19.4) and in the variational characterizations of H' and $\mathbf{H}$. No other property of the pair is used.

Step 3: the constants. Choose θ_Q, and ε_{U} not larger than the threshold ε_0 of Parts II–III, below the thresholds of (A1)–(A11), of Lemma 13.2, and of Lemma 13.5 (that is, ε_*, depending on d, α, γ). Then:

- Theorem 18.2 gives c, C, γ depending on $d, a, \alpha, \dim \mathfrak{g}$, under $(1,\varepsilon)$-regularity.
- Lemma 19.5 and Theorem 19.6 give $\frac{c_g}{2}$ and c_* for $M \ge M_0$. In their proofs M_0 is fixed by $Ce^{-\gamma M_0} \le c_g/8$ and by the absorption of the terms CM_0^{-2}, with C, γ from the previous item; so M_0 is the constant of Part III and depends only on $d, a, \alpha, \dim \mathfrak{g}$. In Step 2 of the proof of Lemma 19.5 the pair at U^u is compared directly with the model pair $(Q_0, \mathbf{Q})$ at $U = 1$, by Definition 31.1(a) and (A7); so Step 3 concerns the model pair only. For Bałaban's pair the pair at $U = 1$ is the model pair anyway (Proposition 22.10(c)).
- Theorem 19.7(a) with $X = \mathbb{T}_\eta$ follows from Lemma 5.7, with c_* and a conjugation constant C_1 built from the scalar decay constants. The rate is $\gamma_0 = \min(\alpha_1, \frac{1}{3}, c_*/(2C_1))$. It does not involve M.
- Theorem 19.7(b) follows from Lemma 18.3, with $\gamma' = \frac{1}{2}\min(\gamma, \gamma_0)$ and a constant depending on $d, \alpha, \gamma, \gamma_0, C_\delta$. Here C_δ comes from the pointwise bounds of Theorem 18.2(g). The letter M in Lemma 18.3 is the L^2 amplitude of Lemma 13.5, not the regularity scale. That lemma is used with its invariant inputs: the small inputs are $|u| + |\nabla u|$ on the cubes $\Lambda_1(c_z)$, with weights $e^{-\gamma|y-z|}$ and the full covariant gradient ($|u| + |Du|$ for scalar fields). Only the small differential terms multiply them; the large smoothing terms use the local L^2 bounds from Combes–Thomas. The terms containing the pair or E in (A5), (A9) and (A11) use their forward and transposed value bounds, not lattice derivatives or Hölder quotients of the averaging kernels. Every composition uses a positive rate no larger than the inverse and perturbation rates (Lemma 13.3).
- Theorem 19.9: $\mathbf{\Gamma}_\pi$ has decay by Lemma 5.5, with the lower bound from (A10). The sum form of the pointwise bound of $\mathbf{G}$ (Theorem 15.4(c), at regular backgrounds) and (A11) give (b). Here γ_H is half the smallest of the rates above, and C_H is a product of the constants above and S_{γ_H} (Lemma 5.2).

Every constant in this chain depends only on $a, \alpha, \dim \mathfrak{g}$. This proves (a) and (b).

Step 4: (c). By Proposition 22.10(a), (b), (d), (f), Bałaban's pair has properties (i), (iii) (with $E = 0$ and $V = W_k$ real) and (iv). An (M,ε)-regular U_0 has curvature at most $C_{\mathrm{curv}}\varepsilon$ (proof of Corollary 22.11), so $U_0 \in \mathfrak{U}_0(C_{\mathrm{curv}}\varepsilon)$. Choose c_Q so that $\varepsilon \le c_Q$ implies $\max(1, C_{\mathrm{curv}})\varepsilon \le c_3$ and the hypotheses of Lemma 21.13. Then, in each gauge of Definition 31.1, Proposition 22.10(e) applies on $\Lambda_4(c_y)$, with ε replaced by $\max(1, C_{\mathrm{curv}})\varepsilon$. It gives the kernel bounds with a constant that depends on L, by the conventions of Part IV. Call the resulting constant C_Q. The kernel bounds give the block norms by the Schur test, and the sup bounds of the transposes by column sums. Hence $\varepsilon_Q \le C_Q\varepsilon$, and $\varepsilon_Q \le \theta_Q$ for $\varepsilon \le \theta_Q/C_Q$. □

Remark 31.4 (Where L and M do enter). (a) Part IV provides only an L-dependent constant C_Q in Lemma 31.3(c). The corrections r_j of Lemma 22.8 are carried by the

averaged relative transports $\lambda_W(c_-,x)$ and by the transports $R(W(c_-,x))$, for all x in the block. All contours of the families $G(c_-,x)$ start at the block centre c_-, so a bond near c_- lies on contours to a large fraction of the L^d points of the block. So the kernel of $\mathbf{Q}_k(U_0)-\mathbf{Q}$ may concentrate near the centres of the scale-j blocks, and its sup norm, which controls the transposes, is bounded only by $O(C_L\varepsilon)$, by the bookkeeping in the proof of Proposition 22.9(c). Whether this is sharp is not settled here. Lemma 31.3 uses the transposes in (A5), (A9) and (A11). This is why (31.3) is stated only for ε small depending on L. This only affects how small the regularity parameter must be, not C_H or γ_H.

(b) Two results of Part III have constants that depend on M. The random-walk expansion (Theorem 20.2(a)) gives decay at a rate of order $M^{-1}\log M$, and Theorem 19.10(b) has the factor e^{CM}. Neither is used in Lemma 31.3: there, decay comes from Combes–Thomas. Since $M\geq 5L$ in Part VI, no bound that must be independent of L may use them.

(c) The Dirichlet operators $\mathfrak{A}^{X,\kappa}$ for $X\neq\mathbb{T}_\eta$ depend on M through the mass profile m_X. Their constants are still independent of $M\geq M_0$ (Theorem 19.6), but Step 0 does not apply to them, because the operators themselves change with M.

Let C_H, γ_H be the constants of Lemma 31.3(b). By Lemma 31.3(c), (31.3) holds at every real (M,ε)-regular background U_0 with $M\geq M_0$ and $\varepsilon\leq\varepsilon_{\mathrm{B}}$, with $\mathbf{H}(U_0)$ the minimizer map for Bałaban's pair $(Q'_k(U_0),\mathbf{Q}_k(U_0))$. Put

$$C_*=\max\Big(1,\ 2dC_H\sum_{w\in\mathbb{Z}^d}e^{-\gamma_H|w|}(|w|_\infty+2)\Big).$$

Then C_* depends only on a, α and G; in particular it does not depend on L.

Proposition 31.5 (Curvature of background fields)**.** *For every $\delta>0$ there are $a_\delta\in(0,a_1]$ and an integer $n_\delta\geq n_0$, both depending on d, L, G, a, α, M and δ, such that for all $k\geq 0$, $n\geq n_\delta$ and $V\in\mathfrak{U}_k(\varepsilon_1)$ with $\varepsilon_1\leq a_\delta$, every element of the minimal orbit satisfies $U_k(V)\in\mathfrak{U}_0((C_*+\delta)\varepsilon_1)$.*

Proof. For $k=0$, $U_0(V)=V$ and $C_*\geq 1$. Let $k\geq 1$ and $U=U_k(V)$. By Theorem 30.3, U is (M,ε_0)-regular and lies in $\mathfrak{U}(\varepsilon_0)$ with $\varepsilon_0=B_3\varepsilon_1$, and U is critical. Fix $R\geq 64(M+1)$ as below and a point $y\in\mathbb{T}$. For ε_0 small enough, depending on R, Steps 1–3 of the proof of Lemma 29.2 apply with this R. This gives a real gauge transformation w and, on the bonds of $\Lambda_{R/4}(y)$, $U^w=e^{\mathrm{i}\eta\tilde A}$ with $\tilde A=R(u)A^\sharp_*$. Here $A^\sharp_*$ is the Landau representative relative to the spliced reference $U^\sharp$, for which $\varepsilon^\sharp+\beta^\sharp\leq K_R\varepsilon_0$. Details

Let p be a plaquette of $\mathbb{T}_\eta$ with base point in $B(y)$. Its curvature satisfies $|F_U(p)|=|F_{e^{\mathrm{i}\eta\tilde A}}(p)|\leq|(\partial\tilde A)(p)|+C|\tilde A|^2$, where $\partial\tilde A(p)=\partial_\mu\tilde A_\nu-\partial_\nu\tilde A_\mu$; this follows from Lemma 21.2(c). By Theorem 25.1(c), $A^\sharp_*=\mathbf{H}^\sharp B^\sharp-\mathbf{C}^\sharp J_{U^\sharp}+\mathcal{R}$ with $|\mathcal{R}|_{C^1}\leq C(K_R\varepsilon_0)^2$, where $\mathbf{H}^\sharp=\mathbf{H}(U^\sharp)$ and $\mathbf{C}^\sharp=\mathbf{C}(U^\sharp)$.

- In the gauge u, $\nabla_{U^\sharp}$ is the ordinary difference near y. The background $U^\sharp$ is real and $(M,\varepsilon^\sharp)$-regular with $\varepsilon^\sharp\leq K_R\varepsilon_0$ (Lemma 28.6). We assume $K_RB_3a_\delta\leq\varepsilon_{\mathrm{B}}$ of Lemma 31.3(c); this is one more condition on a_δ, which already depends on L and M. By (31.3) applied at $U^\sharp$ (Lemma 31.3(c)),

$$|\partial(R(u)\mathbf{H}^\sharp B^\sharp)|_{B(y)}\leq C_H\sum_{y'}e^{-\gamma_H|y-y'|}|B^\sharp(y')|.$$

- For $|y'-y|_\infty\leq R/4-3$, Step 4 of that proof gives $|B^\sharp(y')|\leq d(|y'-y|_\infty+2)\varepsilon_1$. These terms contribute at most $\frac12C_*\varepsilon_1$ to $|\partial\tilde A_\mu|$.

- The other terms, the term $\mathbf{C}^\sharp J_{U^\sharp}$ and the remainder contribute at most $CK_R\,T(R/8)\varepsilon_0 + C(K_R\varepsilon_0)^2$, as in Step 4. Here T is defined as in that proof, with γ_7 replaced by the smaller of γ_H and the decay rate of $\mathbf{C}^\sharp$.

Altogether

$$|F_U(p)| \le C_*\varepsilon_1 + C(K_R T(R/8) + K_R^2\varepsilon_0)B_3\varepsilon_1 + C(K_R\varepsilon_0)^2.$$

Choose R with $CK_R T(R/8)B_3 \le \delta/2$, then $a_\delta \le a_1$ so that the remaining terms are at most $\frac{\delta}{2}\varepsilon_1$ for $\varepsilon_0 = B_3\varepsilon_1 \le B_3 a_\delta$; the bound $a_\delta \le a_1$ is needed for Theorem 30.3. Every plaquette has its base point in some block. The torus must contain the cubes $\Lambda_{R+3}(y)$ without wrapping, and $n \ge n_0$ is needed for Theorem 30.3; this fixes $n_\delta = \max(n_0, 4R+16)$. □

The proposition makes the small-field sets of consecutive steps compatible. Let h and h' be the curvature thresholds of the densities at steps k and $k+1$, each in the units of its own step. If V' has curvature at most h' in the units of step $k+1$, then the curvature of $U_{k+1}(V')^{(k)}$, in the units of step k, is at most $L^{-2}(1+Ch')(C_*+\delta)h'$, by Proposition 21.12. In four dimensions the change of units does not change the coupling, so the thresholds of consecutive steps are comparable. We use only $h' \le 2h$. For thresholds $h = p(g_k)g_k$ with p decreasing and $g \mapsto p(g)g$ increasing, as in Section 49, this holds whenever $g_{k+1} \le 2g_k$. With $h' \le 2h$, $\delta \le 1$ and $Ch' \le 1$, the bound is at most $4(C_*+1)L^{-2}h$, and this is at most $\frac{1}{2}h$ as soon as

$$L^2 \ge 8(C_*+1).$$

Since C_* does not depend on L, this is a condition on L alone. In the flow the thresholds are $h_k = p(g_k)g_k$ (Section 49.1), and $g_{k+1} \le 2g_k$, hence $h_{k+1} \le 2h_k$, holds at every step (Theorem 53.7(c), Theorem 53.12(e)).

31.3 The background of a step

Lemma 31.6 (Composition of background fields)**.** *Let V' be a real configuration on $\mathbb{T}'$ whose curvature, in the units of step $k+1$, is at most $\varepsilon_1' \le \min(a_{\delta=1},\, a_1/(2C_*+2))$, with a_δ from Proposition 31.5, and $(C_*+1)\varepsilon_1' \le c_1$. Let $U' = U_{k+1}(V')$ be an element of its minimal orbit, and $V^0 = U'^{(k)}$. Then:*

(a) $V^0 \in \mathfrak{U}_k(\varepsilon_1)$, in the units of step k, with $\varepsilon_1 = 2(C_*+1)L^{-2}\varepsilon_1' \le a_1$, and $\mathfrak{a}'(V^0) = V'$;

(b) U' belongs to the minimal orbit of the k-step problem for V^0: $U_k(V^0) = U'$;

(c) for every real V with $\mathfrak{a}'(V) = V'$ and $V \in \mathfrak{U}_k(a_1')$, $a_1' = \min(a_1, \theta_0 L^{-3}B_3^{-1})$, we have $S(U_k(V)) \ge S(U')$. Equality holds iff $V = (V^0)^u$ for a real u with $u|_{\mathbb{T}'} = 1$.

Proof. (a) By Proposition 31.5 (with $\delta = 1$), U' has curvature at most $(C_*+1)\varepsilon_1'$ in the units of step $k+1$. By Proposition 21.12 its k-fold average has curvature at most twice this, in the same units, which is the claim in the units of step k. Moreover $\mathfrak{a}'(V^0) = U'^{(k+1)} = V'$. Details

(b) Near U', the constraint manifold $\{U^{(k)} = V^0\}$ is a submanifold of $\{U^{(k+1)} = V'\}$, since $U^{(k)} = V^0$ implies $U^{(k+1)} = \mathfrak{a}'(V^0) = V'$. As sets, $\mathfrak{B}_k(V^0) \subset \mathfrak{B}_{k+1}(V')$ is false: the cap $U \in \mathfrak{U}_0(c_1)$ in the definition of $\mathfrak{B}$ is weaker in the units of step k. Only the local inclusion is used. The configuration U' is a critical point of S on the manifold $\{U^{(k+1)} = V'\}$ near U' and lies on the submanifold $\{U^{(k)} = V^0\}$. So it is critical on $\mathfrak{B}_k(V^0)$. In the units of step k its curvature and current are smaller than in the units of step $k+1$, so $U' \in \mathfrak{U}(\theta_0)$ in the units of step k. By Theorem 30.3(a) at step k, U' lies in the unique critical orbit, which is the minimal orbit.

(c) By Theorem 30.3(b), $U_k(V) \in \mathfrak{U}(B_3 a_1')$ in the units of step k. In the units of step $k+1$ this gives curvature and current at most $L^3 B_3 a_1' \le \theta_0$. Hence $U_k(V) \in \mathfrak{B}_{k+1}(V') \cap \mathfrak{U}(\theta_0)$, and $S(U_k(V)) \ge S(U')$. If equality holds, $U_k(V)$ is a minimizer of the $(k+1)$-step problem, so

$U_k(V) = U'^u$ with $u|_{\mathbb{T}'} = 1$. Then $V = U_k(V)^{(k)} = (V^0)^{u|_{\mathbb{T}}}$. The converse follows from gauge invariance. □

So V^0 is the minimizer of the coarse action $V \mapsto S(U_k(V))$ on the one-step constraint set, and the background of the $(k+1)$-step problem is the background of V^0. We call V^0 the *base configuration* of the step. It depends on the choice of U' in its orbit only through a gauge transformation on $\mathbb{T}$ that is trivial on $\mathbb{T}'$.

In the units of step k, U' is $(M, B_3\varepsilon_1')$-regular and lies in $\mathfrak{U}(B_3\varepsilon_1')$. Indeed, passing from the units of step $k+1$ to those of step k multiplies Lie fields by L^{-1}, curvatures by L^{-2} and currents by L^{-3}, and the cubes of Definition 18.1 become larger, which only helps (Lemma 31.2). Throughout this part we write

$$\varepsilon := B_3\varepsilon_1'$$

for the regularity parameter of U' as a reference configuration for the k-step problem.

31.4 Lattice symmetry

Let $\mathfrak{S}' = \mathfrak{S}_{k+1}$ be the group of the maps $x \mapsto Rx+v$ of $\mathbb{T}_\eta$ with $R \in B_4$ and $v \in \mathbb{T}'$ (Section 21), in the units of step k. It is generated by the translations of $\mathbb{T}'$ and by B_4 about the points of $\mathbb{T}'$, and $\mathfrak{S}' \subset \mathfrak{S}_k$. Its elements map $\mathbb{T}'$, $\mathbb{T}$ and $\mathbb{T}_\eta$ onto themselves, and $B'(y)$ onto $B'(ry)$. They act on configurations by $(rV)(b) = V(r^{-1}b)$, on scalar fields by $(r\lambda)(x) = \lambda(r^{-1}x)$, on functions of configurations by $(r\rho)(V) = \rho(r^{-1}V)$, and on Lie fields relative to a configuration U by r_U (Proposition 22.10(g)). Let $\mathcal{G}'$ be the group of real gauge transformations on $\mathbb{T}$ that are 1 on $\mathbb{T}'$. The following proposition collects the lattice symmetries of the objects constructed so far.

Proposition 31.7 (Lattice symmetry)**.** *(a)* (The map T.) *For $r \in \mathfrak{S}'$ and every density ρ on configurations on $\mathbb{T}$ that vanishes outside* $\operatorname{dom}\mathfrak{a}'$, *$T(r\rho) = r\,T(\rho)$. If ρ is gauge invariant, so is $r\rho$.*

(b) (Background fields.) *Let $r \in \mathfrak{S}_k$ and $V \in \mathfrak{U}_k(\varepsilon_1)$ with $\varepsilon_1 \le a_1$. Then $U_k(rV)$ and $rU_k(V)$ lie in the same $\mathfrak{G}_0$-orbit, and $S(U_k(rV)) = S(U_k(V))$. Let $r \in \mathfrak{S}'$ and let V' be as in Lemma 31.6. Then rV' satisfies the same hypotheses, and the base configurations satisfy $V^0(rV') = (rV^0(V'))^u$ for some $u \in \mathcal{G}'$.*

(c) (Averages and operators.) *Let U_0 be real and (M,ε)-regular with $\varepsilon \le \varepsilon_{\mathrm{B}}$ (Lemma 31.3(c)), and let $r \in \mathfrak{S}_k$. Then rU_0 is (M,ε)-regular, $J_{rU_0} = r_{U_0} J_{U_0}$ and $W_k(rU_0) = rW$, where $W = W_k(U_0)$. Moreover:*

(i) $Q_k'(rU_0)\, r = r\, Q_k'(U_0)$, $\mathbf{Q}_k(rU_0)\, r_{U_0} = r_W\, \mathbf{Q}_k(U_0)$, and $D_{rU_0} r = r_{U_0} D_{U_0}$ on scalar fields;

(ii) $K(rU_0)\, r = r\, K(U_0)$ for the scalar operators $K = \mathfrak{A}', G', \Gamma', M', H', C', P_{\mathcal{V}}, R$ of Parts II and III on the torus; and, for the operators $S(U_0) = G'RD^T$ and $\Pi(U_0) = I - DS(U_0)$ of Section 15, $S(rU_0)\, r_{U_0} = r\, S(U_0)$ and $\Pi(rU_0)\, r_{U_0} = r_{U_0}\, \Pi(U_0)$;

(iii) $\Phi_{rU_0}(r_{U_0}A) = \Phi_{U_0}(A)$ and $\ell\, \mathcal{Q}(e^{i\eta\, r_{U_0}A}; rU_0) = r_W\, \ell\, \mathcal{Q}(e^{i\eta A}; U_0)$, wherever defined; so the derivatives of all orders of these maps at $r_{U_0}A$ and at A are conjugate by r_{U_0} and r_W;

(iv) let $\mathfrak{B} = D^T D + W'$ be the operator of the form q_U in Section 15. If $\mathfrak{B}(rU_0)\, r_{U_0} = r_{U_0}\, \mathfrak{B}(U_0)$, then the operators $\mathfrak{A}_\pi$, $\mathbf{G}$, $\mathbf{\Gamma}_\pi$, $\mathbf{H}$ and $\mathbf{C}$ built with $\mathfrak{B}$ in Sections 15 and 16 are covariant in the same way; for instance $\mathbf{H}(rU_0)\, r_W = r_{U_0}\mathbf{H}(U_0)$ and $\mathbf{C}(rU_0)\, r_{U_0} = r_{U_0}\mathbf{C}(U_0)$. For $W' = 0$ this holds if $r(x) = Rx + v$ with a permutation matrix R, and for all $r \in \mathfrak{S}_k$ if $U_0 = 1$.

Finally, for $r \in \mathfrak{S}'$ and a configuration Y on $\mathbb{T}$ as in Definition 22.6, for the lattices $\mathbb{T}$ and $\mathbb{T}'$: $(\mathfrak{G}_{rY} r_Y B)(y,x) = (\mathfrak{G}_Y B)(r^{-1}y, r^{-1}x)$.

(d) (Landau representatives.) *Let V be as in Theorem 30.3, U_0 an element of its minimal orbit, and $r \in \mathfrak{S}_k$. Then rU_0 is an element of the minimal orbit of rV. In the notation of Corollary 30.5, $A_*(r_V B; rU_0) = r_{U_0} A_*(B; U_0)$, and, for real B, $\omega(r_V B; rU_0) = r_V\, \omega(B; U_0)$.*

Proof. (a) Substitute $V = rY$ in (31.1). The map $Y \mapsto rY$ permutes the bond variables and inverts those on the bonds b for which $r^{-1}b$ is negatively oriented. The Haar measure is invariant under inversion, so $\prod_b dV(b) = \prod_b dY(b)$, and the density becomes $(r\rho)(rY) = \rho(Y)$. By Lemma 21.10(d), $\mathfrak{a}'(rY)$ is defined if and only if $\mathfrak{a}'(Y)$ is, so $r\rho$ vanishes outside $\operatorname{dom}\mathfrak{a}'$, and $\mathfrak{a}'(rY) = r\,\mathfrak{a}'(Y)$. Put $X = \mathfrak{a}'(Y)$ and $\tilde{V}' = r^{-1}V'$. Then $\mathfrak{a}'(rY)(c)\,V'(c)^{-1} = X(r^{-1}c)\,\tilde{V}'(r^{-1}c)^{-1}$. As c runs over the positively oriented bonds of $\mathbb{T}'$, $r^{-1}c$ runs over all bonds of $\mathbb{T}'$ once, up to orientation. For a reversed bond, $X(-c)\tilde{V}'(-c)^{-1} = X(c)^{-1}\tilde{V}'(c)$ is conjugate to $(X(c)\tilde{V}'(c)^{-1})^{-1}$, and δ is invariant under conjugation and inversion. Hence the product of δ-functions is $\prod_c \delta(X(c)\tilde{V}'(c)^{-1})$, and $(T(r\rho))(V') = (T\rho)(r^{-1}V')$. Finally $r(V^u) = (rV)^{ru}$, so $r\rho$ is gauge invariant if ρ is. Details

(b) The first two claims are Theorem 30.3(c). The curvature of rV' equals that of V', since $\mathfrak{U}_{k+1}(\varepsilon_1')$ is invariant under $\mathfrak{S}'$. In the units of step $k+1$, $\mathfrak{S}'$ is the group of the lattice symmetries of the $(k+1)$-step structure. By Theorem 30.3(c) for the $(k+1)$-step problem, $U_{k+1}(rV') = (rU')^u$ with $U' = U_{k+1}(V')$ and a real u on $\mathbb{T}_\eta$ with $u|_{\mathbb{T}'} = 1$. By the covariance of $\mathfrak{a}^{(k)}$ under real gauge transformations and under $\mathfrak{S}_k \supset \mathfrak{S}'$ (Lemma 21.10),

$$V^0(rV') = \big((rU')^u\big)^{(k)} = \big(r\,U'^{(k)}\big)^{u|_{\mathbb{T}}} = \big(rV^0(V')\big)^{u|_{\mathbb{T}}},$$

and $u|_{\mathbb{T}} \in \mathcal{G}'$.

(c) The regularity of rU_0 and the identity for W_k are proved in the proof of Theorem 25.1(h) (Section 27.7), and the identity for J is Lemma 23.2(b). An (M,ε)-regular U_0 with $\varepsilon \le \varepsilon_{\mathrm{B}}$ lies in $\mathfrak{U}_0(c_3)$ (Step 4 of the proof of Lemma 31.3), so Proposition 22.10 applies. (i) The first two identities are Proposition 22.10(g), and the third is shown in Section 27.7. (ii) The maps r and r_{U_0} preserve the bilinear pairings, so transposes are covariant as well. The operators in (ii) are built from D on scalar fields, D^T, Q_k' and $Q_k'^T$ by sums, products, transposes and inverses (Sections 14 and 15), so (i) gives (ii). (iii) The first identity follows from $e^{\mathrm{i}\eta\, r_{U_0} A}\, rU_0 = r(e^{\mathrm{i}\eta A} U_0)$ and $S \circ r = S$; the second is Proposition 22.10(g), together with $\ell((rX)(c)) = (r_W \ell X)(c)$ for relative configurations at scale k. Differentiating, and using that r_{U_0} and r_W preserve the pairings, gives the claim about derivatives. (iv) By definition, $\mathfrak{A}_\pi = \Pi^T \mathfrak{B} \Pi + DRD^T + a\Pi^T \mathbf{Q}^T \mathbf{Q} \Pi$, $\mathbf{G} = \mathfrak{A}_\pi^{-1}$, $\mathbf{Q}_\pi = \mathbf{Q}\Pi$, $\mathbf{\Gamma}_\pi = (\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T)^{-1}$, $\mathbf{H} = \mathbf{G}\mathbf{Q}_\pi^T \mathbf{\Gamma}_\pi$ and $\mathbf{C} = \mathbf{G} - DG'RG'D^T - \mathbf{H}\mathbf{Q}_\pi\mathbf{G}$. With (i) and (ii), each of them is covariant if $\mathfrak{B}$ is. Let $W' = 0$. If R is a permutation matrix, $e_\mu \mapsto e_{\sigma(\mu)}$, then r maps positively oriented bonds onto positively oriented bonds, r_{U_0} only relabels the bonds, and directly from the definition of D on gauge fields,

$$\big(D_{rU_0}\, r_{U_0} A\big)_{\sigma(\mu)\sigma(\nu)}(rx) = \big(D_{U_0} A\big)_{\mu\nu}(x).$$

Hence $\|D_{rU_0} r_{U_0} A\| = \|D_{U_0} A\|$, and polarization gives $\mathfrak{B}(rU_0)\, r_{U_0} = r_{U_0}\, \mathfrak{B}(U_0)$. At $U_0 = 1$, $D = d$ and $(r_1 A)(b) = A(r^{-1}b)$ with $A(-b) = -A(b)$. Then $(dA)(p)$ is the sum of A around ∂p, which does not depend on the corner at which p is based, and $\|d\, r_1 A\| = \|dA\|$ for all $r \in \mathfrak{S}_k$. The last claim is Lemma 22.7(d), applied to the lattices $\mathbb{T}$ and $\mathbb{T}'$.

(d) By Theorem 30.3(c), rU_0 is an element of the minimal orbit of rV. Since $W_k(U_0) = V$, Theorem 25.1(h) gives the rest; see also Corollary 30.5. □

Remark 31.8 (Reflections, and what the step uses). At curved backgrounds and for $W' = 0$, the hypothesis of part (c)(iv) is verified for the translations and the permutations of the axes, but not for the reflections. The covariant derivative of a Lie field on a plaquette is linearized

at its base corner: $(D_U A)_{\mu\nu}(x) = D_\mu A_\nu(x) - D_\nu A_\mu(x)$, with transports from x. A reflection moves the base corner of a plaquette to another corner of its image. Let r be the reflection in the direction μ about a point of $\mathbb{T}$, let $\nu \neq \mu$, $z = r^{-1}x$, $z' = z - \eta e_\mu$, and let $p = p_{\mu\nu}(z')$. A direct computation from the definitions gives

$$\big(D_{rU}\, r_U A\big)_{\mu\nu}(x) = -R(U_\mu(z'))^{-1}\Big[(D_U A)_{\mu\nu}(z') - \eta^{-1}\big(R(U(\partial p)) - 1\big)R(U_\nu(z'))A_\mu(z' + \eta e_\nu)\Big].$$

So $D_{rU} r_U$ and the transported D_U differ by a local operator of norm at most $C\eta \max_p |F_U(p)|$, which vanishes for flat U. Since $\|D\| \le C\eta^{-1}$, the operator $D^T D$ at a background of curvature ε is covariant under reflections up to a perturbation of norm $C\varepsilon$. For $\mathbf{H}$, $\mathbf{C}$ and the other operators of (c)(iv) at curved backgrounds we therefore claim covariance only under the translations and the permutations of the axes. They are tools in the analysis. The quantities that are defined without a choice of gauge are covariant under all of $\mathfrak{S}'$: the map T, by (a); the background fields and the coarse action $V \mapsto S(U_k(V))$, by (b); and the derivatives of the action and of $\mathcal{Q}$, the averaging pair, the scalar operators and the averaged relative transports, by (c). The covariance of the small-field step, $T_{\mathrm{sf}}(r\rho) = r\, T_{\mathrm{sf}}(\rho)$ for $r \in \mathfrak{S}'$, which also involves the gauge-fixing function and the cutoffs, is part (iii) of Lemma 32.6. The proof devices of Parts II–VI, namely the model pair of Section 12, the contours Γ^{co} of Section 28, the spliced configurations and the trivial extension, enter only proofs. They are not covariant, and no statement above depends on them.

32 Gauge fixing and fluctuation coordinates

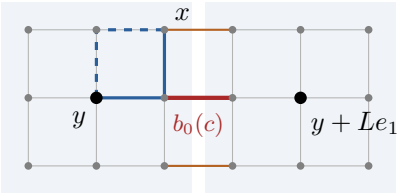
Throughout this section V' is as in Lemma 31.6, $W := V^0$ is the base configuration and $W' = \mathfrak{a}'(W) = V'$. We use the notation of Section 22, with the reference U', so that $W_k = W$ and $W_{k+1} = W'$. Configurations on $\mathbb{T}$ are written $V = XW$, with the relative configuration X, and $X = e^{\mathrm{i}Z}$ with the *fluctuation field* Z; in the units of step k we have $\delta_k = 1$. The integrand of (31.1) is a function of X through $V = XW$. By invariance of the Haar measure, $dV(b) = dX(b)$. We write $g = g_k$. The averaged contour variables of a configuration V on $\mathbb{T}$ are those of the step, $V(y, x)$ for $y \in \mathbb{T}'$ and $x \in B'(y)$ (Definition 21.6). Let $\mathfrak{S}_{k+1}$ be the group of the maps $x \mapsto Rx + v$ with $R \in B_4$ and $v \in \mathbb{T}'$, generated by the translations of $\mathbb{T}'$ and the hyperoctahedral group about the points of $\mathbb{T}'$; it is the group $\mathfrak{S}'$ of Section 31, and $\mathfrak{S}_{k+1} \subset \mathfrak{S}_k$.

32.1 Bonds and blocks

Definition 32.1 (Crossing bonds and the variables $\mathfrak{F}$)**.** We use the following sets of bonds of $\mathbb{T}$.

- The blocks $B'(y)$, $y \in \mathbb{T}'$, are the cubes of L^d points of $\mathbb{T}$ centred at y; they partition $\mathbb{T}$. A bond of $\mathbb{T}$ is *inner* if its endpoints lie in one block, and *crossing* otherwise. A crossing bond $(x, x + e_\mu)$ joins $B'(y)$ to $B'(y + Le_\mu)$.
- For a bond $c = (y, y + Le_\mu)$ of $\mathbb{T}'$, let $b_0(c)$ be the middle bond of the straight contour $[c]$, from $y + \frac{L-1}{2}e_\mu$ to $y + \frac{L+1}{2}e_\mu$. It is the only crossing bond of $[c]$.
- $\mathfrak{F}$ is the set of positively oriented bonds of $\mathbb{T}$ other than the bonds $b_0(c)$.

Every $r \in \mathfrak{S}_{k+1}$ maps blocks to blocks, maps $[c]$ onto $\pm[rc]$ and the middle bond onto the middle bond. So the set of crossing bonds, the set of the $b_0(c)$ and $\mathfrak{F}$ are invariant under $\mathfrak{S}_{k+1}$, as sets of unoriented bonds. The contours of a family $G(y, x)$, $x \in B'(y)$, lie in the box spanned by y and x, hence in $B'(y)$ (Section 21). So they consist of inner bonds and contain no bond $b_0(c)$. See Figure 4.



blue: the two contours of $G(y, x)$

orange: crossing bonds in $\mathfrak{F}$

red: $b_0(c)$, eliminated by the constraint

thin: inner bonds, in $\mathfrak{F}$

Figure 4: Two centred blocks $B'(y)$, $B'(y + Le_1)$ of $\mathbb{T}$ for $d = 2$, $L = 3$. The bond $b_0(c)$ is the middle bond of the straight contour $[c]$ from y to $y + Le_1$. The blocks, the crossing bonds and $b_0(c)$ are invariant under the reflections that map the pair of blocks onto itself; the family $G(y, x)$ is mapped onto the family of the image of x.

32.2 Bałaban's exponential gauge fixing

The integrand of (31.1) is invariant under the group

$$\mathcal{G}' = \{u : \mathbb{T} \to G : \ u|_{\mathbb{T}'} = 1\}$$

of Section 31, by Lemma 21.10(b). Following Bałaban [18, (0.14)–(0.16)], we fix this invariance by exponential weights in the averaged contour variables, not by δ-functions; the latter would define a nonlinear slice. Let

$$\mathfrak{M} = \{V \text{ real configuration on } \mathbb{T} : \ |\ell(V(\partial p))| \le \varepsilon_M \text{ for all plaquettes } p \text{ of } \mathbb{T}\}, \qquad \varepsilon_M = \frac{r_M}{8L^2},$$

be a quantitative form of Bałaban's set of regular configurations [18, p. 254]; in the units of step k, $\mathfrak{M} = \mathfrak{U}_k(\varepsilon_M)$. By Lemma 21.7 and Lemma 21.10(a), for $V \in \mathfrak{M}$ all averaged contour variables $V(y, x)$ are defined and real, their families lie in $\mathcal{D}_{3L^2\varepsilon_M} \subset \mathcal{D}_{r_M/2}$, and $\mathfrak{a}'(V)$ is defined and real. The set $\mathfrak{M}$ is invariant under real gauge transformations of any size and under $\mathfrak{S}_{k+1}$ (Section 21). The base lies in it with a margin: by Lemma 31.6(a), $W \in \mathfrak{U}_k(2(C_* + 1)L^{-2}\varepsilon_1')$, and $(C_* + 1)\varepsilon_1' \le c_1 \le r_M/128$ (Lemma 21.11), so

$$|\ell(W(\partial p))| \le 2(C_* + 1)L^{-2}\varepsilon_1' \le \tfrac{1}{8}\varepsilon_M \qquad \text{for all plaquettes } p. \tag{32.1}$$

Definition 32.2 (Gauge fixing relative to the base)**.** For $u \in G^c$ put

$$\phi(u) = \exp\Big(-\frac{1}{g^2}\big[\mathsf{N} - \tfrac{1}{2}\operatorname{tr}(u + u^{-1})\big]\Big), \qquad z = z(g) = \int_G \phi(u)\, du.$$

For a real configuration $W_\circ \in \mathfrak{M}$ and a configuration V whose averaged contour variables are defined, put

$$\mathrm{gf}_{W_\circ}(V) = \prod_{y \in \mathbb{T}'} \prod_{x \in B'(y)\setminus\{y\}} z^{-1}\, \phi(V(y, x)\, W_\circ(y, x)^{-1}).$$

In this part $W_\circ = W$ is the base, and we write gf_W.

For $u \in G$, $\mathsf{N} - \frac{1}{2}\operatorname{tr}(u + u^{-1}) = \mathsf{N} - \operatorname{Re}\operatorname{tr} u = \frac{1}{2}|u - 1|^2$, and at $u = e^{iH}$ it equals $f(H) := \mathsf{N} - \operatorname{tr}\cos H = \frac{1}{2}\operatorname{tr} H^2 + O(|H|^4)$. This is the normalization of the plaquette terms of the Wilson action in Section 23, and $\operatorname{tr} H^2 = |H|^2$ for $H \in \mathfrak{g}$. The function ϕ is a class function, $0 < \phi \le 1$ on G, and $\phi(u) = 1$ only for $u = 1$. By Laplace's method, $z(g) = \sigma(0)(2\pi g^2)^{\dim \mathfrak{g}/2}(1 + O(g^2))$ as $g \to 0$, with σ the density of the Haar measure in exponential coordinates (Section 32.5). The strength of the gauge fixing is that of the action, as in [18, (0.19)]. For any fixed $a_{\mathrm{gf}} > 0$ one may replace g^{-2} in the weight by $a_{\mathrm{gf}} g^{-2}$ and compute z at that same strength. The normalized Haar-orbit identity is unchanged; the full function Φ_W, its remainder G_3, and the quadratic gauge-fixing term are multiplied by a_{gf}. The

ensuing coercivity and upper-bound constants may depend on this fixed positive strength, and the normalization changes by a background-independent factor. Bałaban's weights contain in addition the restriction $\chi(|U(y,x)-1|<\varepsilon_0)$; here the small-field region is defined by the cutoff of Definition 32.4 instead, on whose support that factor would be 1.

For a complex field Z on the bonds of $\mathbb{T}$ and $x\in B'(y)$ put

$$H_{y,x}(Z)=\ell\big((e^{\mathrm{i}Z}W)(y,x)\,W(y,x)^{-1}\big),$$

whenever the averaged contour variable is defined and $|(e^{\mathrm{i}Z}W)(y,x)\,W(y,x)^{-1}-1|<1$, the domain of ℓ (Section 21.2). The proof of Lemma 32.3(c) verifies both conditions on its small complex Z-ball, by Lemma 21.5(a),(e). For complex fields we use the bilinear pairing $\langle A,B\rangle=\sum_{y\in\mathbb{T}'}\sum_{x\in B'(y)}\operatorname{tr}(A(y,x)B(y,x))$ on functions of the pairs (y,x); for real fields $\langle A,A\rangle=\|A\|^2$. Recall $\mathfrak{G}_W Z(y,x)=\lambda_W(y,x)[Z]$ from Definition 22.6, in the units of step k.

Lemma 32.3 (Exponential gauge fixing)**.** *Let $W_\circ\in\mathfrak{M}$ be real.*

(a) (Faddeev–Popov identity.) *Let f be an integrable function of real configurations on $\mathbb{T}$ with $f(V^u)=f(V)$ for all $u\in\mathcal{G}'$, and $f=0$ outside $\mathfrak{M}$. Then*

$$\int\prod_b dV(b)\;f(V)=\int\prod_b dV(b)\;\mathrm{gf}_{W_\circ}(V)\,f(V).$$

(b) (Invariance.) *For $V\in\mathfrak{M}$ and every real gauge transformation u on $\mathbb{T}$, of any size, $\mathrm{gf}_{W_\circ^u}(V^u)=\mathrm{gf}_{W_\circ}(V)$. For $r\in\mathfrak{S}_{k+1}$, $\mathrm{gf}_{rW_\circ}(rV)=\mathrm{gf}_{W_\circ}(V)$.*

(c) (Expansion.) *For $V\in\mathfrak{M}$, $0<\mathrm{gf}_{W_\circ}(V)\le z^{-|\mathbb{T}\setminus\mathbb{T}'|}$, with equality if and only if $V(y,x)=W_\circ(y,x)$ for all $y\in\mathbb{T}'$, $x\in B'(y)$. There is $c_0>0$, depending only on G, such that for complex Z with $|Z|_\infty\le c_0L^{-1}$ the averaged contour variables of $e^{\mathrm{i}Z}W_\circ$ are defined, and*

$$\mathrm{gf}_{W_\circ}(e^{\mathrm{i}Z}W_\circ)=z^{-|\mathbb{T}\setminus\mathbb{T}'|}e^{-g^{-2}\Phi_{W_\circ}(Z)},$$

$$\Phi_{W_\circ}(Z)=\sum_{y\in\mathbb{T}'}\sum_{x\in B'(y)}f(H_{y,x}(Z))=\tfrac12\langle\mathfrak{G}_{W_\circ}Z,\mathfrak{G}_{W_\circ}Z\rangle+G_3(Z),$$

with $H_{y,x}$ computed at $W_\circ$. Here $G_3=\sum_{y\in\mathbb{T}'}G_{3,y}$, where $G_{3,y}$ depends only on Z on the inner bonds of $B'(y)$, is holomorphic, real for real Z, vanishes to third order at $Z=0$, and satisfies $|G_{3,y}(Z)|\le C_L\max_b|Z(b)|^3$ for $|Z|_\infty\le\frac12c_0L^{-1}$, the maximum over the bonds of $B'(y)$. In particular $\Phi_{W_\circ}$ depends only on Z on inner bonds, and $|G_3(Z)|\le C_L|Z|_\infty\|Z\|^2$.

Proof. (a) For real u of any size and $V\in\mathfrak{M}$, (21.2) gives $V^u(y,x)=u(y)V(y,x)u(x)^{-1}$. For $u\in\mathcal{G}'$ this is $V(y,x)u(x)^{-1}$, since $u(y)=1$. Fix $V\in\mathfrak{M}$. Each $x\in\mathbb{T}\setminus\mathbb{T}'$ lies in exactly one block $B'(y)$, and is not its centre. The map $u\mapsto V(y,x)u^{-1}W_\circ(y,x)^{-1}$ preserves the normalized Haar measure. Hence, with $du=\prod_{x\in\mathbb{T}\setminus\mathbb{T}'}du(x)$, Details

$$\int_{\mathcal{G}'}\mathrm{gf}_{W_\circ}(V^u)\,du=\prod_{y\in\mathbb{T}'}\prod_{x\in B'(y)\setminus\{y\}}z^{-1}\int_G\phi\big(V(y,x)u(x)^{-1}W_\circ(y,x)^{-1}\big)\,du(x)=1.$$

Put $\mathrm{gf}_{W_\circ}=0$ outside $\mathfrak{M}$. Since $\mathfrak{M}$ is $\mathcal{G}'$-invariant, $\mathrm{gf}_{W_\circ}(V^u)f(V^u)=0$ for $V\notin\mathfrak{M}$. The map $V\mapsto V^u$ acts on each bond by a left and a right translation, so it preserves $\prod_b dV(b)$. Since $0\le\mathrm{gf}_{W_\circ}\le z^{-|\mathbb{T}\setminus\mathbb{T}'|}$ and f is integrable, Fubini's theorem applies:

$$\int dV\,\mathrm{gf}_{W_\circ}f=\int_{\mathcal{G}'}du\int dV\,\mathrm{gf}_{W_\circ}(V^u)f(V^u)=\int dV\,f(V)\int_{\mathcal{G}'}\mathrm{gf}_{W_\circ}(V^u)\,du=\int dV\,f(V).$$

No Faddeev–Popov determinant appears, since the $\mathcal{G}'$-orbits are integrated with the Haar measure.

(b) By (21.2), $V^u(y,x)W_\circ^u(y,x)^{-1} = u(y)(V(y,x)W_\circ(y,x)^{-1})u(y)^{-1}$, and ϕ is a class function. For $r \in \mathfrak{S}_{k+1}$, $(rV)(y,x) = V(r^{-1}y, r^{-1}x)$ (Section 21), and r maps the set of pairs (y,x) with $y \in \mathbb{T}'$, $x \in B'(y) \setminus \{y\}$ bijectively onto itself, since it maps $\mathbb{T}'$ onto $\mathbb{T}'$ and $B'(y)$ onto $B'(ry)$.

(c) The first statement follows from $\phi = e^{-|u-1|^2/2g^2}$ on G. Let $\mathbf{U} = \{W_\circ(\Gamma)\}_{\Gamma \in G(y,x)}$, which lies in $\mathcal{D}_{r_M/2}$. For $\Gamma \in G(y,x)$ write $(e^{\mathrm{i}Z}W_\circ)(\Gamma) = e^{\mathrm{i}\xi_\Gamma}W_\circ(\Gamma)$, where $e^{\mathrm{i}\xi_\Gamma} = (e^{\mathrm{i}Z})_{W_\circ}(\Gamma)$ is a product of at most $2(L-1)$ factors $e^{\mathrm{i}R(W_\circ(\cdot))Z(b)}$ with unitary $R(W_\circ(\cdot))$. By Lemma 21.2(c), ξ_Γ is holomorphic in Z with $|\xi_\Gamma| \le 4L|Z|_\infty \le r_M/16$, if c_0 is small. By Lemma 21.5(a),(e) (and its proof), the family $\{e^{\mathrm{i}\xi_\Gamma}W_\circ(\Gamma)\}$ lies in $\mathcal{D}_{r_M}$, the averaged contour variable $(e^{\mathrm{i}Z}W_\circ)(y,x) = \mathrm{Av}(\{e^{\mathrm{i}\xi_\Gamma}W_\circ(\Gamma)\})$ is defined and holomorphic in Z, and

$$|H_{y,x}(Z)| = |\ell(\,\mathrm{Av}(\{e^{\mathrm{i}\xi_\Gamma}W_\circ(\Gamma)\})\,\mathrm{Av}(\mathbf{U})^{-1})| \le C_M \max_\Gamma |\xi_\Gamma| \le 4C_M L|Z|_\infty.$$

For c_0 small this is at most 1. Since $e^{\mathrm{i}\ell(h)} = h$, we have $V(y,x)W_\circ(y,x)^{-1} = e^{\mathrm{i}H_{y,x}(Z)}$ for $V = e^{\mathrm{i}Z}W_\circ$, and $\phi(e^{\mathrm{i}H}) = e^{-g^{-2}f(H)}$. This gives the formula with $\Phi_{W_\circ} = \sum f(H_{y,x})$. The function $H_{y,x}$ depends only on Z on the bonds of the box spanned by y and x, which are inner bonds of $B'(y)$. By Lemma 22.7(a), $H_{y,x}(sZ) = s\,\mathfrak{G}_{W_\circ}Z(y,x) + O(s^2)$. Since $f(H) = \frac{1}{2}\operatorname{tr} H^2 + O(|H|^4)$, the second-order part of $\Phi_{W_\circ}$ is $\frac{1}{2}\langle \mathfrak{G}_{W_\circ}Z, \mathfrak{G}_{W_\circ}Z\rangle$, and

$$G_{3,y}(Z) = \sum_{x \in B'(y)} \Big[f(H_{y,x}(Z)) - \tfrac{1}{2}\operatorname{tr}\left((\mathfrak{G}_{W_\circ}Z(y,x))^2\right)\Big]$$

vanishes to third order. Reality follows from that of $H_{y,x}$ for real Z. For the bound let m be the maximum of $|Z(b)|$ over the bonds of $B'(y)$, and $t \mapsto G_{3,y}(tZ)$ for $|t| \le R := c_0/(Lm)$. There $|H_{y,x}| \le 4C_M c_0 \le 1$ and $|\mathfrak{G}_{W_\circ}(tZ)(y,x)| \le C$ (Lemma 22.7(c)), so $|G_{3,y}(tZ)| \le CL^d$. The maximum principle for $G_{3,y}(tZ)/t^3$ gives $|G_{3,y}(Z)| \le CL^d R^{-3} = CL^{d+3}c_0^{-3}m^3$. Finally $m^3 \le |Z|_\infty \sum_b |Z(b)|^2$, the sum over the bonds of $B'(y)$, and every bond lies in at most one block. □

Part (a) is [18, (0.15)–(0.16)]. Bałaban's weights are relative to 1, i.e. $W_\circ(y,x) = 1$. The real transformation $u \in \mathcal{G}'$ with $u(x) = W(y,x)$ for $x \in B'(y)$ satisfies $W^u(y,x) = 1$, so by (b) $\mathrm{gf}_W(V) = \mathrm{gf}_{W^u}(V^u)$ is Bałaban's weight evaluated at V^u. This is the gauge in which Bałaban expands the fluctuation integral, with the expansion (c) of the gauge-fixing term [18, (2.3)–(2.5)]. Relative to the base the gauge fixing needs no gauge condition on W, so the local gauges of Parts VIII–X remain available.

Definition 32.4 (Fluctuation cutoff)**.** For $w \le \frac{1}{2}c_G$ and a configuration V on $\mathbb{T}$ put $\chi^{\mathrm{w}}(V) = 1$ if for every bond b of $\mathbb{T}$ there is $Z(b) \in \mathfrak{g}$ with $|Z(b)| \le w$ and $V(b) = e^{\mathrm{i}Z(b)}W(b)$, and $\chi^{\mathrm{w}}(V) = 0$ otherwise.

For $w \le \frac{1}{2}c_G$ the field $Z(b) = \ell(V(b)W(b)^{-1})$ is unique (Lemma 21.2(a),(d)). The cutoff is a product over all bonds, including the $b_0(c)$; this is Bałaban's characteristic function [18, (2.9)]. It depends on V' through W. It is not $\mathcal{G}'$-invariant, but it is covariant: $|\ell(V(b)W(b)^{-1})|$ does not change if (V,W) is replaced by (V^u, W^u) for a real gauge transformation u, since then $V(b)W(b)^{-1}$ is conjugated by $u(b_-)$, or by (rV, rW) for $r \in \mathfrak{S}_{k+1}$, since on reversed bonds $V(-b)W(-b)^{-1} = W(b)^{-1}(V(b)W(b)^{-1})^{-1}W(b)$. Let $w \le \frac{1}{8}\varepsilon_M$. On the support of χ^{w}, write $V(\partial p)$ as a product of four conjugates of the $e^{\pm\mathrm{i}Z(b)}$ and of $W(\partial p)$, by moving the factors $W(b)$ to the right. By (32.1) and Lemma 21.2(b), every plaquette satisfies $|\ell(V(\partial p))| \le |\ell(W(\partial p))| + 4w$. Hence

$$\operatorname{supp}\chi^{\mathrm{w}} \subset \mathfrak{U}_k(2(C_*+1)L^{-2}\varepsilon_1' + 4w) \subset \mathfrak{M} \qquad (w \le \tfrac{1}{8}\varepsilon_M). \tag{32.2}$$

32.3 The gauge-fixed step

Definition 32.5 (Gauge-fixed step)**.** For a function f of real configurations on $\mathbb{T}$ that vanishes outside $\mathfrak{M}$ put

$$T^{\mathrm{gf}}(f)(V') = \int \prod_b dV(b)\ \prod_c \delta(\mathfrak{a}'(V)(c)\,V'(c)^{-1})\ \mathrm{gf}_W(V)\,f(V),$$

with W the base of V', and the measures and the δ-function of (31.1).

The integrand may depend on V' through the base W and the reference U'; we then write $f = f_{W,U'}$. We call a function $\chi = \chi_{W,U'}$ of configurations V on $\mathbb{T}$, with $0 \le \chi \le 1$, a *covariant cutoff* if

(C1) $\chi_{W^{u|_{\mathbb{T}}},U'^u}(V^{u|_{\mathbb{T}}}) = \chi_{W,U'}(V)$ for every real gauge transformation u on $\mathbb{T}_\eta$;

(C2) $\chi_{rW,rU'}(rV) = \chi_{W,U'}(V)$ for every $r \in \mathfrak{S}_{k+1}$.

By the paragraph after Definition 32.4, χ^{w} is a covariant cutoff. The identities below between integrals containing δ-functions are first proved with δ replaced by a continuous class function that is invariant under inversion and concentrated near 1, for instance a function of $|\ell(u)|$ supported near 1, and then hold in the limit, as identities of measures. Where $\mathfrak{a}'$ is a submersion they hold also pointwise for the fibre densities of Section 31, at every V' at which the fibre integrals converge absolutely: the substitutions used below are isometries of the bond variables that map fibres of $\mathfrak{a}'$ onto fibres, and the fibre measure is invariant under $\mathcal{G}'$. Both invariances are needed: the proofs use $\delta(a^{-1}b) = \delta(ab^{-1})$, and a class function need not be invariant under $u \mapsto u^{-1}$ (for instance on $SU(3)$).

Lemma 32.6 (Invariance of the gauge-fixed step)**.** *Let ρ be a gauge-invariant function of real configurations on $\mathbb{T}$, let χ be a covariant cutoff, and suppose that $\chi\rho$ vanishes outside $\mathfrak{M}$.*

(i) $T^{\mathrm{gf}}(\chi\rho)(V')$ does not depend on the choice of U' in the minimal orbit of V'.

(ii) $T^{\mathrm{gf}}(\chi\rho)(V'^{u'}) = T^{\mathrm{gf}}(\chi\rho)(V')$ for every real gauge transformation u' on $\mathbb{T}'$.

(iii) For every $r \in \mathfrak{S}_{k+1}$, $T^{\mathrm{gf}}(\chi\cdot r\rho) = r\,T^{\mathrm{gf}}(\chi\rho)$, with the actions $(r\rho)(V) = \rho(r^{-1}V)$ and $(rF)(V') = F(r^{-1}V')$ of Section 31; that is, $T^{\mathrm{gf}}(\chi\cdot r\rho)(rV') = T^{\mathrm{gf}}(\chi\rho)(V')$, where on each side χ is taken at the base and the reference of the argument. In particular, if $\rho(rV) = \rho(V)$ for all V, then $T^{\mathrm{gf}}(\chi\rho)(rV') = T^{\mathrm{gf}}(\chi\rho)(V')$.

(iv) On the set of the V' as in Lemma 31.6, $T^{\mathrm{gf}}(f) = T(f)$ for every integrable f with $f(V^u) = f(V)$ for $u \in \mathcal{G}'$ that vanishes outside $\mathfrak{M}$. Consequently, let $w \le \frac{1}{8}\varepsilon_M$ and $2(C_*+1)L^{-2}\varepsilon_1' + 4w \le \tilde{h}$, so that $\operatorname{supp}\chi^{\mathrm{w}} \subset \mathfrak{M}\cap\mathfrak{U}_k(\tilde{h})$. Then, on the same set, for every covariant cutoff χ and every gauge-invariant $\rho \ge 0$ on $\mathfrak{U}_k(\tilde{h})$,

$$T(\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(\tilde{h})}\rho) - T^{\mathrm{gf}}(\chi^{\mathrm{w}}\chi\rho) = T^{\mathrm{gf}}\big((1-\chi^{\mathrm{w}}\chi)\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(\tilde{h})}\rho\big) \ge 0.$$

Proof. The proofs of (i)–(iii) substitute $V = \tilde{V}^v$ or $V = r\tilde{V}$ in the integral. These substitutions preserve $\prod_b dV(b)$: a gauge transformation acts on each bond by a left and a right translation, and r permutes the bonds, possibly reversing them, which preserves the Haar measure. They also preserve $\mathfrak{M}$. Details

(i) Another element of the minimal orbit is U'^v with v real on $\mathbb{T}_\eta$ and $v|_{\mathbb{T}'} = 1$. By Lemma 21.10(b), its base is $W^{\tilde{v}}$ with $\tilde{v} = v|_{\mathbb{T}} \in \mathcal{G}'$. Substitute $V = \tilde{V}^{\tilde{v}}$. Then $\mathfrak{a}'(V) = \mathfrak{a}'(\tilde{V})$, since $\tilde{v}|_{\mathbb{T}'} = 1$. Moreover $\mathrm{gf}_{W^{\tilde{v}}}(\tilde{V}^{\tilde{v}}) = \mathrm{gf}_W(\tilde{V})$ by Lemma 32.3(b), $\chi_{W^{\tilde{v}},U'^v}(\tilde{V}^{\tilde{v}}) = \chi_{W,U'}(\tilde{V})$ by (C1), and $\rho(\tilde{V}^{\tilde{v}}) = \rho(\tilde{V})$. So the integral with U'^v equals the integral with U'.

(ii) Let u be a real gauge transformation on $\mathbb{T}_\eta$ with $u|_{\mathbb{T}'} = u'$, and $\tilde{u} = u|_{\mathbb{T}}$. The configuration $V'^{u'}$ satisfies the hypotheses of Lemma 31.6, since its curvature is that of V'. By

Theorem 25.1(g) and Theorem 30.3(a) for the $(k+1)$-step problem, U'^u lies in the minimal orbit of $V'^{u'}$, and by Lemma 21.10(b) its base is $W^{\tilde{u}}$. By (i) we may use this representative. Substitute $V = \tilde{V}^{\tilde{u}}$. By Lemma 21.10(b),

$$\mathfrak{a}'(\tilde{V}^{\tilde{u}})(c)\, V'^{u'}(c)^{-1} = u'(c_-)\, \mathfrak{a}'(\tilde{V})(c) V'(c)^{-1}\, u'(c_-)^{-1},$$

and δ is a class function. The other factors are invariant as in (i), now with $\tilde{u}$ in place of $\tilde{v}$.

(iii) By Proposition 31.7(b) and its proof, rV' satisfies the hypotheses of Lemma 31.6, and rU' lies in the minimal orbit of rV'. Since $\mathfrak{S}_{k+1} \subset \mathfrak{S}_k$, Lemma 21.10 gives $(rU')^{(k)} = rW$. By (i) we may use the reference rU' and the base rW at rV'. The hypotheses of the lemma hold with $r\rho$ in place of ρ: $r\rho$ is gauge invariant (Proposition 31.7(a)), and $\chi_{rW,rU'} \cdot r\rho = r(\chi_{W,U'}\rho)$ by (C2), which vanishes outside $r\mathfrak{M} = \mathfrak{M}$. Substitute $V = r\tilde{V}$ in the integral for $T^{\mathrm{gf}}(\chi \cdot r\rho)(rV')$. By Lemma 21.10(d), $\mathfrak{a}'(r\tilde{V})(c) = \mathfrak{a}'(\tilde{V})(r^{-1}c)$, and $(rV')(c) = V'(r^{-1}c)$. As c runs over the positively oriented bonds of $\mathbb{T}'$, $r^{-1}c$ runs over all bonds of $\mathbb{T}'$, each once up to orientation. If $r^{-1}c = -c''$, then $\mathfrak{a}'(\tilde{V})(-c'')V'(-c'')^{-1} = \mathfrak{a}'(\tilde{V})(c'')^{-1}V'(c'')$ by Lemma 21.9, and $\delta(a^{-1}b) = \delta(ab^{-1})$, since δ is invariant under inversion and conjugation. So the product of the δ-functions is unchanged. Moreover $\mathrm{gf}_{rW}(r\tilde{V}) = \mathrm{gf}_W(\tilde{V})$ by Lemma 32.3(b), $\chi_{rW,rU'}(r\tilde{V}) = \chi_{W,U'}(\tilde{V})$ by (C2), and $(r\rho)(r\tilde{V}) = \rho(\tilde{V})$. So the integral equals $T^{\mathrm{gf}}(\chi\rho)(V')$. If $\rho \circ r = \rho$, then $r\rho = \rho$, which gives the last claim.

(iv) Let φ be a bounded Borel function of configurations on $\mathbb{T}'$ that vanishes outside the set of the V' as in Lemma 31.6. The function $V \mapsto \varphi(\mathfrak{a}'(V))f(V)$ is integrable, $\mathcal{G}'$-invariant by Lemma 21.10(b), and vanishes outside $\mathfrak{M}$. The base W, taken at $V' = \mathfrak{a}'(V)$, is constant on $\mathcal{G}'$-orbits, and in the proof of Lemma 32.3(a) the orbit integral of $\mathrm{gf}_{W_\circ}$ equals 1 for every $W_\circ$. So that proof, with $W_\circ$ the base on each orbit, gives $\int \prod_b dV(b)\, \varphi(\mathfrak{a}'(V))\, \mathrm{gf}_W(V) f(V) = \int \prod_b dV(b)\, \varphi(\mathfrak{a}'(V)) f(V)$, that is, $T^{\mathrm{gf}}(f) = T(f)$ as measures on that set. If $\mathfrak{a}'$ is a submersion near the support of f and the fibre integral of $|f|$ at V' is finite, the same argument, run on the fibre, whose measure is $\mathcal{G}'$-invariant, gives the identity of the fibre densities at V'. The inclusion $\operatorname{supp} \chi^{\mathrm{w}} \subset \mathfrak{M} \cap \mathfrak{U}_k(\tilde{h})$ is (32.2). The function $\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(\tilde{h})}\rho$ is $\mathcal{G}'$-invariant, since $\mathfrak{M}$ and $\mathfrak{U}_k(\tilde{h})$ are invariant under real gauge transformations. So $T(\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(\tilde{h})}\rho) = T^{\mathrm{gf}}(\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(\tilde{h})}\rho)$. By the inclusion, $\chi^{\mathrm{w}}\chi\rho = \chi^{\mathrm{w}}\chi\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(\tilde{h})}\rho$. Subtracting gives the identity. The right side is nonnegative, since $\delta \geq 0$, $\mathrm{gf}_W \geq 0$, $0 \leq \chi^{\mathrm{w}}\chi \leq 1$ and $\rho \geq 0$. □

Lemma 32.6 describes the *modified model*. The small-field step is $T^{\mathrm{gf}}(\chi^{\mathrm{w}}\chi_\xi\rho)$, with a white-noise cutoff χ_ξ (Section 48). The cutoffs act on gauge-fixed variables, so this is not of the form $T(\chi\rho)$ with a $\mathcal{G}'$-invariant χ. Parts (i)–(iii) show that it is nevertheless a renormalization group map on gauge-invariant densities that commutes with the lattice symmetries: $T^{\mathrm{gf}}(\chi^{\mathrm{w}}\chi_\xi \cdot r\rho) = r\, T^{\mathrm{gf}}(\chi^{\mathrm{w}}\chi_\xi\rho)$ for $r \in \mathfrak{S}_{k+1}$, by (iii), since $\chi^{\mathrm{w}}\chi_\xi$ is a covariant cutoff (see below). Part (iv) shows that the discarded part, the difference between the step without cutoffs and the small-field step, is nonnegative. In Section 48 the white-noise cutoff is taken bondwise on $\mathfrak{F}$ in the coordinates of this section:

$$\chi_\xi(V) = \prod_{b\in\mathfrak{F}} \mathbf{1}(|\xi_V(b)| \leq p_0), \qquad \xi_V = g^{-1}C_k^{-1/2}Z_V, \quad Z_V(b) = \ell(V(b)W(b)^{-1}),\ b \in \mathfrak{F},$$

for V in the support of χ^{w}, with $C_k = C_k(W, U')$ the covariance of Corollary 34.6. By Corollary 34.6(a),(b), $\xi_{V^{\tilde{u}}} = R(\tilde{u})\xi_V$ and $\xi_{rV} = r_W\xi_V$, with the data transformed accordingly, so $\chi^{\mathrm{w}}\chi_\xi$ is a covariant cutoff.

32.4 Elimination of the constraint

For a complex field Z on the bonds of $\mathbb{T}$ and a bond c of $\mathbb{T}'$ put

$$\ell_c(Z) = \ell(\overline{X}(c)), \qquad \overline{X} = \mathfrak{a}'(e^{\mathrm{i}Z}W)\, W'^{-1},$$

and let l^{pl} be its linearization, $(l^{\mathrm{pl}}Z)(c) = \frac{d}{ds}\big|_{s=0}\ell_c(sZ)$; "pl" refers to the plain average (21.3). The constraint in (31.1) is $\ell_c(Z) = 0$ for all c. Let $\mathsf{a} := 2(C_* + 1)\varepsilon_1'$. By Lemma 31.6 and Proposition 31.5, it bounds the curvature both of W and of the fine reference U', in the units of step $k+1$. The former is what Lemma 22.8 needs: by the paragraph after its proof, it applies with $a = \mathsf{a}$ and $W_{j-1} = W$, which is real and lies in $\mathfrak{U}_k(\mathsf{a}) \subset \mathfrak{U}_k(2\mathsf{a})$. In the units of step $k+1$ we write $p' = p_{k+1} = l_{k+1} - \delta_{k+1}D_{W'}\gamma_{k+1}$ for the operator of Lemma 22.8(f) at the reference U', and $q' = q_{k+1}$, $r' = r_{k+1}$, $\gamma'_{\mathrm{lin}} = \gamma_{k+1}$.

Lemma 32.7 (Local elimination)**.** *There are $C, c > 0$, depending only on d and G, such that the following holds with $\rho_L = cL^{-d-1}$ and $r_L = c^2L^{-2d-1}$, provided $L^d\mathsf{a} \le c$. Moreover $r_L \le \frac{1}{8}\varepsilon_M$.*

(a) ℓ_c is analytic for complex Z with $|Z| \le cL^{-1}$ on the bonds with both endpoints in $B'(c_-) \cup B'(c_+)$, and depends only on these values. Among the bonds $b_0(c')$ it depends only on $Z(b_0(c))$. Moreover $l^{\mathrm{pl}} = Lp'$, and $\ell_c(Z) = (l^{\mathrm{pl}}Z)(c) + O(L^2|Z|^2)$.

(b) $\partial\ell_c/\partial Z(b_0(c)) = L^{1-d}R(W([c_-, b_0(c)_-]))\,(1 + \mathfrak{r}_c(Z))$ with $|\mathfrak{r}_c(Z)| \le C(L^d\mathsf{a} + L^{d+1}|Z|)$.

(c) For complex Z on $\mathfrak{F}$ with $|Z| \le r_L$ there is exactly one $\zeta_c(Z)$ with $|\zeta_c(Z)| \le \rho_L$ such that $\ell_c = 0$ when $Z(b_0(c)) = \zeta_c(Z)$. It is analytic, real for real Z, depends only on Z on the bonds of (a), and satisfies $|\zeta_c(Z)| \le CL^d \max|Z|$, the maximum running over those bonds.

(d) Let $\hat{Z}(Z)$ be the field equal to Z on $\mathfrak{F}$ and to $\zeta_c(Z)$ on $b_0(c)$, and write $\hat{Z} = \hat{Z}^1Z + \hat{Z}^2(Z) + \hat{Z}^{\ge 3}(Z)$ (Taylor expansion). Then $\hat{Z}^1$ is a linear bijection from fields on $\mathfrak{F}$ onto

$$\mathcal{Z} = \ker l^{\mathrm{pl}} = \{B :\ l^{\mathrm{pl}}B = 0\},$$

and $\|Z\| \le \|\hat{Z}^1Z\| \le CL^d\|Z\|$. The maps $\hat{Z}^2$, $\hat{Z}^{\ge 3}$ are supported on the bonds $b_0(c)$ and satisfy $|\hat{Z}^2(Z)(b_0(c))| \le CL^{3d+1}\max|Z|^2$, the maximum over the bonds of (a).

Proof. (a), (b) We apply Lemma 22.8(f) in the units of step $k+1$, with $a = \mathsf{a} \le c_2$; its constants c_2 and C depend only on d and G. There $\delta_k = L^{-1}$, $\delta_{k+1} = 1$, and $e^{\mathrm{i}Z} = e^{\mathrm{i}\delta_k B}$ with $B = LZ$, so Details

$$\ell_c(Z) = (p'B)(c) + c^p_{k+1}(B)(c) = L(p'Z)(c) + c^p_{k+1}(LZ)(c),$$

for $|B| \le c_2$, i.e. $|Z| \le c_2L^{-1}$, with the bounds of Lemma 22.8(c) for c^p_{k+1}. This gives $l^{\mathrm{pl}} = Lp'$, the expansion, the analyticity and the locality. A bond $b_0(c')$ has both endpoints in $B'(c_-) \cup B'(c_+)$ only if $c' = \pm c$.

Next consider the coefficient of $Z(b_0(c))$ in $L(p'Z)(c)$. By Lemma 22.8(f), $\gamma'_{\mathrm{lin}}Z(c_\pm)$ does not depend on $Z(b_0(c))$, and the coefficient of $Z(b_0(c))$ in $(q'Z)(c)$ is $L^{-d}R(W([c_-, b_0(c)_-]))$. The reason is that $[x, x(c)]$ contains $b_0(c)$ only for the L points x on the axis of c, for which $G(c_-, x)$ consists of the straight contour $[c_-, x]$. So the coefficient in $L(q'Z)(c)$ is $L^{1-d}R(W([c_-, b_0(c)_-]))$. The bound on r' in Lemma 22.8(b) and the Lipschitz bound for c^p_{k+1} in (c), applied to $B = LZ$, change it by at most $CL\mathsf{a} + CL^2|Z|$, i.e. relatively by $C(L^d\mathsf{a} + L^{d+1}|Z|)$. The factor L^{d+1} is genuine. The coefficient L^{1-d} is what remains after a cancellation of terms of order one: the L averaged loops through points of the axis are identically 1, each of the other $L^d - L$ loops contributes $-L^{-d}$ through $Y([c])^{-1}$, and the outer factor $Y([c])$ contributes $+1$. The non-abelian corrections act on these terms of order one.

(c) Put $M_c = L^{1-d}R(W([c_-, b_0(c)_-]))$ and consider $\Phi(z) = z - M_c^{-1}\ell_c(Z, z)$, where z is the value on $b_0(c)$. By (b), $|\Phi'(z)| \le \frac{1}{2}$ on $|z| \le \rho_L$ if c is small, since there $L^{d+1}|z| \le c$. By (a), $|\ell_c(Z, 0)| \le CL\max|Z|$, since $|p'Z| \le C\max|Z|$ by Lemma 22.8(b). So $|\Phi(0)| \le C_0L^d\max|Z|$, with C_0 depending only on d and G. For $|Z| \le r_L = c^2L^{-2d-1}$ this is at most $C_0c\,\rho_L \le \frac{1}{2}\rho_L$ for $c \le 1/(2C_0)$. (With the same constant c in ρ_L and r_L the contraction would not close.)

So Φ is a contraction of $\{|z| \le \rho_L\}$ into itself, and its fixed point is the unique zero there. Analyticity follows as in Lemma 26.1. Reality follows from uniqueness, since $\overline{X} \in G$ for real Z. Finally, for $c \le r_M/64$ we have $r_L \le cL^{-2} \le r_M/(64L^2) = \frac{1}{8}\varepsilon_M$.

(d) Differentiating $\ell_c(\hat{Z}(Z)) = 0$ at $Z = 0$ gives $(l^{\mathrm{pl}}\hat{Z}^1 Z)(c) = 0$ for all c, so $\hat{Z}^1 Z \in \mathcal{Z}$. Conversely, $B \in \mathcal{Z}$ is determined by its values on $\mathfrak{F}$: by (a), $(l^{\mathrm{pl}}B)(c)$ involves among the bonds $b_0(\cdot)$ only $b_0(c)$, with the invertible coefficient of (b) at $Z = 0$, so the equation $(l^{\mathrm{pl}}B)(c) = 0$ fixes $B(b_0(c))$. This gives the bijection. The bound on $\hat{Z}^1$ follows from $|\zeta_c^1(Z)|^2 \le CL^{2d} \max |Z|^2 \le CL^{2d} \sum |Z(b)|^2$, the sum over the bonds of (a), since each bond lies in a bounded number of these sets. The bound on $\hat{Z}^2$ follows from Cauchy's estimate, since $|\zeta_c| \le \rho_L$ on $|Z| \le r_L$. □

At the trivial base $W = 1$ (and $U' = 1$) the averaged contour variables linearize to the means of the line integrals (Lemma 22.7(b)), and l^{pl} becomes

$$(q^{\mathrm{pl}}B)(c) = L^{-d} \sum_{x \in B'(c_-)} \Big(\mathfrak{G}_1 B(c_-, x) + B([x, x(c)]) - \mathfrak{G}_1 B(c_+, x(c)) \Big), \tag{32.3}$$

where $B([x, x(c)])$ is the sum of B over the L bonds of the straight segment. This follows by linearizing (21.3) at $Y = 1$ with Lemma 21.5(c), since $L^{-d}\sum_x B([c]) = B([c])$ cancels against the outer factor. Equivalently $q^{\mathrm{pl}} = L(q - D'\tilde{\gamma})$ in the units of step $k+1$, with q the centred tube average, $\tilde{\gamma}B(y) = L^{-d-1}\sum_{x \in B'(y)} \mathfrak{G}_1 B(y, x)$ and $D'\lambda(c) = \lambda(c_+) - \lambda(c_-)$ (Lemma 22.8(f) and the remark after it). The coefficient of $B(b_0(c))$ in $(q^{\mathrm{pl}}B)(c)$ is L^{1-d}.

32.5 The exact formula

Let σ be the density of the normalized Haar measure in exponential coordinates: $d(e^{\mathrm{i}Z}) = \sigma(Z)\, dZ$ for $|Z| < c_G$, where dZ is the Lebesgue measure on $\mathfrak{g}$ given by the trace form. It is analytic, positive and Ad-invariant near 0, and $\sigma(Z)/\sigma(0) = \det\big(\sinh(\frac{\mathrm{i}}{2}\operatorname{ad} Z)/(\frac{\mathrm{i}}{2}\operatorname{ad} Z)\big) = 1 + O(|Z|^2)$. The δ-function of G is then $\delta(e^{\mathrm{i}w}) = \delta(w)/\sigma(w)$ near 1.

Proposition 32.8 (One step in fluctuation coordinates)**.** *Let V' be as in Lemma 31.6, with $L^d\mathsf{a} \le c$, and let $w \le r_L$ (so $w \le \rho_L$ and $w \le \frac{1}{8}\varepsilon_M$). Let ρ be a function of real configurations on $\mathbb{T}$, measurable and bounded on the support of χ^{w}. Then $\chi^{\mathrm{w}}\rho$ vanishes outside $\mathfrak{M}$, which lies in the interior of* $\operatorname{dom}\mathfrak{a}'$*, $\mathfrak{a}'$ is a submersion at every point where $\chi^{\mathrm{w}} = 1$, and the density of $T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)$ at V', defined in Section 31, is*

$$T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(V') = z^{-|\mathbb{T}\setminus\mathbb{T}'|} \int_{(\mathfrak{g})^{\mathfrak{F}}} \prod_{b \in \mathfrak{F}} \sigma(Z(b))\, dZ(b) \;\; \chi^{\mathrm{w}}(\hat{Z}(Z)) \prod_c \mathsf{J}_c(Z)\; e^{-g^{-2}\Phi_W(Z)}\, \rho(e^{\mathrm{i}\hat{Z}(Z)}W),$$

where $\chi^{\mathrm{w}}(\hat{Z}) = \prod_b \mathbf{1}(|\hat{Z}(b)| \le w)$, Φ_W is the function of Lemma 32.3(c), and

$$\mathsf{J}_c(Z) = \frac{\sigma(\zeta_c(Z))}{\sigma(0)\, \det \partial\ell_c/\partial Z(b_0(c))}\,,$$

evaluated at $Z(b_0(c)) = \zeta_c(Z)$. For real Z this determinant is positive (see the proof), so it equals its absolute value, which is what the change of variables produces. The integrand is defined to be 0 if $|Z(b)| > w$ for some $b \in \mathfrak{F}$. On the remaining set $|Z|_\infty \le w \le r_L \le c_0 L^{-1}$, so $\zeta_c(Z)$, $\hat{Z}(Z)$, $\mathsf{J}_c(Z)$, $\Phi_W(Z)$ and $\sigma(Z(b))$ are defined, and the factor $\chi^{\mathrm{w}}(\hat{Z}(Z))$ imposes the cutoff also on the bonds $b_0(c)$. The same convention is used for all integrals over fields on $\mathfrak{F}$ below. Each J_c is an analytic function of Z on the bonds of Lemma 32.7(a), and $\mathsf{J}_c(Z) = L^{(d-1)\dim\mathfrak{g}}(1 + O(L^d\mathsf{a} + L^{2d+1}|Z|_\infty))$, with $|Z|_\infty$ the sup over $\mathfrak{F}$. The value on $b_0(c)$ is ζ_c, with $|\zeta_c| \le CL^d|Z|_\infty$.

Proof. By (32.2), χ^{w} vanishes outside $\mathfrak{M} = \mathfrak{U}_k(\varepsilon_M)$, and $8\varepsilon_M L^2 = r_M$, so $\mathfrak{M}$ lies in the interior of $\operatorname{dom}\mathfrak{a}'$ (Section 31). On its support $V = e^{\mathrm{i}Z}W$ with $Z(b) \in \mathfrak{g}$, $|Z(b)| \le w$ on all bonds, and $dV(b) = \sigma(Z(b))dZ(b)$. Details

Submersion. By Lemma 32.7(a),(b), at such V the derivative of $Z \mapsto (\ell_c(Z))_c$ with respect to the variables $Z(b_0(c))$ is block diagonal, with the invertible blocks $L^{1-d}R(W([c_-, b_0(c)_-]))(1+\mathfrak{r}_c)$; here $|\mathfrak{r}_c| \le \frac{1}{2}$, as in the proof of Lemma 32.7(c), since $|Z| \le w \le \rho_L$. Since $\ell_c(Z) = \ell(\mathfrak{a}'(e^{\mathrm{i}Z}W)(c)\, V'(c)^{-1})$, and the exponential and the logarithm are local diffeomorphisms, $\mathfrak{a}'$ is a submersion at V. So the density at V' is an integral over the fibre $\{\mathfrak{a}'(V) = V'\}$ (Section 31), which we compute with δ-functions. The elimination of $Z(b_0(c))$ below is the change of variables $z \mapsto \ell_c(Z, z)$ in this integral, at fixed variables on $\mathfrak{F}$.

The formula. Since $w \le r_L \le c_0 L^{-1}$ for c small, Lemma 32.3(c) gives $\mathrm{gf}_W(V) = z^{-|\mathbb{T}\setminus\mathbb{T}'|}e^{-g^{-2}\Phi_W(Z)}$. Moreover $\mathfrak{a}'(V)(c)V'(c)^{-1} = \overline{X}(c) = e^{\mathrm{i}\ell_c(Z)}$. By Lemma 32.7(a), the δ-function for c involves, among the variables $Z(b_0(\cdot))$, only $Z(b_0(c))$. The function Φ_W involves none of them, since it depends only on inner bonds (Lemma 32.3(c)). So the integrals over these variables factorize. For fixed variables on $\mathfrak{F}$, $Z(b_0(c)) \mapsto \ell_c$ has at most one zero in $|Z(b_0(c))| \le w$, by Lemma 32.7(c). It is $\zeta_c(Z)$ if $|\zeta_c(Z)| \le w$; otherwise the integrand vanishes, which is accounted for by the factor $\chi^{\mathrm{w}}(\hat{Z}(Z))$. Hence

$$\int \sigma(z)\, dz\ \delta\big(e^{\mathrm{i}\ell_c(Z,z)}\big)\,(\cdots) = \int \sigma(z)\frac{\delta(\ell_c(Z,z))}{\sigma(\ell_c(Z,z))}(\cdots)\, dz = \frac{\sigma(\zeta_c)}{\sigma(0)\,|\det \partial_z \ell_c|}(\cdots)\Big|_{z=\zeta_c}.$$

Positivity of the determinant. By Lemma 32.7(b),

$$\det \partial_z \ell_c = L^{(1-d)\dim\mathfrak{g}} \det R(W([c_-, b_0(c)_-]))\, \det(1+\mathfrak{r}_c).$$

Here $\det R(\cdot) = 1$ on $\mathfrak{g}$, since G is connected (Section 12). In the proof of Lemma 32.7(c), $|\mathfrak{r}_c| \le \frac{1}{2}$ wherever ζ_c is defined, so $t \mapsto \det(1+t\mathfrak{r}_c)$ does not vanish on $[0,1]$, and $\det(1+\mathfrak{r}_c) > 0$ for real Z. Hence $|\det \partial_z \ell_c| = \det \partial_z \ell_c$ for real Z, and J_c is the analytic function of the statement. The bound on J_c follows from Lemma 32.7(b),(c), since $L^{d+1}|\zeta_c| \le CL^{2d+1}|Z|_\infty$ and $\sigma(\zeta_c)/\sigma(0) = 1 + O(|\zeta_c|^2)$. □

No invariance of ρ is used: the gauge fixing has been made by the weight gf_W, and the elimination of the constraint is a change of variables. This is Bałaban's representation [18, (0.16)–(0.19)], [19, (2.3)–(2.7)]: the integration variables are the bonds of $\mathbb{T}$ other than the $b_0(c)$, and the elimination is performed on one bond per coarse bond, so that the Jacobians are local, as in [9, (17), (20)]. The representation is exact. No small-field approximation has been made beyond the insertion of χ^{w}, and by Lemma 32.6(iv) the gauge fixing changes nothing on $\mathcal{G}'$-invariant integrands supported in $\mathfrak{M}$.

33 The coarse effective action

The integrand of Proposition 32.8 depends on the fluctuation in two ways: through the explicit gauge-fixing weight $e^{-g^{-2}\Phi_W}$ of Lemma 32.3, and through the action of the background field, $S(U_k(e^{\mathrm{i}\hat{Z}}W))$. In this section we study the latter as a function of the coarse configuration. Its gradient is the Lagrange multiplier of the variational problem; its derivatives decay along trees; and its Hessian at the base configuration is comparable with the operator $\mathbf{\Gamma}_\pi - a$ of Part II. All statements hold for every unit-lattice field B in the stated balls. None of them uses the one-step constraint or a gauge condition on B, so they apply to the fields $\hat{Z}(Z)$ of Section 32, which satisfy the constraint, and to arbitrary fields on all bonds of $\mathbb{T}$ alike.

Setting. In this section $k \geq 1$: the k-step operators of Parts II–V at U' need at least one averaging step (Section 2). The first step, $k = 0$, is treated in Section 52.2 (Lemma 52.3). Let W be a real configuration on $\mathbb{T}$ in $\mathfrak{U}_k(\varepsilon_1)$, $\varepsilon_1 \leq a_1$, and let U' be an element of its minimal orbit. Let ε be a number such that U' is (M, ε)-regular with $|J_{U'}| \leq \varepsilon$; by Theorem 30.3 one may take $\varepsilon = B_3\varepsilon_1$, and in Section 32 we take $\varepsilon = B_3\varepsilon_1'$ (end of Section 31). We assume $\varepsilon \leq c_6/2$, and we put $r_1 := c_6/4$. In Section 32, W is the base configuration V^0 and U' the $(k+1)$-step background. For complex unit-lattice fields B with $|B|_\infty < 2r_1$, so that $\varepsilon + |B|_\infty < c_6$, let $A_*(B)$ be the Landau representative relative to U' given by Theorem 25.1(f), for the coarse configuration $e^{\mathrm{i}B}W$, and put

$$\mathcal{S}(B) = \Phi(A_*(B)), \qquad \Phi(A) = S(e^{\mathrm{i}\eta A}U').$$

The bounds of Proposition 33.1(c) are stated on the smaller ball $|B|_\infty < r_1$. The margin r_1 between the two balls is used only for Cauchy's estimate in the proof of the third-derivative bound. For real B, $\mathcal{S}(B)$ is the minimum of S on $\mathfrak{B}(e^{\mathrm{i}B}W) \cap \mathfrak{U}(\theta_0)$: by Theorem 25.1(a),(b) the orbit of $\mathsf{U}(A_*(B))$ is the set of minimizers there, and $S(\mathsf{U}(A)) = \Phi(A)$. If moreover $e^{\mathrm{i}B}W \in \mathfrak{U}_k(a_1)$, this orbit is the minimal orbit of Theorem 30.3(a), and $\mathcal{S}(B) = S(U_k(e^{\mathrm{i}B}W))$; this is the case in all uses below. In particular $\mathcal{S}$ depends only on the configuration $e^{\mathrm{i}B}W$, and not on the choice of the reference U'.

For bonds $c_1, \dots, c_m$ of $\mathbb{T}$ let $\tau(c_1, \dots, c_m) = \min_z \sum_i |c_i - z|$; for $m = 2$ this is $|c_1 - c_2|$. It satisfies $\tau(c_1, \dots, c_m) \leq |c_1 - x| + \tau(x, c_2, \dots, c_m)$.

33.1 Gradient and decay of derivatives

Proposition 33.1 (The coarse effective action)**.** *There are $C, \gamma_1 > 0$, independent of k, n, η and W, such that:*

(a) $\mathcal{S}$ is analytic on $\{|B|_\infty < 2r_1\}$ and real for real B.

(b) $\nabla\mathcal{S}(B) = \omega(B)$, where for real B, $\omega(B)$ is the multiplier of Proposition 27.2 for the coarse configuration $e^{\mathrm{i}B}W$; ω is analytic, and $|\omega(B)|_\infty \leq C(\varepsilon + |B|_\infty)$ for $|B|_\infty < 2r_1$.

(c) For $|B|_\infty < r_1$ and all bonds c_1, c_2, c_3:

$$|\partial_{c_1}\partial_{c_2}\mathcal{S}(B)| \leq Ce^{-\gamma_1|c_1-c_2|}, \qquad |\partial_{c_1}\partial_{c_2}\partial_{c_3}\mathcal{S}(B)| \leq Ce^{-\gamma_1\tau(c_1,c_2,c_3)},$$

where ∂_c is the derivative with respect to $B(c)$, the components being suppressed.

(d) Put $\omega_0 = \omega(0)$ and $\Delta^0 = d^2\mathcal{S}(0)$. For $|B|_\infty \leq r_1/2$,

$$\mathcal{S}(B) = \mathcal{S}(0) + \langle\omega_0, B\rangle + \tfrac{1}{2}\Delta^0(B, B) + \mathcal{S}_3(B), \qquad |\mathcal{S}_3(B)| \leq C|B|_\infty\|B\|^2.$$

Proof. (a) A_* is analytic with $|A_*(B)|_{C^1} \leq C_6(\varepsilon + |B|_\infty) \leq c_5$, and Φ is analytic there (Proposition 23.3). Details

(b) For real B, Proposition 27.2 gives $\nabla\Phi(A_*(B)) = d\mathcal{Q}(A_*(B))^T\omega(B)$. Differentiating the constraint $\ell(\mathcal{Q}(e^{\mathrm{i}\eta A_*(B)})) = B$ gives $d\mathcal{Q}(A_*)\, dA_*(B) = I$. Hence

$$d\mathcal{S}(B)h = \langle\nabla\Phi(A_*), dA_*h\rangle = \langle\omega, d\mathcal{Q}(A_*)dA_*h\rangle = \langle\omega, h\rangle.$$

Applying $\mathbf{H}^T$ to the Euler–Lagrange equation gives

$$\omega(B) = \big((I + dC_k^{\mathrm{av}}(A_*)\mathbf{H})^{-1}\big)^T\mathbf{H}^T\nabla\Phi(A_*(B)).$$

This expression is analytic in complex B and defines the continuation of ω. The identities $\nabla\Phi(A_*) = d\mathcal{Q}(A_*)^T\omega$ and $\nabla\mathcal{S} = \omega$ hold for real B, hence for complex B by the identity

theorem: the real ball is a totally real submanifold of maximal dimension. The bound for real B is Proposition 27.2; for complex B it follows from the formula, Lemma 24.1, Proposition 23.3(a), and the identity $\mathbf{H}^T D^T D A_* = (\mathbf{\Gamma}_\pi - a)\mathbf{Q}A_*$. This is the second identity in the proof of Lemma 26.2; it applies since $RD^T A_* = RD^T A'_* = 0$ also for complex B (Lemma 26.1, $A'_* \in \mathbf{H}B + \mathcal{X}$). The bound $|A_*|_{C^1} \le C(\varepsilon + |B|_\infty)$ alone would not bound $D^T D A_*$ pointwise.

(c) *Second derivatives.* We prove the bound on the larger ball $|B|_\infty < 2r_1$. Let $|B|_\infty < 2r_1$. Differentiating $d\mathcal{S}(B)h = \langle \nabla\Phi(A_*), dA_* h\rangle$ once more,

$$d^2\mathcal{S}(B)[h_1,h_2] = d^2\Phi(A_*)[dA_*h_1, dA_*h_2] - \langle \omega, d^2 C_k^{\mathrm{av}}(A_*)[dA_*h_1, dA_*h_2]\rangle,$$

where we used $\langle \nabla\Phi, d^2A_*[h_1,h_2]\rangle = \langle \omega, d\mathcal{Q}\, d^2A_*[h_1,h_2]\rangle$ and the second derivative of the constraint, $d\mathcal{Q}\, d^2A_*[h_1,h_2] + d^2C_k^{\mathrm{av}}(A_*)[dA_*h_1, dA_*h_2] = 0$. Now:

- $d^2\Phi(A) = D^TD + d^2\mathcal{N}(A)$, where $d^2\mathcal{N}(A)$ is a sum of plaquette-local terms with bounded coefficients (proof of Proposition 23.3(c));
- $d^2C_k^{\mathrm{av}}(A)[u,v](c)$ is bounded by $C\|u\|_{S_c}\|v\|_{S_c}$, by polarization of Proposition 22.9(c);
- by Theorem 25.1(f), which holds for complex B with $\varepsilon + |B|_\infty < c_6$, and here $\varepsilon + |B|_\infty < 2r_1 + 2r_1 = c_6$: $|dA_*\delta_{c_i}| + |\nabla dA_*\delta_{c_i}| \le Ce^{-\gamma'|x-c_i|}$.

Hence $|\partial_{c_1}\partial_{c_2}\mathcal{S}(B)| \le C\sum_x e^{-\gamma'(|x-c_1|+|x-c_2|)}\eta^d \le Ce^{-\gamma'|c_1-c_2|/2}$ for $|B|_\infty < 2r_1$.

Third derivatives. Let $|B|_\infty < r_1$, and let e be a unit vector of $\mathfrak{g}$. For complex t with $|t| < 2r_1 - |B|_\infty$ we have $|B + te\,\delta_{c_3}|_\infty < 2r_1$. Since $2r_1 - |B|_\infty > r_1$, the function $t \mapsto \partial_{c_1}\partial_{c_2}\mathcal{S}(B + te\,\delta_{c_3})$ is analytic on a neighbourhood of the closed disc $|t| \le r_1$, and, by the bound just proved on the larger ball, it is bounded there by $Ce^{-\gamma'|c_1-c_2|/2}$. By Cauchy's estimate on the circle $|t| = r_1$,

$$|\partial_{c_1}\partial_{c_2}\partial_{c_3}\mathcal{S}(B)| \le Cr_1^{-1}e^{-\gamma'|c_1-c_2|/2}.$$

The left side is symmetric in c_1, c_2, c_3. So the bound holds with $|c_1 - c_2|$ replaced by the largest distance D between two of the three bonds. Taking $z = c_1$ in the definition of τ gives $\tau(c_1,c_2,c_3) \le |c_2 - c_1| + |c_3 - c_1| \le 2D$. This proves (c) with $\gamma_1 = \gamma'/4$. No bound on the second or higher derivatives of A_* on the fine lattice is used.

(d) Fix B and a bond c, and let $R(s) = \omega(sB)(c)$ for complex s with $|s|\,|B|_\infty < r_1$. By (c), $|R'(s)| = |d\omega(sB)B(c)| \le C\sum_{c'} e^{-\gamma_1|c-c'|}|B(c')| =: K_c$ on this disc. By the Schwarz lemma applied to $R'(s) - R'(0)$, $|R'(s) - R'(0)| \le 4K_c|B|_\infty r_1^{-1}$ for $|s| \le 1$, when $|B|_\infty \le r_1/2$. By (b),

$$\mathcal{S}_3(B) = \int_0^1 \langle \omega(tB) - \omega_0 - t\,d\omega(0)B, B\rangle\, dt, \qquad |\mathcal{S}_3(B)| \le C|B|_\infty \sum_c K_c|B(c)| \le C|B|_\infty\|B\|^2,$$

by the Schur test. □

Part (b) is the discrete form of the statement that the derivative of the minimal action with respect to the constraint is the Lagrange multiplier. Part (c) is what makes the coarse action, and the Gaussian measure it defines, local.

33.2 The Hessian at the base configuration

Recall that $\mathbf{H} = \mathbf{H}(U')$, $\mathcal{X} = \{A : RD^TA = 0,\ \mathbf{Q}A = 0\}$ and $\mathbf{\Gamma}_\pi = \mathbf{\Gamma}_\pi(U')$ are taken at U' with the averaging pair $(Q'_k(U'), \mathbf{Q}_k(U'))$, and that

$$\mathbf{H}^T D^T D\mathbf{H} = \mathbf{\Gamma}_\pi - a. \tag{33.1}$$

This follows from the first identity in the proof of Lemma 26.2 and $S\mathbf{H} = G'RD^T\mathbf{H} = 0$. At $U' = 1$ the right-hand side is the operator $\mathbf{D}$ of Proposition 11.5.

Proposition 33.2 (The Hessian). *(a) For real B,*

$$\Delta^0(B,B) = \min\{q_2(A,A) : \ \mathbf{Q}A = B,\ RD^TA = 0\},$$

$$q_2(A,A) = \|DA\|^2 + d^2\mathcal{N}(0)[A,A] - \langle \omega_0, d^2C_k^{\mathrm{av}}(0)[A,A]\rangle,$$

and the minimum is attained at $A = dA_*(0)B$.

(b) $|\Delta^0(B,B) - \langle B, (\mathbf{\Gamma}_\pi - a)B\rangle| \le C\varepsilon\|B\|^2$.

(c) Δ^0 is real symmetric with decay (C,γ_1).

(d) For every real gauge transformation u on $\mathbb{T}_\eta$, replacing (W,U') by $(W^{u|_{\mathbb{T}}}, U'^u)$ replaces Δ^0 by $R(u)\Delta^0R(u)^{-1}$ and ω_0 by $R(u)\omega_0$, where $R(u)$ acts on unit-lattice fields by $R(u(b_-))$.

(e) For every $r \in \mathfrak{S}_k$, replacing (W,U') by (rW, rU') replaces Δ^0 by $r_W\Delta^0 r_W^{-1}$ and ω_0 by $r_W\omega_0$. Here $(r_WB)(b) = B(r^{-1}b)$, with $B(-b) = -R(W(b))^{-1}B(b)$ on negatively oriented bonds, as in Proposition 22.10(g); r_W is orthogonal.

Proof. (a) Let A^1 satisfy $\mathbf{Q}A^1 = B$ and $RD^TA^1 = 0$. Put $A(t) = tA^1 - \mathbf{H}\mathcal{D}(tA^1)$. By Lemma 26.1, $RD^TA(t) = 0$ and $\ell(\mathcal{Q}(e^{i\eta A(t)})) = tB$. Moreover $\mathcal{D}(tA^1) = t^2C_k^{\mathrm{av},(2)}(A^1) + O(t^3)$, where $C_k^{\mathrm{av},(2)}(A) = \frac{1}{2}d^2C_k^{\mathrm{av}}(0)[A,A]$. For real t near 0 the configuration $\mathsf{U}(A(t))$ lies in $\mathfrak{B}(e^{itB}W) \cap \mathfrak{U}(\theta_0)$. By Theorem 25.1(b), $\mathcal{S}(tB) \le \Phi(A(t))$. Expand Φ with Proposition 23.3: Details

$$\Phi(A(t)) = \mathcal{S}(0) + t\langle J, A^1\rangle + t^2\Big(\tfrac{1}{2}\|DA^1\|^2 + \tfrac{1}{2}d^2\mathcal{N}(0)[A^1,A^1] - \langle J, \mathbf{H}C_k^{\mathrm{av},(2)}(A^1)\rangle\Big) + O(t^3),$$

with $J = J_{U'} = \nabla\Phi(0) = \mathbf{Q}^T\omega_0$. Then $\langle J, A^1\rangle = \langle \omega_0, B\rangle$ and $\langle J, \mathbf{H}C_k^{\mathrm{av},(2)}(A^1)\rangle = \langle \omega_0, C_k^{\mathrm{av},(2)}(A^1)\rangle$, since $\mathbf{QH} = I$. The terms of first order agree with those of $\mathcal{S}(tB)$. So the inequality for $t \to 0$ gives $\Delta^0(B,B) \le q_2(A^1,A^1)$.

For equality, expand $\mathcal{S}(tB) = \Phi(A_*(tB))$ directly. We have $A_*(tB) = tA^1 + t^2A^2 + O(t^3)$ with $A^1 = dA_*(0)B$, both in the slice, and the constraint gives $\mathbf{Q}A^1 = B$, $\mathbf{Q}A^2 + C_k^{\mathrm{av},(2)}(A^1) = 0$. Hence the coefficient of t^2 is $\frac{1}{2}\|DA^1\|^2 + \frac{1}{2}d^2\mathcal{N}(0)[A^1,A^1] + \langle J, A^2\rangle$, and $\langle J, A^2\rangle = \langle \omega_0, \mathbf{Q}A^2\rangle = -\langle \omega_0, C_k^{\mathrm{av},(2)}(A^1)\rangle$. So $\Delta^0(B,B) = q_2(A^1,A^1)$.

(b) Write $A = \mathbf{H}B + h$ with $h \in \mathcal{X}$. Then $\|DA\|^2 = \|D\mathbf{H}B\|^2 + \|Dh\|^2$, since $\langle Dh, D\mathbf{H}B\rangle = 0$ for $h \in \mathcal{X}$, and $\|D\mathbf{H}B\|^2 = \langle B, (\mathbf{\Gamma}_\pi - a)B\rangle$ by (33.1). By Proposition 23.3(c) at $A = 0$, $|d^2\mathcal{N}(0)[A,A]| \le C\varepsilon(\|A\|^2 + \|\nabla A\|^2)$. By Proposition 22.9(c) and $|\omega_0| \le C\varepsilon$, $|\langle \omega_0, d^2C_k^{\mathrm{av}}(0)[A,A]\rangle| \le C\varepsilon\|A\|^2$. Moreover:

- $\|\mathbf{H}B\| + \|\nabla\mathbf{H}B\| \le C\|B\|$, by the decay of $\mathbf{H}$ and $D\mathbf{H}$, the identity $D^T\mathbf{H}B = P_{\mathcal{V}}D^T\mathbf{H}B$, and Lemma 26.4;
- $\|h\|^2 + \|\nabla h\|^2 \le c_X^{-1}\|Dh\|^2$, by Lemma 26.5.

Hence

$$\langle B, (\mathbf{\Gamma}_\pi - a)B\rangle - C\varepsilon\|B\|^2 + (1 - C\varepsilon)\|Dh\|^2 \le q_2(A,A).$$

Taking the minimum over h gives the lower bound; $h = 0$ gives the upper bound.

(c) is Proposition 33.1(c) at $B = 0$. (d) follows from the covariance of the variational problem, Theorem 25.1(g), and $\mathcal{S}$ depending only on $e^{iB}W$: $(e^{iB}W)^{u|_{\mathbb{T}}} = e^{iR(u)B}W^{u|_{\mathbb{T}}}$.

(e) Apply Theorem 25.1(h) with the reference $U_0 = U'$ and $V = W$, so $\beta = 0$; its hypotheses hold since $W_k(U') = W$ and $\varepsilon \le c_6/2$. Its first sentence gives that rU' is real and (M,ε)-regular with $|J_{rU'}| \le \varepsilon$, and that $(rU')^{(k)} = rW$. This needs no bound $\varepsilon \le \varepsilon_{\mathrm{B}}$. By Theorem 30.3(c), $rW \in \mathfrak{U}_k(\varepsilon_1)$, and the minimal orbit of rW is the image under r of the minimal orbit of W; since U' lies in the latter, rU' lies in the former. So the setting applies to (rW, rU'), with the same ε. By the convention on reversed bonds, $e^{ir_WB}(rW) = r(e^{iB}W)$. For real B with

$|B|_\infty < r_1$, Theorem 25.1(h) gives $A_*(r_W B; rU') = r_{U'} A_*(B; U')$, and $\Phi_{rU'}(r_{U'} A) = \Phi_{U'}(A)$ by $e^{\mathrm{i}\eta r_{U'} A} rU' = r(e^{\mathrm{i}\eta A} U')$ and $S \circ r = S$. Hence $\mathcal{S}_{rW}(r_W B) = \mathcal{S}_W(B)$ for real B, where the index denotes the base, and for complex B by the identity theorem. Differentiating at $B = 0$ and using that r_W preserves the pairing gives the claim. Only the coarse action enters. The operators $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$ of Parts II–III are not covariant under reflections at curved backgrounds (Remark 31.8), and (e) is not derived from (b). □

33.3 Dependence on the base configuration

Let $\hat\beta$ be a complex unit-lattice field with $|\hat\beta|_\infty \le r_1/4$, and $\hat W = e^{\mathrm{i}\hat\beta} W$ (bondwise product). For real $\hat\beta$ we have $\mathcal{S}_{\hat W}(B) = \mathcal{S}_W(B \star \hat\beta)$, with $B \star \hat\beta = \ell(e^{\mathrm{i}B} e^{\mathrm{i}\hat\beta})$ bondwise. We use this formula as the definition for complex $\hat\beta$. It defines $\mathcal{S}_{\hat W}$ analytically in $(\hat\beta, B)$ for $|B|_\infty \le r_1/4$, and in particular $\omega_0(\hat W)$ and $\Delta^0(\hat W)$.

Proposition 33.3 (Dependence on the base)**.** *$\omega_0(\hat W)$ and $\Delta^0(\hat W)$ are analytic in $\hat\beta$. Their derivatives with respect to $\hat\beta(c)$ satisfy*

$$|\partial_{\hat\beta(c)} \omega_0(\hat W)(c_1)| \le C e^{-\gamma_1 |c - c_1|}, \qquad |\partial_{\hat\beta(c)} \Delta^0(\hat W)(c_1, c_2)| \le C e^{-\gamma_1 \tau(c, c_1, c_2)}.$$

Proof. $\omega_0(\hat W) = \partial_B \mathcal{S}_W(B \star \hat\beta)|_{B=0}$ and $\Delta^0(\hat W) = \partial_B^2 \mathcal{S}_W(B \star \hat\beta)|_{B=0}$. The map $(B, \hat\beta) \mapsto B \star \hat\beta$ is bondwise analytic. Its derivatives in B at $B = 0$ are bondwise analytic functions of $\hat\beta$. The chain rule expresses the derivatives in terms of $d^m \mathcal{S}_W(\hat\beta)$, $m \le 3$, contracted with bondwise factors. Proposition 33.1(c) at $B = \hat\beta$, with $|\hat\beta|_\infty \le r_1/4 < r_1$, gives the bounds. □ Details

Proposition 33.3 controls the dependence on small changes of the base configuration. Changes that are not small, for example replacing the configuration far away by a different one, require a representation of the operators by terms that depend on the background only locally. This is the random-walk expansion of Section 20, which will be developed for the present operators in the next part.

34 The fluctuation covariance and the step

We return to the setting of Section 32: $W = V^0$ is the base configuration of the step, $U' = U_{k+1}(V')$, and $\varepsilon = B_3 \varepsilon_1'$. The objects of Section 33 are taken at this W and U'.

34.1 The quadratic form of the step

Lemma 34.1 (Expansion in the variables $\mathfrak{F}$)**.** *Let $r_2 = \min(cL^{-2d-2}, cL^{-d} a_1')$ with $c > 0$ small, depending only on d, G, L and $r_1 = c_6/4$ of Proposition 33.1 (the proof needs $2cL^{-2d-2} \le \frac{1}{2} r_L$, $2cL^{-2d-2} \le \frac{1}{2} c_0 L^{-1}$ with c_0 of Lemma 32.3(c), $CL^{-d-2} c \le \frac{1}{4} r_1$ and $Cc \le \frac{1}{2}$), a_1' as in Lemma 31.6(c), and assume $4(C_* + 1) L^{-2} \varepsilon_1' \le a_1'$. For complex fields Z on $\mathfrak{F}$ with $|Z|_\infty \le r_2$ put*

$$F(Z) = \mathcal{S}(\hat Z(Z)) + \Phi_W(Z),$$

with Φ_W as in Lemma 32.3(c). Then F is analytic, $F(Z) = S(U_k(e^{\mathrm{i}\hat Z(Z)} W)) + \Phi_W(Z)$ for real Z, and

$$F(Z) = S(U') + \tfrac{1}{2} \Delta_k(Z, Z) + \mathcal{V}(Z) + G_3(Z),$$

$$\Delta_k(Z, Z) = \Delta^0(\hat Z^1 Z, \hat Z^1 Z) + 2\langle \omega_0, \hat Z^2(Z) \rangle + \|\mathfrak{G}_W Z\|^2.$$

Here $\mathcal{V}$ is the remainder of the coarse action,

$$\mathcal{S}(\hat Z(Z)) = S(U') + \tfrac{1}{2} \Delta^0(\hat Z^1 Z, \hat Z^1 Z) + \langle \omega_0, \hat Z^2(Z) \rangle + \mathcal{V}(Z),$$

G_3 is the remainder of the gauge fixing (Lemma 32.3(c)), and $|\mathcal{V}(Z)|+|G_3(Z)| \le C_L|Z|_\infty\|Z\|^2$. For complex Z, $\|\mathfrak{G}_W Z\|^2$ stands for the bilinear form $\langle \mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle$. It involves only inner bonds, so $\|\mathfrak{G}_W Z\| = \|\mathfrak{G}_W \hat{Z}^1 Z\|$. In particular F has no linear term.

Proof. By Lemma 32.7, $\hat{Z}$ is analytic on $|Z|_\infty \le 2r_2 \le r_L$ with $|\hat{Z}(Z)|_\infty \le CL^d|Z|_\infty \le r_1/2$, so $\mathcal{S}\circ\hat{Z}$ is analytic on the disc of radius $2r_2$ by Proposition 33.1(a). Since $2r_2 \le \frac{1}{2}c_0L^{-1}$ for c small, Φ_W is analytic there by Lemma 32.3(c). For real Z the configuration $e^{\mathrm{i}\hat{Z}(Z)}W$ satisfies the one-step constraint, and its curvature is at most $2(C_*+1)L^{-2}\varepsilon_1' + CL^d r_2 \le a_1'$. Here $|\hat{Z}|_\infty \le CL^d|Z|_\infty$, so $CL^d r_2 \le a_1'/2$ by the choice of r_2, and we assume $4(C_*+1)L^{-2}\varepsilon_1' \le a_1'$. By Lemma 31.6(c), $\mathcal{S}(\hat{Z}(Z)) \ge \mathcal{S}(0) = S(U')$, so $d(\mathcal{S}\circ\hat{Z})(0) = \langle \omega_0, \hat{Z}^1\cdot\rangle = 0$. By Lemma 32.3(c), $\Phi_W(0)=0$ and $d\Phi_W(0)=0$. The second-order term of $\mathcal{S}\circ\hat{Z}$ follows from $\hat{Z}(tZ) = t\hat{Z}^1Z + t^2\hat{Z}^2(Z) + O(t^3)$ and Proposition 33.1(d); that of Φ_W is $\frac{1}{2}\langle \mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle$. For the remainder $\mathcal{V}$, $\nabla(\mathcal{S}\circ\hat{Z})(Z) = d\hat{Z}(Z)^T\omega(\hat{Z}(Z))$. The analyticity of $\hat{Z}$ and the bound $|\hat{Z}(Z)|_\infty \le r_1/2$ hold on the larger disc $|Z|_\infty \le 2cL^{-2d-2}$, since only the first term of the minimum defining r_2 enters them. By Proposition 33.1(c) and the locality of $\hat{Z}$ (range $2L$), $d^2(\mathcal{S}\circ\hat{Z})$ has a kernel bounded by $C_L e^{-\gamma_1|b-b'|}$ on this larger disc. The argument of Proposition 33.1(d), applied to $\mathcal{S}\circ\hat{Z}$ on this disc with $d(\mathcal{S}\circ\hat{Z})(0)=0$, gives the bound on $\mathcal{V}$. Its constant C_L depends on cL^{-2d-2}, but not on a_1'. The bound on G_3 is Lemma 32.3(c). Finally, the contours of the families $G(y,x)$ consist of inner bonds, on which $\hat{Z}^1Z = Z$. □ Details

This is the quadratic form [9, (53)–(54)], [18, (2.5)–(2.8)]. The term $2\langle\omega_0,\hat{Z}^2\rangle$ comes from the curvature of the constraint set and is Bałaban's term $-\langle H_1\hat{D}^{(2)}(A),J\rangle$. The term $\|\mathfrak{G}_W Z\|^2$ is the quadratic part of the gauge fixing, Bałaban's $\sum_{y,x}\frac{1}{2}|B'(y,x)|^2$ in [18, (2.5)], and G_3 is his block-local remainder G_3.

34.2 Coercivity

The lower bound for Δ_k is proved in two steps:

- at the trivial background, a Poincaré inequality for the curl, the linearized averaged contour variables and the linearized constraint, combined with Proposition 11.5;
- at a regular background, an IMS localization on the unit lattice, as in Section 19.

Let q^{pl} be the linearized constraint at the trivial base, (32.3), and $\mathfrak{G}_1$ the linearized averaged contour variables at $W=1$, the means of the line integrals (Lemma 22.7(b)). We use the norms

$$\|\mathfrak{G}_1 B\|^2 = \sum_{y\in\mathbb{T}'}\sum_{x\in B'(y)}|\mathfrak{G}_1B(y,x)|^2, \quad \|q^{\mathrm{pl}}B\|^2 = \sum_c |q^{\mathrm{pl}}B(c)|^2, \quad \|d_1B\|^2 = \sum_p |d_1B(p)|^2,$$

the sums running over the positively oriented bonds c of $\mathbb{T}'$ and the plaquettes p of $\mathbb{T}$. Here $d_1B(p) = B(\partial p)$ is the unit-lattice curl, and $B(\Gamma) = \sum_{b\in\Gamma}\pm B(b)$ is the line integral along a contour.

Lemma 34.2 (Poincaré inequality for one step)**.** *There is C_{P}, depending only on d and L, such that every real field B on the bonds of $\mathbb{T}$ satisfies*

$$\|B\|^2 \le C_{\mathrm{P}}(\|d_1B\|^2 + \|\mathfrak{G}_1B\|^2 + \|q^{\mathrm{pl}}B\|^2).$$

Proof. All estimates are local in a pair of neighbouring blocks. For $y\in\mathbb{T}'$ let $\mathcal{P}(y)$ be the set of plaquettes of $\mathbb{T}$ with all corners in $B'(y)$, and for a bond c of $\mathbb{T}'$ let $\mathcal{P}(c)$ be the set of plaquettes with all corners in $B'(c_-)\cup B'(c_+)$. Put Details

$$g_y = \max_{x\in B'(y)}|\mathfrak{G}_1B(y,x)|, \qquad \delta_y = \max_{p\in\mathcal{P}(y)}|d_1B(p)|, \qquad \delta_c = \max_{p\in\mathcal{P}(c)}|d_1B(p)|.$$

(i) Contours. Let $x \in B'(y)$ and $\Gamma, \Gamma' \in G(y,x)$. By the abelian identity of Lemma 21.7(i) (applied with $Y = 1$), $B(\Gamma) - B(\Gamma')$ is a sum of at most $\frac{3}{2}(L-1)^2$ values $\pm d_1 B(p)$, $p \in \mathcal{P}(y)$. Since $\mathfrak{G}_1 B(y,x)$ is the mean of the $B(\Gamma')$,

$$|B(\Gamma) - \mathfrak{G}_1 B(y,x)| \le \tfrac{3}{2}(L-1)^2 \delta_y.$$

(ii) Inner bonds. Let $b = (x, x')$ with $x, x' \in B'(y)$. As in the proof of Lemma 21.7(ii), there are $\Gamma_0 \in G(y,x)$ and $\Gamma_0' \in G(y,x')$ with $\Gamma_0' = \Gamma_0 b$ in the groupoid of contours. Hence $B(b) = B(\Gamma_0') - B(\Gamma_0)$, and by (i)

$$|B(b)| \le 2g_y + 3(L-1)^2 \delta_y.$$

(iii) Differences of crossing bonds. Let $c = (y, y')$ with $y' = y + Le_\mu$. Let $\mathsf{F} = \{x \in B'(y) : x_\mu = y_\mu + \frac{L-1}{2}\}$ be the face of $B'(y)$ towards y', and $b_x = (x, x + e_\mu)$ for $x \in \mathsf{F}$. These are the L^{d-1} crossing bonds between $B'(y)$ and $B'(y')$, and $b_0(c)$ is one of them. For $x, \tilde{x} \in \mathsf{F}$ choose $\Gamma_x \in G(y,x)$, $\Gamma_{\tilde{x}} \in G(y,\tilde{x})$, $\Gamma' \in G(y', x + e_\mu)$ and $\tilde{\Gamma}' \in G(y', \tilde{x} + e_\mu)$. The closed contour

$$\Gamma_x \, b_x \, \Gamma'^{-1} \, \tilde{\Gamma}' \, b_{\tilde{x}}^{-1} \, \Gamma_{\tilde{x}}^{-1}$$

has at most $8L$ bonds and lies in the box $B'(y) \cup B'(y')$, of side at most $2L$. By Lemma 21.3(iii) it is spanned by at most $16dL^2$ plaquettes of $\mathcal{P}(c)$. The abelian Stokes formula and (i) give

$$\begin{aligned} |B(b_x) - B(b_{\tilde{x}})| &\le |B(\Gamma_x) - B(\Gamma_{\tilde{x}})| + |B(\Gamma') - B(\tilde{\Gamma}')| + 16dL^2\delta_c \\ &\le 2g_y + 2g_{y'} + \big(6(L-1)^2 + 16dL^2\big)\delta_c. \end{aligned}$$

(iv) The mean over the face. Let $m_c = L^{1-d} \sum_{\tilde{x} \in \mathsf{F}} B(b_{\tilde{x}})$. For $x \in B'(y)$ the straight segment $[x, x(c)]$ consists of L bonds in the direction μ. Exactly one of them is crossing, namely $b_{\tilde{x}}$ with $\tilde{x}$ the point of F on the line of x. The other $L-1$ are inner bonds of $B'(y)$ or $B'(y')$. Each $\tilde{x} \in \mathsf{F}$ arises from the L points of $B'(y)$ on its line. Hence

$$L^{-d} \sum_{x \in B'(y)} B([x, x(c)]) = m_c + L^{-d} \sum_{x \in B'(y)} {\sum_b}' B(b),$$

where $\sum'$ runs over the inner bonds of $[x, x(c)]$. Since $x \mapsto x(c)$ maps $B'(y)$ onto $B'(y')$, (32.3) gives

$$m_c = q^{\mathrm{pl}} B(c) - L^{-d} \sum_{x \in B'(y)} \mathfrak{G}_1 B(y,x) + L^{-d} \sum_{x' \in B'(y')} \mathfrak{G}_1 B(y',x') - L^{-d} \sum_{x \in B'(y)} {\sum_b}' B(b).$$

So $|m_c| \le |q^{\mathrm{pl}} B(c)| + g_y + g_{y'} + (L-1)\beta_c$, where β_c is the maximum of $|B|$ over the inner bonds of $B'(y) \cup B'(y')$. By (ii), $\beta_c \le 2\max(g_y, g_{y'}) + 3(L-1)^2\delta_c$.

(v) Conclusion. Since $B(b_x) - m_c = L^{1-d} \sum_{\tilde{x} \in \mathsf{F}} (B(b_x) - B(b_{\tilde{x}}))$, (iii) and (iv) give, for every crossing bond b_x of c,

$$|B(b_x)| \le |q^{\mathrm{pl}} B(c)| + C_L(g_y + g_{y'} + \delta_c).$$

Every bond of $\mathbb{T}$ is inner, or crossing between the blocks of exactly one bond c of $\mathbb{T}'$. Square the bounds of (ii) and (v), and use $g_y^2 \le \sum_{x \in B'(y)} |\mathfrak{G}_1 B(y,x)|^2$ and $\delta_y^2 \le \delta_c^2 \le \sum_{p \in \mathcal{P}(c)} |d_1 B(p)|^2$ for $y = c_\pm$. A block contains at most dL^d bonds, a bond c has L^{d-1} crossing bonds, every $y \in \mathbb{T}'$ is an endpoint of $2d$ bonds of $\mathbb{T}'$, and every plaquette lies in $\mathcal{P}(c)$ for at most $2d$ bonds c. Summing over the bonds of $\mathbb{T}$ gives the claim, with C_{P} depending only on d and L. □

The term $\|\mathfrak{G}_1 B\|^2$ cannot be dropped: if λ is a function on $\mathbb{T}$ with $\lambda|_{\mathbb{T}'} = 0$, then $B = d\lambda$ satisfies $d_1 B = 0$ and, by (32.3), $q^{\mathrm{pl}} B(c) = \lambda(c_+) - \lambda(c_-) = 0$, while $\mathfrak{G}_1 B(y, x) = \lambda(x)$. A numerical computation at $W = 1$, for $d = 4$ and $L = 3$, gives the smallest eigenvalue of the form of the lemma at the momenta tested; the minimum over these samples is about 0.030, with reciprocal about 33 (Appendix C). These finite samples do not certify the minimum over all momenta or a uniform constant of that size. The uniformity of C_{P} in the size of the torus follows from the proof, in which every estimate is local in a pair of neighbouring blocks. Dropping any one of the three terms gives a zero mode.

Lemma 34.3 (Local comparison)**.** *Let $M' \geq \max(M, L)$ be an integer and z a point of the torus (not necessarily of $\mathbb{T}$). Assume $n \geq 80M'$ and $\varepsilon \leq \varepsilon_{\mathrm{cmp}}(M')$, where $\varepsilon_{\mathrm{cmp}}(M') > 0$ depends only on M', d, L, G, a, α and M, and is fixed in the proof. There are a real gauge transformation u on $\Lambda_{40M'}(z)$ and $C_{M'}$ such that, for every real field B on the bonds of $\mathbb{T}$ starting in $\Lambda_{M'}(z)$, with $B^u = R(u)B$:*

$$\big|\langle B, (\mathbf{\Gamma}_\pi - a)B\rangle - \langle B^u, \mathbf{D}B^u\rangle\big| \leq (C_{M'}\varepsilon + Ce^{-\gamma M'})\|B\|^2,$$

$$\big|\|\mathfrak{G}_W B\| - \|\mathfrak{G}_1 B^u\|\big| \leq C_{M'}\varepsilon\|B\|, \qquad \big|\|l^{\mathrm{pl}} B\| - \|q^{\mathrm{pl}} B^u\|\big| \leq C_{M'}\varepsilon\|B\|.$$

Here $\mathbf{D} = \mathbf{\Gamma}(1) - a$ is the operator of Proposition 11.5, and C, γ do not depend on M'.

Proof. The configuration U' is 3ε-regular on every cube $\Lambda_{4M+2}(z'')$ (Section 28). Let $z_0 \in \mathbb{T}_\eta$ with $|z - z_0|_\infty \leq \eta$, so that $\Lambda_{40M'}(z) \subset \Lambda_{40M'+1}(z_0)$. Apply Lemma 28.3 with $r = 4M + 2 \geq 18$, $R = 40M' + 1$ and $\sigma = 3\varepsilon$, on the cube $\Lambda_R(z_0)$. Its hypothesis $C_E(R/r)^p R\sigma \leq 1$ is the first condition defining $\varepsilon_{\mathrm{cmp}}(M')$. The boxes in its proof have sides at most $3R = 120M' + 3 < 80M'L \leq Ln$, the side of $\mathbb{T}$, so they do not wrap around the torus. This gives a real gauge u on $\Lambda_{40M'}(z)$ with $U'^u = e^{i\eta A}$ and $\|A\|_{1,\alpha} \leq C(40M'/M)^p\varepsilon$. Extend u to a real gauge transformation $\tilde{u}$ of $\mathbb{T}_\eta$. Then $\mathbf{\Gamma}_\pi(U'^{\tilde{u}}) = R(\tilde{u})\mathbf{\Gamma}_\pi(U')R(\tilde{u})^{-1}$, by Proposition 22.10(f) and Proposition 16.3.

Let χ be a smooth cutoff equal to 1 on $\Lambda_{20M'}(z)$ and 0 outside $\Lambda_{30M'}(z)$, and put $U^\natural = e^{i\eta\chi A}$. It satisfies (B) with parameter $C_{M'}\varepsilon$. The second condition defining $\varepsilon_{\mathrm{cmp}}(M')$ is that $C_{M'}\varepsilon$ lies below the thresholds of Parts II and III. The configuration $U^\natural$ coincides with $U'^{\tilde{u}}$ on $\Lambda_{20M'}(z)$. By Theorem 19.10(b), the kernels of $\mathbf{\Gamma}_\pi(U'^{\tilde{u}})$ and $\mathbf{\Gamma}_\pi(U^\natural)$ on $\Lambda_{M'}(z)$ differ by at most $C\sum_w e^{-\gamma(|y-w|+|w-y'|)}$, summed over w within $10M$ of the complement of $\Lambda_{20M'}(z)$. These w are at distance at least $9M'$ from $\Lambda_{M'}(z)$, which gives the error $Ce^{-\gamma M'}$ after a Schur bound.

By Part II at the background $U^\natural$, with the averaging pair $(Q'_k(U^\natural), \mathbf{Q}_k(U^\natural))$ (Remark 19.12), $\mathbf{\Gamma}_\pi(U^\natural) - \mathbf{\Gamma}(1)$ has decay $(C_{M'}\varepsilon, \gamma)$. This follows from Theorem 16.1(a) and Lemma 13.3. This proves the first bound.

By Lemma 21.13, whose hypotheses, together with those of Lemma 22.7(e) used below, are the third condition defining $\varepsilon_{\mathrm{cmp}}(M')$, $W^{\tilde{u}} = e^{iB_W}$ on the bonds of the blocks $B'(y)$ that meet $\Lambda_{3M'}(z)$, with $|B_W| \leq C_{M'}\varepsilon$; here we use that U' has curvature at most ε in the units of step k. By Lemma 22.7(d), $\mathfrak{G}_{W^{\tilde{u}}} B^u(y, x) = R(\tilde{u}(y))\,\mathfrak{G}_W B(y, x)$, so $\|\mathfrak{G}_W B\| = \|\mathfrak{G}_{W^{\tilde{u}}} B^u\|$. The value $\mathfrak{G}_{W^{\tilde{u}}} B^u(y, x)$ depends only on B^u on the bonds of $B'(y)$, and vanishes unless $B'(y)$ meets $\Lambda_{M'+1}(z)$. By Lemma 22.7(e), in the units of step k, $|\mathfrak{G}_{W^{\tilde{u}}} B^u(y, x) - \mathfrak{G}_1 B^u(y, x)| \leq CL^2 C_{M'}\varepsilon \max|B^u|$, the maximum over the bonds of $B'(y)$. Summing the squares over y and x gives the second bound.

For the third, the covariance of the averaging (Lemma 21.10(b)) gives $e^{iR(\tilde{u})Z} W^{\tilde{u}} = (e^{iZ}W)^{\tilde{u}}$ and $\mathfrak{a}'((e^{iZ}W)^{\tilde{u}})(c) = \tilde{u}(c_-)\,\mathfrak{a}'(e^{iZ}W)(c)\,\tilde{u}(c_+)^{-1}$. So ℓ_c at the base $W^{\tilde{u}}$ and the field $R(\tilde{u})Z$ is $R(\tilde{u}(c_-))\ell_c(Z)$, and $l^{\mathrm{pl}}(W^{\tilde{u}})B^u(c) = R(\tilde{u}(c_-))\,l^{\mathrm{pl}}(W)B(c)$. Hence $\|l^{\mathrm{pl}} B\| = \|l^{\mathrm{pl}}(W^{\tilde{u}})B^u\|$. In $l^{\mathrm{pl}}(W^{\tilde{u}}) = Lp'(W^{\tilde{u}})$ (Lemma 32.7(a)), the transports $R(W^{\tilde{u}}(\cdot))$ along contours of at most $2L$ bonds, the transports $R(W^{\tilde{u}}(c_-, x))$ and $R(W'^{\tilde{u}}(c))$, and the weights of Lemma 22.7 are within $C_{M'}\varepsilon$ of their values at $W = 1$, by Lemma 21.5(e), Lemma 22.7(e) and Lemma 21.10(a). The

remainder r' is bounded by $CL^2\varepsilon \max|B|$, by Lemma 22.8(b), since $L^2\varepsilon$ bounds the curvature of U' in the units of step $k+1$. So $(l^{\mathrm{pl}}(W^{\tilde{u}}) - q^{\mathrm{pl}})B^u(c)$ is a combination of at most C_L values of B^u on the bonds of $B'(c_-) \cup B'(c_+)$, with coefficients at most $C_{M'}\varepsilon$. This gives the third bound. □

Theorem 34.4 (Coercivity on the constraint set)**.** *There are $c_L > 0$, depending only on d and L, and $M'_L \geq \max(M, L)$ and $\varepsilon_L > 0$, depending only on d, L, G, a, α and M, such that, if $\varepsilon \leq \varepsilon_L$ and $n \geq 80M'_L$,*

$$\langle B, (\mathbf{\Gamma}_\pi - a)B\rangle + \|\mathfrak{G}_W B\|^2 \geq c_L\|B\|^2 \qquad \text{for all real } B \in \mathcal{Z}.$$

One may take $c_L = \frac{1}{2}\min(c_D, 1)\, C_{\mathrm{P}}^{-1}$. The proof does not use that U' is critical.

Proof. Let $\{\chi_j\}$ be the partition of Lemma 17.3 at scale M', restricted to the unit lattice, acting on bond fields through b_-. Put $K = \mathbf{\Gamma}_\pi - a$ and $K_{\mathfrak{G}} = \mathfrak{G}_W^T\mathfrak{G}_W$, both real symmetric operators on fields on all bonds of $\mathbb{T}$. Details

IMS. K is bounded with decay (C, γ) (Theorem 19.9). By Lemma 17.4 and the argument of Lemma 17.5 for kernels, $\sum_j(\chi_j(x) - \chi_j(x'))^2|K(x,x')| \leq CM'^{-2}|x - x'|^2|K(x,x')|$, so

$$\langle B, K\rangle B \geq \sum_j \langle \chi_j B, K\chi_j B\rangle - CM'^{-2}\|B\|^2.$$

The kernel of $K_{\mathfrak{G}}$ vanishes unless the two bonds lie in a common block $B'(y)$, and it is bounded by C_L, since $\mathfrak{G}_W$ is block-local with bounded entries (Lemma 22.7(c)). The kernel of $\sum_j[\chi_j, [\chi_j, K_{\mathfrak{G}}]]$ is $\sum_j(\chi_j(b_-) - \chi_j(b'_-))^2 K_{\mathfrak{G}}(b, b')$, and $\sum_j(\chi_j(x) - \chi_j(x'))^2 \leq C(L/M')^2$ for $|x - x'|_\infty \leq L$, by Lemma 17.3(c),(d). By Lemma 17.4 and the Schur test,

$$\|\mathfrak{G}_W B\|^2 \geq \sum_j \|\mathfrak{G}_W\chi_j B\|^2 - C_L(L/M')^2\|B\|^2.$$

Local bounds. Each $\chi_j B$ is supported on bonds starting in a cube $\Lambda_{M'}(z_j)$. Let u_j be the gauge of Lemma 34.3 at z_j, and $c' = \min(c_D, 1)$. By Lemma 34.3 and $\|\mathfrak{G}_1\| \leq C_L$,

$$\langle \chi_j B, K\chi_j B\rangle + \|\mathfrak{G}_W\chi_j B\|^2 \geq \langle (\chi_j B)^{u_j}, \mathbf{D}(\chi_j B)^{u_j}\rangle + \|\mathfrak{G}_1(\chi_j B)^{u_j}\|^2 - \delta_1\|\chi_j B\|^2,$$

with $\delta_1 = C_{M'}\varepsilon + Ce^{-\gamma M'}$. By Proposition 11.5 and Lemma 34.2, with $\|(\chi_j B)^{u_j}\| = \|\chi_j B\|$, the right side is at least

$$c'(\|d_1(\chi_j B)^{u_j}\|^2 + \|\mathfrak{G}_1(\chi_j B)^{u_j}\|^2) - \delta_1\|\chi_j B\|^2 \geq c'C_{\mathrm{P}}^{-1}\|\chi_j B\|^2 - c'\|q^{\mathrm{pl}}(\chi_j B)^{u_j}\|^2 - \delta_1\|\chi_j B\|^2.$$

Moreover $\|q^{\mathrm{pl}}(\chi_j B)^{u_j}\|^2 \leq 2\|l^{\mathrm{pl}}\chi_j B\|^2 + 2C_{M'}^2\varepsilon^2\|\chi_j B\|^2$.

The constraint. $(l^{\mathrm{pl}}B)(c)$ depends only on B on the bonds of $B'(c_-) \cup B'(c_+)$, with coefficients bounded by C_L, and $|\chi_j(b_-) - \chi_j(c_-)| \leq CL/M'$ for these bonds. Writing $(l^{\mathrm{pl}}\chi_j B)(c) = \chi_j(c_-)(l^{\mathrm{pl}}B)(c) + \sum_b l^{\mathrm{pl}}(c,b)(\chi_j(b_-) - \chi_j(c_-))B(b)$ and using the bounded overlap of the χ_j, we get $\sum_j \|l^{\mathrm{pl}}\chi_j B\|^2 \leq 2\|l^{\mathrm{pl}}B\|^2 + C_L(L/M')^2\|B\|^2$. For $B \in \mathcal{Z}$, $l^{\mathrm{pl}}B = 0$.

Conclusion. Summing over j with $\sum_j \|\chi_j B\|^2 = \|B\|^2$,

$$\langle B, K\rangle B + \|\mathfrak{G}_W B\|^2 \geq \Big(c'C_{\mathrm{P}}^{-1} - CM'^{-2} - C_L(L/M')^2 - Ce^{-\gamma M'} - C_{M'}\varepsilon\Big)\|B\|^2.$$

Choose $M' = M'_L \geq \max(M, L)$ so large that the terms independent of ε are at most $\frac{1}{4}c'C_{\mathrm{P}}^{-1}$. Then choose $\varepsilon_L \leq \varepsilon_{\mathrm{cmp}}(M'_L)$, below the threshold of Corollary 22.11 for Theorem 19.9 (which gives the decay of K used in the IMS step), with $C_{M'_L}\varepsilon_L \leq \frac{1}{4}c'C_{\mathrm{P}}^{-1}$, and put $c_L = \frac{1}{2}c'C_{\mathrm{P}}^{-1}$. Lemma 34.3 applies, since $n \geq 80M'_L$ and $\varepsilon \leq \varepsilon_{\mathrm{cmp}}(M'_L)$. Criticality of U' was not used. □

Remark 34.5 (Gauge directions). At the base of the step, the first two terms of Δ_k vanish in the directions of the $\mathcal{G}'$-orbit of W. Indeed, let λ be a real function on $\mathbb{T}$ with $\lambda|_{\mathbb{T}'} = 0$. For small real t, $W^{e^{it\lambda}} = e^{\mathrm{i}Y_t}W$ with $Y_t = -tD_W\lambda + O(t^2)$, where $D_W\lambda(b) = R(W(b))\lambda(b_+) - \lambda(b_-)$. These configurations satisfy the constraint, so $Y_t = \hat{Z}(Y_t|_{\mathfrak{F}})$ by the uniqueness in Lemma 32.7(c), and $\mathcal{S}(Y_t) = S(U')$ by gauge invariance. Since $d(\mathcal{S}\circ\hat{Z})(0) = 0$, the second derivative in t gives $\Delta^0(\hat{Z}^1Z, \hat{Z}^1Z) + 2\langle\omega_0, \hat{Z}^2(Z)\rangle = 0$ for $Z = D_W\lambda|_{\mathfrak{F}}$. In these directions the coercivity of Δ_k comes from the gauge fixing alone; at $W = 1$, $\mathfrak{G}_1 d\lambda(y,x) = \lambda(x)$. Theorem 34.4 does not distinguish these directions, and the bound $CL^{3d+1}\varepsilon$ on the term with ω_0 below is a perturbation at every base.

34.3 The fluctuation covariance

Corollary 34.6 (Fluctuation covariance)**.** *Let $\varepsilon \le \varepsilon_L$ and $CL^{3d+1}\varepsilon \le \frac{1}{4}c_L$. Then Δ_k, as an operator on real fields on $\mathfrak{F}$, is symmetric with*

$$\tfrac{1}{4}c_L\|Z\|^2 \le \Delta_k(Z,Z) \le C_L\|Z\|^2.$$

It has decay (C_L, γ_1), and $C_k = \Delta_k^{-1}$ has decay (C'_L, γ_C). Moreover:

(a) (Gauge covariance.) *For every real gauge transformation u on $\mathbb{T}_\eta$, replacing (W, U') by $(W^{u|_{\mathbb{T}}}, U'^u)$ replaces Δ_k by $R(u)\Delta_k R(u)^{-1}$, where $R(u)$ acts on fields on $\mathfrak{F}$ bondwise by the orthogonal maps $R(u(b_-))$. Hence C_k and $C_k^{\pm 1/2}$ are replaced by $R(u)C_kR(u)^{-1}$ and $R(u)C_k^{\pm1/2}R(u)^{-1}$.*

(b) (Lattice covariance.) *For every $r \in \mathfrak{S}_{k+1}$, replacing (W, U') by (rW, rU') replaces Δ_k, C_k and $C_k^{\pm1/2}$ by their conjugates by r_W, the map of Proposition 33.2(e), which maps fields on $\mathfrak{F}$ orthogonally onto fields on $\mathfrak{F}$.*

(c) (Complex bases.) *For complex base configurations $\hat{W} = e^{\mathrm{i}\hat{\beta}}W$ with $|\hat{\beta}|_\infty \le c'_L$, where $c'_L \le r_1/4$ is a small constant, define*

$$\Delta_k(\hat{W})(Z,Z) = \Delta^0(\hat{W})(\hat{Z}^1_{\hat{W}}Z, \hat{Z}^1_{\hat{W}}Z) + 2\langle\omega_0(\hat{W}), \hat{Z}^2_{\hat{W}}(Z)\rangle + \langle\mathfrak{G}_{\hat{W}}Z, \mathfrak{G}_{\hat{W}}Z\rangle,$$

with the objects of Proposition 33.3, the elimination of Lemma 32.7 at the base $\hat{W}$, i.e. for $\ell_c(Z) = \ell(\mathfrak{a}'(e^{\mathrm{i}Z}\hat{W})(c)\,\mathfrak{a}'(\hat{W})(c)^{-1})$, and $\mathfrak{G}_{\hat{W}}$ of Definition 22.6. Then:

- *$\Delta_k(\hat{W})$ and $C_k(\hat{W})$ are analytic in $\hat{\beta}$;*
- *$\operatorname{Re}\Delta_k(\hat{W})(Z,\bar{Z}) \ge \frac{1}{8}c_L\|Z\|^2$;*
- *the decay bounds persist, and*

$$|\partial_{\hat{\beta}(c)}\Delta_k(\hat{W})(b,b')| + |\partial_{\hat{\beta}(c)}C_k(\hat{W})(b,b')| \le C_L e^{-\gamma_C\tau(c,b,b')}.$$

Throughout the note, C_k denotes this covariance. The nonlinear remainder of the averaging in Proposition 22.9, which enters Δ^0 through Proposition 33.2(a), is written C_k^{av}.

Proof. Let $B = \hat{Z}^1Z$. Then $B \in \mathcal{Z}$, $\|B\| \ge \|Z\|$, and $\|\mathfrak{G}_W Z\| = \|\mathfrak{G}_W B\|$ (Lemma 34.1). By Proposition 33.2(b) and Theorem 34.4, Details

$$\Delta^0(B,B) + \|\mathfrak{G}_W B\|^2 \ge \langle B, (\mathbf{\Gamma}_\pi - a)B\rangle + \|\mathfrak{G}_W B\|^2 - C\varepsilon\|B\|^2 \ge (c_L - C\varepsilon)\|B\|^2 \ge \tfrac{1}{2}c_L\|Z\|^2,$$

for ε_L small. By Lemma 32.7(d) and $|\omega_0| \le C\varepsilon$, $|2\langle\omega_0, \hat{Z}^2(Z)\rangle| \le CL^{3d+1}\varepsilon\|Z\|^2 \le \frac{1}{4}c_L\|Z\|^2$. The upper bound and the decay follow from Proposition 33.2(c), the locality of $\hat{Z}^1$, $\hat{Z}^2$, and the block-locality of $\mathfrak{G}_W$. Lemma 5.5 gives the decay of C_k.

(a) By Proposition 33.2(d), Δ^0 and ω_0 are covariant. The elimination is covariant: by Lemma 21.10(b), $e^{\mathrm{i}R(\tilde u)\hat Z(Z)}W^{\tilde u} = (e^{\mathrm{i}\hat Z(Z)}W)^{\tilde u}$ satisfies the constraint for $V'^{u|_{\mathbb{T}'}}$, with $\tilde u = u|_{\mathbb{T}}$. Its values are $R(\tilde u)Z$ on $\mathfrak{F}$ and $R(\tilde u(b_0(c)_-))\zeta_c(Z)$ on $b_0(c)$, of the same size. By the uniqueness in Lemma 32.7(c), $\hat Z_{W^{\tilde u}}(R(\tilde u)Z) = R(\tilde u)\hat Z_W(Z)$, so $\hat Z^1$ and $\hat Z^2$ are covariant. By Lemma 22.7(d), $\|\mathfrak{G}_{W^{\tilde u}}R(\tilde u)Z\| = \|\mathfrak{G}_W Z\|$. Hence $\Delta_k(W^{\tilde u}, U'^u)(R(u)Z, R(u)Z) = \Delta_k(W, U')(Z, Z)$. The maps $R(u)$ are orthogonal, so the functional calculus commutes with the conjugation.

(b) The same argument, with Proposition 33.2(e) for Δ^0 and ω_0 (note $\mathfrak{S}_{k+1} \subset \mathfrak{S}_k$), with Lemma 22.7(d) for $\mathfrak{G}$, and with the covariance of the elimination under r. For the latter, $e^{\mathrm{i}r_W\hat Z(Z)}(rW) = r(e^{\mathrm{i}\hat Z(Z)}W)$ satisfies the constraint for $rV' = \mathfrak{a}'(rW)$, by Lemma 21.10(d) and Lemma 21.9. Since r maps $\mathfrak{F}$ onto $\mathfrak{F}$ and the bonds $b_0(c)$ onto the bonds $b_0(rc)$, uniqueness gives $\hat Z_{rW}(r_W Z) = r_W \hat Z_W(Z)$. All ingredients of (b) are intrinsic: the coarse action, the averaging and $\mathfrak{G}$. The comparison with $\mathbf{\Gamma}_\pi$ in the lower bound is not used for (b); $\mathbf{\Gamma}_\pi$ is not covariant under reflections at curved backgrounds (Remark 31.8).

(c) For complex bases, $e^{\mathrm{i}Z}\hat W = e^{\mathrm{i}(Z\star\hat\beta)}W$ with $Z\star\hat\beta = \ell(e^{\mathrm{i}Z}e^{\mathrm{i}\hat\beta})$ bondwise. So ℓ_c at the base $\hat W$ is $\ell(e^{\mathrm{i}\ell_c(Z\star\hat\beta)}e^{-\mathrm{i}\ell_c(\hat\beta)})$, with ℓ_c at the base W, an analytic function of $(Z, \hat\beta)$ on the bonds of $B'(c_-) \cup B'(c_+)$ for $|Z|_\infty + |\hat\beta|_\infty \le cL^{-1}$ (Lemma 32.7(a)). Its derivative with respect to $Z(b_0(c))$ differs from that at $\hat\beta = 0$ by at most $C_L|\hat\beta|_\infty$, by the Cauchy estimate. So for c'_L small the contraction of Lemma 32.7(c) works at $\hat W$, and ζ_c, $\hat Z$, $\hat Z^1$, $\hat Z^2$ at $\hat W$ are analytic in $\hat\beta$, with the bounds of Lemma 32.7. By (32.1), $W \in \mathfrak{U}_k(b)$ with $b = \frac{1}{8}\varepsilon_M$, so $8bL^2 = \frac{1}{8}r_M \le \frac{1}{4}r_M$. Lemma 22.7(f), with $\delta = 1$, $\delta_j = L$ and $\beta = \hat\beta$, therefore applies for $L|\hat\beta|_\infty \le c$: the families of $\hat W$ satisfy the hypothesis of Definition 22.6, and $|\mathfrak{G}_{\hat W}Z(y,x) - \mathfrak{G}_W Z(y,x)| \le CL^2|\hat\beta|_\infty \max|Z|$, the maximum over the bonds of $B'(y)$. The weights and transports of $\mathfrak{G}_{\hat W}$ are holomorphic in $\hat\beta$ (Lemma 22.7(a)), and $\mathfrak{G}_{\hat W}Z(y,x)$ depends only on $\hat\beta$ and Z on the bonds of $B'(y)$. So $\mathfrak{G}_{\hat W} - \mathfrak{G}_W$ is block-local with entries bounded by $CL^2|\hat\beta|_\infty$. Together with Proposition 33.3, this shows that $\Delta_k(\hat W) - \Delta_k(W)$ has decay $(C_L|\hat\beta|_\infty, \gamma_1)$ and derivatives with tree decay; the derivatives of the gauge-fixing term are block-local. The lower bound follows for c'_L small. Lemma 5.5, in the complex form of Section 13, gives the decay of $C_k(\hat W)$. Finally $\partial C_k = -C_k(\partial\Delta_k)C_k$, and $\tau(c, b_1, b_2) + |b - b_1| + |b_2 - b'| \ge \tau(c, b, b')$ gives the tree bound. □

Remark 34.7 (The constants do not depend on M). The constants c_L, C_L, C'_L, γ_1, γ_C and c'_L of Theorem 34.4 and Corollary 34.6, and the constant C_L of Theorem 34.9, do not depend on $M \ge M_0$:

- $c_L = \frac{1}{2}\min(c_D, 1)\, C_{\mathrm{P}}^{-1}$, with the absolute constant c_D of Proposition 11.5 and C_{P} of Lemma 34.2, which depends only on d and L.
- The upper bound and the decay (C_L, γ_1) of Δ_k come from Proposition 33.2(c), that is, Proposition 33.1(c) at $B = 0$, from the locality of $\hat Z^1$ and $\hat Z^2$, and from the block-locality of $\mathfrak{G}_W$ (Lemma 22.7). The constants of Proposition 33.1 come from Theorem 25.1, whose constants may be taken independent of $M \ge M_0$ (the paragraph after the theorem), from Propositions 23.3 and 22.9 (the first assumes only $(1, \varepsilon)$-regularity; the proofs of Sections 21 and 22 do not use M, see Section 46.1, Setting; Lemma 32.7 depends only on d and G), and from $r_1 = c_6/4$. The comparison of Proposition 33.2(b) uses Lemma 26.5, whose constant comes from Theorem 19.6 and does not depend on $M \ge M_0$.
- C'_L and γ_C come from these by Lemma 5.5. The radius $c'_L \le r_1/4$ and the bounds at complex bases, through Proposition 33.3, are of the same kind.
- The constant C_L of Theorem 34.9(d) comes from the proof of Lemma 34.1, on the disc $|Z|_\infty \le 2cL^{-2d-2}$, where c depends only on d, G, L and c_6, none of which involves M, and from Lemma 32.3(c).

The parameter M enters this part only through thresholds and radii: through $M_L' \geq M$; through the constant $C_{M'}$ and the factor e^{CM} of Theorem 19.10(b) in Lemma 34.3, which are absorbed by the choice of M_L' and of $\varepsilon_L \leq \varepsilon_{\mathrm{cmp}}(M_L')$; and through a_1', a_{step} and r_2, which involve the constants a_1, B_3 of Theorem 30.3 and a_δ of Proposition 31.5.

The fluctuation field lives on the unit lattice $\mathbb{T}$, where all operators are bounded. So the decay of the covariance follows from Combes–Thomas directly. In particular we do not need Bałaban's representation of the axial-gauge covariance through Landau-gauge operators on the fine lattice [7, (3.183)–(3.185)]. The fine-lattice Green functions of Parts I–III enter only through Δ^0, i.e. through the coarse effective action. Bałaban's covariance $(C^*\Delta_k C)^{-1}$ on the bonds other than the $b_0(c)$ [19, (2.5)–(2.7)] corresponds to C_k.

34.4 The determinant

The Gaussian integral over the variables on $\mathfrak{F}$ produces the factor $\det(\Delta_k)^{-1/2}$, which depends on the background. We write its logarithm as a sum of local densities.

Proposition 34.8 (Determinant densities)**.** *For real bases let*

$$\log \Delta_k = \int_0^\infty \Big(\frac{1}{1+t} - (\Delta_k + t)^{-1} \Big) dt, \qquad \ell_k(b) = \mathrm{tr}_{\mathfrak{g}} \left[(\log \Delta_k)(b,b) \right], \quad b \in \mathfrak{F}.$$

Then $\log\det \Delta_k = \sum_{b\in\mathfrak{F}} \ell_k(b)$*, and:*

(a) $\log \Delta_k$ *has decay* $(C_L, \gamma_C/2)$*; in particular* $|\ell_k(b)| \leq C_L$*;*

(b) $\ell_k(b)$ *is invariant under real gauge transformations of* (W, U')*;*

(c) for complex bases $\hat W = e^{\mathrm{i}\hat\beta} W$ *as in Corollary 34.6(c), the same integral defines* $\ell_k(b)$ *analytically, and* $|\partial_{\hat\beta(c)} \ell_k(b)| \leq C_L e^{-\gamma_C |b-c|/2}$*;*

(d) for $r \in \mathfrak{S}_{k+1}$, $\ell_k(rb; rW, rU') = \ell_k(b; W, U')$*, where* $\ell_k(-b) := \ell_k(b)$*. In particular, at the trivial base* $W = 1$, $U' = 1$, $\ell_k(b)$ *depends only on the orbit of* b *under the translations of* $\mathbb{T}'$ *and the hyperoctahedral group about the points of* $\mathbb{T}'$*.*

Proof. (a) For $t \geq 0$, $\Delta_k + t$ has decay $(C_L + t, \gamma_1)$ and is bounded below by $\frac14 c_L + t$. By Lemma 5.5 its inverse has decay $(C(c_L/4+t)^{-1}, \gamma_C)$, with γ_C independent of t, since the ratio of the lower bound to the decay constant does not degenerate as $t \to \infty$. The integrand equals $(\Delta_k - 1)(1+t)^{-1}(\Delta_k + t)^{-1}$, which has decay $(C_L(1+t)^{-1}(c_L/4+t)^{-1}, \gamma_C/2)$ and is integrable. For a positive symmetric matrix the integral is the logarithm, and $\mathrm{Tr}\log = \log\det$. Details

(b) By Corollary 34.6(a), $\log\Delta_k$ is conjugated by the orthogonal map $R(u)$, which acts bondwise. So the diagonal blocks are conjugated by orthogonal maps of $\mathfrak{g}$, and their traces are invariant.

(c) For complex bases, $\mathrm{Re}(\Delta_k + t) \geq \frac18 c_L + t$, so the integral converges. It is analytic and agrees with the logarithm for real bases. The derivative is $\int_0^\infty (\Delta_k + t)^{-1} (\partial\Delta_k)(\Delta_k + t)^{-1} dt$. By the tree bound of Corollary 34.6(c) for $\partial\Delta_k$, its diagonal kernel at b is bounded by $C_L \sum_{b_1,b_2} e^{-\gamma_C(|b-b_1|+|b_2-b|)} e^{-\gamma_1 \tau(c,b_1,b_2)} (c_L/8+t)^{-2}$, which integrates to $C_L e^{-\gamma_C|b-c|/2}$.

(d) By Corollary 34.6(b), $\log\Delta_k(rW, rU') = r_W \log\Delta_k(W,U') \, r_W^{-1}$. The map r_W sends a field supported on b to a field supported on rb, acting on the value by an orthogonal map of $\mathfrak{g}$. So the diagonal block at rb is conjugate to that at b, and the traces agree. At the trivial base, $r1 = 1$ for all r. □

Proposition 34.8 is the starting point of the analysis of Bałaban's normalization factor [9, (61)–(63)]. What remains is to show that $\ell_k(b; W) - \ell_k(b; 1)$ is a gauge-invariant function of the background near b, up to exponentially small terms, and that it is small. This requires

the comparison of Δ_k at bases that differ by a large amount far from b. It is part of the localization of the background dependence. This is carried out in Part VIII (Section 38) with Dirichlet approximants, and in Part IX with weakened random walks.

34.5 One step as a Gaussian integral

Theorem 34.9 (One small-field step)**.** *There are $a_{\text{step}} > 0$ and $r_2 > 0$, depending only on d, L, G, a, α and M, and C_L, depending only on d, L, G, a and α, with the following property. Let $k \geq 1$. Let V' be real on $\mathbb{T}'$ with curvature at most $\varepsilon_1' \leq a_{\text{step}}$ in the units of step $k+1$. Let $W = V^0$, U' and the fluctuation coordinates be as above, and let $w \leq r_2$. Let ρ be a density of the form (31.2) on $\mathfrak{U}_k(\tilde{h})$, with $a_1 \geq \tilde{h} \geq 2(C_*+1)L^{-2}\varepsilon_1' + 4w$, and let $g = g_k$. Then*

$$T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(V') = \mathcal{N}_k \exp\Big(-\frac{1}{g^2}S(U') - \frac{1}{2}\sum_{b\in\mathfrak{F}}\ell_k(b) + \sum_c \log \mathsf{j}_c\Big)$$
$$\times \int d\mu_{C_k}(Y)\, \chi^{\mathrm{w}}(\hat{Z}(gY)) \exp\Big(-\frac{(\mathcal{V}+G_3)(gY)}{g^2} + \mathfrak{J}(gY) + \mathfrak{E}(e^{\mathrm{i}\hat{Z}(gY)}W)\Big),$$

where the integrand vanishes unless $g|Y(b)| \leq w$ for all $b \in \mathfrak{F}$, as in Proposition 32.8, and:

(a) $\mathcal{N}_k = z(g)^{-|\mathbb{T}\setminus\mathbb{T}'|}(2\pi g^2)^{|\mathfrak{F}|\dim\mathfrak{g}/2}\,\sigma(0)^{|\mathfrak{F}|}\,L^{(d-1)\dim\mathfrak{g}\,|\mathcal{B}(\mathbb{T}')|}$ does not depend on the background; $|\mathcal{B}(\mathbb{T}')|$ is the number of bonds of $\mathbb{T}'$;

(b) $\mathsf{j}_c = L^{-(d-1)\dim\mathfrak{g}}\mathsf{J}_c(0) = 1 + O(L^d\mathsf{a})$ is a local function of the base near c;

(c) $d\mu_{C_k}$ is the Gaussian measure with the covariance C_k of Corollary 34.6;

(d) $\mathcal{V}$ and G_3 are analytic for complex $|Z|_\infty \leq r_2$, of third order, with $|g^{-2}(\mathcal{V}+G_3)(gY)| \leq C_L g|Y|_\infty\|Y\|^2$; G_3 is a sum of terms localized in single blocks $B'(y)$;

(e) $\mathfrak{J}(Z) = \sum_{b\in\mathfrak{F}}\log(\sigma(Z(b))/\sigma(0)) + \sum_c \log(\mathsf{J}_c(Z)/\mathsf{J}_c(0))$ is a sum of local analytic terms, bounded by $C|Z(b)|^2$ and by $CL^{2d+1}|Z|_\infty$ respectively, with $|Z|_\infty$ over $\mathfrak{F}$ (equivalently by $CL^{d+1}w$ on the support of χ^{w}, where all $|\hat{Z}(b)| \leq w$);

(f) $\mathfrak{E}(e^{\mathrm{i}\hat{Z}(Z)}W) = \mathfrak{E}(e^{\mathrm{i}\eta A_*(\hat{Z}(Z))}U')$, where $\mathfrak{E}(U)$ denotes the gauge-invariant function $\mathfrak{E}_k$ expressed through the background field.

Since $|\mathfrak{F}| = d|\mathbb{T}\setminus\mathbb{T}'|$ and $z(g) = \sigma(0)(2\pi g^2)^{\dim\mathfrak{g}/2}(1+O(g^2))$, the powers of g in $\mathcal{N}_k$ are those of $(d-1)|\mathbb{T}\setminus\mathbb{T}'|\dim\mathfrak{g}$ Gaussian variables. In $d=4$ the action S does not change under the change of units (Section 31), so the exponent $-g_k^{-2}S(U')$ keeps its form in the units of step $k+1$. The coupling g_{k+1} of the next step is not produced by the change of units; it is determined by the extraction of the marginal coefficient β from the effective density, in a later part.

Proof. Choose a_{step} so small that $4(C_*+1)L^{-2}a_{\text{step}} \leq a_1'$, and that Lemma 31.6, Lemma 32.7 (with $L^d\mathsf{a} \leq c$), Theorem 34.4 and Corollary 34.6 apply with $\varepsilon = B_3\varepsilon_1'$. Let r_2 be as in Lemma 34.1; then $r_2 \leq r_L$. By Proposition 32.8, $T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(V')$ is $z^{-|\mathbb{T}\setminus\mathbb{T}'|}$ times an integral over Z on $\mathfrak{F}$, with the weight $e^{-g^{-2}\Phi_W(Z)}$. By (32.2), the configuration $e^{\mathrm{i}\hat{Z}(Z)}W$ has curvature at most $2(C_*+1)L^{-2}\varepsilon_1' + 4w \leq \tilde{h}$ on the support of χ^{w}, which bounds all bonds including $b_0(c)$. So (31.2) applies, and $e^{-g^{-2}\Phi_W(Z)}\rho(e^{\mathrm{i}\hat{Z}}W) = \exp(-g^{-2}F(Z) + \mathfrak{E})$ with F as in Lemma 34.1. Since $\mathsf{j}_c = L^{-(d-1)\dim\mathfrak{g}}\mathsf{J}_c(0)$ by (b), write $\mathsf{J}_c(Z) = L^{(d-1)\dim\mathfrak{g}}\mathsf{j}_c\cdot\mathsf{J}_c(Z)/\mathsf{J}_c(0)$ and $\sigma(Z(b)) = \sigma(0)\,\sigma(Z(b))/\sigma(0)$. The Gaussian integral

$$\int\prod_{b\in\mathfrak{F}} dZ(b)\, e^{-\frac{1}{2g^2}\Delta_k(Z,Z)} f(Z) = (2\pi g^2)^{|\mathfrak{F}|\dim\mathfrak{g}/2}(\det\Delta_k)^{-1/2}\int d\mu_{C_k}(Y) f(gY)$$

Details

and Proposition 34.8 give the formula. (d) follows from Lemma 34.1 and Lemma 32.3(c). (e) follows from $\sigma(Z)/\sigma(0) = 1 + O(|Z|^2)$ and Proposition 32.8. (f) is gauge invariance: by Theorem 25.1(c), $U_k(e^{\mathrm{i}\hat{Z}}W)$ and $e^{\mathrm{i}\eta A_*(\hat{Z})}U'$ differ by a real gauge transformation of $\mathbb{T}_\eta$, which need not lie in $\mathfrak{G}_0$. For the count, $|\mathfrak{F}| = d|\mathbb{T}| - d|\mathbb{T}'|$, since every bond of $\mathbb{T}'$ removes one bond $b_0(c)$. □

Theorem 34.9 is stated for $k \geq 1$, because Sections 33 and 34 use the k-step operators of Parts II–V. For $k = 0$ the corresponding representation, with its own constants $a^{(0)}_{\mathrm{step}}$, $r^{(0)}_2$ and the lower bound $\frac{1}{2}C_{\mathrm{P}}^{-1}$ in place of $\frac{1}{4}c_L$, is Lemma 52.3.

Remark 34.10 (The one-loop term does not depend on the gauge fixing). The prefactor of Theorem 34.9 contains the one-loop term of the step, $-\frac{1}{2}\log\det\Delta_k + \sum_c \log \mathfrak{j}_c$, a function of U'. Up to a constant independent of V', it is the one-loop term of the intrinsic integral $T(\mathbf{1}_{\mathscr{N}} e^{-S(U_k)/g^2})(V')$ over a $\mathcal{G}'$-invariant neighbourhood $\mathscr{N} \subset \mathfrak{M} \cap \mathfrak{U}_k(a_1')$ of the minimal configuration W, which involves no gauge fixing. We take $\mathscr{N}$ inside the $\mathcal{G}'$-saturation of $\{e^{\mathrm{i}Z}W : |Z|_\infty < r_L\}$, as $\mathscr{O}$ in the proof of Lemma 54.5. On this saturation $\mathfrak{a}'$ is a submersion, by the argument of Proposition 32.8 and since $\mathfrak{a}'(V^u) = \mathfrak{a}'(V)$ for $u \in \mathcal{G}'$; so the integral has a density at V' (Section 31). The inclusion in $\mathfrak{U}_k(a_1')$ is needed: there $S(U_k)$ is defined and Lemma 31.6(c) applies, while ε_M is much larger than a_1'. The argument, in outline, is as follows. By Lemma 32.6(iv) the integral equals $T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{N}} e^{-S(U_k)/g^2})(V')$. In the coordinates of Proposition 32.8, the exponent F has the unique, nondegenerate minimum $Z = 0$: by Lemma 31.6(c), $\mathcal{S} \circ \hat{Z}$ is minimal exactly on the $\mathcal{G}'$-orbit of W, and on this orbit Φ_W vanishes only at W, by Lemma 32.3(c) and $W^u(y,x) = W(y,x)u(x)^{-1}$. Laplace's method as $g \to 0$ then gives the prefactor of Theorem 34.9, up to factors $1 + o(1)$ and the background-independent $\mathcal{N}_k$. Off the coordinate patch, the contribution is exponentially small: away from the $\mathcal{G}'$-orbit of W, $S(U_k)$ exceeds $S(U')$ by Lemma 31.6(c); on the orbit, at W^u with $|u(x) - 1| \geq w$ for a fixed radius $w > 0$, the factor of x in gf_W is at most $z^{-1}e^{-w^2/2g^2}$. The two-radius estimate in the proof of Lemma 54.5 carries out this part, and the composite version of this statement is Proposition 54.6. Any other gauge fixing gives the same function of V' up to a constant. A numerical check, for $d = 2$, $G = SU(2)$, $L = 3$ and four coarse fields, compares Laplace's method with the exponential gauge fixing over all bonds to Laplace's method in a tree gauge, which has no Faddeev–Popov determinant either. The two agree up to a constant that comes from the normalizations of the computation, as predicted for a gauge fixing without Faddeev–Popov determinant; Appendix C gives the details.

Remark 34.11 (What Parts VIII–X have to do). Theorem 34.9 is exact on the small-field set. The analysis of its Gaussian integral is the content of Parts VIII–X:

- a cluster expansion, in Part X, of the Gaussian integral of Theorem 34.9 (with the cutoff $\chi^{\mathrm{w}}(\hat{Z}(gY))$ and the exponent $-g^{-2}(\mathcal{V} + G_3)(gY) + \mathfrak{J}(gY) + \mathfrak{E}(e^{\mathrm{i}\hat{Z}(gY)}W)$), which requires a localized (random-walk) representation of C_k and of the coefficients of $\mathcal{V}$; the gauge-fixing remainder G_3 is already localized in single blocks;
- the localization of the dependence on the background of the densities $\ell_k(b)$, of $\mathfrak{j}_c$ and of the new activities;
- the normalization at $U' = 1$, which removes the vacuum energy, and the proof that the remainder is of second order in the curvature of the background.

The inputs proved here are the coercivity and decay of C_k (Corollary 34.6), the tree decay of the derivatives of the coarse action (Proposition 33.1) and of its dependence on the base (Proposition 33.3), and the analyticity of all ingredients in a complex neighbourhood.

Part VIII: Localization of the background dependence

The covariance C_k, the coarse Hessian, the determinant and the background map of the step depend on the background everywhere on the torus, and the polymer expansion of Part X needs them as sums of local pieces. Two kinds of localization occur (Section 39, Summary). In a localization of *radius type*, a piece that depends on a ball of radius m is bounded by $e^{-\gamma m}$. In one of *volume type*, a piece carries a fixed small factor for every cube of its dependence region, as [15, (116)–(122)] requires. A ball of radius m contains of order m^4 cubes, so radius type does not suffice. Part VIII builds the Dirichlet operators, the interior estimates and the graded domains; the localization by telescoping that it also proves is of radius type, and Part IX replaces it by volume type.

What this Part supplies to the main line. The status paragraphs of Sections 37 and 38 list it:

- Dirichlet versions on domains of the averaged operators $\mathbf{\Gamma}^X$, $\mathbf{H}^X$, $\mathbf{C}^X$, of the coarse Hessian and of the fluctuation operator, with uniform coercivity and decay, dependence on the background only near the domain, and comparison of domains with errors concentrated at the seam (Propositions 35.1, 37.4 and 37.6, Lemmas 37.2 and 37.5). Propositions 35.1 and 37.6 are the inputs (W2) of the walk levels (L4) and (L5) of Section 41;
- interior pointwise bounds for the Dirichlet operators, Theorem 35.2(a)–(c), used in Sections 42–47. Part III does not give them, because its bootstrap is global (Remark 35.3). Section 36 proves them by an absorption at all scales for a local elliptic system, obtained by treating two scalar potentials of the slice projection as unknowns (Lemma 36.4, Proposition 36.5);
- the local determinant densities $\ell_Y(b)$, the background and graded activities on the domains $\mathcal{U}(P;\theta_r,\delta_1,\delta_0)$ (Definitions 38.2 and 38.3), and the quadratic bound: a normalized graded activity is bounded by twice its norm times the square of a quantity $\mathfrak{s}$ built from the curvature and the current of the real part and from the complex part of the configuration, whenever $\mathfrak{s} \le \frac{1}{2}$ (Lemma 38.5). These, with Lemma 38.4 and Remarks 38.6(b) and 38.7, are the setting of the activities of Parts IX–XI.

Remark 38.8 fixes the constants of the graded domains, which do not depend on the coupling, the step, the torus or the polymer. It refers to Parts IX and X throughout and is best read with Part X.

Superseded, or off the main line. The radius-type results are not used later: Theorem 37.7 and Corollary 37.8 (Section 37.5), with Lemma 37.1 behind them, are superseded by the weakened walks of Section 41; Proposition 38.1 and Corollary 38.9 by Proposition 41.7; and Section 39 by the weakened background map of Section 45. Theorem 35.2(d) and the identity of Lemma 36.6 serve only Section 39; the main line uses from that lemma the bound on the operator W_0. Of Remark 38.6, the main line uses part (b).

Imports. From Part III: domains, the Dirichlet operators, locality and the comparison of inverses (Theorems 18.2–19.10, Lemma 17.7). From Parts IV–VII: the averaging, the action, the contraction of Section 26, Corollary 30.4, and the covariance and determinant of the step (Corollary 34.6, Proposition 34.8). The clauses on complex perturbations, in

Propositions 35.1(e) and 37.4(c) and Lemma 37.5, use Section 46 and Proposition 47.12, and Remark 38.6(d) uses Lemma 52.1; they are best read after those results.

Details for this Part. The companion, chapter on Part VIII, opens with the list of what the main line uses and a seam calculus for the comparison of domains. It expands the Dirichlet operators, the interior estimates and Theorem 35.2, the quadratic bound and the domain constants, and marks the superseded results.

35 Dirichlet operators for the averaged problem

The objects of Part VII depend on the background field everywhere on the torus. The cluster expansion and the renormalization of the next stage need them as sums of terms that depend on the background only locally, with exponential decay in the size of the localization region. This part constructs such representations. The principle is simple:

- every object is approximated by a version computed with Dirichlet operators on a domain X, which depends only on the background in a fixed neighbourhood of X;
- the approximations on increasing domains are compared, and the differences are telescoped.

The comparison of domains is the content of Theorem 19.10(a), extended here to the averaged operators $\mathbf{\Gamma}$, $\mathbf{H}$, $\mathbf{C}$ (this section), to the coarse Hessian and the fluctuation covariance (Section 37), to the determinant (Section 38), and, with interior pointwise bounds proved in Section 36, to the nonlinear background map and the interaction (Section 39). No truncation of the background itself is needed, so no gauge has to be fixed on large regions. The Dirichlet versions are covariant by construction.

Conventions for Part VIII. We keep the setting of Part VII and work in the units of step k. The background U is a real configuration on $\mathbb{T}_\eta$ that is (M,ε)-regular with $|J_U| \le \varepsilon$, and all operators are taken with the averaging pair $(Q'_k(U), \mathbf{Q}_k(U))$. In applications $U = U'$ is the background of the step, but criticality is not used except where stated. Domains are unions of unit blocks, as in Definition 17.1. For a domain X and $r \ge 0$ let $X_{(r)}$ be the union of the blocks $B(y) \subset X$ with $\operatorname{dist}(y, \mathbb{T}_\eta \setminus X) > r$, and $X^{(r)}$ the union of the blocks within distance r of X. In these definitions, and in the definitions of the seams Σ below, distances are ∞-distances. Put

$$\mathfrak{B}(X) = \{c \text{ bond of } \mathbb{T} : \ B(c_-) \cup B(c_+) \subset X\}.$$

Potentials W are allowed to have range $r_W \le 3$: $(WA)(b)$ depends on A and U on bonds within distance r_W of b. They are real for real U: $W(U)$ maps real fields to real fields. With this change the L^2 statements of Section 19 (coercivity, decay, comparison of domains) hold verbatim, with $\hat{X}$ replaced by $X^{(r_W+1)}$. Their proofs use W only through its norm, its range in A, its decay, its covariance and its reality; its range in U enters only the locality, i.e. the dependence on U only on $X^{(r_W+1)}$. The pointwise statements additionally require Hölder coefficients and are not used for such W. The potential W_2 of Section 37 on the torus depends on U non-locally, through ω_0; this is harmless for torus statements, and on domains we use its Dirichlet version W_2^X, which depends only on U on $\hat{X}$. It has range ≤ 3 in A and is real symmetric at real U, but it depends on U on all of $\hat{X}$. So existence, coercivity and decay hold for it as above, while its comparison of domains is Proposition 35.1(d) with $W \ne W'$, together with Proposition 37.4(d). We write $\hat{X}$ for $X^{(4)}$.

35.1 Definitions

Recall: the Dirichlet operators of Part III. Let X be a domain and U a regular background.

- *Scalar operators* (Section 18.2). $\mathfrak{A}'_X = \mathbf{1}_X\big(D^TD + aQ'(U)^TQ'(U)\big)\mathbf{1}_X$ on $\ell^2_0(X)$, $G'_X = (\mathfrak{A}'_X)^{-1}$, $M'_X = Q'(G'_X)^2Q'^T$, $P_{\mathcal{V},X} = G'_XQ'^T(M'_X)^{-1}Q'G'_X$, $R_X = \mathbf{1}_X - P_{\mathcal{V},X}$, the orthogonal projection onto $\mathcal{W}_X = D_X^TD\mathcal{N}_X$, and $S_X = G'_XR_XD_X^T$. Their bounds are Theorem 18.2.
- *Gauge-field operators* (Section 19.1). $\mathfrak{B}_X = \mathbf{1}_X(D^TD + W)\mathbf{1}_X$ (an operator, not the set $\mathfrak{B}(X)$ of bonds above), $\Pi_X = \mathbf{1}_X - DS_X$, and, on $\ell^2_1(X)$,

$$\mathfrak{A}^X = \Pi_X^T\mathfrak{B}_X\Pi_X + DR_XD_X^T + a\,\Pi_X^T\mathbf{Q}^T\mathbf{Q}\,\Pi_X, \qquad \mathfrak{A}^{X,\kappa} = \mathfrak{A}^X + \mathfrak{A}^{\mathrm{ext}}_X + \kappa\, m_X,$$

  with $\mathfrak{A}^{\mathrm{ext}}_X = D\mathbf{1}_{\mathbb{T}_\eta\setminus X}D^T$ and the mass profile $m_X = \sum_{j\notin J(X)}\chi_j^2$, supported near $\mathbb{T}_\eta \setminus X$. With $\mathfrak{A}^{\mathrm{ext}}_X$, all components of the gauge field satisfy Dirichlet conditions. Coercivity is Theorem 19.6; existence and decay of $\mathbf{G}^X = (\mathfrak{A}^{X,\kappa})^{-1}$ are Theorem 19.7.
- *On the torus* ($X = \mathbb{T}_\eta$, $m_{\mathbb{T}_\eta} = 0$). $\mathbf{H} = \mathbf{G}\mathbf{Q}_\pi^T\mathbf{\Gamma}_\pi$ is the minimizer map, and $\mathbf{C} = \mathbf{G} - DG'RG'D^T - \mathbf{H}\mathbf{Q}_\pi\mathbf{G}$ is the constrained Green function: $\mathbf{C}J$ is the unique A in the tangent space $\mathcal{X}(U) = \{A : \mathbf{Q}A = 0,\ RD^TA = 0\}$ of the constraint set in Bałaban's gauge with $q_U(A', A) = \mathrm{b}(A', J)$ for all $A' \in \mathcal{X}(U)$ (Theorem 16.1(b),(c), and Theorem 19.9 at regular backgrounds). The operators $\mathbf{H}^X$ and $\mathbf{C}^X$ below are their Dirichlet versions.

Let $\mathbf{G}^X = (\mathfrak{A}^{X,\kappa})^{-1}$ be the Dirichlet Green function of Theorem 19.7, built with a potential W supported in X. Let $\mathbf{Q}_X$ be $\mathbf{Q}$ restricted to $\ell^2_1(X)$ followed by the restriction to $\mathfrak{B}(X)$; for $c \in \mathfrak{B}(X)$, $(\mathbf{Q}A)(c)$ depends only on A on the bonds of $B(c_-) \cup B(c_+)$. Define, on $\ell^2(\mathfrak{B}(X))$,

$$\mathbf{\Gamma}^X = (\mathbf{Q}_X\mathbf{G}^X\mathbf{Q}_X^T)^{-1}, \qquad \mathbf{H}^X = \mathbf{G}^X\mathbf{Q}_X^T\mathbf{\Gamma}^X, \qquad \mathbf{C}^X = \mathbf{G}^X - DG'_XR_XG'_XD_X^T - \mathbf{H}^X\mathbf{Q}_X\mathbf{G}^X.$$

When the potential matters we write $\mathbf{\Gamma}^X[W]$, and so on. For $X = \mathbb{T}_\eta$ these are the operators of Theorem 19.9, because $m_{\mathbb{T}_\eta} = 0$ and $\mathbf{Q}_\pi = \mathbf{Q}$ when $E = 0$ (Lemma 15.1(c)).

Since $E = 0$, the slice projection drops out on domains as well: $\mathbf{Q}_XDS_X = D_VQ'S_X = 0$ on $\mathfrak{B}(X)$, because $Q'G'_XR_X = 0$ (Theorem 18.2(d)). The mass term κm_X near $\mathbb{T}_\eta \setminus X$ and the exterior divergence term $\mathfrak{A}^{\mathrm{ext}}_X$ destroy some of the exact identities of Section 15, e.g. $\mathbf{G}^XDR_X = DG'_XR_X$. They are not needed here: the Dirichlet operators serve only as approximants.

35.2 Bounds

Proposition 35.1 (Dirichlet averaged operators)**.** *There are $c, C, \gamma > 0$, independent of X, k, n, η and of $M \geq M_0$, such that for every domain X with $\mathfrak{B}(X) \neq \emptyset$ and every potential W of range ≤ 3, real for real U, with $\|W\| \leq \varepsilon$ and supported in X:*

(a) $c \leq \mathbf{Q}_X\mathbf{G}^X\mathbf{Q}_X^T \leq C$ *(for non-symmetric W the lower bound means* $\mathrm{Re}\langle b, \mathbf{Q}_X\mathbf{G}^X\mathbf{Q}_X^Tb\rangle \geq c\|b\|^2$*, as in (e)); $\mathbf{\Gamma}^X$ has decay (C, γ).*

(b) $\mathbf{Q}_X\mathbf{H}^X = I$*. The operators $\mathbf{H}^X$, $D\mathbf{H}^X$, $\mathbf{C}^X$, $D\mathbf{C}^X$, $\mathbf{C}^XD^T$ and $D\mathbf{C}^XD^T$ have decay (C, γ).*

(c) They depend only on U and W on the bonds of $\hat{X}$, and they are covariant under real gauge transformations on $\hat{X}$.

(d) (Comparison of domains.) *Let $X' \subset X$, let W and W' be potentials supported in X and X', and let Σ be the set of unit points within distance $10M$ of $X \setminus X'$. For $y, y' \in X'_1$*

and each $F \in \{\mathbf{\Gamma}, \mathbf{H}, \mathbf{C}\}$ *(for* $\mathbf{\Gamma}$*, and on the coarse side of* $\mathbf{H}$*, the indices are unit bonds starting at* y*,* y' *and lying in* $\mathfrak{B}(X')$*),*

$$\|\mathbf{1}_y(F^X[W] - F^{X'}[W'])\mathbf{1}_{y'}\| \le C \sum_{w\in\Sigma} e^{-\gamma(|y-w|+|w-y'|)} + \frac{C}{\varepsilon} \sum_{w\in X'_1} e^{-\gamma(|y-w|+|w-y'|)} \|W - W'\|_w,$$

where $\|W - W'\|_w$ *is the norm of* $\mathbf{1}_w(W - W')$*. The same bounds hold for* DF*, and for* $\mathbf{C}D_X^T$ *and* $D\mathbf{C}D_X^T$ *(derivatives on the left, and* D_X^T *on the right). They also hold for* $\mathbf{G}$ *itself and for* $P\mathbf{G}Q$ *with* $P \in \{1, D, D^T, \nabla\}$ *and* $Q \in \{1, D^T, D\}$*, as in Theorem 19.10(a), with the potential* W *on* X *and* W' *on* X'*, for all unit points* y, y'*. For* $\mathbf{H}$*,* $D\mathbf{H}$*,* $\mathbf{C}$ *and their derivatives the bound holds on the fine side for all unit points as well, with the operators extended by zero.*

(e) (Complex perturbations.) *Let* U_δ *be as in Proposition 19.11, with* $\delta \le \delta_0$ *and* $C\delta S_{\gamma/4} \le \frac{c}{2}$*, and let* W *extend holomorphically to* U_δ *as there. Then (a), (b) and (d) hold at* U_δ*, with the lower bound in (a) replaced by* Re*-coercivity with the constant* $\frac{c}{2}$*, and the operators are analytic in* δA*.*

Proof. (a) The upper bound follows from $\|\mathbf{G}^X\| \le C$ and $\|\mathbf{Q}\| \le C$. For the lower bound, let b be a field on $\mathfrak{B}(X)$. The trial fields of Lemma 19.8 for the bonds $c \in \mathfrak{B}(X)$ are supported in tubes inside $B(c_-) \cup B(c_+) \subset X$, so they lie in $\ell^2_1(X)$. The matrix T_U of that proof, restricted to $\mathfrak{B}(X)$, is still diagonally dominant. This gives $\Phi_b \in \ell^2_1(X)$ with $\mathbf{Q}_X\Phi_b = b$ and $\|\Phi_b\|_X \le C\|b\|$. Put $u = \mathbf{G}^X\mathbf{Q}_X^T b$. By Theorem 19.6, $\mathrm{Re}\langle b, \mathbf{Q}_X\mathbf{G}^X\mathbf{Q}_X^T b\rangle = \mathrm{Re}\langle u, \mathfrak{A}^{X,\kappa}u\rangle \ge c_*\|u\|^2_{X,+}$. On the other hand $\|b\|^2 = \langle \mathbf{Q}_X^T b, \Phi_b\rangle = \langle \mathfrak{A}^{X,\kappa}u, \Phi_b\rangle \le C\|u\|_{X,+}\|\Phi_b\|_X \le C\|u\|_{X,+}\|b\|$. Here $\langle \mathfrak{A}^{\mathrm{ext}}_X u, \Phi_b\rangle = \langle \mathbf{1}_{\mathbb{T}_\eta\setminus X} D^T u, \mathbf{1}_{\mathbb{T}_\eta\setminus X} D^T\Phi_b\rangle = 0$, since Φ_b lives on bonds with both endpoints in $B(c_-) \cup B(c_+) \subset X$. Hence $\mathrm{Re}\langle b, \mathbf{Q}_X\mathbf{G}^X\mathbf{Q}_X^T b\rangle \ge c\|b\|^2$, which is the lower bound in (a); the argument does not use the symmetry of W. Decay follows from Lemma 5.5.

(b) The identity is the definition. Decay follows from (a), Theorem 19.7(a), Theorem 18.2(b),(d) and Lemma 5.2.

(c) holds for the ingredients by Sections 18 and 19, and for $\mathbf{Q}_X$ by Proposition 22.10(d),(f).

(d) For $\mathbf{G}$ this is Theorem 19.10(a) when $W = W'$ on X'. A change of the potential is handled by the resolvent identity $\mathbf{G}^{X'}[W] - \mathbf{G}^{X'}[W'] = -\mathbf{G}^{X'}[W]\,\Pi^T_{X'}(W - W')\Pi_{X'}\,\mathbf{G}^{X'}[W']$ and the decay of both factors. For $\mathbf{Q}_X\mathbf{G}^X\mathbf{Q}_X^T$ the kernel bound follows, since $\mathbf{Q}$ is local. For $\mathbf{\Gamma}$ apply Lemma 17.7 on the unit lattice, with $\mathfrak{A} = \mathbf{Q}_X\mathbf{G}^X\mathbf{Q}_X^T$ and $\mathfrak{A}' = \mathbf{Q}_{X'}\mathbf{G}^{X'}\mathbf{Q}_{X'}^T$. The operator Δ of that lemma has two kinds of entries. For $c \in \mathfrak{B}(X')$ they are the entries of $\mathbf{Q}(\mathbf{G}^X - \mathbf{G}^{X'})\mathbf{Q}^T$, which have seam form. For $c \in \mathfrak{B}(X) \setminus \mathfrak{B}(X')$ they are entries $(\mathbf{Q}\mathbf{G}^X\mathbf{Q}^T)(c, c')$ with c within distance 2 of $X \setminus X'$, so again of seam form. For $\mathbf{H}$ and $\mathbf{C}$ use the telescoping argument of Theorem 18.2(e). With derivatives the argument is the same: Theorem 19.10(a) holds with derivatives, the factors $D\mathbf{G}^X$, $\mathbf{G}^X D_X^T$, $D\mathbf{G}^X D_X^T$, DG'_X, $G'_X D_X^T$ have decay, and $\mathbf{C}^X[W]^T = \mathbf{C}^X[W^T]$, where W^T has the properties of a potential that these bounds use (see below). Indeed $\mathfrak{A}^{X,\kappa}[W]^T = \mathfrak{A}^{X,\kappa}[W^T]$, since only $\mathfrak{B}_X$ contains W; here $\mathfrak{B}_X = \mathbf{1}_X(D^T D + W)\mathbf{1}_X$ is the operator of Section 19.1, not the set $\mathfrak{B}(X)$ of bonds. So the bound for $\mathbf{C}^X D_X^T$ follows from that for $D\mathbf{C}^X$ at W^T, since kernel bounds pass to transposes; the bound for $D\mathbf{C}^X D_X^T$ follows by applying D on the left to the telescoping identity for $\mathbf{C}^X D_X^T$. The transpose W^T is real for real U, covariant, supported in X, of norm at most ε and of range ≤ 3 in A, and it extends holomorphically with W; the bounds of (a), (b) and (d) use the potential only through these properties (Conventions). Its range in U can be larger, up to $2r_W$, since an entry $W(b', b)$ depends on U near b'. This range is not used: $\mathbf{C}^X[W^T] = \mathbf{C}^X[W]^T$ depends only on U and W on $\hat{X}$, by (c) for W. For $W \ne W'$, $\|W^T - W'^T\|_w = \|(W - W')\mathbf{1}_w\|$ is bounded by the norms $\|W - W'\|_{w'}$ of the boundedly many blocks w' near w, since $W - W'$

has range ≤ 3 in A. A block $w' \notin X'_1$ among them contributes only if $B(w') \subset X \setminus X'$; then $w' \in \Sigma$ and $\|W - W'\|_{w'} = \|\mathbf{1}_{w'} W\| \le \varepsilon$, so its term joins the first sum. For $P\mathbf{G}Q$ with $W \neq W'$, apply P and Q to the resolvent identity. The factors $P\mathbf{G}^{X'}[W]\Pi^T_{X'}$ and $\Pi_{X'}\mathbf{G}^{X'}[W']Q$ have decay, by Theorem 19.7(a), which holds for these potentials (Conventions), and Theorem 18.2(d) for $S_{X'}$. This gives the bound for $\mathbf{G}$ stated at the end of (d). For $\mathbf{H}$, $\mathbf{C}$ and their derivatives on the fine side, the telescoping uses only Theorem 19.10(a), which holds for all unit points, and the decay of all factors at all blocks; so the bound holds at all fine unit points.

The constants. They do not depend on $M \ge M_0$. The proof uses Theorems 18.2, 19.7(a) and 19.10(a), Lemma 19.8 and the locality of $\mathbf{Q}$. The constants of these results do not depend on $M \ge M_0$: M enters the Dirichlet operators only through the mass profile m_X, whose coefficient κ and coercivity constant are those of Theorem 19.6 (Remark 31.4(c)), and the seam width $10M$ enters only through Σ.

(e) Let U_δ and W be as in (e). The averages at U_δ are those of Proposition 46.4, with the real reference U, and $\mathbf{Q}_X(U_\delta) - \mathbf{Q}_X(U)$ is a local operator of size $C\delta$ (Proposition 46.4(v)). By Proposition 19.11, $\mathbf{G}^X(U_\delta)$ exists, with the decay of Theorem 19.7(a), and $\mathbf{G}^X(U_\delta) - \mathbf{G}^X(U)$ has decay $(C\delta, \gamma/4)$ by Lemma 13.3, with γ the common rate γ_0 there. So $\mathbf{Q}_X\mathbf{G}^X\mathbf{Q}_X^T$ at U_δ is $K_0 + \delta K$, where $K_0 \ge c$ is its value at U, by (a), and δK has decay $(C\delta, \gamma/4)$. By Lemma 13.3 in its unit-lattice form, for $C\delta S_{\gamma/4} \le \frac{c}{2}$, the operator $K_0 + \delta K$ is invertible, with $\mathrm{Re}\langle b, (K_0 + \delta K)b\rangle \ge \frac{c}{2}\|b\|^2$, and its inverse $\mathbf{\Gamma}^X(U_\delta)$ has decay. Then (b) holds at U_δ by the same compositions, and (d) by the same proof, since Theorem 19.10 holds at U_δ (Proposition 19.11) and Lemma 17.7 is algebraic. So (a), (b) and (d) hold at U_δ, with the lower bound in (a) replaced by Re-coercivity with the constant $\frac{c}{2}$. The operators are analytic in δA, as compositions and inverses of analytic families. □

35.3 Interior pointwise bounds

Sections 37 and 38 use Proposition 35.1 only through kernel bounds in block L^2 norms. The nonlinear constructions of Section 39 also need pointwise bounds with derivatives at interior points of a domain. They are provided by the following theorem, which is proved in Section 36. For a field h and a block $B(y)$ put

$$|h|_{1,\alpha;y} = |h|_{B(y)} + |\nabla h|_{B(y)} + [\nabla \tilde{h}_y]_{\alpha,B(y)},$$

where ∇ is the covariant gradient and $\tilde{h}_y$ is h in a real gauge in which (B) holds on $\Lambda_4(c_y)$. For a coarse field, $|B|_{y'}$ is the maximum over the unit bonds starting at y'.

Theorem 35.2 (Interior pointwise bounds)**.** *Let $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, and build the Dirichlet operators on a domain X with the potential $\mathbf{1}_X W \mathbf{1}_X$. There are $\varepsilon_{\mathrm{ID}}, C, \gamma > 0$, independent of X, k, n, η and of $M \ge M_0$, such that the following holds for $\varepsilon \le \varepsilon_{\mathrm{ID}}$, every domain X and every $y \in X_{(4)}$; the sums run over unit points y', and $|f|_{B(y')}$, $|g|_{B(y')}$ are suprema over the bonds, respectively plaquettes, that have a point in $B(y')$.*

(a) $|\mathbf{C}^X f|_{1,\alpha;y} \le C\sum_{y'} e^{-\gamma|y-y'|}|f|_{B(y')}$ *for fine fields* $f \in \ell^2_1(X)$.

(b) $|\mathbf{C}^X D_X^T g|_{1,\alpha;y} \le C\sum_{y'} e^{-\gamma|y-y'|}(|g|_{B(y')} + [\tilde{g}_{y'}]_{\alpha,\Lambda_1(c_{y'})})$ *for 2-forms g, where $D_X^T = \mathbf{1}_X D^T$.*

(c) $|\mathbf{H}^X B|_{1,\alpha;y} \le C\sum_{y'} e^{-\gamma|y-y'|}|B|_{y'}$ *for coarse fields B on $\mathfrak{B}(X)$.*

(d) $|D^T D\mathbf{H}^X Y|_{B(y)} \le C\sum_{y'} e^{-\gamma|y-y'|}|Y(y')|$ *for coarse fields Y on $\mathfrak{B}(X)$.*

(e) (Comparison of domains.) *Let $X' \subset X$, let Σ be the set of unit points within distance $10M$ of $X \setminus X'$, and put $K_\Sigma(y,y') = \sum_{w\in\Sigma} e^{-\gamma(|y-w|+|w-y'|)}$. For $y \in X'_{(4)}$ and sources*

supported on bonds and plaquettes inside X', respectively on $\mathfrak{B}(X')$, the differences between the operators of (a)–(d) on X and on X' satisfy the bounds (a)–(d) with $e^{-\gamma|y-y'|}$ replaced by $K_\Sigma(y,y')$.

Part (d), and (e) for it, use the bound $|J_U| \le \varepsilon$ on the current; the other parts do not. The theorem is stated for the real background U; (d) is not claimed at the complex backgrounds of Proposition 19.11.

What the main line uses. Parts (d) and (e) of Theorem 35.2, the next paragraph, the identity of Lemma 36.6 and the proofs of (d) and (e) in Section 36.4 are used only in Section 39, which is not on the main line. The main line uses (a)–(c), and of Lemma 36.6 only the bound on W_0 (in the proofs of Lemma 37.2 and Proposition 37.4(c)).

For $X = \mathbb{T}_\eta$ there is no mass term, and (c), (d) are contained in Theorem 19.9, Lemma 24.1 and the identity $D^T D\mathbf{H} = \mathbf{Q}^T(\mathbf{\Gamma}_\pi - a) + S^T\mathfrak{F}^T D\mathbf{H}$ of Lemma 26.2; (a) and (b) add Hölder bounds on the gradient. On a proper domain that identity fails. The mass term and the exterior divergence term break $\mathbf{G}^X DR_X = DG'_X R_X$, so $\mathbf{H}^X$ need not lie in the Dirichlet slice, and $S_X\mathbf{H}^X \neq 0$. Then $D^T D\mathbf{H}^X$ contains the term $D^T\mathfrak{F}S_X\mathbf{H}^X$, whose coefficient involves second derivatives of the background. Part (d) is a consequence of gauge invariance of the action, which replaces these second derivatives by the current (Lemma 36.6).

Remark 35.3 (Why Part III does not give Theorem 35.2 directly). The pointwise bounds of Parts II and III are proved by a *global* bootstrap (Lemmas 13.5 and 18.3). The local regularity estimate at a block y involves sup norms of the solution on neighbouring blocks with a small factor ε, and these are absorbed by taking a maximum over *all* blocks. On a domain X the maximum would include blocks adjacent to the Dirichlet cut, where no pointwise estimate is available. Comparison with torus operators (Lemma 17.7) does not help either: it produces sources at the seam that are controlled only in L^2.

Section 36 replaces the global maximum by an absorption inside one cube, at all scales. Two observations make this work:

- The small nonlocal terms of $\mathfrak{A}^{X,\kappa}$, of Riesz type, all come from the slice projection Π_X. They become local once two scalar potentials are treated as unknowns. The other nonlocal terms factor through the unit lattice and are bounded by local L^2 norms; for the averaging term this uses $E = 0$, through $\Pi_X^T\mathbf{Q}^T\mathbf{Q}\Pi_X = \mathbf{Q}^T\mathbf{Q}$.
- For local systems, lower-order terms carry positive powers of the radius, so only the principal perturbations must be small.

Two alternatives are not needed here: Bałaban's treatment of Dirichlet propagators with the random-walk expansion [3, 7], and a canonical truncation of the background in a Coulomb gauge on boxes.

36 Interior regularity for Dirichlet operators

This section proves Theorem 35.2. The pointwise bounds of Parts II and III come from a global bootstrap (Lemma 13.5), which does not work on a domain (Remark 35.3). We replace it by an absorption argument inside a single cube, at all scales. Two observations make this possible.

- The nonlocal terms of $\mathfrak{A}^{X,\kappa}$ are of two kinds. Terms that factor through the unit lattice, such as $DP_{\mathcal{V},X}D_X^T$ and $a\mathbf{Q}^T\mathbf{Q}$, are bounded pointwise by local L^2 norms. The remaining nonlocal terms are small, but of Riesz type; they all come from the slice projection Π_X. They disappear if two scalar potentials, $\sigma = S_X u$ and a dual potential τ, are

treated as additional unknowns. The triple (u, σ, τ) satisfies a *local* elliptic system with L^2-controlled sources (Lemma 36.4).

- For local systems the loss of domain in Theorem 6.3 is harmless. We apply that theorem at a radius proportional to the distance between two nested cubes and absorb with an iteration lemma (Lemma 36.1). This is the absorption technique of elliptic regularity theory, as in the hole-filling arguments of [1, Ch. V]. Lower-order terms carry positive powers of the radius, so only the principal perturbations need to be small.

No estimate near the boundary of X is used.

Conventions. We keep the setting of Part VIII. We may assume $\eta \le \frac{1}{16}$. Otherwise values, differences and Hölder quotients are bounded by block L^2 norms, as in Step 0 of the proof of Theorem 6.3, and Theorem 35.2 follows from the decay in Proposition 35.1(b). For $z \in \mathbb{R}^d$ we write $\Lambda_r = \Lambda_r(z)$. We fix a real gauge in which $U = e^{i\eta A}$ on $\Lambda_4(z)$, with A satisfying (B) there; it exists because U is (M, ε)-regular. All fields are written in this gauge, and the tilde is omitted. Then $D_\mu = \partial_\mu + T_\mu$, $\mathfrak{F}\lambda = c\lambda$ and the normal forms of Lemma 12.2 hold on Λ_3. For a set Q of lattice points, $|h|_Q$ is the supremum, $[h]_{\alpha,Q}$ the Hölder quotient and $\|h\|_Q$ the L^2 norm over Q. An equation holds *on* Λ_r if it holds at every point, respectively at every bond or plaquette starting at a point, of Λ_r. Bounds "on Λ_r" may involve values within distance 2η of Λ_r; we do not indicate this enlargement. A *source in local form* on Λ_r is a field $\mathcal{S} = F + \sum_\mu \partial_\mu^* g_\mu$, componentwise for gauge fields, and we put

$$\mathfrak{n}_r(\mathcal{S}) = |F|_{\Lambda_r} + |g|_{\Lambda_r} + [g]_{\alpha,\Lambda_r}, \qquad \mathfrak{n}_r^0(\mathcal{S}) = |F|_{\Lambda_r} + |g|_{\Lambda_r}.$$

36.1 Absorption at all scales

Lemma 36.1 (Iteration [1, Ch. V, Lemma 3.1])**.** *Let $0 < \rho_0 < R_0$, and let $\Psi \ge 0$ be bounded on $[\rho_0, R_0]$. Suppose there are $\theta \in [0, 1)$ and $A, B, \beta \ge 0$ such that*

$$\Psi(t) \le \theta\, \Psi(t') + A(t' - t)^{-\beta} + B \qquad \text{for all } \rho_0 \le t < t' \le R_0.$$

Then $\Psi(\rho_0) \le C(\theta, \beta)(A(R_0 - \rho_0)^{-\beta} + B)$.

Proof. Choose $\lambda \in (0, 1)$ with $\theta\lambda^{-\beta} < 1$, and put $t_i = \rho_0 + (1 - \lambda^i)(R_0 - \rho_0)$, so that $t_{i+1} - t_i = (1 - \lambda)\lambda^i(R_0 - \rho_0)$. By induction, Details

$$\Psi(\rho_0) \le \theta^k \Psi(t_k) + \sum_{i<k} \theta^i \Big(A(1 - \lambda)^{-\beta} \lambda^{-i\beta} (R_0 - \rho_0)^{-\beta} + B \Big).$$

Since Ψ is bounded, the first term tends to 0 as $k \to \infty$. Summing the geometric series gives the bound. □

Lemma 36.2 (Local regularity at radius s)**.** *Let $\eta \le s \le 1$, $x \in \eta\mathbb{Z}^d$, and write $\Lambda_r = \Lambda_r(x)$ in this lemma. If $-\Delta u = F + \sum_\mu \partial_\mu^* g_\mu$ on Λ_{2s}, then, with C as in Theorem 6.3,*

$$|u|_{\Lambda_s} + s|\partial u|_{\Lambda_s} + s^{1+\alpha}[\partial u]_{\alpha,\Lambda_s} \le C(s^2|F|_{\Lambda_{2s}} + s|g|_{\Lambda_{3s}} + s^{1+\alpha}[g]_{\alpha,\Lambda_{3s}} + s^{-d/2}\|u\|_{\Lambda_{2s}}),$$
$$|u|_{\Lambda_s} + s^\alpha [u]_{\alpha,\Lambda_s} \le C(s^2|F|_{\Lambda_{2s}} + s|g|_{\Lambda_{3s}} + s^{-d/2}\|u\|_{\Lambda_{2s}}).$$

Proof. The function $u_s(x') = u(x + sx')$ lives on the lattice $(\eta/s)\mathbb{Z}^d$, with spacing at most 1. It satisfies $-\Delta u_s = s^2 F_s + \sum_\mu \partial_\mu^*(s g_{\mu,s})$, with lattice derivatives at spacing η/s, where $F_s(x') = F(x + sx')$, and similarly for g. Moreover $\|u_s\|_{\Lambda_2(0)} = s^{-d/2}\|u\|_{\Lambda_{2s}}$, $\partial u_s = s(\partial u)_s$ and $[\partial u_s]_{\alpha,} = s^{1+\alpha}[\partial u]_{\alpha,}$. The proof of Theorem 6.3 uses only that the spacing is at most 1; for spacing larger than $\frac{1}{16}$ Step 0 applies. Apply (d) and (b) of that theorem to u_s. Details □

Proposition 36.3 (Interior estimates for scalar equations)**.** *There is $\varepsilon_S > 0$, depending only on d, α and $\dim \mathfrak{g}$, with the following property. For $0 < \rho < r \le 3$ there is C, depending in addition on $r-\rho$, such that for $\varepsilon \le \varepsilon_S$: if v is a scalar field with $D^T Dv = \mathcal{S}$ on Λ_r, for a source $\mathcal{S}$ in local form, then*

$$|v|_{\Lambda_\rho} + |\partial v|_{\Lambda_\rho} + [\partial v]_{\alpha,\Lambda_\rho} \le C(\mathfrak{n}_r(\mathcal{S}) + \|v\|_{\Lambda_r}),$$
$$|v|_{\Lambda_\rho} + [v]_{\alpha,\Lambda_\rho} \le C(\mathfrak{n}^0_r(\mathcal{S}) + \|v\|_{\Lambda_r}).$$

Proof. If $r-\rho < 40\eta$, both bounds follow from $|v(x)| \le \eta^{-d/2}\|v\|_{\Lambda_r}$, $|\partial v| \le 2\eta^{-1}\sup|v|$, $[w]_\alpha \le 2\eta^{-\alpha}\sup|w|$ and $\eta^{-1} \le 40(r-\rho)^{-1}$, so assume $r-\rho \ge 40\eta$. By Lemma 12.2, $-\Delta v = (F - bv) + \sum_\mu \partial^*_\mu(g_\mu - a_\mu v)$ on Λ_r. Details

First bound. Let $\Psi(t) = \sup_{x\in\Lambda_t}(|v(x)| + |\partial v(x)|)$. Let $\rho \le t < t' \le r$ and $s = (t'-t)/6$. Suppose first $s \ge 3\eta$, so that $\Lambda_{3s}(x)$, enlarged by 2η, lies in $\Lambda_{t'}$ for $x \in \Lambda_t$. Two points of a cube Q of side $\ell \le 6$ are joined by a monotone lattice path in Q, so $[v]_{\alpha,Q} \le C\ell^{1-\alpha}|\partial v|_Q$, and similarly $[a_\mu]_{\alpha,Q} \le C|\partial a_\mu| \le C\varepsilon$. Hence on $\Lambda_{3s}(x)$

$$|bv| + |a_\mu v| + s^\alpha [a_\mu v]_{\alpha,\Lambda_{3s}(x)} \le C\varepsilon\,\Psi(t').$$

The first inequality of Lemma 36.2 on $\Lambda_{3s}(x)$, divided by s, gives

$$|v(x)| + |\partial v(x)| \le C\,\mathfrak{n}_r(\mathcal{S}) + C_S\varepsilon\,\Psi(t') + Cs^{-1-d/2}\|v\|_{\Lambda_r}.$$

If $s < 3\eta$, then $|v(x)|+|\partial v(x)| \le 3\eta^{-1-d/2}\|v\|_{\Lambda_r} \le C(t'-t)^{-1-d/2}\|v\|_{\Lambda_r}$, since $\eta^{-1} \le 18(t'-t)^{-1}$. So $\Psi(t) \le C_S\varepsilon\Psi(t') + C(t'-t)^{-1-d/2}\|v\|_{\Lambda_r} + C\mathfrak{n}_r(\mathcal{S})$ for all $\rho \le t < t' \le r$. For $C_S\varepsilon \le \frac12$, Lemma 36.1 on $[\rho', r]$, $\rho' = \frac12(\rho+r)$, bounds $\Psi(\rho')$. For the Hölder quotient let $s_1 = (r-\rho)/8 \ge \eta$ and $x, x' \in \Lambda_\rho$. If $|x-x'|_\infty \le s_1$, both points lie in $\Lambda_{s_1}(x)$, and Lemma 36.2 on $\Lambda_{3s_1}(x) \subset \Lambda_{\rho'}$ bounds the quotient, with the bound on $\Psi(\rho')$ inserted. Otherwise the quotient is at most $2s_1^{-\alpha}\Psi(\rho)$.

Second bound. The same argument with $\Psi_0(t) = |v|_{\Lambda_t}$ and the second inequality of Lemma 36.2. Now $s^2|bv| + s|a_\mu v| \le C\varepsilon s\Psi_0(t')$, and for $s < 3\eta$ we use $|v(x)| \le \eta^{-d/2}\|v\|_{\Lambda_r}$. □

36.2 The local system

Lemma 36.4 (Local form of the Dirichlet equation)**.** *Let X be a domain, $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, and let $u \in \ell^2_1(X)$ satisfy $\mathfrak{A}^{X,\kappa}u = \mathcal{S}$. Put*

$$\sigma = S_X u, \qquad \tau = G'_X D_X^T \mathfrak{B}_X \Pi_X u,$$
$$\omega_1 = (M'_X)^{-1}Q'G'_X D_X^T u, \qquad \omega_2 = (M'_X)^{-1}Q'G'_X\tau, \qquad \pi_i = G'_X Q'^T\omega_i.$$

Then $Q'\sigma = 0$, and at every point, bond and plaquette at ∞-distance at least 2η from $\mathbb{T}_\eta \setminus X$:

$$D^T Du + DD^T u - D^T\mathfrak{F}\sigma + W(u - D\sigma) - D\tau + \kappa m_X u = \mathcal{S} + D\pi_1 - D\pi_2 - a\mathbf{Q}^T\mathbf{Q}u, \quad (36.1)$$
$$D^T D\sigma = D^T u - \pi_1, \quad (36.2)$$
$$D^T D\tau = \mathfrak{F}^T(Du - \mathfrak{F}\sigma) + D^T(W(u - D\sigma)) - aQ'^T Q'\tau, \quad (36.3)$$
$$D^T D\pi_i = Q'^T\omega_i - aQ'^T Q'\pi_i. \quad (36.4)$$

Moreover there are C, γ, independent of X, k, n, η, such that for every y

$$\|\sigma\|_{B(y)} + \|\tau\|_{B(y)} + \sum_i (|\omega_i(y)| + \|\pi_i\|_{B(y)}) \le C\sum_{y'} e^{-\gamma|y-y'|}(\|u\|_{B(y')} + \|Du\|_{B(y')}),$$

where $\|Du\|_{B(y')}$ is taken over the plaquettes starting in $B(y')$.

Proof. Since $E = 0$, $\mathbf{Q}\Pi_X = \mathbf{Q} - D_V Q' S_X = \mathbf{Q}$, because $Q' S_X = Q' G'_X R_X D_X^T = 0$ by Theorem 18.2(d). So the averaging term of $\mathfrak{A}^X$ is $a\mathbf{Q}^T\mathbf{Q}$. Next, $\Pi_X^T = 1 - S_X^T D_X^T$ with $S_X^T = D R_X G'_X$. Hence Details

$$\Pi_X^T \mathfrak{B}_X \Pi_X u = \mathfrak{B}_X \Pi_X u - D R_X \tau, \qquad R_X \tau = \tau - P_{\mathcal{V},X}\tau = \tau - \pi_2,$$

and $D R_X D_X^T u = D D_X^T u - D\pi_1$. Away from $\mathbb{T}_\eta \setminus X$ the projections $\mathbf{1}_X$ act trivially. There $\mathfrak{B}_X \Pi_X u = D^T D(u - D\sigma) + W(u - D\sigma)$, and $D^T D D\sigma = D^T \mathfrak{F}\sigma$. This gives (36.1). By Lemma 19.2, $\sigma \in \operatorname{ran} S_X \subset \mathcal{N}_X$, so $Q'\sigma = 0$ and $\mathfrak{A}'_X \sigma = D^T D\sigma$ there. Since $\mathfrak{A}'_X \sigma = R_X D_X^T u = D_X^T u - \pi_1$, this gives (36.2). At a point of X, D^T involves only bonds with an endpoint at that point, so $\mathbf{1}_X D^T \mathbf{1}_X D^T = \mathbf{1}_X D^T D^T = \mathbf{1}_X \mathfrak{F}^T$ on 2-forms. Hence

$$D_X^T \mathfrak{B}_X \Pi_X u = \mathbf{1}_X \mathfrak{F}^T D \Pi_X u + D_X^T \mathbf{1}_X W \Pi_X u,$$

with $D\Pi_X u = Du - \mathfrak{F}\sigma$ away from $\mathbb{T}_\eta \setminus X$. Together with $\mathfrak{A}'_X \tau = D_X^T \mathfrak{B}_X \Pi_X u$ this gives (36.3), and (36.4) is $\mathfrak{A}'_X \pi_i = Q'^T \omega_i$. For the bounds, write $\tau = G'_X \mathbf{1}_X \mathfrak{F}^T (Du - \mathfrak{F}\sigma) + G'_X D_X^T \mathbf{1}_X W(u - D\sigma)$. The operators S_X, DS_X, G'_X, $G'_X D_X^T$, $(M'_X)^{-1}$ have decay (Theorem 18.2(b)–(d)). The operators $\mathfrak{F}$, W, Q' are local and bounded, and $|Q'\lambda(y)| \le C\|\lambda\|_{B(y)}$. Compositions of operators with decay have decay (Lemma 5.2). □

Proposition 36.5 (Interior estimates for the local system)**.** *There is $\varepsilon_L > 0$, depending only on d, α, $\dim \mathfrak{g}$ and κ, with the following property. For $0 < \rho < r \le 3$ there is C, depending in addition on $r - \rho$, such that for $\varepsilon \le \varepsilon_L$ the following holds. Let $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, let m be a real function with $|m| \le 1$, and let u be a gauge field and σ, τ scalar fields with*

$$\begin{aligned} D^T Du + DD^T u - D^T \mathfrak{F}\sigma + W(u - D\sigma) - D\tau + \kappa m u &= \mathcal{S}_u, \\ D^T D\sigma - D^T u &= \mathcal{S}_\sigma, \\ D^T D\tau - \mathfrak{F}^T(Du - \mathfrak{F}\sigma) - D^T\big(W(u - D\sigma)\big) &= \mathcal{S}_\tau \end{aligned} \tag{36.5}$$

on Λ_r, for sources in local form. Put $\mathfrak{s} = \mathfrak{n}_r(\mathcal{S}_u) + \mathfrak{n}_r(\mathcal{S}_\sigma) + \mathfrak{n}_r(\mathcal{S}_\tau)$ and $\mathfrak{m} = \|u\|_{\Lambda_r} + \|\sigma\|_{\Lambda_r} + \|\tau\|_{\Lambda_r}$. Then

$$\sum_{v \in \{u,\sigma,\tau\}} \Big(|v|_{\Lambda_\rho} + |\partial v|_{\Lambda_\rho} + [\partial v]_{\alpha,\Lambda_\rho} \Big) \le C(\mathfrak{s} + \mathfrak{m}).$$

Proof. As in Proposition 36.3 we may assume $r - \rho \ge 40\eta$. Details

Step 1: normal form. Let a_μ, b be the coefficients of the normal forms of Lemma 12.2, for scalar fields and, componentwise, for $D^T D + DD^T$ on gauge fields. On 2-forms, $(D^T\omega)_\nu = \sum_\mu (\partial_\mu^* + T_\mu^T)\omega_{\mu\nu}$. Then on Λ_r, componentwise,

$$-\Delta u = F_u + \textstyle\sum_\mu \partial_\mu^* g_{u,\mu}, \qquad -\Delta\sigma = F_\sigma + \sum_\mu \partial_\mu^* g_{\sigma,\mu}, \qquad -\Delta\tau = F_\tau + \sum_\mu \partial_\mu^* g_{\tau,\mu},$$

where, with $\mathcal{S}_\bullet = F_{\mathcal{S}_\bullet} + \sum \partial^* g_{\mathcal{S}_\bullet}$,

$$\begin{aligned} F_u &= F_{\mathcal{S}_u} - bu + T^T(\mathfrak{F}\sigma) - W(u - D\sigma) + D\tau - \kappa m u, & g_{u,\mu} &= g_{\mathcal{S}_u,\mu} - a_\mu u + (\mathfrak{F}\sigma)_{\mu\cdot}, \\ F_\sigma &= F_{\mathcal{S}_\sigma} + T^T u - b\sigma, & g_{\sigma,\mu} &= g_{\mathcal{S}_\sigma,\mu} + u_\mu - a_\mu \sigma, \\ F_\tau &= F_{\mathcal{S}_\tau} + \mathfrak{F}^T(Du - \mathfrak{F}\sigma) + T^T\big(W(u - D\sigma)\big) - b\tau, & g_{\tau,\mu} &= g_{\mathcal{S}_\tau,\mu} + \big(W(u - D\sigma)\big)_\mu - a_\mu \tau. \end{aligned}$$

The coefficients a_μ, b, T, c and those of W are bounded by $C\varepsilon$, and so are their Hölder quotients; moreover $|\partial a_\mu| \le C\varepsilon$ (Lemmas 12.2, 12.3 and the definition of $\mathcal{W}_{\mathrm{loc}}$). Hence on every cube

$Q \subset \Lambda_r$ of side at most 6, with sup norms over Q (enlarged by 2η):

$$\begin{aligned}
|F_u| &\le |F_{\mathcal{S}_u}| + C\varepsilon(|u| + |\sigma| + |\partial\sigma| + |\tau|) + |\partial\tau| + \kappa|u|,\\
|g_u| + [g_u]_{\alpha,Q} &\le |g_{\mathcal{S}_u}| + [g_{\mathcal{S}_u}]_{\alpha,Q} + C\varepsilon(|u| + |\sigma| + [u]_{\alpha,Q} + [\sigma]_{\alpha,Q}),\\
|F_\sigma| &\le |F_{\mathcal{S}_\sigma}| + C\varepsilon(|u| + |\sigma|),\\
|g_\sigma| + [g_\sigma]_{\alpha,Q} &\le |g_{\mathcal{S}_\sigma}| + [g_{\mathcal{S}_\sigma}]_{\alpha,Q} + |u| + [u]_{\alpha,Q} + C\varepsilon(|\sigma| + [\sigma]_{\alpha,Q}),\\
|F_\tau| &\le |F_{\mathcal{S}_\tau}| + C\varepsilon(|u| + |\partial u| + |\sigma| + |\partial\sigma| + |\tau|),\\
|g_\tau| + [g_\tau]_{\alpha,Q} &\le |g_{\mathcal{S}_\tau}| + [g_{\mathcal{S}_\tau}]_{\alpha,Q} + C\varepsilon(|u| + |\sigma| + |\partial\sigma| + |\tau|)\\
&\quad + C\varepsilon([u]_{\alpha,Q} + [\sigma]_{\alpha,Q} + [\partial\sigma]_{\alpha,Q} + [\tau]_{\alpha,Q}).
\end{aligned}$$

On a cube Q of side $\ell \le 6$, $[v]_{\alpha,Q} \le C\ell^{1-\alpha}|\partial v|_Q$ for every v, by joining points with monotone lattice paths in Q.

Step 2: one scale. Suppose first $r \le r_*$, with $r_* \le 3$ fixed in Step 3. For $v \in \{u, \sigma, \tau\}$ let $A_v(t) = \sup_{x\in\Lambda_t}(|v(x)| + |\partial v(x)|)$. Let $\rho \le t < t' \le r$, $s = (t'-t)/12$, and suppose $s \ge 3\eta$. For $x \in \Lambda_t$ the cube $\Lambda_{9s}(x)$, enlarged by 2η, lies in $\Lambda_{t'}$. Apply the first inequality of Lemma 36.2 on $\Lambda_{3s}(x)$ to each equation, and for σ also at scale $3s$ on $\Lambda_{9s}(x)$; divide by s and use Step 1. This gives, with all A_v evaluated at t':

$$\begin{aligned}
&|\sigma(x)| + |\partial\sigma(x)| + s^\alpha[\partial\sigma]_{\alpha,\Lambda_{3s}(x)} \le C_4(A_u + \varepsilon A_\sigma) + C\big(\mathfrak{s} + s^{-1-d/2}\mathfrak{m}\big),\\
&|\tau(x)| + |\partial\tau(x)| \le C_2\varepsilon(A_u + A_\sigma + A_\tau) + C\big(\mathfrak{s} + s^{-1-d/2}\mathfrak{m}\big),\\
&|u(x)| + |\partial u(x)| \le C_3(\varepsilon + \kappa s)A_u + C_3\varepsilon A_\sigma + C_3 s A_\tau + C\big(\mathfrak{s} + s^{-1-d/2}\mathfrak{m}\big).
\end{aligned}$$

In the bound for τ, the Hölder quotient of g_τ contains $\varepsilon[\partial\sigma]_{\alpha,\Lambda_{3s}(x)}$, which is bounded by the first line. In the bound for u, the terms $D\tau$ and $\kappa m u$ belong to F_u, which enters with the factor s. The constants $C_2, C_3, C_4 \ge 1$ depend only on d, α, $\dim \mathfrak{g}$. If $s < 3\eta$, then $|v(x)| + |\partial v(x)| \le 3\eta^{-1-d/2}\|v\|_{\Lambda_r} \le C(t'-t)^{-1-d/2}\mathfrak{m}$ for each v.

Step 3: weights and absorption. Let $s_* = \big(16C_3(1+\kappa)\big)^{-1}$, $r_* = \min(3, 12s_*)$, $w_\sigma = (8C_4)^{-1}$, $w_\tau = 4C_3 s_*$, and

$$\Psi(t) = A_u(t) + w_\sigma A_\sigma(t) + w_\tau A_\tau(t).$$

For $r \le r_*$ we have $s \le s_*$, so $C_3\kappa s \le \frac{1}{16}$ and $C_3 s \le \frac{1}{4}w_\tau$. Adding the bounds of Step 2 with the weights,

$$\begin{aligned}
\Psi(t) \le \Big(C_3\varepsilon + \tfrac{3}{16} + w_\tau C_2\varepsilon\Big)A_u + (C_3 + C_4 w_\sigma + C_2 w_\tau)\varepsilon A_\sigma + \Big(\tfrac{1}{4} + C_2\varepsilon\Big)w_\tau A_\tau\\
+ C\big(\mathfrak{s} + (t'-t)^{-1-d/2}\mathfrak{m}\big).
\end{aligned}$$

For $\varepsilon \le \varepsilon_L$ small this is at most $\frac{1}{2}\Psi(t') + C(\mathfrak{s} + (t'-t)^{-1-d/2}\mathfrak{m})$. The same holds if $s < 3\eta$, by Step 2, since $w_\sigma, w_\tau \le 1$. Lemma 36.1 on $[\rho', r]$, $\rho' = \frac{1}{2}(\rho + r)$, gives $\Psi(\rho') \le C(\mathfrak{s} + \mathfrak{m})$. Since the weights are fixed positive numbers, $A_u + A_\sigma + A_\tau$ is bounded on $\Lambda_{\rho'}$.

Step 4: Hölder quotients. With these sup bounds, Step 1 bounds $F_\bullet$, $g_\bullet$ and the Hölder quotients of g_u, g_σ on $\Lambda_{\rho'}$. Lemma 36.2 at the fixed scale $s_1 = (r-\rho)/40 \ge \eta$ then bounds $[\partial\sigma]_\alpha$, on cubes $\Lambda_{s_1}(x)$, $x \in \Lambda_{(\rho+\rho')/2}$, then $[g_\tau]_\alpha$, and then $[\partial\tau]_\alpha$, and $[\partial u]_\alpha$, on cubes $\Lambda_{s_1}(x)$, $x \in \Lambda_\rho$. Pairs of points of Λ_ρ at distance larger than s_1 are handled by the sup bounds, as in Proposition 36.3.

Step 5: general r. If $r > r_*$, apply Steps 2–4 on the cubes $\Lambda_{r_1}(x)$, $x \in \Lambda_\rho$, $r_1 = \min(r_*, r-\rho)$, which lie in Λ_r. Combine the resulting bounds on $\Lambda_{r_1/2}(x)$ as in Step 4. □

The small cubes in Step 3 are needed only because of the mass term, which is not small: its contribution $\kappa s|u|$ is made small by the radius. The coupling of σ to u is not small either, but it is one-directional: σ feeds back into u and τ only with a factor ε, and this is what the weights express.

36.3 A consequence of gauge invariance

The bound on $D^T D\mathbf{H}^X$ needs pointwise control of $D^T\mathfrak{F}\lambda$. Its coefficient $\partial^* c$ involves second derivatives of the background, which are not bounded uniformly in η. Gauge invariance of the action replaces it by the current. The bound on W_0 in the next lemma is used on the main line; its identity serves only Theorem 35.2(d),(e), which are used only in Section 39.

Lemma 36.6 (Divergence of the curvature term)**.** *Let U be real and $(1,\varepsilon)$-regular with $\varepsilon \le c_5$. Let $W_0 = \nabla^2\mathcal{N}(0)$, where $\mathcal{N}$ is the remainder of Proposition 23.3 with $U_0 = U$. Then $(W_0 h)(b)$ depends only on h on bonds within distance 2η of b, and $|W_0 h(b)| \le C\varepsilon \max_{b'} |h(b')|$ over these bonds. For every scalar field λ,*

$$D^T\mathfrak{F}\lambda = -W_0 D\lambda + \mathcal{J}_U(\lambda), \qquad \mathcal{J}_U(\lambda)(b) = \tfrac{\mathrm{i}}{2}\big[J_U(b),\, \lambda(b_-) + R(U(b))\lambda(b_+)\big].$$

In particular $|D^T\mathfrak{F}\lambda(b)| \le C\varepsilon \max_{b'} |D\lambda(b')| + |J_U(b)|\big(|\lambda(b_-)| + |\lambda(b_+)|\big)$.

Proof. The operator W_0. By Proposition 23.3, $\Phi(A) = S(e^{\mathrm{i}\eta A}U)$ satisfies $\nabla\Phi(0) = J_U$ and $\nabla^2\Phi(0) = D^T D + W_0$. Differentiate the formula for $\mathcal{N}$ in the proof of that proposition twice at $A = 0$. There $\rho(0) = 0$, $d\rho(0) = \rho^{(1)}$ and $d^2\rho(0) = d^2\rho^{(2)}(0)$. The cubic part of $\mathcal{R}$ does not contribute, and the quadratic part of $\mathcal{R}$ has coefficients $O(\varepsilon^2\eta^4)$. Hence W_0 is the sum of: Details

- the operator of the form $h \mapsto \langle d^2\rho^{(2)}(0)[h,\cdot], F^s\rangle$, bounded by $C\varepsilon$ since $|F^s| \le C\varepsilon$;
- $D^T\rho^{(1)} + \rho^{(1)T}D$ and $\rho^{(1)T}\rho^{(1)}$, where $\rho^{(1)}$ is local with coefficients bounded by $C\varepsilon\eta$ and D, D^T cost at most $C\eta^{-1}$;
- the Hessian of the quadratic part of $\mathcal{R}$, bounded by $C\varepsilon^2\eta^4 \cdot \eta^{-2}$.

All are local, with the stated range.

The identity. Let λ be real and $g_t = e^{\mathrm{i}t\lambda}$. For small t and A define A_t by $e^{\mathrm{i}\eta A_t}U = (e^{\mathrm{i}\eta A}U)^{g_t}$, i.e. bondwise

$$e^{\mathrm{i}\eta A_t(b)} = e^{tP} e^{\mathrm{i}\eta A(b)} e^{-tQ}, \qquad P = \mathrm{i}\lambda(b_-), \quad Q = \mathrm{i}R(U(b))\lambda(b_+).$$

By gauge invariance of the Wilson action, $\Phi(A_t) = \Phi(A)$. Let $X_\lambda(A) = \frac{d}{dt}A_t|_{t=0}$. The Baker–Campbell–Hausdorff formula gives, to first order in t and in $Y = \mathrm{i}\eta A(b)$,

$$\log\big(e^{tP}e^{Y}e^{-tQ}\big) = Y + t(P-Q) + \tfrac{t}{2}[P+Q, Y] + O(t^2) + O(t|Y|^2).$$

Hence $X_\lambda(0) = -D\lambda$ and $dX_\lambda(0)h(b) = \frac{\mathrm{i}}{2}[\lambda(b_-) + R(U(b))\lambda(b_+), h(b)]$. Differentiating $\langle \nabla\Phi(A), X_\lambda(A)\rangle = 0$ at $A = 0$ in the direction h gives

$$-\langle (D^T D + W_0)h, D\lambda\rangle + \langle J_U, dX_\lambda(0)h\rangle = 0 \qquad \text{for all real } h.$$

Since $\mathrm{tr}(J[\mu,h]) = \mathrm{tr}([J,\mu]h)$, this says $(D^T D + W_0)D\lambda = \mathcal{J}_U(\lambda)$. Finally $D^T D D\lambda = D^T\mathfrak{F}\lambda$. Both sides are complex linear in λ, so the identity holds for complex λ. The bound follows from $|[X,Y]| \le 2|X||Y|$ and the bound on W_0. □

In the continuum this is the identity $D^*[F,\lambda] = [D^*F,\lambda] + \dots$: the divergence of the curvature is the current.

36.4 Proof of Theorem 35.2

The constants come from Proposition 35.1, Lemma 36.4, Propositions 36.3, 36.5 and 22.10(d),(e), and, for (d), Lemma 36.6; none of them depends on $M \ge M_0$. Fix a domain X and $y \in X_{(4)}$, and let $z = c_y$. Since the ∞-distance from y to $\mathbb{T}_\eta \setminus X$ exceeds 4, the cube $\Lambda_3(z)$ lies in X at ∞-distance more than $\frac{1}{2} \ge 8\eta$ from $\mathbb{T}_\eta \setminus X$. So the equations of Lemma 36.4 hold on $\Lambda_3(z)$.

We use concentric cubes $\Lambda_3 = \Lambda_{r_0} \supset \Lambda_{r_1} \supset \cdots \supset \Lambda_{r_{12}} = \Lambda_1 \supset B(y)$, with $r_j = 3 - j/6$. Each application of Proposition 36.3 or 36.5 passes from one cube to the next, and we do not record the indices. We write

$$\mathfrak{E}(f) = \sum_{y'} e^{-\gamma|y-y'|} |f|_{B(y')},$$

and similarly for 2-forms and coarse fields. Here γ may decrease from line to line, but remains independent of X, k, n, η. Two facts are used repeatedly:

- $|h|_{\Lambda_3} \le C \sum_{|y'-y|_\infty \le 4} |h|_{B(y')}$ and $\|h\|_{\Lambda_3} \le C \sum_{|y'-y|_\infty \le 4} \|h\|_{B(y')}$. Every point of $\Lambda_3 \subset X$ lies in a block $B(y') \subset X$, and these blocks have $|y' - y|_\infty \le 4$;
- for Bałaban's averaging, $|\mathbf{Q}A(c)| \le C\|A\|_{B(c_-)\cup B(c_+)}$, $|Q'\lambda(y')| \le C\|\lambda\|_{B(y')}$, and $\mathbf{Q}^T$, Q'^T are bounded from sup norms to sup norms (Proposition 22.10(d),(e)).

The basic estimate. Let $u \in \ell^2_1(X)$ satisfy $\mathfrak{A}^{X,\kappa} u = \mathcal{S}$, with $\mathcal{S}$ in local form on Λ_3, and put $\mathfrak{L}(u) = \sum_{y'} e^{-\gamma|y-y'|}(\|u\|_{B(y')} + \|Du\|_{B(y')})$. By Lemma 36.4, the L^2 norms of σ, τ, π_i on Λ_3 and the values $|\omega_i|$ at the unit points near y are bounded by $C\mathfrak{L}(u)$. By (36.4) and Proposition 36.3, $|\pi_i| + |D\pi_i| \le C\mathfrak{L}(u)$ on the next cube, and $|a\mathbf{Q}^T\mathbf{Q}u| + |aQ'^TQ'\tau| \le C\mathfrak{L}(u)$ there. So (u, σ, τ) satisfies (36.5) with $m = m_X$ and

$$\mathcal{S}_u = \mathcal{S} + D\pi_1 - D\pi_2 - a\mathbf{Q}^T\mathbf{Q}u, \qquad \mathcal{S}_\sigma = -\pi_1, \qquad \mathcal{S}_\tau = -aQ'^TQ'\tau.$$

Proposition 36.5 gives

$$\sum_{v \in \{u,\sigma,\tau\}} \Big(|v| + |\partial v| + [\partial v]_{\alpha,} \Big)_{\Lambda_{r_j}} \le C\big(\mathfrak{n}_{r_{j-2}}(\mathcal{S}) + \mathfrak{L}(u)\big). \tag{36.6}$$

In the local gauge, $\nabla u = \partial u + Tu$ with $|T|$ and $[T]_\alpha$ bounded by $C\varepsilon$. So (36.6) bounds $|u|_{1,\alpha;y}$.

(c) Let $u = \mathbf{H}^X B$. Then $\mathfrak{A}^{X,\kappa} u = \mathbf{Q}_X^T \mathbf{\Gamma}^X B$, a source with $g = 0$ and $|F|_{\Lambda_3} \le C\mathfrak{E}(B)$ by Proposition 35.1(a). Also $\mathfrak{L}(u) \le C\mathfrak{E}(B)$ by the decay of $\mathbf{H}^X$ and $D\mathbf{H}^X$ (Proposition 35.1(b)). Apply (36.6).

(a) Let $u = \mathbf{C}^X f$, and put $h_0 = G'_X D_X^T f$ and $\sigma_2 = G'_X R_X h_0$. Then $\sigma_2 \in \mathcal{N}_X$ and $D_X^T D\sigma_2 = R_X h_0$ (Theorem 18.2(d)), so $\mathfrak{A}^X D\sigma_2 = DR_X h_0$ by the domain version of Lemma 15.3(a) (Lemma 19.2). With $\mathfrak{A}^{X,\kappa}\mathbf{G}^X = 1$ and $\mathfrak{A}^{X,\kappa}\mathbf{H}^X = \mathbf{Q}_X^T\mathbf{\Gamma}^X$ this gives

$$\mathfrak{A}^{X,\kappa}\mathbf{C}^X f = f - DR_X h_0 - \kappa m_X D\sigma_2 - \mathfrak{A}_X^{\mathrm{ext}} D\sigma_2 - \mathbf{Q}_X^T\mathbf{\Gamma}^X\mathbf{Q}_X\mathbf{G}^X f.$$

The term $\mathfrak{A}_X^{\mathrm{ext}} D\sigma_2$ lives on the bonds crossing ∂X, so it vanishes on Λ_3 and does not enter the estimate. We put the remaining terms in local form.

- $D^T D h_0 = D^T f - aQ'^TQ'h_0$ in the interior, with $D^T f = \partial^* f + T^T f$. The second bound of Proposition 36.3 gives $|h_0| + [h_0]_{\alpha,} \le C\mathfrak{E}(f)$, using $\|h_0\|_{B(y')} \le C \sum e^{-\gamma|y'-y''|} |f|_{B(y'')}$ by the decay of $G'_X D_X^T$.
- $Dh_0 = \partial h_0 + Th_0$ and $\partial_\mu h_0 = -\partial_\mu^*(S_\mu h_0)$. So Dh_0 is a source in local form, with g bounded by $|h_0|$ and $[h_0]_{\alpha,}$.
- $R_X h_0 = h_0 - \pi_3$ with $\pi_3 = G'_X Q'^T (M'_X)^{-1} Q' G'_X h_0$. As for π_i, $|\pi_3| + |D\pi_3| \le C\mathfrak{E}(f)$.
- $D^T D\sigma_2 = h_0 - \pi_3$, since $Q'\sigma_2 = 0$. The first bound of Proposition 36.3 gives $|D\sigma_2| \le C\mathfrak{E}(f)$, using $\|\sigma_2\|_{B(y')} \le C \sum e^{-\gamma|y'-y''|} |f|_{B(y'')}$ by the decay of $G'_X R_X G'_X D_X^T$.
- $|\mathbf{Q}_X^T\mathbf{\Gamma}^X\mathbf{Q}_X\mathbf{G}^X f| \le C\mathfrak{E}(f)$, by the decay of $\mathbf{\Gamma}^X$ and $\mathbf{G}^X$.

Also $\mathfrak{L}(\mathbf{C}^X f) \le C\mathfrak{E}(f)$ by the decay of $\mathbf{C}^X$ and $D\mathbf{C}^X$. Apply (36.6).

(b) Let $u = \mathbf{C}^X D_X^T g$. In the same way,

$$\mathfrak{A}^{X,\kappa} u = D_X^T g - DR_X h_3 - \kappa m_X D\sigma_3 - \mathfrak{A}_X^{\mathrm{ext}} D\sigma_3 - \mathbf{Q}_X^T\mathbf{\Gamma}^X\mathbf{Q}_X\mathbf{G}^X D_X^T g,$$

where again $\mathfrak{A}^{\mathrm{ext}}_X D\sigma_3$ vanishes on Λ_3, with $h_3 = G'_X D_X^T D_X^T g = G'_X \mathbf{1}_X \mathfrak{F}^T g$ and $\sigma_3 = G'_X R_X h_3$. In the interior $D_X^T g = \partial^* g + T^T g$, a source in local form whose norm involves $[g]_{\alpha,\Lambda_3}$. Two points of Λ_3 at ∞-distance at most $\frac{1}{2}$ lie in a common cube $\Lambda_1(c_{y'})$, so $[g]_{\alpha,\Lambda_3} \le C \sum_{|y'-y|_\infty \le 4} (|g|_{B(y')} + [\tilde{g}_{y'}]_{\alpha,\Lambda_1(c_{y'})})$. Here the gauge of z and that of y' differ on $\Lambda_1(c_{y'})$ by a gauge transformation whose differences are bounded by $C\varepsilon$, which changes Hölder quotients by at most $C\varepsilon|g|$. Next, $D^T D h_3 = \mathfrak{F}^T g - a Q'^T Q' h_3$ is bounded, so $|h_3| + |Dh_3| \le C\mathfrak{E}(g)$ by Proposition 36.3, with the L^2 input from the decay of $G'_X \mathbf{1}_X \mathfrak{F}^T$ (and of $G'_X R_X$ for σ_3), and $DR_X h_3$ is a bounded source. The remaining terms are as in (a), with the decay of $\mathbf{G}^X D_X^T$, $\mathbf{C}^X D_X^T$ and $D\mathbf{C}^X D_X^T$.

(d) Let $u = \mathbf{H}^X Y$, and let $\sigma, \tau, \pi_i, \omega_i$ be as in Lemma 36.4. By (c) and (36.6), u, σ, τ and their first derivatives are bounded by $C\mathfrak{E}(Y)$ on Λ_{r_2}. Put $\theta = D_X^T u$, so that $DD^T u = D\theta$ in the interior. Apply D^T to (36.1). By $D^T D^T = \mathfrak{F}^T$, $D^T(D^T \mathfrak{F}\sigma) = \mathfrak{F}^T \mathfrak{F}\sigma$, and by (36.3),

$$\mathfrak{F}^T D u - \mathfrak{F}^T \mathfrak{F}\sigma + D^T(W(u - D\sigma)) - D^T D\tau = a Q'^T Q' \tau.$$

The curvature terms and the potential terms cancel exactly. With $E = 0$, $D^T \mathbf{Q}^T = (\mathbf{Q}D)^T = Q'^T D_V^T$. Together with (36.4) this gives

$$D^T D\theta = Q'^T D_V^T(\mathbf{\Gamma}^X Y - a\mathbf{Q}u) + Q'^T(\omega_1 - \omega_2) - a Q'^T Q'(\pi_1 - \pi_2 + \tau) - \kappa D^T(m_X u).$$

Here $D^T(m_X u) - m_X \theta$ consists of terms $(\partial^* m_X) \cdot (\text{shifted } u)$ and $(\text{differences of } m_X) \cdot T^T(\text{shifted } u)$, so it is bounded by $C(|u| + |\nabla u|)$, since $|\partial m_X| \le C$. So the right-hand side is bounded by $C\mathfrak{E}(Y)$ on Λ_{r_2}. Proposition 36.3, with $\|\theta\|_{\Lambda_{r_2}} \le C|\nabla u|_{\Lambda_{r_2}}$, gives $|D\theta| \le C\mathfrak{E}(Y)$ on Λ_{r_3}. Now solve (36.1) for $D^T D u$:

$$D^T D u = \mathbf{Q}^T \mathbf{\Gamma}^X Y + D\pi_1 - D\pi_2 - a\mathbf{Q}^T \mathbf{Q} u - D\theta + D^T \mathfrak{F}\sigma - W(u - D\sigma) + D\tau - \kappa m_X u.$$

Every term is bounded by $C\mathfrak{E}(Y)$ on Λ_{r_3}. For $D^T \mathfrak{F}\sigma$ this is Lemma 36.6, with $|J_U| \le \varepsilon$. This proves (d).

(e) Let $X' \subset X$, $y \in X'_{(4)}$ and $z = c_y$. Then $\Lambda_3(z)$ is interior to both domains. Consider one of the operators, say $u = \mathbf{C}^X f$ and $u' = \mathbf{C}^{X'} f$ for f supported in X', and let δ denote differences of corresponding quantities on X and X': $\delta u = u - u'$, $\delta\sigma$, $\delta\tau$, $\delta\pi_i$, $\delta\omega_i$, δh_0, $\delta\sigma_2$, and so on. Write K_Σ-sums for bounds of the form $C \sum_{y'} K_\Sigma(y, y') |f|_{B(y')}$.

L^2 differences. Every operator in the construction satisfies the comparison of domains in block L^2 norms, with kernels of seam form, also when composed with D on the left. This covers G', $G'D^T$, $DG'D^T$, S, R, $P_{\mathcal{V}}$, $(M')^{-1}$ (Theorem 18.2(e)), $\mathbf{G}$ (Theorem 19.10(a)), and $\mathbf{\Gamma}$, $\mathbf{H}$, $\mathbf{C}$ (Proposition 35.1(d)). The same holds with D on the left and D_X^T on the right. The local operators $\mathbf{1}_X$, D_X^T, $\mathfrak{B}_X$ differ from their versions on X' only within distance 1 of $X \setminus X'$. These extensions are recorded after Proposition 35.1. For blocks $B(y') \not\subset X'$ we have $\delta u = u$ there, and y' lies in Σ, so the decay of the operators on X alone gives seam form. By telescoping, $\mathfrak{L}(\delta u)$, the L^2 norms of all δ-quantities near y, and $|\delta\omega_i|$ near y are K_Σ-sums.

The difference system. On $\Lambda_3(z)$ the equations of Lemma 36.4 for X and for X' have the same local operators, except for the mass terms. Hence $(\delta u, \delta\sigma, \delta\tau)$ satisfies (36.5) with $m = m_X$. Its sources are the differences of the sources, together with $-\kappa(m_X - m_{X'})u'$. The scalar differences satisfy local equations: for instance $D^T D\delta h_0 = -a Q'^T Q' \delta h_0$, because f is the same, and $D^T D\delta\pi_i = Q'^T \delta\omega_i - a Q'^T Q' \delta\pi_i$. So Proposition 36.3 bounds them pointwise, with derivatives, by K_Σ-sums. The only new terms are the mass differences $\kappa(m_X - m_{X'})u'$ and, in (a), $\kappa(m_X - m_{X'})D\sigma'_2$. We have $m_X - m_{X'} = -\sum_{j \in J(X) \setminus J(X')} \chi_j^2$. For such j, $\Lambda_j \subset X$ but $\Lambda_j \not\subset X'$, so Λ_j meets $X \setminus X'$; here Λ_j is the union of the blocks that meet $\{x : |x - z_j|_\infty \le 3M\}$ (Section 17.2), not a cube $\Lambda_r(z)$ of radius r. Its ∞-diameter is at most $6M + 2$, so $m_X - m_{X'}$ vanishes outside the $8M$-neighbourhood of $X \setminus X'$. At a point $x \in \Lambda_3(z)$ where it does not vanish, there is $w \in \Sigma$ with $|x - w| \le \sqrt{d}$, and $|u'(x)| + |D\sigma'_2(x)| \le C\mathfrak{E}(f)$ by the bounds

already proved for X', applied on $\Lambda_{r_6}(z)$; the difference argument then runs on the cubes $\Lambda_{r_6} \supset \cdots \supset \Lambda_{r_{12}}$. Since $|y-x| \le 4\sqrt{d}$,

$$e^{-\gamma|y-y'|} \le e^{2\gamma(|y-x|+|x-w|)} e^{-\gamma(|y-w|+|w-y'|)},$$

so these terms are K_Σ-sums as well. Proposition 36.5, applied to the differences on the inner cubes, proves (e) for (a). The other operators are treated in the same way; for (d), apply D^T to the difference system as in the proof of (d), and use Lemma 36.6 for $D^T \mathfrak{F} \delta\sigma$. □

37 Localization of the Hessian and of the fluctuation covariance

Status. The localization of this section and of Section 39 is of radius type (Section 39, Summary). The polymer expansion of Part X needs volume type, which Part IX provides. The main line uses from this section Lemma 37.2, Definition 37.3, Proposition 37.4, Lemma 37.5 and Proposition 37.6. Theorem 37.7 and Corollary 37.8 (Section 37.5) are superseded by the weakened representations of Section 41 and are not used later. Lemma 37.1 serves only these results and the radius-type results of Sections 38 and 39, so it is not on the main line either.

37.1 The telescoping principle

For a finite set S of unit points and $m \ge 0$ let $\Lambda_m(\mathsf{S})$ be the union of the blocks $B(y)$ with $\operatorname{dist}(y, \mathsf{S}) \le m$, and let m_{S} be the least m with $\Lambda_m(\mathsf{S}) = \mathbb{T}_\eta$.

Lemma 37.1 (Telescoping)**.** *Let F^X be numbers attached to the domains $X = \Lambda_m(\mathsf{S})$, $0 \le m \le m_{\mathsf{S}}$, with $F^{\mathbb{T}_\eta} = F$ and $|F^{\Lambda_m(\mathsf{S})} - F^{\Lambda_{m-1}(\mathsf{S})}| \le K e^{-\gamma m}$. Then*

$$F = \sum_{m=0}^{m_{\mathsf{S}}} F_m, \qquad F_0 = F^{\Lambda_0(\mathsf{S})}, \quad F_m = F^{\Lambda_m(\mathsf{S})} - F^{\Lambda_{m-1}(\mathsf{S})}, \qquad |F_m| \le K e^{-\gamma m} \ (m \ge 1).$$

If each F^X is a function of the background that depends only on its restriction to a set $\mathcal{R}(X)$, is analytic, and is invariant (or covariant) under real gauge transformations, and if $\mathcal{R}(\Lambda_{m-1}(\mathsf{S})) \subset \mathcal{R}(\Lambda_m(\mathsf{S}))$ for $1 \le m \le m_{\mathsf{S}}$, then F_m has the same properties with $\mathcal{R}(\Lambda_m(\mathsf{S}))$.

The proof is the identity of the telescoping sum. By the inclusion, $F^{\Lambda_m(\mathsf{S})}$ and $F^{\Lambda_{m-1}(\mathsf{S})}$ both depend only on the background on $\mathcal{R}(\Lambda_m(\mathsf{S}))$, and so does F_m. Theorem 37.7, Proposition 38.1 and Theorem 39.4 are applications of this lemma; Corollary 39.5 uses the same telescoping identity directly. In these three applications $\mathcal{R}(X)$ is $\widehat{X^+}$ (Theorem 37.7, Proposition 38.1) or $\hat{X}$ together with the bonds of $\mathfrak{B}(X)$ for the coarse field (Theorem 39.4); these sets increase with X, so the inclusion holds. The estimates needed are comparison bounds for approximants on nested domains, whose differences have seam form.

37.2 The coarse Hessian as an operator of Part II

In this section W has two meanings, each fixed where it occurs: the base configuration $W = U^{(k)}$ of the step (Section 31.3), and a potential, as in $\mathbf{G}[W]$, W_2 and W_2^X.

Let $\mathbf{H}_0 = \mathbf{H}[0]$ be the minimizer map without potential, and for a regular background U put

$$\omega_0 = \mathbf{H}_0^T J_U, \qquad \langle A, W_2 A\rangle = d^2\mathcal{N}(0)[A,A] - \langle \omega_0, d^2 C_k^{\mathrm{av}}(0)[A,A]\rangle,$$

with $\mathcal{N}$ the nonlinear part of the action at U (Proposition 23.3) and C_k^{av} that of the gauge-adapted averaging (Proposition 22.9).

Lemma 37.2 (The Hessian is $\mathbf{\Gamma}_\pi[W_2] - a$)**.** *$W_2$ is a real symmetric bounded operator with $\|W_2\| \le C\varepsilon$ and $\|\mathbf{1}_y W_2 \mathbf{1}_{y'}\| \le C\varepsilon e^{-\gamma|y-y'|}$. If U is critical, i.e. $U = U'$ is the background of the k-step problem for $W = U^{(k)}$, then ω_0 is its multiplier and*

$$\Delta^0 = \mathbf{\Gamma}_\pi[W_2] - a,$$

where Δ^0 is the Hessian of Proposition 33.2 and $\mathbf{\Gamma}_\pi[W_2]$ is the operator of Theorem 19.9 with the potential W_2.

Proof. At $A = 0$ the second derivative $d^2\mathcal{N}(0)$ is a sum of plaquette-local bilinear forms. By the first part of Lemma 36.6, it is a local operator with $|d^2\mathcal{N}(0)[h,h']| \le C\varepsilon\|h\|\,\|h'\|$; its proof is the computation of the proof of Proposition 23.3(c) at $A = 0$, with the lattice trick (23.2). By Proposition 22.9(c), $|d^2C_k^{\mathrm{av}}(0)[h,h](c)| \le C\|h\|^2_{S_c}$. By Cauchy–Schwarz on blocks, as in the proof of Proposition 37.4(b), $|\omega_0(c)| \le \sum_{y'} \|\mathbf{1}_{y'}\mathbf{H}_0\mathbf{1}_c\|\,\|J_U\|_{B(y')} \le C\varepsilon$, by the block decay of $\mathbf{H}_0$ (Theorem 19.9) and $|J_U| \le \varepsilon$. Hence the bounds. The coefficients $\omega_0(c)$ depend on U through the decaying kernel of $\mathbf{H}_0^T$. Details

If U is critical, the Euler–Lagrange equation at $B = 0$ gives $J = \mathbf{Q}^T\omega$. So $\mathbf{H}_0^T J = (\mathbf{Q}\mathbf{H}_0)^T\omega = \omega$. By Proposition 33.2(a), $\Delta^0(B,B) = \min\{\|DA\|^2 + \langle A, W_2A\rangle : \mathbf{Q}A = B,\ RD^TA = 0\}$. By Theorem 16.1(b) with the potential W_2, the minimizer is $\mathbf{H}[W_2]B$. On the constraint set $\langle A, \mathfrak{A}_\pi A\rangle = \|DA\|^2 + \langle A, W_2A\rangle + a\|B\|^2$. The minimum of $\langle A, \mathfrak{A}_\pi A\rangle$ over $\{\mathbf{Q}A = B\}$ is $\langle B, \mathbf{\Gamma}_\pi B\rangle$ and is attained at $\mathbf{H}[W_2]B$. Only the L^2 properties of W_2 enter: the coercivity of Theorem 19.6 uses W only through $\|W\|$, and the variational characterization is algebraic. □

37.3 Dirichlet Hessians

Definition 37.3 (Dirichlet Hessian)**.** For a domain X put

$$\omega_0^X = (\mathbf{H}_0^X)^T\mathbf{1}_X J_U \quad \text{on } \mathfrak{B}(X),$$
$$\langle A, W_2^X A\rangle = d^2\mathcal{N}(0)[\mathbf{1}_X A, \mathbf{1}_X A] - \langle \omega_0^X, d^2C_k^{\mathrm{av}}(0)[A,A]\rangle_{\mathfrak{B}(X)},$$

for $A \in \ell^2_1(X)$, and

$$\Delta^{0,X} = \mathbf{\Gamma}^X[W_2^X] - a \qquad \text{on } \ell^2(\mathfrak{B}(X)).$$

Proposition 37.4 (Dirichlet Hessians)**.** *(a) For $X = \mathbb{T}_\eta$: $\omega_0^{\mathbb{T}_\eta} = \omega_0$ and $\Delta^{0,\mathbb{T}_\eta} = \mathbf{\Gamma}_\pi[W_2] - a$. This is Δ^0 if U is critical.*

(b) $|\omega_0^X| \le C\varepsilon$, $\|W_2^X\| \le C\varepsilon$, and $\Delta^{0,X}$ has decay (C,γ), uniformly in X.

(c) ω_0^X, W_2^X and $\Delta^{0,X}$ depend only on U on $\hat{X}$ and are covariant under real gauge transformations. They are analytic under complex perturbations of U as in Proposition 19.11. The potential W_2^X is real symmetric at real U and has range ≤ 3 in A, but it depends on U on all of $\hat{X}$. It satisfies the hypotheses of Proposition 19.11 for existence, Re*-coercivity and decay; its comparison of domains is (d). The analyticity and the changes under complex perturbations come from:*

- *$d^2\mathcal{N}(0)$ and the kernel of $d^2C_k^{\mathrm{av}}(0)$ are local and analytic in the background (Propositions 23.3 and 22.9, and Proposition 46.4 at complex references);*
- *$\langle b, \omega_0^X\rangle = \langle F_U^s, D\mathbf{H}_0^X b\rangle$, with $D\mathbf{H}_0^X$ analytic by Proposition 19.11 at $W = 0$;*
- *the changes of size $C\delta$ follow from Cauchy's estimate on a ball of twice the radius.*

(d) For $X' \subset X$ and $c, c' \in \mathfrak{B}(X')$,

$$|\omega_0^X(c) - \omega_0^{X'}(c)| \le C\varepsilon \sum_{w\in\Sigma} e^{-\gamma|c-w|},$$
$$|\Delta^{0,X}(c,c') - \Delta^{0,X'}(c,c')| \le C \sum_{w\in\Sigma} e^{-\gamma'(|c-w|+|w-c'|)},$$

with Σ as in Proposition 35.1(d) and some $\gamma' > 0$.

Proof. (a)–(c) follow from Proposition 35.1 and Lemma 37.2. The bound $|\omega_0^X| \le C\varepsilon$ needs no pointwise estimate. By Cauchy–Schwarz on blocks, $|\omega_0^X(c)| \le \sum_{y'} \|\mathbf{1}_{y'}\mathbf{H}_0^X\mathbf{1}_c\|\,\|J_U\|_{B(y')} \le C\varepsilon\sum_{y'} e^{-\gamma|c-y'|}$, using only the block kernel bounds of $\mathbf{H}_0^X$ and $|J_U| \le \varepsilon$. (d) The bound for ω_0 follows from Proposition 35.1(d) for $\mathbf{H}_0$, since J_U is local and $|J_U| \le \varepsilon$. Then $W_2^X - W_2^{X'}$ on X' is bounded near w by $C\varepsilon\sum_{s\in\Sigma} e^{-\gamma|w-s|}$. The $d^2\mathcal{N}$ parts differ only on bonds within distance 1 of $X \setminus X'$. Insert this into Proposition 35.1(d) for $\mathbf{\Gamma}$. Since $\sum_w e^{-\gamma(|c-w|+|w-c'|)}e^{-\gamma|w-s|} \le Ce^{-\gamma'(|c-s|+|s-c'|)}$, the result has seam form. Details

Complex perturbations in (c). Let U_δ be as in Proposition 19.11, of size $\delta < 2\delta_*$, with δ_* fixed at the end of this paragraph. In a local real gauge of U, $U_\delta = e^{i\eta A_0}$ with A_0 satisfying (B) with parameter $C(\varepsilon+\delta)$ (Section 46). This uses the bound on $\partial\delta A$: with a sup bound alone, $H_p - 1$ would only be $O(\eta\delta)$. Hence $|H_p(U_\delta) - 1| \le C\eta^2(\varepsilon+\delta)$, $\|R(U_\delta(b))^{\pm1}\| \le 2$ and $|F^s_{U_\delta}| \le C(\varepsilon+\delta)$ (step (v) of the proof of Proposition 46.7). The proof of the bound on W_0 in Lemma 36.6 uses only these three bounds, the transports being bounded by 2 instead of 1. So $d^2\mathcal{N}_{U_\delta}(0)$ is a local operator of L^2 norm at most $C(\varepsilon+\delta)$. Read ω_0^X through the identity $\langle b, \omega_0^X\rangle = \langle F^s, D\mathbf{H}_0^X b\rangle$ of (c), and use Proposition 46.4(iii),(vi) for $d^2C_k^{\mathrm{av}}(0)$. Then $W_2^X(U_\delta)$ is holomorphic in δA, of range ≤ 3, with $\|W_2^X(U_\delta)\| \le C(\varepsilon+\delta)$, for every δA of size $< 2\delta_*$. The Schwarz lemma on the complex lines $t \mapsto U_{t\,\delta A}$, $|t| < 2\delta_*/\delta$, gives $\|W_2^X(U_\delta) - W_2^X(U)\| \le C(\varepsilon+2\delta_*)\,\delta/\delta_*$. For $\delta \le \delta_*$ this is at most $C(\varepsilon+2\delta_*)$. So W_2^X satisfies the range-3 clause of Proposition 19.11 at these perturbations once $C'(\varepsilon+2\delta_*) \le c$, with c the stability threshold of Lemma 13.2. Here $\delta_* > 0$ is fixed once, depending on d, L, G, α and M, for all ε with $\varepsilon \le \frac12\min(\varepsilon_0, c_3')$, $C\varepsilon \le \frac12 c_5$ and $C'\varepsilon \le \frac12 c$. It satisfies $2C\delta_* \le \frac12\min(\varepsilon_0, c_3')$, $2C\delta_* \le \frac12 c_5$, $2C'\delta_* \le \frac12 c$, and the radius conditions of Propositions 19.11 (with $W = 0$), 46.4 and 35.1(e) at size $2\delta_*$. With $C'\varepsilon \le \frac12 c$, a threshold of this section, the clause holds for all $\delta \le \delta_*$. Moreover $\|W_2^X(U_\delta) - W_2^X(U)\| \le C_W\delta$ with $C_W = C(1+2\delta_*)/\delta_*$, a fixed constant. The constant c_L'' of Lemma 37.5 is chosen with $C_{\mathrm{cmp}}c_L'' \le \delta_*$, with $C_{\mathrm{cmp}} = 4$ of Section 46. □

37.4 The fluctuation operator on domains

We return to the step of Part VII. Its operator Δ_k acts on fields on the set $\mathfrak{F}$ of bonds of $\mathbb{T}$ other than the bonds $b_0(c)$ (Definition 32.1). By Lemma 34.1 it consists of two parts:

- the coarse-action part $\Delta^0(\hat{Z}^1\cdot, \hat{Z}^1\cdot) + 2\langle\omega_0, \hat{Z}^2\rangle$, built with the elimination maps $\hat{Z}^1$, $\hat{Z}^2$ of Lemma 32.7. These depend only on $W = U^{(k)}$ within distance $2L$, and $\hat{Z}^1\delta_b$ is supported within distance $2L$ of b;
- the gauge-fixing part $\langle\mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle$, with $\mathfrak{G}_W$ the linearized averaged contour variables at the base (Definition 22.6). By Lemma 22.7(a), $\mathfrak{G}_W Z(y,x)$ depends only on Z and W on the inner bonds of the block $B'(y)$. So the kernel of $\mathfrak{G}_W^T\mathfrak{G}_W$ on fields on $\mathfrak{F}$ vanishes unless b and b' are inner bonds of a common block; it is block-diagonal, with range at most L.

For a domain X let $\mathbb{T}'(X)$ be the set of the $y \in \mathbb{T}'$ with $B'(y) \subset X$, and put

$$\langle Z, K^X Z'\rangle = \sum_{y\in\mathbb{T}'(X)} \sum_{x\in B'(y)} \langle\mathfrak{G}_W Z(y,x), \mathfrak{G}_W Z'(y,x)\rangle, \qquad \text{i.e.} \quad K^X = \mathfrak{G}_W^T\mathbf{1}_{\mathbb{T}'(X)}\mathfrak{G}_W,$$

on fields on $\mathfrak{F}$, with the bilinear pairing of Section 32. A block contributes to K^X if and only if it lies in X; blocks that meet $\mathbb{T}_\eta \setminus X$ contribute nothing. For $X = \mathbb{T}_\eta$, $K^{\mathbb{T}_\eta} = \mathfrak{G}_W^T \mathfrak{G}_W$. For a domain X and $b, b' \in \mathfrak{F}$ define the *Dirichlet entries*

$$\Delta_k^{(X)}(b,b') = \Delta^{\mathrm{c,X}}(b,b') + K^X(b,b'),$$

where $\Delta^{\mathrm{c,X}}(b,b') = \Delta^{0,X}(\hat{Z}^1\delta_b, \hat{Z}^1\delta_{b'}) + (\text{polarization of } Z \mapsto 2\langle \omega_0^X, \hat{Z}^2(Z)\rangle)(b,b')$ if $\hat{Z}^1\delta_b$ and $\hat{Z}^1\delta_{b'}$ are supported in $\mathfrak{B}(X)$, and $\Delta^{\mathrm{c,X}}(b,b') = 0$ otherwise. For $X = \mathbb{T}_\eta$ these are the entries of Δ_k (Lemma 34.1), at critical U (Proposition 37.4(a)). The entries $K^X(b,b')$ depend only on U on $\hat{X}$, are covariant under real gauge transformations (Lemma 22.7(d)), and are analytic under the complex perturbations of U considered below (Lemma 22.7(a),(f)).

Fix a margin $R_0 \geq 2L+2$. For a domain Y let $Y^+ = Y^{(R_0)}$, let $\mathfrak{F}(Y)$ be the set of bonds of $\mathfrak{F}$ starting in Y, and define the *local fluctuation operator*

$$\Delta_Y = \mathbf{1}_{\mathfrak{F}(Y)} \Delta_k^{(Y^+)} \mathbf{1}_{\mathfrak{F}(Y)} \qquad \text{on } \ell^2(\mathfrak{F}(Y)).$$

An inner bond starting in Y lies in a block $B'(y) \subset Y^{(L)} \subset Y^+$. Hence the gauge-fixing part of Δ_Y is the compression of that of the torus,

$$\mathbf{1}_{\mathfrak{F}(Y)} K^{Y^+} \mathbf{1}_{\mathfrak{F}(Y)} = \mathbf{1}_{\mathfrak{F}(Y)} \mathfrak{G}_W^T \mathfrak{G}_W \mathbf{1}_{\mathfrak{F}(Y)}, \tag{37.1}$$

and only the coarse-action part of Δ_Y depends on the Dirichlet construction.

For a regular background U that is not necessarily critical, we *define* $\Delta_k = \Delta_k^{(\mathbb{T}_\eta)}$ and $C_k = \Delta_k^{-1}$, with the elimination at the base $W = U^{(k)}$. This agrees with Part VII when U is critical (Proposition 37.4(a)).

Lemma 37.5 (Coercivity at general backgrounds)**.** *Under the smallness hypotheses of Corollary 34.6, but for any real regular U with $|J_U| \leq \varepsilon$ (critical or not), $\frac{1}{4}c_L \leq \Delta_k \leq C_L$ on $\ell^2(\mathfrak{F})$. There is $c_L'' > 0$, depending on L and M, with $C_{\mathrm{AB}} c_L'' \leq c_L'$ for the threshold c_L' of Corollary 34.6(c), such that $\operatorname{Re}\Delta_k \geq \frac{1}{8}c_L$ for complex perturbations of size at most c_L'' in the local gauges, if moreover $\varepsilon \leq \varepsilon_{\mathrm{cc}}$ and ε is below the thresholds of the proof of Proposition 37.4(c).*

Proof. Real backgrounds. Let Z be a field on $\mathfrak{F}$ and $B = \hat{Z}^1 Z$. Then $B \in \mathcal{Z} = \ker l^{\mathrm{pl}}$, $\|B\| \geq \|Z\|$, and $\|\mathfrak{G}_W Z\| = \|\mathfrak{G}_W B\|$ (Lemmas 32.7(d) and 34.1). Theorem 34.4 and Lemma 34.3 do not use criticality. Their proofs use only the (M,ε)-regularity of the reference, through Lemma 28.3, which needs $\varepsilon \leq \varepsilon_{\mathrm{cmp}}(M_L')$ (part of the smallness hypotheses of Corollary 34.6), and Lemma 21.13, and the expansions of Part IV at the base $W = U^{(k)}$, which hold at every real base of small curvature (Section 22). So they give Details

$$\langle B, (\mathbf{\Gamma}_\pi[0] - a)B\rangle + \|\mathfrak{G}_W B\|^2 \geq c_L \|B\|^2 \qquad (B \in \mathcal{Z}).$$

By Lemma 37.2, $\|W_2\| \leq C\varepsilon$. Lemma 13.3, applied to $\mathbf{Q}\mathbf{G}[W]\mathbf{Q}^T$, gives $\|\mathbf{\Gamma}_\pi[W_2] - \mathbf{\Gamma}_\pi[0]\| \leq C\varepsilon$. Since $\Delta^{0,\mathbb{T}_\eta} = \mathbf{\Gamma}_\pi[W_2] - a$ (Proposition 37.4(a)),

$$\Delta^{0,\mathbb{T}_\eta}(B,B) + \|\mathfrak{G}_W B\|^2 \geq (c_L - C\varepsilon)\|B\|^2 \geq \tfrac{1}{2}c_L \|Z\|^2$$

for ε small. The term $2\langle \omega_0, \hat{Z}^2(Z)\rangle$ is bounded by $CL^{3d+1}\varepsilon\|Z\|^2$, since $|\omega_0| \leq C\varepsilon$ (Lemma 32.7(d)); this is at most $\frac{1}{4}c_L\|Z\|^2$ by the threshold $CL^{3d+1}\varepsilon \leq \frac{1}{4}c_L$ of Corollary 34.6. This gives the lower bound. The upper bound follows from Proposition 37.4(b), the locality of $\hat{Z}^1$, $\hat{Z}^2$, and $\|\mathfrak{G}_W\| \leq C_L$ (Lemma 22.7(c)).

Complex perturbations. Let $U = e^{i\eta A'} U_r$ be a complex reference of size at most c_L'' (Section 46). Since $C_{\mathrm{cmp}} c_L'' \leq \delta_*$, it is a perturbation U_δ of U_r as in Proposition 19.11 with $\delta \leq \delta_*$. The changes of the terms of $\Delta_k(Z, \bar{Z})$ are bounded as follows.

- The operators of Parts II, III and VIII change by $O(c_L'')$ in norm (Proposition 19.11, and for W_2 the proof of Proposition 37.4(c)); so $\Delta^{0,\mathbb{T}_\eta}$ changes by $O(c_L'')$.
- $|\omega_0| \le C(\varepsilon + c_L'')$ (Proposition 47.12(b)), so the term with ω_0 changes by at most $CL^{3d+1}c_L''\|Z\|^2$.
- The base $W_k(U)$ is $e^{\mathrm{i}\beta}W$ bondwise, with $W = W_k(U_r)$ and $|\beta|_\infty \le C_{\mathrm{AB}}c_L''$ (Proposition 46.4(i)). Since $C_{\mathrm{AB}}c_L'' \le c_L'$, the threshold of Corollary 34.6(c), the elimination at this base is analytic in β, with the bounds of Lemma 32.7, by the argument in the proof of that corollary. So $\hat{Z}^1$ and $\hat{Z}^2$ change by local maps of size $C_L c_L''$.
- The gauge-fixing term. W has curvature at most $2C_{\mathrm{curv}}\varepsilon$ in the units of step k (proof of Corollary 22.11, and Proposition 21.12). For ε and c_L'' small depending on L, Lemma 22.7(f) applies in the units of step k, i.e. with $\delta = 1$. It gives $|(\mathfrak{G}_{W_k(U)} - \mathfrak{G}_W)Z(y,x)| \le CL^2|\beta|_\infty \max|Z|$, the maximum over the bonds of $B'(y)$. Summing over the L^d points $x \in B'(y)$, the block-local operator $\mathfrak{G}_{W_k(U)} - \mathfrak{G}_W$ has norm at most $CL^{d/2+2}|\beta|_\infty$, while $\|\mathfrak{G}_W\| \le CL^{d/2+1}$ by Lemma 22.7(c). So $\langle \mathfrak{G}_{W_k(U)}Z, \mathfrak{G}_{W_k(U)}\bar{Z}\rangle$ differs from $\|\mathfrak{G}_W Z\|^2$ by at most $CL^{d+3}C_{\mathrm{AB}}c_L''\|Z\|^2$.

Hence $\operatorname{Re}\Delta_k(Z,\bar{Z}) \ge (\frac{1}{4}c_L - C_L''c_L'')\|Z\|^2$, with a constant $C_L'' \ge C_L$ absorbing C_W of the proof of Proposition 37.4(c). The largest explicit power of L comes from the term with ω_0. So c_L'' is chosen with $C_{\mathrm{AB}}c_L'' \le c_L'$, with $C_{\mathrm{cmp}}c_L'' \le \delta_*$ (proof of Proposition 37.4(c)), and so small that the sum $C_L''c_L''$ of the four changes is at most $\frac{1}{8}c_L$; the term with ω_0 has the largest power L^{3d+1}, in $d = 4$ a factor L^{-13}. Then $\operatorname{Re}\Delta_k \ge \frac{1}{8}c_L$. □

Proposition 37.6 (Local fluctuation operators)**.** *There is $R_0(L,M) \le C_L M$ such that for $R_0 \ge R_0(L,M)$, under the hypotheses of Lemma 37.5:*

(a) $\Delta_{\mathbb{T}_\eta} = \Delta_k$, and $\|\Delta_Y - \mathbf{1}_{\mathfrak{F}(Y)}\Delta_k\mathbf{1}_{\mathfrak{F}(Y)}\| \le C_L e^{-\gamma'(R_0-2L-10M)/2}$, with γ' of Proposition 37.4(d), taken valid also at the complex perturbations of Lemma 37.5. Hence $\Delta_Y \ge \frac{1}{8}c_L$, and $C_Y = \Delta_Y^{-1}$ has decay (C_L,γ_C) uniformly in Y.

(b) Δ_Y and C_Y depend only on U on $\widehat{Y^+}$. They are covariant, and analytic under the complex perturbations of Lemma 37.5, with $\operatorname{Re}\Delta_Y \ge \frac{1}{16}c_L$. At complex perturbations the current J_U is not bounded uniformly in η. Wherever ω_0 enters, write $\langle b,\omega_0^X\rangle = \langle F_U^s, D\mathbf{H}_0^X b\rangle$ and use the decay of $D\mathbf{H}_0^X$, as in Section 41, level (L4); the proofs of Proposition 37.4(b),(c),(d) and Lemma 37.5 are to be read this way. In the proof of Proposition 37.4(d) this gives $\langle b,\omega_0^X - \omega_0^{X'}\rangle = \langle F_U^s, D(\mathbf{H}_0^X - \mathbf{H}_0^{X'})b\rangle$, which is of seam form, at all fine unit points, by Proposition 35.1(d),(e) for $D\mathbf{H}_0$ at the complex perturbation.

(c) For $Y \subset Y'$ and $b,b' \in \mathfrak{F}(Y)$,

$$|(C_{Y'} - C_Y)(b,b')| + |(\Delta_{Y'} - \Delta_Y)(b,b')| \le C_L \sum_{w\in\Sigma_Y} e^{-\gamma''(|b-w|+|w-b'|)},$$

where Σ_Y is the set of unit points within distance $R_0 + 10M + 2L$ of $Y' \setminus Y$.

(d) (Uniformity in $r \ge 0$.) *$\operatorname{Re}(\Delta_Y + r) \ge \frac{1}{16}c_L + r$, and $(\Delta_Y + r)^{-1}$ has decay $(C_L(\frac{1}{16}c_L + r)^{-1}, \gamma_C)$ uniformly in r and Y. Also (c) holds for $(\Delta_Y + r)^{-1}$, with an extra factor $(\frac{1}{16}c_L + r)^{-1}$ on the right. Indeed, r cancels from the operator of Lemma 17.7, and conjugation does not see it, since $(K + r)_\alpha - (K + r) = K_\alpha - K$. This is the form used at level (L5) of Section 41 and in Proposition 38.1.*

Proof. (a) For $b,b' \in \mathfrak{F}(Y)$ the supports of $\hat{Z}^1\delta_b$, $\hat{Z}^1\delta_{b'}$ lie at distance at least $R_0 - 2L$ Details

from $\mathbb{T}_\eta \setminus Y^+$. The gauge-fixing parts of $\Delta_k(b,b')$ and $\Delta_k^{(Y^+)}(b,b')$ agree, by (37.1). By Proposition 37.4(d) with $X = \mathbb{T}_\eta$ and $X' = Y^+$, together with the locality of $\hat{Z}^1$, $\hat{Z}^2$,

$$|\Delta_k(b,b') - \Delta_k^{(Y^+)}(b,b')| \le C_L e^{-\gamma'(R_0-2L-10M)/2} e^{-\gamma'|b-b'|/2}.$$

Here the sum over the points w of the seam is taken with half of the rate in the distance to the seam, which absorbs the number of these points. A Schur bound gives the norm estimate, with a constant C_{seam}. By the same proof, read as in (b), the bound holds also at the complex perturbations of Lemma 37.5; we take C_{seam} and γ' valid at real U and at these perturbations. Put $R_0(L,M) := 2L + 10M + \frac{2}{\gamma'} \log^+ \frac{16 C_{\mathrm{seam}}}{c_L}$. For $R_0 \ge R_0(L,M)$ the norm in (a) is at most $\frac{1}{16} c_L$, at real U and at these complex perturbations. The constants C_{seam}, c_L and γ' do not depend on M, so $R_0(L,M) \le C_L M$ for $M \ge 1$; and $R_0(L,M) \ge 2L+2$. Then $\mathbf{1}_{\mathfrak{F}(Y)} \Delta_k \mathbf{1}_{\mathfrak{F}(Y)} \ge \frac{1}{4} c_L$, as a compression of Δ_k (Lemma 37.5). This gives the lower bound, since $\frac{1}{4} - \frac{1}{16} \ge \frac{1}{8}$, and Lemma 5.5 gives the decay.

(b) follows from Proposition 37.4(c), the locality of the elimination, the properties of K^X stated before the proposition, and (a) for complex perturbations: there $\operatorname{Re} \mathbf{1}_{\mathfrak{F}(Y)} \Delta_k \mathbf{1}_{\mathfrak{F}(Y)} \ge \frac{1}{8} c_L$ by Lemma 37.5, so $\operatorname{Re} \Delta_Y \ge \frac{1}{8} c_L - \frac{1}{16} c_L$.

(c) Apply Lemma 17.7 with $\mathfrak{A} = \Delta_{Y'}$ and $\mathfrak{A}' = \Delta_Y$. The operator $\Delta = \Delta_{Y'} \mathbf{1}_{\mathfrak{F}(Y)} - \mathbf{1}_{\mathfrak{F}(Y)} \Delta_Y \mathbf{1}_{\mathfrak{F}(Y)}$ has two kinds of entries. For $b \in \mathfrak{F}(Y)$ they are $\Delta_k^{(Y'^+)}(b,b') - \Delta_k^{(Y^+)}(b,b')$. Their gauge-fixing parts cancel, since both equal $(\mathfrak{G}_W^T \mathfrak{G}_W)(b,b')$: the block of an inner bond starting in Y lies in $Y^+ \subset Y'^+$. The remaining coarse-action parts are of seam form by Proposition 37.4(d), since $Y'^+ \setminus Y^+$ lies within distance R_0 of $Y' \setminus Y$. For $b \in \mathfrak{F}(Y') \setminus \mathfrak{F}(Y)$ they are entries $\Delta_k^{(Y'^+)}(b,b')$ with b starting in $Y' \setminus Y$, bounded by $C_L e^{-\gamma_1 |b-b'|}$, since the gauge-fixing part has range at most L. Both have seam form, and the decay of $C_{Y'}$ and C_Y gives the bound for C. The bound for Δ is the first kind of entry. □

37.5 Local expansion of the covariance

This subsection is not on the main line (see the status paragraph at the beginning of the section).

Theorem 37.7 (Localization of the fluctuation covariance)**.** *For bonds $b, b' \in \mathfrak{F}$ let $\mathsf{S} = \{b_-, b'_-\}$ and $\Lambda_m = \Lambda_m(\mathsf{S})$. Then*

$$C_k(b,b') = \sum_{m \ge 0} C_{k,m}(b,b'), \qquad \Delta_k(b,b') = \sum_{m \ge 0} \Delta_{k,m}(b,b'),$$

with $C_{k,0} = C_{\Lambda_0}(b,b')$ and $C_{k,m} = C_{\Lambda_m}(b,b') - C_{\Lambda_{m-1}}(b,b')$ for $m \ge 1$, and similarly for Δ_k. Each $C_{k,m}(b,b')$ and $\Delta_{k,m}(b,b')$ depends only on U on $\widehat{\Lambda_m^+}$. They are covariant, analytic under complex perturbations, and bounded by

$$|C_{k,m}(b,b')| + |\Delta_{k,m}(b,b')| \le C'_L e^{-\gamma_2(m+|b-b'|)}.$$

Proof. Lemma 37.1 with Proposition 37.6(c) for $Y = \Lambda_{m-1} \subset Y' = \Lambda_m$. The points $w \in \Sigma_Y$ have distance at least $m - 1 - R_0 - 10M - 2L$ from S. So Details

$$\sum_{w \in \Sigma_Y} e^{-\gamma''(|b-w|+|w-b'|)} \le C_L e^{-\gamma''(m-R'_0)/2} e^{-\gamma''|b-b'|/2},$$

with $R'_0 = R_0 + 10M + 2L + 1$. The factor $e^{\gamma'' R'_0/2}$ is absorbed into C'_L, and $\gamma_2 = \gamma''/2$. □

Corollary 37.8 (Locality under large changes of the background)**.** *Let U_1, U_2 be two backgrounds satisfying the hypotheses above, which coincide on the bonds of $\hat{Y}$ for a domain Y. Then for bonds $b, b' \in \mathfrak{F}$ starting in Y, with $\delta = \operatorname{dist}(\{b_-, b'_-\}, \mathbb{T}_\eta \setminus Y)$,*

$$|C_k(U_1)(b,b') - C_k(U_2)(b,b')| + |\Delta_k(U_1)(b,b') - \Delta_k(U_2)(b,b')| \le C_L e^{-\gamma_2(\delta - R'_0)}.$$

Proof. The terms with $\widehat{\Lambda_m^+} \subset \hat{Y}$, i.e. $m \le \delta - R'_0$, coincide. The others sum to at most $C_L e^{-\gamma_2(\delta - R'_0)}$. □ Details

Corollary 37.8 compares two arbitrary regular backgrounds that agree on a region. They may differ by large amounts elsewhere, which Proposition 33.3 does not cover. For backgrounds of two renormalization group steps, $C_k(U_i)$ is the true fluctuation covariance of Part VII.

Remark 37.9 (Use in the cluster expansion). Theorem 37.7 is the input for a cluster expansion with an interpolated covariance, e.g. the Brydges–Kennedy forest formula. That formula needs only the decay of the kernel and the positivity of the interpolated covariance. The entries of C_k then appear in the activities, and expanding each entry by the theorem makes the dependence on the background local. This localization is of radius type, which is not sufficient for the polymer expansion (Section 39). The main line uses instead the weakened random walks of Part IX, with the operators of this section as local inverses; Proposition 37.6 is used there at level (L5). Bałaban's random-walk expansion [7] is another alternative. It can be built from the local inverses C_Y, with the parametrix of Section 20 and the local entries $\Delta_{k,m}$ in the remainder, but it is not needed here.

38 Localization and normalization of the determinant

Status. The main line uses from this section the local densities $\ell_Y(b)$ defined below (through Corollary 41.6), the background activities (Definition 38.2), the graded domains (Definition 38.3) with (38.1), Lemmas 38.4 and 38.5, Remark 38.7, and Remarks 38.8 and 38.6(b). Proposition 38.1, the telescoping of the densities, and Corollary 38.9 are of radius type and are not used later; they are superseded by Proposition 41.7, which gives the one-loop source in polymer form at the complex points of the graded domains. Remark 38.6(b) records the scaling L^{-2m} of Remark 38.8(i) at later scales; at those scales the main line uses Lemma 51.1. Parts (c) and (d) of that remark are comments on Lemma 38.5, and the wrapping polymers of part (d) are treated by Proposition 41.7(d) and Lemma 52.1.

38.1 Local densities

For a domain Y and $b \in \mathfrak{F}(Y)$ let

$$\ell_Y(b) = \operatorname{tr}_{\mathfrak{g}} \left[(\log \Delta_Y)(b,b) \right], \qquad \log \Delta_Y = \int_0^\infty \Big(\frac{1}{1+t} - (\Delta_Y + t)^{-1} \Big) dt,$$

with Δ_Y the local fluctuation operator of Proposition 37.6. For $Y = \mathbb{T}_\eta$ this is the density $\ell_k(b)$ of Proposition 34.8.

Proposition 38.1 (Localization of the determinant)**.** *For $b \in \mathfrak{F}$ let $\Lambda_m = \Lambda_m(\{b_-\})$. Then*

$$\ell_k(b) = \sum_{m \ge 0} \ell_{k,m}(b), \qquad \ell_{k,0}(b) = \ell_{\Lambda_0}(b), \quad \ell_{k,m}(b) = \ell_{\Lambda_m}(b) - \ell_{\Lambda_{m-1}}(b),$$

where $\ell_{k,m}(b)$ depends only on U on $\widehat{\Lambda_m^+}$. It is invariant under real gauge transformations and analytic under complex perturbations of U, and $|\ell_{k,m}(b)| \le C_L e^{-\gamma_3 m}$.

Proof. For $Y \subset Y'$ and $t \geq 0$, Lemma 17.7 applied to $\Delta_{Y'} + t$ and $\Delta_Y + t$ gives, as in Proposition 37.6(c), Details

$$|((\Delta_{Y'} + t)^{-1} - (\Delta_Y + t)^{-1})(b, b)| \leq C_L (c_L/16 + t)^{-2} \sum_{w \in \Sigma_Y} e^{-\gamma'' |b-w|}.$$

The lower bound and the decay constants of $\Delta_Y + t$ are uniform in t, as in Proposition 34.8(a). Integrating in t bounds $|\ell_{Y'}(b) - \ell_Y(b)|$ by the same sum without the factor $(c_L/16 + t)^{-2}$. Lemma 37.1 gives the expansion and the bound. The trace of the diagonal kernel of an operator that is covariant under $R(u)$ is invariant. Analyticity follows from Proposition 37.6(b). □

At the trivial background $U = 1$ the same construction gives numbers $\ell_{k,m}(b; 1)$. They satisfy $\sum_{b,m} \ell_{k,m}(b; 1) = \log \det \Delta_k(1)$, which is all that is used. The sums $\ell_k(b; 1) = \sum_m \ell_{k,m}(b; 1)$ depend only on the orbit of b under the translations of $\mathbb{T}'$ and the hyperoctahedral group about the points of $\mathbb{T}'$ (Proposition 34.8(d)). The individual pieces need not: the mass profiles of the Dirichlet operators are built on the grid $\check{M}\mathbb{Z}^d$ of Section 17, which is not invariant under these maps, and the domains $\Lambda_m(\{b_-\})$ are cubes about the initial point of b. The symmetry of the step is a property of such sums, not of the pieces.

38.2 Normalized background activities

Definition 38.2 (Background activities)**.** Let P be a domain and $\varepsilon_a, \delta_a > 0$. A *background activity at* P is a function $F(U)$ with the following properties:

- it is defined for the complex configurations $U = e^{\mathrm{i}\eta\,\delta A} U_r$ such that U_r is real, (M, ε_a)-regular and satisfies $|J_{U_r}| \leq \varepsilon_a$, and $|\delta A|, |\partial \delta A|, [\partial \delta A]_\alpha \leq \delta_a$ in the local gauges of U_r;
- it depends only on U on the bonds of P;
- it is analytic and invariant under real gauge transformations.

It is *normalized* if $F(1) = 0$. We write $\|F\| = \sup |F|$ over the domain of definition.

The activities of the renormalization group are evaluated on a smaller, graded domain. It allows a real part with small curvature and current, and a complex part that is small in sup norm and smaller still in its derivatives. Only a neighbourhood of the domain P matters.

For a domain P let $P^{(3)}$ be the set of points at unit-lattice ℓ^∞-distance at most 3 from P, with the bonds joining them; so $\Lambda_3(x) \subset P^{(3)}$ for $x \in P$. For a real configuration U_r and a $\mathfrak{g}^c$-valued field A on the bonds of a set Y, let $|\nabla A|_Y$ be the maximum of $|D_{U_r,\nu} A_\mu(x)|$ over the differences that involve only points of Y. Below, $\nabla A(x)$ denotes the components at x with this property. Let

$$[\nabla A]_{\alpha, Y} = \sup \Big\{ \frac{|\nabla A(x) - R(U_r(\Gamma_{x,x'})) \nabla A(x')|}{|x - x'|^\alpha} : \; x \neq x' \in \Lambda_3(z') \subset Y \Big\},$$

where $\Gamma_{x,x'}$ is the lattice path from x to x' that changes the coordinates in the order $1, \dots, d$. Here $|\cdot|$ of a collection of components is the maximum over the components; in the quotient, the components defined at both x and x' are compared. Both quantities are invariant under real gauge transformations $(U_r \mapsto U_r^u,\ A(b) \mapsto R(u(b_-))A(b))$. In a real gauge on a box Λ' in which $U_r = e^{\mathrm{i}\eta A_r}$ with $\|A_r\|_{1,\alpha;\Lambda'} \leq \sigma \leq 3$, the field $\tilde{A}$ that represents A satisfies, on Λ',

$$|\nabla A - \partial \tilde{A}| \leq C\sigma |A|, \qquad |[\nabla A]_\alpha - [\partial \tilde{A}]_\alpha| \leq C\sigma (|A| + |\partial \tilde{A}|). \tag{38.1}$$

Indeed $D_{U_r,\nu} \tilde{A}_\mu - \partial_\nu \tilde{A}_\mu = \eta^{-1} (R(e^{\mathrm{i}\eta A_{r,\nu}}) - 1) \tilde{A}_\mu(\cdot + \eta e_\nu)$, whose Hölder quotients are bounded by $C\sigma(|A| + |\partial \tilde{A}|)$, since $|A_r| + |\partial A_r| \leq \sigma$ and $|x - x'|^{1-\alpha} \leq C$ on the cubes. Moreover $\|R(U_r(\Gamma_{x,x'})) - 1\| \leq C|x - x'|\sigma$ for x, x' in a cube $\Lambda_3(z')$.

Definition 38.3 (Graded domains and activities)**.** Let P be a domain and $\theta_r, \delta_1, \delta_0 > 0$. The set $\mathcal{U}(P) = \mathcal{U}(P; \theta_r, \delta_1, \delta_0)$ consists of the complex configurations U on $\mathbb{T}_\eta$ for which there are

- a real configuration U_r on $\mathbb{T}_\eta$ with $|F_{U_r}| < \theta_r$ and $|J_{U_r}| < \theta_r$ everywhere, and
- a $\mathfrak{g}^c$-valued field A on the bonds of $P^{(3)}$ with $U = e^{i\eta A} U_r$ there, $|A| < \delta_0$, and $|\nabla A|_{P^{(3)}} + [\nabla A]_{\alpha, P^{(3)}} < \delta_1$, with ∇ taken with respect to U_r.

The set is invariant under real gauge transformations, and it contains 1. A *graded activity at* P is a function F of $U|_P$ that is defined and holomorphic on an open set containing $\{U|_P : U \in \mathcal{U}(P)\}$ and invariant under real gauge transformations on $\mathcal{U}(P)$. It is *normalized* if $F(1) = 0$. We write $\|F\| = \sup_{\mathcal{U}(P)} |F|$.

In four dimensions the constants $\theta_r, \delta_1, \delta_0$ do not depend on the coupling; they are fixed in Remark 38.8.

Lemma 38.4 (Complex gauge invariance)**.** *Let F be a graded activity at P, let $U \in \mathcal{U}(P)$, and let λ_1 be a $\mathfrak{g}^c$-valued scalar field on $\mathbb{T}_\eta$ such that $U^{e^{it\lambda_1}} \in \mathcal{U}(P)$ for all $t \in [0,1]$. Then $F(U^{e^{i\lambda_1}}) = F(U)$.*

Proof. Let $\mathcal{O}$ be the open set on which F is holomorphic, and W the set of $\mathfrak{g}^c$-valued scalar fields λ with $U^{e^{i\lambda}}|_P \in \mathcal{O}$. Since $\lambda \mapsto U^{e^{i\lambda}}|_P$ is holomorphic, W is open and $\Phi(\lambda) = F(U^{e^{i\lambda}})$ is holomorphic on W. Let W_0 be the connected component of W that contains the segment $\{t\lambda_1 : t \in [0,1]\}$. Since $\mathcal{U}(P)$ is invariant under real gauge transformations, W_0 contains a ball of real fields around 0, and Φ is constant on it. This ball is a nonempty open subset of the real points, which form a totally real subspace of maximal dimension. By the identity theorem Φ is constant on W_0, so $\Phi(\lambda_1) = \Phi(0)$. □ Details

Why a quadratic bound. Background activities (Definition 38.2) are the form in which the constructions produce local functions of the background, for instance the one-loop source (proof of Proposition 41.7). Graded activities (Definition 38.3) are the activities of the renormalization group, and Remark 38.7 passes from the first to the second. The next lemma bounds a normalized graded activity by $2\mathfrak{s}^2\|F\|$ when $\mathfrak{s} \le \frac{1}{2}$ (for semisimple G and a polymer that does not wrap), where $\mathfrak{s}$ is built from the derivatives $\mathfrak{m}$ of the complex part A', from the bound θ on the curvature and the current of the real part, and from $|A'|$. Without the affine complex gauge of its proof, $|A'|$ would enter $\mathfrak{s}$ linearly. On the domain of scale k, read in the units of scale $j < k$, $|A'| \le L^{-(k-j)}\delta_0$, while $\mathfrak{m}$ and θ are $O(L^{-2(k-j)})$ (Remark 38.8(i)). So $\mathfrak{s}$ would be of order $L^{-(k-j)}$ instead of $L^{-2(k-j)}$. In $d = 4$ a cube of scale k contains $L^{4(k-j)}$ polymers of scale j, so already the orders ≥ 3 of an old activity would then not be summable over the scales. Remark 41.8(a) gives this argument, and Remark 41.8(b) explains why the quadratic part that remains is marginal and must be extracted as a coupling.

Lemma 38.5 (Quadratic bound for normalized activities)**.** *There are $C_0 \ge 1$ and $c_{\mathrm{q}} > 0$, depending only on d, α and G (C_0 is fixed first, then $c_{\mathrm{q}} \le c(C_0)$), with the following property. Let G be semisimple, $n \ge n_0$, $0 < \theta_r \le a_2$ (a_2 from Corollary 30.4) and $0 < \delta_1 \le \delta_0$. Let P be a connected domain, and let F be a normalized graded activity at P. Let $U = e^{i\eta A'} U_r$ on $P^{(3)}$ be a representation as in Definition 38.3, with $|F_{U_r}|, |J_{U_r}| \le \theta < \theta_r$. Let $\Lambda = \Lambda_R(z) \supset P^{(3)}$ with $2R < n$, and put*

$$B_P = 3C_E B_5 \max\Big(1, \frac{R}{4M+2}\Big)^p, \qquad \mathfrak{m} = |\nabla A'|_{P^{(3)}} + [\nabla A']_{\alpha, P^{(3)}},$$

with C_E, p from Lemma 28.3 and B_5 from Corollary 30.4. Let D_N be the diameter of the set of bonds of $P^{(3)}$ for the following distance between parallel bonds: the least unit-lattice length

of a chain of parallel bonds of $P^{(3)}$ in which each bond is the translate of the previous one by one fine lattice step, in any direction. Put

$$c_* = \delta_0 + (1 + D_N)(|A'| + B_P\theta),$$

and assume $RB_P\theta \le 1$, $c_* \le c_{\mathrm{q}}$ and $|A'| < \frac{1}{2}\delta_0$. Put

$$\mathfrak{m}_1 = (1 + C_0 c_*)\,\mathfrak{m} + C_0|A'|(|A'| + (1 + D_N)B_P\theta), \qquad \mathfrak{s} = (1 + C_0 c_*)\Big(1 + \frac{D_N\delta_1}{\delta_0}\Big)\frac{\mathfrak{m}_1 + B_P\theta}{\delta_1}.$$

If $\mathfrak{s} \le \frac{1}{2}$, then

$$|F(U)| \le 2\,\mathfrak{s}^2\,\|F\|.$$

Proof. The proof uses two cancellations. At the base point $x_0 \in P$, an affine complex gauge removes the value of the complex field (Step 2). What remains are differences of the field. They are summed along chains of parallel bonds and bounded by the covariant quantity $\mathfrak{m}$, through the derivative estimate (38.2). At the configuration 1, normalization and real gauge invariance remove the terms of order 0 and 1 of an interpolation (Step 3). Details

Constants. Let $c_{\mathrm{an}} \in (0, 1]$, depending only on d and G, be so small that the following expansions hold whenever $c_* \le c_{\mathrm{an}}$, with constants depending only on d and G: (24.1) for complex λ, with its bound on $\mathfrak{b}$, and the power series of $e^{-\mathrm{i}\,\mathrm{ad}\,X}$ and $\mathfrak{d}(X)$, for the fields of Step 2; the Baker–Campbell–Hausdorff formula, for the fields of Step 3. Such a c_{an} exists, since these fields are bounded by Cc_* under the hypotheses of the lemma, with C depending only on d and G (Steps 2 and 3). We assume $c_{\mathrm{q}} \le c_{\mathrm{an}}$; in particular $c_* \le 1$. Below, C denotes a constant that depends only on d, α and G, through (38.1), (23.2) and these expansions. Each such C is valid for every $c_* \le c_{\mathrm{an}}$, and none depends on C_0 or on c_{q}. By the definition of c_*,

$$B_P\theta \le c_*, \qquad D_N B_P\theta \le c_*, \qquad (1 + D_N)|A'| \le c_*, \qquad \delta_0 \le c_*.$$

Moreover $D_N \ge 1$. Indeed, $P^{(3)}$ contains every bond at a point of P together with its translates by one unit, and a chain from a bond to such a translate has length at least 1.

Step 1: a real gauge on Λ. By Corollary 30.4, U_r is $(M, B_5\theta)$-regular, hence $3B_5\theta$-regular on every cube $\Lambda_{4M+2}(z')$. If $R \ge 4M + 2$, Lemma 28.3 with $r = 4M + 2$ applies, since $RB_P\theta \le 1$. If $R < 4M + 2$, Λ lies in a cube $\Lambda_{4M+2}(z)$. In both cases there is a real gauge u on Λ, extended arbitrarily to $\mathbb{T}_\eta$, with $U_r^u = e^{\mathrm{i}\eta A_r}$ on the bonds of Λ and $\|A_r\|_{1,\alpha;\Lambda} \le B_P\theta$. Let $\tilde{A}'(b) = R(u(b_-))A'(b)$ represent A' in this gauge. Then $U^u = e^{\mathrm{i}\eta\tilde{A}'}U_r^u$ on $P^{(3)}$, and $|\tilde{A}'| = |A'|$. For a field B on the bonds of $P^{(3)}$, let $|\partial B|$ and $[\partial B]_\alpha$ be the quantities $|\nabla B|_{P^{(3)}}$ and $[\nabla B]_{\alpha,P^{(3)}}$ for the real configuration 1. They are built from the flat differences $\partial_\mu B_\nu(y) = \eta^{-1}(B_\nu(y + \eta e_\mu) - B_\nu(y))$ that involve only points of $P^{(3)}$. By (38.1), with $\sigma = B_P\theta \le 1$, we have the derivative estimate

$$|\partial\tilde{A}'| + [\partial\tilde{A}']_\alpha \le (1 + CB_P\theta)\,\mathfrak{m} + CB_P\theta|A'|. \tag{38.2}$$

Real gauge invariance gives $F(U) = F(U^u)$.

Step 2: an affine complex gauge. Fix $x_0 \in P$, use the coordinates of Λ, and put

$$\lambda_1(x) = -\sum_\mu (x - x_0)_\mu\,\tilde{A}'_\mu(x_0) \quad (x \in \Lambda), \qquad \lambda_1 = 0 \text{ off } \Lambda.$$

Here $\tilde{A}'_\mu(x_0)$ is the value of $\tilde{A}'$ on the bond $(x_0, x_0 + \eta e_\mu)$, which lies in $P^{(3)}$ since $x_0 \in P$. Then $\partial_\nu\lambda_1 = -\tilde{A}'_\nu(x_0)$ and $\partial\partial\lambda_1 = 0$ on Λ. On $P^{(3)}$, $|\lambda_1| \le (1 + D_N)|A'| \le c_*$, since $|x - x_0|_1$ is at most the length of a chain as in the definition of D_N, plus 1. Moreover $D_{U_r^u}\lambda_1 - \partial\lambda_1 = O(|A_r||\lambda_1|)$, and its derivatives are $O((|\partial A_r| + [\partial A_r]_\alpha)|\lambda_1| + |A_r||\partial\lambda_1|)$. In

particular $|D_{U_r^u}\lambda_1| \le |A'| + CB_P\theta(1+D_N)|A'| \le C|A'|$ on $P^{(3)}$. By (24.1), which holds for complex λ by analytic continuation, $(U^u)^{e^{-it\lambda_1}} = e^{i\eta B_t} U_r^u$ on $P^{(3)}$ for $t \in [0,1]$, with

$$B_t(b) = e^{-it\,\mathrm{ad}\,\lambda_1(b_-)}\tilde{A}'(b) + t\,\mathfrak{d}\big(t\,\mathrm{ad}\,\lambda_1(b_-)\big) D_{U_r^u}\lambda_1(b) + \eta\,\mathfrak{b}\big(t\lambda_1(b_-), \tilde{A}'(b), tD_{U_r^u}\lambda_1(b)\big).$$

The cancellation at the base point. Let b be a bond of $P^{(3)}$ in direction ν at x. Split $B_t(b) = M_t(b) + E_t(b)$. The main part is

$$M_t(b) = \tilde{A}'(b) + t\,\partial_\nu\lambda_1(x) = \tilde{A}'_\nu(x) - t\tilde{A}'_\nu(x_0) = (1-t)\,\tilde{A}'_\nu(x) + t\big(\tilde{A}'_\nu(x) - \tilde{A}'_\nu(x_0)\big),$$

and the rest is

$$\begin{aligned} E_t(b) = {} & \big(e^{-it\,\mathrm{ad}\,\lambda_1(x)} - 1\big)\tilde{A}'(b) + t\big(\mathfrak{d}(t\,\mathrm{ad}\,\lambda_1(x)) - 1\big) D_{U_r^u}\lambda_1(b) \\ & + t\big(D_{U_r^u}\lambda_1 - \partial\lambda_1\big)(b) + \eta\,\mathfrak{b}\big(t\lambda_1(x), \tilde{A}'(b), tD_{U_r^u}\lambda_1(b)\big). \end{aligned}$$

The gauge is chosen so that $t\,\partial_\nu\lambda_1(x_0) = -t\tilde{A}'_\nu(x_0)$ cancels the value of the field at the base point. At $t=1$ the main part is the difference $\tilde{A}'_\nu(x) - \tilde{A}'_\nu(x_0)$, which vanishes at $x = x_0$. So the sup norm $|A'|$ enters B_1 only through E_1. By the bounds on λ_1 and $D_{U_r^u}\lambda_1$ above and the bound on $\mathfrak{b}$ in (24.1), the four terms of E_t are at most $C(1+D_N)|A'|^2$, $C(1+D_N)|A'|^2$, $C(1+D_N)B_P\theta|A'|$ and $C|A'|^2$. Hence

$$|E_t(b)| \le C(1+D_N)|A'|\big(|A'| + B_P\theta\big).$$

The summation along a chain. Let b^0 be the bond in direction ν at x_0. It lies in $P^{(3)}$. By the definition of D_N there is a chain $b^0, b^1, \dots, b^m = b$ of bonds of $P^{(3)}$ in direction ν, with $b^{i+1} = b^i \pm \eta e_{\mu_i}$ and $m\eta \le D_N$. Then

$$\tilde{A}'_\nu(x) - \tilde{A}'_\nu(x_0) = \sum_{i=0}^{m-1} \big(\tilde{A}'(b^{i+1}) - \tilde{A}'(b^i)\big),$$

and each term is $\pm\eta\,\partial_{\mu_i}\tilde{A}'_\nu(y_i)$. Here y_i is the initial point of that one of b^i, b^{i+1} whose translate by ηe_{μ_i} is the other. This difference involves only the endpoints of b^i and b^{i+1}, which lie in $P^{(3)}$. It is a covariant difference up to a transport. Indeed, in the gauge u,

$$\eta\,\partial_\mu\tilde{A}'_\nu(y) = \eta\,R(u(y))\big(D_{U_r,\mu}A'_\nu\big)(y) - \big(R(e^{i\eta A_{r,\mu}(y)}) - 1\big)\tilde{A}'_\nu(y + \eta e_\mu),$$

and the last term is at most $C\eta B_P\theta|A'|$. This is the first inequality of (38.1), which enters (38.2). Summing over the chain,

$$|\tilde{A}'_\nu(x) - \tilde{A}'_\nu(x_0)| \le m\eta\,|\partial\tilde{A}'| \le D_N|\partial\tilde{A}'|,$$

and (38.2) bounds $|\partial\tilde{A}'|$ by $\mathfrak{m}$ and $B_P\theta|A'|$. Since $|\tilde{A}'_\nu(x)| \le |A'|$, we get $|M_t(b)| \le (1-t)|A'| + tD_N|\partial\tilde{A}'|$. With the bound on E_t, for $t \in [0,1]$,

$$\begin{aligned} |B_t| &\le |A'| + D_N|\partial\tilde{A}'| + C(1+D_N)|A'|\big(|A'| + B_P\theta\big), \\ |B_1| &\le D_N|\partial\tilde{A}'| + C(1+D_N)|A'|\big(|A'| + B_P\theta\big). \end{aligned} \tag{38.3}$$

Derivatives. Since $\partial\partial\lambda_1 = 0$, the differences of the main part are those of $\tilde{A}'$: the gauge does not change them. Differentiating E_t with the chain rule in all four terms (for the term with η, $\eta|\partial f| \le |\partial f|$ and $\eta[\partial f]_\alpha \le 2\eta^{1-\alpha}|\partial f| \le 2|\partial f|$, with $|\partial\mathfrak{b}|$ given by the chain rule),

$$|\partial B_t| + [\partial B_t]_\alpha \le |\partial\tilde{A}'| + [\partial\tilde{A}']_\alpha + C\big(|A'|^2 + D_N|A'|\big(|\partial\tilde{A}'| + [\partial\tilde{A}']_\alpha\big) + (1+D_N)B_P\theta|A'|\big). \tag{38.4}$$

The constant C_0. The field B_t is given in the gauge u. By (38.1) with $\sigma = B_P\theta$,

$$|\nabla B_t|_{P^{(3)}} + [\nabla B_t]_{\alpha,P^{(3)}} \le (1 + CB_P\theta)(|\partial B_t| + [\partial B_t]_\alpha) + CB_P\theta|B_t|,$$

with ∇ taken with respect to U_r^u. Insert (38.4) and (38.3), and then (38.2). Use $1+D_N \le 2D_N$, the bounds of $B_P\theta$, $D_NB_P\theta$ and $D_N|A'|$ by c_*, and $(1+Cc_*)(1+C'c_*) \le 1+(C+C'+CC')c_*$, which holds for $c_* \le 1$. For instance, the coefficient of $|\partial\tilde{A}'| + [\partial\tilde{A}']_\alpha$ is at most $(1+CB_P\theta)(1+CD_N|A'|) + CD_NB_P\theta \le 1 + C'c_*$. This gives a constant C_2, depending only on d, α and G, such that for all $t \in [0,1]$

$$|\nabla B_t|_{P^{(3)}} + [\nabla B_t]_{\alpha,P^{(3)}} \le \mathfrak{m}_2, \qquad |B_t| \le |A'| + D_N\mathfrak{m}_2, \qquad |B_1| \le D_N\mathfrak{m}_2, \tag{38.5}$$

where $\mathfrak{m}_2 = (1 + C_2c_*)\,\mathfrak{m} + C_2|A'|\big(|A'| + (1+D_N)B_P\theta\big)$. We take $C_0 \ge C_2$. Then $\mathfrak{m}_2 \le \mathfrak{m}_1$, and (38.5) gives, for all $t \in [0,1]$,

$$|\nabla B_t|_{P^{(3)}} + [\nabla B_t]_{\alpha,P^{(3)}} \le \mathfrak{m}_1, \qquad |B_t| \le |A'| + D_N\mathfrak{m}_1, \qquad |B_1| \le D_N\mathfrak{m}_1.$$

Since $1+C_0c_* \ge 1$ and $1+D_N\delta_1/\delta_0 \ge \max(1, D_N\delta_1/\delta_0)$, we have $\mathfrak{s} \ge \mathfrak{m}_1/\delta_1$ and $\mathfrak{s} \ge D_N\mathfrak{m}_1/\delta_0$. So $\mathfrak{s} \le \frac12$ gives $\mathfrak{m}_1 \le \frac12\delta_1$ and $D_N\mathfrak{m}_1 \le \frac12\delta_0$. Together with $|A'| < \frac12\delta_0$, this gives $|B_t| < \delta_0$, and $\mathfrak{m}_1 \le \frac12\delta_1$ gives $|\nabla B_t|_{P^{(3)}} + [\nabla B_t]_{\alpha,P^{(3)}} < \delta_1$. Hence $(U^u)^{e^{-it\lambda_1}} \in \mathcal{U}(P)$ for $t \in [0,1]$. Its real part is U_r^u, whose curvature and current have the same size as those of U_r, hence are below θ_r. By Lemma 38.4, $F(U^u) = F\big((U^u)^{e^{-i\lambda_1}}\big)$.

Step 3: a perturbation of 1. On the bonds of $P^{(3)}$ write $e^{i\eta B_1}e^{i\eta A_r} = e^{i\eta A_{\mathrm{tot}}}$. By the Baker–Campbell–Hausdorff formula, $A_{\mathrm{tot}} = B_1 + A_r + \eta\,\mathfrak{c}$ with $|\mathfrak{c}| \le C|B_1||A_r|$. With (23.2) and (38.1), there is a constant C_3, depending only on d, α and G, such that

$$|A_{\mathrm{tot}}| \le (1 + C_3c_*)(D_N\mathfrak{m}_1 + B_P\theta), \qquad |\partial A_{\mathrm{tot}}| + [\partial A_{\mathrm{tot}}]_\alpha \le (1 + C_3c_*)(\mathfrak{m}_1 + B_P\theta).$$

Here the corrections are bounded by $|B_1||A_r| + B_P\theta|B_1| \le c_*\mathfrak{m}_1$ and by $|B_1|B_P\theta \le \delta_0B_P\theta$. From Step 2 these bounds use only $|\nabla B_1|_{P^{(3)}} + [\nabla B_1]_{\alpha,P^{(3)}} \le \mathfrak{m}_1$ and $|B_1| \le D_N\mathfrak{m}_1 \le \frac12\delta_0$, and they are linear in $\mathfrak{m}_1$. So C_3 does not depend on C_0. We take $C_0 \ge C_3$. For complex s let $V_s = e^{i\eta sA_{\mathrm{tot}}}$ on the bonds of $P^{(3)}$ and $V_s = 1$ elsewhere. With real part 1 and the field sA_{tot}, for which $\nabla = \partial$, the configuration V_s lies in $\mathcal{U}(P)$ for $|s| < s_0$, where

$$s_0^{-1} = \max\Big(\frac{|A_{\mathrm{tot}}|}{\delta_0},\ \frac{|\partial A_{\mathrm{tot}}| + [\partial A_{\mathrm{tot}}]_\alpha}{\delta_1}\Big) \le \mathfrak{s}.$$

Indeed, $C_3 \le C_0$ and $\delta_1 \le \delta_0$ give $|A_{\mathrm{tot}}|/\delta_0 \le (1+C_0c_*)\big((D_N\delta_1/\delta_0)\,\mathfrak{m}_1/\delta_1 + B_P\theta/\delta_1\big) \le \mathfrak{s}$, and the second quotient is at most $(1 + C_0c_*)(\mathfrak{m}_1 + B_P\theta)/\delta_1 \le \mathfrak{s}$. So $f(s) = F(V_s)$ is holomorphic for $|s| < s_0$, with $s_0 \ge 2$, and bounded there by $\|F\|$. Moreover:

- $f(1) = F\big((U^u)^{e^{-i\lambda_1}}\big) = F(U)$, since the two configurations agree on P;
- $f(0) = F(1) = 0$;
- $f'(0) = 0$. Indeed, for a constant real gauge transformation v and small complex fields B on P, real gauge invariance gives $F(e^{i\eta R(v)}B) = F(e^{i\eta B})$. Differentiating at $B = 0$, the gradient $\operatorname{grad} F(1)(b) \in \mathfrak{g}^c$ is $\mathrm{Ad}(G)$-invariant for every bond b. For semisimple G the only such vector is 0 [9, (31)–(32)].

Cauchy's estimate gives $|f(1)| \le \|F\| \sum_{j\ge2} s_0^{-j} \le 2s_0^{-2}\|F\| \le 2\mathfrak{s}^2\|F\|$.

The order of the constants, and the slack. The proof needs $C_0 \ge \max(1, C_2, C_3)$. The constants C_2 and C_3 come from (38.1), (23.2) and the expansions fixed with c_{an}. They depend only on d, α and G, and they hold for every $c_* \le c_{\mathrm{an}}$. Neither depends on C_0 or on c_{q}. So $C_0 = \max(1, C_2, C_3)$ is fixed first. Then c_{q} is any number in $(0, c_{\mathrm{an}}]$. No condition links c_{q} to

C_0, so $c(C_0) = c_{\mathrm{an}}$ in the statement, and decreasing c_{q} does not change C_0. The hypothesis $c_* \le c_{\mathrm{q}}$ is used only to apply the expansions and, through $c_* \le 1$, to turn products of small factors into terms linear in c_*. The slack lies in the hypotheses. The factors $1 + C_0 c_*$ in $\mathfrak{m}_1$ and $\mathfrak{s}$ absorb the factors $1 + C_2 c_*$ and $1 + C_3 c_*$. The condition $\mathfrak{s} \le \frac{1}{2}$ gives $D_N \mathfrak{m}_1 \le \frac{1}{2}\delta_0$, and $|A'| < \frac{1}{2}\delta_0$ takes the other half of δ_0; this keeps the path of Step 2 in $\mathcal{U}(P)$. The same condition gives $s_0 \ge 2$, so the Cauchy sum is at most twice its first term. □

Remark 38.6 (Later scales). (a) *A form of the bound.* In the setting of Lemma 38.5 (G semisimple, $n \ge n_0$, P connected, $\Lambda = \Lambda_R(z) \supset P^{(3)}$ with $2R < n$), let F be a normalized graded activity at P on $\mathcal{U}(P; \theta_r, \delta_1, \delta_0)$, with $\theta_r \le a_2$ and $\delta_1 \le \delta_0 \le \frac{1}{2} c_{\mathrm{q}}$. Let $U = e^{\mathrm{i}\eta A'} U_r$ on $P^{(3)}$ be a representation as in Definition 38.3 with $|F_{U_r}|, |J_{U_r}| \le \theta' < \theta_r$, $\mathfrak{m} \le \delta_1'$ and $|A'| \le \delta_0' < \frac{1}{2}\delta_0$. Suppose $R B_P \theta' \le 1$ and $(1 + D_N)(\delta_0' + B_P \theta') \le \frac{1}{2} c_{\mathrm{q}}$. Then $c_* \le c_{\mathrm{q}}$, and, since $\delta_1 \le \delta_0$,

$$\mathfrak{s} \le (1 + C_0 c_{\mathrm{q}})(1 + D_N) \, \frac{(1 + C_0 c_{\mathrm{q}})\delta_1' + C_0 \delta_0' (\delta_0' + (1 + D_N) B_P \theta') + B_P \theta'}{\delta_1}.$$

If $\mathfrak{s} \le \frac{1}{2}$, Lemma 38.5 gives $|F(U)| \le 2\mathfrak{s}^2 \|F\|$. There is no splitting into real and imaginary parts and no current of a complex configuration: the interpolation runs through perturbations of 1.

(b) *Later scales.* Let U be a point of the domain $m \ge 1$ scales later, read in the units of the scale of P, with $L \ge 3$. By Remark 38.8(i), $(\theta', \delta_1', \delta_0') = (L^{-2m}\theta_r, L^{-2m}\delta_1, L^{-m}\delta_0)$, and $\delta_0 \le 1$ gives

$$\mathfrak{s} \le C(D_N, B_P) \, L^{-2m} \Big(1 + \frac{\delta_0^2}{\delta_1} + \frac{\theta_r}{\delta_1} \Big).$$

The hypotheses of (a) and $\mathfrak{s} \le \frac{1}{2}$ hold for $m \ge m_q$, with m_q depending on D_N, B_P, R, L, δ_0^2/δ_1 and θ_r/δ_1, but not on g. The leading constant is not $1 + o(1)$. In four dimensions nothing is contracted, and only the power of L^{-m} is used. At later scales the main line uses Lemma 51.1, which gives the sharper rate $L^{-(4+\alpha)m}$ for the remainder after the part quadratic in the curvature.

(c) The term $C_0 |A'|^2$ in $\mathfrak{m}_1$ cannot be dropped. Take $U_r = 1$ and A' constant with $[A'_\mu, A'_\nu] \ne 0$. Then $\mathfrak{m} = 0$, but U has curvature of order $|A'|^2$ and is not complex-gauge equivalent to 1.

(d) The lemma needs a cube $\Lambda \supset P^{(3)}$ with $2R < n$, so it does not apply to polymers that wrap around the torus. These need a separate treatment. They are wrapping in the sense of Section 52.1, and are bounded by their supremum, with the decay of the activities in the size of the polymer; for the one-loop pieces this is Proposition 41.7(d). Under the torus hypotheses (T0)–(T2) they are bounded uniformly, at their own scale and at all later scales. A wrapping polymer of scale j has $d_M(X) \ge n_j/(4M)$. For activities bounded by $A e^{-\lambda d_M(X)}$, with $\lambda \ge \kappa_0 + 1$, the wrapping polymers of all scales $j \le k$ contribute at most $2K_0 A e^{-(\lambda - \kappa_0) n_k/(4M)}$ per grid cube of scale k (Lemma 52.1(c)–(e) and the paragraph after it).

Remark 38.7 (Background activities restrict to graded activities). Let F be a background activity at P in the sense of Definition 38.2. Suppose $\theta_r \le a_2$, $\max(1, B_5)\theta_r \le \varepsilon_a$ and $C_{\mathrm{ext}}(\delta_0 + \delta_1) < \delta_a$, with $C_{\mathrm{ext}} \ge 1$ the constant below. Then F defines a graded activity at P with $\|F\|$ not larger. Indeed, for $U \in \mathcal{U}(P)$ let χ be a cutoff at unit scale, equal to 1 at unit distance at most 1 from P and to 0 at distance at least 2, and put $\tilde{U} = e^{\mathrm{i}\eta\chi A} U_r$, with $\tilde{U} = U_r$ off $P^{(3)}$. By Corollary 30.4, U_r is $(M, B_5\theta_r)$-regular, and $|J_{U_r}| \le \theta_r$. By (38.1), χA obeys the bounds of Definition 38.2 with $C_{\mathrm{ext}}(\delta_0 + \delta_1)$: $|\chi A| < \delta_0$, and its covariant differences

and Hölder quotients are at most $C_{\text{ext}}(\delta_0+\delta_1)$, with C_{ext} depending only on d, α and the cutoff. Here (38.1) is applied with $\sigma \le 3\varepsilon_a$. Since F depends only on the bonds of P, and $\tilde U = U$ there, $F(\tilde U)$ is a function of $U|_P$. It is holomorphic near every point $U_0|_P$, $U_0 \in \mathcal{U}(P)$. Indeed, fix (U_r, A) for U_0. For W near $U_0|_P$, let $\tilde U_W$ equal W on P and $\tilde U_0$ elsewhere. Since $C_{\text{ext}}(\delta_0+\delta_1) < \delta_a$ is strict, $\tilde U_W$ stays in the domain of F for W near $U_0|_P$. So $G(W) = F(\tilde U_W)$ is holomorphic, and it equals the restricted activity.

How to read the next remark. Remark 38.8 fixes the constants of the graded domains and checks the conditions on them. Most of these conditions concern results of Parts IX and X, so the remark is best read with Part X. The order in which the parameters of Parts VII–X are chosen is Remark 49.8, and the guide collects the order of all parameters in Table 8 (Section G11).

Remark 38.8 (Domain constants). In four dimensions the activities of the renormalization group live on graded domains $\mathcal{U}(X;\theta_r,\delta_1,\delta_0)$, read in the units of their own scale, whose constants do not depend on the coupling, the step, the torus or the polymer, as in [18, (1.11)–(1.16)]. The constants and the conditions on them are as follows.

- *A cap and a threshold.* After M and M' are fixed, fix a cap $\bar\delta \le \min(\delta_{AB}, 1/C_{\text{cx}})$ on the size of the complex references, with δ_{AB} of Proposition 46.4 and C_{cx} of Proposition 46.7. This is not circular. C_{cx} may depend on M and M', but it enters only $\bar\delta$, which is chosen after them. The thresholds $M_{\mathfrak{N}}$ of Proposition 44.11 and $M_1' \ge M_{\mathfrak{N}}$ of Theorem 44.13 are lower bounds for M, so they are fixed before $\bar\delta$: $M_{\mathfrak{N}}$ with the multiplicative constants of Part IX in M_1 doubled, and M_1' with θ' replaced by $2\theta'$; the coercivity constants are the fixed ones of Proposition 46.7, and every decay bound that enters is taken at the fixed rate of the differences there, a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3; neither depends on δ_c, so the same numbers serve at real and at complex references (Proposition 46.7, step (vii) of the proof). By Proposition 46.7 and step (vii) of its proof, at complex references of size $\delta_c < \delta_P \le \bar\delta$ every multiplicative constant changes by at most a factor $1 + C_{\text{cx}}\delta_c \le 2$, and the inputs of θ' by at most $1+\sigma_{\text{cx}}$, where $\sigma_{\text{cx}} = 2^{1/3}-1$, so that $(1+\sigma_{\text{cx}})^3 \le 2$. The threshold of Corollary 34.6 in the list below already contains $\varepsilon \le \varepsilon_{\text{cmp}}(M_L')$ of Lemma 34.3. Let $\varepsilon_{\max}$ be half the smallest of the regularity thresholds listed below. Each is computed by its real-reference proof with every multiplicative constant doubled, K_H included (so that $R_P = \varrho_P/(2K_H)$), and with the coercivity constants and rates of Proposition 46.7. These numbers do not depend on $\bar\delta$ or δ_P, so $\varepsilon_{\max}$ is fixed before δ_P. By Proposition 46.7, they bound the true thresholds at complex references of size $\delta_c < \delta_P$. Since the parameter ε of Definition 18.1 lies in $(0,1]$, $\varepsilon_{\max} \le \frac12$. These thresholds are:
  - ε_P of Theorem 45.5;
  - ε_{ID} of Theorem 35.2, and the thresholds of Sections 36, 41 and 43;
  - ε_0 and c_3'/C of Section 46.1, and ε_{cc} of Section 46;
  - c_5/C of Proposition 23.3;
  - the threshold $C'\varepsilon \le \frac12 c$ of the proof of Proposition 37.4(c);
  - the threshold of Corollary 34.6 and Lemma 37.5, which contains $CL^{3d+1}\varepsilon \le \frac14 c_L$, i.e. $CL^{13}\varepsilon \le \frac14 c_L$ in $d=4$;
  - $\frac{1}{20}$, the bound on $\varepsilon + C\delta_c$ in the proof of Proposition 47.5(b);
  - from Parts IV and VII: ε_{B} of Lemma 31.3(c) at $M = M_0$, the smallness of Corollary 22.11, and ε_* of Lemma 24.1.

The thresholds of Section 44 are those of the second item (Proposition 44.11), and the first item, ε_P, contains them (proof of Theorem 45.5). Then $\delta_P \le \bar\delta$ is chosen as in Proposition 46.7, whose smallness conditions are stated with $\varepsilon_{\max}$.

- *Conditions.* With $c_{\rm q}$ of Lemma 38.5, $C_{\rm ext}$ of Remark 38.7, c of Lemma 32.7, c_a of Propositions 41.7 and 47.8, r_M of Lemma 21.5, and a_2 of Corollary 30.4:

(DC1) $\theta_r \le c_a$ and $\delta_1 \le \delta_0 \le c_a$, with $c_a \le \min(\frac12 c_{\rm q}, a_2, 1)$;

(DC2) $\max(1, B_5)\theta_r \le \varepsilon_{\max}$;

(DC3) $2L^d\theta_r \le c$, i.e. $2L^4\theta_r \le c$;

(DC4) $2C_{\rm ext}(L^{-1}\delta_0 + L^{-2}\delta_1) < \min(\delta_P, \delta_{P,0})$, with $\delta_{P,0}$ of Proposition 52.2;

(DC5) $c_a \le c_{\rm DC}\, r_M L^{-2}$ and $\delta_P \le c_{\rm DC}\, r_M L^{-1}$, with $c_{\rm DC} > 0$ depending only on d and G;

(DC6) $C_{\rm sym}\,\theta_r\,\delta_0 \le \delta_1$, with $C_{\rm sym} = 2C_{\rm refl}$ and $C_{\rm refl}$ of Lemma 50.6, which depends only on d; it is used only for the symmetrization of the generations (Section 50.1).

Under these conditions the following hold.

(i) *Later scales.* Let $j < k$ (for $j = k$ this is the monotonicity of Definition 38.3), let $X \subset Y$ be polymers of the scales j and k, and let $L \ge 3$. Let U be a point of $\mathcal{U}(Y; \theta_r, \delta_1, \delta_0)$, in the units of scale k. In the units of scale j, U lies in $\mathcal{U}(X; L^{-2(k-j)}\theta_r, L^{-2(k-j)}\delta_1, L^{-(k-j)}\delta_0)$, with the same real part. For $k = j+1$ the slack is at least $\frac12\delta_1$ in both complex budgets and $(1-L^{-2})\theta_r$ in the real one. The same holds for the domains $\mathcal{U}^\flat = \mathcal{U}(\cdot; \theta_r, \delta_1^\flat, \delta_0)$, $\delta_1^\flat = \frac14\delta_1$, of Section 50, with $\delta_1^\flat$ in place of δ_1 on both sides. Moreover, for $k = j+1$, a point of $\mathcal{U}(Y)$ of scale k lies in $\mathcal{U}^\flat(X)$ of scale j with slack at least $(\frac14 - L^{-2})\delta_1 \ge \frac18\delta_1$ in the gradient budget. All of this holds, with the same reason, for every domain $X \subset Y$ of scale j in place of a polymer.

(ii) *References.* Let U' be a point of the domain of step $k+1$ at a polymer Y', extended as in Remark 38.7 and read in the units of step k. Then:

- U' is a complex reference (Section 46) of size $\delta_c \le 2C_{\rm ext}(L^{-1}\delta_0 + L^{-2}\delta_1) < \delta_P$, and $< \delta_{P,0}$ for the step from scale 0;
- its real part is $(M, B_5\theta_r)$-regular, with current at most $L^{-3}\theta_r \le \varepsilon_{\max}$;
- the curvature parameter of Lemma 32.7 at the base $W = U'^{(k)}$ satisfies $\mathsf{a} \le 2\theta_r$;
- the averaged contour variables of the step and the transports $\mathfrak{G}$ are defined and holomorphic at the base W, and at the configurations $e^{\mathrm{i}Z}W$ with $|Z|_\infty \le r_I$ (Section 47).

So Theorem 45.5(a), (b), (d) and Propositions 46.7, 47.12 and 47.7 apply at U', and Lemma 32.7 at its base.

(iii) *The coupling.* The remaining uses of the domain are conditions on the couplings of the step, with the small-field parameters $p_k = p(g_k)$, $p_{0,k} = p_0(g_k)$ of Section 49:

- $(1 + C_J)(C_* + 1)\, p_{k+1} g_{k+1} < \theta_r$, so that the background fields $U_{k+1}(V')$ lie in the domain (Section 45);
- $\tau \le C_P L^d g_k p_{0,k}/\delta_1 < \frac12$, which is (Q3) of Section 48 (Section 47.5); for the activities carried by the flow, on the domains $\mathcal{U}^\flat$, the slack of Proposition 47.10 is $\frac18\delta_1$ (Remark 50.9(c)), which only enlarges C_P. Up to constants it also gives the two radii of Section 45.

Reasons. (i) The configuration lives on the bonds of $\mathbb{T}_\eta$ at every scale. In the units of scale j the spacing is $\eta_j = L^{k-j}\eta_k$, so the field of a bond, $U = e^{\mathrm{i}\eta_k A}U_r = e^{\mathrm{i}\eta_j A_j}U_r$, is $A_j = L^{-(k-j)}A$. Covariant differences gain a further factor $L^{-(k-j)}$, and the Hölder quotients a factor $L^{-(2+\alpha)(k-j)}$ in all. Curvatures gain $L^{-2(k-j)}$ and currents $L^{-3(k-j)}$, everywhere on $\mathbb{T}_\eta$. Since $X \subset Y$ and $3L^{-(k-j)} \le 1$, we have $X^{(3)} \subset Y^{(3)}$. A cube $\Lambda_3(z')$ of scale j in $X^{(3)}$ has radius at most 1 in the units of scale k, and its centre lies within distance 1 of Y. Take a point z'' of Y on the lattice of scale k with $|z'' - z'| \le \frac{3}{2}$. Then the cube lies in $\Lambda_3(z'') \subset Y^{(3)}$. So the quantities of Definition 38.3 at X are bounded by those at Y, times the factors above. For $k = j+1$ we have $(1-L^{-2})\delta_1 \ge \frac{1}{2}\delta_1$ and $(1-L^{-1})\delta_0 \ge \frac{1}{2}\delta_0 \ge \frac{1}{2}\delta_1$.

(ii) In the units of step $k+1$ the extension of Remark 38.7 has $|\chi A| < \delta_0$, and covariant differences and Hölder quotients at most $C_{\mathrm{ext}}(\delta_0+\delta_1)$. In the units of step k these are multiplied by L^{-1}, L^{-2} and $L^{-2-\alpha}$. This gives the bound on δ_c, since $L^{-2}\delta_0 \le L^{-1}\delta_0$; the comparison with the local real gauges of the definition of a complex reference is (38.1), whose constant is absorbed in C_{ext}. The real part has curvature and current below θ_r in the units of step $k+1$, hence below $L^{-2}\theta_r$ and $L^{-3}\theta_r$ in the units of step k. So Corollary 30.4, applied at step k (with $\theta_r \le c_a \le a_2$), makes it $(M, B_5 L^{-2}\theta_r)$-regular, a fortiori $(M, B_5\theta_r)$-regular. The parameter a is a curvature in the units of step $k+1$ (Section 47, Setting). By Proposition 21.12, the curvatures of U'_r and of $(U'_r)^{(k)}$ are at most $2\theta_r$. The conditions (DC1)–(DC4) and the choice of $\varepsilon_{\max}$ give the hypotheses of the results listed, except for the last item, which is (DC5):

- *Real parts.* A point of the domain of scale j, read in its own units, has a real part with curvature below $\theta_r \le c_a$. The configuration it induces on the unit lattice of scale j has curvature at most $2c_a$ (Proposition 21.12). By the first condition of (DC5) with $c_{\mathrm{DC}} \le \frac{1}{128}$, $2c_a \le \frac{1}{8}\varepsilon_M$, with $\varepsilon_M = r_M/(8L^2)$ of Section 32. So the real parts lie in the regular set $\mathfrak{M}$ of the step at that scale, with the margin of (32.1), and all their averaged contour variables are defined and real (Lemma 21.7(i)). The real base $W_r = (U'_r)^{(k)}$ of (ii) has curvature at most $2L^{-2}\theta_r$ in the units of step k, so its margin is larger.

- *Complex parts.* By Proposition 46.4(i) and Lemma 46.3(a), in the units of step k the base is $W = e^{\mathrm{i}\beta}W_r$ bondwise, with $|\beta|_\infty \le C_{\mathrm{AB}}\delta_c < C_{\mathrm{AB}}\delta_P$. A configuration $e^{\mathrm{i}Z}W$ with $|Z|_\infty \le r_I$ is $e^{\mathrm{i}(Z\star\beta)}W_r$, where $Z\star\beta = \ell(e^{\mathrm{i}Z}e^{\mathrm{i}\beta})$ bondwise, and $|Z\star\beta|_\infty \le 2(r_I + C_{\mathrm{AB}}\delta_P)$ by Lemma 21.2(c). Since $r_I \le \frac{1}{32}c_0 L^{-1}$ (Section 47.2), the second condition of (DC5), with c_{DC} small depending on d and G, gives $|Z\star\beta|_\infty \le \frac{1}{8}c_0 L^{-1}$, with c_0 of Lemma 32.3(c). It also gives $L\,C_{\mathrm{AB}}\delta_P \le c$ with the constant c of Lemma 22.7(f). So Lemma 32.3(c) at the real base W_r, which lies in $\mathfrak{M}$, makes the averaged contour variables of $e^{\mathrm{i}Z}W$ defined and holomorphic in (Z,β). Lemma 22.7(f) makes $\mathfrak{G}_W$ defined and holomorphic in β, with entries within $CL^2C_{\mathrm{AB}}\delta_c$ of those of $\mathfrak{G}_{W_r}$. Since β is holomorphic in the complex part of U' (Lemma 46.3(b)), all these objects are holomorphic in the reference.

(iii) By Proposition 31.5, the curvature and the current of $U_{k+1}(V')$ are at most $(1 + C_J)(C_* + \delta)p_{k+1}g_{k+1}$ in the units of step $k+1$ (Section 45). The bound on τ is that of Section 47.5.

Lemma 38.5 converts the size of the activity into a bound that is quadratic in the gauge-invariant curvature and current, and in the derivatives of the complex part. The polynomial factors in the diameter are absorbed by the exponential decay of the activities.

Corollary 38.9 (The one-loop vacuum polarization)**.** *Define* $\mathfrak{e}_m(b;U) = -\frac{1}{2}(\ell_{k,m}(b;U) - \ell_{k,m}(b;1))$. *Then*

$$-\tfrac{1}{2}\log\det\Delta_k(U) = -\tfrac{1}{2}\log\det\Delta_k(1) + \sum_{b\in\mathfrak{F}}\sum_{m\ge 0}\mathfrak{e}_m(b;U),$$

and each $\mathfrak{e}_m(b;\cdot)$ *is a normalized background activity at* $\widehat{\Lambda_m^+}$*, with* $\|\mathfrak{e}_m(b)\| \le C_L e^{-\gamma_3 m}$*, for*

constants ε_a, δ_a depending on L and M. For real $U \in \mathfrak{U}(\theta)$ with θ small,

$$\sum_{m \geq 0} |\mathfrak{e}_m(b; U)| \leq C_L(\theta^2 + e^{-cn}).$$

Proof. The identity follows from Proposition 38.1, at U and at 1, and Proposition 34.8. Details
Normalization holds by construction. The properties of a background activity follow from Proposition 38.1. The domain of definition consists of the regular, not necessarily critical, backgrounds and their complex perturbations. On it, Δ_Y is coercive with uniform constants by Lemma 37.5 and Proposition 37.6; here $\varepsilon_a \leq \varepsilon_{\max}$ and $C_{\mathrm{cmp}}\delta_a \leq c_L''$, so that every configuration of Definition 38.2 is a complex perturbation as in Lemma 37.5 (first item of Section 46). Hence the sup bound $\|\mathfrak{e}_m(b)\| \leq C_L e^{-\gamma_3 m}$ holds on the whole domain. For the bound, take $\theta_r = \delta_1 = \delta_0 = c_{\mathrm{ol}}$ in Remark 38.7, with $c_{\mathrm{ol}} > 0$ so small that $c_{\mathrm{ol}} \leq \min(\frac{1}{2}c_{\mathrm{q}}, a_2)$, $\max(1, B_5)c_{\mathrm{ol}} \leq \varepsilon_a$ and $2C_{\mathrm{ext}}c_{\mathrm{ol}} < \delta_a$, so that $\mathfrak{e}_m(b; \cdot)$ is a normalized graded activity at $\widehat{\Lambda_m^+}$ with sup at most $C_L e^{-\gamma_3 m}$. Apply Lemma 38.5 with $U_r = U$, $A' = 0$ and a cube $\Lambda \supset (\widehat{\Lambda_m^+})^{(3)}$ of side at most $C(1+m+R_0)$. Then $\mathfrak{m} = 0$, $D_N \leq C(1+m+R_0)$ and $B_P \leq C_L(1+m+R_0)^p$, so $\mathfrak{s} \leq C_L(1+m+R_0)^{p+1}\theta$. When this is at most a small constant, all hypotheses hold, and the lemma gives $C_L(1+m+R_0)^{2p+2}\theta^2 e^{-\gamma_3 m}$. This requires $C(1+m+R_0) \leq n$ as well. For larger m use $|\mathfrak{e}_m| \leq C_L e^{-\gamma_3 m}$, together with $e^{-\gamma_3 m/2} \leq C\theta^2$ when $m \geq c'\theta^{-1/(p+1)}$, and $e^{-\gamma_3 m} \leq e^{-cn}$ when $C(1+m+R_0) > n$. Summing over m gives the bound.

The term e^{-cn} cannot be dropped. A flat background with non-trivial holonomy around the torus has $\theta = 0$ but is not gauge equivalent to 1, and $\ell_k(b; U) - \ell_k(b; 1)$ is then generically nonzero, of order e^{-cn}. It reflects polymers that wrap around the torus. On the graded domains of Remark 38.8 the one-loop pieces are of order one, so these sources are of the same order, and they need no comparison with the coupling. In polymer form (Proposition 41.7(d)) they are bounded at all later scales under (T0)–(T2) by Lemma 52.1(c)–(e). □

This bounds the one-loop activities at real backgrounds, under a global small-field bound. On the main line it is superseded by Proposition 41.7: the pieces here are of radius type $(e^{-\gamma_3 m})$ rather than polymer type, and their size at later scales is not derived here. Since Lemma 38.5 allows complex points, and Remark 38.7 makes $\mathfrak{e}_m(b; \cdot)$ a graded activity, the argument above also bounds $\mathfrak{e}_m(b; U)$ at complex U, provided U has the form of Definition 38.3 on $(\widehat{\Lambda_m^+})^{(3)}$. The polymer bounds on the graded domains, at complex points and at later scales, are Proposition 41.7(c),(d) and Remark 41.8(b). The vacuum part $-\frac{1}{2}\log\det\Delta_k(1)$ is independent of the background and enters the vacuum energy. The normalized activities are of second order in the curvature and current. The coefficient C_L does not contain g_k: this is a one-loop effect.

39 Localization of the background map and of the interaction

Status. This section is not on the main line. Its localization is of radius type (Summary below). It is superseded by the weakened background map of Section 45 and the step quantities of Section 47. Only the scaling of the plaquette terms in the proof of Theorem 39.4 is used later, in Proposition 47.5.

The interaction $\mathcal{V}$ of Theorem 34.9, and the old activities composed with the background map $Z \mapsto e^{i\eta A_*(\hat{Z}(Z))}U$, depend on the fluctuation and on the background through the nonlinear map $B \mapsto A_*(B)$ of Part V. That map is the fixed point of a contraction. We approximate it by fixed points of maps built with Dirichlet operators and with cut-off sources, so that every nonlinear expression is evaluated in the interior of a domain. The contraction needs pointwise bounds with derivatives for the Dirichlet operators at interior points. These are provided by Theorem 35.2, proved in Section 36.

39.1 Comparison of fixed points

For fields h write $|h|_1(x) = |h(x)| + |\nabla h(x)|$, the gradient involving only neighbours of x, and for a unit point y put $a_y(h) = \sup_{x \in B(y)} |h|_1(x)$. The comparison is made in these block quantities. Theorem 35.2 gives bounds from block suprema of the data to block quantities of the output; it gives no bound on a fine-lattice kernel, and none is used.

Lemma 39.1 (Comparison of fixed points)**.** *Let Ω be a domain, let χ be a function with $0 \le \chi \le 1$, $\chi = 1$ on $\Omega_{(2)}$, $\chi = 0$ on $\mathbb{T}_\eta \setminus \Omega_{(1)}$, $|\nabla \chi| \le 1$, and let $\Sigma \supset \Omega \setminus \Omega_{(2)}$. Let $\mathcal{T}$ be a map of fields on $\mathbb{T}_\eta$ with fixed point A, and $\mathcal{T}_\Omega$ a map of fields on Ω with fixed point A^Ω. Assume, for the unit points y with $B(y) \subset \Omega$:*

(i) for fields A_1, A_2 in a ball containing A and $\tilde{A} = \chi A^\Omega + (1-\chi)A$,

$$a_y(\mathcal{T}(A_1) - \mathcal{T}(A_2)) \le \sum_{y'} K(y,y')\, a_{y'}(A_1 - A_2), \qquad \sup_y \sum_{y'} K(y,y') e^{\gamma' |y-y'|} \le \tfrac{1}{4};$$

(ii) $a_y(\mathcal{T}(\tilde{A}) - \mathcal{T}_\Omega(A^\Omega)) \le \sigma e^{-\gamma' \operatorname{dist}(y,\Sigma)}$.

Then $a_y(A - A^\Omega) \le 2\sigma e^{-\gamma' \operatorname{dist}(y,\Sigma)}$ for $B(y) \subset \Omega$.

Proof. Let $\delta = A - A^\Omega$ on Ω, and $\|\delta\|_w = \sup_y a_y(\delta) e^{\gamma' \operatorname{dist}(y,\Sigma)}$, the supremum over the y with $B(y) \subset \Omega$; it is finite. Then Details

$$\delta = (\mathcal{T}(A) - \mathcal{T}(\tilde{A})) + (\mathcal{T}(\tilde{A}) - \mathcal{T}_\Omega(A^\Omega)), \qquad A - \tilde{A} = \chi\delta.$$

Since χ vanishes on the outer layer $\Omega \setminus \Omega_{(1)}$, the field $\chi\delta$ vanishes off $\Omega_{(1)}$, its gradient involves δ only on Ω, and $|\nabla(\chi\delta)| \le |\nabla\delta| + |\nabla\chi||\delta|$. So $a_{y'}(\chi\delta) \le 2a_{y'}(\delta)$ if $B(y') \subset \Omega$, and $a_{y'}(\chi\delta) = 0$ otherwise. Since $e^{-\gamma' \operatorname{dist}(y',\Sigma)} \le e^{\gamma'|y-y'|} e^{-\gamma' \operatorname{dist}(y,\Sigma)}$, assumption (i) bounds a_y of the first term by $\frac{1}{2}\|\delta\|_w e^{-\gamma' \operatorname{dist}(y,\Sigma)}$. Assumption (ii) bounds the second. Hence $\|\delta\|_w \le \frac{1}{2}\|\delta\|_w + \sigma$. □

The hybrid field $\tilde{A}$ is what allows a comparison of the torus map with a map defined only on Ω, without boundary estimates for the latter.

39.2 Dirichlet background maps

Let U be a background as in Section 35, and let B be a complex unit-lattice field with $|B|_\infty \le r_1$. Recall from Section 26 that $A_*(B) = A' - \mathbf{H}\mathcal{D}(A')$ with $A' = A_1 + \mathbf{H}B$, where A_1 is the fixed point of

$$\mathcal{T}(A_1) = -\mathbf{C}(J + \nabla\mathcal{V}(A_1 + \mathbf{H}B)).$$

The gradient $\nabla\mathcal{V}$ is the sum of the terms listed in the proof of Lemma 26.2(a). Write each of them with the derivatives outside, e.g. $\mathbf{H}^T D^T D A' = (D\mathbf{H})^T (DA')$ and $D^T D\mathbf{H}\mathcal{D} = D^T(D\mathbf{H}\mathcal{D})$. Then it is a composition of:

- the operators $\mathbf{H}$, $\mathbf{H}^T$, $D\mathbf{H}$, $(D\mathbf{H})^T$;
- the derivatives $d\mathcal{D}$, $d\mathcal{D}^T$;
- local functions of A' and $\nabla A'$ (from $\nabla\mathcal{N}$);
- one outer D^T, absorbed by $\mathbf{C}D^T$.

Definition 39.2 (Dirichlet background map)**.** Fix $R_1 \ge 10$. For a domain X let $\Omega_X = X_{(4)}$. Let ζ be a smooth function on $\mathbb{T}_\eta$ with $\zeta = 1$ on $X_{(R_1)}$, $\zeta = 0$ off $X_{(R_1-2)}$, and bounded derivatives (up to second order and Hölder), and let $\hat{\zeta}$ be the indicator of the unit bonds in $\mathfrak{B}(X_{(R_1)})$. Fine fields on Ω_X are extended by 0. Every pairing over X that involves the field A', such as $(D\mathbf{H}^X)^T(DA')$, is evaluated with $\zeta A'$ in place of A', so that the extension by zero creates no jump. Define:

- $\mathcal{D}^X(A')$ as the fixed point of $Y \mapsto \hat{\zeta} C_k^{\mathrm{av}}(A' - \mathbf{H}^X Y)$;
- $\nabla\mathcal{V}^X$ by replacing, in the expression for $\nabla\mathcal{V}$:
  - every operator by its Dirichlet version on X (Proposition 35.1) with $W = 0$;
  - $\mathcal{D}$ by $\mathcal{D}^X$;
  - J and $\nabla\mathcal{N}$ by ζJ and $\zeta\nabla\mathcal{N}$;

  and multiplying the result by ζ;
- $\mathcal{T}^X(A_1) = -\mathbf{C}^X\zeta(J + \nabla\mathcal{V}^X(A_1 + \mathbf{H}^X\hat{\zeta}B))$, restricted to Ω_X;
- $A_*^X(B) = A'^X - \mathbf{H}^X\hat{\zeta}\mathcal{D}^X(A'^X)$ on Ω_X, where $A'^X = A_1^X + \mathbf{H}^X\hat{\zeta}B$ and A_1^X is the fixed point of $\mathcal{T}^X$.

Every nonlinear expression is evaluated on $X_{(R_1-2)} \subset \Omega_X$, where Theorem 35.2 gives pointwise bounds with derivatives. By construction:

- $A_*^X(B)$ depends only on U on $\hat{X}$ and on B on $\mathfrak{B}(X)$;
- it is covariant under real gauge transformations on $\hat{X}$;
- it is analytic in (B, U);
- for $X = \mathbb{T}_\eta$ it equals $A_*(B)$.

Proposition 39.3 (Comparison of background maps)**.** *There are $r_4, C, \gamma_4 > 0$ such that for $\varepsilon + |B|_\infty \le r_4$:*

(a) $\mathcal{T}^X$ is a contraction on $\{|A_1|_1 \le \rho_2 \text{ on } \Omega_X\}$, uniformly in X, and $|A_*^X(B)|_1 \le C(\varepsilon + |B|_\infty)$ on Ω_X;

(b) the Hölder quotients $[\nabla A_*^X(B)]_{\alpha,B(y)}$, for blocks in Ω_X, are bounded by $C(\varepsilon + |B|_\infty)$, by the Hölder parts of Theorem 35.2;

(c) for domains $X' \subset X$ and $x \in \Omega_{X'}$,

$$|A_*^X(B) - A_*^{X'}(B)|_1(x) \le C(\varepsilon + |B|_\infty)\, e^{-\gamma_4 \operatorname{dist}(x,\Sigma')},$$

where Σ' is the set of points within distance $R_1 + 10M + 4$ of $(\mathbb{T}_\eta \setminus X') \cup (X \setminus X')$. The Hölder quotients $[\nabla(A_*^X(B) - A_*^{X'}(B))_y^\sim]_{\alpha,B(y)}$, for blocks $B(y) \subset \Omega_{X'}$, obey the same bound with $x = y$.

Proof. (a) The proof of Proposition 26.3 applies on Ω_X, with Lemma 24.1 replaced by Theorem 35.2. The sources are cut off to $X_{(R_1)}$, so they are evaluated where the interior bounds hold. The map $\mathcal{D}^X$ is defined by the contraction of Lemma 26.1: $C_k^{\mathrm{av}}(A)(c)$ for $c \in \mathfrak{B}(X_{(R_1)})$ depends on A on $S_c \subset \Omega_X$, and only the sup norm of $\mathbf{H}^X Y$ there enters. Details

(b) follows from the Hölder parts of Theorem 35.2 as in (a). (c) Apply Lemma 39.1 with $\mathcal{T} = \mathcal{T}^X$ (regarded as a map of fields on Ω_X), $\mathcal{T}_\Omega = \mathcal{T}^{X'}$, $\Omega = \Omega_{X'}$ and $\Sigma = \Sigma' \cap \Omega_{X'}$. Choose r_4 so that the bound of (a) gives $|A_1^X|_1, |A_1^{X'}|_1 \le \frac{1}{6}\rho_2$. Since $|\nabla\chi| \le 1$, $|\chi a + (1-\chi)b|_1 \le 3\max(|a|_1, |b|_1)$, so the hybrid field $\tilde{A}$ also lies in $\{|A_1|_1 \le \frac{1}{2}\rho_2\}$, where (i) is proved. The lemma gives $a_y(A_1^X - A_1^{X'}) \le 2\sigma e^{-\gamma' \operatorname{dist}(y,\Sigma)}$. For $x \in B(y)$, $\operatorname{dist}(x,\Sigma') \ge \operatorname{dist}(y,\Sigma') - 1$, which gives the pointwise form of (c).

Hypothesis (i). The derivative of $\mathcal{T}^X$ in A_1 is $-\mathbf{C}^X\zeta\, d\nabla\mathcal{V}^X(A'^X)$. On the torus, (27.2) in the proof of Proposition 27.5 bounds $d\nabla\mathcal{V}$ in block form, with the factor $C(\varepsilon + \rho_2)$. Its proof uses the block bounds of $\mathbf{H}$, $D\mathbf{H}$, their transposes and $D^T D\mathbf{H}$, the kernels of $d\mathcal{D}$ and $d\mathcal{D}^T$, the local bound for $d\nabla\mathcal{N}$, and the mixed bound (26.1) of Lemma 26.1. The same proof

applies to $\nabla\mathcal{V}^X$, with Theorem 35.2 in place of Lemma 24.1. Indeed $\hat\zeta$ restricts $d^2C_k^{\mathrm{av}}$ to the coarse bonds of $X_{(R_1)}$, whose sets S_c are interior, and ζ restricts the sources and the output to $X_{(R_1-2)}$. So $\mathbf{H}^X$, $D\mathbf{H}^X$, their transposes and $D^TD\mathbf{H}^X$ are needed only at interior blocks, where Theorem 35.2(c),(d) gives block bounds with exponential decay. In particular

$$\big|(d^2\mathcal{D}^X[h,\cdot])^T\omega\big|_{B(y)} \le C|\omega|_\infty \sum_{y'} e^{-\gamma|y-y'|}a_{y'}(h),$$
$$\big|d\nabla\mathcal{V}^X(A')h\big|_{B(y)} \le C(\varepsilon+\rho_2)\sum_{y'} e^{-\gamma|y-y'|}a_{y'}(h),$$

with $|f|_{B(y)}$ the supremum of $|f|$ on $B(y)$. By Theorem 35.2(a),(b), $a_y(\mathbf{C}^X\zeta f) \le C\sum_{y'} e^{-\gamma|y-y'|}|f|_{B(y')}$. Composing, (i) holds with $K(y,y') = C_{\mathrm{map}}(\varepsilon+\rho_2)e^{-\gamma_{\mathrm{map}}|y-y'|/2}$. Then $\sum_{y'} K(y,y')e^{\gamma'|y-y'|} \le \frac14$ for $\gamma' \le \gamma_{\mathrm{map}}/4$, by the choice of θ_0 in Section 27: $\varepsilon+|B|_\infty \le r_4 \le \theta_0$, where we take $r_4 \le \theta_0$, and $\rho_2 \le (8B_1 + C_H + K)\theta_0$, with B_1, C_H and K as there (this K is the constant $4C_C(1+C_H)$ of Section 26, not the kernel above). No bound on a fine-lattice kernel of $d\mathcal{T}^X$ is used.

Hypothesis (ii). $\mathcal{T}^X(\tilde A) - \mathcal{T}^{X'}(A^{X'})$ is a sum of three kinds of terms:

- differences of Dirichlet operators on X and X', which have seam form by the pointwise comparison of Theorem 35.2(e) and are applied to fields bounded by $C(\varepsilon+|B|_\infty)$;
- differences of the cutoffs $\zeta_X - \zeta_{X'}$ and $\hat\zeta_X - \hat\zeta_{X'}$, supported within distance R_1 of $\mathbb{T}_\eta \setminus X'$;
- terms in which $\tilde A$ and $A^{X'}$ differ, i.e. outside $\Omega_{X'(2)}$, which lie in Σ'.

Each kind contributes at most $C(\varepsilon+|B|_\infty)e^{-\gamma\,\mathrm{dist}(y,\Sigma')}$ to a_y, after composition with the decaying block kernels. Since M may be much larger than R_1, the mass terms of $\mathfrak{A}^{X,\kappa}$ and $\mathfrak{A}^{X',\kappa}$ are active on parts of $X_{(R_1)}$; their difference is part of the comparison of domains and is included in Σ'. The same comparison applies to $\mathcal{D}^X$ and to $A_*^X = A'^X - \mathbf{H}^X\hat\zeta\mathcal{D}^X(A'^X)$.

Hölder quotients in (c). Write $\delta = A_1^X - A_1^{X'}$ on $\Omega_{X'}$ as $(\mathcal{T}^X(A_1^X) - \mathcal{T}^X(\tilde A)) + (\mathcal{T}^X(\tilde A) - \mathcal{T}^{X'}(A_1^{X'}))$. The derivative of $\mathcal{T}^X$ is $\mathbf{C}^X\zeta$ applied to $d\nabla\mathcal{V}^X$, and the block bound for $d\nabla\mathcal{V}^X$ in the proof of hypothesis (i) maps the block quantities $a_{y'}$ to block suprema, with the same exponential weights. Theorem 35.2(a),(c) therefore bound the Hölder quotients of the gradient of the first term by C times the weighted bound on $a_{y'}(\chi\delta)$ just proved. The second term is bounded by the Hölder parts of Theorem 35.2(e), as in the proof of hypothesis (ii). The same applies to $\mathbf{H}^X\hat\zeta\mathcal{D}^X$, which gives the Hölder part of (c). □

39.3 Localization of the interaction

Write $\Phi(A) - \Phi(0) = \sum_p s_p(A)$, where, with $H = U(\partial p)$ and $H_A = (e^{i\eta A}U)(\partial p)$,

$$s_p(A) = \tfrac12 \operatorname{tr}\left(H + H^{-1} - H_A - H_A^{-1}\right)$$

is the difference of the plaquette terms of the analytic Wilson action of Section 23; in $d=4$ the action carries no power of η. It depends only on A on the bonds of p and is holomorphic in complex A; for real A it equals $\operatorname{Re}\operatorname{tr}(H - H_A)$. The real part itself is not holomorphic, and it is not used at complex fields. Then $|s_p(A) - s_p(A')| \le C\eta^d(\varepsilon + |A|_1 + |A'|_1)\,|A - A'|_1$ near p, by the expansion of Proposition 23.3.

Theorem 39.4 (Localization of the interaction)**.** *Let $r_5 = \min\left(r_2, \frac12 r_L, r_4/(1+2CL^d)\right)$, with r_2 of Lemma 34.1, r_L and C of Lemma 32.7(c), and r_4 of Proposition 39.3; and let $\gamma_5 = \gamma_4$ of Proposition 39.3. In the setting of Theorem 34.9, with $U = U'$ and $\varepsilon \le r_5$, there are pieces $\mathcal{V}_{p,m}$, indexed by plaquettes p of $\mathbb{T}_\eta$ and $m \ge 0$, such that for complex Z with $|Z|_\infty \le r_5$:*

$$\mathcal{V}(Z) = \sum_p \sum_{m\ge 0} \mathcal{V}_{p,m}(Z),$$

where:

- $\mathcal{V}_{p,m}$ *depends only on* Z *and* U *in* $\widehat{\Lambda_m(\{p\})}$ *(enlarged by* $2L$ *for* Z*);*
- *it is analytic and invariant under real gauge transformations;*
- *it is of third order in* Z*, with* $|\mathcal{V}_{p,m}(Z)| \le C_L |Z|^3_{\Lambda_m(\{p\})} e^{-\gamma_5 m}\eta^d$.

Here $|Z|_\Lambda$ *is the supremum of* Z *on the bonds of* $\mathfrak{F}$ *within distance* $2L$ *of* Λ.

Proof. For a domain X containing p in $X_{(R_1+1)}$, put $F_p^X(B) = s_p(A_*^X(B)) - s_p(A_*^X(0))$. For $X = \mathbb{T}_\eta$ this is $s_p(A_*(B)) - s_p(A_*(0))$, and $A_*(0) = 0$ since U' is critical. By Proposition 39.3(c), for $X = \Lambda_m(\{p\})$ and $X' = \Lambda_{m-1}(\{p\})$, $m \ge m_1 = R_1 + 10M + 5$, Details

$$|F_p^{\Lambda_m}(B) - F_p^{\Lambda_{m-1}}(B)| \le C(\varepsilon + |B|_\infty)^2 e^{-\gamma_4(m-m_1)}\eta^d,$$

since s_p is of order $\eta^4 = \eta^d$ in the fields. Lemma 37.1, started at m_1 (the pieces with $m < m_1$ are 0, and $F_{p,m_1} = F_p^{\Lambda_{m_1}}$), gives $\mathcal{S}(B) - \mathcal{S}(0) = \sum_p \sum_m F_{p,m}(B)$, with pieces depending only on B and U in $\widehat{\Lambda_m(\{p\})}$, and $|F_{p,m}(B)| \le C e^{\gamma_4 m_1}(\varepsilon + |B|_\infty)^2 e^{-\gamma_5 m}\eta^d$. For $|Z|_\infty \le 2r_5$ we have $|\hat Z(Z)|_\infty \le CL^d|Z|_\infty$ (Lemma 32.7(c)), so $\varepsilon + |\hat Z(Z)|_\infty \le r_4$, and Proposition 39.3 applies at $B = \hat Z(Z)$. Put $G_{p,m}(Z) = F_{p,m}(\hat Z(Z))$. These are analytic, vanish at $Z = 0$, and by Lemma 32.7 depend on Z only within distance $2L$ of $\Lambda_m(\{p\})$.

Let T_2G denote the Taylor polynomial of degree 2 at $Z = 0$. Since the sums converge absolutely,

$$\sum_{p,m} T_2 G_{p,m} = T_2(\mathcal{S}(\hat Z) - \mathcal{S}(0)) = \tfrac12 \Delta_k(Z,Z) - \tfrac12 \langle \mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle$$

by the second display of Lemma 34.1: the coarse action contributes the part $\frac12\Delta^0(\hat Z^1 Z, \hat Z^1 Z) + \langle \omega_0, \hat Z^2(Z)\rangle$ of $\frac12\Delta_k$, and the gauge-fixing term $\frac12\langle \mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle$ comes from Φ_W, not from $\mathcal{S}$. The linear term vanishes. Hence $\mathcal{V}_{p,m} = G_{p,m} - T_2 G_{p,m}$ sum to $\mathcal{V}$. For the bound fix Z with $0 < |Z|_\infty \le r_5$, and put $g(t) = \mathcal{V}_{p,m}(tZ)$. The comparison bounds hold on the complex ball of radius $2r_5$, where $\sup|G_{p,m}| \le C(\varepsilon + 2r_5)^2 e^{-\gamma_5 m}\eta^d$. So g is analytic for $|t| < 2r_5/|Z|_\infty$. By Cauchy's estimate each Taylor term of $t \mapsto G_{p,m}(tZ)$ of degree at most 2 obeys the same bound there, so $|g(t)| \le 4C(\varepsilon + 2r_5)^2 e^{-\gamma_5 m}\eta^d$. Since g vanishes to third order at $t = 0$, the maximum principle for $g(t)/t^3$ on the discs $|t| \le \varrho$, $\varrho < 2r_5/|Z|_\infty$, and the limit $\varrho \to 2r_5/|Z|_\infty$ give

$$|\mathcal{V}_{p,m}(Z)| \le 4C(\varepsilon + 2r_5)^2 (2r_5)^{-3} |Z|^3_\infty e^{-\gamma_5 m}\eta^d \le \tfrac92 C r_5^{-1} |Z|^3_\infty e^{-\gamma_5 m}\eta^d,$$

using $\varepsilon \le r_5$. The factor r_5^{-1} is part of C_L. Since $G_{p,m}$ depends on Z only within distance $2L$ of $\Lambda_m(\{p\})$, $|Z|_\infty$ may be replaced by $|Z|_{\Lambda_m(\{p\})}$. Gauge invariance follows from the covariance of A_*^X and the invariance of s_p. □

The same argument localizes the old activities composed with the background map. An activity has its own domain, so the radius depends on it. Let P be a domain, let m_1 be the least m with $P^{(2)} \subset \Omega_{\Lambda_m(P)}$, and let χ_P be a smooth function, equal to 1 on $P^{(1)}$ and to 0 off $P^{(2)}$, with bounded derivatives. For $m \ge m_1$ the field $\chi_P A_*^{\Lambda_m(P)}(\hat Z(Z))$ is defined on $\mathbb{T}_\eta$; the cutoff, not an extension by zero, keeps its derivatives bounded. By Proposition 39.3, Lemma 32.7(c) and (38.1), there is C_{a}, depending only on d, α, G, L, M and R_1, such that the following holds in the local gauges of U', for complex Z with $|Z|_\infty \le r_5$. The sup norm, the gradient and the Hölder quotient of $\chi_P A_*^{\Lambda_m(P)}(\hat Z(Z))$ are at most $C_{\mathrm{a}}(\varepsilon + L^d|Z|_\infty)$. For $m > m_1$, those of $\chi_P(A_*^{\Lambda_m(P)} - A_*^{\Lambda_{m-1}(P)})(\hat Z(Z))$ are at most $C_{\mathrm{a}}(\varepsilon + L^d|Z|_\infty)e^{-\gamma_4 m}$. For the latter, the points of $P^{(2)}$ are at distance at least $m - R_1 - 10M - 8$ from the set Σ' of Proposition 39.3(c), and the factor $e^{\gamma_4(R_1+10M+8)}$ is part of C_{a}.

Corollary 39.5 (Composed activities)**.** *Let F be a background activity at a domain P, with parameters ε_a, δ_a (Definition 38.2). Assume $\varepsilon \le \varepsilon_a$ and $C_{\mathrm{a}}\varepsilon \le \frac{1}{4}\delta_a$, and let $0 < r_F \le r_5$ with $C_{\mathrm{a}}L^d r_F \le \frac{1}{4}\delta_a$. Then, for complex Z with $|Z|_\infty \le r_F$,*

$$F\big(e^{\mathrm{i}\eta A_*(\hat{Z}(Z))}U'\big) = \sum_{m \ge m_1} F_m(Z),$$

where F_m depends only on U' in $\widehat{\Lambda_m(P)}$ and on Z within distance $2L$ of $\Lambda_m(P)$, is analytic and gauge invariant, and satisfies $|F_m| \le 2e^{\gamma_4 m_1}\|F\|e^{-\gamma_4 m}$.

Proof. For $m \ge m_1$ let $A_m = \chi_P A_*^{\Lambda_m(P)}(\hat{Z}(Z))$ and $V_m = e^{\mathrm{i}\eta A_m}U'$. On $P^{(1)}$, V_m equals $e^{\mathrm{i}\eta A_*^{\Lambda_m(P)}(\hat{Z}(Z))}U'$, so by the locality of F the value $F(V_m)$ does not depend on the choice of χ_P. If $\Lambda_m(P) = \mathbb{T}_\eta$, then $A_*^{\Lambda_m(P)} = A_*$ and $F(V_m)$ is the left side. Put $F_{m_1} = F(V_{m_1})$ and $F_m = F(V_m) - F(V_{m-1})$ for $m > m_1$; the sum is finite. The reference U' is real, (M, ε_a)-regular and $|J_{U'}| \le \varepsilon_a$, since $\varepsilon \le \varepsilon_a$. By the hypotheses and the choice of C_{a}, A_{m-1} has sup norm, gradient and Hölder quotient at most $\frac{1}{2}\delta_a$, and $A_m - A_{m-1}$ at most $\frac{1}{2}\delta_a e^{-\gamma_4 m}$. So $f(t) = F\big(e^{\mathrm{i}\eta(A_{m-1} + t(A_m - A_{m-1}))}U'\big)$ is analytic and bounded by $\|F\|$ for $|t| < e^{\gamma_4 m}$, and the Schwarz lemma for $f - f(0)$ gives $|F_m| = |f(1) - f(0)| \le 2\|F\|e^{-\gamma_4 m}$. Also $|F_{m_1}| \le \|F\|$. The dependence on U' and Z follows from that of A_*^X (on U on $\hat{X}$ and on B on $\mathfrak{B}(X)$) and from the range $2L$ of $\hat{Z}$. Analyticity follows from that of A_*^X, $\hat{Z}$ and F, and gauge invariance from the covariance of A_*^X and $\hat{Z}$ and the invariance of F; χ_P is scalar. □

Details

39.4 Summary

The background-dependent ingredients of one renormalization group step have the following representations as sums of local pieces with exponential decay:

- the fluctuation covariance and the quadratic form, entrywise (Theorem 37.7);
- the determinant, as a background-independent constant plus normalized activities of second order in the curvature and current (Corollary 38.9);
- the interaction, as pieces of third order in the fluctuation (Theorem 39.4), and the old activities composed with the background map (Corollary 39.5).

The first two use only kernel bounds. The last two also use the interior pointwise bounds of Theorem 35.2.

All four are of *radius type*. A piece that depends on the fields and on the background in a ball Λ_m is bounded by $e^{-\gamma m}$. The polymer expansion of the next stage needs *volume type*, i.e. a fixed small factor for every cube of the dependence region, as in [15, (116)–(122)]. A ball of radius m contains of order m^d cubes, so the bound $e^{-\gamma m}$ does not suffice. Bałaban and Dimock obtain volume-type localization from random-walk expansions with weakening parameters. For the operators of Parts II and III these require the nested walks of Section 41. The Dirichlet operators of Part III and Theorem 35.2 are the local propagators of such walks. The results of this part are therefore inputs to that construction, but they do not replace it. The local Jacobians $\mathfrak{J}$, the factors $\mathfrak{j}_c$ and the gauge-fixing remainder G_3 of Theorem 34.9 are local already (G_3 is a sum of terms localized in single blocks $B'(y)$). With the pieces above, they are the background-dependent inputs of the step, in radius-type form. The later parts proceed as follows.

- Part IX constructs volume-type localizations by random walks with weakening, with the Dirichlet operators of Parts III and VIII as local propagators (Sections 40–45).

- Part X performs the cluster expansion of the Gaussian integral with these volume-type activities, in place of the radius-type ones above, and normalizes the resulting activities at the trivial background (Lemma 38.5).
- In four dimensions the old activities are not reblocked, rescaled or contracted: each generation is bounded in the norm of its creation scale, and its marginal part is extracted at the later scales (Sections 50, 51 and 53, Theorem 53.12).

Part IX: Weakened random walks

The localization of Part VIII is of radius type. The polymer expansion of Part X needs volume type: a fixed small factor for every cube of the dependence region (introduction to Part VIII). Part IX provides it, as Bałaban and Dimock do, by random walks with weakening parameters. A product paving of the torus into boxes $\square$ of side between $M/4$ and M is fixed, with one parameter $s_\square$ per box, and $s = (s_\square)$ ranges over $\mathbb{D}_\kappa = \{|s_\square| \le e^\kappa\}$ (Section 40).

What weakening buys. Let $\kappa \ge 1$, let $f(s)$ be analytic in the interior of $\mathbb{D}_\kappa$ and continuous on it, with $|f| \le F$, and suppose that it decouples: if $s_\square = 0$ outside a union of boxes, then f depends only on the parameters and the data near the component that contains a given polymer Y. Lemma 47.2 then writes $f(1)$ as a sum over polymers $Z \supset Y$ of pieces that depend on the data only near Z and are bounded by $Fe^{-(\kappa-1)|Z\setminus Y|}$: a factor $e^{-(\kappa-1)}$ for every box. The proof has two steps. Write $f(1)$ by $g(1) = g(0) + \int_0^1 g'(t)\,dt$ in each variable $s_\square$ outside Y; the terms in which a differentiated box is not connected to Y vanish by the decoupling. Each derivative is bounded by Cauchy's formula on a circle of radius $e^\kappa - 1$, which stays in $\mathbb{D}_\kappa$; this gives the factor per box. So volume-type localization reduces to one task: to build every ingredient of the step as a decoupling function of s, analytic and bounded uniformly on $\mathbb{D}_\kappa$. Part IX does this, and Part X applies Lemma 47.2 (Section 47).

What this Part proves.

- *Walks in L^2* (Sections 40 and 41). A weakenable family is a sum over polymers X_ω of monomials s^{n_ω} times operators that depend on the data only near X_ω (Definition 40.2). On every domain, the inverse of an operator with local propagators and a factorized non-local part is such a family, for M large, built by a walk in which every step carries the monomial of the boxes near its local domain (Theorem 41.4). The walk runs at five nested levels (L1)–(L5), from the scalar operators to the fluctuation operator. Its outputs are $C_k^{1/2}(s)$ and the determinant densities (Corollary 41.6), and the one-loop source as normalized graded activities with volume decay (Proposition 41.7). The quadratic part of this source is marginal in four dimensions and must be extracted as a coupling, not bounded (Remark 41.8(b)).
- *Pointwise bounds* (Sections 42–44). The background map needs pointwise $C^{1,\beta}$ bounds for $\mathbf{H}(s)$, $\mathbf{C}(s)$ and $\mathbf{C}(s)D^T$, uniformly on $\mathbb{D}_\kappa$, with $\beta = \min(\alpha, \frac{1}{2})$. Interior estimates with L^p data (Theorem 43.2, Corollary 43.8) and regularity tiers give them (Theorems 44.13 and 44.15). In four dimensions L^2 data lie at the endpoint $p = \frac{d}{2}$, so the tiers run $L^2 \to L^4 \to L^8 \to C^{1,\beta}$ (Lemma 44.6). No regularity at the Dirichlet cuts is used.
- *The weakened background map* (Section 45). $A_*(s; B)$ is the fixed point of a weakened contraction. It is bounded uniformly on $\mathbb{D}_\kappa$, analytic in (s, B), equal to $A_*(B)$ at $s = 1$ at critical references, and it factorizes when $s_\square = 0$ outside a union of boxes (Theorem 45.5).

Imports and exports. Part IX builds on the Dirichlet operators, locality and complex perturbations of Part III, on Propositions 35.1 and 37.6, Theorem 35.2(a)–(c), Section 36 and Lemma 38.5 of Part VIII, and on the contraction of Section 26. Its clauses on complex references (in Propositions 41.7 and 44.11 and Theorem 44.13) use Proposition 46.7, and Section 41.5 uses Parts X and XI; both are best read with Part X. Corollary 41.6, Proposition 41.7 and Theorem 45.5 are the inputs of the step in Part X (Theorem 48.9), and the torus independence of the one-loop pieces (Proposition 41.14) is used in Parts X and XI. Theorem 41.4 also completes the gauge-field walk of Theorem 20.2(b). Off the main line: Section 42 is a discussion, whose Route 1 (boundary regularity) is not carried out, and Proposition 43.5 is not used later.

Details for this Part. The companion, chapter on Part IX, begins with the main line through this Part and the δ-budget of the walks, and then expands each of its steps, from the weakenable families to the weakened background map. Of Section 42 it checks only the chain of exponents.

40 Weakenable families

The localization of Part VIII is of radius type: a piece that depends on the data in a ball of radius m is bounded by $e^{-\gamma m}$. The polymer expansion of the next stage needs volume type, a fixed small factor for every cube of the dependence region, as in [15, (116)–(122)]. Bałaban and Dimock obtain this from random walks with weakening parameters $s_\square$, with analyticity for $|s_\square| \le e^{\kappa}$. This part constructs such weakened walks for all the operators of the step, in L^2 norms. Section 42 explains what is missing for pointwise norms.

Setting. We keep the conventions of Part VIII. The background U is real and (M,ε)-regular, or a complex perturbation of such a background as in Proposition 19.11. Here ρ is a fixed number, chosen in Section 41, independent of M. The torus side satisfies $n \ge C(\rho + C_v)M$, with $C \ge 8$ (used in Section 41.5), so that the sets below do not wrap around. We use the grid points z_j, the functions χ_j, $h_j = \chi_j^2$ and the sets K_j, Λ_j of Section 17, and the enlarged domains

$$\Lambda_j^v = \text{the union of the blocks meeting } \{x : |x - z_j|_\infty \le \rho M\}.$$

For $\rho = 9$ these are the domains Λ_j^w of Section 20. We have $|h_j(x) - h_j(x')| \le CM^{-1}|x - x'|$, and every point lies in at most 3^d supports K_j.

Boxes and polymers. Fix a *product paving* $\mathcal{P} = \{I^1_{a_1} \times \cdots \times I^d_{a_d}\}$ of $\mathbb{T}_\eta$. Here each $\{I^\mu_a\}_a$ is a partition of the circle $\mathbb{R}/n\mathbb{Z}$ into intervals of integer length between $M/4$ and M. Its elements are called *boxes*; they are rectangular unions of unit blocks. A *polymer* is a $*$-connected union X of boxes: any two boxes of X are joined by a chain of boxes of X, consecutive ones sharing at least a corner. Because the paving is a product, two boxes that do not touch are separated in some direction by a whole interval, so they are at distance at least $M/4$. At most 2^d boxes lie within distance $r < M/8$ of a point, and each box touches at most 3^d boxes. Write $|X|$ for the number of boxes of X, and $X^{(r)}$ for the r-neighbourhood in lattice units. For each j let $X(j)$ be the union of the boxes that meet $(\Lambda_j^v)^{(R_v)}$. Here $R_v \le C_v M$ is a fixed margin, chosen at each level to contain the dependence region of the local propagators. The local operators on a domain Y depend on the background only on $\hat{Y}$, and those of the fluctuation operator only on $\widehat{Y^+}$, with $Y^+ = Y^{(R_0)}$ and the margin $R_0 \ge R_0(L,M)$ of Proposition 37.6(b). We take $R_v \ge R_0 + 4 + 2r_{\mathrm{loc}}$, where r_{loc} is the largest range of the local operators of the step; then $(\Lambda_j^v)^{(R_v)}$ contains these sets for $Y = \Lambda_j^v$, together with their $2r_{\mathrm{loc}}$-neighbourhoods. Then $X(j)$ is a polymer, and $|X(j)| \le \mathsf{N}_{\mathrm{b}}$, with N_{b} depending only on d, ρ and C_v. (The subscript

Margin	Value, and where it is fixed	Role
$\hat{X}$	$X^{(4)}$ (Conventions of Part VIII, Section 35)	the operators on a domain X depend on U only on $\hat{X}$ (Proposition 35.1(c))
interior points	$y \in X_{(4)}$ (Theorem 35.2)	where the interior pointwise bounds hold
seam of the Dirichlet operators	unit points within distance $10M$ of $X \setminus X'$ (Proposition 35.1(d))	contains the region within distance $8M$ of $X \setminus X'$ where the mass terms of X and X' differ (proof of Theorem 35.2(e))
R_0, $Y^+ = Y^{(R_0)}$	$R_0 \geq 2L+2$ (Section 37.4), $R_0 \geq R_0(L,M) = 2L + 10M + \frac{2}{\gamma'}\log^+ \frac{16 C_{\mathrm{seam}}}{c_L} \leq C_L M$ (Proposition 37.6)	the elimination maps have range $2L$; the local fluctuation operator Δ_Y is built on Y^+, and differs from the compression of Δ_k by at most $\frac{1}{16}c_L$ in norm (proof of Proposition 37.6(a))
seam of the fluctuation operator	$R_0 + 10M + 2L$ (Proposition 37.6(c); level (L5) of Section 41)	comparison of Δ_Y, C_Y on two domains
boxes of the paving	integer sides between $M/4$ and M (this section)	boxes that do not touch are at distance at least $M/4$
margin r of a family	$r \leq M/16$ (Definition 40.2)	$2r < M/4$, so products of families are indexed by polymers (Lemma 40.4)
Λ_j^v, ρ	blocks meeting $\{\lvert x - z_j\rvert_\infty \leq \rho M\}$ (this section); $(\rho-1)M - w \geq \frac{1}{4}(\rho-1)M$ at every level, with w the seam width, e.g. $\rho \geq 2R_0/M + 30$ (41.1)	the local domains of the walk; the seam stays at distance at least $\frac{1}{4}(\rho-1)M$ from K_j
R_v	$R_0 + 4 + 2r_{\mathrm{loc}} \leq R_v \leq C_v M$ (this section)	$X(j)$ contains the dependence regions of the local operators on Λ_j^v

Table 10: The margins of Parts VIII and IX.

distinguishes this number of boxes from the matrix size N of $G \subset U(\mathsf{N})$.) Table 10 collects the margins of Parts VIII and IX.

Monomials. For $s = (s_\square)_{\square \in \mathcal{P}} \in \mathbb{C}^{\mathcal{P}}$ and a function $n : \mathcal{P} \to \{0,1,2,\dots\}$ with finite support, put $s^n = \prod_\square s_\square^{n(\square)}$, with $\deg s^n = \sum_\square n(\square)$ and $\operatorname{supp} s^n = \{\square : n(\square) \geq 1\}$. For a union of boxes X, s^X denotes the monomial with $n = \mathbf{1}_X$. For $\kappa > 0$ let $\mathbb{D}_\kappa = \{s : \lvert s_\square\rvert \leq e^\kappa \text{ for all } \square\}$. Repeated factors in a monomial are harmless; only the support and the degree enter.

40.1 Decay norms

The field spaces are those of Parts II–VII: scalar fields, gauge fields and 2-forms on $\mathbb{T}_\eta$, and unit-lattice fields on $\mathbb{T}$, including fields on the set $\mathfrak{F}$ of Definition 32.1 (the positively oriented bonds of $\mathbb{T}$ other than the bonds $b_0(c)$). In this part $\mathfrak{F}$ also denotes the curvature operator DD of Lemma 12.3, called the curvature factor; the word or the argument tells the two apart. All of them decompose over unit blocks, with projections $\mathbf{1}_y$, $y \in \mathbb{T}$. For an operator F between two such spaces and $\gamma \geq 0$ put

$$\lvert F\rvert_\gamma = \max\Big\{ \sup_y \sum_{y'} e^{\gamma\lvert y-y'\rvert} \lVert \mathbf{1}_y F \mathbf{1}_{y'}\rVert,\ \sup_{y'} \sum_y e^{\gamma\lvert y-y'\rvert} \lVert \mathbf{1}_y F \mathbf{1}_{y'}\rVert \Big\}.$$

Lemma 40.1 (Decay norms). *(a)* $\lVert F\rVert \leq \lvert F\rvert_0 \leq \lvert F\rvert_\gamma$, *and* $\lvert FG\rvert_\gamma \leq \lvert F\rvert_\gamma \lvert G\rvert_\gamma$.

(b) If F has decay (C,γ') with $\gamma'>\gamma$, then $|F|_\gamma \le C S_{\gamma'-\gamma}$, with $S_\beta=\sum_y e^{-\beta|y|}$.

(c) $|[F,h_j]|_{\gamma/2}\le C(\gamma M)^{-1}|F|_\gamma$ for $0<\gamma\le 1$, with C depending only on d.

Proof. (a) The first inequality is the Schur test. For the second, use Details

$$\|\mathbf{1}_y F G \mathbf{1}_{y''}\| \le \sum_{y'} \|\mathbf{1}_y F \mathbf{1}_{y'}\|\,\|\mathbf{1}_{y'} G \mathbf{1}_{y''}\|, \qquad e^{\gamma|y-y''|}\le e^{\gamma|y-y'|}e^{\gamma|y'-y''|}.$$

(b) is immediate. (c) Write

$$\mathbf{1}_y[F,h_j]\mathbf{1}_{y'} = \mathbf{1}_y F \mathbf{1}_{y'}\big(h_j-h_j(c_y)\big) - \big(h_j-h_j(c_y)\big)\mathbf{1}_y F\mathbf{1}_{y'}.$$

On $B(y)$ we have $|h_j-h_j(c_y)|\le CM^{-1}$, and on $B(y')$ we have $|h_j-h_j(c_y)|\le CM^{-1}(1+|y-y'|)$. Hence $\|\mathbf{1}_y[F,h_j]\mathbf{1}_{y'}\|\le CM^{-1}(1+|y-y'|)\|\mathbf{1}_y F\mathbf{1}_{y'}\|$. Finally $(1+t)e^{-\gamma t/2}\le C\gamma^{-1}$. □

On the fine lattice D is not bounded uniformly in η. As in Parts II and III, operators are therefore estimated together with the versions carrying derivatives. For an operator F on fine fields put

$$|F|^D_\gamma = |F|_\gamma + |DF|_\gamma + |FD^T|_\gamma + |DFD^T|_\gamma,$$

where D stands for the covariant derivative appropriate to the spaces involved. For scalar operators followed by D on gauge fields, and so on, the obvious variants are meant.

40.2 Weakenable families

Definition 40.2 (Weakenable family)**.** Let $r\le M/16$. A *weakenable family with margin r* is a collection $(X_\omega, n_\omega, O_\omega)_{\omega\in\Omega}$ with the following properties:

- each X_ω is a polymer;
- n_ω is a function on $\mathcal{P}$ with $\operatorname{supp} s^{n_\omega}$ equal to the set of boxes of X_ω;
- each O_ω is an operator whose kernel is supported in $X_\omega^{(r)}\times X_\omega^{(r)}$, and which depends on the background, and on any other data, only on $X_\omega^{(r)}$.

We write $O(s)=\sum_\omega s^{n_\omega}O_\omega$, $\deg\omega=\deg s^{n_\omega}$, and

$$\|O\|_{\kappa,\gamma} = \sup_{\square\in\mathcal{P}}\sum_{\omega:\,\square\subset X_\omega} e^{\kappa\deg\omega}|O_\omega|_\gamma, \qquad \|O\|^D_{\kappa,\gamma} = \text{the same with } |O_\omega|^D_\gamma.$$

We say that $O(s)$ *represents* an operator O if $O(1)=O$, the series converging absolutely.

The data dependence is part of the definition. It is what makes the weakening also localize the dependence on the background. A family with margin r can always be converted to one with margin 0, on the level of boxes. Replace X_ω by $X_\omega^\sharp$, the union of the boxes meeting $X_\omega^{(r)}$, and multiply the monomial by $s^{X_\omega^\sharp\setminus X_\omega}$. Degrees grow by at most a factor 3^d, and every term then depends on the data, and acts on fields, only inside $X_\omega^\sharp$. We use this conversion whenever dependence inside a union of boxes, not a neighbourhood of it, is needed. This is the case for the factorization over disjoint polymers in the cluster expansion.

Additive local terms are the terms of a family that come neither from a walk (Theorem 41.4), nor from a product of families (Lemma 40.4), nor from a local operator applied to a family (Lemma 40.5). In Section 41 they are the identities in $R_X=\mathbf{1}_X-P_{\mathcal{V},X}$ and $\Pi_X=I-DS_X$ (level (L2)), the local part $d^2\mathcal{N}(0)|_X$ of W_2^X (level (L4)), and the constant $-a$ in $\Delta^{0,X}=\mathbf{\Gamma}^X[W_2^X]-a$ (the repetition of level (L4)). The constant r of level (L5) belongs to the local part of the operator there and is not a term of a family. Such a term T, a multiple of the identity or a local operator of range at most r, is split canonically, box by box. A bond, a

plaquette or a unit bond is based at its initial point, and a point at itself. For a box $\square$ let $\mathbf{1}_\square$ be the projection onto the field components at the sites based in $\square$. Then

$$T = \sum_{\square \in \mathcal{P}} T\mathbf{1}_\square,$$

and $T\mathbf{1}_\square$ is a term with polymer $\square$ and monomial $s_\square$; it has margin at most r. Since the boxes partition $\mathbb{T}_\eta$, this is a decomposition of T. It uses only the paving and the base points, so it involves no choice; in particular it commutes with the translations that preserve the paving, and with the transfers between tori of Section 41.5. The split is used in the L^2 norms of this part. A split by indicators does not preserve Hölder tiers, so no tier bound is claimed for an additive term standing alone (Section 44). The constant $\frac{1}{1+t}$ in the determinant densities is split by the partition of unity h_j (proof of Corollary 41.6), which is canonical as well.

Lemma 40.3 (Basic properties)**.** *Let O be a weakenable family with $\|O\|_{\kappa,\gamma} < \infty$.*

(a) For $s \in \mathbb{D}_\kappa$ the series $O(s)$ converges absolutely, $|O(s)|_\gamma \le 2^d \|O\|_{\kappa,\gamma}$, and $s \mapsto O(s)$ is analytic in the interior of $\mathbb{D}_\kappa$. The same holds with $|\cdot|^D$.

(b) (Localization.) *Let Z be a union of boxes, and let $s_\square = 0$ for all $\square \not\subset Z$. Then only the terms with $X_\omega \subset Z$ contribute. $O(s)$ is block diagonal with respect to the $*$-components Z_i of Z, i.e. $\mathbf{1}_{Z_i^{(r)}} O(s) \mathbf{1}_{Z_{i'}^{(r)}} = 0$ for $i \neq i'$, and $O(s)$ vanishes off $Z^{(r)} \times Z^{(r)}$. The block on Z_i depends on the data only on $Z_i^{(r)}$, and on the weakening parameters only through $s_\square$, $\square \subset Z_i$.*

(c) (Covariance.) *If every O_ω is covariant under real gauge transformations, so is $O(s)$.*

Proof. (a) A term contributes to $\mathbf{1}_y O(s) \mathbf{1}_{y'}$ only if $y \in X_\omega^{(r)}$. Then X_ω contains one of the at most 2^d boxes within distance r of y. Hence Details

$$\sum_{y'} e^{\gamma|y-y'|} \|\mathbf{1}_y O(s) \mathbf{1}_{y'}\| \le \sum_{\square \text{ near } y} \sum_{\omega : \square \subset X_\omega} |s^{n_\omega}|\, |O_\omega|_\gamma \le 2^d \|O\|_{\kappa,\gamma},$$

and similarly for the column sums. The convergence is uniform on $\mathbb{D}_\kappa$, so the sum is analytic. (b) If $X_\omega \not\subset Z$, then $s^{n_\omega} = 0$. A polymer $X_\omega \subset Z$ lies in one component Z_i, and its monomial involves only the boxes of X_ω. Distinct components are at distance at least $M/4 > 2r$, so $X_\omega^{(r)}$ meets no other $Z_{i'}^{(r)}$. (c) is clear. $\square$

Lemma 40.4 (Products)**.** *Let $F_1, \dots, F_n$ be weakenable families with margins at most r, and let $0 < \delta \le 1$. Index the product $F_1 \cdots F_n$ by the tuples $(\omega_1, \dots, \omega_n)$ for which $X_{\omega_i}^{(r)}$ and $X_{\omega_{i+1}}^{(r)}$ meet for every i. All other products vanish, and this index set is determined by the geometry alone, not by the background. Give each tuple the domain $\bigcup_i X_{\omega_i}$ and the monomial $\prod_i s^{n_{\omega_i}}$. This is a weakenable family with margin r, and*

$$\|F_1 \cdots F_n\|_{\kappa,\gamma} \le n \prod_{i=1}^{n} (C_d \delta^{-2} \|F_i\|_{\kappa+\delta,\gamma}).$$

Proof. If the product of the terms does not vanish, consecutive kernel supports meet. So Details $X_{\omega_i}^{(r)} \cap X_{\omega_{i+1}}^{(r)} \neq \emptyset$, and the boxes involved are within distance $2r < M/4$. Hence they touch, and the union is a polymer. The monomial has the right support, and the product of the terms has kernel in the union of the supports and depends only on the data there. By Lemma 40.1(a), $|\prod_i F_{i,\omega_i}|_\gamma \le \prod_i |F_{i,\omega_i}|_\gamma$.

Fix a box $\square$ and let i be the first index with $\square \subset X_{\omega_i}$. Given ω_i, sum over ω_{i+1}: it must meet $X_{\omega_i}^{(2r)}$, which is covered by at most $3^d |X_{\omega_i}|$ boxes. So the sum is at most

$3^d|X_{\omega_i}|\,\|F_{i+1}\|_{\kappa,\gamma}$. Continue to the right, and in the same way to the left of i. Each factor receives at most two factors $|X_{\omega_l}| \le \deg\omega_l$, from its neighbours. Since $\sum_\omega e^{\kappa\deg\omega}(\deg\omega)^2|F_\omega| \le C\delta^{-2}\sum_\omega e^{(\kappa+\delta)\deg\omega}|F_\omega|$, each factor contributes at most $C_d\delta^{-2}\|F_l\|_{\kappa+\delta,\gamma}$. Summing over i gives the factor n. □

Lemma 40.5 (Local factors and commutators)**.** *Let O be a weakenable family with margin r.*

(a) Let $\mathcal{L}$ be a bounded operator of range r_0 that depends on the data only within distance r_0 of its support. Examples are $\mathbf{Q}$, $\mathbf{Q}^T$, Q', Q'^T, D_V, the curvature factor $\mathfrak{F}$, $W \in \mathcal{W}_{\mathrm{loc}}$, multiplication by h_j or m_X, the elimination maps $\hat{Z}^1$, $\hat{Z}^2$, and the linearized averaged contour variables $\mathfrak{G}_W$ and their transposes. Then $\mathcal{L}O$ and $O\mathcal{L}$, with the same domains and monomials, are weakenable with margin $r+2r_0$, and $\|\mathcal{L}O\|_{\kappa,\gamma} \le |\mathcal{L}|_\gamma\,\|O\|_{\kappa,\gamma}$, with $|\mathcal{L}|_\gamma \le e^{\gamma r_0'}|\mathcal{L}|_0$, where r_0' is the range in block units. The block Schur norm $|\mathcal{L}|_0$ is at most $C(d,r_0')\max_{y,y'}\|\mathbf{1}_y\mathcal{L}\mathbf{1}_{y'}\|$. It can exceed the operator norm $\|\mathcal{L}\|$, so the latter cannot be used here.

(b) $[O,h_j]$, with the same domains and monomials, is weakenable with $\|[O,h_j]\|_{\kappa,\gamma/2} \le C(\gamma M)^{-1}\|O\|_{\kappa,\gamma}$. The same holds with $|\cdot|^D$, up to an additional term $CM^{-1}\|O\|^D_{\kappa,\gamma}$ coming from $[D,h_j]$.

Proof. (a) Apply Lemma 40.1(a) term by term. The supports grow by the range of $\mathcal{L}$. (b) Apply Lemma 40.1(c) term by term. For the derivative versions use $D[O_\omega,h_j] = [DO_\omega,h_j] - [D,h_j]O_\omega$, where $[D,h_j]$ is a multiplication by differences of h_j, of size CM^{-1}. □ Details

41 Weakened inverses by nested walks

In this section domains are arbitrary unions of blocks, together with the empty set. The family of domains is closed under intersection with the Λ_j^v. For domains $X' \subset X$ and a width $w > 0$, let $\Sigma_w(X,X')$ be the set of blocks within distance w of $X \setminus X'$, the *seam*. The width depends on the level of the construction: $w = 10M$ at the levels (L1)–(L4) below, and $w = R_0 + 10M + 2L$ at the level (L5), with $R_0 = R_0(L,M)$ from Proposition 37.6. We fix ρ so large that

$$(\rho-1)M - w \ge \tfrac{1}{4}(\rho-1)M \qquad \text{at every level,} \tag{41.1}$$

e.g. $\rho \ge 2R_0/M + 30$. This is possible because $R_0(L,M) \le C_L M$. The number N_{b} of Section 40 then depends on L, but not on M.

41.1 Seam norms

For an operator F, a set of blocks Σ and $\gamma \ge 0$, let $|F|_{\gamma,\Sigma}$ be the decay norm of Section 40, with the weight $e^{\gamma|y-y'|}$ replaced by $e^{\gamma(|y-y'|+d_\Sigma(y)+d_\Sigma(y'))}$, where $d_\Sigma = \mathrm{dist}(\cdot,\Sigma)$. A kernel of seam form,

$$\|\mathbf{1}_y F \mathbf{1}_{y'}\| \le C\sum_{w\in\Sigma} e^{-\gamma'(|y-w|+|w-y'|)},$$

has $|F|_{\gamma'/4,\Sigma} \le CS^2_{\gamma'/2}$. Indeed $|y-y'| \le |y-w|+|w-y'|$, $d_\Sigma(y) \le |y-w|$ and $d_\Sigma(y') \le |w-y'|$. Conversely, $|F|_{\gamma,\Sigma} < \infty$ gives $\|\mathbf{1}_A F \mathbf{1}_{A'}\| \le |F|_{\gamma,\Sigma} e^{-\gamma(d_\Sigma(A)+d_\Sigma(A'))}$. If F is supported on $\Sigma\times\mathbb{T}_\eta$ or $\mathbb{T}_\eta\times\Sigma$, then $|F|_{\gamma,\Sigma} \le |F|_{2\gamma}$. Since $d_\Sigma(y) \le |y-y_1| + d_\Sigma(y_1)$,

$$|AFB|_{\gamma,\Sigma} \le |A|_{2\gamma}\,|F|_{\gamma,\Sigma}\,|B|_{2\gamma}. \tag{41.2}$$

For weakenable families, $\|O\|_{\kappa,\gamma,\Sigma}$ is defined with $|O_\omega|_{\gamma,\Sigma}$. By (41.2), Lemma 40.4 holds with one factor measured in the seam norm and the others in $|\cdot|_{2\gamma}$.

Factorized families. On the fine lattice the non-local parts of the operators contain derivatives, and are unbounded uniformly in η. As in Parts II and III, they are written as finite sums of products with derivatives outside. Here we write them as $\sum_i X_i^T M_i Y_i$ with $X_i^T \in \{1, D^T\}$ and $Y_i \in \mathfrak{D} \cup \{1\}$. A left gradient D, where present, is made part of the middle factor M_i. A *factorized weakenable family* is such a sum in which each M_i is a weakenable family. Its norm is $\sum_i \|M_i\|_{\kappa,\gamma}$: the middle factors are measured without further derivatives. The derivatives X_i^T, Y_i are kept outside the terms. A left D^T is absorbed by the factor to its left in a product, and Y_i by the local propagator to its right, whose derivative norms are controlled by (W2) below.

Derivative norms of the middle factors are neither used nor available in general. At level (L3) the middle factor $M = WDG'R$ has DM containing the derivatives of the coefficients of W, which are only Hölder continuous: for $W = w_q I$ with $w_q(x) = c\varepsilon q^{-\alpha}\sin(qx_1)$ and c small, a member of $\mathcal{W}_{\mathrm{loc}}(\varepsilon)$, $|DM|$ is at least of order $\varepsilon q^{1-\alpha}$ for $1 \ll q \lesssim \eta^{-1}$. In the same way, derivatives of the curvature factor $\mathfrak{F} = DD$ (Lemma 12.3) are not controlled at complex references.

Definition 41.1 (Compatible representations)**.** Factorized weakenable representations $O^X(s) = \sum_i X_i^T M_i^X(s) Y_i$ of operators O^X on $\ell^2(X)$, one for each domain X, are *compatible*, with constants (C_O, γ) and seam width w, if:

- the index sets do not depend on X, matched terms carry the same domain and monomial, and a term vanishes on X if one of its local domains does not meet X;
- $\|M_i^X\|_{\kappa,2\gamma} \le C_O$ for all X and i;
- for $X' \subset X$, the differences $\mathbf{1}_{X'} M_i^X(s) \mathbf{1}_{X'} - M_i^{X'}(s)$ satisfy $\|\cdot\|_{\kappa,\gamma,\Sigma_w(X,X')} \le C_O$.

The last condition is the weakened form of the comparison of domains (Theorem 18.2(e), Theorem 19.10(a), Proposition 35.1(d), Proposition 37.6(c)). On 2-forms, $\mathbf{1}_X$ denotes the projection onto the fields vanishing on the plaquettes with no corner in X.

41.2 The walk

Let $\mathfrak{A}^X$ be operators on $\ell^2(X)$, one for each domain X, with $\mathfrak{A} = \mathfrak{A}^{\mathbb{T}_\eta}$, satisfying:

(W1) *Structure.* $\mathfrak{A}^X = \mathfrak{A}^X_{\mathrm{loc}} + \kappa_m m_X + \mathfrak{N}^X$, where:

- $\mathfrak{A}^X_{\mathrm{loc}}$ is a local operator of range r_0, possibly of second order and containing a block-local averaging term, which depends on X only through cut terms at ∂X. For example, $\mathbf{1}_X DD^T \mathbf{1}_X$ differs from $D\mathbf{1}_X D^T$ by a cut term.
- $\kappa_m m_X$ is the mass term of Section 19 in the gauge case, and 0 otherwise.
- $\mathfrak{N}^X$ is the non-local part.

(W2) *Local propagators.* For every j and every domain X, $\mathfrak{A}^{\Lambda_j^v \cap X}$ is invertible on $\ell^2(\Lambda_j^v \cap X)$. Its inverse G_j^X, extended by zero, depends on the data only on $(\Lambda_j^v \cap X)^\wedge$, satisfies $|G_j^X|^D_{\gamma_0} \le C_G$, and obeys the comparison of domains in seam form with width w, with derivatives, and constants (C_G, γ_0). Here, and in the derivative norms of the outputs below, D stands for every first-order derivative that occurs as a factor Y_i, X_i^T or as an external derivative. In the gauge case these are D and the divergence $D^T_{X_j}$ of the local domain $X_j = \Lambda_j^v \cap X$. The bounds for $D^T_{X_j} G_j^X$ and $D^T_{X_j} G_j^X D^T$ come from the norm $\|\!|\cdot|\!\|_{X_j,+}$ (Theorem 19.7(a)). In the gauge case (W2) also includes the full gradient ∇G_j^X, $\nabla G_j^X D^T$, and the comparison of domains for these and for the torus divergence (Theorem 19.10(a)). With the Dirichlet conditions of Section 19, even the torus divergence

$D^T G_j^X$ is bounded. The regrouping in the proof of Lemma 41.3 is kept, since it needs only the divergence of the local domain.

(W3) *Non-local part.* $\mathfrak{N}^X = \sum_i X_i^T \mathfrak{N}_i^X Y_i$ has compatible factorized representations $\mathfrak{N}^X(s)$ with constants $(C_{\mathfrak{N}}, \gamma_0)$ at $\kappa + 2\delta$. Lemma 41.3 at κ' uses $\mathfrak{N}$ at $\kappa' + \delta$, and Theorem 41.4 at κ uses Lemma 41.3 at $\kappa + \delta$. So each walk costs 2δ in the weakening rate.

Lemma 41.2 (Parametrix on domains)**.** *Under (W1), (W2), with* $\mathcal{K}^X = \sum_j h_j G_j^X$ *on* $\ell^2(X)$*,*

$$\mathfrak{A}^X \mathcal{K}^X = I - \sum_j \mathcal{R}_j^X,$$
$$\mathcal{R}_j^X = -\Big([\mathfrak{A}_{\mathrm{loc}}^X, h_j] + [\mathfrak{N}^X, h_j] + h_j (\mathfrak{N}^X \mathbf{1}_{\Lambda_j^v \cap X} - \mathbf{1}_{\Lambda_j^v \cap X} \mathfrak{N}^{\Lambda_j^v \cap X} \mathbf{1}_{\Lambda_j^v \cap X}) \Big) G_j^X.$$

Proof. As in Lemma 20.1, $\mathfrak{A}^X h_j G_j^X = h_j \mathfrak{A}^X G_j^X + [\mathfrak{A}^X, h_j] G_j^X$, and m_X commutes with h_j. Details

- *Local parts.* On $\operatorname{supp} h_j$ the local parts of $\mathfrak{A}^X$ and $\mathfrak{A}^{\Lambda_j^v \cap X}$ agree on fields supported in $\Lambda_j^v \cap X$. The only possible difference is a cut term at $\partial \Lambda_j^v$, which is at distance at least $(\rho - 1)M > r_0$ from $\operatorname{supp} h_j \subset K_j$.
- *Mass profiles.* These agree on K_j. If $K_i \cap K_j \neq \emptyset$, then $\Lambda_i \subset \Lambda_j^v$ for $\rho \geq 6$. So $\Lambda_i \subset X$ iff $\Lambda_i \subset \Lambda_j^v \cap X$, i.e. $i \in J(X)$ iff $i \in J(\Lambda_j^v \cap X)$, and only such i contribute to m on K_j.

Hence $h_j \mathfrak{A}^X G_j^X = h_j + h_j (\mathfrak{N}^X \mathbf{1}_{\Lambda_j^v \cap X} - \mathbf{1} \mathfrak{N}^{\Lambda_j^v \cap X} \mathbf{1}) G_j^X$. Sum over j, using $\sum_j h_j = 1$. □

Since $h_j \mathbf{1}_{X \setminus (\Lambda_j^v \cap X)} = 0$, the seam term equals $h_j \mathbf{1}_{X'} (\mathfrak{N}^X - \mathfrak{N}^{X'}) \mathbf{1}_{X'} G_j^X$ with $X' = \Lambda_j^v \cap X$.

The weakened step. Insert the representations of (W3), and give each step the monomial $s^{X(j)}$. Each term of $\mathcal{R}_j^X$ is then of the form $D^T A + B$, with the left derivative coming from X_i^T, and

$$\begin{aligned} \mathcal{R}^X(s) &= -\sum_j s^{X(j)} \Big([\mathfrak{A}_{\mathrm{loc}}^X, h_j] + [\mathfrak{N}^X(s), h_j] + h_j \mathbf{1}_{X'} (\mathfrak{N}^X(s) - \mathfrak{N}^{X'}(s)) \mathbf{1}_{X'} \Big) G_j^X \\ &= D^T \mathcal{A}^X(s) + \mathcal{B}^X(s), \end{aligned}$$

with weakenable families $\mathcal{A}^X$ and $\mathcal{B}^X$ ($\mathcal{A}^X = 0$ at the scalar levels). Each term has domain $X(j) \cup X_\omega$. This is a polymer, since X_ω meets $X(j)$ for every non-vanishing term. Put

$$G^X(s) = \sum_{n \geq 0} \Big(\sum_j s^{X(j)} h_j G_j^X \Big) \mathcal{R}^X(s)^n.$$

In a product, every D^T in front of $\mathcal{A}$ is absorbed by the factor on its left, which is one of $h_j G_j^X D^T$, $\mathcal{A}^X D^T$, $\mathcal{B}^X D^T$. So the products are products of the bounded factors

$$\mathcal{F}_0 = \sum_j s^{X(j)} h_j G_j^X, \quad \mathcal{F}_0 D^T, \quad \mathcal{A}^X, \quad \mathcal{A}^X D^T, \quad \mathcal{B}^X, \quad \mathcal{B}^X D^T.$$

This replaces the use of the identity of Lemma 36.6. That identity needs a real background, while the weakened walks are needed for complex perturbations as well.

Lemma 41.3 (Step operators)**.** *Assume (W1)–(W3) and*

$$|[\mathfrak{A}_{\mathrm{loc}}^X, h_j] G_j^X|_{\gamma_0} + |[\mathfrak{A}_{\mathrm{loc}}^X, h_j] G_j^X D^T|_{\gamma_0} \leq C_G C_{\mathrm{loc}} M^{-1}.$$

Then, with $\gamma = \gamma_0/8$ and $\gamma_0 \le 1$,

$$\|\mathcal{A}^X\|_{\kappa,\gamma} + \|\mathcal{A}^X D^T\|_{\kappa,\gamma} + \|\mathcal{B}^X\|_{\kappa,\gamma} + \|\mathcal{B}^X D^T\|_{\kappa,\gamma} \le \theta,$$
$$\theta := C\, e^{\kappa \mathsf{N}_{\mathrm{b}}} \Big(C_G C_{\mathrm{loc}} M^{-1} + C\delta^{-1} (\gamma_0 M)^{-1} C_{\mathfrak{N}} C_G + C\delta^{-1} C_{\mathfrak{N}} C_G e^{-\gamma_0(\rho-1)M/16} \Big).$$

The constant C includes the number of j with $\square \subset X(j)$, which is bounded independently of M.

Proof. The local commutator. This is the hypothesis, times $e^{\kappa \mathsf{N}_{\mathrm{b}}}$ from the monomial. Details

Regrouping. Let $X' = \Lambda_j^v \cap X$. Consider a term $X_i^T M_i^X Y_i^X$ of $\mathfrak{N}^X$ with $Y_i^X = D_X^T$; the matching term of $\mathfrak{N}^{X'}$ has $Y_i^{X'} = D_{X'}^T$. Put $\sigma_j = (D_X^T - D_{X'}^T) G_j$. It is supported at points of $X \setminus X'$, so $h_j \sigma_j = 0$, and $h_j D_X^T G_j = h_j D_{X'}^T G_j$. Then the non-local commutator and the seam term of this term are

$$[X_i^T M_i^X D_X^T, h_j] G_j = [X_i^T M_i^X, h_j] D_{X'}^T G_j + X_i^T M_i^X [D_X^T, h_j] G_j - h_j X_i^T M_i^X \sigma_j,$$
$$h_j (X_i^T M_i^X D_X^T - X_i^T M_i^{X'} D_{X'}^T) G_j = h_j X_i^T (M_i^X - M_i^{X'}) D_{X'}^T G_j + h_j X_i^T M_i^X \sigma_j,$$

and the terms with σ_j cancel exactly. We use the steps in this regrouped form. Then only $D_{X'}^T G_j$ and $D_{X'}^T G_j D^T$ enter, which (W2) bounds, and $[D_X^T, h_j]$ is a multiplication by differences of h_j.

The non-local commutator. It is bounded by Lemma 40.5(b), applied to the factors M_i, together with the commutators $[X_i^T, h_j]$, $[Y_i, h_j]$, which are of size CM^{-1}. The derivatives Y_i are absorbed by G_j^X, using $|G_j^X|^D$. In the terms $X_i^T [M_i, h_j] Y_i G_j$ and $X_i^T M_i [Y_i, h_j] G_j$ a left D^T stays in front and gives the form $D^T \mathcal{A}$; then $\mathcal{A} D^T$ contains $Y_i G_j D^T$, again controlled by (W2). So only the norms $\|M_i\|$ enter, never derivatives of M_i.

The seam term. By Definition 41.1, the differences $\mathbf{1}_{X'} (\mathfrak{N}_i^X - \mathfrak{N}_i^{X'}) \mathbf{1}_{X'}$ are of seam form for $\Sigma_w(X, X')$. By (41.1), every block of this seam lies at distance at least $\frac{1}{4}(\rho - 1)M$ from K_j. The factor h_j restricts the output to K_j, so the seam norm and (41.2) give the factor $e^{-\gamma_0(\rho-1)M/16}$.

Counting. The terms of the non-local commutator and of the seam term built from a term ω of $\mathfrak{N}$ carry the monomial $s^{X(j)} s^{n_\omega}$, of degree $|X(j)| + \deg \omega \le \mathsf{N}_{\mathrm{b}} + \deg \omega$. So the factor $e^{\kappa \mathsf{N}_{\mathrm{b}}}$ multiplies them as well. For a fixed term ω of $\mathfrak{N}$, the number of j for which the commutator or the seam term is non-zero is at most $C|X_\omega| \le C \deg \omega$. This is absorbed by evaluating $\|\mathfrak{N}\|$ at $\kappa + \delta$, at the cost of δ^{-1}. The versions with D^T on the right follow from (W2) and (W3) in the same way.

The power of C_G. Each term of $\mathcal{A}^X$, $\mathcal{B}^X$, $\mathcal{A}^X D^T$ and $\mathcal{B}^X D^T$ contains exactly one factor of the form $P G_j^X P'$ with $P, P' \in \{1, D, D^T\}$, which (W2) bounds by C_G; so θ is linear in C_G. This matters at level (L5), where $C_G = C_L(\frac{1}{16} c_L + r)^{-1} < 1$ for large r. □

Theorem 41.4 (Weakened inverses)**.** *Assume (W1)–(W3) and the hypothesis of Lemma 41.3. For (b) assume also the comparison of domains of the local commutators: for domains $X' \subset X$ the zero extensions of $[\mathfrak{A}_{\mathrm{loc}}^X, h_j] G_j^X - [\mathfrak{A}_{\mathrm{loc}}^{X'}, h_j] G_j^{X'}$ and of $[\mathfrak{A}_{\mathrm{loc}}^X, h_j] G_j^X D^T - [\mathfrak{A}_{\mathrm{loc}}^{X'}, h_j] G_j^{X'} D^T$ are of seam form for $\Sigma_w(X, X')$, with constants $(C_G C_{\mathrm{loc}}, \gamma_0)$. Let $\delta \in (0, 1]$. If $M \ge M_1(\kappa, \delta, C_G, C_{\mathrm{loc}}, C_{\mathfrak{N}}, \gamma_0, \rho, C_v)$, then for every domain X:*

(a) $G^X(s)$ is a weakenable family with margin at most $r + 2r_0$ that represents $(\mathfrak{A}^X)^{-1}$, with

$$\|G^X\|_{\kappa,\gamma_0/8}^D \le C(\kappa, \delta) e^{(\kappa+\delta)\mathsf{N}_{\mathrm{b}}} C_G;$$

(b) the representations of $G^X(s)$, $DG^X(s)$, $G^X(s) D^T$ and $DG^X(s) D^T$ are compatible with constants $(C_{\mathrm{cpt}}, \gamma_0/32)$ and seam width w. Here C_{cpt} depends only on κ, δ, γ_0, C_G, C_{loc}, $C_{\mathfrak{N}}$ and on ρ and C_v (through N_{b}); it may be taken independent of $M \ge M_1$, of X and of k, n, η;

(c) *each term is covariant under real gauge transformations if the* G_j^X *and the terms of* $\mathfrak{R}^X(s)$ *are.*

Proof. (a) After the regrouping of the derivatives, the n-th term is a sum of 2^n products of $n+1$ bounded factors from the list above. By Lemma 40.4 and Lemma 41.3 at $\kappa+\delta$, Details

$$\left\| \mathcal{F}_0 \mathcal{R}^X(s)^n \right\|_{\kappa,\gamma} \leq (n+1)\big(C\delta^{-2} e^{(\kappa+\delta)\mathsf{N}_\mathfrak{b}} C_G\big)\big(2C\delta^{-2}\theta\big)^n .$$

The series converges if $2C\delta^{-2}\theta \leq \frac{1}{2}$, which holds for M large, by the form of θ. With derivatives, D falls on the first factor, which is controlled by $|DG_j^X|$ and $[D, h_j]$. D^T falls on the last factor, controlled by $\|\mathcal{A}^X D^T\|$, $\|\mathcal{B}^X D^T\|$ or $|G_j^X D^T|$. At $s=1$ the series is $\mathcal{K}^X \sum_n (\sum_j \mathcal{R}_j^X)^n$, which by Lemma 41.2 is a right inverse of $\mathfrak{A}^X$. The space $\ell^2(X)$ is finite-dimensional, since the torus is finite. So this right inverse of $\mathfrak{A}^X$ is its inverse. No coercivity or positivity of any operator at $s \neq 1$ is used: the series is the definition of $G^X(s)$. Data locality and the domains hold term by term.

(b) Use the same index set on X and on $X' \subset X$. On X' a term vanishes if one of its local domains $\Lambda_j^v \cap X'$ is empty. The sets $X(j)$ and the monomials do not depend on the domain, so matched terms have the same domain and monomial. On each field space let $P = \mathbf{1}_{X'}$ be the orthogonal projection onto the fields of X' (Definition 17.1; on 2-forms, the fields vanishing on the plaquettes with no corner in X'), and $Q = \mathbf{1}_X - P$. On gauge fields Q is not $\mathbf{1}_{X \setminus X'}$: a bond from X' to $X \setminus X'$ is a field of both domains. The fields in the range of Q lie within one lattice step of $X \setminus X'$. After the regrouping, a term on X is a product of bounded pieces: the h_j; the local propagators G_j^X with the derivatives of (W2), among them the external D of the first factor and the external D^T of the last; the local commutators $[\mathfrak{A}_{\mathrm{loc}}^X, h_j] G_j^X$, also with D^T on the right; the middle factors of the non-local parts, also commuted with h_j; and the multiplications $[X_i^T, h_j]$, $[Y_i, h_j]$ and, with an external D, $[D, h_j]$. Every factor of a word ends with a local propagator, possibly with a derivative of (W2) on its left, so the left D^T of the next step is absorbed into it, as in $G_j^X D^T$ or $[\mathfrak{A}_{\mathrm{loc}}^X, h_j] G_j^X D^T$. Insert $P+Q$ between the pieces of a term on X, and telescope, using $P\Phi' P = \Phi'$ for the pieces Φ' on X'. The difference $P(\text{term on } X)P - (\text{term on } X')$ is a sum of products, each containing one of the following factors:

- a difference of local propagators $P(G_j^X - G_j^{X'})P$, with the derivatives applied before the comparison, of seam form by (W2), since $\Lambda_j^v \cap X' \subset \Lambda_j^v \cap X$ and $(\Lambda_j^v \cap X) \setminus (\Lambda_j^v \cap X') \subset X \setminus X'$;
- a difference of non-local parts, on X and X' or on $\Lambda_j^v \cap X$ and $\Lambda_j^v \cap X'$, of seam form by (W3), also after commutation with h_j, which commutes with P;
- a difference of local commutators $P[\mathfrak{A}_{\mathrm{loc}}^X, h_j] G_j^X P - [\mathfrak{A}_{\mathrm{loc}}^{X'}, h_j] G_j^{X'}$, also with D^T on the right, of seam form by the hypothesis of (b);
- a cross term $P\Phi Q$ of a piece Φ, a difference $P\Phi P - \Phi'$ of the multiplications $[X_i^T, h_j]$, $[Y_i, h_j]$, $[D, h_j]$, or a piece containing G_j^X whose domain $\Lambda_j^v \cap X'$ is empty. These are bounded and supported on the seam, where $|\cdot|_{\gamma,\Sigma} \leq |\cdot|_{2\gamma}$.

By (41.2) and Lemma 40.4, each such product is bounded in the seam norm. No bare derivative stands next to P or Q: every derivative is part of a bounded piece. A local commutator $[\mathfrak{A}_{\mathrm{loc}}, h_j]$ never stands without its propagator: at (L1) and (L3) it is a differential operator, not bounded uniformly in η. A term with n steps has at most $C(n+1)$ pieces, so the sum over the position of the distinguished factor costs a factor $C(n+1)$, which is absorbed. The steps are taken in the regrouped form of the proof of Lemma 41.3 on both domains. On X' the divergences of the local domains agree at the points of X', so their differences are

again of seam form: $P\mathbf{1}_{\Lambda_j^v\cap X}D^TG_j^XP-\mathbf{1}_{\Lambda_j^v\cap X'}D^TG_j^{X'}=\mathbf{1}_{\Lambda_j^v\cap X'}D^T(G_j^X-G_j^{X'})P$, which (W2) bounds through the torus divergence. The constant $C_{\rm cpt}$ collects the bound of (a), which is proportional to C_G, and the seam constants of (W2), (W3) and of the local commutators. It may be taken as its value at $M=M_1$, since the bounds of the series decrease in M. It cannot be taken independent of C_G: for $\mathfrak{A}^X=tI_X$ on unit-lattice fields and $\mathfrak{N}=0$ the family is $t^{-1}\sum_j s^{X(j)}h_j$, whose norm is at least $2^{-d}t^{-1}$ by Lemma 40.3(a).

(c) is clear. □

41.3 The levels

Theorem 41.4 is applied at five levels, and (L3), (L4) are applied twice, without and with the potential W_2. At each level the hypotheses follow from the previous levels and from Parts III, VII and VIII. The order is

$$\text{(L1)}\to\text{(L2)}\to\text{(L3), (L4) with } W=0\to\omega_0,\ W_2\to\text{(L3), (L4) with } W_2\to\text{(L5)}.$$

Every use of Lemma 40.4 costs δ in κ, every use of Theorem 41.4 costs 2δ, each costs a fixed factor in γ, and there are boundedly many of them. The parameters are chosen in the order L, then κ (see below), then ρ by (41.1), then M large depending on all of these, and finally ε small depending on M. The constants C_G, $C_{\rm loc}$, $C_{\mathfrak{N}}$ and the rates γ_0 of all levels do not depend on M: for $M\ge M_0$, and, at a level whose non-local parts come from Theorem 41.4(b) at earlier levels, for M at least the thresholds M_1 of those levels, with $C_{\rm cpt}$ taken at these thresholds. So the thresholds are fixed level by level, in the order above, and M_1 denotes the largest of them. At (L1)–(L3) this holds for the torus operators by Lemma 31.2 and Step 0 of the proof of Lemma 31.3, for the Dirichlet operators by Theorem 19.6 (Remark 31.4(c)), and for the non-local parts by Theorem 41.4(b) at the earlier levels. At (L4) and (L5) the inputs are Propositions 35.1, 37.4 and 37.6. They are built from these Dirichlet operators, from the averaging operators, whose proofs do not use M (Section 46.1, Setting), and from the constants c_L, C_L, C'_L, γ_1 and γ_C of Part VII, which do not depend on $M\ge M_0$ (Remark 34.7). Theorem 19.10(b), the one result of Part III whose constant contains e^{CM}, does not enter the constants of (W2): (W2) uses Theorem 19.10(a), and (b) enters only through the coercivity of Part VII, in Lemma 34.3, where it is absorbed by the choice of M'_L and ε_L. The margin $R_0\le C_LM$ enters only ρ and the seam width. So each threshold M_1 is a number, fixed before M. The smallness of ε is another matter. The thresholds of Parts II and III do not depend on M, and $\varepsilon_{\rm B}$ of Lemma 31.3(c) may be taken at $M=M_0$ (Lemma 31.2). The thresholds of Parts VI–VIII depend on M, for instance a_1 of Theorem 30.3 (whose constant B_3 also depends on M), a_δ of Proposition 31.5 and $\varepsilon_L\le\varepsilon_{\rm cmp}(M'_L)$ with $M'_L\ge M$ (Theorem 34.4); this is why ε is chosen last.

(L1) *Scalar operators.*

- $\mathfrak{A}'^X=\mathbf{1}_X(D^TD+aQ'^TQ')\mathbf{1}_X$, with $\mathfrak{N}=0$, since Q'^TQ' is block-local and is counted in $\mathfrak{A}_{\rm loc}$.
- (W2) is Theorem 18.2(b),(e), on the domains $\Lambda_j^v\cap X$.
- The hypothesis of Lemma 41.3 is $|[\mathfrak{A}',h_j]G'_j|_{\gamma_0}\le CM^{-1}$, in the decay norm, with D^T on the right, from Theorem 20.2(a), whose proof applies verbatim on the domains $\Lambda_j^v\cap X$. The comparison of the local commutators in Theorem 41.4(b) holds as well. Since h_j commutes with $\mathbf{1}_X$ and $\mathbf{1}_X\mathbf{1}_{X'}=\mathbf{1}_{X'}$, the difference is $\mathbf{1}_X[\mathfrak{A}',h_j](G'_{\Lambda_j^v\cap X}-G'_{\Lambda_j^v\cap X'})+(\mathbf{1}_X-\mathbf{1}_{X'})[\mathfrak{A}',h_j]G'_{\Lambda_j^v\cap X'}$, with $\mathfrak{A}'=D^TD+aQ'^TQ'$. By the discrete Leibniz rule, $[\mathfrak{A}',h_j]u$ is a sum of multiplications of size CM^{-1} applied to u and to Du, up to shifts by η, and of $a[Q'^TQ',h_j]u$. So the first term is

of seam form by Theorem 18.2(e) for G' and DG', also with D^T on the right. The second is bounded by the same expansion and Theorem 18.2(b), and supported at the points of $X \setminus X'$.

This gives weakened $G'^X(s)$, together with $DG'^X(s)$, $G'^X(s)D^T$ and $DG'^X(s)D^T$. The walk has only local steps, as in [15, Lemma 6].

(L2) *Scalar averaged operators.*

- $Q'G'^XQ'^T$ and $M'^X = Q'(G'^X)^2Q'^T$ on $\ell^2(X_1)$, with $\mathfrak{A}_{\mathrm{loc}} = 0$.
- Their weakened representations come from (L1), Lemma 40.4 and Lemma 40.5(a). Compatibility follows from (L1).
- (W2) holds for the Dirichlet versions on $\Lambda_j^v \cap X$, by Theorem 18.2(c),(e).

Theorem 41.4 gives $\Gamma'^X(s)$ and $(M'^X)^{-1}(s)$. Products then give H'^X, $P_{\mathcal{V}}^X$, R_X, S_X, DS_X and Π_X, with derivatives.

(L3) *Gauge-field operators.*

- $\mathfrak{A}^{X,\kappa_m}$, with $\mathfrak{A}_{\mathrm{loc}}^X = \mathbf{1}_X(D^TD + DD^T + W + a\mathbf{Q}^T\mathbf{Q})\mathbf{1}_X$ on $\ell_1^2(X)$, the Dirichlet restriction of the torus operator, which includes the exterior divergence term $\mathfrak{A}_X^{\mathrm{ext}}$, and mass term $\kappa_m m_X$.
- Using $E = 0$ as in Lemma 36.4,

$$\mathfrak{N}^X = -DP_{\mathcal{V},X}D_X^T - \mathfrak{B}_X DS_X - (\mathfrak{B}_X DS_X)^T + S_X^T D_X^T \mathfrak{B}_X DS_X,$$

with $\mathfrak{B}_X D = D^T\mathfrak{F} + WD$. This is a factorized family with $X_i^T \in \{1, D^T\}$. Its terms are listed in the proof of Lemma 44.10. For example, $D^T\mathfrak{F}S_X$ has $X_i^T = D^T$, $M_i = \mathfrak{F}G'_X R_X$ and $Y_i = D_X^T$; $-DP_{\mathcal{V},X}D_X^T$ has $X_i^T = 1$, $M_i = DP_{\mathcal{V},X}$ and $Y_i = D_X^T$. Every middle factor is bounded in the plain norm by (L1), (L2) with derivatives, by the sup bound on $\mathfrak{F}$, and by the norm of W: the sup bound of its coefficients for $W \in \mathcal{W}_{\mathrm{loc}}$, the family norm for W_2. A left gradient always falls on a scalar propagator, never on the coefficients of $\mathfrak{F}$ or W, so no derivative of these coefficients is needed. The compatibility of the middle factors follows from Theorem 41.4(b) at (L1), (L2), which includes the derivatives of G'^X and $P_{\mathcal{V}}^X$, and from the locality of $\mathfrak{F}$ and W.
- (W2) is Theorem 19.7(a) and Theorem 19.10(a) on the domains $\Lambda_j^v \cap X$.
- The hypothesis of Lemma 41.3 holds on every domain. The local commutator is $[\mathfrak{A}_{\mathrm{loc}}^X, h_j] = \mathbf{1}_X[D^TD + DD^T + W + a\mathbf{Q}^T\mathbf{Q}, h_j]\mathbf{1}_X$, since h_j commutes with $\mathbf{1}_X$. The terms $D^T[D, h_j]G_j$ and $D[D^T, h_j]G_j$ need the full gradient of G_j on the support of ∂h_j, also where it meets ∂X. This is bounded by Lemma 19.3 and Theorem 19.7(a). The same holds with D^T on the right, and at complex perturbations. On the torus one may also use the localized Hodge identity at interior blocks, as in the proof of Lemma 44.5. The comparison of the local commutators in Theorem 41.4(b) follows from the same expansion. With $\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta} = D^TD + DD^T + W + a\mathbf{Q}^T\mathbf{Q}$, the difference is $\mathbf{1}_X[\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j](G_j^X - G_j^{X'}) + (\mathbf{1}_X - \mathbf{1}_{X'})[\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j]G_j^{X'}$. In the first term $[\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j]$ acts through D, the torus divergence D^T, ∇, $\mathbf{Q}$ and W, with coefficients and local maps of size CM^{-1}; Theorem 19.10(a) compares these derivatives of the local propagators, also with D^T on the right, in seam form. In the second term $\mathbf{1}_X - \mathbf{1}_{X'}$ projects onto the bonds within one lattice step of $X \setminus X'$, and $[\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j]G_j^{X'}$ is bounded as above.

The result is $\mathbf{G}^X(s)$, together with $D\mathbf{G}^X(s)$, $\mathbf{G}^X(s)D^T$ and $D\mathbf{G}^X(s)D^T$.

(L4) *Averaged gauge operators and the Hessian.*

- $\mathbf{Q}_X\mathbf{G}^X\mathbf{Q}_X^T$ on $\ell^2(\mathfrak{B}(X))$, with $\mathfrak{A}_{\rm loc}=0$.
- (W2) is Proposition 35.1(a),(d), and (e) at complex perturbations of the background.

This gives $\mathbf{\Gamma}^X(s)$, hence $\mathbf{H}^X(s)$, $\mathbf{C}^X(s)$ and $\omega_0^X(s)=\mathbf{H}_0^X(s)^T\mathbf{1}_X J_U$. Then

$$W_2^X(s)=d^2\mathcal{N}(0)\big|_X-\omega_0^X(s)\cdot d^2C_k^{\rm av}(0)$$

is a weakenable family. Its first term is local, and the second is $\omega_0^X(s)$ times the bounded local kernel of $d^2C_k^{\rm av}(0)$ (Proposition 22.9(c)).

For complex perturbations of the background, J_U is not bounded uniformly in η. There one writes $\langle b,\omega_0\rangle=\langle J_U,\mathbf{H}_0 b\rangle$ with $J_U=D^TF_U^s$ (Lemma 23.2), and moves the derivative onto $\mathbf{H}_0 b$, using $|D\mathbf{H}_0|$. The same remark applies to Proposition 37.4(c).

In the repetition of (L3) the potential $W_2^X(s)$ is put into the non-local part $\mathfrak{N}$, with $X_i=Y_i=1$, not into $\mathfrak{A}_{\rm loc}$. It depends on X globally, through ω_0^X, so it does not satisfy the requirement of (W1) on local parts. Its plain norm and its compatibility follow from (L3), (L4) with $W=0$ and Theorem 41.4(b). In this repetition the local propagators are the inverses of the Dirichlet operators with the potential $W_2^{X_j}$, $X_j=\Lambda_j^v\cap X$, which depends on U on all of $\hat X_j$. So (W2) is not Theorems 19.7(a) and 19.10(a) literally. Existence, Re-coercivity and decay come from Proposition 19.11, whose hypotheses $W_2^{X_j}$ satisfies (Proposition 37.4(b),(c)); the comparison of domains comes from Propositions 35.1(d), which allows different potentials on the two domains, and 37.4(d), as stated in Section 19. These comparisons include $P\in\{D,D^T,\nabla\}$, so the comparison of the local commutators in Theorem 41.4(b) follows as at the first pass of (L3); the local part is the same. Repeating (L3) and (L4) with the potential W_2^X gives $\mathbf{\Gamma}^X[W_2^X](s)$ and the Dirichlet Hessians $\Delta^{0,X}(s)$. Through the local maps $\hat Z^1$, $\hat Z^2$ it gives the coarse-action parts $\Delta^{{\rm c},X}(s)$ of the operators $\Delta_k^{(X)}$, and hence $\Delta_Y^{\rm c}(s)=\mathbf{1}_{\mathfrak{F}(Y)}\Delta^{{\rm c},Y^+}(s)\mathbf{1}_{\mathfrak{F}(Y)}$, the coarse-action part of Δ_Y (Section 37). The gauge-fixing part of Δ_Y is the block-local operator $K_Y^{\rm gf}=\mathbf{1}_{\mathfrak{F}(Y)}\mathfrak{G}_W^T\mathfrak{G}_W\mathbf{1}_{\mathfrak{F}(Y)}$ of (37.1). It needs no walk and is not weakened; it enters at level (L5) as a local part.

(L5) *The fluctuation operator.*

- $\mathfrak{A}^X=\Delta_X+r$ on $\ell^2(\mathfrak{F}(X))$, for $r\ge 0$, with

$$\mathfrak{A}_{\rm loc}^X=r+K_X^{\rm gf},\qquad K_X^{\rm gf}=\mathbf{1}_{\mathfrak{F}(X)}\mathfrak{G}_W^T\mathfrak{G}_W\mathbf{1}_{\mathfrak{F}(X)},\qquad \mathfrak{N}^X=\Delta_X^{\rm c}\text{ from (L4)},$$

  so that $\Delta_X=\Delta_X^{\rm c}+K_X^{\rm gf}$ by (37.1). Since $\Delta_X^{\rm c}$ is built from $\Delta^{{\rm c},X^+}$, the representations of (L4) are used on the domains $X^+=X^{(R_0)}$.

- The local part $K_X^{\rm gf}$ is block-diagonal: its kernel vanishes unless the two bonds are inner bonds of a common block $B'(y)$, so it has range at most L, with entries bounded by C_L (Lemma 22.7(c)). It depends on X only through the restriction to $\mathfrak{F}(X)$, i.e. through cut terms at ∂X, as (W1) requires. At real references $K_X^{\rm gf}\ge 0$. It is analytic in the reference: at complex references $\mathfrak{G}_W$ is taken at the complex base, and $K_X^{\rm gf}$ changes by a block-local operator of size $C_L\delta_c$ (Lemma 22.7(f) and Remark 38.8, (DC5)).

- (W2) is Proposition 37.6(c),(d), with the seam width $w=R_0+10M+2L$. The local propagators on $Y=\Lambda_j^v\cap X$ depend on the background on $\widehat{Y^+}$ (Proposition 37.6(b)),

not only on $\hat{Y}$; this set lies in $(\Lambda_j^v)^{(R_v)}$ by the choice of R_v in Section 40, which is the form of the data dependence of (W2) used at this level.

- The local operators satisfy $\mathrm{Re}(\Delta_{\Lambda_j^v \cap X} + r) \geq \frac{1}{16} c_L + r$. So the local propagators $(\Delta_{\Lambda_j^v \cap X} + r)^{-1}$ have norm at most $(\frac{1}{16} c_L + r)^{-1}$ and decay $(C_L(\frac{1}{16} c_L + r)^{-1}, \gamma_C)$, uniformly in r (Proposition 37.6(d)).
- The hypothesis of Lemma 41.3. Since $[r, h_j] = 0$, the local commutator is $[K_X^{\mathrm{gf}}, h_j]$, with h_j acting on fields on $\mathfrak{F}$ through the initial points of the bonds. Its entries are $K_X^{\mathrm{gf}}(b, b')(h_j(b'_-) - h_j(b_-))$, with $|b_- - b'_-|_\infty \leq L$, and $|h_j(x) - h_j(x')| \leq C M^{-1} |x - x'|$. So $|[K_X^{\mathrm{gf}}, h_j]|_{\gamma_0} \leq C_L L M^{-1}$, with C_L including $e^{\gamma_0 L}$, and $|[K_X^{\mathrm{gf}}, h_j] G_j^X|_{\gamma_0} \leq C_G C_L L M^{-1}$. This is the hypothesis with $C_{\mathrm{loc}} = C_L L$. It is harmless, since M is chosen after L. There are no derivatives at this level. The comparison of the local commutators in Theorem 41.4(b) holds as well: $[K_X^{\mathrm{gf}}, h_j] = \mathbf{1}_{\mathfrak{F}(X)} [\mathfrak{G}_W^T \mathfrak{G}_W, h_j] \mathbf{1}_{\mathfrak{F}(X)}$, so the difference is $\mathbf{1}_{\mathfrak{F}(X)} [\mathfrak{G}_W^T \mathfrak{G}_W, h_j](G_j^X - G_j^{X'}) + (\mathbf{1}_{\mathfrak{F}(X)} - \mathbf{1}_{\mathfrak{F}(X')}) [\mathfrak{G}_W^T \mathfrak{G}_W, h_j] G_j^{X'}$. The first term is of seam form by Proposition 37.6(c),(d); entries with an index on a bond starting in $X \setminus X'$ lie in the seam. The second term lives on such bonds. The constants are $C\, C_G\, C_L L M^{-1}$, uniformly in r.
- The margin R_v of $X(j)$ is at least $R_0 + 4 + 2 r_{\mathrm{loc}}$ (Section 40).

The constant $C_G = C_L(\frac{1}{16} c_L + r)^{-1}$ of (W2) is at most $16 C_L / c_L$, and $C_{\mathrm{loc}} = C_L L$ and $C_{\mathfrak{N}}$ do not depend on r. So M_1 and the constants of Theorem 41.4 are uniform in r, and the bound of (a) carries the factor C_G. It gives $R_r(s)$, a representation of $(\Delta_k + r)^{-1}$, with $\|R_r\|_{\kappa,\gamma} \leq C(c_L + r)^{-1}$.

Remark 41.5 (Boundary conditions of the gauge-field operators). The gauge-field Dirichlet operators of Section 19 contain the exterior divergence term $\mathfrak{A}_X^{\mathrm{ext}}$, so that all components of the gauge field satisfy Dirichlet conditions. The walks on proper domains need this.

- *Relative boundary conditions do not suffice.* With the interior divergence D_X^T alone, the normal components on the bonds crossing ∂X would be free. These are relative boundary conditions, and the normal trace of $\mathbf{G}^X g$ is then of order 1. The local commutator $[D \mathbf{1}_X D^T, h_j]$ produces, on a crossing bond at the inner point x, the term $-\eta^{-1}([D^T, h_j]u)(x)$, whose main part is $\eta^{-1}(\partial_n h_j) u_n$. In L^2 this is of order $\eta^{-1/2} M^{-1}$. So for such operators the hypothesis of Lemma 41.3 fails uniformly in η on every proper domain.
- *Dirichlet conditions for all components.* With $\mathfrak{A}_X^{\mathrm{ext}}$, the local part of the operators is the Dirichlet restriction of the torus operator, so all components satisfy Dirichlet conditions. Coercivity holds in $\|\!| \cdot |\!\|_{X,+}$, and the full gradient is controlled up to ∂X (Lemma 19.3). The local commutators are then bounded, and the cutoff proof of the comparison of domains (Theorem 19.10(a)) leaves no boundary terms.
- *What the term does not affect.* The torus operators, since $\mathfrak{A}_{\mathbb{T}_\eta}^{\mathrm{ext}} = 0$, and hence Parts IV–VII; the scalar operators and the slice objects S_X, R_X, $P_{\mathcal{V},X}$. Enlarging the scalar domain is not needed, and would not restore exact identities on domains, since gradients of scalars supported at exterior points do not lie in $\ell_1^2(X)$. The mass term is kept, because the exterior divergence alone gives no L^2 control near the boundary.

Corollary 41.6 (Weakened covariance). *For $M \geq M_1(\kappa, L)$, the formula*

$$C_k^{1/2}(s) = \frac{1}{\pi} \int_0^\infty \frac{dr}{\sqrt{r}}\, R_r(s)$$

defines a weakenable family representing $C_k^{1/2}$. Its norm $\|C_k^{1/2}\|_{\kappa',\gamma'}$ is finite for $\kappa' = \kappa - C\delta$, and some $\gamma' > 0$ depending on L. Here $C \le 20$. The walks of (L1)–(L5) and the repetitions of (L3), (L4) with W_2, seven in all, cost at most 2δ each. The products between them, at most six, cost δ each. Its terms are covariant and depend on the background only on their domains. Its margin is that of the families $R_r(s)$, independent of r and of M; it is at most $M/16$ for M large, as the definition of weakenable families requires (Section 40). The norm is bounded uniformly in the background, real or complex as in Section 40; the terms are analytic in the complex perturbation of the background, and real for real backgrounds and real s. In the same way the diagonal entries, for $b \in \mathfrak{F}$,

$$\ell_k(b) = \sum_j h_j(b)\, \ell_{\Lambda_j^v}(b) - \int_0^\infty \operatorname{tr}_{\mathfrak{g}} \left[R_t^{(\ge 1)}(b,b)\right] dt$$

are represented by weakenable families of numbers $(X_\omega, n_\omega, O_\omega)_{\omega\in\Omega_b}$. Here $R_t^{(\ge 1)}$ collects the walk terms of $(\Delta_k + t)^{-1}$ with at least one step, and $\ell_{\Lambda_j^v}$ are the densities of Section 38. Moreover $O_\omega = 0$ unless $b \in X_\omega^{(r)}$, and

$$\|\ell_k(b)\|_{\kappa'} := \sup_{\square} \sum_{\omega\in\Omega_b:\, \square\subset X_\omega} e^{\kappa' \deg \omega} |O_\omega| \le C,$$

uniformly in b and in the background, real or complex as in Section 40.

Proof. By (L5), $\|R_r\|_{\kappa',\gamma'} \le C(c_L + r)^{-1}$, which is integrable against $r^{-1/2}dr$. Interchanging the sum over terms with the integral gives a representation whose terms are r-integrals of terms of R_r, with the same domains. At $s = 1$ it is $C_k^{1/2}$, by $\frac{1}{\pi}\int_0^\infty r^{-1/2}(\lambda + r)^{-1}dr = \lambda^{-1/2}$, which also holds for operators with positive real part [15, (192)–(193)]. The bound of (L5) is uniform in the background and in r, and the terms of $R_r(s)$ are analytic in the background and real at real backgrounds and real s, since they are built from real operators by sums, products and resolvents at real r; the r-integral preserves these properties. Details

For the determinant, the walk terms of $(\Delta_k + t)^{-1}$ with no steps are $\sum_j h_j(\Delta_{\Lambda_j^v} + t)^{-1}$. Since $\sum_j h_j(b) = 1$, the constant $\frac{1}{1+t}$ in the formula of Proposition 34.8 splits accordingly. So these terms contribute $\sum_j h_j(b)\ell_{\Lambda_j^v}(b)$. The terms with $n \ge 1$ steps contain $n+1$ resolvents and are $O((c_L + t)^{-2})$, hence integrable in t. More precisely, by (L5), $\|R_t^{(\ge 1)}\|_{\kappa',\gamma'} \le C(c_L + t)^{-2}$, uniformly in t and in the background. Since $|\operatorname{tr}_{\mathfrak{g}} O(b,b)| \le C|O|_{\gamma'}$, integration in t gives the norm bound for the terms with steps. The terms without steps are bounded by $|\ell_{\Lambda_j^v}(b)| \le C_L$, and their monomials $s^{X(j)}$ have degree at most $C(\rho)$. The index set does not depend on t, so the t-integral commutes with the sum over terms. □

By Lemma 40.3(b), at $s_\square = 0$ outside a union of boxes Z, all these unit-lattice objects split into independent blocks, one for each $*$-component of Z. After the conversion to margin 0 of Section 40, each block acts on fields and depends on the background only inside its component.

- *Scope.* This is the volume-type localization of the fluctuation covariance, of the coarse Hessian and of the determinant. Dimock's localization step [15, (225)–(233)] uses in addition sup bounds on the field $W_k(s)$ [15, Lemma 16, (100)]. Its Yang–Mills analogue needs the fine-lattice field $\mathbf{H}(s)Z$ pointwise, which is the subject of Section 42.
- *The first cube.* Unlike [15, (102)], the first cube of a walk is included in its monomial. So $G^X(0) = 0$, and the background dependence is cut together with the field; the price is a factor $e^{\kappa \mathsf{N}_{\mathrm{b}}}$ per step, which is harmless.
- *The order of the parameters.* In [15] the analyticity radius is $M^{1/2}$, i.e. $\kappa_1 = \frac{1}{2}\log M$. Here κ is fixed first. The rate $\kappa'_w = \kappa - C\delta$ must satisfy the rate conditions collected in

Remark 49.8 and those of Proposition 48.5. These involve κ_n, κ_0, $c_\sharp$, $c'_\sharp$, $c_\natural$, $c_\flat$ and C_d, not L. Choosing κ after L and before M is consistent with this. The commutator of the gauge-fixing part at level (L5) is $C_L L M^{-1}$, so it also requires M to be chosen after L.

41.4 The one-loop source in polymer form

Corollary 38.9 bounds the one-loop term by pieces of radius type. The polymer expansion needs pieces of volume type, at the complex points of the graded domains (Definition 38.3). Both follow from Corollary 41.6 and Lemma 38.5.

For $b \in \mathfrak{F}$ let $(X_\omega, n_\omega, O_\omega)_{\omega\in\Omega_b}$ be the weakenable family of numbers of Corollary 41.6 that represents $\ell_k(b)$, converted to margin 0, and write $X^\sharp_\omega$ for its polymers; $\deg\omega$ denotes the degree before the conversion. The index set, the polymers and the monomials are determined by the geometry of the walks (Lemma 40.4), not by the background; only the numbers $O_\omega = O_\omega(U)$ depend on it. For a polymer X put

$$\mathfrak{e}(X,U) = -\frac{1}{2}\sum_{b\in\mathfrak{F}} \sum_{\omega\in\Omega_b:\, X^\sharp_\omega = X} \big(O_\omega(U) - O_\omega(1)\big).$$

We write κ' for the rate of Corollary 41.6, $\kappa_2 = 3^{-d}\kappa'$, and C for constants that may depend on L, M, κ and ρ.

Proposition 41.7 (One-loop source in polymer form)**.** *Let G be semisimple and $q_1 = 2p+6$, with p from Lemma 28.3. There is $c_a > 0$ such that the following holds.*

(a) $-\frac{1}{2}\log\det\Delta_k(U) = -\frac{1}{2}\log\det\Delta_k(1) + \sum_X \mathfrak{e}(X,U)$*, with absolute convergence, for real U in the setting of Section 40, and for complex U in the ball $\mathcal{B}$ of the proof of Proposition 46.7.*

(b) Each $\mathfrak{e}(X,\cdot)$ depends only on U on the bonds of X, and is a normalized graded activity at X on $\mathcal{U}(X;c_a,c_a,c_a)$ (Definition 38.3). Moreover

$$\sup_{\mathcal{U}(X;c_a,c_a,c_a)} |\mathfrak{e}(X,\cdot)| \le C|X|^2 e^{-\kappa_2|X|}.$$

(c) Let X have diameter at most $n/2-8$, and let $U = e^{i\eta A'}U_r \in \mathcal{U}(X;c_a,c_a,c_a)$ with $|F_{U_r}|, |J_{U_r}| \le \theta$. Put

$$\mathfrak{t} = |\nabla A'|_{X^{(3)}} + [\nabla A']_{\alpha,X^{(3)}} + |A'|^2 + \theta.$$

If $C(1+|X|)^{q_1}(\mathfrak{t}+|A'|) \le 1$, then

$$|\mathfrak{e}(X,U)| \le C(1+|X|)^{q_1}\,\mathfrak{t}^2\, e^{-\kappa_2|X|}.$$

(d) Let $\theta_r, \delta_1, \delta_0 \le c_a$, and put $\mathfrak{t}_0 = \delta_1 + \delta_0^2 + \theta_r$. For every polymer X of diameter at most $n/2-8$, and every $U \in \mathcal{U}(X;\theta_r,\delta_1,\delta_0)$,

$$|\mathfrak{e}(X,U)| \le C\Big(\mathfrak{t}_0^2 + \exp\big(-c(\mathfrak{t}_0+\delta_0)^{-1/q_1}\big)\Big)e^{-(\kappa_2-1)|X|},$$

and in any case $|\mathfrak{e}(X,U)| \le Ce^{-(\kappa_2-1)|X|}$. For polymers of larger diameter, $|\mathfrak{e}(X,U)| \le C|X|^2e^{-\kappa_2|X|} \le Ce^{-(\kappa_2-1)|X|}e^{-cn/M}$. The constants C and c depend only on L, M, κ and ρ, not on $\theta_r, \delta_1, \delta_0$ or g.

Proof. (a) For real U: by Proposition 34.8, $\log\det\Delta_k = \sum_{b\in\mathfrak{F}}\ell_k(b)$, at U and at 1. By Corollary 41.6, $\ell_k(b) = \sum_{\omega\in\Omega_b} O_\omega$, with absolute convergence. Group the terms by $X^\sharp_\omega$. For complex U, both sides are holomorphic in the perturbation on the convex ball $\mathcal{B}$, and they agree at its real points; the identity theorem applies as in step (iii) of the proof of Proposition 46.7. Details

(b) Each O_ω depends on the background only on $X^{(r)}_\omega \subset X^\sharp_\omega$. It is invariant under real gauge transformations, since it is the diagonal trace of a covariant kernel or a term $h_j(b)\ell_{\Lambda^v_j}(b)$. It is analytic under the complex perturbations of Section 40. These are those of Proposition 19.11. The averaging and the averaged backgrounds at complex references are given by Proposition 46.4, and the remaining inputs by Proposition 46.7. A configuration of Definition 38.2 with $\varepsilon_a \le \varepsilon_{\max}$ is a complex reference of size at most $C_{\mathrm{cmp}}\delta_a$, with $C_{\mathrm{cmp}} = 4$ of the first item of Section 46; so this needs $\varepsilon_a \le \varepsilon_{\max}$ and $C_{\mathrm{cmp}}\delta_a < \delta_P$. So $\mathfrak{e}(X,\cdot)$ is a normalized background activity at X in the sense of Definition 38.2, for some $\varepsilon_a, \delta_a > 0$. Remark 38.7 makes it a graded activity on $\mathcal{U}(X; c_a, c_a, c_a)$, if $c_a \le a_2$, $\max(1, B_5)c_a \le \varepsilon_a$ and $2C_{\mathrm{ext}}c_a < \delta_a$. For the bound, let ω satisfy $X^\sharp_\omega = X$. Every box of X touches a box of X_ω, so $\deg\omega \ge |X_\omega| \ge 3^{-d}|X|$. Hence

$$\sum_{\omega\in\Omega_b:\, X^\sharp_\omega = X} |O_\omega| \le \sum_{\square\subset X}\ \sum_{\omega:\,\square\subset X_\omega} e^{\kappa'\deg\omega}|O_\omega|\, e^{-\kappa' 3^{-d}|X|} \le C|X| e^{-\kappa_2|X|},$$

by the norm bound of Corollary 41.6, uniformly in b and in the background. At most $CM^d|X|$ bonds $b\in\mathfrak{F}$ contribute.

(c) Apply Lemma 38.5 with $P = X$, $\theta_r = \delta_1 = \delta_0 = c_a \le \frac12 c_{\mathrm{q}}$, and a cube $\Lambda \supset X^{(3)}$ of side $2R = 2\,\mathrm{diam}(X) + 8 < n$. Since $\mathrm{diam}(X) \le M|X|$, $R/(4M+2) \le C(1+|X|)$, so $B_P \le C(1+|X|)^p$. Also $D_N \le CM(1+|X|)$, since X is a $*$-connected union of boxes of side at most M. Hence

$$\mathfrak{m}_1 \le C(\mathfrak{m} + |A'|^2 + (1+|X|)^{p+1}|A'|\theta), \qquad \mathfrak{s} \le C(1+|X|)^{p+2}\mathfrak{t},$$

and $c_* \le \frac12 c_{\mathrm{q}} + C(1+|X|)^{p+1}(|A'|+\theta)$, $RB_P\theta \le C(1+|X|)^{p+1}\theta$. For a suitable C, the condition $C(1+|X|)^{q_1}(\mathfrak{t}+|A'|) \le 1$ implies all hypotheses of the lemma and $\mathfrak{s} \le \frac12$. The lemma and (b) give $|\mathfrak{e}(X,U)| \le 2\mathfrak{s}^2 C|X|^2 e^{-\kappa_2|X|} \le C(1+|X|)^{2p+6}\mathfrak{t}^2 e^{-\kappa_2|X|}$.

(d) The conditions of Definition 38.3 are monotone in the three constants, so $\mathcal{U}(X;\theta_r,\delta_1,\delta_0) \subset \mathcal{U}(X;c_a,c_a,c_a)$, and (b) gives the second bound, since $|X|^2 e^{-|X|} \le C$. For the first, take a representation $U = e^{i\eta A'}U_r$ on $X^{(3)}$ as in Definition 38.3, so that $\mathfrak{t} \le \mathfrak{t}_0$ and $|A'| < \delta_0$. If $C(1+|X|)^{q_1}(\mathfrak{t}_0+\delta_0) \le 1$, (c) gives $|\mathfrak{e}(X,U)| \le C(1+|X|)^{q_1}\mathfrak{t}_0^2 e^{-\kappa_2|X|} \le C\mathfrak{t}_0^2 e^{-(\kappa_2-1)|X|}$, since $(1+|X|)^{q_1}e^{-|X|} \le C$. Otherwise $1+|X| > (C(\mathfrak{t}_0+\delta_0))^{-1/q_1}$, and (b) gives

$$|\mathfrak{e}(X,U)| \le C|X|^2 e^{-\kappa_2|X|} \le Ce^{-(\kappa_2-1)|X|}e^{-|X|/2} \le Ce^{-(\kappa_2-1)|X|}\exp\big(-c(\mathfrak{t}_0+\delta_0)^{-1/q_1}\big).$$

For polymers of diameter larger than $n/2-8$, $|X| \ge cn/M$. □

Remark 41.8 (Use in the renormalization group). (a) *Why the affine gauge matters.* Without the complex gauge step of Lemma 38.5, the sup norm $|A'|$ would enter $\mathfrak{s}$ linearly. On the domain of scale k, read in the units of scale $j<k$, $|A'| \le L^{-(k-j)}\delta_0$, while $\mathfrak{m}$ and θ are $O(L^{-2(k-j)})$ (Remark 38.8(i)). So $\mathfrak{s}$ would be of order $L^{-(k-j)}$ instead of $L^{-2(k-j)}$. In $d=4$ there are $L^{4(k-j)}$ polymers of scale j per cube of scale k, so already the orders ≥ 3 of an old activity, of size $L^{-3(k-j)}$ per polymer, would not be summable over the scales. With the affine gauge, $|A'|$ enters only through $|A'|^2$ and the smallness condition.

(b) *The next scale.* By Remark 38.8(i), (d) gives $|\mathfrak{e}(X,U)| \le Ce^{-(\kappa_2-1)|X|}$ on the domain of step $k+1$, read in the units of step k, with C independent of g and of the domain constants. So the one-loop source is a term of order one of the new activity. On that domain $\mathfrak{t} \le L^{-2}\mathfrak{t}_0$ and $|A'| \le L^{-1}\delta_0$, so (c) gives a factor L^{-4} per polymer of step k that

satisfies its smallness condition. In $d = 4$ a grid cube of step $k+1$ contains L^4 grid cubes of step k, so this factor is exactly compensated: the quadratic part is marginal. It must be extracted as a coupling constant, not bounded.

(c) *What is achieved.* All polymers, wrapping ones included, are normalized by construction: $\mathfrak{e}(X, 1) = 0$. Each $\mathfrak{e}(X, \cdot)$ depends on the background only inside X. Joint analyticity in the weakening parameters is not needed, since the determinant lies outside the fluctuation integral (Theorem 34.9) and $\mathfrak{e}$ is extracted at $s = 1$. The pieces of the polymers that do not wrap are the same functions on every torus, and all pieces are covariant under the translations by $M\mathbb{Z}^4$ (Proposition 41.14).

(d) *The domain.* Part (d) uses the graded domains of Definition 38.3, with the complex part controlled on $X^{(3)}$, i.e. with a domain margin of at least 3, not 0. (In Section 48, r_E denotes the margin of dependence of an activity; the domain margin is then $r_E + 3$.)

(e) *Inputs from the flow.* Four further points are needed to use (d) in a flow. They are supplied by Sections 49, 52 and 53 (Lemma 53.4(b) for the reblocking):

- *The decay rate.* $\kappa_2 - 1 > 0$ needs $\kappa' > 3^d$. Matching the rate κ_n of the activity norm needs $(3^{-d}\kappa' - 1)/c_\sharp \geq \kappa_n + \kappa_0 + 2$, with a factor 3^d from the conversion of margins. The rate conditions are collected in Remark 49.8(1)–(3).
- *The polymer scale.* The polymers here are unions of boxes of the paving, with sides between $M/4$ and M, while an activity norm uses Dimock's tree length d_M of the polymer grid (49.2). The paving is matched to the polymer grid of Section 49.
- *Reblocking and the background.* The polymers are reblocked into polymers of step $k+1$ by Lemma 49.3 and Remark 49.4. The background U is related to the next coarse configuration through the base $W = U'^{(k)}$ (Section 47, Setting).
- *Wrap-around.* The wrapping polymers, of diameter larger than $n/2-8$ (Section 52.1), give the sources $Ce^{-(\kappa_2-1)|X|}e^{-cn/M}$ of (d). On the domains of Remark 38.8 these are of the order of the other one-loop pieces. Under the torus hypotheses (T0)–(T2) of Lemma 52.1 they are bounded uniformly: at later scales by their supremum, and, summed over all scales, they contribute at most a constant times $e^{-cn_k/M}$ per grid cube of scale k (Lemma 52.1(c)–(e)).

41.5 Torus independence of the one-loop pieces

The pieces $\mathfrak{e}(X)$ of Proposition 41.7 and the Jacobian terms $\log \mathfrak{j}_c$ of Proposition 47.8 are constructed on one torus. Part XI compares them on tori of different sides (Section 58). This subsection shows that the pieces of the polymers that do not wrap, and the terms $\log \mathfrak{j}_c$, are the same functions of the configuration on every torus, and that all of them are covariant under the translations by the grid $M\mathbb{Z}^4$ (Proposition 41.14). Their common value defines the infinite-volume pieces. No walks on the infinite lattice are constructed. The proof carries every term of the construction of this section from one torus to another (Lemma 41.13), by induction over the levels (L1)–(L5). It rests on an inventory of the ingredients (Lemma 41.11): each of them is given by a local rule in the lattices, the grid and the configuration, and none of them depends on the torus as a whole.

Setting. Fix $k \geq 0$ and use the units of step k; the fine spacing is $\eta = L^{-k}$ on every torus. A side n is *admissible* if it satisfies the torus hypotheses (T0)–(T2) of Section 52.1 at step k. For an admissible n let $\mathbb{T}_\eta[n] = \eta\mathbb{Z}^4/n\mathbb{Z}^4$, with the projection $\pi_n : \eta\mathbb{Z}^4 \to \mathbb{T}_\eta[n]$, and let $\mathbb{T}^{(i)}[n]$ be the image of $L^{i-k}\mathbb{Z}^4$, for $i \leq k+1$; so $\mathbb{T}[n] = \mathbb{T}^{(k)}[n]$ and $\mathbb{T}'[n] = \mathbb{T}^{(k+1)}[n]$. The *grid* is the image of $M\mathbb{Z}^4$. All these lattices contain the origin. Since M is a power of L (Section 49.2)

and $M > L$ (Section 47, Setting), $M\mathbb{Z}^4 \subset L^{i-k}\mathbb{Z}^4$ for $i \le k+1$, and since $LM \mid n$ by (T1), all of them project to $\mathbb{T}_\eta[n]$. On every $\mathbb{T}_\eta[n]$ the constructions of Section 40 and of this section at step k (for $k = 0$, those of Proposition 52.2(c)) are made with the polymer grid of scale k as paving (Section 48.2), and with the same parameters (Lemma 41.11(b)). An index $[n]$ marks the torus where needed, as in $\mathfrak{e}[n](X, U)$. Put $R_{\mathrm{g}} = (\rho + C_v + 8)M$. We take the constant in the condition $n \ge C(\rho + C_v)M$ of Section 40, which is part of (T0), with $C \ge 8$. Since $\rho \ge 30$, this gives $4R_{\mathrm{g}} < n$.

Definition 41.9 (Grid transfers)**.** Let n, n' be admissible and $a \in M\mathbb{Z}^4$. A *grid transfer* from $\mathbb{T}_\eta[n]$ to $\mathbb{T}_\eta[n']$ is a bijection $\iota : P \to P'$ from a set P of points of $\mathbb{T}_\eta[n]$ onto a set P' of points of $\mathbb{T}_\eta[n']$, of one of two kinds:

(i) *(Grid translations.)* $n' = n$, $P = P' = \mathbb{T}_\eta[n]$, and $\iota = \tau_a$, where $\tau_a(\pi_n x) = \pi_n(x + a)$;

(ii) *(Transfers of lifts.)* $P = \pi_n(\tilde{P})$ for a set $\tilde{P} \subset \eta\mathbb{Z}^4$ of ℓ^∞-diameter less than $\frac{1}{2}\min(n, n')$, and $\iota(\pi_n x) = \pi_{n'}(x + a)$ for $x \in \tilde{P}$.

A bond, a plaquette or a unit bond is *based* at its initial point (at p_- for a plaquette p), and a point, fine or unit, is based at itself. For such a site x based at a point of P, ιx is the site of the same kind and the same directions based at the image of that point. Configurations U on $\mathbb{T}_\eta[n]$ and U' on $\mathbb{T}_\eta[n']$ *agree on* P if $U'(\iota b) = U(b)$ for every bond b of P, i.e. with both endpoints in P. Two domains X of $\mathbb{T}_\eta[n]$ and X' of $\mathbb{T}_\eta[n']$, or empty sets, are *related* if $X = \mathbb{T}_\eta[n]$ and $X' = \mathbb{T}_\eta[n']$, or if $X \subset P$ and $X' = \iota(X)$. For an operator O whose kernel vanishes unless both sites are based in P, $\iota_* O$ is the operator with the kernel $(\iota_* O)(\iota x, \iota y) = O(x, y)$, which vanishes at the other pairs of sites; the fibres $(\mathbb{C}, \mathfrak{g}^c, \dots)$ at x and at ιx are identified.

For a grid translation every domain X is related to $\tau_a X$, and agreement on P means $U' = \tau_a U$, with $(\tau_a U)(b) = U(b - a)$ as in (G2) of Definition 50.1.

Lemma 41.10 (Geometry of grid transfers)**.** *Let $\iota : P \to P'$ be a grid transfer from $\mathbb{T}_\eta[n]$ to $\mathbb{T}_\eta[n']$, with the vector $a \in M\mathbb{Z}^4$.*

(a) ι preserves the ℓ^∞-distances of the tori, and $\iota(x + \eta e_\mu) = \iota x + \eta e_\mu$ if $x, x + \eta e_\mu \in P$. If $S \subset P$, $R \le R_{\mathrm{g}}$ and $S^{(R)} \subset P$, then $\iota(S^{(R)}) = \iota(S)^{(R)}$. The inverse ι^{-1} is a grid transfer from $\mathbb{T}_\eta[n']$ to $\mathbb{T}_\eta[n]$, with the vector $-a$.

(b) Let $x \in P$, and let B and B' be the closed ℓ^∞-balls of radius R_{g} about x in $\mathbb{T}_\eta[n]$ and about ιx in $\mathbb{T}_\eta[n']$. The translation by a of the lifts induces an isometric bijection $\iota_x : B \to B'$. It agrees with ι on $B \cap P$, maps $Y \cap B$ onto $\iota(Y) \cap B'$ for every set $Y \subset P$, maps the points of the lattices $\mathbb{T}^{(i)}$, $i \le k+1$, and of the grid in B onto those in B', and maps lattice steps to lattice steps.

Consequently, let an object be defined on every admissible torus by the same rule, which at a site based at y uses only the lattices $\mathbb{T}^{(i)}$, $i \le k+1$, the grid, the ℓ^∞-distance and possibly a domain Y, all within distance R_{g} of y. Then, for related domains Y and Y', the object of Y at a site x based at a point of P equals the object of Y' at ιx.

Proof. For a grid translation all claims are clear, with $\iota_x = \tau_a$; the translation by $a \in M\mathbb{Z}^4$ maps $L^{i-k}\mathbb{Z}^4$, $i \le k+1$, and $M\mathbb{Z}^4$ onto themselves. Let ι be a transfer of a lift $\tilde{P}$. Details

(a) Let $p, q \in P$, with lifts $\tilde{p}, \tilde{q} \in \tilde{P}$. Since $|\tilde{p} - \tilde{q}| < \frac{1}{2}n$, the distance of p and q on $\mathbb{T}_\eta[n]$ is $|\tilde{p} - \tilde{q}|$; since $|\tilde{p} - \tilde{q}| < \frac{1}{2}n'$, it is also the distance of $\iota p = \pi_{n'}(\tilde{p} + a)$ and ιq. If $q = p + \eta e_\mu$, then $\tilde{q} - \tilde{p} - \eta e_\mu \in n\mathbb{Z}^4$ has norm less than n, so $\tilde{q} = \tilde{p} + \eta e_\mu$ and $\iota q = \iota p + \eta e_\mu$. Let $S^{(R)} \subset P$ with $R \le R_{\mathrm{g}}$. The inclusion $\iota(S^{(R)}) \subset \iota(S)^{(R)}$ holds since ι preserves distances. Conversely, let q' be within distance R of ιs, $s \in S$, $s = \pi_n(\tilde{s})$ with $\tilde{s} \in \tilde{P}$, and choose w with $\pi_{n'}(w) = q'$ and $|w - \tilde{s} - a| \le R$. The point $q = \pi_n(w - a)$ lies within distance R of s, so $q \in P$ and $q = \pi_n(\tilde{q})$

with $\tilde q \in \tilde P$. Then $|\tilde q - (w-a)| \le |\tilde q - \tilde s| + |\tilde s - (w-a)| < \frac12 n + R_{\mathrm g} < n$, so $\tilde q = w - a$ and $\iota q = q'$. Finally $P' = \pi_{n'}(\tilde P + a)$, and $\tilde P + a$ has the diameter of $\tilde P$.

(b) Let $x = \pi_n(\tilde x)$ with $\tilde x \in \tilde P$, and let $\tilde B$ be the ball of radius $R_{\mathrm g}$ about $\tilde x$ in $\eta\mathbb{Z}^4$. Since $4R_{\mathrm g} < \min(n, n')$, π_n maps $\tilde B$ isometrically onto B, and $\pi_{n'}$ maps $\tilde B + a$ isometrically onto B'. Put $\iota_x(\pi_n y) = \pi_{n'}(y + a)$ for $y \in \tilde B$. Let $p \in B \cap P$, $p = \pi_n(\tilde p)$ with $\tilde p \in \tilde P$, and $p = \pi_n(y)$ with $y \in \tilde B$. Then $|\tilde p - y| \le |\tilde p - \tilde x| + |\tilde x - y| < \frac12 n + R_{\mathrm g} < n$, so $\tilde p = y$ and $\iota_x p = \iota p$. If $Y \subset P$, this gives $\iota_x(Y \cap B) \subset \iota(Y) \cap B'$. Conversely, let $p \in Y$ with $\iota p \in B'$, and write $\iota p = \pi_{n'}(y + a)$ with $y \in \tilde B$. Since $|\tilde p - y| < \frac12 n' + R_{\mathrm g} < n'$ and $\tilde p - y \in n'\mathbb{Z}^4$, $\tilde p = y$, so $p \in B \cap Y$. The translation by a maps the lattices $L^{i-k}\mathbb{Z}^4$, $i \le k+1$, the lattice $M\mathbb{Z}^4$ and the lattice steps onto themselves.

For the consequence, apply (b) at the base point y of x. The map ι_y carries the lattices, the grid and the distances in the ball about y onto those in the ball about ιy, and Y onto Y' there: for $Y \subset P$ by (b), and for $Y = \mathbb{T}_\eta[n]$ since ι_y is onto. So the rule gives the same result. □

Lemma 41.11 (Inventory of the ingredients)**.** *The pieces of Proposition 41.7 and the terms of Proposition 47.8 at step k are built from the following ingredients; Steps 2 and 5 of the proof of Lemma 41.13 list how.*

(a) (Geometry.) *Each of the following is defined on every admissible torus by the same rule, as in the consequence of Lemma 41.10, with range at most $R_{\mathrm g}$:*

- *the unit blocks $B(y)$, the blocks $B_i(y)$ and the contour families $G(y,x)$ of the averagings (Section 21), the blocks $B'(y)$ of $\mathbb{T}'$, the inner and the crossing bonds, the bonds $b_0(c)$ and the set $\mathfrak{F}$ (Definition 32.1), and the plaquettes;*
- *the grid cubes of scale k, which are the boxes of the paving, the polymers, the numbers $|X|$, the neighbourhoods $X^{(r)}$, and the margin conversion $X \mapsto X^\sharp$, with the split of the additive local terms box by box (Section 40);*
- *the grid points z_j, the functions χ_j and $h_j = \chi_j^2$ (Lemma 17.3), the sets K_j, Λ_j, Λ_j^v and $X(j)$; for a domain Y, the set $J(Y)$ and the mass profile m_Y (Section 19), the sets Y_1, $\mathfrak{B}(Y)$, $\hat Y$, $\mathbb{T}'(Y)$, $\mathfrak{F}(Y)$ and $Y^+ = Y^{(R_0)}$ (Sections 17 and 37).*

(b) (Constants.) *L, M, κ, δ, ρ, C_v, R_v, $R_0 = R_0(L,M)$, the seam widths w, the mass coefficient κ_m, the averaging parameter a, the margins r and r', the decay rates, the thresholds M_1, and the bounds ε, δ_P, ε_a, δ_a, c_a on the references. They are fixed in the order of Remarks 49.8 and 38.8, before the torus. The torus enters only through the lower bounds (T0), (T2) and the divisibility (T1). The IMS scale M' and its partition of unity (Theorem 34.4, Proposition 52.2(b)), the local gauges of Definition 18.1 and the model configurations of the proofs enter only proofs of bounds, not the definitions of the operators.*

(c) (Local operators.) *Each of the following operators $\mathcal{L}$ has a range $r_{\mathcal L} \le R_{\mathrm g}$. Its kernel $\mathcal{L}(x,x')$ vanishes unless the base points of x and x' are within distance $r_{\mathcal L}$, and it is given by the same formula on every torus, in the objects of (a), in the domain Y where one appears, and in the configuration on the bonds of the ball of radius $r_{\mathcal L}$ about the base point of x'. At complex references it is the holomorphic continuation of the same formula.*

- *The covariant derivative $D = D_U$, D^T, $D_Y^T = \mathbf{1}_Y D^T$, the projections $\mathbf{1}_Y$, $\mathbf{1}_{\mathfrak{F}(Y)}$ and $\mathbf{1}_\square$, the exterior divergence term $\mathfrak{A}_Y^{\mathrm{ext}} = D\mathbf{1}_{\mathbb{T}_\eta \setminus Y} D^T$ (Section 19, Remark 41.5), the mass term $\kappa_m m_Y$, and multiplication by h_j.*
- *The averaged backgrounds $W_j = U^{(j)}$, $j \le k$, and among them the base $W = U^{(k)}$, built by the one-step averagings from the averaged contour variables and the group*

average (Section 21, Lemma 21.10(c); at complex references Proposition 46.4(i)); the conditions of the averaging domain (Definition 22.1), which are local; and the linearized averages $Q' = Q'_k(U)$, $\mathbf{Q} = \mathbf{Q}_k(U)$ *and their restrictions* $\mathbf{Q}_Y$ *(Proposition 22.10(d)).*

- *The curvature* F_U *and its symmetrization* F_U^s*, the curvature factor* $\mathfrak{F}$ *(Lemma 12.3), the current* $J_U = D^T F_U^s$ *(Lemma 23.2), the second variations* $d^2\mathcal{N}(0)$ *(Proposition 23.3; it is a sum of plaquette-local forms) and* $d^2 C_k^{\mathrm{av}}(0)$ *(Proposition 22.9(a),(c)). The potential* W *of the local part at level (L3) is* 0*; the potential* W_2 *enters the non-local part (level (L4)).*
- *The elimination at the base:* ℓ_c, ζ_c, $\hat{Z}^1$, $\hat{Z}^2$ *and* $\mathfrak{r}_c$ *(Lemma 32.7(a)–(d)); the linearized averaged contour variables* $\mathfrak{G}_W$ *(Definition 22.6, Lemma 22.7(a)),* $K^Y = \mathfrak{G}_W^T \mathbf{1}_{\mathbb{T}'(Y)} \mathfrak{G}_W$ *and* $K_Y^{\mathrm{gf}} = \mathbf{1}_{\mathfrak{F}(Y)} \mathfrak{G}_W^T \mathfrak{G}_W \mathbf{1}_{\mathfrak{F}(Y)}$ *(Section 37).*
- *At* $k = 0$*: the Hessian* Δ^0 *of* $B \mapsto S(e^{\mathrm{i}B}U')$ *at* 0*,* $\omega_0 = J_{U'}$*, and the operator* Δ_0 *of Proposition 52.2(b).*

(d) (Jacobians.) *For a bond* c *of* $\mathbb{T}'$ *put* $P_c = \bigcup_{y \in B'(c_-) \cup B'(c_+)} B(y)$*, a set of fine points. Then* $\log \mathrm{j}_c = -\log\det(1 + \mathfrak{r}_c(0))$ *(Proposition 47.8(b)) is given by the same formula on every torus in the configuration on the bonds of* P_c *and the objects of (a) in* P_c*.*

Consequently, let $\iota : P \to P'$ *be a grid transfer, and let* U *and* U' *be references that agree on* P*; primes denote the objects at* U' *on* $\mathbb{T}_\eta[n']$*. For every operator* $\mathcal{L}$ *of (c), with related domains* Y*,* Y' *in its definition, and every set* S *with* $S^{(r_\mathcal{L})} \subset P$*,* $\mathcal{L}' \mathbf{1}_{\iota S} = \iota_*(\mathcal{L} \mathbf{1}_S)$*; if* $S^{(2r_\mathcal{L})} \subset P$*, also* $\mathbf{1}_{\iota S} \mathcal{L}' = \iota_*(\mathbf{1}_S \mathcal{L})$*. If* $P_c \subset P$*, then* ιc *is a bond of* $\mathbb{T}'[n']$*,* $\iota(P_c) = P_{\iota c}$*, and* $\log \mathrm{j}'_{\iota c}(U') = \log \mathrm{j}_c(U)$*.*

Proof. (a) The lattices and the grid are those of the Setting. The blocks $B_i(y)$ and $B'(y)$ are the cubes of L^d points of the next finer lattice centred at the points y of the coarser one, L being odd, and $B(y)$ is the set of the points of $\mathbb{T}_\eta$ at distance at most $\frac{1}{2}(1-\eta)$ from $y \in \mathbb{T}$. The contours of $G(y,x)$ are the coordinate paths from y to x in the box spanned by y and x (Section 21). A bond is crossing if its endpoints lie in different blocks B', and a positively oriented bond b of direction μ is a bond $b_0(c)$ if and only if $b_- - \frac{1}{2}(L-1)e_\mu \in \mathbb{T}'$ (Definition 32.1). The grid cubes of scale k are the unions of the blocks $B(y)$ with $|y-c|_\infty \le \frac{1}{2}(M-1)$, c in the grid, and they are the boxes of the paving (Sections 49.2 and 48.2). Polymers, $|X|$, $X^{(r)}$ and $X^\sharp$ are defined from the boxes and the distance, and the split of Section 40 uses only the boxes and the base points. Since $M \mid n$, $\check{M} = M$, and the z_j are the points of the grid (Sections 17 and 49.2). Let ϕ be the function of the proof of Lemma 17.3. Since $n \ge 8M$, the periodization of ϕ_j does not overlap, and $\chi_j(y) = \phi_j(y)\big(\sum_i \phi_i(y)^2\big)^{-1/2}$ with $\phi_i(y) = \prod_\mu \phi((y_\mu - z_{i,\mu})/M)$, the differences read on lifts at distance less than M, for the grid points z_i at distance less than M from y, and $\phi_i(y) = 0$ for the others; this is a rule of range M. The sets K_j, Λ_j, Λ_j^v are defined by the distance to z_j, at most $\rho M + 1$, and $X(j)$ is the union of the boxes that meet $(\Lambda_j^v)^{(R_v)}$, within distance $(\rho + C_v + 2)M$ of z_j. For a domain Y, $J(Y) = \{j : \Lambda_j \subset Y\}$ and $m_Y = \sum_{j \notin J(Y)} \chi_j^2$; at a point y only the j with $|z_j - y| < M$ contribute, and their Λ_j lie within distance $4M+1$ of y. The sets Y_1, $\mathfrak{B}(Y)$, $\hat{Y}$, $\mathbb{T}'(Y)$, $\mathfrak{F}(Y)$ and Y^+ are defined from Y, the blocks and the distance, within distance $R_0 + L \le R_{\mathrm{g}}$.

Details

(b) This is the order of the choices in Remarks 49.8 and 38.8, and in Sections 40 and 41; none of these choices involves n. The proofs of the lower bounds (Theorem 34.4, Lemma 37.5, Proposition 52.2(b)) use the IMS partition and the local gauges; the operators themselves are defined without them.

(c) The kernels of D, D^T and of the projections are explicit in $U(b)$ for the bonds at the site; D_Y^T, $\mathbf{1}_Y$, $\mathfrak{A}_Y^{\mathrm{ext}}$ and m_Y involve Y only near the site. By Lemma 21.10(c), the one-step average of a bond c depends only on the bonds of $B_j(c_-) \cup B_j(c_+)$, so W_j on a bond of $\mathbb{T}^{(j)}$ depends only on U on the bonds of the unit blocks at its endpoints; the averaged contour

variables and the group average are defined by the contour families and the configuration on them (Section 21). The kernels of $\mathbf{Q}_k$ and Q'_k are given by Proposition 22.10(d) and the formula before it. At complex references the averaged backgrounds and the averages are the continuations of Proposition 46.4, given by the same formulas. The curvature factor is the multiplication by the coefficients $c_{\mu\nu}$ of Lemma 12.3 composed with a shift by 2η, and F_U, F_U^s, J_U are plaquette-local. The second variation $d^2\mathcal{N}(0)$ is a sum of plaquette-local forms (proof of Lemma 37.2), and $d^2C_k^{\mathrm{av}}(0)[A,A](c)$ depends only on A and U on the bonds of $B(c_-)\cup B(c_+)$ (Proposition 22.9(a)). By Lemma 32.7(a),(c), ℓ_c and ζ_c depend only on the variables and on the base on the bonds of P_c, and $\hat{Z}^1\delta_b$, $\hat{Z}^2$ are supported within distance $2L$ of b and depend only on the base there (Section 37). By Lemma 22.7(a), $\mathfrak{G}_W Z(y,x)$ depends only on Z and W on the bonds of the box spanned by y and x. At $k=0$, Δ^0 has a kernel that vanishes unless the two bonds lie in a common plaquette, and Δ_0 has range $2L+2$ (Proposition 52.2(b)). Every formula uses the lattices and the configuration only through the objects of (a), and all ranges are at most $4L+2\le R_{\mathrm{g}}$.

(d) By Proposition 47.8(b) and Lemma 32.7(b), $\mathfrak{r}_c(0)$ is determined by the derivative of ℓ_c at $Z=0$ with respect to $Z(b_0(c))$, at the base W; by (c), this is a formula in U on the bonds of P_c.

The consequence follows from Lemma 41.10. The kernel of $\mathcal{L}\mathbf{1}_S$ is supported on pairs of sites based in $S^{(r_{\mathcal{L}})}\times S$, and its entries use the configuration on the bonds of $S^{(r_{\mathcal{L}})}\subset P$, on which U and U' agree; for $\mathbf{1}_S\mathcal{L}$ the corresponding set is $S^{(2r_{\mathcal{L}})}$. For the Jacobians, $\iota(P_c)=P_{\iota c}$ by Lemma 41.10(b), and the formula of (d) gives the same value. □

Remark 41.12 (Global objects). The following objects depend on the torus as a whole, and they do not enter the pieces as such.

- *Torus operators.* The inverses on the whole torus, such as Δ_k, C_k, $\mathbf{G}$, $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$, $P_{\mathcal{V}}$ and R on $\mathbb{T}_\eta$, enter $\mathfrak{e}$ only through the terms of their weakened representations, i.e. through the families on the domain $\mathbb{T}_\eta$. Lemma 41.13 carries these terms one by one: the whole torus is related to the whole torus.
- *Constants.* The normalizations $\mathcal{N}_k$, N_χ and $\sum_b \ell_k(b;1)$ and the vacuum energy depend on $|\mathbb{T}|$. They are not part of $\mathfrak{e}$, which is normalized term by term, $O_\omega(U)-O_\omega(1)$, nor of $\log \mathfrak{j}_c$ (Section 52.3).
- *Criticality.* At every reference Δ_k is defined by the formula of Section 37 (Proposition 47.12). The minimizer of the variational problem enters only through the reference itself.
- *Extensions.* The value of $\mathfrak{e}(X,\cdot)$ at a point of a graded domain is computed at the cutoff extension of Remark 38.7; it does not depend on the extension, since $\mathfrak{e}(X,\cdot)$ depends only on the bonds of X. The same holds for $\log \mathfrak{j}_c$ and P_c.

Lemma 41.13 (Transfer between tori). *Let $\iota : P\to P'$ be a grid transfer from $\mathbb{T}_\eta[n]$ to $\mathbb{T}_\eta[n']$, and let U and U' be references on $\mathbb{T}_\eta[n]$ and $\mathbb{T}_\eta[n']$, in the setting of Section 40, that agree on P; for $k\ge 1$ this includes the points of the domain of Definition 38.2 used in the proof of Proposition 41.7(b). Primes denote the objects on $\mathbb{T}_\eta[n']$ at U'.*

(a) (Dirichlet operators.) *Let $Y\subset P$ be a domain with $Y^{(R_v)}\subset P$; for a grid translation, let Y be any domain. Then the Dirichlet operators of Sections 18, 19, 35 and 37 on the domains Y and Y^+, listed in Step 2 of the proof, and their inverses are carried by ι. For instance $\mathbf{\Gamma}'^{\iota Y}=\iota_*\mathbf{\Gamma}^Y$ and $(\Delta'_{\iota Y}+t)^{-1}=\iota_*(\Delta_Y+t)^{-1}$ for $t\ge 0$. Moreover $\ell'_{\iota Y}(\iota b)=\ell_Y(b)$ for $b\in\mathfrak{F}(Y)$.*

(b) (Weakened families.) *Let $\mathcal{O}$ be one of the following families: a family of one of the levels (L1)–(L4), or of the repetitions of (L3), (L4) with the potential W_2, on a domain X (for*

a factorized family, each of its middle factors); the families $R_r(s)$, $r \geq 0$, and $C_k^{1/2}(s)$ on $\mathbb{T}_\eta[n]$; or the family of numbers that represents $\ell_k(b)$, for a bond $b \in \mathfrak{F}$ based in P (Corollary 41.6). Let $\mathcal{O}'$ be the same family on $\mathbb{T}_\eta[n']$, on a domain X' related to X, respectively for the bond ιb, and let $r_{\mathcal{O}}$ be its margin. Then there is a bijection $\omega \mapsto \iota(\omega)$ from the terms ω of $\mathcal{O}$ with $X_\omega^{(r_{\mathcal{O}})} \subset P$ onto the terms ω' of $\mathcal{O}'$ with $X_{\omega'}^{(r_{\mathcal{O}})} \subset P'$, such that

$$X_{\iota(\omega)} = \iota(X_\omega), \qquad n_{\iota(\omega)}(\iota\square) = n_\omega(\square) \text{ for the boxes } \square \subset P, \qquad O'_{\iota(\omega)} = \iota_* O_\omega,$$

and with $O'_{\iota(\omega)} = O_\omega$ for the families of numbers. The same holds for the families $D\mathcal{O}$, $\mathcal{O}D^T$ and $D\mathcal{O}D^T$, where they are defined. The bijection acts on the index data of the terms; it is the same for all related domains, and it maps matched terms to matched terms.

(c) (Margin 0.) *After the conversion to margin 0 of Section 40, (b) holds with the condition $X_\omega^\sharp \subset P$ in place of $X_\omega^{(r_{\mathcal{O}})} \subset P$, and $X_{\iota(\omega)}^\sharp = \iota(X_\omega^\sharp)$.*

For $k = 0$ the same holds for the families of Proposition 52.2(c), with (a) for the operators $\mathbf{1}_Y(\Delta_0 + t)\mathbf{1}_Y$, their inverses and the densities $\ell_Y(b)$.

Proof. The map ι_* is conjugation by the relabelling of the sites. So it commutes with sums, products, transposes and inverses of operators on ℓ^2-spaces of sites based in P, with absolutely convergent integrals over parameters, and with the diagonal traces $\mathrm{tr}_{\mathfrak{g}} O(b,b)$. We use Lemmas 41.10 and 41.11 without further mention. For a grid translation the conditions "$\subset P$" are void, and the argument below applies to all terms. Details

Step 1: domains. Let X, X' be related, and let j be a grid point with $X(j) \subset P$; write ιj for the index of the grid point ιz_j. By Lemma 41.10(b) at $x = z_j$, $\iota(\Lambda_j^v) = \Lambda_{\iota j}^v$ and $\iota(X(j)) = X(\iota j)$, and the domains $\Lambda_j^v \cap X$ and $\Lambda_{\iota j}^v \cap X'$ are related; their R_v-neighbourhoods lie in $X(j)$ and $X(\iota j)$. If $Y \subset P$ and $Y^{(R_0)} \subset P$, then $\iota(Y^+) = \iota(Y)^+$, so Y^+ and $\iota(Y)^+$ are related; and $\mathbb{T}_\eta[n]^+ = \mathbb{T}_\eta[n]$.

Step 2: Dirichlet operators. Let Y be as in (a). By Sections 18, 19, 35, 37 and 38, the Dirichlet operators on Y are obtained, in the following order, by finitely many sums, products, transposes and inverses on the finite-dimensional spaces $\ell_0^2(Y)$, $\ell_1^2(Y)$, $\ell^2(Y_1)$, $\ell^2(\mathfrak{B}(Y))$ and $\ell^2(\mathfrak{F}(Y))$, from the local operators of Lemma 41.11(c) on the domains Y and Y^+:

- $\mathfrak{A}'_Y = \mathbf{1}_Y(D^TD + aQ'^TQ')\mathbf{1}_Y$ and $G'_Y = (\mathfrak{A}'_Y)^{-1}$; $Q'G'_YQ'^T$, $M'^Y = Q'(G'_Y)^2Q'^T$ and their inverses $\mathbf{\Gamma}'^Y$, $(M'^Y)^{-1}$; H'^Y, $P_{\mathcal{V},Y}$, $R_Y = \mathbf{1}_Y - P_{\mathcal{V},Y}$, S_Y and Π_Y;
- $\mathfrak{A}^{Y,\kappa_m} = \mathfrak{A}^Y + \mathfrak{A}_Y^{\mathrm{ext}} + \kappa_m m_Y$ (Section 19), whose local part is the Dirichlet restriction $\mathbf{1}_Y(D^TD + DD^T + a\mathbf{Q}^T\mathbf{Q})\mathbf{1}_Y$, including $\mathfrak{A}_Y^{\mathrm{ext}}$, and whose non-local part $\mathfrak{N}^Y$ is that of level (L3); its inverse $\mathbf{G}^Y$, $\mathbf{\Gamma}^Y = (\mathbf{Q}_Y\mathbf{G}^Y\mathbf{Q}_Y^T)^{-1}$, $\mathbf{H}^Y$ and $\mathbf{C}^Y$;
- ω_0^Y, W_2^Y, the same operators with the potential W_2^Y in the non-local part, $\mathbf{\Gamma}^Y[W_2^Y]$, and $\Delta^{0,Y} = \mathbf{\Gamma}^Y[W_2^Y] - a$ (Definition 37.3);
- $\Delta^{\mathrm{c},Y}$, $\Delta_k^{(Y)} = \Delta^{\mathrm{c},Y} + K^Y$, the local fluctuation operator $\Delta_Y = \mathbf{1}_{\mathfrak{F}(Y)}\Delta_k^{(Y^+)}\mathbf{1}_{\mathfrak{F}(Y)}$ and its resolvents $(\Delta_Y + t)^{-1}$;
- $\log \Delta_Y$, an absolutely convergent integral in t, and the density $\ell_Y(b)$, a diagonal trace.

At complex references the same formulas are used, with the continued local operators (Propositions 19.11 and 46.7). These operators depend on the configuration only on the bonds of $\hat{Y}$, respectively of $\widehat{Y^+}$ for Δ_Y and the objects built from it (Sections 18, 19, 35, Propositions 37.4(c) and 37.6(b)). The margin $R_v \geq R_0 + 4 + 2r_{\mathrm{loc}}$ of Section 40, with $r_{\mathrm{loc}} \geq r_{\mathcal{L}}$ for the operators $\mathcal{L}$ of Lemma 41.11(c), contains these sets together with the margins $2r_{\mathcal{L}}$ of the consequence

of Lemma 41.11. So the configuration enters only on the bonds of $Y^{(R_v)} \subset P$, where U and U' agree. They involve the domain only through Y and Y^+, and $\iota(Y^+) = \iota(Y)^+$ by Step 1. So the local operators that enter are carried by ι, and so are the results, since ι_* commutes with the operations listed. This is (a). For a grid translation the same holds for every domain, including $\mathbb{T}_\eta[n]$.

Step 3: one walk. Consider one application of Theorem 41.4, on related domains X and X'. Assume that the local parts are built from the operators of Lemma 41.11(c) and constants, on related domains, and that the terms of the representations $\mathfrak{N}^Z(s)$ of the non-local parts satisfy the conclusion of (b) for all related domains Z, Z'. We show that the terms of $G^X(s)$, $DG^X(s)$, $G^X(s)D^T$ and $DG^X(s)D^T$ satisfy the conclusion of (b).

By the construction of the walk (the weakened step and the proof of Theorem 41.4), the terms of $G^X(s)$ are the products $\Phi_0\Phi_1\cdots\Phi_m$, $m \geq 0$, in which $\Phi_0 = h_{j_0}G^X_{j_0}$ and each Φ_i, $i \geq 1$, is a term of $\mathcal{R}^X(s)$, in the regrouped form of the proof of Lemma 41.3 and with the derivatives regrouped as after the definition of $G^X(s)$. A term of $\mathcal{R}^X(s)$ is given by a grid point j and its kind, one of the finitely many regrouped pieces listed in the proofs of Lemma 41.3 and of the weakened step, with the placement of the derivatives. The kinds are of three types: the local commutator $[\mathfrak{A}^X_{\mathrm{loc}}, h_j]G^X_j$; the commutator of h_j with a term ω of $\mathfrak{N}^X(s)$ (with the outer derivatives of the factorization), times G^X_j; or the seam term $h_j\mathbf{1}_{Z_j}(O^X_\omega - O^{Z_j}_\omega)\mathbf{1}_{Z_j}G^X_j$, $Z_j = \Lambda^v_j \cap X$, with a pair of matched terms of $\mathfrak{N}^X(s)$ and $\mathfrak{N}^{Z_j}(s)$. Its polymer is $X(j)$, respectively $X(j) \cup X_\omega$, and its monomial is $s^{X(j)}$, respectively $s^{X(j)}s^{n_\omega}$. A product is a term if the r-neighbourhoods of the polymers of consecutive factors meet (Lemma 40.4); its polymer is the union and its monomial the product. So a term Ω of $G^X(s)$ is given by its index data: the grid points $j_0, \dots, j_m$, the kinds of the steps and the terms ω_i of the non-local parts. These data, and the condition that consecutive neighbourhoods meet, do not depend on the configuration, and the index set is the same on all domains (Theorem 41.4(b)). The map $\Omega \mapsto \iota(\Omega)$ defined below acts on the index data; so it is the same on all domains, and it maps matched terms to matched terms.

Let Ω be a term with $X^{(r)}_\Omega \subset P$, r its margin. The polymers of its factors lie in X_Ω, and their margins are at most r, since margins only grow under Lemmas 40.4, 40.5 and Theorem 41.4. So $X(j_i) \subset P$ for all i, and the terms ω_i satisfy $X^{(r)}_{\omega_i} \subset P$. Let $\iota(\Omega)$ be the term with the index data $\iota j_0, \dots, \iota j_m$, the same kinds, and the terms $\iota(\omega_i)$ of the non-local parts on X' and on $\Lambda^v_{\iota j_i} \cap X'$; these exist by hypothesis, since the domains are related by Step 1. By Step 1 the polymers of the factors are carried by ι, and so is the condition that consecutive neighbourhoods meet. Hence $\iota(\Omega)$ is a term, with polymer $\iota(X_\Omega)$ and the carried monomial. Its factors are carried by ι: $h_{\iota j}$ and $\mathfrak{A}^{X'}_{\mathrm{loc}}$ by Lemma 41.11, since in the factor of j they act on fields supported in $\Lambda^v_j \cap X$ and their ranges are less than R_v; the local propagator $G^{X'}_{\iota j}$ by Step 2, since G^X_j depends on X only through $\Lambda^v_j \cap X$ and $(\Lambda^v_j)^{(R_v)} \subset X(j) \subset P$; the terms of the non-local parts by hypothesis. The derivatives D^T_X, $D^T_{Z_j}$ of the regrouping are local operators on related domains. So $O'_{\iota(\Omega)} = \iota_* O_\Omega$. The same construction for the grid transfer ι^{-1} inverts $\Omega \mapsto \iota(\Omega)$, which is therefore a bijection. The versions with derivatives follow by composing with the local operators D and D^T.

Step 4: products, local factors and additive terms. The products of Lemma 40.4 are indexed by the tuples of terms whose consecutive r-neighbourhoods meet, and those of Lemma 40.5(a) have the polymers and monomials of the factor. If the factors satisfy the conclusion of (b), so do the products, by the argument of Step 3; the margins of the factors are at most that of the product. The additive local terms are split box by box (Section 40): the term $T\mathbf{1}_\square$ has polymer $\square$ and monomial $s_\square$, and $T'\mathbf{1}_{\iota\square} = \iota_*(T\mathbf{1}_\square)$ for the boxes $\square$ with $\square^{(r)} \subset P$. Integrals in r or t of families whose index sets do not depend on the parameter are carried term by term.

Step 5: the levels. We follow the order of Section 41,

$$\text{(L1)} \to \text{(L2)} \to \text{(L3), (L4) with } W=0 \to \omega_0,\ W_2 \to \text{(L3), (L4) with } W_2 \to \text{(L5)},$$

and check at each level the hypotheses of Step 3 for all related domains.

- (L1): the local part $\mathbf{1}_X(D^TD + aQ'^TQ')\mathbf{1}_X$ is local, and $\mathfrak{N}=0$. Step 3 gives $G'^X(s)$ and its derivative versions.
- (L2): $Q'G'^X(s)Q'^T$ and $M'^X(s) = Q'G'^X(s)G'^X(s)Q'^T$ are products (Step 4); with $\mathfrak{A}_{\rm loc}=0$, Step 3 gives $\mathbf{\Gamma}'^X(s)$ and $(M'^X)^{-1}(s)$, and Step 4 the products H'^X, $P^X_{\mathcal{V}}$, R_X, S_X, DS_X and Π_X, the identities in R_X and Π_X being split box by box.
- (L3): the local part, including $\mathfrak{A}^{\rm ext}_X$ and $\kappa_m m_X$, is local on related domains. The middle factors of $\mathfrak{N}^X$ are products of the families of (L1), (L2) with the local operators D, D^T_X and $\mathfrak{F}$ (Step 4). Step 3 gives $\mathbf{G}^X(s)$ and its derivative versions.
- (L4): $\mathbf{Q}_X\mathbf{G}^X(s)\mathbf{Q}^T_X$ is a product; Step 3 gives $\mathbf{\Gamma}^X(s)$, and Step 4 gives $\mathbf{H}^X(s)$, $\mathbf{C}^X(s)$, $\omega^X_0(s)$ and $W^X_2(s) = d^2\mathcal{N}(0)|_X - \omega^X_0(s)\cdot d^2C^{\rm av}_k(0)$, with $d^2\mathcal{N}(0)|_X$ split box by box.
- The repetitions of (L3), (L4), with $W^X_2(s)$ in the non-local part: as above, they give $\mathbf{\Gamma}^X[W^X_2](s)$, $\Delta^{0,X}(s) = \mathbf{\Gamma}^X[W^X_2](s) - a$, with $-a$ split box by box, and, by composition with $\hat{Z}^1$, $\hat{Z}^2$ (Step 4), $\Delta^{\rm c,X}(s)$ and $\Delta^{\rm c}_Y(s) = \mathbf{1}_{\mathfrak{F}(Y)}\Delta^{\rm c,Y^+}(s)\mathbf{1}_{\mathfrak{F}(Y)}$, the domains Y^+ being related by Step 1.
- (L5), on the domain $\mathbb{T}_\eta[n]$: the local part $r + K^{\rm gf}_{\mathbb{T}_\eta}$ is local. The non-local part is $\Delta^{\rm c}_{\mathbb{T}_\eta}(s)$, and the seam terms use $\Delta^{\rm c}_{\Lambda^v_j}(s)$, built on the related domains $(\Lambda^v_j)^+$ and $(\Lambda^v_{\iota j})^+$. The local propagators $(\Delta_{\Lambda^v_j} + r)^{-1}$ are carried by Step 2. Step 3 gives $R_r(s)$, for every $r \ge 0$, with an index set that does not depend on r.

Step 6: Corollary 41.6. $C^{1/2}_k(s)$ is the integral of $\pi^{-1}r^{-1/2}R_r(s)$ over r, term by term. The family of numbers of $\ell_k(b)$ consists of the terms without steps, $h_j(b)\ell_{\Lambda^v_j}(b)$ with polymer $X(j)$ and monomial $s^{X(j)}$, carried by Step 2 with $Y = \Lambda^v_j$, and of the terms $-\int_0^\infty \mathrm{tr}_{\mathfrak{g}}\, O_\omega(t)(b,b)\,dt$, where $O_\omega(t)$ runs over the terms of $R_t(s)$ with at least one step, carried by Step 5 and the remark at the beginning of the proof. A term vanishes unless $b \in X^{(r)}_\omega$; so the terms with $X^{(r)}_\omega \subset P$ belong to bonds b based in P, and $\iota(\omega)$ is a term for ιb. This proves (b).

Step 7: margin 0. If $X^\sharp_\omega \subset P$, then $X^{(r)}_\omega \subset P$, since every point of $X^{(r)}_\omega$ lies in a box that meets $X^{(r)}_\omega$. The boxes are carried by ι, and a box $\square \subset P$ meets $X^{(r)}_\omega$ if and only if $\iota\square$ meets $\iota(X^{(r)}_\omega) = X^{(r)}_{\iota(\omega)}$. A box of $\mathbb{T}_\eta[n']$ that meets $X^{(r)}_{\iota(\omega)}$ contains a point ιp, $p \in X^{(r)}_\omega$, and it is the image of the box of $\mathbb{T}_\eta[n]$ that contains p, which lies in $X^\sharp_\omega \subset P$. So $X^\sharp_{\iota(\omega)} = \iota(X^\sharp_\omega)$, and the monomials $s^{X^\sharp_\omega \setminus X_\omega}$ are carried as well. With the inverse transfer this gives (c).

The case $k=0$. By Proposition 52.2(c) there is one walk, with $\mathfrak{A}^X = \mathbf{1}_X(\Delta_0 + r)\mathbf{1}_X$ on $\ell^2(\mathfrak{F}(X))$, $\mathfrak{N}=0$, no mass term, and the densities $\ell_{\Lambda^v_j}(b)$ of the operators $\mathbf{1}_\Lambda\Delta_0\mathbf{1}_\Lambda$, $\Lambda = \Lambda^v_j$. The operator Δ_0 is local (Lemma 41.11(c)), so Step 2 holds for $\mathbf{1}_Y(\Delta_0 + t)\mathbf{1}_Y$, and Steps 3, 6 and 7 apply without change. □

Proposition 41.14 (One-loop locality)**.** *Let $k \ge 0$, use the units of step k, and let c_a be as in Propositions 41.7 and 47.8(c). At $k=0$ the pieces $\mathfrak{e}(X)$ are those of Proposition 52.2(c).*

(a) (Torus independence.) *Let n, n' be admissible, let $\tilde{X}$ be a polymer of scale k of the infinite lattice (Section 58.3) with* $\operatorname{diam}\tilde{X} \le \frac{1}{2}\min(n,n') - 8$, *let $a \in M\mathbb{Z}^4$, and let ι be the transfer of the lift $\tilde{X}$, from $X = \pi_n(\tilde{X})$ onto $X' = \pi_{n'}(\tilde{X} + a)$. Then X and X' are polymers of scale k that are not wrapping, and*

$$\mathfrak{e}[n'](X', U') = \mathfrak{e}[n](X, U)$$

for all $U \in \mathcal{U}(X; c_a, c_a, c_a)$ on $\mathbb{T}_\eta[n]$ and $U' \in \mathcal{U}(X'; c_a, c_a, c_a)$ on $\mathbb{T}_\eta[n']$ that agree on X. Every polymer X of scale k of $\mathbb{T}_\eta[n]$ that is not wrapping is $\pi_n(\tilde{X})$ for a polymer $\tilde{X}$ of the infinite lattice with $\operatorname{diam}\tilde{X} = \operatorname{diam} X$, a lift of X.

(b) (Grid covariance.) *For every admissible n, every polymer X of scale k of $\mathbb{T}_\eta[n]$, wrapping or not, every $a \in M\mathbb{Z}^4$ and every $U \in \mathcal{U}(X; c_a, c_a, c_a)$,*

$$\mathfrak{e}(X + a, \tau_a U) = \mathfrak{e}(X, U), \qquad (\tau_a U)(b) = U(b - a).$$

The same holds at every reference U in the setting of Section 40.

(c) (Jacobians.) *Let c be a positively oriented bond of $\mathbb{T}'[n]$, $\tilde{c}$ a bond of $L\mathbb{Z}^4$ with $\pi_n(\tilde{c}) = c$, $a \in M\mathbb{Z}^4$ and $c' = \pi_{n'}(\tilde{c} + a)$, and let ι be the transfer of the lift of P_c that contains $\tilde{c}$ (Lemma 41.11(d)). Then $\log \mathfrak{j}_{c'}[n'](U') = \log \mathfrak{j}_c[n](U)$ for all $U \in \mathcal{U}(P_c; c_a, c_a, c_a)$ and $U' \in \mathcal{U}(P_{c'}; c_a, c_a, c_a)$ that agree on P_c, and at all references in the setting of Section 40 that agree on P_c. Moreover $\log \mathfrak{j}_{c+a}(\tau_a U) = \log \mathfrak{j}_c(U)$.*

(d) (Infinite-volume pieces.) *Let $\tilde{X}$ be a polymer of scale k of the infinite lattice. Let $\mathcal{U}_{\mathrm{fl}}(\tilde{X})$ be the set of the $\mathfrak{g}^c$-valued fields B on the bonds of $\tilde{X}^{(3)}$ with $|B| < c_a$ and $|\nabla B|_{\tilde{X}^{(3)}} + [\nabla B]_{\alpha, \tilde{X}^{(3)}} < c_a$, where ∇ and the Hölder quotient are those of Definition 38.3 with the real part 1. For an admissible n with $\operatorname{diam}\tilde{X} \le \frac{1}{2}n - 8$ let $V_B[n]$ be the configuration on $\mathbb{T}_\eta[n]$ equal to $e^{\mathrm{i}\eta B}$ on the images of the bonds of $\tilde{X}^{(3)}$ and to 1 on the other bonds. Then $V_B[n] \in \mathcal{U}(\pi_n \tilde{X}; c_a, c_a, c_a)$, and*

$$E_k^\infty(\tilde{X}, B) := \mathfrak{e}[n]\big(\pi_n(\tilde{X}), V_B[n]\big), \qquad B \in \mathcal{U}_{\mathrm{fl}}(\tilde{X}),$$

does not depend on n. The function $E_k^\infty(\tilde{X}, \cdot)$ is holomorphic on $\mathcal{U}_{\mathrm{fl}}(\tilde{X})$, depends only on B on the bonds of $\tilde{X}$, vanishes at $B = 0$, is real for real B, and is invariant under small real gauge transformations in the sense of Section 58.3. For $a \in M\mathbb{Z}^4$,

$$E_k^\infty(\tilde{X} + a, \tau_a B) = E_k^\infty(\tilde{X}, B), \qquad |E_k^\infty(\tilde{X}, B)| \le C_{\mathfrak{e}} e^{-(\kappa_2 - 1)|\tilde{X}|},$$

with $(\tau_a B)(b) = B(b - a)$ and $C_{\mathfrak{e}}$ of Lemma 53.4(b). For every admissible n, every polymer X of scale k of $\mathbb{T}_\eta[n]$ that is not wrapping, every lift $\tilde{X}$ of X, and every $U \in \mathcal{U}(X; c_a, c_a, c_a)$ that agrees on X with $V_B[n]$ for some $B \in \mathcal{U}_{\mathrm{fl}}(\tilde{X})$,

$$\mathfrak{e}[n](X, U) = E_k^\infty(\tilde{X}, B).$$

In the same way, for a positively oriented bond $\tilde{c}$ of $L\mathbb{Z}^4$, with the lift $\tilde{P}_{\tilde{c}}$ of P_c that contains it, $J_k^\infty(\tilde{c}, B) := \log \mathfrak{j}_{\pi_n \tilde{c}}[n](V_B[n])$, for the fields B on the bonds of $\tilde{P}_{\tilde{c}}^{(3)}$ with the same bounds, does not depend on n, has the same properties with $|J_k^\infty| \le C_{\mathrm{jac}}$ (Lemma 53.4(b)), and $J_k^\infty(\tilde{c} + a, \tau_a B) = J_k^\infty(\tilde{c}, B)$.

Proof. (a) Since $\operatorname{diam}\tilde{X} < \frac{1}{2}n$, π_n is injective on $\tilde{X}$ and preserves the ℓ^∞-distances there. It maps the grid cubes of $\tilde{X}$ onto distinct grid cubes of $\mathbb{T}_\eta[n]$, since $M \mid n$, and $*$-adjacent cubes onto $*$-adjacent cubes. So X is a polymer with $|X| = |\tilde{X}|$ and $\operatorname{diam} X = \operatorname{diam}\tilde{X} \le \frac{1}{2}n - 8$, i.e. it does not wrap; the same holds for X'. Conversely, a polymer X that is not wrapping has a lift of $X^{(3)}$ on which the projection preserves the distances (Section 50.2, Lifts and linear fields). The preimage $\tilde{X}$ of X in it is a union of grid cubes of $\eta\mathbb{Z}^4$, $*$-connected, with $\operatorname{diam}\tilde{X} = \operatorname{diam} X \le \frac{1}{2}n - 8$. Details

Let $\tilde{U}$ and $\tilde{U}'$ be the cutoff extensions of Remark 38.7 of U and U', for some representations as in Definition 38.3. By the proof of Proposition 41.7(b) (Proposition 52.2(c) for $k = 0$), they are references at which the families of Section 41 are defined, and $\mathfrak{e}[n](X, U) = \mathfrak{e}[n](X, \tilde{U})$,

$\mathfrak{e}[n'](X', U') = \mathfrak{e}[n'](X', \tilde{U}')$. The extensions equal U and U' on the bonds of X and X', so they agree on X. By definition,

$$\mathfrak{e}[n](X, \tilde{U}) = -\frac{1}{2} \sum_{b \in \mathfrak{F}} \sum_{\omega \in \Omega_b \colon X^\sharp_\omega = X} (O_\omega(\tilde{U}) - O_\omega(1)).$$

A term with $X^\sharp_\omega = X$ vanishes unless $b \in X^{(r)}_\omega \subset X$, so only the bonds b based in X contribute. Apply Lemma 41.13(b),(c) with $P = X$, at the references $(\tilde{U}, \tilde{U}')$ and $(1, 1)$; the configurations 1 agree on every set. The map $b \mapsto \iota b$ is a bijection of the bonds of $\mathfrak{F}$ based in X onto those of $\mathfrak{F}'$ based in X' (Lemma 41.11(a)), and $(b, \omega) \mapsto (\iota b, \iota(\omega))$ is a bijection of the pairs with $X^\sharp_\omega = X$ onto the pairs with $X^\sharp_{\omega'} = X'$, with equal values at both references. The sums converge absolutely (proof of Proposition 41.7(b)), so they are equal.

(b) The set $\mathcal{U}(X; c_a, c_a, c_a)$ is mapped onto $\mathcal{U}(X + a; c_a, c_a, c_a)$ by τ_a (Section 50, after Definition 50.1), and the translate of a cutoff extension of U is a cutoff extension of $\tau_a U$. Apply Lemma 41.13(b),(c) to the grid translation τ_a, for which every term is carried and $X^\sharp_{\tau_a(\omega)} = X^\sharp_\omega + a$, and argue as in (a). At a reference U of the setting the lemma applies directly, at the references U and $\tau_a U$.

(c) By Proposition 47.8(c), $\log \mathfrak{j}_c(U)$ is the value of the formula of Lemma 41.11(d) at the configuration on the bonds of P_c. The lift of P_c that contains $\tilde{c}$ has diameter less than $2L + 1 < \frac{1}{2} \min(n, n')$, so its transfer is a grid transfer, and the consequence of Lemma 41.11 gives the first claim. The second is the case of the grid translation τ_a.

(d) Since $\operatorname{diam} \tilde{X}^{(3)} < \frac{1}{2} n$, the projection maps $\tilde{X}^{(3)}$ isometrically onto $(\pi_n \tilde{X})^{(3)}$, and the differences and Hölder quotients of Definition 38.3 on $\tilde{X}^{(3)}$ are those on $(\pi_n \tilde{X})^{(3)}$. With the real part 1, whose curvature and current vanish, and the field B, $V_B[n] \in \mathcal{U}(\pi_n \tilde{X}; c_a, c_a, c_a)$. For two admissible sides n, n', the configurations $V_B[n]$ and $V_B[n']$ agree on $\pi_n \tilde{X}$ under the transfer of $\tilde{X}$ with $a = 0$, so (a) shows that $E^\infty_k(\tilde{X}, B)$ does not depend on n. The map $B \mapsto V_B[n]$ is holomorphic, and $\mathfrak{e}(X, \cdot)$ is holomorphic on an open set that contains $\{U|_X : U \in \mathcal{U}(X; c_a, c_a, c_a)\}$, so $E^\infty_k(\tilde{X}, \cdot)$ is holomorphic. It depends only on B on the bonds of $\tilde{X}$, since $\mathfrak{e}(X, \cdot)$ depends only on U on the bonds of X; it vanishes at $B = 0$ since $\mathfrak{e}(X, 1) = 0$. It is real for real B by the argument for reality in the proof of Lemma 53.4(a), applied to $\mathfrak{e}(X, \cdot)$, whose terms are real at real references (Corollary 41.6). Let $B, B' \in \mathcal{U}_{\mathrm{fl}}(\tilde{X})$ with $e^{\mathrm{i}\eta B'(b)} = e^{\mathrm{i}\lambda(b_-)} e^{\mathrm{i}\eta B(b)} e^{-\mathrm{i}\lambda(b_+)}$ on the bonds of $\tilde{X}$, for a small real function λ. Extend $\lambda \circ \pi_n^{-1}$ from $\pi_n \tilde{X}$ to a real function on $\mathbb{T}_\eta[n]$ and put $u = e^{\mathrm{i}\lambda}$. Then $V_B[n]^u \in \mathcal{U}(\pi_n \tilde{X}; c_a, c_a, c_a)$, since the graded domains are invariant under real gauge transformations, and $V_B[n]^u$ agrees with $V_{B'}[n]$ on the bonds of $\pi_n \tilde{X}$. Since $\mathfrak{e}$ depends only on these bonds and is invariant under real gauge transformations, $E^\infty_k(\tilde{X}, B') = E^\infty_k(\tilde{X}, B)$. The grid covariance follows from (b), since $V_{\tau_a B}[n] = \tau_a V_B[n]$ and $\pi_n(\tilde{X} + a) = \pi_n(\tilde{X}) + a$, and the bound from Lemma 53.4(b). If U agrees with $V_B[n]$ on $X = \pi_n \tilde{X}$, then $\mathfrak{e}[n](X, U) = \mathfrak{e}[n](X, V_B[n])$, since $\mathfrak{e}(X, \cdot)$ depends only on the bonds of X. For the Jacobians the same arguments apply, with (c) in place of (a) and (b), and with Proposition 47.8(b),(c). □

Remark 41.15 (Use in Part XI). (a) *The one-loop part of the new generation.* Let Y' be a polymer of scale $k + 1$ that is not wrapping, and let $E'_{\mathrm{ol}}(Y', \cdot)$ be the sum of the pieces $\mathfrak{e}(X)$, $\bar{X} = Y'$, and of the terms $\log \mathfrak{j}_c$, $Y'_c = Y'$ (Section 58). These X and P_c lie in Y'. A configuration V_B of the flat domain of Y' of scale $k + 1$ (Section 58.3), read in the units of step k, restricts on $X^{(3)}$ to a field in $\mathcal{U}_{\mathrm{fl}}(\tilde{X})$, by Remark 38.8(i) and (DC1), and likewise on $P_c^{(3)}$. So, by Proposition 41.14(d), $E'_{\mathrm{ol}}(Y', V_B)$ is the sum of the values of E^∞_k and J^∞_k at the lifts. It is the same function on every torus on which Y' does not wrap, with no error term. Moreover E'_{ol} has the grid covariance (G2) of scale $k + 1$ on every torus, by (b) and (c): a translation by $M\mathbb{Z}^4$ in the units of scale $k + 1$ is a translation by

$LM\mathbb{Z}^4 \subset M\mathbb{Z}^4$ in those of step k, and it maps the grids of both scales onto themselves, so $\overline{X+a} = \bar{X} + a$ and $Y'_{c+a} = Y'_c + a$.

(b) *Scope.* The pieces of wrapping polymers are bounded by Proposition 41.7(d) and Lemma 52.1; they are not compared between tori. Covariance of the single pieces under the reflections of $\mathfrak{S}_k^{\mathrm{grid}}$ is not proved: the partition of unity is not assumed symmetric, and the operators $\mathbf{\Gamma}_\pi$, $\mathbf{H}$ of the levels (L3), (L4) are not shown to be covariant under reflections at curved backgrounds (Remark 31.8). The sums over the polymers are invariant under $\mathfrak{S}_{k+1}$ (Theorem 48.9).

42 Pointwise norms: the difficulty and two routes

The covariance, the coarse Hessian and the determinant enter the step through L^2 bounds, and for them Section 41 is complete. The background field is different. To localize the interaction and the composed activities, the weakened background map $A_*(s)$ must be built with weakened operators $\mathbf{H}(s)$, $\mathbf{C}(s)$, analytic on $\mathbb{D}_\kappa$. Its construction is a contraction in sup norms with derivatives (Proposition 26.3), or in $C^{1,\alpha}$. So, for every $s \in \mathbb{D}_\kappa$, it needs bounds of the form

$$|\mathbf{C}(s)f|_{1,\alpha;y} \le C \sum_{y'} e^{-\gamma|y-y'|} |f|_{B(y')}, \tag{42.1}$$

together with the same bounds for $\mathbf{H}(s)$ and $\mathbf{C}(s)D^T$.

Where the difficulty is. The simplest way to obtain (42.1) is to make the step operators $\mathcal{R}(s)$ bounded from sup norms to sup norms. Then every walk term is bounded term by term, as in [15, (97)–(99)]. At level (L3) the interior estimates alone do not give this. The argument has three steps.

- *The seam term must be weakened on both sides.* With $X' = \Lambda_j^v$ and G_j the local propagator on X', the step $\mathcal{R}_j$ of the walk contains the seam term

$$h_j \mathbf{1}_{X'} (\mathfrak{N}(s) - \mathfrak{N}^{X'}(s)) \mathbf{1}_{X'} G_j,$$

  the difference of the non-local parts on the torus and on X' (Lemma 41.2). It is small only through the cancellation between its two parts. Weakening only one of them destroys the cancellation: for $|s_\square| = e^\kappa$, the terms without steps are multiplied by $e^{\kappa \mathsf{N}_\mathrm{b}}$. So both parts are weakened, compatibly, and the terms of $\mathfrak{N}^{X'}(s)$ act on $u = G_j f$ throughout Λ_j^v, up to its boundary.

- *Near the boundary only L^2 is known.* The interior estimate (Proposition 36.5) bounds u pointwise at distance at least 4 from $\partial\Lambda_j^v$; near $\partial\Lambda_j^v$ only its L^2 norm is controlled.

- *L^2 inputs do not give pointwise outputs in four dimensions.* The non-local part $\mathfrak{N}^{X'}(s)$ is built from the weakened scalar walks of the levels (L1) and (L2), whose steps contain the commutators $[\mathfrak{A}_{\mathrm{loc}}, h_{j'}]$, of first order (Section 41, Lemma 44.10). Its terms that start near the boundary are products whose factors need pointwise information on their inputs. In $d = 4$, interior regularity with L^2 sources does not even control values, since two derivatives in L^2 do not embed into L^∞ in four dimensions; it controls gradients still less.

So, in the boundary layer of Λ_j^v, the terms of $\mathfrak{N}^{X'}(s)$ act on a function of which only the L^2 norm is controlled, and the interior estimates give no bound of $\mathcal{R}(s)$ from sup norms to sup norms. Section 44 overcomes this with two observations, stated at its beginning: the outputs of a walk term are interior, and interior factors raise the integrability.

Note, however, that absolute convergence does not force all factors to be sup-to-sup. It suffices that $\mathcal{R}(s)$ is a contraction in some Banach space adapted to the geometry. This leaves two routes.

Route 1: boundary regularity. This route is not taken in this note (see the Summary below); it is described for comparison. Prove regularity up to the boundary for the Dirichlet operators on the boxes of the construction:

> *(BR)* For a box X, the operators G'_X, $G'_X D_X^T$, $\mathbf{G}^X$, $\mathbf{G}^X D_X^T$, $\mathbf{H}^X$ and $\mathbf{C}^X$ satisfy the bounds of Theorem 35.2 and Proposition 36.5 at all blocks of X, not only at interior ones.

Then all factors are sup-to-sup, and the construction of Section 41 carries over to (42.1). Bałaban proves estimates of this kind for his Dirichlet propagators [2, 3, 7].

For the scalar Dirichlet Laplacian on a lattice box, odd reflection across the plane of the exterior layer gives a problem on the full lattice. At edges and corners, the reflections across the faces commute. The absorption argument of Section 36 can then be repeated on half-cubes and quarter-cubes at the boundary:

- the small perturbations become sources for the reflected equation. Their reflected coefficients may jump across the reflection plane, which is harmless only because u vanishes on the exterior layer;
- the averaging terms are block-local and the boxes are unions of blocks, so they remain L^2-controlled sources;
- the mass term is of order zero.

The gauge case is not covered by this sketch. On $\ell_1^2(X)$ the bonds crossing ∂X are kept, and since the exterior divergence is part of the operator, all components satisfy Dirichlet conditions for $D^T D + D D^T$ (Section 19). The reflection then has to respect the lattice structure of the crossing bonds. L^2 regularity of second order on boxes is classical in the continuum, but the pointwise statement on the lattice, with these operators, has not been carried out.

Route 2: a mixed norm, without boundary estimates. At the top level $X = \mathbb{T}_\eta$, every pointwise output is needed only on the supports K_j. Every Dirichlet cut involved lies at distance at least $\frac{1}{4}(\rho - 1)M$ from these supports, by (41.1). Two observations suggest that a norm mixing L^2 and pointwise control suffices:

- pieces of a kernel that jump a distance of at least 2 need only L^2 input, by interior regularity on a cube where the source vanishes;
- a chain of walk factors from a cut to K_j passes through many interior steps. Each step of the form commutator times propagator gains one derivative, and each step of order zero times propagator gains two. In $d = 4$ one interior propagator takes block-L^2 data to values and gradients in L^4. A second one takes L^4 data, which lie in $L^{8/3}$ on unit cubes, to C^β values with gradients in L^8. A third one takes L^8 data to $C^{1,\beta}$, with $\beta = \min(\alpha, \frac{1}{2})$. So a bounded number of interior steps take block-L^2 data to $C^{1,\beta}$.

The norm would be block-L^2 within a few units of the cuts and pointwise elsewhere, with exponential weights. Its analytic input is a set of interior estimates with L^p data, variants of Proposition 36.5, together with the off-diagonal smoothing just described. These estimates are the subject of Section 43 (Corollary 43.8). No boundary regularity is needed. The step bounds are the subject of Section 44, in block L^p and Hölder norms rather than a mixed norm. At the endpoint $p = \frac{d}{2}$, Section 44 uses one more tier, block L^4 (Lemma 44.6), and it bounds

the non-local parts of the steps at level (L3) in Proposition 44.11. This gives the bounds (42.1) for $\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$ and $\mathbf{C}(s)D^T$ on the torus, with $W = 0$ at level (L3) and $\alpha \le \frac{1}{2}$ (Theorems 44.13 and 44.15, Remark 44.12). A trace inequality on rectangles (Lemma 44.7) handles the Dirichlet cuts. $\mathbf{C}(s)$ is represented through the torus identity $\mathbf{C} = \mathbf{G}\Pi^T - \mathbf{HQG}$.

Summary. Sections 40 and 41 give volume-type localization of everything that enters the step through L^2 bounds: the fluctuation covariance and its square root, the coarse Hessian, and the determinant. The background map, and with it the interaction and the composed activities, need in addition pointwise control of the weakened operators, (42.1). Route 2, which stays closer to the interior methods of this note, provides it (Sections 43 and 44). Route 1 is Bałaban's; it is not carried out here.

43 Interior estimates with L^p data

Route 2 of Section 42 needs interior estimates for the local propagators whose sources are controlled only in L^p norms. Sections 6 and 36 assume bounded sources. This section supplies the missing estimates:

(I1) sources in L^p, $p > d$, give $C^{1,\beta}$ solutions, $\beta \le 1 - d/p$ (Theorem 43.2(a), Propositions 43.4(a) and 43.7(a));

(I2) sources $F + \partial^* g$ with $F \in L^p$, $\frac{d}{2} < p < d$, and $g \in L^{p^*}$ give bounded C^β solutions with gradients in L^{p^*}, $\beta \le 2 - d/p$ (Theorem 43.2(b),(c), Propositions 43.4(b) and 43.7(b)). In $d = 4$ the case $p = 2$ of L^2 sources is the endpoint $p = \frac{d}{2}$. It is excluded for the values. The flat gradient estimate, Theorem 43.2(c), holds at $p = 2$; Propositions 43.4(b) and 43.7(b) are not stated there. For the scalar local propagators Lemma 44.6 derives the covariant version of this gradient bound from it, without absorption. The case used in Section 44 for the values is $p = \frac{8}{3}$, with $p^* = 8 = 2d$ and $\beta \le \frac{1}{2}$;

(I3) the Calderón–Zygmund bound behind the gradient part of (I2) (Lemma 43.1(d));

(I4) the $C^{2,\beta}$ Schauder estimate for scalar equations without averaging term (Proposition 43.5); it and Remark 43.6 are not used on the main line, which uses the torus identity (44.1) instead.

The proofs follow Sections 6 and 36. The only new analytic inputs are the Hardy–Littlewood–Sobolev and the Calderón–Zygmund inequalities on $\eta\mathbb{Z}^d$, both uniform in η by scaling. For the scalar local propagators themselves, (I4) fails because of the averaging term (Remark 43.6). The consequences for the weakened operator $\mathbf{C}(s)$ are discussed at the end of the section. (I2) with L^2 data fails at the endpoint $p = 2 = \frac{d}{2}$. Section 44 handles it with one more tier, L^4, and one more interior factor (Lemma 44.6).

Notation. For a set Q of lattice points and $1 \le p < \infty$ let $\|h\|_{p,Q} = (\eta^d \sum_{x \in Q} |h(x)|^p)^{1/p}$, so that $\|h\|_{2,Q} = \|h\|_Q$; for gauge fields and 2-forms the sum also runs over the components. We write p' for the dual exponent and, for $1 < p < d$, $p^* = dp/(d-p)$, so that $1/p - 1/p^* = 1/d$. For $0 < \beta < 1$, $[h]_{\beta,Q}$ is the Hölder quotient with exponent β; thus $[h]_{\alpha,Q} = [h]_{\alpha,Q}$. Where Hölder quotients of the coefficients enter, namely in part (a) of Propositions 43.4 and 43.7, in Proposition 43.5 and in Corollary 43.8(a), we assume $\beta \le \alpha$, the exponent of hypothesis (B). On a cube Q of side at most 6, $[h]_{\beta,Q} \le C[h]_{\alpha,Q}$. Hence the coefficients of Lemmas 12.2 and 12.3, and those of $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, have β-Hölder quotients bounded by $C\varepsilon$ on such cubes. Constants depend on d, $\dim \mathfrak{g}$, α, β and the exponents p, q, but not on η, k, n.

43.1 Two inequalities on the fine lattice

Lemma 43.1 (Kernel bounds)**.** *Let $1 < p < \infty$, $x \in \eta\mathbb{Z}^d$, and let h be a field on $\eta\mathbb{Z}^d$.*

(a) If $0 \le s < d/p'$ and $R \ge \eta$: $\sum_{|x'-x|\le R} \eta^d (|x-x'|+\eta)^{-s} |h(x')| \le C R^{d/p'-s} \|h\|_p$.

(b) If $s > d/p'$ and $r \ge \sqrt{d}\,\eta$: $\sum_{|x'-x|>r} \eta^d |x-x'|^{-s} |h(x')| \le C r^{d/p'-s} \|h\|_p$.

(c) (Hardy–Littlewood–Sobolev.) *Let $0 < s < d$ and $1 < p < q < \infty$ with $1/q = 1/p - (d-s)/d$, and $k_s(x) = (|x|+\eta)^{-s}$. Then $\|k_s \star h\|_q \le C\|h\|_p$.*

(d) (Calderón–Zygmund.) *Let $K = \partial_\nu \partial_\mu^* g_\eta$. Then $\|K \star h\|_p \le C\|h\|_p$ for finitely supported h.*

Here $\|\cdot\|_p$ is the norm on all of $\eta\mathbb{Z}^d$, and C depends only on d, s, p, q.

Proof. (a) and (b) follow from Hölder's inequality and Lemma 6.2(a), respectively (c), with the exponent sp', which is smaller, respectively larger, than d. Details

(c) For $x' \in \eta\mathbb{Z}^d$ let $W(x') = x' + [-\eta/2, \eta/2)^d$ and $H = \sum_{x'} h(x') \mathbf{1}_{W(x')}$, so that $\|H\|_{L^p(\mathbb{R}^d)} = \|h\|_p$. For $w \in W(x)$ and $z \in W(x')$ we have $|w-z| \le |x-x'| + \sqrt{d}\eta \le (1+\sqrt{d})(|x-x'|+\eta)$, so $k_s(x-x') \le C|w-z|^{-s}$. With the Riesz potential $I_s F(w) = \int_{\mathbb{R}^d} |w-z|^{-s} F(z)\,dz$ we get, for every $w \in W(x)$,

$$(k_s \star |h|)(x) = \sum_{x'} \int_{W(x')} k_s(x-x') |H(z)|\,dz \le C\,(I_s|H|)(w).$$

Averaging the q-th power over $w \in W(x)$ and summing over x gives $\|k_s \star h\|_q \le C \| I_s|H| \|_{L^q(\mathbb{R}^d)}$. The Hardy–Littlewood–Sobolev theorem [12, Ch. V, §1.2, Theorem 1] bounds the right-hand side by $C\|H\|_{L^p}$.

(d) By the proof of Proposition 6.1, $\eta^d K(x-x') = K_1(x/\eta - x'/\eta)$ with $K_1 = \partial_\nu \partial_\mu^* g_1$ on $\mathbb{Z}^d$, and both sides of the inequality scale by $\eta^{d/p}$. So we may take $\eta = 1$. On $\ell^2(\mathbb{Z}^d)$, convolution with K_1 is the Fourier multiplier

$$\frac{(e^{\mathrm{i}\xi_\nu} - 1)(e^{-\mathrm{i}\xi_\mu} - 1)}{\sum_\lambda 4\sin^2(\xi_\lambda/2)}, \qquad \xi \in [-\pi, \pi]^d,$$

of modulus at most 1, since $|e^{\mathrm{i}t} - 1| = 2|\sin(t/2)|$ and $4|\sin(\xi_\nu/2)\sin(\xi_\mu/2)| \le 2\sin^2(\xi_\nu/2) + 2\sin^2(\xi_\mu/2)$. So it is bounded on ℓ^2 with norm at most 1. By Proposition 6.1, $|K_1(z)| \le C(1+|z|)^{-d}$, and the first differences of K_1 are bounded by $C(1+|z|)^{-d-1}$. Let $y \ne y'$ and $|x-y| \ge 2|y-y'|$. A monotone lattice path from y to y' has at most $\sqrt{d}|y-y'|$ steps, and its points w satisfy $|x-w| \ge \frac{1}{2}|x-y|$. Hence $|K_1(x-y) - K_1(x-y')| \le C|y-y'||x-y|^{-d-1}$. Lemma 6.2(c), together with a bounded number of points with $|x-y| < \sqrt{d}$, gives Hörmander's condition

$$\sup_{y \ne y'} \sum_{|x-y| \ge 2|y-y'|} |K_1(x-y) - K_1(x-y')| \le C.$$

The same holds for $K_1(-\cdot)$. With counting measure and Euclidean distance, $\mathbb{Z}^d$ is a space of homogeneous type. The Calderón–Zygmund theorem on spaces of homogeneous type [13, Ch. I] gives weak type $(1,1)$. Marcinkiewicz interpolation gives the bound for $1 < p \le 2$, and duality, applied to the kernel $K_1(-\cdot)$, for $2 \le p < \infty$. □

43.2 The flat estimates

Theorem 43.2 (Interior regularity with L^p data)**.** *Let $\eta \le 1$ be any lattice spacing, with Λ_r as in Theorem 6.3, and let $u, F : \Lambda_3 \to \mathbb{R}$ and $g : \Lambda_3 \to \mathbb{R}^d$ satisfy $-\Delta u = F + \sum_\mu \partial_\mu^* g_\mu$ at every point of Λ_2.*

(a) (I1) *If $p > d$ and $0 < \beta \le 1 - d/p$, then $|u|_{\Lambda_1} + |\partial u|_{\Lambda_1} + [\partial u]_{\beta,\Lambda_1} \le C(\|F\|_{p,\Lambda_2} + |g|_{\Lambda_3} + [g]_{\beta,\Lambda_3} + \|u\|_{\Lambda_2})$.*

(b) (I2), values. *If $\frac{d}{2} < p < d$, $q > d$ and $0 < \beta \le \min(2 - d/p, 1 - d/q)$, then $|u|_{\Lambda_1} + [u]_{\beta,\Lambda_1} \le C(\|F\|_{p,\Lambda_2} + \|g\|_{q,\Lambda_3} + \|u\|_{\Lambda_2})$.*

(c) (I2), gradients. *If $1 < p < d$, then $\|\partial u\|_{p^*,\Lambda_1} \le C(\|F\|_{p,\Lambda_2} + \|g\|_{p^*,\Lambda_3} + \|u\|_{\Lambda_2})$.*

In $d = 4$, (c) applies with $p = 2$ and gives gradients in L^4 for sources $F + \partial^* g$ with $F \in L^2$ and $g \in L^4$, while (b) needs $p > 2$, so it gives no bound on the values for L^2 sources. With $p = \frac{8}{3}$ and $q = p^* = 8$, (b) and (c) give, for sources $F + \partial^* g$ with $F \in L^{8/3}$ and $g \in L^8$, solutions in $C^{1/2}$ with gradients in L^8.

Proof. We use the proof of Theorem 6.3. Step 0 is unchanged, since $\|h\|_{p,\Lambda_1} \le C|h|_{\Lambda_1}$. So assume $\eta \le \frac{1}{16}$. The representation (6.2) holds: Details

$$u = \mathcal{G}\Phi_1 + \mathcal{G}\Phi_2 + \sum_\mu (\partial_\mu^* g_\eta) \star (\chi g_\mu) + \sum_\mu (\partial_\mu^* g_\eta) \star m_\mu \qquad \text{on } \Lambda_{1+2\eta},$$

with $\Phi_1 = \chi F - \sum_\mu (\partial_\mu^* \chi) S_\mu^{-1} g_\mu$. Step 2 does not involve the sources. The terms with Φ_2 and m_μ have two derivatives bounded by $C\|u\|_{\Lambda_2}$ on Λ_1, and they are Lipschitz together with their first derivatives. So they satisfy all three bounds. Moreover $\|\Phi_1\|_{p,\Lambda_2} \le \|F\|_{p,\Lambda_2} + C\|g\|_{p,\Lambda_3}$. Since Λ_3 has bounded volume, $\|g\|_{p,\Lambda_3}$ is bounded by $C|g|_{\Lambda_3}$ in (a), by $C\|g\|_{q,\Lambda_3}$ in (b), where $q > p$, and by $C\|g\|_{p^*,\Lambda_3}$ in (c). Recall $|g_\eta(y)| \le C(|y| + \eta)^{2-d}$, and that k-th differences of g_η are bounded by $C(|y| + \eta)^{2-d-k}$, $k \le 3$ (Proposition 6.1). All points below lie in $\Lambda_{1+2\eta}$, and sums run over the support of Φ_1, respectively of χg, which lies in Λ_2.

(a) The term $\sum_\mu (\partial_\mu^* g_\eta) \star (\chi g_\mu)$ is treated as in Step 5 of the proof of Theorem 6.3, with β in place of α. For $\mathcal{G}\Phi_1$, Lemma 43.1(a) with $s = d-2$ and $s = d-1$ gives $|\mathcal{G}\Phi_1| + |\partial\mathcal{G}\Phi_1| \le C\|\Phi_1\|_p$; here $s < d/p'$ because $p > d$. For the Hölder quotient of $k \star \Phi_1$, $k = \partial_\mu g_\eta$, let $x, x' \in \Lambda_1$ and $\eta \le \delta = |x - x'| \le 1$, and split the sum at $|x - x''| = 4\sqrt{d}\delta$, as in Step 3 of that proof. On the near part, Lemma 43.1(a) with $s = d - 1$ and $R = (4\sqrt{d} + 1)\delta$ gives the bound $C\delta^{d/p'-d+1}\|\Phi_1\|_p = C\delta^{1-d/p}\|\Phi_1\|_p$. On the far part, $|k(x - x'') - k(x' - x'')| \le C\delta|x - x''|^{-d}$, by the path argument of Step 3. Lemma 43.1(b) with $s = d > d/p'$ and $r = 4\sqrt{d}\delta$ gives $C\delta \cdot \delta^{d/p'-d}\|\Phi_1\|_p = C\delta^{1-d/p}\|\Phi_1\|_p$. Since $\delta \le 1$ and $\beta \le 1 - d/p$, this is at most $C\delta^\beta\|\Phi_1\|_p$. Pairs with $\delta > 1$ are covered by the sup bound.

(b) By Lemma 43.1(a), $|\mathcal{G}\Phi_1(x)| \le C\|\Phi_1\|_p$, since $d - 2 < d/p'$ for $p > \frac{d}{2}$, and $|(\partial_\mu^* g_\eta) \star (\chi g_\mu)(x)| \le C\|g\|_q$, since $d - 1 < d/q'$ for $q > d$. For the Hölder quotient of $\mathcal{G}\Phi_1$ split as in (a). The near part is bounded by $C\delta^{d/p'-d+2}\|\Phi_1\|_p = C\delta^{2-d/p}\|\Phi_1\|_p$. On the far part the differences of g_η are bounded by $C\delta|x - x''|^{1-d}$, and Lemma 43.1(b) with $s = d - 1 > d/p'$ (since $p < d$) gives $C\delta \cdot \delta^{d/p'-d+1}\|\Phi_1\|_p = C\delta^{2-d/p}\|\Phi_1\|_p$. The Hölder quotient of $(\partial_\mu^* g_\eta) \star (\chi g_\mu)$ is bounded as in (a), with $(q, \chi g_\mu)$ in place of (p, Φ_1). This gives $C\delta^{1-d/q}\|g\|_q$.

(c) We have $\partial\mathcal{G}\Phi_1 = (\partial g_\eta) \star \Phi_1$ with $|\partial g_\eta| \le C k_{d-1}$. Lemma 43.1(c) with $s = d-1$, for which $1/q = 1/p - 1/d = 1/p^*$, gives $\|\partial\mathcal{G}\Phi_1\|_{p^*} \le C\|\Phi_1\|_p$. Next, $\partial_\nu((\partial_\mu^* g_\eta) \star (\chi g_\mu)) = (\partial_\nu \partial_\mu^* g_\eta) \star (\chi g_\mu)$, and Lemma 43.1(d) bounds its L^{p^*} norm by $C\|g\|_{p^*,\Lambda_2}$. □

Lemma 43.3 (Rescaled estimates)**.** *Let $\eta \le s \le 1$, $x \in \eta\mathbb{Z}^d$, $\Lambda_r = \Lambda_r(x)$, and $-\Delta u = F + \sum_\mu \partial_\mu^* g_\mu$ on Λ_{2s}. Under the conditions of Theorem 43.2(a), (b), (c), respectively,*

$$\begin{aligned}
|u|_{\Lambda_s} + s|\partial u|_{\Lambda_s} + s^{1+\beta}[\partial u]_{\beta,\Lambda_s} &\le C(s^{2-d/p}\|F\|_{p,\Lambda_{2s}} + s|g|_{\Lambda_{3s}} + s^{1+\beta}[g]_{\beta,\Lambda_{3s}} + s^{-d/2}\|u\|_{\Lambda_{2s}}),\\
|u|_{\Lambda_s} + s^\beta[u]_{\beta,\Lambda_s} &\le C(s^{2-d/p}\|F\|_{p,\Lambda_{2s}} + s^{1-d/q}\|g\|_{q,\Lambda_{3s}} + s^{-d/2}\|u\|_{\Lambda_{2s}}),\\
\|\partial u\|_{p^*,\Lambda_s} &\le C(\|F\|_{p,\Lambda_{2s}} + \|g\|_{p^*,\Lambda_{3s}} + s^{-1-d/2+d/p^*}\|u\|_{\Lambda_{2s}}).
\end{aligned}$$

Proof. As for Lemma 36.2. On the lattice $(\eta/s)\mathbb{Z}^d$ we have $\|h_s\|_{p,\Lambda_r(0)} = s^{-d/p}\|h\|_{p,\Lambda_{rs}}$ for $h_s(x') = h(x+sx')$, and $\partial u_s = s(\partial u)_s$. For the third bound, $\|\partial u_s\|_{p^*,\Lambda_1(0)} = s^{1-d/p^*}\|\partial u\|_{p^*,\Lambda_s}$, while $\|s^2 F_s\|_{p,\Lambda_2(0)} = s^{2-d/p}\|F\|_{p,\Lambda_{2s}}$ and $\|sg_s\|_{p^*,\Lambda_3(0)} = s^{1-d/p^*}\|g\|_{p^*,\Lambda_{3s}}$. Since $d/p - d/p^* = 1$, the two data terms carry the same power s^{1-d/p^*} as the left-hand side. □ Details

43.3 Scalar equations

We keep the conventions of Section 36: the local gauge on $\Lambda_4(z)$, sources in local form $\mathcal{S} = F + \sum_\mu \partial_\mu^* g_\mu$, and bounds on Λ_r that may involve values within 2η of Λ_r. For a given representation of $\mathcal{S}$ put

$$\mathfrak{n}_r^{p,\beta}(\mathcal{S}) = \|F\|_{p,\Lambda_r} + |g|_{\Lambda_r} + [g]_{\beta,\Lambda_r}, \qquad \mathfrak{m}_r^{p,q}(\mathcal{S}) = \|F\|_{p,\Lambda_r} + \|g\|_{q,\Lambda_r}.$$

Proposition 43.4 (Scalar interior estimates with L^p data)**.** *There is $\varepsilon_S' > 0$ with the following property. For $0 < \rho < r \le 3$ there is C, depending in addition on $r-\rho$, such that for $\varepsilon \le \varepsilon_S'$: if v is a scalar field with $D^T D v = \mathcal{S}$ on Λ_r, then:*

(a) if $p > d$ and $\beta \le 1 - d/p$: $|v|_{\Lambda_\rho} + |\partial v|_{\Lambda_\rho} + [\partial v]_{\beta,\Lambda_\rho} \le C\big(\mathfrak{n}_r^{p,\beta}(\mathcal{S}) + \|v\|_{\Lambda_r}\big)$;

(b) if $\frac{d}{2} < p < d$ and $\beta \le 2 - d/p$: $|v|_{\Lambda_\rho} + [v]_{\beta,\Lambda_\rho} + \|\partial v\|_{p^*,\Lambda_\rho} \le C\big(\mathfrak{m}_r^{p,p^*}(\mathcal{S}) + \|v\|_{\Lambda_r}\big)$.

Proof. As in Proposition 36.3, we may assume $r - \rho \ge 40\eta$, and we write $-\Delta v = (F - bv) + \sum_\mu \partial_\mu^*(g_\mu - a_\mu v)$ on Λ_r. Details

(a) Repeat the proof of the first bound of Proposition 36.3, with exponent β and with the first inequality of Lemma 43.3 in place of that of Lemma 36.2. The perturbation bv is part of F, and $\|bv\|_{p,\Lambda_{2s}(x)} \le C\varepsilon s^{d/p}|v|_{\Lambda_{t'}}$. So it enters, as before, with the factor s^2. After division by s the source enters as $s^{1-d/p}\|F\|_{p,\Lambda_r} \le \|F\|_{p,\Lambda_r}$. The rest is unchanged.

(b) Let $A(t) = |v|_{\Lambda_t} + \|\partial v\|_{p^*,\Lambda_t}$ and $\beta_0 = 2 - d/p$; note $1 - d/p^* = \beta_0$. Let $\rho \le t < t' \le r$, $s = (t'-t)/6$, and suppose $s \ge 3\eta$. For $x \in \Lambda_t$, the second inequality of Lemma 43.3 with $q = p^*$ gives

$$|v(x)| \le C\big(s^{\beta_0}\|F - bv\|_{p,\Lambda_{2s}(x)} + s^{\beta_0}\|g - av\|_{p^*,\Lambda_{3s}(x)} + s^{-d/2}\|v\|_{\Lambda_r}\big).$$

Since $s \le 1$, $\|bv\|_{p,\Lambda_{2s}(x)} + \|av\|_{p^*,\Lambda_{3s}(x)} \le C\varepsilon|v|_{\Lambda_{t'}}$, and this is at most $C\mathfrak{m}_r^{p,p^*}(\mathcal{S}) + C\varepsilon A(t') + Cs^{-d/2}\|v\|_{\Lambda_r}$. For the gradient, cover Λ_t by the cubes $\Lambda_s(x_i)$, with x_i the points of Λ_{t+s} on a grid of lattice points of spacing $h = \eta\lfloor s/\eta\rfloor$. Since $s \ge 3\eta$, $\frac12 s \le h \le s$. Every point of Λ_t is within $\frac12 h$ of a grid point, in each coordinate, so these cubes cover Λ_t, and Lemma 43.3 applies at each $x_i \in \eta\mathbb{Z}^d$. The cubes $\Lambda_{3s}(x_i)$ lie in $\Lambda_{t+4s} \subset \Lambda_{t'}$ and overlap at most 13^d times. Apply the third inequality of Lemma 43.3 on each cube, raise it to the power p^* and sum over i. For the data use $\sum_i a_i^{p^*} \le (\sum_i a_i^p)^{p^*/p}$ and $\sum_i a_i^{p^*} \le (\sum_i a_i^2)^{p^*/2}$, valid since $p^* > p$ and $p^* > d > 2$. This gives

$$\|\partial v\|_{p^*,\Lambda_t} \le C\big(\mathfrak{m}_r^{p,p^*}(\mathcal{S}) + \|bv\|_{p,\Lambda_{t'}} + \|av\|_{p^*,\Lambda_{t'}} + s^{-\theta}\|v\|_{\Lambda_r}\big), \qquad \theta = 1 + \tfrac{d}{2} - \tfrac{d}{p^*},$$

and the perturbations are bounded by $C\varepsilon|v|_{\Lambda_{t'}}$. If $s < 3\eta$, then $A(t) \le C\eta^{-1-d/2}\|v\|_{\Lambda_r} \le C(t'-t)^{-1-d/2}\|v\|_{\Lambda_r}$. So $A(t) \le C_S'\varepsilon A(t') + C(t'-t)^{-1-d/2}\|v\|_{\Lambda_r} + C\mathfrak{m}_r^{p,p^*}(\mathcal{S})$, and for $C_S'\varepsilon \le \frac12$ Lemma 36.1 on $[\rho', r]$, $\rho' = \frac12(\rho + r)$, bounds $A(\rho')$. The Hölder quotient on Λ_ρ follows from the second inequality of Lemma 43.3 at the scale $s_1 = (r-\rho)/8$, with the sup bound inserted, as in Proposition 36.3. □

Proposition 43.5 (Schauder estimate, (I4))**.** *There is $\varepsilon_S'' > 0$ with the following property. For $0 < \rho < r \le 3$ there is C, depending in addition on $r-\rho$, such that for $\varepsilon \le \varepsilon_S''$: if v is a scalar field with $D^T D v = F$ on Λ_r, then*

$$\max_{\mu,\nu}\big(|\partial_\mu\partial_\nu v|_{\Lambda_\rho} + [\partial_\mu\partial_\nu v]_{\beta,\Lambda_\rho}\big) \le C\big(|F|_{\Lambda_r} + [F]_{\beta,\Lambda_r} + \|v\|_{\Lambda_r}\big).$$

The same holds for the covariant second derivatives $D_\mu D_\nu v$.

Proof. We may assume $r-\rho \geq 40\eta$. Otherwise $\eta^{-1} < 40(r-\rho)^{-1}$, and $|v(x)| \leq \eta^{-d/2}\|v\|_{\Lambda_r}$ gives the bounds: second differences, flat or covariant (the transports are unitary), are at most $4\eta^{-2-d/2}\|v\|_{\Lambda_r}$, and their β-Hölder quotients at most $8\eta^{-2-\beta-d/2}\|v\|_{\Lambda_r}$, since distinct lattice points are at distance at least η. So the proposition holds with $C = 12\big(40/(r-\rho)\big)^{2+\beta+d/2}$ in this case. Let $\rho_1 = \frac{1}{2}(\rho+r)$ and $\mathfrak{N} = |F|_{\Lambda_r} + [F]_{\beta,\Lambda_r} + \|v\|_{\Lambda_r}$. The first bound of Proposition 36.3, with exponent β and $g=0$, gives $|v| + |\partial v| + [\partial v]_\beta \leq C\mathfrak{N}$ on Λ_{ρ_1}. Write $-\Delta v = F - bv - \sum_\mu \partial_\mu^*(a_\mu v)$, fix ν and put $w = \partial_\nu v$. Since $\partial_\nu h = -\partial_\nu^*(S_\nu h)$ and ∂_ν commutes with ∂_μ^*,

$$-\Delta w = \sum_\mu \partial_\mu^* G_\mu, \qquad G_\mu = -\delta_{\mu\nu} S_\nu(F - bv) - \partial_\nu(a_\mu v) \qquad \text{on } \Lambda_{\rho_1}.$$

Here a_μ is a multiplication operator composed with a shift by at most η, so $\partial_\nu(a_\mu v)$ is a sum of terms (differences of the coefficients) $\cdot$ (shifted v) and (coefficients) $\cdot$ (shifted ∂v). By Lemma 12.2, $|a_\mu| + |\partial a_\mu| + |b| \leq C\varepsilon$, and the β-Hölder quotients of ∂a_μ and b are bounded by $C\varepsilon$. Hence $|G| + [G]_\beta \leq |F| + [F]_\beta + C\varepsilon\big(|v| + |\partial v| + [\partial v]_\beta\big) \leq C\mathfrak{N}$ on Λ_{ρ_1}. Let $s_1 = (\rho_1 - \rho)/4 \geq \eta$. For $x \in \Lambda_\rho$, the first inequality of Lemma 36.2 on $\Lambda_{3s_1}(x) \subset \Lambda_{\rho_1}$, with exponent β, $F=0$ and $\|w\|_{\Lambda_{2s_1}(x)} \leq C|\partial v|_{\Lambda_{\rho_1}}$, bounds $|\partial w| + [\partial w]_\beta$ on $\Lambda_{s_1}(x)$ by $C\mathfrak{N}$. Pairs of points at distance larger than s_1 are handled by the sup bound. Finally, $D_\mu D_\nu v - \partial_\mu \partial_\nu v$ is a sum of terms (coefficient) $\cdot$ (shifted v or ∂v), with coefficients bounded by $C\varepsilon$ whose β-Hölder quotients are bounded by $C\varepsilon$. They include $\partial_\mu \tau_\nu$, which is only C^α by Lemma 12.2, and $\beta \leq \alpha$ suffices. □

Remark 43.6 (The averaging term and (I4)). Proposition 43.5 needs a Hölder source. For the scalar Dirichlet propagator, $v = G'_X f$ satisfies $D^T D v = f - aQ'^T Q' v$ in the interior. The function $Q'^T Q' v$ jumps across the faces of the blocks, by a times the difference of the averages of v on adjacent blocks, up to parallel transport. This jump is not small. A jump across a single face already destroys the Hölder bound on second derivatives: for $-\Delta v = \mathbf{1}_{\{x_1>0\}}$ in the continuum, $\partial_1^2 v$ is discontinuous, and on the lattice $[\partial_1\partial_1 v]_\beta$ is of order $\eta^{-\beta}$. Jumps across the faces meeting at an edge destroy even the sup bound. If $-\Delta v = \sum_{i=1}^4 c_i \mathbf{1}_{Q_i}$ for the four quadrants Q_i around an edge, then $\partial_1\partial_2 v$ grows like $(c_1 - c_2 + c_3 - c_4)\log(1/|x|)$ at the edge. On the lattice this becomes $\log(1/\eta)$. Here the coefficient is a mixed second difference of the block averages, which is generically non-zero. So (I4) does not hold for the scalar local propagators G'_Y themselves. Proposition 43.5 applies when the averaging term vanishes, i.e. to fields σ with $Q'\sigma = 0$. For instance, $\sigma = G'_X R_X h$ satisfies $D^T D\sigma = R_X h = h - P_{\mathcal{V},X} h$ in the interior (Theorem 18.2(d)). Here $P_{\mathcal{V},X} h = G'_X Q'^T (M'_X)^{-1} Q' G'_X h$ is $C^{1,\beta}$ there, by Proposition 36.3. Hence $|\partial\partial\sigma| + [\partial\partial\sigma]_\beta$ is bounded at $y \in X_{(4)}$ by $|h| + [h]_\beta$ nearby plus weighted L^2 norms.

43.4 The local system

Recall: the local system of Section 36. For $u \in \ell^2_1(X)$ with $\mathfrak{A}^{X,\kappa} u = \mathcal{S}$, Lemma 36.4 puts $\sigma = S_X u$, $\tau = G'_X D_X^T \mathfrak{B}_X \Pi_X u$, $\omega_1 = (M'_X)^{-1} Q' G'_X D_X^T u$, $\omega_2 = (M'_X)^{-1} Q' G'_X \tau$ and $\pi_i = G'_X Q'^T \omega_i$. At distance at least 2η from $\mathbb{T}_\eta \backslash X$, (u,σ,τ) then satisfies the local system (36.5) of Proposition 36.5,

$$\begin{aligned} D^T D u + D D^T u - D^T \mathfrak{F}\sigma + W(u - D\sigma) - D\tau + \kappa m u &= \mathcal{S}_u, \\ D^T D\sigma - D^T u &= \mathcal{S}_\sigma, \\ D^T D\tau - \mathfrak{F}^T(Du - \mathfrak{F}\sigma) - D^T\big(W(u - D\sigma)\big) &= \mathcal{S}_\tau, \end{aligned}$$

with $m = m_X$, $\mathcal{S}_u = \mathcal{S} + D\pi_1 - D\pi_2 - a\mathbf{Q}^T\mathbf{Q}u$, $\mathcal{S}_\sigma = -\pi_1$ and $\mathcal{S}_\tau = -aQ'^T Q'\tau$ (proof of Theorem 35.2, the basic estimate). There, with $\mathfrak{L}(u) = \sum_{y'} e^{-\gamma|y-y'|}\big(\|u\|_{B(y')} + \|Du\|_{B(y')}\big)$,

the L^2 norms of σ, τ, π_i on $\Lambda_3(c_y)$ and the values $|\omega_i|$ near y are at most $C\mathfrak{L}(u)$. The proof of Proposition 36.5 runs in five steps: the normal form of the equations (Step 1), the estimates at one scale (Step 2), weights and absorption (Step 3), Hölder quotients (Step 4), and general radii (Step 5).

Proposition 43.7 (System interior estimates with L^p data)**.** *There is $\varepsilon'_L > 0$, depending in addition on the mass parameter κ_m of Section 19, with the following property. For $0 < \rho < r \le 3$ there is C, depending in addition on $r - \rho$, such that for $\varepsilon \le \varepsilon'_L$ the following holds. Let W, m, (u, σ, τ) and the sources $\mathcal{S}_u, \mathcal{S}_\sigma, \mathcal{S}_\tau$ be as in Proposition 36.5, with (36.5) on Λ_r, and let $\mathfrak{m} = \|u\|_{\Lambda_r} + \|\sigma\|_{\Lambda_r} + \|\tau\|_{\Lambda_r}$.*

(a) If $p > d$ and $\beta \le 1 - d/p$:

$$\sum_{v \in \{u,\sigma,\tau\}} \big(|v| + |\partial v| + [\partial v]_\beta\big)_{\Lambda_\rho} \le C\Big(\sum_\bullet \mathfrak{n}_r^{p,\beta}(\mathcal{S}_\bullet) + \mathfrak{m}\Big).$$

(b) If $\frac{d}{2} < p < d$ and $\beta \le 2 - d/p$:

$$\sum_{v \in \{u,\sigma,\tau\}} \big(|v|_{\Lambda_\rho} + [v]_{\beta,\Lambda_\rho} + \|\partial v\|_{p^*,\Lambda_\rho}\big) \le C\Big(\sum_\bullet \mathfrak{m}_r^{p,p^*}(\mathcal{S}_\bullet) + \mathfrak{m}\Big).$$

Proof. *(a)* Repeat the proof of Proposition 36.5 with exponent β and with Lemma 43.3 in place of Lemma 36.2. The perturbations in F_u, F_σ, F_τ are bounded pointwise, as in Step 1 there. On $\Lambda_{2s}(x)$ their L^p norms are at most $Cs^{d/p}$ times their sup norms, so they enter with the same powers of s as before. The sources enter as $s^{1-d/p}\|F_{\mathcal{S}_\bullet}\|_p \le \|F_{\mathcal{S}_\bullet}\|_p$. In Step 2 the Hölder quotient of $\partial\sigma$ at scale $3s$ is bounded by the first inequality of Lemma 43.3, which gives $s^\beta[\partial\sigma]_\beta \le \cdots + s^{1-d/p}\|F_\sigma\|_p$. The coefficients of W have β-Hölder quotients bounded by $C\varepsilon$, so Step 1 holds with β in place of α. Steps 3–5 are unchanged. Details

(b) Replace $A_v(t)$ in Steps 2 and 3 of the proof of Proposition 36.5 by $A_v(t) = |v|_{\Lambda_t} + \|\partial v\|_{p^*,\Lambda_t}$. Values are bounded at each $x \in \Lambda_t$ by the second inequality of Lemma 43.3, with $q = p^*$. The L^{p^*} norms of gradients are bounded by the third inequality on the cubes $\Lambda_s(x_i)$ of the proof of Proposition 43.4(b), summed in ℓ^{p^*}. The perturbations of Step 1 there are estimated cube by cube:

- terms bounded pointwise by $|v|$, in $F_\bullet$: their L^p norm on a cube of side $4s$ is at most $Cs^{d/p}$ times the sup, and the ℓ^{p^*} sum over the Cs^{-d} cubes gives the factor $s^{d/p-d/p^*} = s$; for values the factor is $s^{\beta_0+d/p} = s^2$;
- gradient terms in $F_\bullet$ ($D\tau$ in F_u; $\varepsilon\partial u$ and $\varepsilon\partial\sigma$ in F_τ and F_u): by Hölder on a cube of side $4s$, $\|\partial v\|_{p,\Lambda_{2s}(x_i)} \le Cs\|\partial v\|_{p^*,\Lambda_{2s}(x_i)}$, which gives the factor s (and $s^{1+\beta_0}$ for values);
- terms in $g_\bullet$: their L^{p^*} norms are bounded by $C\varepsilon$ times $|v|$ or $\|\partial v\|_{p^*}$, except for u_μ in g_σ, which has coefficient 1. The potential W is local with bounded coefficients, so $\|Wh\|_{p^*,Q} \le C\varepsilon\|h\|_{p^*,Q'}$ for a slightly larger cube Q'.

Hölder quotients of the coefficients are not needed. So Step 2 gives inequalities of the same form as there:

$$A_\sigma(t) \le C_4(A_u + \varepsilon A_\sigma) + R, \qquad A_\tau(t) \le C_2\varepsilon(A_u + A_\sigma + A_\tau) + R,$$
$$A_u(t) \le C_3(\varepsilon + \kappa_m s)A_u + C_3\varepsilon A_\sigma + C_3 s A_\tau + R,$$

with the A_v on the right evaluated at t' and $R = C(\sum_\bullet \mathfrak{m}_r^{p,p^*}(\mathcal{S}_\bullet) + (t'-t)^{-\theta'}\mathfrak{m})$ for some $\theta' > 0$. Step 3, with the same weights and small cubes, and Lemma 36.1 bound $A_u + A_\sigma + A_\tau$

on $\Lambda_{\rho'}$. The Hölder quotients of the values follow at a fixed scale from the second inequality of Lemma 43.3. In Step 5, for $r > r_*$, cover Λ_ρ by the cubes $\Lambda_{r_1/2}(x_i)$, with x_i on a grid of spacing $r_1/2$. There are at most $C(r_1)$ of them. Sum the p^*-th powers of the bounds on these cubes. □

In Proposition 36.5 the bound for τ needed the Hölder quotient of $\partial\sigma$, through the term $WD\sigma$ in g_τ. In (b) only $\|\partial\sigma\|_{p^*}$ enters, so this step disappears.

43.5 Consequences for the Dirichlet operators

For a block y and fields on $\eta\mathbb{Z}^d$ write $\mathfrak{E}^p_y(F) = \sum_{y'} e^{-\gamma|y-y'|}\|F\|_{p,B(y')}$. For a 2-form or a gauge field g write $\mathfrak{E}^\beta_y(g) = \sum_{y'} e^{-\gamma|y-y'|}(|g|_{B(y')} + [\tilde g_{y'}]_{\beta,\Lambda_1(c_{y'})})$, as in Theorem 35.2(b).

Corollary 43.8 (Dirichlet propagators with L^p data)**.** *In the setting of Theorem 35.2 there are $\varepsilon', C, \gamma > 0$, independent of X, k, n, η, such that the following holds for $\varepsilon \le \varepsilon'$, every domain X and every $y \in X_{(4)}$. Let O be one of the operators G'_X, $\mathbf{G}^X$, and let $f = F + D^T_X g$, with F a scalar, respectively gauge, field and g a gauge field, respectively 2-form, supported in X.*

(a) If $p > d$ and $\beta \le 1 - d/p$: $|Of|_{1,\beta;y} \le C(\mathfrak{E}^p_y(F) + \mathfrak{E}^\beta_y(g))$.

(b) If $\frac{d}{2} < p < d$ and $\beta \le 2 - d/p$: $|Of|_{B(y)} + [\widetilde{Of}_y]_{\beta,\Lambda_1(c_y)} + \|\nabla Of\|_{p^*,B(y)} \le C(\mathfrak{E}^p_y(F) + \mathfrak{E}^{p^*}_y(g))$.

Here $|\cdot|_{1,\beta;y}$ is $|\cdot|_{1,\alpha;y}$ of Section 35 with exponent β. In particular $\nabla G'_X D^T_X$ and $\nabla \mathbf{G}^X D^T_X$ map L^q to L^q locally, with exponential decay, for $d < q < \infty$: take $p = dq/(d+q)$, so that $p^* = q$. This is the Calderón–Zygmund input (I3); the case used in Section 44 is $q = 2d = 8$. The corollary gives gradient bounds in L^q only for $q = p^* \in (d, \infty)$. The endpoint $p = \frac{d}{2}$, $q = d$, i.e. L^2 data in $d = 4$, is not covered by it. For such data the flat estimate, Theorem 43.2(c) with $p = 2$, gives gradients in L^4; Lemma 44.6 uses it for the scalar propagators.

Proof. For $O = \mathbf{G}^X$ let $u = \mathbf{G}^X f$, so that $\mathfrak{A}^{X,\kappa_m} u = f$. Proceed as in the basic estimate of the proof of Theorem 35.2. By Lemma 36.4, (u, σ, τ) satisfies (36.5) on $\Lambda_3(c_y)$, with Details

$$\mathcal{S}_u = f + D\pi_1 - D\pi_2 - a\mathbf{Q}^T\mathbf{Q}u, \qquad \mathcal{S}_\sigma = -\pi_1, \qquad \mathcal{S}_\tau = -aQ'^T Q'\tau.$$

In the local gauge, $D^T_X g = \partial^* g + T^T g$ is in local form, with $T^T g$ part of F. The terms $D\pi_i$, $a\mathbf{Q}^T\mathbf{Q}u$ and $aQ'^TQ'\tau$ are bounded pointwise by $C\mathfrak{L}(u)$, as in the proof of Theorem 35.2. By the L^2 decay of $\mathbf{G}^X$, $D\mathbf{G}^X$, $\mathbf{G}^X D^T_X$ and $D\mathbf{G}^X D^T_X$ (Theorem 19.7(a)), together with $\|h\|_{B(y')} \le \|h\|_{p,B(y')}$ for $p \ge 2$, we have $\mathfrak{L}(u) \le C(\mathfrak{E}^p_y(F) + \mathfrak{E}^q_y(g))$ for $p, q \ge 2$. Here $\mathfrak{E}^q_y(g)$ is defined like $\mathfrak{E}^p_y(F)$, and $\mathfrak{E}^q_y(g) \le \mathfrak{E}^\beta_y(g)$. The Hölder quotient of g is converted to the local gauge as in the proof of Theorem 35.2(b). Proposition 43.7(a), respectively (b) with the given p, gives the claim. For $O = G'_X$, $v = G'_X f$ satisfies $D^T Dv = f - aQ'^TQ'v$ in the interior. Here $|Q'v(y')| \le C\|v\|_{B(y')}$, and G'_X, DG'_X, $G'_X D^T_X$, $DG'_X D^T_X$ have L^2 decay (Theorem 18.2(b)). Apply Proposition 43.4. □

Scope. By Corollary 43.8, the Dirichlet propagators G'_Y and $\mathbf{G}^Y$ with potentials $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$ satisfy (I1) and (I2) for data in L^p with $p > \frac{d}{2}$; the levels (L1)–(L3) are compositions of these with bounded block-local operators. The levels with the potential W^X_2 (range up to 3, non-local through ω_0) are not covered: the cube bound $\|Wh\|_{p^*,Q} \le C\varepsilon\|h\|_{p^*,Q'}$ of Proposition 43.7(b) fails at scales $s \ll 3$. The background map needs pointwise bounds only for operators built with $W \in \mathcal{W}_{\mathrm{loc}}$. For $C^{1,\beta}$ outputs from sources in L^p, $p > d$, the exponent restriction is $\beta \le \min(\alpha, 1 - \frac{d}{p})$; for the tier $L^{2d} = L^8$ of Section 44 this is $\beta \le \min(\alpha, \frac{1}{2})$. In $d = 4$, L^2 data sit at the endpoint $p = \frac{d}{2}$, and Section 44 passes from L^2 to L^8 with one more interior factor (Lemma 44.6).

- *(I4) and* $\mathbf{C}(s)$. By Remark 43.6, second derivatives are available only for fields annihilated by Q'. In a weakened product $DG'(s)R(s)G'(s)D^T$ the identity $Q'G'R = 0$ does not hold term by term. So the term-by-term argument for $\mathbf{C}(s)$ from its defining formula does not work. Applying the unweakened operator to $\mathbf{C}(s)f$ and treating the weakening errors as sources does not help either, for the same reason (Remark 44.14). Section 44 uses the torus identity $\mathbf{C} = \mathbf{G}\Pi^T - \mathbf{HQG}$ instead, which needs no second derivatives of scalar fields.

- *Step bounds.* The step bounds of route 2 are carried out in Section 44, in block L^4, L^8 and Hölder norms. The bound for the non-local part of the steps, where L^2 data near the Dirichlet cuts must be lifted to L^8, is Proposition 44.11 there (with Lemma 44.9).

44 Regularity tiers and step bounds

This section carries out the main step of route 2 of Section 42: the pointwise bounds (42.1) for the weakened operators $\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$ and $\mathbf{C}(s)D^T$ on the torus, uniformly on $\mathbb{D}_\kappa$. The analytic inputs are the interior estimates of Section 43 and the L^2 theory of Section 41. No regularity at the Dirichlet cuts is used. The construction rests on two observations.

- *Outputs are interior.* Every term of a walk ends, on the left, with a first factor h_jG_j, evaluated on K_j. Every local step $[\mathfrak{A}_{\mathrm{loc}}, h_j]G_j$ is evaluated on $\operatorname{supp}\nabla h_j \subset K_j$. Both sets lie at ∞-distance at least $(\rho-1)M-1$ from the cut $\partial\Lambda_j^v$. Near the cuts, only L^2 information enters, and it enters only as an input.

- *Interior factors raise the integrability.* Pointwise bounds need data in L^p with $p > d$ (Corollary 43.8(a)). In $d = 4$, data in L^2 sit at the endpoint $p = \frac{d}{2}$ of (I2): one interior propagator takes them to gradients in L^4, but not to bounded values. So the integrability must be raised along $L^2 \to L^4 \to L^8 \to C^{1,\beta}$ by the last interior factors of a walk term, while the rest of the term is estimated in L^2. At the gauge level, by the regrouping of the derivatives in Section 41, a factor carrying a D^T on the right always receives its input from a factor of type $\mathcal{A}$, which has already raised it.

The intermediate norms are block L^4 and L^8 norms and block Hölder norms, which compose exactly like the L^2 decay norms. A source space of the form $L^p + D^T(C^\beta)$ is not needed as such. Its divergence part is carried by the factors $\mathcal{A}$ of the step and by the regrouping rule.

For $\mathbf{C}(s)$ the formula $\mathbf{G} - DG'RG'D^T - \mathbf{HQG}$, weakened term by term, does not work: the weakened $DG'(s)R(s)G'(s)D^T$ has no uniform C^1 bound (Remark 44.14). We use instead the torus identity $\mathbf{C} = \mathbf{G}\Pi^T - \mathbf{HQG}$, which involves only first derivatives of scalar fields.

Status in four dimensions. In three dimensions one interior factor takes L^2 data to $C^{1/2}$ values with L^6 gradients. In four dimensions L^2 data sit at the endpoint $p = \frac{d}{2}$ of (I2). One interior factor takes them to gradients in L^4, a second one to C^β values with gradients in L^8, and a third one to $C^{1,\beta}$ (Lemmas 44.6 and 44.5). Only the scalar walks inside the non-local part $\mathfrak{N}$ at level (L3), on $\mathbb{T}_\eta$ and on the local domains, need this lift from L^2 (Lemma 44.9). The gauge propagators act only on data of tier 1 or higher. At level (L3) the bound for the non-local parts of the steps (Proposition 44.11), and with it Theorem 44.13 at level (L3) and Theorem 44.15, are stated for $W = 0$ (Remark 44.12). All results hold with $\beta = \min(\alpha, \frac{1}{2})$. From Section 45 on, $\alpha \le \frac{1}{2}$, so that $\beta = \alpha$ (Remark 44.12(b)).

Setting. We keep the setting of Section 41 at the levels (L1)–(L4), with $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, and of Section 43. We take $\beta = \min(\alpha, \frac{1}{2})$. The level with W_2 is not treated. The background is real: the interior estimates of Sections 36 and 43 are proved for real backgrounds, and the

complex perturbations of the background used for analyticity in Sections 40 and 41 are treated in Proposition 46.7. The weakening parameters s are complex. $X'_{(r)}$ denotes the blocks of X' at distance more than r from $\mathbb{T}_\eta \setminus X'$, as in Part VIII. As there, distances between blocks and sets are ∞-distances: in $X'_{(r)}$, $K_j^{(r)}$ and Y_j, in the seams Σ_w and the seam distance d_Σ, and in the radii of the near parts of Definition 44.1. Only the exponential weights $e^{\gamma|y-y'|}$ use the Euclidean distance. The rate γ_0 of the levels (L1) and (L3) is fixed in the paragraph *Rates* of Section 44.4; Lemmas 44.5 and 44.6 hold with rate $\gamma_1 = \gamma_0/2$, Lemmas 44.9 and 44.10 with rate $2\gamma_0$, and the walks are estimated at rate $\gamma = \gamma_0/8$. The inequalities $d_\Sigma(y) \le |y - w|$ for $w \in \Sigma$ and $d_\Sigma(y) \le |y - y_1| + d_\Sigma(y_1)$ of Section 41 hold with the ∞-distance, since $|\cdot|_\infty \le |\cdot|$.

44.1 Tiers and tier norms

For a block y and a fine field g (scalar, gauge field or 2-form) put

$$\mathfrak{n}_y^0(g) = \|g\|_{B(y)}, \qquad \mathfrak{n}_y^{1/2}(g) = \|g\|_{4,B(y)}, \qquad \mathfrak{n}_y^1(g) = \|g\|_{8,B(y)},$$
$$\mathfrak{n}_y^2(g) = |g|_{B(y)} + [\tilde g_y]_{\beta,\Lambda_1(c_y)}, \qquad \mathfrak{n}_y^3(g) = |g|_{B(y)} + |\nabla g|_{B(y)} + [\nabla \tilde g_y]_{\beta,\Lambda_1(c_y)},$$

with $\tilde g_y$ the field in a real gauge in which (B) holds on $\Lambda_4(c_y)$, as in Theorem 35.2. In $\mathfrak{n}_y^3$ the Hölder quotient is taken on $\Lambda_1(c_y)$, which is larger than $B(y)$, as in $\mathfrak{n}_y^2$. The interior estimates give this directly: Propositions 36.5, 43.4 and 43.7 hold on $\Lambda_\rho(c_y)$ with $\rho = 1$. Tier 3 is only ever an output tier. In Theorems 44.13 and 44.15, $|\cdot|_{1,\beta;y}$ is understood in this sense. For coarse fields all five are $|\omega(y)|$. Since blocks have volume 1, $\mathfrak{n}_y^0 \le \mathfrak{n}_y^{1/2} \le \mathfrak{n}_y^1 \le C\mathfrak{n}_y^2$. Moreover $\mathfrak{n}_y^2(g) \le C \max_{|y'-y|_\infty \le 1} \mathfrak{n}_{y'}^3(g)$, which costs 1 in the radius of the bounds below; we do not record this. For the derivative of a field h, $\mathfrak{n}_y^2(\nabla h) \le C\mathfrak{n}_y^3(h)$. The index $t \in \{0, \frac12, 1, 2, 3\}$ is the *tier*. In words: L^2, L^4, L^8, C^β and $C^{1,\beta}$. The exponent $8 = 2d$ is chosen so that (I1) takes tier 1 to $C^{1,\beta}$ with $\beta \le 1 - \frac{d}{8} = \frac12$ (Corollary 43.8(a)). The tier $L^4 = L^d$ records the endpoint. From L^2 data a single interior factor gives gradients in L^4 only (Theorem 43.2(c) with $p = 2$). So in $d = 4$ it does not lift the gradients of tier-0 data to tier 1; it lifts them to tier $\frac12$. Tier $\frac12$ is used only in the lemmas that lead to Proposition 44.11; no theorem is stated with it. Table 11 lists the tiers and the tier bounds of the interior factors (Lemmas 44.5 and 44.6).

In tier bounds such as $a \to b$, in the norms $|\cdot|^{a\to b}$ and in Lemmas 44.2 and 44.3, the letters a, b, c, d, also with indices, denote tiers. They are unrelated to the averaging coefficient a, to the coefficient b in the proof of Lemma 44.6, and to the dimension $d = 4$.

Definition 44.1 (Tier bounds)**.** Let $a, b \in \{0, \frac12, 1, 2, 3\}$, $\gamma \ge 0$, $R \in [0, \infty]$, and let Y be a set of blocks. An operator F satisfies a *tier bound* $a \to b$ *on* Y *with radius* R and constant c if there are kernels $k, k' \ge 0$ such that for all g and all $y \in Y$

$$\mathfrak{n}_y^b(Fg) \le \sum_{|y'-y|_\infty \le R} k(y,y')\, \mathfrak{n}_{y'}^a(g) + \sum_{y'} k'(y,y')\, \mathfrak{n}_{y'}^0(g), \qquad \sup_{y\in Y} \sum_{y'} e^{\gamma|y-y'|}(k+k')(y,y') \le c.$$

The infimum of such c is $|F|^{a\to b}_{\gamma;R,Y}$. For Y the set of all blocks and $R = \infty$ we write $|F|^{a\to b}_\gamma$, the *global* tier norm; then the second sum can be absorbed in the first.

The first sum is the *near part*: regularity of the input is required only near y. The second sum, the *far part*, uses only L^2. Interior estimates give bounds of this split form directly. Global tier norms are needed for walks of unbounded length. Split forms with finite radius are used only for boundedly many factors.

Lemma 44.2 (Tier calculus)**.** *(a)* $|F|^{0\to 0}_\gamma \le |F|_\gamma$*, the decay norm of Section 40.*

(b) (Global composition.) $|FG|^{a\to c}_\gamma \le |F|^{b\to c}_\gamma |G|^{a\to b}_\gamma$.

Tier	$\mathfrak{n}^t_y(g)$	In words	Tier bounds into this tier from the interior factors
0	$\|g\|_{B(y)}$	L^2	none; the far parts of all tier bounds use only this tier
$\frac{1}{2}$	$\|g\|_{4,B(y)}$	L^4	$0 \to \frac{1}{2}$: $\varphi G'_j$, $\varphi DG'_j$ (Lemma 44.6)
1	$\|g\|_{8,B(y)}$	L^8	$\frac{1}{2} \to 1$: $\varphi DG'_j$ (Lemma 44.6); $0 \to 1$, at every block: $[aQ'^TQ', h_j]G'_j$ (Lemma 44.6); $1 \to 1$: φDG_jD^T (Lemma 44.5)
2	$\|g\|_{B(y)} + [\tilde{g}_y]_{\beta,\Lambda_1(c_y)}$	C^β	$\frac{1}{2} \to 2$ (values): $\varphi G'_j$ (Lemma 44.6); $1 \to 2$: φG_jD^T (values), φDG_j; $2 \to 2$: φDG_jD^T (Lemma 44.5)
3	$\|g\|_{B(y)} + \|\nabla g\|_{B(y)} + [\nabla\tilde{g}_y]_{\beta,\Lambda_1(c_y)}$	$C^{1,\beta}$	$1 \to 3$: φG_j; $2 \to 3$: φG_jD^T (Lemma 44.5). Only an output tier

Table 11: The tiers of Section 44, with $\beta = \min(\alpha, \frac{1}{2})$. In the last column G'_j is a scalar local propagator of level (L1) and G_j a local propagator of level (L1) or (L3), on $Z_j = \Lambda^v_j \cap X$; the bounds hold on the interior sets Y_j of Lemma 44.5, with radius 4 and rate $\gamma_1 = \gamma_0/2$. Lemmas 44.8–44.10, Proposition 44.11 and Theorem 44.13 combine these entries.

(c) (Local composition.) *Suppose* $|F|^{b\to c}_{\gamma;R_F,Y_F} \le c_F$ *and* $|G|^{a\to b}_{\gamma;R_G,Y_G} \le c_G$. *Suppose also that every block within distance* R_F *of* Y_F *lies in* Y_G, *and that* $|G|_\gamma < \infty$. *Then*

$$|FG|^{a\to c}_{\gamma;R_F+R_G,Y_F} \le c_F\,(c_G + |G|_\gamma).$$

(d) (Commutators.) $|[F, h_j]|^{a\to b}_{\gamma/2;R,Y} \le C(\gamma M)^{-1}|F|^{a\to b}_{\gamma;R,Y}$ *for* $0 < \gamma \le 1$.

(e) (Seams.) *With the seam weights of Section 41, define* $|F|^{a\to b}_{\gamma,\Sigma}$ *like* $|F|^{a\to b}_\gamma$, *with* $e^{\gamma|y-y'|}$ *replaced by* $e^{\gamma(|y-y'|+d_\Sigma(y)+d_\Sigma(y'))}$, *and similarly for split forms. Then* $|AFB|^{a\to d}_{\gamma,\Sigma} \le |A|^{c\to d}_{2\gamma}|F|^{b\to c}_{\gamma,\Sigma}|B|^{a\to b}_{2\gamma}$, *and the analogue of (c) holds.*

Proof. (a) $\|\mathbf{1}_yF g\| \le \sum_{y'} \|\mathbf{1}_yF\mathbf{1}_{y'}\|\,\|\mathbf{1}_{y'}g\|$. Details

(b) Insert the bound for G into that for F. The kernels compose as matrices. Weighted row sums are submultiplicative, by $e^{\gamma|y-y''|} \le e^{\gamma|y-y'|}e^{\gamma|y'-y''|}$.

(c) In the near part of F, y' lies within R_F of $y \in Y_F$, hence in Y_G. So the bound for G applies at y', and its near part has radius R_G. In the far part of F use $\mathfrak{n}^0_{y'}(Gg) \le \sum_{y''} \|\mathbf{1}_{y'}G\mathbf{1}_{y''}\|\mathfrak{n}^0_{y''}(g)$. Then compose as in (b).

(d) As in Lemma 40.1(c), write $[F, h_j]g = F(\varphi g) - \varphi Fg$ with $\varphi = h_j - h_j(c_y)$. By Lemma 17.3, $|\partial^k h_j| \le CM^{-k}$. Hence multiplication by φ changes $\mathfrak{n}^t_{y'}$ by at most a factor $CM^{-1}(1 + |y - y'|)$, for every t; for $t = 3$ use the discrete Leibniz rule. It needs $|h|$ and $|\nabla h|$ on all of $\Lambda_1(c_y)$, while $\mathfrak{n}^3_y$ contains them only on $B(y)$; they follow from the values on $B(y)$ and the Hölder quotient on $\Lambda_1(c_y)$, up to the local-gauge correction $C\varepsilon|h|$. Finally use $(1 + t)e^{-\gamma t/2} \le C\gamma^{-1}$.

(e) As (41.2), using $d_\Sigma(y) \le |y - y_1| + d_\Sigma(y_1)$. For the analogue of (c), the same inequality moves the seam weight from the factor in seam form to its neighbours, in the near and in the far parts. This costs the factor 2 in the rate of the neighbours. □

Weakenable families. For a weakenable family put

$$\|O\|^{a\to b}_{\kappa,\gamma} = \sup_{\square} \sum_{\omega:\square\subset X_\omega} e^{\kappa \deg\omega}|O_\omega|^{a\to b}_\gamma,$$

and similarly for split forms and seam norms. Lemma 40.3(a) and Lemma 40.4 hold in these norms, with margins increased by 1: Hölder quotients at y are taken on $\Lambda_1(c_y)$. Their proofs

use only subadditivity of the seminorms, the submultiplicativity of Lemma 44.2(b), and the supports. In products, the tiers must match: each factor is measured from the tier that its right neighbour delivers. For split forms the composition of Lemma 44.2(c) needs, in addition, the decay norms of the factors on the right. For a split form of a single term we therefore put

$$|F|^{\mathrm{sp}} = |F|^{a\to b}_{\gamma;R,Y} + |F|_{\gamma},$$

and $\|O\|^{\mathrm{sp}}_{\kappa,\gamma}$ is defined with $|O_\omega|^{\mathrm{sp}}$. For a family that is measured in the decay norm only, $\|O\|^{\mathrm{sp}}_{\kappa,\gamma} = \|O\|_{\kappa,\gamma}$.

Lemma 44.3 (Products in split tier norms)**.** *Let $F_1, \dots, F_n$ be weakenable families with margins at most r (so that their tier norms are evaluated with margin $r+1 \le M/16$, as in the paragraph above), let $0 < \delta \le 1$ and $1 \le m \le n$. For $i \le m$ suppose that every term of F_i satisfies a split tier bound $a_i \to b_i$ on a set Y_i with radius R_i, where $a_i \le b_{i+1}$ in the order $0 < \frac12 < 1 < 2$, using $\mathfrak{n}^a_y \le C\mathfrak{n}^b_y$ for $a \le b \le 2$, and every block within distance R_i of Y_i lies in Y_{i+1}, for $i < m$. If $m < n$, suppose $a_m = 0$, and measure $F_{m+1}, \dots, F_n$ in the decay norm. Then every term of the product $F_1 \cdots F_n$, indexed as in Lemma 40.4, satisfies the split tier bound $a_m \to b_1$ on Y_1 with radius $R_1 + \cdots + R_m$, and*

$$\|F_1 \cdots F_n\|^{\mathrm{sp}}_{\kappa,\gamma} \le 3n \prod_{i=1}^{n} (C_d \delta^{-2} \|F_i\|^{\mathrm{sp}}_{\kappa+\delta,\gamma}).$$

The same holds in seam form: if one factor is measured in the seam version of its norm at rate γ, and all others at rate 2γ, then the product is bounded in the seam version at rate γ.

Proof. One term. Fix a tuple $(\omega_1, \dots, \omega_n)$ of the index set, write F_i for its terms, $c_i = |F_i|^{a_i\to b_i}_{\gamma;R_i,Y_i}$ for $i \le m$, and $P = F_{m+1} \cdots F_n$, with $P = 1$ if $m = n$. By Lemma 44.2(a) and Lemma 40.1(a), P satisfies the bound $0 \to 0$ at every block, with constant $|P|_\gamma \le \prod_{i>m} |F_i|_\gamma$. We show by induction, for $k = m, m-1, \dots, 1$,

Details

$$|F_k \cdots F_n|^{a_m\to b_k}_{\gamma;R_k+\cdots+R_m,Y_k} \le 2c_k \prod_{k<i\le m} (c_i + |F_i|_\gamma) \prod_{i>m} |F_i|_\gamma.$$

For $k = m$ and $m < n$ this is Lemma 44.2(c) with $F = F_m$ and $G = P$, whose bound holds at every block, with $R_G = \infty$. Since $a_m = 0$, the near part of the composite is of tier 0 and is moved to the far part, so the radius is R_m. For $m = n$ the claim is trivial. For $k < m$ apply Lemma 44.2(c) with $F = F_k$ and $G = F_{k+1} \cdots F_n$. Its inclusion hypothesis is that of the lemma, and $|G|_\gamma \le \prod_{i>k} |F_i|_\gamma$. So the left-hand side is at most c_k times the bound for G plus $|G|_\gamma$, which is at most $2c_k(c_{k+1} + |F_{k+1}|_\gamma) \prod_{k+1<i\le m}(c_i + |F_i|_\gamma) \prod_{i>m} |F_i|_\gamma$. Together with $|F_1 \cdots F_n|_\gamma \le \prod_i |F_i|_\gamma$ this gives $|F_1 \cdots F_n|^{\mathrm{sp}} \le 3 \prod_i |F_i|^{\mathrm{sp}}$, where $|F_i|^{\mathrm{sp}} = |F_i|_\gamma$ for $i > m$.

Summation. The index set, the domains, the monomials and the supports are those of Lemma 40.4. Its proof uses only the product form of the bound for one term, so it applies with $\|F_i\|^{\mathrm{sp}}$ in place of $\|F_i\|$. Two remarks complete the argument. First, the inclusion hypothesis is checked term by term. For a family $\sum_j s^{X(j)} \varphi_j G_j$ of local factors whose j-term satisfies its bound on Y_j and vanishes outside $K^{(2)}_j$, the j-term satisfies it on $Y_j \cup \{y : y \notin K^{(2)}_j\}$. Second, tier norms are weighted row sums only, and compositions need only row sums.

Seams. Replace Lemma 44.2(c) by its seam analogue in (e). The factors at rate 2γ absorb the seam weights, as in (41.2). □

44.2 Interior factors

Lemma 44.4 (Local Sobolev inequality)**.** *For v on Λ_2, $\|v\|_{4,\Lambda_1} \le C(\|\partial v\|_{\Lambda_2} + \|v\|_{\Lambda_2})$.*

Proof. Let χ be the cutoff of Step 1 of the proof of Theorem 6.3, rescaled to Λ_2, and $w = \chi v$, extended by zero. Then $w = \mathcal{G}(-\Delta w) = \sum_\mu (\partial_\mu^* g_\eta) \star (\partial_\mu w)$. The kernel is bounded by Ck_{d-1} (Proposition 6.1), so Lemma 43.1(c) with $s = d-1$, $p = 2$ and $q = \frac{2d}{d-2} = 4$ gives $\|w\|_4 \le C\|\partial w\|_2$. Finally, $\|\partial w\|_2 \le C(\|\partial v\|_{\Lambda_2} + \|v\|_{\Lambda_2})$. □ Details

This is the first step $L^2 \to L^4$ of the chain of the introduction, for the values; for the gradients it is Theorem 43.2(c) with $p = 2$. Both are used in Lemma 44.6.

Lemma 44.5 (Interior factors)**.** *Let G_j be a local propagator of level (L1) or (L3) on a domain $Z_j = \Lambda_j^v \cap X$, i.e. G'_{Z_j} or $\mathbf{G}^{Z_j}$. Let φ be a function supported in $K_j^{(1)}$ with $|\partial^k \varphi| \le C$ for $k \le 2$, and let $Y_j = \{y :$ all blocks within distance 8 of y lie in $(Z_j)_{(4)}\}$. Then the following tier bounds hold on Y_j, with radius 4, rate $\gamma_1 = \gamma_0/2$ and a constant C_T, independent of j, X, M, k, n, η:*

$$\begin{array}{lll} \varphi G_j : & 1 \to 3, & \\ \varphi G_j D^T : & 1 \to 2 \text{ (values)}, & 2 \to 3, \\ \varphi D G_j : & 1 \to 2, & \\ \varphi D G_j D^T : & 1 \to 1, & 2 \to 2. \end{array}$$

In the gauge case, $\varphi \mathbf{G}^{Z_j} D$ satisfies $2 \to 3$ on scalar data supported in Z_j, at distance at least 2η from $\mathbb{T}_\eta \setminus Z_j$. All these operators also have L^2 decay (C, γ_0). Outside $K_j^{(2)}$ they vanish. With $\varphi \equiv 1$ the same bounds hold at every $y \in Y_j$, with the same constants, since the proof below uses only the cube $\Lambda_3(c_y)$; then they need not vanish outside $K_j^{(2)}$.

For the scalar propagators, the entries from tier 0 and from tier $\frac{1}{2}$ are given in Lemma 44.6 below. The gauge propagators are needed only from tier 1. The entries $0 \to 2$ (values) for φG_j and $0 \to 1$ for $\varphi G_j D^T$ and $\varphi D G_j$ are not available at the endpoint: from L^2 data an interior propagator gives gradients in L^4 (Theorem 43.2(c) with $p = 2$), but neither bounded values nor gradients in L^8.

Proof. At $y \in Y_j$ the cube $\Lambda_3(c_y)$ is interior to Z_j. The bounds follow from the proof of Corollary 43.8: Details

- part (a) with $p = 2d = 8$ gives the entry $1 \to 3$ of φG_j, with $\beta \le 1 - \frac{d}{8} = \frac{1}{2}$, and the entry $2 \to 3$ of $\varphi G_j D^T$, with the Hölder data entering through Proposition 36.5;
- part (b) with $p = \frac{8}{3}$, so that $p^* = 8$, gives the entry $1 \to 2$ (values) of $\varphi G_j D^T$, with $\beta \le 2 - \frac{d}{p} = \frac{1}{2}$, and the entry $1 \to 1$ of $\varphi D G_j D^T$; the latter is the Calderón–Zygmund input;
- the entries $1 \to 2$ of $\varphi D G_j$ and $2 \to 2$ of $\varphi D G_j D^T$ follow from those of φG_j and $\varphi G_j D^T$, since $\mathfrak{n}_y^2(\nabla h) \le C\mathfrak{n}_y^3(h)$.

In that proof the local data enter only through their norms on $\Lambda_3(c_y)$, which gives the near part with radius 4. The rest enters through $\mathfrak{L}(u)$, which gives the far part, by the L^2 decay of the propagators with derivatives. Weighted row sums with rate $\gamma_1 < \gamma_0$ are bounded by $CS_{\gamma_0 - \gamma_1}$ (Lemma 40.1(b)). The L^2 inputs used in the far parts are the decay of G_j, DG_j, $G_j D^T$, $DG_j D^T$ (Theorems 18.2(b), 19.7(a)), and, in the gauge case, of $D_{Z_j}^T \mathbf{G}^{Z_j}$ and $D_{Z_j}^T \mathbf{G}^{Z_j} D^T$. The latter follow as in Theorem 19.7(a), since $\|\!|\!| \cdot |\!|\!|_X$ contains $\|D_X^T A\|$. Where the full gradient of $v = G_j g$ or $v = G_j D^T g$ is needed in L^2 near y (in Lemma 44.4, and in Section 41), it is controlled as follows. In the scalar case it is $\|Dv\|$. In the gauge case, for a cutoff χ supported in the interior, summation by parts on the full lattice gives $\|\partial(\chi v)\|^2 = \|d(\chi v)\|^2 + \|d^*(\chi v)\|^2$ in the local gauge, up to $C\varepsilon \|\!|\!| \chi v |\!|\!|^2$. So the full gradient is controlled by $\|Dv\|$, $\|D_{Z_j}^T v\|$ and $\|v\|$ near y. For $\varphi \mathbf{G}^{Z_j} D$, note that $D\lambda = \partial\lambda + T\lambda$ and $\partial_\mu \lambda = -\partial_\mu^*(S_\mu \lambda)$, so $D\lambda$ is a source in local form with Hölder data $S_\mu \lambda$. For λ supported in Z_j away from $\mathbb{T}_\eta \setminus Z_j$, the restriction

of $D\lambda$ to $\ell^2_1(Z_j)$ is $D\mathbf{1}_{Z_j}\lambda$, and the L^2 decay of $\mathbf{G}^{Z_j}D\mathbf{1}_{Z_j}$ is the transpose of that of $D^T_{Z_j}\mathbf{G}^{Z_j}$. Scalar data near the cut are treated in Lemma 44.8. Multiplication by φ preserves all tiers, by the Leibniz rule. □

Lemma 44.6 (Endpoint factors)**.** *Let $G'_j = G'_{Z_j}$ be a scalar local propagator of level (L1) on a domain $Z_j = \Lambda^v_j \cap X$, and let φ and Y_j be as in Lemma 44.5. Then the following tier bounds hold on Y_j, with radius 4, rate $\gamma_1 = \gamma_0/2$ and the constant C_T of Lemma 44.5, enlarged if necessary, independent of j, X, M, k, n, η:*

$$\begin{aligned} \varphi G'_j &: \quad 0 \to \tfrac12, \quad \tfrac12 \to 2 \ (\textit{values}), \\ \varphi DG'_j &: \quad 0 \to \tfrac12, \quad \tfrac12 \to 1. \end{aligned}$$

With $\varphi \equiv 1$ the same bounds hold at every $y \in Y_j$. In the entries from tier 0 both sums of Definition 44.1 involve only $\mathfrak{n}^0$. Moreover, at every block y,

$$|[aQ'^TQ', h_j]G'_j f|_{B(y)} \le CM^{-1}\|G'_j f\|_{B(y)},$$

so $[aQ'^TQ', h_j]G'_j$ maps $0 \to 1$ at every block, with constant CC_GM^{-1}.

Proof. Let $y \in Y_j$, $v = G'_j f$ and $\Lambda_r = \Lambda_r(c_y)$. All blocks within distance 8 of y lie in $(Z_j)_{(4)}$. Details
So Λ_4 lies in Z_j, away from $\mathbb{T}_\eta \setminus Z_j$, and $D^TDv = f - aQ'^TQ'v$ on Λ_4, as in the proof of Corollary 43.8. We work in the real gauge on Λ_4 in which (B) holds, as in Section 36. Two facts are used.

- $|Q'w(y')| \le C\|w\|_{B(y')}$, Q'^T is bounded from sup norms to sup norms (Proposition 22.10(d),(e)), and Q'^TQ' is block-local. Hence $\|aQ'^TQ'v\|_{p,\Lambda_r} \le C\|v\|_{\Lambda_{r+1}}$ for $p \le 4$ and $r \le 3$.
- By the L^2 decay of G'_j and DG'_j (Theorem 18.2(b)), $\|v\|_{\Lambda_4} + \|Dv\|_{\Lambda_4} \le \sum_{y'} k_0(y,y')\mathfrak{n}^0_{y'}(f)$, where $\sum_{y'} e^{\gamma_1|y-y'|}k_0(y,y') \le CC_GS_{\gamma_0-\gamma_1}$ (Lemma 40.1(b)).

The second fact gives the far part of all bounds below.

From tier 0. By Lemma 12.2, $-\Delta v = F + \sum_\mu \partial^*_\mu g_\mu$ on Λ_3, with $F = f - aQ'^TQ'v - bv$ and $g_\mu = -a_\mu v$. Theorem 43.2(c) with $p = 2$, so that $p^* = 4$, gives

$$\|\partial v\|_{4,\Lambda_1} \le C(\|F\|_{\Lambda_2} + \|g\|_{4,\Lambda_3} + \|v\|_{\Lambda_2}).$$

Here $\|F\|_{\Lambda_2} \le \|f\|_{\Lambda_2} + C\|v\|_{\Lambda_3}$, by the first fact and $|b| \le C\varepsilon$, and $\|g\|_{4,\Lambda_3} \le C\varepsilon\|v\|_{4,\Lambda_3}$. Cover Λ_3 by the cubes $\Lambda_1(x_i)$, with x_i on a grid of spacing 1 in Λ_2. There are at most 5^d of them, and $\Lambda_2(x_i) \subset \Lambda_4$. Lemma 44.4 on each of them gives $\|v\|_{4,\Lambda_3} \le C(\|\partial v\|_{\Lambda_4} + \|v\|_{\Lambda_4})$. Next, $D = \partial + T$ with $|\tau_\mu| \le 2e^2\varepsilon$, so $\|\partial v\|_{\Lambda_4} \le \|Dv\|_{\Lambda_4} + C\varepsilon\|v\|_{\Lambda_4}$. For the values, Lemma 44.4 on $\Lambda_1 \supset B(y)$ gives $\|v\|_{4,B(y)} \le C(\|\partial v\|_{\Lambda_2} + \|v\|_{\Lambda_2})$. Finally $\|Dv\|_{4,B(y)} \le \|\partial v\|_{4,\Lambda_1} + C\varepsilon\|v\|_{4,\Lambda_1}$. Together, for $\varepsilon \le 1$,

$$\mathfrak{n}^{1/2}_y(v) + \mathfrak{n}^{1/2}_y(Dv) \le C(\|f\|_{\Lambda_2} + \|v\|_{\Lambda_4} + \|Dv\|_{\Lambda_4}).$$

The first term is at most $\sum_{|y'-y|_\infty \le 2} \mathfrak{n}^0_{y'}(f)$, and the other two are bounded by the second fact. In the entries from tier 0 no absorption is used. The perturbations bv and $a_\mu v$ are bounded by the L^2 norms of v and Dv, which are available, so these entries need no smallness of ε beyond $\varepsilon \le 1$. The entries from tier $\frac12$ use Proposition 43.4(b), which is an absorption argument and needs $\varepsilon \le \varepsilon'_S$, one of the thresholds of Section 43. The averaging term has a coefficient of order 1, but it enters only as an L^2 source.

From tier $\frac12$. Apply the proof of Corollary 43.8(b) with $p = \frac83$, so that $p^* = 8$ and $\beta \le 2 - \frac{d}{p} = \frac12$. There Proposition 43.4(b) on Λ_3, with the source $f - aQ'^TQ'v$, gives

$$|v|_{\Lambda_1} + [v]_{\beta,\Lambda_1} + \|\partial v\|_{8,\Lambda_1} \le C(\|f\|_{8/3,\Lambda_3} + \|v\|_{\Lambda_4}).$$

By Hölder's inequality on Λ_3, which has volume 6^d,

$$\|f\|_{8/3,\Lambda_3} \le C\|f\|_{4,\Lambda_3} \le C \sum_{|y'-y|_\infty \le 3} \mathfrak{n}_{y'}^{1/2}(f).$$

Moreover $\|Dv\|_{8,B(y)} \le \|\partial v\|_{8,\Lambda_1} + C\varepsilon|v|_{\Lambda_1}$. With the second fact this gives the entries $\frac{1}{2} \to 2$ (values) and $\frac{1}{2} \to 1$, in split form with radius 4. The local data enter only through Λ_3, as in Lemma 44.5.

The cutoff. Multiplication by φ preserves the tiers $t \le 2$, by $|\varphi| + |\partial\varphi| \le C$ and the Leibniz rule for the Hölder quotient. With $\varphi \equiv 1$ there is nothing to do. The constants depend only on those of Theorem 43.2, Lemma 44.4, Proposition 43.4 and Theorem 18.2.

The averaging term. Since $Q'^T Q'$ is block-local, on $B(y)$ we have $[aQ'^T Q', h_j]v = aQ'^T Q'(\psi v) - \psi\, aQ'^T Q' v$ with $\psi = h_j - h_j(c_y)$, and $|\psi| \le CM^{-1}$ on $B(y)$. By the first fact, $|[aQ'^T Q', h_j]v|_{B(y)} \le CM^{-1}\|v\|_{B(y)}$. Since blocks have volume 1, $\mathfrak{n}_y^1 \le |\cdot|_{B(y)}$, and the L^2 decay of G'_j gives the tier bound. This uses no interior condition. □

For $X = \mathbb{T}_\eta$ the blocks of $K_j^{(2)}$ lie in Y_j, because K_j is at ∞-distance at least $(\rho-1)M - 1$ from $\partial\Lambda_j^v$. So for the top-level walks the bounds of Lemmas 44.5 and 44.6 hold at every block where the factors do not vanish. In particular they are global tier bounds.

Scalar gradients at the cut. The space $\ell_1^2(Z)$ contains the bonds crossing ∂Z. So for a scalar field λ on $\mathbb{T}_\eta$, the restriction of $D\lambda$ to $\ell_1^2(Z)$ is $D(\mathbf{1}_Z\lambda) + c_Z(\lambda)$, where $c_Z(\lambda)$ is supported on the crossing bonds and has size $\eta^{-1}|\lambda|$ at the exterior endpoint. This surface term is not bounded in L^2 uniformly in η. By duality, $\langle w, \mathbf{G}^Z c_Z(\lambda)\rangle = \mathrm{b}(w^*, \mathbf{G}^Z c_Z(\lambda))$ is bounded by $|\lambda|$ times the L^1 trace $\sum_b \eta^{d-1}|v(b)|$ of $v = \mathbf{G}^Z w^*$ on the crossing bonds (proof of Lemma 44.8). We need the following discrete trace inequality. A *rectangle* is a domain of the form $Z = \{x :\ x_\mu \in I_\mu,\ \mu = 1, \dots, d\}$, where each I_μ is either the whole circle or a lattice arc whose complement has at least two points. The local domains Λ_j^v are rectangles, at every $\eta \le 1$. Indeed, $(\Lambda_j^v)^{(R_v)} \subset X(j)$ with $R_v \ge 1$, and $X(j)$ does not wrap around the torus, by the torus side of Section 40. If in some direction the complement of the interval of Λ_j^v had fewer than $2R_v$ points, then $(\Lambda_j^v)^{(R_v)}$, and with it $X(j)$, would wrap around the torus in that direction. So each such complement has at least $2R_v \ge 2$ points. What the proof uses is the product structure, which rules out re-entrant corners. Write $\mathfrak{B}_{\mathrm{cr}}(y')$ for the bonds that have one endpoint in $Z \cap B(y')$ and the other endpoint outside Z.

Lemma 44.7 (Trace inequality on rectangles)**.** *Let Z be a rectangle, i.e. a product of intervals that is a union of unit blocks, so that each side that does not wrap around the torus has length at least 1; as in the definition of a rectangle, in each direction in which Z does not wrap around, the complement of its interval has at least two points. For a union of unit blocks this holds automatically if $\eta \le \frac{1}{2}$. Let U be a real $(1,\varepsilon)$-regular background with $\varepsilon \le 1$, or a complex perturbation of such a background of size δ_c small (Proposition 19.11). Then for every $A \in \ell_1^2(Z)$ and every block y' meeting ∂Z,*

$$\sum_{b \in \mathfrak{B}_{\mathrm{cr}}(y')} \eta^{d-1}|A(b)| \le C\big(\|A\| + \|DA\| + \|D_Z^T A\|\big)_{B(y')^{(2)}},$$

where the norms are those of $\|\!|\cdot|\!\|_Z$, localized to $B(y')^{(2)}$. Thus $\|DA\|$ runs over all plaquettes with a bond in $\mathfrak{B}(Z)$, including those crossing ∂Z, and $\|D_Z^T A\|$ over the points of Z. The crossing plaquettes carry the boundary condition. With interior plaquettes only, the lemma would be false: the lattice gradient of a harmonic function has normal trace only in $H^{-1/2}$. The constant depends only on d and $\dim \mathfrak{g}$.

The side length matters. For a slab of one fine layer, at $U = 1$, put $A_1 = a$ on the normal bonds crossing both faces and $A = 0$ elsewhere. Then $dA = 0$ and $d_Z^* A = 0$, while the localized norm is $O(\sqrt{\eta})|a|$ and the trace is of order $|a|$. The local domains Λ_j^v, and the rectangles to which the lemma is applied, are unions of unit blocks.

Proof. Step 1: an exact identity at $U = 1$. Let $\mathfrak{B}(Z)$ be the set of bonds with an endpoint in Z, so that $A \in \ell_1^2(Z)$ vanishes off $\mathfrak{B}(Z)$. By Lemma 2.2 on the torus, Details

$$\|dA\|^2 + \|d^*A\|^2 = \|\partial A\|^2 = \sum_{\mu,\nu} \|\partial_\mu A_\nu\|^2,$$

a sum over all pairs of parallel bonds at distance η. Split $d^*A = \mathbf{1}_Z d^*A + \mathbf{1}_{Z^c} d^*A$. Let $x \notin Z$. A bond at x lies in $\mathfrak{B}(Z)$ only if its other endpoint lies in Z. Since Z is a product of intervals, x has at most one neighbour in Z. Indeed, x must differ from Z in exactly one coordinate ν, and by exactly η. So $(d^*A)(x) = \pm\eta^{-1}A(b_x)$, where b_x is the unique bond from x into Z, or $(d^*A)(x) = 0$ if there is none. Let b'_x be the bond parallel to b_x that continues outward from x. It has no endpoint in Z, so $A(b'_x) = 0$, and the pair (b_x, b'_x) contributes exactly $\eta^d \eta^{-2}|A(b_x)|^2$ to $\|\partial A\|^2$. The pairs (b_x, b'_x) are distinct for distinct x. Hence

$$\|dA\|^2 + \|d_Z^* A\|^2 = \|\partial A\|^2 - \|\mathbf{1}_{Z^c} d^*A\|^2 \;\geq\; \|\partial A\|^2_{\mathfrak{B}(Z)},$$

where $\|\partial A\|^2_{\mathfrak{B}(Z)}$ is the sum over the pairs of parallel bonds that *both* lie in $\mathfrak{B}(Z)$. All omitted terms are non-negative. Convexity of the rectangle enters through the uniqueness of b_x.

Step 2: a one-dimensional trace. Let $b \in \mathfrak{B}_{\mathrm{cr}}(y')$ point in the direction $\pm e_\nu$ normal to the face. The parallel bonds $b_m = b \mp m\eta e_\nu$, $0 \leq m < \eta^{-1}$, lie in $\mathfrak{B}(Z)$, since Z is a union of blocks and has side at least 1. Consecutive ones form pairs of $\|\partial A\|_{\mathfrak{B}(Z)}$. Averaging $|A(b)| \leq |A(b_m)| + \sum_{j<m} |A(b_j) - A(b_{j+1})|$ over m and summing over the crossing bonds of $B(y')$ with weight η^{d-1} gives

$$\sum_{b \in \mathfrak{B}_{\mathrm{cr}}(y')} \eta^{d-1}|A(b)| \leq C\big(\|A\| + \|\partial A\|_{\mathfrak{B}(Z)}\big)_{B(y')^{(1)}},$$

by Cauchy–Schwarz on a region of bounded volume.

Step 3: localization. Let χ be a smooth cutoff at unit scale, equal to 1 on $B(y')^{(1)}$ and supported in $B(y')^{(2)}$, and put $(\chi A)(b) = \chi(b_-)A(b)$. Then $\chi A \in \ell_1^2(Z)$. Moreover $d(\chi A) - \chi\, dA$ and $d_Z^*(\chi A) - \chi\, d_Z^* A$ are sums of differences of χ times shifted values of A, bounded by $C\|A\|_{B(y')^{(2)}}$. Apply Steps 1 and 2 to χA.

Step 4: the background. All quantities in the lemma are invariant under real gauge transformations. By regularity, on a cube containing $B(y')^{(3)}$ (the regularity cubes have half-side $4M+2 \geq 6$) there is a real gauge with $U = e^{\mathrm{i}\eta A_0}$ satisfying (B). There, applying Lemma 12.2 in this local gauge, $D = \partial + T$ and $D^T = \partial^* + T^T$, with T local and $|T| \leq C\varepsilon$. So $\|d(\chi A)\| \leq \|D(\chi A)\| + C\varepsilon\|\chi A\|$, and similarly for d_Z^*. Finally, $\|D(\chi A)\| \leq \|DA\|_{B(y')^{(2)}} + C\|A\|_{B(y')^{(2)}}$ by the Leibniz rule for D, and similarly for D_Z^T. For complex perturbations, $D_{U_\delta} = D_U + T_\delta$ with T_δ local of size $C\delta_c$ (Proposition 19.11). □

The continuum analogue is the Gaffney–Friedrichs inequality for fields with vanishing tangential trace on a convex domain. On a rectangle the lattice version is an exact identity. The exterior divergence cancels exactly against the outward normal jumps. The remaining boundary terms, the tangential jumps, are non-negative and are dropped. At re-entrant corners the uniqueness of b_x, and with it the inequality, would fail, as in the continuum.

Lemma 44.8 (Cut term)**.** *Let Z be a rectangle in the sense of Lemma 44.7. Then, for scalar λ on $\mathbb{T}_\eta$,*

$$\|\mathbf{G}^Z c_Z(\lambda)\|_{B(y)} + \|D\mathbf{G}^Z c_Z(\lambda)\|_{B(y)} + \|D_Z^T \mathbf{G}^Z c_Z(\lambda)\|_{B(y)} \leq C \sum_{y'} e^{-\gamma_1|y-y'|}|\lambda|_{B(y')^{(1)}},$$

the sum running over the blocks y' of Z meeting ∂Z. The neighbourhood $B(y')^{(1)}$ is needed because $c_Z(\lambda)$ takes λ at the exterior endpoint of the crossing bonds. Consequently, $\mathbf{1}_{\ell^2_1(Z)}D$ followed by $\varphi\mathbf{G}^Z$ satisfies the bound $2 \to 3$ of Lemma 44.5 at blocks of Y_j, also for $\varphi \equiv 1$, with a far part in $\mathfrak{n}^2$ instead of $\mathfrak{n}^0$ at the blocks meeting ∂Z.

Proof. L^2 input. Theorem 19.7(a), proved with Combes–Thomas in the norm $\|\!|\!| \cdot |\!|\!|_{X,+}$, gives decay (C, γ_0) for all operators $P\,\mathbf{G}^X Q$ with $P \in \{1, D, D_X^T\}$ and $Q \in \{1, D^T, D\mathbf{1}_X\}$. For $Q = D\mathbf{1}_X$ this uses $\langle u, D\mathbf{1}_X g\rangle = \langle D_X^T u, g\rangle \le |\!|\!|u|\!|\!|_X \|g\|$. Details

Proof of the bound. For a field u, $\|u\|_{B(y)}$ is the supremum of $|\langle w, u\rangle| = |\mathrm{b}(w^*, u)|$ over the fields w supported in $B(y)$ with $\|w\| \le 1$. Here w^* is the pointwise adjoint, which has the same support and norm. The operator $\mathbf{G}^Z$ is symmetric for the bilinear form b. At real references this is its symmetry, since b-transposes of real operators are their adjoints on real fields. At complex references the kernel of $\mathbf{G}^Z$ is holomorphic in the perturbation, and the identity $(\mathbf{G}^Z)^T = \mathbf{G}^Z$ extends from the real points by the identity theorem, as in step (iii) of the proof of Proposition 46.7. Hence $\mathrm{b}(w^*, \mathbf{G}^Z c_Z(\lambda)) = \mathrm{b}(\mathbf{G}^Z w^*, c_Z(\lambda))$, and similarly with $D^T w^*$ and $D\mathbf{1}_Z w^*$ for the derivatives. Moreover $|\mathrm{b}(v, c_Z(\lambda))| \le C \sum_b \eta^{d-1} |v(b)|\,|\lambda|$, the sum running over the crossing bonds, with λ taken at their exterior endpoints. Apply Lemma 44.7 to $v = \mathbf{G}^Z w^*$, respectively $\mathbf{G}^Z D^T w^*$ and $\mathbf{G}^Z D\mathbf{1}_Z w^*$. This needs $|\!|\!|v|\!|\!|_Z$ near the crossing blocks, which the L^2 input just stated provides, with exponential decay.

The consequence. Write $\lambda = \lambda_{\mathrm{in}} + \lambda_{\mathrm{cut}}$ with a smooth cutoff, so that λ_{in} vanishes within distance 2 of $\mathbb{T}_\eta \setminus Z$ and λ_{cut} is supported within distance 3 of it. The source $D(\mathbf{1}_Z \lambda_{\mathrm{in}})$ is covered by Lemma 44.5. At $y \in Y_j$, the source $D(\mathbf{1}_Z \lambda_{\mathrm{cut}}) + c_Z(\lambda)$ is far. It enters only through the L^2 decay of $\{1, D, D_Z^T\}\mathbf{G}^Z D\mathbf{1}_Z$ and the bound just proved, i.e. as a far part. That far part is of tier 0 for λ_{cut}, and is bounded by $|\lambda|$ for c_Z. □

A far part in $\mathfrak{n}^2$ is harmless in global tier norms. It is used only in Theorem 44.13(b) and Proposition 44.11, where the input is globally of tier 2.

44.3 The scalar levels on domains: lifting

The middle factors of the non-local part at level (L3) contain the scalar walks of the levels (L1), (L2) on $\mathbb{T}_\eta$ and on the local domains. In the steps they act on fields that are controlled only in L^2 near the cuts. The following lemma bounds these walks in split form: regularity of the input is needed only near the output block, and only for the term without steps.

Lemma 44.9 (Lift lemma)**.** *Let Z be a domain, and let $G'^Z(s)$, $P_{\mathcal{V}}^Z(s)$ and $R_Z(s)$ be the weakened families of the levels (L1), (L2) on Z (Section 41), with $M \ge M_1$ of Theorem 41.4 at these levels, at the weakening rate $\kappa + \delta$, and $(\rho - 1)M \ge 16$. Let $Y = Z_{(24)}$. The following split tier bounds hold on Y, with radius 8, near tier as indicated and far tier 0, rate $2\gamma_0$, and constants $C(\kappa, \delta)$ at κ, the families being evaluated at $\kappa + \delta$:*

(Z1) $G'^Z(s)\colon 1 \to 2$ (values), and $DG'^Z(s)\colon 1 \to 1$. Only the term of the walk without steps uses the near part;

(Z2) $P_{\mathcal{V}}^Z(s)\colon 0 \to 2$ (values), and $DP_{\mathcal{V}}^Z(s)\colon 0 \to 1$;

(Z3) $R_Z(s)\colon t \to t$ for $t \in \{0, \frac{1}{2}, 1, 2\}$. Here, inside products, $R_Z(s) = \mathbf{1}_Z - P_{\mathcal{V}}^Z(s)$ with the literal identity, which needs no monomial. No tier bound is claimed for $R_Z(s)$ standing alone. It is used alone only in L^2, with its identity split box by box (Section 40), and that split does not preserve tiers.

For a domain X', the differences $\mathbf{1}_{X'} O^{\mathbb{T}_\eta}(s) \mathbf{1}_{X'} - O^{X'}(s)$, for the families O of (Z1)–(Z3), satisfy the same bounds in seam form on $X'_{(24)}$, with rate γ_0 and the seam $\Sigma_w(\mathbb{T}_\eta, X')$. Here γ_0 is the rate of level (L3), fixed in the paragraph Rates *below; the lemma is used only at that level. The constants do not depend on M, k, n, η or the background.*

Proof. Structure of the terms. At level (L1), $\mathfrak{N} = 0$ and $\mathcal{A} = 0$. A term of $G'^Z(s)$ with n steps is a product $\Phi_0\Phi_1\cdots\Phi_n$ with $\Phi_0 = \mathcal{F}_0 = \sum_j s^{X(j)} h_j G'_j$, $G'_j = G'_{\Lambda_j^v \cap Z}$, and $\Phi_i = \mathcal{B}$ for $i \geq 1$. As in the proof of Theorem 44.13 (local parts), the derivative in front of $D^T(\varphi G'_j)$ is expanded onto G'_j. So $\mathcal{B}$ is a sum of terms $s^{X(j)}\varphi' G'_j$, $s^{X(j)}\varphi D G'_j$ and $s^{X(j)}[aQ'^T Q', h_j]G'_j$, up to shifts by at most η, with φ', φ supported in $K_j^{(1)}$, $|\partial^k \varphi'| \leq C M^{-2}$ and $|\partial^k \varphi| \leq C M^{-1}$ for $k \leq 2$. Similarly $D\mathcal{F}_0 = \sum_j s^{X(j)}(h_j D G'_j + [D, h_j]G'_j)$, where $[D, h_j]$ is a multiplication by differences of h_j, up to a shift. The rates of the levels (L1), (L2) are at least $32\gamma_0$ (paragraph *Rates*), so all bounds below hold at rate $2\gamma_0$, and their seam forms at rate γ_0.

Interior blocks. Let $y \in Z_{(16)}$ be within distance 2 of K_j. The blocks within distance 8 of y are at distance more than 4 from $\mathbb{T}_\eta \setminus Z$, and, since K_j is at ∞-distance at least $(\rho - 1)M - 1 \geq 15$ from $\mathbb{T}_\eta \setminus \Lambda_j^v$, also from $\mathbb{T}_\eta \setminus \Lambda_j^v$. So $y \in Y_j$ for $Z_j = \Lambda_j^v \cap Z$. At blocks farther from K_j the j-terms of $\mathcal{F}_0$, $D\mathcal{F}_0$ and $\mathcal{B}$ vanish. Hence, by Lemmas 44.5 and 44.6, every term of these families satisfies on $Z_{(16)}$, with radius 5 (the shifts cost 1),

$$\mathcal{F}_0 \colon \tfrac{1}{2} \to 2 \text{ (values)}, \qquad D\mathcal{F}_0 \colon \tfrac{1}{2} \to 1, \qquad \mathcal{B} \colon 0 \to \tfrac{1}{2}.$$

For the averaging terms of $\mathcal{B}$ this holds at every block, by Lemma 44.6 and $\mathfrak{n}^{1/2} \leq \mathfrak{n}^1$. The split norms $\|\cdot\|^{\mathrm{sp}}_{\kappa+\delta,2\gamma_0}$ of $\mathcal{F}_0$ and $D\mathcal{F}_0$ are at most $Ce^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} \max(C_T, C_G)$, and that of $\mathcal{B}$ at most $Ce^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} \max(C_T, C_G)M^{-1} + \theta$, with θ of Lemma 41.3 at level (L1). Here C includes the number of j with $\square \subset X(j)$.

(Z1). The term without steps is $\mathcal{F}_0$, respectively $D\mathcal{F}_0$. It maps $\frac{1}{2} \to 2$ (values), respectively $\frac{1}{2} \to 1$, and hence $1 \to 2$, respectively $1 \to 1$, since $\mathfrak{n}^{1/2} \leq \mathfrak{n}^1$. A term with $n \geq 1$ steps is $\Phi_0\Phi_1(\Phi_2\cdots\Phi_n)$. Apply Lemma 44.3 with $m = 2$: Φ_0 in $\frac{1}{2} \to 2$ (values), respectively $D\Phi_0$ in $\frac{1}{2} \to 1$, on $Z_{(24)}$; Φ_1 in $0 \to \frac{1}{2}$ on $Z_{(16)}$; and $\Phi_2, \dots, \Phi_n$ in the decay norm. The blocks within distance 5 of $Z_{(24)}$ lie in $Z_{(16)}$. The result is a bound $0 \to 2$, respectively $0 \to 1$, with no near part, and

$$\|\Phi_0\cdots\Phi_n\|^{\mathrm{sp}}_{\kappa,2\gamma_0} \leq 3(n+1)\big(C_d\delta^{-2}\|\mathcal{F}_0\|^{\mathrm{sp}}_{\kappa+\delta,2\gamma_0}\big)\big(C_d\delta^{-2}\|\mathcal{B}\|^{\mathrm{sp}}_{\kappa+\delta,2\gamma_0}\big)\big(C_d\delta^{-2}\theta\big)^{n-1}.$$

The sum over n converges under the condition $2C_d\delta^{-2}\theta \leq \frac{1}{2}$ of Theorem 41.4(a), which holds for $M \geq M_1$. The near part enters only through the term without steps, with radius $5 \leq 8$.

(Z2). $P_{\mathcal{V}}^Z(s) = G'^Z(s)Q'^T(M'^Z)^{-1}(s)Q'G'^Z(s)$ is a product of weakenable families and local operators (Section 41, (L2)). The family $Q'G'^Z(s)$ maps L^2 to coarse fields, with a decay norm bounded by (L1), and $(M'^Z)^{-1}(s)$ is a unit-lattice family bounded by (L2). For coarse fields all tiers are $|\omega(y)|$. The operator Q'^T maps them to fields that are bounded on each block, with $\mathfrak{n}^1_y(Q'^T\omega) \leq |Q'^T\omega|_{B(y)} \leq C|\omega(y)|$. So $Q'^T(M'^Z)^{-1}(s)Q'G'^Z(s)$ maps $0 \to 1$ at every block, with no near part. Lemma 44.3 with $m = 2$, with (Z1) for $G'^Z(s)$, respectively $DG'^Z(s)$, on $Z_{(24)}$, gives (Z2).

(Z3). At blocks of $Z_{(24)}$ the identity preserves all tiers, and $P_{\mathcal{V}}^Z(s)$ maps $0 \to 2$, hence $0 \to t$ for $t \leq 2$, by (Z2).

Seams. Telescope the differences as in the proof of Theorem 41.4(b), on $\mathbb{T}_\eta$ and on X'. Each difference is a sum of products with one distinguished factor, and the sum over its position is absorbed as there. Let $y \in X'_{(24)}$.

- Let the distinguished factor be Φ_0 or Φ_1, and a difference of local propagators: $\varphi(G'_{Z_1} - G'_{Z_2})$, $\varphi D(G'_{Z_1} - G'_{Z_2})$ or the averaging commutator of such a difference, with $Z_1 = \Lambda_j^v$, $Z_2 = \Lambda_{j'}^v \cap X'$, between factors $\mathbf{1}_{X'}$. Its argument f is supported in X', so $\mathbf{1}_{Z_1} f = \mathbf{1}_{Z_2} f$. Hence $\delta v = (G'_{Z_1} - G'_{Z_2})f$ satisfies $D^T D\delta v = -aQ'^T Q'\delta v$ at the points of Z_2 away from $\mathbb{T}_\eta \setminus Z_2$. At the blocks within distance 5 of y where $\varphi \neq 0$, the cube Λ_4 around the block consists of such points. Proposition 36.3, first bound, with the source $-aQ'^T Q'\delta v$, bounds $\mathfrak{n}^3$ of δv at such a block y' by $C\sum_{|y''-y'|_\infty \leq 4} \|\delta v\|_{B(y'')}$. By Theorem 18.2(e)

these L^2 norms are of seam form, for a seam contained in $\Sigma_w(\mathbb{T}_\eta, X')$. So this factor maps $0 \to 2$ in seam form, with no near part. The factors to its left are then bounded as in (Z1), by the seam version of Lemma 44.3.

- Let the distinguished factor be a difference at a position $k \geq 2$. It is of seam form in L^2, as in the proof of Theorem 41.4(b). Then $\Phi_1 \colon 0 \to \frac{1}{2}$ and $\Phi_0 \colon \frac{1}{2} \to 2$, respectively $\frac{1}{2} \to 1$, act as in (Z1), and the seam version of Lemma 44.3 composes.

- Let the distinguished factor be a cross term $\mathbf{1}_{X'}\Phi_k\mathbf{1}_{\mathbb{T}_\eta \setminus X'}$ or a factor whose local domain $\Lambda^v_{j'} \cap X'$ is empty. It is supported, on its input or on its output side, within distance $r_0 + 1 \leq 3$ of $\mathbb{T}_\eta \setminus X'$, i.e. in the seam; here $r_0 \leq 2$ is the range of the local part at level (L1), since $Q'^T Q'$ is block-local. On the seam the seam norm at rate γ_0 is bounded by the norm at rate $2\gamma_0$. For $k \geq 2$ the factor is treated as in the previous item. For $k \leq 1$, the blocks within distance 5 of y are at distance more than 16 from this support. So the near part of the bound of Φ_0 at y does not see it, and the bound $0 \to \frac{1}{2}$ of Φ_1 has no near part. Hence only far parts, i.e. L^2 norms, see the support, and the seam weight follows.

For (Z2), (Z3), telescope the product defining $P^Z_{\mathcal{V}}(s)$. The distinguished factor is a difference of $G'^Z(s)$, bounded by the seam part of (Z1) with an input that is of tier 1 at every block, or a difference of $Q'G'^Z(s)$ or $(M'^Z)^{-1}(s)$, of seam form in L^2 by Theorem 41.4(b) at (L1), (L2). The seam version of Lemma 44.3 composes.

Constants. They are products of C_T, C_G, the constants of the walks of (L1), (L2) in Theorem 41.4, $e^{(\kappa+\delta)\mathsf{N}_\mathfrak{b}}$ and powers of δ^{-1}. None of them depends on M; the factors M^{-1} in the norms of $\mathcal{B}$ only help. □

44.4 Step bounds for the top-level walks

We now take $X = \mathbb{T}_\eta$ at the levels (L1) and (L3). Write G_j for the local propagators on Λ^v_j and $u = G_j g$.

Rates. At each of these levels let γ_0 be at most:

- the rate of (W2), from Theorems 18.2 and 19.7;
- the rate γ of Corollary 43.8, which enters the far parts of Lemma 44.5;
- at (L3), $\frac{1}{32}$ of the smaller of the output and compatibility rates of the walks of (L1) and (L2) (Theorem 41.4).

Then Lemmas 44.5 and 44.6 hold with rate $\gamma_0/2$. At level (L3) the inputs from the levels (L1), (L2) have rates at least $32\gamma_0$, so Lemmas 44.9 and 44.10 hold with rate $2\gamma_0$, and γ_0 in seam form, as (W3) requires. The walks are estimated with rate $\gamma = \gamma_0/8$. The rates decrease by fixed factors at each level. There are boundedly many levels, so the final rates are positive constants.

Lemma 44.10 (The non-local part)**.** *Let $Z \in \{\mathbb{T}_\eta\} \cup \{\Lambda^v_j\}$. At level (L3), $\mathfrak{N}^Z(s)$ is the sum of the factorized terms $X_i^T M_i^Z(s) Y_i$ listed in the proof, with $X_i^T \in \{1, D^T\}$ and $Y_i \in \{D_Z^T, D, 1\}$. Each middle factor $M_i^Z(s)$ is a product of the weakened scalar families $G'^Z(s)$, $R_Z(s)$, $P^Z_{\mathcal{V}}(s)$ of (L1), (L2), with the derivatives D, D^T attached to them, and of the local operators $\mathfrak{F}$, $\mathfrak{F}^T$ and W of order 0. A left gradient belongs to the middle factor and falls on a scalar family. The operators $\mathfrak{F}$, $\mathfrak{F}^T$ and W enter only as multiplication operators between such families, so bounds on $M_i^Z(s)$ need only the sup norms and Hölder quotients of their coefficients, never their derivatives.*

Let $W = 0$, and let M satisfy the conditions of Lemma 44.9 at $\kappa + \delta$. Then the middle factors satisfy the following split tier bounds on $Z_{(48)}$, with radius $R_N = 24$, near tier as indicated and far tier 0, rate $2\gamma_0$, and constants $C'_{\mathfrak{N}}(\kappa, \delta)$ at κ:

(N1) if $X_i^T = 1$, then $M_i^Z(s)\colon 1 \to 1$; for the term $-DP_{\mathcal{V}}D^T$, $M_i^Z(s) = DP_{\mathcal{V}}^Z(s)\colon 0 \to 1$;

(N2) if $X_i^T = D^T$, then $M_i^Z(s) = \mathfrak{F}G'^Z(s)R_Z(s)\colon 1 \to 2$;

(N3) for $X' = \Lambda_j^v$, the differences $\mathbf{1}_{X'}M_i^{\mathbb{T}_\eta}(s)\mathbf{1}_{X'} - M_i^{X'}(s)$ satisfy the bounds of (N1), (N2) in seam form on $X'_{(48)}$, with rate γ_0 and the seam $\Sigma_w(\mathbb{T}_\eta, X')$.

In (N2) the β-Hölder quotients of the coefficients of $\mathfrak{F}$ enter; elsewhere only sup norms. The constants $C'_{\mathfrak{N}}(\kappa, \delta)$ do not depend on M, k, n, η or the background.

Proof. By Section 41, (L3), and $(\mathfrak{B}DS)^T = S^T(\mathfrak{F}^T D + D^T W)$ with $S^T = DRG'$, the terms of $\mathfrak{N}$ are the following (all on Z, at s). As in Section 41, a left gradient D is part of the middle factor. Details

(i) $-DP_{\mathcal{V}}D^T$: $X^T = 1$, $M = DP_{\mathcal{V}}$, $Y = D^T$.

(ii) $-D^T\mathfrak{F}S$ and $-WDS$, with $S = G'RD^T$: here $X^T = D^T$, respectively $X^T = 1$, $Y = D^T$, and $M = \mathfrak{F}G'R$, respectively $M = WDG'R$.

(iii) $-DRG'\mathfrak{F}^T D$ and $-DRG'D^T W$: $X^T = 1$, $M = DRG'\mathfrak{F}^T$, respectively $M = DRG'D^T W$, and $Y = D$, respectively $Y = 1$.

(iv) $S^T D^T \mathfrak{B}DS = DRG'(\mathfrak{F}^T\mathfrak{F} + D^T W D)G'RD^T$: $X^T = 1$, $M = DRG'\mathfrak{F}^T\mathfrak{F}G'R + (DRG'D^T)\,W\,(DG'R)$, $Y = D^T$.

In each middle factor the derivatives stand next to the scalar families: $DR = D - DP_{\mathcal{V}}$, DG', $G'D^T$ and $DG'D^T$. The factors $\mathfrak{F}$, $\mathfrak{F}^T$ and W stand between them.

The tier bounds. Let $W = 0$, so that the terms containing W vanish. Inside the middle factors, $R = \mathbf{1}_Z - P_{\mathcal{V}}$ and $DR = D - DP_{\mathcal{V}}$ with the literal identity (Lemma 44.9(Z3)). The operators $\mathfrak{F}$ and $\mathfrak{F}^T$ are multiplications by the coefficients $c_{\mu\nu}$ of Lemma 12.3, or their transposes, composed with shifts by at most 2η. In the local gauge, $|c_{\mu\nu}| \le C\varepsilon$ and $[c_{\mu\nu}]_{\beta,\Lambda_1(c_y)} \le C[c_{\mu\nu}]_{\alpha,\Lambda_3(c_y)} \le C\varepsilon$, since $\beta \le \alpha$. So $\mathfrak{F}$ and $\mathfrak{F}^T$ satisfy the bounds $t \to t$, $t \in \{0, 1\}$, and $\mathfrak{F}$ also $2 \to 2$, at every block, with radius 1, no far part and constant $C\varepsilon$; for $2 \to 2$ use the Leibniz rule for the Hölder quotient. Let $y \in Z_{(48)}$. In each product below the radii of the factors add up to at most 17, so the near blocks of every factor of the walks lie in $Z_{(31)} \subset Z_{(24)}$, where Lemma 44.9 applies. So the inclusions of Lemma 44.3 hold, with the sets $Z_{(48-r)}$, r the sum of the radii of the factors to the left. Local operators are composed term by term with Lemma 44.2(c) and Lemma 40.5(a).

(i) $DP_{\mathcal{V}}\colon 0 \to 1$, by (Z2).

(ii) $\mathfrak{F}G'R = \mathfrak{F}G' - \mathfrak{F}G'P_{\mathcal{V}}$. By (Z1), $G'\colon 1 \to 2$ (values) with radius 8, and by (Z1), (Z2), $G'P_{\mathcal{V}}\colon 0 \to 2$. Then $\mathfrak{F}\colon 2 \to 2$ gives $1 \to 2$ with radius 9. This is (N2).

(iii) $DRG'\mathfrak{F}^T = DG'\mathfrak{F}^T - DP_{\mathcal{V}}G'\mathfrak{F}^T$. By (Z1), $DG'\colon 1 \to 1$ with radius 8, and $\mathfrak{F}^T\colon 1 \to 1$ with radius 1. By (Z2), $DP_{\mathcal{V}}\colon 0 \to 1$, and $G'\mathfrak{F}^T$ is measured in the decay norm. So $M\colon 1 \to 1$ with radius 9.

(iv) $DRG'\mathfrak{F}^T\mathfrak{F}G'R$. Expand both factors R. The term $DG'\mathfrak{F}^T\mathfrak{F}G'$ maps $1 \to 1$ with radius at most 17: the inner $G'\colon 1 \to 2 \subset 1$ (values), $\mathfrak{F}^T\mathfrak{F}\colon 1 \to 1$, and the outer $DG'\colon 1 \to 1$. In the three terms containing $P_{\mathcal{V}}$, either the left factor is $DP_{\mathcal{V}}\colon 0 \to 1$, followed by factors in the decay norm, or the right factor is $G'P_{\mathcal{V}}\colon 0 \to 2$, followed on the left by $\mathfrak{F}^T\mathfrak{F}$ and DG'. These terms map $0 \to 1$.

This gives (N1) and (N2), with radius at most $17 \le R_N$. The decay norms of the middle factors are bounded by (W3). Lemma 44.3 sums the terms of the families. The number of products is bounded, so $C'_{\mathfrak{N}}(\kappa,\delta)$ is a product of constants of Lemma 44.9 and of $C\varepsilon \le C$, and does not depend on M.

Seams. The middle factors on X' are built in the same way from the families on X'. Telescope the differences as in the proof of Theorem 41.4(b). Each product contains one distinguished factor: a difference of one of the families of Lemma 44.9, bounded by its seam part on $X'_{(24)}$, or a cross term $\mathbf{1}_{X'}F(\mathbf{1}_{\mathbb{T}_\eta}-\mathbf{1}_{X'})$, supported on the seam, where the seam norm at rate γ_0 is bounded by the norm at rate $2\gamma_0$. The operators $\mathfrak{F}$, $\mathfrak{F}^T$ have range 1, and they agree on the two domains at the points of X' away from the seam. The seam version of Lemma 44.3 composes, with the other factors at rate $2\gamma_0$. This gives (N3). □

Local and non-local parts. Write the step operator of the walk at level (L3) on $\mathbb{T}_\eta$ as in Section 41,

$$\mathcal{R}(s) = D^T\mathcal{A}(s) + \mathcal{B}(s), \qquad \mathcal{A} = \mathcal{A}^{\mathrm{loc}} + \mathcal{A}^{\mathrm{nl}}, \qquad \mathcal{B} = \mathcal{B}^{\mathrm{loc}} + \mathcal{B}^{\mathrm{nl}}.$$

The local parts collect the terms that come from the local commutators $[\mathfrak{A}_{\mathrm{loc}}, h_j]G_j$. The non-local parts collect those that come from the non-local commutators $[\mathfrak{N}(s), h_j]G_j$ and from the seam terms $h_j\mathbf{1}_{X'}(\mathfrak{N}(s)-\mathfrak{N}^{X'}(s))\mathbf{1}_{X'}G_j$, $X' = \Lambda_j^v$. All of them are taken in the regrouped form of the proof of Lemma 41.3. At level (L1) there are no non-local parts, and $\mathcal{A} = 0$. The local parts are bounded in tier norms by Lemma 44.5 (proof of Theorem 44.13 below).

The non-local parts are where the Dirichlet cuts enter. Their middle factors (Lemma 44.10) contain the scalar walks of (L1), (L2), on $\mathbb{T}_\eta$ and on the local domains, and in the step they act on Y_iG_jg, a field that is controlled only in L^2 near $\partial\Lambda_j^v$. By Lemma 44.10 the middle factors need regularity of their input only near the output block. The commutators with h_j and the seam terms restrict either the output, or the near part of the input, to a neighbourhood of K_j, where Y_iG_jg is regular.

Proposition 44.11 (The non-local parts of the steps)**.** *Let $\beta = \min(\alpha, \frac{1}{2})$ as in the Setting. Let γ_0 be the rate of level (L3), $\gamma = \gamma_0/8$, and let κ, δ and ρ be as in Section 41. Put*

$$\theta_{\mathfrak{N}}(\kappa) = Ce^{\kappa\mathsf{N}_{\mathrm{b}}}\delta^{-1}\big((\gamma_0 M)^{-1} + e^{-\gamma_0(\rho-1)M/16}\big)C'_{\mathfrak{N}}(\kappa+\delta,\delta)\max(C_T, C_G)^2,$$
$$C^D_{\mathfrak{N}}(\kappa) = Ce^{\kappa\mathsf{N}_{\mathrm{b}}}\delta^{-1}C'_{\mathfrak{N}}(\kappa+\delta,\delta)\max(C_T, C_G)^2,$$

with $C'_{\mathfrak{N}}$ of Lemma 44.10, $C_T \ge 1$ of Lemmas 44.5 and 44.6, $C_G \ge 1$ of (W2), and C depending only on d, ρ, C_v and $\dim\mathfrak{g}$. Let

$$M_{\mathfrak{N}} = \max\big(M_1,\ 16(r+1),\ (2R_N+22)/(\rho-1)\big),$$

where M_1 is that of Theorem 41.4 at the levels (L1), (L2), with the rates of Section 41, at the weakening rate $\kappa+3\delta$: Lemma 44.10 at $\kappa+\delta$ uses Lemma 44.9 at $\kappa+2\delta$, whose products use the walks at $\kappa+3\delta$. Further, $R_N = 24$ is the radius of Lemma 44.10, and r is the largest margin of the weakened families used. In M_1 the multiplicative constants of Part IX are multiplied by 2; the coercivity constants are the fixed ones of Proposition 46.7, and every decay bound that enters is taken at the fixed rate of the differences there, a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3; neither depends on δ_c, so the same numbers serve at real and at complex references (Proposition 46.7, step (vii) of the proof). By that proposition the doubled constants bound their worst case over complex references of size $\delta_c < \delta_P$, since $\delta_P \le \bar\delta \le 1/C_{\mathrm{cx}}$ (Remark 38.8). So $M_{\mathfrak{N}}$ does not depend on $\bar\delta$.

Let $M \ge M_{\mathfrak{N}}$, and let the background be real and (M,ε)-regular, with ε below the thresholds of Sections 36, 41 and 43. Let $W = 0$, and let $\mathcal{A}^{\mathrm{nl}}$, $\mathcal{B}^{\mathrm{nl}}$ be the non-local parts of the steps at

level (L3) on $\mathbb{T}_\eta$, with the factorization of Lemma 44.10. Then, in global tier norms,

$$\|\mathcal{A}^{\mathrm{nl}}\|^{1\to2}_{\kappa,\gamma} + \|\mathcal{A}^{\mathrm{nl}}D^T\|^{1\to2}_{\kappa,\gamma} + \|\mathcal{B}^{\mathrm{nl}}\|^{1\to1}_{\kappa,\gamma} + \|\mathcal{B}^{\mathrm{nl}}D^T\|^{1\to1}_{\kappa,\gamma} \le \theta_{\mathfrak{N}}(\kappa),$$

and, on scalar data, $\|\mathcal{A}^{\mathrm{nl}}D\|^{2\to2}_{\kappa,\gamma} + \|\mathcal{B}^{\mathrm{nl}}D\|^{2\to1}_{\kappa,\gamma} \le C^D_{\mathfrak{N}}(\kappa)$. The numbers $C^D_{\mathfrak{N}}(\kappa)$ and $M\theta_{\mathfrak{N}}(\kappa)$ are bounded independently of M, k, n, η and the background. In particular $\theta_{\mathfrak{N}}(\kappa) \to 0$ as $M \to \infty$, for fixed $L, \kappa, \delta, \rho, \alpha$.

Reading the statement. The background of Proposition 44.11 is real. The sentences of its first paragraph from "In M_1 the multiplicative constants" to "does not depend on $\bar\delta$" concern complex references; they fix $M_{\mathfrak{N}}$ so that it serves there as well. Read them after Proposition 46.7; Remark 46.9 explains them.

Proof. Fix j, put $X' = \Lambda^v_j$, and let $G_j = \mathbf{G}^{X'}$. By Lemma 41.2 and the regrouping in the proof of Lemma 41.3, the non-local parts of the step term of j are, for each term $X_i^T M_i Y_i$ of $\mathfrak{N}(s)$ listed in Lemma 44.10, with $M_i = M_i^{\mathbb{T}_\eta}(s)$, Details

$$X_i^T[M_i, h_j]Y_iG_j, \qquad [X_i^T, h_j]M_iY_iG_j, \qquad X_i^TM_i[Y_i, h_j]G_j,$$
$$h_jX_i^T(\mathbf{1}_{X'}M_i\mathbf{1}_{X'} - M_i^{X'}(s))Y_iG_j.$$

Here $Y_i \in \{D^T_{X'}, D\}$, since $W = 0$, and each part carries the monomial $s^{X(j)}$ times that of the term of M_i. Only the term (ii), with $M_i = \mathfrak{F}G'R$, has $X_i^T = D^T$. In its first and third parts the left D^T stays in front. In its fourth part write $h_jD^T = D^Th_j - [D^T, h_j]$. So $\mathcal{A}^{\mathrm{nl}}$ consists of $[M_i, h_j]Y_iG_j$, $M_i[Y_i, h_j]G_j$ and $h_j(\mathbf{1}_{X'}M_i\mathbf{1}_{X'} - M_i^{X'})Y_iG_j$ for the term (ii), and must map $1 \to 2$. $\mathcal{B}^{\mathrm{nl}}$ consists of all the other parts, including the parts with $[D^T, h_j]$ in front, and must map $1 \to 1$. The versions with D^T, respectively D, on the right have G_jD^T, respectively G_jD, in place of G_j. We write $b = 2$ for the term (ii) and $b = 1$ otherwise. The bounds of Steps 1–6 are proved for one term F of M_i, or of the difference in the fourth part. They are summed in Step 7.

Step 1: the input. Let g be of tier 1, i.e. let $\mathfrak{n}^1_{y'}(g)$ be finite at every block, and put $u = Y_iG_jg$.

(a) At every block y' of $(\Lambda^v_j)_{(16)}$, G_jg is of tier 3 and u of tier 2:

$$\mathfrak{n}^3_{y'}(G_jg) + \mathfrak{n}^2_{y'}(u) \le \sum_{y''} k_1(y', y'')\,\mathfrak{n}^1_{y''}(g).$$

This is the entry $1 \to 3$ of Lemma 44.5 with $\varphi \equiv 1$, i.e. Corollary 43.8(a) with $p = 2d$, together with $\mathfrak{n}^2_{y'}(\nabla h) \le C\mathfrak{n}^3_{y'}(h)$. Here $(\Lambda^v_j)_{(16)} \subset Y_j$; u is a linear combination of components of the full gradient of G_jg, which is controlled as in the proof of Lemma 44.5; and the near and far parts are combined into the kernel k_1, since $\mathfrak{n}^0 \le \mathfrak{n}^1$.

(b) At every block y', $\mathfrak{n}^0_{y'}(u) \le \sum_{y''} k_0(y', y'')\,\mathfrak{n}^0_{y''}(g)$ with $k_0(y', y'') = \|\mathbf{1}_{y'}Y_iG_j\mathbf{1}_{y''}\|$, by (W2).

The weighted row sums of k_0 and k_1 at rate $\gamma_0/2$ are at most $C\max(C_T, C_G)$. With D^T on the right, $u = Y_iG_jD^Tg$, (a) holds with $\mathfrak{n}^2_{y'}(G_jD^Tg) + \mathfrak{n}^1_{y'}(u)$ on the left. This follows from the entries $1 \to 2$ (values) of φG_jD^T and $1 \to 1$ of φDG_jD^T with $\varphi \equiv 1$; the latter holds for every first-order derivative, by (W2) and the control of the full gradient in the proof of Lemma 44.5. Part (b) holds with the decay of $Y_iG_jD^T$.

Step 2: inclusions. The set K_j is at ∞-distance at least $(\rho - 1)M - 1 \ge 2R_N + 21$ from $\mathbb{T}_\eta \setminus \Lambda^v_j$. Hence every block within distance $2R_N + 2$ of K_j lies in $(\Lambda^v_j)_{(16)}$, and every block within distance 2 of K_j lies in $(\Lambda^v_j)_{(48)}$.

Step 3: the commutator with M_i. Let F be a term of M_i, and let k, k' be the kernels of its bound $1 \to b$ in Lemma 44.10(N1), (N2), at rate $2\gamma_0$, with radius R_N. For $Z = \mathbb{T}_\eta$ this bound holds at every block. Let y be a block.

- Let y be at distance more than $R_N + 2$ from K_j. Then h_j vanishes on $\Lambda_1(c_y)$ and on the blocks within distance R_N of y. So $[F, h_j]u = F(h_j u)$ on $\Lambda_1(c_y)$, and only the far part of the bound of F enters:

$$\mathfrak{n}^b_y([F, h_j]u) \le \sum_{y'} k'(y, y')\, \mathfrak{n}^0_{y'}(h_j u) \le CM^{-1} \sum_{y'} (1 + |y - y'|) k'(y, y')\, \mathfrak{n}^0_{y'}(u),$$

  since $|h_j(x)| = |h_j(x) - h_j(c_y)| \le CM^{-1}|x - c_y|$.

- Let y be within distance R_N+2 of K_j. Write $[F, h_j]u = F(\psi u) - \psi F u$ with $\psi = h_j - h_j(c_y)$. As in the proof of Lemma 44.2(d), multiplication by ψ changes $\mathfrak{n}^t_{y'}$ by at most a factor $CM^{-1}(1 + |y - y'|)$. The near part of the bound of F involves $\mathfrak{n}^1_{y'}(u)$ at blocks within distance R_N of y, hence within distance $2R_N + 2$ of K_j. By Step 2 these blocks lie in $(\Lambda^v_j)_{(16)}$, where Step 1(a) applies. The far part involves $\mathfrak{n}^0_{y'}(u)$, which Step 1(b) bounds.

With $(1 + t)e^{-\gamma_0 t} \le C\gamma_0^{-1}$, in both cases $[F, h_j]Y_i G_j$ satisfies a global tier bound $1 \to b$ at rate $\gamma_0/2$, with constant

$$C(\gamma_0 M)^{-1}(|F|^{1\to b}_{2\gamma_0; R_N, \mathbb{T}_\eta} + |F|_{2\gamma_0}) \max(C_T, C_G).$$

The same holds with D^T on the right. Only the tier bounds are used; the decay norms in these constants, here and in Steps 4 and 5, are included for uniformity with Lemma 44.3.

Step 4: the commutators with X_i^T and Y_i. The commutators $[D^T, h_j]$ and $[Y_i, h_j]$ are multiplications by differences of h_j, composed with shifts by at most η. They have size CM^{-1} and are supported in $K_j^{(1)}$.

- The field $[Y_i, h_j]G_j g$ is supported in $K_j^{(1)}$, where $G_j g$ is of tier 3, by Steps 1(a) and 2. So it is of tier 1 at every block, with the kernel $CM^{-1}k_1$. The bound $1 \to b$ of F holds at every block, so it is a global bound. Hence $X_i^T F[Y_i, h_j]G_j$ maps $1 \to b$, with constant $CM^{-1}(|F|^{1\to b}_{2\gamma_0; R_N, \mathbb{T}_\eta} + |F|_{2\gamma_0})C_T$. With D^T on the right, $G_j D^T g$ is of tier 2 on $K_j^{(1)}$, and the same holds.

- For the term (ii), $[D^T, h_j]Fu$ vanishes at the blocks farther than 2 from K_j. At the other blocks y, $\mathfrak{n}^1_y([D^T, h_j]Fu) \le CM^{-1} \sum_{|y'-y|_\infty \le 1} \mathfrak{n}^2_{y'}(Fu)$. The near part of the bound of F at y' involves u only within distance $R_N + 3$ of K_j, where Step 1(a) applies. So this part maps $1 \to 1$, with constant $CM^{-1}(|F|^{1\to 2}_{2\gamma_0; R_N, \mathbb{T}_\eta} + |F|_{2\gamma_0}) \max(C_T, C_G)$.

Step 5: the seam term. Let Δ be a term of $\mathbf{1}_{X'} M_i \mathbf{1}_{X'} - M_i^{X'}(s)$, with the seam bound of Lemma 44.10(N3) on $X'_{(48)}$, at rate γ_0, for the seam $\Sigma = \Sigma_w(\mathbb{T}_\eta, X')$; let k_Δ, k'_Δ be its kernels and c_Δ its constant. Because of the factor h_j, and of $[D^T, h_j]$ for the term (ii), only blocks y within distance 2 of K_j contribute. By Step 2 they lie in $X'_{(48)}$. By (41.1), $d_\Sigma(y) \ge (\rho - 1)M - 1 - 2 - w \ge \frac{1}{4}(\rho - 1)M - 3$. So the seam weights give

$$\sum_{y'} e^{\gamma_0 |y - y'|}(k_\Delta + k'_\Delta)(y, y') \le e^{-\gamma_0 d_\Sigma(y)} c_\Delta \le Ce^{-\gamma_0(\rho-1)M/4} c_\Delta.$$

The near blocks lie within distance $R_N + 2$ of K_j, where Step 1(a) applies, and the far part is bounded by Step 1(b). Multiplication by h_j preserves the tiers, up to a constant, and $[D^T, h_j]$ is treated as in Step 4. So the seam parts satisfy a global bound $1 \to b$ at rate $\gamma_0/2$, with constant $Ce^{-\gamma_0(\rho-1)M/4}(c_\Delta + |\Delta|_{\gamma_0, \Sigma}) \max(C_T, C_G)$. Finally $e^{-\gamma_0(\rho-1)M/4} \le e^{-\gamma_0(\rho-1)M/16}$.

Step 6: an external D. Let λ be a scalar field of tier 2, and $u = Y_i G_j D\lambda$. Here $G_j D\lambda = \mathbf{G}^{X'}(D(\mathbf{1}_{X'}\lambda) + c_{X'}(\lambda))$ (paragraph *Scalar gradients at the cut*). Step 1 holds with $\mathfrak{n}^2_{y''}(\lambda)$ in place of $\mathfrak{n}^1_{y''}(g)$ and of $\mathfrak{n}^0_{y''}(g)$:

- on $(\Lambda^v_j)_{(16)}$, $G_j D\lambda$ is of tier 3, by the entry $2 \to 3$ of $\varphi \mathbf{G}^{Z_j} D$ in Lemma 44.5 with $\varphi \equiv 1$, and by Lemma 44.8 for the part of λ near the cut, whose far part is in $\mathfrak{n}^2$;
- at every block, $\mathfrak{n}^0_{y'}(u) \le C \sum_{y''} e^{-\gamma_1 |y'-y''|} \mathfrak{n}^2_{y''}(\lambda)$. This follows from the L^2 decay of $Y_i \mathbf{G}^{X'} D\mathbf{1}_{X'}$ (proof of Lemma 44.8), from the bound of Lemma 44.8 on $Y_i \mathbf{G}^{X'} c_{X'}(\lambda)$, and from $|\lambda|_{B(y'')^{(1)}} \le \max_{|y'''-y''|_\infty \le 1} \mathfrak{n}^2_{y'''}(\lambda)$.

Steps 3–5 then apply verbatim, and give the bounds $2 \to 2$ for the parts of $\mathcal{A}^{\mathrm{nl}} D$ and $2 \to 1$ for those of $\mathcal{B}^{\mathrm{nl}} D$, with the factors of Steps 3–5. These factors are bounded by constants; only a constant is claimed.

Step 7: summation. The bounds of Steps 3–6 hold for each term of M_i, with domain X_ω, and each j. Such a part is non-zero only if $X^{(r)}_\omega$ meets $K^{(1)}_j$. As in the counting of Lemma 41.3, for a fixed ω at most $C \deg \omega$ values of j contribute. This is absorbed by evaluating the middle factors at $\kappa + \delta$, at the cost of $C\delta^{-1}$. The monomial $s^{X(j)}$ costs $e^{\kappa \mathsf{N}_{\mathrm{b}}}$, and for a fixed box at most C values of j have $\square \subset X(j)$, with C depending only on d, ρ and C_v. By Lemma 44.10, the sums over ω of the split and seam norms of the middle factors at $\kappa + \delta$ are at most $C'_{\mathfrak{N}}(\kappa + \delta, \delta)$. Since $\gamma_0 \le 1$ and $C_T, C_G \ge 1$, Steps 3–5 give $\theta_{\mathfrak{N}}(\kappa)$, and Step 6 gives $C^D_{\mathfrak{N}}(\kappa)$. The rate $\gamma_0/2$ of the bounds exceeds $\gamma = \gamma_0/8$.

The thresholds. Lemma 44.10 at $\kappa + \delta$ requires $M \ge M_1$ at $\kappa + 3\delta$ and $(\rho - 1)M \ge 16$, for Lemma 44.9 at $\kappa + 2\delta$. Step 2 requires $(\rho - 1)M \ge 2R_N + 22$. Tier norms increase the margins by 1, and weakenable families need margins at most $M/16$, so $M \ge 16(r + 1)$. These are the conditions in $M_{\mathfrak{N}}$. The constants C, $C'_{\mathfrak{N}}$, C_T, C_G do not depend on M, so $C^D_{\mathfrak{N}}$ and $M\theta_{\mathfrak{N}}$ do not either. □

Remark 44.12 (Scope of Proposition 44.11). (a) *The potential.* Proposition 44.11, and with it Theorem 44.13 at level (L3) and Theorem 44.15, are stated only for $W = 0$ at level (L3), where the terms of $\mathfrak{N}$ that contain W vanish (Lemma 44.10). This is the only case used: these results enter the note only through the weakened background map, whose operators have $W = 0$ (Section 45), and through its extension to complex references (Proposition 46.7(b),(d)). For $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$, $W \ne 0$, a proof would need, in addition, tier bounds for $G'^Z(s)D^T$ on the local domains, for the middle factors $DRG'D^T W$ and $(DRG'D^T)W(DG'R)$, whose coefficients are only Hölder continuous, and the corresponding treatment of the term with $Y_i = 1$. This has not been carried out.

(b) *The exponent.* The tier exponent $\beta = \min(\alpha, \frac{1}{2})$ comes from the interior factors: (I1) at $p = 2d = 8$ gives $\beta \le 1 - \frac{d}{8} = \frac{1}{2}$, and (I2) at $p = \frac{8}{3}$ gives $\beta \le 2 - \frac{d}{p} = \frac{1}{2}$. The Hölder quotients of the coefficients need $\beta \le \alpha$. The tier L^4 carries no Hölder exponent. The results of this section do not need $\alpha \le \frac{1}{2}$. This restriction is imposed downstream, from Section 45 on, so that $\beta = \alpha$ and the outputs carry the exponent α of the graded domains. Lemma 47.9 uses $\alpha \le \frac{1}{2}$ as well.

(c) *The lift and the form of $\theta_{\mathfrak{N}}$.* In $d = 4$ the lift of the scalar walks from tier 0 needs two interior factors, through the tier L^4 (Lemma 44.6). In $\theta_{\mathfrak{N}}$ the factor $(\gamma_0 M)^{-1}$ comes from the commutators, and the exponential from the seams. Its constants do not depend on M.

Theorem 44.13 (Weakened walks in tier norms)**.** *Let $\mathcal{F}_0, \mathcal{A}, \mathcal{B}$ be the factors of Section 41 at the levels (L1) and (L3) on $\mathbb{T}_\eta$. At level (L3) let $W = 0$. Then*

$$\|\mathcal{A}\|^{1\to 2}_{\kappa,\gamma} + \|\mathcal{A}D^T\|^{1\to 2}_{\kappa,\gamma} + \|\mathcal{B}\|^{1\to 1}_{\kappa,\gamma} + \|\mathcal{B}D^T\|^{1\to 1}_{\kappa,\gamma} \le \theta'(\kappa),$$

with $\gamma = \gamma_0/8$ *and*

$$\theta'(\kappa) = Ce^{\kappa \mathsf{N}_\flat} \max(C_T, C_G) C_{\mathrm{loc}} M^{-1} + \theta_{\mathfrak{N}}(\kappa),$$

where $\theta_{\mathfrak{N}} = 0$ *at level (L1) and* $\theta_{\mathfrak{N}}(\kappa)$ *is that of Proposition 44.11 at level (L3). The walk at* κ *uses the step factors at* $\kappa + \delta$*. It converges if* $2C_d\delta^{-2}\theta'(\kappa+\delta) \le \frac{1}{2}$*, which holds for* $M \ge M_1'$*, with* M_1' *depending on* $L, \kappa, \delta, \rho, \alpha$ *and the constants of Part IX, and* $M_1' \ge M_{\mathfrak{N}}$ *at level (L3), with* $M_{\mathfrak{N}}$ *taken at* $\kappa + \delta$*, since the walk uses the proposition at* $\kappa + \delta$*. Such* M_1' *exists, since both terms of* $\theta'(\kappa+\delta)$ *are bounded by a constant independent of* M *times* M^{-1} *(Proposition 44.11). We choose* M_1' *with* θ' *replaced by* $2\theta'$*; the coercivity constants are the fixed ones of Proposition 46.7, and every decay bound that enters is taken at the fixed rate of the differences there, a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3; neither depends on* δ_c*, so the same numbers serve at real and at complex references (Proposition 46.7, step (vii) of the proof). At complex references of size* $\delta_c < \delta_P$ *the inputs of* θ' *change by at most a factor* $1+\sigma_{\mathrm{cx}}$*,* $(1+\sigma_{\mathrm{cx}})^3 \le 2$ *(*$\sigma_{\mathrm{cx}} = 2^{1/3} - 1$*), and* θ' *is a sum of products of at most three of them, so* $2\theta'$ *bounds* θ' *there (Proposition 46.7, step (vii) of the proof). So* M_1' *does not depend on* $\bar{\delta}$*, and it serves at these references as well. Then* $\|G(s)\|^{1\to 3}_{\kappa,\gamma} \le 4C_d\delta^{-2}C_j e^{(\kappa+\delta)\mathsf{N}_\flat} \max(C_T, C_G)$*, with* C_j *the number of* j *with* $\square \subset X(j)$*. For* $M \ge M_1'$ *the walks* $G(s) = \mathcal{F}_0 \sum_n \mathcal{R}(s)^n$ *satisfy, with constants* $C(\kappa, \delta)$*:*

(a) $G(s)\colon 1 \to 3$*;* $G(s)D^T\colon 1 \to 2$ *(values) and* $2 \to 3$*;* $DG(s)\colon 1 \to 2$*;* $DG(s)D^T\colon 1 \to 1$*;*

(b) in the gauge case, $\mathbf{G}(s)D\colon 2 \to 3$ *on scalar data.*

All of these are global tier norms of weakenable families. In particular

$$|\mathbf{G}(s)f|_{1,\beta;y} \le C \sum_{y'} e^{-\gamma|y-y'|} |f|_{B(y')}, \qquad s \in \mathbb{D}_\kappa,$$

and the same holds for $G'(s)$*. For* $\mathbf{G}(s)D^T$ *and* $DG'(s)D^T$ *the analogous statements are the tier bounds of (a). For* $DG'(s)D^T$ *these are* $1 \to 1$ *and* $2 \to 2$ *only: a bound from sup norms to sup norms fails, as the Calderón–Zygmund estimate does at* $p = \infty$*.*

Reading the statement. The sentences of Theorem 44.13 from "We choose M_1' with θ' replaced by $2\theta'$" to "it serves at these references as well" concern complex references. Read them after Proposition 46.7; Remark 46.9 explains them.

Proof. The input of a step. A step acts on $u = G_j g$ with g of tier 1. By Corollary 43.8(a) on Λ_j^v, with $p = 2d$, u is of tier 3, and Du, $D^T u$ are of tier 2, at every block of $(\Lambda_j^v)_{(16)}$. Near the cut only their L^2 norms are controlled. Details

Local parts. The local commutator $[\mathfrak{A}_{\mathrm{loc}}, h_j]G_j$ consists of:

- $D^T(\varphi G_j)$ and $\varphi \nabla G_j$, with $|\partial^k \varphi| \le CM^{-1}$; in the gauge case ∇G_j stands for DG_j and $D^T_{Z_j} G_j$, from $[D^T D + DD^T, h_j]$; by the discrete Leibniz rule, $D[D^T, h_j]G_j$ is a sum of terms $\varphi' G_j$ with $|\varphi'| \le CM^{-2}$ and of terms $\varphi \nabla G_j$ with ∇ the full covariant gradient, up to shifts by at most η, and the latter are bounded like φDG_j, since $\mathfrak{n}_y^2(\nabla h) \le C\mathfrak{n}_y^3(h)$ and (W2) includes ∇G_j;
- φG_j, with $|\varphi| \le CM^{-2}$;
- $[aQ'^T Q', h_j]G_j$, respectively $[a\mathbf{Q}^T\mathbf{Q}, h_j]G_j$, of size CM^{-1}, whose output is bounded on each block by L^2 norms (tier 1; Lemma 44.6 in the scalar case, and the same argument in the gauge case, where $\mathbf{Q}^T\mathbf{Q}$ has range one block by Proposition 22.10(d),(e));

Terms with W do not occur: level (L1) has no potential, and $W = 0$ at level (L3). These terms live on $K_j^{(1)}$, so they require regularity of u only at blocks of $(\Lambda_j^v)_{(16)}$. By Lemma 44.5, the $\mathcal{A}$-parts φG_j map $1 \to 3$, and $\varphi G_j D^T$ maps $1 \to 2$. The $\mathcal{B}$-parts map $1 \to 1$, also with D^T

on the right. All carry the factor CM^{-1}. At the scalar level (L1), $\mathcal{A} = 0$, and the derivative in front of $D^T(\varphi G'_j)$ is expanded onto G'_j, which gives terms $\varphi' G'_j$ and $\varphi \nabla G'_j$ of the same kind. The counting of Lemma 41.3 gives the first term of θ'. At level (L1) these are all the parts of the step.

Non-local parts. At level (L3) they are bounded by Proposition 44.11, which gives the second term of θ'.

The walk. After the regrouping of Section 41, a term with n steps is a product $\Phi_0 \cdots \Phi_n$ with $\Phi_0 \in \{\mathcal{F}_0, \mathcal{F}_0 D^T\}$ and $\Phi_i \in \{\mathcal{A}, \mathcal{A}D^T, \mathcal{B}, \mathcal{B}D^T\}$ for $i \geq 1$. Measure:

- Φ_i, $i \geq 2$, in $1 \to 1$;
- Φ_1 in $1 \to 2$ if $\Phi_0 = \mathcal{F}_0 D^T$, which forces Φ_1 to be of type $\mathcal{A}$, and in $1 \to 1$ otherwise;
- Φ_0 in $2 \to 3$, respectively $1 \to 3$, by Lemma 44.5.

For $X = \mathbb{T}_\eta$ these are global tier norms, by the remark after Lemma 44.5. Lemma 40.4 in tier norms and $2C_d\delta^{-2}\theta'(\kappa + \delta) \leq \frac{1}{2}$ give (a) for $G(s)$. The other statements follow in the same way:

- an external D^T is carried by Φ_n, which maps $1 \to 1$ or $1 \to 2$, and $2 \to 2$ by Lemma 44.5;
- an external D (in (b)) is handled, for $n = 0$, by the entry $\varphi \mathbf{G}^{Z_j} D \colon 2 \to 3$ and by Lemma 44.8, which uses Lemma 44.7. For $n \geq 1$ the last factor $\Phi_n D$ must map $2 \to 1$, or $2 \to 2$ if it is of type $\mathcal{A}$. Its local parts are $\varphi \nabla G_j D$, $\varphi G_j D$ and $[a\mathbf{Q}^T\mathbf{Q}, h_j]G_j D$. They are bounded by Lemma 44.5 for the interior part and by Lemma 44.8, whose L^2 input covers $P \in \{1, D, D_Z^T\}$, for the rest. Its non-local parts are bounded by Proposition 44.11, with the constant $C_{\mathfrak{N}}^D$. Since $\Phi_n D$ occurs once per term, a bound by a constant suffices;
- a left D uses $D\mathcal{F}_0 \colon 1 \to 2$ and $D\mathcal{F}_0 D^T \colon 2 \to 2$.

The pointwise bound is (a) with $\mathfrak{n}^1_{y'}(f) \leq |f|_{B(y')}$. □

44.5 The constrained operators

Remark 44.14 (Why not the defining formula for $\mathbf{C}$). This remark motivates the choice below; no proof uses it. Weaken $\mathbf{C} = \mathbf{G} - DG'RG'D^T - \mathbf{HQG}$ term by term, and put $\sigma_2(s) = G'(s)R(s)G'(s)D^T f$. Then $Q'\sigma_2(s) \neq 0$ for $s \neq 1$, in general. Each first factor $h_j G'_j k_j$ of the walk satisfies an equation with the averaging term, so $D^T D\sigma_2(s)$ is a sum of Hölder terms and of the jump terms $-a\sum_j s^{X(j)} h_j Q'^T Q' G'_j k_j$. These jump across block faces. At $s = 1$ the jumps cancel, since $D^T D\sigma_2 = Rh_0$ is Hölder there. For $s \neq 1$ they do not cancel in general. By Remark 43.6, $\partial\partial\sigma_2(s)$, and hence $\nabla D\sigma_2(s)$, is in general not bounded uniformly in η near block edges. The other two terms of $\mathbf{C}(s)$ are $C^{1,\beta}$ by Theorem 44.13, so the sum is not uniformly C^1. The same obstruction defeats the alternative of applying the unweakened operator to $\mathbf{C}(s)f$. In the equation for $u(s) = \mathbf{C}(s)f$, the term $DRD^T u(s)$ produces the source D applied to these jump terms. This is a divergence of bounded but discontinuous data, and it gives only $u(s) \in C^\beta$.

On the torus, Lemma 15.3(a) gives $\mathbf{G}DR = DG'R$. Since $E = 0$ (Proposition 22.10), $\mathbf{Q}\Pi = \mathbf{Q}$ (Lemma 15.1(c)), so $\mathbf{Q}_\pi = \mathbf{Q}$, $\mathbf{Q}_\pi^T = \mathbf{Q}^T$ and $\mathbf{\Gamma}_\pi = \mathbf{\Gamma}$. Hence

$$\begin{aligned} DG'RG'D^T &= \mathbf{G}\, DRG'D^T, \qquad & \mathbf{C} &= \mathbf{G}\,\Pi^T - \mathbf{HQG}, \\ \Pi^T &= 1 - DRG'D^T, & \mathbf{H} &= \mathbf{GQ}^T\mathbf{\Gamma}. \end{aligned} \tag{44.1}$$

Here $\Pi^T = 1 - S^T D^T$ with $S^T = DRG'$. We define the weakened operators on $\mathbb{T}_\eta$ by

$$\begin{aligned} \mathbf{H}(s) &= \mathbf{G}(s)\mathbf{Q}^T\mathbf{\Gamma}(s), \\ \mathbf{C}(s) &= \mathbf{G}(s) - \mathbf{G}(s)DG'(s)D^T + \mathbf{G}(s)DP_{\mathcal{V}}(s)G'(s)D^T - \mathbf{H}(s)\mathbf{QG}(s), \end{aligned}$$

with $G'(s)$, $R(s)$ from (L1), (L2), $\mathbf{G}(s)$ from (L3) and $\mathbf{\Gamma}(s)$ from (L4). This is $\mathbf{G}(s)\Pi^T(s) - \mathbf{H}(s)\mathbf{Q}\mathbf{G}(s)$ with $R = 1 - P_{\mathcal{V}}$ expanded, so that no identity is weakened on its own. By (44.1) these represent $\mathbf{H}$ and $\mathbf{C}$. They are products of weakenable families, so Lemma 40.4 gives their L^2 bounds, data locality and covariance, as in Section 41. On the torus they replace the representation of $\mathbf{C}$ in (L4).

Theorem 44.15 (Pointwise bounds for $\mathbf{H}(s)$ and $\mathbf{C}(s)$)**.** *Let $W = 0$. For $M \geq M_1'$ and $s \in \mathbb{D}_\kappa$:*

$$|\mathbf{H}(s)B|_{1,\beta;y} \leq C \sum_{y'} e^{-\gamma|y-y'|}|B|_{y'}, \qquad |\mathbf{C}(s)f|_{1,\beta;y} \leq C \sum_{y'} e^{-\gamma|y-y'|}|f|_{B(y')},$$
$$|\mathbf{C}(s)D^T g|_{1,\beta;y} \leq C \sum_{y'} e^{-\gamma|y-y'|}\big(|g|_{B(y')} + [\tilde{g}_{y'}]_{\beta,\Lambda_1(c_{y'})}\big),$$

with $C = C(\kappa, \delta)$. For $\alpha \leq \frac{1}{2}$, so that $\beta = \alpha$, these are (42.1) and Theorem 35.2(a)–(c) for the weakened operators on the torus.

Proof. All tier norms below are global. Theorem 44.13 at level (L1) on $\mathbb{T}_\eta$ gives $G'(s)\colon 1 \to 3$ and $G'(s)D^T\colon 1 \to 2$ (values). The family $P_{\mathcal{V}}(s) = G'(s)Q'^T(M')^{-1}(s)Q'G'(s)$ of (L2) maps $0 \to 3$: $Q'G'(s)$ maps L^2 to coarse fields by the L^2 bounds of (L1), coarse fields are bounded by these L^2 norms, $(M')^{-1}(s)$ is a unit-lattice family, Q'^T maps coarse fields to bounded fields, i.e. to tier 1, and $G'(s)\colon 1 \to 3$. Hence $R(s) = 1 - P_{\mathcal{V}}(s)$, inside the product, maps $2 \to 2$. Details

- $\mathbf{H}(s)$. $\mathbf{\Gamma}(s)$ is a unit-lattice family. $\mathbf{Q}^T$ maps coarse fields to bounded fields, i.e. to tier 1. Apply $\mathbf{G}(s)\colon 1 \to 3$.
- $\mathbf{G}(s)f$. Tier $1 \to 3$, with $\mathfrak{n}^1_{y'}(f) \leq |f|_{B(y')}$.
- $\mathbf{G}(s)DR(s)G'(s)D^T f$. The scalar field $k = R(s)G'(s)D^T f$ is of tier 2. Theorem 44.13(b) gives $\mathbf{G}(s)Dk$ in tier 3.
- $\mathbf{H}(s)\mathbf{Q}\mathbf{G}(s)f$. $\mathbf{Q}\mathbf{G}(s)f$ is a coarse field, bounded by L^2 norms. Apply the bound for $\mathbf{H}(s)$.
- $\mathbf{C}(s)D^T g$. Use $\mathbf{G}(s)D^T\colon 2 \to 3$, and $R(s)G'(s)D^T D^T g = R(s)G'(s)\mathfrak{F}^T g$, where $\mathfrak{F}^T g$ is bounded, so this is of tier ≥ 2. The coarse term is as before.

The Hölder quotient of g is converted between local gauges as in the proof of Theorem 35.2(b). □

44.6 What this gives, and what remains

- *Given.* The linear inputs of the weakened background map on the torus: $\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$, $\mathbf{C}(s)D^T$ in $C^{1,\beta}$, with exponential weights, uniformly on $\mathbb{D}_\kappa$, with $W = 0$ (Theorems 44.13 and 44.15); and the scalar walks $G'(s)$ and $G'(s)D^T$ in tier norms (Theorem 44.13 at level (L1)). Here $\beta = \min(\alpha, \frac{1}{2})$; for $\alpha \leq \frac{1}{2}$, as from Section 45 on, $\beta = \alpha$.
- *The trace inequality.* The scalar-gradient input $\mathbf{G}(s)Dk$, i.e. the gauge part $\mathbf{G}\,DRG'D^T$ of $\mathbf{C}$, needs Lemma 44.7. It holds because the local domains are rectangles.
- *Not needed: $D^T D\mathbf{H}(s)$.* Section 39 bounds $\nabla\mathcal{V}$ with the pointwise bound of Theorem 35.2(d) on $D^T D\mathbf{H}^X$, which has no term-by-term analogue. For the weakened background map it is not needed. In Lemma 26.2, $D^T D\mathbf{H}$ occurs as $D^T(D\mathbf{H}\mathcal{D})$, which is absorbed by $\mathbf{C}(s)D^T$ (tier $2 \to 3$), and in $\mathbf{H}^T D^T DA' = (D\mathbf{H})^T(DA')$ and $\mathbf{H}^T D^T D\mathbf{H}\mathcal{D}$, which are coarse fields bounded by L^2 norms. The remainder $\nabla\mathcal{N}$ is bounded by C^1 norms (Proposition 23.3(a)). So Theorems 44.13 and 44.15 give all linear inputs of the contraction for $A_*(s)$.

- *Not given: the level with W_2, and $W \neq 0$ at level (L3).* Lemma 44.5 needs $W \in \mathcal{W}_{\mathrm{loc}}$, as noted after Corollary 43.8. At level (L3), Proposition 44.11 is stated for $W = 0$ (Remark 44.12(a)). The background map uses only the operators with $W = 0$, so neither is needed for $A_*(s)$.

- *Domains.* Theorems 44.13 and 44.15 are stated on the torus. Bounds on proper domains are not needed for $A_*(s)$. For the scalar levels on a domain Z, Lemma 44.9 gives the values of $G'^Z(s)$ and $P_V^Z(s)$ in tier 2 and their gradients in tier 1, at the blocks of $Z_{(24)}$. Tier-3 outputs on domains would need a third interior factor in the terms with steps, and are not stated. For (L3) on proper domains one would need the lift argument for the top walk as well.

- *Comparison with route 2 of Section 42.* The mixed L^2/pointwise norm near the cuts is replaced by two simpler facts: tier norms compose exactly, and two or three interior factors at the output suffice (Lemmas 44.5, 44.6 and 44.9, Proposition 44.11). For $\mathbf{C}(s)$ the identity (44.1) replaces the term-by-term route discussed at the end of Section 43.

45 The weakened background map

This section constructs the weakened version of the background map $B \mapsto A_*(B)$ of Part V. It is the fixed point of a contraction built with the weakened operators of Section 44. The localization of the step needs three properties of it:

- analyticity in the weakening parameters $s \in \mathbb{D}_\kappa$;
- factorization when $s_\square = 0$ outside a union of boxes;
- a bound in $C^{1,\beta}$ that is linear in B and uniform in s.

The step quantities of Section 47 use them through the localization lemma (Lemma 47.2). Two points of the construction are not obvious.

- *The current is removed from the equation.* At a critical background $\mathbf{C}J = 0$. The weakened $\mathbf{C}(s)J$ does not vanish, and it would produce a field of size $C(\kappa)|J|$ even at $B = 0$. Without the source the fixed point vanishes at $B = 0$ for every s.

- *No second derivatives are needed.* The gradient of the interaction is written with derivatives outside, so that every term is either bounded or the divergence of a Hölder field. These are exactly the inputs of $\mathbf{C}(s)$ and $\mathbf{C}(s)D^T$ in Theorem 44.15. In particular no bound on $D^T D\mathbf{H}(s)$ is used.

Setting. We keep the setting of Section 44, with $W = 0$. We fix $\alpha \leq \frac{1}{2}$, so that $\beta = \min(\alpha, \frac{1}{2}) = \alpha$; the activity domain of the next stage uses the exponent α.

- *The reference.* U_0 is a real (M, ε)-regular background with $|J_{U_0}| \leq \varepsilon$, in the units of step k. It is $(1, \varepsilon)$-regular, so Proposition 23.3 applies. The bound on the current is used only in Theorem 45.5(c). Elsewhere only the regularity of U_0 enters, through $|\mathsf{F}| \leq C\varepsilon$, where $\mathsf{F} = F^s_{U_0}$ is the symmetrized curvature of Lemma 23.2. We write F to avoid a clash with the weakening parameter s. In the renormalization group step, $U_0 = U_{k+1}(V_{k+1})$. That background is critical for the k-step problem (Lemma 31.6), but criticality is used only in Theorem 45.5(c).

- *The operators.* $\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$ are the weakened operators on $\mathbb{T}_\eta$ of Section 44, at U_0, evaluated at $s \in \mathbb{D}_\kappa$.

- *Bilinear forms.* For complex fields, $\langle \cdot, \cdot \rangle$ denotes the bilinear form b, and gradients and transposes are taken with respect to it. On real fields this is the inner product of Part V.
- *Norms.* We write $|A|_* = \sup_y |A|_{1,\beta;y}$ for fine fields, and $|\omega|_\infty$ for coarse fields. For 2-forms,
$$|g|_{(2)} = \sup_y \big(|g|_{B(y)} + [\tilde{g}_y]_{\beta,\Lambda_1(c_y)} \big).$$
- *Constants.* C_P denotes constants that may depend on $\kappa, \delta, L, M, \rho, \alpha$, but not on k, n, η or $s \in \mathbb{D}_\kappa$. We assume $M \geq M_1'$, as in Theorem 44.15, and $M \geq 64$. Then the margin $r \leq M/16$ of the weakened families satisfies $M/4 - 2r - 2 > 2$.
- *Norms and the activity domain.* The Hölder quotients in $|\cdot|_{1,\beta;y}$ are taken on $\Lambda_1(c_y)$, as in the definition of the tiers in Section 44. Pairs of points at distance at most 1 lie in a common cube $\Lambda_1(c_y)$. Between the local gauges of Corollary 30.4 and those of the blocks, the change of gauge alters Hölder quotients by at most $C\varepsilon$ times the C^1 norm, as in the proof of Theorem 35.2(b). So $|A|_*$ bounds the quantities that define the graded activity domains (Definition 38.3: covariant derivatives, Hölder quotients on the cubes Λ_3 and the sup norm of the complex perturbation) up to a constant C_{conv}, not $1 + C\varepsilon$: pairs at distance more than 1 cost a factor 2 (Lemma 47.9).

45.1 The current at critical backgrounds

Lemma 45.1 (Current of critical backgrounds)**.** *Let U' be a real background field of the k-step problem, i.e. an element of the minimal orbit of Theorem 30.3, with $\varepsilon_1 \leq a(L, M)$ small. Let $\mathbf{H}$ be its minimizer map. Then*

$$J_{U'} = \mathbf{Q}^T \omega_0, \qquad \omega_0 = \mathbf{H}^T J_{U'} = (D\mathbf{H})^T F^s_{U'},$$

and $|J_{U'}|_\infty \leq C_J |F_{U'}|_\infty$. The constant C_J depends only on d, G, α and the parameter a of the averaging term. In particular it does not depend on L or M.

Proof. By Corollary 30.5, $A_*(0) = 0$ relative to $U_0 = U'$. The Euler–Lagrange equation (Theorem 25.1(d)) at $A_* = 0$ reads $\nabla\Phi(0) = d\mathcal{Q}(0)^T \omega_0$, with $\nabla\Phi(0) = J_{U'}$ (Proposition 23.3) and $d\mathcal{Q}(0) = \mathbf{Q}$ (Proposition 22.9). This is also the identity used in the proof of Proposition 33.1. Since $\mathbf{Q}\mathbf{H} = I$, $\mathbf{H}^T\mathbf{Q}^T = I$, so $\omega_0 = \mathbf{H}^T J_{U'}$. By Lemma 23.2(a), $J_{U'} = D^T F^s_{U'}$, so $\mathbf{H}^T J_{U'} = (D\mathbf{H})^T F^s_{U'}$. For a coarse bond c,

$$|\omega_0(c)| \leq \sum_y \|\mathbf{1}_c (D\mathbf{H})^T \mathbf{1}_y\| \, \|F^s\|_{B(y)} \leq C \, |F^s|_\infty,$$

by the L^2 decay of $D\mathbf{H}$ (Theorem 16.1(b), at regular backgrounds Theorem 19.9). Moreover $|F^s| \leq 2|F|$ by Lemma 23.2(c), and $|\mathbf{Q}^T\omega|_\infty \leq 2|\omega|_\infty$ once ε_1 is small depending on L. This follows from Proposition 22.10(e): $\mathbf{Q}^T$ differs from the transpose of the flat tube average by an operator of norm at most $C_L\varepsilon$ from sup norms to sup norms. The flat transpose has sup norm at most 1, since its kernel is nonnegative with column sums $\sum_y w_\mu(y, x) = 1$ (Lemma 2.6(a),(c)).

Constants. The constants of Parts II and III depend on the averaging pair only through the properties listed in Remark 19.12, by Lemma 31.3(a). So they do not depend on L, provided the background is regular with ε small depending on L and M, as there. For $M \geq M_0$ they do not depend on M either. □

Details

With Proposition 31.5, the curvature and the current of $U_k(V)$ are both bounded by $(1 + C_J)(C_* + \delta)\varepsilon_1$. These constants do not grow with L or M. In the renormalization group the reference of the step is $U_{k+1}(V')$, with $\varepsilon_1' = p_{k+1} g_{k+1}$ in the units of step $k + 1$. It lies in

the real budget θ_r of the graded domain of step $k+1$ once $(1+C_J)(C_*+1)p_{k+1}g_{k+1} < \theta_r$ (Remark 38.8(iii)), without a condition on L. In the units of step k its curvature and current are smaller by L^{-2} and L^{-3}. These are the domains of fixed size of Remark 38.8.

45.2 The weakened equation

The linearizing map. For a fine field A', let $\mathcal{D}_s(A')$ be the coarse field X that solves

$$X = C_k^{\mathrm{av}}(A' - \mathbf{H}(s)X), \tag{45.1}$$

with C_k^{av} the nonlinear part of the averaging (Proposition 22.9). At $s=1$ this is the equation of Lemma 26.1.

The functional and its gradient. Put

$$\begin{aligned}\mathcal{V}_s(A') = &-\langle \mathbf{H}(s)\mathcal{D}_s(A'), J\rangle - \langle DA', D\mathbf{H}(s)\mathcal{D}_s(A')\rangle \\ &+ \tfrac{1}{2}\langle D\mathbf{H}(s)\mathcal{D}_s(A'), D\mathbf{H}(s)\mathcal{D}_s(A')\rangle + \mathcal{N}(A' - \mathbf{H}(s)\mathcal{D}_s(A')),\end{aligned}$$

with $J = J_{U_0}$ and $\mathcal{N}$ as in Proposition 23.3. At $s=1$ this is the functional $\mathcal{V}$ of Section 26. Write $A = A' - \mathbf{H}(s)\mathcal{D}_s(A')$ and $d\mathcal{D}_s = d\mathcal{D}_s(A')$. Using $J = D^T\mathsf{F}$ (Lemma 23.2(a)), the gradient is

$$\nabla\mathcal{V}_s(A') = n_0(A') + D^T n_2(A'), \tag{45.2}$$

where

$$\begin{aligned}n_2(A') &= -D\mathbf{H}(s)\mathcal{D}_s(A'),\\ n_0(A') &= -d\mathcal{D}_s^T(D\mathbf{H}(s))^T(DA' - D\mathbf{H}(s)\mathcal{D}_s(A') + \mathsf{F}) + \nabla\mathcal{N}(A) - d\mathcal{D}_s^T\mathbf{H}(s)^T\nabla\mathcal{N}(A).\end{aligned}$$

This is the gradient formula of Lemma 26.2, term by term. There are three differences. $D^T D\mathbf{H}\mathcal{D}$ is kept as $D^T(D\mathbf{H}\mathcal{D})$; $\mathbf{H}^T D^T$ is written as $(D\mathbf{H})^T$; and $\mathbf{H}^T J$ is written as $(D\mathbf{H})^T\mathsf{F}$.

The map. For a coarse field B, put

$$\mathcal{T}_s(A_1) = -\mathbf{C}(s)\,\nabla\mathcal{V}_s(A_1 + \mathbf{H}(s)B), \tag{45.3}$$

on fine fields A_1 on $\mathbb{T}_\eta$, without the constraint $A_1 \in \mathcal{X}$. Given a fixed point A_1, put

$$A'_s = A_1 + \mathbf{H}(s)B, \qquad A_*(s;B) = A'_s - \mathbf{H}(s)\mathcal{D}_s(A'_s).$$

Compared with (26.3), the source J is missing. At a critical reference and $s=1$ this changes nothing, because $\mathbf{C}J = 0$ there (Theorem 45.5(c)).

45.3 Bounds

Lemma 45.2 (Linear inputs)**.** *For $s \in \mathbb{D}_\kappa$:*

(a) $|\mathbf{C}(s)f|_* \le C_P|f|_\infty$ *and* $|\mathbf{C}(s)D^T g|_* \le C_P|g|_{(2)}$*;*

(b) $|\mathbf{H}(s)\omega|_* \le C_P|\omega|_\infty$ *and* $|D\mathbf{H}(s)\omega|_{(2)} \le C_P|\omega|_\infty$*;*

(c) $|\mathbf{H}(s)^T f|_\infty \le C_P|f|_\infty$ *and* $|(D\mathbf{H}(s))^T g|_\infty \le C_P|g|_\infty$ *for fine fields f and 2-forms g.*

Proof. (a) and the first part of (b) are Theorem 44.15, summed over y'. Its kernel $e^{-\gamma|y-y'|}$ has $\sum_{y'} e^{-\gamma|y-y'|} = S_\gamma$. For the second part of (b), write $D\mathbf{H}(s)\omega = D\mathbf{G}(s)\mathbf{Q}^T\mathbf{\Gamma}(s)\omega$. Here $\mathbf{Q}^T\mathbf{\Gamma}(s)$ maps coarse fields to bounded fine fields, i.e. to tier 1. Theorem 44.13(a) at level (L3), with $W=0$, gives $D\mathbf{G}(s)\colon 1 \to 2$ in a global tier norm, which is the bound in $|\cdot|_{(2)}$. For (c), Details

the values of a coarse field on the bonds starting at y are bounded by its block norm there. Then

$$\|\mathbf{1}_y \mathbf{H}(s)^T f\| \le \sum_{y'} \|\mathbf{1}_y \mathbf{H}(s)^T \mathbf{1}_{y'}\| \, \|f\|_{B(y')} \le C|\mathbf{H}(s)|_\gamma \, |f|_\infty,$$

because blocks have volume 1 and fields have at most C components and the decay norm $|\cdot|_\gamma$ is invariant under transposition. Finally $|\mathbf{H}(s)|_\gamma + |D\mathbf{H}(s)|_\gamma \le 2^d \|\mathbf{H}\|^D_{\kappa,\gamma}$ (Lemma 40.3(a)), and $\|\mathbf{H}\|^D_{\kappa,\gamma}$ is finite, since $\mathbf{H}(s) = \mathbf{G}(s)\mathbf{Q}^T\mathbf{\Gamma}(s)$ is a product of the output of Theorem 41.4(a) at level (L3), the local factor $\mathbf{Q}^T$ and the output at level (L4) (Lemmas 40.5(a) and 40.4). □

Lemma 45.3 (The weakened linearizing map)**.** *There is $r_P > 0$, depending on the constants C_P, with the following property. For $s \in \mathbb{D}_\kappa$ and complex A' with $|A'|_\infty \le r \le r_P$, equation (45.1) has a unique solution $\mathcal{D}_s(A')$ in $\{|X|_\infty \le r_P\}$. It is analytic in (s, A') for s in the interior of $\mathbb{D}_\kappa$, and*

$$|\mathcal{D}_s(A')|_\infty \le C_P r^2, \qquad |d\mathcal{D}_s(A')^T \omega|_\infty \le C_P r \, |\omega|_\infty.$$

Proof. Let K_H be the constant of Lemma 45.2(b), so that $|\mathbf{H}(s)X|_* \le K_H|X|_\infty$. For $|X|_\infty, |X'|_\infty \le r_P$, Proposition 22.9(b) gives, with $|A' - \mathbf{H}(s)X|_\infty \le (1 + K_H)r_P$, Details

$$|C_k^{\mathrm{av}}(A' - \mathbf{H}(s)X)|_\infty \le C(1 + K_H)^2 r_P^2,$$
$$|C_k^{\mathrm{av}}(A' - \mathbf{H}(s)X) - C_k^{\mathrm{av}}(A' - \mathbf{H}(s)X')|_\infty \le 2C(1 + K_H) r_P K_H |X - X'|_\infty.$$

For r_P small, the map is a contraction of the ball into itself, and $(1 + K_H)r_P \le c_3$. The fixed point is analytic, since the map is analytic and the contraction is uniform. Inserting $|A'|_\infty \le r$ into the first bound gives $|X| \le C(r + K_H|X|)^2$, hence $|X| \le C_P r^2$. Differentiating (45.1) gives $d\mathcal{D}_s = (I + dC_k^{\mathrm{av}}(A)\mathbf{H}(s))^{-1} dC_k^{\mathrm{av}}(A)$, hence

$$d\mathcal{D}_s^T = dC_k^{\mathrm{av}}(A)^T \big(I + \mathbf{H}(s)^T dC_k^{\mathrm{av}}(A)^T\big)^{-1}.$$

By Proposition 22.9(c), $|dC_k^{\mathrm{av}}(A)^T\omega|_\infty \le C|A|_\infty|\omega|_\infty$. By Lemma 45.2(c), the operator norm of $\mathbf{H}(s)^T dC_k^{\mathrm{av}}(A)^T$ on coarse sup norms is at most $C_P C(1 + K_H)r_P$, which is $\le \frac{1}{2}$ for r_P small. The Neumann series gives the bound. □

Lemma 45.4 (The gradient)**.** *There are $r'_P > 0$ and C_P such that for $s \in \mathbb{D}_\kappa$ and complex A' with $|A'|_{C^1} \le r \le 2r'_P$,*

$$|n_0(A')|_\infty \le C_P(\varepsilon + r)r, \qquad |n_2(A')|_{(2)} \le C_P r^2.$$

For $|A'|_{C^1}, |\tilde{A}'|_{C^1} \le r \le r'_P$,

$$|n_0(A') - n_0(\tilde{A}')|_\infty \le C_P(\varepsilon + r)|A' - \tilde{A}'|_{C^1}, \qquad |n_2(A') - n_2(\tilde{A}')|_{(2)} \le C_P r |A' - \tilde{A}'|_{C^1}.$$

Proof. Choose $r'_P \le r_P/2$ with $4r'_P \le c_5$ and $2K_H C_P r'_P \le 1$. By Lemma 45.3, $|\mathcal{D}_s| \le C_P r^2$. So $|A|_{C^1} \le r + K_H C_P r^2 \le 2r$, by Lemma 45.2(b) and $|\cdot|_{C^1} \le |\cdot|_*$. By Lemma 45.2(b), $|n_2|_{(2)} \le C_P|\mathcal{D}_s|_\infty \le C_P r^2$. The terms of n_0 are bounded as follows: Details

- $|DA'|_\infty \le C|A'|_{C^1}$, $|D\mathbf{H}(s)\mathcal{D}_s|_\infty \le C_P r^2$, and $|\mathsf{F}|_\infty \le C\varepsilon$ by regularity of U_0. Lemma 45.2(c) and Lemma 45.3 then bound the first term by $C_P r(r + \varepsilon)$.
- $|\nabla \mathcal{N}(A)|_\infty \le C(\varepsilon + 2r)2r$, by Proposition 23.3(a), for $2r \le c_5$.
- The last term is bounded by $C_P r \cdot C_P(\varepsilon + 2r)2r$.

These bounds hold on the complex ball $|A'|_{C^1} \le 2r'_P$, and n_0, n_2 are analytic there. The Lipschitz bounds follow from the Cauchy estimate in the direction $A' - \tilde{A}'$, as in the proof of Lemma 26.2(a). □

Only $|A'|_{C^1}$ enters, not the Hölder quotients of $\nabla A'$. The sources of $\mathbf{C}(s)$ are bounded, and those of $\mathbf{C}(s)D^T$ are 2-forms of the form $D\mathbf{H}(s)\omega$. The regularity of the output comes entirely from Lemma 45.2(a).

45.4 The theorem

Theorem 45.5 (Weakened background map)**.** *Let $\alpha \le \frac{1}{2}$. There are $\varepsilon_P, \varrho_P > 0$ and C_P, depending on $\kappa, \delta, L, M, \rho, \alpha$, such that, with $R_P = \varrho_P / K_H$, the following holds for $\varepsilon \le \varepsilon_P$, every reference U_0 as in the setting, and every complex coarse field B with $|B|_\infty \le R_P$.*

(a) (Existence and bounds.) *For every $s \in \mathbb{D}_\kappa$, $\mathcal{T}_s$ has a unique fixed point $A_1(s;B)$ in $\{|A_1|_* \le \varrho_P\}$. With $A_*(s;B)$ as above,*

$$|A_1(s;B)|_* \le C_P(\varepsilon + |B|_\infty)|B|_\infty, \qquad |A_*(s;B)|_* \le C_P|B|_\infty, \qquad A_*(s;0) = 0.$$

(b) (Analyticity.) *$A_*(s;B)$ is analytic in (s,B) for s in the interior of $\mathbb{D}_\kappa$ and $|B|_\infty < R_P$, continuous up to the boundary, and real for real s, B.*

(c) (Identification.) *If U_0 is critical, i.e. $U_0 = U_k(V)$ as in Corollary 30.5, then $A_*(1;B) = A_*(B)$ for $|B|_\infty \le \min(R_P, \beta_V)$, real or complex. Here β_V is the radius of Corollary 30.5 and Proposition 26.3, i.e. $\varepsilon + \beta_V \le \min(c_6, c_7)$.*

(d) (Factorization.) *Let Z be a union of boxes with $*$-components Z_i, and let $s_\square = 0$ for all $\square \not\subset Z$. Then $A_*(s;B)$ vanishes outside $Z^{(r)}$. On $Z_i^{(r)}$ it depends only on B on the coarse bonds of $Z_i^{(r)}$, on U_0 on $Z_i^{(r+1)}$, and on the weakening parameters only through $s_\square$, $\square \subset Z_i$. Here $r \le M/16$ is the largest margin of the weakened families $\mathbf{C}$, $\mathbf{H}$, $D\mathbf{H}$ and their transposes, including the range added by local factors (Lemma 40.5(a)).*

Proof. (a) Let $b = |B|_\infty \le R_P$ and $|A_1|_* \le \varrho_P$. Then $r := |A'|_* \le |A_1|_* + K_H b \le 2\varrho_P$. By (45.2), Lemma 45.2(a) and Lemma 45.4, with $2\varrho_P \le r'_P$, Details

$$|\mathcal{T}_s(A_1)|_* \le C_P|n_0|_\infty + C_P|n_2|_{(2)} \le C_P(\varepsilon + r)r, \qquad |\mathcal{T}_s(A_1) - \mathcal{T}_s(\tilde{A}_1)|_* \le C_P(\varepsilon + 2\varrho_P)|A_1 - \tilde{A}_1|_*.$$

Choose ε_P, ϱ_P with $4C_P(\varepsilon_P + 2\varrho_P) \le 1$ and $C(\varepsilon_P + R_P) \le K_H$, with C the constant of the estimate in (c). The threshold ε_P also contains the smallness conditions on ε of the inputs: $\varepsilon \le c_5$ (Proposition 23.3), $a \le c_3$ (Proposition 22.9), the condition "ε small depending on M" of Section 41, and the thresholds of Sections 36 and 43. These include the thresholds of Section 44 (Proposition 44.11). Then $\mathcal{T}_s$ maps the ball $\{|A_1|_* \le \varrho_P\}$ into itself and is a contraction with constant $\frac{1}{4}$. At the fixed point, $|A_1|_* \le \frac{1}{4}(|A_1|_* + K_H b)$, so $|A_1|_* \le K_H b$ and $r \le 2K_H b$. Inserting this once more gives $|A_1|_* \le C_P(\varepsilon + 2K_H b)2K_H b \le C_P(\varepsilon + b)b$. Next, $|A_*|_* \le |A'_s|_* + K_H|\mathcal{D}_s(A'_s)|_\infty \le C_P b$. For $B = 0$:

- $\mathcal{D}_s(0) = 0$, since $X = 0$ solves (45.1) and the solution is unique;
- $\nabla\mathcal{N}(0) = 0$;
- every term of $n_0(0)$ and $n_2(0)$ contains $\mathcal{D}_s(0)$, $d\mathcal{D}_s(0) = dC_k^{\mathrm{av}}(0)\cdots = 0$ or $\nabla\mathcal{N}(0)$.

So $\mathcal{T}_s(0) = 0$, and the fixed point is 0.

(b) $\mathcal{T}_s(A_1)$ is analytic in (s, A_1, B). The weakened operators are analytic in s in the interior of $\mathbb{D}_\kappa$, and bounded uniformly on $\mathbb{D}_\kappa$ (Lemma 40.3(a)). $\mathcal{D}_s$ is analytic by Lemma 45.3. The functions $\nabla\mathcal{N}$ and C_k^{av} are analytic. The iterates $\mathcal{T}_s^m(0)$ are therefore analytic, and they converge uniformly. For real s and B every term of the weakened families is a real operator, so the iterates are real.

(c) At $s = 1$, the weakened families represent the operators of Part V: $\mathbf{C}(1) = \mathbf{C}$ by (44.1), and $\mathbf{H}(1) = \mathbf{H}$. Hence $\mathcal{D}_1 = \mathcal{D}$ and $\nabla\mathcal{V}_1 = \nabla\mathcal{V}$. By Lemma 45.3 with the ball $|X| \le r_P$, $\mathcal{D}_1 = \mathcal{D}$. Since U_0 is critical, $A_*(0) = 0$. At $B = 0$ this forces the fixed point $A_{1,*}$ of Proposition 26.3 to vanish: $A_{1,*} = \mathbf{H}\mathcal{D}(A_{1,*})$ and $\mathbf{Q}A_{1,*} = 0$ give $\mathcal{D}(A_{1,*}) = 0$. Hence $0 = \mathcal{T}(0) = -\mathbf{C}(J + \nabla\mathcal{V}(0)) = -\mathbf{C}J$. So $\mathcal{T}_1$ coincides with the map $\mathcal{T}$ of (26.3). Let B be real with $|B|_\infty \le \min(R_P, \beta_V)$, and let $A_{1,*}$ be the fixed point of Proposition 26.3. The estimate

in its proof, now without the term $C_C\varepsilon$ that came from $\mathbf{C}J$, gives $|A_{1,*}|_{C^1} \le C(\varepsilon+b)b \le K_H b$. By Lemma 45.4, only $|A'|_{C^1}$ enters the bound for $\mathcal{T}_1$. So $|A_{1,*}|_* = |\mathcal{T}_1(A_{1,*})|_* \le C_P(\varepsilon + 2K_H b)2K_H b \le \varrho_P$. Thus $A_{1,*}$ lies in the ball of (a), and by uniqueness $A_{1,*} = A_1(1;B)$. Then $A_*(1;B) = A_{1,*} + \mathbf{H}B - \mathbf{H}\mathcal{D}(A_{1,*}+\mathbf{H}B) = A_*(B)$, by the definition of A_* after Proposition 26.3. Both sides are analytic in B (Corollary 30.5 and (b)), so the identity extends to complex B.

(d) By Lemma 40.3(b), applied to the weakened families $\mathbf{C}$, $\mathbf{H}$ and their transposes and derivatives, the following hold at such s:

- each operator vanishes off $Z^{(r)} \times Z^{(r)}$;
- it is block diagonal over the $Z_i^{(r)}$;
- its block on Z_i depends on U_0 only on $Z_i^{(r)}$.

For $\mathbf{H}(s)$ and $\mathbf{H}(s)^T$ the coarse side consists of the bonds in $Z^{(r)}$.

Locality of the nonlinear maps.

- $C_k^{\mathrm{av}}(A)(c)$ depends only on A on the bonds of $S_c = B(c_-) \cup B(c_+)$ (Proposition 22.9(a)), and on U_0 there (Proposition 22.10(d) and Lemma 21.10). It vanishes if $A = 0$ there, and so does $dC_k^{\mathrm{av}}(A)(c)$.
- $\mathcal{N} = \sum_p n_p$ is a sum of plaquette terms, each a function of A and U_0 on the bonds of p, vanishing to second order at $A|_p = 0$. So $\nabla\mathcal{N}(A)(b)$ depends only on A on the plaquettes containing b, and vanishes if $A = 0$ there.
- F is local in U_0.

The class preserved by the iteration. Let $\mathcal{K}$ be the class of pairs (A_1, X) of fine and coarse fields with the following property. A_1 vanishes outside $Z^{(r)}$ and X outside $Z^{(r+1)}$. Here the coarse bonds of a set are the bonds starting in it. $X = \mathcal{D}_s(A')$ can be non-zero on coarse bonds that end in $Z^{(r)}$ but start outside it, since $C_k^{\mathrm{av}}(\cdot)(c)$ depends on S_c; but $\mathbf{H}(s)$ sees only the coarse bonds starting in $Z^{(r)}$, by Lemma 40.3(b). The restrictions to $Z_i^{(r)}$, respectively $Z_i^{(r+1)}$, depend only on B on the coarse bonds of $Z_i^{(r)}$, on U_0 on $Z_i^{(r+1)}$, and on $s_\square$ for $\square \subset Z_i$. For B this holds because B enters the iterations only through $\mathbf{H}(s)B$ in $A'_s = A_1 + \mathbf{H}(s)B$, and the block of $\mathbf{H}(s)$ on Z_i reads only the coarse bonds starting in $Z_i^{(r)}$ (Lemma 40.3(b)). The blocks of the weakened operators have the last property by Lemma 40.3(b).

- *The inner iteration.* By the locality just described, the iteration of (45.1), started at 0, stays in $\mathcal{K}$. The components are at distance at least $M/4 - 2r - 2 > 2$, so the local maps do not connect them.
- *The outer iteration.* The same holds for n_0 and n_2, including the Neumann series for $d\mathcal{D}_s^T$, whose terms are products of operators of the class. Finally $\mathcal{T}_s$ maps into the range of $\mathbf{C}(s)$. So $\mathcal{T}_s$ preserves the class.

The fixed point is the limit of the iterates from 0, so it lies in the class, and so does $A_*(s;B)$. □

45.5 Remarks

Remark 45.6 (Why the current is removed). Keep J in (45.3), as in (26.3). At a critical reference $\mathbf{C}J = 0$, but for $s \neq 1$ the term $-\mathbf{C}(s)J = -\mathbf{C}(s)\mathbf{Q}^T\omega_0$ does not vanish, so $A_*(s;0) \neq 0$. Then the composed activities of Proposition 47.10 do not vanish at $Z = 0$. The mixed pieces of the cluster expansion would have a part independent of the fluctuation, which destroys both the bound by τ and their vanishing at $Z = 0$. Nor is the term small in general. At complex references J is not bounded uniformly in η (Proposition 46.7(c)), and the source would have to be read as $-\mathbf{C}(s)D^T F^s$, with the operator $\mathbf{C}(s)D^T$ of Theorem 44.15. It is then

of order C_P times the current of the real part and the derivatives of the complex part, and C_P grows with M, which is chosen after L. Removing J is the only change to the equation, and it is invisible at $s = 1$.

Remark 45.7 (References that are not critical). Parts (a), (b), (d) hold for every real regular reference. At a non-critical reference the fixed point at $s = 1$ is the critical point of $\mathcal{J} - \langle \cdot, J \rangle$, not of $\mathcal{J}$. This is the intended definition. The activities of the next step must be analytic on a domain that contains non-critical configurations. There the step formulas are definitions, and they must contain no source term. There is no linear term in the fluctuation. In the background map, the source $-\mathbf{C}(s)J$ is removed. The current still enters through $(D\mathbf{H}(s))^T \mathsf{F}$ in n_0, but only multiplied by $d\mathcal{D}_s^T = O(|A'|)$. At real critical backgrounds they agree with the renormalization group, which is where they are evaluated (Remark 47.13, Theorem 48.9). Complex references, i.e. complex perturbations of U_0, need the complex versions of the interior estimates. They are treated in Section 46 (Proposition 46.7(d)).

Remark 45.8 (Use in the step). In the step, $B = \hat{Z}(Z)$ with Z the fluctuation on the bonds of $\mathfrak{F}$ (Lemma 32.7). With the white-noise representation, $Z = g_k C_k^{1/2}(s)\xi$, $|\xi| \le p_{0,k}$.

- *Two radii.* The large radius R_P, which does not depend on g_k, is used for the Cauchy bounds on the Taylor remainders of order three in Z. The small one, $|B|_\infty \le C_P L^d p_{0,k} g_k$, gives $|A_*(s; B)|_* \le C_P^2 L^d p_{0,k} g_k$. By the norm conversion (Lemma 47.9) this is below half of the slack $\frac{1}{2}\delta_1$ between the graded domains of steps k and $k+1$ (Remark 38.8(i)) once $C_{\mathrm{conv}} C_P^2 L^d p_{0,k} g_k \le \frac{1}{4}\delta_1$. Up to constants this is the condition $\tau < \frac{1}{2}$ of Section 48. The domains are those of fixed size of Remark 38.8, so the slack does not depend on k.
- *Factorization.* The local elimination $\hat{Z}$ has range $2L$ in block units. So, after composition with $\hat{Z}$, the factorization of Theorem 45.5(d) requires $M \ge CL$.

These are the inputs of the localization lemma (Lemma 47.2) for the interaction pieces and the composed activities (Section 47).

Part X: The small-field step and the small-field flow

Part X puts one small-field step into polymer form (Sections 46–48, Theorem L) and iterates it over all scales (Sections 49–53, Theorem N); Remark 49.1 describes the organization. The results concern the modified small-field model on a fixed torus. At every step the integration is restricted to small fields and the rest is discarded, so the partition function of this model is at most that of the lattice theory with the same counterterm E_0 (Remark 52.6). The large-field stage is not treated.

The step in polymer form. The activities must be analytic in the background on complex domains whose constants do not depend on the coupling, the step, the torus or the polymer (Remark 38.8).

- Section 46 extends Parts IV and IX to complex references (Propositions 46.4 and 46.7), and Section 47 turns the ingredients of Theorem 34.9 into functions of the weakening parameters that factorize over polymers (Propositions 47.5–47.12), and localizes them by Lemma 47.2 (introduction to Part IX).
- Section 48 imposes the cutoffs on the fluctuation field and on the white noise (Definition 48.1) and evaluates the fluctuation integral by a Mayer and polymer expansion

(Theorem 48.6). By parity in the coupling there is no term of order g, and the pure sources are of order $g^2 p_0^6$ (Proposition 48.8). The result is Theorem 48.9.

The flow. In four dimensions the coupling is marginal, and the activities are of order one, not small in the coupling (Remark 49.1). So the old activities are not reblocked and contracted; the density keeps Bałaban's generation form [18, (0.22)–(0.30)].

- A generation of scale j is a family of normalized, gauge-invariant activities on the polymers of the grid of scale j, bounded only at that scale (Definition 50.1). The new terms of a step form a raw generation (Lemma 53.4). Its average over the symmetries of its grid (Lemma 50.8) has an exactly isotropic period density, β_E times that of the action (Proposition 50.13), and $g_{k+1}^{-2} = g_k^{-2} - \beta_{k+1}$.
- At every later scale the marginal part of an old generation is extracted: after $m \geq m_0$ scales, its small polymers in a grid cube sum to $\beta_j S_\square$ up to $C\mathsf{K}L^{-\alpha m}$ (Proposition 51.5; K is the bound on the generations, Remark 53.5), and its other polymers carry a factor L^{-m} (Lemma 51.7). Grouped with their counterterms $-\beta_j S$, the old generations form old activities bounded uniformly in the scale (Proposition 53.2), and no contraction is needed.
- Sections 49 and 52 fix the parameters (Remark 49.8), the grids, the torus, the first step and the vacuum energy (Propositions 52.2 and 52.5).

One step preserves the generation form (Theorem 53.7), and Theorem 53.12 iterates it. As long as the couplings stay in $(0, \gamma]$: the density of every scale is in generation form, every generation is bounded at its creation scale, $|\beta_j| \leq C_\beta \mathsf{K}$ and $\frac{3}{4} g_{j-1} \leq g_j \leq 2 g_{j-1}$, with K and C_β independent of the scale, the torus and the couplings; and, for the counterterm E_0 with $\varepsilon_K = 0$, the partition function of the small-field model is the explicit product $\mathcal{Z}_K$ of normalizations times a number between $e^{-c|\mathbb{T}^{(K)}|}$ and $e^{c|\mathbb{T}^{(K)}|}$, with c depending on g_K but not on K, the torus or the earlier couplings (Corollary 53.13). The generations and the β_j depend continuously on the couplings (Proposition 53.10). The theorem assumes no asymptotic behaviour of the couplings and gives no sign of the β_j (Remark 53.14(a); Theorem 53.12(c) bounds only $|\beta_j|$); Part XI derives the behaviour of the couplings from the one-loop coefficient (Theorem M(iv)).

Imports and exports. Part X uses Theorem 34.9, Lemmas 32.7 and 32.6 (Part VII), the graded domains and Lemma 38.5 (Part VIII), Corollary 41.6, Proposition 41.7 and Theorem 45.5 (Part IX), Dimock's scheme [15] and the Kotecký–Preiss criterion [25]. The flow goes to Part XI, which proves its properties (F1)–(F3) (Proposition 58.6) and the ultraviolet behaviour of its couplings (Theorem 58.9). Part XI also uses Lemma 52.3, Proposition 50.13, and Proposition 47.8(d) and condition (Γ4) of Remark 53.5, which serve only Part XI. Remark 50.9(b) is not needed for the flow.

Details for this Part. The companion has two chapters on Part X. The first expands the complex references, the step quantities, the cluster expansion and the reblocking; the second the parameters and the grids, the generations, the period density, the extraction, the first step and the flow.

46 Complex references

The activities of the renormalization group must be analytic in the background on a complex domain (Definition 38.3). So the step evaluates the operators of Part IX, the interior estimates and the weakened background map at *complex* references. Sections 36–45 were written for real

backgrounds. This section records where reality was used, and shows that every statement needed later extends to small complex perturbations. The multiscale averaged backgrounds of Part IV need a separate argument, given in Section 46.1 (Proposition 46.4). The results of Part IX are extended in Section 46.2.

Complex references. Let U_r be a real (M,ε)-regular background. A *complex reference* of size δ_c is $U = e^{\mathrm{i}\eta A'}U_r$, with A' a $\mathfrak{g}^c$-valued field such that, in the local real gauges of Definition 18.1 for U_r,

$$|A'| + |\nabla A'| + \sup_y [\nabla \tilde{A}'_y]_{\alpha,\Lambda_1(c_y)} < \delta_c.$$

- *Relation to Proposition 19.11.* That proposition writes $U\,e^{\mathrm{i}\eta\,\delta A}$ and uses ordinary derivatives. The two forms are equivalent, with $\delta A = R(U_r)^{-1}A'$. The two sizes are comparable: there is $\varepsilon_{\mathrm{cc}} > 0$, depending only on d, α and G, such that for $\varepsilon \le \varepsilon_{\mathrm{cc}}$ the δ of Proposition 19.11 and the size of A' above bound each other up to the factor $C_{\mathrm{cmp}} = 4$. Indeed, in the gauge of Definition 18.1 on a cube $\hat{\Lambda}$, where $U_r = e^{\mathrm{i}\eta A_r}$, the two fields differ by the factor $R(e^{\mathrm{i}\eta A_r})$, which is within $4\eta\varepsilon$ of 1, with difference quotients and Hölder quotients of size $C\eta\varepsilon$. Covariant and flat gradients differ by $C\varepsilon|A'|$, by (38.1), and the change between the gauges of the blocks and of the cubes costs at most $C\varepsilon$ times the C^1 norm, as in the proof of Theorem 35.2(b). Finally, two points at ℓ^∞-distance less than 1 lie in a common cube $\Lambda_1(c_y)$, and pairs at distance at least 1 have Hölder quotients at most twice the sup of the gradient.
- *The local gauge.* In a local gauge of U_r we have $U = e^{\mathrm{i}\eta A_0}$, with $A_0 = A' \oplus A_r$ combined bondwise by the Baker–Campbell–Hausdorff formula. Then A_0 is complex and satisfies (B) with parameter $C(\varepsilon + \delta_c)$.
- *Potentials.* Local potentials at a complex reference are taken in $\mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}$ (Section 19), and satisfy the bounds of $\mathcal{W}_{\mathrm{loc}}(\varepsilon + C\delta_c)$ in the real gauges of U_r. The non-local potential W_2^X satisfies the hypotheses of Proposition 19.11 for existence, Re-coercivity and decay; its comparison of domains is Proposition 37.4(d). Only its L^2 statements are used (Section 37). In Section 45, $W = 0$.
- *Graded domains.* A point of a graded domain $\mathcal{U}(P;\theta_r,\delta_1,\delta_0)$ fixes the complex part only near P. It is extended by U_r outside a neighbourhood, with a smooth cutoff at unit scale (Remark 38.7). This gives a complex reference with $\delta_c \le 3C_{\mathrm{ext}}(\delta_0+\delta_1)$; for a point of the domain of step $k+1$, read in the units of step k, $\delta_c \le 2C_{\mathrm{ext}}(L^{-1}\delta_0 + L^{-2}\delta_1)$ (Remark 38.8(ii)). The real part has only curvature and current below θ_r. Its $(M, B_5\theta_r)$-regularity comes from Corollary 30.4, with B_5 depending on L and M. Its current is at most ε if $\theta_r \le \varepsilon$; Section 47 adds $|J_{U_r}| \le \varepsilon$ to the definition of a complex reference, for the coercivity of the Hessian.

46.1 Averaged backgrounds at complex references

Recall from Part IV. This subsection uses the multiscale averaging of Sections 21 and 22. In the units of step k, the lattice $\mathbb{T}^{(j)}$ has spacing $\delta_j = L^{j-k}$, with $\mathbb{T}^{(0)} = \mathbb{T}_\eta$ and $\mathbb{T}^{(k)} = \mathbb{T}$ (Section 21.1). Here δ_j is a spacing, unrelated to the domain constants δ_0, δ_1 of Remark 38.8.

- *Group average.* ℓ is the logarithm chart of G, of radius c_G, and $\kappa(u) = \|u\|\|u^{-1}\|$ for $u \in G^c$ (Section 21.2). For a family $\mathbf{U}$ in $\mathcal{D}_{r_M}$, where $\mathcal{D}_r = \{\operatorname{diam}\mathbf{U} < r\}$, the group average $\mathrm{Av}(\mathbf{U})$ is the solution of (21.1) given by Lemma 21.5(a); r_M, C_M and the weights ω_Γ of its derivative are those of that lemma.

- *Bałaban's averaging.* $Y(y,x)$ is the averaged contour variable, the group average over the contour family $G(y,x)$, and $\Lambda_{c,x}(Y)$ are the averaged loops (Definition 21.6); their gauge covariance is (21.2). The one-step average $\mathfrak{a}_j$ is (21.3), $\mathfrak{a}^{(j)} = \mathfrak{a}_j \circ \mathfrak{a}^{(j-1)}$ and $U^{(j)} = \mathfrak{a}^{(j)}(U)$ (Definition 21.8). $\mathfrak{U}_j(a)$ is the set of real configurations on $\mathbb{T}^{(j)}$ with curvature at most a (Definition 21.4), and $U^{(j)} \in \mathfrak{U}_j(2a)$ for $U \in \mathfrak{U}_0(a)$ with $a \le c_1$ (Proposition 21.12).
- *Gauge-adapted averaging.* A relative configuration X at scale j stands for XW_j (Section 22.1). τ_X is the block average of its averaged relative contour variables, $\tilde{\mathfrak{a}}_j(X) = \overline{X}^{(\tau_X^{-1})}$ the modified one-step average, $\mathcal{Q}_j = \tilde{\mathfrak{a}}_j \circ \mathcal{Q}_{j-1}$ the gauge-adapted averaging, and w_j the correspondence transformation, built with the extensions $\mathcal{E}^{(i)}$ (Definition 22.4).
- *Linearizations.* $\tilde{\mathfrak{a}}_j(e^{\mathrm{i}\delta_{j-1}B}) = e^{\mathrm{i}\delta_j(l_jB+c_j(B))}$ and $\tau_X = e^{\mathrm{i}\delta_j(\gamma_jB+c_j^\tau(B))}$, with $l_j = q_j + r_j$ and the constant c_2 (Lemma 22.8); the averaged relative transports $\lambda_W(y,x)$ (Definition 22.6); $\mathbf{Q}_j = l_j \cdots l_1$, the nonlinear part C_j^{av} and the constant c_3 (Proposition 22.9); q'_j and Q'_k (Section 22.6, Proposition 22.10).

Below, $h_j = \tau_0 \cdots \tau_{j-1}$ is a *complex* gauge transformation on $\mathbb{T}^{(j)}$, built from the relative configuration of a complex reference (display after the Setting). It is not the partition of unity h_j of Sections 40–45, nor the threshold $h_k = p_k g_k$ of Part X.

Proposition 22.9 and Proposition 22.10 assume a real U_0, with real averaged backgrounds $W_j \in \mathfrak{U}_j(2a)$ (Proposition 21.12). Lemma 22.8 uses this through:

- the unitarity of the transports $R(W(\gamma))$ along contours and $R(W(y,x))$ along averaged contour variables;
- the families of the background in $\mathcal{D}_{6a\delta_j^2}$ (Lemma 21.7(i)), and through them the weights of the averaged relative transports, with $\sum_\Gamma |\omega_\Gamma| \le 1 + Ca\delta_j^2$ (Lemma 22.7(c));
- the bound on the averaged loops h_x and on E_{loop} (Lemma 21.10(a));
- Lemma 21.13.

The underlying lemmas are proved only for real configurations: Lemma 21.2(b), with the bi-invariant metric on G; Lemmas 21.3 and 21.7(i)–(iii); and Lemma 21.11, which uses a real gauge and gauge invariance of $|F|$. These results feed the bounds $|\mathbf{Q}A(c)| \le C\|A\|$ and $|Q'\lambda| \le C\|\lambda\|$, and the bounded transposes, used in Theorem 35.2 and in Part IX. A naive per-scale bound on complex transports would compound to 2^j over the scales.

We therefore work relative to the real reference U_r. Two observations make this work.

- The averaged backgrounds at U are gauge transforms of $X_jW_j(U_r)$, with $X_j = \mathcal{Q}_j(U')$, by explicit complex gauge transformations h_j (Lemma 46.2). This is proved directly from the one-step definitions, with the complex equivariance of the group average (Lemma 21.5(b)). The axial gauge, Lemma 22.3(b),(c) and Theorem 22.5(a),(b),(e),(f) are not used at complex configurations.
- At the points of $\mathbb{T}^{(j)}$, h_j has size $C\delta_j|A'|$, not $C|A'|$. It is a product of the variables τ_{X_i}, $i < j$. Each τ_{X_i} is the exponential of an average of logarithms of averaged relative contour variables. These are group averages of relative holonomies along the contours of the families $G(y,x)$, of length at most $2(L-1)\delta_i < 2\delta_{i+1}$, and by the Lipschitz bound of Lemma 21.5(e) they are of the same size. Moreover $\sum_{i<j} \delta_{i+1} \le 2\delta_j$. The scale-$j$ gradient of h_j carries the factor δ_j^{-1}, so it is bounded by the sup norm. Hence the relative fields are $O(|A'|)$ uniformly in j, and the losses per scale are summable.

Only the sup norm of A' is used, and the smallness of δ_c depends on d, L and G, but not on M.

Setting. Let U_r be real and (M,ε)-regular with $\varepsilon \le \varepsilon_0$. By the proof of Corollary 22.11, $U_r \in \mathfrak{U}_0(a)$ with $a \le C\varepsilon$, where C does not depend on M (Lemma 12.3). The proofs of Sections 21 and 22 do not use M. So c_3 and the constants C, c of this subsection depend only on d, L and G. Let $c_3' \le c_3$ be such a constant, decreased below where needed, and assume $a \le c_3'$. Put $W_j^r = W_j(U_r) = U_r^{(j)}$. These are real and lie in $\mathfrak{U}_j(2a)$ by Proposition 21.12.

Let $U = U'U_r$ (bondwise), with $U' = e^{\mathrm{i}\eta A'}$ and A' a $\mathfrak{g}^c$-valued field. Under a real gauge transformation u, $U_r \mapsto U_r^u$ and $A'(b) \mapsto R(u(b_-))A'(b)$. So $|A'|_\infty$ is gauge invariant, and the condition $|A'| < \delta_c$ in the local gauges of U_r is the condition $|A'|_\infty < \delta_c$. The derivative bounds on A' in the definition of complex references are not used in this subsection. Let $W_j(U) = U^{(j)} = \mathfrak{a}^{(j)}(U)$, whenever defined.

We use the gauge-adapted averaging of Section 22 relative to the *real* reference U_r, so there $W_j = W_j^r$. Applied to the relative configuration U' it gives

$$X_j = \mathcal{Q}_j(U') = e^{\mathrm{i}\delta_j B_j}, \qquad \tau_i = \tau_{X_i} \ \ (\text{on } \mathbb{T}^{(i+1)}), \qquad h_j(y) = \tau_0(y)\cdots\tau_{j-1}(y) \ \ (y \in \mathbb{T}^{(j)}),$$

with $h_0 = 1$. By Theorem 22.5(c), which holds for complex relative configurations (remark after that theorem), $h_j = \mathsf{w}_j(U')^{-1}$ at the points of $\mathbb{T}^{(j)}$. Recall that $\tilde{\mathfrak{a}}_j(X) = \overline{X}^{(\tau_X^{-1})}$ (Definition 22.4); on the domain of Lemma 22.8 this is $\overline{X^{(g_X)}}$, by Lemma 21.10(b).

Norms of linear maps of $\mathfrak{g}^c$, and of matrices acting on $\mathbb{C}^{\mathsf{N}}$, are operator norms, written $\|\cdot\|$. The constants

$$C_{\mathrm{AB}} = 14d + 3, \qquad \kappa_{\mathrm{AB}} = 16 C_M C_{\mathrm{AB}},$$

with $C_M \ge 1$ of Lemma 21.5, and C_{loop} of Lemma 46.3(d) depend only on d and G. The contours used in the step from scale $j-1$ to scale j, namely those of the families $G(y,x)$, $y \in \mathbb{T}^{(j)}$, the segments $[x, x(c)]$ and $[c]$, and their initial pieces, have at most $2L$ bonds of $\mathbb{T}^{(j-1)}$ (Section 21). Recall $\delta_j = L^{j-k}$, so that

$$\sum_{i \le k} \delta_i \le 2, \qquad \sum_{i<j} \delta_{i+1} = \delta_j \sum_{m<j} L^{-m} \le 2\delta_j. \tag{46.1}$$

Lemma 46.1 (Covariance under complex gauge transformations)**.** *Let $j \ge 1$ and let Y be a configuration on $\mathbb{T}^{(j-1)}$ such that all families $\{Y(\Gamma)\}_{\Gamma \in G(y,x)}$, $y \in \mathbb{T}^{(j)}$, $x \in B_j(y)$, lie in $\mathcal{D}_{r_M/2}$, and $|\Lambda_{c,x}(Y) - 1| \le \frac{1}{2} c_G$ for all bonds c of $\mathbb{T}^{(j)}$, of both orientations, and all $x \in B_j(c_-)$. Let $g : \mathbb{T}^{(j-1)} \to G^c$ with $\kappa(g(y)) \le 2$ for $y \in \mathbb{T}^{(j)}$. Then $\mathfrak{a}_j(Y)$ and $\mathfrak{a}_j(Y^g)$ are defined, and $\mathfrak{a}_j(Y^g) = \mathfrak{a}_j(Y)^{g|_{\mathbb{T}^{(j)}}}$.*

Consequently, let X be a relative configuration at scale $j-1$ such that XW_{j-1} satisfies the hypothesis on Y, for some background W_{j-1} with $W_j = \mathfrak{a}_j(W_{j-1})$. Then $\overline{X^{(g)}} = \overline{X}^{(g|_{\mathbb{T}^{(j)}})}$.

Proof. This is the complex case of Lemma 21.10(b). In detail: by Lemma 21.5(b), the families of Y^g lie in $\mathcal{D}_{r_M}$, and (21.2) holds. Since $c_\pm \in \mathbb{T}^{(j)}$, this gives $\Lambda_{c,x}(Y^g) = g(c_-)\Lambda_{c,x}(Y)g(c_-)^{-1}$, and $Y^g([c]) = g(c_-)Y([c])g(c_+)^{-1}$. Moreover $|ghg^{-1} - 1| \le \kappa(g)|h-1| \le c_G$ for $h = \Lambda_{c,x}(Y)$ and $g = g(c_-)$, so Lemma 21.2(d) gives $\ell(ghg^{-1}) = g\,\ell(h)\,g^{-1}$, and $\exp(\mathrm{i}gZg^{-1}) = g\,e^{\mathrm{i}Z}g^{-1}$. Insert this into (21.3). For relative configurations use $X^{(g)}W_{j-1} = (XW_{j-1})^g$ and $\overline{X} = \mathfrak{a}_j(XW_{j-1})W_j^{-1}$. □ Details

This is the form of Lemma 21.10(b) used below. For real g only the definedness of $\mathfrak{a}_j(Y)$ is needed.

Lemma 46.2 (Representation of the averaged backgrounds)**.** *There is $\delta_{AB} > 0$, depending on d, L, G, such that for $a \le c_3'$ and $|A'|_\infty \le \delta_{AB}$:*

(a) $|\ell(\tau_i)| \le 3d\,\delta_{i+1}|A'|_\infty$ on $\mathbb{T}^{(i+1)}$ for $i < k$. The function $\lambda_j = \ell(h_j)$ on $\mathbb{T}^{(j)}$ is defined and satisfies $|\lambda_j| \le 7d\,\delta_j|A'|_\infty$ for $j \le k$.

(b) For every $j \le k$, $W_j(U)$ is defined and

$$W_j(U) = \big(X_j W_j^r\big)^{h_j}, \qquad \text{i.e.} \quad W_j(U)(b) = h_j(b_-)\, X_j(b)\, W_j^r(b)\, h_j(b_+)^{-1}. \tag{46.2}$$

Equivalently, $U'^{[j]} = \mathcal{Q}_j(U')^{(h_j)} = \mathcal{Q}_j(U')^{(\mathsf{w}_j(U')^{-1})}$.

Proof. (a) Proposition 22.9(a) applies to the real reference U_r and the complex field A', and gives $|B_i|_\infty \le 2|A'|_\infty$. By Lemma 22.8, $\ell(\tau_i) = \delta_{i+1}\big(\gamma_{i+1}B_i + c^\tau_{i+1}(B_i)\big)$ with $|\gamma_{i+1}B_i| \le 2d|A'|$ and $|c^\tau_{i+1}(B_i)| \le 4C\delta_{i+1}|A'|^2$. This gives the first bound for $|A'| \le d/(4C)$. By (46.1), $s = \sum_{i<j}|\ell(\tau_i)| \le 6d\delta_j|A'|$. Lemma 21.2(c) gives $|\lambda_j| \le s + Cs^2 \le 7d\delta_j|A'|$ for $|A'|$ small. Details

(b) We use induction on j. For $j = 0$ we have $W_0(U) = U = U'U_r = X_0W_0^r$ and $h_0 = 1$. Let $j \ge 1$, assume (46.2) at $j-1$, and put $Y = X_{j-1}W_{j-1}^r$. By Proposition 22.9(a), $X_{j-1} = e^{\mathrm{i}\delta_{j-1}B_{j-1}}$ with $|B_{j-1}|_\infty \le 2|A'|_\infty \le c_2$, after decreasing δ_{AB}. So Lemma 22.8, at the real background $W_{j-1}^r \in \mathfrak{U}_{j-1}(2a)$ and with the complex Lie field B_{j-1}, shows that X_{j-1} lies in the averaging domain at scale j: the families of Y lie in $\mathcal{D}_{r_M/2}$, and its averaged loops satisfy $|\Lambda_{c,x}(Y) - 1| \le \frac12 c_G$. These loops are built from group averages of families; the proof of that lemma bounds them through the Lipschitz bound of Lemma 21.5(e), and Lemma 21.7(iv) gives a bound of the same kind. In particular $\mathfrak{a}_j(Y)$ is defined. By (a), $h_{j-1} = e^{\mathrm{i}\lambda_{j-1}}$ with $\kappa(h_{j-1}) \le e^{2|\lambda_{j-1}|} \le 2$. By the induction hypothesis $W_{j-1}(U) = Y^{h_{j-1}}$, so Lemma 46.1 shows that $W_j(U) = \mathfrak{a}_j(W_{j-1}(U))$ is defined and

$$W_j(U) = \mathfrak{a}_j(Y)^{h_{j-1}|_{\mathbb{T}^{(j)}}}.$$

By the definitions of the relative average and of $\tilde{\mathfrak{a}}_j$,

$$\mathfrak{a}_j(Y) = \overline{X_{j-1}}\, W_j^r, \qquad \overline{X_{j-1}} = \tilde{\mathfrak{a}}_j(X_{j-1})^{(\tau_{j-1})} = X_j^{(\tau_{j-1})},$$

i.e. $\mathfrak{a}_j(Y) = (X_jW_j^r)^{\tau_{j-1}}$. Since $(Z^g)^{g'} = Z^{g'g}$ and $h_{j-1}\tau_{j-1} = h_j$ on $\mathbb{T}^{(j)}$, (46.2) follows at j. The last statement is (46.2) rewritten with the relative gauge action, together with $h_j = \mathsf{w}_j(U')^{-1}$ on $\mathbb{T}^{(j)}$. □

Lemma 46.3 (Averaged backgrounds at complex references)**.** *Possibly after decreasing δ_{AB} and c_3' (depending on d, L, G), the following holds for $a \le c_3'$, $|A'|_\infty \le \delta_c \le \delta_{AB}$ and $0 \le j \le k$.*

(a) $W_j(U) = e^{\mathrm{i}\delta_j\tilde{B}_j}\, W_j^r$ bondwise, where

$$\tilde{B}_j = B_j - D_{W_j^r}\lambda_j + \beta_j, \qquad |\beta_j|_\infty \le C\delta_j|A'|_\infty^2, \qquad |\tilde{B}_j|_\infty \le C_{\mathrm{AB}}|A'|_\infty. \tag{46.3}$$

Moreover $|\tilde{B}_j - \mathbf{Q}_j(U_r)A' + D_{W_j^r}\mathbf{\Gamma}_jA'|_\infty \le C|A'|_\infty^2$, where $\mathbf{\Gamma}_jA' = \sum_{i<j}\delta_{i+1}\gamma_{i+1}\mathbf{Q}_i(U_r)A'$ is evaluated at the points of $\mathbb{T}^{(j)}$.

(b) $\tilde{B}_j$ is holomorphic in A' on $\{|A'|_\infty < \delta_{AB}\}$, and real for real A'. Under a real gauge transformation u of $\mathbb{T}_\eta$ ($U_r \mapsto U_r^u$, $A'(b) \mapsto R(u(b_-))A'(b)$) we have $\tilde{B}_j(b) \mapsto R(u(b_-))\tilde{B}_j(b)$. In particular (a) holds in every real gauge, e.g. in the local gauges of U_r.

(c) Let γ be a contour of n bonds of $\mathbb{T}^{(j)}$ with $s = C_{\mathrm{AB}}n\delta_j|A'|_\infty \le s_0$, where s_0 depends only on N. Then $W_j(U)(\gamma) = e^{\mathrm{i}Z_\gamma}W_j^r(\gamma)$ with $|Z_\gamma| \le 2s$. (Here Z_γ is the Baker–Campbell–Hausdorff series of $\mathfrak{g}^c$-elements of Lemma 21.2(c), which lies in $\mathfrak{g}^c$; so no radius c_G enters.) Hence

$$\|R(W_j(U)(\gamma))^{\pm1}\| \le e^{4s}, \qquad \|R(W_j(U)(\gamma)) - R(W_j^r(\gamma))\| \le e^{4s} - 1.$$

Moreover, for $j \ge 1$, $y \in \mathbb{T}^{(j)}$ and $x \in B_j(y)$, the averaged contour variable $W_{j-1}(U)(y,x)$ is defined, and

$$W_{j-1}(U)(y,x) = e^{\mathrm{i}Z_{y,x}}\, W_{j-1}^r(y,x), \qquad |Z_{y,x}| \le 4C_MC_{\mathrm{AB}}\delta_j|A'|_\infty.$$

Applied at index $j-1$ to the contours of at most $2L$ bonds of $\mathbb{T}^{(j-1)}$ used in the step $j-1 \to j$, where $4s \le 8C_{\mathrm{AB}}\delta_j\delta_c$, and to the averaged contour variables, where $2|Z_{y,x}| \le 8C_M C_{\mathrm{AB}}\delta_j\delta_c$, this gives these bounds for both kinds of transports with e^{4s} replaced by $e^{\kappa_{\mathrm{AB}}\delta_j\delta_c/2}$. Products of one transport of each kind have norm at most $e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}$.

(d) For $j \ge 1$, a bond c of $\mathbb{T}^{(j)}$ and $x \in B_j(c_-)$, let

$$H_x(U) = \ell(\Lambda_{c,x}(W_{j-1}(U))), \qquad E_{\mathrm{loop}}(U) = L^{-d}\sum_x H_x(U),$$

the logarithms of the averaged loops of (21.3) and their average, so that $W_j(U)(c) = e^{\mathrm{i}E_{\mathrm{loop}}(U)}W_{j-1}(U)([c])$. Then

$$|H_x(U) - H_x(U_r)|, \quad |E_{\mathrm{loop}}(U) - E_{\mathrm{loop}}(U_r)| \le C_{\mathrm{loop}}\delta_j|A'|_\infty,$$
$$|H_x(U)| \le 16a\delta_j^2 + C_{\mathrm{loop}}\delta_j|A'|_\infty, \qquad |E_{\mathrm{loop}}(U)| \le 16a\delta_j^2 + C_{\mathrm{loop}}\delta_j|A'|_\infty.$$

Proof. (a) By Lemma 46.2, $V_j = W_j(U)(W_j^r)^{-1} = X_j^{(h_j)}$, i.e. Details

$$V_j(b) = e^{\mathrm{i}\lambda_j(b_-)}\, e^{\mathrm{i}\delta_j B_j(b)}\, e^{-\mathrm{i}R(W_j^r(b))\lambda_j(b_+)}.$$

The three exponents have total size $s \le (14d+2)\delta_j|A'| \le \frac{1}{4}$, by Lemma 46.2(a) and Proposition 22.9(a); here we used that $R(W_j^r)$ is unitary. By Lemma 21.2(c), $\ell(V_j(b)) = \lambda_j(b_-) + \delta_j B_j(b) - R(W_j^r(b))\lambda_j(b_+) + \delta_j\beta_j(b)$ with $\delta_j|\beta_j| \le Cs^2$. This is (46.3), with $D_{W_j^r}\lambda(b) = \delta_j^{-1}(R(W_j^r(b))\lambda(b_+) - \lambda(b_-))$ as in Section 22. By Lemma 46.2(a), $|D_{W_j^r}\lambda_j| \le \delta_j^{-1}(|\lambda_j(b_+)| + |\lambda_j(b_-)|) \le 14d|A'|$. Hence $|\tilde{B}_j| \le (2 + 14d + C\delta_j|A'|)|A'| \le C_{\mathrm{AB}}|A'|$.

For the linear part, use $B_j = \mathbf{Q}_j A' + C_j^{\mathrm{av}}(A')$ with $|C_j^{\mathrm{av}}(A')| \le C|A'|^2$ (Proposition 22.9(b)). Further, $\lambda_j = \sum_{i<j}\ell(\tau_i) + O(\delta_j^2|A'|^2)$ and $\ell(\tau_i) = \delta_{i+1}\gamma_{i+1}\mathbf{Q}_i A' + O(\delta_{i+1}|A'|^2)$. By (46.1), $|\lambda_j - \mathbf{\Gamma}_j A'| \le C\delta_j|A'|^2$, and $D_{W_j^r}$ costs at most $2\delta_j^{-1}$.

(b) B_j and τ_i are holomorphic in A' by Proposition 22.9 and Lemma 22.8. Then λ_j and $\tilde{B}_j$ are holomorphic, since they are obtained from these by ℓ, products and Lemma 21.2(c). For real A' all configurations involved are real. For covariance, $U^u = e^{\mathrm{i}\eta R(u_-)A'}U_r^u$. Lemma 21.10(b) with real g, applied at each scale, where the averages are defined by Lemma 46.2(b), gives $W_j(U^u) = W_j(U)^{u|_{\mathbb{T}^{(j)}}}$, and likewise $W_j(U_r^u) = (W_j^r)^{u|_{\mathbb{T}^{(j)}}}$. Hence $V_j(b) \mapsto u(b_-)V_j(b)u(b_-)^{-1}$.

(c) $W_j(U)(\gamma) = (V_j)_{W_j^r}(\gamma)\, W_j^r(\gamma)$, and $(V_j)_{W_j^r}(\gamma) = \prod_i R(W_j^r(\gamma_{<i}))(V_j(b_i))$. For negatively oriented bonds, $V_j(-b) = R(W_j^r(b))^{-1}(V_j(b)^{-1})$. Since $R(W_j^r)$ is unitary, every factor is $e^{\mathrm{i}\delta_j Z_i}$ with $|Z_i| \le C_{\mathrm{AB}}|A'|$. Lemma 21.2(c) gives $|Z_\gamma| \le s + Cs^2 \le 2s$ for $s \le s_0 = \min(\frac{1}{4}, C^{-1})$. By (12.1), $\|R(e^{\pm\mathrm{i}Z})\| \le e^{2|Z|}$ and $\|R(e^{\mathrm{i}Z}) - 1\| \le e^{2|Z|} - 1$, and $R(W_j^r(\gamma))$ is unitary.

For the averaged contour variables, the contours $\Gamma \in G(y,x)$ have at most $2(L-1)$ bonds of $\mathbb{T}^{(j-1)}$, with $\delta_{j-1} = \delta_j/L$. By the first part at index $j-1$, $W_{j-1}(U)(\Gamma) = e^{\mathrm{i}Z_\Gamma}W_{j-1}^r(\Gamma)$ with $|Z_\Gamma| \le 4C_{\mathrm{AB}}\delta_j|A'|$. The real family $\{W_{j-1}^r(\Gamma)\}$ lies in $\mathcal{D}_{6a\delta_j^2} \subset \mathcal{D}_{r_M/2}$ (Section 22, setting), and $|Z_\Gamma| \le r_M/16$ after decreasing δ_{AB}. By Lemma 21.5(a),(e) and the proof of (e), the family $\{W_{j-1}(U)(\Gamma)\}$ lies in $\mathcal{D}_{r_M}$, its group average is the averaged contour variable $W_{j-1}(U)(y,x)$, and $|\ell(W_{j-1}(U)(y,x)W_{j-1}^r(y,x)^{-1})| \le C_M \max_\Gamma |Z_\Gamma|$. The bounds on the transports follow from (12.1), since $R(W_{j-1}^r)$ is unitary and $8C_{\mathrm{AB}} \le \frac{1}{2}\kappa_{\mathrm{AB}}$. They need $2C_{\mathrm{AB}}\delta_j\delta_c \le s_0$, which holds after decreasing δ_{AB}.

(d) The averaged loop $\Lambda_{c,x}(W_{j-1}(U))$ is a product of four factors: the averaged contour variables $W_{j-1}(U)(c_-,x)$ and $W_{j-1}(U)(x(c),c_+) = W_{j-1}(U)(c_+,x(c))^{-1}$, and the transports along $[x,x(c)]$ and along $[c]^{-1}$, two straight contours of L bonds of $\mathbb{T}^{(j-1)}$. By (c) at index $j-1$, each factor is $e^{\mathrm{i}\zeta}$ times the corresponding factor at U_r, or that factor times $e^{\mathrm{i}\zeta}$, with $|\zeta| \le 4C_M C_{\mathrm{AB}}\delta_j|A'|$ for the averaged contour variables and $|\zeta| \le 2C_{\mathrm{AB}}\delta_j|A'|$ for the straight

contours. The factors at U_r are real. Moving the factors $e^{\mathrm{i}\zeta}$ to the left conjugates them by real elements, which preserves their norms. So

$$\Lambda_{c,x}(W_{j-1}(U)) = e^{\mathrm{i}\eta_1}e^{\mathrm{i}\eta_2}e^{\mathrm{i}\eta_3}e^{\mathrm{i}\eta_4}\,e^{\mathrm{i}H}, \qquad \sum_i |\eta_i| \le 12 C_M C_{\mathrm{AB}} \delta_j |A'|,$$

with $H = H_x(U_r)$. By Lemma 21.10(a) and Proposition 21.12, $|H| \le 16a\delta_j^2 \le 16c_3'$. Let $\varphi(t) = \ell(e^{\mathrm{i}t\eta_1}\cdots e^{\mathrm{i}t\eta_4}e^{\mathrm{i}H}) - H$. For $|t| \le r_0/\sum_i|\eta_i|$, with r_0 small depending on N (the logarithm is again a Baker–Campbell–Hausdorff series of $\mathfrak{g}^c$-elements, so c_G does not enter), it is holomorphic and bounded by a constant $K_{\mathrm{S}}(\mathsf{N})$, and $\varphi(0) = 0$; this uses $16c_3' \le r_0$. The Schwarz lemma gives $|\varphi(1)| \le (K_{\mathrm{S}}/r_0)\sum_i|\eta_i|$. This is the bound with $C_{\mathrm{loop}} = 12C_MC_{\mathrm{AB}}K_{\mathrm{S}}/r_0$. Average over x for E_{loop}. □

Proposition 46.4 (Averaged backgrounds at complex references, (AB))**.** *There are $\delta_{AB} > 0$ and $c_3' \le c_3$, depending only on d, L, G (through the constants of Part IV), such that the following holds for $a \le c_3'$, every real $U_r \in \mathfrak{U}_0(a)$, and every complex reference $U = e^{\mathrm{i}\eta A'}U_r$ with $|A'|_\infty \le \delta_c < \delta_{AB}$. All bounds are uniform in k, in η and in $j \le k$. Below, relative configurations, τ_X, $\tilde{\mathfrak{a}}_j$, $\mathcal{Q}_j$, l_j, q_j, r_j, γ_j, q_j', $\mathbf{Q}_k$, Q_k' at the reference U are defined as in Section 22, with the complex backgrounds $W_j(U)$ in place of W_j.*

(i) (Backgrounds.) *$W_j(U)$ is defined for $j \le k$, and Lemmas 46.2 and 46.3 hold. In particular, relative to $W_j(U_r)$, $W_j(U)$ is a complex perturbation of size at most $C_{\mathrm{AB}}\delta_c$, in every real gauge.*

(ii) (One step.) *Lemma 22.8 holds at $W = W_{j-1}(U)$, with c_2 replaced by $\frac{1}{2}c_2$. Parts (a), (c), (d) and (f) hold as stated, except the reality statement in (f); (e) is dropped, and (b) is replaced by*

$$|(q_jB)(c)|^2 \le e^{2\kappa_{\mathrm{AB}}\delta_j\delta_c}L^{-d}\sum_x L^{-1}\sum_{b\in[x,x(c)]}|B(b)|^2,$$
$$|(r_jB)(c)| \le C(a\delta_j^2 + \delta_j\delta_c)\max|B|, \qquad |\gamma_jB| \le de^{\kappa_{\mathrm{AB}}\delta_j\delta_c}\max|B|.$$

(iii) (Gauge-adapted averaging.) *Proposition 22.9 holds at the reference U, for complex A with $|A|_\infty \le c_3'$. Parts (a), (b) and (d) hold in the same form, and "real for real A" is dropped. In (c) the kernel bounds become*

$$|K(c,b)| \le (1+Ca)\,e^{C_L\delta_c}, \qquad |K_A(c,b)| \le Ce^{C_L\delta_c}|A|_\infty,$$
$$|d^2C_k^{\mathrm{av}}(A)[h,h](c)| \le Ce^{C_L\delta_c}\|h\|_{S_c}^2,$$

and the consequences of (c) (pointwise bounds by $\|h\|_{S_c}$, sup-norm bounds of the transposes) hold with their constants multiplied by $e^{C_L\delta_c}$.

(iv) (Linearized averages.) *$Q_k'(U)Q_k'(U)^T = I$, with the transpose taken with respect to* b. *Proposition 22.10(d), and (f) for real u, hold as stated. The transports along the chains of averaged contour variables satisfy $\|T_U(y,x)\| \le e^{\kappa_{\mathrm{AB}}\delta_c}$ and $\|T_U(y,x) - T_{U_r}(y,x)\| \le e^{\kappa_{\mathrm{AB}}\delta_c} - 1$. Hence $|Q_k'(U)\lambda(y)| \le e^{\kappa_{\mathrm{AB}}\delta_c}\|\lambda\|_{B(y)}$, and $Q_k'(U)^T$ is bounded on sup norms by $e^{\kappa_{\mathrm{AB}}\delta_c}$. Proposition 22.10(e) holds with ε replaced by $\varepsilon + \delta_c$. More precisely, suppose that on a cube containing $B(c_-)\cup B(c_+)$ we have $U_r = e^{\mathrm{i}\eta A_r}$ in a real gauge, with $|A_r| \le \varepsilon$ and $a \le \varepsilon \le c_3'$. Then, in this gauge, $|K_{\mathbf{Q}_k(U)}(c,b) - K_{\mathbf{Q}}(c,b)| + |K_{Q_k'(U)}(y,x) - K_{Q_0}(y,x)| \le C_L(\varepsilon + \delta_c)$, and the same holds for the transposes.*

(v) (Closeness to the real reference.) *With the supports of Proposition 22.10(d),*

$$|K_{\mathbf{Q}_k(U)}(c,b) - K_{\mathbf{Q}_k(U_r)}(c,b)| \le C_L\delta_c, \qquad |K_{Q_k'(U)}(y,x) - K_{Q_k'(U_r)}(y,x)| \le e^{\kappa_{\mathrm{AB}}\delta_c} - 1,$$
$$\|R(W_k(U)(b)) - R(W_k(U_r)(b))\| \le e^{2C_{\mathrm{AB}}\delta_c} - 1.$$

So $\mathbf{Q}_k(U) - \mathbf{Q}_k(U_r)$, $Q'_k(U) - Q'_k(U_r)$, $D_{W_k(U)} - D_{W_k(U_r)}$ *and their transposes are local operators of size* $C_L\delta_c$.

(vi) (Analyticity.) $W_j(U)$, l_j, $\mathbf{Q}_k(U)$, $Q'_k(U)$ *and* $D_{W_k(U)}$ *are holomorphic in* A' *on* $\{|A'|_\infty < \delta_{AB}\}$. *The nonlinear part* $C_k^{\mathrm{av}}(A)$ *at the reference* U *is holomorphic in* (A', A) *jointly.*

Proof. (i) is Lemmas 46.2 and 46.3. Details

(ii) The proof of Lemma 22.8 uses the reality of W in four places; everything else is algebraic, uses Lemma 22.7(a), which allows complex W, or uses only analyticity and Cauchy estimates.

(1) *Analyticity.* Relative transports along the contours of at most $2L$ bonds of $\mathbb{T}^{(j-1)}$ used in the step are products of at most $2L$ factors $e^{\mathrm{i}\delta R(W(\gamma'))B(b)}$, with γ' an initial piece. By Lemma 46.3(c), $\|R(W(\gamma'))\| \le e^{\kappa_{\mathrm{AB}}\delta_j\delta_c} \le 2$ instead of 1. By the same lemma, $W(\Gamma) = e^{\mathrm{i}Z_\Gamma}W^r_{j-1}(\Gamma)$ with $|Z_\Gamma| \le 4C_{\mathrm{AB}}\delta_j\delta_c$, so the families $\{W(\Gamma)\}$ lie in $\mathcal{D}_{C(a\delta_j^2+\delta_j\delta_c)}$, and the families $\{(XW)(\Gamma)\} = \{X_W(\Gamma)W(\Gamma)\}$ lie in $\mathcal{D}_{C(c_2+a\delta_j^2+\delta_j\delta_c)} \subset \mathcal{D}_{r_M/2}$. The averaged relative contour variables $X_W(y,x) = \mathrm{Av}(\{X_W(\Gamma)W(\Gamma)W(y,x)^{-1}\})$ are bounded by Lemma 21.5(e), as in the real case. The factor 2 is absorbed by $c_2 \to \frac{1}{2}c_2$. By Lemma 46.3(d), the arguments of all logarithms are within $C(c_2 + a\delta_j^2 + \delta_j\delta_c)$ of 1, and $\kappa(\tau_X(y)) \le 2$. The Cauchy estimates for c_j, c_j^τ and c_j^p are then unchanged.

(2) *The loops.* In the linear part, $|H_x|, |E_{\mathrm{loop}}| \le 16a\delta_j^2$ is replaced by $16a\delta_j^2 + C_{\mathrm{loop}}\delta_j\delta_c$, by Lemma 46.3(d). The bounds $|\mathfrak{f}(\operatorname{ad} H_x) - 1| \le C|H_x|$, $|\mathfrak{e}(\operatorname{ad} E_{\mathrm{loop}}) - 1| \le C|E_{\mathrm{loop}}|$ and $|R(e^{\mathrm{i}E_{\mathrm{loop}}}) - R(h_x)| \le C(|E_{\mathrm{loop}}| + |H_x|)$ hold for complex H_x, E_{loop} of size at most 1. Indeed $\mathfrak{e}$ is entire, $\mathfrak{f}$ is analytic for $\|\operatorname{ad}\| < 2\pi$ (Lemma 22.8), and $\|\operatorname{ad} H\| \le 2|H| \le 2$. Moreover $W_j(c) = e^{\mathrm{i}E_{\mathrm{loop}}}W([c])$, so $\|R(h_xW([c])) - R(W_j(c))\| = \|(R(h_xe^{-\mathrm{i}E_{\mathrm{loop}}}) - 1)R(W_j(c))\| \le C(|E_{\mathrm{loop}}| + |H_x|)$, since $\|R(W_j(c))\| \le 2$.

(3) *The tube average and the weights.* In $q_jB(c)$ every value $B(b)$ is multiplied by $R(W(c_-,x))R(W(\gamma'))$, a transport along an averaged contour variable times one along an initial piece γ' of $[x, x(c)]$. By Lemma 46.3(c) it has norm at most $e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}$, and the scalar weights are those at $W = 1$. The Cauchy–Schwarz inequality gives the bound on q_j. The weights of the averaged relative transports at the complex families satisfy $\sum_\Gamma |\omega_\Gamma| \le 1 + C_M \max_\Gamma |H_\Gamma| \le 1 + C(a\delta_j^2 + \delta_j\delta_c)$, by Lemma 21.5(a),(d) and (1), and the transports in $B^W(\Gamma)$ have norm at most $e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}$. So $|\lambda_W(y,x)[B]| \le 2(L-1)(1 + C(a\delta_j^2 + \delta_j\delta_c))e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}\max|B|$, and $|\gamma_jB| \le de^{\kappa_{\mathrm{AB}}\delta_j\delta_c}\max|B|$.

(4) *The remainder* r_j. The algebra of the proof is unchanged: with all factors $\mathfrak{e}, \mathfrak{f}, R(h_x), R(e^{\mathrm{i}E_{\mathrm{loop}}})$ equal to 1 and $R(h_xW([c])) = R(W_j(c))$, only q_jB remains. So r_jB is a sum of terms, each containing one of the differences in (2), multiplied by Ξ, Ξ_x, $B^W([c])$ or $\lambda_W(c_+, x(c))[B]$, which are bounded by $CLe^{\kappa_{\mathrm{AB}}\delta_j\delta_c}\max|B|$ by (3), and by the prefactor L^{-1}. Hence $|r_jB| \le C(a\delta_j^2 + \delta_j\delta_c)\max|B|$.

Part (f) follows as in the real case, from the factorization

$$\overline{X}(c) = \tau_X(c_-)\,\tilde{\mathfrak{a}}_j(X)(c)\,R(W_j(c))(\tau_X(c_+)^{-1}),$$

in which the linear parts add, and from the Cauchy estimates. Its statements on the coefficient of $B(b_0(c))$ are algebraic: for x on the axis of c the family $G(c_-,x)$ has one element, so $W(c_-,x) = W([c_-,x])$ for every W.

(iii) *(a), (b).* By (ii), $|l_jB|_\infty \le (1 + C(a\delta_j^2 + \delta_j\delta_c))|B|_\infty$ and $|c_j(B)|_\infty \le C\delta_j|B|_\infty^2$. The induction of the proof of Proposition 22.9 gives, by (46.1) and $\sum_i \delta_i^2 \le \frac{4}{3}$,

$$|B_j|_\infty \le |A|_\infty \prod_{i\le j}(1 + C(a\delta_i^2 + \delta_i\delta_c) + 2C\delta_i|A|_\infty) \le |A|_\infty e^{C(2a+2\delta_c+4|A|_\infty)} \le 2|A|_\infty$$

for $a, |A|_\infty \le c_3'$ and $\delta_c < \delta_{AB}$. The bounds on C_j^{av} and the locality follow as there.

(c). We run the proof of Proposition 22.9(c) with nonnegative majorants. For a scale-$(m-1)$ field v write $|v|$ for the function $b \mapsto |v(b)|$. In $q_m v(c')$ each value $v(b)$ is multiplied by one transport of norm at most $e^{\kappa_{\mathrm{AB}}\delta_m\delta_c}$, by (ii)(3), and the scalar weights are those at $W=1$. So $|q_m v| \le e^{\kappa_{\mathrm{AB}}\delta_m\delta_c} q_m^{(1)}|v|$ pointwise, where $q_m^{(1)}$ is the tube average at $W=1$. By (ii)(4), $|r_m v| \le r_m^{\mathrm{maj}}|v|$, where r_m^{maj} is a nonnegative kernel with the support of the real case and at most C_L entries per row, each bounded by $C(a\delta_m^2 + \delta_m\delta_c)$; it already accounts for the complex transports inside r_m. Hence

$$|l_j \cdots l_{i+1} v| \le e^{\kappa_{\mathrm{AB}}\delta_c \sum_m \delta_m} (q_j^{(1)} + r_j^{\mathrm{maj}}) \cdots (q_{i+1}^{(1)} + r_{i+1}^{\mathrm{maj}})|v|.$$

The prefactor is at most $e^{2\kappa_{\mathrm{AB}}\delta_c}$, by (46.1). The product of majorants is the real case with unitary transports, except that the coefficients of the insertions are $C(a\delta_m^2 + \delta_m\delta_c)$ instead of $Ca\delta_m^2$. The steps "Transports" and "Insertions" of that proof therefore apply verbatim, with $a\delta_m^2$ replaced by $a\delta_m^2 + \delta_m\delta_c$. They show that the product is of type

$$\prod_m (1 + C_L(a\delta_m^2 + \delta_m\delta_c)) \le \prod_m (1 + C_L a\delta_m^2) \prod_m (1 + C_L\delta_m\delta_c) \le (1 + Ca)\, e^{C_L\delta_c}.$$

In the step "First derivatives", $dc_m(B)$ plays the role of r_m, with coefficients $C\delta_m|A|_\infty$ by the Cauchy estimates of (ii), and the same majorant argument applies. In the step "Second derivatives", the type bounds just proved replace those of the real case. So both steps hold with constants multiplied by $e^{C_L\delta_c}$. The consequences follow from the kernel bounds as before.

(d). The values of w_k at the points of $\mathbb{T}$ are given by Theorem 22.5(c), which is the recursion of Definition 22.4 and holds for complex relative configurations, and by (ii); this gives the same bounds. At fine points, the extensions $\mathcal{E}^{(i)}$ transport along chains of averaged contour variables, with norm at most $e^{\kappa_{\mathrm{AB}}\delta_c}$ instead of 1, by (iv); the averaged relative contour variables $(X_i)_W(y,x)$ are bounded by Lemma 21.5(e), as in the real case.

(iv) For every $Y \in G^c$, $R(Y)^T = R(Y^{-1})$ with respect to b, since $\mathrm{tr}(YXY^{-1}Z) = \mathrm{tr}(XY^{-1}ZY)$. Hence $q_j'^T g(x) = R(W_{j-1}(U)(y,x))^{-1} g(y)$ and $q_j' q_j'^T = I$, as in the proof of Proposition 22.10(a). That proof uses only the volume factors, not unitarity. The Hermitian identity $Q_k' Q_k'^* = I$ need not hold for complex U; the b-transpose identity always holds. Locality (d) holds because Lemma 21.10(c) holds verbatim for complex configurations, and Lemma 22.8(a) holds by (ii). Covariance (f) follows from Lemma 21.10(b) with real g and (21.2).

The transport $T_U(y,x)$ along the chain of averaged contour variables is a product over $m<k$ of $R(W_m(U)(x_{m+1}, x_m))$, where $x = x_0 \in B_1(x_1)$, …, $x_{k-1} \in B_k(y)$ is the chain of x, and $x_k = y$. By Lemma 46.3(c) at index $m+1$, $W_m(U)(x_{m+1},x_m) = e^{\mathrm{i}Z_m} W_m(U_r)(x_{m+1},x_m)$ with $|Z_m| \le 4C_M C_{\mathrm{AB}}\delta_{m+1}\delta_c$. By (12.1), (46.1) and the unitarity of $R(W_m(U_r)(x_{m+1},x_m))$, $\|T_U\| \le \exp\left(8C_M C_{\mathrm{AB}}\delta_c \sum_m \delta_{m+1}\right) \le e^{\kappa_{\mathrm{AB}}\delta_c}$. Telescoping the product gives $\|T_U - T_{U_r}\| \le e^{\kappa_{\mathrm{AB}}\delta_c} - 1$.

For (e), compute in the given gauge, extended arbitrarily to $\mathbb{T}_\eta$; this is allowed by (f). By locality, $\tilde{B}_j$ on the relevant bonds depends only on U_r, A' in the cube. By Lemma 21.13, $W_j(U_r) = e^{\mathrm{i}\delta_j A_{r,j}}$ with $|A_{r,j}| \le \varepsilon + 32a \le C\varepsilon$. By Lemma 46.3(a),(b) and Lemma 21.2(c), $W_j(U) = e^{\mathrm{i}\delta_j A_{0,j}}$ with $|A_{0,j}| \le C(\varepsilon + \delta_c)$. All transports along contours of at most $2L$ bonds of $\mathbb{T}^{(j-1)}$, and, by Lemma 21.5(c), those along the averaged contour variables, are then within $C(\varepsilon+\delta_c)\delta_j$ of 1. By Lemma 22.7(e), which allows complex W, $|\lambda_W(y,x)[B] - \lambda_1(y,x)[B]| \le C(\varepsilon+\delta_c)L\delta_j \max|B|$. With (ii), $l_j - q_j(1)$ is a combination of at most C_L values with coefficients $C_L(\varepsilon+\delta_c)\delta_j$, using $a\delta_j^2 \le \varepsilon\delta_j$. The telescoping of the proof of Proposition 22.10(e), with the type bounds of (iii), gives the kernel bound $\sum_j C_L(\varepsilon+\delta_c)\delta_j e^{C_L\delta_c} \le C_L(\varepsilon+\delta_c)$. The same argument applies to Q_k'.

(v) The bound for Q'_k is the bound on the chain transports in (iv). For the unit-lattice derivative, $W_k(U)(b) = e^{\mathrm{i}\tilde{B}_k(b)} W_k(U_r)(b)$ with $|\tilde{B}_k| \le C_{\mathrm{AB}}\delta_c$, and (12.1) applies. For $\mathbf{Q}_k$ telescope:

$$\mathbf{Q}_k(U) - \mathbf{Q}_k(U_r) = \sum_{j \le k} l_k(U) \cdots l_{j+1}(U) \left(l_j(U) - l_j(U_r)\right) l_{j-1}(U_r) \cdots l_1(U_r).$$

By the formula for l_j in the proof of Lemma 22.8, $l_j B(c)$ is a sum of at most C_L terms. Each is a product of a bounded number of factors of the forms $R(W(\gamma))$, $R(W(y,x))$, ω_Γ, $R(e^{\mathrm{i}E_{\mathrm{loop}}})$, $R(h_x)$, $\mathfrak{e}(\mathrm{ad}\, E_{\mathrm{loop}})$ and $\mathfrak{f}(\mathrm{ad}\, H_x)$, applied to one value $B(b)$, with γ a contour of at most $2L$ bonds. Each factor is bounded by 2. By Lemma 46.3(c),(d), each factor changes by at most $C\delta_j\delta_c$ from U_r to U. For the weights $\omega_\Gamma = (\sum_{\Gamma'} \mathfrak{f}(\mathrm{ad}\, H_{\Gamma'}))^{-1}\mathfrak{f}(\mathrm{ad}\, H_\Gamma)$ this follows from the change of $H_\Gamma = \ell(W(\Gamma)W(y,x)^{-1})$, which is at most $C\delta_j\delta_c$ by Lemma 46.3(c) and the argument of Lemma 46.3(d). So $l_j(U) - l_j(U_r)$ is a combination of at most C_L values with coefficients $C_L\delta_j\delta_c$. With the type bounds of (iii), at U and at U_r, the kernel of the sum is bounded by $\sum_j C_L\delta_j\delta_c(1 + Ca)e^{C_L\delta_c} \le C_L\delta_c$. The supports are those of Proposition 22.10(d). Transposes follow by summing the kernel bounds.

(vi) By Lemma 46.3(b), $W_j(U)$ is holomorphic in A'. On the domain of (ii), $\delta_j^{-1}\ell(\tilde{\mathfrak{a}}_j(e^{\mathrm{i}\delta_{j-1}B})(c))$ is jointly holomorphic in B and in the finitely many bond variables of $W_{j-1}(U)$ on which it depends. Its derivative at $B = 0$, which is l_j, is therefore holomorphic in A'. The rest follows by composition, uniformly in k by the bounds of (iii)–(v). □

Remark 46.5 (Why the naive bound fails). The linear part in Lemma 46.3(a) explains why a per-scale bound fails. The derivative at $A' = 0$ of the plain iterated average is $\mathbf{Q}_j - D_{W_j^r}\mathbf{\Gamma}_j$: the gauge-adapted linearization plus a coarse pure-gauge term. The plain one-step relative average, whose linearization is $p_j = l_j - \delta_j D_{W_j}\gamma_j$ (Lemma 22.8(f)), has a sup-norm loss of up to a factor $2d + 1$, which comes entirely from that pure-gauge term. The term does not accumulate, since $\mathbf{\Gamma}_j$ is a sum of δ_{i+1}-weighted contour averages.

Remark 46.6 (Scope of Proposition 46.4). (1) *Not proved, and not needed below.*

- A curvature bound for $W_j(U)$ of order $a + \delta_c$ in field-strength units for $j < k$, i.e. a complex analogue of Lemma 21.11 and Proposition 21.12. We only have $|H_x(U)| \le 16a\delta_j^2 + C_{\mathrm{loop}}\delta_j\delta_c$. This is weaker by δ_j^{-1} in the δ_c-term, but summable over scales. At the unit scale the two agree: $|F_{W_k(U)}| \le 2a + C\delta_c$, from Lemma 46.3(c) with $n = 4$ and the Schwarz-lemma argument of (d).
- Derivative or Hölder bounds of $\tilde{B}_j$ for $0 < j < k$.
- Theorem 22.5(a),(b),(e),(f) and Lemma 22.3(b),(c) at complex configurations. They are stated for real relative configurations; only Theorem 22.5(c),(d) and Lemma 22.3(a) hold for complex ones.

Lemma 22.8, Proposition 22.9 and Proposition 22.10 use the backgrounds only through transports along contours of bounded length and along averaged contour variables, and through the weights of the averaged relative transports. At complex references Part X uses the averaging only through the objects of Definition 22.4 and their linearizations, which are defined by the recursion; through Theorem 22.5(c), for the values of w_j at the coarse points; and through the exact identities of step (iii) of the proof of Proposition 46.7, which are continued from the real points by the identity theorem. The averaged axial gauge and Theorem 22.5(a),(b) enter Part X only through Part V, at real references (Theorem 45.5(c), Lemma 45.1). So nothing is assumed about the third item. If a later argument needs any of the items above, it must be supplied separately.

(2) *Structure of the proof.* Proposition 46.4(ii)–(v) is a line-by-line modification of the proofs in Section 22, with the type bookkeeping of Proposition 22.9(c). The new inputs, Lemmas 46.1–46.3, are self-contained given Proposition 22.9 at real references.

(3) *Downstream.*

- The bounds $|\mathbf{Q}A(c)| \le C\|A\|_{S_c}$ and $|Q'\lambda(y)| \le C\|\lambda\|_{B(y)}$, and the sup-norm bounds of $\mathbf{Q}^T, Q'^T$, used in Theorem 35.2 and Part IX, follow from (iii) and (iv) with constants $e^{C_L\delta_c} \le 2$.
- The exact identities ($E = 0$, Proposition 22.10(b), and those listed in step (iii) of the proof of Proposition 46.7) are not extended on all of $\{|A'|_\infty < \delta_{AB}\}$: there only the sup norm of A' is controlled, so the real points need not be regular backgrounds. They hold on the ball $\mathcal{B}$ of that proof, whose real points are regular, by (vi) and the identity theorem.

46.2 Extension of the results of Part IX

Conventions at complex references. Proposition 46.7 is read as follows.

- Each result listed in (a)–(d) of the proposition holds at a complex reference of size $\delta_c < \delta_P$ whose real part is (M, ε)-regular with $\varepsilon \le \varepsilon_{\max}$, with ε replaced by $\varepsilon + C\delta_c$.
- The Re-coercivity constants and the decay rates are fixed and do not depend on δ_c: $\frac{c}{2}$ with the c of the real reference, and $\frac{1}{8}c_L$, $\frac{1}{16}c_L$ at level (L5); and the rates that the real-reference proofs give when every input decay bound is taken at a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3.
- Every other constant, computed at the real reference with these coercivity constants and rates, changes by at most a factor $1 + C_{\mathrm{cx}}\delta_c \le 2$.

The constants involved are chosen in the following order (Remark 49.8(4),(5), Remark 38.8):

$$L \to \kappa, \delta \to \rho \to M_1, M_{\mathfrak{N}}, M_1' \to M \to \delta_*, C_W, c_L'' \to M' \to C_{\mathrm{cx}} \to \bar\delta \to \varepsilon_{\max} \to \delta_P.$$

The thresholds M_1, $M_{\mathfrak{N}}$, M_1' are computed with doubled constants, so they serve at complex references and are fixed before M (Remark 46.9). The radius δ_* and the constant $C_W = C(1+2\delta_*)/\delta_*$ are fixed in the proof of Proposition 37.4(c), depending on d, L, G, α and M; the constant c_L'' is fixed in the proof of Lemma 37.5, with $C_{\mathrm{cmp}}c_L'' \le \delta_*$. The constant C_{cx} depends on $L, M, M', \kappa, \delta, \rho, \alpha$ and G, and enters only the cap $\bar\delta \le \min(\delta_{AB}, 1/C_{\mathrm{cx}})$. The threshold $\varepsilon_{\max}$ does not depend on $\bar\delta$ or δ_P, and the conditions on δ_P, among them $\delta_P \le \bar\delta$ and $\delta_P \le c_L''$, are listed at the end of the proof of Proposition 46.7. The domain constants come after δ_P, and (DC4) makes $\delta_c < \delta_P$ at the references of the step (Remark 46.8).

Proposition 46.7 (Extension to complex references)**.** *There is $\delta_P > 0$, depending on $L, M, \kappa, \rho, \alpha$, on the loss parameter δ of Part IX, and on the cap $\bar\delta$ and the threshold $\varepsilon_{\max}$ of Remark 38.8, such that the following holds for $\delta_c < \delta_P$, at complex references of size δ_c whose real part is (M, ε)-regular with $\varepsilon \le \varepsilon_{\max}$. The statements hold with ε replaced by $\varepsilon + C\delta_c$. The* Re*-coercivity constants and the decay rates do not depend on δ_c. The coercivity constants are those of Proposition 19.11, i.e. $\frac{c}{2}$ with c that of the real reference (Lemma 13.2), and at level (L5) $\frac{1}{8}c_L$ and $\frac{1}{16}c_L$ (Lemma 37.5, Proposition 37.6). The decay rates are those that the real-reference proofs give when every input decay bound is taken at the fixed rate of the differences, a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3 (step (vii) of the proof). The other constants, computed at the real reference with these coercivity constants and rates, change by at most a factor $1 + C_{\mathrm{cx}}\delta_c \le 2$, where C_{cx} depends on $L, M, M', \kappa, \delta, \rho, \alpha$ and G, but not on δ_c, $\bar\delta$, $\varepsilon_{\max}$ or δ_P, and $C_{\mathrm{cx}}\bar\delta \le 1$:*

(a) *Theorem 35.2(a)–(c), Corollary 43.8, and Lemmas 44.5 and 44.6;*

(b) *Lemma 44.7, Lemma 44.8, Lemmas 44.9 and 44.10, Proposition 44.11, Theorem 44.13 at the levels (L1) and (L3), and Theorem 44.15, with $W = 0$ at level (L3);*

(c) *Proposition 23.3(a)–(c). Here F^s and $J = D^T F^s$ are the holomorphic continuations, with $P_{\mathfrak{g}}$ replaced by its complex-linear extension $P_{\mathfrak{g}^c}$ and the summand continued without taking real parts. $|F^s| \le C(\varepsilon + \delta_c)$, while J need not be bounded uniformly in η;*

(d) *Theorem 45.5(a), (d), and (b) without the realness statement. The fixed point $A_*(s; B)$ is analytic in A', jointly with (s, B).*

Not extended, and not needed at complex references: Theorem 35.2(d), the identity of Lemma 36.6 (its bound on W_0 extends; proof of Proposition 37.4(c)), Lemma 45.1 and Theorem 45.5(c). The "real for real arguments" statements of Propositions 22.9 and 23.3 and of Lemma 22.8(e),(f) are dropped at complex references as well.

Proof. We go through the places where reality enters. Details

(i) *The L^2 theory.*

- Proposition 19.11 gives coercivity (as Re-coercivity), decay and comparison of domains for the Dirichlet operators of Parts III and VIII. It was proved for the averaging of Section 12; for the gauge-adapted averaging of Part IV it needs property (v) of Remark 19.12, i.e. Proposition 46.4(iv)–(vi) and step (iii) below. The averaged operators $\mathbf{\Gamma}^X$, $\mathbf{H}^X$, $\mathbf{C}^X$ follow by compositions and perturbed inverses.
- Part IX allows complex perturbations from the outset (Section 40, Setting).

(ii) *The local gauge in the interior estimates.* Instead of a real gauge in which U satisfies (B), use the real gauge of U_r. There $U = e^{i\eta A_0}$ with complex A_0 satisfying (B) with parameter $C(\varepsilon + \delta_c)$.

- Lemmas 12.2 and 12.3 allow $\mathfrak{g}^c$-valued A.
- The absorption, iteration and L^p arguments use only sup norms and quotients, no maximum principle.
- The flat estimates (Theorems 6.3, 43.2, Lemmas 36.2, 43.1) apply to complex-valued fields componentwise.

(iii) *Algebraic and analytic identities.* Fix U_r and consider A' in the open ball $\mathcal{B} = \{\text{size}(A') < \delta_P\}$. The size is a norm on the complex fields A', so $\mathcal{B}$ is convex. Real A' give real configurations $U = e^{i\eta A'} U_r$ that are $(M, \varepsilon + C\delta_P)$-regular, with curvature at most $C(\varepsilon + \delta_P)$; this needs $\varepsilon_{\max} + C\delta_P \le \min(\varepsilon_0, c_3')$. At these real points the operators below are those of Parts IV–IX at the real background U: Proposition 46.4 defines all objects at the reference U through $W_j(U) = U^{(j)}$ by the formulas of Section 22, and inverses are unique. So every identity proved in the note for real regular backgrounds of small curvature holds at the real points of $\mathcal{B}$. These form a totally real subspace of maximal dimension of the convex set $\mathcal{B}$. So an identity between two holomorphic functions on $\mathcal{B}$ that holds at the real points holds on all of $\mathcal{B}$, by the identity theorem. This is used in two stages.

(iii-a) Identities between objects of Proposition 46.4 and of the action. These are holomorphic on $\mathcal{B}$ by Proposition 46.4(vi), and since the action is a polynomial in the entries of U and U^{-1}. They are the exact covariance $E = 0$ (Proposition 22.10(b)); $J = D^T F^s$ (Lemma 23.2(a)); and $\langle \rho^{(1)}(A), F^s \rangle = 0$ in the proof of Proposition 23.3.

After this stage the pair $(Q'_k, \mathbf{Q}_k)$ has property (v) of Remark 19.12 on $\mathcal{B}$, and step (i) applies. Stage (iii-a) holds with the same proof on every such ball whose real points satisfy $\varepsilon + C \cdot (\text{radius}) \le \min(\varepsilon_0, c'_3)$; in particular on the ball of radius $2\delta_*$ of the proof of Proposition 37.4(c).

(iii-b) Identities that involve inverses. After step (i), the operators of Parts III, VIII and IX are holomorphic on $\mathcal{B}$, as compositions and inverses of holomorphic families with uniformly bounded inverses, and uniformly convergent series of such. The identities are those of Lemma 19.2 used in Lemma 36.4; $\mathbf{G}DR = DG'R$ (Lemma 15.3(a)) and (44.1); the b-symmetry $(\mathbf{G}^Z)^T = \mathbf{G}^Z$ used in Lemma 44.8; and the identity of Proposition 41.7(a). For the last, $\log\det\Delta_k$ means $\sum_b \ell_k(b)$, with the integral of Proposition 34.8(c). On $\mathcal{B}$, $\operatorname{Re}\Delta_k \ge \frac{1}{8}c_L$ (Lemma 37.5; it needs $|J_{U_r}| \le \varepsilon$, which holds at every reference where Proposition 41.7 is applied, Section 47, Setting; its conditions $\varepsilon \le \varepsilon_{\mathrm{cc}}$ and those of the proof of Proposition 37.4(c) hold since $\varepsilon \le \varepsilon_{\max}$, Remark 38.8), so the spectrum of Δ_k lies in $\{\operatorname{Re} z > 0\}$. There the integral is the principal logarithm: it is holomorphic on $\mathcal{B}$, real at the real points, and its exponential is $\det\Delta_k$.

No connectedness of the set of all complex references is needed. That set need not be connected: if G has a $U(1)$ factor, real regular backgrounds with different fluxes exist.

(iv) *Unitarity of transports and the averaging.* This is Proposition 46.4.

(v) *The action.* The proof of Proposition 23.3 uses a gauge with $|A_0|, |\partial A_0| \le \varepsilon$, $|R(H_p)-1| \le C\varepsilon\eta^2$ and $|F^s| \le C\varepsilon$. These hold with $C(\varepsilon+\delta_c)$, using the holomorphic summand and $P_{\mathfrak{g}^c}$. The current J is not bounded uniformly in η. In the weakened background map it enters only as $(D\mathbf{H}(s))^T\mathsf{F}$ (Section 45).

(vi) *The trace inequality.* Lemma 44.7 is stated for complex perturbations.

(vii) *The tiers.* Corollary 43.8 and Lemmas 44.5 and 44.6, also with $\varphi \equiv 1$, follow from (i), (ii) and (iv) as at real backgrounds. In particular, Lemma 44.4 and Theorem 43.2 are flat estimates, applied componentwise; the normal form holds with complex coefficients of size $C(\varepsilon+\delta_c)$, by (ii); and the bounds $|Q'\lambda(y)| \le C\|\lambda\|_{B(y)}$, the sup bounds of Q'^T and the block-locality of Q'^TQ' hold by (iv). Lemma 44.8 follows from (i) and (vi). Theorem 44.13 at level (L1) has only local steps, which Lemmas 44.5 and 44.6 bound. Lemmas 44.9 and 44.10 and Proposition 44.11 use, besides these:

- the walks of Part IX, which allow complex perturbations from the outset, and the comparison of domains, by (i);
- Proposition 36.3, for the differences of local propagators, by (ii);
- the bounds on Q', Q'^T and Q'^TQ' just listed, by (iv);
- Lemmas 44.7 and 44.8, for an external D, by (vi);
- Lemma 12.3 with $\mathfrak{g}^c$-valued A, which gives $|c_{\mu\nu}| + [c_{\mu\nu}]_\beta \le C(\varepsilon+\delta_c)$ for the coefficients of $\mathfrak{F}$ in Lemma 44.10(N2).

Their proofs apply verbatim, and no new condition on δ_c arises; the thresholds used are those listed below. *The fixed rates.* For the operators of Parts II, III and VIII, Lemma 13.2 gives the hypotheses of Lemma 5.7 with the constants $c/2$, C_1+1 and $\min(\alpha_1, \gamma/2, 1)$, where c, C_1, α_1 are those at the real reference and γ is the rate of the perturbation. So these operators decay at the rate $\gamma_{\mathrm{c}} = \min(\alpha_1, \gamma/2, \frac{1}{3}, c/(4(C_1+1)))$. By Lemma 13.3, their differences from the operators at U_r have decay $(C\delta, \gamma_{\mathrm{c}}/4)$, with $\delta \le C_{\mathrm{cmp}}\delta_c$. At level (L5) the lower bounds are $\frac{1}{8}c_L$ and $\frac{1}{16}c_L$, and the differences again

decay at a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates. Every input decay bound below, at real and at complex references, is taken at these quarter rates. At the rate γ_{c} itself the constants at a complex reference are only bounded by about twice the real ones.

The other constants. In outline: each constant is a nonnegative combination of products of monotone, positively homogeneous functions of finitely many inputs; at a complex reference the inputs change by at most a factor $1 + C_{\mathrm{in}}\delta_c$, and convexity in the scaling factor turns this into a change of every constant by at most $1 + C_{\mathrm{cx}}\delta_c$, level by level. In detail, order the levels of Sections 41 and 44, and the constants of Section 45, as in Section 41: $\ell = 1, \dots, N$. Each multiplicative constant K of level ℓ is $F_K(x_\ell)$, where x_ℓ collects the primitive inputs and the constants of the earlier levels, and F_K is a nonnegative combination of products of nonnegative, nondecreasing, positively 1-homogeneous functions of x_ℓ. These include sums, products, maxima, Neumann and walk series with nonnegative coefficients, and integrals of such, like the resolvent integral of Corollary 41.6. So $t \mapsto F_K((1+t)x_\ell)$ is convex. Put $\sigma_{\mathrm{cx}} = 2^{1/3} - 1$, so that $(1+\sigma_{\mathrm{cx}})^3 \le 2$. For $x' \le (1+t)x$ and $0 \le t \le \sigma_{\mathrm{cx}}$, convexity gives

$$F_K(x') \le F_K(x)\Big(1 + \frac{t}{\sigma_{\mathrm{cx}}}\Big(\frac{F_K((1+\sigma_{\mathrm{cx}})x)}{F_K(x)} - 1\Big)\Big).$$

At a complex reference the primitive inputs change by at most a factor $1 + C_{\mathrm{in}}\delta_c$: the averaging kernels by $e^{C_L\delta_c} \le 1 + 2C_L\delta_c$ (Proposition 46.4(iii)–(v)); the operators of Parts II, III and VIII by the differences above; and W_2^X by $C_{\mathrm{cmp}}C_W\delta_c$ (proof of Proposition 37.4(c)). Each input bound K is taken free of ε (with $\varepsilon \le 1$; e.g. $\|W_2^X\| \le C$), so an additive change $C\delta_c$ is a factor $1 + (C/K)\delta_c$; for inverses, Lemma 13.3 gives the change relative to the bound, also at level (L5), where $C_G < 1$. Put $C^{(\ell)} = (C_{\mathrm{in}}^{(\ell)}/\sigma_{\mathrm{cx}}) \max\big(1, \max_K(F_K((1+\sigma_{\mathrm{cx}})x_\ell)/F_K(x_\ell) - 1)\big)$, the maximum over the constants of level ℓ, with $C_{\mathrm{in}}^{(\ell)} = \max(C_{\mathrm{in}}, C^{(1)}, \dots, C^{(\ell-1)})$. Let $C_{\mathrm{cx}} = \max_\ell C^{(\ell)}$. For $\delta_c \le 1/C_{\mathrm{cx}}$ one has $C_{\mathrm{in}}^{(\ell)}\delta_c \le \sigma_{\mathrm{cx}}$ for every ℓ. So, by induction over ℓ, every input of every level changes by at most a factor $1 + \sigma_{\mathrm{cx}}$, and every constant by at most $1 + C_{\mathrm{cx}}\delta_c \le 2$. The values $F_K((1+\sigma_{\mathrm{cx}})x_\ell)$ are finite. Indeed, the step constants θ of Lemma 41.3 and θ' of Theorem 44.13 are sums of products of at most three inputs of their level, so their values at $(1+\sigma_{\mathrm{cx}})x_\ell$ are at most twice those at x_ℓ. Moreover M_1 is computed with the inputs doubled, and M_1' with θ' replaced by $2\theta'$. So for $M \ge M_1, M_1'$ the walks converge at $(1+\sigma_{\mathrm{cx}})x_\ell$, with ratio at most $\frac{1}{2}$. For the same reason $2\theta'$ bounds θ' at complex references, and the inputs of M_1 at most double there. C_{cx} is built from real-reference constants that hold for all ε below thresholds fixed before $\bar\delta$, such as ε_0 and c_5. So it depends on L, M, M', κ, δ, ρ, α and G, but not on $\bar\delta$, $\varepsilon_{\max}$ or δ_P. The thresholds $M_{\mathfrak{N}}$, M_1 and M_1', fixed in this way before M, serve at every complex reference with $\delta_c < \delta_P$. The proofs of Theorems 44.13 and 44.15 then apply verbatim.

(viii) *Analyticity in A'.* Every input is holomorphic in $A' \in \mathcal{B}$, by (i)–(v), and the contraction of Theorem 45.5 is uniform in A'. The factorization of Theorem 45.5(d), including its statement on the dependence on s, follows from the same proof: Lemma 40.3(b) allows complex backgrounds, the averaging operators are local at complex references (Proposition 46.4(iii),(iv)), and the nonlinear part is local by (c).

(ix) *Not extended.*

- The identity of Lemma 36.6 differentiates real gauge transformations and needs $|J_U| \le \varepsilon$. It enters only Theorem 35.2(d), which is not used after Section 45.

- Lemma 45.1 and Theorem 45.5(c) concern real critical backgrounds, where they are used.

Smallness. δ_P must satisfy the following:

- $\delta_P \le c_L''$ of Lemma 37.5, for its coercivity, used in Section 47. Since $C_{\mathrm{cmp}} c_L'' \le \delta_*$, with $C_{\mathrm{cmp}} = 4$ (first item of Section 46) and δ_* of the proof of Proposition 37.4(c), the perturbations of size $< C_{\mathrm{cmp}}\delta_P$, and the ball of twice that size used there for the Cauchy estimate, lie within the radii fixed with δ_*. These are the radii of Proposition 19.11 for $W = 0$ and for W_2, W_2^X, of Proposition 46.4, of Proposition 23.3, of Proposition 35.1(e), and of $E = 0$ (step (iii)). With $C_{\mathrm{AB}} c_L'' \le c_L'$ the condition gives $C_{\mathrm{AB}}\delta_P \le c_L'$; moreover $C_{\mathrm{AB}}\delta_P \le cL^{-d-1}$, for the elimination of Lemma 32.7 at a complex base (Corollary 34.6(c), with $|\mathfrak{r}_{\mathrm{c}}| \le \frac{1}{2}$);
- $\varepsilon_{\max} + C\delta_P \le \min(\varepsilon_0, c_3', \varepsilon_{\mathrm{ID}}, \varepsilon_{\mathrm{cc}}, \varepsilon_S, \varepsilon_L, \varepsilon_S', \varepsilon_L', \varepsilon', \frac{1}{20})$, with ε_S, ε_L the thresholds of Propositions 36.3 and 36.5, ε_S', ε_L' those of Propositions 43.4 and 43.7, and ε' that of Corollary 43.8; moreover $\varepsilon_{\max} + C\delta_P$ satisfies the smallness condition on ε depending on M of Section 41, and $C(\varepsilon_{\max} + \delta_P) \le 1$, for the normal form of Lemma 12.2, used in Lemmas 44.6 and 44.7;
- $C(\varepsilon_{\max} + \delta_P) \le c_5$ of Proposition 23.3;
- $\varepsilon_{\max} + C\delta_P \le \varepsilon_P$, with ε_P of Theorem 45.5 computed with the multiplicative constants doubled;
- at complex references the condition $(1 + K_H) r_P \le c_3$ in the proof of Lemma 45.3 becomes $(1 + 2K_H) r_P \le c_3'$, with r_P computed with the multiplicative constants doubled;
- $\delta_P \le c_{\mathrm{DC}} r_M L^{-1}$, the condition on δ_P in (DC5) of Remark 38.8. At a complex reference the base of the step is $W = e^{\mathrm{i}\beta} W_r$ bondwise, with $|\beta|_\infty \le C_{\mathrm{AB}}\delta_c$ (Proposition 46.4(i)). For c_{DC} small, (DC5) gives $C_{\mathrm{AB}}\delta_P \le \frac{1}{32} c_0 L^{-1}$, with c_0 of Lemma 32.3(c), and $LC_{\mathrm{AB}}\delta_P \le c$, with c of Lemma 22.7(f). So the averaged contour variables of the step and the averaged relative transports $\mathfrak{G}_W$ are defined and holomorphic at the complex base (Remark 38.8); the analogous condition on the real part is the first condition of (DC5). This is used in Section 47.
- $\delta_P \le \bar{\delta}$, so that $C_{\mathrm{cx}}\delta_P \le 1$ (step (vii)).

At complex references the constant K_H is at most $2K_H$, by step (vii). Later sections use $2K_H$ in place of K_H, and the radii R_P, r_P and the constants C_P, ε_P, ϱ_P of Theorem 45.5 computed with the multiplicative constants doubled. These bound, respectively are valid for, the corresponding quantities at every complex reference with $\delta_c < \delta_P$, and they do not depend on δ_P. Here $\varepsilon_{\max}$ and the cap $\bar{\delta} \ge \delta_P$ are those of Remark 38.8; $\varepsilon_{\max}$ is fixed before δ_P, by the rule stated there, and the proposition holds at every complex reference whose real part is (M, ε)-regular with $\varepsilon \le \varepsilon_{\max}$. Every joint condition above has the form $\varepsilon + C\delta_c \le T$, with T fixed before both constants. The thresholds $M_{\mathfrak{N}}$ and M_1' of Section 44 are lower bounds for M (Remark 49.8(4)). So they are fixed before M, and hence before $\bar{\delta}$, with their constants doubled as in step (vii). All of these are conditions on δ_P after L, M, κ, ρ, $\bar{\delta}$ and $\varepsilon_{\max}$ are fixed. □

Remark 46.8 (Use in the renormalization group). The references of the step are the points of the graded domain of step $k+1$, read in the units of step k and extended as above. Their complex parts have size $\delta_c \le 2C_{\mathrm{ext}}(L^{-1}\delta_0 + L^{-2}\delta_1)$. So $\delta_c < \delta_P$ is the condition (DC4) on the domain constants, which are chosen after δ_P (Remark 38.8); it does not involve the coupling. The domains are those of fixed size of Remark 38.8, and (DC5) makes the averaged contour

variables of the step analytic at these references. By Propositions 46.4 and 46.7, the new activities produced by the step are analytic functions of the complex background, as the norm requires.

Remark 46.9 (Complex references in Section 44). Proposition 44.11 and Theorem 44.13 concern real backgrounds. The paragraphs of their statements on complex references fix how the thresholds $M_{\mathfrak{N}}$ and M_1' are computed, so that the same numbers serve at the complex references of this section. Remark 38.8, Remark 49.8(4) and step (vii) of the proof of Proposition 46.7 use the same rule. It has four parts.

(a) *Fixed coercivity constants and rates.* The coercivity constants are the fixed ones of Proposition 46.7, and every decay bound that enters is taken at the fixed rate of the differences there, a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3. Neither depends on δ_c, so the same numbers serve at real and at complex references.

(b) *Doubled constants.* The threshold M_1 of Theorem 41.4, wherever it enters (Remark 49.8(4), Proposition 44.11), is computed with the multiplicative constants of Part IX multiplied by 2, and M_1' with θ' replaced by $2\theta'$. Likewise $2K_H$ replaces K_H, and the radii R_P, r_P and the constants C_P, ε_P, ϱ_P of Theorem 45.5, and the regularity thresholds that define $\varepsilon_{\max}$, are computed with the multiplicative constants doubled (end of the proof of Proposition 46.7, Remark 38.8).

(c) *Why doubling suffices.* At complex references of size $\delta_c < \delta_P \le \bar\delta \le 1/C_{\mathrm{cx}}$, every multiplicative constant changes by at most a factor $1 + C_{\mathrm{cx}}\delta_c \le 2$, and the inputs of θ' by at most a factor $1 + \sigma_{\mathrm{cx}}$, where $\sigma_{\mathrm{cx}} = 2^{1/3} - 1$, so that $(1 + \sigma_{\mathrm{cx}})^3 \le 2$. Since θ' is a sum of products of at most three inputs, $2\theta'$ bounds θ' there (Proposition 46.7, step (vii) of the proof).

(d) *The order.* The numbers of (a) and (b) do not depend on $\bar\delta$, $\varepsilon_{\max}$ or δ_P. So $M_{\mathfrak{N}}$, M_1 and M_1' are lower bounds for M, fixed before M (Remark 49.8(4)). The constant C_{cx} may depend on M and M', but it enters only the cap $\bar\delta \le \min(\delta_{AB}, 1/C_{\mathrm{cx}})$, which is chosen after them; then come $\varepsilon_{\max}$ and $\delta_P \le \bar\delta$ (Remark 38.8).

47 The weakened step quantities

This section turns the ingredients of one step (Theorem 34.9) into the input of the cluster expansion. That input consists of functions of the weakening parameters $s \in \mathbb{D}_\kappa$, the white-noise field and the background, with three properties. They are analytic with uniform bounds; they factorize when $s_\square = 0$ outside a union of boxes; and they are invariant under real gauge transformations. They are defined by source-free formulas for every background in the domain, not only at critical backgrounds. Section 48 integrates them, and the flow of Section 53 applies the result at every step.

The determinant is treated in Proposition 41.7, and the Gaussian measure through $C_k^{1/2}(s)$ (Corollary 41.6). This section treats the interaction (Section 47.2), the gauge-fixing remainder (Section 47.3), the Jacobians (Section 47.4), the old activities composed with the background map (Section 47.5) and the Hessian away from critical backgrounds (Section 47.6), which Section 37 already defines. Section 47.1 gives the localization lemma that turns these functions of s into pieces of volume type.

The formula of one step. Recall the step of Part VII. The reference $U' = U_{k+1}(V')$ is a background field of the configuration V' of step $k+1$, the base $W = V^0 = U'^{(k)}$ is its average at scale k (Lemma 31.6), and the fluctuation field Z lives on the bonds $\mathfrak{F}$ of Definition 32.1;

the step integrates over the configurations $e^{\mathrm{i}\hat{Z}(Z)}W$ (Proposition 32.8). Theorem 34.9 writes the step as

$$T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(V') = \mathcal{N}_k \exp\Big(-\frac{1}{g^2}S(U') - \frac{1}{2}\sum_{b\in\mathfrak{F}}\ell_k(b) + \sum_c \log \mathfrak{j}_c\Big)$$
$$\times \int d\mu_{C_k}(Y)\,\chi^{\mathrm{w}}(\hat{Z}(gY))\exp\Big(-\frac{(\mathcal{V}+G_3)(gY)}{g^2} + \mathfrak{J}(gY) + \mathfrak{E}(e^{\mathrm{i}\hat{Z}(gY)}W)\Big).$$

Here $\mathcal{N}_k$ does not depend on the background (Theorem 34.9(a)); $\ell_k(b)$ are the determinant densities of Proposition 34.8; $\mathfrak{j}_c$ are the Jacobians of the coarse bonds c (Theorem 34.9(b)); μ_{C_k} is the Gaussian measure with the covariance $C_k = \Delta_k^{-1}$ of Corollary 34.6; χ^{w} is the fluctuation cutoff of Definition 32.4, and $\hat{Z}$ the elimination map of Lemma 32.7; $\mathcal{V}+G_3$ is the interaction, of third order (Theorem 34.9(d)); $\mathfrak{J}$ collects the logarithms of the densities σ and of the Jacobians (Theorem 34.9(e)); and $\mathfrak{E}$ is the sum of the old activities, composed with the background map (Theorem 34.9(f)). In this section the determinant becomes the one-loop pieces of Proposition 41.7, the measure is written with $C_k^{1/2}(s)$, the interaction gives the pieces of Sections 47.2 and 47.3, $\mathfrak{J}$ those of Section 47.4, and $\mathfrak{E}$ the composed activities of Section 47.5.

Setting.

- *The reference.* U' is a reference in the sense of Section 45: real and (M,ε)-regular with $|J_{U'}| \le \varepsilon$, or a complex reference $U' = e^{\mathrm{i}\eta A'}U'_r$ of size $\delta_c < \delta_P$ in the sense of Section 46, whose real part is (M,ε)-regular with $\varepsilon \le \varepsilon_{\max}$ (Proposition 46.7) and satisfies $|J_{U'_r}| \le \varepsilon$. Theorem 45.5(a), (b), (d) hold at it (Proposition 46.7(d)). The norm $|\cdot|_*$ is that of Theorem 45.5, with $\beta = \alpha$, and the Hölder exponent α used below is fixed with $\alpha \le \frac{1}{2}$, as in Section 45. The constant K_H and the radii R_P, r_P of Part IX are those computed with the multiplicative constants doubled (Proposition 46.7, end of the proof); they serve at every complex reference with $\delta_c < \delta_P$ and do not depend on δ_P. The thresholds M_1 and M_1' are those of Remark 49.8(4) and do not depend on δ_P. At a complex reference, $|\cdot|_*$ is taken in the real gauges of U'_r, and covariant derivatives with respect to U'_r; the other reading, with respect to U', differs by a local term of size $C\delta_c|A|$. Within this setting, where regularity is assumed separately, the current bound on U'_r is used at complex references only for coercivity (Proposition 47.12(d)); at real references it is also used in Theorem 45.5(c). On the graded domains it is used in addition for the (M,ε)-regularity of U'_r (Corollary 30.4), for the background-activity domain (Proposition 41.7(b)) and in Lemma 38.5. It holds there by (DC2): in the units of step k, $|J_{U'_r}| < L^{-3}\theta_r \le \varepsilon$ (Remark 38.8(ii)).

- *Extension.* A point of the activity domain of step $k+1$ at a polymer Y' of step $k+1$ fixes U' only near Y'. It is extended to a reference by the cutoff of Remark 38.7, with the same real part U'_r. A piece indexed by a polymer Z is used only when Y' contains the LM-cubes meeting $Z^{(r')}$ (Remark 47.11). By the factorization below, such a piece does not depend on the extension.

- *The base.* For every reference put

$$W := U'^{(k)}, \qquad V' := U'^{(k+1)}.$$

  At critical backgrounds these are V^0 and V' of Lemma 31.6. They are local in U' and covariant, and analytic at complex references by Proposition 46.4(i). The elimination of Lemma 32.7, for the plain constraint ℓ_c of Section 32, is taken at the base W. In that lemma the parameter a is defined through the curvature ε_1' of V'; at a general

reference it is replaced by a bound on the curvatures of the real parts U'_r and $(U'_r)^{(k)}$, which is all the proof uses. On the graded domains these curvatures are at most $2\theta_r$ (Remark 38.8(ii)), so $L^d\mathsf{a} \le c$ is (DC3). Since $c \le r_M/64$ in Lemma 32.7, the real part $W_r = (U'_r)^{(k)}$ of the base then satisfies $|\ell(W_r(\partial p))| \le L^{-2}\mathsf{a} \le \frac{1}{8}\varepsilon_M$, so it lies in the regular set $\mathfrak{M}$ of Section 32 with the margin of (32.1). At complex bases the elimination is analytic in the reference, by the argument in the proof of Corollary 34.6 together with Proposition 46.4(i),(vi). This needs $C_{\mathrm{AB}}\delta_c \le c'_{L}$, which is part of the smallness of δ_P (Proposition 46.7). It gives ζ_c, $\hat{Z}$, $\hat{Z}^1$, $\hat{Z}^2$, with $|\hat{Z}(Z)|_\infty \le CL^d|Z|_\infty$. The averaged contour variables of the step and the transports $\mathfrak{G}_W$ at complex bases are defined and holomorphic by (DC5) (Remark 38.8, reasons for (ii)).

- *The fields.* We write ξ for the white-noise field on $\mathfrak{F}$ (Dimock's W), since in Sections 47–52 W is the base, and $Z(s) = gC_k^{1/2}(s)\xi$. $A_*(s;B)$ is the weakened background map of Theorem 45.5 at the reference U'.

- *Margins.* Let r be the largest margin of the weakened families used: $\mathbf{C}(s)$, $\mathbf{H}(s)$, $D\mathbf{H}(s)$ and their transposes, and $C_k^{1/2}(s)$ (Corollary 41.6). The definition of weakenable families (Section 40) requires $r \le M/16$. The margins are sums of a bounded number of ranges of local operators (Lemma 40.5(a), Theorem 41.4(a), and the box split of the additive local terms, Section 40), independent of M; so $r \le M/16$ is a condition on M, listed in Remark 49.8. A family with margin r is also a family with every larger margin up to $M/16$, since its kernel supports and its data dependence lie in every larger neighbourhood; so we may and do assume $r \ge 3$. Put $r' = r + 2L + 2$. We assume $M > 32(L+1)$, so that $2r' < M/4$. Two polymers that do not touch are then at distance at least $M/4 > 2r'$, so their r'-neighbourhoods are disjoint.

- *Constants.* C_P denotes constants that may depend on $\kappa, \delta, L, M, \rho, \alpha$, but not on k, n, η or s.

Remark 47.1 (Notation in Sections 47–53). Some letters are used here with a meaning that differs from earlier parts of the note. We keep them, and list them once.

- C_T is the bound on $C_k^{1/2}(s)$ of Section 47.2, not the tier constant of Section 44. C_{tor} is the torus constant of (T2).

- C_d, A, ϵ_* are the constants of Theorem 48.6; they are not the constants of Lemma 40.4 or the bootstrap threshold of the earlier parts.

- κ and $\kappa' = \kappa - C\delta$ are the weakening rates of Part IX; Section 49 writes κ'_w for κ', since there κ' is also the reblocked rate of Lemma 49.3. κ_n is the rate of the activity norm, fixed in Remark 49.8(1). The rate κ in Theorem 48.6 is generic. Theorem 45.5 is stated on $\mathbb{D}_\kappa$ with κ the rate of its inputs $\mathbf{H}(s)$, $\mathbf{C}(s)$ at level (L4); that rate is at least κ', and the pieces below are formed on $\mathbb{D}_{\kappa'}$.

- $\ell_k(b;1)$ is the density at the reference $U' = 1$, not at $s = 1$.

- U' is the reference; in Section 46 it was also the relative factor $e^{\mathrm{i}\eta A'}$. W is the base; earlier it also denotes potentials and configurations.

- r is the margin; the resolvent parameter of Corollary 41.6 is unrelated. r_E is the margin of dependence of the old activities (Section 48).

- δ is the loss parameter of Part IX, δ_c the size of a complex reference, and $\theta_r, \delta_1, \delta_0$ (with primes) the budgets of a graded domain. $\bar{\delta}$ is the cap of Remark 38.8, and θ in Section 44 is unrelated; δ_0 of Proposition 19.11 enters only through δ_P.

- B_P, B_3, B_5 are unrelated to the coarse field B; C_{P} (roman) is the Poincaré constant of Lemma 34.2, unrelated to C_P. In Part IX, N_{b} is the box count of Section 40, and N, in Proposition 46.4 and earlier, is the matrix size of G. h_j is a partition of unity, unrelated to $h_k = p_k g_k$. a is the curvature parameter of Lemma 32.7, not the averaging parameter a.
- $\hat{Z}$ is the elimination map acting on fields; applied to a polymer (Remarks 47.3 and 49.4) it denotes the union of the LM-cubes meeting $Z^{(r')}$. The polymer Z and the field $Z(s)$ are distinguished by the argument.
- p is a plaquette in s_p and v_p, and p, $p_0 = p_{0,k}$ bound the white noise; $p_k = p(g_k)$ is the small-field scale. ζ is the bound on $\hat{Z}$, unrelated to ζ_c. τ is the composed-activity ratio of Proposition 47.10. $\varphi_{y,x}$ are the gauge-fixing pieces of Definition 47.6, $G_3(\square; s)$ their sum over a box, and $f(H) = \mathsf{N} - \operatorname{tr}\cos H$; the composed activities are $f(X; s, Z)$, with a polymer as first argument. F (Section 45) and F^s denote the same symmetrized curvature.
- ϵ is, in Section 52, the finest spacing; ε_k is the vacuum energy per volume, unrelated to the thresholds ε_0, ε_1, ε_M and to ε^0 of N_χ. $n = n_k$ is the side of $\mathbb{T}^{(k)}$ (see (T0)), and n_0 is a threshold, not a value at step 0. In Sections 52 and 53, $h_0 = p(g_0)g_0$ is the small-field threshold at step 0; in Section 46.1, h_j with $h_0 = 1$ is a different quantity.
- In Section 53, $E' = E'^{(k+1)}[\mathcal{E}_k, g_k]$ is the raw new generation of a step, Y'_c the polymer of scale $k+1$ that carries the Jacobian term of the bond c, and $\mathcal{E}_k(Y)$ the grouped old activities. B_{new}, $B^{(0)}_{\mathrm{new}}$ bound the new terms of a step, and $\mathsf{K} = 2\max(B_{\mathrm{new}}, B^{(0)}_{\mathrm{new}})$ bounds the generations; K is unrelated to the number K of steps, and B_{new} to B_1 of Theorem 24.2 and to B_3, B_5, B_P. C_{old}, $C_{\mathfrak{e}}$, C_{jac}, C_{rb} and C_{new} are the constants of Proposition 53.2 and Lemma 53.4, and (Γ1)–(Γ4) are the conditions on γ of Remark 53.5; (Γ4) and its constant c_{F} are used only in Part XI.
- In Sections 50–53: $\mathfrak{S}_j^{\mathrm{grid}} \subset \mathfrak{S}_j$ is the group of the symmetries of the grid of scale j, and E^{sym} the average of a raw generation over it (Definition 50.7); it is not the group $\mathfrak{S}_j$ of all lattice symmetries. $\mathcal{U}^\flat$ denotes the domains with $\delta_1^\flat = \frac{1}{4}\delta_1$ in place of δ_1; $\Theta^\flat$ is the quantity Θ of Lemma 51.1 with $\delta_1^\flat$ in place of δ_1 (Remark 49.8(5)). C_{sym} is the constant of (DC6). C_β, C_{ex} and C_{ne} are the constants of Proposition 50.13(b), Proposition 51.5 and Lemma 51.7. $\mathcal{X}_j^{\mathrm{sm}}(\square)$ is the set of the small polymers of scale j based in $\square$, $\Psi^{(j)}_\square$ their sum minus $\beta_j S_\square$, and $E^{(j),\mathrm{ne}}_Y$ the sum over the other polymers of scale j with closure Y (Definition 53.1).
- In Section 53.5: Σ_k is the set of the coupling sequences $\pi = (g_0, \dots, g_k) \in (0, \gamma]^{k+1}$ with $g_{i+1} \le 2g_i$, not a sum, and $E^{(j)}[\pi]$, $\beta_j[\pi]$ are the generations and the coefficients of a free sequence π. B_k is the set of the bare couplings g_0 for which $g_1, \dots, g_k$ are defined and lie in $(0, \gamma]$ (Proposition 53.10, Corollary 53.11); it is unrelated to B_{new}, B_1, B_3, B_5 and B_P. $\mathcal{A}^\flat(Y; U_r)$ and U_A are the fields of the proof of Proposition 53.10. C_{lat}, C_{vac}, C'_{vac} and $\bar{p}_5$ are the constants of Theorem 53.12(d),(f) and its proof, and I_K, c_V, c'_V those of Corollary 53.13 and its proof; c_V and c'_V are unrelated to the coarse configurations V and to the constant c' of Definition 49.5.

47.1 The localization lemma

For a polymer Y, a polymer $Z \supset Y$ and $t \in [0,1]^{Z \setminus Y}$, let $s(Z,t)$ be the point of $\mathbb{D}_\kappa$ with $s_\square = 1$ for $\square \subset Y$, $s_\square = t_\square$ for $\square \subset Z \setminus Y$, and $s_\square = 0$ otherwise.

Lemma 47.2 (Localization)**.** *Let $\kappa \geq 1$, let Y be a polymer, and let f be a function on $\mathbb{D}_\kappa$, analytic in the interior and continuous, with $|f| \leq F$. Suppose that, for every union of boxes $Z' \supset Y$ with $*$-component $Z'_0 \supset Y$, and every s with $s_\square = 0$ for $\square \not\subset Z'$, $f(s)$ depends on s only through $s_\square$, $\square \subset Z'_0$, and on the data only on $(Z'_0)^{(r')}$. Then*

$$f(1) = \sum_{Z \supset Y} f_{\mathrm{loc}}(Y, Z), \qquad f_{\mathrm{loc}}(Y, Z) = \int_{[0,1]^{Z \setminus Y}} \partial_{s_{Z \setminus Y}} f\big(s(Z,t)\big)\, dt,$$

the sum running over the polymers $Z \supset Y$, where:

(a) $f_{\mathrm{loc}}(Y, Z)$ depends on the data only on $Z^{(r')}$;

(b) $|f_{\mathrm{loc}}(Y, Z)| \leq F\, e^{-(\kappa-1)|Z \setminus Y|}$;

(c) if f is invariant under real gauge transformations of the data, or analytic in further parameters with $|f| \leq F$ uniformly, then so is $f_{\mathrm{loc}}(Y, Z)$.

Proof. Write $f(1)$ as a function of the variables $s_\square$, $\square \not\subset Y$, with $s_\square = 1$ on Y. Apply $g(1) = g(0) + \int_0^1 g'(t)\, dt$ in each of these variables. This gives the sum over sets S of boxes outside Y of $\int_{[0,1]^S} \partial_{s_S} f\, dt$, evaluated at $s_\square = 1$ on Y, $s_\square = t_\square$ on S, and $s_\square = 0$ elsewhere. Let $Z' = Y \cup S$, with $*$-component $Z'_0 \supset Y$. By hypothesis, at these points f does not depend on $s_\square$ for $\square \subset S \setminus Z'_0$. So the term vanishes unless $S \subset Z'_0$, i.e. unless $Z = Y \cup S$ is a polymer. This gives the formula. Part (a) follows from the hypothesis at $Z' = Z$. Details

For (b), every point $s(Z,t)$ has $|s_\square| \leq 1$. In each differentiated variable, Cauchy's formula on the circle of radius $e^\kappa - 1$ around $t_\square$ stays in $\mathbb{D}_\kappa$, by continuity up to the boundary. Each variable is differentiated once, so $|\partial_{s_S} f| \leq F(e^\kappa - 1)^{-|S|} \leq F e^{-(\kappa-1)|S|}$ for $\kappa \geq 1$. Part (c) holds because differentiation in s and integration over t preserve both properties. □

Remark 47.3 (Margins). The pieces are indexed by polymers Z and depend on the data on $Z^{(r')}$, not inside Z. In the cluster expansion, polymers that do not touch have disjoint r'-neighbourhoods, so "touching" is the hard-core condition. Margins should not be converted to 0 for the composed activities. Replacing Z by $Z^\sharp$ (Section 40) turns the decay $e^{-(\kappa-1)|Z \setminus Y|}$ into $e^{-3^{-d}(\kappa-1)|Z^\sharp| + (\kappa-1)|Y|}$. For Y a single box, the factor $e^{(\kappa-1)|Y|}$ is a constant. For the composed activities, Y is the old polymer, and the factor would swamp the old decay. The next-step polymer of a piece is the union $\hat{Z}$ of the LM-cubes meeting $Z^{(r')}$. In the reblocking this costs at most $C|\hat{Z}|_{LM}$, counted in LM-cubes, with C independent of L and M, since $r' < M/8$.

47.2 The interaction pieces

Write $S(U) = \sum_p s_p(U)$, where s_p is the term of the plaquette p of $\mathbb{T}_\eta$ in the action (Section 23); the subscript is a plaquette, and s_p is unrelated to the weakening parameter s. Put $r_I = \min\big(r_L, c_I L^{-d} R_P, \frac{1}{32} c_0 L^{-1}\big)$, with r_L from Lemma 32.7, R_P from Theorem 45.5 (computed with the multiplicative constants doubled, see the Setting), c_0 from Lemma 32.3(c), and c_I so small that $|\hat{Z}(Z)|_\infty < R_P$ for $|Z|_\infty < r_I$. (For the constant c of Lemma 32.7 small, $r_L \leq \frac{1}{32} c_0 L^{-1}$, so the third entry does not change the minimum; the second can be smaller than r_L, since R_P depends on κ, δ, M, ρ.)

Definition 47.4 (Interaction pieces)**.** For a plaquette p, $s \in \mathbb{D}_\kappa$ and complex Z on $\mathfrak{F}$ with $|Z|_\infty < r_I$, put

$$v_p(s; Z) = s_p\big(e^{\mathrm{i}\eta A_*(s; \hat{Z}(Z))} U'\big) - T_2\big[s_p\big(e^{\mathrm{i}\eta A_*(s; \hat{Z}(\cdot))} U'\big)\big](Z),$$

where T_2 is the Taylor polynomial of degree 2 at $Z = 0$, for fixed s and U'.

Proposition 47.5 (Interaction pieces). *(a) $v_p(s;Z)$ is analytic in (s,Z) for s in the interior of $\mathbb{D}_\kappa$ and $|Z|_\infty < r_I$, continuous up to the boundary in s, and analytic in the reference.*

(b) $|v_p(s;Z)| \le C_P\,\eta^d\,(|Z|_\infty/r_I)^3$ for $|Z|_\infty \le \frac12 r_I$, uniformly in $s \in \mathbb{D}_\kappa$.

(c) (Identification.) *If U' is real and critical, then $\sum_p v_p(1;Z) = \mathcal{V}(Z)$ for $|Z|_\infty \le \min(\frac12 r_I, r_2)$ with $|\hat Z(Z)|_\infty \le \beta_V$.*

(d) (Factorization.) *Let $s_\square = 0$ for $\square \not\subset Z'$, a union of boxes with $*$-components Z'_i. Then $v_p = 0$ unless p has a point in $Z'^{(r+1)}$. If p has a point in $(Z'_i)^{(r+1)}$, then v_p depends on s only through $s_\square$, $\square \subset Z'_i$, on Z only on $(Z'_i)^{(r+2L)}$, and on U' only on $(Z'_i)^{(r')}$.*

(e) (Invariance.) *v_p is invariant under real gauge transformations u, acting by $U' \mapsto U'^u$ and $Z(b) \mapsto R(u(b_-))Z(b)$.*

Proof. (a) follows from Theorem 45.5(b), the analyticity of $\hat Z$ and of s_p, and Proposition 46.7(d). Details
The Taylor coefficients at $Z=0$ are analytic in s and in the reference.

(b) Fix s and Z, and put $g(t) = v_p(s;tZ)$. It is analytic for $|t| < r_I/|Z|_\infty$ and vanishes to third order at $t=0$. Let $h(t) = s_p(e^{i\eta A_*(s;\hat Z(tZ))}U') - s_p(U')$. Write $s_p(U) = \mathsf{N} - \operatorname{tr}\cos X_p$ with $X_p = \ell(U(\partial p))$; in $d=4$ the action carries no power of η. Then $|s_p(U)| \le C|X_p|^2$ for $|X_p| \le 1$. By Theorem 45.5(a), $|A_*(s;B)|_* \le C_P|B|_\infty \le C_P R_P = C_P\varrho_P/K_H \le \frac18$, since $4C_P(\varepsilon_P + 2\varrho_P) \le 1$ in its proof (with $K_H \ge 1$). So, in the gauge of U'_r, the four bond fields of p at $e^{i\eta A_*}U'$ have sup norm at most $\frac18 + \varepsilon + C\delta_c \le \frac14$, since $\varepsilon + C\delta_c \le \frac{1}{20}$ (Proposition 46.7, smallness). For $k \ge 1$, $\eta = L^{-k} \le \frac13$, so $|U(\partial p) - 1| \le e^\eta - 1 \le \frac12$. At $k=0$, where $\eta = 1$ and $A_* = B$ (Proposition 52.2), $|B|_\infty \le \rho_L$ (Lemma 32.7(c)), so, with $\varepsilon + C\delta_c \le \frac{1}{20}$ and $\rho_L \le \frac{1}{20}$ (Proposition 52.2), the bond fields have sup norm at most $\frac{1}{10}$ and $|U(\partial p) - 1| \le e^{2/5} - 1 \le \frac12$. In both cases $|X_p| \le 1$ (Lemma 21.2(a)). By the expansion of Lemma 12.3, $X_p = \eta^2(F_{U'}(p) + O(|A_*|_*))$, and $|F_{U'}| \le C(\varepsilon + \delta_c)$. So $|s_p| \le C\eta^4 = C\eta^d$ at $e^{i\eta A_*}U'$ and at U', and $|h(t)| \le C_P\eta^d$ for $|t| < r_I/|Z|_\infty$. No expansion in A_* is made, so the current, which is not bounded uniformly in η at complex references, does not enter. By Cauchy's estimate each of the three Taylor terms of h of degree at most 2 is bounded by $C_P\eta^d$ there. So $g = h - T_2 h$ is bounded by $4C_P\eta^d$ there, and vanishes to order three. The maximum principle for $g(t)/t^3$ on the discs $|t| \le \varrho$, $\varrho < r_I/|Z|_\infty$, and the limit $\varrho \to r_I/|Z|_\infty$ give $|g(1)| \le 4C_P\eta^d(|Z|_\infty/r_I)^3$.

(c) By Theorem 45.5(c), $A_*(1;B) = A_*(B)$ for $|B|_\infty \le \min(R_P, \beta_V)$. So, summing over the finitely many plaquettes,

$$\sum_p s_p(e^{i\eta A_*(1;\hat Z(Z))}U') = \mathcal{S}(\hat Z(Z)).$$

By the second display of Lemma 34.1, $\mathcal{S}(\hat Z(Z)) = S(U') + \frac12\Delta^0(\hat Z^1 Z, \hat Z^1 Z) + \langle\omega_0, \hat Z^2(Z)\rangle + \mathcal{V}(Z)$, where $\mathcal{V}$ vanishes to third order. Hence $\sum_p T_2[\cdots] = S(U') + \frac12\Delta_k(Z,Z) - \frac12\langle \mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle$, and $\sum_p v_p(1;Z) = \mathcal{V}(Z)$. The missing quadratic term is that of the gauge fixing, which is treated in Section 47.3.

(d) By Theorem 45.5(d), $A_*(s;B)$ vanishes outside $Z'^{(r)}$. On $(Z'_i)^{(r)}$ it depends on B only on the coarse bonds of $(Z'_i)^{(r)}$, on U' only on $(Z'_i)^{(r+1)}$, and on s only in Z'_i. If p has no point in $Z'^{(r+1)}$, then $A_* = 0$ on the bonds of p. So $s_p(e^{i\eta A_*}U') = s_p(U')$ for all Z, and $v_p = 0$. Otherwise, all bonds of p lie in $(Z'_i)^{(r+2)}$ for exactly one i, since the components are at distance at least $M/4$. The map $\hat Z$ has range $2L$ in block units, and the base $W = U'^{(k)}$ is local. This gives the dependence stated.

(e) $A_*(s)$ is covariant (Lemma 40.3(c) and the construction in Section 45), $\hat Z$ is covariant, and s_p is invariant. The Taylor coefficients of an invariant function are invariant. □

Grouping by boxes, put $V(\square;s)=g^{-2}\sum_p v_p(s;Z(s))$, the sum running over the plaquettes whose base point lies in $\square$. The weakened families of Part IX live on $\mathbb{D}_{\kappa'}$, $\kappa'=\kappa-C\delta$, and $|Z(s)|_\infty\le gC_T|\xi|_\infty$ there, with $C_T=2^d\sqrt{d}\,\|C_k^{1/2}\|_{\kappa',\gamma'}$. Indeed, by Lemma 40.3(a) and Corollary 41.6, $|C_k^{1/2}(s)|_{\gamma'}\le 2^d\|C_k^{1/2}\|_{\kappa',\gamma'}$. The row sums of the block norms bound the operator on sup norms up to the factor $\sqrt{d}$ from $\|\xi\|_{B(y')}\le\sqrt{d}|\xi|_\infty$. By Corollary 41.6, C_T is uniform in the reference. Let $|\xi|_\infty\le p$ with $gC_Tp\le\frac12 r_I$. Then $V(\square;s)$ is defined and analytic on the interior of $\mathbb{D}_{\kappa'}$, continuous up to the boundary. By Corollary 41.6, Lemma 40.3(b) and Proposition 47.5(d), Lemma 47.2 applies to $f(s)=V(\square;s)$ on $\mathbb{D}_{\kappa'}$ with $Y=\square$, and $F=\sup_{\mathbb{D}_{\kappa'}}|V(\square;\cdot)|\le C_Pg^{-2}M^d(gC_Tp/r_I)^3$. With $Z(s)=gC_k^{1/2}(s)\xi$, this is of order $g\,|\xi|^3$. The control of ξ belongs to the cluster expansion.

47.3 The gauge-fixing pieces

The exponent of Theorem 34.9 contains the remainder G_3 of the gauge fixing (Lemma 32.3(c)). It is a sum of terms localized in single blocks $B'(y)$. Since $Z(s)$ is not local in ξ, it is localized like the interaction. We define its pieces at every reference of the Setting, as Taylor remainders.

Recall $f(H)=\mathsf{N}-\operatorname{tr}\cos H$, an even entire function with $f(H)=\frac12\operatorname{tr}H^2+O(|H|^4)$ (Section 32). For $y\in\mathbb{T}'$, $x\in B'(y)$ and a complex field Z on $\mathfrak{F}$, extended by 0 to the bonds $b_0(c)$, put

$$H_{y,x}(Z)=\ell\big((e^{\mathrm{i}Z}W)(y,x)\,W(y,x)^{-1}\big),$$

at the base $W=U'^{(k)}$ of the reference. It involves only the inner bonds of $B'(y)$, which all lie in $\mathfrak{F}$.

Definition 47.6 (Gauge-fixing pieces)**.** For complex Z on $\mathfrak{F}$ with $|Z|_\infty<r_I$ put

$$\varphi_{y,x}(Z)=f\big(H_{y,x}(Z)\big)-\tfrac12\langle\mathfrak{G}_WZ(y,x),\mathfrak{G}_WZ(y,x)\rangle,$$

where $\langle A,B\rangle=\operatorname{tr}(AB)$ is the bilinear form on $\mathfrak{g}^c$, and $\mathfrak{G}_WZ(y,x)=\lambda_W(y,x)[Z]$ (Definition 22.6, in the units of step k).

Proposition 47.7 (Gauge-fixing pieces)**.** *There is C, depending only on d and G, such that the following holds at every reference of the Setting with $\delta_c<\delta_P$.*

(a) $\varphi_{y,x}$ is defined and analytic for $|Z|_\infty<r_I$, and analytic in Z and in the reference jointly. It does not depend on the weakening parameters s.

(b) $|\varphi_{y,x}(Z)|\le C(|Z|_\infty/r_I)^3$ for $|Z|_\infty\le\frac12 r_I$.

(c) $\varphi_{y,x}$ vanishes to third order at $Z=0$: it is $f\circ H_{y,x}$ minus its Taylor polynomial of degree 2. At real references, $\sum_{x\in B'(y)}\varphi_{y,x}=G_{3,y}$ and $\sum_{y,x}\varphi_{y,x}=G_3$, with $G_{3,y}$, G_3 of Lemma 32.3(c) at $W_\circ=W$.

(d) (Block locality.) *$\varphi_{y,x}(Z)$ depends on Z only on the inner bonds of $B'(y)$, and on the reference only through the base on these bonds.*

(e) (Invariance.) *$\varphi_{y,x}$ is invariant under real gauge transformations u, acting by $U'\mapsto U'^u$ and $Z(b)\mapsto R(u(b_-))Z(b)$. It is real at real references for real Z.*

Proof. Definition and size of $H_{y,x}$. By the Setting, the real part $W_r=(U'_r)^{(k)}$ of the base lies in $\mathfrak{M}$. At a real reference $W=W_r$. Since $r_I\le\frac{1}{32}c_0L^{-1}$, Lemma 32.3(c) at $W_\circ=W_r$ shows that $H_{y,x}$ is defined and holomorphic, with $|H_{y,x}(Z)|\le 4C_ML|Z|_\infty\le\frac18$; here and below $c_0\le 1/(4C_M)$, which is part of the choice "c_0 small" in Lemma 32.3(c). At a complex reference, $W=e^{\mathrm{i}\beta}W_r$ bondwise, with $|\beta|_\infty\le C_{\mathrm{AB}}\delta_c$, and β is holomorphic in the complex part of U' (Proposition 46.4(i), Lemma 46.3(a),(b)). By (DC5), $C_{\mathrm{AB}}\delta_P\le\frac{1}{32}c_0L^{-1}$ for

Details

c_{DC} small. Write $e^{\mathrm{i}Z}W = e^{\mathrm{i}(Z\star\beta)}W_r$ with $Z\star\beta = \ell(e^{\mathrm{i}Z}e^{\mathrm{i}\beta})$ bondwise; by Lemma 21.2(c), $|Z\star\beta|_\infty \le 2(|Z|_\infty + |\beta|_\infty) \le \frac{1}{8}c_0L^{-1}$, since $|Z|_\infty + |\beta|_\infty$ is small. Let $H^r_{y,x}$ be the function of Lemma 32.3(c) at $W_\circ = W_r$. Then

$$H_{y,x}(Z) = \ell\big(e^{\mathrm{i}H^r_{y,x}(Z\star\beta)}e^{-\mathrm{i}H^r_{y,x}(\beta)}\big),$$

with $|H^r_{y,x}(Z\star\beta)| \le \frac{1}{8}$ and $|H^r_{y,x}(\beta)| \le \frac{1}{32}$. By Lemma 21.2(c), $H_{y,x}$ is defined and holomorphic in (Z,β), with $|H_{y,x}(Z)| \le C$. By Lemma 22.7(f), whose hypothesis $L\max|\beta| \le c$ is again part of (DC5), W satisfies the hypothesis of Definition 22.6. So $\mathfrak{G}_W Z(y,x) = \frac{d}{dt}\big|_{t=0} H_{y,x}(tZ)$ by Lemma 22.7(a), and $|\mathfrak{G}_W Z(y,x)| \le CL\max|Z|$, the maximum over the bonds of $B'(y)$, by Lemma 22.7(c),(f).

(a) f is entire, and $H_{y,x}$ and $\mathfrak{G}_W$ are holomorphic in Z and β, hence in the reference. The definition does not involve s.

(c) $H_{y,x}(0) = 0$, and the linear part of $H_{y,x}$ is $\mathfrak{G}_W Z(y,x)$. Since $f(H) = \frac{1}{2}\operatorname{tr} H^2 + O(|H|^4)$, the Taylor polynomial of degree 2 of $f\circ H_{y,x}$ at $Z=0$ is $\frac{1}{2}\langle \mathfrak{G}_W Z(y,x), \mathfrak{G}_W Z(y,x)\rangle$, at every reference. At real references the sum over x is the definition of $G_{3,y}$ in the proof of Lemma 32.3(c).

(b) Fix Z with $0 < |Z|_\infty \le \frac{1}{2}r_I$, and put $\phi(t) = \varphi_{y,x}(tZ)$ for $|t| < r_I/|Z|_\infty$. There $|H_{y,x}(tZ)| \le C$ and $|\mathfrak{G}_W(tZ)(y,x)| \le CLr_I \le C$, so $|\phi(t)| \le C$, since f is entire. By (c), $\phi(t)/t^3$ is holomorphic on this disc, and the maximum principle gives $|\varphi_{y,x}(Z)| = |\phi(1)| \le C(|Z|_\infty/r_I)^3$.

(d) The averaged contour variable $(e^{\mathrm{i}Z}W)(y,x)$ depends only on the bonds of the box spanned by y and x, which are inner bonds of $B'(y)$ (Section 32); so does $\lambda_W(y,x)$ (Lemma 22.7(a)).

(e) For a real gauge transformation u the base becomes W^u (Setting), and $e^{\mathrm{i}R(u)Z}W^u = (e^{\mathrm{i}Z}W)^u$ bondwise. By (21.2), which holds for real u whenever the averaged contour variables are defined, $H_{y,x}$ becomes $\ell(u(y)e^{\mathrm{i}H_{y,x}}u(y)^{-1}) = u(y)H_{y,x}u(y)^{-1}$ (Lemma 21.2(d)). By Lemma 22.7(d), $\mathfrak{G}_W Z(y,x)$ becomes $u(y)\,\mathfrak{G}_W Z(y,x)\,u(y)^{-1}$. Both terms of $\varphi_{y,x}$ are invariant under this conjugation. At a real reference and for real Z, $H_{y,x}$ and $\mathfrak{G}_W Z(y,x)$ lie in $\mathfrak{g}$, so $\varphi_{y,x}$ is real. □

The pieces are grouped by boxes like the interaction. Every $y \in \mathbb{T}'$ lies in exactly one box. Put

$$G_3(\square; s) = g^{-2} \sum_{y\in\mathbb{T}'\cap\square} \sum_{x\in B'(y)} \varphi_{y,x}(Z(s)).$$

At real references, $\sum_\square G_3(\square;1) = g^{-2}G_3(Z)$ for $Z = gC_k^{1/2}\xi$, by Proposition 47.7(c) and $C_k^{1/2}(1) = C_k^{1/2}$ (Corollary 41.6). Let $|\xi|_\infty \le p$ with $gC_Tp \le \frac{1}{2}r_I$. Then $G_3(\square;s)$ is defined and analytic on the interior of $\mathbb{D}_{\kappa'}$, continuous up to the boundary. A box contains at most $(2M)^d$ pairs (y,x), so by Proposition 47.7(b)

$$\sup_{\mathbb{D}_{\kappa'}} |G_3(\square;\cdot)| \le C(2M)^d g^{-2}(gC_Tp/r_I)^3 \le C_P g\, p^3.$$

The terms of $G_3(\square;s)$ are local in Z, with range L, but $Z(s)$ is not local in ξ. As for $\mathfrak{J}(Z(s))$ below, Lemma 47.2 applies to $f(s) = G_3(\square;s)$ on $\mathbb{D}_{\kappa'}$ with $Y = \square$, by Corollary 41.6, Lemma 40.3(b) and Proposition 47.7(d); the blocks $B'(y)$, $y\in\square$, lie within distance L of $\square$, which the margin $r' = r + 2L + 2$ covers. The pieces are analytic in the reference and invariant under real gauge transformations, by Proposition 47.7(a),(e) and Lemma 47.2(c).

47.4 The Jacobians

Proposition 47.8 (Jacobians). *(a)* $\mathfrak{J}(Z) = \sum_{b\in\mathfrak{F}} \log(\sigma(Z(b))/\sigma(0)) + \sum_c \log(\mathsf{J}_c(Z)/\mathsf{J}_c(0))$ *is a sum of terms, each analytic for* $|Z|_\infty \le r_L$*, depending on* Z *and on the base only*

within range $2L$ of one bond, vanishing at $Z = 0$ and bounded as in Theorem 34.9(e). The terms are invariant under real gauge transformations. Since Proposition 32.8 is written with det*, which is positive at real references, they are analytic in Z and in the reference jointly, on the complex references with $\delta_c < \delta_P$, and real at real references and real Z.*

(b) By Proposition 32.8, $\mathfrak{j}_c = \det(1 + \mathfrak{r}_c(0))^{-1}$, with $\mathfrak{r}_c$ from Lemma 32.7(b). It is an analytic function of the base near c, invariant under real gauge transformations, and $\mathfrak{j}_c = 1$ at the trivial base.

(c) Let G be semisimple. As a function of U' (in the units of step k) through $W = U'^{(k)}$, $\log \mathfrak{j}_c$ is a normalized graded activity at the blocks $B'(c_-) \cup B'(c_+)$ on $\mathcal{U}(\cdot; c_a, c_a, c_a)$ for $c_a \leq c_J L^{-d-2}$ ($= c_J L^{-6}$ in $d = 4$), with sup at most C. Hence $|\log \mathfrak{j}_c| \leq C$ on $\mathcal{U}(\cdot; \theta_r, \delta_1, \delta_0)$ for $\theta_r, \delta_1, \delta_0 \leq c_a$. Moreover, with $\mathfrak{t}_0$ as in Proposition 41.7(d), $|\log \mathfrak{j}_c| \leq C_P \mathfrak{t}_0^2$ whenever $C_P(\mathfrak{t}_0 + \delta_0) \leq 1$. (For abelian G, $\mathfrak{r}_c \equiv 0$ and $\mathfrak{j}_c \equiv 1$.)

(d) (Lattice covariance.) *Let the base W be real, as in Section 32, and let $r \in \mathfrak{S}_{k+1}$. Then rW is the base of rV' for the background field rU' (Proposition 31.7(b)), and $\mathfrak{j}_{c^r}(rW) = \mathfrak{j}_c(W)$ for every positively oriented bond c of $\mathbb{T}'$, where c^r is the positively oriented bond with $c^r = \pm rc$. Hence $\sum_c \log \mathfrak{j}_c(rW) = \sum_c \log \mathfrak{j}_c(W)$. The same holds at $k = 0$, with $\mathfrak{S}_1$ and the base $W = U'$.*

Part (d) is used only in Part XI, in the proof of Lemma 54.9.

Proof. (a) is Proposition 32.8 and Theorem 34.9(e), with Lemma 32.7(a),(c) for the locality. For invariance, a real gauge transformation u acts on the relative variables by $Z(b) \mapsto R(u(b_-))Z(b)$ and on ℓ_c by $R(u(c_-))$. The function σ is Ad-invariant, and $\det \partial \ell_c / \partial Z(b_0(c))$ changes by $\det R(u(c_-)) \det R(u(b_0(c)_-))^{-1} = 1$. The terms are analytic functions of Z and of the base, through ℓ_c and ζ_c, and the base and the elimination are analytic in the reference (Setting). At real references with $L^d \mathsf{a} \leq c$ and $|Z|_\infty \leq r_L$, $\partial \ell_c / \partial Z(b_0(c))$ is real, and its determinant is positive (Proposition 32.8). So the analytic terms are those of Theorem 34.9(e) there. Details

(b) At $Z = 0$ we have $\zeta_c(0) = 0$, since ℓ_c vanishes at $Z = 0$. By Lemma 32.7(b) and $\det R = 1$ on $\mathfrak{g}$, $\det \partial \ell_c / \partial Z(b_0(c)) = L^{(1-d)\dim \mathfrak{g}} \det(1 + \mathfrak{r}_c(0))$. This gives the formula. At the trivial base, $l^{\mathrm{pl}} = q^{\mathrm{pl}}$ of (32.3), whose coefficient of $Z(b_0(c))$ is L^{1-d} exactly. So $\partial \ell_c / \partial Z(b_0(c)) = L^{1-d}$ and $\mathfrak{r}_c(0) = 0$; in $d = 4$ the coefficient is L^{-3}. Invariance is (a) at $Z = 0$. For complex bases, $\det R(g) = 1$ for $g \in G^c$, since G is connected (Section 12). So the formula is the analytic continuation.

(c) On $\mathcal{U}(P; c_a, c_a, c_a)$, the base $W = U'^{(k)}$ is a complex perturbation of size Cc_a of a real base of curvature at most Cc_a, in the units of step k (Proposition 21.12, Proposition 46.4(i)). In Lemma 32.7, a is a curvature in the units of step $k+1$ (Setting), which is L^2 times the curvature in the units of step k. So $L^d \mathsf{a} \leq C L^{d+2} c_a$ for the real part. With the plain constraint, $l^{\mathrm{pl}} = Lp'$ and $p' = l' - D_{W'} \gamma'$ (Lemma 32.7(a), Lemma 22.8(f)). The contour average γ' involves only inner bonds, so it does not involve $Z(b_0(c))$, and the coefficient of $Z(b_0(c))$ comes from l' alone. The complex part of the base contributes to $\mathfrak{r}_c(0)$ through the remainder r_j of Proposition 46.4(ii) for the step from scale k to scale $k+1$. Proposition 46.4 is stated for $j \leq k$; we apply it with $k+1$ in place of k, in the units of step $k+1$, where the curvature becomes $L^2 C c_a$ and the size of the complex part LCc_a. Its hypotheses hold for c_J small: $L^2 C c_a \leq c_3'$ and $LCc_a < \delta_{AB}$. The remainder is relatively $CL^{d+1} c_a$. Hence $|\mathfrak{r}_c(0)| \leq CL^{d+2} c_a \leq \frac{1}{2}$ for c_J small. Then $\log \mathfrak{j}_c = -\log \det(1 + \mathfrak{r}_c(0))$ is analytic and bounded by C. It is normalized by (b), invariant by (b), and depends on U' only on the bonds of $B'(c_-) \cup B'(c_+)$. The first bound on the smaller domains follows since the domains are monotone in their constants. For the second, apply Lemma 38.5 with $\theta_r = \delta_1 = \delta_0 = c_a$ and a cube Λ of bounded side around the

two blocks, as in the proof of Proposition 41.7(c). Then $\mathfrak{s} \le C_P \mathfrak{t}$, and $C_P(\mathfrak{t}_0 + \delta_0) \le 1$ gives the hypotheses of the lemma and $\mathfrak{s} \le \frac{1}{2}$.

(d) By Proposition 31.7(b), rU' is an element of the minimal orbit of rV', and its base is $(rU')^{(k)} = rW$, by Lemma 21.10. Write $\ell_c(Z;W)$ for ℓ_c at the base W, and $(r_W Z)(b) = Z(r^{-1}b)$, with $Z(-b) = -R(W(b))^{-1}Z(b)$, as in Proposition 33.2(e). By the convention on reversed bonds, $r(e^{\mathrm{i}Z}W) = e^{\mathrm{i}r_W Z}\, rW$ bondwise. By Lemma 21.10(d) and Lemma 21.9, $\mathfrak{a}'(r(e^{\mathrm{i}Z}W)) = r\,\mathfrak{a}'(e^{\mathrm{i}Z}W)$; in particular $\mathfrak{a}'(rW) = rW'$. So the relative average at the base rW is $\mathfrak{a}'(e^{\mathrm{i}r_W Z}rW)\,(rW')^{-1} = r\overline{X}$, with $\overline{X} = \mathfrak{a}'(e^{\mathrm{i}Z}W)W'^{-1}$ as in Section 32 and $(r\overline{X})(c') = \overline{X}(r^{-1}c')$. Let c be a positively oriented bond of $\mathbb{T}'$, and $c^r = \epsilon\, rc$ with $\epsilon = \pm 1$. If $\epsilon = 1$, $(r\overline{X})(c^r) = \overline{X}(c)$ and $\ell_{c^r}(r_W Z; rW) = \ell_c(Z;W)$. If $\epsilon = -1$, Lemma 21.9 gives $(r\overline{X})(c^r) = \mathfrak{a}'(e^{\mathrm{i}Z}W)(-c)\,W'(-c)^{-1} = R(W'(c))^{-1}(\overline{X}(c)^{-1})$, so $\ell_{c^r}(r_W Z; rW) = -R(W'(c))^{-1}\ell_c(Z;W)$, by Lemma 21.2(d). So $\ell_{c^r}(r_W Z; rW) = O_c\,\ell_c(Z;W)$ with an orthogonal map O_c of $\mathfrak{g}$, since W' is real. The map r sends $b_0(c)$ onto $\pm b_0(c^r)$ (Section 32), so $(r_W Z)(b_0(c^r)) = O'_c Z(b_0(c))$ with $O'_c = 1$ or $O'_c = -R(W(b_0(c)))^{-1}$, also orthogonal, and $Z(b_0(c))$ enters $r_W Z$ only there. Differentiating at $Z = 0$ with respect to $Z(b_0(c))$,

$$\frac{\partial \ell_{c^r}}{\partial Z(b_0(c^r))}(0; rW)\, O'_c = O_c \frac{\partial \ell_c}{\partial Z(b_0(c))}(0;W).$$

Orthogonal maps have determinant ± 1, so the two determinants agree up to sign. Both are positive at the real base (Proposition 32.8), so they are equal. Since $\zeta_c(0) = 0$, Proposition 32.8 gives $\mathrm{j}_c = L^{-(d-1)\dim\mathfrak{g}}\mathsf{J}_c(0) = L^{-(d-1)\dim\mathfrak{g}}\,(\det \partial\ell_c/\partial Z(b_0(c)))^{-1}$ at $Z = 0$, and the same at (c^r, rW). So $\mathrm{j}_{c^r}(rW) = \mathrm{j}_c(W)$. As c runs over the positively oriented bonds of $\mathbb{T}'$, so does c^r, which gives the sum. At $k = 0$ the same argument applies: Lemma 21.10(d), Lemma 21.9, Lemma 32.7 and Proposition 32.8 are one-step statements that hold at $k = 0$ (proof of Proposition 52.2(a)), and rU' is an element of the minimal orbit of rV' by Theorem 30.3(c) for the one-step problem. □

Part (d) uses only the intrinsic objects of the step: the averaging, the bonds $b_0(c)$ and the elimination at a real base. It is used for the lattice symmetry of the one-loop term of the step at the background fields (Lemma 54.9(b)).

The terms of $\mathfrak{J}(Z(s))$ are local in Z, but $Z(s)$ is not local in ξ. So $\mathfrak{J}(Z(s))$, grouped by boxes, goes through Lemma 47.2 like $V(\square; s)$.

47.5 The composed activities

The norm $|\cdot|_*$ of Theorem 45.5 and the quantities of Definition 38.3 are compared as follows.

Lemma 47.9 (Norm conversion)**.** *There is C_{conv}, depending only on d, α and G (through Theorem 35.2(b)), with the following property. Let $\alpha \le \frac{1}{2}$, so that the exponent $\beta = \min(\alpha, \frac{1}{2})$ of $|\cdot|_*$ equals α. Let U'_r be real and (M,ε)-regular with $\varepsilon \le \varepsilon_0$, and let A be a $\mathfrak{g}^c$-valued field on the bonds of $\mathbb{T}_\eta$, with $|A|_*$ taken in the real gauges of U'_r (Section 45). Then for every set Y,*

$$|A|_Y + |\nabla A|_Y + [\nabla A]_{\alpha,Y} \le C_{\mathrm{conv}} |A|_*,$$

with $|A|_Y$ the sup norm on the bonds of Y, and ∇ and the Hölder quotient of Definition 38.3 taken with respect to U'_r.

Proof. Every bond lies in a block $B(y)$, and $\mathfrak{n}^3_y(A)$ bounds $|A|_{B(y)}$ and $|\nabla A|_{B(y)}$. If the gradient in $\mathfrak{n}^3_y$ is read in the gauge of y as a flat derivative, (38.1) with $\sigma = C\varepsilon$ converts it, with an error $C\varepsilon|A|$. Consider a quotient in $[\nabla A]_{\alpha,Y}$, with $x \ne x'$ in a cube $\Lambda_3(z') \subset Y$. If $|x - x'| \ge 1$, it is at most $|\nabla A(x)| + |\nabla A(x')| \le 2|\nabla A|_Y$, since $R(U'_r(\Gamma_{x,x'}))$ is orthogonal. If $|x - x'| < 1$, then x, x' lie in a common cube $\Lambda_1(c_y)$ (Section 45, setting). In the real gauge of y, $U'_r = e^{\mathrm{i}\eta A_r}$ with

Details

$\|A_r\|_{1,\alpha} \le C\varepsilon$ on $\Lambda_4(c_y)$ (Theorem 35.2(b)). There (38.1) and $\|R(U_r'(\Gamma_{x,x'})) - 1\| \le C|x-x'|\varepsilon$ bound the quotient by $[\partial \tilde{A}_y]_{\alpha,\Lambda_1(c_y)} + C\varepsilon(|A| + |\partial \tilde{A}_y|) \le C\mathfrak{n}_y^3(A)$. Taking the supremum over y gives the bound. □

In C_{conv}, the pairs at distance at least 1 cost a factor 2.

Proposition 47.10 (Composed activities)**.** *Let X be a connected domain of step k (an old polymer), let $r_E \ge 0$ with $r_E + 3 \le r$, and put $P = X^{(r_E)}$. Let $\mathcal{E}(X,\cdot)$ be a graded activity at P on $\mathcal{U}(P;\theta_r,\delta_1,\delta_0)$ (Definition 38.3), with sup $\|\mathcal{E}(X)\|$; so it depends on the configuration on P, and its domain is controlled on $P^{(3)} = X^{(r_E+3)}$. Let $U' = e^{\mathrm{i}\eta A'}U_r'$ be a reference as in the Setting, and suppose that on $P^{(3)}$ this is a representation as in Definition 38.3, with budgets $\theta_r, \delta_1', \delta_0'$, where $\delta_1' < \delta_1$ and $\delta_0' < \delta_0$. Thus the real part of the representation is the U_r' in whose gauges $|\cdot|_*$ is measured. Let $0 < \zeta \le \frac{1}{2}R_P$, and put*

$$\tau = C_1 C_{\mathrm{conv}} C_P\, \zeta \left(\min(\delta_1 - \delta_1',\ \delta_0 - \delta_0')\right)^{-1}.$$

Then, for $\tau < \frac{1}{2}$, $s \in \mathbb{D}_\kappa$ and $|\hat{Z}(Z)|_\infty \le \zeta$:

(a) $e^{\mathrm{i}\eta A_*(s;\hat{Z}(Z))}U' \in \mathcal{U}(P;\theta_r,\delta_1,\delta_0)$, and

$$f(X;s,Z) = \mathcal{E}(X, e^{\mathrm{i}\eta A_*(s;\hat{Z}(Z))}U') - \mathcal{E}(X,U')$$

is analytic in (s,Z) and in the reference jointly, continuous up to the boundary in s, with $|f(X;s,Z)| \le 4\tau\|\mathcal{E}(X)\|$;

(b) if $s_\square = 0$ outside a union of boxes Z' that contains the boxes meeting X, then $f(X;s,Z)$ depends on s only through the boxes of the $*$-component Z_0' of Z' containing them, and on Z and U' only on $(Z_0')^{(r')}$;

(c) $f(X;s,Z)$ is invariant under real gauge transformations.

Proof. (a) For $|t| < \frac{1}{2\tau}$ consider $e^{\mathrm{i}\eta t A_*}U' = e^{\mathrm{i}\eta(tA_* \oplus A')}U_r'$ on $P^{(3)}$, with $A_* = A_*(s;\hat{Z}(Z))$, combined bondwise by the Baker–Campbell–Hausdorff formula. The real part U_r' is unchanged, so no current of A_* is needed. By Theorem 45.5(a), $|A_*|_* \le C_P\zeta$. By Lemma 47.9, the sup norm, the covariant gradient and the Hölder quotient of A_* on $P^{(3)}$, with respect to U_r', are at most $C_{\mathrm{conv}}C_P\zeta$. By the Baker–Campbell–Hausdorff formula and (23.2), the corresponding quantities of $tA_* \oplus A'$ exceed those of A' by at most $C_1 C_{\mathrm{conv}} C_P|t|\zeta < \frac{1}{2}\min(\delta_1 - \delta_1', \delta_0 - \delta_0')$, for a suitable C_1. So $tA_* \oplus A'$ satisfies the bounds of $\mathcal{U}(P;\theta_r,\delta_1,\delta_0)$ on $P^{(3)}$, strictly. Hence $t \mapsto \mathcal{E}(X, e^{\mathrm{i}\eta t A_*}U') - \mathcal{E}(X,U')$ is analytic for $|t| < \frac{1}{2\tau}$, bounded by $2\|\mathcal{E}(X)\|$, and vanishes at $t = 0$. Since $\tau < \frac{1}{2}$, the point $t = 1$ lies in the disc, and the Schwarz lemma gives $|f| \le 2\|\mathcal{E}(X)\| \cdot 2\tau$ there. Details

Analyticity. $A_*(s;B)$ is analytic in (s,B) and in A' jointly (Theorem 45.5(b), Proposition 46.7(d)). $\hat{Z}$ is analytic in Z and in the base, and the base is analytic in A' (Proposition 46.4(vi), Setting). $\mathcal{E}(X,\cdot)$ is holomorphic on an open set containing $\mathcal{U}(P)|_P$, and the bounds above hold strictly. So f is analytic in (s,Z,A') jointly, as a composition. By (b), it depends on A' only on $(Z_0')^{(r')}$, so it is analytic in the reference near that set. Continuity up to the boundary in s follows from Theorem 45.5(b).

(b) $\mathcal{E}(X,\cdot)$ depends on the configuration on $P = X^{(r_E)} \subset (Z_0')^{(r)}$, and A_* on $(Z_0')^{(r)}$ is determined by the data on $(Z_0')^{(r')}$ and by s in Z_0' (Theorem 45.5(d), and Lemma 32.7 for $\hat{Z}$). (c) follows from the covariance of A_* and $\hat{Z}$ and the invariance of $\mathcal{E}$. □

In the step, $\mathcal{E} = \mathcal{E}_k$ on the graded domain $\mathcal{U}^\flat(X) = \mathcal{U}(X;\theta_r,\delta_1^\flat,\delta_0)$ of step k, $\delta_1^\flat = \frac{1}{4}\delta_1$, on which the activities carried by the flow are bounded (Section 50); the proposition is applied with

$\delta_1^\flat$ in place of δ_1, with $r_E = 0$ and $P = X$, and U' is the extension (Setting) of a point of the domain of step $k+1$, in the units of step k. On $X^{(3)}$ the extension is not felt, since the pieces used have $Z^{(r')}$ inside the polymer of step $k+1$ (Setting), and $X^{(3)} \subset Z^{(r')}$. By Remark 38.8(i), the representation on $X^{(3)}$ has the same real part and budgets $\delta_1' = L^{-2}\delta_1$, $\delta_0' = L^{-1}\delta_0$. So $\min(\delta_1^\flat - \delta_1', \delta_0 - \delta_0') \geq (\frac{1}{4} - \frac{1}{9})\delta_1 \geq \frac{1}{8}\delta_1$, since $L \geq 3$ and $\delta_1 \leq \delta_0$ (Remark 50.9(c)). On the support of the small-field cutoff, ζ is of order $L^d g_k p_{0,k}$ ($\zeta = w$, Section 48). So $\tau \leq C_P L^d g_k p_{0,k}/\delta_1$, which is small when $g_k p_{0,k}$ is, the domain constants being fixed. For $|\xi|_\infty \leq p$ with $CL^d g C_T p \leq \zeta$, the function $s \mapsto f(X; s, Z(s))$ is analytic on the interior of $\mathbb{D}_{\kappa'}$ and bounded by $4\tau\|\mathcal{E}(X)\|$. Lemma 47.2 applies to it on $\mathbb{D}_{\kappa'}$, without conversion of margins (Remark 47.3). Here Y is the set of boxes of X when X is a union of boxes, as for the polymer grid; in general it is the union of the boxes meeting X.

The next remark uses Theorem 48.6(a), the enlarged closure $\hat{Y}$ of Remark 49.4 and, in its last paragraph, the grouped old activities of Definition 53.1. It may be read after Section 49.3.

Remark 47.11 (Extended references). A point of the domain of step $k+1$ at a polymer Y' fixes U' only near Y'. It is extended by the cutoff of Remark 38.7. In the cutoff band, $|\nabla(\chi A)| \sim |A|$, which need not be within the gradient budget δ_1 of the domain of step k. So composed configurations of old polymers meeting the band need not lie in the domain of step k at X. Therefore, before the Mayer expansion, all old activities and all pieces whose polymer is not contained in Y' (in the units of step k) are set to zero. By Theorem 48.6(a), the output $E^{\#}(Y)$ with $\hat{Y} = Y'$ is unchanged, since it depends only on the pieces of polymers inside Y. So the new activity at Y' is defined at every point of the domain of step $k+1$ at Y' and does not depend on the extension. Pieces with $Z \subset Y'$ near $\partial Y'$ may depend on the extension, but they do not enter $E^{\#}(Y)$ with $\hat{Y} = Y'$. At a real critical U', take $U_r = U'$ and $A = 0$, so that no extension is needed. The restricted and the full systems then share every $E(Z)$ with $Z \subset Y$. By Theorem 48.6(a), applied also to $E_{\mathrm{p}}^{\#}$ and to the λ-family of Corollary 48.7, their outputs agree. Since $\sum_{Y'} \sum_{\hat{Y}=Y'} = \sum_Y$, the identity of Theorem 48.9 is not affected.

The collar inclusions. Let Y contribute to the output at Y', i.e. $\hat{Y} = Y'$, with $\hat{Y}$ the enlarged closure of Remark 49.4. Then $Y^{(r')} \subset Y'$, and the following holds.

- By Theorem 48.6(a), the output at Y uses only pieces $E(Z)$ with $Z \subset Y$.
- Each such piece depends on the reference only on $Z^{(r')} \subset Y^{(r')} \subset Y'$.
- A mixed piece uses old activities $\mathcal{E}_k(X)$ with $X \subset Z$. Their domain budget is needed on $X^{(3)} \subset Z^{(r')} \subset Y'$, since $r' = r + 2L + 2 \geq 3$.

So on every set that matters the extension is not felt. The real part there is that of U', and the slack of Remark 38.8(i) gives the hypotheses of Proposition 47.10. The same inclusions hold for the pure system and for the λ-family of Corollary 48.7.

To avoid an implicit definition, the restricted system for an output polymer Y' is defined on the input pieces with $Z \subset Y'$ only. Equivalently, it may be defined on a cutoff extension that equals U' on the 3-neighbourhood of Y', which is the margin of the next domain. Pieces near $\partial Y'$ whose collars cross it may depend on the extension, but they do not enter the outputs with $\hat{Y} = Y'$. At real critical references the restricted and the unrestricted systems share all inputs that enter, so their outputs coincide.

Grouped old activities. In the flow the old activities of step k are the grouped activities $\mathcal{E}_k(Y)$ of Definition 53.1, indexed by the polymers Y of scale k, the polymers of the paving. The activity $\Psi_\square^{(j)}$ of a grid cube $\square$, which contains the counterterm $-\beta_j S_\square$, is carried by the polymer $\square^+$. Each $\mathcal{E}_k(Y)$ is a graded activity at Y itself, on $\mathcal{U}^\flat(Y)$, with $r_E = 0$ and $P = Y$ (Proposition 53.2). So everything above applies with the old polymer $X = Y$. The restricted system for Y' contains $\mathcal{E}_k(Y)$ if and only if $Y \subset Y'$, and $\mathcal{E}_k(Y)$ is set to zero or kept as a whole. In particular the counterterm $-\beta_j S_\square$ is never separated from the small polymers of

$\square$. A mixed piece that uses $\mathcal{E}_k(\square^+)$ has $\square^+ \subset Z$, and the budget of $\mathcal{E}_k(\square^+)$ is needed on $(\square^+)^{(3)} \subset Z^{(r')} \subset Y'$, as in the collar inclusions. The single activities $E^{(j)}(X,\cdot)$ inside $\mathcal{E}_k(Y)$ are functions of the configuration on $X \subset Y$, and their domains are controlled on $X^{(3)} \subset Y^{(3)}$ (Remark 38.8(i)), so they need no extension of their own. The new activity at Y' is the raw new generation of Definition 53.3.

47.6 The Hessian away from critical backgrounds

Section 37 already defines $\Delta_k = \Delta_k^{(\mathbb{T}_\eta)}$ and $C_k = \Delta_k^{-1}$ at regular backgrounds that need not be critical, with the elimination at the base $U^{(k)}$ (Definition 37.3 and the definition following it). We use this definition at every reference. At complex references, J_U is not bounded uniformly in η. So $\omega_0 = \mathbf{H}_0^T J_U$ is continued in the form

$$\omega_0 = (D\mathbf{H}_0)^T F^s_{U'},$$

which agrees with $\mathbf{H}_0^T J_U$ at real references, since $J_U = D^T F^s_U$ (Lemma 23.2). Here $\mathbf{H}_0$ is $\mathbf{H}$ with potential 0, and $F^s_{U'}$ is continued holomorphically. This is the form used at level (L4) of Section 41.

Proposition 47.12 (Hessian at a general reference)**.** *(a) At real critical references, Δ_k agrees with Lemma 34.1 and Corollary 34.6, including the gauge-fixing term $\langle \mathfrak{G}_W \cdot, \mathfrak{G}_W \cdot \rangle$.*

(b) $|\omega_0|_\infty \le C(\varepsilon + \delta_c)$. The operator W_2 is real symmetric at real references, with $\|W_2\| \le C(\varepsilon+\delta_c)$ and L^2 decay. Only the L^2 theory of Part II is used for it, not the pointwise class $\mathcal{W}(\varepsilon,\gamma)$.

(c) Δ_k has decay (C_L,γ) for some $\gamma > 0$.

(d) For $\varepsilon \le \varepsilon_{\max}$ and $\delta_c < \delta_P$, so that $\varepsilon + \delta_c$ is small depending on L (Remark 38.8), $\operatorname{Re}\Delta_k(Z,\bar{Z}) \ge \frac{1}{8} c_L \|Z\|^2$.

(e) Δ_k is analytic in the reference and covariant under real gauge transformations. At $s = 1$ it is represented by the weakened family $\Delta^{\mathrm{c}}_{\mathbb{T}_\eta}(s)$ of level (L4), plus the gauge-fixing part $\mathfrak{G}_W^T \mathfrak{G}_W$, which is not weakened.

Proof. (a) is Lemma 37.2 and Proposition 37.4(a), together with $K^{\mathbb{T}_\eta} = \mathfrak{G}_W^T \mathfrak{G}_W$ (Section 37). Details

(b) At real references this is Proposition 37.4(b) and Lemma 37.2. At complex references use the block L^2 decay of $D\mathbf{H}_0$ and $|F^s_{U'}| \le C(\varepsilon+\delta_c)$ (Proposition 46.7(c)). The bound $\|W_2\| \le C(\varepsilon+\delta_c)$ and the L^2 decay at complex references come from Proposition 37.4(c). There $d^2\mathcal{N}(0)$ is bounded through its plaquette form, with the bounds $|R(H_p)-1| \le C\eta^2(\varepsilon+\delta_c)$ and $|F^s| \le C(\varepsilon+\delta_c)$ of step (v) of the proof of Proposition 46.7, which use the derivative part of δ_c. The L^2 operators at complex references come from Proposition 19.11, through step (i) of the proof of Proposition 46.7.

(c) is Proposition 37.4(b) for $X = \mathbb{T}_\eta$, together with the locality of $\hat{Z}^1$, $\hat{Z}^2$ and the block-locality of $\mathfrak{G}_W$. At complex references the coefficient operators (the averages and the covariant derivatives) change by local operators of size $C_L\delta_c$ (Proposition 19.11, Proposition 46.4(v)), and by Lemma 13.3 the inverses in Δ_k change by $C_L\delta_c$ in the decay norm, and $\mathfrak{G}_W$ by a block-local operator of the same size (Lemma 22.7(f) with (DC5)), which preserves the decay.

(d) This is Lemma 37.5, which does not use criticality, applied at the real part U'_r; it needs $|J_{U'_r}| \le \varepsilon$ (Setting); its conditions $\varepsilon \le \varepsilon_{\mathrm{cc}}$ and those of the proof of Proposition 37.4(c) hold since $\varepsilon \le \varepsilon_{\max}$ (Remark 38.8). The lemma covers complex perturbations of size at most c''_L, and a complex reference with $\delta_c < \delta_P$ is one, since $\delta_P \le c''_L$ (Proposition 46.7, smallness). As required by Proposition 37.6(b), ω_0 is read there in the form $(D\mathbf{H}_0)^T F^s_{U'}$.

(e) Analyticity and covariance are Proposition 37.4(c), and for the gauge-fixing part Lemma 22.7(a),(d),(f). Away from criticality, covariance follows directly from the covariance of $\mathbf{\Gamma}_\pi[W_2]$, $\mathbf{H}_0$, F^s, $\hat{Z}$ and $\mathfrak{G}_W$. The representation is the construction of level (L4) with W_2, and the gauge-fixing part as in level (L5). □

Remark 47.13 (Source-free step). For a general reference, the step quantities are those defined in this section. The exponent is $-g^{-2}\sum_p v_p(s;Z(s)) - g^{-2}\sum_{y,x}\varphi_{y,x}(Z(s))$, with no terms of order 0, 1 or 2 in Z. The Gaussian integral is taken through the white-noise representation $Z = gC_k^{1/2}\xi$, with $C_k = \Delta_k^{-1}$ as in Section 47.6; at complex references this representation, not a Gaussian measure, defines it. The determinant densities are those of Proposition 41.7, the Jacobians those of Proposition 47.8, and the composed activities those of Proposition 47.10. The gauge-fixing pieces are those of Proposition 47.7. At real critical references they reproduce Theorem 34.9, by Propositions 47.5(c), 47.7(c) and 47.12(a), and Theorem 45.5(c) for the composed activities: there $\sum_p T_2[\cdots] + \frac{1}{2}\langle \mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle = S(U') + \frac{1}{2}\Delta_k(Z,Z)$. At a non-critical reference this identity fails. This is harmless, since the exponent is a remainder by definition.

Remark 47.14 (What the cluster expansion receives). (a) *Pieces.* Lemma 47.2 applied to $V(\square;s)$, to $G_3(\square;s)$, to the boxes of $\mathfrak{J}(Z(s))$, and to $f(X;s,Z(s))$ gives pieces indexed by polymers Z with volume-type decay $e^{-(\kappa'-1)|Z\setminus Y|}$. They depend on ξ and U' only on $Z^{(r')}$, are analytic and invariant, and are bounded in terms of $|\xi|$ on the region where $|Z(s)|_\infty \le \frac{1}{2}r_I$ and $|\hat{Z}(Z(s))|_\infty \le \zeta$.

(b) *Constants.* The vacuum terms $\mathcal{N}_k$ and $-\frac{1}{2}\sum_b \ell_k(b;1)$, and the new main term $g^{-2}S(U')$, are passed on unchanged.

(c) *Parameters.* The conditions on the rates are those of Proposition 48.5 and of Remark 49.8, with $\kappa' = \kappa - C\delta$ and the rate κ_n of an activity norm. Further, $M > 32(L+1)$, $c_a \le c_J L^{-d-2}$, (DC5), and $|\xi|_\infty \le p$ with $gC_T p \le \frac{1}{2}r_I$ and $C_w L^d g C_T p \le \zeta$; these give the two bounds in (a), since $|Z(s)|_\infty \le gC_T|\xi|_\infty$ (Section 47.2) and $|\hat{Z}(Z)|_\infty \le C_w L^d |Z|_\infty$ (Lemma 32.7(c),(d)). In Section 48 the bound p is the white-noise cutoff p_0 and $\zeta = w = C_w L^d C_T g p_0$; the first condition is part of (Q1), and the second holds with equality. All of these are collected in Remark 49.8 and Remark 38.8.

(d) *Not done here.* The small-field cutoff and its removal, the Mayer expansion, the bounds in $|\xi|$ and the reblocking to step $k+1$ are in Section 48 and Lemma 53.4; the large-field boundary is not treated.

48 The cluster expansion

This section evaluates the fluctuation integral of one small-field step as a sum of polymer activities. It follows Dimock's small-field scheme [15, Sects. 4.3–4.6, App. B]:

- the small-field condition is imposed on the white noise;
- the integrand is localized in the white noise by the weakening parameters;
- the result is integrated by a Mayer and polymer expansion.

The inputs are those of Section 47 and Part IX. The output, Theorem 48.9, is the step of the flow of Section 53. Notation is as in Section 47: ξ is the white-noise field on the set $\mathfrak{F}$ of Definition 32.1, W the base, U' the reference, and $g = g_k$. The small-field step is the gauge-fixed step T^{gf} of Section 32, with cutoffs.

Parameters. Let $p_0 = p_{0,k} \geq 1$. Put

$$w := C_w L^d C_T g p_0, \qquad \zeta := w,$$

with C_T from Section 47.2 and C_w the constant in $|\hat{Z}(Z)|_\infty \leq C_w L^d |Z|_\infty$ (Lemma 32.7). We assume:

(Q1) $gC_T p_0 \leq \frac{1}{2} r_I$, and $w \leq \min(r_2, r_L, \frac{1}{2} c_G, \beta_V, \frac{1}{2} R_P)$;

(Q2) $2(C_* + 1)L^{-2}\varepsilon_1' + 4w \leq \tilde{h}$, where $\tilde{h}$ is the threshold of the density, together with the other hypotheses of Theorem 34.9 on the thresholds: $\tilde{h} \leq a_1$ and $\varepsilon_1' \leq a_{\text{step}}$;

(Q3) $\tau < \frac{1}{2}$ in Proposition 47.10, for the domains of the old activities.

In the renormalization group $\tilde{h} = p_k g_k$ and $\varepsilon_1' = p_{k+1} g_{k+1}$. On the graded domains of Remark 38.8, (Q3) is the condition $\tau \leq C_P L^d g_k p_{0,k}/\delta_1 < \frac{1}{2}$ of Remark 38.8(iii) (with the slack $\frac{1}{8}\delta_1$ of the domains $\mathcal{U}^b$, Remark 50.9(c)), and (Q1), (Q2) are conditions on the small-field parameters and on L that do not involve the domain. For couplings in $(0, \gamma]$, (Q1) and (Q3) hold by (Γ2) of Remark 53.5, and (Q2) holds for every $\varepsilon_1' \leq \min(2h_k, \gamma p(\gamma))$ (Remark 53.5, consequences), in particular for $\varepsilon_1' = h_{k+1}$ if $g_{k+1} \leq \min(2g_k, \gamma)$. Since $w \leq r_L \leq \frac{1}{8}\varepsilon_M$ by (Q1), (Q2) also gives the hypotheses of Lemma 32.6(iv).

48.1 The small-field step

Definition 48.1 (White-noise cutoff and small-field step)**.** For a configuration V on $\mathbb{T}$ in the support of χ^{w}, let $Z_V(b) = \ell(V(b)W(b)^{-1})$ for $b \in \mathfrak{F}$, the values on $\mathfrak{F}$ of the relative field of V (Definition 32.4), and $\xi_V = g^{-1} C_k^{-1/2} Z_V$, with $C_k = C_k(W, U')$ the covariance of Corollary 34.6 at the reference U'. Put

$$\chi_\xi(V) = \prod_{b \in \mathfrak{F}} \mathbf{1}(|\xi_V(b)| \leq p_0),$$

and define the small-field step by

$$T_{\mathrm{sf}}(\rho) = T^{\mathrm{gf}}(\chi^{\mathrm{w}} \chi_\xi \rho),$$

with the gauge-fixed step T^{gf} of Definition 32.5.

This is the bondwise cutoff introduced after Lemma 32.6, and T_{sf} is the step of Section 31. The cutoffs act on the variables of the gauge fixing relative to the base, so $\chi^{\mathrm{w}}\chi_\xi$ is not invariant under the group $\mathcal{G}'$; the step is a modified model. By Corollary 34.6(a),(b), $\xi_{V^{\tilde{u}}} = R(\tilde{u})\xi_V$ and $\xi_{rV} = r_W \xi_V$, with the data (W, U') transformed accordingly. So $|\xi_V(b)|$ is invariant under real gauge transformations, and $\chi^{\mathrm{w}}\chi_\xi$ is a covariant cutoff in the sense of (C1), (C2) of Section 32. The lattice covariance (C2) rests on Corollary 34.6(b), whose proof uses only intrinsic objects (the coarse action, the averaging and $\mathfrak{G}$), as required by Remark 31.8. Hence, by Lemma 32.6(i)–(iii), for gauge-invariant ρ:

- $T_{\mathrm{sf}}(\rho)(V')$ does not depend on the choice of U' in the minimal orbit of V', and is invariant under real gauge transformations of V';
- if $\rho \circ r = \rho$ for some $r \in \mathfrak{S}_{k+1}$, then $T_{\mathrm{sf}}(\rho)(rV') = T_{\mathrm{sf}}(\rho)(V')$.

So T_{sf} is a renormalization group map on gauge-invariant densities, with a kernel that depends on V' through U'. It is defined for V' small field at step $k+1$, since C_k depends on U' everywhere. This is the small-field step of [15, (206)], with Bałaban's gauge fixing and cutoff [18, (0.14)–(0.16), (2.9)]. By Lemma 32.6(iv), whose hypotheses hold by (Q1), (Q2), the discarded part

$$\mathcal{R}_{\mathrm{LF}} = T(\mathbf{1}_{\mathfrak{M} \cap \mathfrak{U}_k(\tilde{h})} \rho) - T_{\mathrm{sf}}(\rho) = T^{\mathrm{gf}}((1 - \chi^{\mathrm{w}}\chi_\xi)\mathbf{1}_{\mathfrak{M} \cap \mathfrak{U}_k(\tilde{h})} \rho)$$

on the set of the V' as in Lemma 31.6 is nonnegative for $\rho \geq 0$. Outside that set $T_{\mathrm{sf}}(\rho) = 0$, and all of $T(\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(\tilde{h})}\rho)$ is discarded. It belongs to the large-field stage, which is not treated here. Because C_k is global in U', that stage will need a version of χ_ξ that depends on regions.

Lemma 48.2 (White noise)**.** *Assume (Q1). For $|\xi|_\infty \leq p_0$ and $s \in \mathbb{D}_{\kappa'}$, put $Z(s) = gC_k^{1/2}(s)\xi$. Then $|Z(s)|_\infty \leq gC_T p_0$ and $|\hat{Z}(Z(s))|_\infty \leq w$. In particular, on the support of χ_ξ:*

- $\chi^{\mathrm{w}} = 1$;
- *all the step quantities of Section 47 are defined at $Z(s)$, for every $s \in \mathbb{D}_{\kappa'}$;*
- *the composed configurations lie in the domains of the old activities, by (Q3).*

Proof. $|C_k^{1/2}(s)\xi|_\infty \leq C_T|\xi|_\infty$ (Section 47.2), and $C_k^{1/2}(1) = C_k^{1/2}$ (Corollary 41.6), and $|\hat{Z}(Z)|_\infty \leq C_w L^d |Z|_\infty$ (Lemma 32.7(c),(d)). At $s = 1$, $V = e^{\mathrm{i}\hat{Z}(Z(1))}W$ satisfies the constraint, and $\hat{Z}(Z(1))$ is its relative field on all bonds, including the $b_0(c)$. Since $|\hat{Z}(Z(1))|_\infty \leq w$, $\chi^{\mathrm{w}}(V) = 1$. The other statements follow from (Q1), (Q3) and the definitions of r_I and ζ; for the gauge-fixing pieces, $|Z(s)|_\infty \leq \frac{1}{2}r_I$ is what Proposition 47.7 needs. □ Details

Proposition 48.3 (The small-field step)**.** *Assume (Q1)–(Q3), and let ρ be as in Theorem 34.9. Then*

$$T_{\mathrm{sf}}(\rho)(V') = \mathcal{N}_k N_\chi \exp\Big(-\frac{1}{g^2}S(U') - \frac{1}{2}\sum_{b\in\mathfrak{F}}\ell_k(b) + \sum_c \log \mathfrak{j}_c + \mathfrak{E}(U')\Big)\int d\mu^*(\xi)\, e^{\delta\mathcal{E}(\xi)},$$

where $d\mu^* = N_\chi^{-1}\chi_\xi\, d\mu_I$ is the product over $b \in \mathfrak{F}$ of standard Gaussian measures on $\mathfrak{g}$ restricted to $|\xi(b)| \leq p_0$. Further, $N_\chi = \exp(-\varepsilon^0|\mathfrak{F}|)$ with $\varepsilon^0 = O(p_0^{\dim\mathfrak{g}-2}e^{-p_0^2/2})$, and

$$\delta\mathcal{E}(\xi) = -\frac{1}{g^2}(\mathcal{V}(Z) + G_3(Z)) + \mathfrak{J}(Z) + (\mathfrak{E}(e^{\mathrm{i}\hat{Z}(Z)}W) - \mathfrak{E}(U')), \qquad Z = gC_k^{1/2}\xi,$$

with G_3 the remainder of the gauge fixing (Lemma 32.3(c)).

Proof. By (Q2), the hypothesis of Theorem 34.9 holds with the cutoff w. This is the analogue of [15, Lemma 15]: on the support of χ^{w} the configuration $e^{\mathrm{i}\hat{Z}}W$ has curvature at most $2(C_*+1)L^{-2}\varepsilon_1' + 4w \leq \tilde{h}$, as in the proof of Theorem 34.9. In the coordinates of Proposition 32.8, $V = e^{\mathrm{i}\hat{Z}(Z)}W$ and $Z_V = Z$ on $\mathfrak{F}$, so $\chi_\xi(V) = \prod_{b\in\mathfrak{F}}\mathbf{1}(|g^{-1}C_k^{-1/2}Z(b)| \leq p_0)$ is a function of Z on $\mathfrak{F}$. Proposition 32.8 holds for every bounded measurable ρ, so it applies to $\chi_\xi\rho$, and the proof of Theorem 34.9 applies to $\chi^{\mathrm{w}}\chi_\xi\rho$, with χ_ξ carried along in the integrand. Substitute $Y = C_k^{1/2}\xi$, so that $d\mu_{C_k}(Y)$ becomes $d\mu_I(\xi)$. By Lemma 48.2, $\chi^{\mathrm{w}}(\hat{Z}(gY)) = 1$ on the support of χ_ξ. Pull out $\mathfrak{E}(U')$ [15, (211)–(212)]. The value of N_χ is the Gaussian mass of $\{|\xi(b)| \leq p_0\}$ per bond of $\mathfrak{F}$, for $\mathfrak{g}$-valued $\xi(b)$. □ Details

At references that are not real and critical, or outside the small-field region, the right-hand side is the definition of the step (Remark 47.13).

48.2 Localization in the white noise

Write the integrand with the weakening parameters:

$$\delta\mathcal{E}(\xi; s) = -\sum_\square V(\square; s) - \sum_\square G_3(\square; s) + \sum_\square \mathfrak{J}_\square(Z(s)) + \sum_X f(X; s, Z(s)),$$

with $V(\square; s)$ as in Section 47.2, $G_3(\square; s)$ the gauge-fixing pieces grouped by boxes (Section 47.3), $\mathfrak{J}_\square$ the terms of $\mathfrak{J}$ grouped by the box of their bond, and f as in Proposition 47.10. At $s = 1$ this is $\delta\mathcal{E}(\xi)$, by Corollary 41.6 ($C_k^{1/2}(1) = C_k^{1/2}$), Propositions 47.5(c) and 47.7(c), and Theorem 45.5(c).

Definition 48.4 (Decay functional)**.** A *decay functional* is a function $\mathsf{d} \geq 0$ on polymers with constants $c_\flat \in (0,1]$, $c_\sharp \geq 1$, $c_\natural \geq 1$ such that

$$c_\flat |Z| - 1 \leq \mathsf{d}(Z) \leq c_\sharp(|Z| - 1), \qquad \mathsf{d}(Z_1 \cup Z_2) \leq \mathsf{d}(Z_1) + \mathsf{d}(Z_2) + c_\natural$$

for touching polymers Z_1, Z_2. Examples are $|Z| - 1$ (with $c_\flat = c_\sharp = c_\natural = 1$), and Dimock's tree length d_M [15, (119)], with $c_\flat = 3^{-d}$ by [15, (122)], $c_\sharp = \sqrt{d}$ and $c_\natural = 2\sqrt{d}$. Adding the boxes of $Z \setminus X$ one at a time, each touching the previous union, gives $\mathsf{d}(Z) \leq \mathsf{d}(X) + c_\natural |Z \setminus X|$ for $X \subset Z$.

The old activities are indexed by connected unions X of boxes. Each $\mathcal{E}_k(X, \cdot)$ depends on the configuration on $X^{(r_E)}$, where r_E is the margin of dependence, and is a graded activity at $X^{(r_E)}$ in the sense of Proposition 47.10, with its domain controlled on $X^{(r_E+3)}$. Part (b) of Proposition 47.10 needs $r_E \leq r$. We assume $r_E + 3 \leq r$ (for $r_E = 0$ this is $r \geq 3$, Section 47, Setting), which keeps $X^{(r_E+3)}$ inside $Z^{(r')}$ for $X \subset Z$, where the extension of the reference is not felt; for activities that are graded activities at their own polymer, $r_E = 0$. The paving of Section 40 is taken to be the polymer grid, which needs $M \mid n$. They are assumed to satisfy $\sup |\mathcal{E}_k(X, \cdot)| \leq \|\mathcal{E}_k\| e^{-\kappa_n \mathsf{d}(X)}$, the supremum over the graded domain $\mathcal{U}(X^{(r_E)}; \theta_r, \delta_1^\flat, \delta_0)$ ($\mathcal{U}_k^\flat(X)$ for $r_E = 0$; Section 50), on which Proposition 47.10 is applied with $\delta_1^\flat$ in place of δ_1, for a decay functional d and the rate κ_n of the activity norm. For a generation of scale k, with $r_E = 0$ and $\mathsf{d} = d_M$, the smallest such constant $\|\mathcal{E}_k\|$ is the norm of Definition 50.1. In the flow the old activities are the grouped activities $\mathcal{E}_k(Y)$ of Definition 53.1, indexed by the polymers Y of scale k. The activity $\mathcal{E}_k(\square^+)$ contains the small old polymers of $\square$, the counterterms $-\beta_j S_\square$ of all generations $j \leq k$, and the other old polymers with closure $\square^+$. These activities have $r_E = 0$ and $\mathsf{d} = d_M$, and $\|\mathcal{E}_k\| \leq C_{\mathrm{old}}\mathsf{K}$ (Proposition 53.2).

Proposition 48.5 (Localized pieces)**.** *Assume (Q1)–(Q3), $\kappa' - 1 \geq c_\natural \kappa_n + \log(1/c_\flat)$, and the decay of the old activities stated above. On the support of χ_ξ,*

$$\delta\mathcal{E}(\xi) = \sum_Z E_{\mathrm{p}}(Z) + \sum_Z E_{\mathrm{m}}(Z),$$

the sums running over polymers. Each piece depends on ξ and U' only on the bonds of $Z^{(r')}$, and on the old activities only through $\mathcal{E}_k(X, \cdot)$ with $X \subset Z$. Each piece is invariant under real gauge transformations, and real at real references for real ξ. The pure pieces are analytic in the reference on all complex references of Section 46 with $\delta_c < \delta_P$ whose real part is (M, ε)-regular with $\varepsilon \leq \varepsilon_{\max}$. The mixed pieces $E_{\mathrm{m}}(Z)$ are analytic in the reference on the set of references at which the hypothesis of Proposition 47.10 on U' holds for every $X \subset Z$. The pieces satisfy, uniformly for $|\xi|_\infty \leq p_0$,

$$|E_{\mathrm{p}}(Z)| \leq \epsilon_{\mathrm{p}}\, e^{-(\kappa'-2)|Z|}, \qquad |E_{\mathrm{m}}(Z)| \leq \epsilon_{\mathrm{m}}\, e^{-(\kappa_n - 1)\mathsf{d}(Z)},$$

with $\epsilon_{\mathrm{p}} = C_P g p_0^3$ and $\epsilon_{\mathrm{m}} = C_P \tau \|\mathcal{E}_k\|$.

Proof. Apply Lemma 47.2 on $\mathbb{D}_{\kappa'}$ to each term. Its hypotheses hold by Lemma 48.2, Propositions 47.5(a),(d), 47.7(a),(d), 47.8(a) and 47.10(a),(b), and Lemma 40.3(b) and Corollary 41.6 for $C_k^{1/2}(s)$. These give analyticity in the reference on the domains stated, jointly with s. Details

- *Interaction.* With $Y = \square$, the bound $F \leq C_P g^{-2} M^d (g C_T p_0 / r_I)^3 = C_P g p_0^3$ holds (Section 47.2).
- *Gauge fixing.* With $Y = \square$, the bound $F \leq C_P g p_0^3$ holds for $G_3(\square; s)$ (Section 47.3). The gauge-fixing pieces join the pure pieces.
- *Jacobians.* $F \leq C_P(L^{2d+1} g p_0 + g^2 p_0^2) \leq C_P g p_0^3$ (Theorem 34.9(e)), since $p_0 \geq 1$ and C_P may depend on L.

- *Pure pieces.* Put $E_{\mathrm{p}}(Z) = \sum_{\square \subset Z} (f_{\mathrm{loc}}(\square, Z)$ of these terms). Then $|E_{\mathrm{p}}(Z)| \le |Z| F e^{\kappa'-1} e^{-(\kappa'-1)|Z|}$, and $|Z| e^{-|Z|} \le 1$ gives the bound after adjusting C_P by $e^{\kappa'}$.
- *Mixed pieces.* Here $Y = X$. Indeed, $\mathcal{E}_k(X, \cdot)$ depends on the configuration on $X^{(r_E)}$ with $r_E \le r$, and by Theorem 45.5(d) and Lemma 40.3(b), A_* on $(Z_0')^{(r)}$ depends on s only through the boxes of Z_0'. So the hypothesis of Lemma 47.2 holds with Y the set of boxes of X. Proposition 47.10 gives $F \le 4\tau \|\mathcal{E}_k\| e^{-\kappa_n \mathsf{d}(X)}$, so

$$|f_{\mathrm{loc}}(X, Z)| \le 4\tau \|\mathcal{E}_k\| e^{-\kappa_n \mathsf{d}(X) - (\kappa'-1)|Z \setminus X|}.$$

  By Definition 48.4, $\mathsf{d}(Z) \le \mathsf{d}(X) + c_\natural |Z \setminus X|$. So

$$|f_{\mathrm{loc}}(X, Z)| \le 4\tau \|\mathcal{E}_k\| e^{-\kappa_n \mathsf{d}(Z)} e^{-(\kappa'-1-c_\natural \kappa_n)|Z \setminus X|}.$$

  Summing over $X \subset Z$, which over-counts the connected X harmlessly,

$$\sum_{X \subset Z} e^{-(\kappa'-1-c_\natural \kappa_n)|Z \setminus X|} \le \big(1 + e^{-(\kappa'-1-c_\natural \kappa_n)}\big)^{|Z|} \le e^{c_\flat |Z|} \le e^{\mathsf{d}(Z)+1}.$$

  This defines $E_{\mathrm{m}}(Z) = \sum_{X \subset Z} f_{\mathrm{loc}}(X, Z)$, with the bound stated.

Data dependence, analyticity and invariance are Lemma 47.2(a),(c). Reality: at a real reference and for real ξ and real s, all the functions localized are real (Corollary 41.6 for $C_k^{1/2}(s)$, and the realness of s_p, $\hat{Z}$, the gauge-fixing pieces, the Jacobians and the old activities). The localization takes real derivatives and integrals over $t \in [0,1]^{Z \setminus Y}$, so the pieces are real. Margins are not converted (Remark 47.3). □

48.3 The Mayer and polymer expansion

Two polymers are *incompatible* if they touch, i.e. if a box of one shares a point with a box of the other. Non-touching polymers are at distance at least $M/4 > 2r'$ (Section 47, Setting). So pieces of non-touching polymers depend on disjoint sets of bonds of $\mathfrak{F}$, and are independent under the product measure $d\mu^*$.

We use the cluster expansion of Kotecký and Preiss [25], in the following form. Let $\mathcal{K}$ be a set of objects γ ("gas polymers"), each with a support $\operatorname{supp}\gamma$, a polymer, and a complex activity $k(\gamma)$. Call two objects compatible if their supports do not touch. Then

$$\sum_{\{\gamma_1, \dots, \gamma_n\} \text{ compatible}} \prod_i k(\gamma_i) = \exp\Big(\sum_{\mathcal{C}} \phi^T(\mathcal{C}) \prod_{\gamma \in \mathcal{C}} k(\gamma) \Big),$$

the sum running over the clusters $\mathcal{C}$, i.e. the finite multisets of objects whose incompatibility graph is connected, with the Ursell coefficients ϕ^T. This holds with absolute convergence if there are functions $a, b \ge 0$ on $\mathcal{K}$ with

$$\sum_{\gamma' \not\sim \gamma} |k(\gamma')| e^{a(\gamma')+b(\gamma')} \le a(\gamma) \qquad \text{for all } \gamma. \tag{48.1}$$

In that case $\sum_{\mathcal{C} \not\sim \gamma} |\phi^T(\mathcal{C}) \prod_{\gamma' \in \mathcal{C}} k(\gamma')| e^{\sum_{\gamma' \in \mathcal{C}} b(\gamma')} \le a(\gamma)$, the sum running over the clusters containing an object incompatible with γ [25].

Theorem 48.6 (Polymer expansion of the fluctuation integral)**.** *Let* d *be a decay functional. There are* C_d*,* A *and* $\epsilon_* > 0$*, depending only on* d *and the constants of* d*, with the following property. Let* $E(Z)$*, indexed by polymers, be complex functions of* ξ *and of further parameters. Suppose each* $E(Z)$ *depends on* ξ *only on the bonds of* $\mathfrak{F}$ *in* $Z^{(r')}$*, and on the other spatial data,*

such as the reference, only on $Z^{(r')}$; global parameters, such as the coupling or an interpolation parameter, are allowed. Suppose also

$$|E(Z)| \le \epsilon\, e^{-\kappa(\mathsf{d}(Z)+c_\natural)} \qquad \text{for } |\xi|_\infty \le p_0,$$

with $\kappa \ge C_d$ and $\epsilon \le \epsilon_*$. Then

$$\int \exp\Big(\sum_Z E(Z)\Big) d\mu^* = \exp\Big(\sum_Y E^\#(Y)\Big),$$

where Y runs over polymers, and:

(a) $E^\#(Y)$ depends only on the functions $E(Z)$ with $Z \subset Y$, and on the data only on $Y^{(r')}$;

(b) $\sum_{Y \supset \square} |E^\#(Y)| e^{(\kappa - C_d)(\mathsf{d}(Y)+c_\natural)} \le A\epsilon$ for every box $\square$;

(c) if the $E(Z)$ are analytic in a parameter, uniformly, so is $E^\#(Y)$; if they are invariant under real gauge transformations, so is $E^\#(Y)$, since $d\mu^*$ is invariant under the bondwise orthogonal action $\xi(b) \mapsto R(u(b_-))\xi(b)$;

(d) if the $E(Z)$ are even under a joint transformation of a parameter and $\xi \mapsto -\xi$, then $E^\#(Y)$ is even in the parameter;

(e) if the $E(Z)$ are real for real ξ at some value of the parameters, so is $E^\#(Y)$ at that value.

Proof. Mayer expansion. Put $K(Z) = e^{E(Z)} - 1$, so $|K(Z)| \le 2|E(Z)|$ for $\epsilon \le 1$. Expand $\prod_Z (1 + K(Z))$ as a sum over finite sets of distinct polymers. Group each set into its connected components for touching, the clusters of polymers Γ, with supports $\operatorname{supp}\Gamma = \bigcup_{Z\in\Gamma} Z$, which are polymers. Distinct components do not touch, so the integral factorizes: Details

$$\int \prod_Z (1 + K(Z))\, d\mu^* = \sum_{\{\Gamma_1,\dots,\Gamma_n\} \text{ compatible}} \prod_i k(\Gamma_i), \qquad k(\Gamma) = \int \prod_{Z\in\Gamma} K(Z)\, d\mu^*.$$

Then $|k(\Gamma)| \le \prod_{Z\in\Gamma} 2\epsilon e^{-\kappa(\mathsf{d}(Z)+c_\natural)}$.

A tree count. For weights $u(Z) \ge 0$ let $S(\square)$ be the sum of $\prod_{Z\in\Gamma} u(Z)$ over the clusters Γ with a polymer containing $\square$, and $S_* = \sup_\square S(\square)$. Remove from such a Γ a polymer $Z_0 \ni \square$. The rest splits into touching-connected families, each containing a polymer that contains a box $\square'$ touching Z_0. Two such families cannot contain the same $\square'$, since they would touch. There are at most $3^d|Z_0|$ such boxes. Hence $S(\square) \le \sum_{Z_0 \ni \square} u(Z_0)(1 + S_*)^{3^d|Z_0|}$. Restrict first to polymers of size at most N, so that all sums are finite, and let $N \to \infty$. Put $\eta := \sup_\square \sum_{Z\ni\square} u(Z) e^{3^d|Z|}$, and scale all weights by $\lambda \in [0,1]$. Then S_* is a continuous function of λ that vanishes at $\lambda = 0$. While $S_* \le 1$, the inequality gives $S_* \le \lambda\eta$, since $(1 + S_*)^{3^d|Z_0|} \le e^{3^d|Z_0|}$. So S_* never exceeds $\lambda\eta$, and $S_* \le \eta$ whenever $\eta \le 1$.

Polymer expansion. Apply the Kotecký–Preiss theorem to the objects Γ, with $a(\Gamma) = A\epsilon|\operatorname{supp}\Gamma|$ and $b(\Gamma) = \beta(\mathsf{d}(\operatorname{supp}\Gamma) + c_\natural)$, $\beta = \kappa - C_d$. Along a spanning tree of the touching graph of Γ, the subadditivity of d gives

$$\mathsf{d}(\operatorname{supp}\Gamma) + c_\natural \le \sum_{Z\in\Gamma} (\mathsf{d}(Z) + c_\natural), \qquad |\operatorname{supp}\Gamma| \le \sum_{Z\in\Gamma} |Z| \le c_\flat^{-1} \sum_{Z\in\Gamma} (\mathsf{d}(Z) + c_\natural).$$

Objects Γ' incompatible with Γ have a polymer containing a box that touches $\operatorname{supp}\Gamma$, and there are at most $3^d|\operatorname{supp}\Gamma|$ such boxes. So the left side of (48.1) is at most $3^d|\operatorname{supp}\Gamma|\, S_*$, with

$$u(Z) = 2\epsilon\, e^{-(\kappa - \beta - A\epsilon c_\flat^{-1})(\mathsf{d}(Z)+c_\natural)} \le 2\epsilon\, e^{-(C_d - c_\flat^{-1})c_\flat|Z|},$$

for $A\epsilon \le 1$, using $\mathsf{d}(Z) + c_\natural \ge c_\flat|Z|$. The number of polymers $Z \ni \square$ with $|Z| = m$ is at most C_0^m, with C_0 depending on d. Choose $C_d = c_\flat^{-1}(\log C_0 + 3^d + 2)$. Then $\eta \le C_1\epsilon$, so $S_* \le C_1\epsilon$, and (48.1) holds with $A = 3^d C_1$ for $\epsilon \le \epsilon_*$.

The output. Group the clusters $\mathcal{C}$ of objects by $Y = \bigcup_{\Gamma\in\mathcal{C}} \operatorname{supp}\Gamma$, a polymer since $\mathcal{C}$ is connected. Put $E^{\#}(Y) = \sum_{\mathcal{C}:\,\cup\operatorname{supp}=Y} \phi^T(\mathcal{C}) \prod_{\Gamma\in\mathcal{C}} k(\Gamma)$. Then (a) holds, since each $k(\Gamma)$ involves only the $E(Z)$ with $Z \in \Gamma$, and $\bigcup_{Z\in\Gamma} Z^{(r')} \subset Y^{(r')}$. For (b), subadditivity gives $e^{\beta(\mathsf{d}(Y)+c_\natural)} \le \prod_{\Gamma\in\mathcal{C}} e^{b(\Gamma)}$. The clusters with $Y \supset \square$ are incompatible with the single-box object $\{\square\}$, a legitimate object since a box is a polymer. So the Kotecký–Preiss bound gives $\sum_{Y\supset\square} |E^{\#}(Y)| e^{\beta(\mathsf{d}(Y)+c_\natural)} \le a(\{\square\}) = A\epsilon$. For (c), all sums converge uniformly, and $d\mu^*$ is invariant under the bondwise orthogonal action with the symmetric cutoff. For (d), $d\mu^*$ is even, so each $k(\Gamma)$ is even in the parameter, and the combinatorics do not depend on ξ. For (e), $d\mu^*$ is a real measure, so each $k(\Gamma)$ is real, and the Ursell coefficients ϕ^T are real; the series converges absolutely. □

Corollary 48.7 (Two decay classes)**.** *In the situation of Proposition 48.5, put* $c'_\sharp = \max(c_\sharp, c_\natural)$. *Let* $\kappa_n - 1 \ge C_d$, $(\kappa'-2)/c'_\sharp \ge \kappa_n - 1$, $\epsilon_{\mathrm{p}} \le \frac{1}{2}\epsilon_*$ *and* $\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)} \le \frac{1}{4}\epsilon_*$. *Then* $\int e^{\delta\mathcal{E}} d\mu^* = \exp(\sum_Y (E^{\#}_{\mathrm{p}}(Y) + E^{\#}_{\mathrm{m}}(Y)))$, *where:*

- $E^{\#}_{\mathrm{p}}$ *is the output of Theorem 48.6 for the pure pieces alone. It satisfies*

$$\sum_{Y\supset\square} |E^{\#}_{\mathrm{p}}(Y)| e^{((\kappa'-2)/c'_\sharp - C_d)(\mathsf{d}(Y)+c_\natural)} \le A\epsilon_{\mathrm{p}}.$$

- $E^{\#}_{\mathrm{m}} = E^{\#} - E^{\#}_{\mathrm{p}}$ *satisfies*

$$\sum_{Y\supset\square} |E^{\#}_{\mathrm{m}}(Y)| e^{(\kappa_n - 1 - C_d)(\mathsf{d}(Y)+c_\natural)} \le 4A\,\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)}.$$

Proof. Since $\mathsf{d}(Z) + c_\natural \le c'_\sharp |Z|$, the pure pieces satisfy $|E_{\mathrm{p}}(Z)| \le \epsilon_{\mathrm{p}} e^{-((\kappa'-2)/c'_\sharp)(\mathsf{d}(Z)+c_\natural)}$. Theorem 48.6 gives the first bound. The mixed pieces satisfy $|E_{\mathrm{m}}(Z)| \le \epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)} e^{-(\kappa_n-1)(\mathsf{d}(Z)+c_\natural)}$. For the full system at the rate $\kappa_n - 1$, replace E_{m} by λE_{m} with complex $|\lambda| \le R = \epsilon_*/(2\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)}) \ge 2$. Then $\epsilon = \epsilon_{\mathrm{p}} + |\lambda|\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)} \le \epsilon_*$, and Theorem 48.6 applies uniformly in λ; $E^{\#}(Y;\lambda)$ is analytic in λ, and $E^{\#}(Y;0) = E^{\#}_{\mathrm{p}}(Y)$. Cauchy's formula on $|\lambda| = R \ge 2$ gives

$$|E^{\#}(Y;1) - E^{\#}(Y;0)| \le \frac{2}{R} \cdot \frac{1}{2\pi} \oint |E^{\#}(Y;\lambda)| \frac{|d\lambda|}{|\lambda|}.$$

Summing over Y under the integral, with $A\epsilon_*$ from Theorem 48.6(b), gives the second bound. □

Details

The rate of the mixed class is the rate κ_n of the activity norm minus $1 + C_d$. It is recovered by the reblocking gain of Dimock's tree length (Remark 49.4, the rates in Remark 49.8), under $\kappa_n \ge 2(1 + C_d + \kappa_0)$; see Lemma 53.4(b).

48.4 The order g and the size of the pure sources

Replace g by a complex parameter t in the fluctuation only: $Z = tC_k^{1/2}(s)\xi$, and t^{-2} in front of the interaction pieces. (The same device, with the old activities kept, gives (F3) of Proposition 58.6.) The prefactor of Proposition 48.3 and $\mathfrak{E}(U')$ do not depend on t.

Proposition 48.8 (Pure sources)**.** *Let* $r_t = \min\left(c\, r_I^3 (C_P M^d C_T^3 p_0^3)^{-1},\ r_I (2C_T p_0)^{-1}\right)$, *with* c *small, the constant* C_P *including the factors* $e^{\kappa'}$. *For* $|t| \le r_t$ *the pure pieces, with* t *in place of* g, *satisfy the hypotheses of Theorem 48.6 with* $\epsilon \le \frac{1}{2}\epsilon_*$, *as in the proof of Corollary 48.7. The output* $E^{\#}_{\mathrm{p}}(Y;t)$ *is analytic and even in* t, *and* $E^{\#}_{\mathrm{p}}(Y;0) = 0$. *Hence, for* $2g \le r_t$,

$$\sum_{Y\supset\square} |E^{\#}_{\mathrm{p}}(Y;g)| e^{((\kappa'-2)/c'_\sharp - C_d)(\mathsf{d}(Y)+c_\natural)} \le C_P g^2 p_0^6.$$

Proof. For $|t| \le r_t$: $|t|C_T p_0 \le \frac{1}{2} r_I$, by the second entry of the minimum. The pure pieces only need $|Z|_\infty \le \frac{1}{2} r_I$: Propositions 47.5(a),(b), 47.7(a),(b) and 47.8(a). The parts of (Q1) that involve w, Lemma 48.2 and the identification of Proposition 47.5(c) are used only at $t = g$, where they hold by (Q1). So the bounds of Proposition 48.5 for the pure pieces hold with g replaced by $|t|$. That is, $\epsilon_{\mathrm{p}}(t) \le C_P |t| p_0^3$, from the interaction and the gauge-fixing pieces, plus the Jacobian term $C_P |t| p_0$. This is at most $\frac{1}{2}\epsilon_*$ for c small. Analyticity in t follows from Theorem 48.6(c). Every pure piece depends on (t, ξ) only through t^{-2} and $tC_k^{1/2}(s)\xi$: the interaction pieces are Taylor remainders of order three in Z for every s (Definition 47.4), and $\mathfrak{J}$ depends on Z. The gauge-fixing pieces enter as $t^{-2}\varphi_{y,x}(tC_k^{1/2}(s)\xi)$; they do not depend on s otherwise, and they are Taylor remainders of order three as well (Proposition 47.7(c)). Since $t^{-2}v_p(s; tY)$ and $t^{-2}\varphi_{y,x}(tY)$ are analytic at $t = 0$, being remainders of order three, the pieces are analytic and even under $(t, \xi) \mapsto (-t, -\xi)$. The localization takes s-derivatives at fixed ξ. So Theorem 48.6(d) applies, and $E_{\mathrm{p}}^{\#}$ is even in t. At $t = 0$ every pure piece vanishes, so $E_{\mathrm{p}}^{\#}(Y; 0) = 0$. So $E_{\mathrm{p}}^{\#}(Y; t)/t^2$ is analytic on $|t| \le r_t$. Cauchy's formula on $|t| = r_t$ gives, for $2g \le r_t$, $|E_{\mathrm{p}}^{\#}(Y; g)| \le 2(g/r_t)^2 \frac{1}{2\pi} \oint |E_{\mathrm{p}}^{\#}(Y; t)|\, |dt|/r_t$. Summing over Y under the integral, with Theorem 48.6(b) and $r_t^{-2} \le C_P p_0^6$ (both entries of the minimum, $p_0 \ge 1$), gives the bound. In particular there is no term of order g. □

Details

48.5 The step in polymer form

Theorem 48.9 (Small-field step in polymer form)**.** *Let G be semisimple, and let ρ be as in Theorem 34.9. Let $\alpha \le \frac{1}{2}$, and assume (Q1)–(Q3), (DC5), $2g \le r_t$, the decay of the old activities, and the hypotheses of Proposition 48.5 and Corollary 48.7. Then, at real critical references with V' small field,*

$$T_{\mathrm{sf}}(\rho)(V') = \mathcal{N}_k N_\chi \exp\Big(-\frac{1}{g^2} S(U') - \frac{1}{2}\sum_b \ell_k(b; 1) + \sum_X \mathfrak{e}(X, U') + \sum_c \log \mathfrak{j}_c + \mathfrak{E}(U') + \sum_Y \big(E_{\mathrm{p}}^{\#}(Y, U') + E_{\mathrm{m}}^{\#}(Y, U')\big)\Big),$$

with $\mathfrak{e}$ from Proposition 41.7 and $\ell_k(b; 1)$ the density of Proposition 34.8 at $U' = 1$. At other references the right-hand side is the definition of the step (Remark 47.13). Every term depends on U' only near its polymer, is invariant under real gauge transformations, and is real at real references (Theorem 48.6(e)). The terms $\mathfrak{e}(X)$, $\log \mathfrak{j}_c$ and $E_{\mathrm{p}}^{\#}$ are analytic in the reference on the complex references of Section 46 with $\delta_c < \delta_P$ whose real part is (M, ε)-regular with $\varepsilon \le \varepsilon_{\max}$ ($\mathfrak{e}$ and $\log \mathfrak{j}_c$ on the graded domains of Propositions 41.7 and 47.8). The terms $E_{\mathrm{m}}^{\#}(Y)$ and $\mathfrak{E}$ are analytic on the references at which Proposition 47.10 applies to the old polymers inside Y, respectively to all old polymers; Remark 47.11 shows that this suffices for the flow. Moreover:

- *$E_{\mathrm{p}}^{\#}$ is $O(g^2 p_0^6)$, with decay rate $(\kappa' - 2)/c_\sharp' - C_d$ (Proposition 48.8);*
- *$E_{\mathrm{m}}^{\#}$ is $O(\tau \|\mathcal{E}_k\| e^{c_\natural \kappa_n})$, with decay rate $\kappa_n - 1 - C_d$ (Corollary 48.7);*
- *$\mathfrak{e}(X)$ and $\log \mathfrak{j}_c$ are bounded on the graded domains by Propositions 41.7(d) and 47.8(c). On the domains of Remark 38.8 the bounds are $Ce^{-(\kappa_2 - 1)|X|}$ and C, with C independent of g and of the domain constants, for polymers of diameter at most $n/2 - 8$; larger polymers have an extra factor $e^{-cn/M}$. Their quadratic parts are marginal in $d = 4$ (Remark 41.8(b)).*
- Symmetry. *Let $\rho \circ r = \rho$ for some $r \in \mathfrak{S}_{k+1}$, the group generated by the hyperoctahedral group B_4 about the points of $\mathbb{T}'$ and the translations of $\mathbb{T}'$. Then the left side is invariant under $V' \mapsto rV'$ (Lemma 32.6(iii)). So is the sum over polymers in the exponent on the*

right, i.e. $\sum_X \mathfrak{e}(X,U') + \sum_c \log \mathfrak{j}_c + \mathfrak{E}(U') + \sum_Y (E^{\#}_{\mathrm{p}} + E^{\#}_{\mathrm{m}})(Y,U')$*: the term* $g^{-2}S(U')$ *is invariant (Proposition 31.7(b) at step* $k+1$*), the constants do not depend on* V'*, and all terms are real. The single terms* $\mathfrak{e}(X)$*,* $E^{\#}_{\mathrm{p}}(Y)$*,* $E^{\#}_{\mathrm{m}}(Y)$ *need not be covariant. They are indexed by the polymers of the paving of Section 40, and they are built with the grid points* z_j *of the partitions of unity and with the mass profiles. The translations by* $M\mathbb{Z}^4$ *preserve the polymer grid and the grid of the* z_j*, and under them the terms* $\mathfrak{e}(X)$ *and* $\log \mathfrak{j}_c$ *are covariant:* $\mathfrak{e}(X+a,\tau_a U') = \mathfrak{e}(X,U')$ *and* $\log \mathfrak{j}_{c+a}(\tau_a U') = \log \mathfrak{j}_c(U')$ *for* $a \in M\mathbb{Z}^4$ *(Proposition 41.14(b),(c)). For* $E^{\#}_{\mathrm{p}}(Y)$ *and* $E^{\#}_{\mathrm{m}}(Y)$ *this is not claimed, and no single term is claimed to be covariant under the other elements of* $\mathfrak{S}_{k+1}$*.*

Proof. Combine Proposition 48.3, Proposition 48.5, Corollary 48.7, Proposition 48.8, and Proposition 41.7 for the determinant. The locality in U' follows from the data dependence on $Y^{(r')}$ in Theorem 48.6(a). The symmetry statement is the invariance of the left side, which holds by the paragraph after Definition 48.1. The right side equals the left side at every real critical reference with V' small field. The configuration rV' is again small field, and rU' lies in its minimal orbit (Proposition 31.7(b) and the proof of Lemma 32.6(iii)); by Lemma 32.6(i) it may be taken as its reference. □ Details

Remark 48.10 (Normalization and what remains). (a) *Normalization.* The new activities are normalized by subtracting their values at $U' = 1$. The constant $\sum_Y E^{\#}(Y,1)$ goes into the vacuum energy, and $\log \mathcal{N}_k$, $\log N_\chi$ and $-\frac{1}{2}\sum_b \ell_k(b;1)$ into the explicit normalization $\mathcal{Z}_{k+1}$ (Section 52.3). Only bounds per grid cube are needed for the vacuum energy; no periodicity and no Gaussian reference are used.

(b) *The old activities* at U', $\mathfrak{E}(U') = \sum_X \mathcal{E}_k(X,U')$, depend on U' on the margins of their own domains; in the flow these are their own polymers ($r_E = 0$). In four dimensions the old activities are not reblocked, rescaled or contracted. They are the generations of the earlier scales, evaluated at the new background field, and at each step they are regrouped into the activities of Definition 53.1, with the counterterms (Proposition 53.2).

(c) *Where the rest is done.*

- The reblocking of the new terms to polymers of step $k+1$, the comparison of the rates with the activity norm and the recovery of the loss C_d for the mixed class: Lemma 53.4, with the rates of Remark 49.8.
- The decay assumed for the old activities: Proposition 53.2. One step of the induction over the scales: Theorem 53.7, and Proposition 53.8 for the first step. The iteration over all scales: Theorem 53.12.
- The wrap-around polymers at later scales: Lemma 52.1(c)–(e) and Lemma 51.7(c).
- The large-field remainder $\mathcal{R}_{\mathrm{LF}}$ is not treated in the note.

49 Polymers, scales and parameters of the flow

This section and Section 52 collect the parts of the small-field flow that are common to all steps. This section contains the small-field parameters, the polymer grids of all scales and their alignment, the reblocking, the classes of the polymers of an old generation at a later scale, and the conditions on all parameters of Parts VII–X. Section 52 contains the scales of the torus, the first step, the vacuum energy and the bookkeeping of the modified model. Sections 50 and 51 begin the four-dimensional flow itself: the generations of activities, their symmetrization over the grid symmetries and their coefficient β, and the extraction of the marginal part of an old generation at every later scale. Section 53 puts one step into generation form and iterates it over all scales (Theorem 53.12; Remark 49.1).

Remark 49.1 (Plan of the flow). The four-dimensional small-field flow is Theorem 53.12, with Proposition 53.10 and Corollaries 53.11 and 53.13 (Section 53.5); it is summarized in Theorem N. It follows Bałaban [18, (0.22)–(0.30), Sects. 1–5], [19, Sects. 1–2], and is organized as follows.

- *Generations.* The density of step k has the form

$$\mathcal{Z}_k \exp\Big(-\frac{1}{g_k^2} S(U_k) - \varepsilon_k |\mathbb{T}^{(k)}| + \sum_{j \le k} (E^{(j)} - \beta_j S)(U_k) \Big).$$

  The generation j consists of a family $E^{(j)}$ of normalized, gauge-invariant graded activities on the polymers of the grid of scale j (Definition 50.1), and of the counterterm $-\beta_j S$. It is created by the step from scale $j-1$ to scale j, as the symmetrization of the new terms of that step over the symmetries of the grid (Lemma 53.4, Lemma 50.8). Its bounds are stated only at that scale, on the domains $\mathcal{U}^\flat$ of Section 50, slightly smaller than those of Remark 38.8. At later scales only the background field in its argument changes, and this background field is again a background field of scale j (Lemma 50.3).

- *Extraction of β.* At later scales the quadratic order of an old generation in the curvature is marginal in four dimensions (Remark 41.8(b)). It is cancelled by the counterterm $-\beta_j S$: Section 51 extracts the marginal part of an old generation at every later scale (Proposition 51.5) and bounds its other polymers (Lemma 51.7), and Section 53 groups every generation with its counterterm into old activities that are bounded uniformly in the scale (Proposition 53.2). The coefficient β_{j+1} of the new generation is β_E of Proposition 50.13, read off from the period density of the symmetrized generation, and $g_{j+1}^{-2} = g_j^{-2} - \beta_{j+1}$ (Theorem 53.7). Part XI computes the one-loop coefficient. The one-loop part of β_{j+1} is compared with it in (F2) of Proposition 58.6. The comparison, at fixed couplings and as the torus grows, of β_j with the coefficient that Bałaban's vacuum polarization [18, (1.20)–(1.22)] assigns to the generations is Proposition 58.8, through the generation locality of Section 59.

- *Iteration without contraction.* The activities are of order one, not small in the coupling, and the iteration does not contract them. Smallness comes from the fluctuation cutoff [19, (2.41)]. One step in generation form is Theorem 53.7, with Proposition 53.8 for the first step, and the iteration over all scales is Theorem 53.12: as long as the couplings stay in $(0, \gamma]$, the density of every scale has the generation form, with bounds uniform in the scale and in the torus.

From Parts VII–X the flow uses:

- one step, as constructed in Sections 32–48: Theorem 34.9 with Lemma 32.6, the localization of Parts VIII and IX, and the step in polymer form, Theorem 48.9;

- the domains of Remark 38.8, whose constants do not depend on the coupling, the step, the torus or the polymer;

- the interior estimates, the tier results and the weakened background map of Sections 43–45;

- the covariance of the step under the lattice symmetries, Proposition 31.7 and Lemma 32.6(iii); only sums over polymers are covariant, and single activities of a step are not claimed to be covariant (Theorem 48.9); the generations are made covariant under the symmetries of their grid by the symmetrization (Section 50.1);

- this section and Section 52: the small-field parameters, the polymer grids and their alignment, the reblocking (applied to the new generation only), the polymer classes, the

conditions of Remark 49.8, Lemma 52.1, the first step and its formula (Proposition 52.2, Lemma 52.3), the vacuum energy (Section 52.3) and the modified model (Remark 52.6).

Section 53 supplies the raw new generation and its bound (Lemma 53.4), the old activities with a bound uniform in k (Proposition 53.2), the mixed class through them, $g_{k+1} \le 2g_k$ (Theorem 53.7), the growth bound on the couplings needed in Proposition 52.5(c) (Theorem 53.12(f)), and the continuity in the couplings (Proposition 53.10, Corollary 53.11). The properties (F1)–(F3) of the flow that Theorem 58.9 uses are proved in Proposition 58.6. The wrapping polymers are bounded at all later scales by Lemma 52.1(c)–(e). The large-field stage is not treated (Remark 52.6(d)).

Why generations. In four dimensions the Wilson action does not change under the change of units (Section 31), and the new terms of a step are of order one: the one-loop pieces are bounded independently of the coupling (Remark 41.8(b)). Their quadratic order in the curvature is marginal. A point of the domain of scale $k = j + m$, read in the units of scale j, has its curvature budget multiplied by L^{-2m} (Remark 38.8(i)). So the quadratic part of an activity of a polymer of scale j is of order L^{-4m}, while a grid cube of scale k contains L^{4m} grid cubes of scale j (Lemma 49.2(b)). Summed over a cube of scale k, the quadratic part stays of order one at every later scale (Remark 41.8(b), Section 51). Reblocking the activities of a step into activities of the next scale therefore gives no contraction. Instead, the activities created at scale j are kept as a generation, bounded only at scale j and evaluated at the background fields of the later scales (Lemma 50.3). At every later scale their marginal part is cancelled by the counterterm $-\beta_j S$. In a grid cube of scale k, for $m \ge m_0$, the small polymers sum to $\beta_j S_\square$ up to $C_{\mathrm{ex}} \|E\|_j L^{-\alpha m}$, since the rest of their normal form is of order $L^{-(4+\alpha)m}$ per cell (Proposition 51.5, Remark 51.6(a)); the other polymers carry a factor L^{-m} (Lemma 51.7). These gains are summable over m. So the old generations, each grouped with its counterterm, are bounded uniformly in the scale, and no contraction is needed (Remark 51.6(d), Proposition 53.2). The counterterm must stay with the small polymers of its cube, since $\sum_{j \le k} \beta_j = g_0^{-2} - g_k^{-2}$ is not bounded uniformly in k (Remark 51.6(c)). The small factors of a step come from the cutoffs: the fluctuation field is bounded by w, of order $L^d g_k p_{0,k}$ (Remark 49.8, the couplings), so the terms of the step that contain old activities are of order τ times their norm, with $\tau \le C_P L^d g_k p_{0,k}/\delta_1$ (Proposition 47.10 and the paragraph after it, Theorem 48.9); and the pure sources are of order $g_k^2 p_{0,k}^6$, by parity in the coupling (Proposition 48.8).

49.1 Small-field parameters

Let $g_p \in (0, 1]$, and let $p, p_0 : (0, g_p] \to [1, \infty)$ be functions with the following properties:

- p is decreasing, and $g \mapsto g\,p(g)$ is increasing;
- $p_0(g)/p(g) \to 0$ as $g \to 0$;
- $g^\nu p(g) \to 0$ as $g \to 0$, for every $\nu > 0$; so also $g^\nu p_0(g) \to 0$;
- p_0 is continuous and decreasing, and $g \mapsto g\,p_0(g)$ is increasing; that is, p_0 is continuous and

$$p_0(g) \ge p_0(g'), \qquad g\,p_0(g) \le g' p_0(g') \qquad \text{for } 0 < g \le g' \le g_p. \tag{49.1}$$

For instance $p_0(g) = (\log g^{-1})^q$ and $p(g) = (\log g^{-1})^{q'}$ with $0 < q < q'$, on $(0, g_p]$ with $g_p = e^{-\max(1,q')}$; compare [15]. These have all the properties above; for the monotonicity, $g\,p_0(g)$ is increasing for $g < e^{-q}$ and $g\,p(g)$ for $g < e^{-q'}$, and $g_p \le e^{-q'} < e^{-q}$. The continuity of p_0 is used for the continuity of the flow in the couplings (Proposition 53.10, Corollary 53.11), where the

white-noise cutoff is sharp, and (49.1) for the bound on the vacuum energy (Theorem 53.12(f)). At step k put

$$p_k = p(g_k), \qquad p_{0,k} = p_0(g_k), \qquad h_k = p_k g_k.$$

The density of step k is given on the small-field set $\mathfrak{U}_k(h_k)$, i.e. (31.2) holds with the threshold $\tilde{h} = h_k$, and $p_0 = p_{0,k}$ bounds the white noise of step k (Section 48). The couplings stay in $(0, \gamma]$, with $\gamma \le g_p$ fixed last (Remark 49.8(6)). In Sections 49, 52 and 58 the letter γ denotes this bound on the couplings, except in the kernel bound of Lemma 58.4(d), where $\gamma > 0$ is a decay rate; it is not one of the decay rates γ, γ_0 of the operators in Parts I–IX. So $h_k \le \gamma p(\gamma)$, which is small. In four dimensions h_k enters the small-field conditions of the fluctuation and the set on which the sum of a generation is required to be invariant (Definition 50.1), but not the domains of the activities, whose constants are fixed (Remark 38.8).

Consecutive thresholds. If $g_{k+1} \le 2g_k$, then $h_{k+1} \le 2h_k$. Indeed, if $g_{k+1} \le g_k$, then $h_{k+1} \le h_k$, since $g \mapsto gp(g)$ is increasing. Otherwise $p_{k+1} \le p_k$, since p is decreasing, and $h_{k+1} \le p_k \cdot 2g_k$. This is the relation $h' \le 2h$ used in Section 31.

Admissible thresholds. A number $h > 0$ is an *admissible threshold* if:

- $h \le \min\,(a_{\delta=1},\, a_1/(2C_*+2))$ and $(C_*+1)h \le c_1$; these are the hypotheses of Lemma 31.6 on ε_1';
- h satisfies the smallness condition on ε_1 of Lemma 45.1;
- $(1 + C_J)(C_* + 1)h < \theta_r$, with C_J of Lemma 45.1 and θ_r of Remark 38.8.

These are upper bounds on h, so with h every number in $(0, h]$ is an admissible threshold. They involve the parameters up to the domain constants, and not the couplings. By the choice of γ (Remark 49.8(6)), $\gamma p(\gamma)$ is an admissible threshold, and so is h_k for every coupling $g_k \in (0, \gamma]$. The consequences of admissibility are collected in Section 50.

49.2 The polymer grid

Scales and units. The *scale* j, $0 \le j \le K$, is that of the lattice $\mathbb{T}^{(j)}$ of step j (Section 52.1). In the units of scale j, $\mathbb{T}^{(j)}$ is the unit lattice, of side n_j; for $i \ge j$ the lattice $\mathbb{T}^{(i)}$ has spacing L^{i-j}; all these lattices contain the origin; and the fine lattice $\mathbb{T}_\eta$ has spacing $\eta = L^{-j}$. Activities are functions of configurations on $\mathbb{T}_\eta$, which is the same lattice at every scale. A change of units changes the domains and the norms in which they are measured (Remark 38.8(i)), not the functions. Two parities are used. The block factor L is odd (Remark 49.8(2)), and M is a power of L (Remark 49.8(4)), so M is odd as well. Moreover M divides n_j for $j \le K$, since $n_j = L^{K-j} n_K$, and n_K and M are powers of L with $n_K \ge M$ (Section 52.1).

Grid cubes and polymers. Let $j \le K$, and use the units of scale j; distances are ℓ^∞-distances on the torus. For a point $c \in M\mathbb{Z}^4$ of $\mathbb{T}^{(j)}$, the *grid cube of scale j centred at c* is the union of the M^4 blocks $B(y)$, $y \in \mathbb{T}^{(j)}$, with $|y - c|_\infty \le \frac{1}{2}(M-1)$. By Lemma 49.2(a) below, it is the set of the points $x \in \mathbb{T}_\eta$ with $|x - c|_\infty \le \frac{1}{2}(M - \eta)$, and the grid cubes of scale j pave $\mathbb{T}_\eta$. It fills the closed cube $c + [-\frac{1}{2}M, \frac{1}{2}M]^4$ of the continuum torus. These are Bałaban's cubes of size M, with centres at the points of the lattice of spacing M [18, p. 257]. The *polymer grid* of scale j is the paving of Section 40 with all intervals of length M, centred at the points of $M\mathbb{Z}$; this is the paving used at step j (Section 48). A *polymer of scale j* is a $*$-connected union of grid cubes of scale j; two cubes are $*$-adjacent if the closed cubes of the continuum that they fill share a point. For a polymer X, let $d_M(X)$ be the length, in units of M, of the shortest tree in the continuum that meets every cube of X [15, (119)], and $|X|_M$ the number of its cubes. Then [15, (122), (123)]

$$d_M(X) \le \sqrt{d}\,(|X|_M - 1), \qquad |X|_M \le 3^d(1 + d_M(X)), \qquad \sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \le K_0, \tag{49.2}$$

with $\kappa_0 \geq 1$ and K_0 depending only on d. For $d = 4$ the constants are $\sqrt{d} = 2$ and $3^d = 81$. The first inequality holds since the centres of two $*$-adjacent cubes are at distance at most $\sqrt{d}$, in units of M, so the segments along a spanning tree of the adjacency graph form a tree that meets every cube. It replaces $d_M \leq |X|_M$ of [15], which is stated there for connected polymers. There, polymers are connected through common faces [15, Sect. 3.1], and the second and the third inequality hold for $*$-connected polymers as well. For the second, let T be a tree that meets every cube of X, a finite union of segments, of length ℓ in units of M, and lift it to $\mathbb{R}^d$. Every cube of X contains a point t of T, so, in units of M, it is the image of a subset of $t + [-1, 1]^d$, and the union of the cubes lies in the image of $T + [-1, 1]^d$. A tree is obtained from a point by adding segments $[p, p + su]$, with p a point of the tree already built, $s > 0$ and $|u| = 1$. Each such segment enlarges $T + [-1, 1]^d$ by at most s times the $(d-1)$-volume $2^{d-1}|u|_1 \leq 2^{d-1}\sqrt{d}$ of the projection of $[-1, 1]^d$ along u. So $T + [-1, 1]^d$ has volume at most $2^d + 2^{d-1}\sqrt{d}\,\ell$, and the projection to the torus does not increase volume. The cubes have volume 1 and disjoint interiors, so $|X|_M \leq 2^d + 2^{d-1}\sqrt{d}\,\ell \leq 3^d(1 + \ell)$; the infimum over the trees gives the second inequality. In the proof of the third [15, Lemma 25], the polymers $X \supset \square$ with $|X|_M = n$ are counted by paths of length $2(n-1)$ along a spanning tree of the adjacency graph. Under $*$-adjacency a cube has $3^d - 1$ neighbours, so there are at most $(3^d - 1)^{2(n-1)}$ such paths. The proof then uses the second inequality, in the form $d_M(X) \geq 3^{-d}|X|_M - 1$. So the third inequality holds for $*$-connected polymers, with κ_0 and K_0 depending only on d. So d_M is a decay functional (Definition 48.4), with the constants given there: $c_\flat = 3^{-d}$, $c_\sharp = \sqrt{d}$, $c_\natural = 2\sqrt{d}$. For a polymer X of scale j we write $d_j(X)$ for $d_M(X)$, computed with the grid of scale j, in the units of scale j.

Lemma 49.2 (Nesting of the grids)**.** *Let $0 \leq j \leq i \leq K$, and use the units of scale j.*

(a) The grid cube of scale i centred at $c \in L^{i-j}M\mathbb{Z}^4$ is the set of the points $x \in \mathbb{T}_\eta$ with $|x - c|_\infty \leq \frac{1}{2}(L^{i-j}M - \eta)$. The grid cubes of scale i pave $\mathbb{T}_\eta$.

(b) It is the disjoint union of the $L^{4(i-j)}$ grid cubes of scale j centred at the points $c + Mt$, $t \in \mathbb{Z}^4$, $|t|_\infty \leq \frac{1}{2}(L^{i-j} - 1)$. Every grid cube of scale j lies in exactly one grid cube of scale i.

(c) Let X be a polymer of scale j, and let $\mathrm{cl}_i X$ be the union of the grid cubes of scale i that meet X. These are the grid cubes of scale i that contain a cube of X. The set $\mathrm{cl}_i X$ is a polymer of scale i, $\mathrm{cl}_j X = X$, and

$$d_j(X) \geq L^{i-j}\, d_i(\mathrm{cl}_i X).$$

Proof. The integers $\eta^{-1} = L^j$, L and M are odd. We use two facts on an odd positive integer B, as in Section 21. First, every integer s has exactly one representation $s = Bu + r$ with integers u and $|r| \leq \frac{1}{2}(B-1)$. Second, for an odd positive integer q, the integers s with $|s| \leq \frac{1}{2}(qB-1)$ are exactly the sums $Bt+r$ with integers $|t| \leq \frac{1}{2}(q-1)$ and $|r| \leq \frac{1}{2}(B-1)$, each obtained once. Indeed, by the first fact the qB sums are distinct, and they lie in this interval of qB integers. We apply both facts in each coordinate to $s = (x - c)_\mu/\eta$, for points $x \in \mathbb{T}_\eta$ and a point c of $\mathbb{T}_\eta$; this is an integer. Since $L^{i-j}M$ divides n_j, i.e. $M \mid n_i$, the representations are compatible with the periodicity of the torus. Details

(a) In the units of scale i, the cube is the union of the blocks $B(y)$, $y \in \mathbb{T}^{(i)}$, with $|y - c|_\infty \leq \frac{1}{2}(M-1)$. In the units of scale j, these blocks are the sets $\{x \in \mathbb{T}_\eta : |x - y|_\infty \leq \frac{1}{2}(L^{i-j} - \eta)\}$ (Section 21), with $y \in c + L^{i-j}\mathbb{Z}^4$ and $|y - c|_\infty \leq \frac{1}{2}(M-1)L^{i-j}$. The second fact with $B = L^{i-j}/\eta = L^i$ and $q = M$ gives the first claim. The first fact with $B = L^{i-j}M/\eta = ML^i$ and $c = 0$ shows that these sets, for $c \in L^{i-j}M\mathbb{Z}^4$, pave $\mathbb{T}_\eta$: every point x lies in the one centred at $L^{i-j}Mu$, with $x_\mu/\eta = ML^i u_\mu + r_\mu$.

(b) By (a) with $i = j$, the grid cube of scale j centred at $c+Mt$ is $\{x \in \mathbb{T}_\eta : |x-c-Mt|_\infty \le \frac{1}{2}(M-\eta)\}$. The second fact with $B = M/\eta = ML^j$ and $q = L^{i-j}$, for which $qB = ML^i$ as in (a), gives the first claim. The grid cubes of scale j are disjoint, so a grid cube of scale j that meets a grid cube of scale i is one of the cubes of this union. Since the grid cubes of scale i are disjoint and pave $\mathbb{T}_\eta$, every grid cube of scale j lies in exactly one of them.

(c) By (b), a grid cube of scale i that meets X contains a cube of X, and $\mathrm{cl}_j X = X$. The closed cube of the continuum filled by a grid cube of scale j lies in the one filled by the grid cube of scale i that contains it. So two $*$-adjacent cubes of scale j lie in equal or $*$-adjacent cubes of scale i, and $\mathrm{cl}_i X$ is $*$-connected, a polymer of scale i. A tree that meets every cube of X meets every cube of $\mathrm{cl}_i X$, since each of them contains a cube of X. In the units of scale j the cubes of scale i have side $L^{i-j}M$. So $d_i(\mathrm{cl}_i X)$ is at most the length of the tree divided by $L^{i-j}M$, and the infimum over the trees gives $d_j(X) \ge L^{i-j} d_i(\mathrm{cl}_i X)$. □

So the grids of all scales are aligned: every grid cube of scale i is a union of grid cubes of scale $j \le i$, and its centre is the centre of one of them. At step k, the grid points of the partitions of unity of Section 17 at the scale M are the centres of the grid cubes of scale k, since $M \mid n_k$ gives $\check{M} = M$ there; the partition at the IMS scale of Theorem 34.4 enters only proofs of lower bounds.

Neighbourhoods and the action of a cube. For a grid cube $\square$ of scale k, centred at c, let $\square^+$ be the union of the grid cubes of scale k centred at the points of $c + M\{-1,0,1\}^4$, in the units of scale k: the cube $\square$ and the grid cubes that share a corner with it. This is Bałaban's cube $\square^1$ of size $3M$ with the centre of $\square$ [18, p. 257]. By the second fact in the proof of Lemma 49.2, with $q = 3$, $\square^+$ is the set of the points $x \in \mathbb{T}_\eta$ with $|x-c|_\infty \le \frac{1}{2}(3M-\eta)$, in the units of scale k. Write $S = \sum_p s_p$, as in Section 47, the sum running over the positively oriented plaquettes $p = p_{\mu\nu}(x)$ of $\mathbb{T}_\eta$, and call $x = p_-$ the *base point* of p (Definition 21.4). For a grid cube $\square$ of any scale put

$$S_\square(U) = \sum_{p:\, p_- \in \square} s_p(U).$$

Since the grid cubes of one scale pave $\mathbb{T}_\eta$, the sum of $S_\square$ over them is S. Each $S_\square$ is analytic in complex configurations, invariant under real gauge transformations, real at real configurations and zero at $U = 1$. It depends only on U on the bonds of $\square^{(1)}$, since the corners of a plaquette are at distance at most $\eta \le 1$ from its base point. Like S, it does not depend on the units (Section 23). Moreover $S_{\square+a}(\tau_a U) = S_\square(U)$ for every translation a that maps the grid of the scale of $\square$ onto itself, where $(\tau_a U)(b) = U(b-a)$.

Base cubes. Let X be a polymer of scale j. Recall that X is *wrapping* if its diameter, the largest distance between two of its points, is larger than $\frac{1}{2}n_j - 8$ (Section 52.1). Suppose that X is not wrapping. Consider the projection to a coordinate circle $\mathbb{R}/n_j\mathbb{Z}$ of the union of the closed cubes of the continuum that the cubes of X fill. It is connected, since X is $*$-connected. It is not the whole circle: otherwise X would contain two points whose coordinates in this direction differ by at least $\frac{1}{2}n_j - \eta$ on the circle, and $\operatorname{diam} X > \frac{1}{2}n_j - 8$. So it is a closed arc of length less than n_j, and X lies in the product of these four arcs. This product is the injective image of a box in $\mathbb{R}^4$, which is unique up to translations by $n_j\mathbb{Z}^4$. The *base cube* $Q(X)$ is the cube of X whose centre, read in this box, is lexicographically least. If X is wrapping, $Q(X)$ is the cube of X whose centre, with coordinates in $[0, n_j)$, is lexicographically least. For a polymer X that is not wrapping and every $a \in M\mathbb{Z}^4$, the polymer $X + a$ is not wrapping, its arcs are those of X translated by a, and

$$Q(X + a) = Q(X) + a.$$

A polymer X of scale j is *based in* a grid cube $\square$ of scale $k \ge j$ if $Q(X) \subset \square$. By Lemma 49.2(b), every polymer of scale j is based in exactly one grid cube of each scale $k \ge j$. So a sum

over the polymers of scale j is the sum over the grid cubes $\square$ of scale k of the sums over the polymers based in $\square$. For $k = j$, X is based in $\square$ if and only if $Q(X) = \square$.

49.3 Reblocking

For a polymer X of step k let $\bar{X}$ be the union of the LM-cubes that contain a cube of X; equivalently, that meet X, by Lemma 49.2(b). In the notation of Lemma 49.2(c), $\bar{X} = \mathrm{cl}_{k+1} X$. For a family $\{F(X,\cdot)\}$ indexed by the polymers of step k put $(\mathcal{B}F)(Y,\cdot) = \sum_{X:\,\bar{X}=Y} F(X,\cdot)$, and let $\mathcal{B}F$ act on the configurations in the units of step $k+1$ by rescaling. The LM-cubes are the grid cubes of step $k+1$, by (T1) and Lemma 49.2(a), and $d_{LM}(Y)$ in the units of step k is $d_M(Y)$ in those of step $k+1$.

Lemma 49.3 (Reblocking; [15, Lemma 10])**.** *Let $\kappa \geq \kappa_0 + 1$ and $\kappa' = L(\kappa - \kappa_0 - 1)$. For nonnegative numbers $a(X)$ with $a(X) \leq Ae^{-\kappa d_M(X)}$,*

$$\sum_{X:\,\bar{X}=Y} a(X) \leq 3^d K_0 L^d A e^{-\kappa' d_{LM}(Y)}.$$

If $a(X) = 0$ unless $d_M(X) \geq L$, then the same holds with the factor $3^d K_0 L^d e^{-L}$ and the rate $L(\kappa - \kappa_0 - 2)$, for $\kappa \geq \kappa_0 + 2$.

Proof. A minimal tree for X is a tree on the LM-cubes of $\bar{X}$, so $d_M(X) \geq L d_{LM}(\bar{X})$. Then $e^{-\kappa d_M(X)} \leq e^{-(\kappa-\kappa_0)L d_{LM}(Y)} e^{-\kappa_0 d_M(X)}$. Every X with $\bar{X} = Y$ contains a grid cube of Y, so (49.2) gives $\sum_{\bar{X}=Y} e^{-\kappa_0 d_M(X)} \leq K_0 L^d |Y|_{LM} \leq 3^d K_0 L^d e^{d_{LM}(Y)}$. (Dimock writes 9 in place of 3^d in [15, (132)], which is inconsistent with his (122).) For the last statement use $e^{-(\kappa_0+1)d_M} \leq e^{-L} e^{-\kappa_0 d_M}$, with $\kappa - 1$ in place of κ. This is [15, (130)–(132), (139)]. □ Details

Remark 49.4 (Pieces with margins). The pieces of Sections 47 and 48 are indexed by polymers Z and depend on the data on $Z^{(r')}$, with $r' < M/8 < LM$. For them use the enlarged closure $\hat{Z}$, the union of the LM-cubes meeting $Z^{(r')}$. Then $L d_{LM}(\hat{Z}) \leq d_M(Z) + C|\hat{Z}|_{LM}$, with C depending only on d, since $r'/(LM) < 1/(8L)$. Indeed a minimal tree of Z meets every cube of Z, hence every LM-cube of $\bar{Z}$. Every other LM-cube of $\hat{Z}$ contains a point within r' of a cube of Z, so a branch of Euclidean length at most $\sqrt{d}(M + r') \leq 2\sqrt{d}M$ from the tree reaches it, since r' is an ℓ^∞-distance and $r' < M$. One branch per such cube costs at most $2\sqrt{d}/L$ in units of LM. The loss is counted in LM-cubes. With $|\hat{Z}|_{LM} \leq 3^d(1 + d_{LM}(\hat{Z}))$ it is absorbed by the L-gain of Lemma 49.3 once $L \geq C3^d$, at the price of a constant $e^{C\kappa 3^d}$.

49.4 Old polymers at later scales

A generation of scale j (Section 50) is a family of activities indexed by the polymers of scale j. At a later scale k these polymers fall into the following classes. Only the small ones are expanded at scale k (Section 51); the others are bounded through the decay of the generation in the size of the polymer.

Definition 49.5 (Polymer classes)**.** Let $1 \leq j \leq k \leq K$ and $m = k - j$. Let

$$c' = \frac{10 \log L}{\kappa_n - \kappa_0},$$

with κ_n of Remark 49.8(1) and κ_0 of (49.2), and let $m_0 \geq 1$ be the integer of Remark 49.8(5). A polymer X of scale j is, at scale k,

- *recent* if $m < m_0$;
- *small* if $m \geq m_0$ and $d_j(X) \leq c'm$;

- *large* if $m \geq m_0$ and $d_j(X) > c'm$.

It is *wrapping* if it is wrapping at its own scale, i.e. if its diameter, in the units of scale j, is larger than $\frac{1}{2}n_j - 8$ (Section 52.1). For a grid cube $\square$ of scale k, the *small polymers of* $\square$ are the small polymers of scale j based in $\square$.

So, at scale k, every polymer of scale j is exactly one of recent, small and large. The class depends on j and k only through m, and $X + a$ is in the class of X for every translation a that maps the grid of scale j onto itself. Wrapping is a property of X at scale j. By Lemma 49.6(a), a wrapping polymer is large or recent at every scale $k \geq j$.

Lemma 49.6 (Small polymers)**.** *Let $1 \leq j < k \leq K$ with $m = k - j \geq m_0$, and let X be a polymer of scale j with $d_j(X) \leq c'm$, based in the grid cube $\square$ of scale k. In the units of scale j:*

(a) $\operatorname{diam} X \leq M(2 + c'm) \leq \frac{1}{2}n_j - 8$*; so X is not wrapping;*

(b) $X^{(3)} \subset \square^+$.

Proof. (a) By Lemma 52.1(b) at scale j, whose first claim uses no condition on the torus, $\operatorname{diam} X \leq M(2 + d_j(X)) \leq M(2 + c'm)$. By Remark 49.8(5), $L^m \geq 2(2 + c'm) + 1$, so $M(2 + c'm) \leq \frac{1}{2}M(L^m - 1)$. Since $M \mid n_k$, we have $n_j = L^m n_k \geq L^m M$. With $M \geq 64$ (Remark 49.8(4)), Details

$$M(2 + c'm) + 8 \leq \tfrac{1}{2}L^m M - \tfrac{1}{2}M + 8 \leq \tfrac{1}{2}L^m M \leq \tfrac{1}{2}n_j.$$

(b) Let c be the centre of $\square$, and let $x \in X^{(3)}$. There is a point $x' \in X$ with $|x - x'|_\infty \leq 3$. Let q be a point of the base cube $Q(X) \subset \square$. In the units of scale j, Lemma 49.2(a) gives $|q - c|_\infty \leq \frac{1}{2}(L^m M - \eta)$, and $\square^+$ is the set of the points at distance at most $\frac{1}{2}(3L^m M - \eta)$ from c (the description of $\square^+$ above, rescaled by L^m). By (a), $|x' - q|_\infty \leq \frac{1}{2}M(L^m - 1)$. Hence

$$|x - c|_\infty \leq 3 + \tfrac{1}{2}M(L^m - 1) + \tfrac{1}{2}(L^m M - \eta) \leq \tfrac{1}{2}(3L^m M - \eta),$$

since $3 \leq \frac{1}{2}M(L^m + 1)$. So $x \in \square^+$. □

Remark 49.7 (The constants c' and m_0). (a) *Why c' has the factor* 10. An old activity is expanded at a later scale in a gauge chosen for each polymer (Lemma 51.1). That gauge is built from the regularity of scale k on a cube of radius at most 1 in the units of scale k, so the lemma applies only to polymers whose radius $R_X = M(2 + d_j(X)) + 3$ of Section 51 is less than $L^{m/2}$, while the polymers based in a cube of scale k can be of any size. The polynomial factors in its error are absorbed by the decay of the generation. We use the cutoff $d_j(X) \leq c'm$, inside this range, and bound the other polymers through their decay. Let E be a generation of scale j with norm $\|E\|_j$ (Definition 50.1), let $m \geq 1$, and let $\square$ be a grid cube of scale k. Then

$$\sum_{X \text{ based in } \square:\ d_j(X) > c'm} \|E(X)\|\, e^{\frac{1}{2}(\kappa_n - \kappa_0)d_j(X)} \leq K_0 L^{-m} \|E\|_j.$$

Indeed, $\|E(X)\| e^{\frac{1}{2}(\kappa_n - \kappa_0)d_j(X)} \leq \|E\|_j e^{-\kappa_0 d_j(X)} e^{-\frac{1}{2}(\kappa_n - \kappa_0)d_j(X)}$, and the last factor is less than $e^{-\frac{1}{2}(\kappa_n - \kappa_0)c'm} = L^{-5m}$. Each X contains its base cube, one of the L^{4m} grid cubes of scale j in $\square$ (Lemma 49.2(b)), and by (49.2) the sum of $e^{-\kappa_0 d_j(X)}$ over the polymers X that contain a given cube is at most K_0. The factor $e^{\frac{1}{2}(\kappa_n - \kappa_0)d_j(X)}$ kept on the left gives decay at scale k, since $d_j(X) \geq L^m d_k(\mathrm{cl}_k X)$ (Lemma 49.2(c)). So half of the rate $\kappa_n - \kappa_0$ pays for the L^{4m} base cubes and the gain L^{-m}, and the other half is left for the decay at scale k. Hence the factor $10 = 2(4 + 1)$.

(b) *The integer m_0.* It is the least integer $m_0 \geq 1$ such that, for all $m \geq m_0$, $L^m \geq 2(2+c'm)+1$ (used in Lemma 49.6) and the conditions (51.1) of Lemma 51.1 hold with $\Theta^\flat = 4\Theta - 3$ in place of Θ (Section 51.2), with the constant c' above (Remark 49.8(5)). The first condition is implied by (51.1), since $L^m \geq R_m^2 \geq R_m = M(2+c'm)+4$. Each of these conditions bounds a function of m of polynomial growth by a power of L^m, so m_0 is finite. It depends on d, α, G, a, L, κ_n, M and the domain constants, but not on the couplings, the scales j, k or the torus. There is no circularity: c' is fixed before m_0, and the constants in these conditions do not depend on m_0.

49.5 The order of the parameters

Remark 49.8 (Parameters). The parameters of Parts VII–X are chosen in the following order. Each may depend on those chosen before it. The functions p, p_0 on $(0, g_p]$ of Section 49.1, and in Theorem 58.9 the exponent q, are fixed at the outset and depend on no parameter below.

(1) *The rate κ_n.* The activities of the flow are bounded in norms of the form $\sup_X \|\mathcal{E}(X)\| e^{\kappa_n d_M(X)}$, with $\|\cdot\|$ the supremum of Definition 38.3 on the domains $\mathcal{U}^\flat$ of Section 50 (on the domains $\mathcal{U}$ of Remark 38.8 for the raw output of a step); for a generation this is the norm of Definition 50.1. The rate satisfies

$$\kappa_n \geq 2(1 + C_d + \kappa_0),$$

with C_d of Theorem 48.6 for the decay functional d_M, and κ_0 of (49.2).

(2) *The block size.* L is odd, and

$$L \geq C3^d + 3, \qquad L^2 \geq 8(C_* + 1),$$

with C of Remark 49.4 and C_* of Proposition 31.5. The first condition is for the reblocking of pieces with margins, below. The second is the compatibility of the thresholds of consecutive steps (Section 31), given $h_{k+1} \leq 2h_k$. Parts I–VI need only $L \geq 3$. Then $c' = 10 \log L/(\kappa_n - \kappa_0)$ (Definition 49.5).

(3) *The weakening rate.* The rate κ and the loss δ of Part IX, with $\kappa'_w = \kappa - C\delta$ the rate κ' of Corollary 41.6, satisfy

$$\frac{3^{-d}\kappa'_w - 1}{c_\sharp} \geq \kappa_n + \kappa_0 + 2, \qquad \frac{\kappa'_w - 2}{c'_\sharp} - C_d \geq \kappa_n + \kappa_0 + 2, \qquad \kappa'_w - 1 \geq c_\natural \kappa_n + \log(1/c_\flat),$$

with the constants of d_M as a decay functional and $c'_\sharp = \max(c_\sharp, c_\natural)$. The third condition is that of Proposition 48.5. These conditions do not involve L, so κ may be chosen after L and before M (Section 41).

(4) *The scales.* ρ, then M, with $M > 32(L+1)$, M a power of L, and

$$M \geq \max\left(M_0,\, 5L,\, M_1,\, M_1^{(0)},\, M_1',\, 64\right).$$

Here M_0 and $5L$ are the conditions of Theorem 30.3; $M_1 = M_1(\kappa, \delta, C_G, C_{\mathrm{loc}}, C_{\mathfrak{N}}, \gamma_0, \rho, C_v)$ is the largest of the thresholds of Theorem 41.4 at the levels of Section 41, fixed level by level (the paragraph before the levels), and $M_1^{(0)}$ its value for the operators of the first step (Proposition 52.2(c)); M_1' and 64 are the conditions of Section 45, and $M_1' \geq M_{\mathfrak{N}}$ of Proposition 44.11. M_1 is computed with the multiplicative constants of Part IX doubled, and M_1' with θ' replaced by $2\theta'$ (Theorem 44.13); for both, the coercivity constants are the fixed ones of Proposition 46.7, and every decay bound that enters is taken at the fixed rate of the differences there, a quarter of the minimum of the perturbation rate

and the two chosen inverse decay rates in Lemma 13.3; neither depends on δ_c, so the same numbers serve at real and at complex references (Proposition 46.7, step (vii) of the proof). At complex references of size $\delta_c < \delta_P$ the inputs of M_1 change by at most a factor 2, and those of θ' by at most $1+\sigma_{\mathrm{cx}}$ with $(1+\sigma_{\mathrm{cx}})^3 \le 2$. So M_1 and M_1' serve at complex references and do not depend on $\bar\delta$. The constants C_G, C_{loc}, $C_{\mathfrak{N}}$ and γ_0 entering M_1 do not depend on M (Section 41, the paragraph before the levels). Both depend also on the Hölder exponent α, which is fixed at the outset, with $\alpha \le \frac12$ from Section 45 on. The threshold $M_1^{(0)}$ is computed with the bounds of Proposition 52.2(b) and its proof at complex references ($\operatorname{Re}\Delta_0 \ge \frac12 C_{\mathrm{P}}^{-1}$, a norm change of at most $\frac18 C_{\mathrm{P}}^{-1}$, range $2L+2$), which do not depend on $\delta_{P,0}$. Further, $M \ge 16r$, where r is the largest margin of the weakened families used (Section 47, Setting); r does not depend on M, and $M \ge 16r$ is required by the definition of weakenable families (Section 40). Then the scale $M' \ge M_L'$ of Theorem 34.4 and of Proposition 52.2(b). Since L is odd, so is M; the alignment of the grids of different scales uses this (Lemma 49.2).

(5) *The domains.* The cap $\bar\delta$ and the threshold $\varepsilon_{\max}$ of Remark 38.8; then the constants of the complex references, $\delta_P \le \bar\delta$ (Proposition 46.7) and $\delta_{P,0}$ (Proposition 52.2); then $c_a \le cL^{-d-2}$ (Remark 47.14(c)), the domain constants $\delta_1 \le \delta_0$, and then θ_r. All of these are subject to the conditions (DC1)–(DC6) of Remark 38.8; (DC6) bounds θ_r by $\delta_1/(C_{\mathrm{sym}}\delta_0)$. Put $\delta_1^\flat = \frac14\delta_1$ (Section 50). Finally the integer $m_0 \ge 1$ of Definition 49.5: the least integer such that $L^m \ge 2(2+c'm)+1$ and (51.1), with Θ replaced by $\Theta^\flat = 1+(B_5\theta_r+\delta_0^2)/\delta_1^\flat$, hold for all $m \ge m_0$; since $\Theta \le \Theta^\flat$, (51.1) then holds with Θ as well (Remark 49.7(b)); the first condition is implied by (51.1), since $L^m \ge R_m^2$.

(6) *The couplings.* $\gamma \le g_p$, and all couplings lie in $(0,\gamma]$. The bound γ is small depending on all parameters above. In particular $\gamma p(\gamma)$ is an admissible threshold (Section 49.1). The constants of Section 53 are fixed between (5) and (6) (Remark 53.5(i),(ii)). All the conditions on γ in the note are the following:

- (Γ1)–(Γ4) of Remark 53.5. Conditions (Γ1)–(Γ3) are those of the flow (Section 53); they contain the conditions of one step on the coupling, listed below. Condition (Γ4) is used only in Part XI (Proposition 58.6).
- In Theorem 58.9 only, where $p_0(g) = (\log g^{-1})^q$: $\gamma \le e^{-3q}$ and $C_3\gamma^2 p_0(\gamma)^6 \le \frac12\beta_\infty$, with C_3 of Proposition 58.6 and β_∞ of Theorem 57.1.

Each of them is an upper bound on γ by quantities fixed before γ, and it remains true when γ is decreased; for the last one this uses that $g^2p_0(g)^6$ is increasing on $(0, e^{-3q}]$. So all of them hold for γ small enough, and there is no circularity.

The torus satisfies (T0)–(T2) of Section 52.1; its side $n_*(L,M,\rho)$ in (T0) depends on the parameters up to (4).

The rates. The rate conditions in (1)–(3) make every source of the step decay at the rate κ_n after reblocking. Let $a(Z) \ge 0$ be indexed by polymers Z, with data on $Z^{(r')}$, and $a(Z) \le \bar A e^{-\lambda d_M(Z)}$ with $\lambda \ge \kappa_0 + 1$. Every Z with $\hat Z = Y$ contains a grid cube of Y. By Remark 49.4 and the counting in the proof of Lemma 49.3,

$$\sum_{Z:\,\hat Z = Y} a(Z) \le 3^d K_0 L^d e^{C(\lambda-\kappa_0)3^d}\, \bar A\, e^{-[(\lambda-\kappa_0)(L-C3^d)-1]\, d_{LM}(Y)}.$$

For $L \ge C3^d+3$ the rate in brackets is at least $3(\lambda-\kappa_0)-1$. This is at least κ_n if $\lambda \ge \kappa_n+\kappa_0+2$. It is also at least κ_n for $\lambda = \kappa_n - 1 - C_d$, since $\kappa_n \ge 2(1+C_d+\kappa_0)$. The sources are:

- the one-loop pieces, with $|\mathfrak{e}(X,U)| \le Ce^{-(\kappa_2-1)|X|}$ and $\kappa_2 = 3^{-d}\kappa_w'$ (Proposition 41.7(d)). Since $|X| \ge d_M(X)/c_\sharp$, their rate in d_M is at least $(\kappa_2-1)/c_\sharp$;

- the Jacobians $\log \mathsf{j}_c$, each supported on a polymer of bounded size, with at most $C(L, M)$ of them per grid cube; they satisfy the bound for every λ, with $\bar{A}$ depending on λ;
- the pure class, with the rate $(\kappa'_w - 2)/c'_\sharp - C_d$ (Proposition 48.8);
- the mixed class, with the rate $\kappa_n - 1 - C_d$ (Corollary 48.7).

By (3), the first and the third have $\lambda \geq \kappa_n + \kappa_0 + 2$.

The couplings. The conditions of one step on the coupling hold uniformly for couplings in $(0, \gamma]$, with $\tilde{h} = h_k$, $\varepsilon'_1 = h_{k+1}$ and $p_0 = p_{0,k}$, for γ small:

- the thresholds: every h_j is an admissible threshold (Section 49.1), which includes $h_k \leq a_1$ and $(1 + C_J)(C_* + 1)h_{k+1} < \theta_r$ (Remark 38.8, the coupling); $h_{k+1} \leq a_{\text{step}}$; and $h_1 \leq a^{(0)}_{\text{step}}$ of Lemma 52.3. These hold since $h_j \leq \gamma p(\gamma)$;
- (Q1) of Section 48, and $2g_k \leq r_t$ of Proposition 48.8, since $w = C_w L^d C_T g_k p_{0,k}$, $r_t \geq c p_{0,k}^{-3}$ with c fixed before γ, and $g p_0(g)^3 \to 0$;
- (Q3), i.e. $\tau \leq C_P L^d g_k p_{0,k}/\delta_1 < \frac{1}{2}$, since $g p_0(g) \to 0$;
- (Q2), if $g_{k+1} \leq 2g_k$. Then $h_{k+1} \leq 2h_k$ (Section 49.1), so $2(C_* + 1)L^{-2} h_{k+1} \leq \frac{1}{2} h_k$ by (2); and $Cw \leq \frac{1}{2} h_k$, since $2CC_w L^d C_T p_0(g) \leq p(g)$ for $g \leq \gamma$, as $p_0/p \to 0$;
- the hypotheses $\epsilon_{\text{p}} \leq \frac{1}{2}\epsilon_*$ and $\epsilon_{\text{m}} e^{c_\natural(\kappa_n - 1)} \leq \frac{1}{4}\epsilon_*$ of Corollary 48.7. Here $\epsilon_{\text{p}} = C_P g_k p_{0,k}^3$, and $\epsilon_{\text{m}} = C_P \tau \|\mathcal{E}_k\|$ is small if the old activities are bounded by a constant fixed before γ.

With the bound $C_{\text{old}}\mathsf{K}$ on the old activities (Proposition 53.2), these are condition (Γ2) of Remark 53.5, which adds (Γ1), (Γ3) and the constants K, C_{old}, C_β, C_{new} to this list; $g_{k+1} \leq 2g_k$ is Theorem 53.7(c). Along the flow all of them hold at every step (Theorem 53.12(e)).

Three constants fixed in Part VIII. The order above also contains three constants that are fixed inside proofs in Part VIII. The radius δ_* and the constant $C_W = C(1 + 2\delta_*)/\delta_*$ are fixed in the proof of Proposition 37.4(c), depending on d, L, G, α and M. The constant c''_L is fixed in the proof of Lemma 37.5, with $C_{\text{AB}} c''_L \leq c'_L$, $C_{\text{cmp}} c''_L \leq \delta_*$, and $C''_L c''_L \leq \frac{1}{8} c_L$ for a constant C''_L that absorbs C_W; it depends on L and M, and through δ_* and C_W also on d, G and α. All three are fixed after M and before the cap $\bar{\delta}$ of (5): C_W enters the constant C_{cx} of Proposition 46.7 (step (vii) of its proof), hence $\bar{\delta}$, and c''_L enters the condition $\delta_P \leq c''_L$ of that proposition. The conventions paragraph before Proposition 46.7 displays this part of the order. In (6), as everywhere in Sections 49, 52 and 58, γ is the bound on the couplings (Section 49.1), not a decay rate, except in the kernel bound of Lemma 58.4(d). Table 8 in Section G11 of the guide lists the parameters of Parts VII–XI in this order, with their roles.

50 Generations

In four dimensions the effective action of a step is not turned into a single activity of the new scale. As in Bałaban's analysis [18, (0.22)–(0.23)], it is a sum of generations (Remark 49.1). The generation of scale j is created by the step from scale $j - 1$ to scale j. It is a family of activities on the polymers of scale j, bounded on the domains of scale j only. At every later scale k it is evaluated at the background field $U_k(V)$ of that scale, and only this argument changes.

This section defines a generation and its norm (Definition 50.1). The definition does not refer to the flow of the couplings: the couplings enter only through the small-field threshold at which the symmetry of the generation is required (Remark 50.2). Lemma 50.3 shows that a background field of a later scale is a background field of scale j, of a configuration in the

small-field set of scale j. So the symmetry of a generation, a property of its sum at scale j, persists at all later scales. A generation is moreover covariant, as a family, under the translations and reflections of its grid, (G4). The step does not produce covariant families. Section 50.1 averages the family it produces over the symmetries of the grid; this changes neither its sum nor its bound, but it needs a slightly smaller domain. The coefficient β_j of the counterterm $-\beta_j S$ that accompanies a generation in the flow is not part of the definition. It is read off from the generation through its period density, a quadratic form in a constant curvature (Section 50.2). By (G4), the period density of a generation is a multiple β_E of that of the action, with no topological and no anisotropic part (Proposition 50.13).

Setting. The scales are $0 \le j \le K$. The torus and the parameters are as in Sections 52.1 and 49.5, and objects of scale j are read in the units of scale j unless said otherwise. Then $M \mid n_j$, so the polymer grid of scale j is defined (Section 49.2). Moreover $n_j \ge n_K \ge n_*$, and n_* contains the torus conditions of Theorem 30.3, of Proposition 31.5 with $\delta = 1$, and of Part VII for the step from scale $j-1$ to scale j (Section 52.1).

Grid symmetries and the generation domains. In the units of scale j, let $\mathfrak{S}_j^{\mathrm{grid}}$ be the group of the maps $r(x) = Rx + a$ of $\mathbb{T}_\eta$ with $R \in B_4$ and $a \in M\mathbb{Z}^4$. It is finite. It is generated by the translations by the vectors of $M\mathbb{Z}^4$ and by B_4 acting about the centre 0 of a grid cube of scale j, and it contains B_4 acting about the centre $c \in M\mathbb{Z}^4$ of any grid cube, $x \mapsto c + R(x-c)$. It consists of the elements of $\mathfrak{S}_j$ (Section 21) that map the grid cubes of scale j onto grid cubes. Indeed, the grid cube centred at c is the set of the points at ℓ^∞-distance at most $\frac{1}{2}(M-\eta)$ from c (Lemma 49.2(a)), the elements of $\mathfrak{S}_j$ are ℓ^∞-isometries, and for $r(x) = Rx + v$ with $v \in \mathbb{T}^{(j)}$ we have $rc \in M\mathbb{Z}^4$ for all $c \in M\mathbb{Z}^4$ if and only if $v \in M\mathbb{Z}^4$. Every $r \in \mathfrak{S}_j^{\mathrm{grid}}$ extends to an isometry of the continuum torus that maps the closed cubes filled by the grid cubes onto each other. So it maps polymers of scale j onto polymers, and it preserves d_j, the number $|X|_M$ of cubes and the diameter; in particular it maps wrapping polymers onto wrapping polymers, and $(rX)^{(3)} = r(X^{(3)})$. It acts on configurations by $(rU)(b) = U(r^{-1}b)$, with $U(-b) = U(b)^{-1}$, as in Section 21; the translation by a is the map τ_a of (G2) below.

With the domain constants θ_r, δ_1, δ_0 of Remark 38.8 put

$$\delta_1^\flat = \tfrac{1}{4}\delta_1, \qquad \mathcal{U}^\flat(X) = \mathcal{U}(X;\theta_r,\delta_1^\flat,\delta_0) \subset \mathcal{U}(X) = \mathcal{U}(X;\theta_r,\delta_1,\delta_0).$$

Generations are bounded on the domains $\mathcal{U}^\flat(X)$. The step produces families bounded on the domains $\mathcal{U}(X)$, and their averages over $\mathfrak{S}_j^{\mathrm{grid}}$ are bounded on $\mathcal{U}^\flat(X)$ (Section 50.1). The reflections in $\mathfrak{S}_j^{\mathrm{grid}}$ need not map $\mathcal{U}(X)$ into $\mathcal{U}(rX)$, since Definition 38.3 uses forward differences and coordinate-ordered paths, but they map $\mathcal{U}^\flat(X)$ into $\mathcal{U}(rX)$ under a further condition (DC6) on the domain constants (Lemma 50.6 and (50.5)).

Background fields at an admissible threshold. Let h be an admissible threshold (Section 49.1), let $j \le K$, and let $V \in \mathfrak{U}_j(h)$ be real. Since $h \le a_1$, the configuration V has a minimal orbit (Theorem 30.3), a $\mathfrak{G}_0$-orbit of real configurations on $\mathbb{T}_\eta$; we write $U_j(V)$ for any of its elements. By Proposition 31.5 with $\delta = 1$ and Lemma 45.1, $U_j(V)$ has curvature at most $(C_* + 1)h$ and current at most $C_J(C_* + 1)h$, both less than θ_r. So

$$U_j(V) \in \mathcal{U}^\flat(X) \subset \mathcal{U}(X) \qquad \text{for every polymer } X \text{ of scale } j,$$

with real part $U_j(V)$ and $A = 0$ in Definition 38.3; the field $A = 0$ satisfies the bounds of every graded domain. The set $\mathfrak{U}_j(h)$ is invariant under real gauge transformations and under $\mathfrak{S}_j$ (Section 21). Let v be a real gauge transformation on $\mathbb{T}^{(j)}$, and u a real one on $\mathbb{T}_\eta$ with $u|_{\mathbb{T}^{(j)}} = v$. Then $U \mapsto U^u$ maps the minimal orbit of V onto that of V^v. Indeed, it maps

$\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ onto $\mathfrak{B}(V^v) \cap \mathfrak{U}(\theta_0)$, preserves S and maps critical orbits onto critical orbits (the paragraph after Theorem 25.1), and the critical orbit is unique (Theorem 30.3(a)).

Definition 50.1 (Generation)**.** Let $1 \le j \le K$, and let h be an admissible threshold. A *generation of scale j at threshold h* is a family $E = \{E(X,\cdot)\}$, indexed by the polymers X of scale j (Section 49.2), with the following properties, in the units of scale j.

(G1) *(Activities.)* Each $E(X,\cdot)$ is a normalized graded activity at X on $\mathcal{U}^\flat(X) = \mathcal{U}(X;\theta_r,\delta_1^\flat,\delta_0)$ (Definition 38.3), with the domain constants of Remark 38.8 and $\delta_1^\flat = \frac{1}{4}\delta_1$, and $E(X,U)$ is real for real $U \in \mathcal{U}^\flat(X)$. So $E(X,U)$ depends only on U on the bonds of X, it is holomorphic and invariant under real gauge transformations on $\mathcal{U}^\flat(X)$, and $E(X,1) = 0$.

(G2) *(Grid covariance.)* For every $a \in M\mathbb{Z}^4$, every polymer X and every $U \in \mathcal{U}^\flat(X)$,

$$E(X+a,\tau_a U) = E(X,U), \qquad (\tau_a U)(b) = U(b-a).$$

(G3) *(Invariance of the sum.)* The function

$$\Phi_E(V) = \sum_X E(X, U_j(V)), \qquad V \in \mathfrak{U}_j(h),$$

satisfies $\Phi_E(rV) = \Phi_E(V)$ for every $V \in \mathfrak{U}_j(h)$ and every $r \in \mathfrak{S}_j$.

(G4) *(Covariance under the grid symmetries.)* For every $r \in \mathfrak{S}_j^{\mathrm{grid}}$, every polymer X and every $U \in \mathcal{U}^\flat(X)$ with $rU \in \mathcal{U}^\flat(rX)$,

$$E(rX,rU) = E(X,U), \qquad (rU)(b) = U(r^{-1}b).$$

The *norm* of E is

$$\|E\|_j = \sup_X e^{\kappa_n d_j(X)} \|E(X)\|, \qquad \|E(X)\| = \sup_{\mathcal{U}^\flat(X)} |E(X,\cdot)|,$$

with κ_n of Remark 49.8(1) and d_j the tree length of the grid of scale j (Section 49.2).

In (G2), τ_a maps $\mathcal{U}^\flat(X)$ onto $\mathcal{U}^\flat(X+a)$, since translations preserve curvatures, currents, covariant differences and the coordinate-ordered paths of Definition 38.3. So (G2) is the case of the translations in (G4); we keep it separately, since the cell sums of the period density use only (G2) (Lemma 50.12(c)). For the other elements of $\mathfrak{S}_j^{\mathrm{grid}}$ the condition $rU \in \mathcal{U}^\flat(rX)$ in (G4) is not automatic (Lemma 50.6). In (G3) the sum is finite, and Φ_E is well defined by the paragraph before the definition. Each $U_j(V)$ lies in every $\mathcal{U}^\flat(X)$, and $\Phi_E(V)$ does not depend on the element $U_j(V)$ of the minimal orbit, since the orbit is a $\mathfrak{G}_0$-orbit and each $E(X,\cdot)$ is invariant under real gauge transformations. For the same reason $\Phi_E(V^v) = \Phi_E(V)$ for real gauge transformations v on $\mathbb{T}^{(j)}$. By (G1), Φ_E is real. The domains $\mathcal{U}^\flat(X)$ are smaller than the domains $\mathcal{U}(X)$ of the step because of the symmetrization (Section 50.1).

Remark 50.2 (The couplings in the definition). The couplings enter Definition 50.1 only through the threshold h, and only in (G3). In the flow, the generation of scale j is taken at the threshold $h_j = p(g_j)g_j$ of the density of scale j (Section 49.1). Nothing in the definition depends on how the couplings $g_0,\dots,g_j$ are chosen. They may be the couplings of the flow, defined by the recursion $g_{i+1}^{-2} = g_i^{-2} - \beta_{i+1}$, or an arbitrary sequence in $(0,\gamma]$, as for the continuity in $g_0,\dots,g_j$ (Proposition 53.10) required in (F1) of Section 58. The step that produces the generations needs more: (Q2) requires $g_{k+1} \le 2g_k$ (Remark 53.5, consequences), as does Lemma 50.3(c). Continuity in the couplings is therefore to be stated on sequences with $g_{i+1} \le 2g_i$. Since $\mathfrak{U}_j(h') \subset \mathfrak{U}_j(h)$ for $h' \le h$, and both sets are invariant under $\mathfrak{S}_j$, a generation at threshold h is one at every threshold $h' \le h$. The norm does not depend on the threshold.

The following lemma is the input for (G3) at later scales. Part (c) is the only place where the couplings enter.

Lemma 50.3 (Background fields at later scales)**.** *Let $1 \le j \le k \le K$, let h be an admissible threshold, and let $V \in \mathfrak{U}_k(h)$ be a real configuration on $\mathbb{T}^{(k)}$, in the units of scale k. Let $U = U_k(V)$ be an element of its minimal orbit, a configuration on $\mathbb{T}_\eta$, and, for $j \le i \le k$, let $U^{(i)}$ be its average on $\mathbb{T}^{(i)}$ (Definition 21.8). Then, for $j \le i \le k$:*

(a) In the units of scale i, $U^{(i)} \in \mathfrak{U}_i(2(C_*+1)L^{-2(k-i)}h)$ if $i < k$, and $U^{(k)} = V$. In particular $U^{(i)} \in \mathfrak{U}_i(h)$.

(b) U is an element of the minimal orbit of $U^{(i)}$ for the problem of scale i; that is, one may take $U_i(U^{(i)}) = U$.

(c) Let $g_j, \dots, g_k \in (0, \gamma]$ with $g_{i+1} \le 2g_i$ for $j \le i < k$, and let $h = h_k$. Then $2(C_*+1)L^{-2(k-j)}h_k \le h_j$ if $j < k$, and $U^{(j)} \in \mathfrak{U}_j(h_j)$.

(d) Let E be a generation of scale j at an admissible threshold h', with $h' \ge h$ if $j = k$, and $h' \ge 2(C_*+1)L^{-2(k-j)}h$ if $j < k$; under the hypotheses of (c), $h' = h_j$ is such a threshold. Then $U \in \mathcal{U}^\flat(X)$ for every polymer X of scale j, in the units of scale j, and

$$\sum_X E(X, U_k(V)) = \Phi_E(U_k(V)^{(j)}).$$

The left side does not depend on the element $U_k(V)$ of the minimal orbit. As a function of $V \in \mathfrak{U}_k(h)$ it is invariant under real gauge transformations and under $\mathfrak{S}_k$.

Proof. By the Setting, the torus conditions of Theorem 30.3 and Proposition 31.5 hold at every scale $i \le K$, and those of Part VII hold for every step from scale i to scale $i+1 \le K$. Details

(a) Since $U \in \mathfrak{B}(V)$, $U^{(k)} = V$. Let $i < k$. By Proposition 31.5 with $\delta = 1$, which applies since $h \le a_{\delta=1}$, $U \in \mathfrak{U}_0((C_*+1)h)$ in the units of scale k. Since $(C_*+1)h \le c_1$, Proposition 21.12 gives $U^{(i)} \in \mathfrak{U}_i(2(C_*+1)h)$, in the same units. The curvature of a configuration Y on $\mathbb{T}^{(i)}$ is $\delta_i^{-2}\ell(Y(\partial p))$ (Definition 21.4), where the spacing δ_i of $\mathbb{T}^{(i)}$ is L^{i-k} in the units of scale k and 1 in those of scale i. So in the units of scale i the bound is multiplied by $L^{-2(k-i)}$. Finally $2(C_*+1)L^{-2(k-i)} \le 2(C_*+1)L^{-2} \le \frac{1}{4}$, by $L^2 \ge 8(C_*+1)$ (Remark 49.8(2)).

(b) By downward induction on i. For $i = k$ this is the choice of U. Let $j \le i < k$, and suppose that U is an element of the minimal orbit of $U^{(i+1)}$ for the problem of scale $i+1$. Apply Lemma 31.6 to the step from scale i to scale $i+1$, in the units of scale i (so $\mathbb{T} = \mathbb{T}^{(i)}$ and $\mathbb{T}' = \mathbb{T}^{(i+1)}$), with $V' = U^{(i+1)}$, $\varepsilon_1' = h$ and $U' = U$. Its hypotheses hold: V' is real, its curvature in the units of scale $i+1$ is at most h by (a), and h satisfies the conditions on ε_1', since it is admissible. The base configuration of the lemma is $V^0 = U'^{(i)} = U^{(i)}$, and part (b) of the lemma states that U is an element of the minimal orbit of $U^{(i)}$ for the problem of scale i.

(c) By the paragraph on consecutive thresholds in Section 49.1, $h_{i+1} \le 2h_i$ for $j \le i < k$, so $h_k \le 2^m h_j$ with $m = k - j$. The threshold h_k is admissible, since $g_k \in (0, \gamma]$. If $m = 0$, then $U^{(j)} = V \in \mathfrak{U}_j(h_j)$. If $m \ge 1$, then, since $2L^{-2} \le 1$ and $L^2 \ge 8(C_*+1)$,

$$2(C_*+1)L^{-2m}h_k \le 2(C_*+1)(2L^{-2})^m h_j \le 4(C_*+1)L^{-2}h_j \le \tfrac{1}{2}h_j,$$

and (a) gives $U^{(j)} \in \mathfrak{U}_j(\frac{1}{2}h_j) \subset \mathfrak{U}_j(h_j)$.

(d) By (a) and the hypothesis on h', $U^{(j)} \in \mathfrak{U}_j(h')$. Under the hypotheses of (c), h_j is an admissible threshold, since $g_j \in (0, \gamma]$, and it satisfies the hypothesis on h', by the bound of (c). By (b), U is an element of the minimal orbit of $U^{(j)}$. Since h' is admissible, the paragraph before Definition 50.1 gives $U \in \mathcal{U}^\flat(X)$ for every polymer X of scale j, and the definition of Φ_E, with $U_j(U^{(j)}) = U$, gives the identity. Two elements of the minimal orbit of V differ by

a real gauge transformation, and each $E(X,\cdot)$ is invariant under real gauge transformations on $\mathcal{U}^\flat(X)$, which is invariant under them. So the left side does not depend on the element. Let v be a real gauge transformation on $\mathbb{T}^{(k)}$, and u a real one on $\mathbb{T}_\eta$ with $u|_{\mathbb{T}^{(k)}} = v$. By the paragraph before Definition 50.1, at scale k, $V^v \in \mathfrak{U}_k(h)$ and U^u is an element of the minimal orbit of V^v; and $E(X,U^u) = E(X,U)$ for every X. This proves the gauge invariance. Let $r \in \mathfrak{S}_k$. By Theorem 30.3(c), $rV \in \mathfrak{U}_k(h)$, and rU is an element of the minimal orbit of rV. Since $\mathfrak{S}_k \subset \mathfrak{S}_j \subset \mathfrak{S}_i$ for $i \le j$, Lemma 21.10(d), applied to the averagings $\mathfrak{a}_1, \dots, \mathfrak{a}_j$ in the units of scale k, gives $(rU)^{(j)} = rU^{(j)}$. The identity for rV, then (G3) for $U^{(j)} \in \mathfrak{U}_j(h')$ and $r \in \mathfrak{S}_j$, then the identity for V give

$$\sum_X E(X, rU) = \Phi_E\big(rU^{(j)}\big) = \Phi_E\big(U^{(j)}\big) = \sum_X E(X,U). \qquad \square$$

Remark 50.4 (Symmetry of the density at later scales). Let $g_1, \dots, g_k \in (0,\gamma]$ with $g_{i+1} \le 2g_i$ for $i < k$, and let ρ_k have the form of Remark 49.1 on $\mathfrak{U}_k(h_k)$, and vanish outside it, with generations $E^{(j)}$ of scale j at the thresholds h_j, $1 \le j \le k$. Then $\rho_k(rV) = \rho_k(V)$ for every configuration V and every $r \in \mathfrak{S}_k$. Indeed, $\mathfrak{U}_k(h_k)$ is invariant under $\mathfrak{S}_k$, the terms $S(U_k(V))$ are invariant by Theorem 30.3(c), and the sums of the generations by Lemma 50.3(c),(d). Since $\mathfrak{S}_{k+1} \subset \mathfrak{S}_k$, this is the hypothesis $\rho \circ r = \rho$ of the symmetry statement of Theorem 48.9, for every $r \in \mathfrak{S}_{k+1}$.

50.1 Symmetrization

The step produces families of activities whose sum has the symmetry (G3), but not the covariance (G4): single activities need not be covariant (Theorem 48.9). This subsection averages such a family over $\mathfrak{S}_j^{\mathrm{grid}}$. Lemmas 50.5 and 50.6 compare the domain of a polymer with that of its image under an element of $\mathfrak{S}_j^{\mathrm{grid}}$. Definition 50.7 and Lemma 50.8 give the average. Throughout, $1 \le j \le K$, and we use the units of scale j. By Lemma 52.1(a), $n_j \ge 8LM$.

Why the next two lemmas are needed. The symmetrization evaluates the activity of a polymer X at the image of a configuration under a symmetry r of the grid. The graded domains of Definition 38.3 are not invariant under these symmetries. They use forward differences, which a reflection turns into backward ones, and Hölder quotients on the collar $X^{(3)}$, whose pairs of points a symmetry need not map to pairs of the same kind near the outer boundary (Remark 50.9(d)). Lemma 50.6 measures the loss: the gradient budget at most doubles, up to a term $C_{\mathrm{refl}}\theta_r\delta_0'$, and the factor 2 comes from Lemma 50.5(c). Under (DC6) this gives (50.5): a symmetry of the grid maps the domain $\mathcal{U}^\flat(X)$, with $\delta_1^\flat = \frac{1}{4}\delta_1$, into the domain $\mathcal{U}(rX)$, on which the raw output of a step is bounded. This is why the generations are bounded on the domains $\mathcal{U}^\flat$ (the paragraph on (DC6) below).

Lemma 50.5 (Pairs in the collar)**.** *Let X be a polymer of scale j, and let $\Pi \subset \{0,\eta,2\eta\}^4$ be a set of vectors with $0 \in \Pi$. Let $y \ne y'$ be points of $\mathbb{T}_\eta$ with $|y-y'|_\infty \le 6$ such that $y+\Pi$ and $y'+\Pi$ lie in $X^{(3)}$. Then there is a point q of $\mathbb{T}_\eta$ with the following properties.*

(a) For every μ, q_μ lies on the shorter arc of the coordinate circle between y_μ and y'_μ. Hence $|y-q| \le |y-y'|$ and $|q-y'| \le |y-y'|$, for the Euclidean, the ℓ^1- and the ℓ^∞-distance.

(b) $q+\Pi \subset X^{(3)}$.

(c) There are points w, w' of X with $y, q \in \Lambda_3(w)$ and $q, y' \in \Lambda_3(w')$. The cubes $\Lambda_3(w)$ and $\Lambda_3(w')$ lie in $X^{(3)}$.

Part (c) is the point. The pairs (y,q) and (q,y') enter the Hölder quotients of Definition 38.3 on $X^{(3)}$, while the pair (y,y') need not: near the outer boundary of $X^{(3)}$, for instance near a corner at which two cubes of X meet, y and y' need not lie in a common cube $\Lambda_3(w) \subset X^{(3)}$.

Proof. Let $\Gamma_X \subset M\mathbb{Z}^4$ be the set of the centres of the grid cubes of X, and put $R = \frac{1}{2}(M-\eta)+3$. Details
The grid cube centred at c is the set of the points at ℓ^∞-distance at most $\frac{1}{2}(M-\eta)$ from c (Lemma 49.2(a)). So a point x of $\mathbb{T}_\eta$ lies in $X^{(3)}$ if and only if $|x-c|_\infty \le R$ for some $c \in \Gamma_X$. For a point t of a coordinate circle, let $K(t)$ be the set of the points $k \in M\mathbb{Z}$ of the circle at distance at most R from t. Then

$$x \in X^{(3)} \iff \Gamma_X \cap \big(K(x_1)\times K(x_2)\times K(x_3)\times K(x_4)\big) \neq \emptyset. \tag{50.1}$$

The arcs $\{t : |t-k| \le R\}$, $k \in M\mathbb{Z}$, have length $2R = M+6-\eta < 2M$, since $M \ge 64$ (Remark 49.8(4)), and there are $n_j/M \ge 8L$ of them on the circle. So every point lies in one or in two of them. The points that lie in two of them form the *bands* $[k+M-R, k+R]$, $k \in M\mathbb{Z}$; a band is an arc of length $2R-M = 6-\eta$, and two bands are at distance $2M-2R = M+\eta-6$. Off the bands K is constant along any arc that does not meet a band, and on a band it is the union of its values just before and just after the band. The end points of the bands are points of the fine lattice, since $\frac{1}{2}(M-\eta)$ and 3 are multiples of η.

One coordinate. Let a, a' be points of the circle at distance at most 6, let $[s,l]$ be the shorter arc between them, and let $J = [s, l+2\eta]$. We claim that there is a point $b \in [s,l]$ of the fine lattice with

$$K(b+\pi) \supset K(a+\pi) \cup K(a'+\pi) \qquad \text{for } \pi \in \{0,\eta,2\eta\}. \tag{50.2}$$

Since $a+\pi$ and $a'+\pi$ lie in $[s+\pi, l+\pi] \subset J$, it suffices that $K(b+\pi) \supset K(t)$ for all $t \in [s+\pi, l+\pi]$. The arc J has length at most $6+2\eta < M+\eta-6$, since $M \ge 64$ (Remark 49.8(4)), so it meets at most one band. If it meets none, or if it lies in a band, K is constant on J, and we take $b = s$. Otherwise let $B = [\beta, \beta']$ be the band that J meets, so $\beta' = \beta+6-\eta$, and at least one end point of B lies in J. If $\beta \in J$ and $\beta' \notin J$, then K increases (for the inclusion) along J, and we take $b = l$. If $\beta' \in J$ and $\beta \notin J$, then K decreases along J, and we take $b = s$. If $\beta, \beta' \in J$, we take $b = \beta$. Then $s \le \beta \le \beta' - 2\eta \le l$, since $\beta' \le l+2\eta$ and $6-\eta \ge 2\eta$. Moreover $b+\pi \in B$ for $\pi \le 2\eta$, so $K(b+\pi)$ is the largest value of K on J. In all cases (50.2) holds.

The point q. Let q_μ be the point b of the first step for $a = y_\mu$ and $a' = y'_\mu$. This gives (a). Let $\pi \in \Pi$. By (50.2), $K(q_\mu+\pi_\mu) \supset K(y_\mu+\pi_\mu)$ for every μ, and $y+\pi \in X^{(3)}$; so (50.1) gives $q+\pi \in X^{(3)}$, which is (b). For (c), choose $c \in \Gamma_X$ with $c_\mu \in K(y_\mu)$ for all μ; it exists, since $y \in X^{(3)}$. By (50.2) with $\pi = 0$, also $c_\mu \in K(q_\mu)$, so $|y-c|_\infty \le R$ and $|q-c|_\infty \le R$. In each coordinate the arcs $[\max(y_\mu,q_\mu)-3, \min(y_\mu,q_\mu)+3]$ and $[c_\mu - \frac{1}{2}(M-\eta), c_\mu + \frac{1}{2}(M-\eta)]$ then meet, since $|y_\mu - q_\mu| \le 6$, and they meet in a point of the fine lattice. So there is a point w of the grid cube centred at c, hence of X, with $|w-y|_\infty \le 3$ and $|w-q|_\infty \le 3$. The same argument for q and y', with $c' \in \Gamma_X$ chosen for y', gives w'. Since $w, w' \in X$, the cubes $\Lambda_3(w)$ and $\Lambda_3(w')$ lie in $X^{(3)}$. □

Lemma 50.6 (Grid symmetries and graded domains)**.** *There is a constant $C_{\mathrm{refl}} \ge 1$, depending only on d, with the following property. Let $r \in \mathfrak{S}_j^{\mathrm{grid}}$, let X be a polymer of scale j, and let $\theta_r \le 1$ and $0 < \delta'_1 \le \delta'_0 \le 1$. Let $U = e^{i\eta A}U_r$ on the bonds of $X^{(3)}$ be a representation as in Definition 38.3 for $\mathcal{U}(X;\theta_r,\delta'_1,\delta'_0)$, and let rA be the field on the bonds of $(rX)^{(3)}$ given by $(rA)(b) = A(r^{-1}b)$, with $A(-b) = -R(U_r(b))^{-1}A(b)$ on reversed bonds. Then $rU = e^{i\eta\, rA}\, rU_r$ on the bonds of $(rX)^{(3)}$, the configuration rU_r is real with $|F_{rU_r}| < \theta_r$ and $|J_{rU_r}| < \theta_r$ everywhere, $|rA| = |A|$, and, with ∇ taken with respect to U_r and to rU_r,*

$$|\nabla(rA)|_{(rX)^{(3)}} + [\nabla(rA)]_{\alpha,(rX)^{(3)}} \le |\nabla A|_{X^{(3)}} + 2[\nabla A]_{\alpha,X^{(3)}} + \tfrac{1}{2}C_{\mathrm{refl}}\,\theta_r\big(|\nabla A|_{X^{(3)}} + |A|\big). \tag{50.3}$$

Here $|A|$ is the maximum over the bonds of $X^{(3)}$. Consequently

$$r\,\mathcal{U}(X;\theta_r,\delta_1',\delta_0') \subset \mathcal{U}(rX;\theta_r,2\delta_1'+C_{\mathrm{refl}}\theta_r\delta_0',\delta_0').$$

Proof. Write $r(x)=Rx+a$, with $Re_\mu = s_\mu e_{\pi(\mu)}$ for a permutation π and signs s_μ. For each μ put $\mu'=\pi^{-1}(\mu)$ and $\sigma_\mu = s_{\mu'}$, so that $R^{-1}e_\mu = \sigma_\mu e_{\mu'}$. We use $\|R(e^{\mathrm{i}Y})-1\| \le 2e^2|Y|$ for $Y\in\mathfrak{g}$ with $|Y|\le 1$, which follows from $R(e^{\mathrm{i}Y}) = e^{\mathrm{i}\,\mathrm{ad}\,Y}$ and $|[Y,Z]|\le 2|Y||Z|$ (Section 12). In this proof C denotes constants that depend only on d. Details

The real part and the sup norm. The configuration rU_r is real, and $|F_{rU_r}(p)| = |F_{U_r}(r^{-1}p)|$ and $|J_{rU_r}|_\infty = |J_{U_r}|_\infty$, as in the proof of Theorem 25.1(h). With the convention on reversed bonds, $r(e^{\mathrm{i}\eta A}U_r) = e^{\mathrm{i}\eta\, rA}\, rU_r$ bondwise, as there. Since $R(U_r(b))$ is unitary, $|rA|=|A|$. The map r maps the bonds of $X^{(3)}$ onto those of $(rX)^{(3)}$.

Differences. Fix μ,ν and a point x such that the difference $D_\nu(rA)_\mu(x)$, taken with respect to rU_r, involves only points of $(rX)^{(3)}$. These points are $x+\eta(\epsilon_\mu e_\mu+\epsilon_\nu e_\nu)$, $\epsilon\in\{0,1\}^2$, if $\mu\neq\nu$, and x, $x+\eta e_\mu$, $x+2\eta e_\mu$ if $\mu=\nu$. Put $z=r^{-1}x$, $\kappa=\eta\mathbf{1}_{\{\sigma_\mu=-1\}}e_{\mu'}+\eta\mathbf{1}_{\{\sigma_\nu=-1\}}e_{\nu'}$ and $y=z-\kappa$. Then r^{-1} maps the points involved in $D_\nu(rA)_\mu(x)$ onto those involved in $D_{\nu'}A_{\mu'}(y)$; so the latter lie in $X^{(3)}$, and z is one of them. With $(D_\nu\lambda)(x)=\eta^{-1}\big(R(U(x,x+\eta e_\nu))\lambda(x+\eta e_\nu)-\lambda(x)\big)$ (Section 12), the definitions of rA and rU_r give

$$D_\nu(rA)_\mu(x) = \sigma_\mu\sigma_\nu\, T_x\, D_{\nu'}A_{\mu'}(y) + e_{\mu\nu}(x). \tag{50.4}$$

Here $T_x = R(U_r(\gamma_x))^{-1}$ is the transport along a path γ_x of at most two bonds from y to z through the points involved, and $e_{\mu\nu}(x)=0$ unless $\sigma_\mu=-1$ and $\mu\neq\nu$. We give the case $\sigma_\mu=-1$, $\sigma_\nu=1$; the case $\sigma_\mu=\sigma_\nu=-1$ is the same, with $y=z-\eta e_{\mu'}-\eta e_{\nu'}$, and in the other cases the computation gives (50.4) with $e_{\mu\nu}(x)=0$. Let $\sigma_\mu=-1$ and $\sigma_\nu=1$, so $y=z-\eta e_{\mu'}$, and put $\hat y = y+\eta e_{\nu'}$ and $\hat z = z+\eta e_{\nu'}$. Then $(rA)_\mu(x) = -R(U_r(y,z))^{-1}A_{\mu'}(y)$, $(rA)_\mu(x+\eta e_\nu) = -R(U_r(\hat y,\hat z))^{-1}A_{\mu'}(\hat y)$ and $(rU_r)(x,x+\eta e_\nu) = U_r(z,\hat z)$. So

$$\begin{aligned} D_\nu(rA)_\mu(x) &= -R(U_r(y,z))^{-1}D_{\nu'}A_{\mu'}(y) \\ &\quad -\eta^{-1}\Big(R(U_r(z,\hat z))R(U_r(\hat y,\hat z))^{-1} - R(U_r(y,z))^{-1}R(U_r(y,\hat y))\Big)A_{\mu'}(\hat y). \end{aligned}$$

The two products in the last term are the transports from $\hat y$ to z along the two pairs of sides of the plaquette p with corners $y,z,\hat y,\hat z$. They differ by a factor $R(H)$ with H conjugate to $U_r(\partial p)^{\pm1} = e^{\pm\mathrm{i}\eta^2 F_{U_r}(p)}$, and $|\eta^2 F_{U_r}(p)|<\eta^2\theta_r\le 1$, since $|F_{U_r}|<\theta_r$ (Definition 38.3). Hence

$$|e_{\mu\nu}(x)| \le \eta^{-1}\|R(U_r(\partial p))-1\|\,|A| \le 2e^2\eta\,\theta_r|A| \le 2e^2\theta_r|A|.$$

As x runs over the points at which $D_\nu(rA)_\mu$ involves only points of $(rX)^{(3)}$, y runs over the points at which $D_{\nu'}A_{\mu'}$ involves only points of $X^{(3)}$. So (50.4) gives $|\nabla(rA)|_{(rX)^{(3)}} \le |\nabla A|_{X^{(3)}} + 2e^2\theta_r|A|$.

Closed paths. Let γ be a closed lattice path of N bonds in $\mathbb{T}_\eta$ with $N\eta<n_j$. Then $\|R(U_r(\gamma))-1\| \le e^2N^2\eta^2\theta_r$. Indeed, γ lifts to a closed path in $\eta\mathbb{Z}^4$. Exchanging two consecutive bonds of γ along different axes for the other two sides of the plaquette they span changes $R(U_r(\gamma))$ by at most $\|R(U_r(\partial p))-1\|\le 2e^2\eta^2\theta_r$, since all transports are unitary. At most $\frac12 N^2$ such exchanges sort the bonds of γ by their axes. Since γ is closed, its displacement along each axis vanishes; so the sorted path consists of closed paths along the coordinate lines through its initial point. Their transports are 1, since such a path is reduced to the empty path by removing backtracks.

Hölder quotients. Let $x\neq\tilde x$ be points of a cube $\Lambda_3(\bar z)\subset(rX)^{(3)}$, and let μ,ν be such that $D_\nu(rA)_\mu$ involves only points of $(rX)^{(3)}$ at x and at $\tilde x$. Then $|x-\tilde x|_\infty\le 6$, and $\eta\le|x-\tilde x|\le 12$. Let z,y and $\tilde z,\tilde y$ be the points defined above for x and for $\tilde x$. The vector κ is the same for both, so $\tilde y-y = R^{-1}(\tilde x-x)$ and $|\tilde y-y|=|\tilde x-x|$. Let Π be the set of the vectors from y to

the points involved in $D_{\nu'}A_{\mu'}(y)$. Then $\Pi \subset \{0,\eta,2\eta\}^4$, $0 \in \Pi$, and $y+\Pi$, $\tilde y+\Pi$ lie in $X^{(3)}$. Let q, w, w' be the points of Lemma 50.5 for y and $\tilde y$. Write $D(\cdot) = D_{\nu'}A_{\mu'}(\cdot)$ and $\Gamma = \Gamma_{x,\tilde x}$, and let γ be the path from y to $\tilde y$ that runs along γ_x to z, along $r^{-1}\Gamma$ to $\tilde z$, and back along $\gamma_{\tilde x}$ to $\tilde y$. Since $(rU_r)(\Gamma) = U_r(r^{-1}\Gamma)$, (50.4) gives

$$\begin{aligned} D_\nu(rA)_\mu(x) - R\big((rU_r)(\Gamma)\big)D_\nu(rA)_\mu(\tilde x) &= \sigma_\mu\sigma_\nu\, T_x\big(D(y) - R(U_r(\gamma))D(\tilde y)\big) \\ &\quad + e_{\mu\nu}(x) - R\big((rU_r)(\Gamma)\big)e_{\mu\nu}(\tilde x), \end{aligned}$$

and

$$\begin{aligned} D(y) - R(U_r(\gamma))D(\tilde y) &= \big(D(y) - R(U_r(\Gamma_{y,q}))D(q)\big) + R(U_r(\Gamma_{y,q}))\big(D(q) - R(U_r(\Gamma_{q,\tilde y}))D(\tilde y)\big) \\ &\quad + \big(R(U_r(\Gamma_{y,q}))R(U_r(\Gamma_{q,\tilde y})) - R(U_r(\gamma))\big)D(\tilde y). \end{aligned}$$

By Lemma 50.5(b), $D(q)$ involves only points of $X^{(3)}$, and by (c) the pairs (y,q) and $(q,\tilde y)$ lie in cubes $\Lambda_3(w), \Lambda_3(w') \subset X^{(3)}$. So the first two terms are at most $[\nabla A]_{\alpha,X^{(3)}}\big(|y-q|^\alpha + |q-\tilde y|^\alpha\big) \le 2[\nabla A]_{\alpha,X^{(3)}}|x-\tilde x|^\alpha$; a term with $q = y$ or $q = \tilde y$ vanishes. By Lemma 50.5(a), the paths $\Gamma_{y,q}$, $\Gamma_{q,\tilde y}$ and $r^{-1}\Gamma$ have length at most $|x-\tilde x|_1 \le 4|x-\tilde x|$, and γ_x, $\gamma_{\tilde x}$ have length at most 2η. So the closed path $\Gamma_{y,q}\Gamma_{q,\tilde y}\gamma^{-1}$ has N bonds with $N\eta \le 12|x-\tilde x| + 4\eta \le 16|x-\tilde x| < n_j$, and the bound for closed paths makes the last term at most $256e^2\theta_r|x-\tilde x|^2|\nabla A|_{X^{(3)}} \le 256{\cdot}144\,e^2\theta_r|x-\tilde x|^\alpha|\nabla A|_{X^{(3)}}$. Finally $|e_{\mu\nu}(x)| + |e_{\mu\nu}(\tilde x)| \le 4e^2\eta\theta_r|A| \le 4e^2\theta_r|A||x-\tilde x|^\alpha$, since $\eta \le \eta^\alpha \le |x-\tilde x|^\alpha$. Dividing by $|x-\tilde x|^\alpha$ and taking the supremum over the pairs and the components,

$$[\nabla(rA)]_{\alpha,(rX)^{(3)}} \le 2[\nabla A]_{\alpha,X^{(3)}} + C\theta_r\big(|\nabla A|_{X^{(3)}} + |A|\big).$$

Together with the bound on the differences this gives (50.3). If $|\nabla A|_{X^{(3)}} + [\nabla A]_{\alpha,X^{(3)}} < \delta_1' \le \delta_0'$ and $|A| < \delta_0'$, the right side of (50.3) is less than $2\delta_1' + C_{\mathrm{refl}}\theta_r\delta_0'$. This proves the inclusion. □

The condition (DC6). By (DC6) of Remark 38.8, $C_{\mathrm{sym}}\theta_r\delta_0 \le \delta_1$, with $C_{\mathrm{sym}} = 2C_{\mathrm{refl}}$ and C_{refl} of Lemma 50.6. Like (DC1)–(DC3), it is an upper bound on θ_r, here in terms of δ_1/δ_0, with a constant that depends only on d. So it is compatible with them, and with the order of Remark 49.8(5), where θ_r is chosen after δ_1 and δ_0. The quantities that depend on the domain constants through θ_r/δ_1, such as Θ of Lemma 51.1, only decrease. Under (DC6), Lemma 50.6 with $\delta_1' = \delta_1^\flat$ and $\delta_0' = \delta_0$ gives $2\delta_1^\flat + C_{\mathrm{refl}}\theta_r\delta_0 \le \frac12\delta_1 + \frac12\delta_1$, so

$$r\,\mathcal{U}^\flat(X) \subset \mathcal{U}(rX) \qquad \text{for every } r \in \mathfrak{S}_j^{\mathrm{grid}} \text{ and every polymer } X \text{ of scale } j. \tag{50.5}$$

The term $e_{\mu\nu}$ in the proof of Lemma 50.6 is of order $\eta\theta_r|A|$, and it does not vanish at curved real parts. Since $|A|$ is bounded only by δ_0, it has to be absorbed in the budget δ_1 of the differences; this is what (DC6) does. The factor 2 in front of the Hölder quotient comes from Lemma 50.5, and it is the reason for the factor $\frac14$ in $\delta_1^\flat$.

Definition 50.7 (Raw generations and symmetrization)**.** Let h be an admissible threshold. A *raw generation of scale j at threshold h* is a family $E = \{E(X,\cdot)\}$, indexed by the polymers X of scale j, with (G1) on the domains $\mathcal{U}(X)$ in place of $\mathcal{U}^\flat(X)$, and with (G3). That is, each $E(X,\cdot)$ is a normalized graded activity at X on $\mathcal{U}(X) = \mathcal{U}(X;\theta_r,\delta_1,\delta_0)$, real at real configurations, and the function Φ_E of (G3) is invariant under $\mathfrak{S}_j$ on $\mathfrak{U}_j(h)$. Its norm is

$$\|E\|_j^{\mathrm{raw}} = \sup_X e^{\kappa_n d_j(X)} \sup_{\mathcal{U}(X)} |E(X,\cdot)|.$$

Assume (DC6). The *symmetrization* of E is the family $E^{\mathrm{sym}} = \{E^{\mathrm{sym}}(X,\cdot)\}$ with

$$E^{\mathrm{sym}}(X,U) = \frac{1}{|\mathfrak{S}_j^{\mathrm{grid}}|}\sum_{r\in\mathfrak{S}_j^{\mathrm{grid}}} E(r^{-1}X, r^{-1}U), \qquad U \in \mathcal{U}^\flat(X).$$

By (50.5), applied to r^{-1}, $r^{-1}U \in \mathcal{U}(r^{-1}X)$ for $U \in \mathcal{U}^\flat(X)$, so every term is defined. A raw generation need not have (G2). The one-loop part E'_{ol} of a raw new generation, the sum of the pieces $\mathfrak{e}(X)$ and of the terms $\log \mathfrak{j}_c$ in Definition 53.3 (Section 58), does have it, by Proposition 41.14(b),(c) and Remark 41.15(a); for the outputs $E^\#$ of the cluster expansion it is not proved. In the flow, the new activities of the step from scale $j-1$ to scale j, normalized and reblocked, form a raw generation (Lemma 53.4, Theorem 53.7(b)); (G3) for them is the symmetry statement of Theorem 48.9, whose hypothesis is Remark 50.4.

Lemma 50.8 (Symmetrization)**.** *Assume (DC6). Let h be an admissible threshold, and let E be a raw generation of scale j at threshold h.*

(a) (Generation.) *E^{sym} is a generation of scale j at threshold h: it has (G1)–(G4). Moreover $\|E^{\mathrm{sym}}\|_j \le \|E\|_j^{\mathrm{raw}}$.*

(b) (The sum at scale j.) *For every $V \in \mathfrak{U}_j(h)$,*

$$\sum_X E^{\mathrm{sym}}(X, U_j(V)) = \sum_X E(X, U_j(V));$$

that is, $\Phi_{E^{\mathrm{sym}}} = \Phi_E$.

(c) (Later scales.) *Let $j \le k \le K$, and let h'' be an admissible threshold with $h \ge h''$ if $j = k$, and $h \ge 2(C_* + 1)L^{-2(k-j)}h''$ if $j < k$. Let $V \in \mathfrak{U}_k(h'')$ be a real configuration, in the units of scale k, and $U = U_k(V)$ an element of its minimal orbit. Then $U \in \mathcal{U}^\flat(X)$ for every polymer X of scale j, in the units of scale j, and*

$$\sum_X E^{\mathrm{sym}}(X, U) = \sum_X E(X, U) = \Phi_E(U^{(j)}).$$

Under the hypotheses of Lemma 50.3(c), $h = h_j$ and $h'' = h_k$ are such thresholds.

(d) (Complex points.) *Let $j \le k \le K$, let $\mathcal{W}$ be a finite-dimensional real vector space, let $\mathcal{D}$ be a connected open subset of its complexification, and let $B \mapsto U_B$ be a holomorphic map from $\mathcal{D}$ into the complex configurations on $\mathbb{T}_\eta$ such that $U_B \in \mathcal{U}^\flat(X)$ for every $B \in \mathcal{D}$ and every polymer X of scale j. Suppose that $\mathcal{D} \cap \mathcal{W}$ contains a nonempty open subset $\mathcal{D}_{\mathbb{R}}$ of $\mathcal{W}$ such that, for $B \in \mathcal{D}_{\mathbb{R}}$, U_B is an element of the minimal orbit of a real configuration V_B as in (c). Then*

$$\sum_X E^{\mathrm{sym}}(X, U_B) = \sum_X E(X, U_B) \qquad \text{for every } B \in \mathcal{D}.$$

Proof. (a) *(G1).* Fix X and $r \in \mathfrak{S}_j^{\mathrm{grid}}$. Since $(r^{-1}U)(b) = U(rb)$, and rb is a bond of X or the reverse of one for every bond b of $r^{-1}X$, the map $U|_X \mapsto (r^{-1}U)|_{r^{-1}X}$ is holomorphic. By (50.5), it maps $\{U|_X : U \in \mathcal{U}^\flat(X)\}$ into $\{U'|_{r^{-1}X} : U' \in \mathcal{U}(r^{-1}X)\}$, on a neighbourhood of which $E(r^{-1}X, \cdot)$ is holomorphic. So every term of $E^{\mathrm{sym}}(X, \cdot)$ is a function of $U|_X$, holomorphic on a neighbourhood of $\{U|_X : U \in \mathcal{U}^\flat(X)\}$, and so is the finite average. For a real gauge transformation u, $r^{-1}(U^u) = (r^{-1}U)^{u \circ r}$, and $u \circ r$ is a real gauge transformation. Since $\mathcal{U}^\flat(X)$ is invariant under real gauge transformations and $E(r^{-1}X, \cdot)$ is invariant on $\mathcal{U}(r^{-1}X)$, $E^{\mathrm{sym}}(X, U^u) = E^{\mathrm{sym}}(X, U)$ for $U \in \mathcal{U}^\flat(X)$. Further $r^{-1}1 = 1$, so $E^{\mathrm{sym}}(X, 1) = 0$, and $r^{-1}U$ is real for real U. Finally $d_j(r^{-1}X) = d_j(X)$, so Details

$$|E^{\mathrm{sym}}(X, U)| \le \max_r \sup_{\mathcal{U}(r^{-1}X)} |E(r^{-1}X, \cdot)| \le \|E\|_j^{\mathrm{raw}} e^{-\kappa_n d_j(X)} \qquad (U \in \mathcal{U}^\flat(X)).$$

This gives (G1) and the bound on the norm.

(G4) and (G2). Let $s \in \mathfrak{S}_j^{\mathrm{grid}}$, and let $U \in \mathcal{U}^\flat(X)$ with $sU \in \mathcal{U}^\flat(sX)$. Substituting $r = st$,

$$E^{\mathrm{sym}}(sX, sU) = \frac{1}{|\mathfrak{S}_j^{\mathrm{grid}}|} \sum_t E\big(t^{-1}s^{-1}sX, t^{-1}s^{-1}sU\big) = E^{\mathrm{sym}}(X, U).$$

For a translation $s = \tau_a$, $a \in M\mathbb{Z}^4$, the condition $sU \in \mathcal{U}^\flat(sX)$ holds for all $U \in \mathcal{U}^\flat(X)$, and this is (G2). (G3) follows from (b).

(b) Let $V \in \mathfrak{U}_j(h)$, $U = U_j(V)$ and $r \in \mathfrak{S}_j^{\mathrm{grid}}$. Since $\mathfrak{S}_j^{\mathrm{grid}} \subset \mathfrak{S}_j$, Theorem 30.3(c) gives $r^{-1}V \in \mathfrak{U}_j(h)$, and $r^{-1}U$ is an element of the minimal orbit of $r^{-1}V$. The configurations U and $r^{-1}U$ lie in every domain $\mathcal{U}^\flat(X)$ (the paragraph before Definition 50.1). The map $X \mapsto r^{-1}X$ is a bijection of the polymers of scale j. So, by (G3) for E,

$$\sum_X E(r^{-1}X, r^{-1}U) = \sum_X E(X, r^{-1}U) = \Phi_E(r^{-1}V) = \Phi_E(V).$$

Averaging over r gives $\sum_X E^{\mathrm{sym}}(X, U) = \Phi_E(V) = \sum_X E(X, U)$. So $\Phi_{E^{\mathrm{sym}}} = \Phi_E$, which is invariant under $\mathfrak{S}_j$: this is (G3) for E^{sym}.

(c) By Lemma 50.3(a) and the hypothesis on h, $U^{(j)} \in \mathfrak{U}_j(h)$, and by Lemma 50.3(b), U is an element of the minimal orbit of $U^{(j)}$ for the problem of scale j. Since h is admissible, the paragraph before Definition 50.1 gives $U \in \mathcal{U}^\flat(X) \subset \mathcal{U}(X)$ for every X. By the definition of Φ_E, with $U_j(U^{(j)}) = U$, $\sum_X E(X, U) = \Phi_E(U^{(j)})$, and likewise for E^{sym}. Now (b) gives the claim. The last sentence is Lemma 50.3(c).

(d) Each $E(X, \cdot)$ and each $E^{\mathrm{sym}}(X, \cdot)$ is holomorphic on a neighbourhood of $\{U|_X : U \in \mathcal{U}^\flat(X)\}$, the map $B \mapsto U_B|_X$ is holomorphic, and the sums are finite. So both sides are holomorphic on $\mathcal{D}$. By (c) they agree on $\mathcal{D}_{\mathbb{R}}$, a nonempty open subset of the totally real subspace $\mathcal{W}$ of maximal dimension. Since $\mathcal{D}$ is connected, the identity theorem gives the claim, as in the proof of Lemma 38.4. □

Remark 50.9 (The symmetrization in the flow). (a) *The density does not change.* The density of a step depends on a generation only through its sums at the background fields of small-field configurations (Remark 49.1). By Lemma 50.8(b),(c), these are the sums of the raw generation. So the flow may carry E^{sym} in place of E from the creation scale on; only the decomposition into activities changes. In a later step, Theorem 48.9 evaluates the old activities one at a time at the composed configurations $e^{\mathrm{i}\eta A_*(s;\hat{Z}(Z))}U'$ of Proposition 47.10. For $s \neq 1$ these are not background fields, even for real Z, and no identity between the sums of E and E^{sym} is claimed there. None is needed. The values of the old activities at $s \neq 1$, at complex values of the class parameter λ of Corollary 48.7, and at complex references enter Sections 47–48 only through bounds on each activity separately (Propositions 47.10 and 48.5, Corollary 48.7); for E^{sym} these bounds follow from $\|E^{\mathrm{sym}}\|_j \leq \|E\|_j^{\mathrm{raw}}$. The identities that involve the density (Proposition 48.3, Theorem 48.9, Remark 47.11, the normalization at $U' = 1$) are asserted only at real critical references and real ξ. The step is the integral of $e^{\delta\mathcal{E}(\xi)}$, and the localization reproduces $\delta\mathcal{E}(\xi)$ at $s = 1$ (Section 48.2). At a real critical reference and for real ξ, put $B = \hat{Z}(Z)$. The composed configuration at $s = 1$ is $U_B = e^{\mathrm{i}\eta A_*(B)}U'$ (Theorem 45.5(c)). Its real gauge transform $U_B^{w_B}$, $w_B = \mathsf{w}_k(e^{\mathrm{i}\eta A_*(B)})$, is an element of the minimal orbit of $e^{\mathrm{i}B}W$ (Theorem 25.1(c), Corollary 30.5, Theorem 30.3(a)), at which Lemma 50.8(c) applies. The transformation w_B need not lie in $\mathfrak{G}_0$. The domains $\mathcal{U}^\flat(X)$ and the activities of E and E^{sym} are invariant under real gauge transformations, so their sums agree also at U_B. So the step gives the same density with E and with E^{sym}. The new activities it produces depend on which family is carried; the flow uses those produced from the symmetrized families.

(b) *Complex fields at $s = 1$.* This is not needed for the flow, by (a); it shows that the sums of E and E^{sym} agree also at complex points with $s = 1$. Let U' be a real critical reference of the step from scale $k \geq j$ to scale $k+1$, with base $W = U'^{(k)}$, and put $U_B = e^{\mathrm{i}\eta A_*(1;B)}U'$ for coarse fields B on the bonds of $\mathbb{T}^{(k)}$. For real B with $|B|_\infty \leq \min(w, R_P, \beta_V)$, $A_*(1;B) = A_*(B)$ (Theorem 45.5(c)), so $U_B^{w_B}$, with the real gauge transformation $w_B = \mathsf{w}_k(e^{\mathrm{i}\eta A_*(B)})$ of $\mathbb{T}_\eta$, is an element of the critical orbit for $e^{\mathrm{i}B}W$ (Corollary 30.5, Theorem 25.1(c)). Moreover $e^{\mathrm{i}B}W \in \mathfrak{U}_k(h_k)$ by (Q2), as in the proof of Proposition 48.3, so this orbit is its minimal orbit (Theorem 30.3(a)). The transformation w_B need not lie in $\mathfrak{G}_0$. Put $\hat{V}_B = U_B^{(k)}$, the transform of $e^{\mathrm{i}B}W$ by the restriction of w_B^{-1} to $\mathbb{T}^{(k)}$. It is real, and $\hat{V}_B \in \mathfrak{U}_k(h_k)$, since the curvature is gauge invariant. By Theorem 25.1(g) and Theorem 30.3(a), U_B is an element of the minimal orbit of $\hat{V}_B$. For complex B with $|B|_\infty < R_P$ and $C_{\mathrm{conv}}C_P|B|_\infty < \delta_1^\flat$, U_B has real part U' and a field bounded as in Definition 38.3 with the budgets $\delta_1^\flat$ and δ_0 (Theorem 45.5(a),(b) and Lemma 47.9); so it lies in the domains $\mathcal{U}^\flat$ of scale k, and, by Remark 38.8(i), in the domains $\mathcal{U}^\flat(X)$ of scale j. On this polydisc, and under the hypotheses of Lemma 50.3(c) on the couplings, Lemma 50.8(d), with $V_B = \hat{V}_B$ for real B, shows that the sums of E and E^{sym} agree at U_B, at complex B as well.

(c) *The slack.* The newest generation, of scale j, is an old activity of the step from scale j to scale $j+1$, on the domains $\mathcal{U}^\flat(X)$ of scale j. The reference is a point of the domain $\mathcal{U}$ of scale $j+1$, and on $X^{(3)}$ its representation has the budgets $L^{-2}\delta_1$ and $L^{-1}\delta_0$ (Remark 38.8(i)). So Proposition 47.10 applies with $\delta_1^\flat$ in place of δ_1, and

$$\min\left(\delta_1^\flat - L^{-2}\delta_1,\ \delta_0 - L^{-1}\delta_0\right) \geq \left(\tfrac{1}{4} - \tfrac{1}{9}\right)\delta_1 \geq \tfrac{1}{8}\delta_1,$$

since $L \geq 3$ and $\delta_1 \leq \delta_0$. This replaces the slack $\frac{1}{2}\delta_1$ of the domains $\mathcal{U}$ by $\frac{1}{8}\delta_1$; it multiplies τ by 4, and (Q3) remains a condition $\tau \leq C_P L^d g_k p_{0,k}/\delta_1 < \frac{1}{2}$ with a larger C_P. At later scales $k > j$ the generation enters the step through the activities $\Psi_\square$ and E_Y^{ne} of Section 51, which are bounded on the domains $\mathcal{U}^\flat$ of scale k; so the slack is again $\frac{1}{8}\delta_1$. The single activities $E(X,\cdot)$ are defined on the whole domain $\mathcal{U}$ of scale k, since its points lie in $\mathcal{U}(X; L^{-2(k-j)}\theta_r, L^{-2(k-j)}\delta_1, L^{-(k-j)}\delta_0) \subset \mathcal{U}^\flat(X)$ (Remark 38.8(i)).

(d) *Why the domain shrinks.* Definition 38.3 uses forward differences and coordinate-ordered paths, and a reflection turns forward differences into backward ones. At a curved real part a backward difference is a transported forward difference at a shifted point, up to the plaquette term $e_{\mu\nu}$ of Lemma 50.6; and the shifted points of a cube Λ_3 of $(rX)^{(3)}$ need not lie in a cube Λ_3 of $X^{(3)}$ near the outer boundary, which costs the factor 2 of Lemma 50.5. A symmetrized Hölder seminorm in Definition 38.3 would remove the factor 2, but not the plaquette term.

50.2 The period density

The coefficient of the counterterm of a generation is read off from its response to a constant curvature. For each polymer this response is a quadratic form c^X in the curvature. Summed over a period of the grid, with the weights $1/|X|$, it gives the period density $\bar{T}_E$. By (G4) the period density is invariant under B_4, and the B_4-invariant forms are the multiples of the density of the action (Lemma 50.10). The comparison of the sum of a generation over the small polymers of a cube of a later scale with this multiple of the action of the cube is Section 51.

Curvature data and invariant forms. A *curvature datum* $F = (F_{\mu\nu})_{\mu<\nu}$ in $\mathfrak{g}^c$ is extended antisymmetrically, $F_{\nu\mu} = -F_{\mu\nu}$ and $F_{\mu\mu} = 0$, and $|F| = \max_{\mu,\nu}|F_{\mu\nu}|$. For $R \in B_4$ put

$R \cdot F = RFR^{\mathsf T}$, i.e. $(R\cdot F)_{\mu\nu} = \sum_{\kappa,\lambda} R_{\mu\kappa}R_{\nu\lambda}F_{\kappa\lambda}$, and for $v \in G$ let vFv^{-1} be the curvature datum $(vF_{\mu\nu}v^{-1})$. With the bilinear form $\operatorname{tr}(XY)$ on $\mathfrak g^c$ (Section 12) put

$$\operatorname{Iso}(F) = \sum_{\mu<\nu} \operatorname{tr}\left(F_{\mu\nu}^2\right), \qquad \operatorname{Pf}(F) = \operatorname{tr}\left(F_{12}F_{34} - F_{13}F_{24} + F_{14}F_{23}\right).$$

A *quadratic form* on the curvature data is a homogeneous polynomial of degree 2 in their complex coordinates. Let $B_4^+ = \{R \in B_4 : \det R = 1\}$.

Lemma 50.10 (Invariant quadratic forms)**.** *Let G be compact, connected and simple, and let $\mathsf q$ be a quadratic form on the curvature data in $\mathfrak g^c$ with $\mathsf q(vFv^{-1}) = \mathsf q(F)$ for all $v \in G$.*

(a) If $\mathsf q(R\cdot F) = \mathsf q(F)$ for all $R \in B_4$, then $\mathsf q = \beta \operatorname{Iso}$ for a unique $\beta \in \mathbb C$.

(b) If $\mathsf q(R\cdot F) = \mathsf q(F)$ for all $R \in B_4^+$, then $\mathsf q = \beta\operatorname{Iso} + \tau \operatorname{Pf}$ for unique $\beta, \tau \in \mathbb C$. Both forms are invariant under B_4^+ and under conjugation, and $\operatorname{Pf}(R\cdot F) = (\det R)\operatorname{Pf}(F)$ for $R \in B_4$.

(c) If $\mathsf q$ is real on the curvature data in $\mathfrak g$, then β and τ are real.

Proof. Let $\mathsf b$ be the symmetric bilinear form with $\mathsf b(F,F) = \mathsf q(F)$. Write $F = \sum_{\mu<\nu} e_{\mu\nu}\otimes F_{\mu\nu}$, with the basis $e_{\mu\nu} = e_\mu \wedge e_\nu$ of $\Lambda^2\mathbb C^4$; then $R\cdot F$ is the action of $\Lambda^2 R$ on the first factor, $Re_{\mu\nu} = Re_\mu \wedge Re_\nu$. Since G is connected, invariance under conjugation by G is invariance under $\operatorname{ad}\mathfrak g$, hence under $\operatorname{ad}\mathfrak g^c$, as $\mathsf b$ is complex bilinear. So, for basis vectors e, e' of $\Lambda^2\mathbb C^4$, $(X,Y) \mapsto \mathsf b(e\otimes X, e'\otimes Y)$ is an invariant bilinear form on $\mathfrak g^c$. Since $\mathfrak g$ is compact and simple, $\mathfrak g^c$ is simple, and its adjoint representation is irreducible. The form $\operatorname{tr}(XY)$ is invariant, and it is nondegenerate on $\mathfrak g^c$, since $\operatorname{tr}(X^2) = |X|^2 > 0$ for $0 \neq X \in \mathfrak g$. Through it, an invariant bilinear form corresponds to an endomorphism of $\mathfrak g^c$ that commutes with $\operatorname{ad}\mathfrak g^c$, which is a multiple of the identity by Schur's lemma. Hence $\mathsf b(e\otimes X, e'\otimes Y) = Q(e,e')\operatorname{tr}(XY)$ with a bilinear form Q on $\Lambda^2\mathbb C^4$, symmetric since $\mathsf b$ is, and Details

$$\mathsf q(F) = \sum_{\mu<\nu}\sum_{\kappa<\lambda} Q(e_{\mu\nu}, e_{\kappa\lambda}) \operatorname{tr}\left(F_{\mu\nu}F_{\kappa\lambda}\right).$$

So $\mathsf q$ is invariant under R if and only if $Q(Rw, Rw') = Q(w,w')$ for all w, w'. The form Iso corresponds to the identity I, and Pf to the form P with $P(e_{12},e_{34}) = P(e_{14},e_{23}) = \frac12$, $P(e_{13},e_{24}) = -\frac12$, and $P = 0$ on the other pairs of basis vectors.

Sign changes. For $\epsilon \in \{\pm1\}^4$ the diagonal matrix D_ϵ maps $e_{\mu\nu}$ to $\epsilon_\mu\epsilon_\nu e_{\mu\nu}$, so invariance under D_ϵ gives $Q(e_{\mu\nu},e_{\kappa\lambda}) = \epsilon_\mu\epsilon_\nu\epsilon_\kappa\epsilon_\lambda Q(e_{\mu\nu},e_{\kappa\lambda})$. Let $\{\mu,\nu\} \neq \{\kappa,\lambda\}$. If the two pairs share one index σ, let ϱ be an index of one of them that is not in the other, and let $\epsilon_\sigma = \epsilon_\varrho = -1$, $\epsilon = 1$ otherwise. Then $\det D_\epsilon = 1$ and the product of the signs is -1; so $Q(e_{\mu\nu},e_{\kappa\lambda}) = 0$, already under B_4^+. If the pairs are disjoint, the product is $\det D_\epsilon$. Under B_4, a single sign change gives $Q(e_{\mu\nu},e_{\kappa\lambda}) = 0$. Under B_4^+ three entries remain: $p_1 = Q(e_{12},e_{34})$, $p_2 = Q(e_{13},e_{24})$ and $p_3 = Q(e_{14},e_{23})$.

Permutations. A permutation matrix maps $e_{\mu\nu}$ to $\pm e_{\pi(\mu)\pi(\nu)}$, with the indices in either order. Composed with a sign change if its determinant is -1, it lies in B_4^+ and acts on the pairs in the same way, up to signs. Since S_4 acts transitively on the pairs, the diagonal entries $Q(e_{\mu\nu},e_{\mu\nu})$ are all equal, to β say. Under B_4 this gives $Q = \beta I$, which is (a). Under B_4^+, the even permutation matrix with $e_1\mapsto e_1$, $e_2\mapsto e_3$, $e_3\mapsto e_4$, $e_4\mapsto e_2$ maps the pair (e_{12},e_{34}) to $(e_{13},-e_{24})$ and (e_{13},e_{24}) to $(e_{14},-e_{23})$. So $p_1 = -p_2$ and $p_2 = -p_3$, and $Q = \beta I + 2p_1P$, i.e. $\mathsf q = \beta\operatorname{Iso} + \tau\operatorname{Pf}$ with $\tau = 2p_1$.

The two forms. Both are invariant under conjugation. Iso is invariant under B_4, since every element of B_4 maps the basis $(e_{\mu\nu})$ onto itself up to signs. Under a sign change D_ϵ, every term of Pf gets the factor $\epsilon_1\epsilon_2\epsilon_3\epsilon_4 = \det D_\epsilon$. Under the transposition of e_1 and e_2, $(F_{12},F_{34},F_{13},F_{24},F_{14},F_{23})$ becomes $(-F_{12},F_{34},F_{23},F_{14},F_{24},F_{13})$, and Pf changes sign, since

$\mathrm{tr}(XY) = \mathrm{tr}(YX)$; by symmetry the same holds for every transposition. Since B_4 is generated by the sign changes and the transpositions, $\mathrm{Pf}(R \cdot F) = (\det R)\,\mathrm{Pf}(F)$. For the curvature datum with the single entry $F_{12} = X \in \mathfrak{g}$, $X \neq 0$, $\mathrm{Iso}(F) = \mathrm{tr}\, X^2 > 0$ and $\mathrm{Pf}(F) = 0$; for $F_{12} = F_{34} = X$ and the other entries 0, $\mathrm{Iso}(F) = 2\,\mathrm{tr}\, X^2$ and $\mathrm{Pf}(F) = \mathrm{tr}\, X^2$. This gives the uniqueness in (a) and (b), and, evaluating $\mathfrak{q}$ at these two real curvature data, (c). □

Lifts and linear fields. Let X be a polymer of scale j that is not wrapping. As in the construction of the base cubes (Section 49.2), the projection to a coordinate circle of the union of the closed cubes of the continuum filled by the cubes of X is an arc. Let ℓ be its length. The points of the fine lattice in this arc are coordinates of points of X, and the two of them nearest to its ends are at distance $\ell - \eta$ along the arc. If $\ell - \eta > \frac{1}{2}n_j$, two of these points are at distance at least $\frac{1}{2}(n_j - \eta)$ on the circle; if $\ell - \eta \le \frac{1}{2}n_j$, the two end points are at distance $\ell - \eta$ on the circle. Since $\mathrm{diam}\, X \le \frac{1}{2}n_j - 8$, the first case does not occur, and $\ell \le \frac{1}{2}n_j - 8 + \eta$. The four arcs, enlarged by 3 at both ends, have length less than $\frac{1}{2}n_j$, and their product contains $X^{(3)}$. This product is the injective image of a box of $\mathbb{R}^4$ under the projection $\mathbb{R}^4 \to \mathbb{R}^4/n_j\mathbb{Z}^4$, and we call the preimage of $X^{(3)}$ in this box a *lift* of $X^{(3)}$. Lifts are unique up to translations by $n_j\mathbb{Z}^4$, the bonds of $X^{(3)}$ lift to bonds of $\eta\mathbb{Z}^4$, and the ℓ^∞-distance of two points of $X^{(3)}$ is that of their lifts. By Lemma 52.1(b), whose first claim uses no condition on the torus, the points of X are at mutual distance at most $M(2 + d_j(X))$. So every point of a lift of $X^{(3)}$ is at ℓ^∞-distance at most

$$D_X = M\big(2 + d_j(X)\big) + 3 \le 3M\big(1 + d_j(X)\big)$$

from the lift of any point of X.

For a curvature datum F and a point $x_0 \in \mathbb{R}^4$, the *linear field* of F with origin x_0 is the field on the oriented bonds of $\eta\mathbb{Z}^4$

$$A_F(b) = \tfrac{1}{2}\,(m_b - x_0)^{\mathsf{T}} F\, t_b = \tfrac{1}{2}\sum_{\mu,\nu}(m_b - x_0)_\mu F_{\mu\nu}(t_b)_\nu,$$

where m_b is the midpoint of b and $t_b \in \{\pm e_1, \dots, \pm e_4\}$ its direction; so $A_F(-b) = -A_F(b)$. For $b = (b_-, b_- + \eta e_\nu)$, $A_F(b) = \frac{1}{2}\sum_\mu F_{\mu\nu}(b_- - x_0)_\mu$, since $F_{\nu\nu} = 0$; for $x_0 = x_X$ this is the field $A_{\mathrm{lin}}[F]$ of Section 51. Its differences are constant, $\partial_\mu A_{F,\nu} = \frac{1}{2}F_{\mu\nu}$. Let $\tilde r(x) = Rx + \tilde a$ be an affine map of $\mathbb{R}^4$ with $R \in B_4$. The bond $\tilde r^{-1}b$ has midpoint $R^{-1}(m_b - \tilde r x_0) + x_0$ and direction $R^{-1}t_b$, and $R^{-1} = R^{\mathsf{T}}$; so the field $(\tilde r A_F)(b) = A_F(\tilde r^{-1}b)$ is

$$\tilde r A_F = A_{R\cdot F} \qquad \text{with the origin } \tilde r x_0. \tag{50.6}$$

For a lift of $X^{(3)}$, an origin x_0 and $t \in \mathbb{C}$, let $V^X_{t,F}$ be the configuration equal to $e^{\mathrm{i}\eta t A_F}$ on the bonds of $X^{(3)}$, read in the lift, and to 1 on the other bonds.

Definition 50.11 (Period density)**.** Let $E = \{E(X,\cdot)\}$ be a family of normalized graded activities $E(X,\cdot)$ at the polymers X of scale j on the domains $\mathcal{U}^\flat(X)$, real at real configurations; for instance a generation of scale j, or a raw generation (Definition 50.7). For a polymer X of scale j that is not wrapping and a curvature datum F in $\mathfrak{g}^c$ put

$$c^X(F) = \frac{1}{2}\,\frac{d^2}{dt^2}\,E(X, V^X_{t,F})\Big|_{t=0},$$

for a lift of $X^{(3)}$ and an origin x_0. The *period density* of E is

$$\bar T_E(F) = M^{-4}\sum_{X \supset Q}\frac{c^X(F)}{|X|},$$

the sum running over the polymers $X \supset Q$ of scale j that are not wrapping, where Q is a grid cube of scale j and $|X| = |X|_M$ is the number of grid cubes of X. The *period density of the action* is

$$\bar{T}_S(F) = n_j^{-4} \sum_p c^p(F), \qquad c^p(F) = \frac{1}{2} \frac{d^2}{dt^2} s_p(V^p_{t,F})\Big|_{t=0},$$

the sum running over the plaquettes p of $\mathbb{T}_\eta$, where $V^p_{t,F}$ equals $e^{\mathrm{i}\eta t A_F}$ on the bonds of p, in coordinates of a neighbourhood of p, and 1 elsewhere. If $\bar{T}_E = \beta \bar{T}_S$ for a number β, we write $\beta_E = \beta$.

By Lemma 50.12 below, $V^X_{t,F} \in \mathcal{U}^\flat(X)$ for small $|t|$, $c^X(F)$ depends only on X and F, and $\bar{T}_E$ does not depend on Q if E has (G2). The weights $1/|X|$ make the reflections about the centre of Q permute the terms of the sum (proof of Proposition 50.13); the base cube of a polymer is not covariant under them. With (G2), the sum of c^X over the polymers with base cube Q is the same, $M^4 \bar{T}_E$ (Section 51). The factor n_j^{-4} is the inverse of the volume of the torus in the units of scale j, so $\bar{T}_S$ is the density per unit volume; by Lemma 50.12(c), $\bar{T}_E$ is such a density as well.

Lemma 50.12 (The period density)**.** *Let G be semisimple, and let E be as in Definition 50.11.*

(a) Let X be a polymer of scale j that is not wrapping, and $F \neq 0$ a curvature datum in $\mathfrak{g}^c$. Then $V^X_{t,F} \in \mathcal{U}^\flat(X)$ for small $|t|$, and for $|t| < \delta_1^\flat/(dD_X|F|)$ if the origin x_0 is the lift of a point of X. The number $c^X(F)$ does not depend on the lift of $X^{(3)}$ or on the origin x_0. It is a quadratic form in F, $c^X(vFv^{-1}) = c^X(F)$ for $v \in G$, and

$$|c^X(F)| \leq 9d^2M^2(1 + d_j(X))^2(\delta_1^\flat)^{-2} \, \|E(X)\| \, |F|^2, \qquad \|E(X)\| = \sup_{\mathcal{U}^\flat(X)} |E(X, \cdot)|.$$

(b) If E has (G2), then $c^{X+a} = c^X$ for $a \in M\mathbb{Z}^4$. If E has (G4), then $c^{rX}(R \cdot F) = c^X(F)$ for every $r \in \mathfrak{S}_j^{\mathrm{grid}}$, with R the linear part of r.

(c) Let $\|E\|_j = \sup_X e^{\kappa_n d_j(X)} \|E(X)\| < \infty$. The sum defining $\bar{T}_E(F)$ converges absolutely, uniformly in the torus, and

$$|\bar{T}_E(F)| \leq 9d^2K_0M^{-2}(\delta_1^\flat)^{-2} \, \|E\|_j \, |F|^2,$$

with K_0 of (49.2). If E has (G2), then $\bar{T}_E$ does not depend on Q, and

$$\bar{T}_E(F) = n_j^{-4} \sum_X c^X(F),$$

the sum running over all polymers of scale j that are not wrapping.

(d) For $p = p_{\mu\nu}(x)$, $\mu < \nu$, $c^p(F) = \frac{1}{2}\eta^4 \operatorname{tr}(F^2_{\mu\nu})$. Hence

$$\bar{T}_S(F) = \tfrac{1}{2} \sum_{\mu<\nu} \operatorname{tr}\,(F^2_{\mu\nu}) = \tfrac{1}{2} \operatorname{Iso}(F).$$

Let $\mathcal{P}(X)$, X a polymer of scale j, be sets of plaquettes such that every plaquette lies in exactly one of them, $\mathcal{P}(X) = \emptyset$ for wrapping X, and the bonds of the plaquettes of $\mathcal{P}(X)$ are bonds of X. If the family $E(X, \cdot) = \sum_{p \in \mathcal{P}(X)} s_p$ is as in Definition 50.11 and has (G2), then $\bar{T}_E = \bar{T}_S$.

Proof. (a) For a field B on the bonds of a lift of $X^{(3)}$, let V_B equal $e^{\mathrm{i}\eta B}$ on the bonds of $X^{(3)}$, read in the lift, and 1 elsewhere. With the real part 1 and the field B in Definition 38.3, Details

$V_B \in \mathcal{U}^\flat(X)$ if $|B| < \delta_0$ and $|\partial B| + [\partial B]_\alpha < \delta_1^\flat$ on $X^{(3)}$. This holds for $B = tA_F$ with $|t|$ small, since A_F is bounded on the finite lift. If x_0 is the lift of a point of X and $|t| < \delta_1^\flat/(dD_X|F|)$, then $|tA_F| \le \frac{1}{2}(d-1)D_X|F||t| < \delta_1^\flat \le \delta_0$ and $|\partial(tA_F)| = \frac{1}{2}|t||F| < \delta_1^\flat$, and the Hölder quotients vanish. This gives the first claim. So $\Phi_X(B) = E(X, V_B)$ is holomorphic for small B, and $c^X(F) = \frac{1}{2}d^2\Phi_X(0)[A_F, A_F]$ is a quadratic form in F, since A_F is linear in F.

As in Step 3 of the proof of Lemma 38.5, $d\Phi_X(0) = 0$, since G is semisimple: by real gauge invariance under constant gauge transformations $v \in G$, the gradient of Φ_X at 0 is $\mathrm{Ad}(G)$-invariant at every bond, and for semisimple G the only such vector is 0. Let λ be a small real scalar field on $\mathbb{T}_\eta$ and $u = e^{\mathrm{i}\lambda}$. On the bonds of $X^{(3)}$, $V_B^u = V_{B^\lambda}$, where $e^{\mathrm{i}\eta B^\lambda(b)} = e^{\mathrm{i}\lambda(b_-)}e^{\mathrm{i}\eta B(b)}e^{-\mathrm{i}\lambda(b_+)}$, and $B^\lambda = B - \partial\lambda + O(|B||\lambda| + |\lambda|^2)$ is analytic in (B, λ), with $\partial\lambda(b) = \eta^{-1}(\lambda(b_+) - \lambda(b_-))$. Since $E(X, \cdot)$ depends only on the bonds of X and is invariant under real gauge transformations on $\mathcal{U}^\flat(X)$, $\Phi_X(B^\lambda) = \Phi_X(B)$ for small B and λ. Since $d\Phi_X(0) = 0$, the terms of second order in (B, λ) give $d^2\Phi_X(0)[B-\partial\lambda, B-\partial\lambda] = d^2\Phi_X(0)[B, B]$. Replacing B by sB and comparing the terms linear in s, $d^2\Phi_X(0)[B, \partial\lambda] = 0$ for all B and all real λ, hence, by linearity, for complex λ. Changing the origin from x_0 to $x_0 + v$ adds to A_F the field $-\frac{1}{2}v^\mathsf{T}Ft_b = \partial\lambda$ with $\lambda(x) = -\frac{1}{2}v^\mathsf{T}Fx$ on the lift. Hence

$$\tfrac{1}{2}d^2\Phi_X(0)[A_F + \partial\lambda, A_F + \partial\lambda] = \tfrac{1}{2}d^2\Phi_X(0)[A_F, A_F],$$

and $c^X(F)$ does not depend on the origin. Two lifts differ by a translation by a vector of $n_j\mathbb{Z}^4$, which amounts to a change of the origin. For constant $v \in G$, $V^X_{t,vFv^{-1}} = (V^X_{t,F})^v$, since $A_{vFv^{-1}} = vA_Fv^{-1}$ and the configuration 1 is invariant; so real gauge invariance gives $c^X(vFv^{-1}) = c^X(F)$. For the bound, take x_0 the lift of a point of X. Then $f(t) = E(X, V^X_{t,F})$ is holomorphic for $|t| < t_X = \delta_1^\flat/(dD_X|F|)$ and bounded there by $\|E(X)\|$, and $c^X(F) = \frac{1}{2}f''(0)$. Cauchy's estimate gives $|c^X(F)| \le \|E(X)\|t_X^{-2} = d^2D_X^2(\delta_1^\flat)^{-2}\|E(X)\||F|^2$, and $D_X \le 3M(1 + d_j(X))$.

(b) Let $a \in M\mathbb{Z}^4$. Translating the lift and the origin by a, $\tau_a V^X_{t,F} = V^{X+a}_{t,F}$, and both configurations lie in the domains $\mathcal{U}^\flat$ for small $|t|$, by (a). So (G2) gives $E(X + a, V^{X+a}_{t,F}) = E(X, V^X_{t,F})$, and $c^{X+a} = c^X$. Let $r \in \mathfrak{S}_j^{\mathrm{grid}}$, and let $\tilde{r}$ be the affine map of $\mathbb{R}^4$ that lifts r and maps the given lift of $X^{(3)}$ onto a lift of $(rX)^{(3)} = r(X^{(3)})$. The configuration $rV^X_{t,F}$ has real part 1, and on reversed bonds its field is $-A_F$; so (50.6) gives $rV^X_{t,F} = V^{rX}_{t,R\cdot F}$, for this lift and the origin $\tilde{r}x_0$. Since $|R \cdot F| = |F|$, both configurations lie in the domains $\mathcal{U}^\flat$ for small $|t|$, by (a). So (G4) gives $c^{rX}(R \cdot F) = c^X(F)$.

(c) By the bound of (a), with $\|E(X)\| \le \|E\|_j e^{-\kappa_n d_j(X)}$ and $|X| \ge 1$,

$$\sum_{X \supset Q} \frac{|c^X(F)|}{|X|} \le 9d^2M^2(\delta_1^\flat)^{-2}\|E\|_j|F|^2 \sum_{X\supset Q} \big(1 + d_j(X)\big)^2 e^{-\kappa_n d_j(X)}.$$

By Remark 49.8(1), $\kappa_n - \kappa_0 \ge \kappa_0 + 2 \ge 3$, and $(1+t)^2e^{-3t} \le 1$ for $t \ge 0$. So the last sum is at most $\sum_{X\supset Q} e^{-\kappa_0 d_j(X)} \le K_0$, by (49.2), uniformly in the torus. This gives the bound. If E has (G2), the translation by $a \in M\mathbb{Z}^4$ maps the polymers $X \supset Q$ that are not wrapping bijectively onto those that contain $Q + a$, and preserves $|X|$; with (b), the sum for $Q + a$ equals that for Q. Since the translations by $M\mathbb{Z}^4$ act transitively on the grid cubes, $\bar{T}_E$ does not depend on Q. There are $(n_j/M)^4$ grid cubes, and every polymer has $|X|$ of them; so

$$\sum_X c^X(F) = \sum_X \sum_{Q\subset X} \frac{c^X(F)}{|X|} = \sum_Q \sum_{X\supset Q} \frac{c^X(F)}{|X|} = \Big(\frac{n_j}{M}\Big)^4 M^4\, \bar{T}_E(F),$$

all sums running over polymers that are not wrapping.

(d) Let $p = p_{\mu\nu}(x)$, with the bonds $b_1, \dots, b_4$ of Definition 21.4 on the fine lattice, and $a_l = A_F(b_l)$. Since the real part is 1, $V^p_{t,F}(\partial p) = e^{\mathrm{i}\eta t a_1}e^{\mathrm{i}\eta t a_2}e^{-\mathrm{i}\eta t a_3}e^{-\mathrm{i}\eta t a_4}$. Here $a_2 - a_4 =$

$\eta\partial_\mu A_{F,\nu}(x) = \frac12\eta F_{\mu\nu}$ and $a_1 - a_3 = -\eta\partial_\nu A_{F,\mu}(x) = \frac12\eta F_{\mu\nu}$, so $a_1+a_2-a_3-a_4 = \eta F_{\mu\nu}$, whatever the origin. By the Baker–Campbell–Hausdorff formula, $V^p_{t,F}(\partial p) = e^{\mathrm{i}Y(t)}$ with $Y(t) = t\eta^2 F_{\mu\nu} + O(t^2)$. Since $s_p(U) = \mathsf{N} - \frac12\operatorname{tr}(U(\partial p) + U(\partial p)^{-1})$ (Section 23), $s_p(V^p_{t,F}) = \operatorname{tr}(1-\cos Y(t)) = \frac12\operatorname{tr}(Y(t)^2) + O(t^4) = \frac12 t^2\eta^4\operatorname{tr}(F^2_{\mu\nu}) + O(t^3)$. So $c^p(F) = \frac12\eta^4\operatorname{tr}(F^2_{\mu\nu})$. There are $(n_j/\eta)^4$ plaquettes in each plane, which gives $\bar T_S$. For the last claim, $E(X, V^X_{t,F}) = \sum_{p\in\mathcal{P}(X)} s_p(V^X_{t,F})$, and on the bonds of $p\in\mathcal{P}(X)$ the configuration $V^X_{t,F}$ is $e^{\mathrm{i}\eta t A_F}$ in coordinates of a neighbourhood of p; so $c^X_E = \sum_{p\in\mathcal{P}(X)} c^p$. By (c), $\bar T_E = n_j^{-4}\sum_X\sum_{p\in\mathcal{P}(X)} c^p = n_j^{-4}\sum_p c^p = \bar T_S$. □

Proposition 50.13 (Isotropy of the period density)**.** *Let G be compact, connected and simple, and let E be a generation of scale j (Definition 50.1); property (G3) is not used.*

(a) There is a unique number β_E with $\bar T_E = \beta_E \bar T_S$, that is,

$$\bar T_E(F) = \tfrac12\beta_E\sum_{\mu<\nu}\operatorname{tr}(F^2_{\mu\nu}) \qquad \text{for all curvature data } F \text{ in } \mathfrak{g}^c.$$

It is real. So the period density of E has no topological part, a multiple of Pf, and no anisotropic part.

(b) $|\beta_E| \le C_\beta\|E\|_j$, with $C_\beta = 18d^2K_0M^{-2}(\delta_1^\flat)^{-2}$.

(c) Let $E = E'^{\mathrm{sym}}$ for a raw generation E' (Definition 50.7), and put $\bar T^{\mathrm{vol}}_{E'}(F) = n_j^{-4}\sum_X c^X_{E'}(F)$, the sum running over the polymers that are not wrapping. Then

$$\bar T_E(F) = \frac{1}{|B_4|}\sum_{R\in B_4}\bar T^{\mathrm{vol}}_{E'}(R\cdot F), \qquad \beta_E = \frac{1}{3\dim\mathfrak{g}}\sum_{\mu<\nu}\sum_a \bar T^{\mathrm{vol}}_{E'}(F^{\mu\nu,a}),$$

where (T_a) is a basis of $\mathfrak{g}$ with $\operatorname{tr}(T_aT_b) = \delta_{ab}$, and $F^{\mu\nu,a}$ is the curvature datum with the single entry $F_{\mu\nu} = T_a$.

Proof. (a) By (G2) and Lemma 50.12(c), $\bar T_E(F) = n_j^{-4}\sum_X c^X(F)$, a quadratic form in F, invariant under conjugation by Lemma 50.12(a). Let $R\in B_4$, and let $r(x) = Rx$, an element of $\mathfrak{S}^{\mathrm{grid}}_j$. It maps the polymers that are not wrapping bijectively onto themselves, and $c^{rX}(R\cdot F) = c^X(F)$ by (G4) and Lemma 50.12(b). So Details

$$\bar T_E(R\cdot F) = n_j^{-4}\sum_X c^{rX}(R\cdot F) = n_j^{-4}\sum_X c^X(F) = \bar T_E(F).$$

The same follows from the definition with the weights $1/|X|$, for the grid cube Q centred at 0: r maps the polymers $X\supset Q$ onto themselves, since $rQ = Q$, and preserves $|X|$. Lemma 50.10(a) gives $\bar T_E = \beta'\,\mathrm{Iso}$ for a unique β', and $\bar T_S = \frac12\,\mathrm{Iso}$ by Lemma 50.12(d); so $\beta_E = 2\beta'$. By (G1), $t\mapsto E(X, V^X_{t,F})$ is real for real t and real curvature data F, so c^X and $\bar T_E$ are real on real curvature data, and β_E is real by Lemma 50.10(c).

(b) Let $F^* = F^{12,a}$ for some a. Then $|F^*| = |T_a| = 1$, since $|X|^2 = \operatorname{tr}(X^2)$ on $\mathfrak{g}$, and $\bar T_S(F^*) = \frac12$. So $\beta_E = 2\bar T_E(F^*)$, and Lemma 50.12(c) gives the bound.

(c) Let X be a polymer that is not wrapping, $r\in\mathfrak{S}^{\mathrm{grid}}_j$ with linear part R_r, and $\tilde r$ as in the proof of Lemma 50.12(b). As there, $r^{-1}V^X_{t,F} = V^{r^{-1}X}_{t,R_r^{-1}\cdot F}$, for the lift $\tilde r^{-1}$ of the given one and the origin $\tilde r^{-1}x_0$. So, for small $|t|$,

$$E(X, V^X_{t,F}) = \frac{1}{|\mathfrak{S}^{\mathrm{grid}}_j|}\sum_r E'(r^{-1}X, V^{r^{-1}X}_{t,R_r^{-1}\cdot F}), \qquad c^X_E(F) = \frac{1}{|\mathfrak{S}^{\mathrm{grid}}_j|}\sum_r c^{r^{-1}X}_{E'}(R_r^{-1}\cdot F).$$

Summing over X, and using that $X\mapsto r^{-1}X$ permutes the polymers that are not wrapping, $\sum_X c^X_E(F) = |\mathfrak{S}^{\mathrm{grid}}_j|^{-1}\sum_r\sum_X c^X_{E'}(R_r^{-1}\cdot F)$. The linear part R_r runs over B_4, each element

being taken by $(n_j/M)^4$ elements r, and $R \mapsto R^{-1}$ is a bijection of B_4. With Lemma 50.12(c) for E, which has (G2), this is the first formula. Every $R \in B_4$ maps each curvature datum $F^{\mu\nu,a}$ to $\pm F^{\mu'\nu',a}$, permuting the pairs $\{\mu,\nu\}$, and quadratic forms are even. So the first formula gives $\sum_{\mu<\nu,a} \bar{T}_E(F^{\mu\nu,a}) = \sum_{\mu<\nu,a} \bar{T}^{\mathrm{vol}}_{E'}(F^{\mu\nu,a})$. By (a) the left side is $\frac{1}{2}\beta_E \sum_{\mu<\nu,a} \mathrm{tr}(T_a^2) = 3 \dim \mathfrak{g}\, \beta_E$. □

Remark 50.14 (On the period density). (a) *The reflections are needed.* Under B_4^+ alone, Lemma 50.10(b) leaves a multiple $\tau\,\mathrm{Pf}$, the density of the topological charge. The symmetry (G3) of the sum of a family does not exclude a multiple of Pf in its period density, and the note does not show that its coefficient is controlled by the sum. A surviving τ would contribute terms of order $|\tau|$ per cube of every later scale, with no decay in the number of scales. The covariance (G4) under all of $\mathfrak{S}_j^{\mathrm{grid}}$, which includes the reflections, excludes it by Proposition 50.13; this is why the generations are symmetrized over B_4, and not only over B_4^+ or the translations.

(b) *Scales.* The number β_E does not depend on the units: in the units of another scale both $\bar{T}_E$ and $\bar{T}_S$ change by the same factor. It does not depend on the threshold of the generation.

(c) *Relation to Part XI and to Bałaban's coefficient.* The relation of β_E, for the generations of the flow, to the one-loop coefficient of Part XI and to the coefficient that Bałaban's definition [18, (1.20)–(1.22)] assigns to the generation is (F2) of Proposition 58.6 and Proposition 58.8, the latter through the generation locality of Section 59, without a rate; neither is used in Part X. These statements take into account that β_E is a property of the family E, not shown to be determined by its sum Φ_E alone (Section 58.3); for the flow this does not matter, since Proposition 51.5 applies to every generation with (G1)–(G4). Proposition 50.13(c) expresses β_E for a symmetrized generation through the raw family produced by the step: it is the isotropic part of the volume average of the quadratic forms $c^X_{E'}$, and the Pfaffian and anisotropic parts of that average are removed by the symmetrization.

51 Extraction of the marginal part

In four dimensions an old generation does not become small at later scales. Its orders ≥ 3 in the curvature are irrelevant, but its quadratic order is marginal (Remark 41.8(b)), so it has to be extracted. This section does this for one generation E of scale j (Section 50) at one later scale k, with $m = k - j$. For a grid cube $\square$ of scale k, the small polymers of $\square$ (Definition 49.5) are expanded. The sum of their activities is cancelled by the counterterm $-\beta_E S_\square$ of the cube, up to $O(L^{-\alpha m})$; this is Proposition 51.5. The other polymers, the recent, the large and the wrapping ones, are not expanded. They are bounded through the decay of the generation, with a gain L^{-m} (Lemma 51.7). The extraction has four steps.

- *Normal form* (Lemma 51.1). Let U be a point of the domain of scale k, read in the units of scale j, and let X be a polymer of scale j whose size is at most a multiple of m. Then, up to a gauge transformation, U is on $X^{(3)}$ a linear field of constant curvature plus a remainder that is smaller by a factor $L^{-\alpha m}$. The curvature is that of U at one point of X; we take the centre of the base cube of X. Part (d) of the lemma reduces $E(X,U)$ to a quadratic form in this curvature, up to an error $O(L^{-(4+\alpha)m})$.

- *Cell sums* (Lemma 51.3). By the grid covariance (G2), the quadratic forms of the polymers with a common base cube Q add up to M^4 times the period density of E (Definition 50.11), evaluated at the curvature of U at the centre of Q. Summed over the L^{4m} grid cubes of scale j in $\square$, the errors are $O(L^{-\alpha m})$.

- *The action* (Lemma 51.4). The action $S_\square$ of the cube is the sum of a family of the same kind, whose period density is the period density $\bar{T}_S$ of S. So the cell sums apply to it as well.
- *Cancellation.* By Proposition 50.13, $\bar{T}_E = \beta_E \bar{T}_S$ exactly. So the two cell sums cancel.

Remark 51.8 puts the pieces together. At a scale k, the old generation $E - \beta_E S$ is a sum of activities indexed by the grid cubes and by the polymers of scale k, bounded uniformly in k and with the gains $L^{-\alpha m}$ and L^{-m}. How these activities enter the step of scale k is not part of this section.

Bałaban expands in a gauge common to a cube of scale k [18, after (0.27)]. There the field is of order $L^{-(k-j)}$ in the units of scale j, and the expansion runs to the fifth order, with Ward identities for the lower orders. He then compares the vacuum polarization tensor, the second-order kernel summed over the localization domains of the infinite lattice, with β_j times the kernel of the action [18, (4.36)–(4.45)]. Here the gauge is chosen per polymer and is complex, as in Lemma 38.5. The field is then of order $L^{-2(k-j)}$, and the second order suffices. Polymers too large for this gauge, $d_j(X) > c'm$, are not expanded. Since this cutoff is logarithmic in L^m, they carry only the power gain L^{-m} (Lemma 51.7), instead of the gain exponential in L^{k-j} of [18, (0.27)]; this suffices for the sums over the generations (Remark 51.8). The comparison is between period densities on the given torus. The period density of E has no anisotropic and no topological part because of the covariance (G4) of the generation under the symmetries of the grid (Proposition 50.13).

51.1 The normal form

Notation for the lemma. Let X be a polymer of scale j, in the units of scale j, with fine spacing $\eta = \eta_j \le 1$, and let x_X be a point of X on the fine lattice. Put

$$R_X = M(2 + d_j(X)) + 3, \qquad \Lambda_X = \Lambda_{R_X}(x_X),$$

with $d_j(X)$ of Section 49.2. Any two points of X are at ℓ^∞-distance at most $M(2+d_j(X))$, by Lemma 52.1(b), whose first claim uses no condition on the torus. Hence $X^{(3)} \subset \Lambda_X$. Moreover $1 + R_X \le 6M(1 + d_j(X))$.

- *Differences.* For a $\mathfrak{g}^c$-valued field B on the positively oriented bonds of a set $Z \subset \Lambda_X$, let $|\partial B|_Z$ be the maximum of the flat differences $|\partial_\mu B_\nu(x)|$ that involve only points of Z. Let $[\partial B]_{\alpha,Z}$ be the supremum of $|\partial_\mu B_\nu(x) - \partial_\mu B_\nu(x')|/|x - x'|^\alpha$ over μ, ν and the pairs $x \ne x'$ of such differences. These are the quantities of Definition 38.3 for the real part 1, except that the Hölder quotient is not restricted to cubes $\Lambda_3(z')$.
- *The domain norm.* For a field B on the bonds of $X^{(3)}$ put

$$\|B\|_X = \max\Big(\frac{|B|_{X^{(3)}}}{\delta_0},\ \frac{|\partial B|_{X^{(3)}} + [\partial B]_{\alpha,X^{(3)}}}{\delta_1}\Big),$$

  as in Step 3 of the proof of Lemma 38.5. If $\|B\|_X < 1$, the configuration equal to $e^{i\eta B}$ on the bonds of $X^{(3)}$ and to 1 elsewhere lies in $\mathcal{U}(X) = \mathcal{U}(X; \theta_r, \delta_1, \delta_0)$, with real part 1.
- *Curvature at a point.* For a configuration V on $\mathbb{T}_\eta$ and a point x, let $F_V(x) = (F_V(p_{\mu\nu}(x)))_{\mu<\nu}$, with $F_V(p) = \eta^{-2}\ell(V(\partial p))$ as in Definition 21.4, on the fine lattice. It is defined for complex V when $|V(\partial p) - 1| \le c_G$.
- *Linear fields.* For a curvature datum $F = (F_{\mu\nu})_{\mu<\nu}$ in $\mathfrak{g}^c$ (Section 50.2) put $F_{\nu\mu} = -F_{\mu\nu}$, $F_{\nu\nu} = 0$ and $|F| = \max_{\mu,\nu} |F_{\mu\nu}|$. The *linear field* of F is

$$A_{\mathrm{lin}}[F](b) = \frac{1}{2}\sum_\mu F_{\mu\nu}\,(b_- - x_X)_\mu \qquad \text{for the bonds } b = (b_-, b_- + \eta e_\nu) \text{ of } \Lambda_X,$$

in the coordinates of Λ_X. Since $F_{\nu\nu} = 0$, b_- may be replaced by the midpoint of b. The differences of $A_{\mathrm{lin}}[F]$ are constant, $\partial_\mu A_{\mathrm{lin}}[F]_\nu = \frac{1}{2}F_{\mu\nu}$, and $A_{\mathrm{lin}}[F]$ vanishes on the bonds $(x_X, x_X + \eta e_\nu)$. On Λ_X, $|A_{\mathrm{lin}}[F]| \le \frac{1}{2}(d-1)R_X|F|$.

Lemma 51.1 (Later-scale normal form)**.** *There is $C_{\mathrm{nf}} \ge 1$, depending only on d, α, G, M and B_5 (so only on d, L, G, a, α and M), with the following property. Let $\alpha \le \frac{1}{2}$, $0 < \theta_r \le a_2$ with $B_5\theta_r \le 1$, and $0 < \delta_1 \le \delta_0 \le 1$, with a_2, B_5 of Corollary 30.4. These hold for the domain constants of Remark 38.8: there $\theta_r \le c_a \le a_2$ and $\delta_0 \le c_a \le 1$ by (DC1), and $B_5\theta_r \le \varepsilon_{\max} \le 1$ by (DC2). Put*

$$\Theta = 1 + \frac{B_5\theta_r + \delta_0^2}{\delta_1}.$$

Let $c' > 0$ and put $R_m = M(2 + c'm) + 4$. The lemma concerns the integers $m \ge 1$ with

$$R_m \le L^{m/2} \qquad \text{and} \qquad C_{\mathrm{nf}} R_m^2\, \Theta\, L^{-m} \le 1. \tag{51.1}$$

For the constant c' of Definition 49.5 they hold for all $m \ge m_0$, with m_0 of Remark 49.8(5). Let $j < k$ with $m = k - j$ satisfying (51.1), and let the torus have side $n_k \ge n_0$ in the units of scale k, with n_0 of Corollary 30.4. Let X be a polymer of scale j with

$$d_j(X) \le c'm,$$

let x_X be a point of X on the fine lattice, and let $Y \supset X$ be a polymer of scale k. Let $U \in \mathcal{U}(Y; \theta_r, \delta_1, \delta_0)$ in the units of scale k. In the units of scale j, with the notation above, there are a real gauge transformation u_r and $\mathfrak{g}^c$-valued scalar fields λ_1, λ_2 on $\mathbb{T}_\eta$, with $\lambda_1(x_X) = \lambda_2(x_X) = 0$, such that the following holds for $u = e^{-\mathrm{i}\lambda_2}e^{-\mathrm{i}\lambda_1}u_r$ and $W = U^u$.

(a) (Normal form.) *$W = e^{\mathrm{i}\eta A}$ on the bonds of Λ_X, with*

$$A = A_{\mathrm{lin}}[F_X] + R, \qquad F_X = F_W(x_X) = u_r(x_X)\, F_U(x_X)\, u_r(x_X)^{-1}.$$

So F_X is the curvature of U at x_X, rotated by an element of G. Moreover A and R vanish on the bonds $(x_X, x_X + \eta e_\nu)$.

(b) (Bounds.) *$|F_X| \le C_{\mathrm{nf}}\delta_1\Theta L^{-2m}$. On Λ_X,*

$$|R| \le C_{\mathrm{nf}}(1 + d_j(X))^{1+\alpha}\delta_1\Theta L^{-(2+\alpha)m}, \qquad |\partial R|_{\Lambda_X} \le C_{\mathrm{nf}}(1 + d_j(X))^{\alpha}\delta_1\Theta L^{-(2+\alpha)m},$$

and $[\partial R]_{\alpha,\Lambda_X} \le C_{\mathrm{nf}}\delta_1\Theta L^{-(2+\alpha)m}$. Consequently $\|A\|_X \le \frac{1}{4}$, and

$$\|A_{\mathrm{lin}}[F_X]\|_X \le C_{\mathrm{nf}}(1 + d_j(X))\, \Theta\, L^{-2m}, \qquad \|R\|_X \le C_{\mathrm{nf}}(1 + d_j(X))^{1+\alpha}\, \Theta\, L^{-(2+\alpha)m}.$$

(c) (Segments.) *$U \in \mathcal{U}(X)$, and for $t \in [0,1]$ the configurations $U^{e^{-\mathrm{i}t\lambda_1}u_r}$ and $U^{e^{-\mathrm{i}t\lambda_2}e^{-\mathrm{i}\lambda_1}u_r}$ lie in $\mathcal{U}(X)$. For complex s with $|s|\,\|A\|_X < 1$ let V_s equal $e^{\mathrm{i}\eta sA}$ on the bonds of $X^{(3)}$ and 1 elsewhere. Then $V_s \in \mathcal{U}(X)$, and $V_1 = W$ on the bonds of $X^{(3)}$. Hence $F(U) = F(V_1)$ for every graded activity F at X, and $s \mapsto F(V_s)$ is holomorphic for $|s| < \|A\|_X^{-1}$, where $\|A\|_X^{-1} \ge 4$.*

(d) (Second order.) *Let G be semisimple, and let F be a normalized graded activity at X. For fields B, B' on the bonds of $X^{(3)}$ let $d^2F(1)[B, B'] = \partial_s\partial_{s'}F(V_{s,s'})|_{s=s'=0}$, where $V_{s,s'}$ equals $e^{\mathrm{i}\eta(sB+s'B')}$ on the bonds of $X^{(3)}$ and 1 elsewhere. Then $|d^2F(1)[B, B']| \le 4\|F\|\,\|B\|_X\|B'\|_X$. The quadratic form $\mathfrak{q}_F(F') = \frac{1}{2}d^2F(1)[A_{\mathrm{lin}}[F'], A_{\mathrm{lin}}[F']]$ satisfies $\mathfrak{q}_F(vF'v^{-1}) = \mathfrak{q}_F(F')$ for $v \in G$, so $\mathfrak{q}_F(F_X) = \mathfrak{q}_F(F_U(x_X))$. Moreover*

$$|F(U) - \mathfrak{q}_F(F_X)| \le C_{\mathrm{nf}}\,\|F\|\,\Theta^3(1 + d_j(X))^{2+\alpha}L^{-(4+\alpha)m}.$$

So, for semisimple G and up to $O(L^{-(4+\alpha)m})$, a normalized activity of a small polymer depends on a point of the later domain only through the curvature of that point at x_X. It is a quadratic form in this curvature, and the form does not depend on the point. In the lemma the letter R denotes the remainder field, and R_X, R_m are radii; $R(\cdot)$ in the proof is the adjoint representation.

Proof. Conventions. In this proof C denotes constants that depend only on d, α, G, M and B_5. The constant C_{nf} is the largest of those needed below, and it is also the constant in (51.1). We write $\Lambda = \Lambda_X$ and use $\eta \le 1$ and $\alpha \le \frac{1}{2}$. Put Details

$$\vartheta_0 = \delta_0 + B_5\theta_r \le 2, \qquad \vartheta_1 = \delta_1 + B_5\theta_r + \vartheta_0^2.$$

Since $\vartheta_0^2 \le 2\delta_0^2 + 2B_5\theta_r$, we have $\vartheta_1 \le 3\delta_1\Theta$; also $\vartheta_0 \le \delta_0\Theta$. Since $1 + R_X \le R_m$, (51.1) gives

$$(1 + R_X)L^{-m/2} \le 1, \qquad C_{\mathrm{nf}}(1 + R_X)^2\,\Theta\,L^{-m} \le 1. \tag{51.2}$$

The first inequality and $\alpha \le \frac{1}{2}$ give $(1+R_X)L^{-3m} \le L^{-(2+\alpha)m}$ and $(1+R_X)^3L^{-4m} \le L^{-(2+\alpha)m}$. The second makes all fields below small enough for the logarithm, the Baker–Campbell–Hausdorff formula and (24.1), if C_{nf} is large.

Two facts on the cube Λ are used throughout.

(H1) Let g be a function on the points of Λ, or a component B_ν of a field on the bonds of Λ. Two points of Λ, or two bonds of Λ in direction ν, are joined by the coordinate-ordered path in the box they span, which lies in Λ. Summing differences along it, $|g(x) - g(x')| \le |x - x'|_1 |\partial g|_\Lambda$. Since $|x - x'| \le 2\sqrt{d}R_X$, the Hölder quotients of g on Λ are at most $C(1 + R_X)|\partial g|_\Lambda$. Together with $[gh]_\alpha \le [g]_\alpha |h| + |g|[h]_\alpha$, this bounds the Hölder quotients of products of such functions and of differences ∂B.

(H2) For a function h on Λ, the Hölder quotients of $\eta\,\partial h$ are at most $2|\partial h|_\Lambda$, since distinct points are at distance at least η and $\eta \le 1$. This is how terms with a factor η are handled, as with (23.2) in the proof of Lemma 38.5.

Step 1: a gauge of scale k. In the units of scale k, Definition 38.3 gives a real configuration U_r on $\mathbb{T}_\eta$ with $|F_{U_r}|, |J_{U_r}| < \theta_r$, and a field A' on the bonds of $Y^{(3)}$ with $U = e^{\mathrm{i}\eta_k A'} U_r$ there, $|A'| < \delta_0$ and $|\nabla A'|_{Y^{(3)}} + [\nabla A']_{\alpha, Y^{(3)}} < \delta_1$. Since $U_r \in \mathfrak{U}(\theta_r)$, $\theta_r \le a_2$ and $n_k \ge n_0$, Corollary 30.4 at scale k makes U_r $(M, B_5\theta_r)$-regular. Definition 18.1 with $z = x_X$ gives a real gauge transformation u_r on the cube of radius $4M + 2$ about x_X, extended to $\mathbb{T}_\eta$ arbitrarily, with $U_r^{u_r} = e^{\mathrm{i}\eta_k A_r^{(k)}}$ on its bonds, $|A_r^{(k)}|, |\partial A_r^{(k)}| \le B_5\theta_r$, and Hölder quotients of $\partial A_r^{(k)}$ at most $B_5\theta_r$ on the cube $\Lambda_3^{(k)}(x_X)$ of radius 3 about x_X. In the units of scale k, Λ has radius $R_X L^{-m} \le 1$, by (51.2). Since $x_X \in X \subset Y$,

$$\Lambda \subset \Lambda_3^{(k)}(x_X) \subset Y^{(3)}.$$

So both representations hold on the bonds of Λ, and every pair of differences that involve only points of Λ enters the Hölder quotients of Definitions 38.3 and 18.1. In particular Λ does not wrap around the torus.

Now pass to the units of scale j. As in the reasons for Remark 38.8(i), fields are multiplied by L^{-m}, differences by L^{-2m}, and Hölder quotients of differences by $L^{-(2+\alpha)m}$. We keep the names A_r and A' for the rescaled fields, and let $\tilde{A}'(b) = R(u_r(b_-))A'(b)$ be A' in the gauge u_r; here R is the adjoint representation, as in Definition 38.3. Then $U^{u_r} = e^{\mathrm{i}\eta\tilde{A}'} e^{\mathrm{i}\eta A_r}$ on the bonds of Λ. By the inclusions above, with Hölder quotients over all pairs in Λ,

$$|A_r| \le B_5\theta_r L^{-m}, \qquad |\partial A_r|_\Lambda \le B_5\theta_r L^{-2m}, \qquad [\partial A_r]_{\alpha,\Lambda} \le B_5\theta_r L^{-(2+\alpha)m},$$

$$|\tilde{A}'| < \delta_0 L^{-m}, \qquad |\nabla A'|_\Lambda < \delta_1 L^{-2m}, \qquad [\nabla A']_{\alpha,\Lambda} < \delta_1 L^{-(2+\alpha)m}.$$

The last two are taken with respect to U_r, with the transports of Definition 38.3. We compare them with flat differences as in the derivation of (38.1). In the gauge u_r,

$$D_{U_r^{u_r},\nu}\tilde A'_\mu - \partial_\nu \tilde A'_\mu = \eta^{-1}(R(e^{\mathrm{i}\eta A_{r,\nu}})-1)\tilde A'_\mu(\cdot+\eta e_\nu).$$

This is bounded by $C|A_r||\tilde A'|$, and its differences by $C(|\partial A_r||\tilde A'| + |A_r||\partial\tilde A'|)$. Moreover $\|R(U_r^{u_r}(\Gamma_{x,x'}))-1\| \le C|x-x'||A_r|$. By (H1), the Hölder quotients of this term and the transport correction are at most $C(1+R_X)B_5\theta_r\vartheta_0 L^{-3m}$. With (51.2),

$$|\partial\tilde A'|_\Lambda \le (\delta_1 + CB_5\theta_r)L^{-2m}, \qquad [\partial\tilde A']_{\alpha,\Lambda} \le (\delta_1+CB_5\theta_r)L^{-(2+\alpha)m}.$$

Step 2: one field. On the bonds of Λ write $e^{\mathrm{i}\eta\tilde A'}e^{\mathrm{i}\eta A_r} = e^{\mathrm{i}\eta A^\circ}$. By the Baker–Campbell–Hausdorff formula, as in Step 3 of the proof of Lemma 38.5, $A^\circ = \tilde A' + A_r + \eta\mathfrak{c}$ with $|\mathfrak{c}| \le C|\tilde A'||A_r|$. Here $\mathfrak{c}$ is an analytic function of $(\eta\tilde A', \eta A_r)$ that vanishes when one of them vanishes (Lemma 21.2(d)). By the Schwarz lemma in each variable and the chain rule, its differences are at most $C(|\partial\tilde A'||A_r| + |\tilde A'||\partial A_r|) \le CB_5\theta_r\vartheta_0 L^{-3m}$. By (H2), the Hölder quotients of $\partial(\eta\mathfrak{c})$ obey the same bound. Hence, with (51.2),

$$|A^\circ| \le 2\vartheta_0 L^{-m}, \qquad |\partial A^\circ|_\Lambda \le C\vartheta_1 L^{-2m}, \qquad [\partial A^\circ]_{\alpha,\Lambda} \le C\vartheta_1 L^{-(2+\alpha)m}. \tag{51.3}$$

Step 3: the affine gauge. Let $a_\nu = A^\circ(x_X, x_X+\eta e_\nu)$, so that $|a| \le 2\vartheta_0 L^{-m}$, and put

$$\lambda_1(x) = -\sum_\mu (x-x_X)_\mu\, a_\mu \quad (x\in\Lambda), \qquad \lambda_1 = 0 \ \text{ off } \Lambda.$$

On Λ, $\partial_\nu\lambda_1 = -a_\nu$ and $\partial\partial\lambda_1 = 0$, and $|\lambda_1| \le dR_X|a|$, which is small by (51.2). By (24.1) with $U_0 = 1$, which holds for complex λ by analytic continuation (Step 2 of the proof of Lemma 38.5), $U^{e^{-\mathrm{i}t\lambda_1}u_r} = e^{\mathrm{i}\eta A_t^\circ}$ on the bonds of Λ for $t\in[0,1]$, with

$$A_t^\circ(b) = e^{-\mathrm{i}t\,\mathrm{ad}\,\lambda_1(b_-)}A^\circ(b) + t\,\mathfrak{d}\big(t\,\mathrm{ad}\,\lambda_1(b_-)\big)\partial\lambda_1(b) + \eta\,\mathfrak{b}\big(t\lambda_1(b_-), A^\circ(b), t\,\partial\lambda_1(b)\big).$$

Put $B = A_1^\circ$. For $b = (x_X, x_X+\eta e_\nu)$ we have $\lambda_1(b_-) = 0$ and $\lambda_1(b_+) = -\eta a_\nu$, so $U^{e^{-\mathrm{i}\lambda_1}u_r}(b) = e^{\mathrm{i}\eta a_\nu}e^{-\mathrm{i}\eta a_\nu} = 1$ and $B(b) = 0$.

For a bond b in direction ν at x, the main part of $A_t^\circ(b)$ is $A_\nu^\circ(x) - ta_\nu = (A_\nu^\circ(x)-a_\nu) + (1-t)a_\nu$. By (H1), $|A_\nu^\circ(x)-a_\nu| \le dR_X|\partial A^\circ|_\Lambda$. Each other term contains $\lambda_1(b_-)$ or $\partial\lambda_1$, together with A° or $\partial\lambda_1$. They are bounded by $C(1+R_X)\vartheta_0^2 L^{-2m}$. Hence

$$|A_t^\circ| \le 2\vartheta_0 L^{-m} + C(1+R_X)\vartheta_1 L^{-2m}, \qquad |B| \le C(1+R_X)\vartheta_1 L^{-2m}.$$

Differentiating with $\partial\partial\lambda_1 = 0$ gives $\partial A_t^\circ = e^{-\mathrm{i}t\,\mathrm{ad}\,\lambda_1}\partial A^\circ + E_t$. Each term of E_t contains the constant $\partial\lambda_1 = -a$, together with A°, a or ∂A°. So $|E_t| \le C\vartheta_1 L^{-2m}$, and by (H1) and (H2) the Hölder quotients of E_t on Λ are at most $C(1+R_X)\vartheta_0\vartheta_1 L^{-3m}$. The Hölder quotients of $e^{-\mathrm{i}t\,\mathrm{ad}\,\lambda_1}\partial A^\circ$ are at most $(1+C|\lambda_1|)[\partial A^\circ]_{\alpha,\Lambda} + C(1+R_X)|a||\partial A^\circ|_\Lambda$. With (51.3) and (51.2), for $t\in[0,1]$,

$$|\partial A_t^\circ|_\Lambda \le C\vartheta_1 L^{-2m}, \qquad [\partial A_t^\circ]_{\alpha,\Lambda} \le C\vartheta_1 L^{-(2+\alpha)m}. \tag{51.4}$$

The term ϑ_0^2 of ϑ_1 comes from the commutators with a; it is the non-abelian part of the curvature.

Step 4: the quadratic gauge. Let $S_{\mu\nu} = \frac12(\partial_\mu B_\nu + \partial_\nu B_\mu)(x_X)$, so that $|S| \le C\vartheta_1 L^{-2m}$ by (51.4). With $y = x - x_X$ put

$$\lambda_2(x) = -\frac12\sum_{\mu,\rho} S_{\mu\rho}\, y_\mu y_\rho + \frac{\eta}{2}\sum_\mu S_{\mu\mu}\, y_\mu \quad (x\in\Lambda), \qquad \lambda_2 = 0 \ \text{ off } \Lambda.$$

For differences in Λ,

$$\partial_\nu \lambda_2(x) = -\sum_\rho S_{\nu\rho}\, y_\rho, \qquad \partial_\mu \partial_\nu \lambda_2 = -S_{\mu\nu};$$

the term with η removes the shift $\frac{\eta}{2} S_{\nu\nu}$ in the lattice difference of y_ν^2. Moreover $\lambda_2(x_X) = \lambda_2(x_X + \eta e_\mu) = 0$ and $\lambda_2(x_X + \eta e_\mu + \eta e_\nu) = -\eta^2 S_{\mu\nu}$ for all μ, ν, and

$$|\lambda_2| \le C(1+R_X)^2 \vartheta_1 L^{-2m}, \qquad |\partial\lambda_2|_\Lambda \le C(1+R_X)\vartheta_1 L^{-2m}, \qquad |\partial\partial\lambda_2| \le C\vartheta_1 L^{-2m}.$$

As in Step 3, $U^{e^{-\mathrm{i}t\lambda_2} e^{-\mathrm{i}\lambda_1} u_r} = e^{\mathrm{i}\eta B_t}$ on the bonds of Λ for $t \in [0,1]$, with

$$B_t(b) = e^{-\mathrm{i}t \operatorname{ad} \lambda_2(b_-)} B(b) + t\, \mathfrak{d}(t \operatorname{ad} \lambda_2(b_-)) \partial\lambda_2(b) + \eta\, \mathfrak{b}(t\lambda_2(b_-), B(b), t\,\partial\lambda_2(b)).$$

Hence $|B_t| \le |B| + |\partial\lambda_2|_\Lambda + C(1+R_X)^3\vartheta_1^2 L^{-4m} \le C(1+R_X)\vartheta_1 L^{-2m}$. Differentiating gives $\partial B_t = e^{-\mathrm{i}t \operatorname{ad}\lambda_2} \partial B + t\,\partial\partial\lambda_2 + E_t'$. Each term of E_t' contains $\partial\lambda_2$ or $\partial\partial\lambda_2$, together with one of λ_2, $\partial\lambda_2$, B and ∂B. Since $\partial\partial\lambda_2$ is constant, (H1) and (H2) give $|E_t'| \le C(1+R_X)^2\vartheta_1^2 L^{-4m}$, and Hölder quotients of E_t' at most $C(1+R_X)^3\vartheta_1^2 L^{-4m}$. With (51.4) and (51.2), for $t \in [0,1]$,

$$|\partial B_t|_\Lambda \le C\vartheta_1 L^{-2m}, \qquad [\partial B_t]_{\alpha,\Lambda} \le C\vartheta_1 L^{-(2+\alpha)m}. \tag{51.5}$$

Put $A = B_1$. Then $W = U^u = e^{\mathrm{i}\eta A}$ on the bonds of Λ.

Step 5: the curvature at x_X; proof of (a) and (b). The field λ_2 vanishes at x_X and at $x_X + \eta e_\nu$, and B vanishes on $(x_X, x_X + \eta e_\nu)$. So $W(x_X, x_X + \eta e_\nu) = 1$, that is, A vanishes on these bonds. Let $b = (x_X + \eta e_\mu, x_X + \eta e_\mu + \eta e_\nu)$. Then $\lambda_2(b_-) = 0$, $\lambda_2(b_+) = -\eta^2 S_{\mu\nu}$, and $\eta B(b) = \eta^2 \partial_\mu B_\nu(x_X)$. So

$$W(b) = e^{\mathrm{i}\eta^2 \partial_\mu B_\nu(x_X)}\, e^{-\mathrm{i}\eta^2 S_{\mu\nu}}.$$

As in Step 2, $\eta A(b) = \ell(W(b)) = \eta^2(\partial_\mu B_\nu(x_X) - S_{\mu\nu}) + \eta^4 \mathfrak{c}_{\mu\nu}$ with $|\mathfrak{c}_{\mu\nu}| \le C|\partial B|^2$. Since A vanishes on $(x_X, x_X + \eta e_\nu)$,

$$\partial_\mu A_\nu(x_X) = \tfrac{1}{2}(\partial_\mu B_\nu - \partial_\nu B_\mu)(x_X) + \eta^2 \mathfrak{c}_{\mu\nu} \qquad \text{for all } \mu, \nu.$$

For $p = p_{\mu\nu}(x_X)$, $\mu < \nu$, the bonds b_1 and b_4 of Definition 21.4 leave x_X. So $W(\partial p) = e^{\mathrm{i}\eta A_\nu(x_X + \eta e_\mu)} e^{-\mathrm{i}\eta A_\mu(x_X + \eta e_\nu)}$, and in the same way

$$F_W(p) = \eta^{-2}\ell(W(\partial p)) = \partial_\mu A_\nu(x_X) - \partial_\nu A_\mu(x_X) + \eta^2 \mathfrak{c}'_{\mu\nu}, \qquad |\mathfrak{c}'_{\mu\nu}| \le C|\partial A(x_X)|^2.$$

Put $F_X = F_W(x_X)$, extended antisymmetrically. The last two displays give

$$\partial_\mu A_\nu(x_X) = \tfrac{1}{2} F_{X,\mu\nu} + \rho_{\mu\nu} \quad \text{for all } \mu, \nu, \qquad |\rho| \le C|\partial B|_\Lambda^2 \le C\vartheta_1^2 L^{-4m},$$

and $|F_X| \le C\vartheta_1 L^{-2m}$. The transformation u equals u_r at x_X, which lies in G. The holonomy of a closed contour based at x_X is conjugated by $u(x_X)$. Hence $W(\partial p) = u_r(x_X) U(\partial p) u_r(x_X)^{-1}$, so $|U(\partial p) - 1| = |W(\partial p) - 1|$ is small, and $F_X = u_r(x_X) F_U(x_X) u_r(x_X)^{-1}$ by Lemma 21.2(d). This proves (a).

Put $R = A - A_{\mathrm{lin}}[F_X]$ on the bonds of Λ. Then R vanishes on the bonds $(x_X, x_X + \eta e_\nu)$, and $\partial R(x_X) = \rho$. Since the differences of $A_{\mathrm{lin}}[F_X]$ are constant, $[\partial R]_{\alpha,\Lambda} = [\partial A]_{\alpha,\Lambda} \le C\vartheta_1 L^{-(2+\alpha)m}$, by (51.5). Comparing every difference with the one at x_X, at distance at most $\sqrt{d} R_X$,

$$|\partial R|_\Lambda \le |\rho| + (\sqrt{d} R_X)^\alpha [\partial R]_{\alpha,\Lambda} \le C(1+R_X)^\alpha \vartheta_1 L^{-(2+\alpha)m}.$$

By (H1), $|R| \le d R_X |\partial R|_\Lambda$. With $\vartheta_1 \le 3\delta_1\Theta$ and $1 + R_X \le 6M(1 + d_j(X))$, this gives the bounds of (b) on F_X and R. For the domain norms, $|A_{\mathrm{lin}}[F_X]| \le \frac{1}{2}(d-1) R_X |F_X|$, $\partial A_{\mathrm{lin}}[F_X] = \frac{1}{2} F_X$ and $\delta_1 \le \delta_0$ give $\|A_{\mathrm{lin}}[F_X]\|_X \le d(1+R_X)|F_X|/\delta_1 \le C(1+R_X)\Theta L^{-2m}$.

Likewise $\|R\|_X \le C(1+R_X)^{1+\alpha}\Theta L^{-(2+\alpha)m}$, using $\delta_1 \le \delta_0$ for the sup norm. Finally $\|A\|_X \le \|A_{\mathrm{lin}}[F_X]\|_X + \|R\|_X \le C(1+R_X)^2\Theta L^{-2m} \le \frac14$, by (51.2) with $C_{\mathrm{nf}} \ge 4C$.

Step 6: proof of (c). By Remark 38.8(i), $U \in \mathcal{U}(X; L^{-2m}\theta_r, L^{-2m}\delta_1, L^{-m}\delta_0) \subset \mathcal{U}(X)$, and $\mathcal{U}(X)$ is invariant under real gauge transformations. On the bonds of $X^{(3)} \subset \Lambda$, the two families of configurations in (c) are $e^{\mathrm{i}\eta A_t^\circ}$ and $e^{\mathrm{i}\eta B_t}$. In Definition 38.3 take the real part 1, whose curvature and current vanish. Then $\nabla = \partial$, and the Hölder quotients of Definition 38.3 on $X^{(3)}$ are among those on Λ. By Steps 3 and 4, $\vartheta_0 \le \delta_0\Theta$, $\vartheta_1 \le 3\delta_1\Theta$ and (51.2),

$$|A_t^\circ|, |B_t| < \delta_0, \qquad |\partial A_t^\circ|_\Lambda + [\partial A_t^\circ]_{\alpha,\Lambda} < \delta_1, \qquad |\partial B_t|_\Lambda + [\partial B_t]_{\alpha,\Lambda} < \delta_1.$$

So both families lie in $\mathcal{U}(X)$. For $|s|\,\|A\|_X < 1$, the configuration V_s has real part 1 and the field sA on $X^{(3)}$, which satisfies the bounds of Definition 38.3 by the definition of $\|\cdot\|_X$. So $V_s \in \mathcal{U}(X)$. By (a), $V_1 = W$ on the bonds of $X^{(3)}$.

Let F be a graded activity at X. Real gauge invariance gives $F(U) = F(U^{u_r})$. Lemma 38.4, applied at U^{u_r} with $-\lambda_1$ and then at $U^{e^{-\mathrm{i}\lambda_1}u_r}$ with $-\lambda_2$, gives $F(U^{u_r}) = F(W)$. Finally $F(W) = F(V_1)$, since F depends only on the configuration on the bonds of X. The map $s \mapsto V_s|_X$ is holomorphic, and F is holomorphic on an open set containing $\{V|_X : V \in \mathcal{U}(X)\}$. So $s \mapsto F(V_s)$ is holomorphic for $|s| < \|A\|_X^{-1}$.

Step 7: proof of (d). Let B, B' be fields on the bonds of $X^{(3)}$; the bound is trivial if one of them vanishes. For $|s| < 1/(2\|B\|_X)$ and $|s'| < 1/(2\|B'\|_X)$ we have $\|sB + s'B'\|_X < 1$. So $V_{s,s'} \in \mathcal{U}(X)$, and $(s,s') \mapsto F(V_{s,s'})$ is holomorphic on this polydisc and bounded by $\|F\|$. Cauchy's estimate gives $|d^2F(1)[B,B']| \le 4\|F\|\,\|B\|_X\|B'\|_X$. The form $d^2F(1)$ is the second derivative at 0 of the holomorphic function $B \mapsto F(e^{\mathrm{i}\eta B})$, so it is symmetric and bilinear. For a constant $v \in G$, $(e^{\mathrm{i}\eta B})^v = e^{\mathrm{i}\eta vBv^{-1}}$, and real gauge invariance gives $d^2F(1)[vBv^{-1}, vBv^{-1}] = d^2F(1)[B,B]$. Since $A_{\mathrm{lin}}[vF'v^{-1}] = vA_{\mathrm{lin}}[F']v^{-1}$, the form $\mathfrak{q}_F$ is invariant under conjugation by G, and $\mathfrak{q}_F(F_X) = \mathfrak{q}_F(F_U(x_X))$ by (a).

Let $f(s) = F(V_s)$. By (c), f is holomorphic for $|s| < s_0 = \|A\|_X^{-1}$, with $s_0 \ge 4$, and bounded by $\|F\|$. Moreover $f(1) = F(U)$, $f(0) = F(1) = 0$, $f''(0) = d^2F(1)[A,A]$, and $f'(0) = 0$ by the argument in Step 3 of the proof of Lemma 38.5 (G semisimple): by real gauge invariance under constant gauge transformations, $\operatorname{grad} F(1)(b)$ is $\mathrm{Ad}(G)$-invariant for every bond b, hence 0. Cauchy's estimate gives

$$|F(U) - \tfrac12 d^2F(1)[A,A]| \le \|F\| \sum_{n\ge3} s_0^{-n} \le 2\|F\|\,\|A\|_X^3.$$

By bilinearity, $\frac12 d^2F(1)[A,A] - \mathfrak{q}_F(F_X) = d^2F(1)[A_{\mathrm{lin}}[F_X], R] + \frac12 d^2F(1)[R,R]$, which is at most $4\|F\|\,\|R\|_X(\|A_{\mathrm{lin}}[F_X]\|_X + \|R\|_X)$. By Step 5 and (51.2), $\|A\|_X \le C(1+R_X)\Theta L^{-2m}$ and $\|R\|_X(\|A_{\mathrm{lin}}[F_X]\|_X + \|R\|_X) \le C(1+R_X)^{2+\alpha}\Theta^2 L^{-(4+\alpha)m}$. Moreover $(1+R_X)^3 L^{-6m} \le (1+R_X)^{2+\alpha}L^{-(4+\alpha)m}$, by the first inequality of (51.2). With $\Theta \ge 1$ and $1+R_X \le 6M(1+d_j(X))$ this gives (d). □

Remark 51.2 (On Lemma 51.1). (a) *Where the gain comes from.* The real part must be put into a gauge with the regularity of scale k, rescaled, as in Step 1. A gauge from Corollary 30.4 at scale j, enlarged to Λ_X by Lemma 28.3 as in Step 1 of the proof of Lemma 38.5, has Hölder quotients of order L^{-2m} only. Then R would be of order L^{-2m}, and the error in (d) of order L^{-4m} per polymer. Summed over the L^{4m} polymers of scale j in a box of scale k, this is of order one, for every older generation. The affine gauge removes the sup norm of order L^{-m}, as in Remark 41.8(a). The quadratic gauge removes the symmetric part of the gradient, which is a gauge direction to first order.

(b) *Covariance instead of commutators.* The proof computes no commutator: the non-abelian part of the curvature enters F_X through covariance (Step 5). In a direct computation, to leading order in η and in the fields, the term $e^{-\mathrm{i}\,\mathrm{ad}\,\lambda_1}A^\circ$ of A_1° in Step 3 contributes

$\mathrm{i}[a_\mu, a_\nu]$ to $\partial_\mu B_\nu(x_X)$. The Maurer–Cartan term $\mathfrak{d}(\operatorname{ad}\lambda_1)\partial\lambda_1$, with $\mathfrak{d}(z) = 1 - \frac{\mathrm{i}}{2}z + O(z^2)$ (the derivative of the exponential map), contributes $-\frac{\mathrm{i}}{2}[a_\mu, a_\nu]$. The sum is half the commutator term $\mathrm{i}[a_\mu, a_\nu]$ of the curvature, as $\partial_\mu A_{\mathrm{lin}}[F]_\nu = \frac{1}{2}F_{\mu\nu}$ requires. Without the Maurer–Cartan term it would be the whole.

(c) *Where the lemma applies.* The lemma holds at every point of the domain of scale k at Y; the polymer Y need only contain X, not $X^{(3)}$. In particular it applies at the composed configurations $e^{\mathrm{i}\eta A_*}U'$ of Proposition 47.10(a), which lie in the domains $\mathcal{U}^\flat$ of scale k when Proposition 47.10 is applied with $\delta_1^\flat$ (Remark 50.9(c)). So it covers the old generations inside the fluctuation integrals of the later steps. For sums over translates, x_X can be taken to be the centre of the base cube $Q(X)$ (Section 49.2); for polymers that are not wrapping, $x_{X+a} = x_X + a$ for $a \in M\mathbb{Z}^4$. This choice is made in Lemma 51.3.

(d) *The constants c' and m_0.* The lemma holds for every $c' > 0$; only the range (51.1) of m depends on c'. It is applied with the c' of Definition 49.5, whose choice is explained in Remark 49.7(a). Since R_m grows linearly in m, (51.1) holds for all large m, and the integer m_0 of Remark 49.8(5) is finite. It depends only on c', d, α, G, L, M, a and the ratios θ_r/δ_1, δ_0^2/δ_1, not on j, k, the torus or the coupling. It is fixed after the domain constants and before the couplings; C_{nf} and Θ do not depend on it.

51.2 Cell sums

Families and period densities. Let $1 \le j \le K$ and use the units of scale j. The centre $z_Q \in M\mathbb{Z}^4$ of a grid cube Q of scale j is a point of Q on the fine lattice. For a grid cube $\square$ of scale $k \ge j$ let $\mathcal{X}_j^{\mathrm{sm}}(\square)$ be the set of the small polymers of $\square$, the small polymers of scale j based in $\square$ (Definition 49.5); it is empty if $k - j < m_0$. A *family of scale j* is a family $\mathsf{H} = \{\mathsf{H}(X,\cdot)\}$, indexed by the polymers X of scale j, with the following properties: each $\mathsf{H}(X,\cdot)$ is a normalized graded activity at X on $\mathcal{U}^\flat(X) = \mathcal{U}(X;\theta_r,\delta_1^\flat,\delta_0)$ (Section 50), real at real configurations; the family has the grid covariance (G2) of Definition 50.1; and, as in Definition 50.1,

$$\|\mathsf{H}\|_j = \sup_X e^{\kappa_n d_j(X)}\|\mathsf{H}(X)\| < \infty,$$

with $\|\mathsf{H}(X)\| = \sup_{\mathcal{U}^\flat(X)}|\mathsf{H}(X,\cdot)|$. A generation E of scale j with $\|E\|_j < \infty$ is a family of scale j. For a family and a polymer X that is not wrapping, the quadratic form c_{H}^X and the period density $\bar{T}_{\mathsf{H}}$ are those of Definition 50.11:

$$c_{\mathsf{H}}^X(F) = \frac{1}{2}\frac{d^2}{dt^2}\,\mathsf{H}(X, e^{\mathrm{i}\eta t A_F})\Big|_{t=0}, \qquad \bar{T}_{\mathsf{H}}(F) = M^{-4}\sum_{X\supset Q}\frac{c_{\mathsf{H}}^X(F)}{|X|}.$$

Here Q is a grid cube of scale j, the sum runs over the polymers $X \supset Q$ that are not wrapping, and $|X|$ is the number of grid cubes of X. The argument $F = (F_{\mu\nu})_{\mu<\nu}$ is a curvature datum in $\mathfrak{g}^c$, extended antisymmetrically, with $|F| = \max_{\mu,\nu}|F_{\mu\nu}|$. The field $A_F(b) = \frac{1}{2}\sum_\mu F_{\mu\nu}(b_- - x_0)_\mu$, for the bonds b in direction ν, is the linear field of F with an origin x_0, in the coordinates of the lift of $X^{(3)}$ to $\mathbb{R}^4$ (Section 50.2, paragraph on lifts and linear fields). The configuration $e^{\mathrm{i}\eta t A_F}$ is taken on the bonds of $X^{(3)}$ and equal to 1 elsewhere; $\mathsf{H}(X,\cdot)$ sees only the bonds of X. The form c_{H}^X does not depend on the origin x_0 (Lemma 50.12(a)), and we take $x_0 = z_{Q(X)}$. If X is small at a scale k and $x_X = z_{Q(X)}$ in Lemma 51.1, the coordinates of Λ_X lift $X^{(3)}$, and A_F is the linear field $A_{\mathrm{lin}}[F]$ of the lemma on the bonds of $X^{(3)}$. So c_{H}^X is then the form $\mathfrak{q}_{\mathsf{H}(X)}$ of Lemma 51.1(d): both are half the second derivative at $t = 0$ of $\mathsf{H}(X,\cdot)$ along the configurations equal to $e^{\mathrm{i}\eta t A_{\mathrm{lin}}[F]}$ on the bonds of $X^{(3)}$ and to 1 elsewhere.

The normal form on the domains $\mathcal{U}^\flat$. Lemma 51.1 holds for all constants that satisfy its hypotheses. Below it is applied with $(\theta_r, \delta_1^\flat, \delta_0)$ in place of $(\theta_r, \delta_1, \delta_0)$, at both scales; its hypotheses hold, since $\delta_1^\flat \le \delta_0$. Its domains are then the domains $\mathcal{U}^\flat$ of scale j and of scale k, the norm $\|\cdot\|_X$ of Section 51.1 is taken with $\delta_1^\flat$ in place of δ_1, and Θ is replaced by

$$\Theta^\flat = 1 + \frac{B_5\theta_r + \delta_0^2}{\delta_1^\flat} = 4\Theta - 3.$$

The integer m_0 is defined with $\Theta^\flat$ (Remark 49.8(5)), so (51.1) holds with $\Theta^\flat$ for $m \ge m_0$. Since $\delta_1^\flat\Theta^\flat \le \delta_1\Theta$, part (b) gives $|F_X| \le C_{\rm nf}\delta_1^\flat\Theta^\flat L^{-2m} \le C_{\rm nf}\delta_1\Theta L^{-2m}$.

Lemma 51.3 (Cell sums)**.** *Let G be semisimple, let $1 \le j \le K$, and let H be a family of scale j.*

(a) By Lemma 50.12(a) and (b), for every polymer X of scale j that is not wrapping, c_{H}^X is a quadratic form on the curvature data F in $\mathfrak{g}^c$, invariant under the conjugation $F \mapsto vFv^{-1}$ by every $v \in G$, with $c_{\mathsf{H}}^{X+a} = c_{\mathsf{H}}^X$ for $a \in M\mathbb{Z}^4$, and

$$|c_{\mathsf{H}}^X(F)| \le 9d^2M^2\big(1 + d_j(X)\big)^2(\delta_1^\flat)^{-2}\,\|\mathsf{H}(X)\|\,|F|^2. \tag{51.6}$$

For every grid cube Q of scale j and every F,

$$\sum_{X:\,Q(X)=Q} c_{\mathsf{H}}^X(F) = \sum_{X\supset Q} \frac{c_{\mathsf{H}}^X(F)}{|X|} = M^4\,\bar T_{\mathsf{H}}(F), \tag{51.7}$$

both sums running over the polymers that are not wrapping. In particular $\bar T_{\mathsf{H}}$ does not depend on Q.

(b) Let $j < k \le K$ with $m = k - j \ge m_0$, let $\square$ be a grid cube of scale k, and let $U \in \mathcal{U}^\flat(\square^+) = \mathcal{U}(\square^+; \theta_r, \delta_1^\flat, \delta_0)$ in the units of scale k. In the units of scale j, for every grid cube Q of scale j in $\square$ the curvature $F_Q = F_U(z_Q)$ is defined, $|F_Q| \le C_{\rm nf}\delta_1^\flat\Theta^\flat L^{-2m} \le C_{\rm nf}\delta_1\Theta L^{-2m}$, and

$$\Big|\sum_{X\in\mathcal{X}_j^{\rm sm}(\square)} \mathsf{H}(X,U) - M^4\sum_{Q\subset\square}\bar T_{\mathsf{H}}(F_Q)\Big| \le C_{\rm cs}\,(\Theta^\flat)^3\,\|\mathsf{H}\|_j\,L^{-\alpha m}. \tag{51.8}$$

Here Q runs over the L^{4m} grid cubes of scale j in $\square$, $C_{\rm nf}$ is that of Lemma 51.1 and $\Theta^\flat$ is as above, and $C_{\rm cs} = (C_{\rm nf} + 18d^2M^2C_{\rm nf}^2)K_0$, with K_0 of (49.2).

Proof. (a) The properties of c_{H}^X and (51.6) are Lemma 50.12(a) and (b), since H has (G2). Details
Let Q be a grid cube of scale j, and let $\mathfrak{T}$ be the group of the translations of the torus by the vectors of $M\mathbb{Z}^4$. It acts simply transitively on the grid cubes of scale j, it maps polymers that are not wrapping to polymers that are not wrapping, and $Q(X+a) = Q(X) + a$ for these (Section 49.2). In particular $X + a \ne X$ for $a \ne 0$. For a polymer $X \supset Q$ that is not wrapping, let $a_X \in \mathfrak{T}$ be the translation with $Q(X) + a_X = Q$, and put $\pi(X) = X + a_X$. Then $Q(\pi(X)) = Q$, $|\pi(X)| = |X|$ and $c_{\mathsf{H}}^{\pi(X)} = c_{\mathsf{H}}^X$. Conversely, let X' be a polymer with $Q(X') = Q$ that is not wrapping. The polymers $X \supset Q$ with $\pi(X) = X'$ are the polymers $X' - a$ with $a \in \mathfrak{T}$ and $Q + a \subset X'$. Indeed, $X' - a \supset Q$ means $Q + a \subset X'$, and then $Q(X' - a) = Q - a$, so $a_{X'-a} = a$. There is one such a for each of the $|X'|$ cubes of X', and the polymers $X' - a$ are distinct. Hence

$$\sum_{X\supset Q}\frac{c_{\mathsf{H}}^X(F)}{|X|} = \sum_{X':\,Q(X')=Q}|X'|\,\frac{c_{\mathsf{H}}^{X'}(F)}{|X'|},$$

which is (51.7). The left side of (51.7) for $Q + a$ is that for Q, by the covariance of c_{H}^X; so $\bar T_{\mathsf{H}}$ does not depend on Q.

(b) The torus satisfies (T0)–(T2) at every scale $k \le K$ (Lemma 52.1(a)), so $n_k \ge n_0$. Since $m \ge m_0$, m satisfies (51.1) with $\Theta^\flat$ (Remark 49.8(5)). Let Q be a grid cube of scale j in $\square$, and let X be a small polymer of $\square$ with $Q(X) = Q$; for instance $X = Q$, since $d_j(Q) = 0$. By Lemma 49.6, X is not wrapping and $X \subset X^{(3)} \subset \square^+$. So Lemma 51.1, with the constants $(\theta_r, \delta_1^\flat, \delta_0)$ as above, applies to X, with $x_X = z_Q$ and $Y = \square^+$. By its part (a), $F_Q = F_U(z_Q)$ is defined and conjugate to F_X by an element of G; by part (b), $|F_Q| = |F_X| \le C_{\mathrm{nf}} \delta_1^\flat \Theta^\flat L^{-2m}$. By part (d), applied to $\mathsf{H}(X,\cdot)$, with $\mathfrak{q}_{\mathsf{H}(X)} = c_{\mathsf{H}}^X$ and $\mathfrak{q}_{\mathsf{H}(X)}(F_X) = \mathfrak{q}_{\mathsf{H}(X)}(F_Q)$,

$$|\mathsf{H}(X,U) - c_{\mathsf{H}}^X(F_Q)| \le C_{\mathrm{nf}} \, \|\mathsf{H}(X)\| \, (\Theta^\flat)^3 (1 + d_j(X))^{2+\alpha} L^{-(4+\alpha)m}. \tag{51.9}$$

Every polymer of scale j based in $\square$ has its base cube among the L^{4m} grid cubes Q of scale j in $\square$ (Lemma 49.2(b)). A polymer that is not wrapping is small or large at scale k, since $m \ge m_0$. So (51.7) gives

$$\sum_{X \in \mathcal{X}_j^{\mathrm{sm}}(\square)} \mathsf{H}(X,U) - M^4 \sum_{Q \subset \square} \bar{T}_{\mathsf{H}}(F_Q) = \sum_{Q \subset \square} \sum_{X \text{ small: } Q(X) = Q} (\mathsf{H}(X,U) - c_{\mathsf{H}}^X(F_Q)) - \sum_{Q \subset \square} \sum_{X \text{ large: } Q(X) = Q} c_{\mathsf{H}}^X(F_Q),$$

where the last sum runs over the large polymers that are not wrapping. We use $\|\mathsf{H}(X)\| \le \|\mathsf{H}\|_j e^{-\kappa_n d_j(X)}$, $\kappa_n - \kappa_0 \ge \kappa_0 + 2 \ge 3$ (Remark 49.8(1)), and, by (49.2), $\sum_{X \supset Q} e^{-\kappa_0 d_j(X)} \le K_0$ for every Q.

The Taylor remainders. Since $\alpha \le \frac{1}{2}$, $(1+d)^{2+\alpha} e^{-(\kappa_n - \kappa_0)d} \le (1+d)^3 e^{-3d} \le 1$ for $d \ge 0$. By (51.9), the first sums contribute at most

$$C_{\mathrm{nf}} (\Theta^\flat)^3 \|\mathsf{H}\|_j L^{-(4+\alpha)m} \sum_{Q \subset \square} \sum_{X \supset Q} e^{-\kappa_0 d_j(X)} \le C_{\mathrm{nf}} K_0 \, (\Theta^\flat)^3 \|\mathsf{H}\|_j L^{-\alpha m}.$$

The large polymers. For a large polymer, $d_j(X) > c'm$, so $e^{-\frac{1}{2}(\kappa_n - \kappa_0) d_j(X)} \le e^{-\frac{1}{2}(\kappa_n - \kappa_0) c' m} = L^{-5m}$, by the choice of c' (Definition 49.5). Moreover $(1+d)^2 e^{-\frac{1}{2}(\kappa_n - \kappa_0)d} \le (1+d)^2 e^{-d} \le 2$. With (51.6) and $|F_Q| \le C_{\mathrm{nf}} \delta_1^\flat \Theta^\flat L^{-2m}$, the second sums contribute at most

$$9 d^2 M^2 C_{\mathrm{nf}}^2 (\Theta^\flat)^2 L^{-4m} \cdot 2 L^{-5m} \|\mathsf{H}\|_j \sum_{Q \subset \square} \sum_{X \supset Q} e^{-\kappa_0 d_j(X)} \le 18 d^2 M^2 C_{\mathrm{nf}}^2 K_0 \, (\Theta^\flat)^2 \|\mathsf{H}\|_j L^{-5m}.$$

Since $\Theta^\flat \ge 1$ and $L^{-5m} \le L^{-\alpha m}$, the two bounds give (51.8). □

The action as a family. The action of a grid cube $\square$ of scale k is $S_\square = \sum_{p:\, p_- \in \square} s_p$ (Section 49.2), with the plaquette terms $s_p(U) = \mathsf{N} - \frac{1}{2} \operatorname{tr} (U(\partial p) + U(\partial p)^{-1})$ of Section 23. To apply Lemma 51.3 to it, we group the plaquette terms into a family of scale $j \le k$. For a plaquette p of $\mathbb{T}_\eta$ let X_p be the union of the grid cubes of scale j that contain a corner of p, and put

$$S^{(j)}(X,U) = \sum_{p:\, X_p = X} s_p(U)$$

for the polymers X of scale j. The sum is empty unless $X = X_p$ for some plaquette p.

Lemma 51.4 (The action as a family)**.** *Let $1 \le j \le K$, and use the units of scale j.*

(a) For every plaquette p, X_p is a polymer of scale j of at most four grid cubes, with $d_j(X_p) = 0$. It is not wrapping, and $Q(X_p)$ is the grid cube that contains p_-. Hence $\sum_X S^{(j)}(X,U) = S(U)$, and for every grid cube $\square$ of scale $k \ge j$

$$\sum_{X \text{ based in } \square} S^{(j)}(X,U) = S_\square(U).$$

If $k - j \ge m_0$, every polymer based in $\square$ that carries a plaquette term is in $\mathcal{X}_j^{\mathrm{sm}}(\square)$, so the sum on the left is the sum over $\mathcal{X}_j^{\mathrm{sm}}(\square)$.

(b) *Each $S^{(j)}(X,\cdot)$ is a normalized graded activity at X, real at real configurations. It is defined and holomorphic for all G^c-valued configurations, and invariant under all gauge transformations. The family $S^{(j)}$ has the grid covariance (G2), and $S^{(j)}(rX,rU) = S^{(j)}(X,U)$ for every element r of the hyperoctahedral group B_4 acting about the centre of a grid cube of scale j.*

(c) *Let $i \le K$, let P be a domain, and let $U \in \mathcal{U}(P;\theta_r,\delta_1,\delta_0)$ in the units of scale i. For every plaquette p whose corners lie in $P^{(3)}$,*

$$|s_p(U)| \le C_s\,\eta_i^4\,(\theta_r+\delta_1+\delta_0^2)^2,$$

with C_s depending only on d and G. Consequently $\|S^{(j)}\|_j \le 6C_sM^4$, so $S^{(j)}$ is a family of scale j. Moreover $|S_\square(U)| \le 6C_sM^4$ for every grid cube $\square$ of scale $k \le K$ and every $U \in \mathcal{U}(\square^+;\theta_r,\delta_1,\delta_0)$ in the units of scale k.

(d) *For every polymer X of scale j, $c^X_{S^{(j)}}(F) = \frac12\eta^4\sum_{p:\,X_p=X}\operatorname{tr}\left(F^2_{\mu(p)\nu(p)}\right)$, where $\mu(p)<\nu(p)$ are the directions of the plane of p. Hence*

$$\bar T_{S^{(j)}}(F) = \frac12\sum_{\mu<\nu}\operatorname{tr}\left(F^2_{\mu\nu}\right).$$

This is the period density $\bar T_S$ of the action (Definition 50.11); this is also the last claim of Lemma 50.12(d).

Proof. (a) Let $p = p_{\mu\nu}(x)$, $\mu<\nu$, and let $c_0 \in M\mathbb{Z}^4$ be the centre of the grid cube of scale j that contains $x = p_-$, so $|x-c_0|_\infty \le \frac12(M-\eta)$ (Lemma 49.2(a)). The corners of p are $x+\eta(\epsilon_\mu e_\mu+\epsilon_\nu e_\nu)$, $\epsilon\in\{0,1\}^2$. If $(x-c_0)_\mu < \frac12(M-\eta)$, then $x+\eta e_\mu$ lies in the cube centred at c_0; otherwise $(x-c_0)_\mu = \frac12(M-\eta)$, and $x+\eta e_\mu$ lies in the cube centred at c_0+Me_μ. The same holds for ν. So the cubes of X_p are among the four cubes centred at $c_0+M(\epsilon_\mu e_\mu+\epsilon_\nu e_\nu)$, and X_p contains the cube centred at c_0. The closed cubes of the continuum that they fill all contain the point $c_0+\frac12M(e_\mu+e_\nu)$. So they are pairwise $*$-adjacent, X_p is a polymer, and the tree that consists of this point meets all of them; hence $d_j(X_p)=0$. The diameter of X_p is less than $2M \le \frac12 n_j - 8$, since $n_j \ge 8LM$ (Lemma 52.1(a)), so X_p is not wrapping. In a box that lifts X_p, the centres of its cubes are among the points $c_0+M(\epsilon_\mu e_\mu+\epsilon_\nu e_\nu)$, and the lexicographically least of them is c_0. So $Q(X_p)$ is the cube that contains p_-. Details

Each plaquette p belongs to exactly one polymer X_p, so $\sum_X S^{(j)}(X,U) = \sum_p s_p(U) = S(U)$. Since $p_- \in Q(X_p)$ and $Q(X_p)$ lies in exactly one grid cube of scale k (Lemma 49.2(b)), X_p is based in $\square$ if and only if $p_-\in\square$. This gives the identity for $S_\square$. If $k-j \ge m_0$, a polymer that carries a plaquette term has $d_j(X)=0 \le c'(k-j)$, so it is small.

(b) The terms s_p with $X_p = X$ depend only on U on the bonds of p, which join corners of p and so are bonds of X. Each s_p is holomorphic in the G^c-valued configurations, since $U(\partial p)^{-1} = U(b_4)U(b_3)U(b_2)^{-1}U(b_1)^{-1}$ with the bonds of Definition 21.4. It is invariant under all gauge transformations, since $U^g(\partial p) = g(p_-)U(\partial p)g(p_-)^{-1}$; it vanishes at $U=1$; and it is real for real U (Section 23). For $a\in M\mathbb{Z}^4$, $X_{p+a} = X_p + a$ and $s_{p+a}(\tau_a U) = s_p(U)$, which gives (G2). Let $r\in B_4$ act about the centre of a grid cube of scale j, a point of $M\mathbb{Z}^4$ on the fine lattice. Then r maps the fine lattice and the grid cubes of scale j onto themselves, and the plaquettes onto plaquettes; so $X_{rp} = rX_p$. Moreover $(rU)(\partial(rp))$ is conjugate to $U(\partial p)^{\pm1}$, as in Section 23, and $\operatorname{tr}(H+H^{-1})$ is invariant under conjugation and inversion. So $s_{rp}(rU) = s_p(U)$, and summing over the plaquettes p with $X_p = X$ gives $S^{(j)}(rX,rU) = S^{(j)}(X,U)$.

(c) Write $\eta = \eta_i$. Let $p = p_{\mu\nu}(x)$ have its corners in $P^{(3)}$, and let $U = e^{\mathrm{i}\eta A}U_r$ on the bonds of $P^{(3)}$ as in Definition 38.3. Let $b_1,\dots,b_4$ be the bonds of p as in Definition 21.4, $A_l = A(b_l)$, and $H_p = U_r(\partial p) = e^{\mathrm{i}\eta^2 F_{U_r}(p)}$. The identity (23.1) is an identity of products and holds for complex A:

$$U(\partial p) = e^{\mathrm{i}\eta a_1}e^{\mathrm{i}\eta a_2}e^{-\mathrm{i}\eta a_3}e^{-\mathrm{i}\eta a_4}H_p,$$

with $a_1 = A_1$, $a_2 = R(U_r(b_1))A_2$, $a_3 = R(H_p U_r(b_4))A_3$ and $a_4 = R(H_p)A_4$. Here $|a_l| < \delta_0$, since R of a real element preserves the norm, and $|\eta^2 F_{U_r}(p)| < \eta^2\theta_r$. By (DC1) and (DC5), $\theta_r, \delta_0 \le c_a \le c_{\mathrm{DC}} r_M L^{-2} \le \frac{1}{20}$, since $c_{\mathrm{DC}} \le \frac{1}{128}$ and $r_M \le \frac{1}{4}c_G \le \frac{1}{16}$ (Remark 38.8). So the norms of the five exponents add up to at most $s = \eta(4\delta_0 + \theta_r) \le \frac{1}{4}$, and Lemma 21.2(c) gives

$$Y := \ell(U(\partial p)) = \eta(a_1 + a_2 - a_3 - a_4) + \eta^2 F_{U_r}(p) + \beta, \qquad |\beta| \le Cs^2 \le C\eta^2(4\delta_0 + \theta_r)^2.$$

The covariant differences of Definition 38.3 at x are $\eta D_{U_r,\mu} A_\nu(x) = R(U_r(b_1))A_2 - A_4$ and $\eta D_{U_r,\nu} A_\mu(x) = R(U_r(b_4))A_3 - A_1$. So

$$a_1 + a_2 - a_3 - a_4 = \eta(D_{U_r,\mu}A_\nu - D_{U_r,\nu}A_\mu)(x) + (1 - R(H_p))(R(U_r(b_4))A_3 + A_4).$$

The first term is at most $2\eta\delta_1$. For the second, $|(R(H) - 1)Z| \le 2|H - 1||Z|$ for unitary H, and $|H_p - 1| \le 2\eta^2\theta_r$ by Lemma 21.2(a); so it is at most $8\eta^2\theta_r\delta_0$. With $\eta \le 1$, $\delta_0 \le 1$ and $\theta_r^2 \le \theta_r$, this gives $|Y| \le C'\eta^2(\theta_r + \delta_1 + \delta_0^2)$, with C' depending only on d and G. Moreover $|Y| \le s + Cs^2 \le \frac{1}{4} + C$. Finally $U(\partial p) = e^{\mathrm{i}Y}$, so $s_p(U) = \mathrm{tr}(1 - \cos Y)$. The power series gives $|1 - \cos Y| \le \frac{1}{2}\cosh(\frac{1}{4} + C)\,|Y|^2$, and $|\operatorname{tr} Z| \le \sqrt{\mathsf{N}}|Z|$. This proves the bound.

The plaquettes with $X_p = X$ have $p_- \in Q(X)$, by (a). There are at most $6(M/\eta)^4$ of them, since $Q(X)$ contains $(M/\eta)^4$ points of the fine lattice. Their corners lie in $X \subset X^{(3)}$, and $\theta_r + \delta_1 + \delta_0^2 \le 1$. So $\|S^{(j)}(X)\| \le 6C_s M^4$. Since $d_j(X) = 0$ whenever $S^{(j)}(X, \cdot) \ne 0$, $\|S^{(j)}\|_j \le 6C_s M^4$. With (b), $S^{(j)}$ is a family of scale j. In the units of scale k, the plaquettes with $p_- \in \square$ are at most $6(M/\eta_k)^4$, and their corners lie at distance at most $\eta_k \le 1$ from $\square$, so in $\square^+$. This gives the bound on $S_\square$.

(d) Let $p = p_{\mu\nu}(x)$, $\mu < \nu$, with $X_p = X$, and let V_t equal $e^{\mathrm{i}\eta t A_F}$ on the bonds of $X^{(3)}$ and 1 elsewhere. Its real part is 1, so $V_t(\partial p) = e^{\mathrm{i}\eta t a_1} e^{\mathrm{i}\eta t a_2} e^{-\mathrm{i}\eta t a_3} e^{-\mathrm{i}\eta t a_4}$ with $a_l = A_F(b_l)$. Since A_F is linear, $a_1 + a_2 - a_3 - a_4 = \eta(\partial_\mu A_{F,\nu} - \partial_\nu A_{F,\mu})(x) = \eta F_{\mu\nu}$. By Lemma 21.2(c), $Y(t) = \ell(V_t(\partial p)) = t\eta^2 F_{\mu\nu} + O(t^2)$ for small $|t|$, and

$$s_p(V_t) = \mathrm{tr}\,(1 - \cos Y(t)) = \tfrac{1}{2}\,\mathrm{tr}\,(Y(t)^2) + O(|t|^4) = \tfrac{1}{2}t^2\eta^4\,\mathrm{tr}\,(F_{\mu\nu}^2) + O(|t|^3).$$

So half the second derivative of $s_p(V_t)$ at $t = 0$ is $\frac{1}{2}\eta^4\,\mathrm{tr}(F_{\mu\nu}^2)$. Summing over the p with $X_p = X$ gives $c^X_{S^{(j)}}$. By (51.7) for the family $S^{(j)}$ and by (a),

$$M^4 \bar{T}_{S^{(j)}}(F) = \sum_{X\colon Q(X) = Q} c^X_{S^{(j)}}(F) = \sum_{p\colon p_- \in Q} \frac{1}{2}\eta^4\,\mathrm{tr}\,(F^2_{\mu(p)\nu(p)}) = M^4 \cdot \frac{1}{2}\sum_{\mu<\nu}\mathrm{tr}\,(F_{\mu\nu}^2),$$

since each of the $(M/\eta)^4$ points of the fine lattice in Q is the base point of one plaquette in each plane. The period density $\bar{T}_S$ of Definition 50.11 is computed in the same way from the plaquette terms of S, and it is the same quadratic form. □

51.3 The extraction

Proposition 51.5 (Extraction)**.** *Let G be simple, let $1 \le j \le k \le K$ and $m = k - j$, and let E be a generation of scale j with the properties (G1)–(G4) of Definition 50.1. Let β_E be its coefficient, defined by $\bar{T}_E = \beta_E \bar{T}_S$ (Proposition 50.13). For a grid cube $\square$ of scale k put*

$$\Psi_\square(U) = \sum_{X \in \mathcal{X}_j^{\mathrm{sm}}(\square)} E(X, U) - \beta_E\, S_\square(U).$$

Then $\Psi_\square$ is a normalized graded activity at $\square^+$ on $\mathcal{U}^\flat(\square^+)$ in the units of scale k, real at real configurations, and for every $U \in \mathcal{U}^\flat(\square^+) = \mathcal{U}(\square^+; \theta_r, \delta_1^\flat, \delta_0)$ in the units of scale k

$$|\Psi_\square(U)| \le C_{\mathrm{ex}}\|E\|_j\, L^{-\alpha m} \quad \text{if } m \ge m_0, \qquad |\Psi_\square(U)| \le C_{\mathrm{ex}}\|E\|_j \quad \text{if } m < m_0.$$

If $m < m_0$, then $\mathcal{X}_j^{\mathrm{sm}}(\square)$ is empty and $\Psi_\square = -\beta_E S_\square$. Here $C_{\mathrm{ex}} = C_{\mathrm{cs}}(\Theta^\flat)^3(1 + 6C_\beta C_s M^4)$, where C_β is the constant of Proposition 50.13 in $|\beta_E| \le C_\beta\|E\|_j$. It depends only on the parameters fixed in Remark 49.8(1)–(5), not on j, k, $\square$, the torus, the couplings or E.

Proof. By Lemma 49.6(b), every $X \in \mathcal{X}_j^{\mathrm{sm}}(\square)$ lies in $\square^+$. The action $S_\square$ depends only on U on the bonds of $\square^{(1)} \subset \square^+$. By Remark 38.8(i), with $\delta_1^\flat$ in place of δ_1, a point of $\mathcal{U}^\flat(\square^+)$ of scale k lies in $\mathcal{U}^\flat(X)$ of scale j for each such X. So each $E(X,\cdot)$ is holomorphic on an open set that contains the restrictions to $\square^+$ of the points of $\mathcal{U}^\flat(\square^+)$, and $S_\square$ is holomorphic everywhere. The coefficient β_E is real. Indeed, for real F and real t the configurations $e^{\mathrm{i}\eta t A_F}$ are real, so $c_E^X(F)$ is real by (G1); hence $\bar{T}_E$ is real on real F, while $\bar{T}_S$ is positive there (Lemma 51.4(d)). Gauge invariance, normalization and reality now follow from (G1) and Lemma 51.4(b). So $\Psi_\square$ is a normalized graded activity at $\square^+$ on $\mathcal{U}^\flat(\square^+)$, real at real configurations.

Let $U \in \mathcal{U}^\flat(\square^+)$ in the units of scale k. If $m < m_0$, no polymer is small at scale k, so $\Psi_\square = -\beta_E S_\square$. Then Lemma 51.4(c) and $|\beta_E| \le C_\beta \|E\|_j$ give $|\Psi_\square(U)| \le 6 C_\beta C_s M^4 \|E\|_j$.

Let $m \ge m_0$. We may assume $\|E\|_j < \infty$. Then the generation E is a family of scale j, and so is $S^{(j)}$ (Lemma 51.4(c)). By Lemma 51.4(a) and (d),

$$S_\square(U) = \sum_{X \in \mathcal{X}_j^{\mathrm{sm}}(\square)} S^{(j)}(X,U), \qquad \bar{T}_{S^{(j)}} = \bar{T}_S.$$

With the curvatures $F_Q = F_U(z_Q)$ of Lemma 51.3(b), in the units of scale j,

$$\Psi_\square(U) = \Big(\sum_{X \in \mathcal{X}_j^{\mathrm{sm}}(\square)} E(X,U) - M^4 \sum_{Q \subset \square} \bar{T}_E(F_Q) \Big) - \beta_E \Big(S_\square(U) - M^4 \sum_{Q \subset \square} \bar{T}_S(F_Q) \Big) + M^4 \sum_{Q \subset \square} (\bar{T}_E - \beta_E \bar{T}_S)(F_Q).$$

The last sum vanishes by Proposition 50.13: the two quadratic forms are equal on the curvature data in $\mathfrak{g}^c$, and the F_Q are such curvature data. Lemma 51.3(b), applied to E and to $S^{(j)}$, bounds the first two terms by $C_{\mathrm{cs}}(\Theta^\flat)^3 L^{-\alpha m}(\|E\|_j + |\beta_E|\,\|S^{(j)}\|_j)$. With $\|S^{(j)}\|_j \le 6 C_s M^4$ and $|\beta_E| \le C_\beta \|E\|_j$, this is at most $C_{\mathrm{ex}} \|E\|_j L^{-\alpha m}$. □

Remark 51.6 (On Proposition 51.5). (a) *The errors.* For $m \ge m_0$ there are two errors, both explicit in the proof of Lemma 51.3(b). The first is the Taylor remainder of Lemma 51.1(d), at most $C_{\mathrm{nf}}(\Theta^\flat)^3 \|E(X)\|(1+d_j(X))^{2+\alpha} L^{-(4+\alpha)m}$ for each small polymer. Its polynomial factor is absorbed by the decay of the generation, and the L^{4m} cells of $\square$ turn it into $L^{-\alpha m}$. This is the only error of order $L^{-\alpha m}$. The second is the quadratic part of the large polymers, which belongs to the period density but not to the small polymers. By the cutoff c' it is $O(L^{-5m})$ per cube $\square$. For $S_\square$ only the first error occurs, since all plaquette terms are carried by small polymers. The property (G4) enters only through Proposition 50.13; (G3) is not used in this section.

(b) *No boundary layer and no Hölder replacement.* The small polymers of $\square$ are grouped by their base cubes, which pave $\square$, and the terms of $S_\square$ by the base points of the plaquettes, which lie in the same cubes (Lemma 51.4(a)). So the cell sums tile $\square$ exactly, and no layer of polymers near the boundary of $\square$ has to be estimated. The normal form of a polymer is taken at the centre of its base cube Q, which is the point at which the curvature F_Q of the cell is evaluated; so no curvature has to be moved from one point of a polymer to another. The symmetric weights $1/|X|$ of Definition 50.11 enter only through the identity (51.7), which is exact by (G2). If these weights were used in the cell sums themselves, two further errors would occur. One comes from the polymers that meet $\square$ but are not based in it, or are based in it but are not contained in it. The other comes from the comparison of $F_U(z_{Q(X)})$ with $F_U(z_{Q'})$ for the other cubes Q' of X, through the Hölder bound on ∂R in Lemma 51.1(b). The identity (51.7) makes both unnecessary.

(c) *Why the counterterm is grouped with the small polymers.* On the complex domain of scale k the action $S_\square$ is of order $M^4(\theta_r+\delta_1+\delta_0^2)^2$ (Lemma 51.4(c)), which is not small in the coupling. In the flow, $\sum_{j\le k}\beta_j = g_0^{-2}-g_k^{-2}$ by the recursion of Remark 49.1, which is not bounded uniformly in k and in the bare coupling. So the counterterms cannot be bounded one by one in the sum over the generations. The differences $\Psi_\square$ can, by the gain $L^{-\alpha m}$.

(d) *Sums over generations.* Let $E^{(j)}$, $1\le j\le k$, be generations of scale j with (G1)–(G4) and $\|E^{(j)}\|_j\le \mathsf{K}$, and let $\Psi_\square^{(j)}$ be the corresponding functions. Then, for every grid cube $\square$ of scale k and every $U\in\mathcal{U}^\flat(\square^+)$ of scale k,

$$\sum_{j=1}^{k}|\Psi_\square^{(j)}(U)|\le C_{\mathrm{ex}}\mathsf{K}\Big(m_0+\sum_{m\ge m_0}L^{-\alpha m}\Big)\le C_{\mathrm{ex}}\mathsf{K}\Big(m_0+\frac{1}{1-L^{-\alpha}}\Big),$$

uniformly in k, K, the torus and the couplings.

(e) *The domain.* The proposition holds on the domain $\mathcal{U}^\flat$ of scale k at $\square^+$, the domain of the activities carried by the flow (Remark 50.9(c)). By Remark 51.2(c), this includes the composed configurations $e^{\mathrm{i}\eta A_*}U'$ of Proposition 47.10(a), at which the old generations are evaluated inside the fluctuation integrals of the later steps.

51.4 The other polymers

The polymers of scale j that are not small at scale k are the recent ones ($m<m_0$) and the large ones ($m\ge m_0$ and $d_j(X)>c'm$). Every wrapping polymer is among them (Lemma 49.6(a)). They are not expanded. Grouped by their closures $\mathrm{cl}_k X$ (Lemma 49.2(c)), they give activities of scale k with the rate κ_n at scale k and a gain L^{-m}. For a family E indexed by the polymers of scale j, a scale $k\ge j$ and a polymer Y of scale k put

$$E_Y^{\mathrm{ne}}(U)=\sum_{X\text{ not small at scale }k:\ \mathrm{cl}_k X=Y}E(X,U).$$

Lemma 51.7 (The other polymers at later scales)**.** *Let $1\le j\le k\le K$ and $m=k-j$. Let E be a family indexed by the polymers of scale j, such that each $E(X,\cdot)$ is a normalized graded activity at X on $\mathcal{U}^\flat(X)$ of scale j and $\|E\|_j<\infty$, with $\|E(X)\|=\sup_{\mathcal{U}^\flat(X)}|E(X,\cdot)|$; for instance a generation of scale j with $\|E\|_j<\infty$. Put $C_{\mathrm{ne}}=3^dK_0L^{5m_0}$.*

(a) For every polymer Y of scale k, E_Y^{ne} is a normalized graded activity at Y on $\mathcal{U}^\flat(Y)$ in the units of scale k, and

$$\sum_{X\ \textit{not small:}\ \mathrm{cl}_k X=Y}\ \sup_{U\in\mathcal{U}^\flat(Y)}|E(X,U)|\le C_{\mathrm{ne}}\|E\|_j\,L^{-m}e^{-\kappa_n d_k(Y)},$$

where $\mathcal{U}^\flat(Y)=\mathcal{U}(Y;\theta_r,\delta_1^\flat,\delta_0)$ is taken in the units of scale k. In particular $\sup_{\mathcal{U}^\flat(Y)}|E_Y^{\mathrm{ne}}|\le C_{\mathrm{ne}}\|E\|_jL^{-m}e^{-\kappa_n d_k(Y)}$.

(b) For every grid cube $\square$ of scale k,

$$\sum_{X\ \textit{not small:}\ X\cap\square\neq\emptyset}\|E(X)\|\le K_0L^{5m_0}\|E\|_j\,L^{-m}.$$

(c) Call X wrapping at scale k *if X is wrapping, or if its closure $\mathrm{cl}_k X$ is wrapping at scale k. These polymers are not small. If $m\ge1$, then for every polymer Y of scale k*

$$\sum_{X\ \textit{wrapping at scale }k:\ \mathrm{cl}_k X=Y}\|E(X)\|\le 3^dK_0\|E\|_j\,L^{-m}e^{-(\kappa_n-\kappa_0)n_j/(16M)}e^{-\kappa_n d_k(Y)}.$$

If families $E^{(j)}$ as above are given for all $j \le k$, with $\|E^{(j)}\|_j \le \mathsf{K}$, then for every grid cube $\square$ of scale k

$$\sum_{j\le k} \sum_{X \text{ of scale } j,\ \text{wrapping at scale } k:\ X\cap\square\neq\emptyset} \|E^{(j)}(X)\| \le 2K_0\mathsf{K}\, e^{-(\kappa_n-\kappa_0)n_k/(4M)}.$$

(d) *Let $E^{(j)}$, $1 \le j \le k$, be families as above with $\|E^{(j)}\|_j \le \mathsf{K}$, and write $E^{(j),\mathrm{ne}}_Y$ for the activities of (a). Then for every polymer Y and every grid cube $\square$ of scale k*

$$\sum_{j=1}^{k} \sup_{\mathcal{U}^\flat(Y)} |E^{(j),\mathrm{ne}}_Y| \le 2C_{\mathrm{ne}}\mathsf{K}\, e^{-\kappa_n d_k(Y)}, \quad \sum_{j=1}^{k} \sum_{X \text{ not small:}\ X\cap\square\neq\emptyset} \|E^{(j)}(X)\| \le 2K_0L^{5m_0}\mathsf{K},$$

where in the second sum X runs over the polymers of scale j. The bounds are uniform in k, K, the torus and the couplings.

Proof. We use $L \ge 4$, which follows from $L^2 \ge 8(C_*+1)$ and $C_* \ge 1$ (Remark 49.8(2), Proposition 31.5), and $\kappa_n - \kappa_0 \ge \frac12\kappa_n + 1$, which follows from $\kappa_n \ge 2\kappa_0 + 2$ (Remark 49.8(1)). Details

(a) If $m = 0$, then $\mathrm{cl}_j X = X$, and the sum consists of the single term $X = Y$, which is recent. So it is at most $\|E(Y)\| \le \|E\|_j e^{-\kappa_n d_j(Y)}$, and $E^{\mathrm{ne}}_Y = E(Y,\cdot)$. Let $m \ge 1$. Every polymer X with $\mathrm{cl}_k X = Y$ lies in Y. By Remark 38.8(i), with $\delta_1^\flat$ in place of δ_1, a point of $\mathcal{U}^\flat(Y)$ of scale k lies in $\mathcal{U}^\flat(X)$ of scale j. So E^{ne}_Y, a finite sum, is a normalized graded activity at Y on $\mathcal{U}^\flat(Y)$ in the units of scale k, and $\sup_{\mathcal{U}^\flat(Y)} |E(X,\cdot)| \le \|E(X)\| \le \|E\|_j e^{-\kappa_n d_j(X)}$. Every such X contains one of the $L^{4m}|Y|_M$ grid cubes of scale j in Y, where $|Y|_M$ is the number of grid cubes of scale k in Y, and $d_j(X) \ge L^m d_k(Y)$ (Lemma 49.2(c)). By (49.2), at the scales j and k,

$$\sum_{X:\,\mathrm{cl}_k X = Y} e^{-\kappa_0 d_j(X)} \le K_0 L^{4m}|Y|_M \le 3^d K_0 L^{4m} e^{d_k(Y)}.$$

Recent polymers, $1 \le m < m_0$. Here $(\kappa_n-\kappa_0)d_j(X) \ge (\kappa_n-\kappa_0)L^m d_k(Y)$, so the sum is at most $3^d K_0 L^{4m}\|E\|_j e^{-((\kappa_n-\kappa_0)L^m-1)d_k(Y)}$. The rate is at least $4(\kappa_n-\kappa_0)-1 \ge \kappa_n$, and $L^{4m} \le L^{5m_0}L^{-m}$.

Large polymers, $m \ge m_0$. Here $d_j(X) > c'm$, and by the choice of c'

$$(\kappa_n-\kappa_0)d_j(X) \ge \tfrac12(\kappa_n-\kappa_0)c'm + \tfrac12(\kappa_n-\kappa_0)L^m d_k(Y) = 5m\log L + \tfrac12(\kappa_n-\kappa_0)L^m d_k(Y).$$

So the sum is at most $3^d K_0 L^{-m}\|E\|_j e^{-(\frac12(\kappa_n-\kappa_0)L^m-1)d_k(Y)}$, and the rate is at least $2(\kappa_n-\kappa_0)-1 \ge \kappa_n$.

(b) A polymer of scale j that meets $\square$ contains one of the L^{4m} grid cubes of scale j in $\square$ (Lemma 49.2(b)). If $m < m_0$, (49.2) gives the bound $K_0L^{4m}\|E\|_j \le K_0L^{5m_0}L^{-m}\|E\|_j$. If $m \ge m_0$, a large polymer has $\|E(X)\| \le \|E\|_j e^{-\kappa_0 d_j(X)} e^{-(\kappa_n-\kappa_0)c'm} = \|E\|_j e^{-\kappa_0 d_j(X)} L^{-10m}$, and the sum is at most $K_0L^{-6m}\|E\|_j$.

(c) By Lemma 52.1(c), the polymers wrapping at scale k have $d_j(X) \ge n_j/(4M)$. They are not small. For $m < m_0$ no polymer is small. For $m \ge m_0$, a small polymer is not wrapping, and its closure lies in $\square^+$ for the cube $\square$ in which it is based (Lemma 49.6); $\square^+$ has diameter less than $3M \le \frac12 n_k - 8$, so the closure is not wrapping. Let $m \ge 1$. The second bound of Lemma 52.1(d), with $A = \|E\|_j$, $\lambda = \kappa_n$ and $n_j = L^m n_k$, gives

$$\sum_{X \text{ wrapping at scale } k:\ \mathrm{cl}_k X = Y} \|E(X)\| \le 3^d K_0 L^{4m}\|E\|_j e^{-(\kappa_n-\kappa_0)n_j/(8M)} e^{-(\frac12(\kappa_n-\kappa_0)L^m-1)d_k(Y)}.$$

The rate is at least κ_n, as in (a). By (T2) at scale k (Lemma 52.1(a)), $n_k \ge 8LM$. With $\kappa_n - \kappa_0 \ge 3$ and $L^m \ge e\,m\log L$ this gives $(\kappa_n-\kappa_0)n_j/(16M) \ge \frac32 L^{m+1} \ge 5m\log L$. Hence $L^{4m}e^{-(\kappa_n-\kappa_0)n_j/(8M)} \le L^{-m}e^{-(\kappa_n-\kappa_0)n_j/(16M)}$. The sum over $j \le k$ is Lemma 52.1(e), with $a_j(X) = \|E^{(j)}(X)\|$, $A = \mathsf{K}$ and $\lambda = \kappa_n$.

(d) Sum the bounds of (a) and (b) over $m = k - j \ge 0$, with $\sum_{m\ge0} L^{-m} \le 2$. □

Remark 51.8 (An old generation at a later scale). Let G be simple, let E be a generation of scale j with (G1)–(G4), and let $k \geq j$, $m = k - j$. Let U be a configuration that lies, in the units of scale k, in $\mathcal{U}^\flat(\square^+)$ for every grid cube $\square$ and in $\mathcal{U}^\flat(Y)$ for every polymer Y of scale k. For instance U may be a background field $U_k(V)$ at an admissible threshold (Section 50, the paragraph before Definition 50.1, at scale k). A polymer X of scale j is either small at scale k, and then based in exactly one grid cube $\square$ of scale k, or not small, and then $\mathrm{cl}_k X$ is one polymer Y of scale k. The $S_\square$ add up to S. So

$$\sum_X E(X,U) - \beta_E S(U) = \sum_\square \Psi_\square(U) + \sum_Y E_Y^{\mathrm{ne}}(U),$$

the sums running over the grid cubes $\square$ and the polymers Y of scale k. The first sum is indexed by the neighbourhoods $\square^+$, with the bounds of Proposition 51.5. The second is indexed by the polymers of scale k, with the decay $e^{-\kappa_n d_k(Y)}$ and the gain L^{-m} of Lemma 51.7(a). Each polymer X in the second sum meets a grid cube of scale k, so Lemma 51.7(b) bounds the second sum by $K_0 L^{5m_0} \|E\|_j L^{-m}$ per grid cube. For generations $E^{(j)}$, $1 \leq j \leq k$, with (G1)–(G4) and $\|E^{(j)}\|_j \leq \mathsf{K}$, Remark 51.6(d) and Lemma 51.7(d) give

$$\Big|\sum_{j=1}^{k}\Big(\sum_X E^{(j)}(X,U) - \beta_{E^{(j)}} S(U)\Big)\Big| \leq C\mathsf{K}\,\frac{|\mathbb{T}^{(k)}|}{M^4}, \qquad C = C_{\mathrm{ex}}\Big(m_0 + \frac{1}{1-L^{-\alpha}}\Big) + 2K_0 L^{5m_0},$$

where $|\mathbb{T}^{(k)}|/M^4$ is the number of grid cubes of scale k. The constant C does not depend on k.

52 Scales of the torus, the first step and the vacuum

This section collects the parts of the iteration that do not depend on the organization of the four-dimensional flow (Remark 49.1): the scales of the torus and the wrapping polymers, the first step and its exact formula, the split of the constants into normalizations and a vacuum energy, and the bookkeeping of the modified model. The model and the map T are those of Section 31. The finest spacing is $\epsilon = L^{-N}$. Here N is the total number of scales, unrelated to $N = L^k$ of Parts I–VII and to the matrix size N. The initial density is $\rho_0(V) = \exp(-g_0^{-2} S(V) - E_0)$ on the ϵ-lattice $\mathbb{T}^{(0)}$, with $g_0 = g$. The couplings g_k of the later steps are produced by the flow (Section 31); here they are parameters in $(0, \gamma]$. The parameters are chosen as in Remark 49.8. Section 53 uses this section for one step in generation form and for the flow over all scales (Theorem 53.12).

52.1 Scales and the torus

Let the torus have physical side ℓ. In the units of step k the lattice $\mathbb{T}^{(k)}$ has side $n_k = \ell L^{N-k}$. The steps are $k = 0, \dots, K-1$, with $K \leq N$, and the density of step K lives on $\mathbb{T}^{(K)}$, of side $n_K = \ell L^{N-K}$. For instance $K = N$, when the last lattice is the physical unit lattice. The configurations of step k live on the unit lattice $\mathbb{T}^{(k)}$, and the background fields on the fine lattice $\mathbb{T}_\eta$, which has the same side in these units but spacing $\eta = L^{-k}$. For $k \geq 1$, $\eta < 1$ and the two lattices are different sets; at $k = 0$, $\eta = 1$ and $\mathbb{T}_\eta = \mathbb{T}^{(0)}$ (Section 52.2). The side n_k is the n of Parts VI, VIII and IX. In Part VII, n denotes the side of $\mathbb{T}'$, which is $n_{k+1} = n_k/L$.

Torus hypotheses. At step k:

(T0) $n_k \geq n_*(L, M, \rho)$, the largest of the torus conditions of the inputs. These are:

- n_0 of Lemma 38.5 and Corollary 30.4, for n_k; Lemma 38.5 needs only that its cube Λ does not wrap around $\mathbb{T}^{(k)}$, i.e. $2R < n_k$;

- n_δ of Proposition 31.5, for n_{k+1};
- $80M'$ with $M' \geq M'_L$ as in Theorem 34.4, also large enough for Proposition 52.2(b), for n_{k+1};
- $8M_0$ of Lemma 31.3, for n_{k+1};
- $C(\rho + C_v)M$ of Section 40 ($C \geq 8$, used in Section 41.5), for n_k.

The conditions for $n_{k+1} = n_k/L$ are included in n_* with the factor L;

(T1) $LM \mid n_k$;

(T2) $n_k \geq C_{\mathrm{tor}} LM$, with $C_{\mathrm{tor}} = 8$. By Lemma 52.1(b), whose proof uses only (T2) at step k, every polymer with $d_M(X) \leq L$ then has $X^{(3)}$ in a cube of side less than $\frac{1}{2}n_k$, and diameter at most $\frac{1}{2}n_k - 8$. So such polymers are not wrapping in either sense: that of Lemma 38.5, and that of Proposition 41.7.

The *diameter* of a set of points of $\mathbb{T}_\eta$ is the largest ℓ^∞-distance on the torus between two of its points, in the units of step k. *Wrapping* polymers are those with diameter larger than $\frac{1}{2}n_k - 8$. A set P of diameter D lies in the cube $\Lambda_D(z)$ for every $z \in P$, so $P^{(3)} \subset \Lambda_{D+3}(z)$. If P is not wrapping, then $2(D+3) < n_k$, and this is a cube as in Lemma 38.5; this is how the proof of Proposition 41.7(c) uses the diameter.

We assume that ℓL^N and M are powers of L, and that the last torus is not too small:

$$n_K \geq n_{\min} := \max\left(C_{\mathrm{tor}} LM,\ n_*(L, M, \rho)\right).$$

The factor L in the first entry makes (T2) hold at the last scale $k = K$ as well. No step starts there, but the bounds of the flow at scale K use the same geometry as at the earlier scales: the cubes of scale K, and the neighbourhoods $\square^+$, that contain the small old polymers, and the lower bound on the size of wrapping polymers (Lemma 52.1(a),(b)). All bounds below are uniform in N and K; this mirrors [15], whose final torus has side $L^{M+\Delta}$.

Polymers of different scales. Polymers of scale j and their tree lengths $d_j(X)$ are those of Section 49.2. The grids of all scales are aligned: for $j \leq k \leq K$ a grid cube of scale k is the disjoint union of $L^{d(k-j)}$ grid cubes of scale j, and the grid cubes of each scale pave $\mathbb{T}_\eta$ (Lemma 49.2). For a polymer X of scale j and $k \geq j$, the union $\mathrm{cl}_k X$ of the grid cubes of scale k that meet X is a polymer of scale k, with $d_j(X) \geq L^{k-j} d_k(\mathrm{cl}_k X)$ (Lemma 49.2(c)); moreover $\mathrm{cl}_j X = X$ and $\mathrm{cl}_{j+1} X = \bar{X}$ (Section 49.3). Since $n_k = \ell L^{N-k}$, we have $n_j = L^{k-j} n_k$.

Lemma 52.1 (Torus hypotheses and wrapping polymers)**.** *Let $n_K \geq n_{\min}$.*

(a) For every $k \leq K$, (T0)–(T2) hold. In particular, in the units of a step $k \leq K$, every cube $\Lambda_R(z)$ with $R \leq 4M + 8$ has $2R < n_k$. This applies to the union $\square^+$ of a grid cube $\square$ and its $*$-neighbours (Section 49.2), to its 3-neighbourhood, and to the cubes $\Lambda_{4M+2}(z)$ of Definition 18.1.

(b) Let $k \leq K$, and let X be a polymer of scale k. Then $\mathrm{diam}(X) \leq M(d_k(X) + 2)$. If X is wrapping, then $d_k(X) \geq n_k/(4M)$. If $d_k(X) \leq n_k/(8M)$, then X is not wrapping, and $X^{(3)}$ lies in a cube $\Lambda_R(z)$ with $2R < \frac{1}{2}n_k$.

(c) Let $j \leq k \leq K$, $m = k - j$, and let X be a polymer of scale j. If X is wrapping, or if $\mathrm{cl}_k X$ is wrapping at scale k, then $d_j(X) \geq n_j/(4M) = L^m n_k/(4M)$. If $d_j(X) \leq L^{m+1}$, then X is not wrapping, and $X^{(3)}$ lies in a cube $\Lambda_R(z)$ of scale j with $2R < \frac{1}{2}n_j$.

(d) Let j, k, m be as in (c), let $\lambda \geq \kappa_0 + 1$ and $A \geq 0$, and let $a(X) \geq 0$ be numbers indexed by the polymers X of scale j with $d_j(X) \geq n_j/(4M)$, such that $a(X) \leq Ae^{-\lambda d_j(X)}$. Then,

for every grid cube $\square$ and every polymer Y of scale k,

$$\sum_{X\cap\square\neq\emptyset} a(X) \le K_0 L^{dm} A\, e^{-(\lambda-\kappa_0)L^m n_k/(4M)},$$
$$\sum_{\mathrm{cl}_k X=Y} a(X) \le 3^d K_0 L^{dm} A\, e^{-(\lambda-\kappa_0)L^m n_k/(8M)}\, e^{-(\frac12(\lambda-\kappa_0)L^m-1)d_k(Y)},$$

with κ_0, K_0 of (49.2).

(e) If, for every $j\le k$, numbers $a_j(X)$ are given as in (d), with the same A and λ, then for every grid cube $\square$ of scale k

$$\sum_{j\le k}\ \sum_{X \text{ of scale } j:\ X\cap\square\neq\emptyset} a_j(X) \le 2K_0 A\, e^{-(\lambda-\kappa_0)n_k/(4M)}.$$

Proof. (a) For $k\le K$, $n_k = L^{K-k}n_K \ge n_K \ge n_{\min}$. This gives (T0) and (T2). Since n_K and M are powers of L with $n_K \ge LM$, $LM \mid n_K \mid n_k$, which is (T1). By (T2), $n_k \ge 8LM > 8M+16$, since $L\ge 3$ and $M\ge 64$. The set $\square^+$ is a cube of side $3M$, and its 3-neighbourhood lies in a cube $\Lambda_R(z)$ with $R\le \frac32 M+4$. Details

(b) Let T be a tree in the continuum torus that meets every cube of X, of length ℓ_{T} in lattice units. Let $x,x'\in X$, let Q,Q' be cubes of X that contain them, and let $q\in\mathsf{T}\cap Q$ and $q'\in\mathsf{T}\cap Q'$. The path in T from q to q' has length at most ℓ_{T}, so the ℓ^∞-distance of q and q' on the torus is at most ℓ_{T}; and $|x-q|_\infty, |x'-q'|_\infty \le M$. Hence $|x-x'|_\infty \le \ell_{\mathsf{T}}+2M$, and the infimum over T gives $\mathrm{diam}(X)\le M(d_k(X)+2)$. By (T2) at step k, which holds by (a), $n_k\ge 8LM$; the rest of the proof of (b) uses only this, $L\ge3$ and $M\ge64$. If X is wrapping, then $\frac12 n_k - 8 < M(d_k(X)+2)$, so

$$d_k(X) > \frac{n_k}{2M} - 2 - \frac{8}{M} \ge \frac{n_k}{4M},$$

since $n_k/(4M)\ge 2L\ge 2+8/M$. If $d_k(X)\le n_k/(8M)$, then $D=\mathrm{diam}(X)\le \frac18 n_k+2M \le \frac12 n_k-8$, since $\frac38 n_k\ge 3LM\ge 2M+8$; so X is not wrapping. For $z\in X$, $X^{(3)}\subset\Lambda_{D+3}(z)$, and $2(D+3)\le \frac14 n_k+4M+6<\frac12 n_k$, since $\frac14 n_k\ge 2LM>4M+6$.

(c) If X is wrapping, (b) at scale j gives $d_j(X)\ge n_j/(4M)$. If $\mathrm{cl}_k X$ is wrapping at scale k, (b) at scale k gives $d_k(\mathrm{cl}_k X)\ge n_k/(4M)$, hence $d_j(X)\ge L^m n_k/(4M) = n_j/(4M)$, by Lemma 49.2(c). Finally, by (T2) at scale k, $n_j/(8M)=L^m n_k/(8M)\ge L^{m+1}$, and (b) at scale j gives the last claim. Since $L^{m+1}\ge L(1+m\log L)$, it covers every polymer of scale j with $d_j(X)\le c\,m$ and $c\le L\log L$.

(d) Every X that meets $\square$ contains one of the L^{dm} grid cubes Q of scale j in $\square$. For the X in the sum, $d_j(X)\ge n_j/(4M)$, so $a(X)\le Ae^{-(\lambda-\kappa_0)n_j/(4M)}e^{-\kappa_0 d_j(X)}$. By (49.2) at scale j, $\sum_{X\supset Q} e^{-\kappa_0 d_j(X)}\le K_0$. Summing over Q, with $n_j=L^m n_k$, gives the first bound. For the second, every X with $\mathrm{cl}_k X = Y$ contains one of the $L^{dm}|Y|_M$ grid cubes of scale j in Y, where $|Y|_M$ is the number of grid cubes of scale k in Y. By Lemma 49.2(c), $d_j(X)\ge L^m d_k(Y)$, so

$$(\lambda-\kappa_0)d_j(X) \ge \tfrac12(\lambda-\kappa_0)\frac{n_j}{4M} + \tfrac12(\lambda-\kappa_0)L^m d_k(Y).$$

With (49.2) as before and $|Y|_M\le 3^d(1+d_k(Y))\le 3^d e^{d_k(Y)}$ this gives the second bound.

(e) The pairs (j,X) with X of scale j and $a_j(X)$ defined have $d_j(X)\ge n_j/(4M)$. By the first bound of (d), the left side is at most $K_0A\sum_{m\ge0}L^{dm}e^{-xL^m}$, with $x=(\lambda-\kappa_0)n_k/(4M)$; the term m comes from $j=k-m$. By (T2) and $\lambda-\kappa_0\ge1$, $x\ge 2L$. Since $L^m\ge 1+m(L-1)$,

$$\sum_{m\ge0} L^{dm}e^{-xL^m} \le e^{-x}\sum_{m\ge0}\left(L^d e^{-x(L-1)}\right)^m \le 2e^{-x},$$

because $L^d e^{-2L(L-1)}\le\frac12$ for $L\ge3$ and $d=4$. $\square$

Wrapping polymers. The one-loop pieces of wrapping polymers are bounded by $Ce^{-(\kappa_2-1)|X|}e^{-cn_k/M}$, those of the other polymers by $Ce^{-(\kappa_2-1)|X|}$ (Proposition 41.7(d)). On the fixed domains of Remark 38.8 both are of order one, so the wrap-around pieces need not be compared with the other activities: Lemma 52.1 bounds them uniformly at all later scales under (T0)–(T2), with $n_K \geq n_{\min}$. Lemma 38.5 needs a cube $\Lambda \supset X^{(3)}$ with $2R < n_k$, which a polymer winding around the torus does not have. Such polymers are wrapping, since a polymer that is not wrapping lies in such a cube. The activities of wrapping polymers are not expanded, at their own scale or later; they are bounded by their supremum.

Let $\{E(X,\cdot)\}$ be activities indexed by the polymers of scale j, with $\sup|E(X,\cdot)| \leq Ae^{-\lambda d_j(X)}$ on the domains $\mathcal{U}(X;\theta_r,\delta_1',\delta_0)$ of scale j, where $\delta_1' = \delta_1$ or $\delta_1' = \delta_1^\flat$, and $\lambda \geq 2\kappa_0 + 2$. The generations of Section 50 are of this kind, with $\delta_1' = \delta_1^\flat$ and $\lambda = \kappa_n$ (Remark 49.8(1)). At a scale $k \geq j$, the domain of scale k at $\mathrm{cl}_k X$, with the same constants, lies in that of scale j at X (Remark 38.8(i)), so the bound holds there. By Lemma 52.1(c), the wrapping polymers of scale j, and those whose closure $\mathrm{cl}_k X$ wraps at scale k, have $d_j(X) \geq n_j/(4M)$. By (d) and (e), their activities, summed over all scales $j \leq k$, contribute at most $2K_0Ae^{-(\lambda-\kappa_0)n_k/(4M)}$ per grid cube of scale k. Grouped by $\mathrm{cl}_k X = Y$, with $k > j$, they decay in $d_k(Y)$ at the rate $\frac{1}{2}(\lambda-\kappa_0)L^{k-j} - 1 \geq \lambda$, since $\lambda - \kappa_0 \geq \frac{1}{2}\lambda + 1$ and $L \geq 4$; the latter follows from $L^2 \geq 8(C_* + 1)$ (Remark 49.8(2)) and $C_* \geq 1$ (Proposition 31.5), as in the proof of Lemma 51.7. These bounds are uniform in k, K, N and the coupling. So the activities of wrapping polymers are bounded crudely at every later scale, as those of large polymers. Their quadratic parts, which would be marginal (Remark 41.8(b)), are neither expanded nor extracted. Conversely, by Lemma 52.1(c), polymers of scale j with $d_j(X) \leq L^{k-j+1}$ are not wrapping, and by (a) the cubes of side at most $8M + 16$ in the units of a scale $k \leq K$ do not wrap. The grouped bounds are Lemma 51.7(c).

52.2 The first step

At $k = 0$ the configurations V live on the ϵ-lattice, $U_0(V) = V$, and the background of the step is $U' = U_1(V')$, which is the case $k = 1$ of Theorem 30.3. The base is $W = U'^{(0)} = U'$. The general analysis of Sections 47–48 uses the k-step operators at U' with the pair $(Q_k', \mathbf{Q}_k)$. At $k = 0$ these are the degenerate case of a single fine point per block: Lemma 31.3 excludes it, and Theorem 25.1 reduces to $\mathfrak{B}(V) = \{V\}$. The first step is simpler, and is treated directly.

Recall of the step in fluctuation coordinates. The proposition and Lemma 52.3 use the coordinates of Part VII. The configurations of the step are $e^{\mathrm{i}\hat{Z}(Z)}W$, with the base W and the fluctuation field Z on the bonds $\mathfrak{F}$ (Definition 32.1); $\hat{Z}$ is the local elimination of the constrained variables (Lemma 32.7). The gauge fixing relative to the base uses the averaged relative transports $\mathfrak{G}_W$ (Definitions 22.6 and 32.2), and its exponent Φ_W is that of Lemma 32.3(c). The exact step in these coordinates is Proposition 32.8; its quadratic part, the Hessian Δ_k, and the covariance $C_k = \Delta_k^{-1}$ are those of Lemma 34.1 and Corollary 34.6; the cutoffs are χ^{w} (Definition 32.4) and χ_ξ (Definition 48.1). For $k \geq 1$ the resulting formula, Theorem 34.9, is displayed at the start of Section 47 (the formula of one step). At $k = 0$ the operator Δ_0 is explicit, and the formula is Lemma 52.3.

Proposition 52.2 (The first step)**.** *Let $k = 0$, so that $\rho_0 = e^{-E_0}\exp(-g_0^{-2}S)$ and $\mathfrak{E}_0 = 0$, and let V' be small field at step 1. Let the parameters be chosen as in Remark 49.8, with the constants of the first step listed at the end of the proof, and assume (Q1), (Q2) and $2g_0 \leq r_t$ at $k = 0$. Then the construction of Sections 32–48 applies at $k = 0$, with the simplifications (a)–(c) below and the replacements listed in the proof, and $e^{E_0}T_{\mathrm{sf}}(\rho_0)(V')$ is given by Theorem 48.9 with $\mathfrak{E} = 0$ and $E_{\mathrm{m}}^{\#} = 0$: there are no old activities, no composed activities and no mixed class, and (Q3) is void. The step quantities simplify as follows.*

(a) No background map. $\mathcal{S}(B) = S(e^{iB}U')$, *and* A_* *is not used. The interaction pieces are* $v_p(Z) = s_p(e^{i\hat{Z}(Z)}U') - T_2[\cdots](Z)$, *local in* Z *within range* $2L+2$.

(b) The Hessian. *With* Δ^0 *the Hessian of* $B \mapsto S(e^{iB}U')$ *at* 0 *and* ω_0 *the gradient there,*

$$\Delta_0(Z,Z) = \Delta^0(\hat{Z}^1 Z, \hat{Z}^1 Z) + 2\langle \omega_0, \hat{Z}^2(Z)\rangle + \|\mathfrak{G}_{U'} Z\|^2.$$

Δ_0 *is local of range at most* $2L+2$, *and* $\operatorname{Re}\Delta_0(Z,\bar{Z}) \geq \frac{1}{2}C_{\mathrm{P}}^{-1}\|Z\|^2$ *on* $\ell^2(\mathfrak{F})$, *uniformly on the complex references of size* $\delta_c \leq \delta_{P,0}$, *with* C_{P} *of Lemma 34.2.*

(c) The walks. $C_0^{1/2}(s)$ *and the determinant densities are given by Theorem 41.4 and Corollary 41.6, with* $\mathfrak{A} = \Delta_0 + r$, $\mathfrak{N} = 0$ *and no mass term.*

Proof. The density ρ_0 is e^{-E_0} times a density of the form (31.2) with $\mathfrak{E}_0 = 0$, defined on all configurations, so with any threshold $\tilde{h} \leq a_1$. The step is linear in the density, so the factor e^{-E_0} passes through it. Details

(a) For $k = 0$ the k-step variational problem is trivial: $U_0(e^{iB}W) = e^{iB}W$, so $\mathcal{S}(B) = S(e^{iB}U')$. Lemma 32.3, Lemma 32.6, Lemma 32.7 and Proposition 32.8 are one-step statements for the averaging $\mathfrak{a}'$ from $\mathbb{T}$ to $\mathbb{T}'$ at the base $W = U'$, and they hold at $k = 0$. The interaction pieces are those of Definition 47.4 with $A_*(s;B)$ replaced by B. They are local, since s_p and $\hat{Z}$ are. The gauge-fixing pieces of Proposition 47.7 depend only on the base and on Z, and are unchanged.

(b) At $k = 0$, Δ^0 is the second variation of the Wilson action on $\mathbb{T}$ at U'. Its kernel vanishes unless the two bonds lie in a common plaquette, and it is bounded. Lemma 52.3(a),(b) below, whose proof uses only the definitions in (a) and (b) of the present proposition, is Lemma 34.1 at $k = 0$, with $\mathcal{S}$ explicit and with the minimality of Lemma 31.6(c) at $k = 0$; there it is the minimality of $U' = U_1(V')$ among the configurations V near U' with $\mathfrak{a}'(V) = V'$ (Theorem 30.3). It gives Δ_0 as stated, with no linear term, and $\|\mathfrak{G}_{U'} Z\| = \|\mathfrak{G}_{U'}\hat{Z}^1 Z\|$.

Lower bound at real references. Let $B = \hat{Z}^1 Z$. Then $B \in \mathcal{Z} = \ker l^{\mathrm{pl}}$ and $\|B\| \geq \|Z\|$ (Lemma 32.7(d)). We follow the proof of Theorem 34.4, with Δ^0 in place of $\mathbf{\Gamma}_\pi - a$ and the base $W = U'$. Let $\{\chi_j\}$ be the partition of Lemma 17.3 at scale M', acting on bond fields through b_-.

- *IMS.* Since Δ^0 has range 1, Lemma 17.4 gives $\Delta^0(B,B) \geq \sum_j \Delta^0(\chi_j B, \chi_j B) - CM'^{-2}\|B\|^2$. As in the proof of Theorem 34.4, $\|\mathfrak{G}_{U'}B\|^2 \geq \sum_j \|\mathfrak{G}_{U'}\chi_j B\|^2 - C_L(L/M')^2\|B\|^2$.

- *Local gauges.* Each $\chi_j B$ is supported on bonds starting in a cube $\Lambda_{M'}(z_j)$. On $\Lambda_{40M'}(z_j)$ there is a real gauge u_j in which $U' = e^{iA_j}$ with $|A_j| + |\partial A_j| \leq C_{M'}\varepsilon$ (Definition 18.1, and Lemma 28.3, which needs $\varepsilon \leq \varepsilon_{\mathrm{cmp}}(M')$; at $k = 0$, $\eta = 1$). By Proposition 23.3 at $\eta = 1$, which applies since U' is $(1,\varepsilon)$-regular with $\varepsilon \leq c_5$, $\Delta^0(B,B) = \|D_{U'}B\|^2 + d^2\mathcal{N}(0)[B,B]$ with $|d^2\mathcal{N}(0)[B,B]| \leq C\varepsilon(\|B\|^2 + \|\nabla B\|^2) \leq C\varepsilon\|B\|^2$, since $\|\nabla B\|^2 \leq 4d\|B\|^2$ at $\eta = 1$. The covariant curl is gauge covariant, and the plaquettes that meet the bonds of $\chi_j B$ lie in $\Lambda_{40M'}(z_j)$; so Lemma 12.2 in the gauge u_j gives $\|D_{U'}\chi_j B\|^2 \geq \|d_1(\chi_j B)^{u_j}\|^2 - C_{M'}\varepsilon\|\chi_j B\|^2$. Hence

$$\Delta^0(\chi_j B, \chi_j B) \geq \|d_1(\chi_j B)^{u_j}\|^2 - C_{M'}\varepsilon\|\chi_j B\|^2.$$

This replaces the first bound of Lemma 34.3. Its second and third bounds hold at $k = 0$ with the same proofs; since $W = U'$, the gauge u_j gives $W^{u_j} = e^{iA_j}$ directly, in place of Lemma 21.13. With $\|\mathfrak{G}_1\| \leq C_L$ and $M' \geq L$ they give

$$\|\mathfrak{G}_{U'}\chi_j B\|^2 \geq \|\mathfrak{G}_1(\chi_j B)^{u_j}\|^2 - C_{M'}\varepsilon\|\chi_j B\|^2,$$
$$\|q^{\mathrm{pl}}(\chi_j B)^{u_j}\|^2 \leq 2\|l^{\mathrm{pl}}\chi_j B\|^2 + C_{M'}\varepsilon^2\|\chi_j B\|^2.$$

- *Poincaré.* By Lemma 34.2 applied to $(\chi_j B)^{u_j}$, with $\|(\chi_j B)^{u_j}\| = \|\chi_j B\|$,

$$\|d_1(\chi_j B)^{u_j}\|^2 + \|\mathfrak{G}_1(\chi_j B)^{u_j}\|^2 \geq C_{\mathrm{P}}^{-1}\|\chi_j B\|^2 - \|q^{\mathrm{pl}}(\chi_j B)^{u_j}\|^2.$$

  As in the proof of Theorem 34.4, $\sum_j \|l^{\mathrm{pl}}\chi_j B\|^2 \leq 2\|l^{\mathrm{pl}}B\|^2 + C_L(L/M')^2\|B\|^2$, and $l^{\mathrm{pl}}B = 0$.

Summing over j with $\sum_j \|\chi_j B\|^2 = \|B\|^2$, and using $\varepsilon \leq 1$,

$$\Delta^0(B,B) + \|\mathfrak{G}_{U'}B\|^2 \geq (C_{\mathrm{P}}^{-1} - CM'^{-2} - C_L(L/M')^2 - C_{M'}\varepsilon)\|B\|^2.$$

Choose M' so large that $CM'^{-2} + C_L(L/M')^2 \leq \frac{1}{8}C_{\mathrm{P}}^{-1}$, and then ε so small that $C_{M'}\varepsilon \leq \frac{1}{8}C_{\mathrm{P}}^{-1}$. Then $\Delta^0(\hat{Z}^1 Z, \hat{Z}^1 Z) + \|\mathfrak{G}_{U'}Z\|^2 \geq \frac{3}{4}C_{\mathrm{P}}^{-1}\|Z\|^2$, since $\|\hat{Z}^1 Z\| \geq \|Z\|$.

The gradient term. At $k = 0$, $\eta = 1$ and $\omega_0 = \nabla_B S(e^{\mathrm{i}B}U')|_{B=0} = J_{U'}$ exactly. It is the multiplier of the trivial 0-step problem, not of the one-step constrained minimum. That $\langle \omega_0, \hat{Z}^1 Z\rangle = 0$, i.e. the absence of a linear term, follows from Lemma 31.6(c) at $k = 0$, which is Lemma 52.3(a). The term $2\langle \omega_0, \hat{Z}^2(Z)\rangle$ is at most $CL^{3d+1}|J_{U'}|\|Z\|^2$, by Lemma 32.7(d), and $|J_{U'}| \leq \varepsilon$ is small by Theorem 30.3; so it is at most $\frac{1}{8}C_{\mathrm{P}}^{-1}\|Z\|^2$ for ε_1' small; in the flow this holds since $\varepsilon_1' \leq \gamma p(\gamma) \leq a_{\mathrm{step}}^{(0)}$ ((Γ2) of Remark 53.5). (The smallness conditions on ε, namely $\varepsilon \leq \min(\varepsilon_{\mathrm{cmp}}(M'), c_5)$, $C_{M'}\varepsilon \leq \frac{1}{8}C_{\mathrm{P}}^{-1}$ and the condition on the gradient term, are part of the choice of $a_{\mathrm{step}}^{(0)}$ in Lemma 52.3.) Away from criticality put $\omega_0 := J_{U'}$; the bound holds with $|J_{U'}| \leq \varepsilon$. The proof of the lower bound does not use criticality: it holds at every real reference on $\mathbb{T}$ that is (M,ε)-regular with $|J_{U'}| \leq \varepsilon$ and whose base satisfies $L^d\mathsf{a} \leq c$, for ε below the conditions just listed. Besides $U_1(V')$, the first step is used at the real parts of the extended points of the graded domains of scale 1, and of the domains $\mathcal{U}(X; c_a, c_a, c_a)$ of scale 0: in Proposition 41.7 and Section 48 at $k = 0$, hence in Lemma 53.4 and Proposition 53.10 at $k = 0$. These real parts lie in $\mathfrak{U}(c_a)$ in the units of step 0, so they are $(M, \max(1, B_5)c_a)$-regular (Corollary 30.4). For them $L^d\mathsf{a} \leq c$ holds by (DC3), respectively by $c_a \leq c_J L^{-d-2}$ with c_J small (Proposition 47.8(c)). The conditions on ε hold there by the choice of ε_a in (c) below.

Complex references and range. Δ_0 is an explicit local analytic function of the reference: the Hessian of the Wilson action, composed with the elimination of Lemma 32.7 at a complex base, plus the bilinear form $\langle \mathfrak{G}_{\hat{W}}Z, \mathfrak{G}_{\hat{W}}Z\rangle$ at the complex base $\hat{W} = U'$, which is holomorphic in the base (Definition 22.6, Lemma 22.7(a)), as in the proof of Corollary 34.6(c). Complex references of size δ_c change Δ_0 by $O(\delta_c)$ in norm. The range follows from that of $\hat{Z}^1$ and $\hat{Z}^2$, and from the block-locality of $\mathfrak{G}_{U'}$. The IMS scale M' is chosen after L and before γ (Remark 49.8(4)), and $n_1 \geq 80M'$ is included in n_*.

(c) The hypotheses of Section 41 hold with $\mathfrak{A}^X = \mathbf{1}_X(\Delta_0 + r)\mathbf{1}_X$ on $\ell^2(\mathfrak{F}(X))$:

- (W1): $\mathfrak{A}^X$ is local, and $\mathfrak{N} = 0$;
- (W2): the compressions to $\Lambda_j^v \cap X$ have $\mathrm{Re} \geq \frac{1}{2}C_{\mathrm{P}}^{-1} + r$, since the lower bound of (b) holds for all fields on $\mathfrak{F}$, hence decay by Lemma 5.5; the comparison of domains is in seam form, by the resolvent identity and the locality of Δ_0, as in Proposition 37.6(c);
- (W3) is empty.

The operator $\mathfrak{A}^X$ contains the gauge-fixing term $\mathbf{1}_{\mathfrak{F}(X)}\mathfrak{G}_{U'}^T\mathfrak{G}_{U'}\mathbf{1}_{\mathfrak{F}(X)}$. The commutator hypothesis of Lemma 41.3 holds, since Δ_0 has range $2L+2$ and $|\nabla h_j| \leq C/M$. So does the comparison of the local commutators in Theorem 41.4(b): the difference is $\mathbf{1}_X[\Delta_0, h_j](G_j^X - G_j^{X'}) + (\mathbf{1}_X - \mathbf{1}_{X'})[\Delta_0, h_j]G_j^{X'}$, of seam form by (W2) and supported next to $X \setminus X'$, respectively. Then Theorem 41.4 and Corollary 41.6 apply. In the latter, the local densities $\ell_{\Lambda_j^v}$ are now the diagonal traces of $\log(\mathbf{1}_\Lambda \Delta_0 \mathbf{1}_\Lambda)$, $\Lambda = \Lambda_j^v$, by the formula of Proposition 34.8. Proposition 41.7 applies with one change. Its constants ε_a, δ_a at $k = 0$ are chosen so that (b) holds at the references; in particular $\varepsilon_a \leq \min(\varepsilon_{\mathrm{cmp}}(M'), c_5)$, $C_{M'}\varepsilon_a \leq \frac{1}{8}C_{\mathrm{P}}^{-1}$, ε_a satisfies the condition on

the gradient term, $\varepsilon_a \le \varepsilon_{\max}$, and $C_{\mathrm{cmp}}\delta_a \le \delta_{P,0}$; the last two make every configuration of Definition 38.2 a complex reference of size at most $C_{\mathrm{cmp}}\delta_a \le \delta_{P,0}$ (first item of Section 46). With them, c_a satisfies $\max(1, B_5)c_a \le \varepsilon_a$ and $2C_{\mathrm{ext}}c_a < \delta_a$, and, by the rule at the end of this proof, the flow uses the smaller of the values of c_a at $k = 0$ and at $k \ge 1$. Parts (a), (b) obtained analyticity at complex references through Propositions 19.11, 46.4 and 46.7, which concern k-step operators. At $k = 0$ this is replaced by the direct statement in (b): Δ_0 is an explicit local analytic function of the reference, coercive under complex perturbations.

The rest of Sections 47–48 applies with these replacements:

- Proposition 47.5(a)–(c) and (e) with $A_* = B$, so that $|A_*|_* \le C_P|B|$ holds trivially, and Proposition 47.8; in place of Proposition 47.5(d), the factorization of the composed pieces proved below;
- $r_I = r_L$, since Theorem 45.5 and its radius R_P are not used;
- in (Q1), the conditions $w \le \beta_V$ and $w \le \frac{1}{2}R_P$ are dropped: β_V enters only the identification through $A_*(1; B) = A_*(B)$, which is trivial for $A_* = B$, and R_P only through A_*;
- complex references at $k = 0$ are those of size $\delta_c \le \delta_{P,0}$. Here $\delta_{P,0}$ is such that the $O(\delta_c)$ change of Δ_0 in (b) is at most $\frac{1}{8}C_{\mathrm{P}}^{-1}$ in norm, and $\delta_{P,0} \le \min(c'_L, cL^{-d-1})$, with c'_L of Corollary 34.6(c), for the elimination and the averaged contour variables at the complex base $W = U'$ (with $|\mathfrak{r}_c| \le \frac{1}{2}$); moreover $\varepsilon + C\delta_{P,0} \le \frac{1}{20}$ and $\rho_L \le \frac{1}{20}$, for Proposition 47.5(b) at $k = 0$. With the bound on the gradient term this gives $\operatorname{Re}\Delta_0(Z, \bar{Z}) \ge (\frac{3}{4} - \frac{1}{8} - \frac{1}{8})C_{\mathrm{P}}^{-1}\|Z\|^2 = \frac{1}{2}C_{\mathrm{P}}^{-1}\|Z\|^2$;
- Section 48 without composed activities, and with Lemma 52.3 in place of Theorem 34.9 and Corollary 34.6: in Proposition 48.3, whose proof is that of Theorem 34.9 with χ_ξ carried along, and in the covariance (C1), (C2) of the white-noise cutoff χ_ξ, which rests on Lemma 52.3(c).

Factorization at $k = 0$. The first sentence of Proposition 47.5(d), a statement for a fixed field Z, does not hold at $k = 0$: there $v_p(Z)$ does not depend on s, and it need not vanish when all $s_\square$ do. For instance, at $U' = 1$, let Z be supported on one bond $b \in \mathfrak{F}$ of a plaquette p that contains no bond $b_0(c)$. Then $v_p(Z) = f(Z(b)) - T_2[f](Z(b))$ with $f(X) = \mathsf{N} - \operatorname{tr}\cos X$, which is of order $|Z(b)|^4$ and not 0 for small real $Z(b) \ne 0$. What Lemma 47.2 needs is the factorization of the composed pieces $v_p(Z(s))$, $Z(s) = gC_0^{1/2}(s)\xi$, which we now prove, as for the gauge-fixing pieces. For a plaquette p let $B'(p)$ be the union of the blocks $B'(y)$, $y \in \mathbb{T}'$, that meet p. Its ∞-diameter is less than $2L$, since in each coordinate direction the points of p lie in at most two consecutive blocks, and it contains $B'(c_-) \cup B'(c_+)$ for every bond $b_0(c)$ of p. By Lemma 32.7(a),(c),(d) and Lemma 21.10(c), $\hat{Z}(Z)$ on the bonds of p depends on Z and on the base $W = U'$ only on the bonds of $B'(p)$, and so does $v_p(Z)$. If $Z = 0$ on the bonds of $\mathfrak{F}$ in $B'(p)$, then $\hat{Z}(Z) = 0$ on the bonds of p, since $\zeta_c(0) = 0$ by the uniqueness in Lemma 32.7(c); so $t \mapsto s_p(e^{\mathrm{i}\hat{Z}(tZ)}U')$ is constant, and $v_p(Z) = 0$.

Now let $s_\square = 0$ for $\square \not\subset Z'$, a union of boxes with $*$-components Z'_i. By (c), Corollary 41.6 and Lemma 40.3(b), $Z(s)$ vanishes off $Z'^{(r)}$; on $(Z'_i)^{(r)}$ it depends on ξ and on U' only on $(Z'_i)^{(r)}$, and on s only through $s_\square$, $\square \subset Z'_i$. Every term of $C_0^{1/2}(s)$ carries a monomial of positive degree (Definition 40.2), so $Z(0) = 0$. Hence $v_p(Z(s)) = 0$ unless $B'(p)$ meets $Z'^{(r)}$; in particular $v_p(Z(0)) = 0$. If $B'(p)$ meets $Z'^{(r)}$, then it meets $(Z'_i)^{(r)}$ for exactly one i and lies in $(Z'_i)^{(r+2L)}$, since its diameter is less than $2L$ and $2r' < M/4$. Then $v_p(Z(s))$ depends on s only through $s_\square$, $\square \subset Z'_i$, on ξ only on $(Z'_i)^{(r)}$, and on U' only on $(Z'_i)^{(r+2L)} \subset (Z'_i)^{(r')}$. For a box $\square \subset Z'_0$, every plaquette based in $\square$ has a point of $B'(p)$ in Z'_0, so this holds with $i = 0$. So $V(\square; s) = g^{-2}\sum_p v_p(Z(s))$, the sum over the plaquettes based in $\square$, vanishes at

$s = 0$, depends on s only through the boxes of Z'_0, and depends on ξ and U' only on $(Z'_0)^{(r')}$. This is the hypothesis of Lemma 47.2.

Since $\mathfrak{E}_0 = 0$, there is no old activity, hence no composed activity and no mixed class, and (Q3) concerns only the old activities. The torus hypotheses at $k = 0$ hold by Lemma 52.1. The constants of the first step have their own values: $c_L^{(0)} = 2C_{\mathrm{P}}^{-1}$ in place of c_L (Lemma 52.3(c)), and its own C_T, r_I, r_t, κ' and $M_1^{(0)}$ (Remark 49.8(4)). At complex references, (b) gives the lower bound $\frac{1}{2}C_{\mathrm{P}}^{-1}$, which is better than the bound $\frac{1}{8}c_L^{(0)} = \frac{1}{4}C_{\mathrm{P}}^{-1}$ of Corollary 34.6(c) with $c_L^{(0)}$ in place of c_L. Where the flow uses a constant of the step, it uses the worse of its values at $k = 0$ and at $k \geq 1$: the larger constant in an upper bound, the smaller constant in a lower bound, and the smaller smallness threshold. The rate conditions of Remark 49.8(3) hold at $k = 0$ with the same choice of the weakening rate: at $k = 0$ there is a single walk, Theorem 41.4 for $\mathfrak{A} = \Delta_0 + r$ with $\mathfrak{N} = 0$, and $C_0^{1/2}(s)$ is an r-integral of it with no products, so the rate of the first step is at least $\kappa - 2\delta \geq \kappa'_w$; the norms increase with the rate, and the conditions are lower bounds on it. □

The next lemma is the exact representation of the first step, Theorem 34.9 at $k = 0$. It is the *formula for the first step* of Section 54. Its parts (a) and (b) use only the definitions in Proposition 52.2(a),(b); part (c) uses the lower bound proved in Proposition 52.2(b), whose proof uses (a) and (b).

Lemma 52.3 (The formula for the first step)**.** *Let $k = 0$, so that $\eta = 1$ and $\mathbb{T} = \mathbb{T}^{(0)} = \mathbb{T}_\eta$, let the parameters up to (4) of Remark 49.8 be fixed, and let the torus satisfy (T0)–(T2) at $k = 0$ (Lemma 52.1(a)). There are $a_{\mathrm{step}}^{(0)} > 0$, $r_2^{(0)} > 0$ and C_L, depending only on d, L, G, a, α, M and M', with the following property. Let V' be a real configuration on $\mathbb{T}' = \mathbb{T}^{(1)}$ with curvature at most $\varepsilon'_1 \leq a_{\mathrm{step}}^{(0)}$ in the units of step 1, let $U' = U_1(V')$ be an element of its minimal orbit, and put $\varepsilon = B_3\varepsilon'_1$.*

(a) (Composition.) *Lemma 31.6 holds at $k = 0$, with the base configuration $V^0 = U'$. Thus the base $W = U'$ lies in $\mathfrak{U}_0(L^{-2}(C_* + 1)\varepsilon'_1)$, $\mathfrak{a}'(W) = V'$, and every real $V \in \mathfrak{U}_0(a'_1)$ with $\mathfrak{a}'(V) = V'$ satisfies $S(V) \geq S(U')$, with equality if and only if $V = U'^u$ for some $u \in \mathcal{G}'$.*

(b) (The exponent.) *For complex fields Z on $\mathfrak{F}$ with $|Z|_\infty \leq 2r_2^{(0)}$ put $F(Z) = S(e^{\mathrm{i}\hat{Z}(Z)}U') + \Phi_W(Z)$, with $\hat{Z}$ of Lemma 32.7 and Φ_W of Lemma 32.3(c). Then F is analytic, real for real Z, and, for $|Z|_\infty \leq r_2^{(0)}$,*

$$F(Z) = S(U') + \tfrac{1}{2}\Delta_0(Z, Z) + \mathcal{V}(Z) + G_3(Z), \qquad \mathcal{V}(Z) = \sum_p v_p(Z).$$

Here Δ_0 is the form of Proposition 52.2(b); $v_p(Z) = s_p(e^{\mathrm{i}\hat{Z}(Z)}U') - T_2[s_p(e^{\mathrm{i}\hat{Z}(\cdot)}U')](Z)$ are the interaction pieces of Proposition 52.2(a), with T_2 the Taylor polynomial of degree 2 at $Z = 0$; and G_3 is the remainder of Lemma 32.3(c). In particular F has no linear term, and $|\mathcal{V}(Z)| + |G_3(Z)| \leq C_L|Z|_\infty\|Z\|^2$.

(c) (The covariance.) *Δ_0 is real symmetric on real fields on $\mathfrak{F}$, has range at most $2L + 2$, and*

$$\tfrac{1}{2}C_{\mathrm{P}}^{-1}\|Z\|^2 \leq \Delta_0(Z, Z) \leq C_L\|Z\|^2.$$

So Δ_0 and $C_0 = \Delta_0^{-1}$ satisfy the bounds of Corollary 34.6, with c_L replaced by $c_L^{(0)} = 2C_{\mathrm{P}}^{-1}$ and a decay rate $\gamma_C^{(0)} > 0$ of C_0 depending on L, and the covariance (a), (b) of that corollary, with $\mathfrak{S}_1$ in place of $\mathfrak{S}_{k+1}$. The densities $\ell_0(b) = \mathrm{tr}_{\mathfrak{g}}[(\log \Delta_0)(b, b)]$, $b \in \mathfrak{F}$, with $\log \Delta_0$ defined by the integral of Proposition 34.8, satisfy $\log\det \Delta_0 = \sum_{b\in\mathfrak{F}} \ell_0(b)$ and (a), (b), (d) of that proposition.

(d) (The formula.) *Let $w \le r_2^{(0)}$, and let ρ be a density of the form (31.2) at $k=0$, i.e. $\rho(V) = \exp\big(-g^{-2}S(V) + \mathfrak{E}(V)\big)$ for $V \in \mathfrak{U}_0(\tilde h)$, with a gauge-invariant $\mathfrak{E}$ and $a_1 \ge \tilde h \ge 2(C_*+1)L^{-2}\varepsilon_1' + 4w$, measurable and bounded on the support of χ^{w}. Then*

$$T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(V') = \mathcal{N}_0 \exp\Big(-\frac{1}{g^2}S(U') - \frac{1}{2}\sum_{b\in\mathfrak{F}}\ell_0(b) + \sum_c \log \mathfrak{j}_c\Big) \\ \times \int d\mu_{C_0}(Y)\, \chi^{\mathrm{w}}\big(\hat Z(gY)\big)\, e^{-g^{-2}(\mathcal{V}+G_3)(gY)+\mathfrak{J}(gY)+\mathfrak{E}(e^{\mathrm{i}\hat Z(gY)}U')},$$

where $\mathcal{N}_0$, $\mathfrak{j}_c$ and $\mathfrak{J}$ are given by Theorem 34.9(a), (b) and (e) at $k=0$, with their bounds there, $d\mu_{C_0}$ is the Gaussian measure with covariance C_0, and $g^{-2}|(\mathcal{V}+G_3)(gY)| \le C_L g|Y|_\infty \|Y\|^2$ for $g|Y|_\infty \le r_2^{(0)}$.

In particular, the prefactor of the gauge-fixed first step is $\mathcal{N}_0 \exp\big(-g^{-2}S(U') + \Gamma_0^{\mathrm{step}}(U')\big)$, with

$$\Gamma_0^{\mathrm{step}}(U') = -\tfrac{1}{2}\log\det\Delta_0 + \sum_c \log \mathfrak{j}_c,$$

the one-loop term of the step from scale 0 in Section 54.

Proof. Constants. Choose $a_{\mathrm{step}}^{(0)}$ so small that $\varepsilon_1' \le a_{\mathrm{step}}^{(0)}$ implies: the hypotheses of Lemma 31.6, i.e. $\varepsilon_1' \le \min(a_{\delta=1}, a_1/(2C_*+2))$ and $(C_*+1)\varepsilon_1' \le c_1$; the condition $L^d\mathsf{a} \le c$ of Lemma 32.7, with $\mathsf{a} = 2(C_*+1)\varepsilon_1'$; $4(C_*+1)L^{-2}\varepsilon_1' \le a_1'$; and the smallness conditions on $\varepsilon = B_3\varepsilon_1'$ in the proof of Proposition 52.2(b), $\varepsilon \le \min(\varepsilon_{\mathrm{cmp}}(M'), c_5)$, $C_{M'}\varepsilon \le \frac{1}{8}C_{\mathrm{P}}^{-1}$ and $CL^{3d+1}\varepsilon \le \frac{1}{8}C_{\mathrm{P}}^{-1}$. Let C_w be such that $|\hat Z(Z)|_\infty \le C_w L^d|Z|_\infty$ for $|Z|_\infty \le r_L$ (Lemma 32.7(c),(d)). Choose $r_2^{(0)}$ with $2r_2^{(0)} \le \min(r_L, \frac{1}{2}c_0L^{-1})$, $2C_wL^dr_2^{(0)} \le 1$ and $8C_wL^dr_2^{(0)} \le a_1'$. Recall $r_L \le \frac{1}{8}\varepsilon_M$ and $r_L \le c_0L^{-1}$ (Lemma 32.7, proof of Proposition 32.8). Details

(a) By Theorem 30.3 for the one-step problem, in the units of step 1, U' is a minimizer of S on $\mathfrak{B}_1(V') \cap \mathfrak{U}(\theta_0)$, so $\mathfrak{a}'(U') = U'^{(1)} = V'$. The 0-fold average is the identity, so $V^0 = U'^{(0)} = U'$, and $U_0(V^0) = V^0 = U'$ by the case $k=0$ of Theorem 30.3, where $\mathfrak{B}(V) = \{V\}$; this is Lemma 31.6(b). By Proposition 31.5 with $\delta = 1$, U' has curvature at most $(C_*+1)\varepsilon_1'$ in the units of step 1, hence at most $L^{-2}(C_*+1)\varepsilon_1' \le \varepsilon_1 = 2(C_*+1)L^{-2}\varepsilon_1'$ in the units of step 0; this is Lemma 31.6(a), with Proposition 21.12 replaced by the identity. For Lemma 31.6(c), let V be real with $\mathfrak{a}'(V) = V'$ and $V \in \mathfrak{U}_0(a_1')$. By Theorem 30.3(b) at $k=0$, $U_0(V) = V$ lies in $\mathfrak{U}(B_3a_1')$ in the units of step 0. In the units of step 1 its curvature and current are at most $L^3B_3a_1' \le \theta_0$. So $V \in \mathfrak{B}_1(V') \cap \mathfrak{U}(\theta_0)$, and $S(V) \ge S(U')$ by Theorem 30.3(a) at step 1. Equality holds if and only if V lies in the minimal orbit, i.e. $V = U'^u$ with u real and $u|_{\mathbb{T}'} = 1$, which is $u \in \mathcal{G}'$, since $\mathbb{T}_\eta = \mathbb{T}$.

(b) *Analyticity.* For $|Z|_\infty \le 2r_2^{(0)} \le r_L$, $\hat Z(Z)$ is analytic, with $|\hat Z(Z)|_\infty \le 2C_wL^dr_2^{(0)} \le 1$, and real for real Z (Lemma 32.7(c),(d)). The action is a finite sum of the plaquette terms s_p, which are analytic on G^c-valued configurations (Section 23). So $Z \mapsto S(e^{\mathrm{i}\hat Z(Z)}U')$ is analytic, and real for real Z. Since $2r_2^{(0)} \le \frac{1}{2}c_0L^{-1}$, Φ_W is analytic and real there (Lemma 32.3(c)).

No linear term. Let Z be real with $|Z|_\infty \le r_2^{(0)}$. By Lemma 32.7(c), $V = e^{\mathrm{i}\hat Z(Z)}U'$ is real and satisfies $\mathfrak{a}'(V) = \mathfrak{a}'(W) = V'$. As in the paragraph before (32.2), its curvature is at most $L^{-2}(C_*+1)\varepsilon_1' + 4C_wL^dr_2^{(0)} \le a_1'$. By (a), $S(e^{\mathrm{i}\hat Z(Z)}U') \ge S(U')$. So the real-analytic function $Z \mapsto S(e^{\mathrm{i}\hat Z(Z)}U')$ of real Z has a minimum at $Z=0$, and its differential vanishes there. With $\omega_0 = \nabla_B S(e^{\mathrm{i}B}U')|_{B=0} = J_{U'}$, this is $\langle \omega_0, \hat Z^1 Z\rangle = 0$. Moreover $\Phi_W(0) = 0$ and $d\Phi_W(0) = 0$ (Lemma 32.3(c)).

The quadratic part. Put $f_p(Z) = s_p(e^{\mathrm{i}\hat Z(Z)}U')$ and $\mathcal{S}(B) = S(e^{\mathrm{i}B}U')$, with $\Delta^0 = d^2\mathcal{S}(0)$. Since $\hat Z(tZ) = t\hat Z^1 Z + t^2\hat Z^2(Z) + O(t^3)$, the coefficient of t^2 in $\sum_p f_p(tZ) = \mathcal{S}(\hat Z(tZ))$ is $\frac{1}{2}\Delta^0(\hat Z^1 Z, \hat Z^1 Z) + \langle \omega_0, \hat Z^2(Z)\rangle$. That of $\Phi_W(tZ)$ is $\frac{1}{2}\langle \mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle$, and $\Phi_W - \frac{1}{2}\langle \mathfrak{G}_W\cdot, \mathfrak{G}_W\cdot\rangle =$

G_3. Together with the absence of a linear term this gives $\sum_p T_2[f_p](Z) + \Phi_W(Z) = S(U') + \frac{1}{2}\Delta_0(Z,Z) + G_3(Z)$, which is the decomposition of F, with $v_p = f_p - T_2[f_p]$. The contours of the families $G(y,x)$ consist of inner bonds, on which $\hat{Z}(Z) = Z$, so $\|\mathfrak{G}_W Z\| = \|\mathfrak{G}_W \hat{Z}^1 Z\|$.

The remainder. The term $s_p(e^{\mathrm{i}B}U')$ depends on B on the four bonds of p. By Lemma 32.7(a),(c), $\hat{Z}(Z)(b_0(c)) = \zeta_c(Z)$ depends on Z only on the bonds in $B'(c_-) \cup B'(c_+)$. So f_p depends on Z only on a set N_p of at most C_L bonds of $\mathfrak{F}$, within distance $2L+2$ of p, and every bond lies in N_p for at most C_L plaquettes p. For $|B|_\infty \le 1$ we have $\|e^{\mathrm{i}B(b)}\| \le e$, so $|s_p(e^{\mathrm{i}B}U')| \le C$. Let $|Z|_\infty \le r_2^{(0)}$, put $m_p = \max_{b\in N_p} |Z(b)|$, and let $m_p > 0$ (otherwise $v_p(Z) = 0$). Let Z_p equal Z on N_p and 0 elsewhere, so that $f_p(tZ) = f_p(tZ_p)$. Then $t \mapsto f_p(tZ_p)$ is analytic and bounded by C for $|t| \le R_p = 2r_2^{(0)}/m_p \ge 2$, since $|tZ_p|_\infty \le 2r_2^{(0)}$ there. By Cauchy's estimate its Taylor coefficients of order n are at most CR_p^{-n}, so

$$|v_p(Z)| \le \sum_{n\ge 3} CR_p^{-n} \le 2CR_p^{-3} \le C_L m_p^3.$$

Hence $|\mathcal{V}(Z)| \le C_L \sum_p m_p^3 \le C_L |Z|_\infty \sum_p \sum_{b\in N_p} |Z(b)|^2 \le C_L |Z|_\infty \|Z\|^2$. The bound on G_3 is Lemma 32.3(c), since $|Z|_\infty \le \frac{1}{2}c_0 L^{-1}$, and Φ_W depends only on the inner bonds, which lie in $\mathfrak{F}$.

(c) Δ^0 is the Hessian at $B=0$ of the real function $B \mapsto S(e^{\mathrm{i}B}U')$ of real B, so it is real symmetric. The form $Z \mapsto 2\langle \omega_0, \hat{Z}^2(Z)\rangle$ is a real quadratic form, since $\hat{Z}$ is real for real Z, and $\|\mathfrak{G}_W Z\|^2$ is real symmetric. So Δ_0 is real symmetric. Its range is that of Proposition 52.2(b). For the lower bound, the proof of Proposition 52.2(b) at the real reference U', whose smallness conditions on ε hold by the choice of $a_{\mathrm{step}}^{(0)}$, gives $\Delta^0(\hat{Z}^1 Z, \hat{Z}^1 Z) + \|\mathfrak{G}_W Z\|^2 \ge \frac{3}{4} C_{\mathrm{P}}^{-1} \|Z\|^2$ and $|2\langle \omega_0, \hat{Z}^2(Z)\rangle| \le \frac{1}{8} C_{\mathrm{P}}^{-1} \|Z\|^2$ for real Z. Hence $\Delta_0(Z,Z) \ge \frac{5}{8} C_{\mathrm{P}}^{-1} \|Z\|^2$. For the upper bound: $|\Delta^0(B,B)| \le C\|B\|^2$, by Cauchy's estimate for the plaquette terms as in (b); $\|\hat{Z}^1 Z\| \le CL^d \|Z\|$ (Lemma 32.7(d)); and $\|\mathfrak{G}_W Z\|^2 \le C_L \|Z\|^2$, since $|\mathfrak{G}_W Z(y,x)| \le CL \max |Z|$, the maximum over the bonds of $B'(y)$ (Lemma 22.7(c)). These are the bounds of Corollary 34.6 with $\frac{1}{4}c_L^{(0)} = \frac{1}{2} C_{\mathrm{P}}^{-1}$. Since Δ_0 is bounded and of finite range, it has decay $(C_L e^{(2L+2)\gamma'}, \gamma')$ for every $\gamma' > 0$ (Lemma 5.2(d)), and Lemma 5.5 gives the decay of C_0. The proof of Proposition 34.8(a) applies verbatim to $\Delta_0 + t$, with the lower bound $\frac{1}{2} C_{\mathrm{P}}^{-1} + t$, and gives the decay of $\log \Delta_0$, the identity $\log\det \Delta_0 = \sum_b \ell_0(b)$ and $|\ell_0(b)| \le C_L$.

Covariance. At $k=0$, $W = U'$, and a real gauge transformation u of $\mathbb{T}_\eta = \mathbb{T}$ replaces (W, U') by (U'^u, U'^u), the reference of $V'^{u|_{\mathbb{T}'}}$ (proof of Lemma 32.6(ii)). Since $(e^{\mathrm{i}B}U')^u = e^{\mathrm{i}R(u)B}U'^u$ and S is gauge invariant, Δ^0 and ω_0 are replaced by $R(u)\Delta^0 R(u)^{-1}$ and $R(u)\omega_0$. For $r \in \mathfrak{S}_1$, rU' lies in the minimal orbit of rV', by Theorem 30.3(c) for the one-step problem, as in the proof of Proposition 31.7(b); and $r(e^{\mathrm{i}B}U') = e^{\mathrm{i}r_{U'}B}\, rU'$ by the convention on reversed bonds, as in the proof of Lemma 23.2(b); since $S \circ r = S$, Δ^0 and ω_0 are replaced by $r_{U'}\Delta^0 r_{U'}^{-1}$ and $r_{U'}\omega_0$. These are the conclusions of Proposition 33.2(d),(e) at $k=0$. The rest of the proofs of Corollary 34.6(a),(b) and Proposition 34.8(b),(d), namely the covariance of the elimination and of $\mathfrak{G}_W$, uses only Lemma 21.10(b),(d), Lemma 21.9, the uniqueness in Lemma 32.7(c) and Lemma 22.7(d), which are one-step statements and hold at $k=0$.

(d) Proposition 32.8 holds at $k=0$. Indeed, its proof uses Lemma 31.6 only through (32.1), which holds by (a), since $L^{-2}(C_*+1)\varepsilon_1' \le 2(C_*+1)L^{-2}\varepsilon_1' \le \frac{1}{8}\varepsilon_M$; and it uses Lemmas 32.3 and 32.7, which are one-step statements for the averaging $\mathfrak{a}'$ at the base W and hold at $k=0$ (proof of Proposition 52.2(a)); and $w \le r_2^{(0)} \le r_L$. By (32.2), which holds at $k=0$ by (a), the configuration $e^{\mathrm{i}\hat{Z}(Z)}W$ has curvature at most $2(C_*+1)L^{-2}\varepsilon_1' + 4w \le \tilde{h}$ on the support of χ^{w}, on all bonds, including the $b_0(c)$. There $|Z|_\infty \le w \le r_2^{(0)}$. Since $U_0(V) = V$ and $W = U'$,

$$e^{-g^{-2}\Phi_W(Z)} \rho(e^{\mathrm{i}\hat{Z}(Z)}W) = \exp\Big(-\frac{1}{g^2} F(Z) + \mathfrak{E}(e^{\mathrm{i}\hat{Z}(Z)}U') \Big),$$

with F as in (b). Since $\mathsf{j}_c = L^{-(d-1)\dim\mathfrak{g}}\mathsf{J}_c(0)$ (Theorem 34.9(b) at $k=0$, as in the statement of (d)), write $\mathsf{J}_c(Z) = L^{(d-1)\dim\mathfrak{g}}\mathsf{j}_c \cdot \mathsf{J}_c(Z)/\mathsf{J}_c(0)$ and $\sigma(Z(b)) = \sigma(0)\,\sigma(Z(b))/\sigma(0)$; this gives $\mathfrak{J}$ and its bounds, as in Theorem 34.9(e). By (c), Δ_0 is positive, and

$$\int \prod_{b\in\mathfrak{F}} dZ(b)\, e^{-\frac{1}{2g^2}\Delta_0(Z,Z)} f(Z) = (2\pi g^2)^{|\mathfrak{F}|\dim\mathfrak{g}/2} (\det\Delta_0)^{-1/2} \int d\mu_{C_0}(Y) f(gY).$$

With (b) for F, $\log\det\Delta_0 = \sum_b \ell_0(b)$ from (c), and the constants collected into $\mathcal{N}_0$, Proposition 32.8 gives the formula. The bound on $\mathcal{V}+G_3$ is (b) at $Z=gY$. □

Remark 52.4 (Generation 1). The first step creates the generation of scale 1; there is no generation of scale 0, since $\mathfrak{E}_0 = 0$. By Proposition 52.2 and Lemma 52.3, $e^{E_0}T_{\mathrm{sf}}(\rho_0)(V')$ has the form of Theorem 48.9 with $\mathfrak{E}=0$. Its new terms are the one-loop pieces $\mathfrak{e}(X)$, built from Δ_0 (Proposition 52.2(c)), the Jacobian terms $\log\mathsf{j}_c$, and the pure class $E^{\#}_{\mathrm{p}}$. There are no old activities, hence no composed activities and no mixed class: $E^{\#}_{\mathrm{m}}=0$, and $\delta_0^{\mathrm{m}}=0$ in Proposition 52.5(a). Its one-loop part $\sum_X \mathfrak{e}(X,U') + \sum_c \log\mathsf{j}_c$ is $\Gamma_0^{\mathrm{step}}(U')$ of Lemma 52.3 minus the constant $-\frac{1}{2}\sum_b \ell_0(b;1)$, which goes into $\mathcal{Z}_1$ (Proposition 41.7(a) at $k=0$). These terms, normalized and reblocked, form the raw new generation of the first step (Definition 53.3 with $k=0$). The generation $E^{(1)}$, its coefficient $\beta_1 = \beta_{E^{(1)}}$ and the coupling g_1, with $g_1^{-2} = g_0^{-2} - \beta_1$, are constructed in Proposition 53.8, which also gives $\|E^{(1)}\|_1 \le \mathsf{K}$ and, if $g_1 \le \gamma$, the generation form of ρ_1.

52.3 The vacuum energy

Write the density of step k as

$$\rho_k(V) = \mathcal{Z}_k \exp\Big(-\frac{1}{g_k^2} S(U_k(V)) - \varepsilon_k|\mathbb{T}^{(k)}| + \mathfrak{E}_k(V)\Big), \qquad V \in \mathfrak{U}_k(h_k), \tag{52.1}$$

with $\mathcal{Z}_0 = 1$, $\varepsilon_0 = E_0/|\mathbb{T}^{(0)}|$ and $\mathfrak{E}_0 = 0$. Here $\mathfrak{E}_k$ is a gauge-invariant function, normalized by $\mathfrak{E}_k(1)=0$; in the four-dimensional flow it is the sum of the generations (Definition 53.6). Up to the constant $\mathcal{Z}_k e^{-\varepsilon_k|\mathbb{T}^{(k)}|}$ this is the form (31.2), and since T_{sf} is linear, the constant passes through the step. Let the step $k\to k+1$ be given by Theorem 48.9 (by Proposition 52.2 at $k=0$). Its terms at $U'=1$ are split as follows.

- *Explicit normalizations*: $\mathcal{Z}_{k+1} = \mathcal{Z}_k \mathcal{N}_k N_\chi \exp\big(-\frac{1}{2}\sum_{b\in\mathfrak{F}}\ell_k(b;1)\big)$. These do not depend on the activities.
- *Vacuum energy*: $\varepsilon_{k+1}|\mathbb{T}^{(k+1)}| = \varepsilon_k|\mathbb{T}^{(k)}| - \sum_Y \big(E^{\#}_{\mathrm{p}}(Y,1) + E^{\#}_{\mathrm{m}}(Y,1)\big)$.

The other terms vanish at $U'=1$: $S(1)=0$, $\mathfrak{e}(X,1)=0$, $\log\mathsf{j}_c = 0$ at $U'=1$, and $\mathfrak{E}_k(1)=0$. The remaining terms at $U'=U_{k+1}(V')$, with the values of the $E^{\#}(Y)$ at $U'=1$ subtracted, together with $(g_{k+1}^{-2}-g_k^{-2})S(U')$, form $\mathfrak{E}_{k+1}(V')$, which is normalized. Then $\rho_{k+1} = \mathbf{1}_{\mathfrak{U}_{k+1}(h_{k+1})} T_{\mathrm{sf}}(\rho_k)$ has the form (52.1) at step $k+1$, whatever g_{k+1} is. The choice of g_{k+1} by the extraction of β_{k+1}, and the organization of $\mathfrak{E}_{k+1}$ into generations, are Theorem 53.7 (Proposition 53.8 for $k=0$). Since $|\mathbb{T}^{(k)}| = L^d|\mathbb{T}^{(k+1)}|$,

$$\varepsilon_{k+1} = L^d\varepsilon_k + \delta_k, \qquad \delta_k = \delta_k^{\mathrm{m}} + \delta_k^{\mathrm{p}}, \qquad \delta_k^{\mathrm{m}} = -\frac{\sum_Y E^{\#}_{\mathrm{m}}(Y,1)}{|\mathbb{T}^{(k+1)}|}, \quad \delta_k^{\mathrm{p}} = -\frac{\sum_Y E^{\#}_{\mathrm{p}}(Y,1)}{|\mathbb{T}^{(k+1)}|}. \tag{52.2}$$

In the notation of [15], δ_k^{p} is ε_k^* and δ_k^{m} is $\mathcal{L}_1\mathcal{E}_k$; the latter is the mixed class at $U'=1$, which is not linear in the old activities.

Proposition 52.5 (Vacuum energy)**.** *Let $K\ge 1$, and assume that for every $k<K$ the step $k\to k+1$ is given by Theorem 48.9 (by Proposition 52.2 at $k=0$), with its hypotheses, and with the constants split as above. Then:*

(a) *the δ_k do not depend on E_0, and*

$$|\delta_k^{\mathrm{p}}| \le L^d M^{-d} C_P g_k^2 p_{0,k}^6, \qquad |\delta_k^{\mathrm{m}}| \le 4AL^d M^{-d} \epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n - 1)},$$

with C_P of Proposition 48.8, A of Theorem 48.6 and ϵ_{m} of Proposition 48.5 at step k; $\delta_0^{\mathrm{m}} = 0$;

(b) *there is exactly one E_0 with $\varepsilon_K = 0$. For it, $\varepsilon_k = -\sum_{j=k}^{K-1} L^{-d(j-k+1)}\delta_j$ for $k \le K$, and $E_0 = \varepsilon_0 |\mathbb{T}^{(0)}|$;*

(c) *if $|\delta_{k+m}| \le D_k L^{dm/2}$ for $0 \le m < K - k$, then $|\varepsilon_k| \le 2L^{-d} D_k$.*

Proof. (a) The step is linear in the density, so the constant $\mathcal{Z}_k e^{-\varepsilon_k |\mathbb{T}^{(k)}|}$, and with it E_0, enters neither the activities nor the $E^{\#}(Y)$ of later steps. The grid cubes of step k are the boxes of Part IX (Section 48), and there are $M^{-d}|\mathbb{T}^{(k)}| = L^d M^{-d} |\mathbb{T}^{(k+1)}|$ of them. Every polymer Y contains one, so $\sum_Y |E^{\#}(Y,1)| \le \sum_{\square} \sum_{Y \supset \square} |E^{\#}(Y,1)|$. For the pure class the inner sum is at most $C_P g_k^2 p_{0,k}^6$, by Proposition 48.8; its exponential weight is at least 1, since $(\kappa'_w - 2)/c'_\sharp \ge C_d$ by Remark 49.8(3). For the mixed class it is at most $4A\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n - 1)}$, by Corollary 48.7; its weight is at least 1, since $\kappa_n - 1 \ge C_d$ by Remark 49.8(1). At $k = 0$ there is no mixed class. Details

(b) By (52.2), $\varepsilon_k = L^{dk}\varepsilon_0 + \sum_{j<k} L^{d(k-1-j)}\delta_j$. So $\varepsilon_K = 0$ if and only if $\varepsilon_0 = -\sum_{j<K} L^{-d(j+1)}\delta_j$. By (a) the right side does not depend on E_0, so this fixes $E_0 = \varepsilon_0 |\mathbb{T}^{(0)}|$ uniquely. Substituting gives the formula for ε_k.

(c) By (b), $|\varepsilon_k| \le \sum_{m \ge 0} L^{-d(m+1)} D_k L^{dm/2} = L^{-d} D_k (1 - L^{-d/2})^{-1} \le 2L^{-d} D_k$, since $L^{d/2} \ge 2$. This is [15, Theorem 24, (284)–(288)]. □

For the input of (c), part (a) gives $|\delta_k| \le C(g_k^2 p_{0,k}^6 + g_k p_{0,k})$, the mixed part through $\epsilon_{\mathrm{m}} \le C_P \tau C_{\mathrm{old}} \mathsf{K}$ and $\tau \le C_P L^d g_k p_{0,k}/\delta_1$ (Proposition 53.2(d)); the growth bound needed in (c) is supplied by the flow (Theorem 53.12(f)).

Remark 52.6 (The modified model and a Gaussian reference). (a) *The last step.* A comparison of the action $S(U_K(V))$ with a quadratic form on the small-field set is not available, and it is not used. Small-field configurations on a torus include non-trivial holonomy sectors, and nearly commuting constant modes make the difference of order $|\mathbb{T}^{(K)}| p_K^2$. Bounds on the integral of ρ_K are therefore crude bounds per volume, as in [15] (Corollary 53.13); they suffice because the final torus is fixed.

(b) *A Gaussian reference.* A Gaussian reference in Lie-algebra variables has non-compact zero modes on the last lattice: the directions of the gauge transformations of $\mathbb{T}^{(K)}$, and the harmonic modes. The gauge directions inside the blocks of each step are not zero modes, since the Gaussian weight of the gauge fixing fixes them (Lemma 32.3(c)). And $\mathcal{N}_k$ contains the Haar density $\sigma(0)^{|\mathfrak{F}|}$ and the factor $z(g)^{-|\mathbb{T}\setminus\mathbb{T}'|}$. A reference can be defined as the renormalization group of the quadratic density with the same conventions:

- the measure $\sigma(0)\, dZ$ on $\mathfrak{F}$;
- the quadratic gauge-fixing weight $z(g)^{-|\mathbb{T}\setminus\mathbb{T}'|} e^{-\|\mathfrak{G}_1 Z\|^2/2g^2}$;
- the linearized constraint q^{pl};
- χ_ξ with $C_k(1)$;
- a compact final domain.

Its normalizations are expected to agree with $\mathcal{Z}_k$ term by term; the note does not use this reference. Relating it to the free Gaussian would need $\sum_k |\mathfrak{F}_k| e^{-p_{0,k}^2/2} \le C|\mathbb{T}^{(K)}|$, a condition on the growth of $p_{0,k}$. The explicit form of $\mathcal{Z}_K$ avoids this.

(c) *The modified model.* The densities of the small-field model are $\rho_{k+1} = \mathbf{1}_{\mathfrak{U}_{k+1}(h_{k+1})} T_{\mathrm{sf}}(\rho_k)$, with $T_{\mathrm{sf}}(\rho_k) = T^{\mathrm{gf}}(\chi_k^{\mathrm{w}} \chi_{\xi,k} \rho_k)$ and the cutoffs χ_k^{w}, $\chi_{\xi,k}$ of step k. These depend on the configuration V_{k+1} of step $k+1$ through the base W_k and the reference $U_{k+1}(V_{k+1})$ of the step. Write $V_0 = V$ and $V_{k+1} = \mathfrak{a}_{k+1}(V_k)$, the one-step averaging. Since $\int dV'\, \delta(\mathfrak{a}'(V) V'^{-1}) F(V') = F(\mathfrak{a}'(V))$, unrolling the definitions gives

$$\int \rho_K \, dV_K = \int \prod_b dV(b)\ \rho_0(V) \prod_{k<K} \Big[\mathrm{gf}_{W_k}(V_k)\, \chi_k^{\mathrm{w}}(V_k)\, \chi_{\xi,k}(V_k)\, \mathbf{1}_{\mathfrak{U}_{k+1}(h_{k+1})}(V_{k+1}) \Big],$$

with W_k the base of V_{k+1}. Like gf_{W_k}, the cutoffs χ_k^{w} and $\chi_{\xi,k}$ depend on V_{k+1} as well as on V_k. This defines the modified model: the integral of ρ_0 against $\prod_k [\mathrm{gf}_{W_k} \chi_k^{\mathrm{w}} \chi_{\xi,k}]$ and the small-field conditions, pulled back to the configurations of step 0. The cutoffs are not invariant under the gauge transformations $\mathcal{G}'$ of the blocks, so T^{gf} applied to them does not preserve integrals. So what is discarded at a step is not $T((1-\chi^{\mathrm{w}}\chi_\xi)\rho_k)$. Let first $k \geq 1$. Then ρ_k vanishes outside $\mathfrak{U}_k(h_k)$, and $\mathfrak{U}_k(h_k) \subset \mathfrak{M}$ for γ small, so $\rho_k = \mathbf{1}_{\mathfrak{M} \cap \mathfrak{U}_k(h_k)} \rho_k$. By Lemma 32.6(iv), with the threshold $\tilde{h} = h_k$, the discarded part is

$$T(\rho_k) - T_{\mathrm{sf}}(\rho_k) = T^{\mathrm{gf}}\big((1 - \chi_k^{\mathrm{w}} \chi_{\xi,k}) \mathbf{1}_{\mathfrak{M} \cap \mathfrak{U}_k(h_k)} \rho_k\big) \geq 0,$$

on the set of the V' as in Lemma 31.6, together with all of $T(\rho_k)$ outside that set, where $T_{\mathrm{sf}}(\rho_k) = 0$, and with $(1 - \mathbf{1}_{\mathfrak{U}_{k+1}(h_{k+1})}) T_{\mathrm{sf}}(\rho_k)$, the region of V' outside the small-field set. The hypotheses of Lemma 32.6(iv) are $w \leq \frac{1}{8}\varepsilon_M$ and $2(C_*+1)L^{-2}h_{k+1} + 4w \leq h_k$, which is part of (Q2); ρ_k is gauge invariant and nonnegative, and $\chi_{\xi,k}$ is a covariant cutoff (Section 32). Since T preserves integrals and all terms are nonnegative, $\int \rho_{k+1} \leq \int T(\rho_k) = \int \rho_k$. At $k = 0$ the density ρ_0 is positive on all configurations, also outside $\mathfrak{M}$, where the averaging $\mathfrak{a}'$ need not be defined (Lemma 21.10(a)); so T is applied only to $\mathbf{1}_{\mathfrak{M} \cap \mathfrak{U}_0(h_0)} \rho_0$. By Lemma 32.6(iv), which holds at $k = 0$ (proof of Proposition 52.2(a)),

$$\int \rho_1 \leq \int T_{\mathrm{sf}}(\rho_0) \leq \int T(\mathbf{1}_{\mathfrak{M} \cap \mathfrak{U}_0(h_0)} \rho_0) = \int \mathbf{1}_{\mathfrak{M} \cap \mathfrak{U}_0(h_0)} \rho_0 \leq \int \rho_0;$$

what is discarded at the first step includes the configurations of step 0 outside $\mathfrak{M} \cap \mathfrak{U}_0(h_0)$. Hence $\int \rho_K \, dV_K \leq \int \rho_0 \, dV$: the partition function of the small-field model is at most that of the lattice theory with the same counterterm.

(d) *Not treated.* The large-field stage, which starts from the discarded parts of every step, and the removal of the small-field modification.

53 The small-field flow in generation form

This section puts Sections 50 and 51 into the step of Section 48, and iterates the step. The old generations, each with its counterterm $-\beta_j S$, are grouped into old activities that are bounded uniformly in k (Definition 53.1, Proposition 53.2), and the new terms of the step form a raw generation (Lemma 53.4), whose symmetrization is the new generation and whose coefficient β_{k+1} fixes the next coupling. Theorem 53.7 and Proposition 53.8 show that one step preserves the generation form, under the condition on γ of Remark 53.5. Section 53.5 iterates them over all scales on the given torus: the flow theorem (Theorem 53.12), the continuity in the couplings (Proposition 53.10, Corollary 53.11) and the stability bound (Corollary 53.13).

Setting. Here $d = 4$, and G is compact, connected and simple. The parameters are chosen as in Remark 49.8(1)–(5), with the domain conditions (DC1)–(DC6) of Remark 38.8, and $\delta_1^\flat = \frac{1}{4}\delta_1$. The torus has $n_K \geq n_{\min}$, so (T0)–(T2) hold at every scale $k \leq K$ (Lemma 52.1(a)).

Objects of scale k are read in the units of scale k unless said otherwise. For a domain P of scale k, $\mathcal{U}(P) = \mathcal{U}(P;\theta_r,\delta_1,\delta_0)$ and $\mathcal{U}^\flat(P) = \mathcal{U}(P;\theta_r,\delta_1^\flat,\delta_0)$ are the domains of Section 50, taken at scale k. For a family E indexed by the polymers X of scale j and a number β we write $(E-\beta S)(U) = \sum_X E(X,U) - \beta S(U)$.

Recall from Sections 38–51.

- *Domains.* $\mathcal{U}(P;\theta_r,\delta_1,\delta_0)$ are the graded domains of Definition 38.3, with the domain constants and the conditions (DC1)–(DC6) of Remark 38.8. The raw output of a step is bounded on the domains $\mathcal{U}$, the generations on the smaller domains $\mathcal{U}^\flat$, and a symmetry of the grid maps $\mathcal{U}^\flat(X)$ into $\mathcal{U}(rX)$ (50.5).
- *Generations.* A generation E of scale j is a family of normalized graded activities $E(X,\cdot)$ on $\mathcal{U}^\flat(X)$, indexed by the polymers X of scale j, with the grid covariance (G2), the invariance (G3) of its sum Φ_E under the lattice symmetries, and the covariance (G4) under the symmetries of the grid; its norm is $\|E\|_j = \sup_X e^{\kappa_n d_j(X)}\|E(X)\|$ (Definition 50.1). Its coefficient β_E is defined by $\bar{T}_E = \beta_E \bar{T}_S$ (Proposition 50.13).
- *Old polymers at a later scale k.* They are recent, small or large (Definition 49.5). $\mathcal{X}_j^{\rm sm}(\square)$ is the set of the small polymers of scale j based in the grid cube $\square$; their sum minus $\beta_j S_\square$ is extracted in Proposition 51.5, and the other polymers are bounded in Lemma 51.7.
- *The step.* Theorem 48.9 takes old activities $\mathcal{E}_k(X)$, graded activities with a margin of dependence r_E, that satisfy $\sup|\mathcal{E}_k(X,\cdot)| \le \|\mathcal{E}_k\| e^{-\kappa_n \mathsf{d}(X)}$ (the paragraph before Proposition 48.5). Here they are the grouped activities of Definition 53.1, with $r_E = 0$ (Remark 47.11). The enlarged closure $\hat{Y}$ is that of Remark 49.4.

K is the bound on the generations (Remark 53.5); it is unrelated to the number K of steps, and the two meet in the condition $g_0^{-2} \ge \gamma^{-2} + KC_\beta\mathsf{K}$ of Theorem 53.12.

53.1 The old activities

At scale k an old generation $E^{(j)}$, $j \le k$, is split as in Remark 51.8: the small polymers of each grid cube $\square$ of scale k, compared with $\beta_j S_\square$ (Proposition 51.5), and the other polymers, grouped by their closures (Lemma 51.7).

Definition 53.1 (Old activities)**.** Let $1 \le k \le K$. For $1 \le j \le k$ let $E^{(j)}$ be a generation of scale j with (G1)–(G4) and $\|E^{(j)}\|_j < \infty$, and let $\beta_j = \beta_{E^{(j)}}$ be its coefficient (Proposition 50.13). For a grid cube $\square$ and a polymer Y of scale k let

$$\Psi_\square^{(j)} = \sum_{X\in\mathcal{X}_j^{\rm sm}(\square)} E^{(j)}(X,\cdot) - \beta_j S_\square, \qquad E_Y^{(j),\rm ne} = \sum_{X \text{ not small at scale } k:\ \mathrm{cl}_k X = Y} E^{(j)}(X,\cdot)$$

be the activities of Proposition 51.5 and of Section 51.4 for $E = E^{(j)}$. The *old activities of step k* are the functions

$$\mathcal{E}_k(Y) = \sum_{j=1}^{k} \Big(\mathbf{1}_{\{Y=\square^+\}} \Psi_\square^{(j)} + E_Y^{(j),\rm ne} \Big),$$

indexed by the polymers Y of scale k. Here $\mathbf{1}_{\{Y=\square^+\}}\Psi_\square^{(j)}$ is $\Psi_\square^{(j)}$ if $Y = \square^+$ for a grid cube $\square$ of scale k, and 0 otherwise.

The set $\square^+$ is the union of the 3^4 grid cubes of scale k centred at the points of $c + M\{-1,0,1\}^4$, where c is the centre of $\square$ (Section 49.2). These cubes are distinct, since $n_k \ge 8LM$, and they are $*$-connected; so $\square^+$ is a polymer of scale k. It determines $\square$, the cube of $\square^+$ whose centre is the centre of $\square^+$. So the indicator picks at most one cube $\square$.

For the recent generations, $k-j<m_0$, no polymer of scale j is small at scale k (Definition 49.5). Then $\Psi^{(j)}_{\square}=-\beta_j S_{\square}$, and all polymers of $E^{(j)}$ enter the second term. So the counterterms of the recent generations are carried by the polymers $\square^+$ as well. For the newest generation, $j=k$, the closure $\mathrm{cl}_k X$ is X, and $E^{(k),\mathrm{ne}}_Y=E^{(k)}(Y,\cdot)$. The grouping is necessary, and the counterterm $-\beta_j S_{\square}$ is never separated from the small polymers of $\square$ of the same generation: $\sum_{j\le k}\beta_j$ is not bounded uniformly in k, so the counterterms cannot be old activities of their own (Remark 51.6(c)).

Proposition 53.2 (Old activities)**.** *Let $1\le k\le K$ and $\mathsf{K}>0$. For $1\le j\le k$ let $E^{(j)}$ be a generation of scale j with (G1)–(G4) and $\|E^{(j)}\|_j\le\mathsf{K}$, with $\beta_j=\beta_{E^{(j)}}$, and let $\mathcal{E}_k$ be the old activities of Definition 53.1.*

(a) Each $\mathcal{E}_k(Y)$ is a normalized graded activity at Y on $\mathcal{U}^\flat(Y)=\mathcal{U}(Y;\theta_r,\delta_1^\flat,\delta_0)$, in the units of scale k, real at real configurations.

(b) For every polymer Y of scale k,

$$\sup_{\mathcal{U}^\flat(Y)}|\mathcal{E}_k(Y,\cdot)|\le C_{\mathrm{old}}\,\mathsf{K}\,e^{-\kappa_n d_k(Y)},\qquad C_{\mathrm{old}}=C_{\mathrm{ex}}\Big(m_0+\frac{1}{1-L^{-\alpha}}\Big)e^{15\kappa_n}+2C_{\mathrm{ne}},$$

with C_{ex} of Proposition 51.5 and $C_{\mathrm{ne}}=3^dK_0L^{5m_0}$ of Lemma 51.7. The constant C_{old} depends only on the parameters of Remark 49.8(1)–(5). It does not depend on k, K, the torus, the couplings or K.

(c) Let U be a configuration that lies in $\mathcal{U}^\flat(Y)$ for every polymer Y of scale k; for instance a background field $U_k(V)$, $V\in\mathfrak{U}_k(h)$, at an admissible threshold h. Then

$$\sum_Y\mathcal{E}_k(Y,U)=\sum_{j=1}^k(E^{(j)}-\beta_jS)(U),$$

the sum on the left running over the polymers Y of scale k.

(d) (The hypothesis of the step.) *The family $\mathcal{E}_k$ satisfies the hypothesis on the old activities stated before Proposition 48.5, at step k, with $r_E=0$, $P=Y$, $\mathsf{d}=d_k$, the domains $\mathcal{U}^\flat(Y)$, and $C_{\mathrm{old}}\mathsf{K}$ in place of $\|\mathcal{E}_k\|$. Let $Y'\supset Y$ be a polymer of scale $k+1$, and let U' be the extension of a point of the domain $\mathcal{U}(Y')$ of scale $k+1$ (Section 47, Setting). Under (Q1) and (Q3) at step k, Proposition 47.10 applies to $\mathcal{E}_k(Y)$ at U', with $\zeta=w$, with $\delta_1^\flat$ in place of δ_1 and with the slack $\frac18\delta_1$. Then $\tau\le C_PL^dg_kp_{0,k}/\delta_1$, and $\epsilon_{\mathrm{m}}\le C_P\tau\,C_{\mathrm{old}}\mathsf{K}$ in Proposition 48.5.*

Proof. (a) Let Y be a polymer of scale k. Each $E^{(j),\mathrm{ne}}_Y$ is a normalized graded activity at Y on $\mathcal{U}^\flat(Y)$, by Lemma 51.7(a). It is real at real configurations. Indeed, every X in its sum lies in Y; a real point of $\mathcal{U}^\flat(Y)$ of scale k lies in $\mathcal{U}^\flat(X)$ of scale j, by Remark 38.8(i) with $\delta_1^\flat$ in place of δ_1; and $E^{(j)}(X,\cdot)$ is real at the real points of $\mathcal{U}^\flat(X)$, by (G1). If $Y=\square^+$, each $\Psi^{(j)}_{\square}$ is a normalized graded activity at $\square^+$ on $\mathcal{U}^\flat(\square^+)$, real at real configurations, by Proposition 51.5. So $\mathcal{E}_k(Y)$ is a finite sum of functions of $U|_Y$, each holomorphic on an open set that contains $\{U|_Y:U\in\mathcal{U}^\flat(Y)\}$. It is holomorphic on the intersection of these open sets, and it is invariant under real gauge transformations on $\mathcal{U}^\flat(Y)$, normalized and real at real configurations. Details

(b) Let $U\in\mathcal{U}^\flat(Y)$. By Lemma 51.7(d), $\sum_{j=1}^k|E^{(j),\mathrm{ne}}_Y(U)|\le 2C_{\mathrm{ne}}\mathsf{K}\,e^{-\kappa_nd_k(Y)}$. If Y is not of the form $\square^+$, this is the bound. Let $Y=\square^+$. By Remark 51.6(d),

$$\sum_{j=1}^k|\Psi^{(j)}_{\square}(U)|\le C_{\mathrm{ex}}\mathsf{K}\Big(m_0+\frac{1}{1-L^{-\alpha}}\Big).$$

This includes the recent generations, for which $\Psi^{(j)}_{\square} = -\beta_j S_{\square}$ is bounded by $|\beta_j| \le C_\beta \mathsf{K}$ (Proposition 50.13(b)) and by $|S_{\square}| \le 6 C_s M^4$ on $\mathcal{U}(\square^+) \supset \mathcal{U}^\flat(\square^+)$ (Lemma 51.4(c)); these are the bounds in the proof of Proposition 51.5 for $m < m_0$. The closed cube filled by each cube of $\square^+$ contains a corner of the closed cube filled by $\square$, and a path along 15 edges of the latter visits its 16 corners; so $d_k(\square^+) \le 15$. Hence $\sum_j |\Psi^{(j)}_{\square}(U)| \le C_{\mathrm{ex}}(m_0 + (1-L^{-\alpha})^{-1}) e^{15\kappa_n} \mathsf{K}\, e^{-\kappa_n d_k(\square^+)}$, and (b) follows.

(c) This is the decomposition of Remark 51.8, summed over j. A polymer X of scale $j \le k$ is either small at scale k, and then based in exactly one grid cube $\square$ of scale k, or not small, and then $\mathrm{cl}_k X$ is exactly one polymer of scale k (Section 49.2, Lemma 49.2(c)). In the first case $X \subset \square^+$ (Lemma 49.6(b)), and $U \in \mathcal{U}^\flat(\square^+)$ lies in $\mathcal{U}^\flat(X)$ of scale j; in the second case $X \subset \mathrm{cl}_k X$, and the same holds. So every term $E^{(j)}(X, U)$ is defined, and it occurs exactly once on the left. The functions $S_{\square}$ add up to S, and each $\beta_j S_{\square}$ occurs once, in $\mathcal{E}_k(\square^+)$. All sums are finite. The background fields $U_k(V)$ lie in every $\mathcal{U}^\flat(Y)$ by the paragraph before Definition 50.1, at scale k.

(d) The polymers of scale k are the connected unions of boxes of the paving, which is the polymer grid of scale k at step k (Section 48.2); the rest of the hypothesis is (a) and (b). The set $Y^{(3)}$ lies within distance $3L^{-1} \le 1$ of Y' in the units of step $k+1$, where the extension equals the point of $\mathcal{U}(Y')$ (Remark 38.7). So the paragraph after Proposition 47.10 applies with $X = Y$: the slack is at least $\frac{1}{8}\delta_1$, and $\tau \le C_P L^d g_k p_{0,k}/\delta_1$ for $\zeta = w$. The bound on ϵ_{m} is the definition $\epsilon_{\mathrm{m}} = C_P \tau \|\mathcal{E}_k\|$ of Proposition 48.5, with $\|\mathcal{E}_k\| \le C_{\mathrm{old}} \mathsf{K}$. □

The old activities are of order K per grid cube, uniformly in k; in the step they are small only through the composed activities, whose size τ is proportional to $g_k p_{0,k}$. This is Bałaban's mechanism [19, (2.41)]: the activities are not contracted, and smallness comes from the fluctuation cutoff.

53.2 The new generation of a step

Let $0 \le k < K$, and consider the step from scale k to scale $k+1$, in the units of step k (Sections 47–48; Proposition 52.2 for $k = 0$). Its new terms are:

- the one-loop pieces $\mathfrak{e}(X, \cdot)$, for the polymers X of scale k (Proposition 41.7; at $k = 0$ they are built from Δ_0, Proposition 52.2(c));
- the Jacobian terms $\log \mathfrak{j}_c$, for the positively oriented bonds c of $\mathbb{T}^{(k+1)}$ (Proposition 47.8);
- the outputs $E^{\#}(Y) = E^{\#}_{\mathrm{p}}(Y) + E^{\#}_{\mathrm{m}}(Y)$ of Corollary 48.7, for the polymers Y of scale k; at $k = 0$, $E^{\#}_{\mathrm{m}} = 0$.

The paving of the step is the polymer grid of scale k (Section 48.2), so the polymers of the step are the polymers of scale k. The pieces and the outputs are defined at every reference of Section 47 (Remark 47.13). For a polymer Y' of scale $k+1$ and a point U of the domain $\mathcal{U}(Y')$ of scale $k+1$, the outputs $E^{\#}(Y, U)$ with $\hat{Y} = Y'$ are those of the restricted system of Remark 47.11 for Y', at the extension of U; they do not depend on the extension. Here $\hat{Y}$ is the union of the grid cubes of scale $k+1$ that meet $Y^{(r')}$ (Remark 49.4), and $\bar{X} = \mathrm{cl}_{k+1} X$ (Section 49.3). For a bond c of $\mathbb{T}^{(k+1)}$ let Y'_c be the union of the grid cubes of scale $k+1$ that meet $B'(c_-) \cup B'(c_+)$. A grid cube of scale $k+1$ is a union of blocks of $\mathbb{T}^{(k+1)}$ (Section 49.2), so each block lies in exactly one of them, and two neighbouring blocks lie in equal or face-adjacent ones. Hence Y'_c is a polymer of scale $k+1$ with at most two cubes, and $d_{k+1}(Y'_c) = 0$, since a point of a common face meets both closed cubes. Each polymer X, each bond c and each polymer Y of scale k determines exactly one polymer $\bar{X}$, Y'_c, $\hat{Y}$ of scale $k+1$.

Definition 53.3 (The raw new generation)**.** Let g_k be the coupling of the step, and let $\mathcal{E}_k$ be old activities of step k as in Lemma 53.4 below ($\mathcal{E}_k = 0$ if $k = 0$). The *raw new generation* of

the step from scale k to scale $k+1$ is the family $E' = E'^{(k+1)}[\mathcal{E}_k, g_k]$, indexed by the polymers Y' of scale $k+1$, with

$$E'(Y',U) = \sum_{X\colon \hat{X}=Y'} \mathfrak{e}(X,U) + \sum_{c\colon Y'_c=Y'} \log \mathfrak{j}_c(U) + \sum_{Y\colon \hat{Y}=Y'} \big(E^{\#}(Y,U) - E^{\#}(Y,1)\big)$$

for $U \in \mathcal{U}(Y')$, the domain of scale $k+1$. Here $\mathfrak{e}$, $\log \mathfrak{j}_c$ and $E^{\#}$ are the new terms of the step at the coupling g_k, with the old activities $\mathcal{E}_k$. The sums run over the polymers X and Y of scale k and the positively oriented bonds c of $\mathbb{T}^{(k+1)}$. The family E' is read as a family of scale $k+1$, in the units of scale $k+1$.

The family E' is defined by the step at the coupling g_k, with the cutoffs w and $p_{0,k}$ of step k. It does not involve the coupling g_{k+1} or any threshold of scale $k+1$.

Lemma 53.4 (The raw new generation)**.** *Let $0 \le k < K$, and let E' be the raw new generation of the step from scale k to scale $k+1$. For $k \ge 1$, assume the following. The density ρ of the step is as in Theorem 34.9, at the threshold $\tilde{h} = h_k$ and the coupling $g = g_k$, with $\mathfrak{E}(V) = \sum_Y \mathcal{E}_k(Y, U_k(V))$ for $V \in \mathfrak{U}_k(h_k)$. The old activities $\mathcal{E}_k(Y)$ are indexed by the polymers Y of scale k, and they are normalized graded activities at Y on $\mathcal{U}^\flat(Y)$, real at real configurations, with $\sup_{\mathcal{U}^\flat(Y)} |\mathcal{E}_k(Y,\cdot)| \le \|\mathcal{E}_k\| e^{-\kappa_n d_k(Y)}$. The hypotheses of Theorem 48.9 hold for them at step k, except those that involve the curvature bound ε'_1 of V', namely (Q2) and $\varepsilon'_1 \le a_{\mathrm{step}}$, which are assumed in (c); ϵ_{p} and ϵ_{m} are those of Proposition 48.5. For $k = 0$, let $\rho = \exp(-g_0^{-2} S)$, and assume the hypotheses of Proposition 52.2 in the same sense, with ϵ_{p} of the first step and $\epsilon_{\mathrm{m}} = 0$.*

(a) Each $E'(Y',\cdot)$ is a normalized graded activity at Y' on $\mathcal{U}(Y') = \mathcal{U}(Y'; \theta_r, \delta_1, \delta_0)$, in the units of scale $k+1$, and it is real at real configurations. That is, E' has (G1) with the domains $\mathcal{U}(Y')$ in place of $\mathcal{U}^\flat(Y')$.

(b) With $\|\cdot\|^{\mathrm{raw}}_{k+1}$ of Definition 50.7,

$$\|E'\|^{\mathrm{raw}}_{k+1} \le B_{\mathrm{new}} + C_{\mathrm{new}}(\epsilon_{\mathrm{p}} + \epsilon_{\mathrm{m}}), \qquad B_{\mathrm{new}} = \max\big(1,\ 3^d K_0 L^d C_{\mathfrak{e}} + 2dM^4 C_{\mathrm{jac}}\big).$$

Here $C_{\mathfrak{e}}$ is a constant with $|\mathfrak{e}(X,U)| \le C_{\mathfrak{e}} e^{-(\kappa_2 - 1)|X|}$ for all polymers X and all $U \in \mathcal{U}(X; c_a, c_a, c_a)$ (Proposition 41.7(b)), C_{jac} is the bound C on $|\log \mathfrak{j}_c|$ of Proposition 47.8(c), and K_0 is that of (49.2). The constant C_{new}, given in the proof, depends only on the parameters of Remark 49.8(1)–(5). At $k = 0$ the same holds with the constants of the first step (Proposition 52.2, end of the proof); we write $B^{(0)}_{\mathrm{new}}$ and $C^{(0)}_{\mathrm{new}}$ for them.

(c) Let h' be an admissible threshold with $h' \le a_{\mathrm{step}}$ ($h' \le a^{(0)}_{\mathrm{step}}$ if $k = 0$) such that (Q2) holds with $\varepsilon'_1 = h'$ and $\tilde{h} = h_k$. Let $V' \in \mathfrak{U}_{k+1}(h')$ and $U' = U_{k+1}(V')$. Then $U' \in \mathcal{U}(Y')$ for every polymer Y' of scale $k+1$, and

$$T_{\mathrm{sf}}(\rho)(V') = \mathcal{N}_k N_\chi \exp\Big(-\frac{1}{g_k^2} S(U') - \frac{1}{2}\sum_b \ell_k(b;1) + \sum_Y E^{\#}(Y,1) + \mathfrak{E}(U') + \sum_{Y'} E'(Y',U')\Big),$$

with $\mathfrak{E}(U')$ the function $\mathfrak{E}$ expressed through the background field, as in Theorem 34.9(f); $\mathfrak{E} = 0$ if $k = 0$.

(d) In (c), suppose moreover that $\rho(rV) = \rho(V)$ for all V and all $r \in \mathfrak{S}_{k+1}$, and that $V' \mapsto \mathfrak{E}(U_{k+1}(V'))$ is invariant under $\mathfrak{S}_{k+1}$ on $\mathfrak{U}_{k+1}(h')$. Then E' has (G3) at the threshold h': the function $\Phi_{E'}(V') = \sum_{Y'} E'(Y', U_{k+1}(V'))$ is invariant under $\mathfrak{S}_{k+1}$ on $\mathfrak{U}_{k+1}(h')$. So E' is a raw generation of scale $k+1$ at threshold h' (Definition 50.7).

Proof. Let Y' be a polymer of scale $k+1$ and $U \in \mathcal{U}(Y')$, the domain of scale $k+1$, read in the units of step k. By Remark 38.8(i), for every domain $P \subset Y'$ of scale k, U lies in $\mathcal{U}(P; L^{-2}\theta_r, L^{-2}\delta_1, L^{-1}\delta_0)$ with the same real part, hence in $\mathcal{U}(P; c_a, c_a, c_a)$ by (DC1). For $k \geq 1$ and an old polymer $P = Y \subset Y'$, Proposition 53.2(d) applies at the extension of U, with the slack $\frac{1}{8}\delta_1$; its proof uses only the properties of the old activities assumed here, and (Q1), (Q3). The pieces of Proposition 48.5, the outputs of Corollary 48.7, their bounds and the analyticity statements of Theorem 48.9 use (Q1) and (Q3) only, through Lemma 48.2; (Q2) enters only the identification of Proposition 48.3, i.e. part (c). Details

(a) *Locality.* The piece $\mathfrak{e}(X, \cdot)$ depends only on U on the bonds of $X \subset \bar{X}$ (Proposition 41.7(b)). The term $\log \mathfrak{j}_c$ depends only on U on the bonds of $B'(c_-) \cup B'(c_+) \subset Y'_c$ (Proposition 47.8(c)). The output $E^{\#}(Y)$ with $\hat{Y} = Y'$ depends on the reference only on $Y^{(r')} \subset Y'$ (Theorem 48.6(a) and the collar inclusions of Remark 47.11), where the extension equals U. So $E'(Y', U)$ is a function of $U|_{Y'}$.

Holomorphy. The pieces $\mathfrak{e}(X, \cdot)$ and the terms $\log \mathfrak{j}_c$ are graded activities on the domains $\mathcal{U}(P; c_a, c_a, c_a)$, $P = X$ and $P = B'(c_-) \cup B'(c_+)$ (Propositions 41.7(b) and 47.8(c); at $k = 0$ with Proposition 52.2(c)). By the inclusions above they are holomorphic functions of $U|_{Y'}$ near every point of $\{U|_{Y'} : U \in \mathcal{U}(Y')\}$. For the outputs, let $U_0 \in \mathcal{U}(Y')$, fix a representation (U_r, A_0) of U_0 on $Y'^{(3)}$, and let $U|_{Y'}$ vary near $U_0|_{Y'}$, with the same real part U_r. The extension $\tilde{U} = e^{\mathrm{i}\eta\chi A} U_r$ of Remark 38.7 depends holomorphically on $U|_{Y'}$. By Remark 38.8(ii) it is a complex reference of size $\delta_c < \delta_P$, whose real part is $(M, B_5\theta_r)$-regular with current at most $L^{-3}\theta_r$; at $k = 0$, $\delta_c < \delta_{P,0}$ by (DC4). For $k \geq 1$, Proposition 47.10 applies at $\tilde{U}$ to every old activity $\mathcal{E}_k(Y)$ with $Y \subset Y'$, by the first paragraph; these are the only old activities of the restricted system for Y'. By Theorem 48.9, $E^{\#}_{\mathrm{p}}(Y)$ and $E^{\#}_{\mathrm{m}}(Y)$ are analytic in the reference at such references, and by Remark 47.11 the outputs with $\hat{Y} = Y'$ do not depend on the extension. As in Remark 38.7, $U|_{Y'} \mapsto E^{\#}(Y, U)$ is therefore holomorphic near $U_0|_{Y'}$.

Gauge invariance and normalization. Let u be a real gauge transformation. Then $U^u \in \mathcal{U}(Y')$, with the representation $(U^u_r, R(u)A)$, and its extension is $\tilde{U}^u$, since the cutoff χ is scalar. All terms are invariant under real gauge transformations of the reference (Theorem 48.9). Moreover $\mathfrak{e}(X, 1) = 0$ (Proposition 41.7(b)), $\mathfrak{j}_c = 1$ at the trivial base (Proposition 47.8(b)), and the outputs are normalized by the subtraction of $E^{\#}(Y, 1)$. So $E'(Y', 1) = 0$.

Reality. Let $U \in \mathcal{U}(Y')$ be real, with a representation (U_r, A) on $Y'^{(3)}$. Then A is real. Indeed, $e^{\mathrm{i}\eta A(b)} = U(b)U_r(b)^{-1} \in G$, and $|\eta A(b)| < \delta_0 \leq c_a$, which is small by (DC1) and (DC5); so $\eta A(b) = \ell(e^{\mathrm{i}\eta A(b)}) \in \mathfrak{g}$ (Lemma 21.2(a),(d) and the choice of c_G in Section 21). Since the bounds of Definition 38.3 are strict, there is $t_0 > 1$ such that, for $z \in \mathbb{C}$ with $|z| < t_0$, the configuration U_z equal to $e^{\mathrm{i}\eta z A} U_r$ on the bonds of $Y'^{(3)}$ and to U_r elsewhere lies in $\mathcal{U}(Y')$, with the representation (U_r, zA). The function $\varphi(z) = E'(Y', U_z)$ is holomorphic there, by (a). For real z the configuration U_z is real. For real z near 0 its curvature and current are less than θ_r everywhere, since they are at $z = 0$ and the torus is finite. Then $U_z \in \mathcal{U}(Y')$ with the representation $(U_z, 0)$, and its extension is U_z itself, a real reference of Section 47 (Remark 38.8(ii)). At real references all terms of Theorem 48.9 are real, and so is $E^{\#}(Y, 1)$. So φ is real on an interval of real z around 0, hence $\varphi(z) = \overline{\varphi(\bar{z})}$ for $|z| < t_0$, and $\varphi(1) = E'(Y', U)$ is real.

(b) Let $U \in \mathcal{U}(Y')$.

One-loop pieces. By Proposition 41.7(b) and $|X|^2 e^{-|X|} \leq 1$, a constant $C_{\mathfrak{e}}$ as stated exists; at $k = 0$, by Proposition 52.2(c). Here $|X| = |X|_M$. By (49.2), $d_k(X) \leq c_\sharp(|X|_M - 1)$ with $c_\sharp = \sqrt{d}$. By Remark 49.8(3), $(\kappa_2 - 1)/c_\sharp \geq \kappa_n + \kappa_0 + 2$, with $\kappa_2 = 3^{-d}\kappa'_w$. Hence

$$\sup_{\mathcal{U}(\bar{X})} |\mathfrak{e}(X, \cdot)| \leq C_{\mathfrak{e}} e^{-(\kappa_2 - 1)|X|} \leq C_{\mathfrak{e}} e^{-(\kappa_n + \kappa_0 + 2) d_k(X)},$$

the supremum being over the domain of scale $k+1$ at $\bar{X}$. Lemma 49.3 with $\kappa = \kappa_n + \kappa_0 + 2$,

whose reblocked rate is $L(\kappa_n+1)\ge\kappa_n$, gives

$$\sum_{X:\,\bar X=Y'}|\mathfrak{e}(X,U)|\le 3^dK_0L^dC_{\mathfrak{e}}\,e^{-\kappa_n d_{k+1}(Y')}.$$

Jacobians. By Proposition 47.8(c), $|\log \mathfrak{j}_c(U)|\le C_{\rm jac}$. If $Y'_c=Y'$, then c_- lies in one of the at most two cubes of Y', each of which contains M^4 points of $\mathbb{T}^{(k+1)}$, and c has one of d directions. So at most $2dM^4$ bonds c have $Y'_c=Y'$, and they exist only if $d_{k+1}(Y')=0$. Hence $\sum_{c:\,Y'_c=Y'}|\log\mathfrak{j}_c(U)|\le 2dM^4C_{\rm jac}e^{-\kappa_n d_{k+1}(Y')}$.

Outputs. The bounds of Corollary 48.7 hold at the extension of U and at 1, uniformly, since both are references of the Setting with the slack $\frac18\delta_1$ for the old activities. They hold for the restricted system, whose input pieces are the pieces $E(Z)$ with $Z\subset Y'$, defined and bounded at the extension by Proposition 53.2(d), and 0 otherwise. Since $e^{\lambda(d_k(Y)+c_\natural)}\ge e^{\lambda d_k(Y)}$, they give, for every polymer Y of scale k,

$$|E^\#_{\rm p}(Y)|\le A\epsilon_{\rm p}e^{-\lambda_{\rm p}d_k(Y)},\qquad |E^\#_{\rm m}(Y)|\le 4A\epsilon_{\rm m}e^{c_\natural(\kappa_n-1)}e^{-\lambda_{\rm m}d_k(Y)},$$

with A the constant of Theorem 48.6 (not the field of a representation), $\lambda_{\rm p}=(\kappa'_w-2)/c'_\sharp-C_d\ge\kappa_n+\kappa_0+2$ (Remark 49.8(3)) and $\lambda_{\rm m}=\kappa_n-1-C_d\ge\kappa_0+1$ (Remark 49.8(1)). Each $E^\#(Y)$ depends on the data only on $Y^{(r')}$. Apply the reblocking bound of the paragraph on the rates in Remark 49.8, with $\lambda=\kappa_n+\kappa_0+2$ for the pure class and $\lambda=\lambda_{\rm m}$ for the mixed class. In both cases $\lambda-\kappa_0\le\kappa_n+2$, so the constant there is at most $C_{\rm rb}=3^dK_0L^de^{C(\kappa_n+2)3^d}$, with C of Remark 49.4, and the rate after reblocking is at least κ_n. With the factor 2 for the subtraction of $E^\#(Y,1)$,

$$\begin{aligned}\sum_{Y:\,\hat Y=Y'}|E^\#(Y,U)-E^\#(Y,1)|&\le 2C_{\rm rb}A(\epsilon_{\rm p}+4e^{c_\natural(\kappa_n-1)}\epsilon_{\rm m})e^{-\kappa_n d_{k+1}(Y')}\\&\le C_{\rm new}(\epsilon_{\rm p}+\epsilon_{\rm m})e^{-\kappa_n d_{k+1}(Y')},\qquad C_{\rm new}=8C_{\rm rb}A\,e^{c_\natural(\kappa_n-1)}.\end{aligned}$$

Here $d_{LM}(\hat Y)$ in the units of step k is $d_{k+1}(\hat Y)$ (Section 49.3). Adding the three bounds and taking the supremum over U and Y' gives (b). At $k=0$ there is no mixed class, and the constants are those of the first step.

(c) Since h' is admissible, $U'\in\mathcal{U}^\flat(Y')\subset\mathcal{U}(Y')$ for every Y', by the paragraph before Definition 50.1 at scale $k+1$. The hypotheses of Theorem 48.9 on the thresholds hold with $\varepsilon'_1=h'$: (Q2), $h_k\le a_1$ by admissibility, and $h'\le a_{\rm step}$. So the formula of Theorem 48.9 (Proposition 52.2 at $k=0$) holds at the real critical reference U'. At U' no extension is needed, and the outputs of the restricted systems are those of the full system (Remark 47.11). Each polymer X, each bond c and each polymer Y of scale k occurs in exactly one term $E'(Y',U')$. So

$$\sum_X\mathfrak{e}(X,U')+\sum_c\log\mathfrak{j}_c(U')+\sum_Y E^\#(Y,U')=\sum_{Y'}E'(Y',U')+\sum_Y E^\#(Y,1),$$

which gives (c).

(d) The set $\mathfrak{U}_{k+1}(h')$ is invariant under $\mathfrak{S}_{k+1}$. By the symmetry statement of Theorem 48.9 (Proposition 52.2 at $k=0$), since $\rho\circ r=\rho$, the sum

$$\sum_X\mathfrak{e}(X,U')+\sum_c\log\mathfrak{j}_c(U')+\mathfrak{E}(U')+\sum_Y E^\#(Y,U'),\qquad U'=U_{k+1}(V'),$$

is invariant under $V'\mapsto rV'$, $r\in\mathfrak{S}_{k+1}$, on $\mathfrak{U}_{k+1}(h')$. By (c), it equals $\Phi_{E'}(V')+\mathfrak{E}(U_{k+1}(V'))+\sum_Y E^\#(Y,1)$. The last term does not depend on V', and the second is invariant by hypothesis. So $\Phi_{E'}(rV')=\Phi_{E'}(V')$. The function $\Phi_{E'}$ does not depend on the element $U_{k+1}(V')$ of the minimal orbit, since each $E'(Y',\cdot)$ is invariant under real gauge transformations, by (a). With (a), E' is a raw generation of scale $k+1$ at threshold h'. □

53.3 The constants of the flow

Remark 53.5 (The constants of the flow and the condition on γ). The constants introduced in this section are fixed after the parameters (1)–(5) of Remark 49.8 and before the bound γ of (6), in the following order. This completes the order of Remark 49.8.

(i) The constants B_{new} and $B_{\mathrm{new}}^{(0)}$ of Lemma 53.4(b). They depend only on d, G, a, α, L, M, M', κ, δ and ρ, through the one-loop and Jacobian bounds (Propositions 41.7(b), 47.8(c) and 52.2) and through K_0. They do not depend on the domain constants, on m_0, on the couplings or on the torus. Then

$$\mathsf{K} = 2\max\left(B_{\mathrm{new}}, B_{\mathrm{new}}^{(0)}\right).$$

(ii) The constants C_{old} of Proposition 53.2(b), $C_\beta = 18d^2K_0M^{-2}(\delta_1^\flat)^{-2}$ of Proposition 50.13(b), and C_{new}, $C_{\mathrm{new}}^{(0)}$ of Lemma 53.4(b). They depend on the parameters of Remark 49.8(1)–(5), and not on K, the couplings or the torus.

(iii) The bound $\gamma \le g_p$ on the couplings. For $g \in (0,\gamma]$ put $p = p(g)$, $p_0 = p_0(g)$, $w = C_wL^dC_Tgp_0$, $\tau = C_PL^dgp_0/\delta_1$ (the bound of Proposition 53.2(d)), $\epsilon_{\mathrm{p}} = C_Pgp_0^3$ and $\epsilon_{\mathrm{m}} = C_P\tau C_{\mathrm{old}}\mathsf{K}$; let $\epsilon_{\mathrm{p}}^{(0)}$ be ϵ_{p} with the constants of the first step. The bound γ is so small that, for every $g \in (0,\gamma]$:

(Γ1) $C_\beta\mathsf{K}\gamma^2 \le \frac{3}{4}$;

(Γ2) the conditions of Remark 49.8(6) on the couplings hold, for the steps $k \ge 1$ and for the first step, with the bound $C_{\mathrm{old}}\mathsf{K}$ on the old activities. That is: $\gamma p(\gamma)$ is an admissible threshold, and $\gamma p(\gamma) \le \min(a_{\mathrm{step}}, a_{\mathrm{step}}^{(0)})$; (Q1) and $2g \le r_t$ hold; $\tau < \frac{1}{2}$, which is (Q3); $8C_wL^dC_Tp_0 \le p$; $\epsilon_{\mathrm{p}} \le \frac{1}{2}\epsilon_*$; and $\epsilon_{\mathrm{m}}e^{c_\natural(\kappa_n-1)} \le \frac{1}{4}\epsilon_*$;

(Γ3) $C_{\mathrm{new}}(\epsilon_{\mathrm{p}}+\epsilon_{\mathrm{m}}) \le B_{\mathrm{new}}$ and $C_{\mathrm{new}}^{(0)}\epsilon_{\mathrm{p}}^{(0)} \le B_{\mathrm{new}}^{(0)}$;

(Γ4) $2gp_0(g) \le c_{\mathrm{F}}$, where

$$c_{\mathrm{F}} = \min\Big(\frac{R_P}{2C_wL^dC_T},\ \frac{\delta_1}{2C_PL^d},\ \frac{\delta_1\epsilon_*}{4C_P^2L^dC_{\mathrm{old}}\mathsf{K}\,e^{c_\natural(\kappa_n-1)}}\Big)$$

is the largest number such that the conditions $w \le \frac{1}{2}R_P$ (part of (Q1)), $\tau < \frac{1}{2}$ and $\epsilon_{\mathrm{m}}e^{c_\natural(\kappa_n-1)} \le \frac{1}{4}\epsilon_*$ of (Γ2) hold whenever gp_0 is replaced in w, τ, ϵ_{m} by a number less than c_{F}. It is used only in Proposition 58.6, at complex couplings t with $|t|p_0(g) < c_{\mathrm{F}}$.

Theorem 58.9 adds two conditions, for $p_0(g) = (\log g^{-1})^q$: $\gamma \le e^{-3q}$ and $C_3\gamma^2p_0(\gamma)^6 \le \frac{1}{2}\beta_\infty$, with C_3 of Proposition 58.6 and β_∞ of Theorem 57.1. These constants are fixed before γ as well. *Feasibility.* Since $p_0 \ge 1$, $gp_0 \le gp_0^3$; and $r_t \ge cp_0^{-3}$ with c fixed before γ (Proposition 48.8). So each condition in (Γ1)–(Γ4) follows if γ, $\gamma p(\gamma)$, $\sup_{0<g\le\gamma} gp_0(g)^3$ and $\sup_{0<g\le\gamma} p_0(g)/p(g)$ are small enough, depending on the constants of (i), (ii) and on the parameters of Remark 49.8(1)–(5). These four quantities tend to 0 as $\gamma \to 0$, by the properties of p and p_0 (Section 49.1). The constants in (Γ1)–(Γ4) do not depend on γ, so there is no circularity. Condition (Γ3) is the condition on γ under which the new generation is bounded by K: with Lemma 53.4(b) it gives $\|E'\|_{k+1}^{\mathrm{raw}} \le 2B_{\mathrm{new}} \le \mathsf{K}$.

Consequences. Let γ satisfy (Γ1)–(Γ3).

- If $|\beta| \le C_\beta\mathsf{K}$ and $g \in (0,\gamma]$, then $g^{-2} - \beta \ge g^{-2}(1 - C_\beta\mathsf{K}g^2) \ge \frac{1}{4}g^{-2} > 0$, by (Γ1). So the number $g' > 0$ with $g'^{-2} = g^{-2} - \beta$ satisfies $g' \le 2g$.

- Let $g_k \in (0,\gamma]$, and let $h' \in (0, \min(2h_k, \gamma p(\gamma))]$. Then h' is an admissible threshold with $h' \le a_{\text{step}}$, and (Q2) holds at step k with $\varepsilon_1' = h'$ and $\tilde h = h_k$. Indeed, $2(C_*+1)L^{-2}h' \le 4(C_*+1)L^{-2}h_k \le \frac12 h_k$, by $L^2 \ge 8(C_*+1)$, and $4w \le \frac12 h_k$, by $8C_wL^dC_Tp_0 \le p$. In particular $h' = h_k$ is such a threshold. So is $h' = h_{k+1}$ if $g_{k+1} \in (0,\gamma]$ and $g_{k+1} \le 2g_k$, since then $h_{k+1} \le 2h_k$ (Section 49.1) and $h_{k+1} \le \gamma p(\gamma)$. The same holds at $k=0$, with $a_{\text{step}}^{(0)}$.

53.4 One step

Definition 53.6 (Generation form)**.** Let K be as in Remark 53.5. Let $1 \le k \le K$, and let $g_1, \dots, g_k \in (0,\gamma]$ with $g_{i+1} \le 2g_i$ for $1 \le i < k$, with the thresholds $h_i = p(g_i)g_i$. A density ρ_k on the real configurations on $\mathbb{T}^{(k)}$ has the *generation form at scale k* if there are, for $1 \le j \le k$, generations $E^{(j)}$ of scale j at the thresholds h_j (Definition 50.1) with $\|E^{(j)}\|_j \le \mathsf{K}$, and numbers $\mathcal{Z}_k > 0$ and $\varepsilon_k \in \mathbb{R}$, such that, with $\beta_j = \beta_{E^{(j)}}$ (Proposition 50.13),

$$\rho_k(V) = \mathcal{Z}_k \exp\Big(-\frac{1}{g_k^2} S\big(U_k(V)\big) - \varepsilon_k |\mathbb{T}^{(k)}| + \sum_{j=1}^{k} \big(E^{(j)} - \beta_j S\big)\big(U_k(V)\big)\Big) \qquad \text{for } V \in \mathfrak{U}_k(h_k),$$

and $\rho_k(V) = 0$ for the other configurations V.

The right side is well defined. By Lemma 50.3(c),(d), $U_k(V) \in \mathcal{U}^\flat(X)$ for every polymer X of every scale $j \le k$, the sums do not depend on the element $U_k(V)$ of the minimal orbit, and they are invariant under real gauge transformations and under $\mathfrak{S}_k$. The same holds for $S(U_k(V))$, by Theorem 30.3(c). The form is (52.1) with $\mathfrak{E}_k(V) = \sum_{j\le k}(E^{(j)} - \beta_j S)(U_k(V))$, which is normalized, since $U_k(1) = 1$. The definition does not require the recursion $g_j^{-2} = g_{j-1}^{-2} - \beta_j$ for the couplings; the theorem below uses it to define the next coupling. The first density ρ_0 has no generations and is positive on all configurations; it is treated in Proposition 53.8.

Theorem 53.7 (One step in generation form)**.** *Let $\mathsf{K} = 2\max(B_{\text{new}}, B_{\text{new}}^{(0)})$, and let γ satisfy (Γ1)–(Γ3) of Remark 53.5. Let $1 \le k < K$, let $g_1, \dots, g_k \in (0,\gamma]$ with $g_{i+1} \le 2g_i$ for $1 \le i < k$, and let ρ_k have the generation form at scale k, with generations $E^{(1)}, \dots, E^{(k)}$, coefficients $\beta_j = \beta_{E^{(j)}}$ and constants $\mathcal{Z}_k$, ε_k. Let $\mathcal{E}_k$ be the old activities of Definition 53.1, and let $E' = E'^{(k+1)}[\mathcal{E}_k, g_k]$ be the raw new generation (Definition 53.3) of the step from scale k to scale $k+1$.*

(a) (The step.) *$\rho_k(rV) = \rho_k(V)$ for all V and all $r \in \mathfrak{S}_{k+1}$. Up to the constant $\mathcal{Z}_k e^{-\varepsilon_k|\mathbb{T}^{(k)}|}$, ρ_k is a density as in Theorem 34.9 at the threshold $\tilde h = h_k$, with $\mathfrak{E}(V) = \sum_Y \mathcal{E}_k(Y, U_k(V))$ for $V \in \mathfrak{U}_k(h_k)$. The hypotheses of Theorem 48.9 other than (Q2) hold at step k, with the old activities $\mathcal{E}_k$, $\|\mathcal{E}_k\| \le C_{\text{old}}\mathsf{K}$ and $\epsilon_{\text{m}} \le C_P\tau C_{\text{old}}\mathsf{K}$; (Q2) holds with $\varepsilon_1' = h'$ for every threshold h' as in (b).*

(b) (The raw generation.) *E' has (G1) on the domains $\mathcal{U}(Y')$ of scale $k+1$, and*

$$\|E'\|_{k+1}^{\text{raw}} \le B_{\text{new}} + C_{\text{new}}(\epsilon_{\text{p}} + \epsilon_{\text{m}}) \le \mathsf{K}.$$

For every threshold $h' \in (0, \min(2h_k, \gamma p(\gamma))]$, E' has (G3) at threshold h'; so it is a raw generation of scale $k+1$ at threshold h' (Definition 50.7). The family E' does not depend on g_{k+1} or on any threshold of scale $k+1$.

(c) (The new generation and the coupling.) *The symmetrization $E^{(k+1)} = E'^{\text{sym}}$ (Definition 50.7) is a generation of scale $k+1$ at every threshold h' as in (b). It has (G1)–(G4), and $\|E^{(k+1)}\|_{k+1} \le \mathsf{K}$. The coefficient $\beta_{k+1} = \beta_{E^{(k+1)}}$ of Proposition 50.13 is real, with $|\beta_{k+1}| \le C_\beta\mathsf{K}$. The number $g_{k+1} > 0$ with*

$$g_{k+1}^{-2} = g_k^{-2} - \beta_{k+1}$$

satisfies $g_{k+1} \le 2g_k$.

(d) (Generation form at scale $k+1$.) *Suppose that* $g_{k+1} \le \gamma$. *Then (Q2) holds at step* k *with* $\varepsilon_1' = h_{k+1}$, $E^{(k+1)}$ *is a generation of scale* $k+1$ *at threshold* h_{k+1}, *and*

$$\rho_{k+1} = \mathbf{1}_{\mathfrak{U}_{k+1}(h_{k+1})} T_{\mathrm{sf}}(\rho_k)$$

has the generation form at scale $k+1$, *with the generations* $E^{(1)}, \dots, E^{(k+1)}$ *and the constants*

$$\mathcal{Z}_{k+1} = \mathcal{Z}_k \mathcal{N}_k N_\chi \, e^{-\frac{1}{2}\sum_b \ell_k(b;1)}, \qquad \varepsilon_{k+1}|\mathbb{T}^{(k+1)}| = \varepsilon_k|\mathbb{T}^{(k)}| - \sum_Y \big(E_{\mathrm{p}}^{\#}(Y,1) + E_{\mathrm{m}}^{\#}(Y,1)\big).$$

These are the recursions of Section 52.3, and Proposition 52.5(a) bounds the increment δ_k *of (52.2), with* $\epsilon_{\mathrm{m}} \le C_P \tau C_{\mathrm{old}} \mathsf{K}$.

Proof. (a) By Remark 50.4, $\rho_k(rV) = \rho_k(V)$ for all V and all $r \in \mathfrak{S}_k \supset \mathfrak{S}_{k+1}$; it uses the thresholds h_j of the generations and $g_{i+1} \le 2g_i$. The generations satisfy the hypotheses of Proposition 53.2. By its part (c) at $U = U_k(V)$, $V \in \mathfrak{U}_k(h_k)$, and since h_k is admissible, Details

$$\sum_{j \le k} \big(E^{(j)} - \beta_j S\big)(U_k(V)) = \sum_Y \mathcal{E}_k(Y, U_k(V)) = \mathfrak{E}(V).$$

This function is gauge invariant (Definition 53.6 and the paragraph after it). It is bounded on $\mathfrak{U}_k(h_k)$ (Remark 51.8), and continuous there, hence measurable: near every point, Corollary 30.5 gives a continuous choice of $U_k(V)$, and the sums consist of finitely many holomorphic terms and of S. Moreover $h_k \le a_1$ by admissibility. So $\rho_k/(\mathcal{Z}_k e^{-\varepsilon_k|\mathbb{T}^{(k)}|})$ is a density of the form (31.2) on $\mathfrak{U}_k(h_k)$, as in Theorem 34.9. By Proposition 53.2(a),(b),(d), the old activities $\mathcal{E}_k$ satisfy the hypothesis on old activities of Theorem 48.9, with $\|\mathcal{E}_k\| \le C_{\mathrm{old}}\mathsf{K}$ and the slack $\frac{1}{8}\delta_1$. The step uses the identity between $\mathfrak{E}$ and the old activities only at $s = 1$, at real critical references and for real ξ, where the composed configuration $U_B = e^{\mathrm{i}\eta A_*(1;B)}U'$, $B = \hat{Z}(Z)$, is a real gauge transform of an element $U_k(e^{\mathrm{i}\hat{Z}(Z)}W)$ of the minimal orbit of $e^{\mathrm{i}\hat{Z}(Z)}W$, with $e^{\mathrm{i}\hat{Z}(Z)}W \in \mathfrak{U}_k(h_k)$ on the support of the cutoffs (Remark 50.9(a), Theorem 25.1(c), (Q2)). The transformation need not lie in $\mathfrak{G}_0$. The domains $\mathcal{U}^\flat(Y)$ and the old activities are invariant under real gauge transformations (Definition 38.3, Proposition 53.2(a)), so U_B lies in every $\mathcal{U}^\flat(Y)$ and $\mathcal{E}_k(Y, U_B) = \mathcal{E}_k(Y, U_k(e^{\mathrm{i}\hat{Z}(Z)}W))$. So, by the identity above, applied at $e^{\mathrm{i}\hat{Z}(Z)}W$ and at V^0, where $U' = U_k(V^0)$ and $V^0 \in \mathfrak{U}_k(h_k)$ (Lemma 31.6, (Q2)),

$$\sum_Y f(Y; 1, Z) = \mathfrak{E}(e^{\mathrm{i}\hat{Z}(Z)}W) - \mathfrak{E}(U'),$$

with f of Proposition 47.10, which is the term of $\delta\mathcal{E}(\xi)$ in Proposition 48.3. The hypotheses of Theorem 48.9 on the coupling, (Q1)–(Q3), $2g_k \le r_t$, $\epsilon_{\mathrm{p}} \le \frac{1}{2}\epsilon_*$ and $\epsilon_{\mathrm{m}} e^{c_{\natural}(\kappa_n - 1)} \le \frac{1}{4}\epsilon_*$, hold by ($\Gamma$2) at $g = g_k$; for (Q2) see (b). The conditions on the rates in Proposition 48.5 and Corollary 48.7 are Remark 49.8(1),(3), and (DC5) and $\alpha \le \frac{1}{2}$ hold by Remark 49.8(4),(5).

(b) By (a), Lemma 53.4 applies, with $\|\mathcal{E}_k\| \le C_{\mathrm{old}}\mathsf{K}$. Its parts (a) and (b), with (Γ3), give (G1) on the domains $\mathcal{U}(Y')$ and $\|E'\|_{k+1}^{\mathrm{raw}} \le B_{\mathrm{new}} + C_{\mathrm{new}}(\epsilon_{\mathrm{p}} + \epsilon_{\mathrm{m}}) \le 2B_{\mathrm{new}} \le \mathsf{K}$. Let $h' \in (0, \min(2h_k, \gamma p(\gamma))]$. By Remark 53.5, consequences, h' is admissible, $h' \le a_{\mathrm{step}}$, and (Q2) holds with $\varepsilon_1' = h'$. We check the hypotheses of Lemma 53.4(d). The first is (a). For the second, let $V' \in \mathfrak{U}_{k+1}(h')$ and $U' = U_{k+1}(V')$. Then $S(U_{k+1}(rV')) = S(U')$ for $r \in \mathfrak{S}_{k+1}$, by Theorem 30.3(c) at scale $k+1$. For $j \le k$, apply Lemma 50.3(d) with the later scale $k+1$, with h' in place of h and h_j in place of h'. Its hypothesis on h_j holds: for $j = k$, $2(C_* + 1)L^{-2}h' \le \frac{1}{2}h_k$ by the consequences in Remark 53.5; for $j < k$, Lemma 50.3(c) and

$h' \le 2h_k$ give $2(C_*+1)L^{-2(k+1-j)}h' \le 2L^{-2}h_j \le h_j$. So $V' \mapsto \sum_X E^{(j)}(X, U_{k+1}(V'))$ is invariant under $\mathfrak{S}_{k+1}$ on $\mathfrak{U}_{k+1}(h')$. Moreover $\mathfrak{E}(U')$, the function $\mathfrak{E}$ expressed through the background field (Theorem 34.9(f)), is $\mathfrak{E}(V^0)$, since $U' = U_k(V^0)$ (Lemma 31.6(b)). Here $V^0 \in \mathfrak{U}_k(2(C_*+1)L^{-2}h') \subset \mathfrak{U}_k(h_k)$, by Lemma 31.6(a) and (Q2). So the identity in (a) gives

$$\mathfrak{E}(U') = \sum_{j \le k} (E^{(j)} - \beta_j S)(U'),$$

and $V' \mapsto \mathfrak{E}(U_{k+1}(V'))$ is invariant under $\mathfrak{S}_{k+1}$ on $\mathfrak{U}_{k+1}(h')$. Lemma 53.4(d) gives (G3) at threshold h'. The last statement of (b) is the paragraph after Definition 53.3.

(c) Let h' be as in (b). By (b), E' is a raw generation of scale $k+1$ at threshold h', and (DC6) holds. Lemma 50.8(a) shows that $E^{(k+1)} = E'^{\mathrm{sym}}$ is a generation of scale $k+1$ at threshold h', with (G1)–(G4), and that $\|E^{(k+1)}\|_{k+1} \le \|E'\|^{\mathrm{raw}}_{k+1} \le \mathsf{K}$. By Proposition 50.13(a),(b), β_{k+1} is real and $|\beta_{k+1}| \le C_\beta \mathsf{K}$. By the first consequence in Remark 53.5, with $g = g_k$ and $\beta = \beta_{k+1}$, $g_k^{-2} - \beta_{k+1} \ge \frac{1}{4} g_k^{-2} > 0$, so g_{k+1} is defined and $g_{k+1} \le 2g_k$.

(d) Since $g_{k+1} \in (0, \gamma]$ and $g_{k+1} \le 2g_k$, the threshold $h' = h_{k+1}$ is as in (b) (Remark 53.5, consequences). So (Q2) holds with $\varepsilon'_1 = h_{k+1}$, and, by (c), $E^{(k+1)}$ is a generation of scale $k+1$ at threshold h_{k+1}. Let $V' \in \mathfrak{U}_{k+1}(h_{k+1})$ and $U' = U_{k+1}(V')$. Since T_{sf} is linear, Lemma 53.4(c) with $h' = h_{k+1}$ gives $T_{\mathrm{sf}}(\rho_k)(V')$, with the factor $\mathcal{Z}_k e^{-\varepsilon_k |\mathbb{T}^{(k)}|}$. In its formula, $\mathfrak{E}(U') = \sum_{j \le k}(E^{(j)} - \beta_j S)(U')$, as shown in the proof of (b) with $h' = h_{k+1}$; $\sum_{Y'} E'(Y', U') = \sum_{Y'} E^{(k+1)}(Y', U')$, by Lemma 50.8(b) at scale $k+1$ and threshold h_{k+1}; and $-g_k^{-2} S(U') = -g_{k+1}^{-2} S(U') - \beta_{k+1} S(U')$. With the constants $\mathcal{Z}_{k+1}$ and ε_{k+1} as stated, this gives

$$\rho_{k+1}(V') = \mathcal{Z}_{k+1} \exp\Big(-\frac{1}{g_{k+1}^2} S(U') - \varepsilon_{k+1} |\mathbb{T}^{(k+1)}| + \sum_{j=1}^{k+1} (E^{(j)} - \beta_j S)(U') \Big)$$

for $V' \in \mathfrak{U}_{k+1}(h_{k+1})$, and $\rho_{k+1}(V') = 0$ otherwise. The generations $E^{(j)}$, $j \le k$, are unchanged: they are generations of scale j at the thresholds h_j, with norms at most K. With (c), and since $g_1, \dots, g_{k+1} \in (0, \gamma]$ with $g_{i+1} \le 2g_i$, this is the generation form at scale $k+1$. The constant $\mathcal{Z}_{k+1}$ is positive, and ε_{k+1} is real, since the $E^{\#}(Y, 1)$ are real (Theorem 48.9, at the real critical reference $U' = 1$). Proposition 52.5(a) applies to the step, with $\epsilon_{\mathrm{m}} \le C_P \tau C_{\mathrm{old}} \mathsf{K}$ by (a). □

Proposition 53.8 (The first generation)**.** *Let K and γ be as in Theorem 53.7, let $g_0 \in (0, \gamma]$ and $\varepsilon_0 \in \mathbb{R}$, and let $\rho_0 = \exp(-g_0^{-2} S - \varepsilon_0 |\mathbb{T}^{(0)}|)$, so that $\varepsilon_0 |\mathbb{T}^{(0)}| = E_0$. Then (b)–(d) of Theorem 53.7 hold for $k = 0$, with ρ_0 in place of ρ_k, with the raw new generation E' of the first step (Definition 53.3 with $k = 0$), with $B^{(0)}_{\mathrm{new}}$, $C^{(0)}_{\mathrm{new}}$ and $\epsilon^{(0)}_{\mathrm{p}}$ in place of B_{new}, C_{new} and ϵ_{p}, with $\epsilon_{\mathrm{m}} = 0$, and with $\mathcal{Z}_0 = 1$. Here $E^{(1)} = E'^{\mathrm{sym}}$, $\beta_1 = \beta_{E^{(1)}}$ and $g_1^{-2} = g_0^{-2} - \beta_1$; if $g_1 \le \gamma$, then $\rho_1 = \mathbf{1}_{\mathfrak{U}_1(h_1)} T_{\mathrm{sf}}(\rho_0)$ has the generation form at scale 1, with*

$$\mathcal{Z}_1 = \mathcal{N}_0 N_\chi \, e^{-\frac{1}{2}\sum_b \ell_0(b;1)}, \qquad \varepsilon_1 |\mathbb{T}^{(1)}| = \varepsilon_0 |\mathbb{T}^{(0)}| - \sum_Y E^{\#}_{\mathrm{p}}(Y, 1).$$

Proof. The proof of Theorem 53.7(b)–(d) applies, with the following changes. There are no old activities: $\mathfrak{E} = 0$, there are no composed activities and no mixed class, and (Q3) is void (Proposition 52.2). Part (a) of the theorem is replaced by three facts. First, $e^{\varepsilon_0 |\mathbb{T}^{(0)}|} \rho_0$ is of the form (31.2) with $\mathfrak{E}_0 = 0$ on all configurations, so at the threshold $\tilde{h} = h_0 \le a_1$ (proof of Proposition 52.2). Second, $\rho_0(rV) = \rho_0(V)$ for all V and all $r \in \mathfrak{S}_1$, since $S(rV) = S(V)$: r permutes the plaquettes, and $(rV)(\partial(rp))$ is conjugate to $V(\partial p)^{\pm 1}$ (Section 23; proof of Lemma 51.4(b)). Third, the hypotheses of Proposition 52.2, (Q1), (Q2) and $2g_0 \le r_t$ at $k = 0$, hold by (Γ2) and the consequences in Remark 53.5, with $\varepsilon'_1 = h'$ for every threshold $h' \in (0, \min(2h_0, \gamma p(\gamma))]$; in particular $h' \le a^{(0)}_{\mathrm{step}}$. So Lemma 53.4 applies at $k = 0$, with the constants of the first step, and the second hypothesis of its part (d) is void. In (d), $\mathcal{Z}_0 = 1$ and $E^{\#}_{\mathrm{m}} = 0$ give the constants $\mathcal{Z}_1$ and ε_1. □

Details

Remark 53.9 (On Theorem 53.7). (a) *The symmetrization.* The properties (G2) and (G4) of $E^{(k+1)}$ hold by construction, so they need no further check on the step, and the flow carries $E^{(k+1)}$, not E', from scale $k+1$ on (Remark 50.9(a)).

(b) *No circularity in g_{k+1}.* The order is E', $E^{(k+1)}$, β_{k+1}, g_{k+1}, h_{k+1}, and (G3) for E' holds already at the threshold h_k.

(c) *Free coupling sequences.* Parts (a)–(c) do not use the recursion $g_j^{-2} = g_{j-1}^{-2} - \beta_j$; so they define the generations for free coupling sequences (Proposition 53.10(a)).

53.5 The flow over all scales

This subsection iterates Theorem 53.7. Throughout, $\mathsf{K} = 2\max(B_{\mathrm{new}}, B_{\mathrm{new}}^{(0)})$, γ satisfies (Γ1)–(Γ3) of Remark 53.5, and $n_K \geq n_{\min}$, as in the Setting. For $0 \leq k \leq K$ let Σ_k be the set of the sequences $\pi = (g_0, \dots, g_k) \in (0, \gamma]^{k+1}$ with $g_{i+1} \leq 2g_i$ for $0 \leq i < k$.

Proposition 53.10 (Free coupling sequences and continuity)**.** *Let $0 \leq k < K$ and $\pi = (g_0, \dots, g_k) \in \Sigma_k$. Let $E^{(1)} = E'^{\mathrm{sym}}$ with $E' = E'^{(1)}[0, g_0]$, and, for $1 \leq j \leq k$, $E^{(j+1)} = E'^{\mathrm{sym}}$ with $E' = E'^{(j+1)}[\mathcal{E}_j, g_j]$ (Definition 53.3), where $\beta_i = \beta_{E^{(i)}}$ and $\mathcal{E}_j$ are the old activities of step j built from the $E^{(i)}$ and β_i with $i \leq j$ (Definition 53.1). We write $E^{(j)}[\pi]$ and $\beta_j[\pi]$ when the dependence on the couplings matters. The couplings are the entries of π; they are not computed from the β_j.*

(a) (Free sequences.) *The recursion is defined. For $1 \leq j \leq k+1$, $E^{(j)}$ is a generation of scale j at every threshold $h' \in (0, \min(2h_{j-1}, \gamma p(\gamma))]$, with (G1)–(G4) and $\|E^{(j)}\|_j \leq \mathsf{K}$; for $j \leq k$ these thresholds include h_j. The coefficient β_j is real, with $|\beta_j| \leq C_\beta \mathsf{K}$. Both $E^{(j)}$ and β_j depend only on $(g_0, \dots, g_{j-1})$. For $k \geq 1$, $E^{(k+1)}$ is the new generation that Theorem 53.7(c) produces from a density in generation form at scale k with the generations $E^{(1)}, \dots, E^{(k)}$, the coupling g_k and any constants $\mathcal{Z}_k > 0$, $\varepsilon_k \in \mathbb{R}$; for $k = 0$, $E^{(1)}$ is that of Proposition 53.8, for any $\varepsilon_0 \in \mathbb{R}$.*

(b) (Continuity.) *For every polymer X of scale $k+1$ and every $U \in \mathcal{U}^\flat(X)$ of scale $k+1$, the functions $\pi \mapsto E^{(k+1)}[\pi](X, U)$ and $\pi \mapsto \beta_{k+1}[\pi]$ are continuous on Σ_k.*

Proof. (a) By induction on j. For $j = 1$ this is Proposition 53.8 at the coupling g_0, i.e. part (c) of Theorem 53.7 at $k = 0$. Let $1 \leq j \leq k$, and assume the claims for $E^{(1)}, \dots, E^{(j)}$. For $1 \leq i \leq j$, $g_i \leq 2g_{i-1}$ gives $h_i \leq 2h_{i-1}$ (Section 49.1), and $h_i \leq \gamma p(\gamma)$; so $E^{(i)}$ is a generation at the threshold h_i. Let $\mathcal{Z}_j > 0$ and $\varepsilon_j \in \mathbb{R}$. The density of Definition 53.6 with the generations $E^{(1)}, \dots, E^{(j)}$, the coupling g_j and the constants $\mathcal{Z}_j$, ε_j is defined, and has the generation form at scale j, since the conditions of that definition on $g_1, \dots, g_j$ hold for $\pi \in \Sigma_k$. Theorem 53.7(a)–(c) applies at $k = j$; these parts do not use the recursion for the couplings (Remark 53.9(c)). By its part (c), $E^{(j+1)}$ is a generation of scale $j+1$ at every threshold $h' \in (0, \min(2h_j, \gamma p(\gamma))]$, with (G1)–(G4) and norm at most K, and β_{j+1} is real with $|\beta_{j+1}| \leq C_\beta \mathsf{K}$. The raw generation $E'^{(j+1)}[\mathcal{E}_j, g_j]$ is determined by g_j and the old activities $\mathcal{E}_j$ (Definition 53.3 and the paragraph after it), and $\mathcal{E}_j$ is determined by $E^{(1)}, \dots, E^{(j)}$ and $\beta_1, \dots, \beta_j$. So $E^{(j+1)}$ and β_{j+1} depend only on $(g_0, \dots, g_j)$, and not on $\mathcal{Z}_j$, ε_j; this is the last statement for $j = k$.

Details

(b) Let $\pi_0 \in \Sigma_k$. Besides dominated convergence and the Weierstrass test, we use one elementary fact (∗). Let $\mathcal{A} \subset \mathbb{C}^n$ be open, and let f_π, $\pi \in \Sigma_k$, be holomorphic on $\mathcal{A}$ with $|f_\pi| \leq B$, such that $\pi \mapsto f_\pi(A)$ is continuous at π_0 for every $A \in \mathcal{A}$. Then $f_\pi(A_\pi) \to f_{\pi_0}(A_0)$ whenever $\pi \to \pi_0$ and $A_\pi \to A_0 \in \mathcal{A}$: if the closed polydisc of radius $2r$ about A_0 lies in $\mathcal{A}$, Cauchy's estimates give $|f_\pi(A) - f_\pi(A_0)| \leq nB|A - A_0|_\infty / r$ on the polydisc of radius r, uniformly in π. The continuity of p_0 (Section 49.1) is used only in Steps 2 and 3, through the white-noise cutoff, which is sharp. We argue by induction on k. By (a), $E^{(k+1)}[\pi]$ depends on

π through g_k and, for $k \geq 1$, through the old activities $\mathcal{E}_k$, which are built from the $E^{(j)}[\pi]$ and $\beta_j[\pi]$ with $j \leq k$. For $k \geq 1$ we assume that (b) holds with $j-1$ in place of k, for $1 \leq j \leq k$.

Step 1: the old activities ($k \geq 1$). Let Y be a polymer of scale k, and let U_r be a real configuration with $|F_{U_r}|, |J_{U_r}| < \theta_r$. Let $\mathcal{A}^\flat(Y; U_r)$ be the set of the $\mathfrak{g}^c$-valued fields A on the bonds of $Y^{(3)}$ with $|A| < \delta_0$ and $|\nabla A|_{Y^{(3)}} + [\nabla A]_{\alpha, Y^{(3)}} < \delta_1^\flat$, with ∇ taken with respect to U_r. It is open, since the torus is finite and the bounds are strict. For $A \in \mathcal{A}^\flat(Y; U_r)$ let U_A equal $e^{\mathrm{i}\eta A} U_r$ on the bonds of $Y^{(3)}$ and U_r elsewhere; then $U_A \in \mathcal{U}^\flat(Y)$. So $f_\pi(A) = \mathcal{E}_k[\pi](Y, U_A)$ is holomorphic on $\mathcal{A}^\flat(Y; U_r)$ (Definition 38.3), with $|f_\pi| \leq C_{\mathrm{old}} \mathsf{K}$ (Proposition 53.2(b)). By Definition 53.1, $f_\pi(A)$ is a finite sum of values $E^{(j)}[\pi](X, U_A)$, with $X \subset Y$ of scale $j \leq k$, and of numbers $\beta_j[\pi] S_\square(U_A)$; and U_A lies in $\mathcal{U}^\flat(X)$ of scale j, by Remark 38.8(i) with $\delta_1^\flat$ in place of δ_1. By the induction hypothesis, $\pi \mapsto f_\pi(A)$ is continuous on Σ_k. By (∗), $(\pi, A) \mapsto \mathcal{E}_k[\pi](Y, U_A)$ is continuous on $\Sigma_k \times \mathcal{A}^\flat(Y; U_r)$.

Step 2: the pieces. Let g_k^0 be the last entry of π_0. Fix a polymer Y' of scale $k+1$ and a point $U \in \mathcal{U}(Y')$ of scale $k+1$, or $U = 1$, and let $\tilde{U} = e^{\mathrm{i}\eta A'} U_r$ be its extension (Section 47, Setting), which does not depend on π. For $\pi \in \Sigma_k$ consider the pieces $E_{\mathrm{p}}(Z)$ and $E_{\mathrm{m}}(Z)$, $Z \subset Y'$, of the restricted system at $\tilde{U}$ (Proposition 48.5, Remark 47.11), as functions of the white noise ξ with $|\xi|_\infty \leq p_0(g_k)$. Fix ξ with $|\xi|_\infty < p_0(g_k^0)$. Since p_0 is continuous, $|\xi|_\infty < p_0(g_k)$ for π near π_0, so the pieces at π, which depend on ξ only on the bonds of $Z^{(r')}$, are defined at ξ. We show that they converge to those at π_0. By Lemma 47.2, each piece is a finite sum of terms $f_{\mathrm{loc}}(Y_0, Z) = \int_{[0,1]^S} \partial_{s_S} f(s(Z,t))\, dt$, with S the set of boxes of $Z \setminus Y_0$ and f one of the functions localized in Proposition 48.5. The coordinates of $s(Z,t)$ lie in $[0,1]$, and the circles of radius 1 about them lie in $\mathbb{D}_{\kappa'}$, since $e^{\kappa'} \geq e > 2$. Cauchy's formula on these circles writes $f_{\mathrm{loc}}(Y_0, Z)$ as an integral of f over a compact set, against a bounded kernel that does not depend on π. The functions f are bounded uniformly in π, by the bounds of Section 47, Lemma 48.2 and (Γ2) at g_k. By dominated convergence, it suffices to show that $\pi \mapsto f(s)$ is continuous at π_0 for every $s \in \mathbb{D}_{\kappa'}$. Put $Z_\pi(s) = g_k C_k^{1/2}(s) \xi$, with $C_k^{1/2}(s)$ at the reference $\tilde{U}$ (Corollary 41.6); it does not depend on π otherwise, and $|Z_\pi(s)|_\infty \leq g_k C_T p_0(g_k) \leq \frac{1}{2} r_I$ by (Q1).

- The interaction, gauge-fixing and Jacobian functions of a box $\square$ are $g_k^{-2} \sum_p v_p(s; Z_\pi(s))$, $g_k^{-2} \sum_{y,x} \varphi_{y,x}(Z_\pi(s))$ and $\mathfrak{J}_\square(Z_\pi(s))$, with the sums over the plaquettes and the pairs (y,x) of $\square$ (Section 47). Here $v_p(s;\cdot)$, $\varphi_{y,x}$ and the terms of $\mathfrak{J}$ are analytic for $|Z|_\infty < r_I$ (Propositions 47.5(a), 47.7(a) and 47.8(a)), and do not depend on π. So these functions are continuous in π.

- The composed activities are $f(Y; s, Z_\pi(s)) = \mathcal{E}_k[\pi](Y, e^{\mathrm{i}\eta A_*(s; \hat{Z}(Z_\pi(s)))} \tilde{U}) - \mathcal{E}_k[\pi](Y, \tilde{U})$, for the old polymers $Y \subset Y'$. On the bonds of $Y^{(3)}$ the composed configuration is $e^{\mathrm{i}\eta A_\pi} U_r$, with $A_\pi = A_*(s; \hat{Z}(Z_\pi(s))) \oplus A'$ combined bondwise by the Baker–Campbell–Hausdorff formula, and $A_\pi \in \mathcal{A}^\flat(Y; U_r)$ (proof of Proposition 47.10(a), applied with $\delta_1^\flat$ in place of δ_1 as in Proposition 53.2(d)); likewise $A' \in \mathcal{A}^\flat(Y; U_r)$. The maps $A_*(s;\cdot)$ and $\hat{Z}$ at the reference $\tilde{U}$ do not depend on π, and they are continuous. So $A_\pi \to A_{\pi_0}$. Since $\mathcal{E}_k(Y)$ depends only on the bonds of Y, Step 1 gives the continuity of $f(Y; s, Z_\pi(s))$ at π_0.

At $k = 0$ there are no composed activities, and the pieces of the first step are of the same form, with $A_*(s; B)$ replaced by B (Proposition 52.2(a)).

Step 3: the outputs. The outputs $E^\#(Y) = E^\#_{\mathrm{p}}(Y) + E^\#_{\mathrm{m}}(Y)$ of the restricted system, $\hat{Y} = Y'$, are the outputs of Theorem 48.6 for the system of all its pieces (Corollary 48.7 at $\lambda = 1$). By (Γ2), this system satisfies the hypothesis of Theorem 48.6 with $\kappa = \kappa_n - 1 \geq C_d$ and $\epsilon = \epsilon_{\mathrm{p}} + \epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n - 1)} \leq \epsilon_*$, for every $\pi \in \Sigma_k$. By the proof of that theorem, $E^\#(Y) = \sum_{\mathcal{C}} \phi^T(\mathcal{C}) \prod_{\Gamma \in \mathcal{C}} k(\Gamma)$, the sum running over the clusters $\mathcal{C}$ with $\bigcup_{\Gamma \in \mathcal{C}} \operatorname{supp} \Gamma = Y$, with $k(\Gamma) = \int \prod_{Z \in \Gamma} K(Z)\, d\mu^*$ and $K(Z) = e^{E(Z)} - 1$. The measure $d\mu^*$ is the product over the bonds $b \in \mathfrak{F}$ of the standard Gaussian measure on $\mathfrak{g}$, restricted to $|\xi(b)| \leq p_0(g_k)$ and normalized (Proposition 48.3). So $k(\Gamma)$ is an integral over the finitely many bonds of $\mathfrak{F}$ in

$\bigcup_{Z\in\Gamma} Z^{(r')}$, with respect to the standard Gaussian measure, of $\prod_b N_1(p_0(g_k))^{-1}\mathbf{1}(|\xi(b)|\le p_0(g_k))\prod_{Z\in\Gamma}K(Z)$, where $N_1(p)$ is the standard Gaussian mass of $\{|\xi(b)|\le p\}$, continuous in p, and $N_1(p_0)\ge N_1(1)>0$. As $\pi\to\pi_0$ this integrand converges at every ξ with $|\xi(b)|\neq p_0(g_k^0)$ for all these bonds: by Step 2 if $|\xi|_\infty<p_0(g_k^0)$, and since the cutoff vanishes eventually otherwise. The remaining set has measure zero, and the integrand is bounded by $N_1(1)^{-n_\Gamma}(2\epsilon_*)^{|\Gamma|}$, with n_Γ the number of these bonds. By dominated convergence, $k(\Gamma)$ is continuous at π_0. The proof of Theorem 48.6 bounds $|k(\Gamma)|$ by $\bar k(\Gamma)=\prod_{Z\in\Gamma}2\epsilon_* e^{-\kappa(\mathsf{d}(Z)+c_\natural)}$, which does not depend on π, and verifies (48.1) for the activities $\bar k$. So the Kotecký–Preiss theorem gives $\sum_{\mathcal C}|\phi^T(\mathcal C)|\prod_{\Gamma\in\mathcal C}\bar k(\Gamma)<\infty$, the sum running over the clusters that contain an object incompatible with a box of Y. By the Weierstrass test, $E^\#(Y)$ at $\tilde U$ is continuous at π_0.

Step 4: the generation. By Definition 53.3,

$$E'[\pi](Y',U)=\sum_{X\colon \bar X=Y'}\mathfrak{e}(X,U)+\sum_{c\colon Y'_c=Y'}\log \mathfrak{j}_c(U)+\sum_{Y\colon \hat Y=Y'}(E^\#(Y,\tilde U)-E^\#(Y,1)),$$

a finite sum. The one-loop pieces $\mathfrak{e}(X,\cdot)$ and the Jacobian terms $\log\mathfrak{j}_c$ do not depend on π: they are functions of the reference, defined through the fluctuation operator Δ_k (Δ_0 at $k=0$) and the elimination at the base (Propositions 41.7, 47.8(b) and 52.2(b)), which involve neither the coupling, nor the cutoff, nor the old activities. So, by Step 3, $\pi\mapsto E'[\pi](Y',U)$ is continuous on Σ_k, for every polymer Y' of scale $k+1$ and every $U\in\mathcal U(Y')$. Let X be a polymer of scale $k+1$ and $U\in\mathcal U^\flat(X)$. By (50.5), each term of $E^{(k+1)}(X,U)=|\mathfrak S^{\mathrm{grid}}_{k+1}|^{-1}\sum_r E'(r^{-1}X,r^{-1}U)$ is a value of E' at a point of $\mathcal U(r^{-1}X)$. This gives the continuity of $\pi\mapsto E^{(k+1)}[\pi](X,U)$.

Step 5: the coefficient. Let $F^*=F^{12,a}$ be as in the proof of Proposition 50.13(b). By that proof and Lemma 50.12(c), $\beta_{k+1}=2\bar T_{E^{(k+1)}}(F^*)=2n_{k+1}^{-4}\sum_X c^X(F^*)$, a finite sum over the polymers X of scale $k+1$ that are not wrapping. With the origin at the lift of a point of X, $t\mapsto E^{(k+1)}(X,V^X_{t,F^*})$ is holomorphic for $|t|<t_X=\delta_1^\flat/(dD_X)$ and bounded by $\mathsf K$ there (Lemma 50.12(a), $|F^*|=1$). So $c^X(F^*)=(2\pi\mathrm i)^{-1}\oint_{|t|=t_X/2}E^{(k+1)}(X,V^X_{t,F^*})\,t^{-3}\,dt$. By Step 4 and dominated convergence, $c^X(F^*)$, and hence β_{k+1}, is continuous at π_0. □

Corollary 53.11 (Continuity in the bare coupling)**.** *Let $g_0\in(0,\gamma]$. Define g_j by $g_j^{-2}=g_{j-1}^{-2}-\beta_j[(g_0,\dots,g_{j-1})]$, for as long as $g_1,\dots,g_{j-1}\in(0,\gamma]$. For $0\le k\le K$ let $\mathsf B_k$ be the set of the $g_0\in(0,\gamma]$ for which $g_1,\dots,g_k$ are defined and lie in $(0,\gamma]$; so $\mathsf B_0=(0,\gamma]$. Let $1\le k\le K$.*

(a) For $g_0\in\mathsf B_{k-1}$, $(g_0,\dots,g_{k-1})\in\Sigma_{k-1}$, g_k is defined, $g_k\le 2g_{k-1}$, and $|g_j^{-2}-g_0^{-2}|\le jC_\beta\mathsf K$ for $j\le k$. The set $\mathsf B_k$ contains the interval $(0,(\gamma^{-2}+kC_\beta\mathsf K)^{-1/2}]$.

(b) The map $g_0\mapsto g_k$ is continuous on $\mathsf B_{k-1}$. Hence $\mathsf B_k=\{g_0\in\mathsf B_{k-1}:g_k\le\gamma\}$ is closed in $(0,\gamma]$, and $g_0\mapsto(g_1,\dots,g_k)$ is continuous on $\mathsf B_k$.

Proof. (a) By induction on k. Let $g_0\in\mathsf B_{k-1}$. For $1\le i<k$, $g_0\in\mathsf B_{i-1}$, so $g_i\le 2g_{i-1}$ by (a) at i; hence $(g_0,\dots,g_{k-1})\in\Sigma_{k-1}$. So $|\beta_k|\le C_\beta\mathsf K$ by Proposition 53.10(a), and the first consequence in Remark 53.5 gives $g_{k-1}^{-2}-\beta_k\ge\frac14 g_{k-1}^{-2}>0$; so g_k is defined, and $g_k\le 2g_{k-1}$. Moreover $g_j^{-2}-g_0^{-2}=-\sum_{i\le j}\beta_i$ has modulus at most $jC_\beta\mathsf K$. If $g_0^{-2}\ge\gamma^{-2}+kC_\beta\mathsf K$, the same induction gives $g_j^{-2}\ge g_0^{-2}-jC_\beta\mathsf K\ge\gamma^{-2}$, i.e. $g_j\le\gamma$, for $j\le k$; so $g_0\in\mathsf B_k$. Details

(b) By induction on k; assume (b) at every $i<k$. On $\mathsf B_{k-1}$ the map $g_0\mapsto(g_0,\dots,g_{k-1})\in\Sigma_{k-1}$ is continuous, since each g_i, $i<k$, is continuous on $\mathsf B_{i-1}\supset\mathsf B_{k-1}$; and β_k is continuous on Σ_{k-1}, by Proposition 53.10(b) with $k-1<K$ in place of k. So $g_k=(g_{k-1}^{-2}-\beta_k)^{-1/2}$ is continuous on $\mathsf B_{k-1}$, since $g_{k-1}^{-2}-\beta_k\ge\frac14 g_{k-1}^{-2}>0$. Hence $\mathsf B_k=\{g_0\in\mathsf B_{k-1}:g_k\le\gamma\}$ is closed in $\mathsf B_{k-1}$, which is closed in $(0,\gamma]$, and $g_0\mapsto(g_1,\dots,g_k)$ is continuous on $\mathsf B_k$. □

Theorem 53.12 (The four-dimensional small-field flow)**.** *Let $d=4$, let G be compact, connected and simple, let the parameters be chosen as in Remark 49.8, with the constants of Remark 53.5(i),(ii) and with γ satisfying (Γ1)–(Γ3), and let $n_K\ge n_{\min}$, so that the*

torus satisfies (T0)–(T2) at every scale $k \le K$ (Lemma 52.1(a)). Let $g_0 \in (0,\gamma]$, $E_0 \in \mathbb{R}$ and $\rho_0 = \exp(-g_0^{-2}S - E_0)$. Let $E^{(j)}$, β_j and g_j be the generations, coefficients and couplings of Proposition 53.10 and Corollary 53.11, i.e. of the recursion of Proposition 53.8 and Theorem 53.7, and let $\rho_j = \mathbf{1}_{\mathfrak{U}_j(h_j)} T_{\mathrm{sf}}(\rho_{j-1})$ as long as $g_1,\dots,g_j \le \gamma$. Let $1 \le k \le K$ with $g_1,\dots,g_k \le \gamma$. Then (a)–(e) hold at k, whatever the later couplings; (f) holds if $g_1,\dots,g_K \le \gamma$, in particular if $g_0^{-2} \ge \gamma^{-2} + KC_\beta \mathsf{K}$ (Corollary 53.11(a)).

(a) (Generation form; [18, (0.22)–(0.24)].) *For $V \in \mathfrak{U}_k(h_k)$, with $U_k = U_k(V)$,*

$$\rho_k(V) = \mathcal{Z}_k \exp\Big(-\frac{1}{g_k^2} S(U_k) - \varepsilon_k |\mathbb{T}^{(k)}| + \sum_{j=1}^{k} \big(E^{(j)} - \beta_j S\big)(U_k) \Big),$$

and $\rho_k(V) = 0$ for the other configurations V. The density ρ_k is invariant under real gauge transformations and under $\mathfrak{S}_k$.

(b) (Creation scale; [18, (0.25)].) *Each $E^{(j)}$, $j \le k$, is a generation of scale j at the threshold h_j, with (G1)–(G4), and $\|E^{(j)}\|_j \le \mathsf{K}$: $\sup_{\mathcal{U}^\flat(X)} |E^{(j)}(X,\cdot)| \le \mathsf{K} e^{-\kappa_n d_j(X)}$ for every polymer X of scale j, on the domains of scale j, whose constants do not depend on the coupling.*

(c) (The coefficients.) *The period density of $E^{(j)}$ is exactly isotropic,*

$$\bar{T}_{E^{(j)}}(F) = \beta_j \bar{T}_S(F) = \tfrac{1}{2}\beta_j \sum_{\mu<\nu} \mathrm{tr}\,(F_{\mu\nu}^2)$$

for all curvature data F in $\mathfrak{g}^c$, with no topological and no anisotropic part. The number β_j is real, and $|\beta_j| \le C_\beta \mathsf{K}$.

(d) (Later scales; [18, (0.28)–(0.30)].) *Let $j \le k$ and $m = k - j$. For every grid cube $\square$ of scale k and every $U \in \mathcal{U}^\flat(\square^+)$ of scale k,*

$$\Big| \sum_{X \in \mathcal{X}_j^{\mathrm{sm}}(\square)} E^{(j)}(X,U) - \beta_j S_\square(U) \Big| \le \begin{cases} C_{\mathrm{ex}} \mathsf{K}\, L^{-\alpha m}, & m \ge m_0, \\ C_{\mathrm{ex}} \mathsf{K}, & m < m_0, \end{cases}$$

the other polymers satisfy the bound of Lemma 51.7(a), and the old activities that of Proposition 53.2(b). For every $V \in \mathfrak{U}_k(h_k)$,

$$\Big| \sum_{j=1}^{k} \big(E^{(j)} - \beta_j S\big)(U_k(V)) \Big| \le C_{\mathrm{lat}} \mathsf{K}\, |\mathbb{T}^{(k)}|,$$

$$C_{\mathrm{lat}} = M^{-4}\Big(C_{\mathrm{ex}} \Big(m_0 + \frac{1}{1 - L^{-\alpha}} \Big) + 2K_0 L^{5m_0} \Big).$$

(e) (Couplings.) *For $1 \le j \le k$, $|g_j^{-2} - g_{j-1}^{-2}| = |\beta_j| \le C_\beta \mathsf{K}$ and $\frac{3}{4} g_{j-1} \le g_j \le 2 g_{j-1}$. The step from scale $j-1$ to scale j satisfies the hypotheses of Theorem 48.9 (of Proposition 52.2 if $j = 1$), with the old activities $\mathcal{E}_{j-1}$, and (Q1)–(Q3) with $\varepsilon_1' = h_j$.*

(f) (Vacuum.) *$\mathcal{Z}_0 = 1$, $\varepsilon_0 |\mathbb{T}^{(0)}| = E_0$, and for $0 \le k < K$*

$$\mathcal{Z}_{k+1} = \mathcal{Z}_k \mathcal{N}_k N_\chi\, e^{-\frac{1}{2}\sum_b \ell_k(b;1)}, \qquad \varepsilon_{k+1} = L^4 \varepsilon_k + \delta_k, \qquad |\delta_k| \le C_{\mathrm{vac}} (g_k^2 p_{0,k}^6 + g_k p_{0,k}),$$

with δ_k of (52.2), which does not depend on E_0. There is exactly one E_0 with $\varepsilon_K = 0$. For this E_0, $|\varepsilon_k| \le C'_{\mathrm{vac}} g_k p_{0,k}$ for $0 \le k \le K$; in particular $|E_0| \le C'_{\mathrm{vac}} g_0 p_0(g_0) |\mathbb{T}^{(0)}|$.

Here C_β and C_{ex} are the constants of Proposition 50.13(b) and Proposition 51.5, and C_{vac}, C'_{vac} are given in the proof. All of them depend only on the parameters of Remark 49.8(1)–(5) (C'_{vac} also on the function p_0), and not on k, K, N, the torus or the couplings.

Proof. Induction. Let $1 \le k \le K$ with $g_1, \dots, g_k \le \gamma$. We show by induction on $k' \le k$ that $\rho_{k'}$ has the generation form at scale k' (Definition 53.6), with the generations $E^{(1)}, \dots, E^{(k')}$, the couplings $g_1, \dots, g_{k'} \in (0, \gamma]$, $g_{i+1} \le 2g_i$ for $0 \le i < k'$, and the constants $\mathcal{Z}_{k'}$, $\varepsilon_{k'}$ of the recursions in (f). For $k' = 1$, apply Proposition 53.8 with $\varepsilon_0 = E_0/|\mathbb{T}^{(0)}|$. It gives $g_1 \le 2g_0$, and, since $g_1 \le \gamma$, the generation form of ρ_1 with $\mathcal{Z}_1$ and ε_1. Let $1 \le k' < k$, and assume the claim at k'. Then Theorem 53.7 applies to $\rho_{k'}$, and it produces $E^{(k'+1)}$ and $\beta_{k'+1}$ (Proposition 53.10(a)). Its part (c) gives $g_{k'+1} \le 2g_{k'}$. Since $g_{k'+1} \le \gamma$, its part (d) gives the generation form of $\rho_{k'+1}$ at scale $k'+1$, with $\mathcal{Z}_{k'+1}$ and $\varepsilon_{k'+1}$. The couplings after g_k do not enter. The torus hypotheses at all scales are those of the Setting. Details

(a) This is the generation form. The invariance is the paragraph after Definition 53.6 and Remark 50.4.

(b) The generations of Definition 53.6 are generations at the thresholds h_j with norm at most K, and (G1)–(G4) hold by Theorem 53.7(c) and Proposition 53.8. The domain constants are those of Remark 38.8.

(c) Proposition 50.13(a),(b), with $\|E^{(j)}\|_j \le \mathsf{K}$, and Lemma 50.12(d) for $\bar{T}_S$.

(d) The first bound is Proposition 51.5 for $E = E^{(j)}$. Let $V \in \mathfrak{U}_k(h_k)$. Since h_k is admissible, $U_k(V)$ lies in $\mathcal{U}^\flat(Y)$ for every polymer Y of scale k, in particular for $Y = \square^+$ (the paragraph before Definition 50.1, at scale k). So Remark 51.8 applies at $U = U_k(V)$; there are $|\mathbb{T}^{(k)}|/M^4$ grid cubes of scale k, and its bound is the last one.

(e) The recursion holds by construction, and $|\beta_j| \le C_\beta \mathsf{K}$ by (c). The upper bound $g_j \le 2g_{j-1}$ was shown in the induction. By (Γ1) and $g_{j-1} \le \gamma$,

$$g_j^{-2} \le g_{j-1}^{-2} + C_\beta \mathsf{K} \le (1 + C_\beta \mathsf{K} \gamma^2) g_{j-1}^{-2} \le \tfrac{7}{4} g_{j-1}^{-2},$$

so $g_j \ge (\frac{4}{7})^{1/2} g_{j-1} \ge \frac{3}{4} g_{j-1}$. The hypotheses of the step are Theorem 53.7(a),(d) for $j \ge 2$. For $j = 1$ they are the third fact in the proof of Proposition 53.8, with $h' = h_1 \le \min(2h_0, \gamma p(\gamma))$.

(f) The recursions for $\mathcal{Z}_k$ and ε_k are those of Theorem 53.7(d) and Proposition 53.8, i.e. those of Section 52.3. By (e) at every $k \le K$, the hypotheses of Proposition 52.5 hold. By its part (a), the δ_k do not depend on E_0, $|\delta_k^{\mathrm{p}}| \le L^4 M^{-4} C_P g_k^2 p_{0,k}^6$, and $|\delta_k^{\mathrm{m}}| \le 4AL^4 M^{-4} \epsilon_{\mathrm{m}} e^{c_{\natural}(\kappa_n - 1)}$, with $\delta_0^{\mathrm{m}} = 0$. By Theorem 53.7(a) and Proposition 53.2(d), $\epsilon_{\mathrm{m}} \le C_P \tau C_{\mathrm{old}} \mathsf{K}$ with $\tau \le C_P L^4 g_k p_{0,k} / \delta_1$. This gives the bound on δ_k with

$$C_{\mathrm{vac}} = L^4 M^{-4} \max\left(C_P,\ 4AC_P^2 L^4 C_{\mathrm{old}} \mathsf{K}\, \delta_1^{-1} e^{c_{\natural}(\kappa_n - 1)}\right),$$

where C_P is the largest of the constants so denoted in Propositions 48.8, 48.5 and 53.2(d), at $k = 0$ and at $k \ge 1$ (Section 47, Setting). Proposition 52.5(b) gives the unique E_0 with $\varepsilon_K = 0$.

Put $\bar{p}_5 = \sup_{0 < g \le g_p} g p_0(g)^5$. It is finite. Indeed, $g^{1/5} p_0(g) \to 0$ as $g \to 0$ (Section 49.1), so $g p_0(g)^5 \le 1$ for $g \le g'$ with some $g' > 0$; and $g p_0(g)^5 \le g_p p_0(g')^5$ for $g' \le g \le g_p$, since p_0 is decreasing (49.1). So $g^2 p_0^6 = g p_0 \cdot g p_0^5 \le \bar{p}_5\, g p_0$, and $|\delta_j| \le C_{\mathrm{vac}}(1 + \bar{p}_5) g_j p_{0,j}$ for $j < K$. Let $0 \le k < K$ and $0 \le m < K - k$. If $g_{k+m} \le g_k$, then $g_{k+m} p_0(g_{k+m}) \le g_k p_0(g_k)$, since $g p_0(g)$ is increasing. Otherwise $p_0(g_{k+m}) \le p_0(g_k)$, since p_0 is decreasing (both by (49.1)), and $g_{k+m} \le 2^m g_k$ by (e). In both cases

$$g_{k+m} p_{0,k+m} \le 2^m g_k p_{0,k} \le L^{2m} g_k p_{0,k} = L^{dm/2} g_k p_{0,k}.$$

So the hypothesis of Proposition 52.5(c) holds with $D_k = C_{\mathrm{vac}}(1 + \bar{p}_5) g_k p_{0,k}$, and $|\varepsilon_k| \le 2L^{-4} D_k$. This is the bound with $C'_{\mathrm{vac}} = 2L^{-4} C_{\mathrm{vac}}(1 + \bar{p}_5)$; for $k = K$, $\varepsilon_K = 0$. Finally $E_0 = \varepsilon_0 |\mathbb{T}^{(0)}|$. □

The next corollary bounds the partition function of the small-field model per volume. It adapts the argument of [15] (Remark 52.6(a)), with the bound of Theorem 53.12(d) in place of a small activity norm.

Corollary 53.13 (Small-field stability)**.** *In the situation of Theorem 53.12, with* $g_1, \dots, g_K \le \gamma$*, put*

$$I_K = \int_{\mathfrak{U}_K(h_K)} \exp\Big(-\frac{1}{g_K^2} S(U_K(V)) + \sum_{j=1}^{K} \big(E^{(j)} - \beta_j S\big)(U_K(V))\Big) \prod_b dV(b),$$

the integral over the configurations V *on* $\mathbb{T}^{(K)}$*, with the normalized Haar measure on each positively oriented bond. It does not depend on* E_0*. The partition function of the small-field model is*

$$\int \rho_K(V) \prod_b dV(b) = \mathcal{Z}_K e^{-\varepsilon_K |\mathbb{T}^{(K)}|} I_K,$$

which is $\mathcal{Z}_K I_K$*, with the explicit product* $\mathcal{Z}_K$ *of normalizations, for the* E_0 *with* $\varepsilon_K = 0$*. Moreover*

$$e^{-c|\mathbb{T}^{(K)}|} \le I_K \le e^{c|\mathbb{T}^{(K)}|}, \qquad c = C_{\mathrm{lat}}\mathsf{K} + 3(C_*+1)^2 p(g_K)^2 + 4\dim\mathfrak{g}\,\log\frac{1}{c_V h_K},$$

with C_{lat} *of Theorem 53.12(d),* C_* *of Proposition 31.5, and* $c_V \in (0,1]$ *depending only on* G*. The constant* c *depends on* g_K*, through* $p(g_K)$ *and* $h_K = p(g_K)g_K$*, and on the parameters, but not on* N*,* K*, the torus or* $g_0, \dots, g_{K-1}$*.*

Proof. The identity is Theorem 53.12(a) at $k = K$. The generations and the couplings do not depend on E_0, by Proposition 53.10(a) and Corollary 53.11, so neither does I_K. Let $V \in \mathfrak{U}_K(h_K)$, $U = U_K(V)$, and use the units of scale K; the fine lattice has spacing $\eta = L^{-K}$, and $|\mathbb{T}^{(K)}| = n_K^4$. Details

- *Activities.* By Theorem 53.12(d) at $k = K$, $|\sum_{j\le K}(E^{(j)} - \beta_j S)(U)| \le C_{\mathrm{lat}}\mathsf{K}|\mathbb{T}^{(K)}|$.
- *Action.* For real U every plaquette term is $\mathrm{Re}\,\mathrm{tr}(1 - U(\partial p))$ (Section 23), which is nonnegative, since the eigenvalues of $U(\partial p)$ have modulus 1; so $S(U) \ge 0$. Since h_K is admissible, Proposition 31.5 with $\delta = 1$ gives $U \in \mathfrak{U}_0((C_*+1)h_K)$. So for every plaquette p of $\mathbb{T}_\eta$, $Y_p = \ell(U(\partial p)) \in \mathfrak{g}$ satisfies $|Y_p| \le \eta^2(C_*+1)h_K$, and $s_p(U) = \mathrm{tr}(1 - \cos Y_p) \le \frac{1}{2}\mathrm{tr}(Y_p^2) = \frac{1}{2}|Y_p|^2$, since $1 - \cos x \le \frac{1}{2}x^2$ for the eigenvalues of Y_p. There are $6(n_K/\eta)^4$ positively oriented plaquettes. Hence $g_K^{-2}S(U) \le 3(C_*+1)^2 h_K^2 g_K^{-2}|\mathbb{T}^{(K)}| = 3(C_*+1)^2 p(g_K)^2|\mathbb{T}^{(K)}|$.
- *Volume.* The Haar measure of $\mathfrak{U}_K(h_K)$ is at most 1. For the lower bound let σ be the density of the Haar measure in exponential coordinates (Section 32.5), and choose $c_V' \le \frac{1}{8}c_G$ such that $\sigma \ge \frac{1}{2}\sigma(0)$ on $\{|Z| \le c_V'\}$. Let V be a configuration on $\mathbb{T}^{(K)}$ with $|\ell(V(b))| \le c_V' h_K$ for every bond b. By Lemma 21.2(b), $|\ell(V(\partial p))| \le 4c_V' h_K \le h_K$, and $|V(\partial p) - 1| \le 8c_V' h_K \le c_G$ by Lemma 21.2(a); since the spacing of $\mathbb{T}^{(K)}$ is 1, the curvature of V is at most h_K, and $V \in \mathfrak{U}_K(h_K)$. By Lemma 21.2(d), the set of the $u \in G$ with $|\ell(u)| \le c_V' h_K$ is $\{e^{\mathrm{i}Z} : Z \in \mathfrak{g},\ |Z| \le c_V' h_K\}$, whose Haar measure is at least $\frac{1}{2}\sigma(0)\omega(c_V' h_K)^{\dim\mathfrak{g}}$, with ω the volume of the unit ball of $\mathfrak{g}$. With $c_V = c_V' \min(1, (\frac{1}{2}\sigma(0)\omega)^{1/\dim\mathfrak{g}})$ this is at least $(c_V h_K)^{\dim\mathfrak{g}}$. There are $4|\mathbb{T}^{(K)}|$ positively oriented bonds, so the Haar measure of $\mathfrak{U}_K(h_K)$ is at least $(c_V h_K)^{4\dim\mathfrak{g}|\mathbb{T}^{(K)}|}$.

Here $h_K \le 1$, since admissible thresholds are less than $\theta_r \le c_a \le 1$, and so $\log(c_V h_K)^{-1} \ge 0$. The three bounds give $I_K \le e^{C_{\mathrm{lat}}\mathsf{K}|\mathbb{T}^{(K)}|}$ and

$$I_K \ge \exp\Big(-C_{\mathrm{lat}}\mathsf{K}|\mathbb{T}^{(K)}| - 3(C_*+1)^2 p(g_K)^2 |\mathbb{T}^{(K)}|\Big)(c_V h_K)^{4\dim\mathfrak{g}|\mathbb{T}^{(K)}|} = e^{-c|\mathbb{T}^{(K)}|}. \qquad \square$$

Remark 53.14 (On Theorem 53.12). (a) *Bałaban's theorems.* The theorem is the analogue, for the small-field model on the given torus, of Bałaban's Theorem 3 [18], with his hypothesis $0 < g_k \le \gamma$ for $k \le K$; no asymptotic behaviour of the couplings is assumed [18,

p. 259]. Item (d) is his representation (0.28), with the bounds (0.29) and their consequence (0.30). His (0.29) bounds each term of scale j in (0.28) by $O(1)(L^j\eta)^{4+\alpha}e^{-\kappa d_j(X)}$, where $L^k\eta = 1$ in the units of scale k; summed over the $L^{4(k-j)}$ grid cubes of scale j in a grid cube of scale k, this is the gain $L^{-\alpha(k-j)}$ of (d). His Theorem 1 asserts (0.22)–(0.24) with terms satisfying (0.29), but (0.29) is not among the inductive assumptions (1.1)–(1.22) of his Theorem 3. Two features of his inductive assumptions are not reproduced. First, his activities and his coefficients β_{j+1} are smooth, or analytic, functions of a single coupling on $[0,\gamma]$ [18, pp. 263–264]. Here $E^{(j)}$ and β_j depend on the sequence $(g_0,\dots,g_{j-1})$, and only their continuity is proved (Proposition 53.10). Second, his small-field domain has a fixed constant ε_0 [18, p. 259], and his fluctuation cutoff [18, (2.9)] has either a fixed constant ε_1 or, as an alternative that he also carries through, a bound depending on the coupling. Here both the threshold $h_k = p(g_k)g_k$ and the white-noise bound $p_0(g_k)$ depend on the coupling, as in [15]. So the small-field domain of the note shrinks with the coupling, unlike Bałaban's. The domains of (b) and (d) are fixed, as in [18, (1.11)–(1.16)].

(b) *Part XI.* The coefficient β_j is defined by the period density of the symmetrized generation, and (c) holds. Its comparison, at fixed couplings and as the torus grows, with the coefficient that Bałaban's definition [18, (1.20)–(1.22)] assigns to the generations is Proposition 58.8, which rests on Section 59. The hypotheses (F1)–(F3) of Theorem 58.9 are proved in Proposition 58.6: (F1) rests on Proposition 53.10, (F2) on the torus independence of the one-loop parts of the generations (Lemma 58.3), and (F3) on a parity argument.

(c) *The vacuum.* The order $g_k p_{0,k}$ in (f) comes from the mixed class. The parity argument in the proof of (F3) of Proposition 58.6, applied to $\sum_Y E^{\#}(Y,1)$, would give $|\delta_k| = O(g_k^2 p_{0,k}^6)$ under (Γ4); it is not needed here.

(d) *The torus and the model.* All bounds are uniform in N and K, given $n_K \geq n_{\min}$; wrapping polymers are bounded at all later scales by Lemma 52.1(c)–(e). The flow is that of the small-field model (Remark 52.6(c)); the large-field stage is not treated.

Part XI: The one-loop coefficient in four dimensions

The flow of Part X bounds its coefficients, $|\beta_j| \leq C_\beta \mathsf{K}$, with K the bound on the generations (Remark 53.5), but gives no sign and no lower bound for them (Theorem N(iii)), and it assumes no asymptotic behaviour of the couplings (Remark 53.14(a)). The ultraviolet behaviour is the statement that $g_k^{-2} - g_K^{-2}$ grows like $(K-k)\beta_\infty$, with $\beta_\infty > 0$. Part XI computes the one-loop part of the coefficient, summed over the steps, and transfers it to the flow. As in Part X, the statements about couplings concern the small-field flow on a fixed torus; nothing is claimed about large fields or a continuum limit.

Why the one-loop coefficient suffices. The coefficient β_{j+1} of the flow has two parts (Section 58).

- Its one-loop part is the coefficient of the one-loop term $\Gamma_j^{\text{step}} = -\frac{1}{2}\log\det\Delta_j + \sum_c \log \mathfrak{j}_c$ of the step from scale j, a function of the background alone (Section 54, paragraph "The one-loop terms of the steps"), up to an error exponentially small in the side of the torus at scale $j+1$. This is (F2) of Proposition 58.6; it rests on the torus independence of the one-loop pieces (Proposition 41.14, Lemma 58.3).

- The rest of β_{j+1} is of order $g_j^2 p_0(g_j)^6$, by parity in the coupling; this is (F3).

So the couplings follow the one-loop recursion up to errors that add up to a power of $\log(K-k+2)$.

What this Part proves.

- *One function* (Sections 54 and 55). The one-loop term $\Gamma_k^{(1)}$ of the composite k-step map, given by Laplace's method (Proposition 54.3), is the sum of the Γ_j^{step} up to constants (Proposition 54.6). By the symmetry of Bałaban's averaging, the marginal tensors of these functions are multiples of $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$, with no term linear in p (Lemma 54.9). In four dimensions the change of scale preserves marginal tensors, so the tensors of the steps add up to that of $\Gamma_k^{(1)}$ (Corollary 54.18). In the Landau chart, $\Gamma_k^{(1)}$ is given by the determinants of $\mathbf{\Delta}_*$, of $\mathbf{Q}_k \mathbf{G}_* \mathbf{Q}_k^T$ and of a ghost operator (Theorem 55.4).
- *The computation* (Sections 56 and 57). Uniformly in k, the marginal tensor of the vacuum polarization at $U=1$ equals that of the lattice Feynman-gauge operators up to $O(1)$: the averaging adds operators of finite rank per cell, or raises the power-counting degree (Proposition 56.8, Lemmas 56.6 and 56.7). The Feynman-gauge coefficient is $k\beta_\infty + O(1)$, with $\beta_\infty = \frac{11 C_A}{24\pi^2 T_\rho} \log L$ (Theorem 57.1). Hence $\sum_{j<k} T_{j+1}^{(1)} = (k\beta_\infty + O(1))(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$ (Theorem 58.1).
- *The ultraviolet behaviour* (Section 58). With (F1)–(F3), for every small g and every K there is a bare coupling for which the flow stays in $(0,\gamma]$, has $g_K = g$, and satisfies $|g_k^{-2} - g^{-2} - (K-k)\beta_\infty| \le C(\log(K-k+2))^{6q+1}$, where $p_0(g) = (\log g^{-1})^q$ (Theorem 58.9). This is the analogue of Bałaban's (0.31) for the couplings of this flow, except that for the last m_1 steps its lower half holds up to an additive constant (Remark 58.10).

These results form Theorem M.

Off the main line. Section 59 proves that, at fixed couplings and at flat configurations, the generations converge as the torus grows (Theorem 59.12). With it, the β_j converge to the coefficient that Bałaban's definition [18, (1.20)–(1.22)] assigns to the generations of this note (Proposition 58.8). No rate is proved, and Theorem M does not use these results.

Imports. Parts I–VII (in particular the operators at $U=1$ of Part I), Propositions 41.7 and 41.14 and Lemma 41.11 of Part IX, and from Part X Propositions 47.8 and 52.2 and Lemma 52.3; Sections 58 and 59 also use the flow, Section 41.5, Theorem 45.5 and the graded domains of Part VIII (Definition 38.3); Section 59 uses in addition the weakened families of Part IX (Lemmas 40.3 and 40.4, Theorem 41.4, Corollary 41.6).

Details for this Part. The companion has two chapters on Part XI. The first expands the composite one-loop term, the marginal tensors, the infinite-volume operators, the Landau chart and the vacuum polarization; the second the lattice Feynman-gauge coefficient, the period density, the properties (F1)–(F3), the ultraviolet limit and the generation locality.

54 The one-loop term of the composite map

This part prepares the ultraviolet limit of the four-dimensional theory: the growth $k\beta_\infty + O(1)$ of the one-loop coefficient of the running coupling summed over k steps, in particular its positivity for large k. This is the input of our version, for the couplings of the small-field flow, of Bałaban's Theorem 2 [18], which he states for $SU(2)$ and whose perturbative proof

he deferred. This section reduces the one-loop coefficients of all steps to a single object, the one-loop term of the composite k-step map.

Why one coupling, and the plan of Part XI. The couplings g_j of the flow change from step to step, but the object of this Part does not depend on them. The one-loop term $\Gamma_j^{\text{step}} = -\frac{1}{2}\log\det\Delta_j + \sum_c \log \mathfrak{j}_c$ of the step from scale j is a function of the background field alone (paragraph "The one-loop terms of the steps" below), and the one-loop term $\Gamma_k^{(1)}$ of the composite map, obtained by Laplace's method as $g \to 0$, does not depend on g (Proposition 54.3). So a single coupling g serves as the parameter of Laplace's method. The couplings of the flow enter only in Section 58, where the coefficient β_{j+1} of the flow is compared with the one-loop coefficient, through (F2) and (F3) of Proposition 58.6. The chain is the following.

- Laplace's method defines $\Gamma_k^{(1)}$ (Proposition 54.3); the integral over the intermediate scale gives the one-step relation (Lemma 54.5), and induction gives $\Gamma_k^{(1)} = \sum_{j<k} \Gamma_j^{\text{step}}$ up to a constant (Proposition 54.6).
- The lattice symmetry gives the marginal tensor $\beta(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$ of each such function, with no term linear in p (Lemma 54.9). In four dimensions the change of scale preserves marginal tensors (Lemma 54.17), so the tensors of the steps add up to that of $\Gamma_k^{(1)}$ (Corollary 54.18). Section 54.2 builds the infinite-volume operators at $U = 1$ that these statements and Sections 55–58 use.
- Section 55 writes $\Gamma_k^{(1)}$ in the Landau chart; Sections 56 and 57 compute its coefficient, $k\beta_\infty + O(1)$ (Theorem 58.1); Section 58 transfers it to the flow (Proposition 58.6) and proves Theorem 58.9.

The introduction to Part XI describes the same chain.

What is used. The section uses Parts IV–VII: the one-step formula of Theorem 34.9, the exponential gauge fixing of Lemma 32.3, and Lemma 32.6(iv). It also uses Proposition 47.8(c),(d), Proposition 52.2 and Lemma 52.3 of Part X, the latter two for the step from scale 0; the lemma is the formula for the first step. Section 54.2 uses Part I. A single coupling g is kept throughout. This is legitimate in $d = 4$, where the Wilson action of Section 23 does not change under a change of units (Section 31 and the end of Theorem 34.9). The coupling of a step changes only through the extraction of β, which is not used here; the coupling paragraphs of Section 31 are not used either. Apart from the single coupling, the dimension enters the proofs of this section only through the scaling in Lemma 54.17. The orientation in $d = 4$ mentioned in Remark 54.8 is a comment: the proof of Lemma 54.9 uses the point reflection $x \mapsto -x$, which lies in the hyperoctahedral group in every dimension.

Setting. Fix $k \geq 1$. A real configuration V on $\mathbb{T}$ is *small field* if:

- its curvature is so small that the hypotheses of Lemma 31.6 and of Theorem 34.9 hold at every intermediate scale $j < k$, for the configurations $U_k(V)^{(j+1)}$ (at $j = 0$, those of Lemma 52.3);
- the side of the torus at every scale is at least the threshold n_* of (T0), which contains $n_{\delta=1}$ and $80M_L'$.

In Part XI "small field" always has this meaning: a condition on the curvature of V and on the torus, which does not involve the coupling. It is not the small-field set $\mathfrak{U}_k(h_k)$ of Part X, whose threshold $h_k = p(g_k)g_k$ depends on the coupling (Section 49.1).

Recall from Parts IV and V. For a real reference U_0 put $W_k = U_0^{(k)}$, and let $\mathfrak{B}(V) = \{U : U^{(k)} = V\}$ be the constraint set. $\mathfrak{U}(\theta)$ is the set of real configurations with curvature and current at most θ (Section 23), and $\Phi(A) = S(e^{i\eta A}U_0)$. Corollary 24.3 parametrizes the orbits of $\mathfrak{G}_0$ in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$ by fields in the Landau gauge: the set $\mathcal{S}_V$ of the real A with $|A|_\infty \le r_u$, $RD^TA = 0$, $e^{i\eta A}U_0 \in \mathfrak{U}(\theta)$ and $\mathcal{Q}(e^{i\eta A}) = VW_k^{-1}$ is mapped bijectively by $A \mapsto \mathsf{U}(A)$ onto the averaged axial representatives of these orbits (Definition 22.2), and $S(\mathsf{U}(A)) = \Phi(A)$. Here R is the projection $I - P_{\mathcal{V}}$ of Part II, $\mathcal{Q}$ the gauge-adapted averaging and w_k, which enters U, the correspondence transformation (Definition 22.4). Below $U_0 = U_k(V)$ lies in the minimal orbit $\mathscr{M}(V)$ of Theorem 30.3; then $W_k = V$, and the condition reads $\mathcal{Q}(e^{i\eta A}) = 1$.

Let $\mathscr{M}(V)$ be the minimal orbit of V (Theorem 30.3), $U_k(V)$ an element of it, and $\mathfrak{G}_0 = \{u : u|_{\mathbb{T}} = 1\}$. The group $\mathfrak{G}_0$ acts freely on all configurations: if $U^u = U$ and $u(y) = 1$ at a point, parallel transport along any path from y gives $u = 1$.

Take $U_0 = U_k(V)$ as the reference of Part IV, and let $\theta \le \theta_0$ be such that $U_k(V) \in \mathfrak{U}(\theta')$ for some $\theta' < \theta$. The choice $\theta = \theta_0$ is allowed: by the hypotheses of Lemma 31.6 at $j = k-1$ (of Lemma 52.3 if $k = 1$), $V \in \mathfrak{U}_k(\frac{1}{2}a_1)$, so $U_k(V) \in \mathfrak{U}(\frac{1}{2}B_3a_1) \subset \mathfrak{U}(\frac{1}{2}\theta_0)$ by Theorem 30.3(b). The reference is (M, B_3a_1)-regular with $|J_{U_0}|_\infty \le B_3a_1$ and $U_0^{(k)} = V$, and $B_3a_1 \le \frac{1}{2}c_6$ (proof of Theorem 30.3, with $C_L^{\text{units}} \ge 2$ from Lemma 30.2); so Corollary 24.3 applies, with $\varepsilon = B_3a_1$ and $\beta = 0$ (the β of that corollary, unrelated to the coefficients β_j of the flow), since $\theta_0 \le \frac{1}{3}c_4$ and $c_6 \le \frac{1}{3}c_4$ (Section 25.3). It gives a bijection $A \mapsto \mathsf{U}(A)$ from the set $\mathcal{S}_V$ onto the averaged axial representatives of the $\mathfrak{G}_0$-orbits in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$, with $S(\mathsf{U}(A)) = \Phi(A)$. Here $\mathcal{S}_V$ is the set of real A with $|A|_\infty \le r_u$, $RD^TA = 0$, $e^{i\eta A}U_0 \in \mathfrak{U}(\theta)$ and $\mathcal{Q}(e^{i\eta A}) = 1$. For $r > 0$ let $\mathscr{N}_r(V)$ be the union of the $\mathfrak{G}_0$-orbits of the $\mathsf{U}(A)$ with $A \in \mathcal{S}_V$, $|A|_\infty < r$. These definitions use only Theorem 30.3 and Corollary 24.3. So they make sense, by the same argument, for every real $V \in \mathfrak{U}_k(a_1)$, on a torus as in the setting, and every $\theta \le \theta_0$ with $\mathscr{M}(V) \subset \mathfrak{U}(\theta')$ for some $\theta' < \theta$; we call such a pair (V, θ) *admissible*. By Lemma 54.1(b) below, $\mathscr{N}_r(V)$ does not depend on the choice of $U_k(V)$ in $\mathscr{M}(V)$. For $g > 0$ put

$$I_k(V; g, r) = \int \prod_b dU(b) \prod_c \delta(U^{(k)}(c)V(c)^{-1})\ \mathbf{1}_{\mathscr{N}_r(V)}(U)\ e^{-S(U)/g^2},$$

with the normalized Haar measure and the Haar δ-function, as in (31.1).

The neighbourhoods $\mathscr{N}_r(V)$ are defined through a chart whose reference is an element of the minimal orbit. The next lemma shows that they do not depend on this choice, that they are covariant, and that they shrink continuously to the minimal orbit. As in Section 27.7, a subscript or a second argument indicates the reference: $\mathcal{S}_V(U_0)$, U_{U_0}, $\mathcal{Q}(\cdot\,; U_0)$, $\mathsf{w}_k(\cdot\,; U_0)$. For a real gauge transformation u we write $R(u)$ for its action on Lie fields, $(R(u)A)(b) = R(u(b_-))A(b)$, as in Proposition 22.10(f), and on scalar fields, $(R(u)\lambda)(x) = R(u(x))\lambda(x)$. To avoid a clash, in the proof below R_{U_0} denotes the projection $R = I - P_{\mathcal{V}}$ of Part II at the reference U_0.

Lemma 54.1 (The chart neighbourhoods)**.** *(a)* (Gauge covariance of the chart.) *Let U_0, V, θ and β be as in Corollary 24.3, and let u be a real gauge transformation on $\mathbb{T}_\eta$, with $v = u|_{\mathbb{T}}$. Then U_0^u satisfies the hypotheses of Section 24 with the same ε, $W_k(U_0^u) = W_k^v$, and V^v satisfies the hypothesis of Corollary 24.3 relative to U_0^u, with the same β. The map $A \mapsto R(u)A$ maps $\mathcal{S}_V(U_0)$ onto $\mathcal{S}_{V^v}(U_0^u)$, preserves $|\cdot|_\infty$, and*

$$\mathsf{U}_{U_0^u}(R(u)A) = \mathsf{U}_{U_0}(A)^u \qquad (A \in \mathcal{S}_V(U_0)).$$

(b) (The choice of the reference.) *Let (V, θ) be admissible. Then $\mathscr{N}_r(V)$ does not depend on the choice of $U_k(V)$ in $\mathscr{M}(V)$. For every real gauge transformation u on $\mathbb{T}_\eta$ the pair $(V^{u|_{\mathbb{T}}}, \theta)$ is admissible, and $\mathscr{N}_r(V^{u|_{\mathbb{T}}}) = \mathscr{N}_r(V)^u$.*

(c) (The chart radius.) *Let $\theta \le \theta_0$, and let $\mathcal{K}_0$ be a compact set of real configurations on $\mathbb{T}$ such that (V,θ) is admissible for every $V \in \mathcal{K}_0$. For $U \in \mathfrak{U}(\theta)$ with $U^{(k)} \in \mathcal{K}_0$ let $\varrho(U) = |A|_\infty$, where $\mathsf{U}(A)$, $A \in \mathcal{S}_{U^{(k)}}$, is the averaged axial representative of the $\mathfrak{G}_0$-orbit of U, for a reference in $\mathscr{M}(U^{(k)})$. Then ϱ is well defined, invariant under $\mathfrak{G}_0$ and continuous on the compact set $\mathcal{D}_{\mathcal{K}_0} = \{U \in \mathfrak{U}(\theta) : U^{(k)} \in \mathcal{K}_0\}$, and it vanishes exactly on the minimal orbits. For $V \in \mathcal{K}_0$,*

$$\mathscr{N}_r(V) = \{U \in \mathfrak{B}(V) \cap \mathfrak{U}(\theta) :\ \varrho(U) < r\},$$

a Borel set, and every open set of configurations that contains $\mathscr{M}(V)$ contains $\mathscr{N}_r(V)$ for r small.

Proof. (a) *The hypotheses.* The regularity of Definition 18.1 asks for local gauges, so it is invariant under real gauge transformations. Moreover $J_{U_0^u} = R(u)J_{U_0}$ (Lemma 23.2(b)), $W_j(U_0^u) = W_j^{u|_{\mathbb{T}^{(j)}}}$ for $j \le k$ (Lemma 21.10(b)), and $V^v(c)\,W_k^v(c)^{-1} = v(c_-)\,V(c)W_k(c)^{-1}\,v(c_-)^{-1}$, whose logarithm has the same norm (Lemma 21.2(d)). Details

The averaging. For a relative configuration X at scale j, relative to $W_j = W_j(U_0)$, let $R(u)X$, $(R(u)X)(b) = R(u(b_-))X(b)$, be read relative to W_j^u, where u also denotes its restriction to $\mathbb{T}^{(j)}$. Then $(R(u)X)W_j^u = (XW_j)^u$ bondwise. For X at scale $j-1$, (21.2) and Lemma 21.10(b) give

$$(R(u)X)_{W^u}(y,x) = u(y)\,X_W(y,x)\,u(y)^{-1}, \qquad \overline{R(u)X} = R(u)\,\overline{X},$$
$$(R(u)X)^{(ugu^{-1})} = R(u)(X^{(g)}),$$

with the pointwise conjugate ugu^{-1}; the last identity is $((XW)^u)^{ugu^{-1}} = ((XW)^g)^u$. Hence $\tau_{R(u)X} = u\tau_X u^{-1}$, the axial condition is preserved, $\mathcal{E}_j$, computed with W^u, satisfies $\mathcal{E}_j(uhu^{-1}) = u\,\mathcal{E}_j(h)\,u^{-1}$, and $g_{R(u)X} = ug_X u^{-1}$ by the uniqueness in Lemma 22.3(a). So $\tilde{\mathfrak{a}}_j(R(u)X) = R(u)\,\tilde{\mathfrak{a}}_j(X)$, and by induction over j

$$\mathcal{Q}_j(R(u)U_1;U_0^u) = R(u)\,\mathcal{Q}_j(U_1;U_0), \qquad \mathsf{w}_j(R(u)U_1;U_0^u) = u\,\mathsf{w}_j(U_1;U_0)\,u^{-1}.$$

This is the argument of Proposition 22.10(g), with u in place of r, and it proves the nonlinear part of Proposition 22.10(f). For $U_1 = e^{i\eta A}$ with $|A|_\infty \le c_3$, all maps are defined at U_1 and at $R(u)U_1 = e^{i\eta R(u)A}$ (Proposition 22.9).

The gauge condition. Directly from the definition, $D_{U_0^u}R(u) = R(u)D_{U_0}$ on scalar fields, and $Q'_k(U_0^u)R(u) = R(v)Q'_k(U_0)$ (Proposition 22.10(f)). The maps $R(u)$ preserve the pairings. So every operator built from D, D^T, Q'_k and $Q_k'^T$ by sums, products, transposes and inverses is covariant, as in the proof of Theorem 25.1(h); in particular $R_{U_0^u}D^T_{U_0^u}R(u) = R(u)R_{U_0}D^T_{U_0}$.

The chart. Let $A \in \mathcal{S}_V(U_0)$. Then $R(u)A$ is real, $|R(u)A|_\infty = |A|_\infty$, $R_{U_0^u}D^T_{U_0^u}R(u)A = 0$, $e^{i\eta R(u)A}U_0^u = (e^{i\eta A}U_0)^u \in \mathfrak{U}(\theta)$, and $\mathcal{Q}(e^{i\eta R(u)A};U_0^u) = R(v)(VW_k^{-1}) = V^v(W_k^v)^{-1}$. So $R(u)$ maps $\mathcal{S}_V(U_0)$ into $\mathcal{S}_{V^v}(U_0^u)$, and onto it, by the same argument for u^{-1} at the reference U_0^u. With $\mathsf{w} = \mathsf{w}_k(e^{i\eta A};U_0)$,

$$\mathsf{U}_{U_0^u}(R(u)A) = (R(u)e^{i\eta A})^{(uwu^{-1})}U_0^u = R(u)((e^{i\eta A})^{(\mathsf{w})})\,U_0^u = ((e^{i\eta A})^{(\mathsf{w})}U_0)^u = \mathsf{U}_{U_0}(A)^u.$$

(b) Two elements of $\mathscr{M}(V)$ differ by some $u \in \mathfrak{G}_0$, for which $v = 1$. By (a), the configurations $\mathsf{U}(A)$, $A \in \mathcal{S}_V$, $|A|_\infty < r$, for the reference $U_k(V)^u$ are the images under u of those for $U_k(V)$, so they have the same $\mathfrak{G}_0$-orbits. Let now u be any real gauge transformation on $\mathbb{T}_\eta$, and $v = u|_{\mathbb{T}}$. The curvature is covariant, so $V^v \in \mathfrak{U}_k(a_1)$; the minimal orbit of V^v is $\mathscr{M}(V)^u$ (Theorem 25.1(g), Theorem 30.3(a)); and $\mathfrak{U}(\theta')$ is invariant under real gauge transformations. So (V^v,θ) is admissible. With the reference $U_k(V)^u \in \mathscr{M}(V^v)$, (a) shows that the $\mathsf{U}(A')$,

$A' \in \mathcal{S}_{V^v}$, $|A'|_\infty < r$, are the $\mathsf{U}(A)^u$, $A \in \mathcal{S}_V$, $|A|_\infty < r$. Since $(U^w)^u = (U^u)^{uwu^{-1}}$ and $u\mathfrak{G}_0 u^{-1} = \mathfrak{G}_0$, the $\mathfrak{G}_0$-orbits of the $\mathsf{U}(A)^u$ are the images under u of those of the $\mathsf{U}(A)$. Hence $\mathscr{N}_r(V^v) = \mathscr{N}_r(V)^u$.

(c) *Definition.* Let $U \in \mathcal{D}_{\mathcal{K}_0}$, $V = U^{(k)}$ and $U_0 \in \mathscr{M}(V)$. Since $\mathfrak{U}(\theta) \subset \mathfrak{U}_0(c_1)$, the $\mathfrak{G}_0$-orbit of U lies in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$. It has exactly one averaged axial representative relative to U_0 (Lemma 22.3(c)), which is $\mathsf{U}_{U_0}(A)$ for exactly one $A \in \mathcal{S}_V(U_0)$ (Corollary 24.3). For another reference U_0^u, $u \in \mathfrak{G}_0$, the configuration $\mathsf{U}_{U_0}(A)^u = \mathsf{U}_{U_0^u}(R(u)A)$ of (a) lies in the same orbit and is averaged axial relative to U_0^u, by Corollary 24.3. So it is the representative relative to U_0^u, and $|R(u)A|_\infty = |A|_\infty$. Hence ϱ is well defined, and it is constant on $\mathfrak{G}_0$-orbits. Since $\mathsf{w}_k(1) = 1$, $\mathsf{U}_{U_0}(0) = U_0$, and $0 \in \mathcal{S}_V(U_0)$, because $U_0 \in \mathfrak{U}(\theta')$. So $\varrho(U) = 0$ if and only if the representative is U_0, i.e. $U \in \mathscr{M}(V)$. The formula for $\mathscr{N}_r(V)$ restates its definition.

Continuity. The class $\mathfrak{U}(\theta)$ is closed, hence compact, and $U \mapsto U^{(k)}$ is continuous on $\mathfrak{U}_0(c_1)$; so $\mathcal{D}_{\mathcal{K}_0}$ is compact. We use three facts.

(i) The averages $U^{(j)}$, the averaged contour variables and the maps of Definition 22.4 are continuous functions of the configuration and of the reference, where they are defined: they are composed of group operations, ℓ, exp and the group average M, which is holomorphic (Lemma 21.5(a)). So are D_{U_0}, $Q'_k(U_0)$ and $R_{U_0} = I - P_{\mathcal{V}}$, with $P_{\mathcal{V}} = T(T^TT)^{-1}T^T$, $T = G'Q'^T$ and $G' = (D^TD + aQ'^TQ')^{-1}$ (Section 14); the matrices inverted are invertible at every reference considered. The averaged axial representative of the $\mathfrak{G}_0$-orbit of U relative to U_0 is U^g, where $g|_{\mathbb{T}} = 1$ and, scale by scale from $\mathbb{T}$ down to $\mathbb{T}_\eta$,

$$g(x) = W_{i-1}(y,x)^{-1}\, g(y)\, U^{(i-1)}(y,x), \qquad x \in B_i(y),\ y \in \mathbb{T}^{(i)},$$

with the averaged contour variables of $U^{(i-1)}$ and of $W_{i-1} = U_0^{(i-1)}$. This is the formula of Lemma 22.3(a), applied as in the proof of Lemma 22.3(b). So the representative depends continuously on $(U, U_0) \in \mathfrak{U}_0(c_1)^2$.

(ii) The union $\mathcal{R}$ of the minimal orbits $\mathscr{M}(V)$, $V \in \mathcal{K}_0$, is compact, and a limit of elements of $\mathscr{M}(V_n)$ with $V_n \to V$ lies in $\mathscr{M}(V)$. Indeed, let $U_n \in \mathscr{M}(V_n)$ with $V_n \in \mathcal{K}_0$. Pass to a subsequence with $V_n \to V \in \mathcal{K}_0$, and let $U_0 \in \mathscr{M}(V)$. For n large, $V_n = e^{\mathrm{i}B_n}V$ bondwise, with real $B_n \to 0$. Since $B_3 a_1 \le \frac{1}{2}c_6$, Corollary 30.5 applies with $\beta = \frac{1}{2}c_6$: $\mathscr{M}(V_n)$ contains $\mathsf{U}_{U_0}(A_*(B_n))$, and $A_*(B_n) \to A_*(0) = 0$. So $U_n = \mathsf{U}_{U_0}(A_*(B_n))^{u_n}$ with $u_n \in \mathfrak{G}_0$. Along a further subsequence $u_n \to u \in \mathfrak{G}_0$, and $U_n \to U_0^u \in \mathscr{M}(V)$, by the continuity of U_{U_0}.

(iii) Let Σ be the set of the pairs (U_0, A) with $U_0 \in \mathcal{R}$ and $A \in \mathcal{S}_{U_0^{(k)}}(U_0)$. By (i), the conditions that define $\mathcal{S}$ are closed in (U_0, A), so Σ is compact. The map $F(U_0, A) = (U_0, \mathsf{U}_{U_0}(A))$ is continuous, by (i), and injective, by Corollary 24.3. So F is a homeomorphism of Σ onto its image.

Let $U_n \to U$ in $\mathcal{D}_{\mathcal{K}_0}$, and let $U_{0,n} \in \mathscr{M}(U_n^{(k)})$. By (ii), every subsequence has a further subsequence along which $U_{0,n} \to U_0 \in \mathscr{M}(U^{(k)})$. Along it, the representatives $\mathsf{U}_{U_{0,n}}(A_n)$ of the orbits of the U_n converge to the representative $\mathsf{U}_{U_0}(A)$ of the orbit of U, by (i), and $A_n \to A$, by (iii). So $\varrho(U_n) \to \varrho(U)$ along it. Hence ϱ is continuous.

Neighbourhoods. Let $V \in \mathcal{K}_0$. The set $\mathscr{N}_r(V)$ is relatively open in the compact set $\mathfrak{B}(V) \cap \mathfrak{U}(\theta) = \{U \in \mathfrak{U}(\theta) : U^{(k)} = V\}$, hence a Borel set. Let $\mathcal{V}$ be open with $\mathscr{M}(V) \subset \mathcal{V}$. The sets $K_r = \{U \in \mathfrak{B}(V) \cap \mathfrak{U}(\theta) : \varrho(U) \le r\}$ are compact, they decrease as $r \downarrow 0$, and their intersection is $\mathscr{M}(V)$. So $K_r \setminus \mathcal{V}$ is empty for r small, and then $\mathscr{N}_r(V) \subset K_r \subset \mathcal{V}$. □

Lemma 54.2 (Fibre mass)**.** *Let (V, θ_0) be admissible. Then $U \mapsto U^{(k)}$ is a submersion at every point of $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$, $\mathfrak{B}(V)$ is an analytic submanifold near this compact set, and the measure $\prod_b dU(b) \prod_c \delta(U^{(k)}(c)V(c)^{-1})$ gives it a finite mass. If $\mathcal{K}_0$ is a compact set of real*

configurations on $\mathbb{T}$ such that (V,θ_0) is admissible for every $V\in\mathcal{K}_0$, these masses are bounded uniformly for $V\in\mathcal{K}_0$.

Proof. The set $\mathfrak{B}(V)\cap\mathfrak{U}(\theta_0)$ is compact, since $U\mapsto U^{(k)}$ is continuous on $\mathfrak{U}_0(c_1)$. Since (V,θ_0) is admissible, the setting applies with θ_0 in place of θ and a reference $U_0\in\mathscr{M}(V)$, and Corollary 24.3 applies with $\varepsilon=B_3a_1$ and $\beta=0$. By Theorem 24.2 every $A\in\mathcal{S}_V$ satisfies $|A|_{C^1}\le B_1(\varepsilon+\theta_0)=\frac12\rho_1$, so $\mathcal{S}_V$ is a compact subset of the manifold $\mathcal{M}_V$ of Lemma 25.2(a), near which the map $A\mapsto\ell(\mathcal{Q}(e^{\mathrm{i}\eta A}))=\ell(\mathsf{U}(A)^{(k)}W_k^{-1})$ (for every A with $|A|_\infty\le c_3$, by Theorem 22.5(a), as in the proof of Corollary 24.3) on the slice $\{RD^TA=0\}$ is a submersion (proof of Lemma 25.2(a)). Hence $U\mapsto U^{(k)}$ is a submersion at every $\mathsf{U}(A)$, $A\in\mathcal{S}_V$, and, since $(U^u)^{(k)}=U^{(k)}$ for $u\in\mathfrak{G}_0$, at every point of the compact set $\mathfrak{B}(V)\cap\mathfrak{U}(\theta_0)$ (Corollary 24.3). So $\mathfrak{B}(V)$ is an analytic submanifold near this set, the Haar δ-measure has a continuous density on it there, and the mass is finite.

Details

For the uniform bound, the set $\mathcal{D}=\{U\in\mathfrak{U}(\theta_0):U^{(k)}\in\mathcal{K}_0\}$ is compact, as in Lemma 54.1(c), and $U\mapsto U^{(k)}$ is a submersion at each of its points, by the first part. Near each point of $\mathcal{D}$ it is a coordinate projection in suitable analytic coordinates, in which the measures $\prod_b dU(b)\prod_c\delta(U^{(k)}(c)V(c)^{-1})$ on the fibres have bounded densities; finitely many such coordinate neighbourhoods cover $\mathcal{D}$. So the masses of the sets $\mathfrak{B}(V)\cap\mathfrak{U}(\theta_0)$, $V\in\mathcal{K}_0$, are bounded by some m. □

Proposition 54.3 (The one-loop term)**.** *Let $\theta\le\theta_0$. There are $r_0>0$, constants $c_k(g)>0$ independent of V, r and θ, and a function $\Gamma_k^{(1)}(V)$ independent of r, g and θ, such that for $0<r\le r_0$ and every real $V\in\mathfrak{U}_k(a_1)$ for which (V,θ) is admissible*

$$I_k(V;g,r)=c_k(g)\exp\Big(-\frac{1}{g^2}S\big(U_k(V)\big)+\Gamma_k^{(1)}(V)+o(1)\Big)\qquad(g\to0),$$

locally uniformly in V on the set of these V: every point of this set has a neighbourhood in it on which the $o(1)$ is uniform. $\Gamma_k^{(1)}$ is real analytic in V and invariant under gauge transformations of $\mathbb{T}$.

Proof. Plan. Modulo $\mathfrak{G}_0$, the integral I_k runs over a neighbourhood of the minimal orbit in the constraint set, which the chart $\mathcal{S}_V$ parametrizes. On $\mathcal{S}_V$ the action Φ has the unique minimum $A=0$, with Hessian q_2 on the tangent space; Laplace's method there gives $\Gamma_k^{(1)}$. A reference fixed for all V near a given V° gives the uniformity in V.

Details

The constraint set. Near $\mathfrak{B}(V)$, the differential of $U\mapsto U^{(k)}$ in the exponential coordinates is $\mathbf{Q}_k$ plus D_V composed with the linearization of w_k at the points of $\mathbb{T}$ (Lemma 27.1(a)). It is onto: by the chain rule applied to $A\mapsto\mathsf{U}(A)^{(k)}=\mathcal{Q}(e^{\mathrm{i}\eta A})W_k$ (Theorem 22.5(a)), whose differential at 0 on the slice is $\mathbf{Q}_k$, with $\mathbf{Q}_k\mathbf{H}=I$. So $\mathfrak{B}(V)$ is a real analytic submanifold near the minimal orbit, and, by Lemma 54.2, near every point of $\mathfrak{B}(V)\cap\mathfrak{U}(\theta)$. The Haar δ-function disintegrates the Haar measure into a positive analytic measure on it.

The quotient. The group $\mathfrak{G}_0$ is compact and acts freely, so the quotient by $\mathfrak{G}_0$ of a neighbourhood of $\mathfrak{B}(V)\cap\mathfrak{U}(\theta)$ in $\mathfrak{B}(V)$ is a real analytic manifold, and $\mathfrak{B}(V)\cap\mathfrak{U}(\theta)$ modulo $\mathfrak{G}_0$ is a compact subset of it. By Corollary 24.3, $A\mapsto[\mathsf{U}(A)]$ is a bijection from $\mathcal{S}_V$ onto this compact set. It is a chart of the quotient near $A=0$; see the paragraph "The chart at 0" below. Since $U_0\in\mathfrak{U}(\theta')$ with $\theta'<\theta$, and curvature and current are continuous, $\mathcal{S}_V$ contains a neighbourhood of 0 in the manifold of the real A with $RD^TA=0$ and $\mathcal{Q}(e^{\mathrm{i}\eta A})=1$ (Lemma 25.2(a)). In this chart the integrand is $e^{-\Phi(A)/g^2}$, and near $A=0$ the measure has a positive analytic density $\varsigma_V(A)$ with respect to the Euclidean volume on $\mathcal{S}_V$ (paragraph "The chart at 0"). The density includes the Jacobian of $d\mathsf{U}(0)$ and the volume of the orbits. Laplace's method uses ς_V only near $A=0$; away from $A=0$ only the finiteness of the mass is used (Lemma 54.2).

The minimum. By Proposition 26.6, Φ has on $\mathcal{S}_V$ the unique minimum $A = 0$. Its Hessian on the tangent space $\mathcal{X} = \{h : \mathbf{Q}_k h = 0,\ RD^T h = 0\}$ is q_2 (Proposition 33.2): 0 is critical on the constraint set, and the curvature of the constraint gives the multiplier term. It is coercive. Indeed, the reference $U_0 = U_k(V)$ is critical, with $\beta = 0$ and $A_* = 0$, and it is (M, ε)-regular with $|J_{U_0}| \le \varepsilon$ for $\varepsilon = B_3 a_1$ (Theorem 30.3(b) with $\varepsilon_1 = a_1$), where $\varepsilon \le \frac{1}{2}c_6 \le \frac{1}{2}c_7$. On all real fine fields h, $q_2(h,h) = \|Dh\|^2 + d^2\mathcal{N}(0)[h,h] - \langle \omega_0, d^2 C_k^{\mathrm{av}}(0)[h,h]\rangle$, with $|d^2\mathcal{N}(0)[h,h]| \le C\varepsilon(\|h\|^2 + \|\nabla h\|^2)$, $|\langle \omega_0, d^2 C_k^{\mathrm{av}}(0)[h,h]\rangle| \le C\varepsilon\|h\|^2$ and the multiplier ω_0 of Proposition 27.2 (proof of Lemma 26.7). By Lemma 26.7, $q_2(h,h) \ge \frac{1}{2}c_*(\|h\|^2 + \|Dh\|^2)$ for $h \in \mathcal{X}$. Hence $\Phi \ge \Phi(0) + cr^2$ on $\{|A|_\infty = r\} \cap \mathcal{S}_V$ for $r \le r_1$, with $r_1, c > 0$ depending on V. Since $\mathcal{S}_V$ is compact and 0 is the unique minimum, $\Phi - \Phi(0) \ge m_1 > 0$ on $\mathcal{S}_V \cap \{|A|_\infty \ge r_1\}$, so $\Phi \ge \Phi(0) + c'|A|_\infty^2$ on all of $\mathcal{S}_V$, with $c' = \min(c, m_1/r_u^2)$. No smallness of r is used below; we take $r_0 = r_u$.

The chart at 0. Let $h \in \mathcal{X}$, and let $A(t)$ be a curve in the manifold $\{RD^T A = 0,\ \mathcal{Q}(e^{\mathrm{i}\eta A}) = 1\}$ with $A(0) = 0$ and $A'(0) = h$ (Lemma 25.2(a)). Then $\gamma(t) = \mathsf{U}(A(t))$ is a curve in $\mathfrak{B}(V)$ through U_0, $\gamma'(0) = d\mathsf{U}(0)h$, and $S(\gamma(t)) = \Phi(A(t))$, whose second derivative at $t = 0$ is $q_2(h,h)$, since 0 is critical on the constraint set with the multiplier ω_0. Since U_0 minimizes S on $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$ (Theorem 30.3(a)), it is a critical point of $S|_{\mathfrak{B}(V)}$, and the second derivative of S along a curve in $\mathfrak{B}(V)$ through U_0 is the Hessian of $S|_{\mathfrak{B}(V)}$ at U_0, evaluated at the velocity. Since S is constant on the $\mathfrak{G}_0$-orbit of U_0, this Hessian vanishes on the tangent vectors of the orbit. So if $d\mathsf{U}(0)h$ is tangent to the orbit, then $q_2(h,h) = 0$ and $h = 0$. Hence $(h, \xi) \mapsto d\mathsf{U}(0)h + D_{U_0}\xi$, $\xi|_{\mathbb{T}} = 0$, is injective, since $D_{U_0}\xi = 0$ and $\xi|_{\mathbb{T}} = 0$ give $\xi = 0$. The map $(A, u) \mapsto \mathsf{U}(A)^u$, on a neighbourhood of 0 in the constraint manifold times $\mathfrak{G}_0$, is therefore an injective immersion: it is an immersion at the points $(0, u)$, by the $\mathfrak{G}_0$-equivariance, hence near them, since $\mathfrak{G}_0$ is compact; and it is injective by Corollary 24.3, since $\mathfrak{G}_0$ acts freely. Its image contains a neighbourhood of $\mathscr{M}(V)$ in $\mathfrak{B}(V)$ (Lemma 54.1(c) and the uniqueness in the implicit function theorem). So the dimensions agree, since otherwise the image would have measure zero, and the map is an analytic diffeomorphism onto an open set. In particular $A \mapsto [\mathsf{U}(A)]$ is a chart near 0, and the density ς_V is positive.

Laplace's method on the finite-dimensional manifold $\mathcal{S}_V$ gives the expansion, with

$$\Gamma_k^{(1)}(V) = -\tfrac{1}{2}\log\det\left(q_2|_{\mathcal{X}}\right) + \log\varsigma_V(0)$$

up to a constant, and $c_k(g)$ is $(2\pi g^2)^{N/2}$, $N = \dim\mathcal{X}$, times a constant. Both terms are local at the minimum, so they do not depend on r, nor on the choice of θ, and they are analytic in V (paragraph "Uniformity in V", (vi)). For the gauge invariance let u be a real gauge transformation on $\mathbb{T}_\eta$ and $v = u|_{\mathbb{T}}$. By Lemma 54.1(b), (V^v, θ) is admissible and $\mathscr{N}_r(V^v) = \mathscr{N}_r(V)^u$, and $V^v \in \mathfrak{U}_k(a_1)$. The substitution $U \mapsto U^u$ preserves $\prod_b dU(b)$ and S, and $\delta((U^u)^{(k)}, V^v) = \delta(U^{(k)}, V)$, by Lemma 21.10(b) and since δ is a class function. So $I_k(V^v; g, r) = I_k(V; g, r)$. Since $S(U_k(V^v)) = S(U_k(V))$ and $c_k(g)$ does not depend on V, $\Gamma_k^{(1)}(V^v) = \Gamma_k^{(1)}(V)$.

Uniformity in V. So far V was fixed. Let V° be a point of the set of the statement, and fix $U_0 \in \mathscr{M}(V^\circ)$. For V near V° write $V(c) = e^{\mathrm{i}B(c)}V^\circ(c)$; then $B = B(V)$ is real, small and analytic in V. For all these V we use the fixed reference U_0, with its operators R, D, $\mathcal{Q}$, $\mathbf{Q}_k$, its minimizer map $\mathbf{H}$, the space $\mathcal{X}$ and $\Phi(A) = S(e^{\mathrm{i}\eta A}U_0)$. By Corollary 24.3, with $\beta = |B|_\infty$, $A \mapsto \mathsf{U}_{U_0}(A)$ is a bijection from the set $\mathcal{S}_V(U_0)$ of the real A with $|A|_\infty \le r_u$, $RD^T A = 0$, $e^{\mathrm{i}\eta A}U_0 \in \mathfrak{U}(\theta)$ and $\mathcal{Q}(e^{\mathrm{i}\eta A}) = e^{\mathrm{i}B}$ onto the averaged axial representatives of the $\mathfrak{G}_0$-orbits in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$, and $S(\mathsf{U}_{U_0}(A)) = \Phi(A)$. By Corollary 30.5, the minimal orbit of V is the $\mathfrak{G}_0$-orbit of $\mathsf{U}_{U_0}(A_*(B))$, with A_* analytic, $A_*(0) = 0$, $RD^T A_*(B) = 0$ and $\mathcal{Q}(e^{\mathrm{i}\eta A_*(B)}) = e^{\mathrm{i}B}$.

(i) *Coordinates.* Fix a Euclidean norm $|\cdot|$ on $\mathcal{X}$. On the slice $\{RD^T A = 0\}$ the map $f(A) = \ell(\mathcal{Q}(e^{\mathrm{i}\eta A}))$ is analytic, with $f(0) = 0$ and $df(0) = \mathbf{Q}_k$ (Proposition 22.9); $\mathbf{H}$ maps into the slice, and $\mathbf{Q}_k\mathbf{H} = I$ (proof of Lemma 25.2(a)). By the analytic implicit

function theorem, for (x,B) near $(0,0)$ the equation $f(x+\mathbf{H}y)=B$ has a unique small solution $y=y(x,B)$, analytic in (x,B). So $A(x,B)=x+\mathbf{H}y(x,B)$ parametrizes $\mathcal{S}_V(U_0)$ near 0 by a neighbourhood of 0 in the fixed space $\mathcal{X}$, and $x=(I-\mathbf{HQ}_k)A(x,B)$. Put $x_*(B)=(I-\mathbf{HQ}_k)A_*(B)\in\mathcal{X}$; by the uniqueness, $A(x_*(B),B)=A_*(B)$.

(ii) *The phase.* $F(x,B)=\Phi(A(x,B))$ is analytic. Since $A_*(B)$ is a critical point of Φ on the constraint set (Lemma 25.2(b)), $x_*(B)$ is a critical point of $F(\cdot,B)$, and $F(x_*(B),B)=S(U_k(V))$. Its Hessian $H(B)=\partial_x^2F(x_*(B),B)$ is continuous in B, and $H(0)$ is q_2 on $\mathcal{X}$, which is positive definite (paragraph "The minimum"). Since the third derivatives of F are bounded on a compact neighbourhood of $(0,0)$, there are $\delta,c_H>0$ such that, for B small, $F(x,B)-F(x_*(B),B)\ge c_H|x-x_*(B)|^2$ for $|x-x_*(B)|<\delta$.

(iii) *The density.* Let $\Upsilon(x,B,u)=\mathsf{U}_{U_0}(A(x,B))^u$, for (x,B) near $(0,0)$ and $u\in\mathfrak{G}_0$. It is analytic, and $\Upsilon(x,B,u)^{(k)}=e^{\mathrm{i}B}V^\circ$, by Corollary 24.3 and $(U^u)^{(k)}=U^{(k)}$. At $B=0$, $(x,u)\mapsto\Upsilon(x,0,u)$ is a local diffeomorphism into $\mathfrak{B}(V^\circ)$ near $\mathscr{M}(V^\circ)$, by the paragraph "The chart at 0" at V°. If $d\Upsilon$ annihilates a tangent vector at a point $(0,0,u)$, then the B-component of the vector vanishes, since $B\mapsto e^{\mathrm{i}B}V^\circ$ is a local diffeomorphism, and then the vector vanishes. Since $U\mapsto U^{(k)}$ is a submersion near $\mathscr{M}(V^\circ)$, the space of the (x,B,u) has the dimension of the space of configurations, so $d\Upsilon$ is invertible at these points, and, by the compactness of $\mathfrak{G}_0$, on $\mathcal{P}\times\mathfrak{G}_0$ for a neighbourhood $\mathcal{P}$ of $(0,0)$. Moreover Υ is injective there: equal values have equal averages, hence equal B; then they are gauge transforms of averaged axial representatives of the same $\mathfrak{G}_0$-orbit, which are equal (Lemma 22.3(c)), so $A(x,B)$ and x are equal (Corollary 24.3), and so is u, since the action is free. Hence Υ is an analytic diffeomorphism of $\mathcal{P}\times\mathfrak{G}_0$ onto an open set of configurations. Pulled back by Υ, the Haar measure has a positive analytic density $\mathsf{h}(x,B,u)$ with respect to $dx\,dB\,du$, with the Haar measure du of $\mathfrak{G}_0$. Since $\Upsilon^{(k)}=e^{\mathrm{i}B}V^\circ$, the measure $\prod_b dU(b)\prod_c\delta(U^{(k)}(c)V(c)^{-1})$ on the image is the image of $j(B)\mathsf{h}(x,B,u)\,dx\,du$ at $B=B(V)$, with $j(B)=\prod_c\sigma(B(c))^{-1}$, where σ is the density of the normalized Haar measure of G in exponential coordinates (Section 32.5). So the integral of a $\mathfrak{G}_0$-invariant function $\mathcal{F}$ over $\Upsilon(\mathcal{P}\times\mathfrak{G}_0)\cap\mathfrak{B}(V)$ is $\int\mathcal{F}(\mathsf{U}_{U_0}(A(x,B)))\,\varsigma(x,B)\,dx$, with $\varsigma(x,B)=j(B)\int_{\mathfrak{G}_0}\mathsf{h}(x,B,u)\,du$, positive and analytic.

(iv) *Laplace's method with parameters.* For V near V° and $B=B(V)$ let $\mathcal{P}_\delta(V)=\Upsilon(\{x:|x-x_*(B)|<\delta\}\times\{B\}\times\mathfrak{G}_0)\subset\mathfrak{B}(V)$. By (iii), the integral of e^{-S/g^2} over $\mathcal{P}_\delta(V)$, for the measure of I_k, is $\int_{|x-x_*(B)|<\delta}e^{-F(x,B)/g^2}\varsigma(x,B)\,dx=g^Ne^{-F(x_*(B),B)/g^2}\int\phi_g(y,B)\,dy$, where $\phi_g(y,B)=e^{-(F(x_*+gy,B)-F(x_*,B))/g^2}\varsigma(x_*+gy,B)$, with $x_*=x_*(B)$, for $g|y|<\delta$, and $\phi_g(y,B)=0$ otherwise. By (ii), $0\le\phi_g(y,B)\le Ce^{-c_H|y|^2}$. Since $\partial_xF(x_*(B),B)=0$ and F and ς have bounded derivatives on a compact neighbourhood of $(0,0)$, $\phi_g(y,B)\to e^{-\frac12H(B)[y,y]}\varsigma(x_*(B),B)$ as $g\to0$, uniformly for y in compact sets and B small. With the uniform Gaussian bound on the tails, $\int\phi_g(y,B)\,dy\to(2\pi)^{N/2}\det H(B)^{-1/2}\varsigma(x_*(B),B)$, uniformly for B small. So the integral over $\mathcal{P}_\delta(V)$ is

$$(2\pi g^2)^{N/2}e^{-S(U_k(V))/g^2}\det H(B)^{-1/2}\varsigma(x_*(B),B)(1+o(1)),$$

uniformly for V near V°.

(v) *The rest of the integral.* Let $b_0>0$ be small, and let $\mathcal{K}_{V^\circ}$ be the set of the $V=e^{\mathrm{i}B}V^\circ\in\mathfrak{U}_k(a_1)$ with $|B|_\infty\le b_0$. It is compact, and it is a neighbourhood of V° in the set of the statement: for b_0 small, (V,θ) is admissible for every $V\in\mathcal{K}_{V^\circ}$, since its minimal orbit consists of gauge transforms of $e^{\mathrm{i}\eta A_*(B)}U_0$, whose curvature and current are close to those of $U_0\in\mathfrak{U}(\theta')$. Let $\mathcal{D}=\{U\in\mathfrak{U}(\theta):U^{(k)}\in\mathcal{K}_{V^\circ}\}$, a compact set. The function $U\mapsto S(U)-S(U_k(U^{(k)}))$ is continuous on $\mathcal{D}$, since $S(U_k(e^{\mathrm{i}B}V^\circ))=\Phi(A_*(B))$; it is

nonnegative, and it vanishes exactly on the minimal orbits (Proposition 26.6). The set $\mathcal{O}_\delta = \Upsilon\big(\{(x,B) : |x - x_*(B)| < \delta,\ |B|_\infty < 2b_0\} \times \mathfrak{G}_0\big)$ is open and contains these orbits, and a point of $\mathfrak{B}(V)$ in $\mathcal{O}_\delta$ lies in $\mathcal{P}_\delta(V)$, since $\Upsilon^{(k)} = e^{\mathrm{i}B}V^\circ$. The function ϱ of Lemma 54.1(c), with $\mathcal{K}_{V^\circ}$ in place of $\mathcal{K}_0$, is continuous on $\mathcal{D}$ and vanishes exactly on these orbits. So there is $c > 0$ such that $S(U) \geq S(U_k(U^{(k)})) + 2c$ on the compact sets $\mathcal{D} \setminus \mathcal{O}_\delta$ and $\{U \in \mathcal{D} : \varrho(U) \geq r\}$. Moreover (V, θ_0) is admissible for every $V \in \mathcal{K}_{V^\circ}$, since $\theta \leq \theta_0$. So, by Lemma 54.2, the masses of the sets $\mathfrak{B}(V) \cap \mathfrak{U}(\theta) \subset \mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$, $V \in \mathcal{K}_{V^\circ}$, for the measure of I_k, are bounded by some m. For δ and b_0 small, $\mathcal{P}_\delta(V) \subset \mathfrak{B}(V) \cap \mathfrak{U}(\theta)$, as for the minimal orbits: its points are gauge transforms of $e^{\mathrm{i}\eta A(x,B)}U_0$ with $A(x,B)$ small, so on the fixed torus their curvature and current are within $o(1)$ of those of $U_0 \in \mathfrak{U}(\theta')$, $\theta' < \theta$. Now $I_k(V; g, r)$ is the integral over $\mathscr{N}_r(V) = \{U \in \mathfrak{B}(V) \cap \mathfrak{U}(\theta) : \varrho(U) < r\}$ (Lemma 54.1(c)). It differs from the integral over $\mathcal{P}_\delta(V)$ by at most the integrals over $\mathscr{N}_r(V) \setminus \mathcal{P}_\delta(V) \subset \mathcal{D} \setminus \mathcal{O}_\delta$ and over $\mathcal{P}_\delta(V) \setminus \mathscr{N}_r(V) \subset \{U \in \mathcal{D} : \varrho(U) \geq r\}$, hence by at most $m e^{-(S(U_k(V))+2c)/g^2}$. By (iv) this is a relative error $O(g^{-N} e^{-2c/g^2})$, uniformly for $V \in \mathcal{K}_{V^\circ}$.

(vi) *Conclusion.* By (iv) and (v), uniformly for $V \in \mathcal{K}_{V^\circ}$,

$$I_k(V; g, r) = (2\pi g^2)^{N/2} e^{-S(U_k(V))/g^2} \det H(B)^{-1/2} \varsigma(x_*(B), B)(1 + o(1)).$$

At each single $V \in \mathcal{K}_{V^\circ}$ the expansion of the paragraphs "The constraint set"–"Laplace's method" holds as well (it is (iv)–(v) with $V^\circ = V$, $B = 0$), and the terms of order g^0 of the two agree. Since $c_k(g)$ is $(2\pi g^2)^{N/2}$ times a constant, $\Gamma_k^{(1)}(V) = -\frac{1}{2} \log\det H(B(V)) + \log \varsigma(x_*(B(V)), B(V))$ on $\mathcal{K}_{V^\circ}$, up to a constant, and the $o(1)$ of the statement is uniform on $\mathcal{K}_{V^\circ}$.

□

Lemma 54.4 (Choice of the neighbourhood)**.** *Let $V \in \mathfrak{U}_k(a_1)$ and $\theta \leq \theta_0$ be such that (V, θ) is admissible (for instance V small field and θ as above), let $r \leq r_0$, and let $\mathscr{N}$ be a Borel set with $\mathscr{N}_r(V) \subset \mathscr{N} \subset \mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$. Then $I_k(V; g, r)$ and the same integral over $\mathscr{N}$ differ by a factor $1 + O(e^{-c/g^2})$, with $c > 0$. In particular their terms of order g^0 agree.*

No invariance of $\mathscr{N}$ under $\mathfrak{G}_0$ is required.

Proof. Put $S_* = S(U_k(V))$. The set $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ is compact, since $U \mapsto U^{(k)}$ is continuous on $\mathfrak{U}_0(c_1)$. Its part outside $\mathscr{N}_r(V)$ lies in $K_1 \cup K_2$. Here K_1 is the set of its elements whose curvature or current has a value of norm at least θ, and $K_2 = \{U \in \mathfrak{B}(V) \cap \mathfrak{U}(\theta) : \varrho(U) \geq r\}$, with ϱ of Lemma 54.1(c) for $\mathcal{K}_0 = \{V\}$. Both sets are compact and invariant under $\mathfrak{G}_0$, since curvature and current are gauge covariant, and by that lemma. They do not meet $\mathscr{M}(V)$, since $\mathscr{M}(V) \subset \mathfrak{U}(\theta')$ with $\theta' < \theta$, and $\varrho = 0$ on $\mathscr{M}(V)$. By Proposition 26.6, $S > S_*$ on $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ outside $\mathscr{M}(V)$. So $S \geq S_* + 2c$ on $K_1 \cup K_2$, for some $c > 0$, and the integral over $\mathscr{N} \setminus \mathscr{N}_r(V)$ is at most $e^{-(S_*+2c)/g^2} m$, where m is the mass of $K_1 \cup K_2$ for the measure $\prod_b dU(b) \prod_c \delta(U^{(k)}(c) V(c)^{-1})$. This mass is finite, by Lemma 54.2, since the pair (V, θ_0) is admissible and $K_1 \cup K_2 \subset \mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$. On the other hand, by Proposition 54.3, $I_k(V; g, r) \geq \frac{1}{2} c_k(g) \exp(-S_*/g^2 + \Gamma_k^{(1)}(V))$ for g small, and $c_k(g)$ is a constant times a power of g, by Laplace's method. So the two integrals differ by a factor $1 + O(e^{-c/g^2})$. □ Details

The one-loop terms of the steps. For the step from scale j to scale $j+1$, in the units of step j, Theorem 34.9 gives for the gauge-fixed step $T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)$ the prefactor $\mathcal{N}_j \exp(\Gamma_j^{\mathrm{step}}(U'))$, where

$$\Gamma_j^{\mathrm{step}}(U') = -\tfrac{1}{2} \log\det \Delta_j + \sum_c \log \mathfrak{j}_c, \qquad U' = U_{j+1}(V'),$$

with Δ_j the fluctuation operator of the step on the variables $\mathfrak{F}$ (Lemma 34.1, Proposition 34.8) and $\mathfrak{j}_c$ the Jacobians of Theorem 34.9(b). The operator Δ_j contains the quadratic part $\|\mathfrak{G}_W \cdot\|^2$ of the gauge fixing. Both terms are functions of the background U', a configuration on the finest lattice, and $\mathcal{N}_j$ does not depend on it. Up to a constant, Γ_j^{step} does not depend on the gauge fixing. Remark 34.10 sketches this, and the sketch is not used below; Proposition 54.6 below proves the composite version of this statement. The Jacobians are not negligible: $\log \mathfrak{j}_c$ is quadratic in the curvature of scale $j+1$ (Proposition 47.8(c)), so it contributes to the marginal tensor below. For $j=0$ the base is $W=U'$ and the background map is trivial. The formula of Theorem 34.9 then holds with the fluctuation operator Δ_0 of Proposition 52.2(b): this is Lemma 52.3, the *formula for the first step*. It is used by Proposition 54.6 through its case $k=1$, and hence by Corollary 54.18; its part (c) is used by Lemma 54.9(b), and hence (c), for the step from scale 0. Lemma 54.5, which concerns the steps from the scales $j \geq 1$, does not use it. In the flow of Part X the same function appears as the one-loop part of the new generation: $\Gamma_j^{\mathrm{step}}(U') - \Gamma_j^{\mathrm{step}}(1) = \sum_X \mathfrak{e}(X,U') + \sum_c \log \mathfrak{j}_c(U')$ (Proposition 41.7(a); Proposition 52.2(c) for $j=0$) is the sum of the one-loop part E'_{ol} of the raw new generation of the step (Definition 53.3, Section 58). It involves neither the coupling nor the old generations. Its coefficient, read off on the torus, is the one-loop part $\beta_{j+1}^{\mathrm{ol}}$ of β_{j+1}, and Proposition 58.6 compares it with the coefficient of the marginal tensor below.

Lemma 54.5 (The one-step relation)**.** *Let $k \geq 2$, let V be small field, and put $V^0 = U_k(V)^{(k-1)}$, a configuration of scale $k-1$, read in the units of step $k-1$. Then V^0 is small field at scale $k-1$, and, up to a constant independent of V,*

$$\Gamma_k^{(1)}(V) = \Gamma_{k-1}^{(1)}(V^0) + \Gamma_{k-1}^{\mathrm{step}}(U_k(V)).$$

Proof. Plan. Integrate out the configuration of scale $k-1$ ("Factorization"). The inner integral is I_{k-1}, which Proposition 54.3 evaluates at scale $k-1$. In the outer integral, replace the region by the fluctuation cutoff χ^{w}, by a two-radius estimate (the terms R_1, R_2). Then Theorem 34.9 applies to the outer integral, and the result is compared with Proposition 54.3 at scale k. Details

Notation. We use the notation of Sections 32 and 34 for the step from scale $k-1$ to scale k, in the units of step $k-1$; in particular, in this proof $\mathbb{T}$ and $\mathbb{T}'$ denote the lattices of scale $k-1$ and k, and $B'(y)$, $y \in \mathbb{T}'$, the blocks of the step. So $V'=V$, $U'=U_k(V)$, the base is $W=V^0=U'^{(k-1)}$, and $\mathcal{G}'$ is the group of real gauge transformations of the lattice of scale $k-1$ that are 1 on the lattice of scale k. Let c_L be the constant of Theorem 34.4, and r_2, C_L those of Theorem 34.9(d). We take $w \leq \min(r_2, c_L/(16C_L))$, and smaller where stated. We also take $8w \leq a'_1$ and $16B_3 w \leq L^{-3}\theta_0$ (both implied by $w \leq r_2$ if $c \leq \frac{1}{16}$ in Lemma 34.1); then the same holds for $w' < w$. Put $C_1 = 128C_M$, with C_M from Lemma 21.5, and $w' = w/(C_1 L)$.

The region. Let $\mathscr{A}$ be the set of configurations V_{k-1} on $\mathbb{T}$ such that, for every bond b, $V_{k-1}(b) = e^{\mathrm{i}Z(b)}W(b)$ with $Z(b) \in \mathfrak{g}$ and $|Z(b)| < w'$, and let $\mathscr{O} = \{V_{k-1}^u : V_{k-1} \in \mathscr{A},\ u \in \mathcal{G}'\}$ be its $\mathcal{G}'$-saturation. By (32.2), with w' in place of w, $\mathscr{A} \subset \mathfrak{M}$. Since $\mathfrak{M}$ is invariant under real gauge transformations, $\mathscr{O} \subset \mathfrak{M}$, and $\mathscr{O}$ is $\mathcal{G}'$-invariant. A gauge transformation acts on each bond variable by a left and a right translation, which preserve the bi-invariant distance on G. So every point of $\mathscr{O}$ lies within w', bondwise, of the orbit $\{(V^0)^u : u \in \mathcal{G}'\}$. Let $\varepsilon'_1 \leq a_{\mathrm{step}}$ bound the curvature of V in the units of step k, as in Theorem 34.9. By Lemma 31.6(a) and $4(C_*+1)L^{-2}a_{\mathrm{step}} \leq a'_1$ (proof of Theorem 34.9), $W \in \mathfrak{U}_{k-1}(\frac{1}{2}a'_1)$, with a'_1 of Lemma 31.6(c). Let $\mathcal{K}_0$ be the $\mathcal{G}'$-saturation of the set of the $e^{\mathrm{i}Z}W$ with $|Z(b)| \leq w'$ on all bonds. It is compact and contains $\mathscr{O}$, and, by the argument before (32.2), $\mathcal{K}_0 \subset \mathfrak{U}_{k-1}(\frac{1}{2}a'_1 + 4w')$.

Factorization. Integrating out the configuration of scale $k-1$ against the Haar δ-functions

gives, exactly,

$$\int \prod_b dU(b)\,\delta(U^{(k)},V)\,F(U)$$
$$= \int \prod_{b'} dV_{k-1}(b')\;\delta(\mathfrak{a}_k(V_{k-1}),V)\int \prod_b dU(b)\,\delta(U^{(k-1)},V_{k-1})F(U),$$

for every integrable F that vanishes where $U^{(k)}$ is not defined, since $U^{(k)}=\mathfrak{a}_k(U^{(k-1)})$.

The region $\mathscr{N}'$. Put $\theta_{k-1}=L^{-3}\theta_0$. For $V_{k-1}\in\mathcal{K}_0$ the pair (V_{k-1},θ_{k-1}) is admissible at scale $k-1$, in the units of step $k-1$: $V_{k-1}\in\mathfrak{U}_{k-1}(a_1)$, and by Theorem 30.3(b) its minimal orbit lies in $\mathfrak{U}(B_3(\frac12 a_1'+4w'))\subset\mathfrak{U}(\frac34\theta_{k-1})$, since $B_3a_1'\le L^{-3}\theta_0$. Take $F=\mathbf{1}_{\mathscr{N}'}e^{-S/g^2}$, with the region $\mathscr{N}'$ of configurations U for which $V_{k-1}=U^{(k-1)}$ lies in $\mathscr{O}$ and $U\in\mathscr{N}_{r'}(V_{k-1})$, the neighbourhood of the setting at scale $k-1$, with θ_{k-1} in place of θ and $0<r'\le r_0$. This region has two properties:

(N1) its intersection with $\{U:U^{(k)}=V\}$ lies in $\mathfrak{B}(V)\cap\mathfrak{U}(\theta_0)$;

(N2) it is a Borel set, and it contains $\mathscr{N}_r(V)$ for r small.

For (N1), let $U\in\mathscr{N}'$. Then $U\in\mathfrak{U}(\theta_{k-1})$ in the units of step $k-1$. Passing to the units of step k multiplies curvatures by L^2 and currents by L^3 (proof of Lemma 30.2). So $U\in\mathfrak{U}(\theta_0)\subset\mathfrak{U}_0(c_1)$ in the units of step k, and $U\in\mathfrak{B}(V)$ if $U^{(k)}=V$. This is the reason for the choice of θ_{k-1}.

For (N2), let ϱ_{k-1} be the function of Lemma 54.1(c) at scale $k-1$, for θ_{k-1} and $\mathcal{K}_0$. It is continuous on the compact set $\mathcal{D}=\{U\in\mathfrak{U}(\theta_{k-1}):U^{(k-1)}\in\mathcal{K}_0\}$, and $\mathscr{N}'=\{U\in\mathcal{D}:U^{(k-1)}\in\mathscr{O},\ \varrho_{k-1}(U)<r'\}$, by that lemma. This set is relatively open in $\mathcal{D}$, hence a Borel set. Let $\mathcal{D}^\circ$ be the set of the U whose curvature and current, in the units of step $k-1$, are smaller than θ_{k-1}, and for which $U^{(k-1)}\in\mathscr{O}$. It is open, since $\mathscr{O}$ is open and curvature, current and $U\mapsto U^{(k-1)}$ are continuous there, and $\mathcal{D}^\circ\subset\mathcal{D}$. So $\mathscr{N}^\circ=\{U\in\mathcal{D}^\circ:\varrho_{k-1}(U)<r'\}$ is open, and $\mathscr{N}^\circ\subset\mathscr{N}'$. It contains $\mathscr{M}(V)$. Indeed, let $u\in\mathfrak{G}_0$, with restriction $\tilde u\in\mathcal{G}'$ to the lattice of scale $k-1$. By Lemma 31.6(b) and Theorem 25.1(g), $U_k(V)^u$ lies in the minimal orbit, at scale $k-1$, of $(U_k(V)^u)^{(k-1)}=(V^0)^{\tilde u}\in\mathscr{O}$. So $\varrho_{k-1}(U_k(V)^u)=0$, and $U_k(V)^u\in\mathfrak{U}(\frac34\theta_{k-1})$ in the units of step $k-1$, by the admissibility shown above; hence $U_k(V)^u\in\mathscr{N}^\circ$. By Lemma 54.1(c) at scale k, with $\{V\}$ in place of $\mathcal{K}_0$, $\mathscr{N}_r(V)\subset\mathscr{N}^\circ\subset\mathscr{N}'$ for r small.

The integrand on the left side of the factorization vanishes unless $U^{(k)}=V$. By (N1), (N2) and Lemma 54.4, applied to $\mathscr{N}'\cap\mathfrak{B}(V)$, the left side is $I_k(V;g,r)\,(1+O(e^{-c/g^2}))$. On the right side the inner integral is $I_{k-1}(V_{k-1};g,r')$ if $V_{k-1}\in\mathscr{O}$, and 0 otherwise. So, up to a factor $1+O(e^{-c/g^2})$, $I_k(V;g,r)$ is the outer integral $T(\mathbf{1}_{\mathscr{O}}\,I_{k-1}(\cdot\,;g,r'))(V)$, where T is the map (31.1) of the step.

The inner integral. The orbit of V^0 under $\mathcal{G}'$ is compact and consists of small-field configurations of scale $k-1$, since $U_{k-1}(V^0)=U_k(V)$ (Lemma 31.6(b)) and the conditions of the setting are invariant under gauge transformations; this is the first claim. Let $\mathcal{K}_w$ be the $\mathcal{G}'$-saturation of the set of the $e^{\mathrm{i}Z}W$ with $|Z(b)|\le w$ on all bonds. As for $\mathcal{K}_0$, with w in place of w', it is compact, it contains $\mathcal{K}_0\supset\mathscr{O}$ and the closed w-neighbourhood of V^0, bondwise, and $\mathcal{K}_w\subset\mathfrak{U}_{k-1}(\frac12 a_1'+4w)$. By the choice of w, $8w\le a_1'$ and $16B_3w\le L^{-3}\theta_0$. Then, as in the paragraph "The region $\mathscr{N}'$", (V_{k-1},θ_{k-1}) is admissible at scale $k-1$ for every $V_{k-1}\in\mathcal{K}_w$, and $V_{k-1}\in\mathfrak{U}_{k-1}(a_1)$. By Proposition 54.3 at scale $k-1$, with θ_{k-1} in place of θ, which does not change $\Gamma^{(1)}_{k-1}$, the expansion of I_{k-1} holds locally uniformly on the set of these configurations, hence uniformly on the compact set $\mathcal{K}_w$:

$$I_{k-1}(V_{k-1};g,r')=c_{k-1}(g)\exp\Big(-\frac{1}{g^2}S(U_{k-1}(V_{k-1}))+\Gamma^{(1)}_{k-1}(V_{k-1})+o(1)\Big).$$

This uniformity is the paragraph "Uniformity in V" of the proof of Proposition 54.3, which uses only that (V_{k-1},θ_{k-1}) is admissible and $V_{k-1}\in\mathfrak{U}_{k-1}(a_1)$: near a point V° of $\mathcal{K}_w$, with a

reference $U_0 \in \mathscr{M}(V^\circ)$, Corollary 30.5 writes the minimal orbit of $e^{\mathrm{i}B}V^\circ$ as that of $\mathsf{U}_{U_0}(A_*(B))$, so the phase, the constraint and the density of Laplace's method depend analytically on B, with the Hessian bound uniform. Let $\rho(V_{k-1}) = \exp(-g^{-2}S(U_{k-1}(V_{k-1})) + \Gamma^{(1)}_{k-1}(V_{k-1}))$ on $\mathcal{K}_w$. Since T is positive, $T(\mathbf{1}_{\mathscr{O}} I_{k-1}) = c_{k-1}(g)\, e^{o(1)}\, T(\mathbf{1}_{\mathscr{O}}\rho)$.

Gauge fixing. The function ρ is gauge invariant: $S(U_{k-1}(\cdot))$ by Theorem 30.3, $\Gamma^{(1)}_{k-1}$ by Proposition 54.3. So $\mathbf{1}_{\mathscr{O}}\rho$ is $\mathcal{G}'$-invariant, bounded, and vanishes outside $\mathfrak{M}$. By Lemma 32.6(iv), $T(\mathbf{1}_{\mathscr{O}}\rho) = T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho)$.

From the region to the cutoff. We show that

$$T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho)(V) = T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(V)\,\big(1 + O(\varepsilon_1 + \varepsilon_2)\big), \qquad \varepsilon_1 = C|\mathbb{T}|e^{-cw^2/g^2}, \quad \varepsilon_2 = C_{\mathbb{T}}e^{-c'_L w^2/g^2},$$

with $c > 0$ absolute, C depending on G, $c'_L > 0$ depending on L, and $C_{\mathbb{T}}$ depending on L and on the torus; this is the two-radius estimate, with the inner radius w' and the outer radius w. Write $\mathbf{1}_{\mathscr{O}} - \chi^{\mathrm{w}} = \mathbf{1}_{\mathscr{O}}(1-\chi^{\mathrm{w}}) - (1-\mathbf{1}_{\mathscr{O}})\chi^{\mathrm{w}}$, and let R_1, R_2 be the two corresponding terms of $T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho) - T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)$. It suffices to show $0 \le R_1 \le \varepsilon_1 T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho)$ and $0 \le R_2 \le \varepsilon_2 T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)$.

The term R_1: gauge translates leaving the cutoff. Put

$$h(V_{k-1}) = \prod_c \delta\big(\mathfrak{a}_k(V_{k-1})(c)V(c)^{-1}\big)\,\mathbf{1}_{\mathscr{O}}(V_{k-1})\,\rho(V_{k-1}).$$

It is $\mathcal{G}'$-invariant, by Lemma 21.10(b) (with the convention on δ-functions of Section 32). As in the proof of Lemma 32.3(a), substituting V^u_{k-1} for V_{k-1}, $u \in \mathcal{G}'$, and averaging over u gives

$$R_1 = \int \prod_b dV_{k-1}(b)\; h\,\psi, \qquad \psi(V_{k-1}) = \int_{\mathcal{G}'} du\; \mathrm{gf}_W(V^u_{k-1})(1 - \chi^{\mathrm{w}}(V^u_{k-1})),$$

and, in the same way, $T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho) = \int \prod_b dV_{k-1}(b)\, h$, since $\int_{\mathcal{G}'} \mathrm{gf}_W(V^u_{k-1})\,du = 1$ on $\mathfrak{M}$. The function ψ is $\mathcal{G}'$-invariant, so it suffices to bound it on $\mathscr{A}$. Let $V_{k-1} \in \mathscr{A}$, so that $V_{k-1}(b) = e^{\mathrm{i}Z(b)}W(b)$ with $Z(b) \in \mathfrak{g}$, $|Z(b)| < w'$, and let $N = \{e^{\mathrm{i}K} : K \in \mathfrak{g},\ |K| \le w/4\}$.

- If $u(x) \in N$ for all x, then $\chi^{\mathrm{w}}(V^u_{k-1}) = 1$. Indeed,

  $$V^u_{k-1}(b)W(b)^{-1} = u(b_-)\, e^{\mathrm{i}Z(b)}\,\big(W(b)u(b_+)^{-1}W(b)^{-1}\big),$$

  and by Lemma 21.2(b),(d) its logarithm is in $\mathfrak{g}$ and bounded by $\frac{w}{4} + w' + \frac{w}{4} < w$.

- Let $x \in B'(y) \setminus \{y\}$, $y \in \mathbb{T}'$, and $u(x) \notin N$. By (21.2), $V^u_{k-1}(y,x)W(y,x)^{-1} = e^{\mathrm{i}H}v^{-1}$ with $e^{\mathrm{i}H} = V_{k-1}(y,x)W(y,x)^{-1}$ and $v = W(y,x)u(x)W(y,x)^{-1} \notin N$. Since $w' \le c_0 L^{-1}$, the proof of Lemma 32.3(c) gives $|H| \le 4C_M L w' = w/32$. For $v \in G \setminus N$ we have $|v-1| > w/8$: either $|v-1| > c_G \ge w/8$, since $w \le \frac{1}{2}c_G$ (Definition 32.4), or $v = e^{\mathrm{i}\ell(v)}$ with $\ell(v) \in \mathfrak{g}$, $|\ell(v)| > w/4$ and $|\ell(v)| \le 2|v-1|$ (Lemma 21.2(a)). Hence $|e^{\mathrm{i}H}v^{-1} - 1| = |v^{-1} - e^{-\mathrm{i}H}| \ge |v-1| - 2|H| \ge w/16$, and the factor of (y,x) in $\mathrm{gf}_W(V^u_{k-1})$ is at most $z^{-1}e^{-w^2/(512g^2)}$.

Since χ^{w} takes only the values 0 and 1 (Definition 32.4), $1 - \chi^{\mathrm{w}}(V^u_{k-1}) \le \sum_x \mathbf{1}(u(x) \notin N)$. The weight $\mathrm{gf}_W(V^u_{k-1})$ is a product over the points x, and each factor integrates to 1 over $u(x) \in G$. Hence $\psi \le |\mathbb{T}|z^{-1}e^{-w^2/(512g^2)}$ on $\mathscr{A}$. Since $z(g)^{-1} \le Cg^{-\dim\mathfrak{g}}$, this is at most ε_1 for g small.

The term R_2: orbits off the inner region. By Proposition 32.8, applied to the bounded function $(1 - \mathbf{1}_{\mathscr{O}})\rho$,

$$R_2 = z^{-|\mathbb{T}\setminus\mathbb{T}'|}\int \prod_{b\in\mathfrak{F}} \sigma(Z(b))\,dZ(b)\;\chi^{\mathrm{w}}(\hat{Z})\prod_c \mathsf{J}_c\; e^{-g^{-2}F(Z)}\,\big(e^{\Gamma^{(1)}_{k-1}}(1-\mathbf{1}_{\mathscr{O}})\big)\big(e^{\mathrm{i}\hat{Z}(Z)}W\big),$$

with $F = \mathcal{S} \circ \hat{Z} + \Phi_W$ as in Lemma 34.1. On the support, $|\hat{Z}(b)| \le w \le r_2$ on all bonds. By Lemma 34.1, Corollary 34.6 and $C_L w \le c_L/16$,

$$F(Z) \ge S(U') + \tfrac{1}{8} c_L \|Z\|^2 - C_L w \|Z\|^2 \ge S(U') + \tfrac{1}{16} c_L \|Z\|^2.$$

Since $e^{\mathrm{i}\hat{Z}} W \notin \mathscr{A}$ and $\hat{Z}$ is real, some bond has $|\hat{Z}(b)| \ge w'$; with $|\hat{Z}|_\infty \le C L^d |Z|_\infty$ (Lemma 32.7(c)) this gives $\|Z\| \ge w'' := w'/(CL^d)$. The factors σ, J_c and $e^{\Gamma^{(1)}_{k-1}}$ are bounded on the domain by a constant depending on L and the torus. With $n = |\mathfrak{F}| \dim \mathfrak{g}$ and $\int_{\|Z\| \ge w''} e^{-a\|Z\|^2} dZ \le e^{-a w''^2/2} (2\pi/a)^{n/2}$ for $a = c_L/(16 g^2)$, we get

$$R_2 \le C_{\mathbb{T}}\, z^{-|\mathbb{T}\setminus\mathbb{T}'|} (2\pi g^2)^{n/2} e^{-S(U')/g^2} e^{-c_L w''^2/(32 g^2)}.$$

By the next step, $T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho) \ge \frac{1}{2}\mathcal{N}_{k-1} e^{-S(U')/g^2 + \Gamma^{\mathrm{step}}_{k-1}(U') + \Gamma^{(1)}_{k-1}(V^0)}$ for g small, and $\mathcal{N}_{k-1}$ is $z^{-|\mathbb{T}\setminus\mathbb{T}'|}(2\pi g^2)^{n/2}$ times a constant (Theorem 34.9(a)). So $R_2 \le \varepsilon_2 T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)$, with $c_L' = c_L/(32 C^2 C_1^2 L^{2d+2})$.

The outer integral. By Lemma 31.6, the minimal configuration of the step is V^0, and $U_{k-1}(V^0) = U_k(V)$. The density ρ has the form (31.2) with $\mathfrak{E} = \Gamma^{(1)}_{k-1}$, real analytic and gauge invariant, on the support of χ^{w}, which is all that Theorem 34.9 uses. Theorem 34.9, applied to the step from scale $k-1$ to scale k, with w fixed and $g \to 0$, gives

$$\begin{aligned} T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(V) = \mathcal{N}_{k-1} \exp\Big(-\frac{1}{g^2} S(U_k(V)) + \Gamma^{\mathrm{step}}_{k-1}(U_k(V)) \Big) \\ \times \int d\mu_C(Y)\, \chi^{\mathrm{w}}(\hat{Z}(gY))\, e^{-g^{-2}(\mathcal{V}+G_3)(gY) + \mathfrak{J}(gY) + \mathfrak{E}(e^{\mathrm{i}\hat{Z}(gY)} W)}. \end{aligned}$$

As $g \to 0$:

- $g^{-2}|(\mathcal{V} + G_3)(gY)| \le C_L g |Y|_\infty \|Y\|^2$ (Theorem 34.9(d));
- $\mathfrak{J}(gY) = O(g|Y|_\infty)$ (Theorem 34.9(e));
- $\mathfrak{E}(e^{\mathrm{i}\hat{Z}(gY)} W) \to \mathfrak{E}(W) = \Gamma^{(1)}_{k-1}(V^0)$, by continuity.

On the support of χ^{w} the integrand is bounded by $e^{C_L w \|Y\|^2 + C}$, which is integrable against $d\mu_C$, since $C \le 4 c_L^{-1}$ and $C_L w \le c_L/16$. By dominated convergence the integral tends to $e^{\Gamma^{(1)}_{k-1}(V^0)}$.

Conclusion. Collecting the steps,

$$I_k(V; g, r) = c_{k-1}(g) \mathcal{N}_{k-1} \exp\Big(-\frac{1}{g^2} S(U_k(V)) + \Gamma^{(1)}_{k-1}(V^0) + \Gamma^{\mathrm{step}}_{k-1}(U_k(V)) + o(1) \Big).$$

Compare with Proposition 54.3 at scale k. The difference $D(V) = \Gamma^{(1)}_k(V) - \Gamma^{(1)}_{k-1}(V^0) - \Gamma^{\mathrm{step}}_{k-1}(U_k(V))$ does not depend on g, and $D(V) = \log\big(c_{k-1}(g)\mathcal{N}_{k-1}/c_k(g)\big) + o(1)$ as $g \to 0$, where the first term does not depend on V. So $D(V)$ is the limit of this term, a constant independent of V, and $c_k(g) = c_{k-1}(g)\mathcal{N}_{k-1}$ up to a constant factor. Of $\Gamma^{(1)}_{k-1}$ the proof has used only Proposition 54.3, not its relation to the steps below scale $k-1$. □

Proposition 54.6 (Composition of the one-loop terms)**.** *For every $k \ge 1$ and every small-field V, up to a constant independent of V,*

$$\Gamma^{(1)}_k(V) = \sum_{j=0}^{k-1} \Gamma^{\mathrm{step}}_j(U_k(V)).$$

Proof. For $k=1$ the proof of Lemma 54.5 applies to $I_1(V;g,r)=T(\mathbf{1}_{\mathscr{N}_r(V)}e^{-S/g^2})(V)$, with $\rho=e^{-S/g^2}$, i.e. with $\mathfrak{E}=0$ and $\Gamma_0^{(1)}:=0$, and with Lemma 52.3 in place of Lemma 31.6 and Theorem 34.9. Its part (a) replaces Lemma 31.6; its parts (b), (c) replace Lemma 34.1 and Corollary 34.6, with $c_L^{(0)}=2C_{\mathrm{P}}^{-1}$ in place of c_L; and its part (d) replaces Theorem 34.9. Accordingly r_2, C_L and a_{step} are replaced by $r_2^{(0)}$, the C_L of that lemma and $a_{\mathrm{step}}^{(0)}$. At scale 0 the fine lattice is the lattice of scale 0, the base is $W=U_1(V)$ (Lemma 52.3(a)), and the region $\mathscr{N}'$ is replaced by the open set $\mathscr{O}$ itself. It has the properties (N1) and (N2) of that proof, with $\mathscr{O}$ in place of $\mathscr{N}'$. Details

(N1) Let $V_0\in\mathscr{O}$. By Lemma 52.3(a) and the choice of its constants, $W\in\mathfrak{U}_0(L^{-2}(C_*+1)\varepsilon_1')\subset\mathfrak{U}_0(\frac14 a_1')$. By the argument before (32.2), with w' in place of w, $V_0\in\mathfrak{U}_0(\frac14 a_1'+4w')\subset\mathfrak{U}_0(a_1')$. As in the proof of Lemma 52.3(a), Theorem 30.3(b) at $k=0$ gives $V_0\in\mathfrak{U}(B_3a_1')$ in the units of step 0, so its curvature and current are at most $L^3B_3a_1'\le\theta_0$ in the units of step 1. So $V_0\in\mathfrak{B}(V)\cap\mathfrak{U}(\theta_0)$ if $V_0^{(1)}=V$.

(N2) At $k=1$, $\mathfrak{G}_0=\mathcal{G}'$, so $\mathscr{M}(V)=\{W^u:u\in\mathcal{G}'\}$, which lies in the open set $\mathscr{O}$. By Lemma 54.1(c), with $\{V\}$ in place of $\mathcal{K}_0$, $\mathscr{N}_r(V)\subset\mathscr{O}$ for r small.

By Lemma 54.4, applied to $\mathscr{O}\cap\mathfrak{B}(V)$, $I_1(V;g,r)=T(\mathbf{1}_{\mathscr{O}}e^{-S/g^2})(V)\,(1+O(e^{-c/g^2}))$. Proceed as from the step *Gauge fixing* on. This gives $\Gamma_1^{(1)}(V)=\Gamma_0^{\mathrm{step}}(U_1(V))$ up to a constant.

Let $k\ge2$ and assume the claim for $k-1$. By Lemma 54.5, $V^0=U_k(V)^{(k-1)}$ is small field at scale $k-1$ and $\Gamma_k^{(1)}(V)=\Gamma_{k-1}^{(1)}(V^0)+\Gamma_{k-1}^{\mathrm{step}}(U_k(V))$ up to a constant. By the induction hypothesis and Lemma 31.6(b),

$$\Gamma_{k-1}^{(1)}(V^0)=\sum_{j<k-1}\Gamma_j^{\mathrm{step}}(U_{k-1}(V^0))=\sum_{j<k-1}\Gamma_j^{\mathrm{step}}(U_k(V))$$

up to a constant. This gives the claim. □

So the one-loop terms of the steps need not be computed one by one. Their sum is the one-loop term of the composite map, a single determinant on the constraint set modulo the residual gauge group. Section 55 writes it with the operators of Parts II and III.

54.1 Marginal tensors and lattice symmetry

Let Φ be a real analytic function of coarse configurations near 1, on the unit lattice $\mathbb{T}$, invariant under the gauge transformations of $\mathbb{T}$, with vanishing first variation at 1. Let $\Pi_\Phi(x,y)$ be its second variation at 1, i.e. the kernel of $B\mapsto\frac{d^2}{dt^2}\Phi(e^{itB})|_{t=0}$ with respect to the trace pairing $\sum_b\mathrm{tr}(B(b)B'(b))$ of Section 23, with values in $\mathfrak{g}\otimes\mathfrak{g}$. We assume that Π_Φ is invariant under the translations of $\mathbb{T}$ and that, as the side of the torus tends to infinity, its kernels converge with uniform exponential decay, in the sense of Lemma 54.10(e); we use the limit, the infinite-volume second variation, again denoted Π_Φ. We use the *bond-midpoint convention*: the value $B(b)$ sits at the midpoint m_b of b, and $\hat B_\mu(p)=\sum_b e^{-\mathrm{i}p\cdot m_b}B(b)$, the sum over the positively oriented bonds of direction μ. The Fourier transform $\Pi_\Phi(p)_{\mu\nu}$ is analytic near $p=0$. In the convention with $B(b)$ at b_- it is multiplied by the phases $e^{\mathrm{i}(p_\mu-p_\nu)/2}$; if $\Pi_\Phi(p)=O(|p|^2)$ in one convention, this changes it by $O(|p|^3)$, so the tensor below does not depend on the convention. In this part G is simple (Section 1, Scope). By Ad-invariance Π_Φ is then proportional to the identity on $\mathfrak{g}$ [18, (4.33)], and we suppress the Lie algebra indices relative to the trace pairing.

Definition 54.7 (Marginal tensor)**.** If $\Pi_\Phi(p)=O(|p|^2)$ as $p\to0$, the *marginal tensor* of Φ is the tensor T_Φ, symmetric in κ,λ, with $\Pi_\Phi(p)_{\mu\nu}=\sum_{\kappa\lambda}(T_\Phi)_{\mu\nu\kappa\lambda}p_\kappa p_\lambda+O(|p|^3)$. Its *isotropic coefficient* β_Φ is the coefficient of the orthogonal projection of T_Φ, in the Euclidean inner product of tensors, onto the tensor of $\delta_{\mu\nu}|p|^2-p_\mu p_\nu$. With the hypercubic symmetry,

$T_\Phi(p) = \beta_\Phi(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$ (Lemma 54.9(c)), and β_Φ is Bałaban's coefficient [18, (5.37), (5.42)].

Remark 54.8 (The orders 0 and 1 in p). Since Φ is gauge invariant and its first variation at 1 vanishes, $\Pi_\Phi(B, D\lambda) = 0$ for all B and scalar fields λ. In the bond-midpoint convention $\widehat{D\lambda}_\mu(p) = \mathrm{i}\hat{p}_\mu\hat{\lambda}(p)$ with $\hat{p}_\mu = 2\sin(p_\mu/2)$, so Π_Φ is transverse: $\sum_\mu \hat{p}_\mu \Pi_\Phi(p)_{\mu\nu} = 0$. In infinite volume a constant field B is pure gauge to first order, so $\Pi_\Phi(0) = 0$. A term linear in p is not excluded by transversality and reality: $\mathrm{i}c\,\varepsilon_{\mu\nu\lambda\sigma} n_\sigma p_\lambda$, with a lattice vector n, is of Chern–Simons type. It is excluded by the point reflection $x \mapsto -x$, which preserves orientation in $d = 4$. The functions of this part are invariant under it, by Lemma 54.9 below. A nonzero linear term would not be harmless: under the change of scale it would produce relevant terms of order 1 per step.

Let $\mathfrak{S}_k$ be the group of lattice symmetries of Section 2, generated by the translations of $\mathbb{T}$ and by B_4 about the points of $\mathbb{T}$. It acts on configurations by $(rV)(b) = V(r^{-1}b)$.

Lemma 54.9 (Lattice symmetry of the one-loop terms). *(a) Let V be small field and $r \in \mathfrak{S}_k$. Then rV is small field, and $\Gamma_k^{(1)}(rV) = \Gamma_k^{(1)}(V)$.*

(b) Let $j \geq 0$, and let V' be a small-field configuration of scale $j+1$, in the units of step $j+1$. Then $\Gamma_j^{\mathrm{step}}(U_{j+1}(rV')) = \Gamma_j^{\mathrm{step}}(U_{j+1}(V'))$ for every $r \in \mathfrak{S}_{j+1}$. Here, as everywhere in this section, Γ_j^{step} is evaluated at the background fields, the critical references of Theorem 34.9.

(c) Let Φ be one of the functions $\Gamma_k^{(1)}$ or $V' \mapsto \Gamma_j^{\mathrm{step}}(U_{j+1}(V'))$ with $j \geq 0$. Assume that its second variations at 1 on the tori converge with uniform exponential decay as the side tends to infinity (Lemma 54.10(e)). Then, in the bond-midpoint convention, $\Pi_\Phi(Rp) = R\,\Pi_\Phi(p)R^T$ for $R \in B_4$; in particular $\Pi_\Phi(-p) = \Pi_\Phi(p)$. Hence $\Pi_\Phi(p) = O(|p|^2)$, the hypothesis of Definition 54.7 holds, and

$$T_\Phi(p)_{\mu\nu} = \beta_\Phi(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$$

with a single coefficient β_Φ.

The hypothesis of (c) is proved in Lemma 56.2(c) for $\Gamma_k^{(1)}$, and in Lemma 58.5 for the functions Γ_j^{step}.

Proof. (a) By Theorem 30.3(c), rV has the same curvature as V, and its minimal orbit is the image under r of that of V. By Lemma 21.10, $U_k(rV)^{(j+1)}$ is a gauge transform of $r(U_k(V)^{(j+1)})$ for every $j < k$. The conditions of the setting are bounds on curvatures and on the side of the torus, which are invariant under gauge transformations and lattice symmetries. So rV is small field. The class $\mathfrak{U}(\theta')$ is invariant under $\mathfrak{S}_k$ (proof of Theorem 25.1(h)), so the same θ serves for V and for rV. Details

To avoid a clash with the symmetry r, write $r' \in (0, r_0]$ for the radius of the neighbourhoods, and let $\mathscr{N} = \mathscr{N}_{r'}(V) \cup r^{-1}\mathscr{N}_{r'}(rV)$. The map $U \mapsto rU$ maps $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ onto $\mathfrak{B}(rV) \cap \mathfrak{U}(\theta_0)$ and $\mathfrak{G}_0$-orbits onto $\mathfrak{G}_0$-orbits (proof of Theorem 30.3(c)), and it is a homeomorphism. So $\mathscr{N}$ is a Borel subset of $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ that contains $\mathscr{N}_{r'}(V)$, by Lemma 54.1(c), and $r\mathscr{N}$ is one of $\mathfrak{B}(rV) \cap \mathfrak{U}(\theta_0)$ that contains $\mathscr{N}_{r'}(rV)$. By Lemma 54.4, the integrals over $\mathscr{N}$ at V and over $r\mathscr{N}$ at rV equal $I_k(V; g, r')$ and $I_k(rV; g, r')$, respectively, up to factors $1 + O(e^{-c/g^2})$. Substitute $U = r\tilde{U}$ in the integral over $r\mathscr{N}$ at rV:

- the map $\tilde{U} \mapsto r\tilde{U}$ permutes the bond variables and inverts some of them, so it preserves $\prod_b dU(b)$;
- $(r\tilde{U})^{(k)} = r\,\tilde{U}^{(k)}$ by Lemma 21.10, and $\delta(r\tilde{U}^{(k)}, rV) = \delta(\tilde{U}^{(k)}, V)$, since δ is invariant under conjugation and inversion (proof of Proposition 31.7(a));

- $S(r\tilde{U}) = S(\tilde{U})$, and $\mathbf{1}_{r\mathcal{N}}(r\tilde{U}) = \mathbf{1}_{\mathcal{N}}(\tilde{U})$.

So the two integrals are equal, and $I_k(rV; g, r') = I_k(V; g, r')(1 + O(e^{-c/g^2}))$. By Theorem 30.3(c), $S(U_k(rV)) = S(U_k(V))$. Proposition 54.3, at V and at rV, gives $\Gamma_k^{(1)}(rV) - \Gamma_k^{(1)}(V) = o(1)$ as $g \to 0$. Both terms are independent of g, so they are equal.

(b) We use the notation of Sections 32–34 for the step from scale j to scale $j+1$, in the units of step j, with j in place of k: $U' = U_{j+1}(V')$, its base $W = U'^{(j)}$, the operator $\Delta_j = \Delta_j(W, U')$ of Corollary 34.6, and the Jacobians $\mathfrak{j}_c = \mathfrak{j}_c(W)$ of Theorem 34.9(b). For $j = 0$, $W = U'$, and Δ_0 and its properties are those of Lemma 52.3(a)–(c). So

$$\Gamma_j^{\text{step}}(U') = -\tfrac{1}{2} \log \det \Delta_j(W, U') + \sum_c \log \mathfrak{j}_c(W).$$

This does not depend on the choice of U' in its minimal orbit. Indeed, another choice is U'^u, with u real and $u|_{\mathbb{T}'} = 1$, and with the base $W^{u|_{\mathbb{T}}}$. By Corollary 34.6(a) (Lemma 52.3(c) for $j = 0$), Δ_j is then replaced by $R(u)\Delta_j R(u)^{-1}$, with $R(u)$ orthogonal, and the $\mathfrak{j}_c$ are invariant (Proposition 47.8(b)).

Let $r \in \mathfrak{S}_{j+1}$. By (a), rV' is small field. By Proposition 31.7(b), with $j+1$ in place of k, rU' is an element of the minimal orbit of rV'; its base is $(rU')^{(j)} = rW$, by Lemma 21.10, since $\mathfrak{S}_{j+1} \subset \mathfrak{S}_j$. By Corollary 34.6(b) (Lemma 52.3(c) for $j = 0$),

$$\Delta_j(rW, rU') = r_W \, \Delta_j(W, U') \, r_W^{-1},$$

where r_W maps the real fields on $\mathfrak{F}$ orthogonally onto themselves. So $\log \det \Delta_j(rW, rU') = \log \det \Delta_j(W, U')$; equivalently, this is Proposition 34.8(d), summed over $b \in \mathfrak{F}$, since r maps $\mathfrak{F}$ onto $\mathfrak{F}$. By Proposition 47.8(d), $\sum_c \log \mathfrak{j}_c(rW) = \sum_c \log \mathfrak{j}_c(W)$. Hence $\Gamma_j^{\text{step}}(U_{j+1}(rV')) = \Gamma_j^{\text{step}}(rU') = \Gamma_j^{\text{step}}(U')$.

(c) The configuration $V = 1$ is small field. The function Φ is real analytic and gauge invariant. For $\Gamma_k^{(1)}$ this is Proposition 54.3. For $V' \mapsto \Gamma_j^{\text{step}}(U_{j+1}(V'))$ it follows from the identity $\Gamma_j^{\text{step}}(U_{j+1}(V')) = \Gamma_{j+1}^{(1)}(V') - \Gamma_j^{(1)}(V_j(V')) + \kappa_j$ of the paragraph after this proof, with $V_j(V') = U_{j+1}(V')^{(j)}$ and $\Gamma_0^{(1)} = 0$; this identity uses Lemma 54.5 and Proposition 54.6, not this lemma. Indeed, V_j is real analytic up to gauge transformations of $\mathbb{T}^{(j)}$ (Corollary 30.5), on which $\Gamma_j^{(1)}$ does not depend, and $V_j(V'^v)$ is a gauge transform of $V_j(V')$. Its first variation at 1 vanishes, since G is semisimple (Section 31); so it satisfies the assumptions made above Definition 54.7, with the translation invariance given by (a) and (b). For $r(x) = Rx + v$ in $\mathfrak{S}_k$ (respectively $\mathfrak{S}_{j+1}$) and real B, $r(e^{itB}) = e^{it\, r_1 B}$, where $(r_1 B)(b) = B(r^{-1}b)$ and $B(-b) = -B(b)$. By (a) and (b), $\Pi_\Phi(r_1 B, r_1 B) = \Pi_\Phi(B, B)$. The map r sends the bond with midpoint m and direction e_ν to the bond with midpoint $r(m)$ and direction $Re_\nu = \pm e_{\sigma(\nu)}$, reversed if the sign is negative. Hence, regarding B as a vector field on the midpoints, $(r_1 B)(r(m)) = R\, B(m)$, and $\widehat{r_1 B}(p) = e^{-ip\cdot v} R\, \hat{B}(R^{-1}p)$. Since R is orthogonal, the invariance of the form, extended sesquilinearly to complex fields, is $\Pi_\Phi(Rp) = R\, \Pi_\Phi(p) R^T$. For $R = -1$ this is $\Pi_\Phi(-p) = \Pi_\Phi(p)$. The kernel of Π_Φ is real and symmetric, so $\Pi_\Phi(p)^T = \Pi_\Phi(-p) = \overline{\Pi_\Phi(p)}$; with the evenness, $\Pi_\Phi(p)$ is real and symmetric.

Expand $\Pi_\Phi(p) = \Pi^{(1)}(p) + \Pi^{(2)}(p) + O(|p|^3)$ with $\Pi^{(i)}$ homogeneous of degree i; there is no constant term, since $\Pi_\Phi(0) = 0$ (Remark 54.8). Evenness gives $\Pi^{(1)} = 0$, so $\Pi_\Phi(p) = O(|p|^2)$ and $T_\Phi = \Pi^{(2)}$. Write $T_\Phi(p)_{\mu\nu} = \sum_{\kappa\lambda} T_{\mu\nu\kappa\lambda} p_\kappa p_\lambda$ with $T_{\mu\nu\kappa\lambda}$ real and symmetric in κ, λ. Invariance under the reflections of single axes shows that $T_{\mu\nu\kappa\lambda} = 0$ unless each value occurs an even number of times among $\mu, \nu, \kappa, \lambda$; invariance under the permutations of the axes shows that the remaining coefficients depend only on the pattern of equal indices. Hence

$$T_\Phi(p)_{\mu\nu} = t_1\, \delta_{\mu\nu} |p|^2 + t_2\, p_\mu p_\nu + t_3\, \delta_{\mu\nu} p_\mu^2$$

for real t_1, t_2, t_3. The part of order $|p|^3$ of the transversality $\sum_\mu \hat{p}_\mu \Pi_\Phi(p)_{\mu\nu} = 0$ is $\sum_\mu p_\mu T_\Phi(p)_{\mu\nu} = (t_1 + t_2)|p|^2 p_\nu + t_3\, p_\nu^3 = 0$ for all p. So $t_3 = 0$ and $t_2 = -t_1$, and $\beta_\Phi = t_1$. □

The proof of (a) uses only the invariance of the Haar measure and of the δ-function, $S \circ r = S$, the covariance of the averaging (Lemma 21.10(d)), the covariance of the minimal orbits (Theorem 30.3(c), Proposition 31.7(b)) and the freedom in the neighbourhood (Lemma 54.4); it uses no gauge fixing. The proof of (b) uses, besides the covariance of the minimal orbits, the lattice covariance of the coarse action (Proposition 33.2(e)), of the elimination, and of the quadratic part $\|\mathfrak{G}_W \cdot\|^2$ of the gauge fixing (Corollary 34.6(b)); the last is that of the averaged contour variables (Lemma 22.7(d)). Part (b) also follows from (a), by a route that uses no covariance property of the gauge fixing. For $j \geq 1$, Lemma 54.5 at $k = j+1$ gives $\Gamma_j^{\mathrm{step}}(U_{j+1}(V')) = \Gamma_{j+1}^{(1)}(V') - \Gamma_j^{(1)}(V_j(V')) + \kappa_j$, with $V_j(V') = U_{j+1}(V')^{(j)}$ and a constant κ_j; configurations are carried from the units of one step to those of another without change (Section 31). By Theorem 30.3(c) and Lemma 21.10, $V_j(rV')$ is a gauge transform of $rV_j(V')$, and (a) at the scales $j+1$ and j, with the gauge invariance of $\Gamma_j^{(1)}$, gives the claim. For $j = 0$, Proposition 54.6 at $k = 1$ gives $\Gamma_0^{\mathrm{step}}(U_1(V')) = \Gamma_1^{(1)}(V') + \kappa_0$. On this route the gauge fixing enters only through the exact identity of Lemma 32.6(iv) on invariant integrands, through the explicit product form of gf_W and Lemma 32.3 in the bound on R_1, and through the gauge-fixed formula of Theorem 34.9, and of Lemma 52.3 for $j = 0$, which defines Γ_j^{step}. Neither route uses the covariance of the gauge-field operators of Parts II and III, which are covariant under reflections only at flat backgrounds (Remark 31.8).

Part (b) is a statement at the background fields $U' = U_{j+1}(V')$. There Δ_j is the Hessian of the coarse action, composed with the elimination, plus the gauge-fixing term (Lemma 34.1, Proposition 47.12(a)), and its covariance under the reflections in $\mathfrak{S}_{j+1}$ is that of the coarse action, Proposition 33.2(e), which rests on Theorem 25.1(h). At references that are not critical, Δ_j is defined in Section 47.6 through $\mathbf{\Gamma}_\pi[W_2]$ and $\mathbf{H}_0$, which are not shown to be covariant under reflections at curved backgrounds (Remark 31.8). So no covariance under reflections is claimed for Δ_j, for the pieces $\mathfrak{e}(X, U')$ of Proposition 41.7, or for the one-loop part E'_{ol} of the raw new generation, at such references. Nothing below needs it: Section 58 uses (b) only through the second variations at $V' = 1$, i.e. at the background fields $U_{j+1}(V')$ with V' near 1 (proofs of Theorem 58.1 and Proposition 58.6).

The lattice symmetry rests on the symmetry of Bałaban's averaging [18, (0.3)–(0.12)]: with blocks that have the coarse point at a corner, or with a single ordered contour per point, Lemma 21.10(d) fails. The invariance under single reflections, which reverse the orientation, also excludes a topological term: the density $\varepsilon_{\mu\nu\kappa\lambda} \operatorname{tr}(F_{\mu\nu} F_{\kappa\lambda})$ is odd under them, so it cannot occur in a local expansion of $\Gamma_k^{(1)}$ or of the step functions. (At the level of second variations it is invisible anyway, since its quadratic part is a total derivative.)

54.2 Periodic operators at $U = 1$ and infinite volume

Marginal tensors are read off in infinite volume, and the change of scale (Lemma 54.17) applies the minimizer map of Part I to linear fields, which are not periodic. At $U = 1$ the operators of Part I commute with the translations of the unit lattice. This subsection constructs their infinite-volume versions from the operators on the tori, by folding (Lemma 54.10, Corollary 54.11). It transfers their bounds through the Bloch fibres (Lemma 54.12), collects the trace per cell used in Section 56 (Lemma 54.13), and determines the infinite-volume minimizer on linear fields (Lemma 54.14, Proposition 54.15, Corollary 54.16). Only Part I is used.

Throughout, $k \geq 1$ is fixed, and we use the units of scale k: the fine lattice is $\eta\mathbb{Z}^4$, with $\eta = L^{-k} = N^{-1}$, and the unit lattice is $\mathbb{Z}^4 \subset \eta\mathbb{Z}^4$. Constants may depend on η; what matters is that they do not depend on the side n of the torus.

Fields and kernels. A *field space* consists of the fields on the points, on the positively oriented bonds, or on the positively oriented plaquettes of a lattice $\delta\mathbb{Z}^4$, where δ^{-1} is a

divisor of N, with values in a finite-dimensional complex vector space E with a Hermitian product. Examples are $\delta = \eta$ or $\delta = 1$, and $E = \mathbb{C}$ or $E = \mathfrak{g}^c$. These points, bonds or plaquettes are the *sites*, and Σ denotes the set of sites. A site s has a base point $x(s)$, the point itself or the initial point of the bond or plaquette, and $|s - s'| = |x(s) - x(s')|$. The translations by $a \in \mathbb{Z}^4$ act on Σ, $s \mapsto s + a$. Since δ^{-1} is odd, no base point has a half-integer coordinate, so Σ is the disjoint union of the translates of the *cell* Σ_0, the set of the sites with base point in $\{x : |x_\mu| < \frac{1}{2} \text{ for all } \mu\}$. With the weight $w = \delta^d$, the inner product is $\langle f, g\rangle = \sum_s w \langle f(s), g(s)\rangle_E$; on $\eta\mathbb{Z}^4$ and $\mathbb{Z}^4$ it is that of Section 2. A field on the torus of side n is an $n\mathbb{Z}^4$-periodic field. Its inner product is the same sum over the sites of the torus, the classes in $\Sigma/n\mathbb{Z}^4$, and $|s - s'|_n$ is the distance on the torus. The operators of Part I act componentwise in a basis of E. Their kernels are real, so their bounds, proved for real fields, hold for E-valued fields with the same constants. A field f has *polynomial growth* if $|f(s)| \le C(1 + |x(s)|)^m$ for some C and m.

An operator K from the fields on Σ to those on Σ' has a kernel $K(s, s') \in \mathrm{Hom}(E, E')$, with $(Kf)(s) = \sum_{s'} K(s, s')f(s')$. On a torus the sum runs over the sites of the torus, and for $s, s' \in \Sigma$, $K(s, s')$ denotes the kernel at their classes. The weights enter the inner products, not the kernels. The kernel has *kernel decay* (C, γ) if $|K(s, s')| \le Ce^{-\gamma|s-s'|}$, and it is *periodic* if $K(s + a, s' + a) = K(s, s')$ for $a \in \mathbb{Z}^4$. Let $\mathscr{P}$ be the set of the periodic kernels with kernel decay (C, γ) for some $C, \gamma > 0$. For $K \in \mathscr{P}$ and $n \ge 1$, the *periodization* $[K]_n$ is the operator on the torus of side n with the kernel

$$[K]_n(s, s') = \sum_{a \in n\mathbb{Z}^4} K(s, s' + a),$$

which does not depend on the representatives s, s'. Put $\sigma_\gamma = \sup_{t\in\mathbb{R}^4} \sum_s e^{-\gamma|x(s)-t|/2}$, the sum over the sites of a field space; it depends on γ, η and the field space.

Local families. For each $n \ge 8$ let K_n be an operator between field spaces on the torus of side n. The family (K_n) is *local* if there are a periodic kernel K and an r such that $K(s, s') = 0$ for $|s - s'| > r$ and $K_n = [K]_n$ for all $n \ge 8$. The operators of Section 2, namely ∂_μ, ∂^*_μ, d, d^*, d_1, d_1^*, the directional convolutions, ι, ι^*, Q_σ, Q^*_σ, $\mathbf{Q}$ and $\mathbf{Q}^*$, are local, and so is the identity. On every torus each of them is given by one formula, with the points taken modulo the torus, and this is the periodization of the same formula on $\eta\mathbb{Z}^4$. A family is *algebraic in local operators* if it is obtained from local families by finitely many sums, multiplications by scalars, products, adjoints and inverses; the inverse (K_n^{-1}) is formed only if every K_n is invertible.

Lemma 54.10 (Periodic kernels and folding)**.** *(a)* (Calculus.) *A kernel with kernel decay maps the fields of polynomial growth to fields of polynomial growth, and products of such kernels act on them associatively. If K_1 and K_2 have kernel decay (C_1, γ) and (C_2, γ), then K_1K_2 has kernel decay $(C_1C_2\sigma_\gamma, \gamma/2)$, with σ_γ of the intermediate field space. The set $\mathscr{P}$ is closed under sums, products and adjoints. For $K \in \mathscr{P}$ and $n \ge 1$, $[K]_n$ acts on the n-periodic fields as K does; $K \mapsto [K]_n$ respects sums, products and adjoints, and $[I]_n = I$. If K has kernel decay (C, γ), then, with S_γ of Section 5,*

$$\big|[K]_n(s, s') - K(s, s')\big| \le CS_{\gamma/4}\, e^{-\gamma n/2} \qquad \text{for } |s - s'| \le n/4.$$

In particular $K = 0$ if $[K]_n = 0$ for infinitely many n.

(b) (Folding.) *Let $(K_n)_{n\ge8}$ be algebraic in local operators. Then K_n commutes with the translations by the vectors of $\mathbb{Z}^4$. For $8 \le n \mid n'$, $K_{n'}$ maps the n-periodic fields to themselves and acts on them as K_n. Equivalently, $K_n(s, s') = \sum_a K_{n'}(s, s' + a)$, the sum over a set of representatives a of $n\mathbb{Z}^4/n'\mathbb{Z}^4$.*

(c) (Limit.) *Let $(K_n)_{n\geq 8}$ be algebraic in local operators, with kernel decay uniform in n: $|K_n(s,s')| \leq Ce^{-\gamma|s-s'|_n}$ for all $n \geq 8$. Then there is a unique $K_\infty \in \mathscr{P}$ with $[K_\infty]_n = K_n$ for all $n \geq 8$. It has kernel decay (C,γ), and $|K_n(s,s') - K_\infty(s,s')| \leq CS_{\gamma/4}e^{-\gamma n/2}$ for $|s-s'| \leq n/4$.*

(d) (Algebra.) *If (K_n) and (K'_n) are as in (c), so are their sums, products and adjoints, and their limits are $K_\infty + K'_\infty$, $K_\infty K'_\infty$ and K^*_∞. If (K_n) and (K_n^{-1}) are as in (c), then $(K^{-1})_\infty$ is the inverse of K_∞: $K_\infty(K^{-1})_\infty = (K^{-1})_\infty K_\infty = I$. So every identity between such families that holds on all tori holds between their limits, in $\mathscr{P}$ and, by (a), on the fields of polynomial growth.*

(e) (Limits of products.) *Let $\mathfrak{N}$ be an infinite set of sides, and for $n \in \mathfrak{N}$ let X_n be an operator between field spaces on the torus of side n. The family (X_n)* converges with uniform decay *(C,γ) to a kernel X_∞ on the infinite lattice if $|X_n(s,s')| \leq Ce^{-\gamma|s-s'|_n}$ for all $n \in \mathfrak{N}$, and $X_n(s,s') \to X_\infty(s,s')$ for all s,s' as $n \to \infty$ in $\mathfrak{N}$. Then X_∞ has kernel decay (C,γ). If (X_n) and (Y_n) converge with uniform decay (C,γ) to X_∞ and Y_∞, then (X_nY_n) converges with uniform decay $(C^2\sigma_\gamma, \gamma/2)$ to $X_\infty Y_\infty$, and the transposed kernels $X_n^{\mathsf{T}}(s',s) = X_n(s,s')^{\mathsf{T}}$ converge with uniform decay (C,γ) to X_∞^{T}. The families of (c) converge with uniform decay (C,γ) to their limits.*

Proof. (a) Let $|K(s,s')| \leq Ce^{-\gamma|s-s'|}$ and $|f(s')| \leq C'(1+|x(s')|)^m$. Since $1+|x(s')| \leq (1+|x(s)|)(1+|s-s'|)$, Details

$$\sum_{s'} |K(s,s')f(s')| \leq CC'(1+|x(s)|)^m \sum_{s'} (1+|s-s'|)^m e^{-\gamma|s-s'|},$$

and the last sum is bounded uniformly in s, since a unit cube contains the base points of at most $2^d|\Sigma_0|$ sites. This bound, applied twice, justifies the exchange of the sums in $(K_1K_2)f = K_1(K_2f)$. For the product, $|s-s'|+|s'-s''| \geq |s-s''|$ gives $\sum_{s'} e^{-\gamma|s-s'|-\gamma|s'-s''|} \leq e^{-\gamma|s-s''|/2}\sum_{s'} e^{-\gamma|s-s'|/2} \leq \sigma_\gamma e^{-\gamma|s-s''|/2}$. The adjoint of K has the kernel $K^*(s',s) = w(s)w(s')^{-1}K(s,s')^*$. Sums, products and adjoints of periodic kernels are periodic. If K is periodic and f is n-periodic, grouping the sites s' by their classes modulo $n\mathbb{Z}^4$ gives $(Kf)(s) = \sum_{s'}[K]_n(s,s')f(s')$, the sum over the sites of the torus. So $[K]_n$ is the restriction of K to the n-periodic fields, which gives the statements on sums and products and $[I]_n = I$. For adjoints, the same grouping in $\sum_{s,s'} w(s)\langle g(s), K(s,s')f(s')\rangle_E$, with s over the sites of the torus and $s' \in \Sigma$, gives $\langle g, [K]_n f\rangle = \langle [K^*]_n g, f\rangle$ for n-periodic f and g. Finally, for $|s-s'| \leq n/4$ and $0 \neq a \in n\mathbb{Z}^4$ we have $|a| \geq n$ and $|x(s)-x(s')-a| \geq |a| - n/4 \geq \frac{3}{4}|a|$, so

$$|[K]_n(s,s') - K(s,s')| \leq C \sum_{0\neq a\in n\mathbb{Z}^4} e^{-3\gamma|a|/4} \leq Ce^{-\gamma n/2} \sum_{a\in\mathbb{Z}^4} e^{-\gamma|a|/4} = CS_{\gamma/4}e^{-\gamma n/2}.$$

If $[K]_n = 0$ for infinitely many n, letting $n \to \infty$ gives $K = 0$.

(b) We argue along the construction of the family. A local family $([K]_n)$ commutes with the translations, since K is periodic, and by (a) both $[K]_{n'}$ and $[K]_n$ act on the n-periodic fields as K. Sums, multiples and products preserve the two properties. The translations are unitary, so $K^*_{n'}$ commutes with them, and it maps the n-periodic fields, the common fixed points of the translations by the ne_μ, to themselves. On n-periodic fields, the inner product of the torus of side n' is $(n'/n)^d$ times that of the torus of side n. So, for n-periodic f and g, $\langle g, K^*_{n'}f\rangle = (n'/n)^d\langle K_n g, f\rangle = \langle g, K_n^* f\rangle$, both in the inner product of the torus of side n', and hence $K^*_{n'}f = K_n^*f$. If every K_n is invertible and f is n-periodic, then $u = K_n^{-1}f$ is n-periodic and $K_{n'}u = K_nu = f$. So $K_{n'}^{-1}f = K_n^{-1}f$, and $K_{n'}^{-1}$ commutes with the translations, as $K_{n'}$ does. The identity for the kernels follows by applying $K_{n'}$ to the n-periodic field $\sum_a \delta_{s'+a}e$, $e \in E$, which is the field $\delta_{s'}e$ of the torus of side n.

(c) Let $s, s' \in \Sigma$ and $8 \le n \mid n'$. Let $A(n')$ be the set of the $a \in n\mathbb{Z}^4$ with $x(s) - x(s') - a \in (-\frac{n'}{2}, \frac{n'}{2}]^d$. It is a set of representatives of $n\mathbb{Z}^4/n'\mathbb{Z}^4$, and for $a \in A(n')$ the distance on the torus of side n' is $|s - (s' + a)|_{n'} = |x(s) - x(s') - a|$. By (b),

$$K_n(s, s') = \sum_{a \in A(n')} K_{n'}(s, s' + a).$$

If $|s - s'| \le n/4$, then $0 \in A(n')$, and the terms with $a \ne 0$ sum to at most $CS_{\gamma/4} e^{-\gamma n/2}$, as in (a). For arbitrary $n_1, n_2 \ge 8$ with $|s - s'| \le \min(n_1, n_2)/4$, comparing both with $n' = n_1 n_2$ gives $|K_{n_1}(s, s') - K_{n_2}(s, s')| \le 2CS_{\gamma/4} e^{-\gamma \min(n_1, n_2)/2}$. So $K_n(s, s')$ converges as $n \to \infty$; let $K_\infty(s, s')$ be the limit. By (b), $K_n(s + a, s' + a) = K_n(s, s')$ for $a \in \mathbb{Z}^4$, so K_∞ is periodic. Since $|s - s'|_n = |s - s'|$ for n large, $|K_\infty(s, s')| \le Ce^{-\gamma|s-s'|}$. Now fix $n \ge 8$ and s, s', and let $n' = nm$ with $m \to \infty$. The sets $A(n')$ increase to $n\mathbb{Z}^4$, the terms $K_{n'}(s, s' + a)$ converge to $K_\infty(s, s' + a)$, and they are bounded by $Ce^{-\gamma|x(s)-x(s')-a|}$, which is summable over $a \in n\mathbb{Z}^4$. By dominated convergence, $K_n(s, s') = [K_\infty]_n(s, s')$. The uniqueness and the bound follow from (a).

(d) The sums, products and adjoints are algebraic in local operators. Their kernel decay is uniform in n: for products, represent the sites s' of the torus by the $s' \in \Sigma$ with $x(s') - x(s) \in (-\frac{n}{2}, \frac{n}{2}]^d$, so that $|s - s'|_n = |s - s'|$, and argue as in (a); for adjoints, use the formula for K^* in (a). Since $[\cdot]_n$ respects the operations, $[K_\infty K'_\infty]_n = K_n K'_n$ for all $n \ge 8$, so $K_\infty K'_\infty$ is the limit of $(K_n K'_n)$, by the uniqueness in (c); likewise for sums and adjoints. If (K_n^{-1}) is as in (c), then $[K_\infty (K^{-1})_\infty]_n = K_n K_n^{-1} = [I]_n$ for all n, so $K_\infty (K^{-1})_\infty = I$ by (a), and likewise in the other order.

(e) The bound on X_∞ holds since $|s - s'|_n = |s - s'|$ for n large. For the product, fix s and s'', and write $(X_n Y_n)(s, s'') = \sum_{s' \in D_n} X_n(s, s') Y_n(s', s'')$, where D_n is the set of the $s' \in \Sigma$ with $x(s') - x(s) \in (-\frac{n}{2}, \frac{n}{2}]^d$. For $s' \in D_n$ the term is bounded by $C^2 e^{-\gamma|s-s'|}$, which is summable over $s' \in \Sigma$; every s' lies in D_n for n large, and the terms converge to $X_\infty(s, s') Y_\infty(s', s'')$. Dominated convergence gives the limit $(X_\infty Y_\infty)(s, s'')$, and the uniform decay follows as in (d). The statement on transposes is clear, and the last one is (c). □

Corollary 54.11 (The operators of Part I in infinite volume)**.** *The following families, on the tori of all sides $n \ge 8$, are algebraic in local operators and have kernel decay uniform in n: G_σ, $Q_\sigma G_\sigma Q_\sigma^*$, Γ_σ, H_σ, C_σ, $\mathcal{D}_\sigma$, M, M^{-1}, $P_\mathcal{V}$ and R (Sections 7, 8), $\boldsymbol{\Delta}_a$, $\mathbf{G}$, $\mathbf{QGQ}^*$, $\boldsymbol{\Gamma}$, $\mathbf{H}$, $\mathbf{C}$ and $\mathbf{D}$ (Sections 9–11), and their products with the local operators of Section 2. We denote their limits in the sense of Lemma 54.10(c) by the same letters. The identities of Part I between them hold for the limits, in $\mathscr{P}$ and on the fields of polynomial growth. In particular*

$$\boldsymbol{\Delta}_a = d^* d + dRd^* + a\mathbf{Q}^*\mathbf{Q}, \qquad \boldsymbol{\Delta}_a \mathbf{G} = \mathbf{G}\boldsymbol{\Delta}_a = I, \qquad \mathbf{H} = \mathbf{GQ}^*\boldsymbol{\Gamma},$$
$$\mathbf{QH} = I, \qquad Rd^*\mathbf{H} = 0, \qquad d^* d\,\mathbf{H} = \mathbf{Q}^*(\boldsymbol{\Gamma} - a),$$

$\mathbf{Q}d = d_1 Q_0$, $Q_0 Q_0^* = I$, $R = I - P_\mathcal{V}$, $R^2 = R$, $Q_0 G_0 R = 0$, $-\Delta G_0 R = R$, and $P_\mathcal{V} 1 = 1$ for the constant field 1.

Proof. The families are algebraic in local operators by their definitions: $G_\sigma = (-\Delta + aQ_\sigma^* Q_\sigma)^{-1}$, invertible on every torus by Corollary 3.4; $\Gamma_\sigma = (Q_\sigma G_\sigma Q_\sigma^*)^{-1}$ (Proposition 8.1); $H_\sigma = G_\sigma Q_\sigma^* \Gamma_\sigma$; $C_\sigma = G_\sigma - H_\sigma Q_\sigma G_\sigma$; $\mathcal{D}_\sigma = \Gamma_\sigma - a$; $M = Q_0 G_0^2 Q_0^*$, invertible by Proposition 8.8(a); $P_\mathcal{V} = G_0 Q_0^* M^{-1} Q_0 G_0$; $R = I - P_\mathcal{V}$; $\boldsymbol{\Delta}_a = d^* d + dRd^* + a\mathbf{Q}^*\mathbf{Q}$; $\mathbf{G} = \boldsymbol{\Delta}_a^{-1}$ (Theorem 9.6); $\boldsymbol{\Gamma} = (\mathbf{QGQ}^*)^{-1}$ (Proposition 11.2); $\mathbf{H} = \mathbf{GQ}^*\boldsymbol{\Gamma}$; $\mathbf{C} = \mathbf{G} - \mathbf{G}dRd^*\mathbf{G} - \mathbf{HQG}$; $\mathbf{D} = \boldsymbol{\Gamma} - a$. Details

Decay. Part I proves decay in the sense of Definition 5.1, with constants independent of n, for G_σ (Theorem 7.1(a)), $Q_\sigma G_\sigma Q_\sigma^*$ and Γ_σ (Proposition 8.1), H_σ (Theorem 8.4(a)), C_σ (Proposition 8.6(d)), $\mathcal{D}_\sigma$ (Proposition 8.5), M^{-1} and $P_\mathcal{V}$ (Proposition 8.8(e) and its proof), $\mathbf{G}$ (Theorem 10.1(a)), $\mathbf{QGQ}^*$ and $\boldsymbol{\Gamma}$ (Proposition 11.2), $\mathbf{H}$ (Theorem 11.3(c)), $\mathbf{C}$ (Theorem 11.7(c))

and $\mathbf{D}$ (Proposition 11.5). A local family has such decay, with constants depending on η (Lemma 5.2(d)), and products of operators with decay have decay (Lemma 5.2(b)); this covers M, R, $\mathbf{\Delta}_a$ and the products with local operators. Decay in the sense of Definition 5.1 gives kernel decay. Indeed, let s lie in the support of $\mathbf{1}_y$ and s' in that of $\mathbf{1}_{y'}$, so that $|y-y'| \geq |s-s'|_n - \sqrt{d}$. Testing K with $\delta_{s'}e$ and $\delta_s e'$, unit vectors $e, e' \in E$, gives $|K(s,s')| \leq (w(s')/w(s))^{1/2} \|\mathbf{1}_y K \mathbf{1}_{y'}\| \leq \eta^{-d/2} C e^{\gamma\sqrt{d}} e^{-\gamma|s-s'|_n}$.

Identities. On every torus they are proved in Part I: Theorem 11.3(a) for $\mathbf{QH} = I$ and $Rd^*\mathbf{H} = 0$; then $d^*d\mathbf{H} = \mathbf{\Delta}_a\mathbf{H} - dRd^*\mathbf{H} - a\mathbf{Q}^*\mathbf{QH} = \mathbf{Q}^*\mathbf{\Gamma} - a\mathbf{Q}^*$; Lemma 2.6(b),(d) for $Q_0Q_0^* = I$ and $\mathbf{Q}d = d_1Q_0$; Proposition 8.8(b),(d) for the identities of R. Lemma 54.10(d) carries them to the limits. The constant field 1 is n-periodic, and $P_{\mathcal{V}}1 = 1$ on every torus, since the constants lie in $\mathcal{V}$ (Proposition 8.8(c)); so $P_{\mathcal{V}}1 = 1$ for the limit, by Lemma 54.10(a),(c). □

Bloch fibres. For $\theta \in \mathbb{R}^4$ let $\mathcal{H}_\theta(\Sigma)$ be the space of the fields φ on Σ with $\varphi(s+a) = e^{\mathrm{i}\theta\cdot a}\varphi(s)$ for all $a \in \mathbb{Z}^4$, with the inner product $\langle\varphi,\psi\rangle_\theta = \sum_{s\in\Sigma_0} w\,\langle\varphi(s),\psi(s)\rangle_E$. It depends on θ modulo $2\pi\mathbb{Z}^4$. Restriction to the cell identifies it unitarily with $\ell^2(\Sigma_0;E)$, with the weights, for every θ. On the fine lattice, $\mathcal{H}_\theta$ is spanned by the plane waves of the momenta $\theta + 2\pi l$, $l \in \mathbb{Z}^4/N\mathbb{Z}^4$, as in the Bloch decomposition of Section 4. By Lemma 54.10(a), every $K \in \mathscr{P}$ maps $\mathcal{H}_\theta(\Sigma)$ into $\mathcal{H}_\theta(\Sigma')$; the restriction is the *fibre* $\hat{K}(\theta)$ of K. For a field f on Σ of finite support put $f_\theta(s) = \sum_{a\in\mathbb{Z}^4} e^{-\mathrm{i}\theta\cdot a} f(s+a)$, and for an n-periodic field f the same with the sum over $a \in \mathbb{Z}^4/n\mathbb{Z}^4$. Then $f_\theta \in \mathcal{H}_\theta(\Sigma)$, and $(Kf)_\theta = \hat{K}(\theta)f_\theta$, by the substitution $s' \mapsto s'-a$ in the sum defining Kf. Let $\Theta_n = (2\pi/n)\mathbb{Z}^4/2\pi\mathbb{Z}^4$.

Lemma 54.12 (Bloch fibres)**.** *Let $K \in \mathscr{P}$, from the fields on Σ to those on Σ'.*

(a) In the identification with the cells, $\hat{K}(\theta)$ has the matrix $\big(\sum_{a\in\mathbb{Z}^4} K(s,s'+a)e^{\mathrm{i}\theta\cdot a}\big)_{s\in\Sigma'_0,\,s'\in\Sigma_0}$. It is real analytic and $2\pi\mathbb{Z}^4$-periodic in θ. The fibres of sums, products and adjoints are the sums, products and adjoints of the fibres, and $\hat{I}(\theta) = I$.

(b) (Tori.) *For $n \geq 1$, $f \mapsto (f_\theta)_{\theta\in\Theta_n}$ maps the n-periodic fields onto $\bigoplus_{\theta\in\Theta_n}\mathcal{H}_\theta(\Sigma)$, with $f = n^{-d}\sum_\theta f_\theta$ and $\langle f,g\rangle = n^{-d}\sum_\theta\langle f_\theta,g_\theta\rangle_\theta$, and $[K]_n = \bigoplus_{\theta\in\Theta_n}\hat{K}(\theta)$.*

(c) (Infinite volume.) *The map $f \mapsto (f_\theta)$ extends to a unitary map from $\ell^2(\Sigma)$ onto the square integrable functions on $[-\pi,\pi]^d$ with values in $\ell^2(\Sigma_0;E)$, for the measure $(2\pi)^{-d}d\theta$. It carries K to the multiplication by $\hat{K}(\theta)$. Hence $\|K\| = \max_\theta\|\hat{K}(\theta)\|$ on ℓ^2. If $K = K^*$ and $c \in \mathbb{R}$, then $\langle f, Kf\rangle \geq c\|f\|^2$ for all $f \in \ell^2(\Sigma)$ if and only if $\hat{K}(\theta) \geq c$ for all θ.*

(d) (Inverses.) *If $\hat{K}(\theta)$ is invertible for every θ, then K has an inverse in $\mathscr{P}$, with the fibres $\hat{K}(\theta)^{-1}$. This holds if $K = K^* \geq c > 0$ on $\ell^2(\Sigma)$.*

(e) (Transfer.) *Let $\mathfrak{N}$ be an infinite set of sides. Then $\|K\| = \sup_{n\in\mathfrak{N}}\|[K]_n\|$, and, if $K = K^*$, $K \geq c$ on $\ell^2(\Sigma)$ if and only if $[K]_n \geq c$ for all $n \in \mathfrak{N}$.*

(f) (Powers of $\langle\partial\rangle$.) *Let $\langle\partial\rangle = (1-\Delta)^{1/2}$ on the fine scalar or gauge fields, componentwise. For $t \in \mathbb{R}$, $\langle\partial\rangle^t \in \mathscr{P}$, its fibres are the powers $t/2$ of those of $1-\Delta$, and $[\langle\partial\rangle^t]_n = (1-\Delta)^{t/2}$ on the torus of side n.*

(g) (Part I in infinite volume.) *The bounds of Part I of the forms $X \geq c$ and $\|X\| \leq C$, for operators X built from those of Corollary 54.11 and from powers of $\langle\partial\rangle$ by sums and products, hold for the limits on ℓ^2, with the same constants. For instance, with the constants of Corollary 3.4, Theorem 9.6, Propositions 8.1, 11.2 and 8.8(a),*

$$-\Delta + aQ_\sigma^*Q_\sigma \geq c_0\langle\partial\rangle^2, \qquad \mathbf{\Delta}_a \geq c_g\langle\partial\rangle^2, \qquad \|\langle\partial\rangle\mathbf{G}\langle\partial\rangle\| \leq c_g^{-1},$$
$$c_{\mathrm{QGQ}} \leq \mathbf{QGQ}^* \leq 2c_g^{-1}, \qquad M \geq c_{\mathrm{QGQ}}^2.$$

Here c_{QGQ} is the constant c_F of Corollary 4.3, renamed in this part to avoid a clash with the constant c_{F} of (Γ4).

Proof. (a) Every $s' \in \Sigma$ is $s'' + a$ with unique $s'' \in \Sigma_0$ and $a \in \mathbb{Z}^4$, and $\varphi(s''+a) = e^{\mathrm{i}\theta\cdot a}\varphi(s'')$ for $\varphi \in \mathcal{H}_\theta$; this gives the matrix. For $s \in \Sigma_0'$ and $s' \in \Sigma_0$, $|x(s) - x(s') - a| \geq |a| - \sqrt{d}$, so the series converges absolutely and uniformly for $|\operatorname{Im}\theta| \leq \gamma' < \gamma$ if K has kernel decay (C,γ); the matrix is holomorphic there. Sums and products of restrictions are restrictions. For adjoints, the grouping of the proof of Lemma 54.10(a), now modulo the diagonal action of all of $\mathbb{Z}^4$, gives $\langle\varphi, K\psi\rangle_\theta = \langle K^*\varphi, \psi\rangle_\theta$ for $\varphi \in \mathcal{H}_\theta(\Sigma')$, $\psi \in \mathcal{H}_\theta(\Sigma)$; the phases cancel. Details

(b) For n-periodic f, $n^{-d}\sum_{\theta\in\Theta_n} f_\theta(s) = \sum_a f(s+a)\, n^{-d}\sum_\theta e^{-\mathrm{i}\theta\cdot a} = f(s)$. For $\theta, \theta' \in \Theta_n$, the sum over the sites of the torus of $w\langle f_\theta, g_{\theta'}\rangle_E$ is $n^d\delta_{\theta\theta'}\langle f_\theta, g_\theta\rangle_\theta$, which gives the inner product, and the dimensions agree. The last statement is $(Kf)_\theta = \hat{K}(\theta)f_\theta$ together with Lemma 54.10(a).

(c) For f of finite support, $\int \|f_\theta\|_\theta^2\,(2\pi)^{-d}d\theta = \sum_{s\in\Sigma_0} w\sum_a |f(s+a)|^2 = \|f\|^2$, by Parseval's identity for the Fourier series $\theta \mapsto f_\theta(s)$. The map is onto, since every square integrable $\ell^2(\Sigma_0; E)$-valued function on $[-\pi,\pi]^d$ is the sum of its Fourier series. The operator K is bounded on ℓ^2 (Schur's test), and $(Kf)_\theta = \hat{K}(\theta)f_\theta$ for f of finite support, hence for all $f \in \ell^2(\Sigma)$. A multiplication by a continuous function of θ has the norm $\max_\theta \|\hat{K}(\theta)\|$, and it is $\geq c$ if and only if $\hat{K}(\theta) \geq c$ for all θ.

(d) By (a), $\det\hat{K}(\theta)$ is holomorphic for $|\operatorname{Im}\theta| < \gamma$ and $2\pi\mathbb{Z}^4$-periodic, and it does not vanish for real θ. By compactness it does not vanish for $|\operatorname{Im}\theta_\mu| \leq \gamma'$, $\mu = 1,\dots,d$, for some $\gamma' > 0$, and there $G(\theta) = \hat{K}(\theta)^{-1}$ is holomorphic, periodic and bounded by some C'. Put $K'(s', s+a) = \int_{[-\pi,\pi]^d} G(\theta)_{s's} e^{-\mathrm{i}\theta\cdot a}(2\pi)^{-d}d\theta$ for $s' \in \Sigma_0$, $s \in \Sigma_0'$, $a \in \mathbb{Z}^4$, and extend it periodically. Shifting the integration in θ_μ to $\theta_\mu - \mathrm{i}\gamma'\operatorname{sign}(a_\mu)$, which is allowed by the periodicity, gives $|K'(s', s+a)| \leq C'e^{-\gamma'|a|_1}$, so $K' \in \mathscr{P}$. Its fibres are $G(\theta)$, by (a) and the inversion of Fourier series. So KK' and $K'K$ have the fibres I. A kernel in $\mathscr{P}$ is determined by its fibres, its entries being their Fourier coefficients; so $KK' = I$ and $K'K = I$. If $K = K^* \geq c > 0$, the fibres are $\geq c$ by (c), hence invertible.

(e) For every θ and n there is $\theta' \in \Theta_n$ with $|\theta - \theta'| \leq \pi\sqrt{d}/n$, so $\bigcup_{n\in\mathfrak{N}}\Theta_n$ is dense. The claims follow from (b), (c) and the continuity of $\hat{K}$.

(f) In the Fourier variables $q \in [-\pi/\eta, \pi/\eta]^d$ of $\eta\mathbb{Z}^4$, $1-\Delta$ is the multiplication by $\varrho(q) = 1 + \sum_\mu 4\eta^{-2}\sin^2(\eta q_\mu/2)$ (Section 4). For $q_\mu = u_\mu + \mathrm{i}v_\mu$ with $|v_\mu| \leq \frac{1}{4}$,

$$\operatorname{Re} 4\eta^{-2}\sin^2(\eta q_\mu/2) = 2\eta^{-2}(1 - \cos(\eta u_\mu)\cosh(\eta v_\mu)) \geq -2\eta^{-2}(\cosh(\eta v_\mu) - 1) \geq -\tfrac{1}{15},$$

since $\eta \leq \frac{1}{3}$. So $\operatorname{Re}\varrho \geq \frac{1}{2}$ on this strip, and $\varrho^{t/2}$, with the principal branch, is holomorphic and bounded there, and $2\pi\eta^{-1}$-periodic in each q_μ. The kernel of $\langle\partial\rangle^t$ is $(\eta/2\pi)^d\int e^{\mathrm{i}q\cdot(x-x')}\varrho(q)^{t/2}dq$. Shifting the integration in q_μ by $\frac{\mathrm{i}}{4}\operatorname{sign}(x-x')_\mu$ gives the bound $C_{\eta,t}e^{-|x-x'|_1/4}$. The kernel is invariant under the translations of $\eta\mathbb{Z}^4$, so $\langle\partial\rangle^t \in \mathscr{P}$. By (c), the unitary map of (c) carries $\langle\partial\rangle^t = (1-\Delta)^{t/2}$ both to the multiplication by its fibres and to the multiplication by the powers $t/2$ of the fibres of $1-\Delta$. Both are continuous in θ, so they agree, and (b) gives the statement on the tori.

(g) By Corollary 54.11, Lemma 54.10(a) and part (f), the periodizations of such an X are the corresponding operators on the tori of the sides $n \geq 8$, where Part I proves the bounds with constants independent of n. Part (e) carries them to ℓ^2. In the examples, $\langle f, (-\Delta + aQ_\sigma^*Q_\sigma)f\rangle \geq c_0(\|f\|^2 + \|\partial f\|^2) = c_0\langle f, (1-\Delta)f\rangle$, and similarly for $\boldsymbol{\Delta}_a$. On every torus $\boldsymbol{\Delta}_a \geq c_g\langle\partial\rangle^2$ gives $\langle\partial\rangle^{-1}\boldsymbol{\Delta}_a\langle\partial\rangle^{-1} \geq c_g$, whose inverse is $\langle\partial\rangle\mathbf{G}\langle\partial\rangle \leq c_g^{-1}$. □

Trace per cell. Let $\mathscr{B}$ be the set of the bounded operators O from $\ell^2(\Sigma)$ to $\ell^2(\Sigma')$ of the form $(Of)_\theta = \hat{O}(\theta)f_\theta$, with a continuous $2\pi\mathbb{Z}^4$-periodic map $\theta \mapsto \hat{O}(\theta)$ into the linear maps from $\ell^2(\Sigma_0; E)$ to $\ell^2(\Sigma_0'; E')$. By Lemma 54.12(c), $\mathscr{P} \subset \mathscr{B}$. The set $\mathscr{B}$ is closed under sums,

products and adjoints, and for $O \in \mathscr{B}$ and $r > 0$, $|O|^r = (O^*O)^{r/2}$ lies in $\mathscr{B}$, with the fibres $|\hat{O}(\theta)|^r$. Put $[O]_n = \bigoplus_{\theta \in \Theta_n} \hat{O}(\theta)$, as in Lemma 54.12(b). For O from Σ to Σ the *trace per cell* is

$$\operatorname{Tr}_1 O = \int_{[-\pi,\pi]^d} \operatorname{tr} \hat{O}(\theta) \, \frac{d\theta}{(2\pi)^d},$$

and for $O \in \mathscr{B}$ we put $\|O\|_r = (\operatorname{Tr}_1 |O|^r)^{1/r}$, $1 \le r < \infty$, and $\|O\|_\infty = \max_\theta \|\hat{O}(\theta)\|$, the norm of O on ℓ^2. These are the objects of Section 56. For $O \in \mathscr{P}$ with kernel decay (C, γ) and $p \in \mathbb{C}^d$ with $|\operatorname{Im} p| < \gamma$, let $O^{(p)}$ be the kernel $e^{-\mathrm{i}p\cdot(x(s)-x(s'))}O(s,s')$, and let $[\hat{x}_\mu, O]$ be the kernel $(x(s) - x(s'))_\mu O(s,s')$, $\mu = 1, \dots, d$; put $[O, \hat{x}_\mu] = -[\hat{x}_\mu, O]$. On the fields of finite support, $O^{(p)} = e^{-\mathrm{i}p\cdot\hat{x}} O e^{\mathrm{i}p\cdot\hat{x}}$ and $[\hat{x}_\mu, O] = \hat{x}_\mu O - O\hat{x}_\mu$, with the position operator $(\hat{x}f)(s) = x(s)f(s)$.

Lemma 54.13 (Trace per cell). *(a) For $O \in \mathscr{P}$ from Σ to Σ, $\operatorname{Tr}_1 O = \sum_{s\in\Sigma_0} \operatorname{tr}_E O(s,s)$, and, if O has kernel decay (C,γ),*

$$|n^{-d} \operatorname{tr}[O]_n - \operatorname{Tr}_1 O| \le C|\Sigma_0|(\dim E) S_{\gamma/2} e^{-\gamma n/2} \qquad (n \ge 1).$$

For $O \in \mathscr{B}$, $n^{-d}\operatorname{tr}[O]_n \to \operatorname{Tr}_1 O$ as $n \to \infty$. Moreover $\operatorname{Tr}_1(O_1O_2) = \operatorname{Tr}_1(O_2O_1)$ for $O_1, O_2 \in \mathscr{B}$ for which both products are defined.

(b) (Hölder.) *Let $O_1, \dots, O_m \in \mathscr{B}$, with $O_1 \cdots O_m$ defined from Σ to Σ, and let $r_1, \dots, r_m \in [1,\infty]$ with $\sum_i 1/r_i = 1$. Then $|\operatorname{Tr}_1(O_1 \cdots O_m)| \le \prod_i \|O_i\|_{r_i}$.*

(c) (Rank.) *If $\operatorname{rank} \hat{Z}(\theta) \le m$ for all θ, then $\|Z\|_1 \le m\|Z\|_\infty$. This holds with $m = |\Sigma_0''| \dim E''$ if $Z = Z_1Z_2$ with $Z_2 \in \mathscr{B}$ into the fields on Σ'' with values in E''. For the scalar, respectively gauge, fields on $\mathbb{Z}^4$, $m = \dim E''$, respectively $m = d \dim E''$.*

(d) (Commutators with $\hat{x}$.) *Let $O \in \mathscr{P}$ have kernel decay (C,γ). For $|\operatorname{Im} p| < \gamma$, $O^{(p)}$ and $[\hat{x}_\mu, O]$ lie in $\mathscr{P}$. The map $p \mapsto O^{(p)}$ is holomorphic in the norm of operators on ℓ^2, and*

$$\partial_{p_\mu} O^{(p)} = -\mathrm{i}\,[\hat{x}_\mu, O^{(p)}].$$

Moreover $(O_1O_2)^{(p)} = O_1^{(p)}O_2^{(p)}$ and $[\hat{x}_\mu, O_1O_2] = [\hat{x}_\mu, O_1]O_2 + O_1[\hat{x}_\mu, O_2]$; if $O^{-1} \in \mathscr{P}$, then $(O^{-1})^{(p)} = (O^{(p)})^{-1}$ for $|\operatorname{Im} p|$ smaller than the decay rates of O and O^{-1}, and $[\hat{x}_\mu, O^{-1}] = O^{-1}[O, \hat{x}_\mu]O^{-1}$. For real p, $(O^{(p)})^* = (O^*)^{(p)}$, and the fibre of $O^{(p)}$ at θ is unitarily equivalent to $\hat{O}(\theta + p)$; so $\|O^{(p)}\|_r = \|O\|_r$ for $1 \le r \le \infty$, and $\operatorname{Tr}_1 O^{(p)} = \operatorname{Tr}_1 O$.

Proof. (a) By Lemma 54.12(a), $\operatorname{tr} \hat{O}(\theta) = \sum_{s\in\Sigma_0} \sum_a \operatorname{tr}_E O(s, s+a) e^{\mathrm{i}\theta\cdot a}$, and the integral keeps the term $a = 0$. Every site of the torus is the class of a unique $s+b$ with $s \in \Sigma_0$ and $b \in \mathbb{Z}^4/n\mathbb{Z}^4$, so, by periodicity, $\operatorname{tr}[O]_n = n^d \sum_{s\in\Sigma_0} \sum_{a\in n\mathbb{Z}^4} \operatorname{tr}_E O(s, s+a)$. For $s \in \Sigma_0$ and $0 \ne a \in n\mathbb{Z}^4$, $|\operatorname{tr}_E O(s,s+a)| \le (\dim E) C e^{-\gamma|a|} \le (\dim E) C e^{-\gamma n/2} e^{-\gamma|a|/2}$, which gives the bound. For $O \in \mathscr{B}$, $n^{-d}\operatorname{tr}[O]_n = n^{-d} \sum_{\theta\in\Theta_n} \operatorname{tr} \hat{O}(\theta)$ is a Riemann sum of a continuous periodic function. The cyclicity holds fibrewise.

(b) The fibre of $O_1 \cdots O_m$ is $\hat{O}_1(\theta) \cdots \hat{O}_m(\theta)$. By Hölder's inequality for the Schatten norms of linear maps between finite-dimensional Hilbert spaces, $|\operatorname{tr}(\hat{O}_1(\theta) \cdots \hat{O}_m(\theta))| \le \prod_i a_i(\theta)$, with $a_i(\theta) = (\operatorname{tr}|\hat{O}_i(\theta)|^{r_i})^{1/r_i}$ for $r_i < \infty$ and $a_i(\theta) = \|\hat{O}_i(\theta)\|$ for $r_i = \infty$. Hölder's inequality for the probability measure $(2\pi)^{-d}d\theta$ on $[-\pi,\pi]^d$ gives $\int \prod_i a_i \le \prod_i \|O_i\|_{r_i}$.

(c) The trace norm of a linear map of rank at most m is at most m times its norm, since it is the sum of at most m singular values. Apply this fibrewise and integrate. If $Z = Z_1Z_2$, then $\hat{Z}(\theta) = \hat{Z}_1(\theta)\hat{Z}_2(\theta)$ factors through $\ell^2(\Sigma_0''; E'')$, of dimension $|\Sigma_0''| \dim E''$; on $\mathbb{Z}^4$ the cell contains one point, and d positively oriented bonds.

(d) $O^{(p)}$ is periodic and has kernel decay $(C, \gamma - |\operatorname{Im} p|)$, and $[\hat{x}_\mu, O]$ is periodic and has kernel decay $(C_{\gamma'}, \gamma')$ for every $\gamma' < \gamma$. Put $\Delta x = x(s) - x(s')$. The kernel of $O^{(p+h)} - O^{(p)} + \mathrm{i}h\cdot[\hat{x}, O^{(p)}]$ is $O(s,s')e^{-\mathrm{i}p\cdot\Delta x}(e^{-\mathrm{i}h\cdot\Delta x} - 1 + \mathrm{i}h\cdot\Delta x)$, bounded by $C|h|^2|\Delta x|^2 e^{-(\gamma - |\operatorname{Im} p| - |h|)|\Delta x|}$. By Schur's

test its norm on ℓ^2 is $O(|h|^2)$. This gives the derivative; the derivative is again of the same form, so the map is holomorphic. The product rules follow from $\Delta x = (x(s) - x(s')) + (x(s') - x(s''))$ in the kernel of the product. If $O^{-1} \in \mathscr{P}$, then $O^{(p)}(O^{-1})^{(p)} = I^{(p)} = I$, and the product rule applied to $OO^{-1} = I$ gives $[\hat{x}_\mu, O^{-1}] = -O^{-1}[\hat{x}_\mu, O]O^{-1}$. For real p the identity for the adjoint is read off the kernels. For $\varphi \in \mathcal{H}_\theta$, the field $U_p\varphi = e^{\mathrm{i}p\cdot x}\varphi$ lies in $\mathcal{H}_{\theta+p}$, and U_p is unitary from $\mathcal{H}_\theta$ onto $\mathcal{H}_{\theta+p}$. Since $O^{(p)}\varphi = U_p^{-1}OU_p\varphi$, the fibre of $O^{(p)}$ at θ is $U_p^{-1}\hat{O}(\theta+p)U_p$, and that of $|O^{(p)}|^r$ is $U_p^{-1}|\hat{O}(\theta+p)|^r U_p$. The integrals over θ do not change under the shift by p, by periodicity. □

Lemma 54.14 (Uniqueness among fields of polynomial growth)**.** *Let K and K' be kernels with kernel decay, from the fields on Σ to those on Σ' and back, with $K'K = I$. If h is a field of polynomial growth on Σ and $Kh = f$, then $h = K'f$; in particular, $Kh = 0$ implies $h = 0$. This applies to every $K \in \mathscr{P}$ with an inverse in $\mathscr{P}$, for instance to $\mathbf{\Delta}_a$, $\mathbf{Q}\mathbf{G}\mathbf{Q}^*$, $-\Delta + aQ_\sigma^* Q_\sigma$ and M (Corollary 54.11).*

Proof. By Lemma 54.10(a), f has polynomial growth and $K'f = K'(Kh) = (K'K)h = h$. □

Proposition 54.15 (The infinite-volume minimizer)**.** *Let $\mathbf{H}_\infty$ be the limit of the minimizer maps $\mathbf{H}$ of Theorem E, and let $\mathbf{G}$, $\mathbf{\Gamma}$, R, G_0 be the limits of Corollary 54.11; $\mathbf{Q}$ and Q_0 are given on $\eta\mathbb{Z}^4$ by the formulas of Definition 2.5. The fields take values in $\mathfrak{g}^c$, and the operators act componentwise.*

(a) (Convergence.) *The kernels of $\mathbf{H}$ on the tori of the sides $n \geq 8$ are the periodizations of the kernel of $\mathbf{H}_\infty$, an operator from the gauge fields on $\mathbb{Z}^4$ to those on $\eta\mathbb{Z}^4$. They have kernel decay (C, γ) uniform in n, and they converge to it entrywise, with $|\mathbf{H}(s,s') - \mathbf{H}_\infty(s,s')| \leq CS_{\gamma/4}e^{-\gamma n/2}$ for $|s - s'| \leq n/4$.*

(b) (Uniqueness.) *Let A be a gauge field on $\eta\mathbb{Z}^4$ of polynomial growth with $Rd^*A = 0$ and $d^*dA = \mathbf{Q}^*\omega$ for a gauge field ω on $\mathbb{Z}^4$ of polynomial growth. Then $A = \mathbf{H}_\infty\mathbf{Q}A$.*

(c) (Linear fields.) *For every curvature datum F in $\mathfrak{g}^c$ and every $x_0 \in \mathbb{R}^4$, $\mathbf{H}_\infty B_F = A_F$, where A_F and B_F are the linear fields of F with origin x_0 on the bonds of $\eta\mathbb{Z}^4$ and of $\mathbb{Z}^4$ (Section 50.2).*

(d) (Gradients.) *The family $\Psi = Q_0^* - G_0 Rd^* dQ_0^*$, from the scalar fields on the unit torus to those on the fine torus, is algebraic in local operators, with kernel decay uniform in n. On every torus, and for the limits,*

$$Q_0\Psi = I, \qquad Rd^*d\,\Psi = 0, \qquad \mathbf{H}d_1 = d\,\Psi.$$

So $\mathbf{H}_\infty$ maps the coarse gradient $d_1\lambda$ of every scalar field λ on $\mathbb{Z}^4$ of polynomial growth to the fine gradient $d\Psi_\infty\lambda$. If φ is a polynomial on $\mathbb{R}^4$ of degree at most 2, restricted to $\eta\mathbb{Z}^4$, then $\mathbf{H}_\infty d_1 Q_0\varphi = d\varphi$.

(e) (Coarse linear fields.) *Let $p, a \in \mathbb{C}^4$ and $T \in \mathfrak{g}^c$. For an oriented bond b let m_b be its midpoint and $t_b \in \{\pm e_1, \dots, \pm e_4\}$ its direction. Then $\mathbf{H}_\infty$ maps the field $c \mapsto (p \cdot m_c)(a \cdot t_c)T$ on the bonds of $\mathbb{Z}^4$ to the field $b \mapsto (p \cdot m_b)(a \cdot t_b)T$ on the bonds of $\eta\mathbb{Z}^4$, and the constant field $c \mapsto (a \cdot t_c)T$ to the constant field $b \mapsto (a \cdot t_b)T$.*

Proof. (a) By Corollary 54.11 and Lemma 54.10(c). Details

(b) Put $B = \mathbf{Q}A$ and $h = A - \mathbf{H}_\infty B$. By Lemma 54.10(a), B and h have polynomial growth. By Corollary 54.11, $\mathbf{Q}\mathbf{H}_\infty = I$, $Rd^*\mathbf{H}_\infty = 0$ and $d^*d\mathbf{H}_\infty = \mathbf{Q}^*(\mathbf{\Gamma} - a)$. So $\mathbf{Q}h = 0$, $Rd^*h = 0$, and

$$\mathbf{\Delta}_a h = d^*dh + dRd^*h + a\mathbf{Q}^*\mathbf{Q}h = \mathbf{Q}^*\omega', \qquad \omega' = \omega - (\mathbf{\Gamma} - a)B,$$

a field of polynomial growth. By Lemma 54.14 for $\mathbf{\Delta}_a$, with the inverse $\mathbf{G}$, $h = \mathbf{G}\mathbf{Q}^*\omega'$. Then $0 = \mathbf{Q}h = (\mathbf{Q}\mathbf{G}\mathbf{Q}^*)\omega'$, and Lemma 54.14 for $\mathbf{Q}\mathbf{G}\mathbf{Q}^*$, with the inverse $\mathbf{\Gamma}$, gives $\omega' = 0$. Hence $h = 0$.

(c) A_F has linear growth, and $\mathbf{Q}A_F = B_F$ by Lemma 58.2(a). On the bond $b = (x, x+\eta e_\nu)$, $A_F(b) = \frac{1}{2}\sum_\kappa (x + \frac{1}{2}\eta e_\nu - x_0)_\kappa F_{\kappa\nu}$ does not depend on x_ν, since $F_{\nu\nu} = 0$. So $\partial^*_\nu A_{F,\nu} = 0$ for every ν, $d^*A_F = 0$, and $Rd^*A_F = 0$. Moreover $(dA_F)_{\mu\nu} = \partial_\mu A_{F,\nu} - \partial_\nu A_{F,\mu} = F_{\mu\nu}$ is constant (Section 50.2), so $d^*dA_F = 0$. Part (b) with $\omega = 0$ gives $A_F = \mathbf{H}_\infty B_F$.

(d) Let $n \geq 8$. By Lemma 2.6(b), $Q_0Q_0^* = I$, and by Proposition 8.8(b),(d), $Q_0G_0h = 0$ and $-\Delta G_0 h = h$ for h in the range of R. With $d^*d = -\Delta$ on scalar fields,

$$Q_0\Psi = I - Q_0G_0R(-\Delta Q_0^*) = I,$$
$$Rd^*d\Psi = R(-\Delta)Q_0^* - R(-\Delta G_0 R)(-\Delta)Q_0^* = R(-\Delta)Q_0^* - R^2(-\Delta)Q_0^* = 0.$$

Let λ be a scalar field on the unit torus and $A = d\Psi\lambda$. Then $\mathbf{Q}A = d_1Q_0\Psi\lambda = d_1\lambda$ by Lemma 2.6(d), $Rd^*A = 0$ and $dA = 0$. By Theorem 11.3(b), $\mathbf{H}d_1\lambda$ is the unique minimizer of $\|dA\|^2$ on $\{A : \mathbf{Q}A = d_1\lambda,\ Rd^*A = 0\}$, and $d\Psi\lambda$ attains the value 0; so $\mathbf{H}d_1\lambda = d\Psi\lambda$. The family Ψ is algebraic in local operators, with kernel decay uniform in n, by Corollary 54.11 and Lemma 54.10(d), which also carries the three identities to the limits, and Lemma 54.10(a) applies them to the fields of polynomial growth.

Let φ be a polynomial of degree at most 2, and $A = d\varphi$. Then A has polynomial growth and $dA = 0$. On the lattice, $-\Delta x_\mu^2 = -2$, and $-\Delta$ annihilates $x_\mu x_\nu$ for $\mu \neq \nu$, x_μ and 1; so $d^*A = -\Delta\varphi$ is constant. By Corollary 54.11, $R1 = 1 - P_{\mathcal{V}}1 = 0$, so $Rd^*A = 0$. Part (b) with $\omega = 0$ and $\mathbf{Q}d\varphi = d_1Q_0\varphi$ give $d\varphi = \mathbf{H}_\infty d_1Q_0\varphi$.

(e) Put $\varphi(x) = \frac{1}{2}(p\cdot x)(a\cdot x)$ and let F be the curvature datum $F_{\kappa\lambda} = (p_\kappa a_\lambda - p_\lambda a_\kappa)T$, with origin 0. For an oriented bond b of $\eta\mathbb{Z}^4$, $b_\pm = m_b \pm \frac{1}{2}\eta t_b$, so

$$(d\varphi)(b) = \eta^{-1}(\varphi(b_+) - \varphi(b_-)) = \tfrac{1}{2}((p\cdot m_b)(a\cdot t_b) + (a\cdot m_b)(p\cdot t_b)),$$

exactly, while $A_F(b) = \frac{1}{2}m_b^{\mathsf{T}}Ft_b = \frac{1}{2}((p\cdot m_b)(a\cdot t_b) - (a\cdot m_b)(p\cdot t_b))T$. Hence $(p\cdot m_b)(a\cdot t_b)T = A_F(b) + (d\varphi)(b)T$. The same computation on $\mathbb{Z}^4$ gives $(p\cdot m_c)(a\cdot t_c)T = B_F(c) + (d_1\varphi)(c)T$, with φ restricted to $\mathbb{Z}^4$. The blocks are centred, so the average of $\varphi(y+t)$ over the points t of $B(0)$ is $\varphi(y)$ plus the constant $\frac{1}{2}\sum_\mu p_\mu a_\mu \eta^d \sum_{t\in B(0)} t_\mu^2$; hence $d_1Q_0\varphi = d_1\varphi$ on $\mathbb{Z}^4$. By (c) and (d), $\mathbf{H}_\infty$ maps $B_F + d_1\varphi\, T$ to $A_F + d\varphi\, T$. For the constant fields, take $\varphi(x) = a\cdot x$: then $Q_0\varphi = \varphi$ on $\mathbb{Z}^4$, $(d_1\varphi)(c) = a\cdot t_c$ and $(d\varphi)(b) = a\cdot t_b$, and (d) applies. □

In particular $\mathbf{H}_\infty$ maps linear fields to linear fields exactly, not only up to gauge. For the Bloch components this gives the following.

Corollary 54.16 (Umklapp components)**.** *Let δ^{-1} be a divisor of N, and let $\mathbf{K} \in \mathscr{P}$ map the gauge fields on $\mathbb{Z}^4$ to the gauge fields on $\delta\mathbb{Z}^4$, with real kernel, such that for all $p, a \in \mathbb{C}^4$ it maps the field $c \mapsto a\cdot t_c$ to $b \mapsto a\cdot t_b$, and $c \mapsto (p\cdot m_c)(a\cdot t_c)$ to $b \mapsto (p\cdot m_b)(a\cdot t_b)$. For $q \in \mathbb{R}^4$ and $\varepsilon \in \mathbb{C}^4$ let $e_{q,\varepsilon}(c) = e^{\mathrm{i}q\cdot m_c}(\varepsilon\cdot t_c)$ on the bonds of $\mathbb{Z}^4$, and $e'_{q,\varepsilon}$ the same field on the bonds of $\delta\mathbb{Z}^4$. Let $\mathcal{L} = \{0,\dots,\delta^{-1}-1\}^4$. Then, for $p \in \mathbb{R}^4$,*

$$\mathbf{K}e_{p,\varepsilon} = \sum_{\ell\in\mathcal{L}} e'_{p+2\pi\ell,\, k_\ell(p)\varepsilon},$$

a finite sum, with $d\times d$ matrices $k_\ell(p)$ that are real analytic in p, and $k_0(p) = I + O(|p|^2)$, $k_\ell(p) = O(|p|^2)$ for $\ell \neq 0$. By Proposition 54.15(e), this applies to $\mathbf{H}_\infty$, with $\delta = \eta$.

Proof. Fix a direction μ. The functions $m \mapsto e^{\mathrm{i}(p+2\pi\ell)\cdot m}$, $\ell \in \mathcal{L}$, on the midpoints of the bonds of $\delta\mathbb{Z}^4$ of direction μ are p-quasi-periodic under $\mathbb{Z}^4$. On the midpoints $\delta(z + \frac{1}{2}e_\mu)$ of the bonds of direction μ of a cell, with z running over a set of representatives of $\mathbb{Z}^4/\delta^{-1}\mathbb{Z}^4$, they are $e^{\mathrm{i}p\cdot\delta z}$ Details

times the characters $z \mapsto e^{2\pi i \delta\ell\cdot z}$ of $\mathbb{Z}^4/\delta^{-1}\mathbb{Z}^4$, up to nonzero constants, so they are linearly independent and span the p-quasi-periodic functions. Since $\mathbf{K}$ is periodic, $\mathbf{K}e_{p,\varepsilon}$ is p-quasi-periodic, which gives the expansion. The coefficients are linear combinations of the values of $f_p = e^{-ip\cdot m_b}(\mathbf{K}e_{p,\varepsilon})(b) = \sum_c \mathbf{K}(b,c)e^{ip\cdot(m_c-m_b)}(\varepsilon\cdot t_c)$ on one cell, which are holomorphic in p near $\mathbb{R}^4$, by the kernel decay. Write $\chi_\ell(b) = e^{2\pi i\ell\cdot m_b}$, so that $f_p = \sum_\ell \chi_\ell\,(k_\ell(p)\varepsilon)\cdot t_b$, the χ_ℓ being linearly independent. At $p=0$, $f_0 = \mathbf{K}e_{0,\varepsilon} = \varepsilon\cdot t_b$, so $k_\ell(0) = \delta_{\ell 0}I$. Differentiating under the sum, which the kernel decay allows,

$$\frac{d}{d\tau}\Big|_{\tau=0} f_{\tau p}(b) = i\,\mathbf{K}\big[(p\cdot m_c)(\varepsilon\cdot t_c)\big](b) - i(p\cdot m_b)\,\mathbf{K}\big[\varepsilon\cdot t_c\big](b) = i(p\cdot m_b)(\varepsilon\cdot t_b) - i(p\cdot m_b)(\varepsilon\cdot t_b) = 0.$$

So $\sum_\ell \chi_\ell\,((p\cdot\nabla)k_\ell(0)\varepsilon)\cdot t_b = 0$, and $(p\cdot\nabla)k_\ell(0) = 0$ for all ℓ and p. □

54.3 The change of scale

For $j<k$ and a coarse configuration V of scale k, let $V_{j+1}(V) = U_k(V)^{(j+1)}$, the configuration of scale $j+1$ determined by the background. In the units of step $j+1$ it lives on a unit lattice, and $V \mapsto V_{j+1}(V)$ is real analytic and covariant. Put $N' = L^{k-j-1}$. In the units of scale k, the configurations of scale $j+1$ live on $\Lambda' = N'^{-1}\mathbb{Z}^4$, and a configuration $e^{iN'^{-1}B'}$ there has the Lie field B', in the convention of Part IV. Let $\mathbf{Q}^{\mathrm{pl}}_{j+1}$ be the linearization at 1 of the plain average $U \mapsto U^{(j+1)}$, from the fine lattice to Λ', in the units of scale k: $(e^{i\eta tA})^{(j+1)} = e^{iN'^{-1}t\mathbf{Q}^{\mathrm{pl}}_{j+1}A+O(t^2)}$. It is Lemma 58.2(b) with $j+1$ in place of k, read in the units of scale k. In the units of scale $j+1$ the positions are multiplied by N', and the Lie field of this configuration is $N'^{-1}t\mathbf{Q}^{\mathrm{pl}}_{j+1}A+O(t^2)$. The kernel of a second variation, a function of pairs of bonds of Λ', does not change with the units.

Lemma 54.17 (Change of scale)**.** *Let Φ be as above, a function of configurations of scale $j+1$, with $\Pi_\Phi(p) = O(|p|^2)$, and put $\Psi(V) = \Phi(V_{j+1}(V))$. Then Ψ satisfies the assumptions made above Definition 54.7, its infinite-volume second variation is*

$$\Pi_\Psi = N'^{-2}\,\mathbf{K}^{\mathsf{T}}\Pi_\Phi\mathbf{K}, \qquad \mathbf{K} = \mathbf{Q}^{\mathrm{pl}}_{j+1}\mathbf{H}_\infty,$$

$\Pi_\Psi(p) = O(|p|^2)$, and $T_\Psi = T_\Phi$, each in the units of its own scale.

Proof. The linearization. Let $V = e^{iB}$ with B small, on a torus. The reference $U_0 = 1$ is an element of the minimal orbit of the configuration 1, so Corollary 30.5 applies, with $J_{U_0} = 0$ and $W_k = 1$: the minimal orbit of V contains $\mathsf{U}(A_*(B))$, a gauge transform of $e^{i\eta A_*(B)}$, and A_* is analytic in B. By Theorem 25.1(c), $A_*(B) = \mathbf{H}B + O(|B|^2)$, with the minimizer map $\mathbf{H}$ of Part I, since at $U_0 = 1$ the averaging pair is $(Q_0, \mathbf{Q})$ (Proposition 22.10(c)). Indeed, at $U_0 = 1$ the defect E vanishes and $(Q'_k, \mathbf{Q}_k) = (Q_0, \mathbf{Q})$ (Proposition 22.10(b),(c)), so $\mathbf{Q}_\pi = \mathbf{Q}$ (proof of Theorem 16.1(a)); the form of Theorem 25.1 has $W = 0$ (Section 26), so $\mathfrak{A}_\pi = \mathbf{\Delta}_a$ (Section 15, after Lemma 15.1); $P_{\mathcal{V}} = G_0Q_0^*M^{-1}Q_0G_0$; and the minimizer map $\mathbf{G}\mathbf{Q}_\pi^T\mathbf{\Gamma}_\pi$ of Theorem 16.1(b) is the map $\mathbf{H} = \mathbf{G}\mathbf{Q}^*\mathbf{\Gamma}$ of Theorem E. The configuration $U_k(V)$ is a gauge transform of $\mathsf{U}(A_*(B))$, the averaging is covariant under all real gauge transformations (Lemma 21.10(b)), and Φ is gauge invariant. Hence $\Psi(V) = \Phi\big((e^{i\eta A_*(B)})^{(j+1)}\big)$. Since the first variation of Φ at 1 vanishes, only the linearization of $B \mapsto (e^{i\eta A_*(B)})^{(j+1)}$ enters the second variation, and on every torus $\Pi_\Psi = N'^{-2}\mathbf{K}^{\mathsf{T}}\Pi_\Phi\mathbf{K}$ with $\mathbf{K} = \mathbf{Q}^{\mathrm{pl}}_{j+1}\mathbf{H}$; the factor N'^{-2} is the change of units of the Lie fields. Details

Infinite volume. The family $\mathbf{Q}^{\mathrm{pl}}_{j+1}$ is local: $(\mathbf{Q}^{\mathrm{pl}}_{j+1}A)(c')$ depends only on A on the bonds of the blocks of scale $j+1$ of the end points of c' (Lemma 21.10(c)), it is given by one formula on every torus, and it commutes with the translations of Λ' (Lemma 21.10(d)). By Proposition 54.15(a) and Lemma 54.10(a),(c), the maps $\mathbf{K}$ on the tori are the periodizations of $\mathbf{K}_\infty = \mathbf{Q}^{\mathrm{pl}}_{j+1}\mathbf{H}_\infty \in \mathscr{P}$, and they converge to it with uniform decay. By the assumption on Φ

and Lemma 54.10(e), the second variations of Ψ on the tori converge with uniform decay to $N'^{-2}\mathbf{K}_\infty^{\mathsf{T}}\Pi_\Phi\mathbf{K}_\infty$. This kernel is invariant under the translations of $\mathbb{Z}^4$, since $\mathbf{K}_\infty$ commutes with them and Π_Φ with those of $\Lambda' \supset \mathbb{Z}^4$. The function Ψ is real analytic and gauge invariant, as Φ is and V_{j+1} is real analytic and covariant, and its first variation at 1 vanishes. So Ψ satisfies the assumptions above Definition 54.7. From now on $\mathbf{K} = \mathbf{K}_\infty$.

The Umklapp components. As in Definition 54.7, we suppress the Lie algebra indices and use the bond-midpoint convention, with the fields $e_{q,\varepsilon}$ on $\mathbb{Z}^4$ and $e'_{q,\varepsilon}$ on Λ' of Corollary 54.16. The map $\mathbf{K}$ maps the constant fields and the fields $c \mapsto (p \cdot m_c)(a \cdot t_c)$ on $\mathbb{Z}^4$ to the same fields on Λ'. For $\mathbf{H}_\infty$, with $\eta\mathbb{Z}^4$ in place of Λ', this is Proposition 54.15(e). For $\mathbf{Q}^{\mathrm{pl}}_{j+1}$, write the fine field as $A_F + d\varphi$, with $F = p \wedge a$ and $\varphi(x) = \frac{1}{2}(p \cdot x)(a \cdot x)$, as in the proof of Proposition 54.15(e). By Lemma 58.2(b), $\mathbf{Q}^{\mathrm{pl}}_{j+1}A_F$ is the linear field of F on Λ', with no gradient, and by the gauge covariance in its proof $\mathbf{Q}^{\mathrm{pl}}_{j+1}d\varphi = d'\varphi$, the lattice gradient of φ on Λ', $(d'\varphi)(c') = N'(\varphi(c'_+) - \varphi(c'_-))$. Their sum is the field $(p \cdot m)(a \cdot t)$ on Λ', by the same identity on Λ'; constant fields are the case $\varphi(x) = a \cdot x$. So Corollary 54.16 applies, with $\delta = N'^{-1}$: $\mathbf{K}e_{p,\varepsilon} = \sum_{\ell\in\mathcal{L}} e'_{p+2\pi\ell,k_\ell(p)\varepsilon}$, a finite sum over N'^d Umklapp momenta, with $k_0(p) = I + O(|p|^2)$ and $k_\ell(p) = O(|p|^2)$ for $\ell \neq 0$. Since $\mathbf{K}$ is real, $\mathbf{K}e_{-p,\varepsilon} = \sum_\ell e'_{-p-2\pi\ell,\overline{k_\ell(p)}\varepsilon}$.

The tensor. For a $(-p)$-quasi-periodic field g and a p-quasi-periodic field f on the same bonds, put $(g,f)_1 = \sum g(b)f(b)$, the sum over the positively oriented bonds with base point in a cell. For a periodic kernel X, $(g, Xf)_1 = (X^{\mathsf{T}}g, f)_1$, by the grouping of the proof of Lemma 54.10(a). By the definition of the Fourier transform and by this identity, $\varepsilon^{\mathsf{T}}\Pi_\Psi(p)\varepsilon' = (e_{-p,\varepsilon}, \Pi_\Psi e_{p,\varepsilon'})_1 = N'^{-2}(\mathbf{K}e_{-p,\varepsilon}, \Pi_\Phi\mathbf{K}e_{p,\varepsilon'})_1$, the last pairing over the bonds of Λ' with base point in a cell of $\mathbb{Z}^4$. The kernel Π_Φ commutes with the translations of Λ', so $\Pi_\Phi e'_{q,\varepsilon} = e'_{q,\Pi_\Phi(q/N')\varepsilon}$, with the Fourier transform $\Pi_\Phi(\cdot)$ in the units of scale $j+1$. On a cell of $\mathbb{Z}^4$, Λ' has N'^d bonds of each direction, and $(e'_{-p-2\pi\ell,\varepsilon}, e'_{p+2\pi\ell',\varepsilon'})_1 = N'^d\delta_{\ell\ell'}\varepsilon^{\mathsf{T}}\varepsilon'$ for $\ell, \ell' \in \mathcal{L}$, by the orthogonality of the characters. Hence

$$\Pi_\Psi(p) = N'^{d-2}\sum_{\ell\in\mathcal{L}} k_\ell(p)^*\,\Pi_\Phi\big((p+2\pi\ell)/N'\big)\,k_\ell(p).$$

The term $\ell = 0$ is $N'^{d-2}\Pi_\Phi(p/N') + O(|p|^4) = N'^{d-2}T_\Phi(p/N') + O(|p|^3) = N'^{d-4}T_\Phi(p) + O(|p|^3)$, since $\Pi_\Phi(q) = O(|q|^2)$ and T_Φ is homogeneous of degree 2. For $\ell \neq 0$ the momenta $(p+2\pi\ell)/N'$, $|p| \le 1$, lie in a compact set, where Π_Φ is bounded, since its kernel decays exponentially; so these terms are $O(|p|^4)$. Hence $\Pi_\Psi(p) = N'^{d-4}T_\Phi(p) + O(|p|^3)$. In $d = 4$ the factor is 1. □

No gauge argument is needed for the Umklapp components: by Proposition 54.15(e) they vanish to second order in p. The proof uses the infinite-volume minimizer of Proposition 54.15 and the averages of linear fields of Lemma 58.2, both exact.

Corollary 54.18 (The cumulative one-loop tensor)**.** *Let $T^{(1)}_{j+1}$ be the marginal tensor of $V' \mapsto \Gamma^{\mathrm{step}}_j(U_{j+1}(V'))$, a function of the configuration V' of scale $j+1$. Assume that the second variations at 1 of these functions and of $\Gamma^{(1)}_k$ on the tori converge with uniform exponential decay as the side tends to infinity (Lemma 54.10(e)). Then they satisfy the hypothesis of Definition 54.7, each of their marginal tensors is a multiple of $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$, and*

$$\sum_{j<k} T^{(1)}_{j+1} = T_{\Gamma^{(1)}_k}.$$

The hypothesis on the second variations is proved in Lemma 56.2(c) for $\Gamma^{(1)}_k$, and in Lemma 58.5 for the step functions.

Proof. The hypothesis of Definition 54.7 and the form of the tensors are Lemma 54.9(c). By Proposition 54.6 and Lemma 31.6, $\Gamma^{(1)}_k(V) = \sum_{j<k}\Gamma^{\mathrm{step}}_j(U_{j+1}(V_{j+1}(V)))$ up to a constant. Apply Lemma 54.17 to each term; marginal tensors are additive. □ Details

Relation to Bałaban's β-functions. The averaging of Part IV and the gauge fixing of Part VII are Bałaban's [18, (0.3)–(0.19)]. Bałaban reads off β_{j+1} from the components with $\mu \neq \nu$ of the marginal tensor of the new generation, in infinite volume [18, (1.20)–(1.22), (2.14)], and its one-loop part from the term of order g^0 of the new generation. That term is Γ_j^{step}, and by Lemma 54.9 its marginal tensor is $T_{j+1}^{(1)} = \beta_{j+1}^{(1)}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. The flow of Part X reads β_{j+1} off the symmetrized new generation on the given torus, through its period density (Proposition 50.13). The marginal parts of the old generations, which must be cancelled before β is read off, are cancelled there by their counterterms, inside the old activities of the step (Definition 53.1); they do not enter the one-loop part. Section 58 compares the two readings. The one-loop part of β_{j+1} is $\beta_{j+1}^{(1)}$ up to $O(e^{-cn_{j+1}/M})$ ((F2) of Proposition 58.6), the rest of β_{j+1} is $O(g_j^2 p_0(g_j)^6)$ ((F3)), and, at fixed couplings, β_{j+1} converges as the torus grows to the coefficient that Bałaban's definition assigns to the generations of this note (Proposition 58.8, Section 59); no rate is proved for this convergence. The one-loop parts $\beta_{j+1}^{(1)}$, summed over the steps, are the coefficient of the marginal tensor of the single function $\Gamma_k^{(1)}$ (Corollary 54.18 and Theorem 58.1), which the next section writes in the Landau chart.

55 The one-loop term in the Landau chart

The one-loop term $\Gamma_k^{(1)}$ of Proposition 54.3 depends only on the second-order jet of the action and on the density of the measure at the minimal orbit, not on coordinates. This section computes it in Bałaban's Landau chart (Theorem 24.2), where it becomes a combination of determinants of operators that Parts II–V control.

One point needs care. The composite integral has the residual group $\mathfrak{G}_0 = \{u : u|_{\mathbb{T}} = 1\}$. The chart, however, is built on the gauge-adapted averaging $\mathcal{Q}$ (Definition 22.4), whose residual directions are $D_U\lambda$ with $Q_k'(U)\lambda = 0$. It reaches the configurations through the correspondence w_k, which is not in $\mathfrak{G}_0$. The bridge is a smearing over the coarse gauge orbit, which makes the integrand invariant under all gauge transformations. It is carried out in a neighbourhood of the minimal orbit that shrinks as $g \to 0$, at a fixed coarse configuration with a finite stabilizer. The result extends to all small-field V by continuity, since the configurations with a finite stabilizer are dense.

Setting. Let V be small field (Section 54) and $U_0 = U_k(V)$ its minimizer. Up to the paragraph "All V" of the proof of Theorem 55.4, only $V \in \mathfrak{U}_k(a_1)$ and the admissibility of (V, θ_0) are used. As in Section 54, U_0 is the reference of Part IV, averages are relative to U_0, and the minimizer is $A_* = 0$ in the chart. Let $\mathcal{G}$ be the group of real gauge transformations of $\mathbb{T}_\eta$ and $\mathcal{G}_{\mathbb{T}}$ that of $\mathbb{T}$. For $g > 0$ put $\varrho_g = g^{1/2}$, and, for g so small that $\varrho_g \le \varrho_0 := \min(\frac{1}{2}r_u, \frac{1}{2}r_0)$, let

$$\mathcal{D}_{\varrho_g} = \{e^{\mathrm{i}\eta A}U_0 : \ |A|_\infty < 2\varrho_g, \ \text{curvature and current at most } \theta\},$$

the domain of the chart coordinate A, with $\theta = \theta_0$ (so that (V, θ) is admissible and Theorem 24.2 applies). Let

$$\Sigma_{\varrho_g} = \{e^{\mathrm{i}\eta A}U_0 \in \mathcal{D}_{\varrho_g} : \ |A|_\infty < \varrho_g, \ R(U_0)D_{U_0}^T A = 0, \ \mathcal{Q}(e^{\mathrm{i}\eta A}) = 1\},$$

and let $\mathscr{N}_{\varrho_g}$ be the union of the $\mathcal{G}$-orbits of the points of Σ_{ϱ_g}. Then, for g small, $\mathscr{N}_{\varrho_g}(V) \subset \mathscr{N}_{\varrho_g} \cap \mathfrak{B}(V) \subset \mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$, with $\mathscr{N}_{\varrho_g}(V)$ as in Section 54: a configuration $\mathsf{U}(A)$, $A \in \mathcal{S}_V$, $|A|_\infty < \varrho_g$, is a gauge transform of $e^{\mathrm{i}\eta A}U_0 \in \Sigma_{\varrho_g}$, and $\mathfrak{U}(\theta)$ is gauge invariant. The set $\mathscr{N}_{\varrho_g}$ is a Borel set, the image of the σ-compact set $\mathcal{G} \times \Sigma_{\varrho_g}$ under a continuous map. By the proof of Proposition 54.3, $\Phi \ge \Phi(0) + c|A|_\infty^2$ on $\mathcal{S}_V$, with $c = c(V) > 0$. So the integral over $\mathscr{N}_{r_0}(V) \setminus \mathscr{N}_{\varrho_g}(V)$ is at most $Ce^{-(S(U_0)+c\varrho_g^2)/g^2}$, with C the mass of $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ (Lemma 54.2),

while $I_k(V;g,r_0) \geq c' g^N e^{-S(U_0)/g^2}$, with $N = \dim \mathcal{X}$ (Proposition 54.3 and its proof). With Lemma 54.4, applied to $\mathscr{N} = (\mathscr{N}_{\varrho_g} \cap \mathfrak{B}(V)) \cup \mathscr{N}_{r_0}(V)$, the integral over $\mathscr{N}_{\varrho_g} \cap \mathfrak{B}(V)$ is $I_k(V;g,r_0)(1 + O(g^{-N} e^{-c/g}))$, since $\varrho_g^2 = g$. This does not change $\Gamma_k^{(1)}$.

Lemma 55.1 (Smearing over the coarse orbit)**.** *Put $\delta_V^{\mathrm{orb}}(V') = \int_{\mathcal{G}_\mathbb{T}} dh\, \delta(V', V^h)$, with the Haar measure and $\delta(V', V'') = \prod_c \delta(V'(c) V''(c)^{-1})$. For every $\mathcal{G}$-invariant integrable function f that vanishes where $U^{(k)}$ is not defined,*

$$\int dU\, \delta(U^{(k)}, V) f(U) = \frac{1}{\operatorname{vol} \mathcal{G}_\mathbb{T}} \int dU\, \delta_V^{\mathrm{orb}}(U^{(k)}) f(U).$$

On $\mathcal{D}_{\varrho_g}$, $\delta_V^{\mathrm{orb}}(U^{(k)}) = \delta_V^{\mathrm{orb}}(\mathcal{Q}(U U_0^{-1}) V)$.

Proof. Write the right side as $\int dh \int dU\, \delta(U^{(k)}, V^h) f(U)$. For each h choose $\hat{h} \in \mathcal{G}$ with $\hat{h}|_\mathbb{T} = h$ and substitute $U = W^{\hat{h}}$. Then $U^{(k)} = (W^{(k)})^h$ by Lemma 21.10(b). Moreover $\delta((W^{(k)})^h, V^h) = \delta(W^{(k)}, V)$, since δ is a class function, and $f(W^{\hat{h}}) = f(W)$, $dU = dW$. So every h gives the left side. On $\mathcal{D}_{\varrho_g}$, $\mathcal{Q}(U U_0^{-1}) V$ is coarse-gauge-equivalent to $U^{(k)}$ (Lemma 27.1(a)), and δ_V^{orb} is invariant under coarse gauge transformations of its argument. □ Details

So the integrand $F(U) = \delta_V^{\mathrm{orb}}(U^{(k)}) f(U)$, set to 0 where f vanishes, is defined everywhere and invariant under $\mathcal{G}$, since $(U^u)^{(k)} = (U^{(k)})^{u|_\mathbb{T}}$. The gauge-adapted average is used only inside $\mathcal{D}_{\varrho_g}$.

Lemma 55.2 (Stabilizers and the slice)**.** *(a) Restriction to $\mathbb{T}$ is an isomorphism*

$$\mathrm{Stab}_\mathcal{G}(U_0) \to \mathrm{Stab}_{\mathcal{G}_\mathbb{T}}(V).$$

Its elements act on relative configurations by $X(b) \mapsto \mathrm{Ad}_{s(b_-)} X(b)$, and w_k, $\mathcal{Q}$ and the Landau condition are equivariant under them.

(b) Let $U_1, U_2 \in \mathcal{D}_{\varrho_g}$ lie in one $\mathcal{G}$-orbit, with $\mathcal{Q}(U_i U_0^{-1}) = 1$ and $R(U_0) D_{U_0}^T A_i = 0$. Then $U_2 = U_1^s$ for some $s \in \mathrm{Stab}_\mathcal{G}(U_0)$. In particular $|A_2|_\infty = |A_1|_\infty$.

Proof. (a) Restriction is injective, since $\mathfrak{G}_0$ acts freely. For surjectivity, let $h \in \mathrm{Stab}_{\mathcal{G}_\mathbb{T}}(V)$ and $s = \mathcal{E}^{(k)}(h)$, the extension of h by parallel transport along the chains of averaged contour variables of the background (Section 22); it is real. The relative configuration $1^{(s)}$ is averaged axial (Lemma 22.3(b)) and has average 1, since h stabilizes V. It is critical, since s maps the critical orbit to a critical orbit. By Theorem 30.3(a) it lies in the minimal $\mathfrak{G}_0$-orbit, whose only axial representative is 1 (Lemma 22.3(c)). Hence $1^{(s)} = 1$, i.e. $s \in \mathrm{Stab}_\mathcal{G}(U_0)$, with $s|_\mathbb{T} = h$. For s in the stabilizer, the relative action is pointwise conjugation; by (22.1), so is its action on the averaged relative contour variables, $X_W^{(s)}(y,x) = s(y) X_W(y,x) s(y)^{-1}$. Since s is real, the averaged contour variables of $X^{(s)} W_{j-1}$ are defined whenever those of $X W_{j-1}$ are: the averaging domain is invariant under real gauge transformations (Section 22). So the averaged contour variables, τ_X and g_X of Section 22 are equivariant, and hence so are $\mathsf{w}_k(\tilde{U}^{(s)}) = s\, \mathsf{w}_k(\tilde{U})\, s^{-1}$ and $\mathcal{Q}(\tilde{U}^{(s)}) = \mathcal{Q}(\tilde{U})^{(s|_\mathbb{T})}$. This is read off from the definitions; Lemma 27.1 covers only transformations close to 1. Details

(b) The averaged axial representatives U_i' of $U_i U_0^{-1}$ lie in one orbit and are averaged axial, so $U_2' = U_1'^{(\mathcal{E}^{(k)}(h))}$ for some h on $\mathbb{T}$ (Lemma 22.3(b)). Their plain averages are $\mathcal{Q}(U_i U_0^{-1}) = 1$ (Theorem 22.5(a)), so h stabilizes V, and $s = \mathcal{E}^{(k)}(h) \in \mathrm{Stab}_\mathcal{G}(U_0)$ by (a). By equivariance, $U_1^{(s)}$ is a preimage of U_2' under the chart. The uniqueness in Theorem 24.2, which uses the a priori bound of Proposition 24.5 and hence the bound on the current built into $\mathcal{D}_{\varrho_g}$, gives $U_2 = U_1^{(s)}$. Conjugation preserves $|A|_\infty$. □

Recall. The next two results use the linear maps of Section 27.1. For a real fine field A with $|A|_{C^1}$ small, and scalar fields λ on $\mathbb{T}_\eta$ and μ on $\mathbb{T}$,

$$\Xi_A\lambda = \tfrac{d}{dt}\Big|_0 \mathcal{A}_t, \qquad e^{\mathrm{i}\eta\mathcal{A}_t} = (e^{\mathrm{i}\eta A})^{(e^{\mathrm{i}t\lambda})},$$
$$\Theta_A\lambda = -\mathrm{i}\tfrac{d}{dt}\Big|_0 h_{e^{\mathrm{i}\eta A}}(e^{\mathrm{i}t\lambda}), \qquad \Lambda_A\mu = \tfrac{d}{dt}\Big|_0 \ell(\mathcal{Q}(e^{\mathrm{i}\eta A})^{(e^{\mathrm{i}t\mu})}),$$

with $h_{\tilde{U}}(g)$ of Lemma 27.1(b). They satisfy (27.1), $d\mathcal{Q}(A)\,\Xi_A\lambda = \Lambda_A\Theta_A\lambda$, they make sense for small complex A, and $\Theta_0 = Q'$. The subspace $\mathcal{W} = \mathcal{W}(U_0) = D^TD\,\mathcal{N}(Q'(U_0))$ is that of Section 14, with $\mathcal{N}(Q')$ the null space of Q'; the form q_2 is that of Proposition 33.2.

Proposition 55.3 (Faddeev–Popov on the slice)**.** *Let V be such that $\mathrm{Stab}_{\mathcal{G}}(U_0)$ is finite, and g small depending on V. Then, for every $\mathcal{G}$-invariant f supported in $\mathcal{N}_{\varrho_g}$,*

$$\int dU\,\delta(U^{(k)},V)f(U) = c_0\int_{|A|<\varrho_g} dA\;\delta(\mathcal{Q}(e^{\mathrm{i}\eta A}),1)\;\delta(R(U_0)D_{U_0}^TA)\;|\det\mathcal{O}_A|\;\sigma(A)\,f(e^{\mathrm{i}\eta A}U_0),$$

where:

- *c_0 does not depend on V;*
- *$\sigma(A) = \prod_b \sigma(\eta A(b))$ is the density of the Haar measure in the exponential coordinates;*
- *$\mathcal{O}_A\lambda = (\Theta_A\lambda,\, RD^T\Xi_A\lambda)$ is the operator of the proof of Proposition 27.2, and $|\det\mathcal{O}_0| = |\det(RD_{U_0}^TD_{U_0} : \mathcal{N}(Q_k') \to \mathcal{W})|$ is Bałaban's ghost.*

Proof. In this proof, identities between measures are written up to positive factors that do not depend on V: the densities $\sigma(0)^{|\mathbb{T}|}$ and $\sigma(0)^{|\mathbb{T}_\eta|}$ of the Haar measures dh and du at the zeros of the conditions, and the Jacobian of the coordinates A. These factors make up c_0. *The coarse condition.* The orbit of V under $\mathcal{G}_{\mathbb{T}}$ is an embedded submanifold. Choose a function φ near V, vanishing at V, whose differential restricted to the tangent space $\{D_V\mu\}$ of the orbit is invertible. Below the injectivity radius of the orbit at V, which is positive since the stabilizer is finite, the zeros of φ on the orbit are the points V^h with h in the stabilizer, i.e. V itself. On $\mathcal{D}_{\varrho_g}$, $|\ell\mathcal{Q}(e^{\mathrm{i}\eta A})| \le 2|A| < 4\varrho_g$ (Proposition 22.9(a)). On the support of δ_V^{orb} the average lies on the orbit, hence within $4\varrho_g$ of V along the orbit. This lies below the injectivity radius for g small. So, on the support,

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$$\delta_V^{\mathrm{orb}}(V')\,\delta(\varphi(V')) = |\,\mathrm{Stab}_{\mathcal{G}_{\mathbb{T}}}(V)|\;\frac{\delta(V',V)}{|\det(\varphi' D_V)|}.$$

Without the shrinking domain this fails near $V = 1$: there the constant gauge transformations move V only a little, and φ has further zeros on the orbit, e.g. at Weyl-type elements.

Faddeev–Popov for $\mathcal{G}$. Apply the Faddeev–Popov identity for the compact group $\mathcal{G}$ to $F = \delta_V^{\mathrm{orb}}(U^{(k)})f$, with the condition $\chi(U) = (\varphi(\mathcal{Q}(UU_0^{-1})V),\, R(U_0)D_{U_0}^TA)$ on $\mathcal{D}_{\varrho_g}$. Let $U \in \mathcal{N}_{\varrho_g}$. By the previous paragraph and Lemma 55.2(b), the zeros of χ on the orbit of U within $\mathcal{D}_{\varrho_g}$ form a single $\mathrm{Stab}_{\mathcal{G}}(U_0)$-orbit of points of Σ_{ϱ_g}, and all have $|A| < \varrho_g$. So χ has no zeros near the boundary of $\mathcal{D}_{\varrho_g}$. The identity has the weight $1/n(U)$, where $n(U)$ is the number of $u \in \mathcal{G}$ with $U^u \in \mathcal{D}_{\varrho_g}$ and $\chi(U^u) = 0$; it is $\mathcal{G}$-invariant. We show that, for g small depending on V, the zeros of $u \mapsto \chi(U_1^u)$ are exactly the elements of $\mathrm{Stab}_{\mathcal{G}}(U_0)$, for every $U_1 \in \Sigma_{\varrho_g}$.

- Every $s \in \mathrm{Stab}_{\mathcal{G}}(U_0)$ is a zero: $U_1^s \in \Sigma_{\varrho_g}$, by the equivariance of Lemma 55.2(a), and $\varphi(V) = 0$.
- At $U_1 = U_0$, the map $u \mapsto \chi(U_0^u)$ has at $u = s$ the derivative $\mathrm{diag}(-\varphi' D_V, 1) \circ \mathcal{O}_0$, composed with the right translation $\lambda \mapsto e^{\mathrm{i}\lambda}s$, since $U_0^{e^{\mathrm{i}\lambda}s} = (U_0^s)^{e^{\mathrm{i}\lambda}} = U_0^{e^{\mathrm{i}\lambda}}$ (paragraph "The

Faddeev–Popov operator" below). It is invertible: $\mathcal{O}_0$ by the proof of Proposition 27.2, and $\varphi' D_V$ by the choice of φ. The map $(U_1, u) \mapsto \chi(U_1^u)$ is smooth near (U_0, s). By the implicit function theorem there are neighbourhoods N_s of s and W of U_0 such that for $U_1 \in W$ the map $u \mapsto \chi(U_1^u)$ has exactly one zero in N_s. For $U_1 \in \Sigma_{\varrho_g} \cap W$ this zero is s.

- For g small, every zero lies in $\bigcup_s N_s$, and $\Sigma_{\varrho_g} \subset W$. Otherwise there are $g_n \to 0$, $U_n \in \Sigma_{\varrho_{g_n}}$ and zeros $u_n \notin \bigcup_s N_s$. Then $U_n \to U_0$ and $U_n^{u_n} \in \mathcal{D}_{\varrho_{g_n}}$, so $U_n^{u_n} \to U_0$. By the compactness of $\mathcal{G}$, a subsequence of u_n converges to some u_∞, with $U_0^{u_\infty} = U_0$, i.e. $u_\infty \in \mathrm{Stab}_{\mathcal{G}}(U_0) \subset \bigcup_s N_s$, a contradiction.

In particular $\mathrm{Stab}_{\mathcal{G}}(U_1) \subset \mathrm{Stab}_{\mathcal{G}}(U_0)$, and $n(U) = |\mathrm{Stab}_{\mathcal{G}}(U_0)|$ for every $U \in \mathscr{N}_{\varrho_g}$, since every such U is a gauge transform of a point of Σ_{ϱ_g}. So the identity gives the factor $1/|\mathrm{Stab}_{\mathcal{G}}(U_0)|$. This cancels $|\mathrm{Stab}_{\mathcal{G}_{\mathbb{T}}}(V)|$, by Lemma 55.2(a).

The Faddeev–Popov operator. By (27.1), the linearization of χ along the gauge orbit at a point of Σ_{ϱ_g} is $\lambda \mapsto (\varphi' \Lambda_A \Theta_A \lambda,\ RD^T \Xi_A \lambda)$. On Σ_{ϱ_g} the relative average is 1, so $\Lambda_A = -D_V$ in the coordinates of $\mathbb{T}$. So the operator is $\mathrm{diag}(-\varphi' D_V, 1) \circ \mathcal{O}_A$, with determinant $\pm \det(\varphi' D_V) \det \mathcal{O}_A$, and the coarse factors cancel. At $A = 0$, $\mathcal{O}_0 \lambda = (Q_k' \lambda, -RD^T D\lambda)$ (Proposition 27.2). With the orthogonal splitting $\lambda = Q_k'^T \mu + \lambda_0$, $\lambda_0 \in \mathcal{N}(Q_k')$, which holds since $Q_k' Q_k'^T = I$, the operator is block-triangular, and $|\det \mathcal{O}_0| = |\det(RD^T D|_{\mathcal{N}(Q_k')})|$. □

Theorem 55.4 (The one-loop term in the Landau chart)**.** *For every small-field V, with $U_0 = U_k(V)$ and up to a constant independent of V,*

$$\begin{aligned}\Gamma_k^{(1)}(V) = &-\frac{1}{2}\log\det \mathbf{\Delta}_*(U_0) - \frac{1}{2}\log\det\left(\mathbf{Q}_k \mathbf{G}_*(U_0) \mathbf{Q}_k^T\right) \\ &+ \log\left|\det\left(RD_{U_0}^T D_{U_0} : \mathcal{N}(Q_k') \to \mathcal{W}\right)\right|.\end{aligned}$$

Here $\mathbf{\Delta}_* = q_2 + D_{U_0} R D_{U_0}^T + a\mathbf{Q}_k^T \mathbf{Q}_k$ and $\mathbf{G}_* = \mathbf{\Delta}_*^{-1}$. The form q_2 is that of Proposition 33.2, $\mathbf{Q}_k = d\mathcal{Q}(0)$, and all operators are taken at U_0.

Proof. V with a finite stabilizer. Apply Laplace's method to the right side of Proposition 55.3, with $f = \mathbf{1}_{\mathscr{N}_{\varrho_g}} e^{-S/g^2}$. In the coordinates A the constraints are $\mathcal{C}A = 0$ to first order, with $\mathcal{C} = (\mathbf{Q}_k, RD^T)$. The action restricted to $\{\mathcal{Q} = 1,\ RD^T A = 0\}$ is critical at $A = 0$ (Proposition 27.2). Its Hessian there is the Hessian of the Lagrangian, q_2; the curvature of the constraint gives the multiplier term. The factors $|\det \mathcal{O}_A|$ and $\sigma(A)$ enter at $A = 0$. So, up to constants independent of V, the term of order g^0 is the logarithm of

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$$|\det \mathcal{O}_0| \int dA\, \delta(\mathcal{C}A)\, e^{-\frac{1}{2} q_2(A,A)} \ \propto\ |\det \mathcal{O}_0| \left(\det \mathbf{\Delta}_*\right)^{-1/2} \left(\det \mathcal{C}\mathbf{G}_* \mathcal{C}^T\right)^{-1/2}.$$

This is the Gaussian formula for a linear constraint, with $\mathbf{\Delta}_* = q_2 + \mathcal{C}^T \mathrm{diag}(a, 1)\mathcal{C}$, which equals q_2 on $\ker \mathcal{C}$. The operator $\mathbf{\Delta}_*$ is coercive. The reference U_0 is critical, with $\beta = 0$, and (M, ε)-regular with $|J_{U_0}| \le \varepsilon$ for $\varepsilon = B_3 a_1 \le \frac{1}{2} c_7$ (Theorem 30.3(b), as in the paragraph "The minimum" of the proof of Proposition 54.3). So Lemma 26.7 applies: by its statement, $\mathbf{\Delta}_*$ differs from the operator $\mathfrak{A}_\pi$ of Part III by a form bounded by $C'\varepsilon(\|h\|^2 + \|Dh\|^2 + \|D^T h\|^2)$, and $\langle h, \mathbf{\Delta}_* h\rangle \ge \frac{1}{2} c_*(\|h\|^2 + \|Dh\|^2 + \|D^T h\|^2)$ for all real h. It remains to show that $\mathcal{C}\mathbf{G}_*\mathcal{C}^T = \mathbf{Q}_k \mathbf{G}_* \mathbf{Q}_k^T \oplus 1$, on the coarse fields and on $\mathcal{W}$.

Gauge directions are zero modes of q_2. Let $L(A) = \Phi(A) - \langle \omega_*, \ell(\mathcal{Q}(e^{i\eta A}))\rangle$ be the Lagrangian with the multiplier ω_* of Proposition 27.2, so that $\nabla L(0) = 0$ and $\nabla^2 L(0) = q_2$. Gauge invariance of Φ and (27.1) give, for every A and λ,

$$\langle \nabla L(A), \Xi_A \lambda\rangle = -\langle \Lambda_A^T \omega_*, \Theta_A \lambda\rangle.$$

Differentiate in A at $A=0$ in a direction a. Since $\nabla L(0)=0$ and $\Lambda_0^T\omega_*=0$ (Proposition 27.2), what remains is $q_2(a,\Xi_0\lambda)=-\langle(\partial_a\Lambda^T)\omega_*,\Theta_0\lambda\rangle$. For $\lambda\in\mathcal{N}(Q'_k)$ the right side vanishes, since $\Theta_0=Q'_k$. With $\Xi_0\lambda=-D\lambda$, this gives $q_2 D\lambda=0$.

The block structure. For $\lambda\in\mathcal{N}(Q'_k)$, $a\mathbf{Q}_k^T\mathbf{Q}_k D\lambda=a\mathbf{Q}_k^T D_V Q'_k\lambda=0$ by exact covariance (Proposition 22.10(b)). Hence $\boldsymbol{\Delta}_* D\lambda=D(RD^TD\lambda)$, i.e. $\mathbf{G}_* D\mu=D\lambda$ for $\mu=RD^TD\lambda\in\mathcal{W}$. So $RD^T\mathbf{G}_*DR\mu=\mu$ and $\mathbf{Q}_k\mathbf{G}_*DR\mu=D_VQ'_k\lambda=0$ for $\mu\in\mathcal{W}$. Combining the determinants gives the formula; $\sigma(0)$ is a constant.

All V. The proofs of Proposition 55.3 and of the paragraphs above use of V only that $V\in\mathfrak{U}_k(a_1)$ and that (V,θ_0) is admissible: through Lemma 54.4, Theorem 24.2, Lemma 55.2, Theorem 30.3(a),(b), Proposition 54.3 and Lemma 54.2 (Setting), Proposition 27.2 and Lemma 26.7, all of which hold for such V. So the formula holds for every such V whose stabilizer is finite. Let V° be small field. Then $V^\circ\in\mathfrak{U}_k(\frac12 a_1)$ (Setting of Section 54), so there is a neighbourhood $\mathcal{B}$ of V° in the configurations on $\mathbb{T}$ with $\mathcal{B}\subset\mathfrak{U}_k(\frac34 a_1)$. For $V\in\mathcal{B}$ the minimal orbit lies in $\mathfrak{U}(\frac34 B_3a_1)\subset\mathfrak{U}(\frac34\theta_0)$ (Theorem 30.3(b), and $B_3a_1\le\theta_0$), so (V,θ_0) is admissible on $\mathcal{B}$.

- *Density.* By Lemma 55.2(a), the stabilizer of $U_k(V)$ is finite if and only if $\mathrm{Stab}_{\mathcal{G}_\mathbb{T}}(V)$ is. This is a closed subgroup of the compact group $\mathcal{G}_\mathbb{T}$, with Lie algebra $\ker D_V$ on the coarse scalar fields; so it is finite if and only if $\det(D_V^TD_V)\neq0$. This determinant is a real polynomial in the real and imaginary parts of the matrix entries of the $V(c)$, since $R(V(c))=V(c)\otimes\overline{V(c)}$. Its restriction to the connected real-analytic manifold $G^{\{\text{bonds}\}}$ is real analytic and, by the example below, not identically zero; so its zero set has empty interior. Fix a spanning tree of $\mathbb{T}$, put $V=1$ on its bonds, and choose two bonds c_1,c_2 off the tree. If $V(c_1)=h_1$, $V(c_2)=h_2$ and $V=1$ on the other bonds, then $\ker D_V$ consists of the constants $\lambda\equiv X$ with $\mathrm{Ad}_{h_1}X=\mathrm{Ad}_{h_2}X=X$. Take $h_1=e^{\mathrm{i}X_1}$ with $X_1\in\mathfrak{g}$ regular and small, so that the Lie algebra of the centralizer of h_1 is the Cartan subalgebra $\mathfrak{t}$ containing X_1, and $h_2=e^{\mathrm{i}X_2}$ with $X_2\in\mathfrak{g}$ such that no nonzero element of $\mathfrak{t}$ commutes with X_2 (for instance a small multiple of $\sum_\alpha(E_\alpha+E_\alpha^*)$, summed over the simple roots α of $\mathfrak{t}$, with root vectors $E_\alpha\in\mathfrak{g}^c$; an element $Y\in\mathfrak{t}$ commuting with it has $\alpha(Y)=0$ for all simple roots, so $Y=0$), and X_2 small, so that the centralizer of h_2 in $\mathfrak{g}$ is that of X_2. Then $\ker D_V=0$. So the $V\in\mathcal{B}$ with finite stabilizer are dense in $\mathcal{B}$.
- *The left side* is continuous on $\mathcal{B}$, by Proposition 54.3.
- *The right side.* Fix $U_0^\circ\in\mathscr{M}(V^\circ)$. By Corollary 30.5, the minimal orbit of $e^{\mathrm{i}B}V^\circ$ contains $\mathsf{U}_{U_0^\circ}(A_*(B))$, analytic in B. The operators of the formula are analytic in the reference near U_0°: by Proposition 46.4(vi) with $U_r=U_0^\circ$, which applies since $U_0^\circ\in\mathfrak{U}(B_3a_1)\subset\mathfrak{U}_0(B_3a_1)$ (Theorem 30.3(b)) and $B_3a_1\le c_3'$ (choice of a_1 in the proof of Theorem 30.3), and composition, as in the proof of Proposition 46.7, for the references $e^{\mathrm{i}\eta A_*(B)}U_0^\circ$, this covers $\mathbf{Q}_k$, Q'_k, D and $d^2C_k^{\mathrm{av}}(0)$; the gauge transformation w_k relating $e^{\mathrm{i}\eta A_*(B)}U_0^\circ$ to $\mathsf{U}_{U_0^\circ}(A_*(B))$ conjugates them by orthogonal maps; and $d^2\mathcal{N}(0)$ is polynomial in the background (proof of Lemma 55.6(a)). The multiplier is $\omega_0=(\mathbf{Q}_k\mathbf{Q}_k^T)^{-1}\mathbf{Q}_kJ_{U_0}$, which is continuous as well, since $\mathbf{Q}_k^T$ is injective ($\mathbf{Q}_k\mathbf{H}=I$). The right side does not depend on the element of the minimal orbit, since $u\in\mathfrak{G}_0$ conjugates $\boldsymbol{\Delta}_*$, $\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T$, $\mathfrak{A}'$ and M' by orthogonal maps, as in the proof of Lemma 55.6(a). At every $U_k(V)$, $V\in\mathcal{B}$, the operators are invertible: $\boldsymbol{\Delta}_*\ge\frac12 c_*$, by Lemma 26.7 as in the paragraph "V with a finite stabilizer"; $\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T>0$, since $\mathbf{G}_*>0$ and $\mathbf{Q}_k^T$ is injective; and the ghost is $\log\det\mathfrak{A}'+\frac12\log\det M'$ (Corollary 55.5, whose proof does not use this theorem), with $\mathfrak{A}'>0$ and $M'>0$ (Theorem 18.2(a),(c), via Corollary 22.11).

So the identity holds at V°, for every small-field V°, including $V^\circ=1$, where the vacuum polarization is taken. □

The ghost enters to the first power, and the determinant of $\mathbf{Q}_k\mathbf{Q}_k^T$ does not appear. At $U_0 = 1$ every operator is that of Part I. There $\boldsymbol{\Delta}_* = -\Delta - dP_{\mathcal{V}}d^* + a\mathbf{Q}^*\mathbf{Q}$, a Feynman-type operator up to the bounded term $dP_{\mathcal{V}}d^*$ (Section 9), and the ghost operator is $-\Delta : \mathcal{N}(Q_0) \to \mathcal{W}$. Both are regulated at the block scale. A Feynman gauge for $\mathfrak{G}_0$ itself would not be: its constraint is imposed at the points of $\mathbb{T}$, which have no capacity in the continuum limit in four dimensions.

Corollary 55.5 (The ghost on fixed spaces)**.** *With the operators of Section 14, $\mathfrak{A}'(U) = D_U^T D_U + aQ'^T Q'$, $G' = \mathfrak{A}'^{-1}$ and $M' = Q'G'^2Q'^T$,*

$$\log|\det(RD_U^T D_U : \mathcal{N}(Q') \to \mathcal{W})| = \log\det\mathfrak{A}'(U) + \tfrac{1}{2}\log\det M'(U),$$

the determinant on the left being taken in orthonormal frames. So, up to a constant,

$$\Gamma_k^{(1)}(V) = -\tfrac{1}{2}\log\det\boldsymbol{\Delta}_* - \tfrac{1}{2}\log\det(\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T) + \log\det\mathfrak{A}' + \tfrac{1}{2}\log\det M',$$

all operators acting on fixed spaces, at $U_0 = U_k(V)$.

Proof. Let $X = D^T D$ and $P = 1 - Q'^T Q'$, the orthogonal projection onto $\mathcal{N}(Q')$, since $Q'Q'^T = I$. Put $K = RX|_{\mathcal{N}(Q')}$. Since $\mathcal{W} = X\mathcal{N}(Q')$ (Section 14), $RXP = XP$, so $K^*K = PX^2P$ on $\mathcal{N}(Q')$. Moreover $\mathfrak{A}'P = XP$, since $Q'P = 0$, hence $PX^2P = P\mathfrak{A}'^2P$. By Jacobi's identity for complementary minors, applied to the positive operator $\mathfrak{A}'^2$ and the orthogonal decomposition into $\mathcal{N}(Q')$ and $\operatorname{ran} Q'^T$, on which Q'^T is an isometry, Details

$$\det(P\mathfrak{A}'^2P|_{\mathcal{N}(Q')}) = \det(\mathfrak{A}'^2)\cdot\det(Q'\mathfrak{A}'^{-2}Q'^T) = (\det\mathfrak{A}')^2\det M'.$$

Taking $\frac{1}{2}\log$ gives the claim. The left side does not depend on a, and neither does the right. □

For the vacuum polarization the formula is needed at backgrounds near 1 that are not minimizers. There the multiplier is not defined, and we treat it as an independent variable.

Lemma 55.6 (The multiplier at the trivial background)**.** *Fix the torus. For a real configuration $U = e^{i\eta A}$ near 1, with all averages relative to U, and a real coarse field ω, let*

$$\boldsymbol{\Delta}_*(U,\omega) = q_2^0(U) - \mathcal{C}_U[\omega] + D_U R D_U^T + a\mathbf{Q}_k^T\mathbf{Q}_k, \qquad \mathbf{G}_*(U,\omega) = \boldsymbol{\Delta}_*(U,\omega)^{-1}.$$

Here $q_2^0(U) = D_U^T D_U + d^2\mathcal{N}_U(0)$ is the Hessian of the Wilson action at U in the chart $e^{i\eta h}U$ (Proposition 23.3), i.e. q_2 of Proposition 33.2 with the multiplier set to 0, and $\mathcal{C}_U[\omega]$ is the operator of the bilinear form $(h,h') \mapsto \langle\omega, d^2C_k^{\mathrm{av}}(0)[h,h']\rangle$, with the nonlinear part C_k^{av} of $\ell(\mathcal{Q}(e^{i\eta h}))$ relative to U (Proposition 22.9). With the operators of Corollary 55.5 put

$$\widehat{\Gamma}(U,\omega) = -\tfrac{1}{2}\log\det\boldsymbol{\Delta}_*(U,\omega) - \tfrac{1}{2}\log\det(\mathbf{Q}_k\mathbf{G}_*(U,\omega)\mathbf{Q}_k^T) + \log\det\mathfrak{A}'(U) + \tfrac{1}{2}\log\det M'(U),$$

and $\widehat{\Gamma}(U) = \widehat{\Gamma}(U,0)$. Subscripts A and ω denote derivatives with respect to A and ω at $(A,\omega) = (0,0)$.

(a) $\widehat{\Gamma}$ is defined and real analytic on a neighbourhood $\mathcal{U}$ of $(0,0)$ in the variables (A,ω). Let u be a real gauge transformation of $\mathbb{T}_\eta$, and let $R(u|_{\mathbb{T}})$ act on coarse fields by $R(u(c_-))$. If (U,ω) and $(U^u, R(u|_{\mathbb{T}})\omega)$ both correspond to points of $\mathcal{U}$, then $\widehat{\Gamma}(U^u, R(u|_{\mathbb{T}})\omega) = \widehat{\Gamma}(U,\omega)$.

(b) $\widehat{\Gamma}_A = 0$ and $\widehat{\Gamma}_\omega = 0$. Moreover $\widehat{\Gamma}_{A\omega}[d\lambda,\omega] = 0$ for every scalar field λ on $\mathbb{T}_\eta$ and every coarse field ω.

(c) *For small real coarse fields B let $A_*(B)$ be the Landau representative of Theorem 25.1(c) at the reference 1, put $U_B = e^{\mathrm{i}\eta A_*(B)}$, and let*

$$\omega_B = ((\mathbf{Q}_k(U_B)\mathbf{H})^T)^{-1}\mathbf{H}^T J_{U_B},$$

with the minimizer map $\mathbf{H}$ of Part I. Then ω_B is the multiplier of Proposition 27.2 at the reference U_B, and, up to a constant independent of B,

$$\Gamma_k^{(1)}(e^{\mathrm{i}B}) = \widehat{\Gamma}(U_B, \omega_B).$$

Moreover $A_*(B) = \mathbf{H}B + O(|B|^2)$ and $\omega_B = \mathbf{D}B + O(|B|^2)$, with $\mathbf{D} = \mathbf{\Gamma} - a$ of Proposition 11.5.

(d) *For all coarse fields B and B',*

$$\frac{\partial^2}{\partial s\,\partial t}\Gamma_k^{(1)}\big(e^{\mathrm{i}(sB+tB')}\big)\Big|_{s=t=0} = \widehat{\Gamma}_{AA}[\mathbf{H}B, \mathbf{H}B'] + \widehat{\Gamma}_{A\omega}[\mathbf{H}B, \mathbf{D}B'] + \widehat{\Gamma}_{A\omega}[\mathbf{H}B', \mathbf{D}B] + \widehat{\Gamma}_{\omega\omega}[\mathbf{D}B, \mathbf{D}B'].$$

The first term is the second variation at $B = 0$ of $B \mapsto \widehat{\Gamma}(U_B)$.

Proof. (a) On the fixed torus all operators are finite matrices. The entries of D_U, of its transpose and of $d^2\mathcal{N}_U(0)$ are polynomials in those of the $U(b)^{\pm 1}$, by (23.1) and the proof of Proposition 23.3 (explicitly, (57.5)), so they are entire in A. By Proposition 46.4(vi) with $U_r = 1$, $\mathbf{Q}_k(U)$, $Q_k'(U)$ and the bilinear map $d^2C_k^{\mathrm{av}}(0)$ relative to U are holomorphic in A for $|A|_\infty < \delta_{AB}$, and $\mathcal{C}_U[\omega]$ is linear in ω. At $(A,\omega) = (0,0)$, $D_1 = d$, $d^2\mathcal{N}_1(0) = 0$ (Proposition 23.3), $R(1) = R$, $\mathbf{Q}_k(1) = \mathbf{Q}$ and $Q_k'(1) = Q_0$ (Proposition 22.10(c)). So $\mathbf{\Delta}_*(1,0) = \mathbf{\Delta}_a$, $\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T = \mathbf{Q}\mathbf{G}\mathbf{Q}^*$, $\mathfrak{A}'(1) = -\Delta + aQ_0^*Q_0$ and $M'(1) = M$, which are positive definite (Theorem 9.6, Proposition 11.2, Corollary 3.4, Proposition 8.8(a)). Hence the four operators are positive definite, with analytic inverses, for real (A,ω) near $(0,0)$, and $\widehat{\Gamma}$ is real analytic there. Details

For the covariance, let u be real, and let $R(u)$ act on fine fields by $R(u(b_-))$, respectively $R(u(x))$. Since $(e^{\mathrm{i}\eta h}U)^u = e^{\mathrm{i}\eta R(u)h}U^u$ bondwise and S is gauge invariant, $S(e^{\mathrm{i}\eta h}U^u) = S(e^{\mathrm{i}\eta R(u)^{-1}h}U)$; as $R(u)$ is orthogonal, $q_2^0(U^u) = R(u)q_2^0(U)R(u)^{-1}$. The covariant derivatives satisfy $D_{U^u}R(u) = R(u)D_U$, on scalar and on gauge fields. By Proposition 22.10(f), $Q_k'(U^u)R(u) = R(u|_{\mathbb{T}})Q_k'(U)$ and $\mathbf{Q}_k(U^u)R(u) = R(u|_{\mathbb{T}})\mathbf{Q}_k(U)$, so $\mathfrak{A}'$, G', M', $P_{\mathcal{V}}$ and R are covariant. By Proposition 22.10(f), whose nonlinear part is proved in the paragraph "The averaging" of the proof of Lemma 54.1(a) and uses only that the reference is real and regular, $\mathcal{Q}(e^{\mathrm{i}\eta R(u)h}; U^u) = R(u|_{\mathbb{T}})\mathcal{Q}(e^{\mathrm{i}\eta h}; U)$, where the second argument is the reference. Indeed, the relative configuration $R(u)X$ at the reference U^u represents $(XU)^u$, and $W_j(U^u) = W_j(U)^{u|_{\mathbb{T}^{(j)}}}$. By (21.2) and Lemma 21.10(b), the averaged relative contour variables, the τ_X, the g_X and the relative averages at scale j are conjugated by u at the points of the lattice of scale j, and by induction so is $\mathcal{Q}_j$. Hence $C_k^{\mathrm{av}}(R(u)h; U^u) = R(u|_{\mathbb{T}})C_k^{\mathrm{av}}(h; U)$, and $\mathcal{C}_{U^u}[R(u|_{\mathbb{T}})\omega] = R(u)\mathcal{C}_U[\omega]R(u)^{-1}$, since $R(u|_{\mathbb{T}})$ is orthogonal. So $\mathbf{\Delta}_*(U^u, R(u|_{\mathbb{T}})\omega) = R(u)\mathbf{\Delta}_*(U,\omega)R(u)^{-1}$, $\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T$ and M' are conjugated by $R(u|_{\mathbb{T}})$, and $\mathfrak{A}'$ by $R(u)$. The four determinants do not change. The neighbourhood $\mathcal{U}$ may depend on the torus. Only the derivatives at $(0,0)$ are used below, and their kernels are given by formulas that do not depend on the side (Section 56, paragraph "Background derivatives").

(b) Let $u \equiv g$ be constant. Then $1^u = 1$, $(e^{\mathrm{i}\eta A})^u = e^{\mathrm{i}\eta\,\mathrm{Ad}_g A}$ and $R(u|_{\mathbb{T}})\omega = \mathrm{Ad}_g\,\omega$. By (a), the linear form $(A,\omega) \mapsto \widehat{\Gamma}_A[A] + \widehat{\Gamma}_\omega[\omega]$ is invariant under Ad_g for all $g \in G$. Write $\widehat{\Gamma}_A[A] = \sum_b \mathrm{tr}(J(b)A(b))$ with $J(b) \in \mathfrak{g}$, which is possible since tr is a nondegenerate pairing on $\mathfrak{g}$. Then $\mathrm{Ad}_g\,J(b) = J(b)$ for all g, so $J(b)$ lies in the centre of $\mathfrak{g}$, which is 0 since G is simple. Likewise $\widehat{\Gamma}_\omega = 0$. For the mixed derivative let λ be real and $u_s = e^{-\mathrm{i}s\lambda}$. Then

$1^{u_s}(b) = e^{-\mathrm{i}s\lambda(b_-)}e^{\mathrm{i}s\lambda(b_+)} = e^{\mathrm{i}\eta A_s(b)}$, with A_s analytic in s and $A_s = s\,d\lambda + O(s^2)$. By (a), $\widehat{\Gamma}(1^{u_s}, R(u_s|_{\mathbb{T}})\omega') = \widehat{\Gamma}(1,\omega')$ for s and ω' near 0. Differentiating in ω' at $\omega' = 0$ in the direction ω gives $\partial_\omega\widehat{\Gamma}(A_s,0)[R(u_s|_{\mathbb{T}})\omega] = \widehat{\Gamma}_\omega[\omega] = 0$. Differentiating in s at $s = 0$ gives $\widehat{\Gamma}_{A\omega}[d\lambda,\omega] + \widehat{\Gamma}_\omega[\partial_s(R(u_s|_{\mathbb{T}})\omega)|_{s=0}] = 0$, and the second term vanishes. By linearity the identity extends to complex λ and ω.

(c) By Theorem 25.1(a)–(c), $\mathsf{U}(A_*(B)) = U_B^w$, with the real gauge transformation $w = \mathsf{w}_k(U_B)$, lies in the minimal orbit of $e^{\mathrm{i}B}$. By Theorem 25.1(g), U_B lies in the minimal orbit of $V_B = (e^{\mathrm{i}B})^h$, $h = w^{-1}|_{\mathbb{T}}$. For B small, V_B is small field, since the conditions of the Setting of Section 54 are invariant under gauge transformations, and $\Gamma_k^{(1)}(V_B) = \Gamma_k^{(1)}(e^{\mathrm{i}B})$ (Proposition 54.3). Apply Theorem 55.4 and Corollary 55.5 at V_B, with $U_0 = U_B$. There $q_2 = q_2^0(U_B) - \mathcal{C}_{U_B}[\omega_*]$ (Proposition 33.2(a)), with the multiplier ω_* of Proposition 27.2 at the reference U_B, where $A_* = 0$; so $J_{U_B} = \nabla\Phi(0) = \mathbf{Q}_k(U_B)^T\omega_*$. Applying $\mathbf{H}^T$ gives $(\mathbf{Q}_k(U_B)\mathbf{H})^T\omega_* = \mathbf{H}^T J_{U_B}$. At $B = 0$, $\mathbf{Q}_k(1)\mathbf{H} = \mathbf{Q}\mathbf{H} = I$ (Theorem 11.3(a)), so for B small this operator is invertible and $\omega_* = \omega_B$. This proves the formula. By Theorem 25.1(c),(f) at the reference 1, $A_*(B) = \mathbf{H}B + O(|B|^2)$, as in the proof of Lemma 54.17; here the minimizer map of Theorem 25.1(c) is that of the form $\|DA\|^2$, i.e. $W = 0$ (Section 26); at the reference 1, with the pair $(Q_0, \mathbf{Q})$ (Proposition 22.10(c)), it is the unique minimizer of $\|dA\|^2$ on $\{\mathbf{Q}A = B,\ Rd^*A = 0\}$ (Theorem 16.1(b)), i.e. the map $\mathbf{H}$ of Theorem 11.3(b). By Proposition 23.3(a),(d) at the reference 1, where $J = 0$ and $\varepsilon = 0$, $J_{e^{\mathrm{i}\eta A}} = d^*dA + O(|A|^2)$. Hence $J_{U_B} = d^*d\mathbf{H}B + O(|B|^2)$ and $\omega_B = \mathbf{H}^T d^*d\mathbf{H}B + O(|B|^2) = \mathbf{D}B + O(|B|^2)$, since $\langle B, \mathbf{D}B'\rangle = \langle d\mathbf{H}B, d\mathbf{H}B'\rangle$ (Proposition 11.5).

(d) Put $A(s,t) = A_*(sB + tB')$ and $\omega(s,t) = \omega_{sB+tB'}$. By (c) and the chain rule, at $s = t = 0$,

$$\begin{aligned}\partial_s\partial_t\Gamma_k^{(1)} = {}& \widehat{\Gamma}_{AA}[\partial_sA, \partial_tA] + \widehat{\Gamma}_A[\partial_s\partial_tA] + \widehat{\Gamma}_{A\omega}[\partial_sA, \partial_t\omega] \\ & + \widehat{\Gamma}_{A\omega}[\partial_tA, \partial_s\omega] + \widehat{\Gamma}_{\omega\omega}[\partial_s\omega, \partial_t\omega] + \widehat{\Gamma}_\omega[\partial_s\partial_t\omega].\end{aligned}$$

By (b) the second and the last term vanish, and by (c) $\partial_sA = \mathbf{H}B$, $\partial_tA = \mathbf{H}B'$, $\partial_s\omega = \mathbf{D}B$ and $\partial_t\omega = \mathbf{D}B'$. The last claim is the same computation for $B \mapsto \widehat{\Gamma}(U_B) = \widehat{\Gamma}(U_B, 0)$: the chain rule gives $\widehat{\Gamma}_{AA}[\partial_sA, \partial_tA] + \widehat{\Gamma}_A[\partial_s\partial_tA]$, and the second term vanishes by (b). □

So the multiplier enters the second variation of $\Gamma_k^{(1)}$ at $V = 1$ only through $\mathbf{D}$, and the continuation of the multiplier off the minimizers (Section 47.6) is not needed. In infinite volume the last three terms of (d) do not contribute to the marginal tensor: $\mathbf{D}$ is of order $|p|^2$ (Proposition 11.5), and the kernel of $\widehat{\Gamma}_{A\omega}$ annihilates the fine constant fields. These are the gradients of linear functions; the identity of (b), proved on every torus for periodic λ, passes to the infinite-volume kernel as $\Pi^{A\omega}d = 0$, and hence to the gradients of the fields of polynomial growth (Lemma 54.10(a); proof of Lemma 56.2(c)). The fine constant fields are the images under $\mathbf{H}_\infty$ of the coarse ones (Proposition 54.15(e)). So these terms are $O(|p|^3)$ (Lemma 56.2(c)). For the marginal tensor one may therefore set $\omega = 0$. The function $\widehat{\Gamma}(\cdot, 0)$ is gauge invariant on all backgrounds near 1, critical or not.

Remark 55.7 (A numerical check). We tested Theorem 55.4 numerically for $d = 2$, $G = SU(2)$ and $k = 1$. This is outside the hypothesis $d = 4$ of the note, but the identity is algebraic. For Bałaban's symmetric averaging of Part IV, transcribed to $d = 2$, with $L = 3$ on a coarse torus of 2×2 points, the difference of the two sides is the same for the four coarse fields tested, and equals the exact constant $-\frac{d}{2}\dim\mathfrak{g}\,|\mathbb{T}|\log L = -12\log 3$ to within $5 \cdot 10^{-11}$. The tests are described in Appendix C.

56 The vacuum polarization at the trivial background

This section reduces the marginal tensor of $\Gamma_k^{(1)}$ to that of a lattice one-loop term in the background Feynman gauge, up to a bounded error. Section 57 computes the latter. Throughout, $d = 4$, $U_0 = 1$, $\eta = L^{-k}$, and the constants of the norm bounds (Lemmas 56.1, 56.4, 56.6 and 56.7, Proposition 56.8) do not depend on k, η or the side n of the torus. The kernel decay of the operators in infinite volume (Section 54.2), and the errors $O(|p|^3)$ in Lemmas 56.2 and 56.3, which are identities at fixed k, may depend on η. We use the units of scale k: the fine lattice is $\eta\mathbb{Z}^4$, and the unit lattice is $\mathbb{Z}^4$.

Strategy. At $U = 1$ the operators of Theorem 55.4 commute with the translations of the unit lattice, so the vacuum polarization is a one-loop lattice integral, written in infinite volume with the tools of Section 54.2. The marginal tensor of $\Gamma_k^{(1)}$, a function of the coarse field, is that of a function $\widehat{\Gamma}$ of the fine field (Lemma 56.3): $T_{\Gamma_k^{(1)}} = T_{\text{fine}}$. The leading part of $\widehat{\Gamma}$ is the background Feynman-gauge operator $\boldsymbol{\Delta}_{\text{F}}$ with the fixed mass $m = 1$ (Section 56.2). Its ultraviolet cutoff is the fine spacing η; in the infrared the mass regularizes it, as the averaging term $a\mathbf{Q}_k^T\mathbf{Q}_k$ regularizes $\boldsymbol{\Delta}_*$, and at $U = 1$ the two operators are $\langle\partial\rangle^2$ and $\boldsymbol{\Delta}_a$. Its p^2-coefficient grows like $\mathsf{b}\log(1/\eta) = k\mathsf{b}\log L$ (Theorem 57.1). The rest is bounded uniformly in k: the averaging terms are operators of finite rank per cell (Lemma 56.7), the mass term raises the power-counting degree of the p^2-coefficient to at least 6 (Lemma 56.6 and the proof of Proposition 56.8), and the coarse determinants act on fields with finitely many components per cell. So $T_{\text{fine}} = T_{\text{F}} + O(1)$ (Proposition 56.8), and with Theorem 57.1 the coefficient of $T_{\Gamma_k^{(1)}}$ is $k\beta_\infty + O(1)$ (Theorem 58.1).

56.1 The function and its second variation

Let $\widehat{\Gamma}(U) = \widehat{\Gamma}(U, 0)$ be the function of Lemma 55.6,

$$\widehat{\Gamma}(U) = -\tfrac{1}{2}\log\det\boldsymbol{\Delta}_*(U) - \tfrac{1}{2}\log\det\left(\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T\right)(U) + \log\det\mathfrak{A}'(U) + \tfrac{1}{2}\log\det M'(U).$$

Here $\boldsymbol{\Delta}_*(U) = q_2^0(U) + D_U R D_U^T + a\mathbf{Q}_k^T\mathbf{Q}_k$, where q_2^0 is q_2 with the multiplier set to 0. The function $\widehat{\Gamma}$ is defined, analytic and gauge invariant on the backgrounds near 1, critical or not, and its first variation at $U = 1$ vanishes (Lemma 55.6(a),(b)). By Corollary 55.5 and Lemma 55.6(d), the second variation of $\Gamma_k^{(1)}$ at $V = 1$ is that of $V \mapsto \widehat{\Gamma}(U_k(V))$ plus three terms that come from the multiplier. In infinite volume these terms are of order $|p|^3$ (Lemma 56.2(c)), so the marginal tensor of $\Gamma_k^{(1)}$ is that of $V \mapsto \widehat{\Gamma}(U_k(V))$ (Lemma 56.3).

Infinite volume. We use the framework of Section 54.2. At $U = 1$ every operator below is a local operator of Section 2, an operator of Corollary 54.11, a power of $\langle\partial\rangle = (1 - \Delta)^{1/2}$ on the fine scalar or gauge fields (Lemma 54.12(f)), or a product of such operators. We denote their infinite-volume limits, which lie in the class $\mathscr{P}$ of periodic kernels with kernel decay, by the same letters; on the torus of side $n \geq 8$ the operators are the periodizations of these limits. The Bloch fibres, the trace per cell Tr_1, the norms $\|\cdot\|_r$, Hölder's inequality, the rank bound, the conjugates $O^{(p)} = e^{-\mathrm{i}p\cdot\hat{x}}Oe^{\mathrm{i}p\cdot\hat{x}}$ and the commutators $[\hat{x}_\mu, O]$ are those of Lemmas 54.12 and 54.13. The position $x(b)$ of a bond b is its initial point b_-. Momenta $p \in \mathbb{R}^4$ refer to the unit lattice. For $q \in \mathbb{R}^4$ put $|\hat{q}|^2 = \sum_\mu 4\eta^{-2}\sin^2(\eta q_\mu/2)$ and $\langle q\rangle = (1 + |\hat{q}|^2)^{1/2}$. On the fine Brillouin zone $[-\pi/\eta, \pi/\eta]^4$, $\frac{2}{\pi}|q| \leq |\hat{q}| \leq |q|$ (Lemma 4.1(a)), and $\eta\langle q\rangle \leq 1 + 2\sqrt{d}$ for all q, since $\eta \leq 1$ and $|\hat{q}| \leq 2\sqrt{d}/\eta$.

Fourier multipliers. An operator $O \in \mathscr{P}$ between spaces of fine scalar or gauge fields that commutes with the translations of $\eta\mathbb{Z}^4$ is a *Fourier multiplier*: $O(e^{\mathrm{i}q\cdot x}\phi) = e^{\mathrm{i}q\cdot x}o(q)\phi$ for $q \in \mathbb{R}^4$, with a matrix $o(q)$, its *symbol*. For gauge fields, $e^{\mathrm{i}q\cdot x}\phi$ with $\phi = (\phi_\mu)$ denotes the field $b \mapsto e^{\mathrm{i}q\cdot b_-}\phi_\mu$ on the bonds of direction μ. By Fourier analysis on $\eta\mathbb{Z}^4$ (the Fourier transform is unitary from $\ell^2(\eta\mathbb{Z}^4)$ onto $L^2([-\pi/\eta,\pi/\eta]^4)$, and the kernel of O is $(\eta/2\pi)^4 \int e^{\mathrm{i}q\cdot(x-y)}o(q)\,d^4q$; multiplying it by $(x-y)_\mu$ and integrating by parts in q_μ gives the symbol of $[\hat{x}_\mu, O]$), $\|O\| = \sup_q \|o(q)\|$; products of multipliers have the products of the symbols as symbols; $\langle\partial\rangle^t$ has the symbol $\langle q\rangle^t$; $O^{(p)}$ has the symbol $o(q+p)$; and $[\hat{x}_\mu, O]$ has the symbol $\mathrm{i}\partial_{q_\mu}o(q)$. Hence, if $\|o(q)\| \le C\langle q\rangle^s$ for all q, then

$$\|\langle\partial\rangle^{-a}O\langle\partial\rangle^{-b}\| \le C \qquad \text{for all real } a,b \text{ with } a+b=s. \tag{56.1}$$

We then say that O has *degree* s.

Lemma 56.1 (Kernel bounds for the averaging)**.** *Write* $(\mathbf{Q}_k(U)h)(c) = \eta^d\sum_b K_U(c,b)h(b)$ *and* $(Q'_k(U)\lambda)(y) = \eta^d\sum_x T_U(y,x)\lambda(x)$*, as in Propositions 22.9(c) and 22.10, and let* $U = e^{\mathrm{i}\eta A}$*, with a complex fine field* A*.*

(a) $K_U(c,b)$ *vanishes unless* b *starts in* $S_c = B(c_-)\cup B(c_+)$*, and it depends only on the values of* A *on the bonds of* S_c*, by one formula on all tori of sides* $n \ge 8$ *and on* $\eta\mathbb{Z}^4$*. It is holomorphic in* A *for* $|A|_\infty < \delta_{AB}$*, and bounded by* C *there. The same holds for* $T_U(y,x)$*, with* $x \in B(y)$ *and the bonds of* $B(y)$*. The nonlinear part* $C_k^{\mathrm{av}}(h)$ *relative to* U *is, at the bond* c*, an analytic function of* h *and* A *on the bonds of* S_c*, given by one formula.*

(b) The first and second derivatives of $K_U(c,b)$ *at* $A=0$ *in the directions* A_1*,* A_2 *are bounded by* $C\prod_i \max_{S_c}|A_i|$*, the maximum over the bonds of* S_c*, uniformly in* k*; likewise those of* $T_U(y,x)$*, with* $\max_{B(y)}$*.*

(c) Let Z *map the fine gauge fields to the coarse gauge fields, with* $(Zh)(c) = \eta^d\sum_b Z(c,b)h(b)$*, where* $Z(c,b) = 0$ *unless* b *starts in* S_c*, and* $|Z(c,b)| \le C'$*. Then* $\|Z^T\| = \|Z\| \le 2dC'$*, in the inner products of Section 2. The same holds, with the bound* C'*, for maps from the fine to the coarse scalar fields whose kernels are supported on the pairs* (y,x) *with* $x \in B(y)$*.*

Proof. (a) The supports and the locality are Propositions 22.10(d) and 22.9(a), which hold at complex references by Proposition 46.4(iii),(iv). The averaging is composed of the local one-step averages (Lemma 21.10(c)), given by the same formula on every torus. Holomorphy and the bounds are Proposition 46.4(iii),(iv),(vi), at the real reference $U_r = 1$, with constants uniform in k. The same induction as in the proof of Proposition 22.9(a), with the reference as a second argument, shows that $B_k(c)$, hence $C_k^{\mathrm{av}}(h)(c)$, depends on (h,A) only on the bonds of S_c: at scale j the one-step maps use W_{j-1}, W_j and the averaged contour variables of W_{j-1} only on the blocks of the bonds involved (Lemma 21.10(c)), and these lie in $B^{(j)}(c_-)\cup B^{(j)}(c_+)$. At complex references the same holds by Proposition 46.4(ii),(iii). Details

(b) Only the values on S_c enter, so we may replace A_i by $A_i\mathbf{1}_{S_c}$, whose sup norm is $\max_{S_c}|A_i|$. If $\max_{S_c}|A_i| \le 1$, the function $(z_1,z_2) \mapsto K_{e^{\mathrm{i}\eta(z_1A_1+z_2A_2)}}(c,b)$ is holomorphic and bounded by C for $|z_1|+|z_2| < \delta_{AB}$, and Cauchy's estimate bounds its derivatives of orders 1 and 2 at 0. Likewise for T_U.

(c) By the Cauchy–Schwarz inequality, $|(Zh)(c)|^2 \le (\eta^d\sum_b |Z(c,b)|^2)\|h\|^2_{S_c} \le 2dC'^2\|h\|^2_{S_c}$, since $2dN^d$ bonds start in S_c. Each fine bond starts in at most $2d$ of the sets S_c, so $\sum_c \|h\|^2_{S_c} \le 2d\|h\|^2$, and $\|Z\| \le 2dC'$. For scalar fields the blocks contain N^d points and partition $\mathbb{T}_\eta$. □

Background derivatives. Let $X(U)$ be a family of operators between field spaces, given on every torus of side $n \ge 8$ and analytic in $U = e^{\mathrm{i}\eta A}$ near 1. Its *background derivatives* at

$U=1$ are the derivatives $\partial_b X$ and $\partial_b\partial_{b'}X$ at $A=0$ with respect to the values $A(b)$ and $A(b')$ on positively oriented bonds; they are linear, respectively bilinear, in the elements of $\mathfrak{g}^c$ to which they are applied. We call *elementary* the families D_U on scalar fields, the covariant curl D_U, $\mathcal{F}_U = d^2\mathcal{N}_U(0)$, $\mathbf{Q}_k(U)$, $Q'_k(U)$, their transposes, and the operators $\mathcal{C}^c_U$ defined by $\mathcal{C}_U[\omega] = \sum_c \mathcal{C}^c_U[\omega(c)]$ (Lemma 55.6). They are local: there is an absolute r_0 such that, on every torus and on $\eta\mathbb{Z}^4$, the entry of an elementary family at (s,s') vanishes unless $|s-s'|\le r_0$, and it is an analytic function of the values of A on the bonds within distance r_0 of $x(s)$, given by one formula for all $n\ge 8$. The entries of $\mathcal{C}^c_U$ vanish unless s and s' start in S_c, and they depend only on A on the bonds of S_c (Lemma 41.11(c), Proposition 22.9(a), and the proof of Lemma 56.1(a)). For D_U, the curl and $\mathcal{F}_U$ the entries are polynomials in those of the $U(b)^{\pm1}$ (proof of Lemma 55.6(a)); for the other families this is Lemma 56.1(a). So the background derivatives of an elementary family are periodic kernels of finite range, the same on all tori and on $\eta\mathbb{Z}^4$, which vanish unless s, s', b and b' lie within distance $2r_0$ of each other.

The operators $\mathbf{\Delta}_*$, $\mathbf{G}_*$, $\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T$ and its inverse, $\mathfrak{A}'$, $G'=\mathfrak{A}'^{-1}$, M', M'^{-1}, $P_{\mathcal{V}}$ and R, the operators $\mathbf{\Delta}_{\mathrm{F}}$, $\mathfrak{A}'_{\mathrm{F}}$ of Section 56.2 and their inverses, and the operators $\mathbf{\Delta}_*(U,\omega)$, $\mathbf{G}_*(U,\omega)$, $\mathbf{Q}_k\mathbf{G}_*(U,\omega)\mathbf{Q}_k^T$ of Lemma 55.6, are obtained from elementary families by finitely many sums, products and inverses. At $U=1$ (and $\omega=0$) each of them is an operator of Corollary 54.11, a sum of products of local operators of Section 2, or $\langle\partial\rangle^{\pm2}$, so on the tori it is the periodization of its limit in $\mathscr{P}$. We call such families *admissible*. By the Leibniz rule and $\partial_b(X^{-1}) = -X^{-1}(\partial_b X)X^{-1}$, the background derivatives of an admissible family are finite sums, with a number of terms independent of n, of products

$$O_0\, L\, O_1 \qquad \text{and} \qquad O_0\, L\, O_1\, L'\, O_2, \tag{56.2}$$

where the O_i are products of values at $U=1$ of admissible and elementary families, and L, L' are background derivatives of elementary families: $L=\partial_b E$ and $L'=\partial_{b'}E'$, or $L=\partial_b\partial_{b'}E$ in the first form. The derivatives with respect to the values $\omega(c)$ have the same form, with $\partial_{\omega(c)}\mathbf{\Delta}_*(U,\omega) = -\mathcal{C}^c_U$ in place of L or L'. On $\eta\mathbb{Z}^4$ the same expressions, with the limits in $\mathscr{P}$, define $\partial_b X$ and $\partial_b\partial_{b'}X$. By Lemma 54.10(a), their entries at (s,s') are bounded by $Ce^{-\gamma(|x(s)-b_-|+|b_--x(s')|)}$, respectively by $Ce^{-\gamma(|x(s)-b_-|+|b_--b'_-|+|b'_--x(s')|)}$. So, for bounded fields A and A', the sums

$$\delta X[A] = \sum_b \partial_b X\, A(b), \qquad \delta^2 X[A,A'] = \sum_{b,b'} \partial_b\partial_{b'}X\, A(b)A'(b')$$

converge absolutely, entrywise, to kernels with kernel decay. They satisfy the Leibniz rule and $\delta(X^{-1}) = -X^{-1}(\delta X)X^{-1}$.

The background field. For $p\in\mathbb{R}^4$ and $\varepsilon\in(\mathfrak{g}^c)^4$ let $A_{p,\varepsilon}(b) = e^{\mathrm{i}p\cdot b_-}\varepsilon_\mu$ on the bonds b of direction μ, $\bar\varepsilon = (\varepsilon^*_\mu)$ and $|\varepsilon| = \max_\mu|\varepsilon_\mu|$. Since $\mathfrak{g}$ consists of Hermitian matrices, $A_{p,\varepsilon} + A_{-p,\bar\varepsilon}$ is real. The *vertices* of an admissible family X are

$$\mathsf{V}_1(p) = e^{-\mathrm{i}p\cdot\hat x}\,\delta X[A_{p,\varepsilon}], \qquad \mathsf{V}_1(-p) = e^{\mathrm{i}p\cdot\hat x}\,\delta X[A_{-p,\bar\varepsilon}], \qquad \mathsf{V}_2(p) = \delta^2 X[A_{-p,\bar\varepsilon}, A_{p,\varepsilon}],$$

where $\hat x$ is the position on the target space of X; so $\mathsf{V}_1(-p)$ is formed with $\bar\varepsilon$. We write V_i^X when the family must be named. Since $A_{p,\varepsilon}(b+a) = e^{\mathrm{i}p\cdot a}A_{p,\varepsilon}(b)$ for $a\in\mathbb{Z}^4$, the vertices are periodic, so they lie in $\mathscr{P}$. Their entries are absolutely convergent sums of the entries of $\partial_b X$ and $\partial_b\partial_{b'}X$ times phases $e^{\pm\mathrm{i}p\cdot(b_- - x(s))}$. So they are analytic in p, in the norm of operators, and their p-derivatives have the same form, with polynomial factors. By the Leibniz rule,

$$\mathsf{V}_1^{YZ}(p) = \mathsf{V}_1^Y(p)Z + Y^{(p)}\mathsf{V}_1^Z(p), \qquad \mathsf{V}_1^{Y^{-1}}(p) = -(Y^{-1})^{(p)}\mathsf{V}_1^Y(p)Y^{-1},$$

and similarly for V_2. For an admissible family X whose inverse is admissible put, with $X=X(1)$,

$$S_X(p) = \mathrm{Tr}_1\left(X^{-1}\mathsf{V}_2(p)\right) - \mathrm{Tr}_1\left(X^{-1}\mathsf{V}_1(-p)^{(p)}(X^{-1})^{(p)}\mathsf{V}_1(p)\right). \tag{56.3}$$

By Lemma 56.2(a) below, this is the second variation of $\log\det X$ per unit cell in the directions $A_{-p,\bar\varepsilon}$ and $A_{p,\varepsilon}$. Its p^2-coefficient is obtained by differentiating twice in p at $p=0$. Each derivative falls on a vertex or produces a commutator with $\hat x$, since $\partial_p(O^{(p)}) = -\mathrm{i}[\hat x, O^{(p)}]$ and $[\hat x_\mu, X^{-1}] = X^{-1}[X,\hat x_\mu]X^{-1}$ (Lemma 54.13(d)).

Lemma 56.2 (The infinite-volume limit at the trivial background)**.** *Let $\mathfrak{N}$ be the set of the sides $n \geq 8$ of the tori on which $\Gamma_k^{(1)}$ is defined (Section 54); it contains all large n. Kernels are taken with respect to the unweighted pairings $\sum_b \mathrm{tr}(A(b)A'(b))$ over the positively oriented fine, respectively coarse, bonds (on the coarse lattice this is the trace pairing of Section 23), and the Lie algebra indices are suppressed as in Definition 54.7.*

(a) Let X be an admissible family whose inverse is admissible. The kernels of the second variations at $U=1$ of $\log\det X$ on the tori of the sides $n \in \mathfrak{N}$ converge with uniform decay (Lemma 54.10(e)) to a periodic kernel Π_X on the bonds of $\eta\mathbb{Z}^4$ with kernel decay. For $p \in \mathbb{R}^4$,

$$\sum_b \sum_{b'} \Pi_X(b,b')\,\mathrm{tr}\,\big(A_{-p,\bar\varepsilon}(b)A_{p,\varepsilon}(b')\big) = S_X(p),$$

where the first sum runs over the positively oriented bonds b with $x(b)$ in a cell.

(b) The second variations of $\widehat\Gamma$ at $U=1$ converge with uniform decay to the real symmetric periodic kernel $\Pi = -\frac12\Pi_{\boldsymbol\Delta_*} - \frac12\Pi_{\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T} + \Pi_{\mathfrak{A}'} + \frac12\Pi_{M'}$, which has kernel decay. It annihilates the gradients $d\lambda$ of the scalar fields λ of polynomial growth, in particular the constant fields. The vacuum polarization *of $\widehat\Gamma$ at $U=1$, its second variation per cell as in (a), is $-\frac12 S_{\boldsymbol\Delta_*} - \frac12 S_{\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T} + S_{\mathfrak{A}'} + \frac12 S_{M'}$. The same holds for the function $\widehat\Gamma_{\mathrm F}$ of Section 56.2, with the vacuum polarization $-\frac12 S_{\boldsymbol\Delta_{\mathrm F}} + S_{\mathfrak{A}'_{\mathrm F}}$.*

(c) The second variations of $\Gamma_k^{(1)}$ at $V=1$ on the tori of the sides $n \in \mathfrak{N}$ converge with uniform decay to $\mathbf{H}_\infty^{\mathsf T}\Pi\mathbf{H}_\infty + \Xi$. Here Ξ is a real symmetric periodic kernel on the bonds of $\mathbb{Z}^4$ with kernel decay, and its Fourier transform, in the convention of Definition 54.7, satisfies $\Xi(p) = O(|p|^3)$. In particular $\Gamma_k^{(1)}$ satisfies the assumptions made above Definition 54.7, and the hypothesis of Lemma 54.9(c) holds for $\Phi = \Gamma_k^{(1)}$.

Proof. (a) On the torus of side n, the second variation of $\log\det X$ at $U=1$ has the kernel $\mathrm{Tr}(X^{-1}\partial_b\partial_{b'}X) - \mathrm{Tr}(X^{-1}\partial_b X\, X^{-1}\partial_{b'}X)$, with Tr the trace of operators on the torus. By (56.2) and the cyclicity of the trace, it is a finite sum, with a number of terms independent of n, of traces $\mathrm{Tr}(O_0 L O_1 L')$ and $\mathrm{Tr}(O_0 L)$. Here the O_i are periodizations of products of elements of $\mathscr{P}$; in the first form L is local at b and L' at b', and in the second $L = \partial_b\partial_{b'}E$ is local at both. Written out, each trace is a sum over the sites s_1, s_2 within distance $2r_0$ of b and s_3, s_4 within distance $2r_0$ of b' of products of entries of the O_i and of L and L' at these sites. The number of these sites is bounded independently of n; the entries of L and L' do not depend on n; and the entries of the O_i are bounded by $Ce^{-\gamma|s-s'|_n}$ and converge as $n\to\infty$ (Lemma 54.10(a),(c)). Hence the kernels are bounded by $Ce^{-\gamma|b-b'|_n}$ and converge entrywise: they converge with uniform decay to the kernel Π_X given by the same sums with the limits in $\mathscr{P}$. It is periodic, since its constituents are. Details

For the second variation per cell, write $\Pi_X(b,b') = \mathrm{Tr}(X^{-1}\partial_b\partial_{b'}X) - \mathrm{Tr}(X^{-1}\partial_b X\, X^{-1}\partial_{b'}X)$ on $\eta\mathbb{Z}^4$, with the background derivatives on $\eta\mathbb{Z}^4$ defined above. The sums over b in the cell, over b' and over the sites of the traces converge absolutely, by the kernel decay. Each summand is invariant under the simultaneous translation of all bonds and sites by $a \in \mathbb{Z}^4$, since the phases of $A_{-p,\bar\varepsilon}(b)$ and $A_{p,\varepsilon}(b')$ cancel. So the sum may be taken over all b, with the first site of the trace, the row index of the first factor X^{-1}, in the cell Σ_0. Summing first over b and b' gives the kernels of $\delta X[A_{-p,\bar\varepsilon}] = e^{-\mathrm{i}p\cdot\hat x}\mathsf{V}_1(-p)$ and $\delta X[A_{p,\varepsilon}] = e^{\mathrm{i}p\cdot\hat x}\mathsf{V}_1(p)$, respectively of

$\delta^2 X[A_{-p,\bar\varepsilon}, A_{p,\varepsilon}] = \mathsf{V}_2(p)$. The result is $\sum_{s\in\Sigma_0} \operatorname{tr} W(s,s)$, with $W = X^{-1}\mathsf{V}_2(p)$ and with

$$W = X^{-1}e^{-\mathrm{i}p\cdot\hat x}\mathsf{V}_1(-p)X^{-1}e^{\mathrm{i}p\cdot\hat x}\mathsf{V}_1(p) = X^{-1}\mathsf{V}_1(-p)^{(p)}(X^{-1})^{(p)}\mathsf{V}_1(p),$$

operators in $\mathscr{P}$. By Lemma 54.13(a) this is $S_X(p)$.

(b) The convergence and the formula for the vacuum polarization follow from (a), since $\widehat\Gamma$ is the sum of the four terms. On each torus the second variation of the real analytic function $\widehat\Gamma$ is real and symmetric, and so is the limit. On each torus $\widehat\Gamma$ is gauge invariant with vanishing first variation at 1 (Lemma 55.6(a),(b)), so its second variation annihilates the gradients $d\lambda$ (Remark 54.8): $\Pi_n d = 0$ for the kernels Π_n on the tori. The scalar gradient d is local, so $(\Pi_n d)(b,x)$ is a finite sum, which converges to $(\Pi d)(b,x)$; hence $\Pi d = 0$, and $\Pi(d\lambda) = (\Pi d)\lambda = 0$ for λ of polynomial growth, by Lemma 54.10(a). The constant field with the components ε_μ is $d\lambda$ with $\lambda(x) = \varepsilon\cdot x$. For $\widehat\Gamma_{\mathrm F}$ the same argument applies: it is gauge invariant with vanishing first variation, by the argument of Lemma 55.6(a),(b), since D_U and $\mathcal{F}_U$ are covariant.

(c) Let $n \in \mathfrak{N}$. By Lemma 55.6(d), the second variation of $\Gamma_k^{(1)}$ at $V = 1$ has the kernel

$$\mathbf{H}^{\mathsf T}\Pi_n\mathbf{H} + \mathbf{D}\Pi_n^{A\omega}\mathbf{H} + \mathbf{H}^{\mathsf T}(\Pi_n^{A\omega})^{\mathsf T}\mathbf{D} + \mathbf{D}\Pi_n^{\omega\omega}\mathbf{D},$$

where Π_n is the second variation of $\widehat\Gamma$, $\widehat\Gamma_{A\omega}[A,\omega] = \sum_{c,b}\Pi_n^{A\omega}(c,b)\operatorname{tr}(\omega(c)A(b))$ and $\widehat\Gamma_{\omega\omega}[\omega,\omega'] = \sum_{c,c'}\Pi_n^{\omega\omega}(c,c')\operatorname{tr}(\omega(c)\omega'(c'))$, all on the torus of side n, and $\mathbf{D}$ is real symmetric. By (b), $\Pi_n \to \Pi$ with uniform decay. The kernels $\Pi_n^{A\omega}$ and $\Pi_n^{\omega\omega}$ are sums of traces as in (a), with the local insertions $-\mathcal{C}_U^c$ at the coarse bonds c, so by the argument of (a) they converge with uniform decay to periodic kernels $\Pi^{A\omega}$ and $\Pi^{\omega\omega}$ with kernel decay. The operators $\mathbf{H}$ and $\mathbf{D}$ are the periodizations of $\mathbf{H}_\infty$ and $\mathbf{D} \in \mathscr{P}$ (Proposition 54.15(a), Corollary 54.11), so they converge with uniform decay. By Lemma 54.10(e), the kernels converge with uniform decay to $\mathbf{H}_\infty^{\mathsf T}\Pi\mathbf{H}_\infty + \Xi$, with

$$\Xi = \mathbf{D}\Pi^{A\omega}\mathbf{H}_\infty + \mathbf{H}_\infty^{\mathsf T}(\Pi^{A\omega})^{\mathsf T}\mathbf{D} + \mathbf{D}\Pi^{\omega\omega}\mathbf{D},$$

which is real, symmetric and periodic, with kernel decay.

For a kernel $Y \in \mathscr{P}$ on the coarse gauge fields, $Ye_{p,\varepsilon} = e_{p,Y(p)\varepsilon}$, with the fields $e_{p,\varepsilon}$ of Corollary 54.16, since a cell of $\mathbb{Z}^4$ contains one bond of each direction. So $(YY')(p) = Y(p)Y'(p)$, and $Y^{\mathsf T}(p) = Y(p)^*$ for real Y. Hence

$$\Xi(p) = \mathbf{D}(p)\,(\Pi^{A\omega}\mathbf{H}_\infty)(p) + (\Pi^{A\omega}\mathbf{H}_\infty)(p)^*\,\mathbf{D}(p) + \mathbf{D}(p)\Pi^{\omega\omega}(p)\mathbf{D}(p).$$

- $\mathbf{D}(p) = O(|p|^2)$. On every torus $0 \le \langle B, \mathbf{D}B\rangle \le C_D\|d_1 B\|^2$ (Proposition 11.5). By Lemma 54.12(e), $0 \le \mathbf{D} \le C_D d_1^* d_1$ on $\ell^2(\mathbb{Z}^4)$, and by Lemma 54.12(c) the same holds for the fibres. In the bond-midpoint convention the fibre at p and the Fourier transform at p are conjugate by a diagonal unitary matrix of phases, and the Fourier transform of $d_1^* d_1$ is $|\hat p|^2\delta_{\mu\nu} - \hat p_\mu \hat p_\nu \le |p|^2$, with $\hat p_\mu = 2\sin(p_\mu/2)$. So $\|\mathbf{D}(p)\| \le C_D|p|^2$.
- $(\Pi^{A\omega}\mathbf{H}_\infty)(p) = O(|p|)$. By Lemma 55.6(b), $\Pi_n^{A\omega} d = 0$ on every torus; as in (b), $\Pi^{A\omega} d = 0$, and $\Pi^{A\omega}$ annihilates the fine constant fields. By Proposition 54.15(e), $\mathbf{H}_\infty$ maps the coarse constant fields to the fine ones, so $(\Pi^{A\omega}\mathbf{H}_\infty)(0) = 0$. The function $(\Pi^{A\omega}\mathbf{H}_\infty)(p)$ is analytic, by the kernel decay.
- $\Pi^{\omega\omega}(p)$ is bounded near $p = 0$, by the kernel decay.

So $\Xi(p) = O(|p|^3)$. Finally, $\Gamma_k^{(1)}$ is real analytic and gauge invariant (Proposition 54.3), with vanishing first variation at 1, and the limit of its second variations is invariant under the translations of $\mathbb{Z}^4$. So the assumptions made above Definition 54.7 hold, and so does the hypothesis of Lemma 54.9(c). □

From the coarse field to the fine field. We use the bond-midpoint convention of Section 54.1 for the coarse and for the fine fields, with the fields $e_{q,\varepsilon}$ on the bonds of $\mathbb{Z}^4$ and $e'_{q,\varepsilon}$ on the bonds of $\eta\mathbb{Z}^4$ of Corollary 54.16, $\varepsilon \in \mathbb{C}^4$, and the pairing $(g,f)_1$ over the bonds with base point in a cell of the proof of Lemma 54.17. For a periodic kernel Π on the fine bonds with kernel decay and $q'-q \in 2\pi\mathbb{Z}^4$ let

$$\varepsilon^{\mathsf{T}}\Pi(q,q')\varepsilon' = (e'_{-q,\varepsilon}, \Pi e'_{q',\varepsilon'})_1,$$

so that Π couples only momenta with $q'-q \in 2\pi\mathbb{Z}^4$. For the second variation Π of $\widehat{\Gamma}$, $\Pi(p,p)$ is the vacuum polarization of Lemma 56.2(b) in the bond-midpoint convention. In the convention of $A_{p,\varepsilon}$, with the values at the initial points of the bonds, it is $P(p)^*\Pi(p,p)P(p)$, with $P(p) = \mathrm{diag}(e^{-\mathrm{i}\eta p_\mu/2})$. If $\Pi(p,p)$ vanishes at $p=0$ and has no term linear in p, the two have the same terms of order at most 2 in p.

Lemma 56.3 (The coarse map)**.** *Let Π be a real symmetric periodic kernel on the bonds of $\eta\mathbb{Z}^4$, with kernel decay, that annihilates the constant fields. Then $\Pi(p,p)$ is real analytic, $\Pi(0,0)=0$, and $\Pi(p,p+2\pi\ell) = O(|p|)$ for every $\ell \in \mathbb{Z}^4$. The kernel $\mathbf{H}_\infty^{\mathsf{T}}\Pi\mathbf{H}_\infty$ on the bonds of $\mathbb{Z}^4$ is real, symmetric and periodic, with kernel decay, and*

$$(\mathbf{H}_\infty^{\mathsf{T}}\Pi\mathbf{H}_\infty)(p) = \Pi(p,p) + O(|p|^3).$$

If moreover (a) $\Pi(p,p) = O(|p|^2)$, i.e. there is no term linear in p, then $\Pi(p,p) = T_{\mathrm{fine}}(p) + O(|p|^3)$ with a tensor T_{fine}, the p^2-tensor of $\Pi(p,p)$, and $(\mathbf{H}_\infty^{\mathsf{T}}\Pi\mathbf{H}_\infty)(p) = T_{\mathrm{fine}}(p) + O(|p|^3)$. For the second variation Π of $\widehat{\Gamma}$ at $U=1$ the hypothesis on the constant fields and (a) both hold, and the marginal tensor of $\Gamma_k^{(1)}$ is $T_{\Gamma_k^{(1)}} = T_{\mathrm{fine}}$.

Proof. The analyticity follows from the kernel decay. By Corollary 54.16, which applies to $\mathbf{H}_\infty$ by Proposition 54.15(e), $\mathbf{H}_\infty e_{p,\varepsilon} = \sum_{\ell\in\mathcal{L}} e'_{p+2\pi\ell,k_\ell(p)\varepsilon}$, a finite sum over $\mathcal{L} = \{0,\dots,N-1\}^4$, with $k_0(p) = I + O(|p|^2)$ and $k_\ell(p) = O(|p|^2)$ for $\ell \neq 0$. Since $\mathbf{H}_\infty$ is real, $\mathbf{H}_\infty e_{-p,\varepsilon} = \sum_\ell e'_{-p-2\pi\ell,\overline{k_\ell(p)}\varepsilon}$. By the definition of the Fourier transform, $\varepsilon^{\mathsf{T}}(\mathbf{H}_\infty^{\mathsf{T}}\Pi\mathbf{H}_\infty)(p)\varepsilon' = (\mathbf{H}_\infty e_{-p,\varepsilon}, \Pi\mathbf{H}_\infty e_{p,\varepsilon'})_1$, so Details

$$(\mathbf{H}_\infty^{\mathsf{T}}\Pi\mathbf{H}_\infty)(p) = \sum_{\ell,\ell'\in\mathcal{L}} k_\ell(p)^*\,\Pi(p+2\pi\ell, p+2\pi\ell')\,k_{\ell'}(p),$$

a finite sum. For a periodic kernel Y, $(g, Yf)_1 = (Y^{\mathsf{T}}g, f)_1$, by the grouping of the proof of Lemma 54.10(a). Since Π is symmetric and annihilates the constant fields $e'_{0,\varepsilon}$, it follows that $\Pi(0,2\pi\ell)=0$ and $\Pi(2\pi\ell,0)=0$ for all ℓ. So $\Pi(p,p+2\pi\ell)$ and $\Pi(p+2\pi\ell,p)$ are $O(|p|)$, and in particular $\Pi(0,0)=0$. Hence the term $\ell=\ell'=0$ of the sum is $\Pi(p,p) + O(|p|^2)O(|p|) = \Pi(p,p) + O(|p|^3)$, the terms with exactly one of ℓ,ℓ' equal to 0 are $O(|p|)O(|p|^2)$, and the terms with $\ell,\ell' \neq 0$ are $O(|p|^4)$. This proves the first claim. Under (a), $\Pi(p,p)$ has a term of order 2, $T_{\mathrm{fine}}(p)$, and the second claim follows.

For the second variation Π of $\widehat{\Gamma}$, the kernel is real, symmetric and periodic with kernel decay, and it annihilates the constant fields (Lemma 56.2(b)); hypothesis (a) holds by the paragraph after this lemma. By Lemma 56.2(c), $\Pi_{\Gamma_k^{(1)}}(p) = (\mathbf{H}_\infty^{\mathsf{T}}\Pi\mathbf{H}_\infty)(p) + \Xi(p) = T_{\mathrm{fine}}(p) + O(|p|^3)$. □

Hypothesis (a) holds for the second variation Π of $\widehat{\Gamma}$. The function $\widehat{\Gamma}$ is invariant under $\mathfrak{S}_k$ on the real regular backgrounds near 1: for $r \in \mathfrak{S}_k$, the operators at rU that enter $\widehat{\Gamma}$ are conjugate to those at U by the orthogonal maps r_U, r and r_W of Proposition 31.7(c). Indeed, $q_2^0(U)$ is the Hessian of Φ_U at 0, which is covariant by Proposition 31.7(c)(iii); D_U on scalar fields and $\mathbf{Q}_k$ are covariant by (c)(i); and R, $\mathfrak{A}'$ and M' by (c)(ii). Hence $\boldsymbol{\Delta}_*$, $\mathbf{G}_*$ and $\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T$ are conjugated by r_U and r_W, and the four determinants do not change. Only

operators that are covariant under all of $\mathfrak{S}_k$ at curved backgrounds enter; the gauge-field operators of Proposition 31.7(c)(iv), such as $\mathbf{H}$ and $\mathbf{C}$, are not used, since they are covariant under reflections only at $U = 1$ (Remark 31.8). So the second variations of $\widehat{\Gamma}(e^{i\eta A})$ at $A = 0$ on the tori are invariant under $\mathfrak{S}_k$, and so is their limit Π, in particular under the point reflection about a point of $\mathbb{Z}^4$. In the bond-midpoint convention for the fine bonds this gives $\Pi(-p,-p) = \Pi(p,p)$, by the computation in the proof of Lemma 54.9(c). Together with $\Pi(0,0) = 0$ (Lemma 56.3), the term linear in p vanishes. Alternatively, (a) follows from Lemma 54.9(c), whose hypothesis holds by Lemma 56.2(c): $\Pi_{\Gamma_k^{(1)}}(p) = O(|p|^2)$, and $\Pi(p,p) = \Pi_{\Gamma_k^{(1)}}(p) - \Xi(p) + O(|p|^3)$ by the first part of Lemma 56.3. In the convention of $A_{p,\varepsilon}$, $\Pi(p,p)$ is multiplied by the phases $e^{\pm i\eta(p_\mu - p_\nu)/2}$, which change it only by $O(|p|^3)$. No summability over the Umklapp components is needed: at fixed η there are N^4 of them, and the conclusion of Lemma 56.3 is an identity at fixed k.

56.2 The splitting

Write

$$\mathbf{\Delta}_* = \mathbf{\Delta}_{\mathrm{F}} + \mathbf{Y} - m^2, \qquad \mathbf{\Delta}_{\mathrm{F}}(U) = D_U^T D_U + D_U D_U^T + \mathcal{F}_U + m^2, \qquad \mathbf{Y} = -D_U P_{\mathcal{V}} D_U^T + a\mathbf{Q}_k^T \mathbf{Q}_k,$$

with a fixed mass $m = 1$. As in Proposition 33.2, D_U denotes the covariant curl in the first term and the covariant gradient of scalars in the second. $D_U^T D_U + \mathcal{F}_U$ is the Hessian of the Wilson action at the background U in the chart $e^{i\eta h}U$, with $\mathcal{F}_U = d^2\mathcal{N}_U(0)$ and the nonlinear part $\mathcal{N}_U$ of Proposition 23.3, lattice artefacts included. $D_U D_U^T$ is the Feynman gauge-fixing term, and $D_U R D_U^T = D_U D_U^T - D_U P_{\mathcal{V}} D_U^T$. At $U = 1$ the Hodge identity is exact (Lemma 2.2), and $\mathcal{F}_1 = 0$ (Proposition 23.3). So $\mathbf{\Delta}_{\mathrm{F}}(1) = -\Delta + m^2 = \langle\partial\rangle^2$, and $\mathbf{\Delta}_*(1)$ is the operator $\mathbf{\Delta}_a$ of Part I. In the same way,

$$\mathfrak{A}' = \mathfrak{A}'_{\mathrm{F}} + \mathbf{Y}' - m^2, \qquad \mathfrak{A}'_{\mathrm{F}}(U) = D_U^T D_U + m^2, \qquad \mathbf{Y}' = aQ'^T Q',$$

with $\mathfrak{A}'_{\mathrm{F}}(1) = \langle\partial\rangle^2$ on scalar fields. Let

$$\widehat{\Gamma}_{\mathrm{F}}(U) = -\tfrac{1}{2}\log\det\mathbf{\Delta}_{\mathrm{F}}(U) + \log\det\mathfrak{A}'_{\mathrm{F}}(U),$$

the one-loop term of the lattice theory in the background Feynman gauge with mass m, and T_{F} the p^2-tensor of its vacuum polarization at $U = 1$ (Lemma 56.2(b)).

The vertices of $\mathbf{\Delta}_{\mathrm{F}}$ and $\mathfrak{A}'_{\mathrm{F}}$ are local at the fine scale. Their bounds are degree-sharp: each p-derivative and each commutator with $\hat{x}$ gains one factor $\langle\partial\rangle^{-1}$.

Lemma 56.4 (The vertices of the Feynman operators)**.** *Let $X \in \{\mathbf{\Delta}_{\mathrm{F}}, \mathfrak{A}'_{\mathrm{F}}\}$, $p \in \mathbb{R}^4$ with $|p| \le \pi$, and $|\varepsilon| \le 1$. Then $X(1)$, $[X(1), \hat{x}_\mu]$ and the vertices $\mathsf{V}_1(p)$, $\mathsf{V}_2(p)$ of X are Fourier multipliers, with the symbols $x(q) = (|\hat{q}|^2 + m^2)\,I$, $-\mathrm{i}\partial_{q_\mu} x(q)$, $v_1(q;p,\varepsilon)$ and $v_2(q;p,\varepsilon)$.*

(a) $v_1(q;0,\varepsilon) = \sum_\mu 2\eta^{-1}\sin(\eta q_\mu)\,\mathrm{ad}\,\varepsilon_\mu$, acting on each component.

(b) For multi-indices $\alpha, \beta \in \mathbb{N}^4$ with $|\alpha| + |\beta| \le 2$, with C depending only on G and ρ,

$$|\partial_p^\alpha \partial_q^\beta v_1(q;p,\varepsilon)| \le C\langle q\rangle^{1-|\alpha|-|\beta|}, \qquad |\partial_p^\alpha \partial_q^\beta v_2(q;p,\varepsilon)| \le C\langle q\rangle^{-|\alpha|-|\beta|},$$

and $|\partial_q^\beta x(q)| \le C\langle q\rangle^{2-|\beta|}$ for $1 \le |\beta| \le 2$.

Consequently, with $\mathsf{D} = \partial_p^\alpha\,\mathrm{ad}_{\hat{x}}^\beta$, where $\mathrm{ad}_{\hat{x}}^\beta = \mathrm{ad}_{\hat{x}_1}^{\beta_1}\cdots\mathrm{ad}_{\hat{x}_4}^{\beta_4}$ and $\mathrm{ad}_{\hat{x}_\mu} O = [\hat{x}_\mu, O]$, the operators $\mathsf{DV}_1(p)$, $\mathsf{DV}_2(p)$ and $\mathrm{ad}_{\hat{x}}^\beta[X, \hat{x}_\mu]$ ($|\beta| \le 1$) have the degrees $1 - |\alpha| - |\beta|$, $-|\alpha| - |\beta|$ and $1 - |\beta|$; by (56.1), for all real a, b,

$$\begin{aligned} \|\langle\partial\rangle^{-a}\mathsf{DV}_1(p)\langle\partial\rangle^{-b}\| &\le C \quad (a + b = 1 - |\alpha| - |\beta|), \\ \|\langle\partial\rangle^{-a}\mathsf{DV}_2(p)\langle\partial\rangle^{-b}\| &\le C \quad (a + b = -|\alpha| - |\beta|), \\ \|\langle\partial\rangle^{-a}\,\mathrm{ad}_{\hat{x}}^\beta[X, \hat{x}_\mu]\langle\partial\rangle^{-b}\| &\le C \quad (a + b = 1 - |\beta|,\ |\beta| \le 1). \end{aligned} \tag{56.4}$$

In particular $\|\mathsf{V}_1\langle\partial\rangle^{-1}\| + \|\langle\partial\rangle^{-1}\mathsf{V}_1\| + \|\mathsf{V}_2\| + \|\partial_p\mathsf{V}_1\| + \|\langle\partial\rangle\partial_p^2\mathsf{V}_2\langle\partial\rangle\| + \|[X,\hat{x}]\langle\partial\rangle^{-1}\| \le C$.

Proof. Multipliers. S is invariant under the translations of $\eta\mathbb{Z}^4$, so D_U and $\mathcal{F}_U$ are covariant under them, and so is X. At $U=1$, $X(1) = -\Delta + m^2$ componentwise (Lemma 2.2, $\mathcal{F}_1 = 0$), whose symbol is $x(q)$; $[X,\hat{x}_\mu] = -[\hat{x}_\mu, X]$ has the symbol $-\mathrm{i}\partial_{q_\mu}x(q)$. For the translation τ_a by $a \in \eta\mathbb{Z}^4$, $\tau_a \delta X[A]\tau_a^{-1} = \delta X[A(\cdot - a)]$ and $\tau_a e^{-\mathrm{i}p\cdot\hat{x}}\tau_a^{-1} = e^{\mathrm{i}p\cdot a}e^{-\mathrm{i}p\cdot\hat{x}}$, while $A_{p,\varepsilon}(b-a) = e^{-\mathrm{i}p\cdot a}A_{p,\varepsilon}(b)$. The phases cancel, so $\mathsf{V}_1(p)$ commutes with the translations of $\eta\mathbb{Z}^4$; likewise $\mathsf{V}_2(p)$. The vertices have finite range, so they are Fourier multipliers. Details

Scaling. By the proof of Lemma 57.2, after the positions are rescaled by η^{-1}, $X(U)$ is η^{-2} times the operator $X_{\mathrm{V}}(U)$ or $X_{\mathrm{S}}(U)$ of Section 57.2 on $\mathbb{Z}^4$, with $\mathrm{m} = m\eta$, and $e^{\mathrm{i}\eta t A_{p,\varepsilon}} = e^{\mathrm{i}t\eta A'}$, where A' is the field $A_{\eta p,\varepsilon}$ on the bonds of $\mathbb{Z}^4$. A fine plane wave of momentum q is a plane wave of momentum ηq on $\mathbb{Z}^4$. Hence

$$v_1(q;p,\varepsilon) = \eta^{-1}w_1(\eta q;\eta p,\varepsilon), \qquad v_2(q;p,\varepsilon) = w_2(\eta q;\eta p,\varepsilon),$$

where w_1, w_2 are the vertices $v_1^{X_{\mathrm{V}}}$, $v_2^{X_{\mathrm{V}}}$ or $v_1^{X_{\mathrm{S}}}$, $v_2^{X_{\mathrm{S}}}$ of Section 57.2. These are trigonometric polynomials in $(Q,P) \in \mathbb{R}^4 \times \mathbb{R}^4$, linear, respectively sesquilinear, in ε, and they do not depend on η.

(a) *First-order gauge covariance.* Let ε be real, $\lambda(x) = \sum_\mu x_\mu\varepsilon_\mu$, and $u_s = e^{-\mathrm{i}s\lambda}$. Then $1^{u_s}(b) = e^{-\mathrm{i}s\lambda(b_-)}e^{\mathrm{i}s\lambda(b_+)} = e^{\mathrm{i}\eta A_s(b)}$, with $A_s(b)$ analytic in s and $A_s = sA_{0,\varepsilon} + O(s^2)$ on each bond, since $\lambda(b_+) - \lambda(b_-) = \eta\varepsilon_\mu$ for b of direction μ. The families D_U and $\mathcal{F}_U$ are covariant, $X(U^u) = R(u)X(U)R(u)^{-1}$, with $R(u)$ acting by $R(u(x(s)))$; for $\mathcal{F}_U$ this follows from $S(e^{\mathrm{i}\eta h}U^u) = S(e^{\mathrm{i}\eta R(u)^{-1}h}U)$, as in the proof of Lemma 55.6(a). Every entry of $X(e^{\mathrm{i}\eta A})$ depends on finitely many values of A. So, differentiating $X(1^{u_s}) = R(u_s)X(1)R(u_s)^{-1}$ entrywise at $s=0$,

$$\mathsf{V}_1(0) = \delta X[A_{0,\varepsilon}] = -\mathrm{i}\big[\operatorname{ad}\lambda(\hat{x}), X(1)\big] = -\mathrm{i}\sum_\mu \operatorname{ad}\varepsilon_\mu\,[\hat{x}_\mu, X(1)],$$

since $X(1)$ acts as the identity on $\mathfrak{g}^c$. Its symbol is $\sum_\mu \operatorname{ad}\varepsilon_\mu\,\partial_{q_\mu}x(q)$, which is (a). By linearity (a) extends to complex ε. This is consistent with Lemma 57.4.

(b) For $|\alpha|+|\beta| \ge 1$, $\partial_p^\alpha\partial_q^\beta v_1(q;p,\varepsilon) = \eta^{|\alpha|+|\beta|-1}(\partial_P^\alpha\partial_Q^\beta w_1)(\eta q;\eta p,\varepsilon)$, which is bounded by $C\eta^{|\alpha|+|\beta|-1}$, since the derivatives of a trigonometric polynomial are bounded on $\mathbb{R}^8$. As $\eta \le (1+2\sqrt{d})\langle q\rangle^{-1}$, this is at most $C\langle q\rangle^{1-|\alpha|-|\beta|}$. For $\alpha=\beta=0$, by (a), $|v_1(q;0,\varepsilon)| \le \sum_\mu |2\eta^{-1}\sin(\eta q_\mu)|\,|\operatorname{ad}\varepsilon_\mu| \le C|\hat{q}|$, since $|2\sin(2\theta)| \le 4|\sin\theta|$, and $|v_1(q;p,\varepsilon) - v_1(q;0,\varepsilon)| \le |p|\sup|\partial_p v_1| \le C\pi$; so $|v_1| \le C\langle q\rangle$. In the same way $|\partial_p^\alpha\partial_q^\beta v_2| \le C\eta^{|\alpha|+|\beta|} \le C\langle q\rangle^{-|\alpha|-|\beta|}$. Finally $\partial_{q_\mu}x(q) = 2\eta^{-1}\sin(\eta q_\mu)$, bounded by $2|\hat{q}_\mu|$, and $\partial_{q_\mu}\partial_{q_\nu}x(q) = 2\cos(\eta q_\mu)\delta_{\mu\nu}$. The operator bounds follow from (56.1), since p-derivatives and commutators with $\hat{x}$ of multipliers are the multipliers with the derivatives of the symbols (with factors i). □

At $U=1$ the vertex $\mathsf{V}_1(0)$ is thus a commutator with a linear function. There is no term of order η^{-1} without a derivative, for instance from the lattice artefacts of $\mathcal{F}_U$; this is the content of the degree-sharpness of V_1.

Lemma 56.5 (Integrability)**.** $\|\langle\partial\rangle^{-1}\|_r$ *is bounded uniformly in* η *if and only if* $r > 4$.

Proof. By Lemma 54.13(a), $\mathrm{Tr}_1\langle\partial\rangle^{-r}$ is the sum over the cell of the diagonal entries. The cell contains η^{-4} fine points, and the diagonal entry of a multiplier with symbol $\langle q\rangle^{-r}$ is $\eta^4\int\langle q\rangle^{-r}d^4q/(2\pi)^4$, the integral over the fine Brillouin zone $[-\pi/\eta,\pi/\eta]^4$. So $\mathrm{Tr}_1\langle\partial\rangle^{-r}$ is the number of components times $\int(1+|\hat{q}|^2)^{-r/2}\,d^4q/(2\pi)^4$. Here $|\hat{q}| \ge \frac{2}{\pi}|q|$, and $|\hat{q}| \le |q|$. The integral is bounded uniformly in η if and only if $r > 4$. □ Details

Lemma 56.6 (Power counting)**.** *Let* $F_1,\dots,F_m \in \mathscr{B}$ *be operators between field spaces, with* $F_1\cdots F_m$ *defined. Let* $\langle\partial\rangle$ *act on the fine fields as above and as the identity on the coarse*

fields. Suppose that $F_j = \langle\partial\rangle^{a_j} B_j \langle\partial\rangle^{b_j}$, *with* $B_j \in \mathscr{B}$, $\|B_j\|_\infty \le C_0$, *and integers* $a_j, b_j \le 1$, *where* $a_j = 0$ *if the target of* F_j *consists of coarse fields and* $b_j = 0$ *if its source does; and that* $b_j + a_{j+1} \le 0$ *for* $1 \le j < m$.

(a) Then $\|\langle\partial\rangle^{-a_1} F_1 \cdots F_m \langle\partial\rangle^{-b_m}\|_\infty \le C_0^m$.

(b) Let $F_1 \cdots F_m$ *map a space of fine fields to itself, let* $b_m + a_1 \le 0$, *and put* $D = -\sum_{j=1}^m (b_j + a_{j+1})$, *with* $a_{m+1} = a_1$. *If* $D > 4$, *then* $|\operatorname{Tr}_1(F_1 \cdots F_m)| \le C_0^m \max \operatorname{Tr}_1 \langle\partial\rangle^{-D} \le K_D C_0^m$, *the maximum over the fine field spaces involved, with* K_D *independent of* η.

(c) Let $F_1 \cdots F_m$ *map a field space to itself, let* $b_m + a_1 \le 0$, *and let some* B_j *have rank at most* r *per cell. Then* $|\operatorname{Tr}_1(F_1 \cdots F_m)| \le r C_0^m$.

Proof. At fixed η all operators involved are bounded, and the powers of $\langle\partial\rangle$ lie in $\mathscr{P} \subset \mathscr{B}$ (Lemma 54.12(f)). (a) We have Details

$$F_1 \cdots F_m = \langle\partial\rangle^{a_1} B_1 \langle\partial\rangle^{b_1 + a_2} B_2 \cdots \langle\partial\rangle^{b_{m-1} + a_m} B_m \langle\partial\rangle^{b_m},$$

and $\|\langle\partial\rangle^e\|_\infty \le 1$ for $e \le 0$.

(b) By the cyclicity of Tr_1 (Lemma 54.13(a)),

$$\operatorname{Tr}_1(F_1 \cdots F_m) = \operatorname{Tr}_1 \left(B_1 \langle\partial\rangle^{-e_1} B_2 \cdots B_m \langle\partial\rangle^{-e_m}\right),$$

with $e_j = -(b_j + a_{j+1}) \ge 0$ and $\sum_j e_j = D$. Apply Hölder's inequality (Lemma 54.13(b)), with the exponent ∞ for the B_j and D/e_j for the factors with $e_j > 0$. Since $\|\langle\partial\rangle^{-e_j}\|_{D/e_j} = (\operatorname{Tr}_1 \langle\partial\rangle^{-D})^{e_j/D}$, the product of these norms is at most $\max \operatorname{Tr}_1 \langle\partial\rangle^{-D}$, which is bounded uniformly in η by Lemma 56.5, since $D > 4$.

(c) By cyclicity, $\operatorname{Tr}_1(F_1 \cdots F_m) = \operatorname{Tr}_1(B_j R)$, with $R = \langle\partial\rangle^{b_j} F_{j+1} \cdots F_m F_1 \cdots F_{j-1} \langle\partial\rangle^{a_j}$. The exponents of $\langle\partial\rangle$ at the ends of R are $b_j + a_{j+1} \le 0$ and $b_{j-1} + a_j \le 0$, so $\|R\|_\infty \le C_0^{m-1}$, as in (a). By Hölder's inequality and the rank bound (Lemma 54.13(b),(c)), $|\operatorname{Tr}_1(B_j R)| \le \|B_j\|_1 \|R\|_\infty \le r \|B_j\|_\infty \|R\|_\infty$. □

Lemma 56.7 (Properties of the remainders)**.** *Let* $|\varepsilon| \le 1$, *put* $\mathsf{V}_0^X = X(1)$, *and let* $\mathsf{D} = \partial_p^\alpha \operatorname{ad}_{\hat{x}}^\beta$ *with* $|\alpha| + |\beta| \le 2$, *the* p*-derivatives taken at* $p = 0$.

(a) For $X \in \{\mathbf{Y}, \mathbf{Y}'\}$ *and* $i = 0, 1, 2$, *the operators* DV_i^X *lie in* $\mathscr{P}$, *have rank at most* C *per cell, i.e. fibres of rank at most* C, *and norm at most* C.

(b) For $X \in \{\mathbf{Q}_k, Q_k', \mathbf{Q}_k^T, Q_k'^T, \mathbf{Q}_k \mathbf{G}_* \mathbf{Q}_k^T, M'\}$ *and* $i = 0, 1, 2$, *the operators* DV_i^X *lie in* $\mathscr{P}$ *and have norm at most* C. *Moreover* $\|(\mathbf{Q}\mathbf{G}\mathbf{Q}^*)^{-1}\| + \|M^{-1}\| \le C$.

The constants do not depend on k.

Proof. Membership in $\mathscr{P}$. The values at $U = 1$ and the vertices of admissible and elementary families lie in $\mathscr{P}$, and $\mathscr{P}$ is closed under sums, products, $O \mapsto O^{(p)}$ and $[\hat{x}_\mu, \cdot\,]$ (Lemmas 54.10(a) and 54.13(d)); the p-derivatives of the vertices have kernels of the same form, with polynomial factors. This uses the exponential decay at fixed η; the bounds below use only the algebra and Part I. Details

The factors. Write $\mathbf{Y} = -(D_U G' Q'^T) M'^{-1} (Q' G' D_U^T) + a \mathbf{Q}_k^T \mathbf{Q}_k$, with D_U on scalar fields, and $\mathbf{Y}' = a Q'^T Q'$. We use the following factors at $U = 1$ and $p = 0$, with $\mathsf{D}' = \partial_p^{\alpha'} \operatorname{ad}_{\hat{x}}^{\beta'}$, $|\alpha'| + |\beta'| \le 2$.

(i) *Propagators*: G' and $\mathbf{G}$. By Lemma 54.12(g), $\|\langle\partial\rangle G' \langle\partial\rangle\| \le c_0^{-1}$ and $\|\langle\partial\rangle \mathbf{G} \langle\partial\rangle\| \le c_g^{-1}$.

(ii) *Multipliers*: D and D^T on scalar fields, of degree 1, since the symbol of D_μ is $\eta^{-1}(e^{\mathrm{i}\eta q_\mu} - 1)$, of modulus $|\hat{q}_\mu|$. Their operators $\mathsf{D}'\mathsf{V}_i$ with $i + |\alpha'| + |\beta'| \ge 1$ have degree at most 0: $\delta D_U[A]\lambda(b) = \mathrm{i} \operatorname{ad} A(b)\, \lambda(b_+)$ and $\delta^2 D_U[A, A']\lambda(b) = -\frac{\eta}{2}\{\operatorname{ad} A(b), \operatorname{ad} A'(b)\}\lambda(b_+)$, so V_1^D

and V_2^D are bounded multipliers that do not depend on p, and the commutators with $\hat{x}$ differentiate the symbols, which gives bounded multipliers. Further, the operators $\mathsf{D}'\mathsf{V}_i^Y$ and $\mathsf{D}'[Y,\hat{x}_\mu]$ of $Y \in \{\mathfrak{A}'_{\mathrm{F}}, \mathbf{\Delta}_{\mathrm{F}}\}$, of degree at most 1 by Lemma 56.4.

(iii) *Coarse factors*: the operators $\mathsf{D}'\mathsf{V}_i$, $i = 0,1,2$, of $\mathbf{Q}_k$, Q'_k and their transposes, and M^{-1}. By Lemma 56.1, their kernels have the form of Lemma 56.1(c), with C' independent of k: the p-derivatives and the commutators with $\hat{x}$ multiply the kernels by bounded factors such as $(b'_- - c_-)^{\alpha'}$ and $(c_- - b_-)^{\beta'}$, with b, b' in S_c, and the vertices are the derivatives of the kernels in the directions $e^{\mathrm{i}p\cdot(b'_- - c_-)}\varepsilon \mathbf{1}_{S_c}$, of sup norm at most 1. So they are bounded by C. Moreover $M \geq c_{\mathrm{QGQ}}^2$ (Lemma 54.12(g)).

(a) By the Leibniz rule for the vertices, $\partial_p(O^{(p)}) = -\mathrm{i}[\hat{x}, O^{(p)}]$, the product rule for $[\hat{x}_\mu, \cdot\,]$ and $[\hat{x}_\mu, O^{-1}] = O^{-1}[O, \hat{x}_\mu]O^{-1}$ (Lemma 54.13(d)), each operator $\mathsf{D}\mathsf{V}_i^{\mathbf{Y}'}$ is a finite sum of coarse factors and their products, and each operator $\mathsf{D}\mathsf{V}_i^{\mathbf{Y}}$ is a finite sum, with a number of terms independent of k, of such products and of products

$$F_0\, G'\, I_1\, G'\, I_2 \cdots I_{l-1}\, G'\, F_l.$$

Here every I_j is a single multiplier or a product of coarse factors, possibly empty, and F_0, F_l are multipliers derived from D, respectively D^T. Indeed, the operations act on the factors by the Leibniz rule. On a propagator G' they produce $-G'\mathsf{V}_i^{\mathfrak{A}'}G'$, $G'[\mathfrak{A}', \hat{x}_\mu]G'$, or sums of products $G'IG'IG'$ and $G'IG'$, with $\mathfrak{A}' = \mathfrak{A}'_{\mathrm{F}} + \mathbf{Y}' - m^2$, whose vertices and commutators are multipliers or products of coarse factors. On M'^{-1} they produce $-M^{-1}(\mathsf{D}'\mathsf{V}_i^{M'})M^{-1}$ and products of such terms, where $M' = Q'G'G'Q'^T$ expands in the same way, $G'G'$ being two propagators with an empty insertion. On the other factors they produce factors of the same kind. Give each G' the exponents $(-1,-1)$; a multiplier of degree 1 the exponent 1 on the side of a neighbouring propagator and 0 on the other side; and all other factors the exponents $(0,0)$. Every multiplier of degree 1 has a propagator as a neighbour, and no two of them are adjacent. So all gaps $b_j + a_{j+1}$ are at most 0, and the exponents at the ends are 0. Lemma 56.6(a) bounds each product by C. Every term contains a factor that maps into the coarse fields, namely one of the operators $\mathsf{D}'\mathsf{V}_i$ of Q' or of $\mathbf{Q}_k$, because the operations do not remove these factors. So it factors through the coarse fields, whose cell carries $\dim \mathfrak{g}$, respectively $d \dim \mathfrak{g}$, dimensions, and Lemma 54.13(c) gives the rank bound.

(b) For $\mathbf{Q}_k$, Q'_k and their transposes these are the coarse factors. For $\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T$ and $M' = Q'G'^2Q'^T$ the same expansion gives products $F_0PI_1P\cdots PF_l$ with propagators $P = \mathbf{G}$, respectively $P = G'$, insertions I_j that are single multipliers, coarse factors, or operators of (a) for $\mathbf{Y}$ and $\mathbf{Y}'$, which enter through $\mathbf{\Delta}_* = \mathbf{\Delta}_{\mathrm{F}} + \mathbf{Y} - m^2$ and $\mathfrak{A}' = \mathfrak{A}'_{\mathrm{F}} + \mathbf{Y}' - m^2$, and coarse factors F_0, F_l; for M' the factor G'^2 is two adjacent propagators. The same assignment of exponents and Lemma 56.6(a) bound them by C. Finally $\|(\mathbf{QGQ}^*)^{-1}\| \leq c_{\mathrm{QGQ}}^{-1}$ and $\|M^{-1}\| \leq c_{\mathrm{QGQ}}^{-2}$, by Lemma 54.12(g). □

So the multiscale structure of the averaging enters only through operators of finite rank per cell, and contributes nothing that grows with k.

56.3 The reduction

Proposition 56.8 (Reduction to the Feynman part)**.** *The p^2-tensor of the vacuum polarization of $\widehat{\Gamma}$ at $U = 1$ is $T_{\mathrm{F}} + O(1)$, uniformly in k: $T_{\mathrm{fine}} = T_{\mathrm{F}} + O(1)$.*

Proof. In the bond-midpoint convention the vacuum polarizations of $\widehat{\Gamma}$ and of $\widehat{\Gamma}_{\mathrm{F}}$ vanish at $p = 0$, since their kernels annihilate the constant fields (Lemma 56.2(b)), and they have no term linear in p: for $\widehat{\Gamma}$ by the paragraph after Lemma 56.3, and for $\widehat{\Gamma}_{\mathrm{F}}$ by Remark 56.9 Details

below. So T_{fine} and T_{F} are their terms of order 2 in p also in the convention of $A_{p,\varepsilon}$, in which Lemma 56.2(b) gives them. Hence $T_{\text{fine}} - T_{\text{F}}$ is the term of order 2 in p of

$$-\tfrac{1}{2}(S_{\boldsymbol{\Delta}_*} - S_{\boldsymbol{\Delta}_{\text{F}}}) + (S_{\mathfrak{A}'} - S_{\mathfrak{A}'_{\text{F}}}) - \tfrac{1}{2} S_{\mathbf{Q}_k \mathbf{G}_* \mathbf{Q}_k^T} + \tfrac{1}{2} S_{M'},$$

and it suffices to bound the second derivatives of these four terms in p at $p = 0$, for $|\varepsilon| \le 1$, uniformly in k. At fixed k the fibres are finite-dimensional, so Tr_1 is continuous in the norm of operators, and all functions below are smooth in p and t; the derivatives may be taken under Tr_1 and under the integral over t. Only the values at $p = 0$ enter, where $O^{(0)} = O$; so no bound on the conjugates $O^{(p)}$ for $p \neq 0$ is needed.

The differences of determinants. For $t \in [0,1]$ let

$$X_t = (1-t)\boldsymbol{\Delta}_{\text{F}} + t\boldsymbol{\Delta}_* = \boldsymbol{\Delta}_{\text{F}} + t(\mathbf{Y} - m^2),$$

with the vertices $\mathsf{V}_i^{\boldsymbol{\Delta}_{\text{F}}} + t\mathsf{V}_i^{\mathbf{Y}}$. At $U = 1$, $X_0 = \langle\partial\rangle^2$, since $m = 1$, and $X_1 = \boldsymbol{\Delta}_a$. So $X_t \ge c_1\langle\partial\rangle^2$, with $c_1 = \min(1, c_g)$ (Lemma 54.12(g)). Hence $X_t^{-1} \in \mathscr{P}$ (Lemma 54.12(d)), $\|\langle\partial\rangle X_t^{-1}\langle\partial\rangle\| \le c_1^{-1}$, and we let S_{X_t} be the right side of (56.3), with $X_t^{-1} \in \mathscr{P}$ and the vertices $\mathsf{V}_i^{\boldsymbol{\Delta}_{\text{F}}} + t\mathsf{V}_i^{\mathbf{Y}}$; it is an expression in infinite volume and is not identified with a limit of tori. Since $\partial_t X_t^{-1} = -X_t^{-1}(\mathbf{Y} - m^2)X_t^{-1}$ and $\partial_t \mathsf{V}_i^{X_t} = \mathsf{V}_i^{\mathbf{Y}}$,

$$S_{\boldsymbol{\Delta}_*} - S_{\boldsymbol{\Delta}_{\text{F}}} = \int_0^1 \partial_t S_{X_t}\, dt,$$

where $\partial_t S_{X_t}(p)$ is obtained from (56.3) for X_t by replacing, in all possible ways, one factor X_t^{-1} or $(X_t^{-1})^{(p)}$ by $-X_t^{-1}(\mathbf{Y} - m^2)X_t^{-1}$ or its conjugate, or one vertex by the corresponding vertex of $\mathbf{Y}$. Differentiate twice in p at $p = 0$, with $\partial_p(O^{(p)}) = -\mathrm{i}[\hat{x}, O^{(p)}]$, $[\hat{x}_\mu, X_t^{-1}] = X_t^{-1}[X_t, \hat{x}_\mu]X_t^{-1}$ and $[X_t, \hat{x}_\mu] = [\boldsymbol{\Delta}_{\text{F}}, \hat{x}_\mu] + t[\mathbf{Y}, \hat{x}_\mu]$. Every resulting term is a numerical multiple, bounded independently of k, of Tr_1 of a cyclic product $P_1 I_1 P_2 I_2 \cdots P_l I_l$ with $P_j = X_t^{-1}$, in which every *insertion* I_j is one of

(i) m^2, or a multiplier $\mathsf{DV}_i^{\boldsymbol{\Delta}_{\text{F}}}$ $(i = 1, 2)$ or $\mathsf{D}[\boldsymbol{\Delta}_{\text{F}}, \hat{x}_\mu]$, with D as in Lemma 56.4;

(ii) an operator $\mathsf{DV}_i^{\mathbf{Y}}$, $i = 0, 1, 2$, of Lemma 56.7(a);

and at least one I_j is m^2 or of kind (ii). The number of terms does not depend on k. Give each P_j the exponents $(-1, -1)$, m^2 and the operators of kind (ii) the exponents $(0, 0)$, and a multiplier of degree $s = 1, 0, -1, -2$ (Lemma 56.4) the exponents $(1, 0)$, $(0, 0)$, $(0, -1)$, $(-1, -1)$, respectively. All exponents are at most 1, and every gap is at most 0. Each insertion of degree s_j contributes $(1 - a_j) + (1 - b_j) = 2 - s_j$ to D in Lemma 56.6(b): $D = \sum_j (2 - s_j)$.

Terms with an insertion of kind (ii). By Lemma 56.7(a) it has rank at most C per cell and norm at most C. Lemma 56.6(c) bounds the term by C.

The other terms. Here exactly one insertion is m^2, produced by ∂_t, and all other insertions are multipliers of kind (i). The second variation (56.3) has $D = 2$: the tadpole has one insertion V_2 of degree 0, and the bubble two insertions V_1 of degree 1. The derivative ∂_t adds a propagator and the insertion m^2, so $D = 4$. Each p-derivative either lowers the degree of an insertion by one, by Lemma 56.4, or replaces a propagator P by $P[\boldsymbol{\Delta}_{\text{F}}, \hat{x}_\mu]P$, which adds an insertion of degree 1; a second commutator with the same propagator gives $P[\boldsymbol{\Delta}_{\text{F}}, \hat{x}]P[\boldsymbol{\Delta}_{\text{F}}, \hat{x}]P$ or $P[\hat{x}_\nu, [\boldsymbol{\Delta}_{\text{F}}, \hat{x}_\mu]]P$, with insertions of degree 1 and 0. Either way D grows by at least one per derivative. So $D \ge 6$, and Lemma 56.6(b) bounds the term by C. Five factors would suffice: if the second p-derivatives and commutators of V_1 and the first of V_2 were only bounded, and the second of V_2 only of degree -1, every term would still have $D \ge 5 > 4$.

The scalar pair is treated in the same way, with $X'_t = \mathfrak{A}'_{\text{F}} + t(\mathbf{Y}' - m^2)$, $X'_0 = \langle\partial\rangle^2$, $X'_1 = -\Delta + aQ_0^* Q_0 \ge c_0\langle\partial\rangle^2$, the vertices of $\mathfrak{A}'_{\text{F}}$ (Lemma 56.4) and Lemma 56.7(a) for $\mathbf{Y}'$.

The coarse determinants. Let $X \in \{\mathbf{Q}_k \mathbf{G}_* \mathbf{Q}_k^T, M'\}$. At $U = 1$, X^{-1} is $(\mathbf{Q}\mathbf{G}\mathbf{Q}^*)^{-1}$ or M^{-1}, which lie in $\mathscr{P}$. The second derivative of S_X at $p = 0$ is a finite sum of numerical multiples of Tr_1 of products of X^{-1}, of $[\hat{x}_\mu, X^{-1}] = X^{-1}[X, \hat{x}_\mu]X^{-1}$ and $[\hat{x}_\nu, [\hat{x}_\mu, X^{-1}]]$, and of the operators DV_i^X, $i = 0, 1, 2$. By Lemma 56.7(b) all of them are bounded by C. They act on the coarse fields, whose cell carries at most $d \dim \mathfrak{g}$ dimensions, so $|\mathrm{Tr}_1 O| \le d \dim \mathfrak{g} \, \|O\|$ (Lemma 54.13(c)). Hence these terms are $O(1)$. □

Remark 56.9 (Symmetry). $\widehat{\Gamma}_{\mathrm{F}}$ is built from the Wilson action and the covariant derivative of scalar fields only: $\mathbf{\Delta}_{\mathrm{F}} = q_2^0 + D_U D_U^T + m^2$ and $\mathfrak{A}'_{\mathrm{F}} = D_U^T D_U + m^2$, with D_U on scalar fields. The argument of Proposition 31.7(c)(i),(iii) for these two objects uses only $S \circ r = S$ and the definition of D_U, so it applies to every lattice symmetry of the fine lattice. Hence $\widehat{\Gamma}_{\mathrm{F}}$ is invariant under the hypercubic symmetries of the fine lattice, and, by the tensor algebra in the proof of Lemma 54.9(c), with the transversality of the kernels given by Lemma 56.2(b), $T_{\mathrm{F}}(p) = \beta_{\mathrm{F}}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, with no term linear in p. The full function $\widehat{\Gamma}$ is invariant under $\mathfrak{S}_k$ on the real regular backgrounds near $U = 1$, by the paragraph after Lemma 56.3. By this invariance and the same tensor algebra, the p^2-tensor of $\widehat{\Gamma}$ is isotropic as well, and so is the remainder $O(1)$ of Proposition 56.8, the difference of two isotropic tensors. This agrees with Lemma 54.9, which gives the isotropy of $T_{\Gamma_k^{(1)}}$ directly.

57 The lattice Feynman-gauge coefficient

This section computes the isotropic coefficient β_{F} of the tensor T_{F} of Section 56.2. It is the p^2-coefficient of the vacuum polarization, at $U = 1$, of

$$\widehat{\Gamma}_{\mathrm{F}}(U) = -\tfrac{1}{2} \log\det\left(D_U^T D_U + D_U D_U^T + \mathcal{F}_U + m^2\right) + \log\det\left(D_U^T D_U + m^2\right)$$

on the fine lattice of spacing $\eta = L^{-k}$, with $m = 1$. Let G be compact, connected and simple, with the matrix trace of a representation ρ in the action, $\mathrm{tr}_\rho(T^a T^b) = T_\rho \delta^{ab}$ for Hermitian generators T^a, and C_A the Casimir of the adjoint representation, $\sum_a \mathrm{ad}\, T^a \,\mathrm{ad}\, T^a = C_A$ on $\mathfrak{g}$. Traces without index are tr_ρ, as in Section 23. Put $\kappa_\rho = C_A / T_\rho$. Then

$$\mathrm{tr}_{\mathfrak{g}^c}\left(\mathrm{ad}\,\xi \,\mathrm{ad}\,\zeta\right) = \kappa_\rho \,\mathrm{tr}(\xi\zeta), \qquad \xi, \zeta \in \mathfrak{g}^c. \tag{57.1}$$

Indeed, on $\mathfrak{g}$ both sides are Ad-invariant symmetric bilinear forms, so they are proportional, since $\mathfrak{g}$ is simple. The factor is κ_ρ, since $\sum_a \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}\, T^a \,\mathrm{ad}\, T^a) = C_A \dim \mathfrak{g}$ and $\sum_a \mathrm{tr}(T^a T^a) = T_\rho \dim \mathfrak{g}$. Both sides are complex bilinear, so the identity extends from $\mathfrak{g}$ to $\mathfrak{g}^c$.

Theorem 57.1 (The logarithm)**.** *There is a constant $c_{\log}$, depending only on G and ρ, such that, as $k \to \infty$,*

$$\beta_{\mathrm{F}} = \frac{11 C_A}{24\pi^2 T_\rho} \log\frac{1}{\eta} + c_{\log} + o(1) = k\,\beta_\infty + c_{\log} + o(1), \qquad \beta_\infty = \frac{11 C_A}{24\pi^2 T_\rho} \log L.$$

In particular $\beta_{\mathrm{F}} = k\beta_\infty + O(1)$, uniformly in $k \ge 1$. For $SU(N)$ in the fundamental representation, $\beta_\infty = \frac{11N}{12\pi^2} \log L$.

Normalization of the coupling. The density of the note is $\exp(-g^{-2}S)$, and the continuum limit of the Wilson action is $\frac{T_\rho}{4} \sum_{\mu\nu} F^a F^a$ (Remark 57.8). So g is related to the coupling g_{std} of the continuum action $\frac{1}{4g_{\mathrm{std}}^2} \sum_{\mu\nu} F^a F^a$ by $g_{\mathrm{std}}^2 = g^2 / T_\rho$; for $SU(N)$ in the fundamental representation, $T_\rho = \frac{1}{2}$ and $g_{\mathrm{std}}^2 = 2g^2$. In the continuum normalization the coefficient of Remark 57.8 changes g_{std}^{-2} by $\frac{11 C_A}{24\pi^2} \log L = 2b_0 \log L$ per factor L of the cutoff, with the usual one-loop coefficient $b_0 = \frac{11 C_A}{48\pi^2}$. Hence $\beta_\infty = 2b_0 \log L / T_\rho$: per step, g^{-2} changes by $1/T_\rho$ times the change of g_{std}^{-2}. In $\frac{11}{3} = 4 - \frac{1}{3}$ the 4 is the spin part of the vector determinant and $-\frac{1}{3}$ the orbital part of both determinants (proof of Lemma 57.7).

57.1 Reduction to the unit lattice

Lemma 57.2 (Scaling). *$\beta_{\mathrm{F}} = f(m\eta)$, where $f(\mathrm{m})$ is the isotropic p^2-coefficient, at $U = 1$, of*

$$\widehat{\Gamma}^{(\mathrm{m})}_{\mathrm{F}}(U) = -\tfrac{1}{2}\log\det\left(D_U^T D_U + D_U D_U^T + \mathcal{F}_U + \mathrm{m}^2\right) + \log\det\left(D_U^T D_U + \mathrm{m}^2\right)$$

on the lattice $\mathbb{Z}^4$ of unit spacing.

Proof. Rescale positions by η^{-1}. A fine configuration $e^{\mathrm{i}\eta A}$ becomes a configuration $e^{\mathrm{i}A'}$ on $\mathbb{Z}^4$ with $A' = \eta A$. The difference operators scale by η^{-1}, so the operators in $\widehat{\Gamma}_{\mathrm{F}}$ become η^{-2} times their unit-lattice versions, with m^2 replaced by $(m\eta)^2$. The factor η^{-2} does not change the second variations of $\log\det$: on a torus $\log\det(cX) - \log\det X$ does not depend on U, and in infinite volume $S_{cX} = S_X$ in (56.3). After the rescaling, a unit cell contains η^{-4} sites of $\mathbb{Z}^4$, and the field $A_{p,\varepsilon}$ becomes $A' = \eta A_{p,\varepsilon}$, which is the field $A_{\eta p,\eta\varepsilon}$ on the bonds of $\mathbb{Z}^4$. The part of second order in the momentum of the second variation per site at the momentum ηp and the amplitude $\eta\varepsilon$ is $\eta^2\cdot\eta^2$ times its value at p and ε. With the factor η^{-4} for the sites of a cell, the p^2-coefficient per unit cell is that of the unit lattice with the mass $m\eta$. The same computation applies to the action. In $d = 4$, with the pairing $\eta^d\sum\mathrm{tr}(AB)$ of Section 23, the action does not change under the rescaling, and its marginal tensor is $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$ on both lattices. So the normalization relative to S does not change either. □

Details

So Theorem 57.1 is the statement

$$f(\mathrm{m}) = \frac{11C_A}{24\pi^2 T_\rho}\log\frac{1}{\mathrm{m}} + c + o(1) \qquad (\mathrm{m}\to 0), \tag{57.2}$$

with $c_{\log} = c$, since $m\eta = L^{-k}$. It is proved at the end of this section.

57.2 The lattice integral

On $\mathbb{Z}^4$ we use the notation of Sections 12 and 23 with $\eta = 1$. A configuration is $U = e^{\mathrm{i}A}$, with a $\mathfrak{g}^c$-valued gauge field A; it is real if A is. The operators

$$X_{\mathrm{V}}(U) = D_U^T D_U + D_U D_U^T + \mathcal{F}_U + \mathrm{m}^2, \qquad X_{\mathrm{S}}(U) = D_U^T D_U + \mathrm{m}^2$$

act on $\mathfrak{g}^c$-valued gauge fields and scalar fields, respectively, with D_U as in Section 56.2. Here $D_U^T D_U + \mathcal{F}_U = \mathrm{Hess}_U$ is the Hessian of the Wilson action in the chart $e^{\mathrm{i}h}U$: the operator of the bilinear form $\frac{d^2}{ds^2}S(e^{\mathrm{i}sh}U)|_{s=0}$ in the pairing b of Section 12. For real U, $\mathcal{F}_U = d^2\mathcal{N}_U(0)$ is the operator of Section 56.2, by Proposition 23.3. The matrix entries of D_U, D_U^T and Hess_U are polynomials in the matrix entries of $U(b)^{\pm 1}$ for the bonds b at one site, respectively of one plaquette; for Hess_U this is (57.5) below. So the matrix entries of $X_{\mathrm{V}}(e^{\mathrm{i}A})$ and $X_{\mathrm{S}}(e^{\mathrm{i}A})$ are entire functions of finitely many values of A, and they vanish between sites at distance more than 2. At $A = 0$, $X_{\mathrm{V}}(1) = X_{\mathrm{S}}(1) = -\Delta + \mathrm{m}^2$ componentwise (Lemma 2.2; $\mathcal{F}_1 = 0$). Background derivatives at $U = 1$ are derivatives of $A \mapsto X(e^{\mathrm{i}A})$ at $A = 0$; on real directions they are those of the real configurations, and they extend complex-linearly.

Symbols. An operator Y *transfers the momentum* p if it maps plane waves, $Y(e^{\mathrm{i}q\cdot x}\phi) = e^{\mathrm{i}(q+p)\cdot x}y(q)\phi$, with a matrix $y(q)$, its symbol. Translation-invariant operators of finite range transfer the momentum 0, and their symbols are trigonometric polynomials in q. If Y and Y' transfer p and p', with symbols y and y', then $Y'Y$ transfers $p + p'$, with the symbol $y'(q + p)y(q)$. The symbol of $-\Delta + \mathrm{m}^2$ is $|\hat{q}|^2 + \mathrm{m}^2$, with $|\hat{q}|^2 = \sum_\nu 4\sin^2(q_\nu/2)$. For $\mathrm{m} > 0$ we put $G_{\mathrm{m}}(q) = (|\hat{q}|^2 + \mathrm{m}^2)^{-1}$, the symbol of $X(1)^{-1}$. We write Tr for the trace of a linear map of the fibre, $(\mathfrak{g}^c)^4$ or $\mathfrak{g}^c$, to distinguish it from $\mathrm{tr} = \mathrm{tr}_\rho$; this is the trace tr of the fibres in Lemma 54.13.

Vertices. For $p \in \mathbb{R}^4$ and $\varepsilon \in (\mathfrak{g}^c)^4$ let $A_{p,\varepsilon}(b) = e^{\mathrm{i}p\cdot b_-}\varepsilon_\mu$ on the bonds b of direction μ, as in Section 56, and $\bar\varepsilon = (\varepsilon_\mu^*)$. Since $\mathfrak{g}$ consists of Hermitian matrices, $A_{p,\varepsilon} + A_{-p,\bar\varepsilon}$ is real. For $X = X_{\mathrm{V}}, X_{\mathrm{S}}$ the background derivatives at $U = 1$ act by

$$\delta X[A_{p,\varepsilon}]\, e^{\mathrm{i}q\cdot x}\phi = e^{\mathrm{i}(q+p)\cdot x}\, v_1^X(q;p,\varepsilon)\phi, \qquad \delta^2 X[A_{-p,\bar\varepsilon}, A_{p,\varepsilon}]\, e^{\mathrm{i}q\cdot x}\phi = e^{\mathrm{i}q\cdot x}\, v_2^X(q;p,\varepsilon)\phi.$$

The *vertices* v_1^X and v_2^X are trigonometric polynomials in (q,p), i.e. finite sums $\sum_j c_j e^{\mathrm{i}(z_j\cdot q + z_j'\cdot p)}$ with $z_j, z_j' \in \mathbb{Z}^4$ and matrices c_j; v_1^X is linear and v_2^X is sesquilinear in ε. Indeed, the matrix entry of $\delta X[A_{p,\varepsilon}]$ between two sites is a finite sum of derivatives at $A = 0$ of the local entire functions above, times values $e^{\mathrm{i}p\cdot b_-}\varepsilon_\mu$ on nearby bonds; likewise for $\delta^2 X$. They extend to entire functions of complex (q,p).

The integrand. On $\mathbb{Z}^4$ the unit cell is a single site. By Fourier inversion, the trace per cell of a translation-invariant operator with symbol $y(q)$ is $\int_{[-\pi,\pi]^4} \operatorname{Tr} y(q)\, d^4q/(2\pi)^4$. In (56.3) both operators are translation invariant, with the symbols $G_{\mathrm{m}}(q)v_2^X(q;p,\varepsilon)$ and $G_{\mathrm{m}}(q)v_1^X(q+p;-p,\bar\varepsilon)G_{\mathrm{m}}(q+p)v_1^X(q;p,\varepsilon)$, by the composition rule. So the second variation of $\widehat\Gamma_{\mathrm{F}}^{(\mathrm{m})}$ at $U = 1$, per site, in the directions $A_{-p,\bar\varepsilon}$ and $A_{p,\varepsilon}$, is the integral over $q \in [-\pi,\pi]^4$, against $d^4q/(2\pi)^4$, of $P = -\frac12 P_{X_{\mathrm{V}}} + P_{X_{\mathrm{S}}}$, where

$$P_X(q,p,\mathrm{m}) = G_{\mathrm{m}}(q)\operatorname{Tr} v_2^X(q;p,\varepsilon) - G_{\mathrm{m}}(q)G_{\mathrm{m}}(q+p)\operatorname{Tr}\left(v_1^X(q+p;-p,\bar\varepsilon)\, v_1^X(q;p,\varepsilon)\right). \quad (57.3)$$

By Lemma 56.2(a),(b), rescaled as in Lemma 57.2, the integral is the second variation per site of the infinite-volume limit. On a torus of side n it is replaced by the average over $q \in (2\pi/n)\mathbb{Z}^4 \cap [-\pi,\pi)^4$, for p on the same grid, which converges to it exponentially fast in n at fixed $\mathrm{m} > 0$. Fix $\xi_0 \in \mathfrak{g}$ with $\operatorname{tr}\xi_0^2 = 1$, and take $\varepsilon = c \otimes \xi_0$, i.e. $\varepsilon_\mu = c_\mu\xi_0$, with $c \in \mathbb{C}^4$. Then P is sesquilinear in c, and we write $P = \sum_{\mu\nu}\bar c_\mu c_\nu P_{\mu\nu}(q,p,\mathrm{m})$. In the convention of Definition 54.7, which suppresses the Lie algebra indices relative to the trace pairing, the vacuum polarization is

$$\Pi^{(\mathrm{m})}(p)_{\mu\nu} = \int_{[-\pi,\pi]^4} P_{\mu\nu}(q,p,\mathrm{m})\,\frac{d^4q}{(2\pi)^4}.$$

For $\mathrm{m} > 0$, P is a real-analytic function of $(q,p) \in \mathbb{R}^4\times\mathbb{R}^4$, 2π-periodic in q, since $|\hat q|^2 + \mathrm{m}^2 \geq \mathrm{m}^2$. So $\Pi^{(\mathrm{m})}$ is smooth, and its Taylor coefficients at $p = 0$ are the integrals of those of P. Put

$$\mathcal{I}(q,\mathrm{m})(p)_{\mu\nu} = \sum_{\kappa\lambda}\mathcal{I}_{\mu\nu\kappa\lambda}(q,\mathrm{m})\,p_\kappa p_\lambda, \qquad \mathcal{I}_{\mu\nu\kappa\lambda}(q,\mathrm{m}) = \frac12\,\partial_{p_\kappa}\partial_{p_\lambda}P_{\mu\nu}(q,p,\mathrm{m})\big|_{p=0}. \quad (57.4)$$

Then the p^2-tensor of $\Pi^{(\mathrm{m})}$ is

$$T^{(\mathrm{m})} = \int_{[-\pi,\pi]^4}\mathcal{I}(q,\mathrm{m})\,\frac{d^4q}{(2\pi)^4}.$$

The function $\mathcal{I}$ is a rational function of the $e^{\mathrm{i}q_\nu}$ and of m^2, whose denominators are powers of $|\hat q|^2 + \mathrm{m}^2$. So it is defined and continuous also at $\mathrm{m} = 0$, for $q \notin 2\pi\mathbb{Z}^4$. For a tensor $T(p)_{\mu\nu} = \sum_{\kappa\lambda}T_{\mu\nu\kappa\lambda}p_\kappa p_\lambda$, symmetric in κ,λ, let $\operatorname{iso} T$ be its isotropic coefficient (Definition 54.7). It is linear in T, and $\operatorname{iso} T = \frac13(2t_1 - t_2)$ for $T(p)_{\mu\nu} = t_1\delta_{\mu\nu}|p|^2 + t_2 p_\mu p_\nu$. Hence

$$f(\mathrm{m}) = \operatorname{iso} T^{(\mathrm{m})} = \int_{[-\pi,\pi]^4}\operatorname{iso}\mathcal{I}(q,\mathrm{m})\,\frac{d^4q}{(2\pi)^4}.$$

By Remark 56.9, rescaled as in Lemma 57.2, $T^{(\mathrm{m})}(p) = f(\mathrm{m})(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. The proof of Theorem 57.1 does not use this; Lemma 57.7 gives the transversality of the leading term directly. On the lattice the integral is invariant under shifts of q, so the routing of the loop momentum does not matter for $T^{(\mathrm{m})}$.

57.3 The vertices and their naive continuum limit

The vector operator splits by the lattice Weitzenböck identity. It is derived in the proof of Lemma 26.4 as an identity of quadratic forms at real backgrounds; we restate it for operators and complex configurations.

Lemma 57.3 (Lattice Weitzenböck identity)**.** *Let U be a G^c-valued configuration on $\eta\mathbb{Z}^4$ or on a torus of spacing η. On $\mathfrak{g}^c$-valued gauge fields,*

$$D_U^T D_U + D_U D_U^T = \nabla_U^T \nabla_U + \mathsf{W}_U, \qquad (\mathsf{W}_U h)_\mu = \sum_{\nu \neq \mu} [D_\mu, D_\nu^T] h_\nu.$$

Here D_U is the covariant curl in the first term and the covariant gradient of scalar fields in the second, $\nabla_U^T \nabla_U = \sum_\nu D_\nu^T D_\nu$ acts on each component h_μ as on a scalar field, and

$$[D_\mu, D_\nu^T] f(x) = \eta^{-2} \Big(R\big(U_\mu(x) U_\nu(x + \eta e_\mu - \eta e_\nu)^{-1}\big) \\ - R\big(U_\nu(x - \eta e_\nu)^{-1} U_\mu(x - \eta e_\nu)\big) \Big) f(x + \eta e_\mu - \eta e_\nu)$$

for all μ, ν. In particular $[D_\mu, D_\mu^T] = 0$, and $\mathsf{W}_1 = 0$.

Proof. Transposes are taken in the bilinear pairing b. Since $\mathrm{tr}\,\big((R(u)\xi)\zeta\big) = \mathrm{tr}\,\big(\xi\, R(u)^{-1}\zeta\big)$ for $u \in G^c$, Details

$$(D_\nu^T g)(x) = \eta^{-1} \big(R(U_\nu(x - \eta e_\nu))^{-1} g(x - \eta e_\nu) - g(x) \big).$$

For the gradient of scalar fields, $D_U^T h = \sum_\mu D_\mu^T h_\mu$. For the curl and antisymmetric F,

$$\mathrm{b}(F, D_U h) = \tfrac{1}{2} \sum_{\mu\nu} \mathrm{b}(F_{\mu\nu}, D_\mu h_\nu - D_\nu h_\mu) = \sum_{\mu\nu} \mathrm{b}(F_{\mu\nu}, D_\mu h_\nu),$$

so $(D_U^T F)_\nu = \sum_\mu D_\mu^T F_{\mu\nu}$. Hence

$$(D_U^T D_U h + D_U D_U^T h)_\nu = \sum_\mu D_\mu^T (D_\mu h_\nu - D_\nu h_\mu) + D_\nu \sum_\mu D_\mu^T h_\mu = \sum_\mu D_\mu^T D_\mu h_\nu + \sum_\mu [D_\nu, D_\mu^T] h_\mu.$$

Expanding $D_\mu D_\nu^T$ and $D_\nu^T D_\mu$, all terms cancel except those at $x + \eta e_\mu - \eta e_\nu$, which gives the formula. This is the computation in the proof of Lemma 26.4. For $\mu = \nu$ the two transports are $U_\mu(x) U_\mu(x)^{-1} = 1$ and $U_\mu(x - \eta e_\mu)^{-1} U_\mu(x - \eta e_\mu) = 1$, so $[D_\mu, D_\mu^T] = 0$; this is the cancellation of the diagonal terms there. At $U = 1$ all transports are 1. □

For $\varepsilon \in (\mathfrak{g}^c)^4$ and $p \in \mathbb{C}^4$ let $\mathcal{K}(p, \varepsilon)$ be the operator on $(\mathfrak{g}^c)^4$ with the entries

$$\mathcal{K}(p, \varepsilon)_{\mu\nu} = \mathrm{ad}\,(p_\mu \varepsilon_\nu - p_\nu \varepsilon_\mu), \qquad \text{i.e.} \quad (\mathcal{K}(p, \varepsilon)\phi)_\mu = \sum_\nu \big[p_\mu \varepsilon_\nu - p_\nu \varepsilon_\mu, \phi_\nu\big].$$

Lemma 57.4 (The naive continuum limit of the vertices)**.** *Let $\varepsilon \in (\mathfrak{g}^c)^4$, and put*

$$v_{\mathrm{S}}^0(q; p, \varepsilon) = \sum_\nu (2q + p)_\nu \,\mathrm{ad}\,\varepsilon_\nu, \qquad v_{\mathrm{V}}^0(q; p, \varepsilon) = v_{\mathrm{S}}^0(q; p, \varepsilon) \otimes 1 + 2\mathcal{K}(p, \varepsilon),$$

where $\otimes 1$ means the action on each component. Then $v_1^{X_{\mathrm{S}}}(0; 0, \varepsilon) = 0$ and $v_1^{X_{\mathrm{V}}}(0; 0, \varepsilon) = 0$. The functions $\lambda^{-1} v_1^{X_{\mathrm{S}}}(\lambda q; \lambda p, \varepsilon)$ and $\lambda^{-1} v_1^{X_{\mathrm{V}}}(\lambda q; \lambda p, \varepsilon)$, $\lambda \neq 0$, extend to entire functions of $(\lambda, q, p) \in \mathbb{C} \times \mathbb{C}^4 \times \mathbb{C}^4$, with the values $v_{\mathrm{S}}^0(q; p, \varepsilon)$ and $v_{\mathrm{V}}^0(q; p, \varepsilon)$ at $\lambda = 0$. In v_{V}^0, one term $\mathcal{K}(p, \varepsilon)$ comes from W_U and the other from $\mathcal{F}_U$.

These are the vertices of the continuum operators $-D^2$ on scalar fields and $-D^2\delta_{\mu\nu}+\mathsf{E}_{\mu\nu}$ on gauge fields, in the background $A_{p,\varepsilon}$, to first order in ε. Here $D_\mu=\partial_\mu+\mathrm{i}\,\mathrm{ad}\,A_\mu$, $F_{\mu\nu}=\partial_\mu A_\nu-\partial_\nu A_\mu+\mathrm{i}[A_\mu,A_\nu]$, so that $[D_\mu,D_\nu]=\mathrm{i}\,\mathrm{ad}\,F_{\mu\nu}$, and

$$\mathsf{E}_{\mu\nu}=2\mathcal{K}_{\mu\nu},\qquad \mathcal{K}_{\mu\nu}=-\mathrm{i}\,\mathrm{ad}\,F_{\mu\nu}=[D_\mu,D_\nu^T]\quad (D_\nu^T=-D_\nu).$$

Indeed, $-D^2=-\partial^2-\mathrm{i}\sum_\nu(\partial_\nu\,\mathrm{ad}\,A_\nu+\mathrm{ad}\,A_\nu\,\partial_\nu)+O(\varepsilon^2)$ maps $e^{\mathrm{i}q\cdot x}\phi$ to $e^{\mathrm{i}(q+p)\cdot x}v_{\mathrm{S}}^0(q;p,\varepsilon)\phi+O(\varepsilon^2)$, and $F_{\mu\nu}=\mathrm{i}(p_\mu\varepsilon_\nu-p_\nu\varepsilon_\mu)e^{\mathrm{i}p\cdot x}+O(\varepsilon^2)$, so $\mathcal{K}_{\mu\nu}=e^{\mathrm{i}p\cdot x}\mathcal{K}(p,\varepsilon)_{\mu\nu}+O(\varepsilon^2)$. In the continuum, $D^TD+DD^T=-D^2+\mathcal{K}$ (the Weitzenböck identity), and the expansion of $\frac12\|F(A+h)\|^2$ contains $\langle F,\mathrm{i}[h,h]\rangle=\frac12\mathrm{b}(h,\mathcal{K}h)$, the second factor $\mathcal{K}$. The lemma says that the lattice operators reproduce both, with the same sign, which is the background-field result [21]. The sign of E does not enter the coefficient below; the factor 2 does.

Proof. Throughout, $U_t=e^{\mathrm{i}tA_{p,\varepsilon}}$. Each vertex is a trigonometric polynomial $v=\sum_j c_j e^{\mathrm{i}(z_j\cdot q+z_j'\cdot p)}$. If $v(0;0)=\sum_j c_j=0$, then Details

$$\lambda^{-1}v(\lambda q;\lambda p)=\sum_j c_j\,\mathrm{i}(z_j\cdot q+z_j'\cdot p)\,\varphi(\mathrm{i}\lambda(z_j\cdot q+z_j'\cdot p)),\qquad \varphi(w)=\frac{e^w-1}{w},$$

which is entire, with the value $\mathrm{i}\sum_j c_j(z_j\cdot q+z_j'\cdot p)$ at $\lambda=0$: the linear part of v at $(0;0)$. So it suffices to show $v_1^X(0;0,\varepsilon)=0$ and to compute the linear parts. We use: if a product of trigonometric polynomials has one factor vanishing at $(0;0)$, its linear part is the linear part of that factor times the values at $(0;0)$ of the others; if two factors vanish at $(0;0)$, its linear part is 0.

The scalar vertex. At $U=1$ the symbols of D_μ and D_μ^T are $e^{\mathrm{i}q_\mu}-1$ and $e^{-\mathrm{i}q_\mu}-1$. Since $\frac{d}{dt}R(U_{t,\mu}(x))|_{t=0}=\mathrm{i}\,\mathrm{ad}\,A_\mu(x)$, the operators $\delta D_\mu[A_{p,\varepsilon}]$ and $\delta D_\mu^T[A_{p,\varepsilon}]=(\delta D_\mu[A_{p,\varepsilon}])^T$ transfer p, with the symbols $\mathrm{i}e^{\mathrm{i}q_\mu}\,\mathrm{ad}\,\varepsilon_\mu$ and $-\mathrm{i}e^{-\mathrm{i}(q+p)_\mu}\,\mathrm{ad}\,\varepsilon_\mu$. By the composition rule, $\delta(D_\mu^TD_\mu)=\delta D_\mu^T\,D_\mu+D_\mu^T\,\delta D_\mu$ has the symbol

$$\big[-\mathrm{i}e^{-\mathrm{i}(q+p)_\mu}(e^{\mathrm{i}q_\mu}-1)+\mathrm{i}(e^{-\mathrm{i}(q+p)_\mu}-1)e^{\mathrm{i}q_\mu}\big]\,\mathrm{ad}\,\varepsilon_\mu=2e^{-\mathrm{i}p_\mu/2}\sin\big(q_\mu+\tfrac{p_\mu}{2}\big)\,\mathrm{ad}\,\varepsilon_\mu.$$

Summed over μ this is $v_1^{X_{\mathrm{S}}}(q;p,\varepsilon)$. It vanishes at $(0;0)$, and its linear part is v_{S}^0.

The vector vertex. By Lemma 57.3, $X_{\mathrm{V}}(U)=X_{\mathrm{S}}(U)\otimes 1+\mathsf{W}_U+\mathcal{F}_U$, where $X_{\mathrm{S}}(U)\otimes 1=\nabla_U^T\nabla_U+\mathrm{m}^2$ acts on each component. So $v_1^{X_{\mathrm{V}}}=v_1^{X_{\mathrm{S}}}\otimes 1+w+f$, with the vertices w of W and f of $\mathcal{F}$. For a plaquette $\square=p_{\mu\nu}(x)$, $\mu<\nu$ (here $\square$ is a plaquette, not a grid cube),

$$(dA_{p,\varepsilon})(\square)=A_\nu(x+e_\mu)-A_\nu(x)-A_\mu(x+e_\nu)+A_\mu(x)=e^{\mathrm{i}p\cdot x}\mathfrak{f}_{\mu\nu}(p,\varepsilon),$$

with $\mathfrak{f}_{\mu\nu}(p,\varepsilon)=(e^{\mathrm{i}p_\mu}-1)\varepsilon_\nu-(e^{\mathrm{i}p_\nu}-1)\varepsilon_\mu$. We extend $\mathfrak{f}$ antisymmetrically to all μ,ν. It vanishes at $p=0$, and its linear part is $\mathrm{i}(p_\mu\varepsilon_\nu-p_\nu\varepsilon_\mu)$.

The Weitzenböck vertex. Differentiate the formula of Lemma 57.3 at $t=0$, with $\eta=1$: $\frac{d}{dt}R(U_{t,\mu}(x)U_{t,\nu}(y)^{-1})=\mathrm{i}\,\mathrm{ad}\,(A_\mu(x)-A_\nu(y))$. For $\mu\neq\nu$ this gives

$$\delta[D_\mu,D_\nu^T]f(x)=\mathrm{i}\,\mathrm{ad}\,\big(A_\mu(x)-A_\nu(x+e_\mu-e_\nu)+A_\nu(x-e_\nu)-A_\mu(x-e_\nu)\big)f(x+e_\mu-e_\nu),$$

and with $A=A_{p,\varepsilon}$ the argument of ad is $-e^{\mathrm{i}p\cdot(x-e_\nu)}\mathfrak{f}_{\mu\nu}(p,\varepsilon)$. So

$$w_{\mu\nu}(q;p,\varepsilon)=-\mathrm{i}e^{\mathrm{i}(q_\mu-q_\nu-p_\nu)}\,\mathrm{ad}\,\mathfrak{f}_{\mu\nu}(p,\varepsilon),\qquad w_{\mu\mu}=0.$$

It vanishes at $p=0$, and its linear part is $-\mathrm{i}\cdot\mathrm{i}\,\mathrm{ad}(p_\mu\varepsilon_\nu-p_\nu\varepsilon_\mu)=\mathcal{K}(p,\varepsilon)_{\mu\nu}$.

The Wilson vertex. For a plaquette $\square$ with bonds $b_1,\dots,b_4$ as in Definition 21.4, holonomy $H_\square=U(\partial\square)$ and a gauge field h, let $a_1,\dots,a_4$ be as in (23.1), with $\eta=1$, let $\sigma=(1,1,-1,-1)$, and put

$$\alpha_\square=\sum_i\sigma_i a_i,\qquad \beta_\square=\mathrm{i}\sum_{i<j}\sigma_i\sigma_j[a_i,a_j],\qquad H_\square^+=\tfrac12(H_\square+H_\square^{-1}),\qquad H_\square^-=\tfrac{1}{2\mathrm{i}}(H_\square-H_\square^{-1}).$$

Then, for every configuration U,

$$\frac{d^2}{ds^2} S(e^{ish}U)\Big|_{s=0} = \sum_{\square} \Big(\operatorname{tr}\left(\alpha_\square^2 H_\square^+\right) + \operatorname{tr}\left(\beta_\square H_\square^-\right)\Big). \tag{57.5}$$

Indeed, by (23.1), $(e^{ish}U)(\partial\square) = e^{\mathrm{i}Z(s)} H_\square$ with $Z(s) = \ell(e^{isa_1}e^{isa_2}e^{-isa_3}e^{-isa_4}) = s\alpha_\square + \frac{s^2}{2}\beta_\square + O(s^3)$, by the Baker–Campbell–Hausdorff formula $\log(e^{Y_1}\cdots e^{Y_4}) = \sum_i Y_i + \frac{1}{2}\sum_{i<j}[Y_i, Y_j] + O(|Y|^3)$. The summand of S is $\mathfrak{s}(Z, H_\square)$, and $\mathfrak{s}(Z,H) - \mathfrak{s}(0,H) = \operatorname{tr}(ZH^-) + \frac{1}{2}\operatorname{tr}(Z^2H^+) + O(|Z|^3)$, as in the proof of Proposition 23.3, with $H^\pm$ formed from H as $H_\square^\pm$ from $H_\square$. Differentiating twice at $s=0$ gives (57.5).

At $U=1$: $H_\square = 1$, $H_\square^+ = 1$, $H_\square^- = 0$ and $\alpha_\square = (dh)(\square)$. Since $\frac{d}{dt}H_\square = \mathrm{i}(dA)(\square)$ at $t=0$, $\frac{d}{dt}H_\square^+ = 0$ and $\frac{d}{dt}H_\square^- = (dA)(\square)$. By the definition of a_3, a_4, and since $(D_U h)(\square) = a_1 + a_2 - R(U(b_4))h(b_3) - h(b_4)$,

$$\alpha_\square = (D_U h)(\square) - \big(R(H_\square) - 1\big)\big(R(U(b_4))h(b_3) + h(b_4)\big).$$

The operator $D_U^T D_U$ is that of $\sum_\square \operatorname{tr}((D_U h)(\square)^2)$. Subtracting it from (57.5) and differentiating at $t=0$, with $\frac{d}{dt}R(H_\square) = \mathrm{i}\operatorname{ad}(dA)(\square)$, gives the quadratic form of $\delta\mathcal{F}[A]$:

$$\frac{d}{dt}\,\mathrm{b}\big(h, \mathcal{F}_{U_t} h\big)\Big|_{t=0} = \sum_{\square} \Big(\operatorname{tr}\left(\beta_\square^0\,(dA)(\square)\right) - 2\mathrm{i}\operatorname{tr}\left((dh)(\square)\big[(dA)(\square), h(b_3)+h(b_4)\big]\right)\Big), \tag{57.6}$$

where $\beta_\square^0 = \mathrm{i}\sum_{i<j}\sigma_i\sigma_j[h(b_i), h(b_j)]$. The operator $\delta\mathcal{F}[A_{p,\varepsilon}]$ is symmetric, and its symbol $f(q;p,\varepsilon)$ is read off from the polarization of (57.6). For $h = e^{\mathrm{i}q\cdot x}\phi$ and $h' = e^{-\mathrm{i}(q+p)\cdot x}\psi$ it equals $\mathrm{b}(\psi, f(q;p,\varepsilon)\phi)$ per site, where the sum over plaquettes per site runs over $\mu<\nu$. Each term contains the factor $\mathfrak{f}_{\mu\nu}(p,\varepsilon)$ of $(dA)(\square)$, so $f(q;0,\varepsilon) = 0$. The second term contains in addition the symbol of dh or of dh', which vanishes at $(0;0)$; so it contributes nothing to the linear part. In the first term the other factors are evaluated at $q = p = 0$, where all phases are 1. There $h(b_1), \dots, h(b_4)$ become $\phi_\mu, \phi_\nu, \phi_\mu, \phi_\nu$, and the polarization of $\beta_\square^0$ becomes $\mathrm{i}\big([\psi_\mu, \phi_\nu] - [\psi_\nu, \phi_\mu]\big)$, the polarization of $2\mathrm{i}[h_\mu, h_\nu]$. So the linear part of $\mathrm{b}(\psi, f\phi)$ is

$$\sum_{\mu<\nu} \operatorname{tr}\Big(\mathrm{i}\big([\psi_\mu,\phi_\nu] - [\psi_\nu,\phi_\mu]\big)\,\mathrm{i}\big(p_\mu\varepsilon_\nu - p_\nu\varepsilon_\mu\big)\Big) = \sum_{\mu\nu} \operatorname{tr}\Big(\psi_\mu\big[p_\mu\varepsilon_\nu - p_\nu\varepsilon_\mu, \phi_\nu\big]\Big) = \mathrm{b}(\psi, \mathcal{K}(p,\varepsilon)\phi),$$

using $\operatorname{tr}([a,b]Y) = -\operatorname{tr}(a[Y,b])$ and the antisymmetry in μ,ν. Hence the linear part of f is $\mathcal{K}(p,\varepsilon)$.

Altogether $v_1^{X_\mathrm{V}}(0;0,\varepsilon) = 0$, and the linear part of $v_1^{X_\mathrm{V}}$ is $v_\mathrm{S}^0 \otimes 1 + \mathcal{K} + \mathcal{K} = v_\mathrm{V}^0$. □

57.4 The continuum integrand

Let P_0 be the function obtained from P by replacing G_m by $G_\mathrm{m}^0(q) = (|q|^2 + \mathrm{m}^2)^{-1}$, each v_1^X by v_X^0, and each $v_2^X(q;p,\varepsilon)$ by the constant $v_2^X(0;0,\varepsilon)$, and let $\mathcal{I}_0(q,\mathrm{m})$ be obtained from P_0 as $\mathcal{I}$ from P in (57.4).

Lemma 57.5 (Comparison with the continuum)**.** *The tensor $\mathcal{I}_0(q,\mathrm{m})$ is a rational function of q and m^2, with denominators powers of $|q|^2 + \mathrm{m}^2$, homogeneous of degree -4 in (q,m). Explicitly,*

$$P_{0,\mu\nu}(q,p,\mathrm{m}) = \kappa_\rho\, G_\mathrm{m}^0(q)\, G_\mathrm{m}^0(q+p)\Big((2q+p)_\mu(2q+p)_\nu + 4\big(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu\big)\Big) + t_{\mu\nu}(q,\mathrm{m}), \tag{57.7}$$

where $t_{\mu\nu}$ does not depend on p. Moreover there is C such that

$$|\mathcal{I}(q,\mathrm{m}) - \mathcal{I}_0(q,\mathrm{m})| \le C\,(|q|^2 + \mathrm{m}^2)^{-3/2} \qquad \text{for } |q| \le 1,\ 0 \le \mathrm{m} \le 1,\ (q,\mathrm{m}) \neq 0.$$

So $\mathcal{I}_0$ is the integrand of the p^2-tensor of the continuum one-loop term $-\frac12 \log\det(-D^2\delta_{\mu\nu} + \mathsf{E}_{\mu\nu} + \mathrm{m}^2) + \log\det(-D^2+\mathrm{m}^2)$ in the background Feynman gauge, with the routing of the lattice integral. The first term of (57.7) is the orbital part, the second the spin part. The tadpoles give only $t_{\mu\nu}$, since the continuum seagull vertex does not depend on p.

Proof. The formula. With $\varepsilon = c\otimes\xi_0$, Lemma 57.4 gives $v^0_{\mathrm{S}}(q;p,\varepsilon) = ((2q+p)\cdot c)\operatorname{ad}\xi_0$ and $v^0_{\mathrm{V}}(q;p,\varepsilon) = \operatorname{ad}\xi_0 \otimes M(q,p,c)$, with the 4×4 matrix $M(q,p,c) = ((2q+p)\cdot c)1 + 2N(p,c)$, $N(p,c)_{\mu\nu} = p_\mu c_\nu - p_\nu c_\mu$. Since $\xi_0^* = \xi_0$ and $2(q+p)-p = 2q+p$, the vertices at $(q+p;-p,\bar\varepsilon)$ are $((2q+p)\cdot\bar c)\operatorname{ad}\xi_0$ and $\operatorname{ad}\xi_0\otimes(((2q+p)\cdot\bar c)1 - 2N(p,\bar c))$. By (57.1), $\operatorname{tr}_{\mathfrak{g}^c}(\operatorname{ad}\xi_0)^2 = \kappa_\rho$. Moreover $\operatorname{Tr} N = 0$ and $\operatorname{Tr}\big(N(p,\bar c)N(p,c)\big) = -2\big(|p|^2\,\bar c\cdot c - (p\cdot\bar c)(p\cdot c)\big)$ for real p. So the traces in the bubbles of (57.3) are $\kappa_\rho|(2q+p)\cdot c|^2$ for the scalar field and $\kappa_\rho\big(4|(2q+p)\cdot c|^2 + 8(|p|^2|c|^2 - |p\cdot c|^2)\big)$ for the vector field. With the factors $-\frac12$ and 1 of the two determinants and the sign of the bubble, this is (57.7); the tadpoles $G^0_{\mathrm{m}}(q)\operatorname{Tr} v_2^X(0;0,\varepsilon)$ give $t_{\mu\nu}$. Since $P_0(sq,sp,s\mathrm{m}) = s^{-2}P_0(q,p,\mathrm{m})$ for $s>0$, $\mathcal{I}_0$ is homogeneous of degree -4. The derivatives in p of $G^0_{\mathrm{m}}(q+p)$ at $p=0$ are polynomials in q times powers of $G^0_{\mathrm{m}}(q)$, which gives the form of $\mathcal{I}_0$.

Scaling. For $\lambda>0$ put $\Psi(\lambda;q,p,\mathrm{m}) = \lambda^2 P(\lambda q,\lambda p,\lambda\mathrm{m})$. Since $\partial_p = \lambda\partial_{p'}$ for $p'=\lambda p$,

$$\lambda^4\,\mathcal{I}(\lambda q,\lambda\mathrm{m}) = \tfrac12\,\partial_p^2\Psi(\lambda;q,p,\mathrm{m})\big|_{p=0}.$$

In Ψ write $\lambda^2 G_{\lambda\mathrm{m}}(\lambda q) = \big(\sum_\nu s_\lambda(q_\nu)^2 + \mathrm{m}^2\big)^{-1}$, with $s_\lambda(t) = 2\lambda^{-1}\sin(\lambda t/2)$, and the same at $q+p$; the bubbles contain $\lambda^2 G\cdot\lambda^2 G\cdot\lambda^{-1}v_1\cdot\lambda^{-1}v_1$, and the tadpoles $\lambda^2 G\cdot v_2$. The function $s_\lambda(t)$ is entire in (λ,t), with $s_0(t)=t$. By Lemma 57.4, the scaled vertices $\lambda^{-1}v_1^X(\lambda\cdot;\lambda\cdot)$, at the arguments $(q;p,\varepsilon)$ and $(q+p;-p,\bar\varepsilon)$, and $v_2^X(\lambda q;\lambda p,\varepsilon)$ are entire in (λ,q,p), with the values v_X^0 and $v_2^X(0;0,\varepsilon)$ at $\lambda=0$. So Ψ extends to $\lambda=0$, with $\Psi(0;q,p,\mathrm{m}) = P_0(q,p,\mathrm{m})$, and it is real-analytic wherever the two denominators do not vanish.

Compactness. Let $K = \{(q,p,\mathrm{m}) : |q|^2+\mathrm{m}^2 = 1,\ \mathrm{m}\ge 0,\ |p|\le\frac14\}$, a compact set. For $|t|\le\pi/\lambda$ we have $|s_\lambda(t)|\ge\frac{2}{\pi}|t|$. For $(q,p,\mathrm{m})\in K$ and $0\le\lambda\le\sqrt2$, $\lambda|q_\nu|\le\sqrt2<\pi$ and $\lambda|(q+p)_\nu|\le\frac54\sqrt2<\pi$. So the first denominator is at least $\frac{4}{\pi^2}(|q|^2+\mathrm{m}^2) = \frac{4}{\pi^2}$, and the second at least $\frac{4}{\pi^2}(|q+p|^2+\mathrm{m}^2)\ge\frac{1}{\pi^2}$: if $|q|\ge\frac12$, then $|q+p|\ge\frac12|q|$, and otherwise $\mathrm{m}^2\ge\frac34$. By continuity the denominators do not vanish on a neighbourhood of $[0,\sqrt2]\times K$. There Ψ is real-analytic, so $\partial_\lambda\partial_p^2\Psi$ is bounded on $[0,\sqrt2]\times K$, by some $2C$. Hence

$$\big|\lambda^4\mathcal{I}(\lambda q,\lambda\mathrm{m}) - \mathcal{I}_0(q,\mathrm{m})\big| \le C\lambda \qquad \text{for } 0<\lambda\le\sqrt2,\ |q|^2+\mathrm{m}^2=1,\ \mathrm{m}\ge 0.$$

Now let $|q|\le 1$, $0\le\mathrm{m}\le 1$, and $\lambda = (|q|^2+\mathrm{m}^2)^{1/2}\in(0,\sqrt2]$. Apply this to $(q,\mathrm{m})/\lambda$ and use the homogeneity of $\mathcal{I}_0$: $|\mathcal{I}(q,\mathrm{m}) - \mathcal{I}_0(q,\mathrm{m})|\le C\lambda^{-3}$. □

Numerically the bound is not sharp: the difference is of order $(|q|^2+\mathrm{m}^2)^{-1}$ (Remark 57.10). We do not need this. The coefficient below does not depend on the routing either. Replacing $P_0(q,p,\mathrm{m})$ by $P_0(q+sp,p,\mathrm{m})$ adds to each component of $\mathcal{I}_0(q,0)(p)$ the divergence in q of a field W homogeneous of degree -3, smooth on $\mathbb{R}^4\setminus 0$, namely $W(q) = s\,p\,\big(p\cdot\partial_{p'}P_0(q,p',0)\big)|_{p'=0} + \frac{s^2}{2}\,p\,\big(p\cdot\nabla_q P_0(q,0,0)\big)$, by Taylor's formula in sp. For such W, $\nabla\cdot W$ is homogeneous of degree -4, and by the divergence theorem

$$\log R\int_{|\omega|=1}\nabla\cdot W\,d\sigma = \int_{1<|q|<R}\nabla\cdot W\,d^4q = \int_{|q|=R} W\cdot\frac{q}{|q|}\,d\sigma - \int_{|q|=1} W\cdot q\,d\sigma = 0,$$

since the flux through $|q|=R$ is $R^{-3}R^3$ times that through $|q|=1$. So the spherical average of $\nabla\cdot W$ vanishes.

57.5 The infrared asymptotics and the continuum coefficient

Let $d\sigma$ be the surface measure on the unit sphere $S^3 \subset \mathbb{R}^4$, and $B = \{q \in \mathbb{R}^4 : |q| \le 1\}$.

Lemma 57.6 (The infrared asymptotics)**.** *As* $\mathrm{m} \to 0$,

$$T^{(\mathrm{m})} = \mathsf{B} \log \frac{1}{\mathrm{m}} + \mathsf{C} + o(1), \qquad \mathsf{B} = \frac{1}{(2\pi)^4} \int_{|\omega|=1} \mathcal{I}_0(\omega, 0)\, d\sigma(\omega),$$

with the tensor

$$(2\pi)^4\, \mathsf{C} = \int_{[-\pi,\pi]^4 \setminus B} \mathcal{I}(q,0) + \int_B (\mathcal{I}(q,0) - \mathcal{I}_0(q,0)) + \int_B \mathcal{I}_0(q,1) + \int_{\mathbb{R}^4 \setminus B} (\mathcal{I}_0(q,1) - \mathcal{I}_0(q,0)), \tag{57.8}$$

the integrals taken against d^4q *and absolutely convergent. Hence* $f(\mathrm{m}) = \mathsf{b} \log(1/\mathrm{m}) + c + o(1)$, *with* $\mathsf{b} = \operatorname{iso} \mathsf{B}$ *and* $c = \operatorname{iso} \mathsf{C}$.

Proof. Write $(2\pi)^4 T^{(\mathrm{m})}$ as the sum of the integrals of $\mathcal{I}(q,\mathrm{m})$ over $[-\pi,\pi]^4 \setminus B$, of $\mathcal{I}(q,\mathrm{m}) - \mathcal{I}_0(q,\mathrm{m})$ over B, and of $\mathcal{I}_0(q,\mathrm{m})$ over B. Details

On $[-\pi,\pi]^4 \setminus B$, $|\hat{q}|^2 \ge \frac{4}{\pi^2}|q|^2 \ge \frac{4}{\pi^2}$. So $\mathcal{I}$ is continuous on the compact set $([-\pi,\pi]^4 \setminus \operatorname{int} B) \times [0,1]$, and its integral converges to that of $\mathcal{I}(q,0)$ as $\mathrm{m} \to 0$.

On $B \setminus 0$, $\mathcal{I}(q,\mathrm{m}) - \mathcal{I}_0(q,\mathrm{m}) \to \mathcal{I}(q,0) - \mathcal{I}_0(q,0)$ pointwise, and by Lemma 57.5 it is bounded by $C|q|^{-3}$, which is integrable in four dimensions. By dominated convergence its integral converges to the second term of (57.8).

For the last integral substitute $q = \mathrm{m}q'$ and use the homogeneity of $\mathcal{I}_0$:

$$\begin{aligned}\int_B \mathcal{I}_0(q,\mathrm{m})\, d^4q &= \int_{|q'|\le 1/\mathrm{m}} \mathcal{I}_0(q',1)\, d^4q' \\ &= \int_B \mathcal{I}_0(q',1) + \int_{1<|q'|\le 1/\mathrm{m}} (\mathcal{I}_0(q',1) - \mathcal{I}_0(q',0)) + \int_{1<|q'|\le 1/\mathrm{m}} \mathcal{I}_0(q',0).\end{aligned}$$

The first term is finite, since $\mathcal{I}_0(\cdot,1)$ is smooth. For $|q'| = r \ge 1$, $\mathcal{I}_0(q',1) - \mathcal{I}_0(q',0) = r^{-4}(\mathcal{I}_0(\omega,1/r) - \mathcal{I}_0(\omega,0))$ with $\omega = q'/r$, and $\mathcal{I}_0(\omega,s)$ is a rational function of ω and s^2 with denominators powers of $1+s^2$. So this difference is at most Cr^{-6}, which is integrable on $\mathbb{R}^4 \setminus B$, and the second term converges to the last term of (57.8). In polar coordinates the third term is $\int_1^{1/\mathrm{m}} r^3\, r^{-4}\, dr \int_{|\omega|=1} \mathcal{I}_0(\omega,0)\, d\sigma = (2\pi)^4 \mathsf{B} \log(1/\mathrm{m})$. This proves the expansion of $T^{(\mathrm{m})}$, and that of f follows, since iso is linear. □

Lemma 57.7 (The classical coefficient)**.**

$$\mathsf{B} = \frac{11\kappa_\rho}{24\pi^2}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu), \qquad \textit{hence} \qquad \mathsf{b} = \frac{11 C_A}{24\pi^2 T_\rho}.$$

This is the coefficient relative to the Wilson action: in the trace pairing, the marginal tensor of S *is* $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$, *with isotropic coefficient* 1.

Proof. We compute the full tensor B, without assuming transversality. Let $|\omega| = 1$ and $\mathrm{m} = 0$. Then $G_0^0(\omega) = 1$ and Details

$$G_0^0(\omega + p) = (1 + 2\omega\cdot p + |p|^2)^{-1} = 1 - 2\omega\cdot p - |p|^2 + 4(\omega\cdot p)^2 + O(|p|^3).$$

By (57.7), $\mathcal{I}_0(\omega,0)(p)$ is the part of second order in p of $\kappa_\rho((2\omega+p)_\mu(2\omega+p)_\nu + 4(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu))G_0^0(\omega+p)$:

$$\kappa_\rho^{-1}\mathcal{I}_0(\omega,0)(p)_{\mu\nu} = p_\mu p_\nu - 4(\omega_\mu p_\nu + p_\mu\omega_\nu)(\omega\cdot p) - 4\omega_\mu\omega_\nu|p|^2 + 16\,\omega_\mu\omega_\nu(\omega\cdot p)^2 + 4(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu).$$

On S^3 the averages of the monomials are

$$\langle \omega_\alpha \omega_\beta \rangle = \tfrac{1}{4}\delta_{\alpha\beta}, \qquad \langle \omega_\alpha \omega_\beta \omega_\gamma \omega_\delta \rangle = \tfrac{1}{24}(\delta_{\alpha\beta}\delta_{\gamma\delta} + \delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma}), \tag{57.9}$$

by the symmetry under $O(4)$, the normalization $\sum_\alpha \omega_\alpha^2 = 1$, and $\langle \omega_1^4 \rangle = 3\langle \omega_1^2 \omega_2^2 \rangle$, which follows from the invariance under the rotation by $\pi/4$ in the $(1,2)$-plane. Hence the averages of the orbital terms are

$$\begin{aligned} &\langle p_\mu p_\nu - 4(\omega_\mu p_\nu + p_\mu \omega_\nu)(\omega \cdot p) - 4\omega_\mu \omega_\nu |p|^2 + 16\, \omega_\mu \omega_\nu (\omega \cdot p)^2 \rangle \\ &\quad = p_\mu p_\nu - 2p_\mu p_\nu - \delta_{\mu\nu}|p|^2 + \tfrac{2}{3}(\delta_{\mu\nu}|p|^2 + 2p_\mu p_\nu) = -\tfrac{1}{3}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu), \end{aligned}$$

and the spin term does not depend on ω. So

$$\langle \mathcal{I}_0(\omega, 0) \rangle = \kappa_\rho(-\tfrac{1}{3} + 4)(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu) = \frac{11\kappa_\rho}{3}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu).$$

In the form $a\,\delta_{\mu\nu}\delta_{\kappa\lambda} + b\,(\delta_{\mu\kappa}\delta_{\nu\lambda} + \delta_{\mu\lambda}\delta_{\nu\kappa})$ this is $a = \frac{11\kappa_\rho}{3}$ and $b = -\frac{11\kappa_\rho}{6}$; transversality, $a = -2b$, comes out of the computation. The orbital part splits into $+\frac{1}{3}\kappa_\rho$ from the scalar determinant and $-\frac{2}{3}\kappa_\rho$ from the vector determinant; with the spin part this is the split $\frac{11}{3} = 4 - \frac{1}{3}$. Since $|S^3| = 2\pi^2$, $(2\pi)^{-4}|S^3| = \frac{1}{8\pi^2}$, and $\mathsf{B} = \frac{11\kappa_\rho}{24\pi^2}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. Its isotropic coefficient is $\mathsf{b} = \frac{11\kappa_\rho}{24\pi^2} = \frac{11 C_A}{24\pi^2 T_\rho}$. For $SU(N)$ in the fundamental representation, $C_A = N$ and $T_\rho = \frac{1}{2}$, so $\mathsf{b} = \frac{11N}{12\pi^2}$.

Normalization. All second variations are taken in the trace pairing b of Section 23, and $T^{(\mathrm{m})}$ is the p^2-tensor of the vacuum polarization in it (Definition 54.7). On $\mathbb{Z}^4$ at $U = 1$, $J_1 = 0$ and $\mathcal{F}_1 = 0$, so the second variation of S is $\frac{d^2}{dt^2} S(e^{ith})|_{t=0} = \|dh\|^2$ for real gauge fields h (Proposition 23.3). Its kernel is $d^T d$, whose Fourier transform in the bond-midpoint convention is $|\hat{p}|^2 \delta_{\mu\nu} - \hat{p}_\mu \hat{p}_\nu$, with $\hat{p}_\mu = 2\sin(p_\mu/2)$ (Remark 54.8). So the marginal tensor of S is $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$, with isotropic coefficient 1, and b is the coefficient of $\log(1/\mathrm{m})$ relative to S.

Sign. The one-loop term enters the density as $\exp(-g^{-2}S + \Gamma^{(1)})$. So the effective coefficient of $-S$ is $g^{-2} - \mathsf{b}\log(1/\mathrm{m})$, which decreases towards the infrared: this is asymptotic freedom. □

Remark 57.8 (The heat-kernel coefficient). The value of b agrees with the heat-kernel computation of the logarithmic divergence of the continuum one-loop term; we do not use it in the proof. For an operator $-D^2 + \mathsf{E}$ on a bundle over $\mathbb{R}^4$ with curvature $\Omega_{\mu\nu} = [D_\mu, D_\nu]$, the heat-kernel coefficient a_4 contains $(4\pi)^{-2} \int \mathrm{Tr}\,(\frac{1}{2}\mathsf{E}^2 + \frac{1}{12}\Omega_{\mu\nu}\Omega_{\mu\nu})$, up to total derivatives [24] (there the operator is written $-(D^2 + E)$, so $E = -\mathsf{E}$), and $\log\det(-D^2 + \mathsf{E} + \mathrm{m}^2)$, with a cutoff Λ, contains $-\log(\Lambda^2/\mathrm{m}^2)$ times this coefficient. In the adjoint representation $\Omega_{\mu\nu} = \mathrm{i}\,\mathrm{ad}\,F_{\mu\nu} = -\mathcal{K}_{\mu\nu}$, and $\mathrm{tr}_{\mathrm{ad}}(\Omega_{\mu\nu}\Omega_{\mu\nu}) = -C_A F^a_{\mu\nu} F^a_{\mu\nu}$, summed over μ, ν, for $F = F^a T^a$.

- The scalar operator $-D^2$ gives $\frac{1}{12}\,\mathrm{Tr}\,\Omega\Omega = -\frac{1}{12}C_A F^2$. With the factor $+1$, the ghost contributes $+\frac{1}{12}C_A F^2 \cdot \frac{\log(\Lambda^2/\mathrm{m}^2)}{16\pi^2}$.
- The vector operator has curvature $\Omega \otimes 1$, so $\frac{1}{12}\,\mathrm{Tr}\,\Omega\Omega = -\frac{4}{12}C_A F^2$, and $\mathsf{E} = 2\mathcal{K}$ on the vector index, so $\frac{1}{2}\,\mathrm{Tr}\,\mathsf{E}^2 = 2C_A F^2$. With the factor $-\frac{1}{2}$, it contributes $\frac{1}{2}(2 - \frac{1}{3})C_A F^2 \cdot \frac{\log(\Lambda^2/\mathrm{m}^2)}{16\pi^2}$.

The total is $\frac{11}{12}C_A F^2 \cdot \frac{\log(\Lambda^2/\mathrm{m}^2)}{16\pi^2} = \frac{11 C_A}{96\pi^2}\log\frac{\Lambda}{\mathrm{m}} \sum_{\mu\nu} F^a F^a$. The continuum limit of S is $\frac{T_\rho}{4}\sum_{\mu\nu} F^a F^a$, since $\mathsf{N} - \mathrm{Re}\,\mathrm{tr}\,e^{\mathrm{i}F_p} = \frac{1}{2}\,\mathrm{tr}\,F_p^2 + \dots$. So the one-loop term contains $\frac{11 C_A}{24\pi^2 T_\rho}\log\frac{\Lambda}{\mathrm{m}}$ times S, in agreement with Lemma 57.7. To use this as a proof one would have to show that the logarithm of Lemma 57.6 is the heat-kernel logarithm; the direct computation makes this unnecessary.

Proof of Theorem 57.1. By Lemma 57.2 with $m = 1$, $\beta_{\mathrm{F}} = f(\eta)$, $\eta = L^{-k}$. By Lemmas 57.6 and 57.7, $f(\eta) = \frac{11C_A}{24\pi^2 T_\rho} \log \frac{1}{\eta} + c + o(1)$ as $\eta \to 0$, and $\log(1/\eta) = k \log L$. This is the claim, with $c_{\log} = c$, which depends only on G and ρ. The function f is continuous on $(0, 1]$, and $f(\mathrm{m}) - \mathsf{b} \log(1/\mathrm{m})$ has a limit as $\mathrm{m} \to 0$; so it is bounded on $(0, 1]$, and $\beta_{\mathrm{F}} = k\beta_\infty + O(1)$ for all $k \geq 1$. □ Details

Remark 57.9 (Numerical evaluation of f). For $G = SU(2)$ in the fundamental representation, $\mathsf{b} = \frac{11}{6\pi^2} = 0.185756$. We evaluated $f(\mathrm{m})$ numerically for eight masses $0.0625 \leq \mathrm{m} \leq 1$, with a real background along a Cartan generator; the method and the values are in Appendix C. Three conclusions:

(i) *Ward identity.* $\Pi(0)$ vanishes for all m, and the longitudinal part at $p_1 = 0.5$, $\mathrm{m} = 1$, to the accuracy of the computation, while the tadpole and bubble terms are separately of order 0.3.

(ii) *The logarithm and its sign.* Fits of f give b to within $1.2 \cdot 10^{-4}$, and to within $2 \cdot 10^{-5}$ for the fits that use only $\mathrm{m} \leq \frac{1}{4}$ and for those with six parameters. The function f increases as $\mathrm{m} \to 0$. This is the sign of asymptotic freedom.

(iii) *The constant.* With b fixed, the fitted constant c agrees to 10^{-5} with the value 0.563267 predicted by the one-loop relation between the bare Wilson coupling and the $\overline{\mathrm{MS}}$ coupling [23, 22], together with the $\overline{\mathrm{MS}}$ value of the continuum term of Lemma 57.5 (Appendix C).

So the lattice integral of Lemma 57.2 reproduces a known lattice constant, independently of the arguments of this section. In particular, the normalization of Lemma 57.7 is confirmed for $SU(2)$.

Remark 57.10 (Numerical checks of the integrand and of the tensor). Two further checks, described in Appendix C:

(i) *The spherical average, in exact arithmetic.* Computed in rational arithmetic from the vertices of Lemma 57.4, the four-index tensor $\langle \mathcal{I}_0(\omega, 0) \rangle$ has the form $a\,\delta_{\mu\nu}\delta_{\kappa\lambda} + b\,(\delta_{\mu\kappa}\delta_{\nu\lambda} + \delta_{\mu\lambda}\delta_{\nu\kappa})$, with $(a, b) = (\frac{11}{3}, -\frac{11}{6})\kappa_\rho$; the scalar determinant, the orbital and the spin parts of the vector determinant give $(\frac{1}{3}, -\frac{1}{6})\kappa_\rho$, $(-\frac{2}{3}, \frac{1}{3})\kappa_\rho$ and $(4, -2)\kappa_\rho$. So $a = -2b$ holds for each part separately, and the isotropic coefficient is $\frac{11}{3}\kappa_\rho$, as in the proof of Lemma 57.7.

(ii) *The scaling of the lattice integrand.* For $G = SU(2)$, one random polarization and one random direction of p, the largest difference between $\lambda^4 \mathcal{I}(\lambda\omega)$ and $\mathcal{I}_0(\omega)$ over 200 random directions ω decreases like $4\lambda^2$ for the scale factors $\lambda = 2^{-j}$, $0 \leq j \leq 12$, faster than the bound $C\lambda$ of Lemma 57.5. The vertices of this test come from the Taylor coefficients of the plaquette, not from the proof of Lemma 57.4. So it checks the limits of the vertices at the level of the integrand, including the factor 2 of the spin term and the weight of the ghost, independently of the fits of Remark 57.9.

58 The cumulative one-loop coefficient and the ultraviolet limit

This section assembles Sections 54–57 and combines the result with the flow of Part X. Theorem 58.1 is the cumulative one-loop coefficient. Proposition 58.6 derives from the flow (Theorem 53.12) the properties (F1)–(F3) of the couplings that the ultraviolet limit uses, and Theorem 58.9 is our version of the ultraviolet limit, Bałaban's Theorem 2 [18], for the couplings g_k of the small-field flow (see the paragraph after it for the comparison with his (0.31)). Bałaban's couplings come from his own generations, a different construction (Remark 53.14(a) and the statement of Proposition 58.8). The flow reads the coefficients β_j off the generations on the given torus (Proposition 50.13), while Part XI works with marginal tensors in infinite volume. The two are compared through Lemma 58.4, with the torus independence of the

one-loop parts of the generations (Lemma 58.3) and the infinite-volume limits of the step functions (Lemma 58.5); this comparison is (F2). Proposition 58.8 compares β_j only with the coefficient that Bałaban's definition [18, (1.20)–(1.22)] assigns to the generations of this note, at fixed couplings and as the torus grows; it rests on the generation locality of Section 59 and is proved without a rate. Remark 58.10 lists what remains.

Theorem 58.1 (The cumulative one-loop coefficient)**.** *Let G be compact, connected and simple, with $\beta_\infty = \frac{11C_A}{24\pi^2 T_\rho}\log L$ as in Theorem 57.1. Then every $T^{(1)}_{j+1}$ is a multiple $\beta^{(1)}_{j+1}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, and, uniformly in k,*

$$\sum_{j<k} T^{(1)}_{j+1} = T_{\Gamma^{(1)}_k} = \big(k\beta_\infty + O(1)\big)\big(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu\big),$$

where $O(1)$ is a number bounded uniformly in k.

Proof. The second variations at 1 of $\Gamma^{(1)}_k$ and of the step functions $\Gamma^{\mathrm{step}}_j \circ U_{j+1}$, $j<k$, on the tori converge with uniform exponential decay as the side tends to infinity: for $\Gamma^{(1)}_k$ by Lemma 56.2(c), and for the step functions by Lemma 58.5 below, whose proof uses the polymer decomposition of Part X with its parameters; the step functions themselves do not depend on it. The limits are taken along the tori whose side at scale k lies in $\mathfrak{N}_k$. By Lemma 52.1(a), with $K=k$, (T0)–(T2) hold at every scale $\le k$, so these tori satisfy the setting of Section 54. For $j<k$ the same torus, read with $K=j+1$, satisfies the Setting of Section 50 at the scales $\le j+1$. Its conditions at the scales $i \le j+1$ are among those at the scales $\le k$. The bound on the last side holds because the side at scale $j+1$ is L^{k-j-1} times the side at scale k, hence at least $n_{\min}$. So the sides at scale $j+1$ lie in $\mathfrak{N}_{j+1}$. The limit of the second variation of $\Gamma^{(1)}_k$ (Lemma 56.2(c)) is the same along this subsequence. So Corollary 54.18 applies. By Lemma 54.9(c), these functions have no term linear in p, and their marginal tensors are multiples of $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$; the first equality is Corollary 54.18. For the second:

Details

- by Lemma 56.3, $T_{\Gamma^{(1)}_k} = T_{\mathrm{fine}}$, the p^2-tensor of the fine vacuum polarization of $\widehat{\Gamma}$;
- by Proposition 56.8, $T_{\mathrm{fine}} = T_{\mathrm{F}} + O(1)$, uniformly in k;
- by Remark 56.9, $T_{\mathrm{F}} = \beta_{\mathrm{F}}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$; by Lemma 54.9, $T_{\Gamma^{(1)}_k}$ is a multiple of the same tensor, so the remainder $O(1)$ is too, and its coefficient is bounded uniformly in k;
- by Theorem 57.1, $\beta_{\mathrm{F}} = k\beta_\infty + O(1)$.

□

58.1 Averages of linear fields

The proof of Lemma 54.17 uses the averages of the linear fields of Section 50.2. They are exact.

Lemma 58.2 (Averages of linear fields)**.** *Let $k \ge 1$, and use the units of scale k, so that the fine lattice $\eta\mathbb{Z}^4$ has spacing $\eta = L^{-k}$ and the coarse lattice is $\mathbb{Z}^4$. Let F be a curvature datum in $\mathfrak{g}^c$ and $x_0 \in \mathbb{R}^4$, and let A_F and B_F be the linear fields of F with origin x_0 (Section 50.2) on the bonds of $\eta\mathbb{Z}^4$ and of $\mathbb{Z}^4$.*

(a) $\mathbf{Q}A_F = B_F$*, with the gauge-field average $\mathbf{Q}$ of Definition 2.5, given on $\eta\mathbb{Z}^4$ by the same formula.*

(b) Let $\mathbf{Q}^{\mathrm{pl}}$ be the linearization at $U=1$ of the average $U \mapsto U^{(k)}$ of Section 21: $U^{(k)}(c) = e^{\mathrm{i}t(\mathbf{Q}^{\mathrm{pl}}A)(c)+O(t^2)}$ for $U = e^{\mathrm{i}\eta t A}$. By Lemma 21.10(c), $(\mathbf{Q}^{\mathrm{pl}}A)(c)$ depends only on A on the bonds of $B(c_-)\cup B(c_+)$, so $\mathbf{Q}^{\mathrm{pl}}$ acts on fields on $\eta\mathbb{Z}^4$. Then $\mathbf{Q}^{\mathrm{pl}}A_F = B_F$.

This is the input on linear fields of the proof of Lemma 54.17: the blocks of Part IV are centred, and the average of A_F to scale $j+1$ is the linear field of F on the lattice of scale $j+1$, with no gradient ((b) with $j+1$ in place of k); in particular the coarse gradient of Lemma 22.8(f) vanishes on linear fields.

Proof. For a constant field, $A(b) = v_\mu$ on the bonds of direction μ, both $\mathbf{Q}$ and $\mathbf{Q}^{\mathrm{pl}}$ give the coarse field with the same components: for $\mathbf{Q}$ since $Q_\mu 1 = 1$ (Lemma 2.6(b)), for $\mathbf{Q}^{\mathrm{pl}}$ by the gauge covariance below with $\lambda(x) = v \cdot x$. Changing the origin from x_0 to $y \in \mathbb{Z}^4$ adds to A_F and to B_F the same constant field, with components $\frac{1}{2}\sum_\kappa (y - x_0)_\kappa F_{\kappa\mu}$. So it suffices to show, for every coarse point y and direction μ, that the average of A_F with origin y vanishes on $c = (y, y + e_\mu)$; and $B_F(c) = \frac{1}{2}(m_c - y)^{\mathsf{T}} F e_\mu = \frac{1}{4} e_\mu^{\mathsf{T}} F e_\mu = 0$ for this origin. Details

(a) For the bonds b of direction μ starting at $x + i\eta e_\mu$, $0 \le i < \eta^{-1}$, $A_F(b) = \frac{1}{2}\sum_{\kappa\neq\mu}(x - y)_\kappa F_{\kappa\mu}$, since $F_{\mu\mu} = 0$. This is linear in $x - y$. The block $B(y)$ is centred at y, so $\sum_{x\in B(y)}(x - y) = 0$, and Definition 2.5 gives $(\mathbf{Q}A_F)_\mu(y) = 0$.

(b) *Covariance.* Let λ be a $\mathfrak{g}$-valued function and $u = e^{-\mathrm{i}t\lambda}$. Then $(e^{\mathrm{i}\eta t A})^u = e^{\mathrm{i}\eta t(A+\partial\lambda)+O(t^2)}$ bondwise, with $\partial\lambda(b) = \eta^{-1}(\lambda(b_+) - \lambda(b_-))$, and Lemma 21.10(b) gives $\mathbf{Q}^{\mathrm{pl}}(A + \partial\lambda) = \mathbf{Q}^{\mathrm{pl}}A + \partial'\lambda$, with $\partial'\lambda(c) = \lambda(c_+) - \lambda(c_-)$ on the coarse bonds. For $r \in \mathfrak{S}_k$, $r(e^{\mathrm{i}\eta t A}) = e^{\mathrm{i}\eta t\, r_1 A}$ with $(r_1 A)(b) = A(r^{-1}b)$ and $A(-b) = -A(b)$, and Lemma 21.10(d) gives $\mathbf{Q}^{\mathrm{pl}} r_1 = r_1 \mathbf{Q}^{\mathrm{pl}}$. These identities are stated on the torus; by the locality, each of them at a coarse bond involves the fields only on the blocks of that bond and of its image, so they hold on $\eta\mathbb{Z}^4$ for all fields, after the fields are replaced near these blocks by periodic fields on a large torus (for the second identity, by an r-invariant cut-off). Both extend to complex fields by linearity.

Reflections. Let $\nu \neq \mu$, and let r be the reflection of the ν-th coordinate about y, an element of $\mathfrak{S}_k$ with linear part R_ν. It maps c onto itself, with its orientation, and $r_1 A_F = A_{R_\nu \cdot F}$, with origin $ry = y$, by (50.6). So $\ell(F) = (\mathbf{Q}^{\mathrm{pl}}A_F)(c)$ satisfies $\ell(R_\nu \cdot F) = (r_1 \mathbf{Q}^{\mathrm{pl}} A_F)(c) = (\mathbf{Q}^{\mathrm{pl}}A_F)(r^{-1}c) = \ell(F)$. Write $\ell(F) = \sum_{\kappa<\lambda} \ell_{\kappa\lambda}(F_{\kappa\lambda})$, with linear maps $\ell_{\kappa\lambda}$ on $\mathfrak{g}^c$. The map $F \mapsto R_\nu \cdot F$ changes the sign of $F_{\kappa\lambda}$ if exactly one of κ, λ equals ν, and keeps the other entries. Every pair $\kappa < \lambda$ contains an index $\nu \neq \mu$, and the other index of the pair is not ν. So $\ell_{\kappa\lambda} = -\ell_{\kappa\lambda}$ for every pair, and $\ell = 0$. □

58.2 The coefficients of the flow

Let the parameters and the torus be as in Section 53 (Setting), with $n_K \ge n_{\min}$, and let γ satisfy (Γ1)–(Γ3) of Remark 53.5. For $\pi = (g_0, \dots, g_j) \in \Sigma_j$ (Section 53.5) the generations $E^{(i)} = E^{(i)}[\pi]$ and the coefficients $\beta_i = \beta_i[\pi]$, $i \le j+1$, are those of Proposition 53.10; so β_{j+1} is a function of $(g_0, \dots, g_j)$. The couplings of the flow are those of Corollary 53.11: $g_{j+1}^{-2} = g_j^{-2} - \beta_{j+1}$ as long as $g_0, \dots, g_j \in (0, \gamma]$, and then $g_{j+1} \le 2g_j$. In this section the coupling g is that of the last scale, $g_K = g$, and the bare coupling is g_0; in Sections 31 and 52, by contrast, $g = g_0$ denotes the bare coupling, and in Sections 54–57 g is the single coupling of the Laplace asymptotics.

What Part X provides. The density of every scale has the generation form of Remark 49.1. A generation of scale j is a family of normalized, gauge-invariant graded activities on the polymers of scale j, defined on the domains $\mathcal{U}^\flat(X)$ with $\delta_1^\flat = \frac{1}{4}\delta_1$, with the properties (G1)–(G4) and the norm $\|E\|_j = \sup_X e^{\kappa_n d_j(X)}\|E(X)\|$ (Definition 50.1). A step produces a raw generation E' (Definition 53.3); its average over the symmetries of the grid is the new generation (Definition 50.7). The coefficient of a generation is read off from its period density, the quadratic part of its activities at the linear fields A_F of constant curvature, per grid cube: $\bar{T}_E = \beta_E \bar{T}_S$ (Definition 50.11, Proposition 50.13). Along the flow, $\|E^{(j)}\|_j \le \mathsf{K}$ and $|\beta_j| \le C_\beta \mathsf{K}$ (Theorem 53.12). As in Section 49, γ is the bound on the couplings, not a decay rate; (Γ1)–(Γ4) are the conditions on it of Remark 53.5.

The volume coefficient. Let E be a family of normalized graded activities at the polymers of scale j, on domains containing the $\mathcal{U}^\flat(X)$; for instance a family as in Definition 50.11, or a raw generation. We do not require it to be real. Put $\|E\| = \sup_X e^{\kappa_n d_j(X)} \sup_{\mathcal{U}^\flat(X)} |E(X,\cdot)|$, and, with (T_a) and $F^{\mu\nu,a}$ as in Proposition 50.13(c),

$$\bar{T}^{\mathrm{vol}}_E(F) = n_j^{-4} \sum_X c^X(F), \qquad \beta^{\mathrm{vol}}_E = \frac{1}{3\dim \mathfrak{g}} \sum_{\mu<\nu} \sum_a \bar{T}^{\mathrm{vol}}_E(F^{\mu\nu,a}), \tag{58.1}$$

the first sum running over the polymers that are not wrapping. The definition of c^X and Lemma 50.12(a), (c) use neither the reality of E nor (G2): the bound of (c) holds for each grid cube Q, and $\bar{T}^{\mathrm{vol}}_E(F) = n_j^{-4} \sum_Q \sum_{X \supset Q} c^X(F)/|X|$ is an average over the Q. So β^{vol}_E is linear in E, and, as in the proof of Proposition 50.13(b),

$$|\beta^{\mathrm{vol}}_E| \le C_\beta \|E\|, \tag{58.2}$$

with C_β of Proposition 50.13(b). If E has (G2) and (G4), then $\beta^{\mathrm{vol}}_E = \beta_E$, by Lemma 50.12(c) and Proposition 50.13(a). If E' is a raw generation, then $\beta_{E'^{\mathrm{sym}}} = \beta^{\mathrm{vol}}_{E'}$, by Proposition 50.13(c).

The one-loop part. Let E' be the raw new generation of the step from scale k to scale $k+1$ (Definition 53.3). Write $E' = E'_{\mathrm{ol}} + E'_{\#}$, with

$$E'_{\mathrm{ol}}(Y',U) = \sum_{X:\, \bar{X}=Y'} \mathfrak{e}(X,U) + \sum_{c:\, Y'_c = Y'} \log \mathfrak{j}_c(U),$$
$$E'_{\#}(Y',U) = \sum_{Y:\, \hat{Y}=Y'} \big(E^{\#}(Y,U) - E^{\#}(Y,1)\big),$$

the sums as in Definition 53.3. The one-loop part E'_{ol} involves neither the coupling, nor the old activities, nor the white-noise cutoff: it is a function of the reference alone, through the fluctuation operator and the base (Propositions 41.7 and 47.8; Proposition 52.2(c) at $k=0$). The proof of Lemma 53.4(a), (b), restricted to its first two terms, shows that E'_{ol} has (G1) on the domains $\mathcal{U}(Y')$ of scale $k+1$, with $\|E'_{\mathrm{ol}}\|^{\mathrm{raw}}_{k+1} \le \mathsf{K}$. The *one-loop part* of β_{k+1} is

$$\beta^{\mathrm{ol}}_{k+1} = \beta^{\mathrm{vol}}_{E'_{\mathrm{ol}}}.$$

It depends on k, on the torus and on the parameters fixed before γ, not on the couplings. Since $\beta_{k+1} = \beta^{\mathrm{vol}}_{E'}$ and β^{vol} is linear, $\beta_{k+1} = \beta^{\mathrm{ol}}_{k+1} + \beta^{\mathrm{vol}}_{E'_{\#}}$.

The properties of the flow used in the ultraviolet limit. With $\beta^{(1)}_{j+1}$ the coefficient of $T^{(1)}_{j+1}$ (Theorem 58.1):

(F1) If $g_0, \dots, g_k \in (0,\gamma]$, the densities $\rho_1, \dots, \rho_k$ of the flow have the generation form (Definition 53.6), with $g_{j+1}^{-2} = g_j^{-2} - \beta_{j+1}$. The coefficient β_{j+1} is a continuous function of $(g_0, \dots, g_j)$ on Σ_j, and $|\beta_{j+1}| \le C_\beta \mathsf{K}$.

(F2) $|\beta^{\mathrm{ol}}_{j+1} - \beta^{(1)}_{j+1}| \le C_2 e^{-c_2 n_{j+1}/M}$ for $j < K$, with $C_2, c_2 > 0$ independent of j, K and the torus.

(F3) $|\beta_{j+1} - \beta^{\mathrm{ol}}_{j+1}| \le C_3 g_j^2 p_0(g_j)^6$, uniformly in j and in the earlier couplings, for $(g_0, \dots, g_j) \in \Sigma_j$. Here p_0 is the white-noise cutoff of Section 49.

In Bałaban's formulation of (F2), the marginal parts of the old generations are cancelled before the one-loop part is read off. Here the cancellation is built into the step (Remark 58.7(a)).

58.3 Period density and vacuum polarization

Flat domains and the infinite lattice. Use the units of scale j. For a polymer X of scale j that is not wrapping and a field B on the bonds of a lift of $X^{(3)}$, let V_B be the configuration equal to $e^{\mathrm{i}\eta B}$ on the bonds of $X^{(3)}$, read in the lift, and to 1 elsewhere, as in the proof of Lemma 50.12(a). The *flat domain* $\mathcal{U}_1^\flat(X)$ is the set of the V_B with $|B| < \delta_0$ and $|\nabla B| + [\nabla B]_\alpha < \delta_1^\flat$ on $X^{(3)}$, the differences and quotients taken with the real part 1. By Definition 38.3 with the real part 1, $\mathcal{U}_1^\flat(X) \subset \mathcal{U}^\flat(X)$. The flat domain is defined by conditions on the bonds of $X^{(3)}$ alone, so it is the same set on every torus on which X does not wrap. The numbers $c^X(F)$ and the second variations at 1 use $E(X,\cdot)$ only on $\mathcal{U}_1^\flat(X)$ (proof of Lemma 50.12(a)).

A *polymer of scale j of the infinite lattice* is a finite $*$-connected union of cubes of side M centred at the points of $M\mathbb{Z}^4$, read on $\eta\mathbb{Z}^4$, $\eta = L^{-j}$. The tree length d_j, the number $|X|$ of cubes, $X^{(3)}$, the flat domains and (49.2) are as on the torus, and these polymers count as not wrapping. A family E^∞ of *flat activities* indexed by them consists of holomorphic functions $E^\infty(X,\cdot)$ on $\mathcal{U}_1^\flat(X)$ that depend only on B on the bonds of X, with $E^\infty(X,1) = 0$, real for real B, and invariant under small real gauge transformations: $E^\infty(X, V_{B'}) = E^\infty(X, V_B)$ whenever $V_B, V_{B'} \in \mathcal{U}_1^\flat(X)$ and $e^{\mathrm{i}\eta B'(b)} = e^{\mathrm{i}\lambda(b_-)} e^{\mathrm{i}\eta B(b)} e^{-\mathrm{i}\lambda(b_+)}$ with a small real λ. For such a family with (G2) and $\|E^\infty\| = \sup_X e^{\kappa_n d_j(X)} \sup_{\mathcal{U}_1^\flat(X)} |E^\infty(X,\cdot)| < \infty$, the numbers $c^X(F)$ are defined as in Definition 50.11, and we put, for a grid cube Q,

$$\bar{T}_{E^\infty}(F) = M^{-4} \sum_{X \supset Q} \frac{c^X(F)}{|X|}, \qquad \beta^\infty_{E^\infty} = \frac{1}{3 \dim \mathfrak{g}} \sum_{\mu<\nu} \sum_a \bar{T}_{E^\infty}(F^{\mu\nu,a}).$$

By (G2) and the proofs of Lemma 50.12(a), (c), which use only these properties, the sums converge absolutely and do not depend on Q. Let $\widetilde{\Pi}$ be the kernel, in the trace pairing $\sum_{b,b'} \widetilde{\Pi}(b,b') \operatorname{tr}(A(b)A(b'))$, of the second variation at $A = 0$ of $A \mapsto \sum_X E^\infty(X, e^{\mathrm{i}\eta A})$, for small fields A of finite support; only the polymers that contain a bond of the support of A contribute, since $E^\infty(X,\cdot)$ depends only on the bonds of X and $E^\infty(X,1) = 0$, and the sum converges absolutely by (49.2). The kernel is invariant under the translations by $M\mathbb{Z}^4$, and $\widetilde{\Pi}^{\mathrm{av}} = M^{-4} \sum_a \tau_a \widetilde{\Pi} \tau_{-a}$, the average over $a \in \{0, \dots, M-1\}^4$, is invariant under those by $\mathbb{Z}^4$.

Torus-local families. Let $\mathfrak{N}_j$ be the set of the sides $\tilde{n}$ for which the torus whose lattice of scale j has side $\tilde{n}$ satisfies the hypotheses of the Setting of Section 50 at the scales $\le j$ (with $K = j$). A collection $(E[\tilde{n}])_{\tilde{n} \in \mathfrak{N}_j}$, where $E[\tilde{n}]$ is a family of normalized graded activities at the polymers of scale j of the torus of side $\tilde{n}$, on domains containing the $\mathcal{U}^\flat(X)$, real at real configurations, with $\sup_{\tilde{n}} \|E[\tilde{n}]\| < \infty$, is *torus-local* if there are functions $E^\infty(X,\cdot)$ on the flat domains of the polymers X of scale j of the infinite lattice, with $E^\infty(X + a, \tau_a U) = E^\infty(X, U)$ for $a \in M\mathbb{Z}^4$, and constants $C, c > 0$, such that

$$|E[\tilde{n}](X,U) - E^\infty(\tilde{X}, \tilde{U})| \le C e^{-c\tilde{n}/M} e^{-\kappa_n d_j(X)} \tag{58.3}$$

for every $\tilde{n} \in \mathfrak{N}_j$, every polymer X of the torus of side $\tilde{n}$ that is not wrapping, every lift $\tilde{X}$ of X (Section 50.2) and every $U \in \mathcal{U}_1^\flat(X)$, with $\tilde{U} \in \mathcal{U}_1^\flat(\tilde{X})$ its lift. We call E^∞ its *limit*.

One-loop locality. The one-loop parts of the raw new generations form torus-local families, with no error term. This rests on the torus independence of the one-loop pieces and of the Jacobian terms (Section 41.5).

Lemma 58.3 (One-loop locality)**.** *Let $k \ge 0$. For $\tilde{n} \in \mathfrak{N}_{k+1}$ let $E'_{\mathrm{ol}}[\tilde{n}]$ be the one-loop part of the raw new generation of the step from scale k to scale $k+1$ on the torus whose lattice of*

scale $k+1$ has side $\tilde{n}$. Then the collection $(E'_{\mathrm{ol}}[\tilde{n}])_{\tilde{n}\in\mathfrak{N}_{k+1}}$ is torus-local, and $\|E'_{\mathrm{ol}}[\tilde{n}]\| \le \mathsf{K}$ for every $\tilde{n}$. Its limit E^∞ satisfies

$$E'_{\mathrm{ol}}[\tilde{n}](Y',U) = E^\infty(\tilde{Y}',\tilde{U})$$

for every $\tilde{n}\in\mathfrak{N}_{k+1}$, every polymer Y' of scale $k+1$ of the torus of side $\tilde{n}$ that is not wrapping, every lift $\tilde{Y}'$ of Y' and every $U\in\mathcal{U}_1^\flat(Y')$, with its lift $\tilde{U}$. So the left side of (58.3) vanishes, and the bound holds for all $C,c>0$.

Proof. Let $\tilde{n}\in\mathfrak{N}_{k+1}$. By Lemma 52.1(a), the torus satisfies (T0)–(T2) at step k, so its side at scale k, which is $L\tilde{n}$, is admissible in the sense of Section 41.5, and Proposition 41.14 applies to it (with the pieces of Proposition 52.2(c) if $k=0$). The activities $E'_{\mathrm{ol}}[\tilde{n}](Y',\cdot)$ are normalized and real at real configurations, on the domains $\mathcal{U}(Y')\supset\mathcal{U}^\flat(Y')$ of scale $k+1$, and $\|E'_{\mathrm{ol}}[\tilde{n}]\| \le \|E'_{\mathrm{ol}}[\tilde{n}]\|^{\mathrm{raw}}_{k+1} \le \mathsf{K}$ (paragraph "The one-loop part" of Section 58.2), with K independent of the torus. Details

The closure $\bar{X}=\mathrm{cl}_{k+1}X$ (Section 49.3) and the polymer Y'_c, the union of the grid cubes of scale $k+1$ that meet $B'(c_-)\cup B'(c_+)$ (the paragraph before Definition 53.3), are given by rules that are local in the grids of the scales k and $k+1$. So they are defined on the infinite lattice, and on sets that do not wrap they commute with the projection. For a polymer $\tilde{Y}'$ of scale $k+1$ of the infinite lattice and $\tilde{U}=V_B\in\mathcal{U}_1^\flat(\tilde{Y}')$ let $E^\infty(\tilde{Y}',\tilde{U})$ be the sum of the values $E_k^\infty(\tilde{X},\cdot)$ over the polymers $\tilde{X}$ of scale k of the infinite lattice with $\mathrm{cl}_{k+1}\tilde{X}=\tilde{Y}'$, and of the values $J_k^\infty(\tilde{c},\cdot)$ over the positively oriented bonds $\tilde{c}$ of $L\mathbb{Z}^4$ with $\tilde{Y}'_{\tilde{c}}=\tilde{Y}'$, at the restrictions of the field B, read in the units of step k (Proposition 41.14(d)). These sums are finite, since $\tilde{X}$ and $P_{\tilde{c}}$ lie in $\tilde{Y}'$, and the restrictions lie in the domains of E_k^∞ and J_k^∞, by Remark 41.15(a). Let Y' be a polymer of scale $k+1$ of the torus that is not wrapping, and $\tilde{Y}'$ a lift. The polymers X with $\bar{X}=Y'$ and the sets P_c of Lemma 41.11(d) with $Y'_c=Y'$ lie in Y', so lifting into $\tilde{Y}'$ is a bijection from them onto the $\tilde{X}$ and $\tilde{c}$ of the sums above. By Proposition 41.14(d), each term of $E'_{\mathrm{ol}}[\tilde{n}](Y',U)$ is the value of E_k^∞ or J_k^∞ at the lift, so $E'_{\mathrm{ol}}[\tilde{n}](Y',U)=E^\infty(\tilde{Y}',\tilde{U})$ (Remark 41.15(a)). Finally, a translation by $a\in M\mathbb{Z}^4$ in the units of scale $k+1$ is a translation by $La\in M\mathbb{Z}^4$ in those of step k; it commutes with the closure and with $c\mapsto Y'_c$, so the grid covariance of E_k^∞ and J_k^∞ in Proposition 41.14(d) gives $E^\infty(\tilde{Y}'+a,\tau_a\tilde{U})=E^\infty(\tilde{Y}',\tilde{U})$. □

For a bond b let m_b be its midpoint and $\mu(b)$ its direction.

Lemma 58.4 (Period density and vacuum polarization)**.** *Let G be compact, connected and simple, let $1\le j\le K$, and use the units of scale j.*

(a) (A single polymer.) *Let E be a family as in Definition 50.11, and X a polymer that is not wrapping. There is a unique real symmetric kernel κ_X on the positively oriented bonds of a lift of X such that $\frac12\frac{d^2}{dt^2}E(X,V_{tB})|_{t=0}=\sum_{b,b'}\kappa_X(b,b')\,\mathrm{tr}(B(b)B(b'))$ for all $\mathfrak{g}^c$-valued B. It satisfies $\sum_{b'}\kappa_X(b,b')\,\partial\varphi(b')=0$ for every scalar function φ and every b. Put, for $p\in\mathbb{R}^4$,*

$$m_X(p)_{\mu\nu}=-\frac14\sum_{b,b'}\kappa_X(b,b')\big(p\cdot(m_b-m_{b'})\big)^2\big(\delta_{\mu(b)\mu}\delta_{\mu(b')\nu}+\delta_{\mu(b)\nu}\delta_{\mu(b')\mu}\big).$$

Then, for $p,a\in\mathbb{R}^4$ and $T\in\mathfrak{g}^c$, with the curvature datum $(p\wedge a\otimes T)_{\mu\nu}=(p_\mu a_\nu-p_\nu a_\mu)T$,

$$c^X(p\wedge a\otimes T)=a^{\mathsf{T}}m_X(p)\,a\;\mathrm{tr}(T^2).$$

(b) (The fine tensor.) *Put $T_E^{\mathrm{f}}(p)=2n_j^{-4}\sum_X m_X(p)$, the sum over the polymers that are not wrapping. Then:*

(i) $\bar{T}_E^{\mathrm{vol}}(p\wedge a\otimes T)=\frac12 a^{\mathsf{T}}T_E^{\mathrm{f}}(p)\,a\;\mathrm{tr}(T^2)$;

(ii) the isotropic coefficient of T^{f}_E, in the sense of Definition 54.7, is β^{vol}_E.

(c) (The coarse tensor.) *Let E^∞ be a family of flat activities of scale j of the infinite lattice with (G2) and $\|E^\infty\| < \infty$, and put $T^{\mathrm{f}}_{E^\infty}(p) = 2M^{-4}\sum_{X\supset Q} m_X(p)/|X|$. Then (a) and (b) hold with $\bar T_{E^\infty}$, $T^{\mathrm{f}}_{E^\infty}$, $\beta^\infty_{E^\infty}$ in place of $\bar T^{\mathrm{vol}}_E$, T^{f}_E, β^{vol}_E, and $T^{\mathrm{f}}_{E^\infty}(p)$ is the coefficient of order $|p|^2$ of $\widetilde\Pi^{\mathrm{av}}(p,p)$, the diagonal Bloch component per unit volume in the bond-midpoint convention. Let $\mathbf{H}_\infty$ be the infinite-volume minimizer map of Proposition 54.15, and suppose that $\Pi_\Phi = \mathbf{H}_\infty^{\mathsf T}\widetilde\Pi\mathbf{H}_\infty$ is invariant under the translations of $\mathbb{Z}^4$ and under the point reflection $x\mapsto -x$. Then $\Pi_\Phi(p) = O(|p|^2)$, and the marginal tensor of Π_Φ (Definition 54.7) is $T^{\mathrm{f}}_{E^\infty}$. In particular its isotropic coefficient is $\beta^\infty_{E^\infty}$.*

(d) (Torus-local families.) *Let $(E[\tilde n])_{\tilde n\in\mathfrak{N}_j}$ be torus-local, with limit E^∞, and $\mathsf K' = \sup_{\tilde n}\|E[\tilde n]\|$. Then E^∞ is a family of flat activities with (G2) and $\|E^\infty\|\le \mathsf K'$, and there are $C', c' > 0$, depending only on d, M, δ_1, $\mathsf K'$ and the constants of (58.3), with*

$$\big|\beta^{\mathrm{vol}}_{E[\tilde n]} - \beta^\infty_{E^\infty}\big| \le C' e^{-c'\tilde n/M} \qquad (\tilde n\in\mathfrak{N}_j).$$

Moreover, the kernels of the second variations at $A=0$ of $A\mapsto\sum_X E[\tilde n](X,e^{\mathrm{i}\eta A})$ on the tori converge to $\widetilde\Pi$ with uniform decay, in the sense of Lemma 54.10(e): entrywise as $\tilde n\to\infty$, and with a bound $C_\eta e^{-\gamma|m_b - m_{b'}|_{\tilde n}/M}$, uniform in $\tilde n$, where $|\cdot|_{\tilde n}$ is the distance on the torus and C_η, $\gamma>0$ may depend on η.

The tensor T^{f}_E is the p^2-tensor of the total second variation of the pieces of E in the fine field, before the composition with the background field. By (b) it determines the isotropic part β^{vol}_E of the period density, and $\bar T^{\mathrm{vol}}_E$ on the data $p\wedge a\otimes T$. Since a quadratic form on $\Lambda^2\mathbb{R}^4$ that vanishes on the decomposable bivectors is a multiple of the form of Pf (the Plücker relation), it determines $\bar T^{\mathrm{vol}}_E$ up to a multiple of Pf, the part that the symmetrization removes (Remark 50.14(a)). So the caveat of Remark 50.14(c), that β_E is a property of the family, does not matter for the one-loop part: $\beta^{\mathrm{ol}}_{k+1}$ is the isotropic part of $\bar T^{\mathrm{vol}}_{E'_{\mathrm{ol}}}$ (Proposition 50.13(c)), determined by the fine tensor of the one-loop pieces, and by Lemma 58.3 and (d) it is, up to $O(e^{-n_{k+1}/(4M)})$, the coefficient $\beta^\infty_{E^\infty}$ of the infinite-volume limit E^∞ of the one-loop parts. For the plaquette family of Lemma 50.12(d), $\bar T^{\mathrm{vol}} = \bar T_S = \frac12\,\mathrm{Iso}$, and (b)(i) gives $T^{\mathrm{f}}(p)_{\mu\nu} = \delta_{\mu\nu}|p|^2 - p_\mu p_\nu$, the marginal tensor of S in the trace pairing (proof of Lemma 57.7). So the coefficients β_E, defined relative to $\bar T_S$, and the isotropic coefficients of Part XI are normalized alike, with conversion factor 1.

Proof. (a) As in the proof of Lemma 50.12(a), $\Phi_X(B) = E(X,V_B)$ is holomorphic for small B, depends only on B on the bonds of X, and $d\Phi_X(0)=0$. For constant $v\in G$ near 1, $V_{vBv^{-1}} = (V_B)^v$ on the bonds of $X^{(3)}$, so $\frac12 d^2\Phi_X(0)$ is a quadratic form in $(B(b))_b$ invariant under the simultaneous adjoint action of these v, hence of G, which is connected. Each block of its polar form, $(T,T')\mapsto k_{bb'}(T,T')$, is then an invariant bilinear form on $\mathfrak{g}^c$, and, as in the proof of Lemma 50.10, $k_{bb'}(T,T') = \kappa_X(b,b')\,\mathrm{tr}(TT')$, with κ_X symmetric, and real since E is real at real configurations and $\mathrm{tr}(T^2)>0$ for $0\ne T\in\mathfrak{g}$. The proof of Lemma 50.12(a) gives $d^2\Phi_X(0)[B,\partial\lambda]=0$ for all B and λ; with B supported on one bond and $\lambda=\varphi T$, this is the second identity. Details

Let $F = p\wedge a\otimes T$, and take the origin 0 in the lift. On a positively oriented bond b of direction ν, $A_F(b) = \frac12\sum_\kappa (m_b)_\kappa(p_\kappa a_\nu - p_\nu a_\kappa)T = \frac12((p\cdot m_b)a_\nu - (a\cdot m_b)p_\nu)T$. Let $\varphi(x) = \frac12(p\cdot x)(a\cdot x)$. Since $b_\pm = m_b \pm \frac12\eta e_\nu$,

$$\partial\varphi(b) = \eta^{-1}(\varphi(b_+) - \varphi(b_-)) = \tfrac12((p\cdot m_b)a_\nu + (a\cdot m_b)p_\nu)$$

exactly. Hence $A_F = (B_1 - \partial\varphi)T$, with $B_1(b) = (p\cdot m_b)a_{\mu(b)}$. Write $\kappa_X(B,B') = \sum_{b,b'}\kappa_X(b,b')B(b)B'(b')$ for scalar fields. By the second identity, $c^X(F) = \frac12 d^2\Phi_X(0)[A_F,A_F] =$

$\kappa_X(B_1,B_1)\,\mathrm{tr}(T^2)$. Let $B_0(b)=a_{\mu(b)}=\partial\psi(b)$ with $\psi(x)=a\cdot x$, and $B_2(b)=(p\cdot m_b)^2 a_{\mu(b)}$. Expanding the square,

$$\sum_{b,b'}\kappa_X(b,b')\,a_{\mu(b)}a_{\mu(b')}\big(p\cdot m_b-p\cdot m_{b'}\big)^2=\kappa_X(B_2,B_0)-2\kappa_X(B_1,B_1)+\kappa_X(B_0,B_2)$$
$$=-2\kappa_X(B_1,B_1),$$

by the second identity and the symmetry of κ_X. By the definition of m_X, the left side is $-2a^{\mathsf T}m_X(p)a$. So $a^{\mathsf T}m_X(p)a=\kappa_X(B_1,B_1)$, which gives (a). The kernel and m_X do not depend on the lift, since only differences of midpoints enter.

(b) (i) Sum (a) over X. For (ii): by Lemma 50.12(a), $\bar T_E^{\mathrm{vol}}$ is a quadratic form, invariant under conjugation, so, by the first part of the proof of Lemma 50.10, $\bar T_E^{\mathrm{vol}}(F)=\sum_{\mu<\nu}\sum_{\kappa<\lambda}Q(e_{\mu\nu},e_{\kappa\lambda})\,\mathrm{tr}(F_{\mu\nu}F_{\kappa\lambda})$ with a symmetric bilinear form Q on $\Lambda^2\mathbb{C}^4$; for $w\in\Lambda^2\mathbb{R}^4$, $\bar T_E^{\mathrm{vol}}(w\otimes T)=Q(w,w)\,\mathrm{tr}(T^2)$. By (i), $Q(p\wedge a,p\wedge a)=\frac12 a^{\mathsf T}T_E^{\mathrm f}(p)a$ for real p,a, and, polarizing in a,

$$T_E^{\mathrm f}(p)_{\mu\nu}=2Q(p\wedge e_\mu,p\wedge e_\nu).\tag{58.4}$$

(ii) For $R\in B_4$ put $Q^R(w,w')=Q(Rw,Rw')$, with R acting on Λ^2, and $T^R(p)=R^{\mathsf T}T(Rp)R$ for p^2-tensors T (in this part of the proof T is a tensor, not the element $T\in\mathfrak g^c$ of the statement). By (58.4), the tensor built from Q^R is $(T_E^{\mathrm f})^R$, since $2Q(Rp\wedge Re_\mu,Rp\wedge Re_\nu)=\sum_{\alpha,\beta}R_{\alpha\mu}R_{\beta\nu}\,2Q(Rp\wedge e_\alpha,Rp\wedge e_\beta)$. The map $T\mapsto T^R$ is the action of R on the indices of the coefficient tensor, which is orthogonal for the Euclidean inner product, and it fixes the tensor of $\delta_{\mu\nu}|p|^2-p_\mu p_\nu$. So the isotropic coefficient of $T_E^{\mathrm f}$ is that of the tensor built from $\bar Q=|B_4|^{-1}\sum_R Q^R$. The form $\bar Q$ defines a quadratic form on the curvature data that is invariant under B_4 and under conjugation, so $\bar Q=\beta' I$ by Lemma 50.10(a), with $I(w,w')=\sum_{\mu<\nu}w_{\mu\nu}w'_{\mu\nu}$, the form of Iso. Its tensor is $2\beta' I(p\wedge e_\mu,p\wedge e_\nu)=2\beta'(\delta_{\mu\nu}|p|^2-p_\mu p_\nu)$, so the isotropic coefficient of $T_E^{\mathrm f}$ is $2\beta'$. On the other hand, every $R\in B_4$ permutes the $e_{\mu\nu}$ up to signs, so $\sum_{\mu<\nu}Q(e_{\mu\nu},e_{\mu\nu})=\sum_{\mu<\nu}\bar Q(e_{\mu\nu},e_{\mu\nu})=6\beta'$, and, with $\mathrm{tr}(T_a^2)=1$, $\beta_E^{\mathrm{vol}}=(3\dim\mathfrak g)^{-1}\dim\mathfrak g\cdot 6\beta'=2\beta'$.

(c) Part (a) concerns one polymer and holds verbatim, since its proof uses $E(X,\cdot)$ only on the flat domain and its invariance under small gauge transformations. Part (b) holds with the cell sums: (i) follows from (a), summed with the weights $M^{-4}|X|^{-1}$, and (ii) is a statement about the forms $\bar T_{E^\infty}$ and $T^{\mathrm f}_{E^\infty}$ alone. The sums converge absolutely: by Cauchy's estimate, as in Lemma 50.12(a), $|\kappa_X(b,b')|\le C_\eta\|E^\infty\|e^{-\kappa_n d_j(X)}$ with C_η depending on η and the domain constants, a polymer that contains b and b' has $d_j(X)\ge|m_b-m_{b'}|/M-2$ (as in Lemma 52.1(b)), and (49.2) applies. In particular $\widetilde\Pi=2\sum_X\kappa_X$ has exponential decay. The diagonal Bloch component per unit volume is $\widetilde\Pi^{\mathrm{av}}(p,p)_{\mu\nu}=M^{-4}\sum_{b,b'}e^{-\mathrm{i}p\cdot(m_b-m_{b'})}\widetilde\Pi(b,b')$, the sum over the bonds b of direction μ with midpoints in an M-cell and the bonds b' of direction ν. By (G2), every orbit of the translations by $M\mathbb{Z}^4$ on the polymers contains exactly $|X|$ polymers that contain Q. Hence

$$\widetilde\Pi^{\mathrm{av}}(p,p)_{\mu\nu}=2M^{-4}\sum_{X\supset Q}\frac{1}{|X|}\sum_{b,b'}e^{-\mathrm{i}p\cdot(m_b-m_{b'})}\kappa_X(b,b')\,\delta_{\mu(b)\mu}\delta_{\mu(b')\nu},$$

the inner sum over all pairs of bonds of X. Expand the exponential. The term of order 0 vanishes by the second identity of (a), since a constant field is a gradient. The term of order 1 is antisymmetric in μ,ν, by the symmetry of κ_X. The term of order 2 is symmetric in μ,ν and equals $T^{\mathrm f}_{E^\infty}(p)$. So $\widetilde\Pi^{\mathrm{av}}(p,p)=\mathsf l(p)+T^{\mathrm f}_{E^\infty}(p)+O(|p|^3)$, with $\mathsf l$ linear and antisymmetric.

The operator $\mathbf H_\infty$ lies in $\mathscr P$ (Proposition 54.15(a)), so it commutes with the translations of $\mathbb{Z}^4$; since Π_Φ is invariant under them, $\mathbf H_\infty^{\mathsf T}\widetilde\Pi^{\mathrm{av}}\mathbf H_\infty=\Pi_\Phi$. The kernel $\widetilde\Pi^{\mathrm{av}}$ is real and symmetric, since the κ_X are, and it is invariant under the translations of $\mathbb{Z}^4$, with exponential decay. It annihilates the constant fields, by the second identity of (a), since a constant field is the

gradient of a linear function. So Lemma 56.3, with j in place of k, applies to $\widetilde{\Pi}^{\mathrm{av}}$. Its first part gives $\Pi_\Phi(p) = \widetilde{\Pi}^{\mathrm{av}}(p,p) + O(|p|^3)$; its proof uses Proposition 54.15(e) and Corollary 54.16, and the Umklapp sum is finite. The point reflection gives $\Pi_\Phi(-p) = \Pi_\Phi(p)$, by the computation in the proof of Lemma 54.9(c). So $\mathsf{I} = 0$, $\Pi_\Phi(p) = O(|p|^2)$, and the marginal tensor of Π_Φ is $T^{\mathrm{f}}_{E^\infty}$. Its isotropic coefficient is $\beta^\infty_{E^\infty}$, by (b)(ii) in the form just proved.

(d) Let X be a polymer of the infinite lattice. For $\tilde{n}$ large, depending on X, its projection to the torus of side $\tilde{n}$ is a polymer that is not wrapping, with the same cubes and the same tree length, since the projection is isometric on sets of diameter less than $\frac{1}{2}\tilde{n}$, and X is a lift of it. By (58.3), $E^\infty(X,\cdot)$ is the uniform limit on $\mathcal{U}_1^\flat(X)$ of the functions $E[\tilde{n}]$ of the projections, read on the lift. So it is holomorphic, depends only on B on the bonds of X, and is normalized, real and invariant under small real gauge transformations, as these functions are; and $|E^\infty(X,\cdot)| \le \mathsf{K}' e^{-\kappa_n d_j(X)}$. Property (G2) for E^∞, the covariance $E^\infty(X+a,\tau_a U) = E^\infty(X,U)$ for $a \in M\mathbb{Z}^4$, is part of the definition of a torus-local family.

Let $\tilde{n} \in \mathfrak{N}_j$, let Q be a grid cube of the torus and $\tilde{Q}$ a lift. Lifting with $Q \mapsto \tilde{Q}$ is a bijection from the non-wrapping polymers $X \supset Q$ of the torus onto the polymers $\tilde{X} \supset \tilde{Q}$ of the infinite lattice of diameter at most $\frac{1}{2}\tilde{n} - 8$. It preserves $|X|$. For such X and $|F| \le 1$, the function $t \mapsto E[\tilde{n}](X, V^X_{t,F}) - E^\infty(\tilde{X}, V^{\tilde{X}}_{t,F})$, with the origin at the lift of a point of X, is holomorphic for $|t| < \delta_1^\flat/(dD_X)$ and bounded there by $Ce^{-c\tilde{n}/M}e^{-\kappa_n d_j(X)}$, by Lemma 50.12(a) and (58.3). Cauchy's estimate gives

$$|c^X_{E[\tilde{n}]}(F) - c^{\tilde{X}}_{E^\infty}(F)| \le 9d^2M^2(\delta_1^\flat)^{-2}\big(1 + d_j(X)\big)^2 Ce^{-c\tilde{n}/M}e^{-\kappa_n d_j(X)}.$$

The polymers $\tilde{X} \supset \tilde{Q}$ of larger diameter have $d_j(\tilde{X}) \ge \tilde{n}/(4M)$, as in Lemma 52.1(b), and $|c^{\tilde{X}}_{E^\infty}(F)| \le 9d^2M^2(\delta_1^\flat)^{-2}(1+d_j(\tilde{X}))^2\mathsf{K}'e^{-\kappa_n d_j(\tilde{X})}$. Since $(1+t)^2e^{-(\kappa_n-\kappa_0)t} \le e^{-(\kappa_n-\kappa_0-2)t}$ and $\kappa_n - \kappa_0 \ge 3$, (49.2) gives

$$\Big|\sum_{X\supset Q}\frac{c^X_{E[\tilde{n}]}(F)}{|X|} - \sum_{\tilde{X}\supset\tilde{Q}}\frac{c^{\tilde{X}}_{E^\infty}(F)}{|\tilde{X}|}\Big| \le 9d^2M^2(\delta_1^\flat)^{-2}K_0\big(Ce^{-c\tilde{n}/M} + \mathsf{K}'e^{-\tilde{n}/(4M)}\big),$$

uniformly in Q. The second sum is $M^4\bar{T}_{E^\infty}(F)$, and $\bar{T}^{\mathrm{vol}}_{E[\tilde{n}]}(F)$ is M^{-4} times the average over Q of the first. So $|\bar{T}^{\mathrm{vol}}_{E[\tilde{n}]}(F) - \bar{T}_{E^\infty}(F)| \le 9d^2K_0M^{-2}(\delta_1^\flat)^{-2}(C+\mathsf{K}')e^{-c'\tilde{n}/M}$ for $|F| \le 1$, with $c' = \min(c, \frac{1}{4})$. Since $|F^{\mu\nu,a}| = 1$ and there are $6\dim\mathfrak{g}$ such data, the bound on $\beta^{\mathrm{vol}}_{E[\tilde{n}]} - \beta^\infty_{E^\infty}$ follows, with $C' = 18d^2K_0M^{-2}(\delta_1^\flat)^{-2}(C+\mathsf{K}')$. The kernels are compared in the same way: the torus kernel at two bonds is the sum, over the polymers that contain both, of the kernels of the second variations of $E[\tilde{n}](X, e^{\mathrm{i}\eta A})$ ($2\kappa_X$ for X not wrapping); the wrapping ones, by the same Cauchy estimate with $d_j(X) \ge \tilde{n}/(4M)$ (Lemma 52.1(b)), and the polymers of the infinite lattice of diameter larger than $\frac{1}{2}\tilde{n} - 8$, contribute $O(e^{-c'\tilde{n}/M})$, and the others differ by $O(e^{-c\tilde{n}/M})$, by Cauchy's estimate, with constants depending on η. For the bound, by the same Cauchy estimate each of these kernels is at most $C_\eta\mathsf{K}'e^{-\kappa_n d_j(X)}$ at b, b', and a polymer of the torus that contains b and b' has $d_j(X) \ge |m_b - m_{b'}|_{\tilde{n}}/M - 2$, as in Lemma 52.1(b). With (49.2), the torus kernel is at most $C_\eta K_0\mathsf{K}'e^{2(\kappa_n-\kappa_0)}e^{-(\kappa_n-\kappa_0)|m_b-m_{b'}|_{\tilde{n}}/M}$, uniformly in $\tilde{n}$. □

The step functions in infinite volume. For $j \ge 0$ let $\Phi_j(V') = \Gamma_j^{\mathrm{step}}(U_{j+1}(V'))$, a function of the configurations V' of scale $j+1$ near 1 (Section 54). Its marginal tensor is $T^{(1)}_{j+1}$.

Lemma 58.5 (Infinite-volume limits of the step functions)**.** *Let G be compact, connected and simple, let $j \ge 0$, and use the units of scale $j+1$, with $\eta = L^{-j-1}$. Let E^∞ be the limit of the torus-local family of Lemma 58.3 with $k = j$, $\widetilde{\Pi}$ the kernel of its second variation (Section 58.3), and $\mathbf{H}_\infty$ the infinite-volume minimizer map of Proposition 54.15 for the averaging to scale $j+1$. On the torus whose lattice of scale $j+1$ has side $\tilde{n} \in \mathfrak{N}_{j+1}$, the second variation*

of Φ_j at $V'=1$ is $\mathbf{H}^{\mathsf{T}}\Pi_{\tilde n}\mathbf{H}$, where $\Pi_{\tilde n}$ is the kernel of the second variation at $A=0$ of $A\mapsto\sum_{Y'}E'_{\mathrm{ol}}[\tilde n](Y',e^{\mathrm{i}\eta A})$. As $\tilde n\to\infty$ in $\mathfrak{N}_{j+1}$, these second variations converge with uniform decay (Lemma 54.10(e)) to

$$\Pi_{\Phi_j}=\mathbf{H}_\infty^{\mathsf{T}}\widetilde{\Pi}\mathbf{H}_\infty.$$

In particular the hypothesis of Lemma 54.9(c), and the hypothesis of Corollary 54.18 on the step functions, hold for Φ_j.

Proof. Let $\tilde n\in\mathfrak{N}_{j+1}$. On this torus, for small-field V' and $U'=U_{j+1}(V')$, Details

$$\Gamma_j^{\mathrm{step}}(U')-\Gamma_j^{\mathrm{step}}(1)=\sum_X\mathfrak{e}(X,U')+\sum_c\log\mathfrak{j}_c(U')=\sum_{Y'}E'_{\mathrm{ol}}[\tilde n](Y',U'),$$

by Proposition 41.7(a) and Proposition 47.8(b) (Proposition 52.2(c) at $j=0$), regrouped as in Definition 53.3. Let $V'=e^{\mathrm{i}B}$ with B small. As in the proof of Lemma 54.17, $U_{j+1}(V')$ is a gauge transform of $e^{\mathrm{i}\eta A_*(B)}$, with A_* analytic and $A_*(B)=\mathbf{H}B+O(|B|^2)$ (Corollary 30.5, Theorem 25.1(c)); by the proof of Lemma 54.17, $\mathbf{H}$ is the minimizer map of Part I. The function $A\mapsto\sum_{Y'}E'_{\mathrm{ol}}[\tilde n](Y',e^{\mathrm{i}\eta A})$ is invariant under real gauge transformations, since its activities are, and its first variation at $A=0$ vanishes, since it is invariant under the constant gauge transformations and G is semisimple. So only the linearization of A_* enters, and the second variation of Φ_j at $V'=1$ is $\mathbf{H}^{\mathsf{T}}\Pi_{\tilde n}\mathbf{H}$.

By Lemma 58.3, the family $(E'_{\mathrm{ol}}[\tilde n])_{\tilde n\in\mathfrak{N}_{j+1}}$ is torus-local, with limit E^∞. By Lemma 58.4(d), the kernels $\Pi_{\tilde n}$ converge to $\widetilde{\Pi}$ with uniform decay. By Proposition 54.15(a), the maps $\mathbf{H}$ on the tori converge with uniform decay to $\mathbf{H}_\infty$. By Lemma 54.10(e), applied to the transposes and to the two products, $\mathbf{H}^{\mathsf{T}}\Pi_{\tilde n}\mathbf{H}$ converges with uniform decay to $\mathbf{H}_\infty^{\mathsf{T}}\widetilde{\Pi}\mathbf{H}_\infty$. □

58.4 The properties (F1)–(F3)

Proposition 58.6 (The properties (F1)–(F3))**.** *Let G be compact, connected and simple, let the parameters and the torus be as in Section 53, and let γ satisfy (Γ1)–(Γ4) of Remark 53.5. Then (F1), (F2) and (F3) hold. In (F2),*

$$C_2=18d^2K_0M^{-2}(\delta_1^\flat)^{-2}\mathsf{K},\qquad c_2=\tfrac14,$$

the constants of Lemma 58.4(d) for the one-loop parts. In (F3),

$$C_3=C_\beta C_{\mathrm{new}}\epsilon_*(C_{\mathrm{pure}}+c_{\mathrm{F}}^{-2}),$$

where C_{pure} is a constant with $r_t^{-2}\le C_{\mathrm{pure}}p_0^6$ for $p_0\ge1$, with r_t of Proposition 48.8 (such a constant is given in the proof of that proposition, where it is written C_P), and c_{F} is that of (Γ4); for the first step the constants of the first step are used, and C_3 is the larger of the two values.

Proof. (F1). By Theorem 53.12(a), at every k with $g_1,\dots,g_k\le\gamma$, the densities of the flow have the generation form, with the recursion and $g_{j+1}\le2g_j$ of Theorem 53.12(e). The coefficients $\beta_{j+1}[\pi]$ are defined on Σ_j, with $|\beta_{j+1}|\le C_\beta\mathsf{K}$, by Proposition 53.10(a), and they are continuous there by Proposition 53.10(b); p_0 is continuous (Section 49.1). Details

(F3). In outline: replace the coupling in the fluctuation by a complex parameter t. The coefficient $\phi(t)$ of the part $E'_\#$ of the raw generation beyond its one-loop part is then holomorphic and even in t, vanishes at $t=0$, and is bounded for $|t|<r_{\mathrm{F}}$, where $r_{\mathrm{F}}^{-2}\le(C_{\mathrm{pure}}+c_{\mathrm{F}}^{-2})p_0^6$; the maximum principle gives $|\phi(g)|\le Cg^2r_{\mathrm{F}}^{-2}$. In detail, fix k and the sequence $g_0,\dots,g_k$, let $g=g_k$, $p_0=p_0(g)$, and let $\mathcal{E}_k$ be the old activities of step k (Definition 53.1; none if $k=0$). As in Section 48.4, replace g by a complex parameter t in the fluctuation only, $Z(s)=tC_k^{1/2}(s)\xi$, with t^{-2} in front of the interaction and gauge-fixing pieces, and keep the reference, the old

activities and $d\mu^*$. Let $r_{\rm F} = \min(r_t, c_{\rm F}/p_0)$, with r_t of Proposition 48.8 and $c_{\rm F}$ of (Γ4); then $2g \le r_{\rm F}$, by (Γ2) and (Γ4).

Let $|t| < r_{\rm F}$ and $|\xi|_\infty \le p_0$. The pure pieces satisfy the bounds of Proposition 48.5 with $\epsilon_{\rm p}(t) \le \frac{1}{2}\epsilon_*$ (Proposition 48.8). The proof of Lemma 48.2, which needs here only $|t|C_T p_0 \le \frac{1}{2} r_I \le r_L$ (the second entry of r_t), gives $|\hat{Z}(Z(s))|_\infty \le \zeta(t) = C_w L^d C_T |t| p_0$, and $\zeta(t) \le \frac{1}{2} R_P$ and $\tau(t) = C_P L^d |t| p_0/\delta_1 < \frac{1}{2}$ (with the C_P of Proposition 53.2(d)), by the definition of $c_{\rm F}$ in (Γ4). So Proposition 47.10 applies to every old activity with $\zeta = \zeta(t)$, as in Proposition 53.2(d), and the mixed pieces satisfy the bounds of Proposition 48.5 with $\epsilon_{\rm m}(t) = C_P \tau(t) C_{\rm old} \mathsf{K}$, where $\epsilon_{\rm m}(t) e^{c_{\natural}(\kappa_n - 1)} \le \frac{1}{4}\epsilon_*$. The pieces are analytic in t, jointly with the reference: the composed activities $f(X; s, Z)$ are analytic in Z, which is linear in t, and Lemma 47.2(c) keeps the analyticity. So Corollary 48.7 applies for every such t, with its bounds at $\epsilon_{\rm p}(t)$, $\epsilon_{\rm m}(t)$, at the real critical reference 1 and at the extensions of the points of the domains $\mathcal{U}(Y')$, through the restricted systems of Remark 47.11. Let $E'_{\#}(t)$ be the family $E'_{\#}$ built from these outputs. By Theorem 48.6(c) and the proof of Lemma 53.4(a), (b), $E'_{\#}(t)(Y', U)$ is holomorphic in $(t, U|_{Y'})$, normalized and gauge invariant, and

$$\|E'_{\#}(t)\|^{\rm raw}_{k+1} \le C_{\rm new}\big(\epsilon_{\rm p}(t) + \epsilon_{\rm m}(t)\big) \le C_{\rm new}\epsilon_* \qquad (|t| < r_{\rm F}).$$

Parity. Every piece depends on (t, ξ) only through t^{-2} and $Z(s)$: the pure pieces by the proof of Proposition 48.8, and the mixed pieces since $f(X; s, Z(s))$ depends only on $Z(s)$. So they are even under $(t, \xi) \mapsto (-t, -\xi)$, and $E'_{\#}(t)$ is even in t, by Theorem 48.6(d). At $t = 0$ every piece vanishes: the pure pieces by Proposition 48.8, and the composed activities since $Z = 0$ gives $\hat{Z}(0) = 0$, $A_*(s; 0) = 0$ (Theorem 45.5(a)), and so $f(X; s, 0) = \mathcal{E}_k(X, U') - \mathcal{E}_k(X, U') = 0$. Then every $K(Z) = e^{E(Z)} - 1$ in the proof of Theorem 48.6 vanishes, and $E'_{\#}(0) = 0$.

Cauchy. Put $\phi(t) = \beta^{\rm vol}_{E'_{\#}(t)}$. Each $c^X_{E'_{\#}(t)}(F)$ is a Cauchy integral, in the variable of $V^X_{\cdot,F}$, of a function holomorphic in t, and the series (58.1) converges uniformly in t, by the bound of Lemma 50.12(c). So ϕ is holomorphic on $|t| < r_{\rm F}$, even, with $\phi(0) = 0$, and $|\phi(t)| \le C_\beta C_{\rm new}\epsilon_*$ there, by (58.2). Hence $\phi(t)/t^2$ is holomorphic, and the maximum principle gives $|\phi(g)| \le C_\beta C_{\rm new}\epsilon_* (g/r_{\rm F})^2$. Moreover $r_{\rm F}^{-2} \le r_t^{-2} + p_0^2 c_{\rm F}^{-2} \le (C_{\rm pure} + c_{\rm F}^{-2}) p_0^6$, since $r_t^{-2} \le C_{\rm pure} p_0^6$ and $p_0 \ge 1$. At $t = g$ the outputs are those of the step, so $E'_{\#}(g) = E' - E'_{\rm ol}$ and $\beta_{k+1} - \beta^{\rm ol}_{k+1} = \phi(g)$. This is (F3). At $k = 0$ there is no mixed class, $r_{\rm F} = r_t$, and the constants are those of the first step.

(F2). Fix $k < K$. By Lemma 58.3, the one-loop parts $E'_{\rm ol}[\tilde{n}]$, $\tilde{n} \in \mathfrak{N}_{k+1}$, form a torus-local family, real at real configurations, with $\|E'_{\rm ol}[\tilde{n}]\| \le \mathsf{K}$, for which the left side of (58.3) vanishes; let E^∞ be its limit. The torus of the flow has its side n_{k+1} at scale $k+1$ in $\mathfrak{N}_{k+1}$. By Lemma 58.4(d),

$$|\beta^{\rm ol}_{k+1} - \beta^\infty_{E^\infty}| \le C_2 e^{-c_2 n_{k+1}/M}.$$

Indeed, by Lemma 58.3 the left side of (58.3) vanishes. So, in the proof of Lemma 58.4(d), the polymers of diameter at most $\frac{1}{2}\tilde{n} - 8$ contribute nothing to the difference, and only the polymers of the infinite lattice of larger diameter remain. With $\mathsf{K}' \le \mathsf{K}$ they give the bound with $C_2 = 18 d^2 K_0 M^{-2} (\delta_1^\flat)^{-2} \mathsf{K}$ and $c_2 = \frac{1}{4}$.

By Lemma 58.5 with $j = k$, the second variations at $V' = 1$ of $V' \mapsto \Gamma^{\rm step}_k(U_{k+1}(V'))$ on the tori of the sides in $\mathfrak{N}_{k+1}$ converge with uniform decay to $\Pi_\Phi = \mathbf{H}^{\mathsf{T}}_\infty \widetilde{\Pi} \mathbf{H}_\infty$. This is the infinite-volume second variation of Definition 54.7, so $T^{(1)}_{k+1}$ is the marginal tensor of Π_Φ. By Lemma 54.9(b), these functions are invariant under the translations and under B_4 about the coarse points, so their second variations at 1 are, and so is Π_Φ. Lemma 58.4(c) gives $T^{(1)}_{k+1} = T^{\rm f}_{E^\infty}$, with isotropic coefficient $\beta^\infty_{E^\infty}$. By Lemma 54.9(c), $T^{(1)}_{k+1} = \beta^{(1)}_{k+1}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, so $\beta^{(1)}_{k+1} = \beta^\infty_{E^\infty}$. With the bound above this is (F2). The constants C_2, c_2 depend only on d, M, κ_0 (through K_0), $\delta_1^\flat$ and K, not on k, K or the torus. □

Remark 58.7 (On (F1)–(F3)). (a) *The old generations.* The old generations enter the step only through the old activities $\mathcal{E}_k$, in which every counterterm $-\beta_j S_\square$ is grouped with the small polymers of its generation in the cube $\square$ (Definition 53.1). So their marginal parts are cancelled, for the generations with $k-j \geq m_0$ (Proposition 51.5), or grouped with their counterterm in bounded old activities, for the recent ones (Proposition 53.2), before the new generation is formed. The old activities do not enter E'_{ol}, and they enter β_{k+1} only through the mixed class in $E'_{\#}$, which (F3) bounds by $O(g_k^2 p_{0,k}^6)$.

(b) *Sizes.* Without the parity, Corollary 48.7 would give only $O(\epsilon_{\mathrm{p}}+\epsilon_{\mathrm{m}}) = O(g p_0^3)$ in (F3), which is not summable along the flow. The errors of (F2) are summable over the scales: $\sum_{j<K} e^{-c_2 n_{j+1}/M} \leq \sum_{m\geq 0} e^{-c_2 L^m}$, since $n_{j+1} = L^{K-j-1} n_K$ and $n_K \geq M$.

The next proposition uses Section 59. That section uses Parts IV–X, Lemma 54.10, and Sections 58.2 and 58.3 with Lemmas 58.3 and 58.4, but none of the results below.

Proposition 58.8 (The coefficients and Bałaban's β)**.** *Let G be compact, connected and simple, let $j \geq 1$, let γ satisfy (Γ1)–(Γ3), and let $(g_0, \dots, g_{j-1}) \in \Sigma_{j-1}$. For $\tilde{n} \in \mathfrak{N}_j$ let $E'[\tilde{n}]$, $E^{(j)}[\tilde{n}] = E'[\tilde{n}]^{\mathrm{sym}}$ and $\beta_j[\tilde{n}] = \beta_{E^{(j)}[\tilde{n}]}$ be the raw generation, the generation and the coefficient of scale j constructed with these couplings on the torus whose lattice of scale j has side $\tilde{n}$ (Proposition 53.10, with $K = j$). Then:*

(a) the second variations at $V = 1$ of $\Phi_{E^{(j)}[\tilde{n}]}$ converge as $\tilde{n} \to \infty$ in $\mathfrak{N}_j$, with uniform decay (Lemma 54.10(e)), to a kernel Π_Φ with exponential decay, invariant under the translations and under B_4, as Definition 54.7 assumes;

(b) its marginal tensor is $\beta_j^{\mathrm{B}}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, where $\beta_j^{\mathrm{B}} = -\partial_{p_\mu}\partial_{p_\nu}\Pi_\Phi(p)_{\mu\nu}|_{p=0}$, $\mu \neq \nu$, is the coefficient that Bałaban's definition [18, (1.20)–(1.22)] assigns to the family $(E^{(j)}[\tilde{n}])_{\tilde{n}\in\mathfrak{N}_j}$ built with these couplings, normalized relative to S (Definition 54.7). It equals β_j^∞ of Theorem 59.12, and it depends only on $(g_0, \dots, g_{j-1})$. Bałaban's own generation is a different construction, which depends only on g_{j-1};

(c) $\beta_j[\tilde{n}] \to \beta_j^{\mathrm{B}}$ as $\tilde{n} \to \infty$ in $\mathfrak{N}_j$.

Proof. Let h' be a threshold at which $E'[\tilde{n}]$ is a raw generation (Theorem 53.7(b) with $k = j-1$; for $j = 1$, Proposition 53.8). By Lemma 50.8(b), $\Phi_{E^{(j)}[\tilde{n}]} = \Phi_{E'[\tilde{n}]}$ on $\mathfrak{U}_j(h')$, which contains a neighbourhood of $V = 1$, so their second variations at $V = 1$ agree. As in the proof of Lemma 58.5, with $\Phi_{E'[\tilde{n}]}(V) = \sum_Y E'[\tilde{n}](Y, U_j(V))$ in place of Φ_j, the second variation is $\mathbf{H}^{\mathsf{T}}\Pi_{\tilde{n}}\mathbf{H}$, where $\Pi_{\tilde{n}}$ is the kernel of the second variation at $A = 0$ of $A \mapsto \sum_Y E'[\tilde{n}](Y, e^{\mathrm{i}\eta A})$. By Theorem 59.12(a), the collection $(E'[\tilde{n}])$ converges on the flat domains $\mathcal{U}_1$ and has the covariance of Lemma 59.11(b); its members are as in Lemma 59.11(c), with $\|E'[\tilde{n}]\| \leq \|E'[\tilde{n}]\|_j^{\mathrm{raw}} \leq \mathsf{K}$. So, by Lemma 59.11(c), the restriction of its limit $E'^{(j),\infty}$ to the domains $\mathcal{U}_1^\flat$ is a family of flat activities with (G2), and the kernels $\Pi_{\tilde{n}}$ converge with uniform decay to its kernel $\widetilde{\Pi}$. By Proposition 54.15(a), the maps $\mathbf{H}$ on the tori converge with uniform decay to $\mathbf{H}_\infty$. By Lemma 54.10(e), applied to the transposes and to the two products, the second variations converge with uniform decay to $\Pi_\Phi = \mathbf{H}_\infty^{\mathsf{T}}\widetilde{\Pi}\mathbf{H}_\infty$. By (G3) for $E'[\tilde{n}]$ (Theorem 53.7(b), and Proposition 53.8 for $j = 1$), $\Phi_{E'[\tilde{n}]}$ is invariant under $\mathfrak{S}_j$ near $V = 1$, so the kernels and Π_Φ are invariant under the translations and under B_4. By Lemma 58.4(c), the marginal tensor is $T^{\mathrm{f}}_{E'^{(j),\infty}}$, and by the tensor algebra in the proof of Lemma 54.9(c), with the transversality of Π_Φ from Remark 54.8 ($\Phi_{E'[\tilde{n}]}$ is invariant under real gauge transformations, by Lemma 53.4(a) and the paragraph before Definition 50.1, and its first variation at 1 vanishes, since G is semisimple), it is isotropic, hence equal to $\beta^\infty_{E'^{(j),\infty}}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. For $\mu \neq \nu$ the entry $\mu\nu$ of this tensor is $-\beta^\infty_{E'^{(j),\infty}} p_\mu p_\nu$, and $\Pi_\Phi(p)$ differs from the tensor by $O(|p|^3)$; so $\beta_j^{\mathrm{B}} = \beta^\infty_{E'^{(j),\infty}} = \beta_j^\infty$. The limit depends only on the couplings, by Theorem 59.12. Finally (c) is Theorem 59.12(c). □

Details

The comparison is made with the counterterm normalized by S itself. In the trace pairing of Section 23, the marginal tensor of S has isotropic coefficient 1 (proof of Lemma 57.7), so β_j^{B} is also the ratio of the coefficient of $\Phi_{E^{(j)}}$ to that of S; in this form the comparison does not depend on the trace convention. Theorem 59.12 plays, for the generations of this note, the role of the existence of the limit that Bałaban asserts for (1.20) "by the localized representation (1.7)". It is qualitative: no rate is proved, and the convergence is not shown to be uniform in the couplings (Remark 59.13).

58.5 The ultraviolet limit

Theorem 58.9 (The ultraviolet limit)**.** *Let G be compact, connected and simple, let the parameters and the torus be as in Section 53, with $n_K \geq n_{\min}$, and let $p_0(g) = (\log g^{-1})^q$. Let γ satisfy (Γ1)–(Γ4), and moreover $\gamma \leq e^{-3q}$ and $C_3\gamma^2 p_0(\gamma)^6 \leq \frac{1}{2}\beta_\infty$, with β_∞ of Theorem 58.1 and C_3 of Proposition 58.6. There are $g_{\mathrm{uv}} > 0$ and C, independent of K and of the torus, such that for $0 < g \leq g_{\mathrm{uv}}$ and every K there is a bare coupling g_0, depending on K, g and the torus, for which the flow has $g_k \in (0, \gamma]$ for $k \leq K$ and $g_K = g$, and, for every $k \leq K$,*

$$\Big|\frac{1}{g_k^2} - \frac{1}{g^2} - (K-k)\beta_\infty\Big| \leq C\Big(1 + \sum_{j=k}^{K-1} g_j^2 p_0(g_j)^6\Big) \leq C\big(\log(K-k+2)\big)^{6q+1}.$$

In particular, there is m_1, independent of K, g and the torus, such that $\frac{1}{2}(K-k)\beta_\infty \leq g_k^{-2} - g^{-2} \leq 2(K-k)\beta_\infty$ for $K-k \geq m_1$, and $g_k^{-2} - g^{-2} \leq C(K-k)$ for all $k \leq K$.

Bałaban's Theorem 2 [18], stated for $d = 4$ and $G = SU(2)$, asserts that for g small there is a bare coupling for which his couplings satisfy $g_k \in]0, \gamma]$, $g_K = g$ and, with constants $0 < \beta \leq \beta'$,

$$\frac{1}{g^2} + \beta \log(L^k\varepsilon)^{-1} \leq \frac{1}{g_k^2} \leq \frac{1}{g^2} + \beta' \log(L^k\varepsilon)^{-1},$$

his (0.31) [18, p. 259]. With $\varepsilon = L^{-K}$ the finest spacing in the units of the last scale, $\log(L^k\varepsilon)^{-1} = (K-k)\log L$. So the last statement of Theorem 58.9 is the analogue of (0.31) for the couplings of the small-field flow, with $\beta = \beta_\infty/(2\log L)$ and $\beta' = 2\beta_\infty/\log L$, which satisfy $0 < \beta \leq \beta'$, for $K-k \geq m_1$, and its upper half for all k, with $\beta' = \max(2\beta_\infty, C_\beta \mathsf{K})/\log L$. The lower half for the last steps, $1 \leq K-k < m_1$, would need positivity of the single coefficients β_{j+1}, $K - m_1 \leq j < K$. Theorem 58.1 controls only the cumulative sums $c_k = \sum_{j<k}\beta_{j+1}^{(1)}$, and it does not show that a single coefficient $\beta_{j+1}^{(1)} = c_{j+1} - c_j = \beta_\infty + O(1)$ is positive; nor is the error of (F2) small at $j = K-1$ unless n_K is large. There the lower bound holds with an additive constant, $g_k^{-2} - g^{-2} \geq \frac{1}{2}(K-k)\beta_\infty - C$ (by (58.5) below). A possible route to the exact form is described in Remark 58.10(c).

Proof. Cumulative coefficients. Put $c_k = \sum_{j<k}\beta_{j+1}^{(1)}$. By Theorem 58.1, $c_k = k\beta_\infty + r_k$ with $|r_k| \leq C$. Here C does not depend on K or on the torus, since the tensors $T_{j+1}^{(1)}$ are infinite-volume quantities (Definition 54.7). In particular each $\beta_{j+1}^{(1)}$ is bounded, though it may be negative for small j; this does no harm. Details

Localized error. For $0 \leq k \leq K$ and couplings $g_k, \dots, g_{K-1}$ in $(0, \gamma]$, (F2) and (F3) give

$$\sum_{j=k}^{K-1} |\beta_{j+1} - \beta_{j+1}^{(1)}| \leq C_3 \sum_{j=k}^{K-1} g_j^2 p_0(g_j)^6 + C_4 \leq \tfrac{1}{2}(K-k)\beta_\infty + C_4,$$

with $C_4 = C_2 \sum_{m\geq 0} e^{-c_2 L^m}$ (Remark 58.7(b)). The second inequality uses that $g \mapsto g^2 p_0(g)^6$ increases on $(0, e^{-3q}]$, and $C_3\gamma^2 p_0(\gamma)^6 \leq \frac{1}{2}\beta_\infty$. Hence, along any flow that stays in $(0, \gamma]$ from

scale k to scale K,

$$\frac{1}{g_k^2} = \frac{1}{g_K^2} + \sum_{j=k}^{K-1} \beta_{j+1} \geq \frac{1}{g_K^2} + (c_K - c_k) - \tfrac{1}{2}(K-k)\beta_\infty - C_4 \geq \frac{1}{g_K^2} + \tfrac{1}{2}(K-k)\beta_\infty - C_5, \quad (58.5)$$

with $C_5 = 2C + C_4$.

Exit argument. By Corollary 53.11, the set B_K of the $g_0 \in (0,\gamma]$ for which $g_1, \dots, g_K \in (0,\gamma]$ is closed in $(0,\gamma]$ and contains an interval $(0,\bar{g}]$, the map $g_0 \mapsto g_K$ is continuous on it, and $g_K \to 0$ as $g_0 \to 0$, since $|g_K^{-2} - g_0^{-2}| \leq KC_\beta \mathsf{K}$. Let $I = (0, g_0^*]$ be the connected component of B_K that contains $(0,\bar{g}]$; it is closed in $(0,\gamma]$, since B_K is. Some g_j, $j \leq K$, equals γ at g_0^*. If $g_0^* = \gamma$, this holds with $j = 0$. Otherwise there are $g_0 \downarrow g_0^*$ outside B_K, since I is a component of the closed set B_K. At each of them a first coupling g_j, $1 \leq j \leq K$, exceeds γ, and $g_0 \in \mathsf{B}_{j-1}$; the index j is the same along a subsequence. Since g_j is continuous on $\mathsf{B}_{j-1} \ni g_0^*$ (Corollary 53.11(b)), $g_j(g_0^*) \geq \gamma$, hence $g_j(g_0^*) = \gamma$, since $g_0^* \in \mathsf{B}_K$. The localized estimate (58.5) from j to K then gives

$$\frac{1}{g_K^2} \leq \frac{1}{\gamma^2} + C_5,$$

so $g_K \geq \gamma(1 + C_5\gamma^2)^{-1/2} > g$ for $g \leq g_{\mathrm{uv}}$ small. By the intermediate value theorem, some $g_0 \in I$ gives $g_K = g$.

The bound. For this flow and every $k \leq K$,

$$\frac{1}{g_k^2} - \frac{1}{g^2} = \sum_{j=k}^{K-1} \beta_{j+1} = (K-k)\beta_\infty + r_K - r_k + \sum_{j=k}^{K-1} (\beta_{j+1} - \beta_{j+1}^{(1)}),$$

and the last sum is at most $C_3 \sum_{j=k}^{K-1} g_j^2 p_0(g_j)^6 + C_4$ in absolute value, by (F2) and (F3). This gives the first inequality. For the second, (58.5) gives $g_j^2 \leq 4/((K-j)\beta_\infty)$ for $K - j \geq 4C_5/\beta_\infty$, and then, by the monotonicity, $g_j^2 p_0(g_j)^6 \leq C(\log(K-j+2))^{6q}/(K-j)$. The other terms are at most $\gamma^2 p_0(\gamma)^6$, and there are at most $4C_5/\beta_\infty$ of them. Summing over j gives $C(\log(K-k+2))^{6q+1}$. The last statement follows, since $(\log(m+2))^{6q+1}/m \to 0$ as $m \to \infty$, and since $|\beta_{j+1}| \leq C_\beta \mathsf{K}$ for the upper bound when $K - k < m_1$. □

Remark 58.10 (What remains). (a) *The flow.* The flow is Part X, and the properties (F1)–(F3) it gives the ultraviolet limit are proved in Proposition 58.6. Property (F2), the comparison of the one-loop part of β_{j+1}, read off on the torus, with the coefficient of Part XI, read off in infinite volume, rests on the torus independence of the one-loop pieces and of the Jacobian terms (Proposition 41.14, Lemma 58.3), on the infinite-volume minimizer (Proposition 54.15) and on the infinite-volume limits of the second variations (Lemma 58.5). The comparison with Bałaban's (0.31) is the paragraph after Theorem 58.9. A smooth small-field cutoff, as in [18, (2.9)], should give $O(g^2)$ in (F3) and the bound $O(\log(K-k+2))$. Theorem 58.9 concerns the small-field model (Remark 53.14(d)).

(b) *The linear term.* With Bałaban's symmetric averaging, adopted in Part IV, the functions $\Gamma_k^{(1)}$ and $\Gamma_j^{\mathrm{step}} \circ U_{j+1}$ are invariant under the translations and under B_4 (Lemma 54.9). So their vacuum polarizations have no term linear in p, and their marginal tensors are isotropic. The symmetry needs only the averaging: Lemma 54.9(a), and the route to (b) through (a), use no covariance property of the gauge fixing of Part VII; the direct proof of (b) uses only the covariance of its quadratic part, which is that of the averaged contour variables.

(c) *What remains.*

- *A rate for generation locality.* Generation locality is proved in Section 59, qualitatively, so Proposition 58.8 holds without hypothesis. Not proved: a rate in $\beta_j[\tilde{n}] \to \beta_j^{\mathrm{B}}$, and uniformity in the couplings (Remark 59.13). Proposition 58.8 is not used by Theorem 58.9.
- *The last steps.* For the last m_1 steps the lower half of the analogue of Bałaban's (0.31) holds only up to an additive constant (the paragraph after Theorem 58.9). The exact form would need the single coefficients β_{j+1}, $K - m_1 \le j < K$, to be positive. A possible route, for K large, is the convergence to β_∞ of $\beta_{j+1}^{(1)} = c_{j+1} - c_j$, where c_k is the coefficient of $T_{\Gamma_k^{(1)}}$ (Theorem 58.1). By Lemma 56.3 and Proposition 56.8, c_k is β_{F} at $\eta = L^{-k}$ plus the coefficient r_k^{fine} of the remainder $T_{\mathrm{fine}} - T_{\mathrm{F}}$, which is bounded uniformly in k. By Theorem 57.1, $\beta_{\mathrm{F}} - k\beta_\infty$ converges as $k \to \infty$, so the differences of the first terms converge to β_∞. What is missing is the convergence of r_k^{fine} as $k \to \infty$, a continuum limit at $U = 1$ of explicit operators. In addition, the errors of (F2) and (F3) at the last steps must be small compared with β_∞, which needs a lower bound on n_K and a condition on γ. This is not carried out here.
- *The large-field stage*, which is not treated (Remark 53.14(d)).

(d) *Numerical checks.* Theorem 55.4 (Remark 55.7), the constant of Remark 34.10 and the invariance of Lemma 54.9, all three in $d = 2$; the coefficient b, its sign and the lattice constant of Dashen and Gross, for $SU(2)$ (Remark 57.9); and the continuum tensor and, for $SU(2)$, the scaling of the integrand (Remark 57.10) were checked numerically. No proof uses these checks. Appendix C describes them. The scaling check of the lattice integrand (Remark 57.10(ii)) and the earlier checks are independent of the proofs; the spherical average (Remark 57.10(i)) recomputes that of Lemma 57.7 in exact arithmetic from the vertices of Lemma 57.4. They test the formula of Theorem 55.4, the invariance of Lemma 54.9, and the lattice vertices, the joint expansion and the normalization of Section 57. They say nothing about the remainders of Proposition 56.8.

59 Generation locality

This section shows that the generations of Part X, built with fixed couplings on tori of growing side, converge at the flat configurations, and that their coefficients converge (Theorem 59.12). This is what Proposition 58.8 uses. The proof has two parts.

- For a polymer that does not wrap and at a flat configuration, the new terms of one step are a single functional, defined on the infinite lattice, of the old activities at flat configurations (Proposition 59.8). This extends the torus independence of the one-loop pieces (Proposition 41.14) to the whole step at fixed old data.
- The old activities converge as the torus grows, by induction over the scales, and the functional is continuous in them. The induction follows the proof of Proposition 53.10(b), with the side of the torus in place of the couplings.

The generations themselves are not shown to be independent of the torus, and they are not expected to be: the coefficients β_i are averages over the polymers that do not wrap (58.1), and they enter the later generations through the old activities. No rate of convergence is proved (Remark 59.13).

The result in words, and its place. Fix couplings $(g_0, \dots, g_{j-1}) \in \Sigma_{j-1}$, with γ as in (Γ1)–(Γ3) and G compact, connected and simple, and let the side of the torus grow. At the flat configurations, which equal $e^{\mathrm{i}\eta B}$ near a polymer and 1 elsewhere (Section 59.1), the raw

generation and the generation of every scale $i \le j$ converge to activities of the infinite lattice, the coefficients β_i converge to the coefficients β_i^∞ of the limits, and the limits depend only on $(g_0, \dots, g_{i-1})$ (Theorem 59.12). The reason is that every construction of a step is given by local rules: on a polymer that does not wrap, the new terms are a functional of local data, which are the same on all large tori (Proposition 59.8). Theorem 59.12 is stated at the end of the section, after the transfer lemmas. This section is off the main line: Theorem 58.9, and so Theorem M(iv), does not use it. Its only use is Proposition 58.8, the comparison of the β_j with the coefficients that Bałaban's definition assigns to the generations.

59.1 The setting

Setting. Let G be compact, connected and simple. The parameters are those of Section 53 (Setting), and γ satisfies (Γ1)–(Γ3) of Remark 53.5. Let $j \ge 1$ and $\pi = (g_0, \dots, g_{j-1}) \in \Sigma_{j-1}$. For $i \le j$ and $m \in \mathfrak{N}_i$ let $E'^{(i)}[m]$, $E^{(i)}[m] = E'^{(i)}[m]^{\mathrm{sym}}$ and $\beta_i[m] = \beta_{E^{(i)}[m]}$ be the raw generation, the generation and the coefficient of scale i constructed with $(g_0, \dots, g_{i-1})$ on the torus whose lattice of scale i has side m (Proposition 53.10, with $K = i$).

This is legitimate. The Setting of Section 50, which defines $\mathfrak{N}_i$, includes that of Section 52.1, and so $n_K \ge n_{\min}$ with $K = i$. So the Setting of Section 53 holds with $K = i$, and Proposition 53.10(a) applies. By that proposition, $E^{(l)}$ and β_l, $l \le i$, depend only on $(g_0, \dots, g_{l-1})$ and on the torus; the steps $0, \dots, l-1$ do not use the scales above l. By the proof of Theorem 58.1, the side of this torus at scale l is $L^{i-l}m \in \mathfrak{N}_l$. So the generations of scale l on the torus of side m at scale i are the $E^{(l)}[L^{i-l}m]$. For $1 \le k < j$ we write $\mathcal{E}_k[m']$ for the old activities of step k (Definition 53.1) on the torus whose lattice of scale $k+1$ has side $m' \in \mathfrak{N}_{k+1}$. They are built from the $E^{(l)}[L^{k+1-l}m']$ and the $\beta_l[L^{k+1-l}m']$, $l \le k$.

Flat configurations. Use the units of scale i. For a polymer X of scale i of the infinite lattice, or of a torus on which it does not wrap, $\mathcal{U}_1^\flat(X)$ is the flat domain of Section 58.3. Let $\mathcal{U}_1(X)$ be the set defined in the same way with δ_1 in place of $\delta_1^\flat$. It consists of the configurations V_B equal to $e^{i\eta B}$ on the bonds of $X^{(3)}$, read in a lift, and to 1 elsewhere, with $|B| < \delta_0$ and $|\nabla B| + [\nabla B]_\alpha < \delta_1$ on $X^{(3)}$; the differences and quotients are taken with the real part 1. By Definition 38.3 with the real part 1,

$$\mathcal{U}_1^\flat(X) \subset \mathcal{U}_1(X) \subset \mathcal{U}(X).$$

Their points are called *flat*. As in Section 58.3, a flat point is identified with its field B, and on the torus whose lattice of scale i has side m we write $V_B[m]$. The conditions are imposed on the bonds of $X^{(3)}$ only. So they are the same on the infinite lattice and on every torus on which X does not wrap. As sets of fields B, $\mathcal{U}_1(X)$ and $\mathcal{U}_1^\flat(X)$ are open and convex in a finite-dimensional space, since the bounds are strict and are bounds on seminorms of B.

The infinite lattice. Polymers of scale i of the infinite lattice, their tree lengths and their numbers of cubes are those of Section 58.3. Their closures cl_k at the scales $k \ge i$, the sets $\square^+$, the base cubes and the classes of Definition 49.5 are defined by the same rules as on the torus. The base cube of a polymer of the infinite lattice is read in the box spanned by its cubes. These rules are local in the grids. So for a polymer that does not wrap they commute with the projection to the torus (proof of Lemma 58.3; Section 49, base cubes).

Lemma 59.1 (Flat points)**.** *(a) Let $l \le k$, let Y be a polymer of scale k and $X \subset Y$ a polymer of scale l, both not wrapping or both of the infinite lattice. A point of $\mathcal{U}_1^\flat(Y)$ of scale k is, on $X^{(3)}$ and in the units of scale l, a point of $\mathcal{U}_1^\flat(X)$ of scale l. Let Y' be a polymer of scale $k+1$, not wrapping or of the infinite lattice. A point of $\mathcal{U}_1(Y')$ of scale $k+1$ lies, on $X^{(3)}$ for every polymer $X \subset Y'$ of scale k, in $\mathcal{U}_1^\flat(X)$ of scale k, with budgets $L^{-2}\delta_1$ and $L^{-1}\delta_0$.*

(b) *Let Y' be a polymer of scale $k+1$ that does not wrap, and $V = V_B \in \mathcal{U}_1(Y')$. With the representation $(1, B)$, the cutoff extension of Remark 38.7 is $U' = e^{\mathrm{i}\eta\chi B}$. It has real part 1, it equals V on the bonds within unit distance of Y', in the units of scale $k+1$, and it equals 1 off $Y'^{(3)}$. Read in the units of step k, it is a reference of the Setting of Section 47.*

(c) *Let $r \in \mathfrak{S}_i^{\mathrm{grid}}$, and let X be a polymer of scale i that does not wrap. Then $r\,V_B = V_{rB}$, with $(rB)(b) = B(r^{-1}b)$ and $B(-b) = -B(b)$, and $r\,\mathcal{U}_1^\flat(X) \subset \mathcal{U}_1(rX)$.*

Proof. (a) This is Remark 38.8(i), with $\delta_1^\flat$ in place of δ_1 for the first statement. Its proof keeps the real part, here 1. Details

(b) This is Remark 38.7 with $U_r = 1$. The cutoff χ equals 1 at unit distance at most 1 from Y' and 0 at distance at least 2. The last statement is Remark 38.8(ii): the real part 1 is regular, with curvature and current 0.

(c) Apply Lemma 50.6 with $U_r = 1$, $\delta_1' = \delta_1^\flat$ and $\delta_0' = \delta_0$. Then $R(U_r(b)) = 1$, so rA is as stated, and $rU_r = 1$. By (DC6), $2\delta_1^\flat + C_{\mathrm{refl}}\theta_r\delta_0 \le \delta_1$, as in the derivation of (50.5). So $rV_B = e^{\mathrm{i}\eta\, rB}$ on the bonds of $(rX)^{(3)}$, with the budgets of $\mathcal{U}_1(rX)$, and $rV_B = 1$ on the other bonds. □

Admissible sides. Let $k \ge 0$, and let $\tilde{Y}'$ be a polymer of scale $k+1$ of the infinite lattice. A side n at step k is *admissible for* $\tilde{Y}'$ if $n/L \in \mathfrak{N}_{k+1}$ and $\operatorname{diam}\tilde{Y}' \le \frac{1}{2}(n/L) - 8$ in the units of scale $k+1$. Then the projection $Y' = \pi_n\tilde{Y}'$ to the torus $\mathbb{T}_\eta[n]$ of step k does not wrap at scale $k+1$. By Lemma 52.1(a), n is admissible at step k in the sense of Section 41.5, as in the proof of Lemma 58.3. In the units of step k the lift $\tilde{Y}'$ has diameter at most $\frac{1}{2}n - 8L$. So, for two sides n, n' admissible for $\tilde{Y}'$ and every $a \in M\mathbb{Z}^4$, the transfer of the lift $\tilde{Y}'$ with the vector a is a grid transfer from $\mathbb{T}_\eta[n]$ to $\mathbb{T}_\eta[n']$ (Definition 41.9(ii)), with $P = Y'$; here $n' = n$ is allowed. Every $\tilde{Y}'$ has admissible sides, since $\mathfrak{N}_{k+1}$ is unbounded.

59.2 Transfer of one step

This subsection uses the setting of Section 41.5 at step k, with its grid transfers and the consequence of Lemma 41.10. Lemma 41.11(c) lists the linear operators of the step, the elimination ℓ_c, ζ_c, $\hat{Z}^1$, $\hat{Z}^2$ (hence the map $\hat{Z}$ of Lemma 32.7(d)), the transport $\mathfrak{G}_W$, the curvatures F_U, F_U^s and the current J_U, and the averaged backgrounds W_j, at complex references through Proposition 46.4(i). The step also uses the following nonlinear maps.

Lemma 59.2 (Nonlinear local maps)**.** *Each of the following is given on every admissible torus by the same rule. At a site based at y the rule uses only the objects of Lemma 41.11(a) and the configuration and the fields on the bonds within distance $4L + 2 \le R_{\mathrm{g}}$ of y. At complex references it is the holomorphic continuation of the same formula.*

(i) *The nonlinear part C_k^{av} of the gauge-adapted averaging. The value $C_k^{\mathrm{av}}(A)(c)$ is a function of A and of the reference on the bonds of $S_c = B(c_-) \cup B(c_+)$.*

(ii) *The plaquette terms s_p of the action, and the plaquette terms n_p of $\mathcal{N}$ (Proposition 23.3). Each is a function of the configuration, respectively of A and of the reference, on the bonds of p. Hence $\nabla\mathcal{N}(A)(b)$ is a function of A and of the reference on the plaquettes that contain b.*

(iii) *The functions $H_{y,x}$ and $f \circ H_{y,x}$ of Definition 47.6, functions of Z and of the base on the inner bonds of $B'(y)$.*

(iv) *The terms $\log(\sigma(Z(b))/\sigma(0))$ and $\log(\mathsf{J}_c(Z)/\mathsf{J}_c(0))$ of $\mathfrak{J}$ (Proposition 47.8(a)), functions of Z and of the base within distance $2L$ of one bond.*

(v) *The bondwise operations ℓ and $\oplus$, the latter the Baker–Campbell–Hausdorff combination of the proof of Proposition 47.10.*

Consequently, let $\iota : P \to P'$ be a grid transfer, and let the references and the fields agree under ι on P. Then each of these objects has the same value at a site x based in P whose range lies in P and at ιx.

Proof. (i) By Proposition 22.9, $C_k^{\mathrm{av}}(A) = \delta_k^{-1}\ell(\mathcal{Q}_k(e^{i\eta A})) - \mathbf{Q}_k A$, with the gauge-adapted averaging $\mathcal{Q}_k$ of Section 22. It is defined by the same formulas on every torus, through the relative one-step averages at the backgrounds W_j of the reference. By Proposition 22.9(a), Proposition 22.10(d) and Lemma 21.10, as in the proof of Theorem 45.5(d), $C_k^{\mathrm{av}}(A)(c)$ depends only on A and on the reference on the bonds of S_c. At complex references, $W_j(U) = \mathfrak{a}_j(W_{j-1}(U))$ is the same one-step recursion (proof of Lemma 46.2(b)), and Proposition 46.4(iii) gives Proposition 22.9 there. Details

(ii) The term s_p is explicit in $U(\partial p)$ (Section 23). The plaquette terms of $\mathcal{N} = \sum_p n_p$ are functions of A and of U_0 on the bonds of p (proof of Theorem 45.5(d)). At complex references both are the continuations of Proposition 46.7(c).

(iii) This is Proposition 47.7(d), with the averaged contour variables of the box spanned by y and x, which are defined at complex bases by (DC5) (Remark 38.8(ii)).

(iv) This is Proposition 47.8(a). The function σ is explicit, and

$$\mathsf{J}_c(Z) = \frac{\sigma(\zeta_c(Z))}{\sigma(0)\,\det \partial\ell_c/\partial Z(b_0(c))} \qquad \text{at } Z(b_0(c)) = \zeta_c(Z)$$

(Proposition 32.8) is built from the functions ℓ_c and ζ_c of Lemma 41.11(c).

(v) These are bondwise.

The ranges are at most $4L+2$, as in the proof of Lemma 41.11(c). The consequence follows from Lemma 41.10. □

Lemma 59.3 (Transfer of the weakened background map)**.** *Let $\iota : P \to P'$ be a grid transfer from $\mathbb{T}_\eta[n]$ to $\mathbb{T}_\eta[n']$ at step k, and let U_0, U_0' be references as in Section 45, real or complex as in Proposition 46.7(d), that agree on P. Let Z be a union of boxes with $Z^{(r+4)} \subset P$, let $s \in \mathbb{D}_\kappa$ with $s_\square = 0$ for $\square \not\subset Z$, and let s' be given by $s'_{\iota\square} = s_\square$ for $\square \subset Z$ and $s'_{\square'} = 0$ for the other boxes $\square'$. Let B, B' be coarse fields with $|B|_\infty, |B'|_\infty \le R_P$ that agree under ι on the coarse bonds of $Z^{(r+1)}$. Then $A_1(s';B')$ and $A_*(s';B')$, at U_0', vanish off $\iota(Z)^{(r)}$, and on its bonds they are the transfers by ι of $A_1(s;B)$ and $A_*(s;B)$, at U_0.*

Proof. Write ι_* for the transfer of fields and operators (Definition 41.9). We follow the proof of Theorem 45.5(d) on both tori. Its class $\mathcal{K}$ is defined on each torus with Z, respectively $\iota(Z)$. Details

The operators. The operators $\mathbf{C}(s)$, $\mathbf{H}(s)$, $D\mathbf{H}(s)$ and their transposes are sums of products, as in Lemma 40.4, of the families of the levels (L1)–(L4) on the domain $\mathbb{T}_\eta$, with $W = 0$ at (L3), and of the local operators D, D^T, $\mathbf{Q}$, $\mathbf{Q}^T$ (Section 44, the definitions of $\mathbf{H}(s)$ and $\mathbf{C}(s)$ on the torus). By Lemma 40.3(b), at s only their terms with $X_\omega \subset Z$ contribute, and at s' only the images of these. Since $X_\omega^{(r)} \subset Z^{(r)} \subset P$, Lemma 41.13(b) carries the factors, and Step 4 of its proof carries the products: $\iota(\omega)$ is a term with polymer $\iota(X_\omega)$ and the carried monomial, and $O'_{\iota(\omega)} = \iota_* O_\omega$. The local operators are carried by Lemma 41.11(c). Hence $\mathbf{C}'(s') = \iota_*\mathbf{C}(s)$, and likewise for $\mathbf{H}$, $D\mathbf{H}$ and the transposes, on the fields supported in $Z^{(r)}$ and the coarse fields on the coarse bonds of $Z^{(r)}$.

The linearizing map. Let A' be a fine field that vanishes off $Z^{(r)}$, and let A'' vanish off $\iota(Z)^{(r)}$ and agree with $\iota_* A'$ there. The iterates of (45.1) from 0, $X_{p+1} = C_k^{\mathrm{av}}(A' - \mathbf{H}(s)X_p)$, stay in $\mathcal{K}$, so they vanish off $Z^{(r+1)}$. By induction on p, $X_p'' = \iota_* X_p$. The step uses the operators above and Lemma 59.2(i) at the coarse bonds starting in $Z^{(r+1)}$, whose sets S_c lie in $Z^{(r+3)} \subset P$. The iterates converge to $\mathcal{D}_s(A')$, since (45.1) is a contraction of a ball (proof of Lemma 45.3).

So $\mathcal{D}'_{s'}(A'') = \iota_* \mathcal{D}_s(A')$. The derivative $d\mathcal{D}_s^T = dC_k^{\mathrm{av}}(A)^T (I + \mathbf{H}(s)^T dC_k^{\mathrm{av}}(A)^T)^{-1}$ is carried term by term in its Neumann series (proof of Lemma 45.3). Its terms are products of carried operators, since $dC_k^{\mathrm{av}}(A)$ is the derivative of a carried local map.

The fixed point. The iterates $A_{1,p+1} = \mathcal{T}_s(A_{1,p})$, $A_{1,0} = 0$, stay in $\mathcal{K}$. By (45.3) and (45.2), $\mathcal{T}_s(A_1) = -\mathbf{C}(s)\nabla \mathcal{V}_s(A_1 + \mathbf{H}(s)B)$. The gradient is built from $\mathcal{D}_s$ and $d\mathcal{D}_s^T$ at $A_1 + \mathbf{H}(s)B$, the operators above, $\nabla\mathcal{N}$ (Lemma 59.2(ii)) and $\mathsf{F} = F^s_{U_0}$ (Lemma 41.11(c)). The field $\mathbf{H}(s)B$ depends on B only on the coarse bonds of $Z^{(r)}$, where B and B' agree. So, by induction on p, $A''_{1,p} = \iota_* A_{1,p}$. The iterates converge in $|\cdot|_*$, hence bondwise, to the fixed point $A_1(s;B)$, since $\mathcal{T}_s$ is a contraction of the ball of Theorem 45.5(a). Then $A_*(s;B) = A'_s - \mathbf{H}(s)\mathcal{D}_s(A'_s)$, $A'_s = A_1(s;B) + \mathbf{H}(s)B$, is carried. At complex references the operators are those of Proposition 46.7(d), and the class $\mathcal{K}$ is preserved by the same proof (proof of Proposition 46.7, (viii)). □

Lemma 59.4 (Composed activities at flat references)**.** *Let $k \geq 1$, and let X be an old polymer of step k that does not wrap. Let $\mathcal{E}_X$ be a holomorphic function on $\mathcal{U}_1^\flat(X)$ of step k, as a function of the field on the bonds of $X^{(3)}$, which depends only on the bonds of X and is bounded by $\|\mathcal{E}_X\|$. Let $U' = e^{\mathrm{i}\eta A'}$ be a reference of the Setting of Section 47 with real part 1, such that the field A' has, on $X^{(3)}$, the budgets $\delta'_1 < \delta_1^\flat$ and $\delta'_0 < \delta_0$. Let $0 < \zeta \leq \frac{1}{2}R_P$, let τ be as in Proposition 47.10 with $\delta_1^\flat$ in place of δ_1, and assume $\tau < \frac{1}{2}$. Then for $s \in \mathbb{D}_\kappa$ and $|\hat{Z}(Z)|_\infty \leq \zeta$:*

(a) on $X^{(3)}$, the configuration $e^{\mathrm{i}\eta A_*(s;\hat{Z}(Z))}U'$ is a point of $\mathcal{U}_1^\flat(X)$. The function

$$f(X;s,Z) = \mathcal{E}_X(e^{\mathrm{i}\eta A_*(s;\hat{Z}(Z))}U') - \mathcal{E}_X(U')$$

is analytic in (s,Z) and in A' jointly, continuous up to the boundary in s, and $|f(X;s,Z)| \leq 4\tau\|\mathcal{E}_X\|$;

(b) if $s_\square = 0$ outside a union of boxes Z' that contains the boxes meeting X, then $f(X;s,Z)$ depends on s only through the boxes of the $*$-component Z'_0 of Z' containing them, and on Z and U' only on $(Z'_0)^{(r')}$.

Proof. The proof of Proposition 47.10 applies with $P = X$, $r_E = 0$ and $U'_r = 1$. It evaluates $\mathcal{E}_X$ only at the configurations $e^{\mathrm{i}\eta t A_*}U' = e^{\mathrm{i}\eta(tA_* \oplus A')}$ on $X^{(3)}$, $|t| < \frac{1}{2\tau}$, whose real part is $U'_r = 1$, and it shows that they satisfy the bounds of $\mathcal{U}(X;\theta_r,\delta_1^\flat,\delta_0)$ strictly, with the representation $(1, tA_* \oplus A')$. These are points of $\mathcal{U}_1^\flat(X)$. Lemma 47.9 is applied with $U'_r = 1$. □ Details

Lemma 59.5 (Transfer of the pieces)**.** *Let $k \geq 0$, and let $\iota : P \to P'$ be a grid transfer from $\mathbb{T}_\eta[n]$ to $\mathbb{T}_\eta[n']$ at step k. Let U' and U'' be references of the Setting of Section 47 on the two tori that agree on P, at the same coupling $g \in (0,\gamma]$. Let Z be a polymer of step k with $Z^{(r')} \subset P$, and let ξ, ξ' be white-noise fields with $|\xi|_\infty, |\xi'|_\infty \leq p_0(g)$ and $\xi'(\iota b) = \xi(b)$ for the bonds $b \in \mathfrak{F}$ in $Z^{(r')}$.*

(a) The pure pieces of Proposition 48.5 satisfy $E'_{\mathrm{p}}(\iota Z)(\xi') = E_{\mathrm{p}}(Z)(\xi)$. At $k = 0$ this holds for the pure pieces of the first step (Proposition 52.2).

(b) Let $k \geq 1$, and let U' and U'' have real part 1. For every old polymer $X \subset Z$ let $\mathcal{E}_X$ and $\mathcal{E}'_{\iota X}$ be functions as in Lemma 59.4, with $\mathcal{E}'_{\iota X}(\iota_* A) = \mathcal{E}_X(A)$ for the flat fields A on the bonds of $X^{(3)}$, and let U' satisfy the hypothesis of Lemma 59.4 on $X^{(3)}$. Let $E_{\mathrm{m}}(Z) = \sum_{X \subset Z} f_{\mathrm{loc}}(X,Z)$ be built from the functions $f(X;s,Z(s))$ of Lemma 59.4 as in Proposition 48.5, and $E'_{\mathrm{m}}(\iota Z)$ likewise. Then $E'_{\mathrm{m}}(\iota Z)(\xi') = E_{\mathrm{m}}(Z)(\xi)$.

Proof. (a) Let $\square \subset Z$, and let f be one of the functions localized in the proof of Proposition 48.5 with $Y = \square$: $V(\square;s)$, $G_3(\square;s)$, or the terms of $\mathfrak{J}(Z(s))$ grouped in $\square$. Let f' be the same function on $\mathbb{T}_\eta[n']$ for the box $\iota\square$. By Lemma 47.2, $f_{\mathrm{loc}}(\square, Z) = \int_{[0,1]^{Z\setminus\square}} \partial_{s_{Z\setminus\square}} f(s(Z,t))\, dt$, Details

and the points $s(Z,t)$ vanish off Z. We show $f'(s') = f(s)$ for every $s \in \mathbb{D}_{\kappa'}$ that vanishes off Z, with s' as in Lemma 59.3. Then f and f' are the same analytic functions of the variables $s_\square$, $\square \subset Z$, so $f'_{\mathrm{loc}}(\iota\square, \iota Z) = f_{\mathrm{loc}}(\square, Z)$, and summing over $\square \subset Z$ gives (a).

Let s vanish off Z. By Lemma 40.3(b), only the terms of $C_k^{1/2}(s)$ with $X_\omega \subset Z$ contribute, and the operator vanishes off $Z^{(r)} \times Z^{(r)}$. Lemma 41.13(b) carries these terms (the family $C_k^{1/2}(s)$ on $\mathbb{T}_\eta$, Corollary 41.6), since $X_\omega^{(r)} \subset Z^{(r)} \subset P$. So $Z'(s') = gC_k'^{1/2}(s')\xi'$ vanishes off $\iota(Z)^{(r)}$, $Z(s)$ vanishes off $Z^{(r)}$, and $Z'(s')(\iota b) = Z(s)(b)$ for the bonds b of P.

Interaction. $V(\square; s) = g^{-2}\sum_p v_p(s; Z(s))$, the sum over the plaquettes p based in $\square$. By Proposition 47.5(d), $v_p = 0$ unless p has a point in $Z^{(r+1)}$, and then p lies in $Z^{(r+2)}$. For fields Y on $\mathfrak{F}$ that vanish off $Z^{(r)}$, with $|Y|_\infty < r_I$, put $h(Y) = s_p(e^{\mathrm{i}\eta A_*(s;\hat{Z}(Y))}U')$.

- The field $\hat{Z}(Y)$ equals Y on $\mathfrak{F}$ and $\zeta_c(Y)$ on $b_0(c)$. Here $\zeta_c(Y)$ is a function of Y and of the base on the bonds of $B'(c_-) \cup B'(c_+)$ (Lemma 32.7(c)), carried by Lemma 41.11(c). So $\hat{Z}'(\iota_* Y)$ and $\hat{Z}(Y)$ agree under ι on the coarse bonds of $Z^{(r+1)}$. Moreover $|\hat{Z}(Y)|_\infty < R_P$, by the choice of r_I (Section 47.2).
- By Lemma 59.3, applicable since $Z^{(r+4)} \subset Z^{(r')} \subset P$ (as $r' = r + 2L + 2 \geq r + 8$), $A_*(s'; \hat{Z}'(\iota_* Y))$ is the transfer of $A_*(s; \hat{Z}(Y))$.
- By Lemma 59.2(ii), $h'(\iota_* Y) = h(Y)$.

The Taylor polynomial T_2 at $Y = 0$, evaluated at a field that vanishes off $Z^{(r)}$, uses only the derivatives of h at 0 in the directions of these fields. So $v'_{\iota p}(s'; \iota_* Y) = v_p(s; Y)$, and with $Y = Z(s)$, $V'(\iota\square; s') = V(\square; s)$.

Gauge fixing. $G_3(\square; s) = g^{-2}\sum_{y \in \mathbb{T}' \cap \square} \sum_{x \in B'(y)} \varphi_{y,x}(Z(s))$. The function $\varphi_{y,x} = f \circ H_{y,x} - \frac{1}{2}\langle \mathfrak{G}_W \cdot, \mathfrak{G}_W \cdot \rangle$ is carried by Lemma 59.2(iii) and Lemma 41.11(c). The blocks $B'(y)$, $y \in \square$, lie within distance L of $\square$.

Jacobians. The terms of $\mathfrak{J}(Z(s))$ grouped in $\square$ are carried by Lemma 59.2(iv).

The first step. At $k = 0$ there is no background map. The interaction pieces are $v_p(Z) = s_p(e^{\mathrm{i}\hat{Z}(Z)}U') - T_2[\cdots](Z)$, local within range $2L + 2$, and $C_0^{1/2}(s)$ is given by the walk of Theorem 41.4 with $\mathfrak{A} = \Delta_0 + r$ and $\mathfrak{N} = 0$ (Proposition 52.2(a)–(c)). Lemma 41.13 carries it in its case $k = 0$. The same argument applies, with the factorization of the composed pieces in the proof of Proposition 52.2 in place of Proposition 47.5(d).

(b) Let $X \subset Z$ be an old polymer. In the proof of Proposition 48.5, $f(X; s, Z(s))$ is localized by Lemma 47.2 with Y the set of boxes of X. Let s vanish off Z. By Lemma 59.4(b), $f(X; s, Z(s))$ depends on $Z(s)$ and on U' only on $Z^{(r')} \subset P$. By the first part of the proof, $Z(s)$ is carried. By Lemma 59.3, so is $A_*(s; \hat{Z}(Z(s)))$, and the operation $\oplus$ is bondwise. Since $X^{(3)} \subset Z^{(r')}$, the composed flat field on $(\iota X)^{(3)}$ is the transfer of that on $X^{(3)}$. By the hypothesis on $\mathcal{E}_X$, $f'(\iota X; s', Z'(s')) = f(X; s, Z(s))$. As in (a), $f'_{\mathrm{loc}}(\iota X, \iota Z) = f_{\mathrm{loc}}(X, Z)$. The map $X \mapsto \iota X$ is a bijection of the old polymers in Z onto those in ιZ, and (b) follows. □

59.3 The step on the infinite lattice

Definition 59.6 (Flat old data)**.** Let $k \geq 1$, and let $\tilde{Y}'$ be a polymer of scale $k+1$ of the infinite lattice, read in the units of step k. *Flat old data on* $\tilde{Y}'$ is a family $\mathcal{E}^{\mathrm{fl}} = (\mathcal{E}^{\mathrm{fl}}(\tilde{Y}, \cdot))$, indexed by the polymers $\tilde{Y} \subset \tilde{Y}'$ of scale k of the infinite lattice. Each $\mathcal{E}^{\mathrm{fl}}(\tilde{Y}, \cdot)$ is a holomorphic function on $\mathcal{U}_1^\flat(\tilde{Y})$ of scale k, depends only on the field on the bonds of $\tilde{Y}$, vanishes at 0, and satisfies

$$\sup_{\mathcal{U}_1^\flat(\tilde{Y})} |\mathcal{E}^{\mathrm{fl}}(\tilde{Y}, \cdot)| \leq C_{\mathrm{old}} \mathsf{K}\, e^{-\kappa_n d_k(\tilde{Y})},$$

with C_{old} of Proposition 53.2. We write $\mathfrak{D}_k(\tilde{Y}')$ for the set of such families. For $a \in LM\mathbb{Z}^4$ in the units of step k, that is, a vector of $M\mathbb{Z}^4$ in the units of scale $k+1$, $(\tau_a \mathcal{E}^{\mathrm{fl}})(\tilde{Y}+a, \tau_a B) = \mathcal{E}^{\mathrm{fl}}(\tilde{Y}, B)$,

with $(\tau_a B)(b) = B(b-a)$, defines $\tau_a \mathcal{E}^{\mathrm{fl}} \in \mathfrak{D}_k(\tilde{Y}' + a)$.

Let n be a side admissible for $\tilde{Y}'$, and let $\mathcal{E}$ be old activities of step k on $\mathbb{T}_\eta[n]$ as in Lemma 53.4, with $\|\mathcal{E}\| \le C_{\mathrm{old}}\mathsf{K}$. Their *flat restriction* to $\tilde{Y}'$, also written $\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}[\mathcal{E}]$, is

$$\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}(\tilde{Y}, B) = \mathcal{E}(\pi_n \tilde{Y}, V_B[n]), \qquad \tilde{Y} \subset \tilde{Y}',\ B \in \mathcal{U}_1^\flat(\tilde{Y}).$$

Since $\pi_n \tilde{Y}'$ does not wrap, neither does $\pi_n \tilde{Y}$. The configuration $V_B[n]$ lies in $\mathcal{U}_1^\flat(\pi_n \tilde{Y}) \subset \mathcal{U}^\flat(\pi_n \tilde{Y})$. The old activities are normalized graded activities at their polymers on these domains, with the stated bound. Moreover $d_k(\pi_n \tilde{Y}) = d_k(\tilde{Y})$, since $\operatorname{diam} \tilde{Y} \le \frac{1}{2}n - 8$; this is the first part of the proof of Lemma 59.11(a), which uses only the geometry of polymers. So $\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'} \in \mathfrak{D}_k(\tilde{Y}')$.

Definition 59.7 (The step on the infinite lattice)**.** Let $k \ge 0$ and $g \in (0, \gamma]$. Let $\tilde{Y}'$ be a polymer of scale $k+1$ of the infinite lattice, let $\mathcal{E}^{\mathrm{fl}} \in \mathfrak{D}_k(\tilde{Y}')$ if $k \ge 1$, and let $B \in \mathcal{U}_1(\tilde{Y}')$, of scale $k+1$. Choose a side n admissible for $\tilde{Y}'$, put $Y' = \pi_n \tilde{Y}'$, and let U' be the cutoff extension of $V_B[n]$ with the representation $(1, B)$ (Lemma 59.1(b)), read in the units of step k. Consider the pieces

$$E(Z) = E_{\mathrm{p}}(Z) + E_{\mathrm{m}}(Z), \qquad E_{\mathrm{m}}(Z) = \sum_{X \subset Z} f_{\mathrm{loc}}(X, Z),$$

indexed by the polymers Z of step k with $Z^{(r')} \subset Y'$, at the reference U' and the coupling g, and put $E(Z) = 0$ for the other polymers. Here:

- $E_{\mathrm{p}}(Z)$ is the pure piece of Proposition 48.5;
- the sum runs over the old polymers $X \subset Z$;
- $f_{\mathrm{loc}}(X, Z)$ is given by Lemma 47.2 on $\mathbb{D}_{\kappa'}$, with Y the set of boxes of X, applied to $s \mapsto f(X; s, Z(s))$ of Lemma 59.4, with $\mathcal{E}_X(V) = \mathcal{E}^{\mathrm{fl}}(\tilde{X}, \tilde{B}_V)$, where $\tilde{X}$ is the lift of X in $\tilde{Y}'$ and $\tilde{B}_V$ is the lift of the flat field of V on $X^{(3)}$.

At $k = 0$ there are only the pure pieces of the first step (Proposition 52.2), with the constants of the first step. Let $E^{\#}(Y; U')$ be the outputs of Theorem 48.6 for this system, with $\kappa = \kappa_n - 1$ and $\mathsf{d} = d_k$, and let $E^{\#}(Y; 1)$ be the outputs of the same construction at $B = 0$, where $U' = 1$. Put

$$\mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}; B) = \sum_{\tilde{X}\colon \mathrm{cl}_{k+1}\tilde{X} = \tilde{Y}'} E^\infty_k(\tilde{X}, B_{\tilde{X}}) + \sum_{\tilde{c}\colon \tilde{Y}'_{\tilde{c}} = \tilde{Y}'} J^\infty_k(\tilde{c}, B_{\tilde{c}}) + \sum_{Y\colon \hat{Y} = Y'} (E^{\#}(Y; U') - E^{\#}(Y; 1)),$$

where E^∞_k and J^∞_k are the infinite-volume pieces of Proposition 41.14(d), at the restrictions $B_{\tilde{X}}$, $B_{\tilde{c}}$ of B, read in the units of step k. The first two sums are those of the proof of Lemma 58.3. At $k = 0$ we write $\mathscr{S}_{0,\tilde{Y}'}(g; B)$.

The construction is defined.

- By Lemma 59.1(a),(b), U' is a reference of the Setting of Section 47. On $X^{(3)}$ its field has budgets $L^{-2}\delta_1$ and $L^{-1}\delta_0$, for every old polymer $X \subset Y'$. As in the paragraph after Proposition 47.10, the slack is at least $\frac{1}{8}\delta_1$. So Lemma 59.4 applies with $\zeta = w$ and $\tau \le C_P L^d g p_0(g)/\delta_1 < \frac{1}{2}$, by ($\Gamma$2).
- By (Γ2) at g and the proof of Proposition 48.5, which uses the old activities only through the bounds of Lemma 59.4, the pieces satisfy $|E_{\mathrm{p}}(Z)| \le \epsilon_{\mathrm{p}} e^{-(\kappa'-2)|Z|}$ and $|E_{\mathrm{m}}(Z)| \le \epsilon_{\mathrm{m}} e^{-(\kappa_n - 1)d_k(Z)}$, with $\epsilon_{\mathrm{m}} = C_P \tau C_{\mathrm{old}}\mathsf{K}$. They depend on ξ only on the bonds of $\mathfrak{F}$ in $Z^{(r')}$, and on U' only on $Z^{(r')}$.

- So, as in Step 3 of the proof of Proposition 53.10(b), the system satisfies the hypothesis of Theorem 48.6 with $\kappa = \kappa_n - 1 \geq C_d$ and $\epsilon = \epsilon_{\mathrm{p}} + \epsilon_{\mathrm{m}} e^{c_{\mathrm{q}}(\kappa_n - 1)} \leq \epsilon_*$.
- At $k = 0$ there are only the pure pieces of the first step. They satisfy the bound of Proposition 48.5 with $\epsilon_{\mathrm{p}}^{(0)} \leq \frac{1}{2}\epsilon_*$, by (Γ2) for the first step and the end of the proof of Proposition 52.2, and the hypotheses of Theorem 48.6 hold with the constants of the first step.
- The restrictions $B_{\tilde{X}}$, $B_{\tilde{c}}$ lie in $\mathcal{U}_{\mathrm{fl}}(\tilde{X})$ and in the corresponding set for $\tilde{c}$, by Remark 38.8(i) and (DC1), as in the first paragraph of the proof of Lemma 53.4.

If $\hat{Y} = Y'$, then $Y^{(r')} \subset Y'$ (Remark 47.11, collar inclusions). By the formula for $E^{\#}(Y)$ in the proof of Theorem 48.6, the output at Y involves only the pieces $E(Z)$ with $Z \subset Y$, and these have $Z^{(r')} \subset Y'$. The restriction to the polymers with $Z^{(r')} \subset Y'$ keeps every datum of the system inside Y'. Its lift has diameter less than half the side.

Proposition 59.8 (The step on the infinite lattice)**.** *In the setting of Definition 59.7:*

(a) (Torus independence.) $\mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}; B)$ *does not depend on the admissible side* n*.*

(b) (Covariance.) *Let* $a \in M\mathbb{Z}^4$ *in the units of scale* $k+1$*, and let* $a_k = La$ *be the same vector in the units of step* k*. Then*

$$\mathscr{S}_{k,\tilde{Y}'+a}(g; \tau_{a_k}\mathcal{E}^{\mathrm{fl}}; \tau_a B) = \mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}; B).$$

(c) (The raw generation.) *Let* $\tilde{n} \in \mathfrak{N}_{k+1}$ *with* $\operatorname{diam} \tilde{Y}' \leq \frac{1}{2}\tilde{n} - 8$*, and consider the step from scale* k *to scale* $k+1$ *on the torus whose lattice of scale* $k+1$ *has side* $\tilde{n}$*, at the coupling* g*. If* $k \geq 1$*, let* $\mathcal{E}_k$ *be old activities of step* k *on it with the hypotheses of Lemma 53.4 and* $\|\mathcal{E}_k\| \leq C_{\mathrm{old}}\mathsf{K}$*; for instance those of Theorem 53.7(a). Let* $E' = E'^{(k+1)}[\mathcal{E}_k, g]$ *be the raw new generation (Definition 53.3). Then, with* $\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}$ *the flat restriction of* $\mathcal{E}_k$*,*

$$E'(\pi_{\tilde{n}}\tilde{Y}', V_B[\tilde{n}]) = \mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}; B) \qquad \text{for every } B \in \mathcal{U}_1(\tilde{Y}').$$

(d) (Grid covariance.) *In (c), if* $k \geq 1$*, suppose that* $\mathcal{E}_k(Y + a, \tau_a U) = \mathcal{E}_k(Y, U)$ *for every polymer* Y *of scale* k *that does not wrap, every* $a \in M\mathbb{Z}^4$ *in the units of scale* k*, and every* $U \in \mathcal{U}^{\flat}(Y)$*. Then* $E'(Y' + a, \tau_a V) = E'(Y', V)$ *for every polymer* Y' *of scale* $k+1$ *that does not wrap, every* $a \in M\mathbb{Z}^4$ *in the units of scale* $k+1$*, and every* $V \in \mathcal{U}_1(Y')$*. The old activities of Definition 53.1 built from generations with (G2) have this covariance.*

Proof. (a) Let n, n' be admissible for $\tilde{Y}'$, and let ι be the transfer of the lift $\tilde{Y}'$ with the vector 0, a grid transfer with $P = Y'$ (Section 59.1). Details

- *References.* By Lemma 59.1(b), both references equal V_B on the bonds of P, read in the lift. So they agree on P.
- *Pieces.* The flat data $\mathcal{E}_X$ of Definition 59.7 on the two tori are values of the same $\mathcal{E}^{\mathrm{fl}}$ at the lifts, so they agree under ι. The map $Z \mapsto \iota Z$ is a bijection of the polymers of step k in P onto those in $\iota(P)$. It preserves the numbers of boxes and the touching relation, and $Z^{(r')} \subset Y'$ if and only if $(\iota Z)^{(r')} \subset \iota(Y')$, since $r' < R_{\mathrm{g}}$ (Lemma 41.10(a), Lemma 41.11(a)). By Lemma 59.5(a),(b), $E'(\iota Z)(\iota_* \xi) = E(Z)(\xi)$ for every such Z. The bonds of $\mathfrak{F}$ in P are mapped onto those of $\mathfrak{F}'$ in $\iota(P)$ (Lemma 41.11(a)).
- *Outputs.* By the proof of Theorem 48.6,

$$E^{\#}(Y) = \sum_{\mathcal{C}:\, \bigcup_{\Gamma\in\mathcal{C}} \operatorname{supp}\Gamma = Y} \phi^T(\mathcal{C}) \prod_{\Gamma\in\mathcal{C}} k(\Gamma), \qquad k(\Gamma) = \int \prod_{Z\in\Gamma} K(Z)\, d\mu^*,$$

with $K(Z) = e^{E(Z)} - 1$. The clusters $\mathcal{C}$ consist of objects Γ, finite sets of polymers connected for touching, with supports in Y. So for $Y \subset P$ the bijection above induces bijections of the objects and of the clusters, with the same incompatibility graphs and hence the same Ursell coefficients. The integrand of $k(\Gamma)$ depends on ξ only on the bonds of $\mathfrak{F}$ in $\bigcup_{Z\in\Gamma} Z^{(r')} \subset P$, and $d\mu^*$ is the product over these bonds of the standard Gaussian measures restricted to $|\xi(b)| \le p_0(g)$ and normalized. So $k'(\iota\Gamma) = k(\Gamma)$. The series converge absolutely, by the Kotecký–Preiss bound in the proof of Theorem 48.6. Hence $E'^{\#}(\iota Y; U'') = E^{\#}(Y; U')$, and in the same way at the reference 1.

- *Index sets.* The rule $Y \mapsto \hat{Y}$ is local (Remark 49.4), so ι maps the polymers Y with $\hat{Y} = Y'$ onto those with $\hat{Y} = \iota(Y')$. The first two sums in Definition 59.7 are defined on the infinite lattice.

(b) The same argument applies to the transfer of the lift $\tilde{Y}'$ with the vector a_k, from $\pi_n\tilde{Y}'$ onto $\pi_n(\tilde{Y}' + a_k)$ on the same torus $\mathbb{T}_\eta[n]$. The reference of $\tau_a B$ is τ_{a_k} of that of B, and the flat data $\tau_{a_k}\mathcal{E}^{\mathrm{fl}}$ agree with $\mathcal{E}^{\mathrm{fl}}$ under this transfer. For the one-loop part, Proposition 41.14(d) gives the covariance of E_k^∞ and J_k^∞, and the closure commutes with the translation, as in the proof of Lemma 58.3.

(c) By Definition 53.3 and Lemma 53.4, with $V = V_B[\tilde{n}]$,

$$E'(Y', V) = \sum_{\bar{X}=Y'} \mathfrak{e}(X, V) + \sum_{Y'_c=Y'} \log \mathfrak{j}_c(V) + \sum_{\hat{Y}=Y'} (E^{\#}(Y, V) - E^{\#}(Y, 1)),$$

where the outputs $E^{\#}(Y, V)$ are those of the restricted system of Remark 47.11 for Y', at an extension of V. They do not depend on the extension (Remark 47.11). We take the representation $(1, B)$ and its cutoff extension U'.

- *One-loop part.* The polymers X with $\bar{X} = Y'$ lie in Y'. By Remark 38.8(i) and (DC1), V agrees on X with $V_{B_{\tilde{X}}}$, where $B_{\tilde{X}} \in \mathcal{U}_{\mathrm{fl}}(\tilde{X})$. So $\mathfrak{e}(X, V) = E_k^\infty(\tilde{X}, B_{\tilde{X}})$ by Proposition 41.14(d), and likewise for the Jacobian terms. Lifting is a bijection of the index sets, as in the proof of Lemma 58.3.

- *Outputs.* The restricted system for Y' consists of the pieces $E(Z)$, $Z \subset Y'$. By the remark after Definition 59.7, the outputs with $\hat{Y} = Y'$ involve only the pieces with $Z^{(r')} \subset Y'$. These depend on the reference only on $Z^{(r')}$ (Propositions 47.5(d), 47.7(d), 47.10(b), Lemma 47.2(a); at $k = 0$ the factorization of the composed pieces in the proof of Proposition 52.2 replaces Proposition 47.5(d)), where U' equals V. Their pure parts are those of Definition 59.7 on the torus of side $L\tilde{n}$ at step k. Their mixed parts are built from $f(X; s, Z) = \mathcal{E}_k(X, e^{\mathrm{i}\eta A_*(s;\hat{Z}(Z))}U') - \mathcal{E}_k(X, U')$ (Proposition 47.10). In the proof of that proposition and in Lemma 47.2, $\mathcal{E}_k(X, \cdot)$ is evaluated only at configurations with real part 1, that is, at points of $\mathcal{U}_1^\flat(X)$ (proof of Lemma 59.4). There it equals $\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}(\tilde{X}, \cdot)$ at the lift. So the two systems have the same pieces with $Z^{(r')} \subset Y'$, and the outputs with $\hat{Y} = Y'$ coincide.

- *Normalization.* At the real critical reference 1, the restricted and the full systems have the same outputs (Remark 47.11), so $E^{\#}(Y, 1)$ is the output of Definition 59.7 at $B = 0$.

Hence $E'(Y', V)$ is the value of Definition 59.7 computed with $n = L\tilde{n}$, which is admissible for $\tilde{Y}'$. By (a), this value is $\mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}; B)$.

(d) Let Y' not wrap, let $\tilde{Y}'$ be a lift, $V = V_B[\tilde{n}]$, and $a \in M\mathbb{Z}^4$ in the units of scale $k+1$, with $a_k = La$. Then $Y' + a$ does not wrap, $\tilde{Y}' + a$ is a lift of it, and $\tau_a V = V_{\tau_a B}[\tilde{n}]$. For $\tilde{Y} \subset \tilde{Y}'$, the hypothesis gives

$$\mathcal{E}_k(\pi\tilde{Y} + a_k, \tau_{a_k} V_{B''}) = \mathcal{E}_k(\pi\tilde{Y}, V_{B''}),$$

since $a_k \in M\mathbb{Z}^4$ in the units of scale k. So the flat restriction of $\mathcal{E}_k$ to $\tilde{Y}' + a$ is $\tau_{a_k}\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}$. By (c) and (b), $E'(Y' + a, \tau_a V) = E'(Y', V)$.

For the last statement, let $a \in M\mathbb{Z}^4$ in the units of scale k, and let Y not wrap. In the units of scale $l \le k$, $a \in L^{k-l}M\mathbb{Z}^4 \subset M\mathbb{Z}^4$. So (G2) gives $E^{(l)}(X + a, \tau_a U) = E^{(l)}(X, U)$ for $U \in \mathcal{U}^\flat(X)$.

- The translation by a maps the grid cubes of every scale $l \le k$ onto grid cubes and $\square^+$ onto $(\square + a)^+$. It preserves d_l, the classes of Definition 49.5 and the polymers that do not wrap, and $\mathrm{cl}_k(X + a) = \mathrm{cl}_k X + a$.
- For the polymers that do not wrap, the base cubes satisfy $Q(X + a) = Q(X) + a$ (Section 49, base cubes). The polymers $X \subset Y$ do not wrap, and the small polymers do not wrap (Lemma 49.6(a)).
- $S_{\square+a}(\tau_a U) = S_\square(U)$ (Section 49), and τ_a maps $\mathcal{U}^\flat(Y)$ onto $\mathcal{U}^\flat(Y + a)$ (Section 50).

So $\Psi^{(l)}_{\square+a}(\tau_a U) = \Psi^{(l)}_\square(U)$ and $E^{(l),\mathrm{ne}}_{Y+a}(\tau_a U) = E^{(l),\mathrm{ne}}_Y(U)$. Hence $\mathcal{E}_k(Y + a, \tau_a U) = \mathcal{E}_k(Y, U)$. □

Lemma 59.9 (Continuity in the old data)**.** *Let $k \ge 1$ and $g \in (0, \gamma]$. Let $\tilde{Y}'$ be a polymer of scale $k + 1$ of the infinite lattice, and let $\mathcal{E}^{\mathrm{fl}}_l \in \mathfrak{D}_k(\tilde{Y}')$ for $l \in \mathbb{N} \cup \{\infty\}$. Suppose that $\mathcal{E}^{\mathrm{fl}}_l(\tilde{Y}, B'') \to \mathcal{E}^{\mathrm{fl}}_\infty(\tilde{Y}, B'')$ as $l \to \infty$, for every $\tilde{Y} \subset \tilde{Y}'$ and every $B'' \in \mathcal{U}^\flat_1(\tilde{Y})$. Then*

$$\mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}_l; B) \to \mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}_\infty; B) \qquad \text{for every } B \in \mathcal{U}_1(\tilde{Y}').$$

Proof. Fix an admissible side n and compute every $\mathscr{S}$ on $\mathbb{T}_\eta[n]$ (Proposition 59.8(a)). Only the mixed pieces depend on l. The reference U', the fields $Z(s)$ and $\hat{Z}(Z(s))$, and $A_*(s; \hat{Z}(Z(s)))$ do not. We follow Steps 2–4 of the proof of Proposition 53.10(b), with the couplings fixed. Details

Pieces. Fix $s \in \mathbb{D}_{\kappa'}$, ξ with $|\xi|_\infty \le p_0(g)$, and an old polymer $X \subset Z$. Then $f_l(X; s, Z(s)) = \mathcal{E}^{\mathrm{fl}}_l(\tilde{X}, \tilde{B}_1) - \mathcal{E}^{\mathrm{fl}}_l(\tilde{X}, \tilde{B}_0)$, where $\tilde{B}_1$ and $\tilde{B}_0$ are the lifts of the flat fields, on $X^{(3)}$, of the composed configuration and of U'. They lie in $\mathcal{U}^\flat_1(\tilde{X})$ by Lemma 59.4(a), and they do not depend on l. So $f_l(X; s, Z(s)) \to f_\infty(X; s, Z(s))$, and $|f_l| \le 4\tau C_{\mathrm{old}}\mathsf{K}$. As in Step 2 of that proof, Cauchy's formula on the circles of radius 1 about the coordinates of the points $s(Z, t)$ writes $f_{l,\mathrm{loc}}(X, Z)$ as an integral of f_l over a compact set, against a bounded kernel that does not depend on l. By dominated convergence, $f_{l,\mathrm{loc}}(X, Z) \to f_{\infty,\mathrm{loc}}(X, Z)$, so the mixed pieces converge, for every ξ with $|\xi|_\infty \le p_0(g)$.

Outputs. The number $k_l(\Gamma) = \int \prod_{Z\in\Gamma} K_l(Z)\, d\mu^*$ is an integral over the finitely many bonds of $\mathfrak{F}$ in $\bigcup_{Z\in\Gamma} Z^{(r')}$. Its integrand converges at every ξ in the support of $d\mu^*$, and it is bounded by $\prod_{Z\in\Gamma} 2\epsilon_* e^{-\kappa(d_k(Z)+c_\natural)}$, uniformly in l. By dominated convergence, $k_l(\Gamma) \to k_\infty(\Gamma)$. The proof of Theorem 48.6 bounds $|k_l(\Gamma)|$ by $\bar{k}(\Gamma) = \prod_{Z\in\Gamma} 2\epsilon_* e^{-\kappa(d_k(Z)+c_\natural)}$, which does not depend on l, and the Kotecký–Preiss theorem gives $\sum_{\mathcal{C}} |\phi^T(\mathcal{C})| \prod_{\Gamma\in\mathcal{C}} \bar{k}(\Gamma) < \infty$, the sum over the clusters with union Y. By the Weierstrass test, $E^\#_l(Y; U') \to E^\#_\infty(Y; U')$, and in the same way at the reference 1. The sum in Definition 59.7 is finite. □

59.4 Convergence over the scales

Definition 59.10 (Convergence at flat points)**.** Let $i \ge 1$, and let $\mathcal{U}_*$ be $\mathcal{U}_1$ or $\mathcal{U}^\flat_1$. For $m \in \mathfrak{N}_i$ let $E[m]$ be a family indexed by the polymers of scale i of the torus whose lattice of scale i has side m. Suppose that for every such X that does not wrap, $E[m](X, \cdot)$ is a holomorphic function on $\mathcal{U}_*(X)$ of the flat field that depends only on the bonds of X, and that

$$\mathsf{K}' = \sup_m \sup_X e^{\kappa_n d_i(X)} \sup_{\mathcal{U}_*(X)} |E[m](X, \cdot)| < \infty,$$

the second supremum over the polymers that do not wrap. The collection $(E[m])_{m\in\mathfrak{N}_i}$ *converges on* $\mathcal{U}_*$ if, for every polymer $\tilde{X}$ of scale i of the infinite lattice and every $B \in \mathcal{U}_*(\tilde{X})$, the numbers

$E[m](\pi_m \tilde{X}, V_B[m])$ converge as $m \to \infty$ in $\mathfrak{N}_i$. These numbers are defined for the m with $\operatorname{diam} \tilde{X} \le \frac{1}{2}m - 8$. The limit $E^\infty(\tilde{X}, B)$ is the *limit* of the collection.

This is a qualitative counterpart of (58.3), on $\mathcal{U}_1$ or $\mathcal{U}_1^\flat$; (58.3) itself is not changed.

Lemma 59.11 (Limits of convergent families)**.** *Let $(E[m])_{m \in \mathfrak{N}_i}$ converge on $\mathcal{U}_*$, with limit E^∞.*

(a) Each $E^\infty(\tilde{X}, \cdot)$ is holomorphic on $\mathcal{U}_*(\tilde{X})$, depends only on B on the bonds of $\tilde{X}$, and satisfies $|E^\infty(\tilde{X}, \cdot)| \le \mathsf{K}' e^{-\kappa_n d_i(\tilde{X})}$. The convergence is uniform on the compact subsets of $\mathcal{U}_*(\tilde{X})$. If the functions $E[m](X, \cdot)$ vanish at 0, are real for real B, or are invariant under small real gauge transformations in the sense of Section 58.3, then so is $E^\infty(\tilde{X}, \cdot)$.

(b) If $E[m](X + a, \tau_a V) = E[m](X, V)$ for every m, every polymer X that does not wrap, every $a \in M\mathbb{Z}^4$ and every $V \in \mathcal{U}_*(X)$, then $E^\infty(\tilde{X} + a, \tau_a B) = E^\infty(\tilde{X}, B)$.

(c) Suppose, in addition to the hypothesis of (b), that every $E[m]$ is a family of normalized graded activities at the polymers of scale i, on domains containing the $\mathcal{U}^\flat(X)$, real at real configurations and invariant under real gauge transformations, with $\mathsf{K}'' = \sup_m \|E[m]\| < \infty$ (the norm of Section 58.2, over all polymers). Then:

- *the restriction of E^∞ to the domains $\mathcal{U}_1^\flat$ is a family of flat activities with (G2), and $\|E^\infty\| \le \mathsf{K}''$;*
- *$\beta^{\mathrm{vol}}_{E[m]} \to \beta^\infty_{E^\infty}$ as $m \to \infty$ in $\mathfrak{N}_i$;*
- *the kernels of the second variations at $A = 0$ of $A \mapsto \sum_X E[m](X, e^{\mathrm{i}\eta A})$ converge to the kernel $\widetilde{\Pi}$ of E^∞ (Section 58.3) with uniform decay, in the sense of Lemma 54.10(e).*

Proof. (a) Let $\operatorname{diam} \tilde{X} \le \frac{1}{2}m - 8$. Then $d_i(\pi_m \tilde{X}) = d_i(\tilde{X})$. Details

- Projecting a tree of $\mathbb{R}^4$ that meets the cubes of $\tilde{X}$ gives a tree of the same length that meets those of $\pi_m \tilde{X}$. So $d_i(\pi_m \tilde{X}) \le d_i(\tilde{X})$.
- Conversely, the projections to the coordinate circles of the union of the closed cubes of $\pi_m \tilde{X}$ are arcs (Section 49, base cubes), of length at most $\operatorname{diam} \tilde{X} + \eta < \frac{1}{2}m$. Map each circle onto its arc by the identity on the arc, and on the complement, which is longer than the arc, by a 1-Lipschitz map onto the arc that fixes the two endpoints. This map is 1-Lipschitz. The product of these maps is a 1-Lipschitz retraction of the torus onto a box that contains the cubes and lifts isometrically.
- It maps a tree that meets the cubes onto a connected finite union of rectifiable curves, of no greater length, that meets the same cubes and contains a tree that meets them. So $d_i(\tilde{X}) \le d_i(\pi_m \tilde{X})$.

The functions $B \mapsto E[m](\pi_m \tilde{X}, V_B[m])$ are then holomorphic on the open connected set $\mathcal{U}_*(\tilde{X})$ and bounded by $\mathsf{K}' e^{-\kappa_n d_i(\tilde{X})}$. By Montel's theorem every subsequence has a further subsequence that converges uniformly on compact subsets to a holomorphic function, and this function is the pointwise limit $E^\infty(\tilde{X}, \cdot)$. So the whole sequence converges uniformly on compact subsets, and $E^\infty(\tilde{X}, \cdot)$ is holomorphic. The bound, the dependence on the bonds of $\tilde{X}$, the vanishing at 0, the reality and the invariance pass to the pointwise limit.

(b) $\pi_m(\tilde{X} + a) = \pi_m \tilde{X} + a$ and $V_{\tau_a B}[m] = \tau_a V_B[m]$.

(c) *Flat activities.* The torus functions vanish at 0 and are real for real B. They are invariant under small real gauge transformations, as in the proof of Proposition 41.14(d): extend the gauge function from $\pi_m \tilde{X}$ to the torus, and use the invariance of $E[m](X, \cdot)$ under real gauge transformations and its dependence on the bonds of X only. By (a) and (b), the

restriction of E^∞ to $\mathcal{U}_1^\flat$ is a family of flat activities with (G2). Moreover $\|E^\infty\| \le \min(\mathsf{K}', \mathsf{K}'')$, since $\mathcal{U}_1^\flat(X) \subset \mathcal{U}^\flat(X) \cap \mathcal{U}_*(X)$ and $d_i(\pi_m \tilde{X}) = d_i(\tilde{X})$ by (a).

The coefficient. For a polymer X that does not wrap and small $|t|$, the configurations $V_{t,F}^X$ of Lemma 50.12(a) are points of $\mathcal{U}_1^\flat(X)$, and $\tau_a V_{t,F}^X = V_{t,F}^{X+a}$ (proof of Lemma 50.12(b)). So the hypothesis of (b) gives $c_{E[m]}^{X+a} = c_{E[m]}^X$. The sums $\sum_{X \supset Q} c_{E[m]}^X(F)/|X|$ over the polymers that do not wrap are then the same for all grid cubes Q, and the paragraph "The volume coefficient" of Section 58.2 gives

$$\bar{T}_{E[m]}^{\mathrm{vol}}(F) = M^{-4} \sum_{X \supset Q} \frac{c_{E[m]}^X(F)}{|X|}$$

for one fixed cube Q. Lifting with $Q \mapsto \tilde{Q}$ is a bijection from the polymers $X \supset Q$ that do not wrap onto the polymers $\tilde{X} \supset \tilde{Q}$ of the infinite lattice of diameter at most $\frac{1}{2}m - 8$ (proof of Lemma 58.4(d)). For fixed $\tilde{X}$ and m large:

- $c_{E[m]}^{\pi_m \tilde{X}}(F)$ is a Cauchy integral over a fixed circle in t, as in Step 5 of the proof of Proposition 53.10(b). By the uniform convergence on compact sets in (a), it converges to $c_{E^\infty}^{\tilde{X}}(F)$.
- By Lemma 50.12(a), the terms are bounded by $9d^2M^2(1+d_i(\tilde{X}))^2(\delta_1^\flat)^{-2}\mathsf{K}'' e^{-\kappa_n d_i(\tilde{X})}|F|^2$. This is summable over $\tilde{X} \supset \tilde{Q}$ by (49.2), since $(1+t)^2 e^{-(\kappa_n - \kappa_0)t} \le e^{-(\kappa_n - \kappa_0 - 2)t}$ and $\kappa_n - \kappa_0 \ge 3$.

By dominated convergence, $\bar{T}_{E[m]}^{\mathrm{vol}}(F) \to \bar{T}_{E^\infty}(F)$ for every F, and so $\beta_{E[m]}^{\mathrm{vol}} \to \beta_{E^\infty}^\infty$ by (58.1) and the definition of $\beta_{E^\infty}^\infty$.

The kernels. Fix two bonds of the infinite lattice and their projections b, b'. The torus kernel at (b, b') is the sum, over the polymers X of the torus that contain both, of the kernels of the second variations of $E[m](X, e^{i\eta A})$. Only small fields A enter, so only points of $\mathcal{U}_1^\flat(X)$ for the polymers that do not wrap. These are:

- *Polymers that do not wrap.* Their kernels are $2\kappa_X(b, b')$, in the notation of Lemma 58.4(a), and they correspond to the polymers $\tilde{X}$ of the infinite lattice of diameter at most $\frac{1}{2}m - 8$ that contain the lifts. For fixed $\tilde{X}$ and m large, $\kappa_{\pi_m \tilde{X}} \to \kappa_{\tilde{X}}$, by Cauchy's estimate and the uniform convergence on compact sets.
- *All polymers.* By the same Cauchy estimate, each kernel is bounded by $C_\eta \mathsf{K}'' e^{-\kappa_n d_i(X)}$, and this is summable by (49.2).
- *Wrapping polymers.* They have $d_i(X) \ge m/(4M)$ (Lemma 52.1(b)), so they contribute at most $C_\eta \mathsf{K}'' K_0 e^{-(\kappa_n - \kappa_0)m/(4M)} \to 0$.

By dominated convergence, the torus kernels converge entrywise to $\widetilde{\Pi} = 2\sum_{\tilde{X}} \kappa_{\tilde{X}}$. The bound $C_\eta K_0 \mathsf{K}'' e^{2(\kappa_n - \kappa_0)} e^{-(\kappa_n - \kappa_0)|m_b - m_{b'}|_m/M}$, uniform in m, is proved as at the end of the proof of Lemma 58.4(d), and it uses only K''. □

Theorem 59.12 (Generation locality)**.** *Let G be compact, connected and simple, let γ satisfy (Γ1)–(Γ3), let $j \ge 1$ and $\pi = (g_0, \dots, g_{j-1}) \in \Sigma_{j-1}$, and use the notation of Section 59.1. Then for every $i \le j$:*

(a) For every $m \in \mathfrak{N}_i$, $E'^{(i)}[m]$ has the covariance of Lemma 59.11(b) on $\mathcal{U}_1$. The collection $(E'^{(i)}[m])_{m \in \mathfrak{N}_i}$ converges on $\mathcal{U}_1$, with $\mathsf{K}' \le \mathsf{K}$. Its limit $E'^{(i),\infty}$ is covariant under the translations by $M\mathbb{Z}^4$, vanishes at 0, is real for real B, and is invariant under small real gauge transformations.

(b) *The collection* $(E^{(i)}[m])_{m\in\mathfrak{N}_i}$ *converges on* $\mathcal{U}_1^\flat$*, with the limit*

$$E^{(i),\infty}(\tilde X,B)=\frac{1}{|B_4|}\sum_{R\in B_4}E'^{(i),\infty}(R^{-1}\tilde X,R^{-1}B),$$

with B_4 *acting about the origin.*

(c) $\beta_i[m]\to\beta_i^\infty:=\beta^\infty_{E'^{(i),\infty}}$ *as* $m\to\infty$ *in* $\mathfrak{N}_i$*.*

(d) *If* $i<j$*, then, for every polymer* $\tilde Y'$ *of scale* $i+1$ *of the infinite lattice, the flat restrictions* $\mathcal{E}^{\mathrm{fl}}_{\tilde Y'}$ *of the old activities* $\mathcal{E}_i[m']$ *converge, as* $m'\to\infty$ *in* $\mathfrak{N}_{i+1}$*, at every point of the domains* $\mathcal{U}_1^\flat(\tilde Y)$*,* $\tilde Y\subset\tilde Y'$*, to the flat old data* $\mathcal{E}_i^\infty\in\mathfrak{D}_i(\tilde Y')$*. These data are built by Definition 53.1 on the infinite lattice from the* $E^{(l),\infty}$ *and the* β_l^∞*,* $l\le i$*.*

The limits depend only on $(g_0,\dots,g_{i-1})$*.*

Proof. By induction on i. We show that (a) at all scales $l\le i$ gives (b) and (c) at the scale i, and (d) at the scale i if $i<j$. Then we show (a) at $i=1$, and (a) at $i=k+1$ from (b)–(d) at all scales $\le k$. Details

(a) at scale i *gives (b).* Let X be a polymer of scale i of the torus of side m that does not wrap, and let $U=V_B[m]\in\mathcal{U}_1^\flat(X)$. By Definition 50.7,

$$E^{(i)}[m](X,U)=|\mathfrak{S}_i^{\mathrm{grid}}|^{-1}\sum_r E'^{(i)}[m](r^{-1}X,r^{-1}U).$$

- Write $r(x)=Rx+a$ with $R\in B_4$, acting about the origin, and $a\in M\mathbb{Z}^4$. Then $r^{-1}X=R^{-1}X-R^{-1}a$ and $r^{-1}U=\tau_{-R^{-1}a}(R^{-1}U)$, with $R^{-1}a\in M\mathbb{Z}^4$.
- The polymer $R^{-1}X$ does not wrap, and $R^{-1}U\in\mathcal{U}_1(R^{-1}X)$ (Lemma 59.1(c)). So the covariance in (a) gives $E'^{(i)}[m](r^{-1}X,r^{-1}U)=E'^{(i)}[m](R^{-1}X,R^{-1}U)$.
- Each R is the linear part of $(m/M)^4$ elements r (proof of Proposition 50.13(c)). So $E^{(i)}[m](X,U)=|B_4|^{-1}\sum_R E'^{(i)}[m](R^{-1}X,R^{-1}U)$.

Let $\tilde X$ be a polymer of the infinite lattice and $B\in\mathcal{U}_1^\flat(\tilde X)$. For m large, $\pi_m\tilde X$ and the $\pi_mR^{-1}\tilde X$ do not wrap, and $R^{-1}V_B[m]=V_{R^{-1}B}[m]$ with $R^{-1}B\in\mathcal{U}_1(R^{-1}\tilde X)$ (Lemma 59.1(c)). By (a), each term converges, which gives (b). The bound $\mathsf{K}'\le\mathsf{K}$ on $\mathcal{U}_1^\flat$ holds since $\|E^{(i)}[m]\|_i\le\mathsf{K}$ (Proposition 53.10(a)).

(a) at scale i *gives (c).* By Proposition 50.13(c) and the paragraph "The volume coefficient" of Section 58.2, $\beta_i[m]=\beta^{\mathrm{vol}}_{E'^{(i)}[m]}$. The raw generations are normalized graded activities on the domains $\mathcal{U}(X)\supset\mathcal{U}^\flat(X)$, real at real configurations and gauge invariant (Lemma 53.4(a)), with $\|E'^{(i)}[m]\|\le\|E'^{(i)}[m]\|_i^{\mathrm{raw}}\le\mathsf{K}$ (Theorem 53.7(b); Proposition 53.8 for $i=1$). Lemma 59.11(c) applies on $\mathcal{U}_1$, with the covariance of (a), and gives (c). Applied on $\mathcal{U}_1^\flat$ to the generations $E^{(i)}[m]$, which have (G2) and (G4), so that $\beta_{E^{(i)}[m]}=\beta^{\mathrm{vol}}_{E^{(i)}[m]}$ (Section 58.2), it also gives $\beta_i^\infty=\beta^\infty_{E^{(i),\infty}}$.

(a)–(c) at the scales $l\le i$ *give (d).* Let $\tilde Y'$ be a polymer of scale $i+1$, $\tilde Y\subset\tilde Y'$ a polymer of scale i, and $B''\in\mathcal{U}_1^\flat(\tilde Y)$. On the torus whose lattice of scale $i+1$ has side m', the side at scale l is $L^{i+1-l}m'\in\mathfrak{N}_l$. By Definition 53.1,

$$\mathcal{E}_i[m'](\pi\tilde Y,V_{B''})=\sum_{l=1}^{i}\Big(\mathbf{1}_{\{\pi\tilde Y=\square^+\}}\Psi^{(l)}_\square(V_{B''})+E^{(l),\mathrm{ne}}_{\pi\tilde Y}(V_{B''})\Big).$$

- *Finite sums.* The polymers X in these sums lie in $\pi\tilde Y$: the small polymers based in $\square$ lie in $\square^+$ (Lemma 49.6(b)), and the others have $\mathrm{cl}_i\,X=\pi\tilde Y$. So the sums are finite.

- *Classes.* For m' large, $\pi\tilde{Y}$ does not wrap. The polymers $X \subset \pi\tilde{Y}$ of scale l then correspond, under lifting, to the polymers $\tilde{X} \subset \tilde{Y}$ of the infinite lattice, with the same classes, closures and base cubes (Section 59.1, the infinite lattice).
- *Terms.* By Lemma 59.1(a), $V_{B''}$ is on $\tilde{X}^{(3)}$ a point of $\mathcal{U}_1^\flat(\tilde{X})$ of scale l. So each term $E^{(l)}[L^{i+1-l}m'](X, V_{B''})$ converges by (b) at the scale l, along a subsequence of $\mathfrak{N}_l$. Each $\beta_l[L^{i+1-l}m']$ converges by (c), and $S_\square(V_{B''})$ does not depend on m'.

So $\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}(\tilde{Y}, B'')$ converges, and the limit is the value of the data $\mathcal{E}_i^\infty$ built by Definition 53.1 on the infinite lattice. These data are finite sums of holomorphic functions on $\mathcal{U}_1^\flat(\tilde{Y})$, by Lemma 59.1(a). They vanish at 0, since the $E^{(l),\infty}$ do (Lemma 59.11(a)) and $S_\square(1) = 0$. They satisfy the bound $C_{\mathrm{old}}\mathsf{K}e^{-\kappa_n d_i(\tilde{Y})}$, as limits of the values of the torus activities, which satisfy it (Proposition 53.2(b)). So $\mathcal{E}_i^\infty \in \mathfrak{D}_i(\tilde{Y}')$.

(a) at $i = 1$. Let $\tilde{X}$ be a polymer of scale 1 of the infinite lattice and $B \in \mathcal{U}_1(\tilde{X})$. For $m \in \mathfrak{N}_1$ with $\operatorname{diam}\tilde{X} \le \frac{1}{2}m - 8$, Proposition 59.8(c) at $k = 0$ gives $E'^{(1)}[m](\pi_m\tilde{X}, V_B[m]) = \mathscr{S}_{0,\tilde{X}}(g_0; B)$, which does not depend on m. The covariance on $\mathcal{U}_1$ is Proposition 59.8(d), with no old activities. The bound $\mathsf{K}' \le \mathsf{K}$ holds since $\mathcal{U}_1(X) \subset \mathcal{U}(X)$ and $\|E'^{(1)}[m]\|_1^{\mathrm{raw}} \le \mathsf{K}$ (Proposition 53.8). The other properties follow from Lemma 59.11(a),(b), since $E'^{(1)}[m]$ has (G1) on $\mathcal{U}(X)$ (Lemma 53.4(a)) and is invariant under small real gauge transformations, as in the proof of Lemma 59.11(c).

(a) at $i = k + 1 \le j$. Assume (a)–(c) at the scales $l \le k$, so that (d) holds at k. Let $\tilde{Y}'$ be a polymer of scale $k + 1$ of the infinite lattice and $B \in \mathcal{U}_1(\tilde{Y}')$. For $m \in \mathfrak{N}_{k+1}$ with $\operatorname{diam}\tilde{Y}' \le \frac{1}{2}m - 8$, the raw generation $E'^{(k+1)}[m]$ is that of Proposition 53.10(a), built at the coupling g_k with the old activities $\mathcal{E}_k[m]$. These satisfy the hypotheses of Lemma 53.4, with $\|\mathcal{E}_k[m]\| \le C_{\mathrm{old}}\mathsf{K}$, by Theorem 53.7(a), applied as in the proof of Proposition 53.10(a). Proposition 59.8(c) gives

$$E'^{(k+1)}[m](\pi_m\tilde{Y}', V_B[m]) = \mathscr{S}_{k,\tilde{Y}'}(g_k; \mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}[\mathcal{E}_k[m]]; B).$$

By (d) at k and Lemma 59.9, the right side converges to $\mathscr{S}_{k,\tilde{Y}'}(g_k; \mathcal{E}_k^\infty; B)$. So the collection converges on $\mathcal{U}_1$. The bound $\mathsf{K}' \le \mathsf{K}$ holds since $\|E'^{(k+1)}[m]\|_{k+1}^{\mathrm{raw}} \le \mathsf{K}$ (Theorem 53.7(b)).

The old activities $\mathcal{E}_k[m]$ are built from the generations $E^{(l)}[\cdot]$, which have (G2) (Lemma 50.8(a)). So Proposition 59.8(d) gives the covariance on $\mathcal{U}_1$. The remaining properties of the limit follow from Lemma 59.11(a),(b), as at $i = 1$.

All the constructions use only $(g_0, \dots, g_{i-1})$. □

Remark 59.13 (What is not proved). (a) *No rate.* The convergence in Theorem 59.12 has no rate, and it is not shown to be uniform in the couplings. A rate, such as $|\beta_j[\tilde{n}] - \beta_j^\infty| \le C'e^{-\tilde{n}/(4M)}$, is expected to follow from three ingredients:

- a Lipschitz bound of $\mathscr{S}_{k,\tilde{Y}'}$ in the flat old data, with a constant proportional to τ, by an interpolation in the mixed pieces as in Corollary 48.7;
- a bound on the errors of the old activities;
- for constants uniform in j, a further condition on γ.

Uniformity in the couplings would need these constants to be uniform on Σ_{j-1}. This is not carried out here.

(b) *The couplings of the flow.* In the flow of Theorem 53.12 the couplings are computed on the given torus, so they may depend on it. Theorem 59.12 gives $\beta_j[\tilde{n}] \to \beta_j^\infty$ at fixed couplings. It gives no bound on $\beta_j - \beta_j^\infty$ on a given torus, and, without uniformity in the couplings, it does not show that this difference tends to 0 along the flows on growing tori.

(c) *Comparison with Bałaban.* Bałaban's generation $E^{(j+1)}$ and his β_{j+1} are functions of g_j alone, and he takes the infinite-volume limit of (1.20) "by the localized representation (1.7)" [18, (1.18), (1.20)–(1.22)]. The generations of this note depend on $(g_0, \dots, g_{j-1})$ through the old activities and their counterterms β_i. This is why their convergence needs the induction over the scales.

Conclusion

60 Outlook

What has been done. The dimension is $d = 4$ throughout.

- *Part I* treats the trivial background $U = 1$ on the torus, with blocks centred at the points of the unit lattice.
- *Part II* extends the results to backgrounds close to 1 in a global gauge, by perturbation.
- *Part III* replaces the global hypothesis by local regularity: on every cube of size of order M there is a gauge in which the background is close to 1. It provides:
  - Dirichlet operators on unions of blocks;
  - a local proof of coercivity by IMS localization;
  - locality of all operators in the domain and in the background;
  - Bałaban's random-walk expansion for the scalar Green function; for the gauge-field Green function it is completed in factorized form in Part IX (Section 41).
- *Part IV* constructs Bałaban's symmetric nonlinear averaging in four dimensions, with centred blocks and group averages over the coordinate-ordered contours, the averaged axial gauge and the gauge-adapted averaging, and proves the axial-to-Landau chart. The linearized averaging is exactly covariant ($E = 0$), it is covariant under the translations and the hyperoctahedral group B_4, and Parts II and III apply to it: the real statements under properties (i)–(iv) of Remark 19.12, the analytic ones with property (v) in addition.
- *Part V* solves the variational problem relative to a regular reference configuration. It gives existence, uniqueness and minimality of the background field, second-order and Hölder bounds, the Euler–Lagrange equation, analyticity in the coarse field, exponential decay of the derivative, and the lattice symmetry of the solution.
- *Part VI* carries out the induction over the number of steps. This proves Bałaban's variational theorem on the torus with constants independent of k, including its lattice symmetry (Theorem 30.3). As a consequence, every real configuration with small curvature and current is regular (Corollary 30.4). So the hypothesis of Part III is a consequence of gauge-invariant conditions.
- *Part VII* writes one small-field renormalization group step, with Bałaban's exponential gauge fixing, exactly as a Gaussian integral around the background field (Lemma 32.3, Theorem 34.9). It proves the linear input: coercivity, uniformly in k, and decay of the fluctuation covariance (Theorem 34.4, Corollary 34.6), analyticity and tree decay of the coarse effective action, and local determinant densities. The step commutes with the translations and with B_4 (Proposition 31.7, Lemma 32.6).
- *Part VIII* localizes the dependence on the background, including large changes, by Dirichlet approximants and telescoping:

  - for the fluctuation covariance and the coarse Hessian (Theorem 37.7, Corollary 37.8);
  - for the determinant, which becomes a constant plus normalized one-loop activities of second order in the curvature (Corollary 38.9);
  - for the interaction and the composed activities (Theorem 39.4, Corollary 39.5). This uses interior pointwise bounds with derivatives for the Dirichlet operators (Theorem 35.2). They are proved by absorption at all scales inside a single cube, for a local elliptic system in which two scalar potentials of the slice projection are additional unknowns (Section 36).

  This localization is of radius type. On the main line it is superseded by Parts IX and X. From Part VIII the main line keeps the interior bounds, the Dirichlet Hessians and local fluctuation operators, the local determinant densities, the graded domains and the quadratic bound for normalized activities (Lemma 38.5).

- *Part IX* gives the volume-type localization by random walks with weakening parameters: for the covariance, its square root, the coarse Hessian and the determinant in L^2 norms (Section 41); interior estimates with L^p data (Section 43); pointwise $C^{1,\beta}$ bounds for the weakened propagators through tier norms (Section 44); and the weakened background map (Theorem 45.5).

- *Part X* carries out the small-field step in Dimock's scheme, and iterates it over all scales in Bałaban's four-dimensional organization:

  - complex references, including Bałaban's averaged backgrounds (Proposition 46.4, Proposition 46.7);
  - the weakened step quantities, including the gauge-fixing pieces, and the localization lemma (Section 47);
  - the cutoffs on the fluctuation field and the white noise, and the Mayer and polymer expansion; by parity there is no term of order g (Section 48);
  - the step in polymer form (Theorem 48.9);
  - the parts of the flow that do not depend on its four-dimensional organization: the small-field parameters, the polymer grid and the reblocking, the conditions on the parameters (Remark 49.8), the scales of the torus, the first step and its formula (Proposition 52.2, Lemma 52.3), the vacuum energy (Proposition 52.5) and the bookkeeping of the modified model (Remark 52.6);
  - the generations of activities, their symmetrization over the symmetries of their grid, and their coefficient β, read off from a period density that has no topological and no anisotropic part (Section 50, Proposition 50.13);
  - the extraction of the marginal part of an old generation at every later scale (Section 51, Proposition 51.5);
  - the old activities, bounded uniformly in the scale, and one step in generation form, with $g_{k+1}^{-2} = g_k^{-2} - \beta_{k+1}$ (Proposition 53.2, Theorem 53.7);
  - the flow over all scales on a fixed torus, as long as the couplings stay in $(0, \gamma]$, with the vacuum energy and a single counterterm (Theorem 53.12); the continuity in the couplings (Proposition 53.10, Corollary 53.11); and the bounds per volume on the partition function of the small-field model (Corollary 53.13).

  Theorems K and L summarize the step, and Theorem N the flow.

- *Part XI* computes the one-loop coefficient. It gives the relation between the one-loop terms of the composite maps of k and $k-1$ steps (Lemma 54.5) and, with the formula for the first step (Lemma 52.3), the composition of the one-loop terms of the steps (Proposition 54.6); the formula of the composite one-loop term in the Landau chart (Theorem 55.4); its exact invariance under the translations and B_4, and that of the one-loop terms of the steps (Lemma 54.9); and from this invariance, with the infinite-volume limits of the second variations (Lemmas 56.2 and 58.5), a single coefficient β and no term linear in p. The operators at $U = 1$ are constructed in infinite volume by folding the torus operators (Section 54.2), and the one-loop pieces of the polymers that do not wrap, and the Jacobian terms, are the same functions on all admissible tori (Proposition 41.14). It also gives the logarithm on the fine lattice (Theorem 57.1) and the cumulative coefficient $k\beta_\infty + O(1)$ (Theorem 58.1). With the flow of Part X it gives the properties (F1)–(F3) of the couplings (Proposition 58.6) and the ultraviolet limit, with the bounds on $g_k^{-2} - g_K^{-2}$ at every scale k (Theorem 58.9). The comparison of the coefficients β_j, at fixed couplings and as the torus grows, with the coefficient that Bałaban's definition [18, (1.20)–(1.22)] assigns to the generations (Proposition 58.8) rests on the generation locality of Section 59: the raw generations built with fixed couplings converge at the flat configurations as the torus grows, and so do their coefficients. Theorem M summarizes Part XI.

How the induction closes. The reference at step k is the solution of the $(k-1)$-step problem for the trivial extension of V, and it has exactly the prescribed averages. Theorem 25.1 then solves the problem at step k. Its regularity bound is not good enough to continue, because the change of units from δ_{k-1} to δ_k costs a power of L. The local improvement (Lemma 29.2) uses only Theorem 25.1 itself, applied with a different reference. This reference is spliced to the trivial configuration near a given point and coincides with the solution far away. Near the point, the solution is then **H** applied to the coarse field in the gauge of the coordinate-ordered contours from the point, up to an error that is exponentially small in the splicing radius times the a priori bound, plus a quadratic error. The coarse field in this gauge is controlled by the curvature of V. Two features make this work on the torus:

- the uniformity of Theorem 25.1 in the reference;
- the polynomial loss in the patching of local gauges (Lemma 28.3), which is beaten by the exponential decay of **H** and **C**.

Why the global hypothesis was not enough. Applied up to a real gauge transformation, hypothesis (B) is a condition on the real gauge orbit, so it does not break gauge invariance. But it is not local, and it is restrictive. By the isoperimetric argument of Section 12, every Wilson loop obeys a perimeter bound. Field strengths of size ε that are coherent over many blocks, and backgrounds with non-zero flux through a 2-cycle, are excluded. Moreover the renormalization group needs small-field conditions that factorize over local regions, and operators whose dependence on distant changes of the background is weak. Part III provides both: regularity is a local condition, and Theorem 19.10 gives the locality. Part VI shows that the local condition is implied by Bałaban's gauge-invariant small-field conditions.

A naive comparison of $G(U)$ and $G(U')$ through the resolvent identity would produce the difference $D_U - D_{U'}$, whose norm can be of order η^{-1} where the backgrounds differ. The proof of Theorem 19.10 avoids this. It compares each background with a Dirichlet operator on the region where they agree. Changing the domain at a fixed background produces cut couplings and differences of domain-dependent ingredients, including mass profiles and non-local terms, which are supported near the cut or exponentially small away from it. Derivative factors may occur at one or both ends of these terms; the estimates use the corresponding bounded undifferentiated or differentiated propagators.

Relation to Bałaban's construction. Bałaban works with Dirichlet operators on domains [7] and with local regularity on cubes [7, (3.35)–(3.36)], and builds the global operators by random walks. His positivity comes from [7, Theorem 3.11] and local axial-gauge estimates [4, Lemma 2.4]. Our route differs in several technical points:

- coercivity comes from IMS localization together with the coercivity of Part I;
- the local gauge-field propagators carry a mass term near the boundary of their domain;
- the normalization of the gauge transformation from the averaged axial gauge to the Landau gauge is built into a recursive correspondence map (Theorem 22.5). Bałaban's nonlinear averages of gauge transformations then appear only through the identity for pure gauges;
- the second-order regularity of the minimizer is derived from the Euler–Lagrange equation, which follows from gauge invariance and a transversality argument. We do not use the alternative equation [8, (143)];
- the local improvement of the regularity constants is obtained on the torus, by changing the reference configuration, instead of on nested domains (Lemma 29.2). The local regularity of general small-field configurations is derived from that of the background fields, through the chart (Corollary 30.4).

In Parts I–VIII the random-walk expansion is used only for the explicit localized representation, not for the estimates. In Part IX its weakened form (Section 41) also gives the bounds on the weakened operators, uniformly for $s \in \mathbb{D}_\kappa$, and their analyticity in the interior of $\mathbb{D}_\kappa$ (Theorem 41.4, Corollary 41.6, Lemma 40.3(a)). The polymer expansion of Part X uses these bounds; the bounds on the derivatives in the weakening parameters come from Cauchy's estimate in the proof of Lemma 47.2.

For Parts VII–XI, Section G13 of the guide compares the note with Bałaban's and Dimock's papers item by item (Table 9). The main differences are these.

- The generations and the coefficients β_j depend on the whole sequence of the earlier couplings, and only their continuity in the couplings is proved; Bałaban's are smooth functions of a single coupling (Remark 53.14(a), Proposition 53.10).
- The small-field threshold $h_k = p(g_k)g_k$ and the white-noise bound $p_0(g_k)$ depend on the coupling, as in [15], so the small-field domain shrinks with the coupling (Remark 53.14(a)).
- The marginal part of an old generation is extracted in a complex gauge chosen for each polymer, to second order, with a power gain for the polymers that are not expanded; Bałaban expands in a gauge common to a cube, to fifth order, with Ward identities (Section 51).
- The later-scale bounds, which Bałaban's Theorem 1 asserts, are proved in the induction (Theorem 53.12(d), Remark 53.14(a)).
- The ultraviolet behaviour of the couplings of the small-field flow is proved for every compact, connected and simple group, while Bałaban states it for $SU(2)$. For the last m_1 steps its lower half holds up to an additive constant (Theorem 58.9 and the paragraph after it).

What is not proved. The flow results of Parts VII–X and Theorem M(iv) concern the small-field model on a fixed torus (Remark 52.6(c)); Part XI also uses infinite-volume limits of the one-loop terms (Theorem M(ii), (iii)). The large-field stage is not treated (Remark 52.6(d)), so nothing is proved about the partition function or the correlation functions of the lattice

theory itself. No continuum limit is taken: Theorem M(iv) concerns the couplings of the small-field flow, and no limit of the measure or of correlation functions as the lattice spacing tends to 0 is considered. The other omissions are technical: a rate in Proposition 58.8, the exact lower half of the analogue of Bałaban's (0.31) for the last steps, linear bounds for the gauges on large cubes, gauge groups with abelian factors, and dimensions other than 4 (Section 1.5).

Next steps.

(1) *A check by a human expert.* The note has not been checked by a human expert. The companion volume expands the proofs of Parts VII–XI step by step, and its chapter on the parameter ledger records the constants and the order in which they are chosen. Section G14 of the guide names the places to check first.

(2) *What remains of Part XI* (Remark 58.10(c)). Generation locality is proved without a rate (Section 59). A rate, uniform in the couplings, is expected to follow from a Lipschitz bound of the step on the infinite lattice in the old activities, a bound on the errors of the old activities, and a condition on γ (Remark 59.13(a)). For the exact lower half of the analogue of (0.31) at the last steps, a possible route is the convergence of the single one-loop coefficients, through a continuum limit at $U = 1$ of the remainder of Proposition 56.8, together with conditions on n_K and γ. A smooth small-field cutoff, as in [18, (2.9)], should improve (F3) to $O(g^2)$ (Remark 58.10(a)).

(3) *Large fields and nested domains,* where Dirichlet operators on irregular domains are essential. This is the analogue of [16, 17]. It must control what the small-field model discards at every step (Remark 52.6), and so remove the modification. Since the covariance C_k depends on the background everywhere, the white-noise cutoff of Section 48 will need a version that depends on regions. Part III treats arbitrary unions of blocks, so this is the natural setting for them. The splicing of Part VI is a torus device. On nested domains the reference no longer has exactly the prescribed averages, and the analysis on domains of Part III must be extended to the nonlinear problem.

(4) *Gauges on cubes of size up to* $c\,\varepsilon_1^{-1}$ with bounds linear in the size (Remark 30.6).

(5) *Gauge groups with abelian factors.* The quadratic bound for normalized activities uses semisimplicity (Lemma 38.5). For abelian factors, gauge-invariant terms linear in the curvature can occur, and they must be treated separately.

A Decay of the lattice Green function and its derivatives

In this appendix g_1 is the Green function of $-\Delta$ on $\mathbb{Z}^d$, $d = 4$, as in Section 6. For $\mu = 1, \dots, d$ let $D_\mu f(x) = f(x + e_\mu) - f(x)$ and $D_\mu^* f(x) = f(x - e_\mu) - f(x)$. A *word* ∇^β of length m is a composition of m operators from $\{D_\mu, D_\mu^*\}$.

Theorem A.1. *For every word ∇^β of length $m \le 3$ there is C, depending only on β, such that*

$$|\nabla^\beta g_1(x)| \le C(1 + |x|)^{2-d-m}, \qquad x \in \mathbb{Z}^d.$$

Write $\mathbb{T}^d = \mathbb{R}^d / 2\pi\mathbb{Z}^d$, identified with $[-\pi, \pi]^d$. We have

$$\nabla^\beta g_1(x) = \frac{1}{(2\pi)^d} \int_{\mathbb{T}^d} e^{\mathrm{i}p\cdot x}\, \psi(p)\, dp, \qquad \psi(p) = \frac{\sigma_\beta(p)}{\hat\Delta_1(p)}, \qquad \hat\Delta_1(p) = \sum_\mu 4 \sin^2 \frac{p_\mu}{2},$$

where σ_β is the product of the symbols $e^{\mathrm{i}p_\mu} - 1$ (for D_μ) and $e^{-\mathrm{i}p_\mu} - 1$ (for D_μ^*) of the letters of β. The theorem follows from the next two lemmas, applied with $s = m - 2$ and $K = d + m$.

Lemma A.2 (Symbol estimates)**.** *For every multi-index γ there is C_γ such that $|\partial_p^\gamma \psi(p)| \le C_\gamma |p|^{m-2-|\gamma|}$ for $p \in [-\pi,\pi]^d \setminus \{0\}$.*

Proof. Let $f = \hat\Delta_1$. By Lemma 4.1(a) with $\eta = 1$, $f(p) \ge \frac{4}{\pi^2}|p|^2$. Moreover $|\partial_\mu f| = 2|\sin p_\mu| \le 2|p|$, and all higher derivatives of f are bounded. Since $|p| \le \pi\sqrt{d}$, this gives $|\partial^\gamma f| \le C|p|^{2-|\gamma|}$ for all γ. By the Faà di Bruno formula, $\partial^\gamma(1/f)$ is a linear combination of terms $f^{-1-j}\prod_{i=1}^{j}\partial^{\gamma_i} f$ with $\gamma_1 + \cdots + \gamma_j = \gamma$. Each such term is bounded by $C|p|^{-2-2j}\prod_i |p|^{2-|\gamma_i|} = C|p|^{-2-|\gamma|}$. Each letter of σ_β is bounded by $|p_\mu| \le |p|$, and its derivatives are bounded. So by the Leibniz rule $|\partial^\gamma \sigma_\beta| \le C|p|^{\max(m-|\gamma|,0)} \le C'|p|^{m-|\gamma|}$. One more application of the Leibniz rule concludes the proof. □

Lemma A.3 (Decay of Fourier coefficients)**.** *Let ψ be smooth on $\mathbb{T}^d\setminus\{0\}$ and satisfy $|\partial^\gamma\psi(p)| \le C|p|^{s-|\gamma|}$ on $[-\pi,\pi]^d \setminus \{0\}$ for all $|\gamma| \le K$, where $s > -d$ and $K > d + s$. Then*

$$\Big|\int_{\mathbb{T}^d} e^{\mathrm{i}p\cdot x}\psi(p)\,dp\Big| \le C'(1+|x|)^{-d-s}, \qquad x \in \mathbb{Z}^d.$$

Proof. For $|x| \le 1$ the integral is bounded by $\int|\psi| < \infty$, because $s > -d$. Let $R = |x| > 1$, and choose j with $|x_j| \ge R/\sqrt{d}$. Fix $\zeta \in C_c^\infty(\mathbb{R}^d)$ with $\zeta = 1$ on $\{|p| \le 1\}$ and $\zeta = 0$ on $\{|p| \ge 2\}$. Since $R > 1$, $\zeta(R\,\cdot)$ is supported in $\{|p| \le 2/R\} \subset (-\pi,\pi)^d$, so it defines a smooth function on the torus. Split $\psi = \psi_1 + \psi_2$ with $\psi_1(p) = \zeta(Rp)\psi(p)$. Then

$$\Big|\int e^{\mathrm{i}p\cdot x}\psi_1\Big| \le C\int_{|p|\le 2/R}|p|^s\,dp = C''R^{-d-s}.$$

The function ψ_2 vanishes near 0, so it is smooth on the torus. Integrating by parts K times in p_j, without boundary terms by periodicity, gives

$$\int e^{\mathrm{i}p\cdot x}\psi_2 = \Big(\frac{\mathrm{i}}{x_j}\Big)^K\int e^{\mathrm{i}p\cdot x}\,\partial_j^K\psi_2.$$

By the Leibniz rule, $\partial_j^K\psi_2$ is a sum of terms $\partial_j^{K-i}\psi\cdot\partial_j^i[1-\zeta(R\,\cdot)]$. For $i = 0$ the term is bounded by $C|p|^{s-K}$ and vanishes for $|p| < 1/R$. For $i \ge 1$ it vanishes unless $1/R \le |p| \le 2/R$, where $R^i \le 2^i|p|^{-i}$; so it is again bounded by $C|p|^{s-K}$. Hence

$$\Big|\int e^{\mathrm{i}p\cdot x}\psi_2\Big| \le C|x_j|^{-K}\int_{1/R\le|p|\le\pi\sqrt{d}}|p|^{s-K}dp \le C|x_j|^{-K}R^{K-s-d} \le C'R^{-d-s},$$

since $K - s > d$. □

Proof of Theorem A.1. By Lemma A.2 the hypotheses of Lemma A.3 hold with $s = m-2 > -d$ and every K. Take $K = d + m$. □

B Notation

This appendix collects the notation that is used across several Parts, with the meaning of each symbol and, in parentheses, the place of its definition. The list "Letters with several meanings" says how the meanings of a letter are kept apart, and the last table gives the former names of the renamed symbols. Remark 47.1 lists the notation of Sections 47–53, and Section G2 the scales and units.

Scales and parameters

d, L	dimension $d = 4$; block factor, odd, $L \geq 3$ (§2).
k, η, N	Parts I–VII: number of averaging steps, fine spacing $\eta = L^{-k}$, $N = L^k = 1/\eta$ (§2).
$\mathbb{T}$, $\mathbb{T}_\eta$, n	unit torus of side n; fine torus of spacing η (§2).
$\mathbb{T}^{(j)}$, n_j, ℓ	lattice of scale j; its side $n_j = \ell L^{N-j}$ in its own units; physical side (§52.1).
ϵ, N, K	in Section 31.1 and from Section 52 on: finest spacing $\epsilon = L^{-N}$, number of scales N, number of steps $K \leq N$ (§52.1).
a, α, β	averaging coefficient; Hölder exponent; in Part IX $\beta = \min(\alpha, \frac{1}{2})$ (§1, §44).
M, M_0, M'	regularity scale and grid size; its threshold in Part III; localization scale of the step (Def. 18.1, Thm. 19.6, Rem. 49.8(4)).
κ_m	mass coefficient of the Dirichlet operators (§19).
κ, δ, s, $\mathbb{D}_\kappa$	weakening rate and loss; weakening parameters $s = (s_\square)$ in $\mathbb{D}_\kappa = \{\lvert s_\square\rvert \leq e^\kappa\}$ (§40, Rem. 49.8(3)).
ρ	scale of the walk domains, subject to (41.1) (§40).
κ_n, κ_0, d_M, d_j	rate of the activity norms; constant and tree length of the polymer grid (Rem. 49.8(1), §49.2).
θ_r, δ_1, δ_0, $\delta_1^\flat$	domain constants: curvature and current of the real part; gradient and size of the complex part; $\delta_1^\flat = \frac{1}{4}\delta_1$ (Def. 38.3, Rem. 38.8, §50).
$\bar{\delta}$, $\varepsilon_{\max}$, δ_P	cap on the complex references; threshold on the regularity of their real parts; radius of the complex references (Rem. 38.8, Prop. 46.7).
g, g_k, γ	coupling; coupling of scale k; bound on the couplings (from Section 49 on) (§49.1).
p, p_0, q, h_k	small-field functions; $p_0(g) = (\log g^{-1})^q$ in Thm. 58.9; threshold $h_k = p(g_k)g_k$ (§49.1).
w	fluctuation cutoff, $w = C_w L^d C_T g\, p_0$ in the flow (Def. 32.4, Rem. 53.5).
a_1, B_3, a_2, B_5	thresholds and regularity constants: of background fields (a_1, B_3; Thm. 30.3); of configurations of small curvature and current (a_2, B_5; Cor. 30.4).
C_*	curvature constant of background fields, independent of L (Prop. 31.5).
K, C_β, C_{old}	bound of the generations; $\lvert\beta_j\rvert \leq C_\beta \mathsf{K}$; bound of the old activities (Rem. 53.5, Prop. 53.2).
m_0, m_1	age from which the small polymers are extracted, the polymers being recent before it; number of last steps in Thm. 58.9 (Def. 49.5, Thm. 58.9).
β_j, β_E	coefficient of the generation $E^{(j)}$; of a generation E (Thm. N, Prop. 50.13).
β_∞, $\beta_{j+1}^{(1)}$, β_{F}	one-loop coefficient per step; coefficient of the marginal tensor $T_{j+1}^{(1)}$ of the step from j to $j+1$, in infinite volume; fine-lattice Feynman-gauge coefficient (Thms. 57.1, 58.1, §57).
β_j^{B}	the coefficient that Bałaban's definition assigns to the generations (Prop. 58.8).
C_A, T_ρ	adjoint Casimir; $\text{tr}_\rho(T^a T^b) = T_\rho \delta^{ab}$ (§57).

Configurations, fields and symmetries

$U = e^{\text{i}\eta A}$	configuration on the fine lattice, with Lie field A (§1, §21).
$B(y)$, $B'(y)$	block of $\mathbb{T}_\eta$ centred at $y \in \mathbb{T}$; block of L^4 points of $\mathbb{T}$ centred at $y \in \mathbb{T}'$ (§2, Def. 32.1).
$U^{(k)}$, $\mathfrak{a}'$	k-fold average; one-step averaging (Def. 21.8, §31).
$\mathfrak{B}(V)$	constraint set: the real U with $U^{(k)} = V$ and curvature at most c_1 (§25).
$U_k(V)$	background field: an element of the minimal orbit (Thm. 30.3).
V', U', W	coarse configuration of the step; its background field; the base $W = U'^{(k)}$ (Lemma 31.6).
$\mathfrak{F}$, $b_0(c)$	fluctuation bonds: all bonds of $\mathbb{T}$ except the middle bonds $b_0(c)$ (Def. 32.1).
Z, ξ	fluctuation field, $V = e^{\text{i}Z}W$; white noise, $Z = gC_k^{1/2}\xi$ on $\mathfrak{F}$ (Def. 32.4, Def. 48.1).
$F_Y(p)$, J_U	curvature; current (Def. 21.4, Def. 23.1).

$\mathfrak{U}_j(a)$, $\mathfrak{U}(\theta)$	configurations with curvature at most a; with curvature and current at most θ (Def. 21.4, Def. 23.1).
$\mathfrak{M}$	Bałaban's set of regular configurations (§32).
(M,ε)-regular	on every cube $\Lambda_{4M+2}(z) = \{\lvert x-z\rvert_\infty \le 4M+2\}$ a gauge with $\lvert A\rvert$, $\lvert\partial A\rvert$ and the Hölder quotients of ∂A at most ε (Def. 18.1).
$\mathfrak{G}_0$, $\mathcal{G}'$	residual gauge group $\{u : u\lvert_{\mathbb{T}} = 1\}$; gauge transformations of $\mathbb{T}$ trivial on $\mathbb{T}'$ (§21, §31).
B_4, $\mathfrak{S}_k$, $\mathfrak{S}_j^{\mathrm{grid}}$	hyperoctahedral group; group of translations and B_4 about the points of $\mathbb{T}^{(k)}$; symmetries of the grid of scale j (Lemma 21.10, §50).
$\mathcal{Q}$, $\mathbf{Q}_k$, Q'_k	gauge-adapted averaging; linearized averages (Def. 22.4, Prop. 22.10).
C_j^{av}	averaging remainder: $\mathcal{Q}_j(e^{\mathrm{i}\eta A}) = e^{\mathrm{i}\delta_j(\mathbf{Q}_j A + C_j^{\mathrm{av}}(A))}$ (Prop. 22.9).
$\mathrm{Av}(\mathbf{U})$	group average (Lemma 21.5).
$E(U)$	defect of a model averaging pair, $\mathbf{Q}(U)D_U = D_V Q'(U) + E(U)$ (§12, Lemma 12.5).
$\mathfrak{G}_W$, gf_W	averaged transports; exponential gauge-fixing function (Def. 22.6, Def. 32.2).
χ^{w}, χ_ξ	fluctuation cutoff; white-noise cutoff (Def. 32.4, Def. 48.1).
T, T^{gf}, T_{sf}	renormalization group map; gauge-fixed step; small-field step ((31.1), Def. 32.5, Def. 48.1).

Operators

The parentheses give the results that bound each operator; the kinds of bound (coercivity, exponential decay, pointwise bounds) are those stated there.

G_σ	scalar Green functions $(-\Delta + aQ_\sigma^* Q_\sigma)^{-1}$, $\sigma = 0,\dots,d$ (Thm. 7.1).
$(Q_\sigma G_\sigma Q_\sigma^*)^{-1}$	inverse of the averaged Green function (Prop. 8.1).
H_σ, C_σ, $\mathcal{D}_\sigma$	minimizer map; constrained Green function; effective action (Thm. 8.4, Props. 8.6, 8.5).
$R = I - P_{\mathcal{V}}$	projection defining Bałaban's Landau-type gauge $Rd^*A = 0$ (Prop. 8.8).
$\mathbf{\Delta}_a$, $\mathbf{G}$	$\mathbf{\Delta}_a = d^*d + dRd^* + a\mathbf{Q}^*\mathbf{Q}$ and $\mathbf{G} = \mathbf{\Delta}_a^{-1}$ (Thms. 9.6, 10.1).
$\mathbf{H}$, $\mathbf{D}$, $\mathbf{C}$	$\mathbf{H} = \mathbf{G}\mathbf{Q}^*(\mathbf{Q}\mathbf{G}\mathbf{Q}^*)^{-1}$; effective action; $\mathbf{C} = \mathbf{G} - \mathbf{G}dRd^*\mathbf{G} - \mathbf{H}\mathbf{Q}\mathbf{G}$ (Thm. 11.3, Prop. 11.5, Thm. 11.7).
$G'(U)$, $\Gamma'(U)$, $H'(U)$, $C'(U)$, $R(U)$	the scalar operators at a background: $G' = (D^TD + aQ'^TQ')^{-1}$, Γ' the inverse of $Q'G'Q'^T$ (Thm. 14.1, Prop. 14.2; Thm. 18.2).
$\Pi(U)$	projection onto Bałaban's gauge slice $\{R(U)D_U^T A = 0\}$ (Lemma 15.1).
$\mathbf{G}(U)$, $\mathbf{H}(U)$, $\mathbf{C}(U)$	the gauge-field operators at a background (Thms. 15.4, 16.1; Thms. 19.7, 19.9).
G'_X, $\mathbf{G}^X$, …	Dirichlet versions on unions of blocks X (Thm. G(ii), §§18–19).
Δ^0, Δ_k, C_k	coarse Hessian; fluctuation operator on $\mathfrak{F}$; covariance $C_k = \Delta_k^{-1}$ (Prop. 33.2, Thm. 34.4, Cor. 34.6).
$\Delta^{0,X}$, Δ_Y	Dirichlet Hessian; local fluctuation operators (Def. 37.3, Prop. 37.6).
$\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$, $C_k^{1/2}(s)$	weakened operators, analytic in $s \in \mathbb{D}_\kappa$ (Thm. 41.4, Cor. 41.6, Thm. 44.15).
$A_*(B)$, $A_*(s;B)$	Landau representative of the background map; its weakened version (Thm. 25.1, Thm. 45.5).
$\Gamma_k^{(1)}$, Γ_j^{step}	one-loop term of the composite k-step map; of the step from scale j (Prop. 54.3, §54).
Π_Φ, T_Φ	vacuum polarization; marginal tensor (Def. 54.7).

Activities and the flow

$\square$, X, $\square^+$	grid cube of side M; polymer, a $*$-connected union of grid cubes; $\square$ and its $*$-neighbours (§49.2).
$\mathrm{cl}_k X$	union of the grid cubes of scale k that meet X (Lemma 49.2).

small, recent, large, wrapping
: classes of a polymer of scale j at scale k (Def. 49.5).

$\mathcal{U}(X)$, $\mathcal{U}^\flat(X)$
: graded domains $\mathcal{U}(X;\theta_r,\delta_1,\delta_0)$ and $\mathcal{U}(X;\theta_r,\delta_1^\flat,\delta_0)$ (Def. 38.3, §50).

graded activity, normalized
: holomorphic on $\mathcal{U}(X)$, invariant under real gauge transformations; $F(1)=0$ (Def. 38.3).

$E^{(j)}$
: generation of scale j, with (G1)–(G4) (Def. 50.1).

E', E'^{sym}
: raw generation; its symmetrization over $\mathfrak{S}_j^{\mathrm{grid}}$ (Def. 50.7).

$\bar{T}_E$
: period density of E (Def. 50.11).

$\mathfrak{e}(X)$, $\mathfrak{j}_c$
: one-loop pieces; local Jacobians (Prop. 41.7, Prop. 47.8).

$E^{\#}_{\mathrm{p}}$, $E^{\#}_{\mathrm{m}}$
: pure and mixed new sources (Prop. 48.8, Cor. 48.7).

$\mathfrak{E}$, $\mathcal{E}_k$
: old activities in a step; old activities of step k, generations grouped with their counterterms (Thm. 48.9, Def. 53.1).

τ, ϵ_{p}, ϵ_{m}
: size of the composed activities; sizes of the pure and mixed pieces (Prop. 47.10, Prop. 48.5).

$\mathcal{Z}_k$, ε_k, E_0, δ_k
: normalization; vacuum energy per site; counterterm; increment of the vacuum energy (Prop. 52.5, Thm. 53.12(f)).

Σ_j
: the sequences $(g_0,\dots,g_j)\in(0,\gamma]^{j+1}$ with $g_{i+1}\le 2g_i$ (§53.5).

Named conditions

(B)
: the background is close to 1 in one global gauge: $|A|,|\partial A|,[\partial A]_\alpha\le\varepsilon$ (§12).

(i)–(v)
: properties of an averaging pair under which Parts II and III apply (Rem. 19.12).

(A1)–(A11)
: the places where the averaging pair enters the proofs of Parts II and III (proof of Lemma 31.3).

(C1), (C2)
: covariance of a cutoff under real gauge transformations and under $\mathfrak{S}_{k+1}$ (§32).

(W1)–(W3)
: structure, local propagators and non-local part of the operators of a walk (§41).

(L1)–(L5)
: the five levels of the nested walks: scalar operators, scalar averaged operators, gauge-field operators, averaged gauge operators and the Hessian, the fluctuation operator (§41).

(BR)
: regularity up to the boundary for Dirichlet operators on boxes; not used (§42).

(I1)–(I4)
: interior estimates with L^p data; (I4) is not used on the main line (§43).

(N1)–(N3)
: tier bounds for the middle factors of the non-local parts (Lemma 44.10).

(N1), (N2)
: properties of the neighbourhoods used for Laplace's method (§54).

(DC1)–(DC6)
: conditions on the domain constants (Rem. 38.8).

(Q1)–(Q3)
: conditions of the small-field step on the coupling, the thresholds and the old activities (§48).

(G1)–(G4)
: a generation: activities; grid covariance; invariance of the sum; covariance under $\mathfrak{S}_j^{\mathrm{grid}}$ (Def. 50.1).

(T0)–(T2)
: conditions on the side of the torus at each scale (§52.1).

(Γ1)–(Γ4)
: conditions on the bound γ of the couplings (Rem. 53.5).

(F1)–(F3)
: continuity of β_{j+1}; one-loop part against the infinite-volume coefficient; the rest is $O(g_j^2 p_0(g_j)^6)$ (§58, Prop. 58.6).

Letters with several meanings

γ
: decay rates in Parts I–IX; from Section 49 on also the bound on the couplings (Section 49.1).

β
: the coefficients β_j, β_∞, β_E, β_{F}; the Hölder exponent of Part IX; the radius $|B|\le\beta$ of Theorem 25.1.

ρ
: densities ρ_k; the scale of the walk domains (Part IX); the representation in tr_ρ, T_ρ (Part XI); the Lipschitz weight of the Combes–Thomas method (Section 5).

R
: the projection $R(U)$; the adjoint action $R(u)$; elements $R\in B_4$ (Proposition 31.7, Lemma 54.9).

Π	the slice projection $\Pi(U)$ (Part II); the vacuum polarization Π_Φ (Part XI).
T	the torus $\mathbb{T}$ (blackboard bold); the map T; the tensors T_Φ, $T^{(1)}_{j+1}$; T_ρ; the generators T^a.
B	the blocks $B(y)$; a coarse field B, with $\lvert B\rvert_{y'}$ the maximum of $\lvert B\rvert$ over the unit bonds starting at y'; the group B_4; hypothesis (B); the constants B_1, B_3, B_5.
W	the potential $W \in \mathcal{W}(\varepsilon, \gamma_W)$ of Part II; the base $W = U'^{(k)}$ of the step (Remark 47.1).
E	the defect $E(U)$ (Part II, Part IV: $E = 0$); the generations $E^{(j)}$, E'; the sources $E^{\#}$; the old activities $\mathcal{E}_k$, $\mathfrak{E}$; the loop average is E_{loop}.
N	$N = L^k$ in Parts I–VII, except Section 31.1; the number of scales in Section 31.1 and from Section 52 on; the matrix size N.
K	the number of steps; the bound K (sans serif); the gauge-fixing form K^X (Sections 37, 41); the parametrices are $\mathcal{K}^X$, $\mathcal{K}^v$.
a	the averaging coefficient; curvature bounds in $\mathfrak{U}_j(a)$, a_1, a_2, a_{step} (Section 21, conventions).
C_k	the covariance $C_k = \Delta_k^{-1}$; the averaging remainder is C_k^{av}.
$\mathfrak{F}$	the fluctuation bonds (Definition 32.1); the curvature operator of Lemma 12.3; the functional of Section 26 is $\mathcal{J}$.
M	the regularity scale; the operator $M = Q_0 G_0^2 Q_0^*$ (Section 8); the group average is Av.
δ	the spacings $\delta_j = L^{j-k}$ (Part IV); the domain constants δ_0, δ_1; the radius $\delta_0(W)$ of Proposition 19.11; the loss δ of Part IX; δ_P, $\bar{\delta}$, δ_*; the increments δ_k of the vacuum energy.
ε, ϵ	the regularity ε; the spacing $\epsilon = L^{-N}$; the vacuum energy ε_k; ϵ_{p}, ϵ_{m}, ϵ_* (Remark 47.1).
κ	the mass coefficient κ_m; the weakening rate κ; $\kappa(u) = \lVert u\rVert\lVert u^{-1}\rVert$ (Section 21); κ_n, κ_0, κ_2, $\kappa_\rho = C_A/T_\rho$.
n	the side of $\mathbb{T}$; in Part VII the side of $\mathbb{T}'$; the sides n_k (Section 52.1).
p	plaquettes; the small-field functions p, p_0; the momentum p (Part XI) (Remark 47.1).
Δ	$\Delta = -d^*d \le 0$ (Section 2); $\Delta_{U_0} = D^TD + DD^T \ge 0$ (Theorem 24.2); the Hessians Δ^0, Δ_k.
$\hat{X}$	the 1-neighbourhood of a domain (Definition 17.1); other hats are defined where used (Remark 47.1).

Former names

The following symbols carry new names. The former names are those of earlier versions of the note.

Name	Former name	Meaning	Where
C_j^{av}, dC_k^{av}, $C_k^{\mathrm{av},(2)}$	C_j, dC_k, $C_k^{(2)}$	averaging remainder	Prop. 22.9
$\mathcal{J}(A')$	$\mathfrak{F}(A')$	the functional of Section 26	Prop. 26.3
$\lvert B\rvert_{y'}$	$\lvert B(y')\rvert$	maximum of $\lvert B\rvert$ over the unit bonds starting at y'	Thm. E, Thm. 11.3
$\mathrm{Av}(\mathbf{U})$	$M(\mathbf{U})$	group average	Lemma 21.5
$\mathcal{K}^X$, $\mathcal{K}^v$	K^X, K^v	parametrices of the walks	Lemma 41.2, Thm. 20.2
E_{loop}	E	loop average $L^{-d}\sum_x H_x$	§22, Lemma 46.3
r_2^{V}	r_2	radius of the linearizing map	Lemma 26.1
C_P^{patch}	C_P	patching constant	Lemma 28.2
C_L^{units}	C_L	constant of the change of units	Lemma 30.2
γ'_{lin}	γ'	the operator γ_{k+1} of the one-step expansion at step $k+1$	§32, Lemma 22.8
(W1)–(W3)	(A1)–(A3)	hypotheses of the walks of Section 41	§41
rectangle	box	product of intervals in Section 44	Lemma 44.7

C Numerical checks

This appendix describes the numerical checks mentioned in the text. No proof uses them. Each paragraph names the place in the text that refers to it. The programs and their outputs are in the folder `numerics` next to the note, and `numerics/README.md` describes them.

The averaging of Part IV (Remark 22.12). The algebra of Sections 21 and 22 was checked numerically for $G = SU(2)$ and $L = 3$: in $d = 2$ on small tori, with $k = 1$ and $k = 2$, and in $d = 4$ on a single coarse bond, where the families have up to 24 contours. The following hold to rounding error:

- for real families of 2 to 24 elements, Lemma 21.5(a), (b) with real u, v, (c), and $\sum_i \omega_i = I$ with $|\omega_i - \frac{1}{n} I|$ proportional to the diameter;
- (22.1) for real gauge transformations that are not small;
- Lemma 21.9, Lemma 21.10(b) for real g, and Lemma 21.10(d) for all 384 elements of B_4;
- Theorem 22.5(e),(f) and Proposition 22.10(a),(b),(f);
- at $U_0 = 1$, Lemma 22.8(d),(f), with the coefficient L^{1-d}, and Proposition 22.10(c).

Two controls fail, as they should. With a single contour order per family, the covariance under the permutations of the axes fails. With the mean of the logarithms of the contour variables in place of Av, (22.1) and $E = 0$ fail. Not tested numerically: the complex equivariance in Lemma 21.5(b), the formula for the weights in (d), the Lipschitz bound (e), complex g in Lemma 21.10(b), and Lemma 22.7(e),(f).

The constant of the one-step Poincaré inequality (Section 34). Finite samples of the Bloch spectrum relevant to the constant C_{P} of Lemma 34.2 were computed numerically at $W = 1$. By Bloch's theorem over $\mathbb{T}'$ the form $\|d_1 B\|^2 + \|\mathfrak{G}_1 B\|^2 + \|q^{\mathrm{pl}} B\|^2$ reduces to finite matrices, one for each momentum. For $d = 4$ and $L = 3$ its smallest eigenvalue lies in $[0.030, 0.035]$ over the momenta tested, with the minimum among those samples about 0.030 at (π, π, π, π); the reciprocal of this sampled minimum is approximately 33. These finite samples do not certify the minimum over all Bloch momenta or a uniform Poincaré constant of that

size. The existence of a constant uniform in the size of the torus follows from the exact proof of Lemma 34.2, independently of these numerical values. In $d = 2$ the corresponding sampled smallest eigenvalues are about 0.12 for $L = 3$ and 0.08 for $L = 5$.

Gauge fixing and the one-loop term (Remark 34.10). For $d = 2$, $G = SU(2)$, $L = 3$ and four coarse fields, Laplace's method with the exponential gauge fixing over all bonds was compared with Laplace's method in a tree gauge, which has no Faddeev–Popov determinant either. The two differ by the constant $\frac{n_\lambda}{2} \log 2$, where $n_\lambda = 3|\mathbb{T} \setminus \mathbb{T}'|$ is the number of gauge directions, to $2.5 \cdot 10^{-10}$; this is the exact prediction for a gauge fixing without Faddeev–Popov determinant. The constant comes from the normalizations of the program, which omits the factor z^{-1} of gf_W and uses the coordinates $h = e^{\mathrm{i}H}$, $H = \frac{1}{2}\sum_a x_a \tau_a$ with the Pauli matrices τ_a, so that $\operatorname{tr} H^2 = \frac{1}{2}|x|^2$. In these coordinates the exponential weight at the coupling of the action is a Gaussian of variance $2g^2$ in each eliminated gauge direction, which contributes $\frac{1}{2}\log 2$ relative to the tree gauge. Two exact representations of one integral have the same Laplace asymptotics. The program is `symavg.py` in the folder `numerics` next to the note.

The one-loop term in the Landau chart (Remark 55.7). We tested Theorem 55.4 numerically for $d = 2$, $G = SU(2)$ and $k = 1$. This is outside the hypothesis $d = 4$ of the note, but the identity is algebraic. The first test, with the program `chart.py`, was made for an earlier, simpler averaging, with blocks that have the coarse point at a corner and a single coordinate-ordered contour per point, on three lattices: $L = 2$ on coarse tori of 2×2 and 3×3 points, and $L = 3$ on a coarse torus of 2×2 points. The second test is described at the end of the remark. Each coarse field V was the average $U^{(1)}$ of a random fine configuration. For each V:

- A critical point U_0 of S on the fibre $\{U^{(1)} = V\}$ was computed by a damped Newton method for the Lagrangian, which accepts only steps that decrease S. It was computed in a tree gauge, where $U = 1$ on the bonds of a spanning tree of each block, formed by coordinate-ordered contours from the coarse point.
- The left side was computed by Laplace's method in the same gauge. The tree gauge is a computational device of the programs and plays no role in the note: by the invariance of the Haar measure, the variables on the tree bonds can be fixed one at a time, so the Faddeev–Popov identity for this gauge has Jacobian 1. The left side is $-\frac{1}{2}\log\det$ of the Hessian of the Lagrangian on the tangent space of the fibre, minus $\frac{1}{2}\log\det$ of the Gram matrix of the differential of the constraint.
- The right side was computed from the gauge-adapted averaging $\mathcal{Q}$ of the program, relative to U_0. In it, q_2 includes the multiplier term, R is the projection onto $\mathcal{W} = D^T D\,\mathcal{N}(Q_1')$, and the ghost is $\frac{1}{2}\log\det(PX^2P|_{\mathcal{N}(Q_1')})$, as in the proof of Corollary 55.5.

For the averaging of the first test the tree slice is the axial slice. On it, $\mathcal{Q}$ coincides with the constraint $U^{(1)}V^{-1}$, as in Theorem 22.5(d), so the two sides share the constraint and the multiplier. What the first test checks is the passage from the axial slice to the Landau slice: the Faddeev–Popov factors, the power of the ghost, the term $\mathbf{Q}_1\mathbf{G}_*\mathbf{Q}_1^T$, and the multiplier term in the directions off the axial slice. It does not test Theorem 24.2, which only justifies the chart.

First derivatives were taken by complex steps on the holomorphic extensions, which give them to machine precision. Second derivatives combine a complex step with a central difference, with accuracy about 10^{-10}.

Results:

- On each lattice the difference of the two sides was independent of V to within $5 \cdot 10^{-10}$.

Meanwhile each side varied by up to 2.1 between the coarse fields: seven of them for $L = 2$ on the 2×2 torus, and two on each of the other lattices. The difference is in fact exactly $-\frac{d}{2} \dim \mathfrak{g}\, |\mathbb{T}| \log L$, i.e. $-12 \log 2$, $-27 \log 2$ and $-12 \log 3$. The program imposes the coarse condition in unnormalized Lie coordinates, in which the block Θ_0 of $\mathcal{O}_0$ in Proposition 55.3 is $L^{-d/2} Q_1'$ instead of Q_1'. So the test checks an exact identity, not only the independence of V.

- The right side did not depend on a.
- The identities used in the proof hold to within $1.2 \cdot 10^{-9}$: $\nabla\Phi(0) = \mathbf{Q}_1^T \omega_*$ (Proposition 27.2), and $q_2 D\lambda = 0$ and $\mathbf{Q}_1 D\lambda = 0$ for $\lambda \in \mathcal{N}(Q_1')$.
- The identity of Corollary 55.5 holds to 10^{-13}, at a random background and for an averaging with $Q' Q'^T = I$.

The test can detect errors. Wrong variants of the formula give differences that vary with V by $1.9 \cdot 10^{-2}$ to 1.7: without the ghost, with its square, without the multiplier term in q_2, and without $-\frac{1}{2} \log \det(\mathbf{Q}_1 \mathbf{G}_* \mathbf{Q}_1^T)$. The programs and their outputs are in the folder `numerics` next to the note. The program `symavg.py` there repeats the test for Bałaban's symmetric averaging in four dimensions, with centred blocks and averaged contour variables, transcribed to $d = 2$; this is the averaging of Part IV, with the averaged axial gauge and the gauge-adapted averaging of Section 22. It finds the same identity with the same constant: for $L = 3$ on a coarse torus of 2×2 points and four coarse fields, between which each side varies by 0.64, the difference of the two sides is $-12 \log 3$ to within $5 \cdot 10^{-11}$. Here the tree slice and the averaged axial slice differ, so this test checks the identity as a whole. With a single ordered contour in place of the averaged contour variables in the gauge, the right side is the same to $5 \cdot 10^{-11}$, as it must be for any admissible chart. With logarithmic means of the contour variables in place of the group average, for which the exact covariance $E = 0$ fails, the difference misses the prediction by $2 \cdot 10^{-4}$ to $7 \cdot 10^{-4}$.

Numerical evaluation of f (Remark 57.9). For $G = SU(2)$ in the fundamental representation, $\mathsf{b} = \frac{11}{6\pi^2} = 0.185756$. We evaluated $f(\mathrm{m})$ numerically, with a real background along a Cartan generator, $A_2(x) = \zeta\, e^{\mathrm{i}p_1 x_1} + \bar{\zeta}\, e^{-\mathrm{i}p_1 x_1}$ and $A_\mu = 0$ for $\mu \neq 2$. The vacuum polarization is the coefficient of $|\zeta|^2$, per site.

- The vertices of the Wilson term come from the exact Taylor coefficients of the plaquette summand in the chart $e^{\mathrm{i}h} e^{\mathrm{i}A}$, up to order $h^2 A^2$. These were computed by Cauchy's formula on the holomorphic extension, to accuracy 10^{-10}. The vertices of $D_U D_U^T$ and $D_U^T D_U$ come from the expansion of $R(e^{\mathrm{i}A})$. In the component pairing the propagators are $2/(|\hat{q}|^2 + \mathrm{m}^2)$ for the vector field and $1/(|\hat{q}|^2 + \mathrm{m}^2)$ for the scalar field.
- The p_1^2-coefficient was taken as a second difference in p_1 at the level of the integrand, with steps 10^{-3} and $4 \cdot 10^{-3}$. Richardson extrapolation removed the bias of order $(\text{step})^2$, which was up to $4 \cdot 10^{-6}$ at the smallest m.
- The integral over $[-\pi, \pi]^4$ was computed by the trapezoidal rule with N^4 points, $N\mathrm{m} \geq 14$. For smooth periodic integrands this rule converges exponentially. Comparisons of two values of N, made for three of the masses, put the remaining errors of the values below at less than 10^{-6}.

The results are:

m	1	0.5	0.354	0.25	0.177	0.125	0.088	0.0625
$f(\mathrm{m})$	0.50315	0.67048	0.74416	0.81396	0.88141	0.94750	1.01281	1.07770

Three conclusions:

(i) *Ward identity.* $\Pi(0)$ vanishes to 10^{-9} for all m. The longitudinal part $\Pi_{11}(p_1)$ at $p_1 = 0.5$ and $\mathrm{m} = 1$ vanishes to 10^{-10}, the accuracy of the Taylor coefficients. The tadpole and bubble terms are separately of order 0.3. Since each of the two determinants is gauge invariant on its own, this checks the Wilson and gauge-fixing vertices against each other, and the ghost vertices separately. The relative weight of the ghost is checked by (ii) and (iii).

(ii) *The logarithm and its sign.* Fits of $f(\mathrm{m}) = \mathtt{b}\log(1/\mathrm{m}) + c + \mathrm{m}^2(d_2\log(1/\mathrm{m}) + e_2)$, with or without terms of order m^4, give $\mathtt{b}$ between 0.18564 and 0.18579, against 0.185756. The fits that use only $\mathrm{m} \le \frac{1}{4}$, and the fits with six parameters, give 0.18574 to 0.18577. The function f increases as $\mathrm{m} \to 0$. This is the sign of asymptotic freedom.

(iii) *The constant.* With $\mathtt{b}$ fixed, the fits give c between 0.563268 and 0.563274. Dashen and Gross, and Hasenfratz and Hasenfratz, computed the one-loop relation between the bare Wilson coupling and the $\overline{\mathrm{MS}}$ coupling [23, 22]. For $SU(N)$ it reads $\Lambda_{\overline{\mathrm{MS}}}/\Lambda_L = 38.852704\, e^{-3\pi^2/(11N^2)}$, that is, $g^{-2}_{\overline{\mathrm{MS}}}(a^{-1}) = g_0^{-2} - d_1$ with $d_1 = 0.16995599N - \frac{1}{8N}$, in the normalization $\beta = 2N/g_0^2$. Combined with the $\overline{\mathrm{MS}}$ value of the continuum term of Lemma 57.5, it predicts

$$c = \frac{d_1}{T_\rho} + \frac{C_A}{48\pi^2 T_\rho} = 0.563267 \qquad (SU(2)).$$

The second term comes from the trace over the vector index, which is $d = 4 - 2\varepsilon$ rather than 4. So the pole of the orbital term leaves a finite remainder. The agreement is to 10^{-5}.

So the lattice integral of Lemma 57.2 reproduces a known lattice constant, independently of the arguments of this section. In particular, the normalization of Lemma 57.7 is confirmed for $SU(2)$. The programs are in the folder `numerics` next to the note.

The integrand and the tensor (Remark 57.10). Two further checks are in `numerics/y5c` (`numerics/README.md`, Section 5).

(i) *The spherical average, in exact arithmetic.* The program computes $\langle \mathcal{I}_0(\omega, 0)\rangle$ in rational arithmetic, from the vertices of Lemma 57.4 and the moments $\langle \prod_\alpha \omega_\alpha^{2b_\alpha}\rangle = \prod_\alpha (2b_\alpha - 1)!!/(4\cdot 6\cdots(2n+2))$, $n = \sum_\alpha b_\alpha$, which contain (57.9). All 256 components of the four-index tensor have the form $a\,\delta_{\mu\nu}\delta_{\kappa\lambda} + b\,(\delta_{\mu\kappa}\delta_{\nu\lambda} + \delta_{\mu\lambda}\delta_{\nu\kappa})$, with $(a, b) = (\frac{1}{3}, -\frac{1}{6})\kappa_\rho$ from the scalar determinant, $(-\frac{2}{3}, \frac{1}{3})\kappa_\rho$ from the orbital part of the vector determinant, $(4, -2)\kappa_\rho$ from the spin part, and $(\frac{11}{3}, -\frac{11}{6})\kappa_\rho$ in total. So $a = -2b$ holds for each part separately, and the isotropic coefficient is $\frac{11}{3}\kappa_\rho$, as in the proof of Lemma 57.7.

(ii) *The scaling of the lattice integrand.* For $G = SU(2)$, with the lattice vertices of Remark 57.9, the program compares $\lambda^4 \mathcal{I}(\lambda\omega)$ with $\mathcal{I}_0(\omega)$, both contracted with a random real unit polarization c and a random unit direction of p, for 200 random directions ω on the closed half-sphere $\{|q|^2 + \mathrm{m}^2 = 1,\ \mathrm{m} \ge 0\}$ and the scale factor $\lambda = 2^{-j}$, $0 \le j \le 12$. The second derivative in p is a second difference with steps $10^{-3}\lambda$ and $4\cdot 10^{-3}\lambda$, with Richardson extrapolation. The largest difference over the directions is 4.3 at $\lambda = 1$ and $2.4\cdot 10^{-7}$ at $\lambda = 2^{-12}$; it decreases like $4\lambda^2$, faster than the bound $C\lambda$ of Lemma 57.5. The vertices of this test come from the Taylor coefficients of the plaquette, not from the proof of Lemma 57.4. So it checks the limits of the vertices at the level of the integrand, including the factor 2 of the spin term and the weight of the ghost, independently of the fits of Remark 57.9.

Summary (Remark 58.10(d)). The checks of Part XI, with the accuracies reached, are the following.

- Theorem 55.4 was tested on small two-dimensional lattices, where it holds to $5 \cdot 10^{-10}$ (Remark 55.7). This first test, made for a simpler averaging, checks the passage from the axial slice to the Landau slice.
- Bałaban's symmetric averaging of Part IV was transcribed to $d = 2$, with $L = 3$ (`numerics/README.md`, Section 3). Theorem 55.4 holds on the averaged axial slice with the same exact constant $-12 \log 3$, to $5 \cdot 10^{-11}$. Laplace's method with the exponential gauge fixing of Lemma 32.3 differs from Laplace's method in a tree gauge by the constant $\frac{n_\lambda}{2} \log 2$ predicted by the absence of a Faddeev–Popov determinant, to $2.5 \cdot 10^{-10}$ (Remark 34.10). The one-loop term is invariant under a rotation, a reflection and the transposition of the axes, to $2.2 \cdot 10^{-10}$, as Lemma 54.9 predicts. The algebra of the averaging, including its covariance under all 384 elements of B_4 and the exact covariance $E = 0$, was also checked in $d = 4$ on a single coarse bond; there is no Laplace computation in $d = 4$.
- $f(\mathrm{m})$ was evaluated for $SU(2)$ (Remark 57.9). It confirms the coefficient b to $2 \cdot 10^{-5}$ (to $2 \cdot 10^{-4}$ for every fit), the sign, transversality, and the known lattice constant of Dashen and Gross.
- The spherical average of the full continuum tensor was computed in exact rational arithmetic, and the scaling of the lattice integrand to the continuum integrand was checked for $SU(2)$ (Remark 57.10; `numerics/README.md`, Section 5). The first gives the coefficients $\frac{11}{3}$ and $-\frac{11}{6}$ (in units of C_A/T_ρ) of the two invariant tensors, with transversality for the scalar, orbital and spin parts separately; in the second the difference decreases like λ^2 in the scale factor λ.

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